

FANUC MT-LINK*i*

AdminTool OPERATOR'S MANUAL

NOTE

- 1 This manual describes the operation of FANUC MT-LINK*i* AdminTool (system administration software).
- 2 The following table lists the manuals related to MT-LINK*i*. These manuals are bundled into the DVD. This manual is indicated by an asterisk (*).

Manual Title	Contents	Specification Number	
FANUC MT-LINK<i>i</i>			
Startup MANUAL	This manual describes basic functions and operation procedures.	A-40827 -00001EN	
Install MANUAL	This manual describes how to install, update and uninstall MT-LINK <i>i</i> .	A-40828 -00001EN	
AdminTool OPERATOR'S MANUAL	This manual describes the operation of FANUC MT-LINK <i>i</i> AdminTool (system administration software).	A-40829 -00001EN	*
WebUI OPERATOR'S MANUAL	This manual describes the operation of WebUI.	A-40830 -00001EN	
Hardware Key Management Tools OPERATOR'S MANUAL	This manual describes the operation of Hardware Key Management Tools.	A-40871 -00001EN	
Database Schema	This manual describes the data configuration of the database in both MT-LINK <i>i</i> and MT-LINK <i>i</i> Integration Server.	A-40950 -00001EN	
Expanding Database Guide	This manual describes how to expand the databases for MT-LINK <i>i</i> and MT-LINK <i>i</i> Integration Server by using external storage.	A-41621 -00001EN	
Custom WebUI MANUAL	This manual describes how to create Custom WebUI.	A-42146 -00066EN	
Troubleshooting MANUAL	This manual describes how to address troubles which have occurred while running MT-LINK <i>i</i> / Integration Server system.	A-42146 -00187EN	
FANUC MT-LINK<i>i</i> Integration Server			
Integration Server Startup MANUAL	This manual describes basic functions and operation procedures of MT-LINK <i>i</i> Integration Server.	A-41086 -00001EN	
Integration Server Install MANUAL	This manual describes how to install, update and uninstall MT-LINK <i>i</i> Integration Server.	A-41087 -00001EN	
Integration Server AdminTool OPERATOR'S MANUAL	This manual describes the operation of FANUC MT-LINK <i>i</i> Integration Server AdminTool (system administration software).	A-41088 -00001EN	
Integration Server WebUI OPERATOR'S MANUAL	This manual describes the operation of Integration Server WebUI.	A-41089 -00001EN	

* Windows is a trademark or a registered trademark of Microsoft Corporation in the United States and other countries.

* Other company names or product names stated in this document are trademarks or registered trademarks of their respective owners.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	FANUC CORPORATION	
				Apprv.		
				Inoue	Sheet.	1/868

SAFETY PRECAUTIONS

DEFINITION OF WARNING, CAUTION, AND NOTE

This manual includes precautions preventing erroneous operation and settings for MT-LINK*i* from causing problematic impacts on it. Those precautions are classified into Warning and Caution according to severity level. In addition, supplementary information is described as Note. Read the Warnings, Cautions, and Notes thoroughly prior to using MT-LINK*i*.



WARNING

This is applied when severe impacts on system can be expected if the user fails to observe required procedures for operation or settings.



CAUTION

This is applied when light to moderate impacts on system can be expected if the user fails to observe required procedures for operation or settings.

NOTE

This is used to provide supplementary information which is not applied to Warning or Caution.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 2/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

WARNINGS AND CAUTIONS ON USING MT-LINK*i*

Following are CAUTIONs on using MT-LINK*i*.

Connection with FANUC CNC

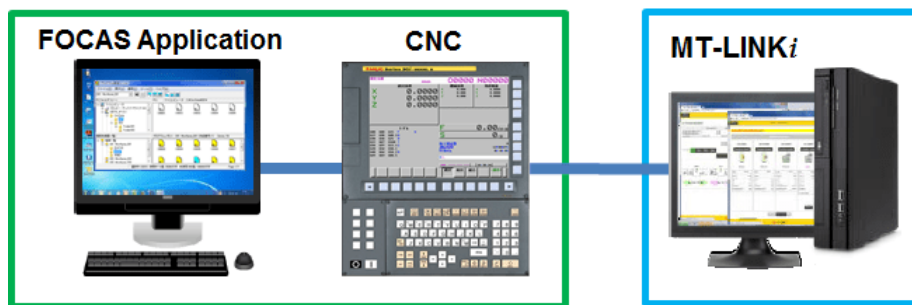
CAUTION

If the CNC which is targeted for connection is communicating with external application software using Ethernet, or using application software utilizing FOCAS function, perform sufficient testing before connecting to MT-LINK*i*. The error process of the FOCAS function may not be executed properly in the external application software, causing an error to occur on that software by connecting MT-LINK*i*.

Causes and countermeasures of troubles

Introduction of MT-LINK*i* increases the number of accesses to CNCs for their CNC information. In turn, this access increase may enforce mutual exclusion to ensure consistency in their CNC-internal information. In such a case, if customized software, such as FOCAS applications and ladder programs, running on existing equipment are not designed to properly address exclusive control, unexpected troubles may occur more likely than not on the system or machines which have been operating without troubles before the introduction of MT-LINK*i*.

Existing Equipment



Possible trouble cases:

- The existing system stops caused by an error.
- A machine stops unexpectedly.
- Erroneous machining occurs.
- Processing defects occur.

■ Causes

Introduction of MT-LINK*i* increases the number of accesses to CNCs for their CNC information. Due to this access increase, customized software may be troubled when using the following libraries to acquire CNC information:

- PC applications
FOCAS1/2 Library for FANUC CNC
- Ladder programs
CNC/PMC Data Window Library

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	3/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	4/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

Applying Updated Machine Settings

 WARNING

When updating machine settings with the [Machine Setting] window of AdminTool, the following steps must be performed:

- (1) Send updated setting data to the database.
- (2) Create Collector setting files and send them to Collector.

If updated machine settings are not sent to the database and Collector and thus they cannot reflect the updates, such as deletion or addition of signals, unexpected troubles may occur because the database and Collector will continue to refer to the older settings.

For the details, refer to the "[APPLYING SETTINGS] - APPLICATION OF SETTINGS" chapter.

Setting Macro Variable Getting Condition Definition of FanucCNC

 WARNING

MT-LINK i controls trigger macro variables using a handshaking process that waits for a CNC to turn a trigger macro variable to "1" and sets the variable to "0."

If trigger macro variables reside in areas that are used on the CNC or by other applications, make sure before setting trigger macro variables that no problem should occur.

For the details, refer to the "**Register Macro Variable Getting Condition Definition of FanucCNC**" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	5/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

Setting Variable History Retrieval Condition Definition of FanucCNC



WARNING

MT-LINK*i* controls trigger variables using a handshaking process that waits for a CNC to turn a trigger variable to "1" and sets the variable to "0."

If trigger variables reside in areas that are used on the CNC or by other applications, make sure before setting trigger variables that no problem should occur.

For the details, refer to the "[Setting Variable History Retrieval Condition Definition of FanucCNC](#)" subsection.

System Setting Files



WARNING

- 1 System setting files may have non-editable items. Items are non-editable if marked with "NO" in the [Edit] column of their list. Do not modify or delete these items.
- 2 If an invalid value is set on the system setting file, MT-LINK*i* system will not be able to start.
Before editing the file, copy the original system setting file and save the copied file with a different file name as a backup.

For the details, refer to the "[SYSTEM SETTING FILES](#)" section.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	6/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

Influence Caused by Changing the System Date and Time

CAUTION

If, while MT-LINK*i* is in operation, the system date and time for the Server PC is set backward, the following problems may occur.

To avoid those problems, make sure before starting MT-LINK*i* that the system date and time is set to the current date and time.

- (1) Stopping MongoDB service after the system date and time has been set backward may prompt the wrong message, "Windows could not stop MongoDB service on Local Computer," though the service has actually stopped.
- (2) If MongoDB service or Server PC is restarted in the period after the system date/time has been set backward and before the newly-set system date/time reaches the time at which the change was made, inconsistency may occur in the database.
- (3) If, during the period described in (2), neither MongoDB service nor Server PC is restarted, no data will be stored in the database. However the database will start storing after the newly-set system date/time reaches the time at which the change was made.
- (4) If the inconsistency described above in (2) occurs in the database for an MT-LINK*i* Server PC which is managed by Integration Server, the Integration Server may malfunction as its database reflects the inconsistency.

Stopping MT-LINK*i*

CAUTION

When turning off the power of Server PC, make sure that the system is shut down before doing so.

For the details, refer to the "[Start/Stop System](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION		Sheet. 7/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

Machine Setting for FANUC Robot Controller



CAUTION

For FANUC Robot controllers for the following FANUC Robot Series robots, if the machine setting is set by using duplicated IP address/host name (if the IP address/host name already in use is being used for a new machine setting), all the duplicated connections will be disconnected.

- R-30iA / R-30iA Mate
- R-J3iB / R-J3iB Mate

For the details, refer to the "[Set Protocol Setting of FanucRobot](#)" subsection.

Setting Variable History Retrieval Condition Definition of FanucRobot



WARNING

When using handshaking method, MT-LINK*i* waits for a FanucRobot type machine to turn a trigger variable to the specified value and then sets the variable to "0."

If trigger variables reside in areas that are used on the FanucRobot type machine or by other applications, make sure before setting trigger variables that no problem should occur.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	8/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

SERVO VIEWER Linkage

CAUTION

For the collection of servo data from a single unit of machine (CNC), only one method of collection can be executed at the same time.

When MT-LINK*i* is collecting servo data from a single unit of machine (CNC), other methods to collect servo data cannot be used. If you attempt to collect data by using two or more methods at the same time, the measurement commands of the methods interfere with each other, causing failure in the normal retrieval of the servo data.

Therefore, the following applications cannot be used simultaneously with the collection of servo data by MT-LINK*i*. When executing automatic collection of servo data by MT-LINK*i*, such applications as shown below should not be running at the same time.

- SERVO VIEWER (PC version)
- SERVO GUIDE (PC version)
- SERVO VIEWER (iHMI version)
- SERVO GUIDE MATE (CNC screen)
- Application for C language executor using high-speed sampling function for servo/spindle data
- MTB application using high-speed sampling FOCAS function for servo/spindle data
- On-machine measurement system using continuous position data output function (CNC option A02B-xxxx-J989)
(Example: Renishaw plc, SPRINT™ system)

When using the above applications, make sure that the MT-LINK*i* is stopped, or either disable or temporarily stop the SERVO VIEWER linkage functionality.

For the details on how to stop the MT-LINK_i system, refer to the "**Start/Stop System**" subsection.

For details on how to stop SERVO VIEWER linkage service (MT-LINK*i* Collector SERVO VIEWER Linkage), refer to the "**Start/Stop System**" subsection.

For details on how to disable the SERVO VIEWER linkage functionality, refer to the "Deleting Measurement Data and Disabling SERVO VIEWER Linkage Function" subsection of the "FANUC MT-LINK*i* Startup MANUAL."

For details on how to temporarily stop the SERVO VIEWER linkage functionality, refer to the "Stop Using SERVO VIEWER Linkage Function Temporarily" subsection of the "FANUC MT-LINK*i* Startup MANUAL."

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description						FANUC CORPORATION		Sheet.	9/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

LIMITATIONS ON USING MT-LINK*i*

Following are limitations on using MT-LINK*i*.

FANUC CNC FOCAS Client

- MT-LINK*i* connects to each CNC via Ethernet, and operates as a FOCAS2 / Ethernet client.

Following are processes and services that operate as a FOCAS2 / Ethernet client:

No.	Number of connections	Process / Service (Type)	Function	Connection	Length of occupying FOCAS tasks
1	1	MT-LINKiCollector (service)	Collecting signals from FanucCNC / FanucRobodrill	Always	Always
2			AdminTool Machine Connection Test*1	During execution	Seconds to minutes
3	1	MT-LINKiCollectorSERVO VIEWERLinkage (service)	SERVO VIEWER linkage	During execution	During execution by schedule
4	1	MT-LINKiFTPServer*2 (service)	WebUI File Transfer	During execution	-
5			AdminTool Machine Data Storing Schedule	During execution	-
6			WebUI Maintenance Data Aggregation	During execution	-
7	1	AdminTool (process)	AdminTool Easy Setting	During execution	Seconds to minutes
8			AdminTool Machine Setting - Signal Check	During execution	Seconds to minutes
9			AdminTool Machine Setting - Retrieve the Number of Tool Groups	During execution	Seconds to minutes
10	1	SignalCheckTool (process)	Signal Check Tool	During execution	Seconds to minutes

*1: Connectivity will be checked through MT-LINKiCollector service.

*2: After connected with a machine, the client always occupies FOCAS tasks. To release it, stop FTPServer service on each Collector Server by using Start/Stop System functionality of AdminTool.

*3: Connections will be shared between No. 1 and No. 2, between No. 4, No. 5, and No. 6, and between No. 7 and No. 9.

(Example: Even when the services listed in No.1 and No.2 rows run at the same time, each of the numbers of connections with CNCs will be 1.)

As shown below, there is a limitation in the connectable number of FOCAS2 / Ethernet clients for each CNC.

	Fast Ethernet Board	Ethernet Board	Embedded Ethernet*5
Number of connectable clients (simultaneous)	20 in maximum	10 in maximum*4	5 in maximum

*4: In case of 6561/03 or 6561/04 (for Series 16*i* /18*i* /21*i*, Power Mate *i*), 3 in maximum.

*5: Communication using the embedded Ethernet is handled by the CPU of the CNC. Due to this, there is a possibility that the operating state of the CNC affects the performance of communication of the embedded Ethernet, or vice versa.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	10/868	
Ed.	Date	Design	Description									
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.						Inoue

- The embedded Ethernet operates on a lower priority compared to such operations as automatic operation process or manual operation, which is executed per axis. In case of automatic operation or when there are many axes, the communication rate may become slow.
- The embedded Ethernet operates on a higher priority compared to such operations as screen display process of the CNC, C language executor (excluding high level tasks), or macro executor (excluding execution macro). Due to this, using the embedded Ethernet for communication may lower the performance of these processes.

Check how many FOCAS2 / Ethernet clients are used for other software applications, and then select an appropriate Ethernet device for it.

There are cases when Fast Ethernet / Ethernet is required depending on the conditions of usage.

2 The maximum numbers of paths / servo axes /spindles*⁶ that can be collected on MT-LINK*i* are as follows:

- Number of paths: 15 (maximum number of paths for system)
- Number of feed axes: 96 (maximum number of controlled servo axes for system)
- Number of spindles: 24 (maximum number of controlled spindles for system)

*6: The numbers may differ by system, even for the same type of CNC.

Setting for Collection of Signals

MT-LINK*i* collects and stores signals each as a document into the database when it detects a change in value of those signals.

A large amount of documents will be stored in the database if signals which can frequently change their values, such as AbsPos, RelPos, and McnPos for FanucCNC type machines, should be set for signal collection at 500 msec collection interval. In such a case, a long running of MT-LINK*i* could cause a huge increase in the size of the database.

Prior to starting signal setting, take enough time to check that you should not select unnecessary signals and that you should not specify collection intervals that can cause an increase in the size of the database.

Before deleting data in the database, or conducting maintenance of it, refer to the following subsections:

- [DB Data Retention Period Setting](#)
- [DB Cleanup](#)
- [DB Optimization](#)

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	11/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

Software to Use

- Install the following version of MongoDB (the database) bundled with MT-LINK*i*. MT-LINK*i* may malfunction when another version of MongoDB runs with it.

Software	Version
MongoDB	4.0.25

MongoDB will be installed upon the installation of MT-LINK*i*.

- Install the following version of Node.js (Web Server) bundled with MT-LINK*i*. MT-LINK*i* may malfunction when another version of Node.js runs with it.

Software	Version
Node.js	16.13.2

For the details on how to install Node.js, refer to the "INSTALLATION OF REQUIRED SOFTWARE" chapter of the "FANUC MT-LINK*i* Install MANUAL."

- Install the following version of .NET Framework bundled with MT-LINK*i*.

Software	Version
.NET Framework	4.7.2

For the details on how to install .NET Framework, refer to the "INSTALLATION OF REQUIRED SOFTWARE" chapter of the "FANUC MT-LINK*i* Install MANUAL."

- Install the following license related software that are bundled with MT-LINK*i*, which are used with software key licenses for MT-LINK*i*.

Software	Version
FANUC License Manager	1.00
CodeMeter Runtime kit	7.40

For the details on how to install FANUC License Manager and CodeMeter Runtime kit, refer to the "INSTALLATION OF REQUIRED SOFTWARE" chapter of the "FANUC MT-LINK*i* Install MANUAL."

- When creating/editing Excel or CSV format files (such as report templates) for MT-LINK*i*, use the following software.

Software	Version
LibreOffice Calc	7.3.7

- When using the SERVO VIEWER linkage function, use the following software.

Software	Version
SERVO VIEWER	Edition 03.50 or later
Linkage tool for MT-LINK <i>i</i>	V2.4 (* bundled with SERVO VIEWER listed above)

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description						FANUC CORPORATION		Sheet.	12/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

Prohibited items when using software key license in a virtual machine environment (VM)

When using the software key license in a virtual machine environment (VM), do not perform operations similar to copying the virtual machine environment, such as backing up and restoring the virtual machine environment (ex. creating/restoring snapshots) while the license is activated.

If the above operation is performed, unauthorized use of the license will be detected and the software license will be invalidated.

When the above operation is required, such as moving the virtual machine environment, please deactivate the license once and re-activate after moving.

If the above operation invalidates the license, you will need to purchase the license again for a fee to restore the license.

Influence of System Workload

Due to the workload on the system, the following events may occur.

(1) Server PC (Receiver)

- The rate of operation will slow down, if the usage rate of CPU becomes high or free memory space is insufficient.
- A delay occurs in the storing of the collected data, and when the delayed amount of data reaches a certain level, some of the data will not be stored in the database.
- If the free space on the disk is under 5GB, the Receiver will be stopped.
- If the free disk space of the disk drive specified as the output target is under 5GB, the output process will be suspended.
- When there is overload in work due to the access process of MongoDB with the usage of an application developed originally by the user, Receiver stops, and the collected data cannot be stored to the database.

(2) Collector PC (Collector)

- The rate of operation will slow down, if the usage rate of CPU becomes high or free memory space is insufficient.
- The Collector stops due to the delay in the data processing, and the collected data cannot be stored in the database.
- The interval for obtaining the signal data becomes longer than 500 msec.

By utilizing the [Utility - System Diagnosis] screen, the trend of the workload on the system can be grasped.

Mail Delivery Function

The mail delivery function provided by MT-LINK*i* does not guarantee real time and fully covered notification of the events (alarm occurrence, detection of value surpassing threshold, etc.) due to the following reasons.

- If the mail client is based on POP protocol, the mail receipt interval may become longer.
- As anti-spam measures, mail delivery requests will be subjected to flow control.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	13/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

Confirmation of Operation when UNC Path Is Specified as Output Path

As for the operation in the case when UNC path is specified as the output path of report, signal data, or CSV file, it has been confirmed that the operation is executed in a normal manner with the following combinations.

(Example: Output source Windows 10 => Output destination Windows Server 2019)

This function may not be executed properly if the combination is not stated below. Make sure to perform sufficient testing before using such a combination.

OS of Output Source (Server PC)	OS of Output Destination (UNC path)
Windows 10	Windows 10
Windows Server 2016	Windows Server 2016
Windows Server 2019	Windows Server 2019
Windows 11	Windows 11
Windows Server 2022	Windows Server 2022
	QTS 4.2.2 (Linux based OS for NAS by QNAP) (NOTE)

NOTE

When using NAS with Linux based OS, use samba V2.0 or later.

Name Resolution by Host Name

The host name of TCP/IP can be specified in the [IP Address/Host Name] entry column of Collector, Receiver, and machine.

When using the host name, an environment is required which enables the name resolution by the host name in the following order.

- 1 Name resolution by hosts file
- 2 Name resolution by DNS

However, name resolution by NetBIOS name (computer name) is not supported. NetBIOS name cannot be specified in the [IP Address/Host Name] entry column.

NOTE

When the IP address is changed for the host name that will be resolved by the name, the communication between the Receiver and the Collector may be disabled. In order to update the IP address of the Collector registered in the Receiver, restart the Receiver. For the details, refer to the "**START/STOP SYSTEM**" section

[IP Address/Host Name] entry locations are shown below.

Setting Target for IP Address/Host Name	Setting Location in AdminTool
Collector	Machine Setting (Collector Setting) Import (Input file: Machine Setting – Collector Setting)
Receiver	Machine Setting (Receiver Setting) Import (Input file: Machine Setting – Receiver Setting)
Machine (FanucCNC)	Machine Setting (Protocol Setting) Import (Input file: Machine Setting – Network Setting)
Machine (FanucRobot)	Machine Setting (Protocol Setting)
Machine (MTConnectAgent)	Machine Setting (Machine Selection)
MongoDB	System Setting (DB Connecting String)
Integration Server	Integration Server Setting (Add/Edit Integration Server)
LDAP Server	LDAP Authentication Setting (LDAP Server Setting)

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 14/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

About the Entry of IP Address

When entering the IP address in the [IP Address/Host Name] entry column for the Collector, the Receiver, or the machine, do not enter "0" (zero) as the first number in each part of the IP address delimited by "." (dot).

When a number starting with "0" is entered, it will be converted into an octal number, and it may not be recognized as the intended IP address.

■ Examples

Setting Value of IP Address	Recognized Value of IP Address (in Decimal Number)	Remarks
192.168.001.001	192.168.1.1	
192.168.001.008	Invalid IP address	"008" does not exist as an octal number
192.168.001.010	192.168.1.8	Octal number "010" is "8" in decimal number.

Inconsistency between Settings of AdminTool and Data Displayed on the Web Screen

In the Web screen, there are items that are displayed based on the current settings of AdminTool. By changing the settings of AdminTool, there may be inconsistencies between the data acquired before the change and the settings (such as Web names) after the change. Also, there may be inconsistencies with the downloaded CSV files. When changing Web names, please consider the above inconsistency, or add a new setting to fulfill your needs.

■ Example 1: [Diagnosis - Macro Variable History] screen

When changing the macro number retrieved and the Web name of the registered MacroID, and displayed the data retrieved on the [Diagnosis - Macro Variable History] screen, specifying a period including the data of both before and after the change

<Settings of AdminTool>

- Before change

MacroID (Web name): Get temperature

Macro number retrieved: 505 (Variable 1)

Macro number retrieved: 506 (Variable 2)

Web name: Temperature-1

Web name: Temperature-2

- After change

MacroID (Web name): Get velocity

Macro number retrieved: 510 (Variable 1)

Macro number retrieved: 511 (Variable 2)

Web name: Velocity-1

Web name: Velocity-2

<Web screen display>

- Before change

MacroID Name:

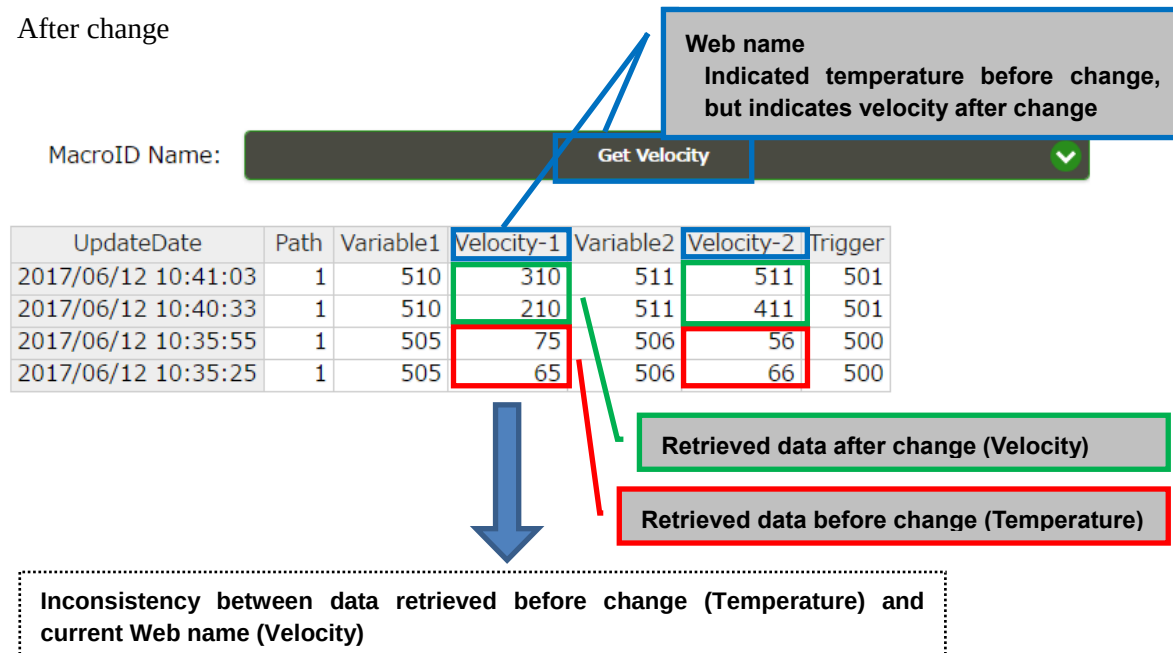
Get Temperature



UpdateDate	Path	Variable1	Temperature-1	Variable2	Temperature-2	Trigger
2017/06/12 10:35:55	1	505	75	506	56	500
2017/06/12 10:35:25	1	505	65	506	66	500

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 15/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

- After change

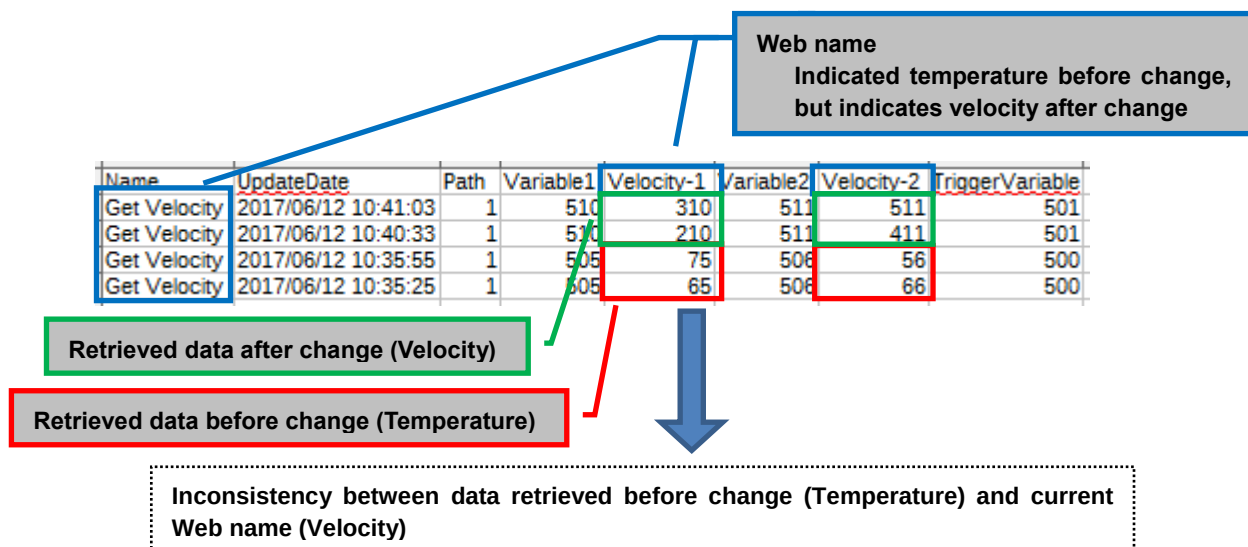


<Download CSV>

- Before change

Name	UpdateDate	Path	Variable1	Temperature-1	Variable2	Temperature-2	TriggerVariable
Get Temperature	2017/06/12 10:35:55	1	505	75	506	56	500
Get Temperature	2017/06/12 10:35:25	1	505	65	506	66	500

- After change



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	16/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

TABLE OF CONTENTS

SAFETY PRECAUTIONS.....	2
DEFINITION OF WARNING, CAUTION, AND NOTE	2
WARNINGS AND CAUTIONS ON USING MT-LINK <i>i</i>	3
Connection with FANUC CNC	3
Applying Updated Machine Settings.....	5
Setting Macro Variable Getting Condition Definition of FanucCNC	5
Setting Variable History Retrieval Condition Definition of FanucCNC	6
System Setting Files.....	6
Influence Caused by Changing the System Date and Time.....	7
Stopping MT-LINK <i>i</i>	7
Machine Setting for FANUC Robot Controller.....	8
Setting Variable History Retrieval Condition Definition of FanucRobot.....	8
SERVO VIEWER Linkage.....	9
LIMITATIONS ON USING MT-LINK <i>i</i>	10
FANUC CNC FOCAS Client.....	10
Setting for Collection of Signals.....	11
Software to Use.....	12
Prohibited items when using software key license in a virtual machine environment (VM)	13
Influence of System Workload	13
Mail Delivery Function.....	13
Confirmation of Operation when UNC Path Is Specified as Output Path	14
Name Resolution by Host Name	14
About the Entry of IP Address.....	15
Inconsistency between Settings of AdminTool and Data Displayed on the Web Screen.....	15
1 INTRODUCTION	25
2 BASIC OPERATION	28
2.1 HOW TO START AND END	28
2.1.1 How to Start/End.....	28
2.1.2 AdminTool Menu Window	35
2.2 CONFIGURATION STEPS.....	41
2.3 ABOUT [!] MARK TO INDICATE INPUT ERROR.....	43
3 [SETTING DATA] - EASY OPERATION.....	44
3.1 EASY SETTING.....	45
3.1.1 Overview of Easy Setting.....	45
3.1.2 Steps for Easy Setting.....	47
3.1.3 Easy Setting Window	48
3.1.4 Acquire Machine Information	52
3.1.5 Start Easy Setting	60
3.1.5.1 Descriptions of the Settings Created by Easy Setting.....	64
3.2 EASY APPLYING SETTINGS	65
4 [SETTING DATA] - SPECIFIC DATA SETTING	69
4.1 MACHINE SETTING.....	71
4.1.1 Steps for Machine Setting	75
4.1.2 Add Collector	75

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				Draw	A-40829-00001EN				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7									
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	FANUC CORPORATION			Sheet.	17/868

4.1.3	Detailed Setting of Machine.....	79
4.1.4	Set Outlier Detection of Signal.....	80
4.1.5	Narrow down Signal List.....	83
4.1.6	Checking the Signals	85
4.1.6.1	Signal Check per Machine.....	87
4.1.6.2	Signal Check per Collector	88
4.1.7	Set External Links	90
4.1.8	Copy/Cut/Paste/Delete Machine Setting	93
4.1.9	Import Machine Setting.....	96
4.1.10	Save Machine Setting.....	99
4.1.11	Apply Machine Setting.....	99
4.1.12	Referenced Error	100
4.2	MACHINE SETTING - FanucCNC.....	101
4.2.1	Steps for the Machine Setting of FanucCNC	101
4.2.2	Add FanucCNC	102
4.2.3	Confirm/Edit Setting of FanucCNC	104
4.2.4	Set Protocol Setting of FanucCNC.....	105
4.2.5	Set Signal of FanucCNC	108
4.2.6	Set Custom Signal of FanucCNC	110
4.2.6.1	Add Custom Signal.....	113
4.2.6.2	Edit Custom Signal	118
4.2.7	Set Tool Life Signal of FanucCNC	123
4.2.7.1	Set Numbers of Tool Groups.....	125
4.2.7.2	Retrieve Path-wise Numbers of Tool Groups	127
4.2.7.3	Messages on Failure to Retrieve Path-wise Numbers of Tool Groups	128
4.2.8	Set Laser Control Data Signal of FanucCNC.....	129
4.2.9	Register Macro Variable Getting Condition Definition of FanucCNC	131
4.2.9.1	Setting Macro Variable Getting Condition Definition of FanucCNC.....	132
4.2.9.2	Setting File Output Function When Retrieving Macro Variable of FanucCNC	139
4.2.9.3	Format of Output File When Retrieving Macro Variable of FanucCNC.....	140
4.2.9.4	If File Output Fails When Retrieving Macro Variable of FanucCNC	141
4.2.10	Setting Variable History Retrieval Condition Definition of FanucCNC.....	142
4.2.10.1	Setting Variable History Retrieval Condition of FanucCNC.....	143
4.2.10.2	Setting File Output Function When Retrieving Variable History of FanucCNC	154
4.2.10.3	Format of Output File When Retrieving Variable History of FanucCNC	155
4.2.10.4	If File Output Fails When Retrieving Variable History of FanucCNC.....	157
4.3	MACHINE SETTING - OPCUAServer	158
4.3.1	Steps for the Machine Setting of OPCUAServer	158
4.3.2	Add OPC UA Server	159
4.3.3	Select a Machine Connected to the OPCUAServer.....	160
4.3.4	Confirm/Edit Setting of OPCUAServer	165
4.3.5	Edit Endpoint of OPCUAServer	166
4.3.6	Set Protocol Setting of OPCUAServer.....	168
4.3.7	Set Signal of OPCUAServer	169
4.3.7.1	Set OPCUAServer Tags as Signals	171
4.3.7.2	Set OPCUAServer Methods as Signals	176
4.3.8	Set Alarm of OPCUAServer	182
4.3.8.1	Set OPCUAServer DA Alarms.....	182
4.4	MACHINE SETTING - FanucRobot.....	198
4.4.1	Steps for the Machine Setting of FanucRobot.....	198
4.4.2	Add FanucRobot.....	198
4.4.3	Confirm/Edit Setting of FanucRobot.....	200
4.4.4	Set Protocol Setting of FanucRobot	201

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue
FANUC CORPORATION				Sheet.	18/868		

4.10.2	Preparing and Reading Product Plan Data	316
4.10.3	Updating Product Plan Data	317
4.10.4	Preparing and Reading Work Comments Data.....	317
4.10.5	Reading Manual Data	318
4.10.6	Exporting Product Plan Data	320
4.10.7	Exporting Work Comments Data	321
4.10.8	Exporting Manual Data	322
4.10.9	Deleting Product Plan Data	324
4.10.10	Deleting Work Comments Data	325
4.10.11	Deleting Manual Data	326
4.10.12	File Format for Manual Data	328
4.10.13	Registration Conditions for Product Plan Data	330
4.10.14	Updating Conditions for Product Plan Data	331
4.10.15	Registration Conditions for Work Comments Data	333
4.10.16	Errors on Reading Manual Data File.....	334
4.10.17	Errors during Exporting of Manual Data File	335
4.10.18	Errors during Deletion of Manual Data File.....	335
4.11	AUTHORITY SETTING	336
4.11.1	Steps for Authority Setting.....	337
4.11.2	Add/Edit Role of Authority	338
4.11.3	Add/Edit User.....	341
4.11.4	Set LDAP Authentication.....	344
	4.11.4.1 Setting Example for Active Directory	349
	4.11.4.2 Setting Example for OpenLDAP	353
4.12	SIGN-IN SETTING.....	356
4.12.1	Set Title Image	356
4.12.2	Set Title Memo	357
4.13	ASSET MANAGEMENT	359
4.13.1	Steps for Setting Asset Management Information.....	362
4.13.2	Steps for Setting Basic Information of Asset Management	363
4.13.3	Setting Image of the Machine.....	363
4.13.4	Setting Asset Management Items	365
4.13.5	Importing / Exporting Asset Management Items	367
	4.13.5.1 Importing Asset Management Items.....	367
	4.13.5.2 Exporting Asset Management Items.....	370
	4.13.5.3 Input / Output File Format of Asset Management Items	372
4.14	POWER CONSUMPTION SETTING	374
4.14.1	Setting CO2 Emissions and Power Consumption for External Devices	376
	4.14.1.1 Add/Edit Device	378

5 [SETTING DATA] - SETTING CHECK.....380

5.1	CHECK MACHINE SETTING	381
5.1.1	Check Connection with the Machine	381
5.1.2	Check Results for Collector.....	384
5.1.3	Check Results for FanucCNC Type	386
5.1.4	Check Results for OPCUAServer Type	392
5.1.5	Check Results for FanucRobot Type.....	393
5.1.6	Check Results for MTConnectAgent Type	396
5.1.7	Check Results for FanucRobodrill Type	398
5.2	DATA VALIDATION.....	399

6 [APPLYING SETTINGS] - APPLICATION OF SETTINGS401

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	20/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

6.1	SEND TO DB.....	402
6.1.1	Send Setting Data to DB.....	403
6.1.2	Recognize Transferred Item	406
6.2	CREATE COLLECTOR SETTING FILES	407
6.2.1	Apply Created AdminTool Setting to Collector.....	407
7	[CONTROLLING SYSTEM] - SYSTEM CONTROL	413
7.1	START/STOP SYSTEM	414
7.1.1	Start/Stop System	414
7.1.2	Start Services Manually.....	417
7.1.3	Stop Services Manually	418
7.1.4	Disable Automatic Start of Services.....	419
8	[SETTING SYSTEM FUNCTIONS] - SYSTEM SETTING	421
8.1	SYSTEM SETTING	421
8.1.1	Change Language Used in MT-LINK <i>i</i>	423
8.1.2	Change Language for Web Names	424
8.1.3	Database Setting	427
8.1.3.1	Settings Required for Connecting to the Database	427
8.1.3.2	Data Storing Setting of Database	429
8.1.3.3	DB Data Retention Period Setting	437
8.1.3.4	DB Cleanup	444
8.1.3.5	DB Optimization.....	445
8.1.4	Set Port Number of Web Server	448
8.1.5	Show Footer Text and Set Background Color of Header/Footer	449
8.1.6	Set Linkage with Other Systems	451
8.1.7	Report Path Setting.....	453
8.1.8	Setting File Output Path of Macro Variable History	454
8.1.9	Setting File Output Path of Variable History	456
8.1.10	Setting File Output Path of Product Plan Setting	457
8.1.11	Display Order Setting of Operational Results Information Display Area	459
8.1.12	Load Additional Language Resource	460
8.1.13	Reset All Values to Default.....	464
9	[SETTING SYSTEM FUNCTIONS] - BATCH SETTING	466
9.1	EXPORT	467
9.1.1	Exporting Machine Setting Information.....	468
9.1.2	Export File of Export Function.....	469
9.2	IMPORT	487
9.2.1	Configuring Input File.....	490
9.2.2	Importing Machine Setting Information.....	500
9.2.3	Settings after Importing Input File	503
9.2.4	Editing Pattern Setting File	504
10	[SETTING SYSTEM FUNCTIONS] - OUTPUT SCHEDULE SETTING.....	521
10.1	REPORT SCHEDULE SETTING	522
10.1.1	PREPARE REPORT TEMPLATE	524
10.1.1.1	Use Default Report Template	526
10.1.1.2	Customize the Report Template	530
10.1.1.3	Template Filename and Destination Folder	531
10.1.1.4	About Report Keyword.....	532
10.1.1.5	About the Prefixes	533
10.1.1.6	About the Suffixes	535

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	21/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

	10.1.1.7	About the Counts Obtained	536
	10.1.1.8	About FULL_TIME	537
	10.1.1.9	About the Template Samples	538
	10.1.2	Steps for Report Schedule Setting	542
	10.1.3	Scheduling Daily Report	543
	10.1.4	Scheduling Weekly Report.....	546
	10.1.5	Scheduling Monthly Report	548
	10.1.6	Steps for Manual Output	550
	10.1.7	Manual Output Daily Report.....	553
	10.1.8	Manual Output Weekly Report	554
	10.1.9	Manual Output Monthly Report	555
	10.1.10	Report Output File.....	557
	10.1.11	When Report Output Fails.....	558
	10.1.12	Customizing the Report Schedule Function	558
10.2		MACHINE DATA STORING SCHEDULE	559
	10.2.1	Steps for Registering the Machine Data Storing Schedule.....	566
	10.2.2	Steps for Storing Machine Data Using Manual Output.....	577
	10.2.3	When Failing to Store the Machine Data	581
10.3		SIGNAL DATA OUTPUT	582
	10.3.1	Creating and Registering the Output Target Data of Signal Data Output	583
	10.3.2	Unregistering the Registered Output Target Data of Signal Data Output	589
	10.3.3	Registering the Output Schedule of Signal Data Output.....	590
	10.3.4	Unregistering the Registered Schedule of Signal Data Output	596
	10.3.5	Manual Output of Signal Data.....	598
	10.3.6	Output File Format of Signal Data Output	602
	10.3.7	When Signal Data Output was Unsuccessful	604
	10.3.8	Customizing the Signal Data Output Function.....	605
11		[SETTING SYSTEM FUNCTIONS] - MAIL SETTING	606
	11.1	MAIL FUNCTION SETTING	607
	11.2	SIGNAL DETECTION MAIL DELIVERY SETTING	612
	11.2.1	Steps for Setting Signal Detection Mail Delivery	613
	11.2.2	Adding Delivery Condition for Signal Detection Mail Delivery	613
	11.2.3	Deleting Delivery Condition of Signal Detection Mail Delivery	618
	11.2.4	Examples of Delivery Condition Settings for Signal Detection Mail Delivery ...	619
	11.2.5	Mail Format for Signal Detection Mail Delivery	621
	11.3	ALARM MAIL DELIVERY SETTING	624
	11.3.1	Steps for Setting Alarm Mail Delivery.....	625
	11.3.2	Adding Delivery Condition for Alarm Mail Delivery.....	625
	11.3.3	Deleting Delivery Condition of Alarm Mail Delivery	629
	11.3.4	Examples of Delivery Condition Settings for Alarm Mail Delivery	630
	11.3.5	Mail Format for Alarm Mail Delivery.....	633
12		[SETTING SYSTEM FUNCTIONS] - INTEGRATION SERVER SETTING.....	636
	12.1	INTEGRATION SERVER SETTING	636
	12.1.1	Steps for Integration Server Setting	637
	12.1.2	Add, Edit, or Delete Integration Server.....	638
	12.1.3	Check Connection with the Integration Server.....	642
	12.1.4	Register Facility Information to the Integration Server.....	645
13		[BACKING UP SETTINGS] - SETTINGS BACKUP.....	648

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION				Sheet. 22/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

13.1	BACKUP SETTINGS	649
13.1.1	Backup Setting Data	649
13.2	RESTORE SETTINGS	651
13.2.1	Restore Backup Setting	652
13.3	RECEIVE FROM DB	657
13.3.1	Receive Setting Data from DB	658
14	OTHERS	660
14.1	LIST OF FanuCNC SIGNALS	660
14.1.1	Basic Signals of FanuCNC	660
14.1.2	Custom Signals of FanuCNC	682
14.1.3	Tool Life Signals of FanuCNC	686
14.1.4	Laser Control Data Signals of FanuCNC	689
14.2	LIST OF OPCUAServer SIGNALS	702
14.2.1	Basic Signals of OPCUAServer	702
14.3	LIST OF FanucRobot SIGNALS	703
14.3.1	Basic Signals of FanucRobot	703
14.3.2	Custom Signals of FanucRobot	704
14.4	LIST OF MTConnectAgent SIGNALS	707
14.4.1	Basic Signals of MTConnectAgent	707
14.5	LIST OF FanucRobodrill SIGNALS	708
14.5.1	Basic Signals of FanucRobodrill	709
14.5.2	Custom Signals of FanucRobodrill	711
14.5.3	ROBODRILL Specific Signal	712
14.5.3.1	Robodrill Signals of FanucRobodrill	712
14.5.3.2	Tool Life Signals of FanucRobodrill	715
14.5.3.3	Periodic Maintenance Signals of FanucRobodrill	717
14.6	OPC UA CLIENT FUNCTIONALITY	723
14.6.1	OPC UA Summary	723
14.6.2	System Configuration with OPC UA Client Functionality	723
14.6.3	OPC UA Data Types	726
14.6.4	OPC UA Supported Information Models	727
14.6.5	OPC UA Supported Security Models	727
14.6.6	Local Discovery Server	728
14.6.7	Ethernet I/O Converter	729
14.7	MTConnect CLIENT FUNCTIONALITY	734
14.7.1	MTConnect Summary	734
14.7.2	System Configuration Using MTConnect Client Functionality	734
14.8	SERVO VIEWER LINKAGE	738
14.8.1	SERVO VIEWER Linkage Summary	739
14.8.2	Measure Sequence of SERVO VIEWER Linkage	740
14.8.3	MT-LINKi Settings Required for SERVO VIEWER Linkage	744
14.9	SYSTEM DIAGNOSIS	745
14.9.1	System Diagnosis Summary	745
14.9.2	Configuration of System Diagnosis	746
14.9.3	Utility - System Diagnosis Screen	747
14.9.3.1	Utility - System Diagnosis Screen (System Diagnosis Information)	747
14.9.3.2	Utility - System Diagnosis (Collector) Screen	751
14.9.4	CSV File Output of System Diagnosis Information	753
14.9.5	Operation of System Diagnosis and Countermeasures against System Anomaly	756
14.10	CUSTOMIZING THE WEB SCREEN	757

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	23/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

14.11	BACKING UP AND RESTORING DATABASE.....	759
14.11.1	Backing up Database	759
14.11.2	Restoring Database.....	760
14.11.2.1	Restoring It to the Same Server PC as It Was Backed up from	760
14.11.2.2	Restoring It to a Different Server PC.....	760
14.12	LDAP AUTHENTICATION	761
14.12.1	Overview on LDAP Authentication	761
14.12.2	Settings for MT-LINKi to Activate LDAP Authentication.....	762
14.12.3	Operational Flow of Using LDAP Authentication Functionality.....	763
14.12.4	How Sign-in from Web Browser Is Authenticated	765
14.13	COLLECT PRODUCTION RESULTS/PRODUCT NAME AUTOMATICALLY.....	767
14.13.1	Assigning CNC Parameter and Program Comment Signal to Basic Signals	769
14.13.2	Assigning FanucCNC / FanucRobodrill Custom Signal to ProductResultNumber	771
14.13.3	Assigning FanucCNC / FanucRobodrill Signal to ProductName.....	775
14.14	DISPLAY OF PRODUCTION RESULTS IN PRODUCTION RESULTS SCREEN	777
14.14.1	Calculation of Production Results	778
14.14.2	Display Example of Production Results	779
14.14.2.1	When Product was Changed during Operation.....	780
14.14.2.2	When Product Was Not Changed While Communication Was Disconnected ..	783
14.14.2.3	When Product Was Changed while Communication Was Disconnected	786
15	APPENDIX.....	789
15.1	SYSTEM SETTING FILES	789
15.1.1	AdminTool.ini	790
15.1.2	receiver.ini	796
15.1.3	collector.ini.....	801
15.1.4	ServoViewer.ini.....	807
15.1.5	FTPServer.ini	811
15.1.6	Setting.js.....	813
15.2	PORT NUMBERS USED IN MT-LINKi/INTEGRATION SERVER	834
15.2.1	System Configuration.....	834
15.2.2	List of Port Numbers in Use.....	835
15.2.3	Windows Firewall Setting	837
15.3	SIGNAL CHECK TOOL	846
15.3.1	Overview of Signal Check Tool	846
15.3.2	Functionalities of Signal Check Tool	847
15.3.3	Installation Folder of Signal Check Tool	848
15.3.4	Steps for Operating Signal Check Tool.....	848
15.3.5	Signal Check Tool Window	850
15.3.5.1	Details of Path Information Window	854
15.3.5.2	Details of Signal Setting Window	855
15.3.6	Machine Setting Import Process.....	859
15.3.7	Acquisition of Machine Information.....	860
15.3.8	Check Signal.....	865
15.3.9	Machine Setting Export.....	868

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 24/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

1

INTRODUCTION

This manual describes how to setup MT-LINK*i* by AdminTool.

In this manual, the following abbreviations may be used:

- AdminTool as short for MT-LINK*i* AdminTool
- Collector as short for MT-LINK*i* Collector
- Receiver as short for MT-LINK*i* Receiver

■ Abbreviations of CNC models

This document refers to FANUC CNC models as "Series 30*i*," "Series 16*i*," or "Series 15*i*":

(1) Series 30*i*

30*i* /31*i* /32*i*-B, 30*i* /31*i* /32*i*-B Plus, 31*i*-B5, 31*i*-B5 Plus, 35*i*-B, PM*i*-A, 0*i*-F, 0*i*-F Plus, 30*i* /31*i* /32*i*-A, 31*i*-A5, 0*i*-D, 30*i* /31*i*-PB, 30*i* /31*i*-LB, 31*i*-WB, 31*i*-WA, 31*i*-WA5, 0*i*-LF Plus, 0*i*-PF, 0*i*-PD

(2) Series 16*i*

16*i* /18*i* /21*i*-B, 0*i*-B, 0*i*-C, 16*i* /18*i* /21*i*-A, PM*i*-D, PM*i*-H, 0*i*-PB, 0*i*-PC, 16*i* /18*i* -PA, 16*i* /18*i*-PB, 16*i*-LA, 16*i* /18*i*-LB, 16*i*-LPA, 16*i*-LPB

(3) Series 15*i*

15*i*-A, 15*i*-B

■ Installation folder

The default installation folders stated in this document are as follows.

<MT-LINK*i* installation folder>: C:\FANUC\MT-LINK*i*
 <MT-LINK*i* Collector installation folder>: C:\FANUC\MT-LINK*i*
 <Integration Server installation folder>: C:\FANUC\MT-LINK*i*-IntegrationServer
 <Node.js installation folder>: C:\Program Files\nodejs
 <MT-LINK*i* FTP Client installation folder>: C:\FANUC\MT-LINK*i*\FileTransfer\FTPClient
 <MT-LINK*i* FTP Server installation folder>: C:\FANUC\MT-LINK*i*\FileTransfer\FTPServer (NOTE 1)
 <MT-LINK*i* Collector Service Manager installation folder>:
 C:\FANUC\MT-LINK*i*\CollectorPCMgr (NOTE 2)

NOTE

- 1 The installation folder for MT-LINK*i* FTP Server will be as follows:
 < Folder one level upper from the installation folder of MT-LINK*i* Collector >
 \FileTransfer\FTPServer
- 2 The installation folder for Collector Service Manager will be as follows:
 <Folder one level upper from the installation folder of MT-LINK*i* Collector >
 \CollectorServiceMgr

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	25/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

■ License

This document uses the term "license" as a collective word for a hardware key and a software key.

(1) Hardware key

This licensing term denotes a hardware key used to verify a license for MT-LINK*i* Version 3.10 or earlier.

(2) Software key

This licensing term denotes a software key for MT-LINK*i* Version 3.11 or later.

NOTE

Hardware keys and software keys can be used in combination on the same Server PC.

■ How to operate Windows

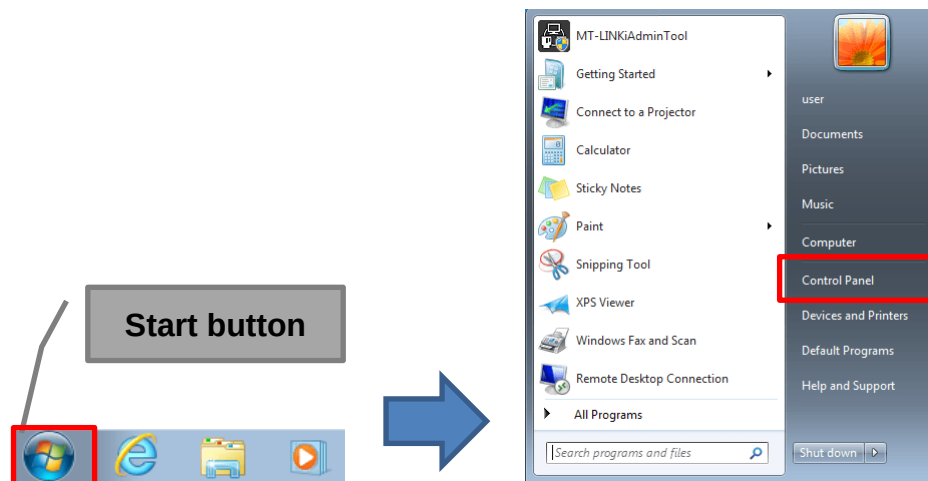
This document may omit how to operate Windows in cases when it would be appropriate.

See below for how to operate operating systems.

(1) How to open [Control Panel]

- Windows Server 2016

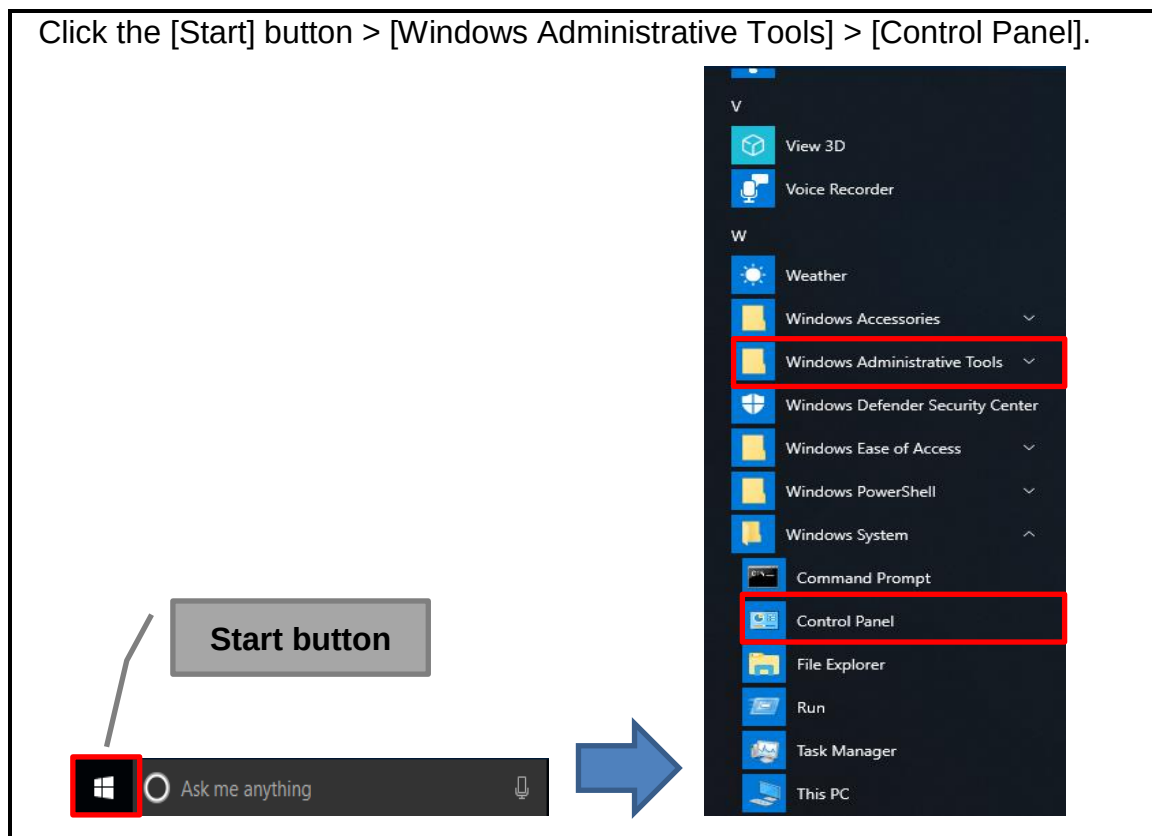
Click the [Start] button > [Control Panel].



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	26/868	
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue		

- Windows 10 / Windows Server 2019 / Windows 11 / Windows Server 2022

Click the [Start] button > [Windows Administrative Tools] > [Control Panel].



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	27/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

2

BASIC OPERATION

This chapter describes the basic operation of AdminTool.

2.1 HOW TO START AND END

This section describes how to start and end AdminTool.

2.1.1 How to Start/End

Only users with administrator permission can start AdminTool.

(1) License verification

■ When use a hardware key

Connect the hardware key to a USB port on the Server PC.

For the details on how to review the optional information of the hardware keys that are connected to the Server PC. For the details, refer to the "FANUC MT-LINK*i* Hardware Key Management Tools OPERATOR'S MANUAL."

■ When use a software key

Activate the software key license on the Server PC, in which MT-LINK*i* components are installed.

For the details on how to verify a software key, refer to the "How to Activate Software Key License" subsection of the "FANUC MT-LINK*i* Install MANUAL."

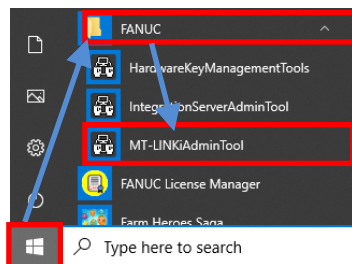
For the details on how to check the status of a software key license, refer to the "FANUC License Manager" subsection of the "FANUC MT-LINK*i* Install MANUAL."

NOTE

- 1 MT-LINK*i* tries to recognize licenses at any time when needed. When it fails to recognize a license for a Machine number Option even after AdminTool started, it will fail to verify the license and thus its features cannot be used until it recognizes the license. Also, if no licenses for Function Options are recognized, no functions to be supported by those options will be available.
- 2 AdminTool cannot start if a license for a related product, ROBODRILL-LINK*i*, is installed. This occurs as well when multiple licenses including for the related product are installed. In this case, Receiver stops and cannot start either.
- 3 The operability of hardware keys is not ensured in a virtual environment (VM).
- 4 The behavior (ON/OFF) of the LED lamp embedded in the hardware key changes as follows:
The lamp turns ON when either the AdminTool or the HardwareKeyManagementTools is running, and turns OFF when neither is running.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 28/868

(2) Click the [Start] button > [FANUC] > [MT-LINKiAdminTool].

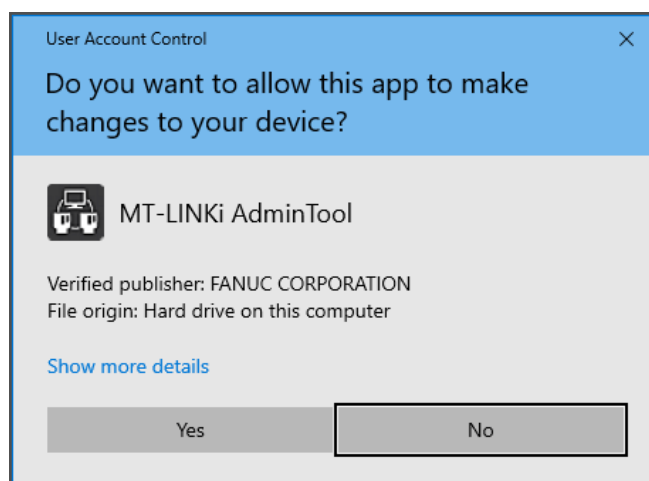


Or, double-click the [MT-LINKiAdminTool] icon on the desktop.

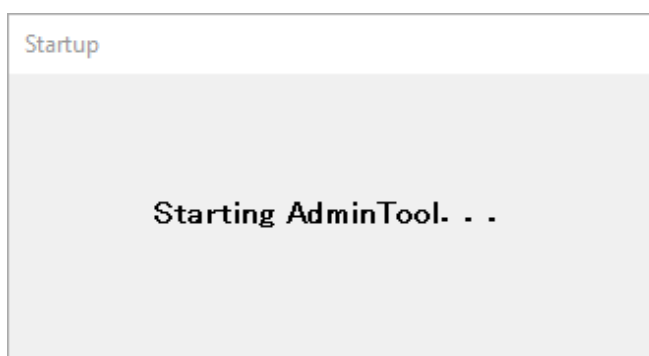


* The [MT-LINKiAdminTool] icon should be created on the desktop when MT-LINKi components are installed with the [Create icon on desktop] check box selected.

(3) When a user with administrator permission starts it, the following window appears depending on the setting of UAC (User Account Control). Click [Yes] to start AdminTool, or click [No] to terminate AdminTool.



(4) The Startup message appears.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	29/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

- (5) When a case listed in the following table is met, AdminTool displays a message and will terminate. Perform the countermeasure that corresponds to the Countermeasure No. of the case.

Message to appear	Case	Countermeasure No.
AdminTool has already started.	AdminTool has already started.	-
To start the AdminTool, you must have administrator permission.	A user without administrator permission tried to start the AdminTool.	1
AdminTool cannot start because no license key has been installed.	No license is installed on the Server PC, or, licenses for Function Options are installed on the Server PC but no license for Machine number Options is installed	2
AdminTool cannot start because the license key includes wrong product information.	A license for a related product, ROBODRILL-LINKi is installed.	2
AdminTool cannot start because license verification failed. (the rest of this message is omitted)	License managing software malfunctions in license verification.	2
Cannot connect MongoDB	AdminTool cannot connect to the database (MongoDB).	3
<Other messages>	Such cases as when the folder or files required to start AdminTool are missing.	4

Countermeasure No.	Countermeasure / Troubleshooting section to refer to
1	Use an administrator account when starting the AdminTool.
2	Fix the trouble by referring to the following subsection of the "FANUC MT-LINKi / Integration Server Troubleshooting MANUAL": The "License Verification Fails" subsection in the "TROUBLES RELATED TO LICENSE" section in the "MT-LINKi TROUBLESHOOTING" chapter
3	Register and restart the database service (MongoDB service) by referring to the following subsection of the "FANUC MT-LINKi / Integration Server Troubleshooting MANUAL": The "Unable to Connect to Database (MongoDB)" subsection in the "TROUBLES RELATED TO DATABASE (MongoDB)" section in the "MT-LINKi TROUBLESHOOTING" chapter
4	Fix the trouble by referring to the following chapter of the "FANUC MT-LINKi / Integration Server Troubleshooting MANUAL": The "HOW TO CONTACT US FOR TROUBLESHOOTING" chapter

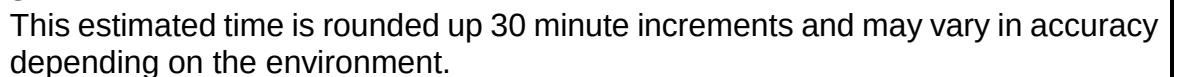
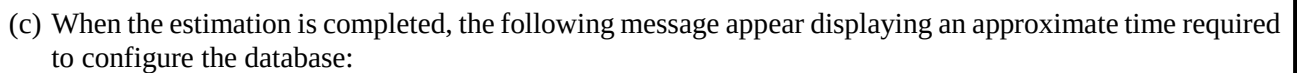
- (6) On the first session after an update installation, AdminTool executes database configuration.
(a) The following message appears and starts estimating the time required to configure a database.

An estimation is in progress of the time required to configure a database.

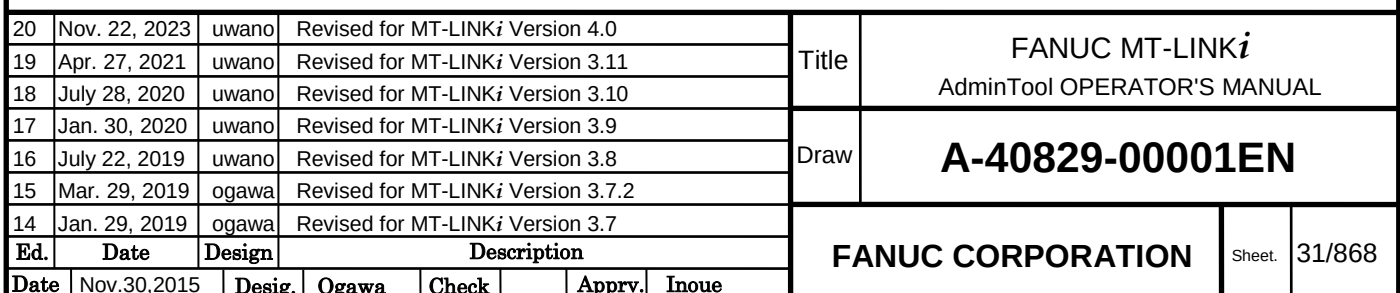
Please wait ..

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 30/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

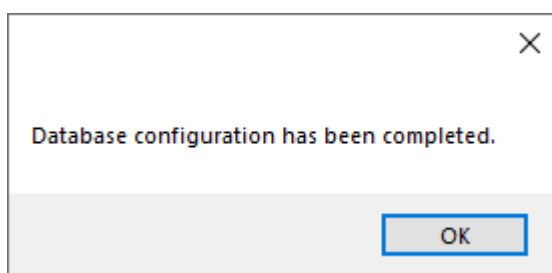
- The "Estimation of Required Time to Configure Database (MongoDB) Takes a Long Time" subsection in the "TROUBLES RELATED TO DATABASE (MongoDB)" section in the "MT-LINK*i* TROUBLESHOOTING" chapter



- The following message appears displaying the elapsed time and progress rate (%):



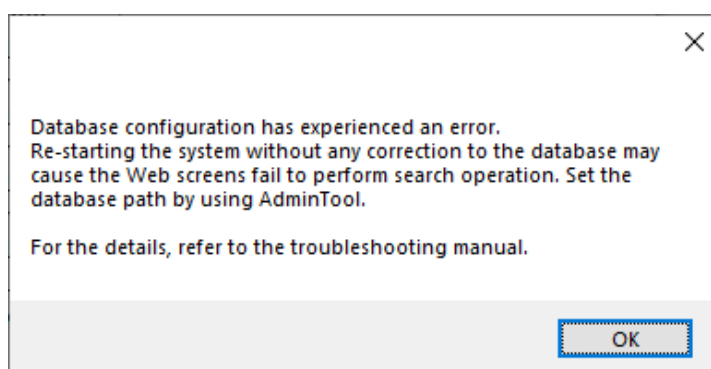
(e) When database configuration is completed, the following message will appear:



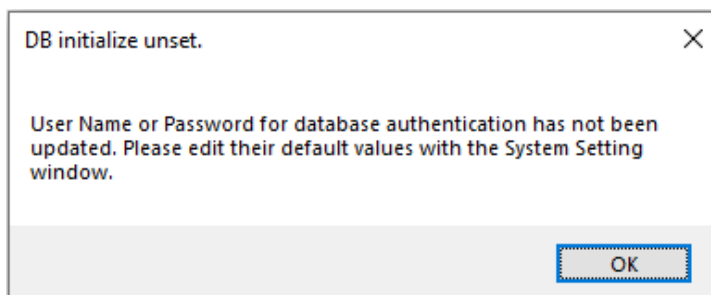
(f) If database configuration fails, the following message will appear:

Fix the problem by referring to the following subsection of the "FANUC MT-LINK*i* / Integration Server Troubleshooting MANUAL":

Case (1) in the "Configuration of Database (MongoDB) Terminates with an Error" subsection in the "TROUBLES RELATED TO DATABASE (MongoDB)" section in the "MT-LINK*i* TROUBLESHOOTING" chapter



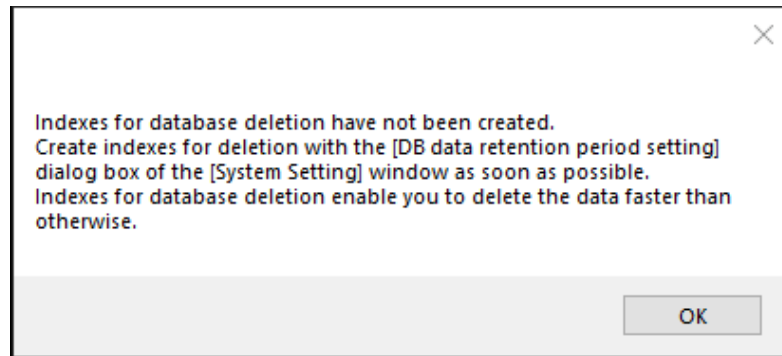
(7) Immediately after a fresh or update installation, the username and password for the database are set to the default value of "admin." If you leave these fields as is, the following message appears. Click [OK] and specify new username and password on the [System Setting] window.



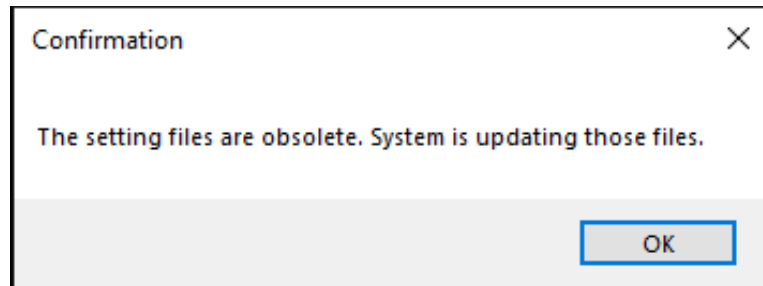
If indexes for database deletion have not been created, the following message will appear. After starting AdminTool, open the [System Setting] window and create indexes for database deletion by using the [DB data retention period setting] dialog box.

For the details, refer to the "[DB Data Retention Period Setting](#)" subsection.

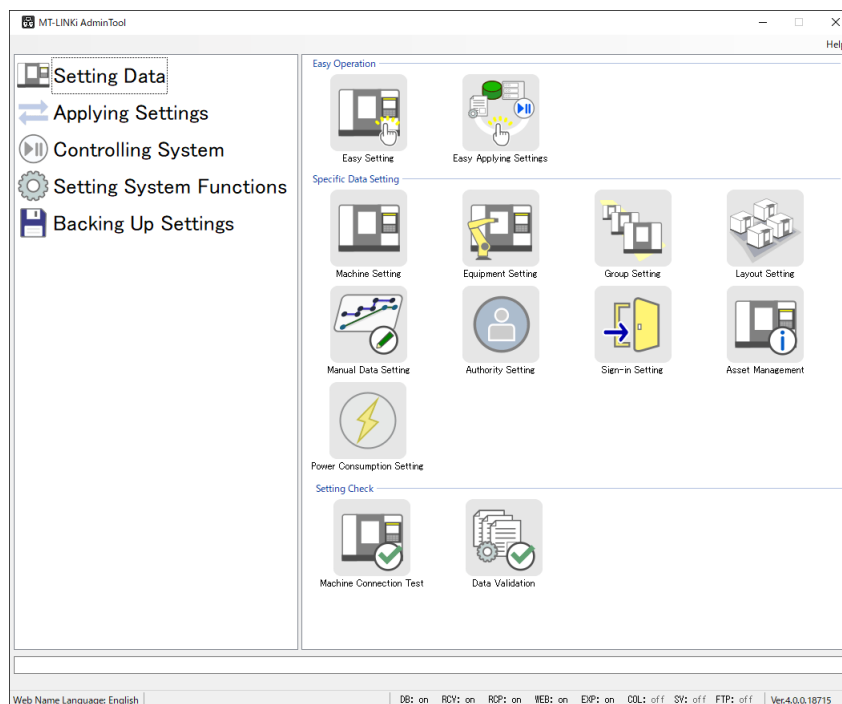
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 32/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	



(8) On the first session after an update installation, the following message may appear.



(9) The AdminTool Menu window ([MT-LINKi AdminTool] window, hereafter referred to as the "AdminTool Menu") appears.
For the details about the window, refer to the "[AdminTool Menu Window](#)" subsection.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	33/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

- (10) If a case listed in the following table is met after the AdminTool started, it displays the corresponding message and will terminate.

Message to appear	Case	Countermeasure No.
Verification of the license key failed. The application will terminate.	No license is recognized, or only licenses for Function Options are recognized, but no license for Machine number Option is recognized.	1
The license key does not include product information about the application. The application will terminate.	A license for a related product, ROBODRILL-LINK <i>i</i> , is installed after AdminTool started.	1
AdminTool will terminate because license verification failed. (the rest of this message is omitted)	License managing software malfunctions in license verification.	1

Countermeasure No.	Countermeasure / Troubleshooting section to refer to
1	Fix the trouble by referring to the following subsection of the "FANUC MT-LINK <i>i</i> / Integration Server Troubleshooting MANUAL": The "License Verification Fails" subsection in the "TROUBLES RELATED TO LICENSE" section in the "MT-LINK <i>i</i> TROUBLESHOOTING" chapter

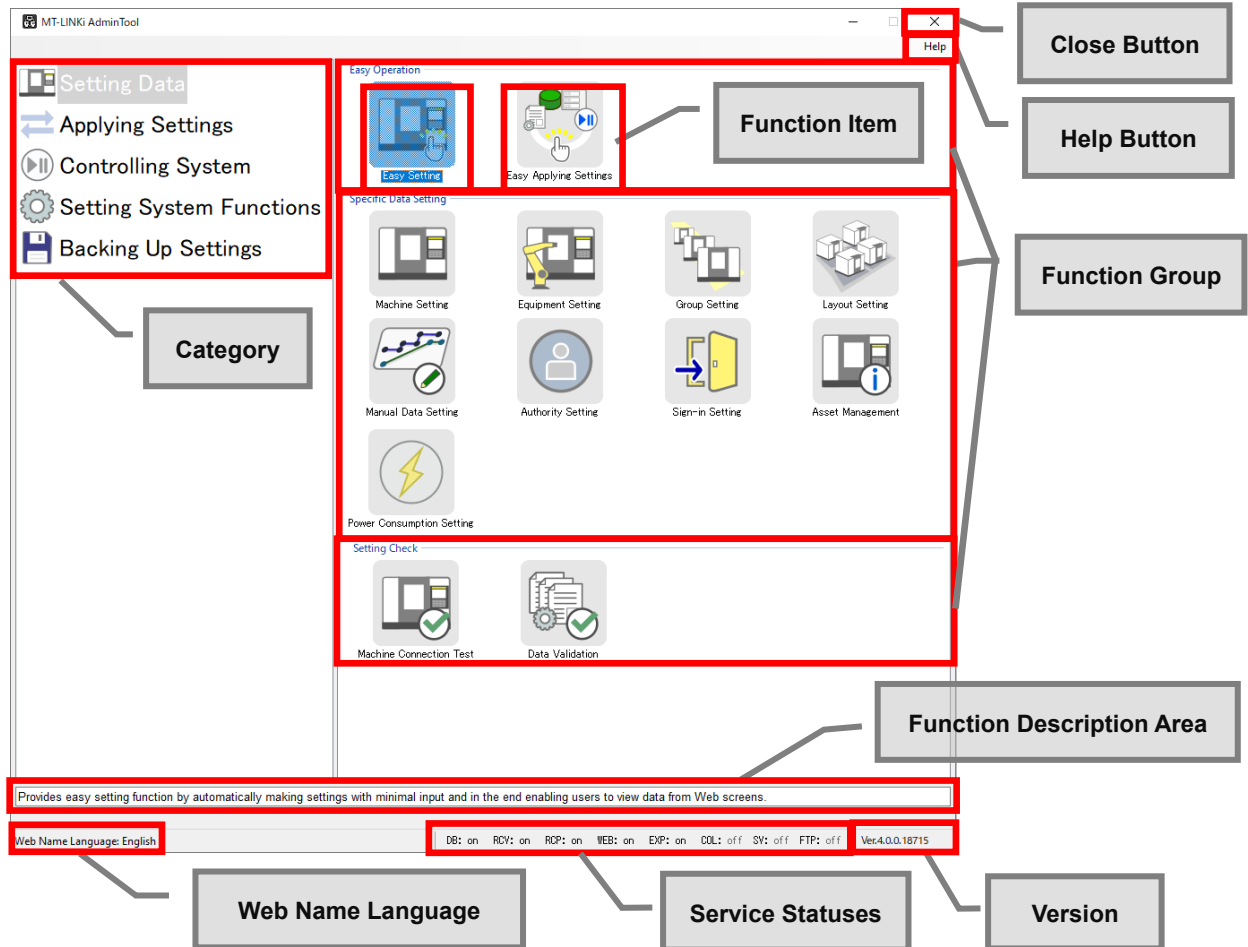
- (11) To end AdminTool, click the [x] (Close) button at the upper-right corner of the AdminTool Menu.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 34/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

2.1.2 AdminTool Menu Window

This subsection describes the AdminTool Menu (the [MT-LINK*i* AdminTool] window).

- (1) Move the mouse over the left side pane of the AdminTool Menu and select a category in the pane. The right side pane in the AdminTool Menu will display the function groups and items belonging to the selected category.
- (2) Select a function item in the right side pane to open the window for the selected menu.



■ The [MT-LINK*i* AdminTool] window

- **Category**
This menu lists categories of functions that AdminTool provides. A category consists of menu groups, each of which is called a function group.
- **Function group**
In turn, a function group consists of menu items, each of which is called a function item. Function items in a function group correspond to AdminTool functions of the same function type as the function group denotes.
- **Function item**
Each of AdminTool functions corresponds a function item which is arranged into a function group in the same category.
- **Function description area**
This area displays description of the selected function item.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	35/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

- Web name language

This field displays the language ("English," "日本語," or "中文 (簡体)") that is set for the [Web name language] drop-down list box in the [System Setting] window.

If a local language is set that is added by loading additional language resources, this field displays the local language. For the details, refer to the "[Change Language for Web Names](#)" and the "[Load Additional Language Resource](#)" subsections.

- Version

This field displays the version information of AdminTool.

- Service status

This field shows statuses of the services composing MT-LINK*i* system in the following format:

<Abbreviation> : <Status>

Server PC / Collector PC	Service Name	Abbr.	Service status
Server PC	Database	DB	on: the service is running on the Server PC off: the service is stopped on the Server PC
	Receiver	RCV	
	Receiver Cache Persister	RCP	
	Web service	WEB	
	Exporter	EXP	
Collector PC	Collector	COL	on: the service is running on all Collector PCs on which it is able to run off: the service is stopped on all Collector PCs on(<x>/<z>): the service is running on some but not all Collector PCs* -: this service is set disabled for all of the Collector PCs (Disabled)
	SERVOVIEWER	SV	
	FTP Server	FTP	

* When a service is running on some but not all Collector PCs, this field displays the status of the service as follows:

on(<x>/<z>), where <x> is the number of the service (COL, SV, or FTP) that is currently running on Collector PCs, and <z> is the number of the Collector PCs on which the service is able to run.

Example 1) (Server PC) Database service is running but the other services are stopped
(Collector PC) one Collector is registered and all services are stopped on the Collector PC

DB: on RCV: off RCP: off WEB: off EXP: off COL: off SV: off FTP: off

Example 2) (Server PC) all services are running on the Server PC
(Collector PC) one Collector is registered and SEVOVIEWER service is set disabled for the Collector PC but the other services are running on it

DB: on RCV: on RCP: on WEB: on EXP: on COL: on SV: - FTP: on

Example 3) (Server PC) all services are running on the Server PC
(Collector PC) three Collectors are registered and some services are not running on one of the three Collector PCs as shown in the figure below

No	Collector Name	Collector	SERVOVIEWER	FTP Server	Communication Status
1	DESKTOP-GHBI2GK	☒	☒	☒	ONLINE
2	DESKTOP-GHBI2GH	☒	☒	☒	ONLINE
3	DESKTOP-GHBI4TK	☐	Disabled	☐	OFFLINE

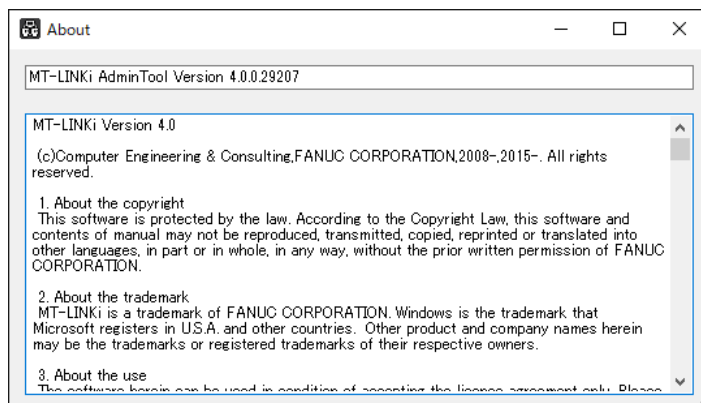
DB: on RCV: on RCP: on WEB: on EXP: on COL: on(2/3) SV: on FTP: on(2/3)

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 36/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

For the details on how to check service statuses and how to start and stop those services, refer to the "**START/STOP SYSTEM**" section.

- [Help]

Selecting the [About] command in the [Help] drop-down list opens the [About] window reading the version and software license information of MT-LINK*i*.



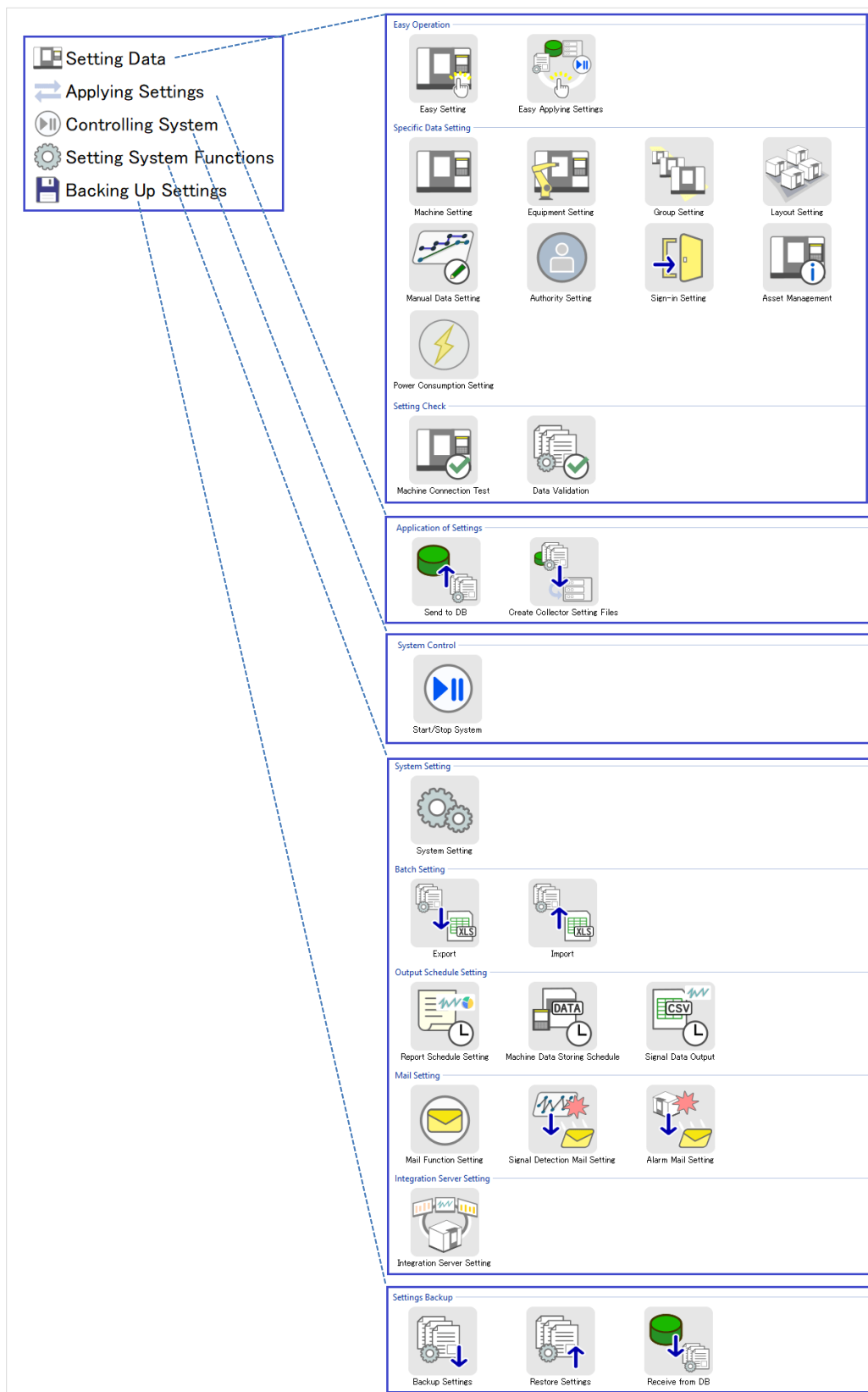
- [x]

Clicking this button closes the AdminTool Menu.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 37/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

The following are menus (category, function groups, and function items) to be displayed on the AdminTool Menu.

- The following figure shows relationships among categories, function groups, and function items (i.e., selecting which category displays which function groups and which function items).



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	38/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

■ The following figure shows selecting which category displays which function groups and function items.

Category	Function group	Function item	Description
Setting Data	Easy Operation (EASY OPERATION)	Easy Setting (EASY SETTING)	This menu provides easy setting function by automatically making settings with minimal input and in the end enabling users to view data from Web screens.
		Easy Applying Settings (EASY APPLYING SETTINGS)	This menu enables users to easily send setting data to the database and Collectors and to apply the setting data to them at once.
	Specific Data Setting (SPECIFIC DATA SETTING)	Machine Setting (MACHINE SETTING)	This menu registers Collectors and machines of a variety of types, and sets up signals to retrieve from them.
		Equipment Setting (EQUIPMENT SETTING)	This menu assigns machines to equipment to connect with and sets equipment icons and statuses that the [Monitoring - Equipment Monitoring] Web screen displays.
		Group Setting (GROUP SETTING)	This menu puts equipment in groups and sets group icons and group statuses that the [Monitoring - Group Monitoring] Web screen displays.
		Layout Setting (LAYOUT SETTING)	This menu provides a way to customize layouts and the background image of equipment to display on the [Monitoring - Overlook] Web screen.
		Manual Data Setting (MANUAL DATA SETTING)	This menu sets plans, product names and work comments to complement history information that the [Results - Production Results] Web screen displays.
		Authority Setting (AUTHORITY SETTING)	This menu sets user accounts and view permission of Web screens, and configures LDAP authentication function.
		Sign-in Setting (SIGN-IN SETTING)	This menu sets the title image and additional information that the sign-in Web screen to display and notify to Web users.
		Asset Management (ASSET MANAGEMENT)	This menu sets machine specific information and images to display on the [Monitoring - Equipment - Monitoring] Web screen.
		Power Consumption Setting (POWER CONSUMPTION SETTING)	This menu sets CO2 emissions and power consumption for external device to display on the [Results - Power Consumption Monitoring] Web screen.
	Setting Check (SETTING CHECK)	Machine Connection Test (CHECK MACHINE SETTING)	This menu checks whether machines set up through Machine Setting are connectable, and enables users to verify the settings.
		Data Validation (DATA VALIDATION)	This menu checks settings for users to review the coherence of them before sending them to the database.
Applying Settings	Application of Settings (APPLICATION OF SETTINGS)	Send to DB (SEND TO DB)	This menu sends setting data to the database and applies them to the system.
		Create Collector Setting Files (CREATE COLLECTOR SETTING FILES)	This menu creates setting files for Collectors and applies them to the Collectors.
Controlling System	System Control (SYSTEM CONTROL)	Start/Stop System (START/STOP SYSTEM)	This menu starts and stop services, such as Receiver and Collector, that compose MT-LINKi system.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	39/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

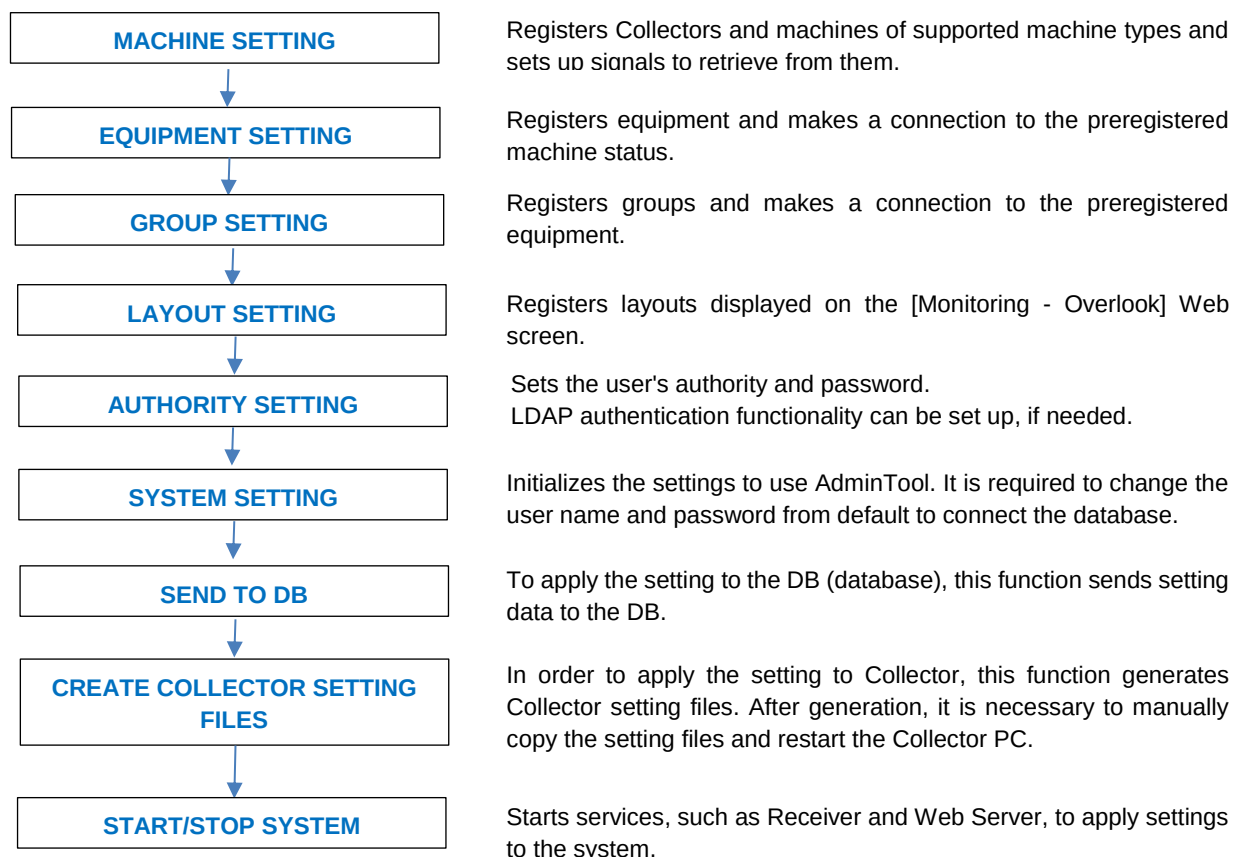
Category	Function group	Function item	Description
Setting System Functions	System Setting (SYSTEM SETTING)	System Setting (SYSTEM SETTING)	This menu makes settings of various systems composing MT-LINKi.
	Batch Setting (BATCH SETTING)	Export (EXPORT)	This menu exports current setting data in batch setting files.
		Import (IMPORT)	This menu imports setting data from batch setting files.
	Output Schedule Setting (OUTPUT SCHEDULE SETTING)	Report Schedule Setting (REPORT SCHEDULE SETTING)	This menu sets schedules to create reports (daily/weekly/monthly) of equipment operation statuses.
		Machine Data Storing Schedule (MACHINE DATA STORING SCHEDULE)	This menu sets schedules to periodically store manage machine programs and data for managing change history of them.
		Signal Data Output (SIGNAL DATA OUTPUT)	This menu sets schedules to chronologically output signal data in CSV files.
	Mail Setting (MAIL SETTING)	Mail Function Setting (MAIL FUNCTION SETTING)	This menu makes primary settings for signal detection and alarm mail delivery, such as settings of mail server and mail delivery destination users.
		Signal Detection Mail Setting (SIGNAL DETECTION MAIL DELIVERY SETTING)	This menu makes settings of delivery conditions and target equipment to send mail when signals are detected that meet the conditions.
		Alarm Mail Setting (ALARM MAIL DELIVERY SETTING)	This menu makes settings of delivery conditions and target equipment to send mail when alarms occur on them.
	Integration Server Setting (INTEGRATION SERVER SETTING)	Integration Server Setting (ALARM MAIL DELIVERY SETTING)	This menu sets data about the MT-LINKi system for Integration Servers to manage it.
Backing Up Settings	Settings Backup (SETTINGS BACKUP)	Backup Settings (BACKUP SETTINGS)	This menu backs up current settings of AdminTool as backup data.
		Restore Settings (RESTORE SETTINGS)	This menu restores setting data of AdminTool that were backed up through Backup Settings.
		Receive from DB (RECEIVE FROM DB)	This menu retrieves stored setting data from the database and overwrites the current settings of AdminTool.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	40/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

2.2 CONFIGURATION STEPS

This section describes the setting steps at the first time (i.e., no settings have been made yet after installation).

- To easily make settings for basic functions of MT-LINK*i*
Start AdminTool and select [Setting Data] > [Easy Operation] > [Easy Setting] menu items to open the [Easy Setting] window. Make settings with the window. For the details on how to make settings, refer to the "**EASY SETTING**" section.
- To make specific settings for functions of MT-LINK*i* according to the operating environment
Start AdminTool, select function items in the same order as they are described below to open the corresponding setting window or dialog box, and make specific settings for them with the opened window or dialog box. For the details, refer to the subsequent chapters.

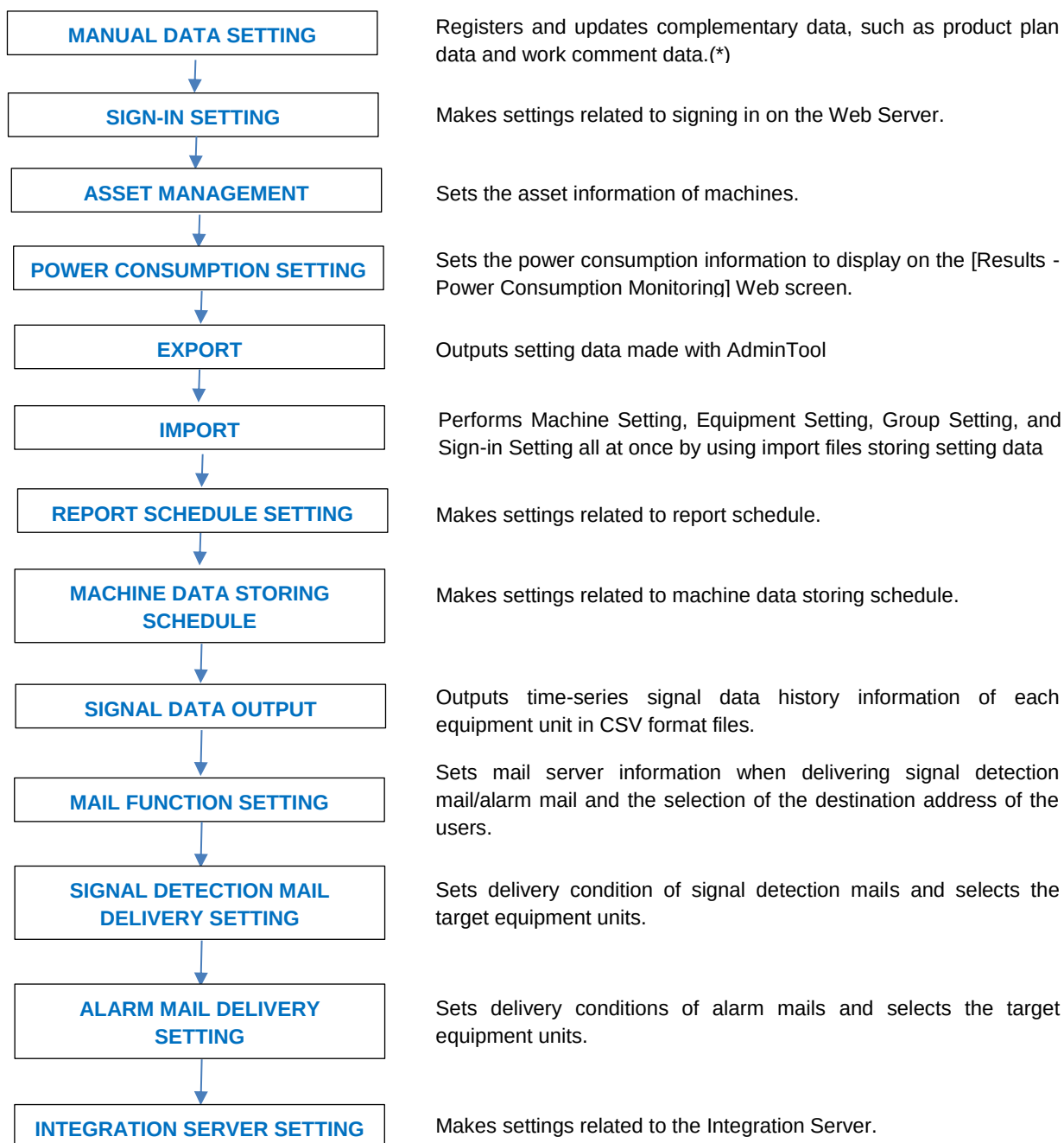


On a subsequent session to modify those settings except Collector settings, the menus "SEND TO DB", "CREATE COLLECTOR SETTING FILES" and "START/STOP SYSTEM" can be performed by using Easy Applying Settings function with ease at once.

To use Easy Applying Settings function, select [Setting Data] > [Easy Operation] > [Easy Applying Settings] menu items to open the [Easy Applying Settings] window. Apply the settings with the window. For the details on how to make settings, refer to the "**EASY APPLYING SETTINGS**" section.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	41/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

Make settings for the following function items if needed:



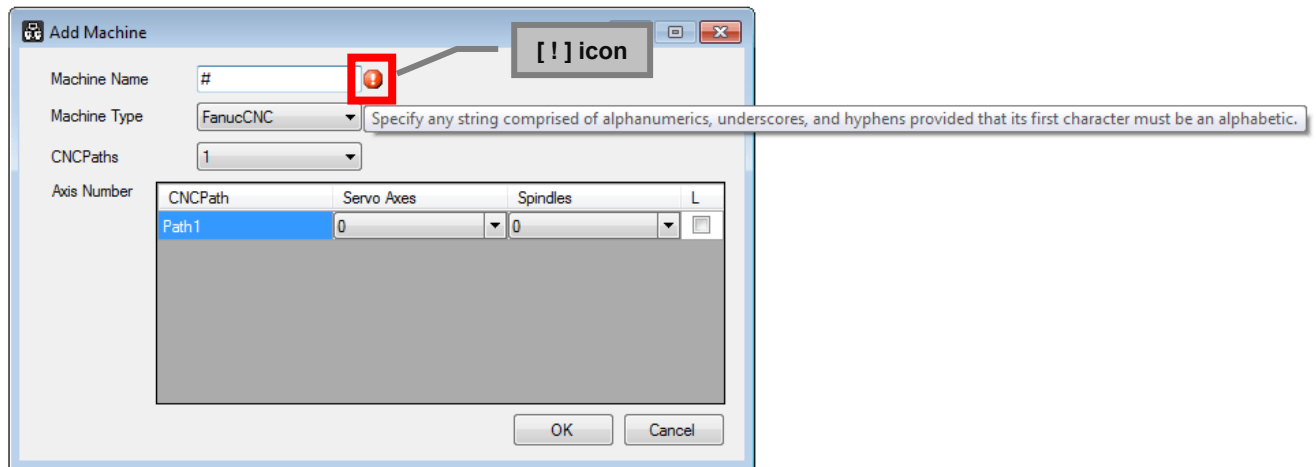
* The [Utility - System Setting] Web screen can be used to make the same settings as described above as well. For the details, refer to the "UTILITY – SYSTEM SETTING SCREEN " section of the "FANUC MT-LINKi WebUI OPERATOR'S MANUAL."

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	42/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

2.3

ABOUT [!] MARK TO INDICATE INPUT ERROR

If AdminTool detects any errors while settings are being made, it notifies you of them by displaying the [!] icon. To see the detailed error information, move the mouse cursor on the icon.



Example:

Here is the example using invalid input for [Machine Name].

* Once an input error occurs, this dialog disables the [OK] button. In such a case, correct your input to continue setting, or click [Cancel] or [x] (Close) at the upper-right corner of the window to discard the settings.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	43/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

3

[SETTING DATA] - EASY OPERATION

This chapter describes how to set data in AdminTool for easily operating MT-LINK*i*.

The AdminTool Menu has the following menu items in the [Setting Data] - [Easy Operation] group:

(1) Easy Setting

This menu provides easy setting function by automatically making settings with minimal input and in the end enabling users to view data from Web screens.

(2) Easy Applying Settings

This menu enables users to easily send setting data to the database and Collectors and to apply the setting data to them at once.

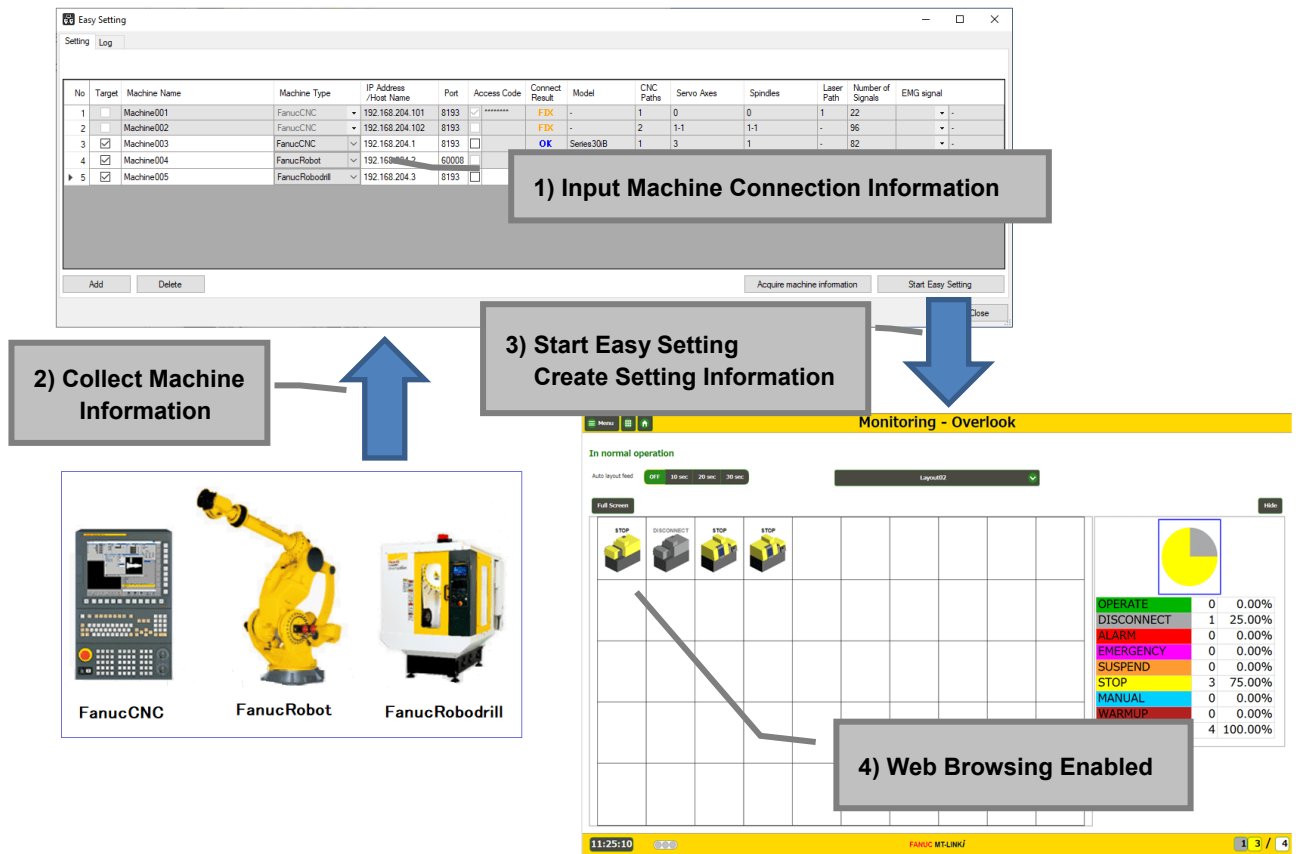
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 44/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

3.1 EASY SETTING

This section explains Easy Setting.

Easy Setting is a functionality which enables simple and quick viewing of Web screen with the least possible inputs.

Just by acquiring the machine information via the inputs of the machine connection information, AdminTool creates various setting information, and automatically makes the settings to enable browsing of the Web screen.



NOTE

This function is available with FanucCNC, FanucRobodrill, and FanucRobot type machines.

ROBODRILL Option license is required to support FanucRobodrill type machines.

3.1.1 Overview of Easy Setting

Easy Setting consists of the following features.

- Connects to each machine based on the machine connection information input (IP address / host name, port number, access code), and acquires information such as CNC path number, axis number, and whether signals are set to be collected.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	45/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- Using the input information and the information acquired by connecting to the machine, creates various setting information of AdminTool (Collector setting, machine setting, signal setting, group setting, overlook layout setting, authority setting).
- Restarts all the services, and automatically makes settings to enable the browsing of the Web screen.
- The maximum number of machines that can be registered differs depending on the options of the license. For the details, refer to the "**MACHINE SETTING**" section.

Machine Setting List														
No	Target	Machine Name	Machine Type	IP Address / Host Name	Port	Access Code	Connect Result	Model	CNC Paths	Servo Axes	Spindles	Laser Path	Number of Signals	EMG signal
1	☐	Machine001	FanucCNC	192.168.204.101	8193	*****	FDX	-	1	0	0	1	22	-
2	☐	Machine002	FanucCNC	192.168.204.102	8193		FDX	-	2	1-1	1-1	-	96	-
3	☐	Machine003	FanucCNC	192.168.204.103	8193		FDX	-	1	0	0	1	22	-
4	☐	Machine004	OPCUAServer	-	-		FDX	-	-	-	-	-	-	-
5	☐	Machine005	FanucRobot	192.168.204.105	60008		FDX	R-30B	-	-	-	-	6	DO 99
6	☐	Machine006	MTConnectAgent	192.168.204.106	5000		FDX	-	-	-	-	-	-	-
7	☐	Machine007	FanucRobodrill	192.168.204.107	8193		FDX	-	1	3	1	-	150	-
8	☒	Machine008	FanucCNC	192.168.204.108	8193		OK	Series30B	1	3	1	-	82	-
9	☒	Machine009	FanucRobodrill	192.168.204.109	8193		OK	Series31B	1	8	0	-	223	-
10	☒	Machine010	FanucRobot	192.168.204.110	60008		NG	-	-	-	-	-	-	DO 99

NOTE

- 1 FanucCNC, FanucRobot, and FanucRobodrill type machines are targeted for Easy Setting. If there are machines, including the other machine types, which are already registered by machine setting, those machines will also be listed. However, the settings of these machines cannot be changed or deleted.
- 2 ROBODRILL Option license is required to support FanucRobodrill type machines.
- 3 The machine targeted for machine information acquisition with Easy Setting must have the power turned ON and must be in a connectable status.
- 4 In the Easy Setting, the machines are assigned to the Collector on the Server PC without an exception. If you wish to assign the machine to a Collector that is not on the Server PC, use [Cut Machine] – [Paste Machine] processes in the Machine Setting after the completion of the Easy Setting procedures. For the details, refer to the "**Copy/Cut/Paste/Delete Machine Setting**" subsection.
- 5 The machines set on the Machine Setting window cannot be deleted by Easy Setting. Use the [Delete Machine] process in the Machine Setting to delete the machine. For the details, refer to the "**Copy/Cut/Paste/Delete Machine Setting**" subsection.
- 6 When the Collector is installed in the Server PC, the name of the Server PC must not include multi-byte characters. For the details, refer to the "**Descriptions of the Settings Created by Easy Setting**" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 46/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

3.1.2 Steps for Easy Setting

Following are the steps for Easy Setting.

- (1) Select [Easy Setting] in the AdminTool Menu to display the [Easy Setting] window ([Setting] tab).
For the details on the items on the [Easy Setting] window, refer to the "[Easy Setting Window](#)" subsection.
- (2) Click the [Add] button to add a new row to the machine setting list. Enter [Machine Name], [Machine Type], [IP Address/Host Name], [Port], and [Access code], and make sure that the [Target] check box is selected. When adding machine settings of two or more machines, repeat this step for each machine.
- (3) Click the [Acquire machine information] button to establish connection with the machine.
If there are machines with "NG" shown in the [Connect Result] column, display the [Log] tab on the [Easy Setting] window to check the detail of the error.
For the details, refer to the "[Acquire Machine Information](#)" subsection.
- (4) Easy Setting can be performed for the machines with the "OK" displayed in the [Connect Result] column.
Select the [Target] check box of the machine targeted for Easy Setting, and click the [Start Easy Setting] button.
Confirm the result of the process displayed on the [Log] tab.
For the details, refer to the "[Start Easy Setting](#)" subsection.
- (5) Return to [Setting] tab and confirm that "FIX" is displayed on the [Connect Result] column of the machines that performed Easy Setting.
- (6) Click the [Close] button to close the [Easy Setting] window.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 47/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

3.1.3 Easy Setting Window

The items on the [Easy Setting] window are explained in the following.

■ [Easy Setting] window ([Setting] tab)

No	Target	Machine Name	Machine Type	IP Address	Port	Access Code	Connect Result	Model	CNC Path	Servo Axes	Spindles	Laser Path	Number of Spindle	EMG signal
1	☐	Machine001	FanucCNC	192.168.204.101	8193	*****	FIX	-	1	0	0	1	22	-
2	☐	Machine002	FanucCNC	192.168.204.102	8193		FIX	-	2	1-1	1-1	-	96	-
3	☐	Machine003	FanucCNC	192.168.204.103	8193		FIX	-	1	0	0	1	22	-
4	☐	Machine004	OPCUAServer	-	-		FIX	-	-	-	-	-	-	-
5	☐	Machine005	FanucRobot	192.168.204.105	60008		FIX	R-30B	-	-	-	-	6	DO 99
6	☐	Machine006	MTConnectAgent	192.168.204.106	5000		FIX	-	-	-	-	-	-	-
7	☐	Machine007	FanucRobodrill	192.168.204.107	8193		FIX	-	1	3	1	-	150	-

Machine setting list

On the machine setting list, the existing machine settings are displayed in a list.

If there are no machines registered, nothing will be shown on the machine setting list.

NOTE

If ROBODRILL Option is neither installed yet nor recognized as installed, no FanucRobodrill type machine will be listed in the machine setting list.

- **Target**
A check box used for selecting the target machine for acquiring machine information and making settings using the Easy Setting.
When the check box is selected, the machine becomes the target for machine information acquisition / Easy Setting.
For the machines with [FIX] displayed in the [Connect Result] column, the check box is fixed to cleared.
* By selecting multiple rows, right-clicking, and selecting [ON] or [OFF] on the menu displayed, the [Target] check boxes on these rows can be selected or cleared at once.
- **Machine Name (entry required)**
Enter the machine name. Half-width alphabet can be used for the first character of the machine name. Half-width alphanumerics, hyphens ("-"), and underscores ("_") can be used for the characters that follow. This field can be up to 32 bytes long. Existing machine name cannot be entered.
For the machines with [FIX] displayed in the [Connect Result] column, the machine name that is already set will be displayed, and it cannot be changed.
- **Machine Type (entry required)**
Select machine type on the drop-down list. FanucCNC, FanucRobot, or FanucRobodrill can be selected.
For the machines with [FIX] displayed in the [Connect Result] column, the machine type already set will be displayed, and it cannot be changed.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 48/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

- **IP Address / Host Name (entry required)**
Enter the IP address or the host name of the machine. In the case of a host name, half-width alphabet can be used for the first character. Half-width alphanumerics, hyphens ("-"), and underscores ("_") can be used for the characters that follow. This field can be up to 253 bytes long.
For the machines with [FIX] displayed in the [Connect Result] column, the IP address or the host name already set will be displayed, and it cannot be changed. If the machine type is OPCUAServer, "-" will be displayed.
- **Port (entry required)**
Enter the port number used for the communication with the machine. A value between "0" and "65535" can be used for the port number.
If the machine type is FanucCNC or FanucRobodrill, "8193" is set as the default value.
If the machine type is FanucRobot, "60008" is set as the default value.
For the machines with [FIX] displayed in the [Connect Result] column, the port number already set will be displayed, and it cannot be changed. If the machine type is OPCUAServer, "-" will be displayed.
- **Access Code**
For FanucCNC and FanucRobodrill type machines on which an access code is set, select the check box and set an access code for this column. Enter less than nine numeric characters. The entered access code will be masked with asterisks ("*").
For machines of other machine types, this column populates the symbol "-" and does not allow to modify.

NOTE

To set an access code on a FanucCNC and FanucRobodrill type machine, CNC system software must be installed that supports a security function to use an access code.

- **Connect Result (cannot be changed)**
Result of connection with the machine when acquiring machine information, is displayed.
"OK": State when the connection with the machine is confirmed
"NG" (for "not good"): State when the connection with the machine cannot be established
"FIX": State when the machine setting is completed
- **Model (cannot be changed)**
If the machine type is FanucCNC or FanucRobodrill, the model will be displayed.
If the machine type is FanucRobot, its robot controller model will be displayed.
For the other machine types, "-" will be displayed.
* For machines of FanucCNC and FanucRobodrill machine type, when [FIX] is displayed in the [Connect Result], "-" may be displayed.
- **CNC Paths (cannot be changed)**
The number of CNC paths for a FanucCNC or FanucRobodrill type machine is displayed. For the other machine types, "-" is displayed.
- **Servo Axes (cannot be changed)**
The number of servo axis for a FanucCNC or FanucRobodrill type machine is displayed. Displayed in the order of the path, delimited by "-." (Example: 2-3-3)
For the other machine types, "-" will be displayed.
- **Spindles (cannot be changed)**
The number of spindle axis for a FanucCNC or FanucRobodrill type machine is displayed. Displayed in the order of the path, delimited by "-." (Example: 1-1-2)
For the other machine types, "-" will be displayed.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION				Sheet. 49/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

- The path of the machine type FanucCNC which has the laser oscillator connected is displayed. When multiple paths exist, they are displayed delimited by "/" (Example: 1/3)
If the laser oscillator is not connected, or if the machine type is other than FanucCNC, "-" will be displayed.

- The IO (DO)* number of EMG signal for a FanucRobot type machine is displayed. (Example: DO99)
By default, this column is set to 0. If the correct address and IO number of EMG signal are verified
before acquisition of them, it is allowed to set this column to them.

- Number of Signals (cannot be changed)

The number of effective signals for a FanucCNC, FanucRobot, or FanucRobodrill type machine is displayed. For the other machine types, "-" will be displayed.

- [Add] button

[Target]: selected

[Machine Name]: "NewMachine<Sequential number>"

<Sequential number> starts from 001 (example:

NewMachine001)

```
[Machine Type]: "FanucCNC"
```

```
[Port]: "8193"
```

- Deletes the row selected on the machine setting list. Multiple rows can be selected and deleted at once. When you click [OK] on the confirmation message, the selected row will be deleted. The rows with [FIX] displayed in the [Connect Result] cannot be deleted.

- Establishes connection for the machine with the [Target] check box selected based on the entered information (IP address / host name, port, and access code), and displays "OK"/"NG (meaning 'Not Good')" in the [Connect Result] column.

When "OK" is displayed in the [Connect Result] column, displays the acquired machine information in the machine setting list ([Machine Type] (see NOTE) / [Model] / [CNC Paths] / [Servo Axes] / [Spindles] / [Laser Path] / [Number of Signals]).

The log during the acquisition of machine information is displayed in the [Log] tab of the [Easy Setting] window. For the machines with "NG" displayed in the [Connect Result] column, refer to the log and check the detail of the error.

- For the machines with "OK" displayed in the [Connect Result] column and the [Target] check box selected, Easy Setting is applied. After the setting is completed, the display in the [Connect Result] column changes from "OK" to "FIX."

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	50/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

NOTE

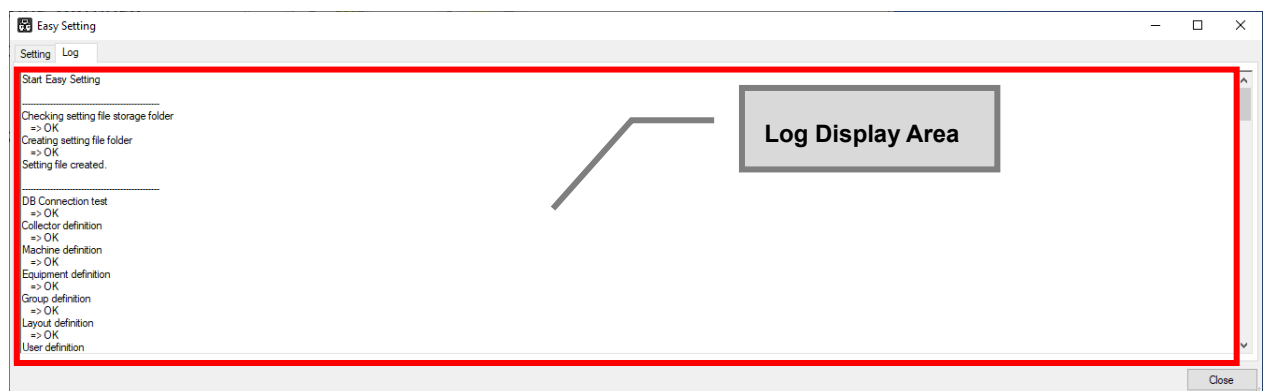
When the Easy Setting process is being executed, the tab is switched to [Log] tab on the [Easy Setting] window, and the log of the setting process is displayed in real time.

- [Close] button

Closes the [Easy Setting] window.

If there are machine settings with information other than [FIX] is displayed in the [Connect Result] column, a confirmation message is displayed. By clicking the [OK] button in the confirmation message, all the entries of the machine settings displaying information other than [FIX] will be aborted.

■ [Easy Setting] window ([Log] tab)



Log display area

When the [Acquire machine information] button in the [Setting] tab is clicked, the log of the result of machine information acquisition is displayed.

When the [Start Easy Setting] button in the [Setting] tab is clicked, the log of the Easy Setting process is displayed.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	51/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

3.1.4 Acquire Machine Information

The acquisition process of the machine information is explained in the following.

- For FanucCNC / FanucRobodrill type machines

No	Target	Machine Name	Machine Type	IP Address / Host Name	Port	Access Code	Connect Result	Model	CNC Paths	Servo Axes	Spindles	Laser Path	Number of Signals	EMG signal
1	☐	Machine001	FanucCNC	192.168.204.101	8193	*****	FDX	-	1	0	0	1	22	-
2	☐	Machine002	FanucCNC	192.168.204.102	8193		FDX	-	2	1-1	1-1	-	96	-
3	☐	Machine003	FanucCNC	192.168.204.103	8193		FDX	-	1	0	0	1	22	-
4	☐	Machine004	OPCUAServer	-	-		FDX	-						-
5	☐	Machine005	FanucRobot	192.168.204.105	60008		FDX	-						-
6	☐	Machine006	MTConnectAgent	192.168.204.106	5000		FDX	-						-
7	☐	Machine007	FanucRobodrill	192.168.204.107	8193		FDX	-						-
8	☒	Machine008	FanucCNC	192.168.204.108	8193									-
9	☒	Machine009	FanucRobodrill	192.168.204.109	8193									-
10	☐	Machine010	FanucRobot	192.168.204.110	60008									-

Buttons: Add, Delete, Acquire machine information, Start Easy Setting, Close

- (1) Click the [Add] button to add a row in the machine setting list, and enter [Machine Name], [Machine Type], [IP Address / Host Name], [Port], and [Access code]. When registering two or more machines, repeat this step for each machine.
- (2) Make sure that the [Target] check box is selected, and click the [Acquire machine information] button.
- (3) By clicking [OK] on the confirmation message for the acquisition, the acquisition process begins. During the machine information acquisition process, a progress bar showing the status (<number of completions> / <number of targets>) is displayed.

Acquiring machine information

1/2 units completed

- (4) When the acquisition of machine information is stopped, or there is a machine displaying "NG" (for "not good") in the [Connect Result] column after the completion of acquisition, the following message is displayed.

Machine information cannot be acquired from a machine.
Review connection or delete unnecessary row.

Check error details of acquisition failure by referring to Log tab.

OK

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 52/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

- (5) If "OK" is displayed in the [Connect Result] column, the machine information ([Model], [CNC Paths], [Servo Axes], [Spindles], [Laser Path], and [Number of Signals]) is displayed, and the acquisition process is completed.

For the machines with "NG" displayed in the [Connect Result] column, click the [Log] tab to check the acquisition result and the details of the error, and review or delete the settings.

NOTE

When the selected machine type in the [Machine Type] column and the acquired machine type are different from each other, the column is set to as follows:

- If "FanucRobodrill" is selected and the acquired machine type is "FanucCNC," the column is set to "FanucCNC"
- If "FanucCNC" is selected and the acquired machine type is "FanucRobodrill," the column is left as "FanucCNC"

The screenshot shows the 'Easy Setting' window with a table of machine settings. The table has columns: No, Target, Machine Name, Machine Type, IP Address / Host Name, Port, Access Code, Connect Result, Model, CNC Paths, Servo Axes, and Machine Information Acquisition Information. The 'Connect Result' column shows 'OK' for Machine008 and Machine009. A red box highlights the 'Machine Information Acquisition Information' section for Machine008 and Machine009. A callout box points to the 'Machine Information Acquisition Information' section.

No	Target	Machine Name	Machine Type	IP Address / Host Name	Port	Access Code	Connect Result	Model	CNC Paths	Servo Axes	Machine Information Acquisition Information
1	☐	Machine001	FanucCNC	192.168.204.101	8193	*****	FDX	-	1	0	-
2	☐	Machine002	FanucCNC	192.168.204.102	8193		FDX	-	2	1-1	-
3	☐	Machine003	FanucCNC	192.168.204.103	8193		FDX	-	0	0	-
4	☐	Machine004	OPCUAServer	-	-		FDX	-	-	-	-
5	☐	Machine005	FanucRobot	192.168.204.105	60008		FDX	R-30B	-	-	6 DO 99
6	☐	Machine006	MTConnectAgent	192.168.204.106	5000		FDX	-	-	-	-
7	☐	Machine007	FanucRobodrill	192.168.204.107	8193		FDX	-	1	3	150
8	☒	Machine008	FanucCNC	192.168.204.108	8193		OK	Series30B	1	3	82
9	☒	Machine009	FanucRobodrill	192.168.204.109	8193		OK	Series31iB	1	8	223
10	☐	Machine010	FanucRobot	192.168.204.110	60008			-	-	-	DO 99

Following are log examples of machine information acquisition process.

Machine type FanucCNC: where "OK" is displayed in the [Connect Result] column

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 53/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

```
Check Machine : Machine001
Machine Type : FanucCNC
IP Address / Host Name : 192.168.0.101
Port No : 8193
Network type ... TCP/IP
Ping executing [192.168.0.101] ... OK
Connected to the CNC ... OK
```

```
< Machine Information >
-- CNC Model = Series30iB
-- Max Path   = 1
-- Laser Paths = 1/3
-- O Number Digits = 4
-- Registration Program = 74
```

```
< Path1 Information >
-- CNC Type    = M
-- Axis        = 3
-- Spindle     = 1
```

Check Machine : Machine002

Check Process finished.

```
Check Machine : Machine001
Machine Type : Unknown
IP Address / Host Name : 192.168.0.101
Port No : 8193
Ping executing [192.168.2.1] ... OK
Connected to the CNC ... NG
Investigate CNC power supply, Ethernet cable and I/F board.[cnc_allclibhnd3(192.168.2.1, 8193, 30, ...)]
(err=-16)(EW_SOCKET)
```

Check Process finished.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description						FANUC CORPORATION		Sheet.	54/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

■ For FanucRobot type machines

No	Target	Machine Name	Machine Type	IP Address / Host Name	Port	Access Code	Connect Result	Model	CNC Paths	Servo Axes	Spindles	Laser Path	Number of Signals	EMG signal
1	☐	Machine001	FanucCNC	192.168.204.101	8193	*****	FDX	-	1	0	0	1	22	-
2	☐	Machine002	FanucCNC	192.168.204.102	8193		FDX	-	2	1-1	1-1	-	96	-
3	☐	Machine003	FanucCNC	192.168.204.103	8193		FDX	-	1	0	0	1	22	-
4	☐	Machine004	OPCUAServer	-	-		FDX	-	-	-	-	-	-	-
5	☐	Machine005	FanucRobot	192.168.204.105	60008			-	-	-	-	-	6	DO 99
6	☐	Machine006	MTConnectAgent	192.168.204.106	5000			-	-	-	-	-	-	-
7	☐	Machine007	FanucRobotdrill	192.168.204.107	8193			-	-	-	-	-	150	-
8	☐	Machine008	FanucCNC	192.168.204.108	8193		OK	Series30B	1	3	1	-	82	-
9	☐	Machine009	FanucRobotdrill	192.168.204.109	8193		OK	Series31iB	1	8	0	-	223	-
10	☒	Machine010	FanucRobot	192.168.204.110	60008			-	-	-	-	-	DO	99

Buttons: Add, Delete, Acquire machine information, Start Easy Setting, Close

NOTE

On setting the emergency stop signal on FANUC Robot controller

If the emergency stop signal of a robot is not set up on the robot controller, the emergency stop status of the robot cannot be retrieved.

Use the System configuration menu on the controller to specify a DO number (an IO number in DO)*¹ to which the emergency stop signal will be output. When an emergency stop occurs (TP external emergency stop, Operation Panel emergency stop), the emergency stop signal will be output to the DO number that is specified on the robot controller.

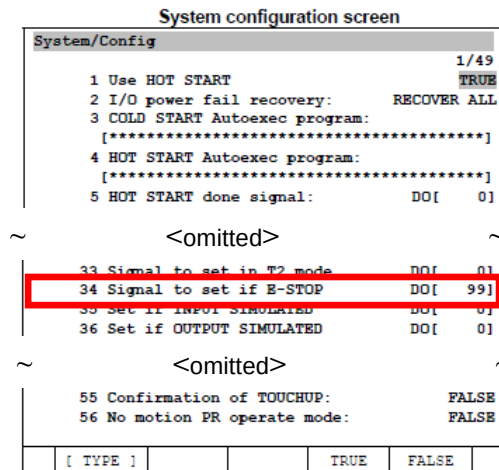
If the DO number is left as the default value ("0"), the emergency stop status cannot be retrieved from the robot (in this case, Easy Setting will not work). To work this function, change the DO number to an appropriate number, power off the robot once, and power on it again.

When assigning a DO number*¹ to the emergency stop signal on a robot controller, make sure that no programs running on the FANUC Robot use it.

(*¹ For robots in R-J3iB or R-J3iB Mate Series, the emergency stop signal is output to an IO number in SDO instead)

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	55/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

The following figure is an example of the [System/Config] screen of a robot controller. This example screen shows the emergency stop signal is set as "DO[99]" on the robot controller.



For the details, refer to the following manuals of the respective robot controllers.

"FANUC Robot series R-30iB/ R-30iB Mate/ R-30iB Plus/ R-30iB Mate Plus CONTROLLER OPERATOR'S MANUAL (Basic Operation) (B-83284EN/05)"

"FANUC Robot series R-30iA CONTROLLER HANDLING TOOL (For series 7DA4/7DA5) OPERATOR'S MANUAL (B-83124EN-2/01)"

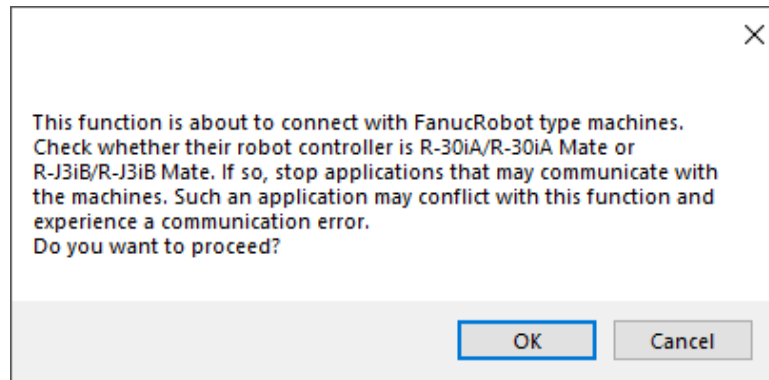
"FANUC Robot series R-30iA Mate CONTROLLER LR HANDLING TOOL (For series 7DA4/7DA5/7DA7) OPERATOR'S MANUAL (B-83134EN-1/01)"

"FANUC Robot series R-J3iB CONTROLLER HANDLING TOOL OPERATOR'S MANUAL (B-81464EN-2/02)"

"FANUC Robot series R-J3iB Mate CONTROLLER LR HANDLING TOOL OPERATOR'S MANUAL (B-81524EN/02)"

- (1) Click the [Add] button to add a row in the machine setting list, and enter the [Machine Name], [Machine Type], [IP Address / Host Name], and [Port] columns. By default, the column is set to 0. Though the address and number of EMG signal (for example, DO99) will be set for the [EMG Signal] column automatically, it is allowed to set it manually if they are verified as correct. When registering two or more machines, repeat this step for each machine.
- (2) Make sure that the [Target] check box is selected, and click the [Acquire machine information] button.
- (3) When the following warning message appears, read the content and click [OK].

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	56/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				



NOTE

Before acquiring machine information from the following FANUC Robot Series machines, stop applications that may communicate with those machines. If such an application is running, it may conflict with this function and experience a communication error:

- R-30iA / R-30iA Mate
- R-J3iB / R-J3iB Mate

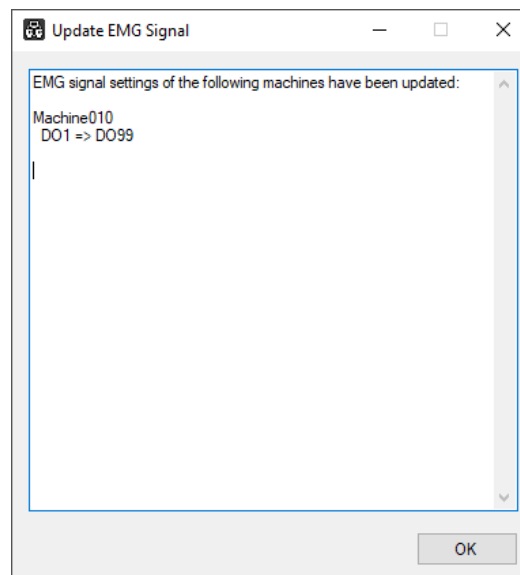
- (4) By clicking [OK] on the confirmation message for the acquisition, the acquisition process begins. During the machine information acquisition process, a progress bar showing the status (<number of completions> / <number of targets>) is displayed.

* In the same manner as with FanucCNC and FanucRobodrill type machines

- (5) When the acquisition of machine information is stopped, or there is a machine displaying "NG" (for "not good") in the [Connect Result] column after the completion of acquisition, an error message is displayed.

* In the same manner as with FanucCNC and FanucRobodrill type machines

- (6) When the acquired EMG signal is different from the value set in the [EMG Signal] column in the [Easy Setting] window, the [Update EMG Signal] dialog box appears as follows:



When acquisition of the IO number of an EMG signal fails, the following error messages will be displayed per errored machine on the [Update EMG Singnal] dialog box. Fix the problem by referring to the relevant countermeasures.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	57/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

However, if the [EMG Signal] column in the [Easy Setting] window is left at the default value 0, Easy Setting will not work any further.

Message	Cause / Countermeasures
<Machine name> This machine does not provide data about the IO number assigned to an EMG signal. Get the DO number or the SDO number assigned to the EMG signal on the robot controller and then set it in the [Easy Setting] window.	This message appears when the DO number or the SDO number assigned to the EMG signal is not acquired. Set the [EMG Signal] column in the [Easy Setting] window to the DO number or the SDO number assigned to the EMG signal by the robot controller (<Machine name>).
<Machine name> The DO number or the SDO number assigned to the EMG signal is invalid. Check the IO settings made on the robot controller.	This message appears when neither the DO number nor the SDO number is assigned to the EMG signal or when the assigned number is invalid (other than 1 to 5000). Check the IO settings made on the robot controller (<Machine name>).

- (7) If "OK" is displayed in the [Connect Result] column, the machine information ([Model], [EMG Signal]), and [Number of Signals]) is displayed, and the acquisition process is completed.

No	Target	Machine Name	Machine Type	IP Address / Host Name	Port	Access Code	Connect Result	Model	CNC Paths	Servo Axes	Spindles	Laser Path	Number of Signals	EMG signal
1	☐	Machine001	FanucCNC	192.168.204.101	8193	*****	FIX	-	1	0	-	-	-	-
2	☐	Machine002	FanucCNC	192.168.204.102	8193		FIX	-	2	1-1	-	-	-	-
3	☐	Machine003	FanucCNC	192.168.204.103	8193		FIX	-	1	0	-	-	-	-
4	☐	Machine004	OPCUAServer	-	-		FIX	-	-	-	-	-	-	-
5	☐	Machine005	FanucRobot	192.168.204.105	60008		FIX	R-30B	-	-	-	-	-	99
6	☐	Machine006	MTConnectAgent	192.168.204.106	5000		FIX	-	-	-	-	-	-	-
7	☐	Machine007	FanucRobodrill	192.168.204.107	8193		FIX	-	1	3	1	-	150	-
8	☐	Machine008	FanucCNC	192.168.204.108	8193		OK	Series30B	1	3	1	-	82	-
9	☐	Machine009	FanucRobodrill	192.168.204.109	8193		OK	Series31B	1	8	0	-	223	-
10	☒	Machine010	FanucRobot	192.168.204.110	60008		OK	R-30B	-	-	-	-	6	DO 99

- (8) For the machines with "NG" displayed in the [Connect Result] column, click the [Log] tab to check the acquisition result and the details of the error, and review or delete the settings.

Following is a log example of machine information acquisition process.

Machine type FanucRobot: where "OK" is displayed in the [Connect Result] column

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 58/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

```

Check Machine : Machine002
Machine Type : FanucRobot
IP Address / Host Name : 192.168.0.102
Port No : 60008
Network type ... TCP/IP
Ping executing [192.168.0.102] ... OK
Connected to the Machine ... OK

```

Acquisition of Machine Information

```
-- Connected Robot = R-30iB
-- Connected Robot Version = V9.10204
-- EMG Signal = DO3 (Previous Value : 0)
```

Acquisition of EMG Signal

```
[0001] -- BATAL1 ... OK (Value:False, Elapse:0 ms)
[0002] -- Mode ... OK (Value:Auto, Elapse:0 ms)
```

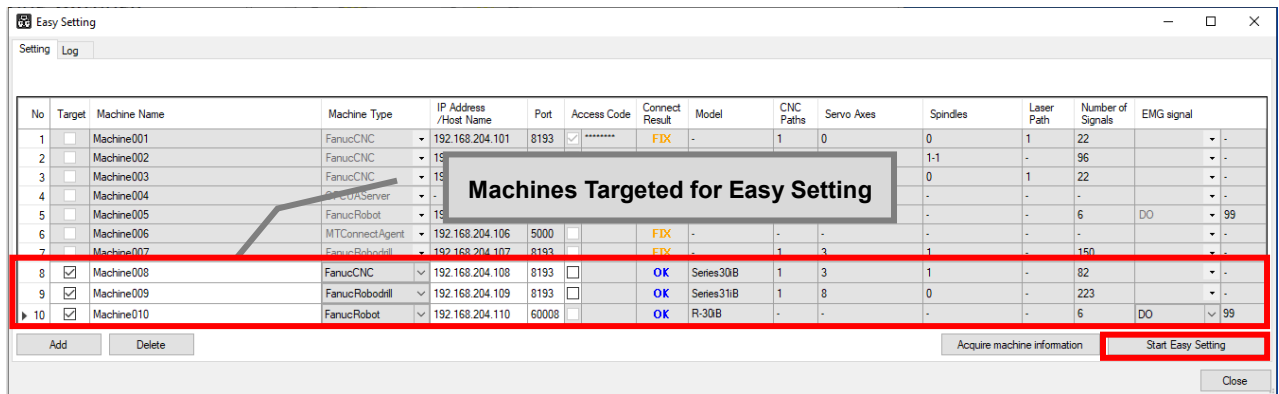
Acquisition of Signal

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					FANUC CORPORATION			Sheet.	59/868
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

3.1.5 Start Easy Setting

In this subsection, the steps to start Easy Setting are explained.

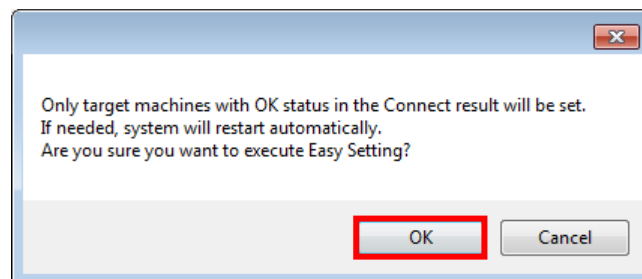
For the machines with "OK" displayed in the [Connect Result] column, settings to browse the Web screen can be set with Easy Setting.



- (1) Among the machines with "OK" displayed in the [Connect Result] column, make sure that the [Target] check box is selected for the machine intended to perform Easy Setting, and click the [Start Easy Setting] button.

Easy Setting will not be performed for the machine which has "NG" displayed in the [Connect Result] column, even if the [Target] check box is selected.

- (2) By clicking the [OK] button in the confirmation message, Easy Setting begins.
During the Easy Setting process, MT-LINK*i* system will be restarted automatically, if required.



- (3) The display switches to the [Log] tab on the [Easy Setting] window, and the log of the Easy Setting process is displayed.

Following processes are executed by Easy Setting.

- 1) Create settings for MT-LINK*i*
- 2) The display in the [Connect Result] column on the machine setting list changes from "OK" to "FIX"
- 3) Send to DB
- 4) Restart Receiver service
- 5) Restart Exporter service
- 6) Restart Web Server service

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 60/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

Following processes are executed after this when the Server PC and the Collector PC are on the same PC (Collector is installed in the Server PC).

- 7) Create settings for the Collector
 - * If SERVO VIEWER Linkage setting is enabled, data for SERVO VIEWER Linkage is created also.
- 8) Restart Collector service
 - * If SERVO VIEWER Linkage setting is enabled, Collector service for SERVO VIEWER Linkage is restarted also.

NOTE

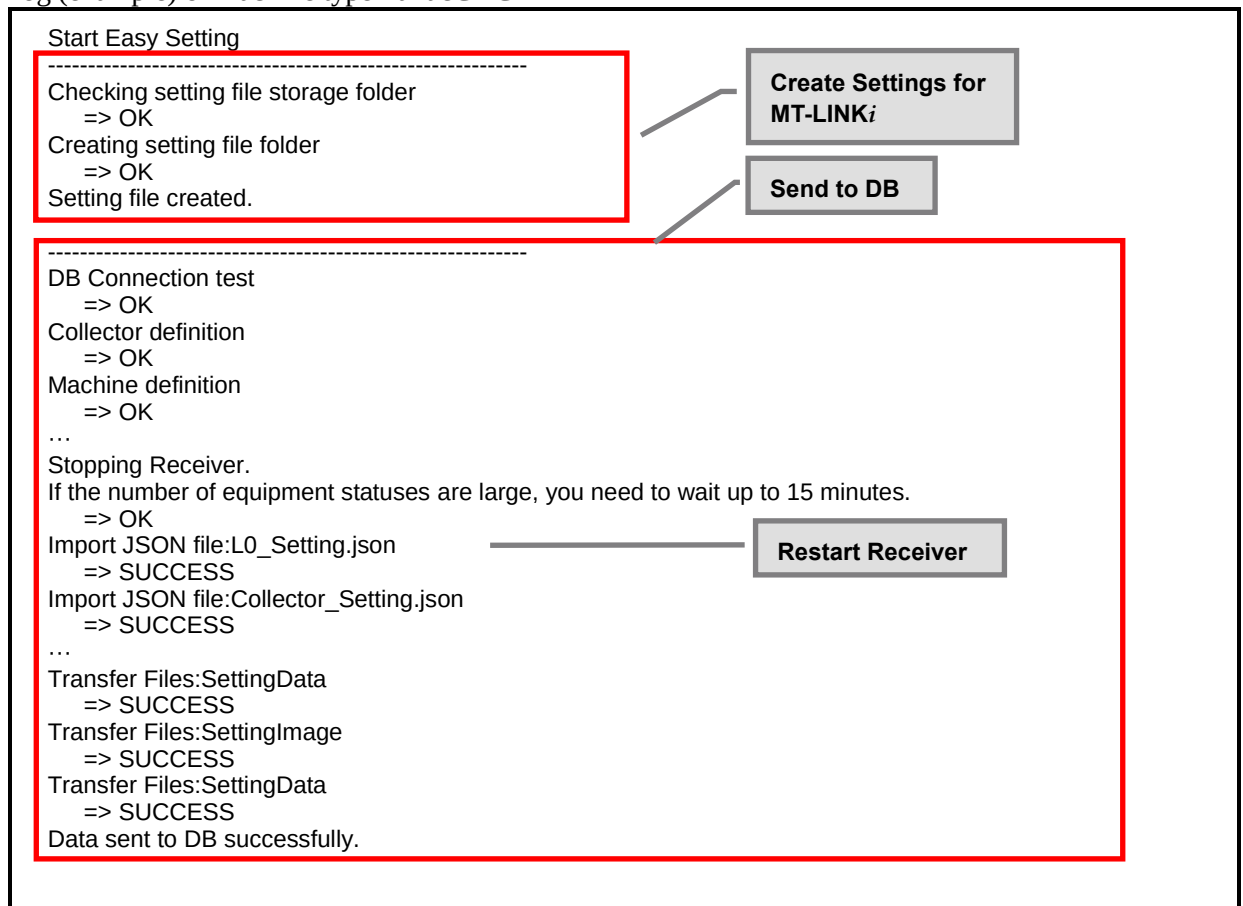
When an error occurs during the Easy Setting process, the process stops. After referring to the log and checking the details of the error, execute the Easy Setting process again.

If the settings are already created, the process for creating the settings will not be executed.

- (4) When "Easy Setting completed" is displayed on the last line of the log, the Easy Setting process is completed.

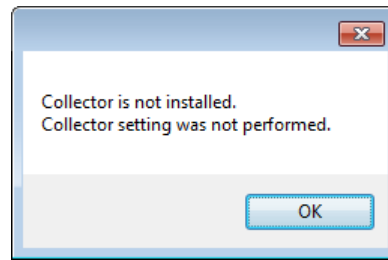
Following is an output example for the log of the Easy Setting process.

Log (example) of machine type FanucCNC



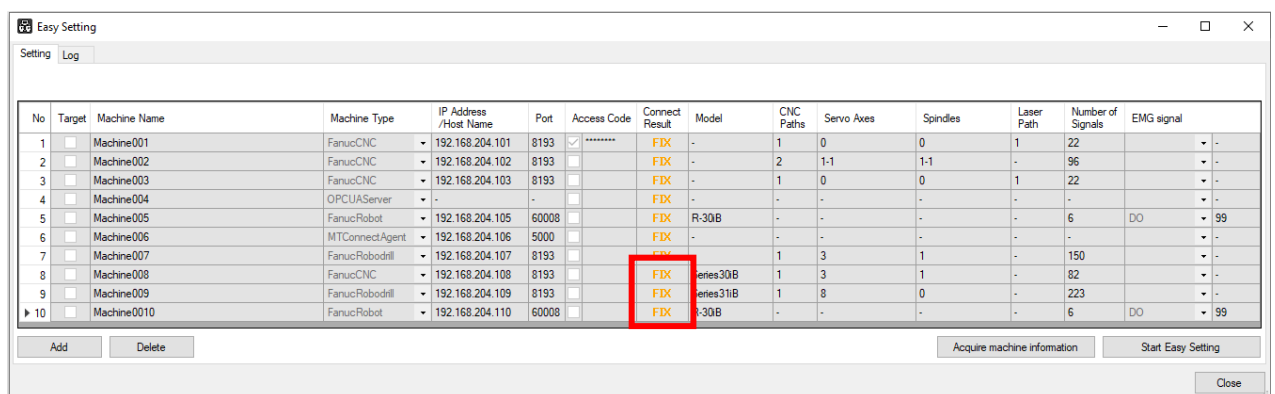
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description						FANUC CORPORATION		Sheet.	61/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

Following message will be displayed if the Collector is not installed on the Server PC.

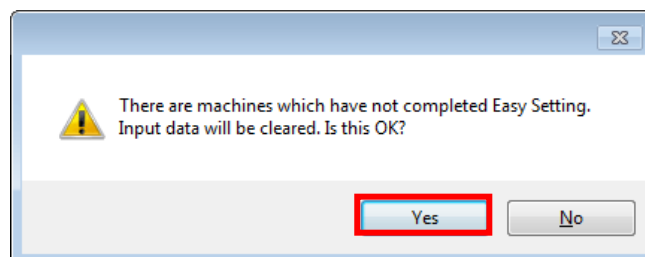


In such a case, Collector setting is not performed. Follow the steps below.

- 1) Install the Collector on the Server PC.
 - 2) On the [Setting] tab of the [Easy Setting] window, click the [Start Easy Setting] button. Collector setting is performed, and the Collector service is restarted.
- (6) Click the [Setting] tab, and confirm that the display in the [Connect Result] column is changed from "OK" to "FIX."



- (7) Click the [Close] button to close the [Easy Setting] window.
- When there are any machine settings with the [Connect Result] other than "FIX," a confirmation message is displayed. By clicking the [OK] button, the entry of the machine settings with the [Connect Result] other than "FIX" will be cleared.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 63/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

3.1.5.1 Descriptions of the Settings Created by Easy Setting

Following are the descriptions of the settings created by Easy Setting.

Settings for MT-LINK*i*

Setting	Description	Remarks
Collector setting	Collector setting is created with the following settings. - Collector name: Computer name of the Server PC (see NOTE) - IP address / host name: IP address of the Server PC	If the Collector setting on the left already exists, it will not be created.
Machine setting	Machine setting is created with the following settings. - Machine name, IP address / host name, port entered All the machines are assigned in the above Collector setting.	-
Signal setting	Signal setting is created with the following settings. <u>For FanucCNC or FanucRobodrill type machines</u> - Basic signals: Created based on the CNC path number / servo axes number / spindle axes number of the machine information acquired - Laser control data signals: Create based on the laser path of the machine information acquired <u>For FanucRobot type machines</u> - Basic signals: Created based on the machine information acquired - Custom signal (EMG signal): Created based on the machine information acquired	Among the signals which are set to be collected, only the signals that can be acquired are set as enabled. However, mandatory signals are set as enabled regardless of the availability.
Equipment setting	For each machine added in the Easy Setting, equipment setting is created as one machine to one equipment unit. - Equipment name: Created with the same name as the machine name (If an equipment unit with the same name is already registered, created with "<sequential number>" added in the end)	Cannot be set if the number of equipment units including the ones existing exceeds the maximum number (100).
Group setting	When the group is unregistered, the group is registered with the following setting. - Group name: "NewGroup0001" All the equipment units added with the above equipment setting is registered to the group displayed as the first group on the group tree.	-
Layout setting	If the layout with automatic alignment is not registered, a layout with automatic alignment is created with the following setting. - Layout name: "Layout<sequential number>"	Not created if the number of registered layouts is the maximum number (20).
Authority setting	When the role is unregistered, role is created with default setting. - Role name: "NewRole0001" If the user is unregistered, user is created with the role setting displayed on the top of the role list. - User name: "NewUser0001"	-

Settings for the Collector

Setting	Description	Remarks
Collector setting	Created when the Server PC and the Collector PC are installed on the same PC (i.e., Collector is installed in the Server PC) (see NOTE). The settings of the Collector on the Server PC, and the machines under the Collector are created.	-
Machine setting		
System setting		

NOTE

When the Collector is installed in the Server PC, the name of the Server PC must not include multi-byte characters. Also, its length is up to 32 bytes.
It must start with an alphanumeric character and the subsequent characters can be alphanumerics, hyphens ("-") and underscores ("_").

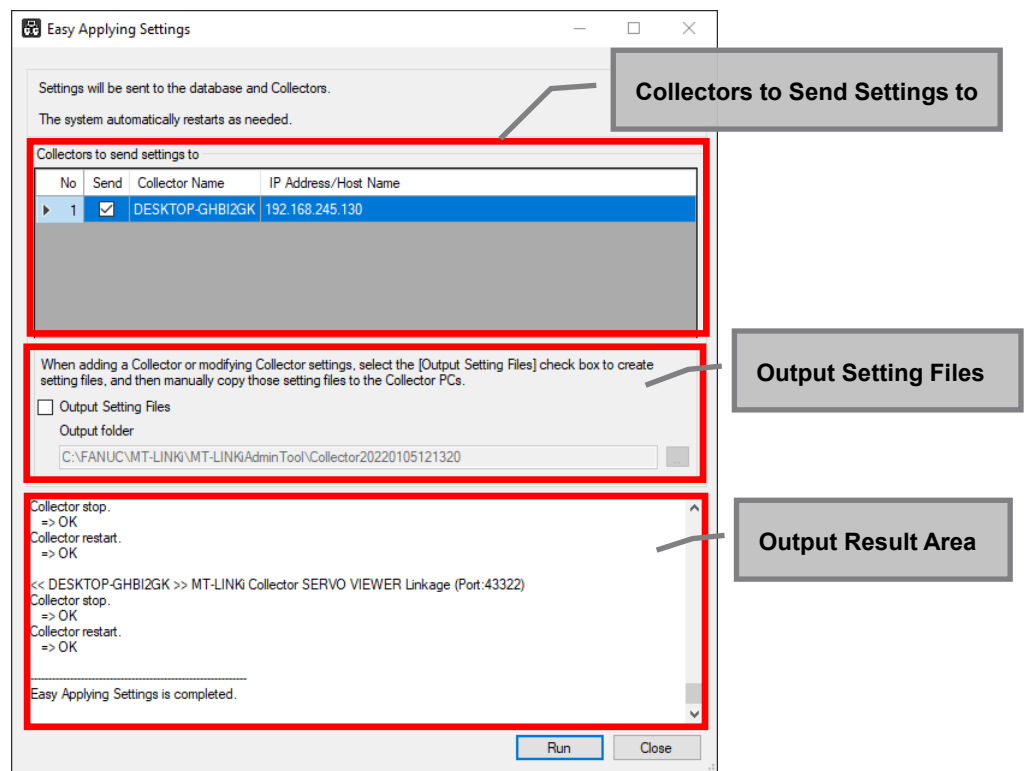
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION				Sheet. 64/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

3.2 EASY APPLYING SETTINGS

Easy Applying Settings provides a way to send setting data changed with AdminTool to the database and Collectors at once, and automatically starts the service of the MT-LINK*i* system.

Easy Applying Settings enables users to perform at once the following operations required for applying setting data to the system and Collectors:

- [Applying Settings] - Application of Settings - [Send to DB]
- [Applying Settings] - Application of Settings - [Create Collector Setting Files]
- [Controlling System] – System Control - [Start/Stop System]



■ Perform the following steps to use Easy Applying Settings

- (1) In the [Collectors to send settings to] group box of the [Easy Applying Settings] window, select the [Send] check boxes for Collectors to which to send their Collector setting files.
- (2) Click the [Run] button to start an easy applying settings process.
- (3) Verify the [Output results area] field. If the field displays "OK" for all tasks in the process, it is completed successfully.

- * If adding a Collector or modifying existing Collector settings, select the [Output Setting Files] check box and the manual copy operation of the output file is required.
For the details, refer to the "[Apply Created AdminTool Setting to Collector](#)" section.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 65/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

■ **The [Easy Applying Settings] window**
Collectors to send settings to

- Send
Select this check box for each Collector to send Collector setting files to.
- Collector Name
This column displays the computer names of Collector PCs.
- IP Address/Host Name
This column displays the IP addresses or host names of Collector PCs.

Output Setting Files

- Output Setting Files
Use this check box to specify whether to create and output Collector setting files.
- Output folder
Specify a folder to output Collector setting files to.

NOTE

When adding a Collector or modifying existing Collector settings, select the [Output Setting Files] check box to create setting files, and then manually copy those setting files to the Collector PCs by performing the following steps:

- 1) Copy all the folders (including their subfolders and files) under each folder that resides under the folder specified for the [Output folder] field and has the name of a Collector whose [Send] check box is selected in the [Collectors to send settings to] group box.
- 2) For each Collector with its [Send] check box selected in the [Collectors to send settings to] group box, store its Collector setting files, which should be copied in the step 1, into the following folder on the Collector PC corresponding to the Collector. If files already reside in the folder, overwrite them with the copied files:
 <MT-LINK*i* Collector installation folder>\Setting
- 3) Restart those Collector PCs or restart the following services on them to apply the updated settings:
 - MT-LINK*i* Collector
 - MT-LINK*i* Collector SEVO VIEWER Linkage
 (* when the [Enabled] check box is selected in the [SERVO VIEWER Linkage Setting] group box in the [Machine Setting] window of AdminTool)

For the details on how to restart those services, refer to the "**START/STOP SYSTEM**" section.

Output results area

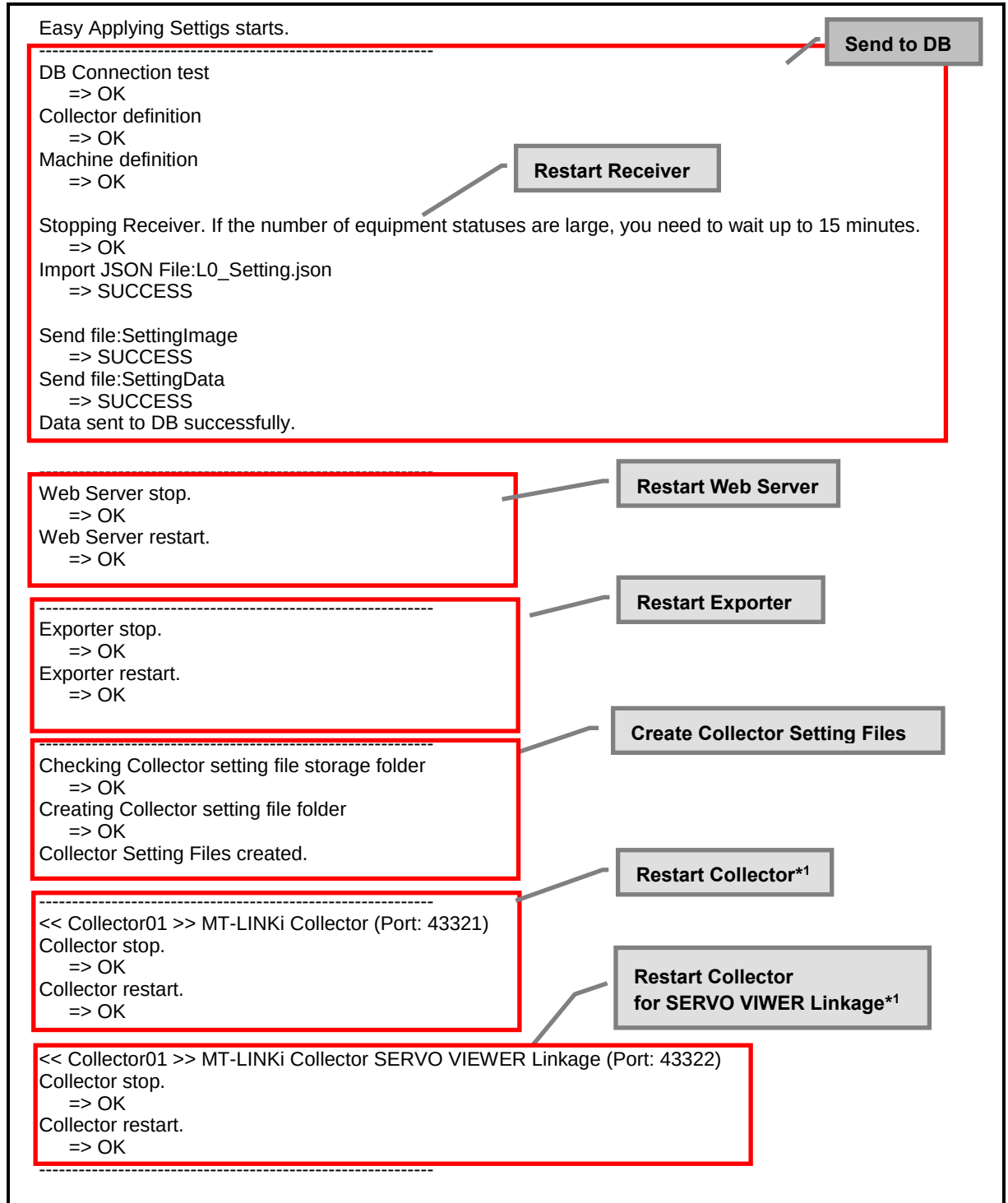
This area displays execution logs of Easy Applying Settings.

Buttons

- Run
Clicking this button execute an easy ppplying settings process.
- Close
Clicking this button closes the [Easy Applying Settings] window.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	66/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

■ Example logs after Easy Applying Settings is completed successfully



*1 when an identical computer is used as the Server PC and the Collector PC

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	67/868	
Ed.	Date	Design	Description									
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.						Inoue

Example logs after Easy Applying Settings is completed successfully (continued from the previous page)

```

-----
<< Collector02 >> MT-LINKi Collector (Port:43321)
Network check
=> OK
Stop Request
=> OK
Collector stop -- Disconnecting. [1/1]
Collector stop -- Finalize.
Collector stop.
=> OK
Send file: Collector_Setting.json
=> OK
Ssnd file: L0_Setting.json
=> OK
Send file: System.json
=> OK
Collector restart.
=> OK

```

Restart Collector*2

**Restart Collector
for SERVO VIEWER Linkage *2**

```

-----
<< Collector02 >> MT-LINKi Collector SERVO VIEWER Linkage (Port:43322)
Network check
=> OK
Stop Request
=> OK
Collector stop -- Finalize
Collector stop -- Finalize
Collector stop.
=> OK
Collector restart.
=> OK

```

Easy Applying Settigs is completed.

*2 when separated computers are used as the Server PC and the Collector PC

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 68/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

4

[SETTING DATA] - SPECIFIC DATA SETTING

This chapter describes how to set specific data in AdminTool.

The AdminTool Menu has the following menu items in the [Setting Data] - [Specific Data Setting] group:

(1) Machine Setting

This menu registers Collectors and machines of a variety of types, and sets up signals to retrieve from them. Machines of the following machine types are supported:

- FanuCNC
- OPCUAServer
- FanucRobot
- MTConnectAgent
- FanucRobodrill

(2) Equipment Setting

This menu assigns machines to equipment to connect with and sets equipment icons and statuses that the [Monitoring - Equipment Monitoring] Web screen displays.

(3) Group Setting

This menu puts equipment in groups and sets group icons and group statuses that the [Monitoring - Group Monitoring] Web screen displays.

(4) Layout Setting

This menu provides a way to customize layouts and the background image of equipment to display on the [Monitoring - Overview] Web screen.

(5) Manual Data Setting

This menu sets plans, product names and work comments to complement history information that the [Results - Production Results] Web screen displays.

(6) Authority Setting

This menu sets user accounts and view permission of Web screens, and configures LDAP authentication function.

(7) Sign-in Setting

This menu sets the title image and additional information that the sign-in Web screen to display and notify to Web users.

(8) Asset Management

This menu sets machine specific information and images to display on the [Monitoring - Equipment - Monitoring] Web screen.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION				Sheet. 69/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

NOTE

- 1 Web names, set in setting windows, including the [Machine Setting], [Equipment Setting], and [Group Setting] windows, are names to be displayed on the Web screens.

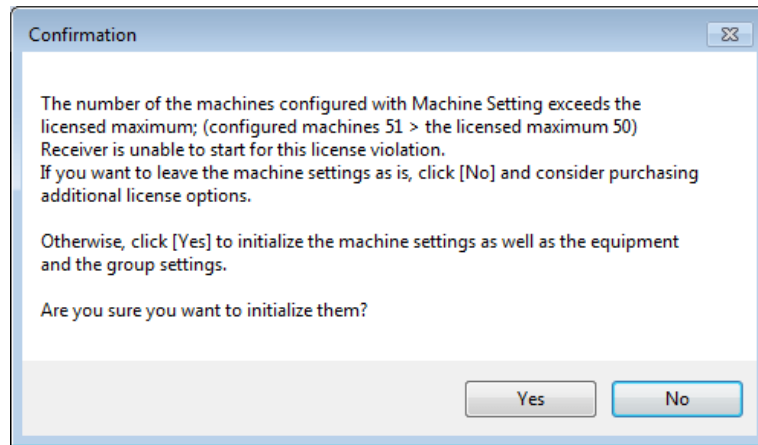
You can specify Web names in the language that you will set in the [Web name language] in the [System Setting] window. If needed, you can change the language in which Web names are displayed. For the details, refer to the "[Change Language for Web Names](#)" subsection.

- 2 When the number of setting items is considerably large, the setting process may fail. If the number of equipment units/axes is very large, change the machine names, Web names, etc. to shorter names.

For the details, refer to the following subsection of the "FANUC MT-LINK*i* / Integration Server Troubleshooting MANUAL":

The "Error Occurs Notifying that the Data Amount is Excessive" subsection in the "TROUBLES RELATED TO ADMINTOOL SETTING" section in the "MT-LINK*i* TROUBLESHOOTING" chapter

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	70/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	



- 3 The number of equipment units and groups that can be registered is 100 in maximum regardless of license option.
- 4 When purchasing a Function option type license, the user must purchase one or more Machine number Options as well.

- To select machines and register their settings. The following five machine types are available. For the details on the Web screens supported for each machine type, refer to the "FANUC MT-LINK*i* Startup MANUAL."
 - FanucCNC
 - OPCUAServer
 - FANUC Robot
 - MTConnectAgent
 - FanucRobodrill
- To apply the following setting functionalities to FanucCNC type machines:
 - Machine settings, such as machine name ([Add Machine])
 - Network settings, such as IP address/host name ([Protocol Setting])
 - Setting basic signals ([Set Machine Signal])
 - Defining custom signals ([Custom Signal])
 - Setting laser control data signals ([Laser control data signal setting])
 - Setting tool life signals ([Set the Number of Tool Groups]) (see NOTE)
 - Setting conditions when and how to get macro values ([Macro Variable Getting Condition Definition])
 - Setting conditions when and how to get variable history ([Variable History Retrieval Condition Definition])
 - Checking whether the signals that are set can be acquired or not

NOTE

Update Option license is required to set up tool life signals of FanucCNC type machines.

- To apply the following setting functionalities to OPCUAServer type machines:
 - Configuring end-points and security for communication with OPC UA Servers
 - Assigning a machine with a tag and methods managed on an OPC UA Server
 - Associating signals with tags and methods
 - Associating signals with alarms retrieved from OPC UA Servers ([DA Alarm Setting])

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	72/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- To apply the following setting functionalities to FanucRobot type machines:
 - Machine settings, such as machine name ([Add Machine])
 - Network settings, such as IP address/host name ([Protocol Setting])
 - Setting basic signals ([Set Machine Signal])
 - Defining custom signals to retrieve I/O signals and numeric/string register data ([Custom Signal])
 - Setting conditions when and how to get variable history ([Variable History Retrieval Condition Definition])
 - Setting position register data to retrieve ([Position Register Setting])
- To apply the following setting functionalities to MTConnectAgent type machines:
 - Searching for MTConnect Agent and selecting the Devices
 - Confirming IP address/host name and other protocol settings, and setting time out
 - Assigning Device information to signals
- To apply the following setting functionalities to FanucRobodrill type machines:
 - Machine settings, such as machine name ([Add Machine])
 - Network settings, such as IP address/host name ([Protocol Setting])
 - Setting basic signals ([Custom Signal])
 - Setting ROBODRILL specific signals ([Set Machine Signal])
 - Defining custom signals ([Custom Signal Setting])
 - Setting conditions when and how to get macro values ([Macro Variable Getting Condition Definition])
 - Setting conditions when and how to get variable history ([Variable History Retrieval Condition Definition])

NOTE

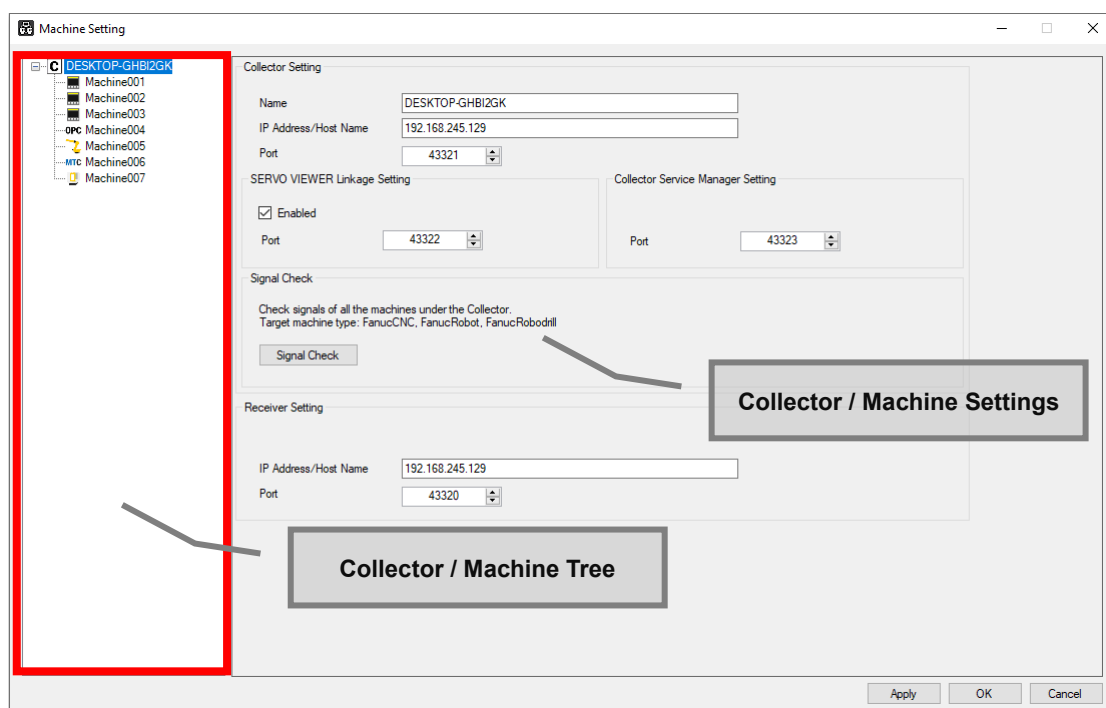
ROBODRILL Option license is required to support FanucRobodrill type machines.

NOTE

When signals are added/deleted in the machine setting, use the [Reload] or [Preset] button on the [Equipment Setting] window, and the [Send to DB] menu in the AdminTool Menu to store those changes in the database. For details of the Preset function, refer to the "[Set Equipment Status](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	73/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

To make machine settings, select [Machine Setting] in the AdminTool Menu. The [Machine Setting] window appears. It has the Collector/Machine tree in its left pane. Once you select a Collector or machine in the tree, its setting values are populated on the right pane.



The Collector/Machine tree displays Collectors and machines each with its type icon placed on its left side. The following table lists type icons for Collector and machines.

Item type	Machine type	Type icon
Collector	-	
Machine	FanucCNC	
	OPCUAServer	
	FanucRobot	
	MTConnectAgent	
	FanucRobodrill	

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	74/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

4.1.1 Steps for Machine Setting

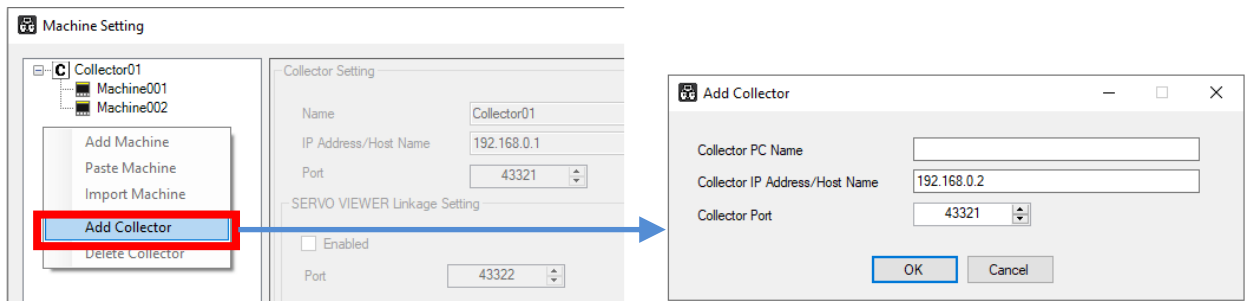
Follow the steps below.

- (1) Select [Machine Setting] in the AdminTool Menu. This displays the [Machine Setting] window.
- (2) Add Collector. For the details, refer to the "[Add Collector](#)" subsection.
- (3) Perform detailed settings of the machine.
- (4) Add machines to be set.
- (5) Set protocol/connection setting for the machines.
- (6) Set signal settings for the machines.
- (7) Check whether the signals that are set can be acquired or not. For the details on the checking methods, refer to the "[Checking the Signals](#)" subsection.
- (8) Save the information about machine settings, and apply them to the system.

4.1.2 Add Collector

- (1) Add Collector

Move the mouse cursor to the Collector tree, right-click it and select [Add Collector].
The [Add Collector] window appears. Set the settings of a Collector PC to add.



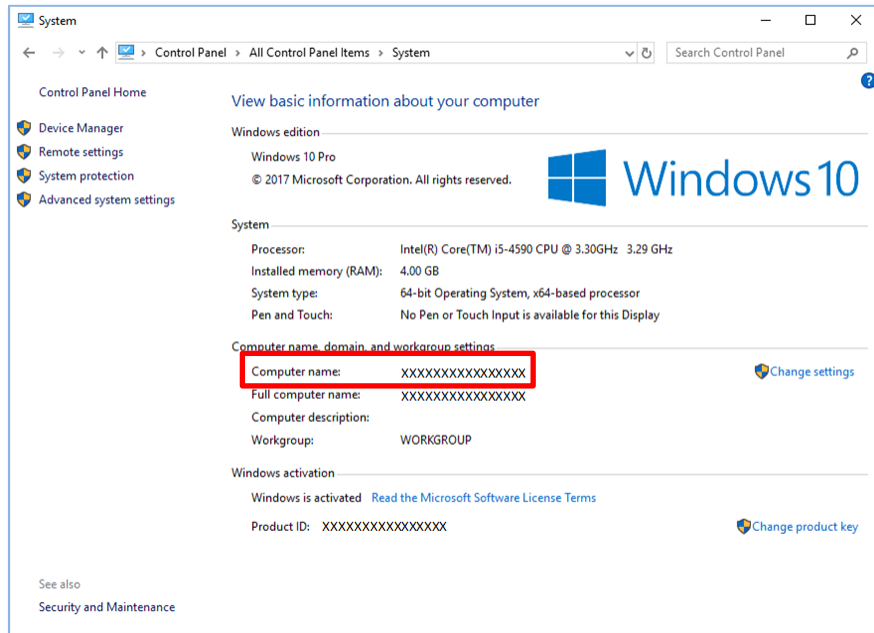
■ The [Add Collector] window

- **Collector PC Name**
Set a name of the Collector PC to this field (See NOTE). Collector PC Name can start with a half-width alphanumeric character and the subsequent characters can be any combination of half-width alphabets, numerics, hyphens ("-") and underscores ("_"). This field can be up to 32 bytes long.
- **Collector IP Address/Host Name**
Set an IP address or the host name of the Collector PC.
When entering the host name, the name can start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and periods ("."). This field can be up to 253 bytes long.
- **Collector Port**
Set a port number for the Collector PC. The port number should be between 1024 and 65535. The default value is set as "43321." If this default value is used in other application, change the setting.
- **The [OK] button**
Verify the settings and close the window. Clicking this button shows a new Collector in the Collector/Machine tree.
- **The [Cancel] button**
Discard the settings and close the window.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 75/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

NOTE

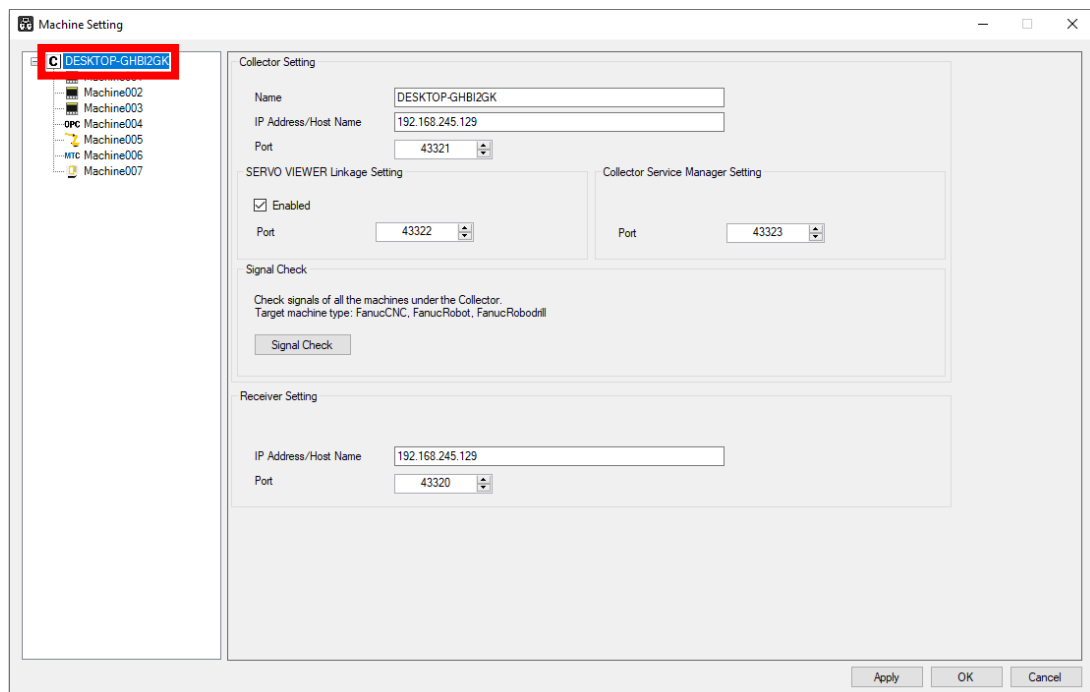
To display the computer name for Collector PC, click [Control Panel] > [System]. Following figure shows an example of the [System] window in Windows 10.



When saving machine settings after adding a Collector, you will receive a warning. For the details, refer to the "**Save Machine Setting**" subsection.

(2) Confirm and edit Collector Setting

The settings of the selected Collector or machine are populated in the right pane of the [Machine Setting] window. Selecting Collector name in the Collector tree will show you Collector setting and Receiver setting, so that you can set and change them.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	76/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- Name

- Collector IP Address/Host Name

When entering the host name, the name can start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and periods (".").

- Port

The default value is set at "43321." If this default value is used for other application, change the setting.

- Enabled

- Port

NOTE

- Port

- [Signal Check] button

NOTE

- | | | | | | | | | | | | | |
|------|---------------|--------|--|-------|--|--------|-------|-------------------|---|--|--------|--------|
| 20 | Nov. 22, 2023 | uwano | Revised for MT-LINK <i>i</i> Version 4.0 | | | | | Title | FANUC MT-LINK <i>i</i>
AdminTool OPERATOR'S MANUAL | | | |
| 19 | Apr. 27, 2021 | uwano | Revised for MT-LINK <i>i</i> Version 3.11 | | | | | | | | | |
| 18 | July 28, 2020 | uwano | Revised for MT-LINK <i>i</i> Version 3.10 | | | | | | | | | |
| 17 | Jan. 30, 2020 | uwano | Revised for MT-LINK <i>i</i> Version 3.9 | | | | | Draw | A-40829-00001EN | | | |
| 16 | July 22, 2019 | uwano | Revised for MT-LINK <i>i</i> Version 3.8 | | | | | | | | | |
| 15 | Mar. 29, 2019 | ogawa | Revised for MT-LINK <i>i</i> Version 3.7.2 | | | | | | | | | |
| 14 | Jan. 29, 2019 | ogawa | Revised for MT-LINK <i>i</i> Version 3.7 | | | | | FANUC CORPORATION | | | Sheet. | 77/868 |
| Ed. | Date | Design | Description | | | | | | | | | |
| Date | Nov.30.2015 | Desig. | Ogawa | Check | | Apprv. | Inoue | | | | | |

■ Receiver Setting

- IP Address/Host Name

Set IP address or the host name of Server PC.

When entering the host name, the name can start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and periods (".").

This field can be up to 253 bytes long.

- Port

Set port number for Server PC. The port number should be between 1024 and 65535.

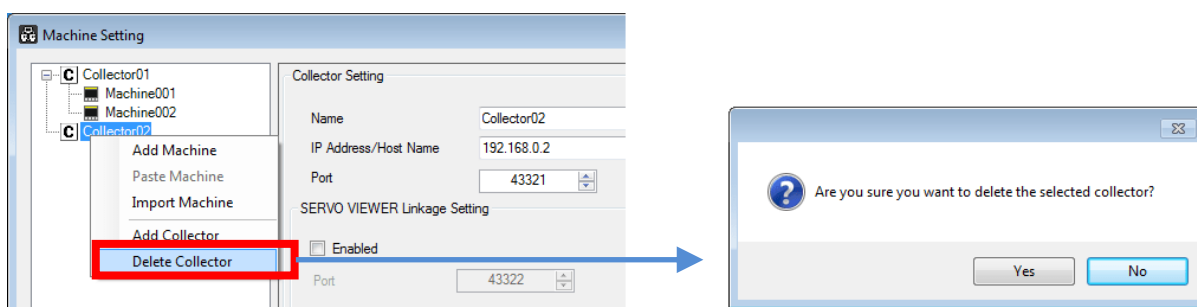
The default value is set to "43320." If this default value is used for other application, change the setting.

NOTE

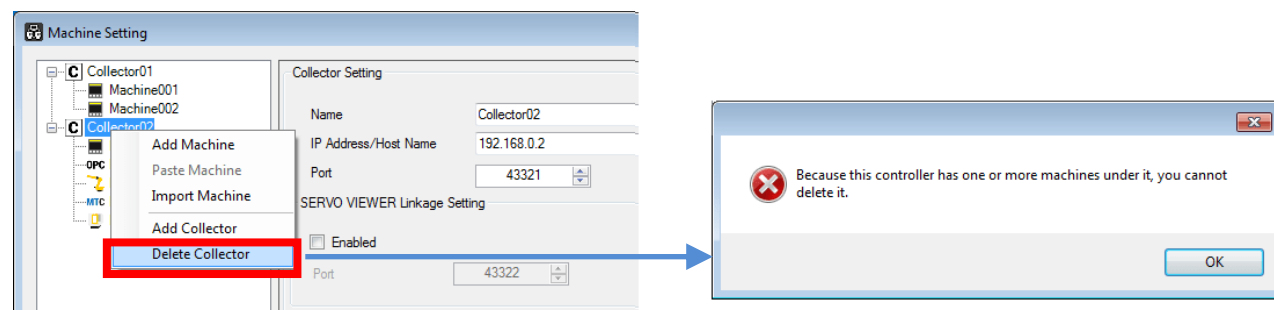
When saving machine settings after adding a Collector, you will receive a warning.
For the details, refer to the "[Save Machine Setting](#)" subsection

(3) Delete Collector

Move cursor on to the Collector tree, right-click the target Collector and select [Delete Collector].



Note that when the Collector has machines registered under it, it cannot be removed. To delete such a Collector, delete all the machines registered under it.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	78/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

4.1.3 Detailed Setting of Machine

(1) Detailed Setting of Machine

Perform detailed setting of the machine.

As for the details of the setting process by machine type, refer to the following.

■ Detailed Settings by Machine Type

For the details on setting of FanucCNC type machines, refer to the "**MACHINE SETTING - FanucCNC**" section.

For the details on setting of OPCUAServer type machines, refer to the "**MACHINE SETTING - OPCUAServer**" section.

For the details on setting of FanucRobot type machines, refer to the "**MACHINE SETTING - FanucRobot**" section.

For the details on setting of MTConnectAgent type machines, refer to the "**MACHINE SETTING - MTConnectAgent**" section.

For setting of FanucRobodrill type machines, refer to the "**MACHINE SETTING - FanucRobodrill**" section.

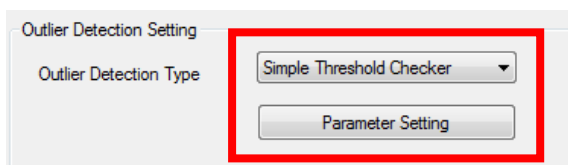
NOTE

The steps for adding the machine, setting the protocol and connection, and setting the signals are different for each machine type.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	79/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

4.1.4 Set Outlier Detection of Signal

Outlier detection can be set for basic signals, custom signals, laser control data signals, and tool life signals. This functionality works only for the signals whose [Outlier Detection] check boxes in the [Machine Setting] window are selected (and whose data types are Int32, UInt32, or Double). By setting the outlier detection, MT-LINK*i* can acquire outlier detection value of the filter and the decision result, besides the original value of the signal.



Follow the steps below for the setting.

- (1) Follow the steps below to display the [Outlier Detection Setting] dialog box.
 - Signal setting for FanucCNC
Select the [Basic Signal] tab, and double-click the rows with their [Outlier Detection] check box selected. For the details, refer to the "[Set Signal of FanucCNC](#)" subsection.
 - Custom signal setting for FanucCNC
Select the [Custom Signal] tab, and click the [Add] or [Edit] button to open the [Signal Setting] dialog box. For the details, refer to the "[Set Custom Signal of FanucCNC](#)" subsection.
 - Laser control data signal setting for FanucCNC
Select the [Laser control data signal setting] tab, and double-click the rows with their [Outlier Detection] check box selected. For the details, refer to the "[Set Laser Control Data Signal of FanucCNC](#)" subsection.
 - Tool life signal setting for FanucCNC
Select the [Tool Life Signal] tab, and double-click the rows with their [Outlier Detection] check box selected. For the details, refer to the "[Set Tool Life Signal of FanucCNC](#)" subsection.
 - Signal setting of OPCUA Server
After selecting the machine, double-click the rows with their [Outlier Detection] check box selected in the signal setting area on the right part of the [Machine Setting] window. For the details, refer to the "[Set Signal of OPCUAServer](#)" subsection.
 - Custom signal setting for FanucRobot
Select the [Custom Signal] tab, and click the [Add] or [Edit] button to open the [Signal Setting] dialog box. For the details, refer to the "[Set Custom Signal of FanucRobot](#)" subsection.
 - Position register setting for FanucRobot
Select the [Position Register] tab, and double-click the rows with their [Outlier Detection] check box selected in the [Position Register Signal Details] dialog box. For the details, refer to the "[Set Position Register of FanucRobot](#)" subsection.
 - Signal setting of MTConnectAgent
After selecting the [Signal Setting] tab, click the [Setting] button to open the [Signal Setting] dialog box. On the [Signal Setting] dialog box, double-click the rows with their [Outlier Detection] check box selected. For the details, refer to the "[Set Signal of MTConnectAgent](#)" subsection.
 - Signal setting for FanucRobodrill
Select the [Basic Signal] tab, and double-click the rows with their [Outlier Detection] check box selected. For the details, refer to the "[Set Signal of FanucCNC](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	80/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

- Custom signal setting for FanucRobodrill
Select the [Custom Signal] tab, and click the [Add] or [Edit] button to open the [Signal Setting] dialog box. For the details, refer to the "[Set Custom Signal of FanucCNC](#)" subsection.
 - Robodrill signal setting for FanucRobodrill
Select the [Robodrill Signal Setting] tab, and double-click the rows with their [Outlier Detection] check box selected. For the details, refer to the "[Set FanucRobodrill Specific Signal](#)" subsection.
- (2) Select [Simple Threshold Checker] for the outlier detection type shown on the [Outlier Detection Setting] dialog box, and click the [Parameter setting] button.
 - (3) Enter the value of parameter in the [Simple Threshold Checker Parameter Setting] dialog box, and click [OK] to complete parameter setting.

■ The [Simple Threshold Checker Parameter] dialog box

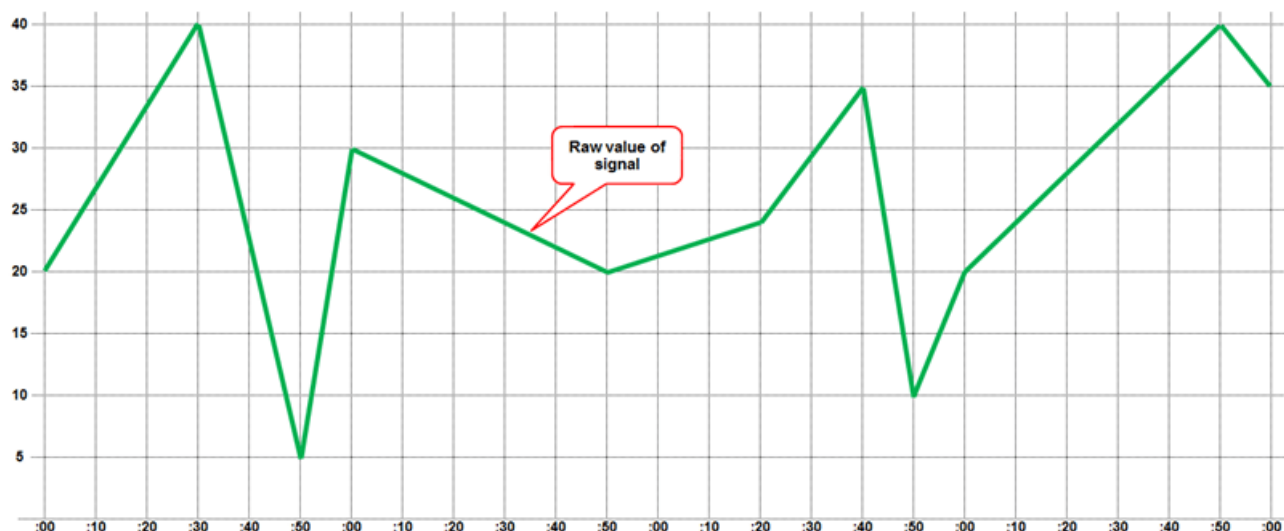
This is a window for setting the Simple Threshold parameters.

- Warning range
Set upper and lower limit values of the warning threshold. When the value exceeds the threshold, it is determined as a warning.
If not determining the warning range, select the [No setting] check box.
However, both upper and lower limit values cannot be set as [No setting].
- Error range
Set upper and lower limit value of the error threshold. When the value exceeds the threshold, it is determined as an error.
If not determining the error range, select the [No setting] check box.
However, both upper and lower limit values cannot be set as [No setting].
- Number of smoothing data
Set number of data used for smoothing the graph.
When determining directly with Simple threshold, it might become too sensitive to noise. For this reason, the number of smoothing data is set to level certain data, and to enable reduction of the noise effect. The average value for the historical data delimited into specified number of pieces is plotted.
If the value for the [Number of smoothing data] is set as "1" smoothing will not be executed.

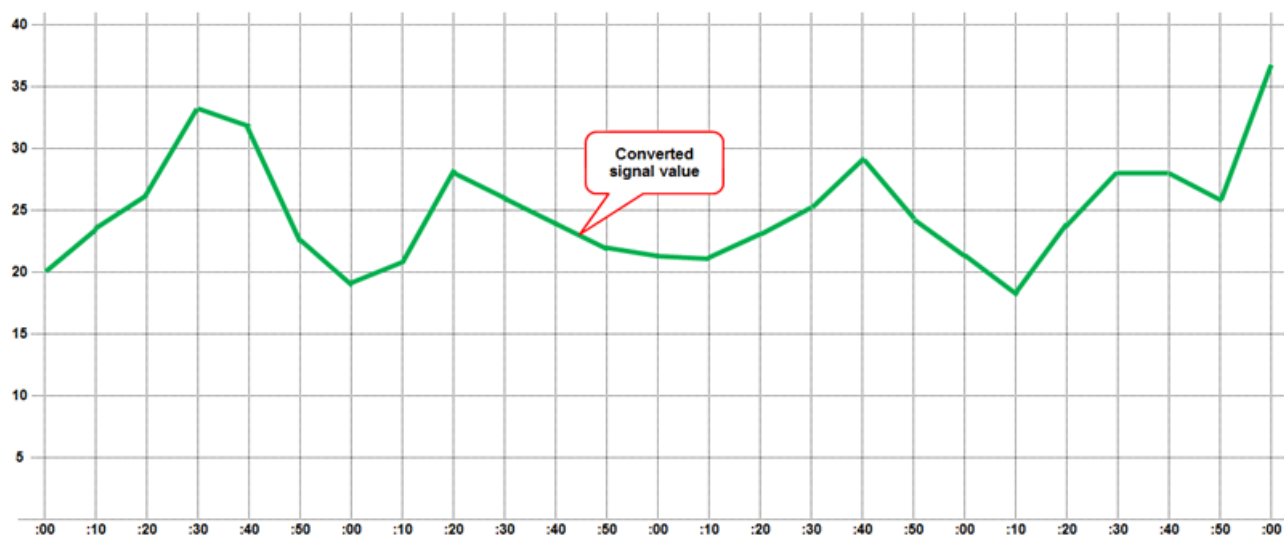
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	81/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

The graphs on the [Monitoring - Signal Monitoring] / [Diagnosis - Signal History] screen of the WebUI based on smoothing data pieces setting will be displayed as shown below.

When number of smoothing data pieces is set at [1]



When number of smoothing data pieces is set at [3]



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	82/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

4.1.5 Narrow down Signal List

The each signal list in the [Machine Setting] window can be narrowed down by setting criteria so that it can list only specific signals to review.

The following criteria can be specified. The items that can be set differ depending on each signal setting screen.

Machine Type	Targeted Signal	Setting Tab / Dialog Box	Criteria Setting Items
FanucCNC	Basic signals	[Basic Signal] tab	- Description - Enable - Reading Cycle
	Tool life signals	[Tool Life Signal] tab	
	Laser control data signals	[Laser control data signal setting] tab	
	Custom signals	[Custom Signal] tab	- Type - Address/Number - Reading Cycle
OPCUAServer	Signals	[Signal Setting] tab	- Signal(*1) - Reading Cycle
	Method signals	[Method Signal Setting] tab	
FanucRobot	Basic signals	[Basic Signal] tab	- Description - Enable - Reading Cycle
	Custom signals	[Custom Signal] tab	- Signal(*2) - I/O
MTConnectAgent	Signals	[Signal Setting] tab	- Signal(*1) - Enable - Reading Cycle
FanucRobodrill	Basic signals	[Basic Signal] tab	- Description - Enable - Reading Cycle
	ROBODRILL specific signals	[Robodrill Signal Setting] tab	
	Custom signals	[Custom Signal] tab	- Type - Address/Number - Reading Cycle

- Description** Set this text box to part of or entire the Web name of a signal to narrow down the signal list. The list will list signals whose Web names in the [Description] column match or include the specified text. This text box is case insensitive.
- Enable** Select [Enable] or [Disable] from this drop down list to narrow down the signal list. The list will list signals that are set to be collected (when [Enable] is selected) or are set not to be collected (when [Disable] is selected) by the Collector.
- Reading Cycle** Select a reading cycle from this drop down list to narrow down the signal list.
- Type** Select a type of signals from this drop down list to narrow down the custom signal list.
- Address/Number** Set this text box to part of or entire the address or number of a signal to narrow down the custom signal list. This text box is case insensitive.
- Signal (*1)** Set this text box to part of or entire the name of a signal to narrow down the signal list. The list will list signals whose names in the [Signal] column match or include the specified text. This text box is case insensitive. (*1: OPCUAServer / MTConnectAgent)
- Signal (*2)** Select a type of signals from this drop down list to narrow down the custom signal list. (*2: FanucRobot)
- I/O** Set this text box to part of or entire the address or number of a signal to narrow down the custom signal list. This text box is case insensitive.

NOTE

The signal list can be narrowed down with any number of criteria. It is not required to set all the criteria. However, the more the criteria are set, the further the list will be narrowed down.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 83/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

Follow the steps below to set criteria and narrow down the signal list.

(1) Set the criteria.

■ Example of Setting Area for Criteria (In case of FanucCNC [Basic Signal] tab)

Protocol Basic Signal Custom Signal Tool Life Signal Macro Variable Getting Condition Definition Variable History Retrieval Condition Definition

Signal Setting

Narrow down

Description Enable ☐ Reading Cycle

Setting Area for Criteria

No	Signal	Description	Axis	CNCPath	Enable	Outlier Detection	Reading Cycle	Check
1	SigOP	Automatic operation signal	1	1	☐	☐	500msec	☐
2	SigSTL	Cycle start lamp signal	1	1	☐	☐	500msec	☐
3	SigSPL	Feed hold lamp signal	1	1	☐	☐	500msec	☐
4	SigAL	Alarm signal	1	1	☐	☐	500msec	☐
5	EMG	Emergency stop signal	1	1	☐	☐	500msec	☐
6	Mode	Mode	1	1	☒	☐	500msec	☐
7	MainProgram	Main program	1	1	☒	☐	500msec	☐
8	ActProgram	Executing program	1	1	☒	☐	500msec	☐
9	Sequence	Sequence number	1	1	☐	☒	500msec	☐
10	MainComment	Program comment (main program)	1	1	☒	☐	500msec	☐
11	ActComment	Program comment (executing program)	1	1	☒	☐	500msec	☐
12	ActS	Speed S [actual]	1	1	☐	☒	500msec	☐
13	ActF	Feed rate F [actual]	1	1	☐	☒	500msec	☐
14	ActFdec	Feed rate F [actual]	1	1	☐	☒	500msec	☐
15	ModalS	Speed S [command]	1	1	☐	☐	500msec	☐
16	ModalF	Feed rate F [command]	1	1	☐	☐	500msec	☐
17	ModalT	T code	1	1	☐	☐	500msec	☐
18	ModalM	1st M code	1	1	☐	☐	500msec	☐

(2) Click the [Apply] button.

(3) Verify that the list is narrowed down as expected.

Clicking the [Discard] button disables the criteria and lists all the registered signals in the signal list.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 84/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

4.1.6 Checking the Signals

For the signals that are set in each signal setting tab or dialog box of machine setting, whether it is possible to acquire the signals or not can be checked.

There are two methods for checking the signals as shown below.

- Signal Check per machine
Check the signals for the selected machine.
- Signal Check per Collector
Check the signals for all the machines under the selected Collector.

Shown below is a list of signals that are targeted for Signal Check.

Machine Type	Targeted Signal	Setting Tab / Dialog Box	Remarks
FanucCNC	Basic signals	[Basic Signal] tab	(NOTE 6)
	Custom signals	[Custom Signal] tab	-
	Laser control data signals	[Laser control data signal setting] tab	(NOTE 6)
	Tool life signals	[Tool Life Signal] tab	(NOTE 6)
	Macro variables	[Macro Variable Getting Condition Definition] tab	-
		[Macro Variable getting condition definition Setting] dialog box	
	Variable histories	[Variable History Retrieval Condition Definition] tab	-
		[Variable History Retrieval Condition Setting] dialog box	
FanucRobot (NOTE 2)	Basic signals	[Basic Signal] tab	(NOTE 6)
	Custom signals	[Custom Signal] tab	-
	Variable histories	[Variable History Retrieval Condition Definition] tab	-
		[Variable History Retrieval Condition Setting] dialog box	
	Position register signal	[Position Register] tab	-
		[Position Register Signal Details] dialog box	
FanucRobodrill	Basic signals	[Basic Signal] tab	(NOTE 6)
	Custom signals	[Custom Signal] tab	-
	Macro variables	[Macro Variable Getting Condition Definition] tab	-
		[Macro Variable getting condition definition Setting] dialog box	
	Variable histories	[Variable History Retrieval Condition Definition] tab	-
		[Variable History Retrieval Condition Setting] dialog box	
	ROBODRILL specific signals	[Robodrill Signal Setting] tab	(NOTE 6)

The results of Signal Check is displayed as follows in the [Check] column of each signal setting tab or dialog box.

- "OK": Signal checked, and signal acquired
- "NG" (for "not good"): Signal checked, and failed to acquire signal
- "-" (hyphen): Not a target of Signal Check
- "" (blank): Signal Check unexecuted (initial state, machine connection failure, etc.)

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	85/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

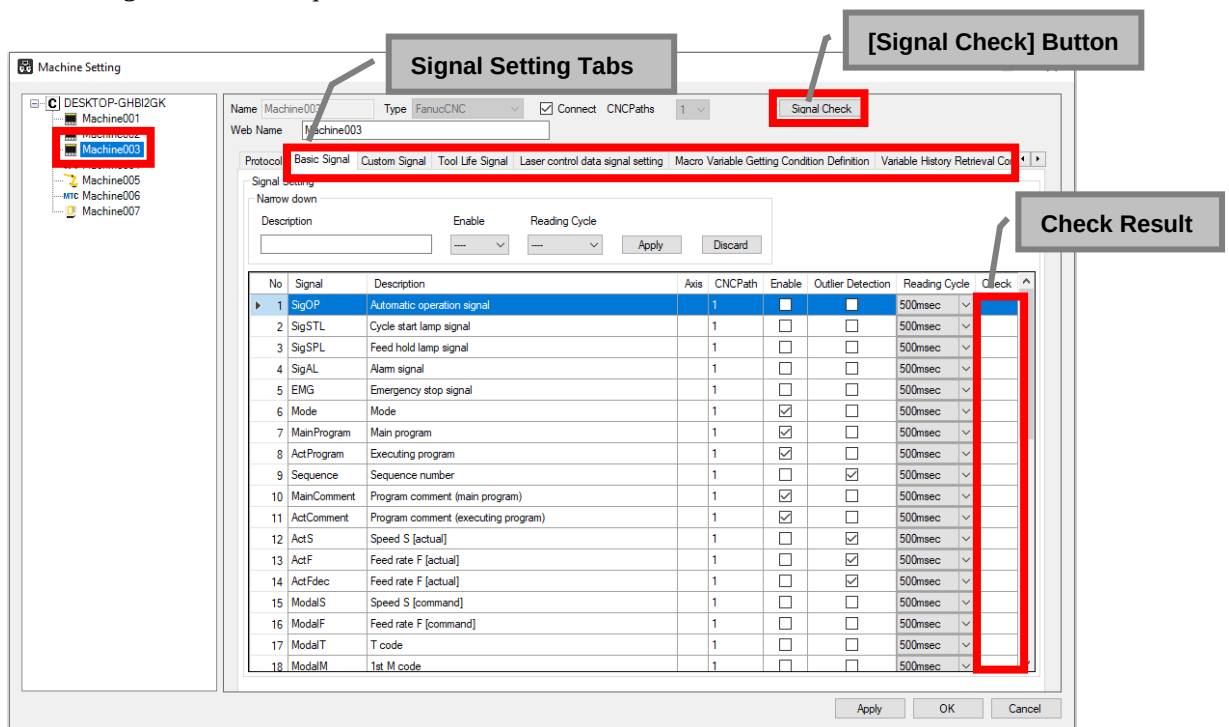
NOTE

- 1 The machines targeted for Signal Check are of FanucCNC, FanucRobot, or FanucRobodrill machine type. Machines of other machine types are not targeted. However, when using Signal Check per Collector method, Signal Check checks FanucRobot type machines only when their robot controller is R-30iB.
 - 2 Before conducting Signal Check of the following FANUC Robot Series machines, stop applications that may communicate with those machines. If such an application is running, it may conflict with Signal Check function and experience a communication error:
 - R-30iA / R-30iA Mate
 - R-J3iB / R-J3iB Mate
 - 3 The machine targeted for Signal Check must have the power turned ON and must be in a connectable status.
 - 4 Signal Check per Collector is not executed for the machines which have the [Connect] check box cleared in the [Machine Setting] window.
 - 5 Signal Check per machine is executed even when the [Connect] check box is cleared in the [Machine Setting] window.
 - 6 For the signals that have the [Enable] column, the [Enable] check box is cleared when they are not targeted.
 - 7 The results of Signal Check are cleared when the [Machine Setting] window is closed.
 - 8 Signals with "NG" displayed as the check result cannot be acquired. Either disable (Clear the check box in the [Enable] column), or review the settings.
 - 9 If you wish to look into the cause for the signal which has "NG" displayed as the check result, use the "Signal Check Tool."
- For the details, refer to the "**SIGNAL CHECK TOOL**" section.

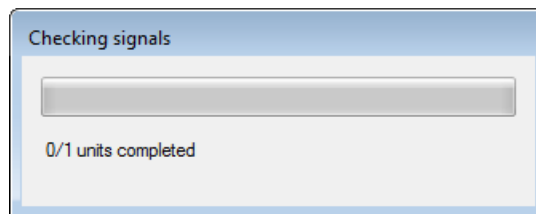
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	86/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

4.1.6.1 Signal Check per Machine

Connects to the machine selected on the Collector/Machine tree, and checks whether the signals set in each signal setting tab or dialog box can be acquired or not.



- Select the targeted machine on the Collector/Machine tree, and click the [Signal Check] button. The Signal Check per machine process is executed.
During the signal checking process, a progress bar showing the status (<number of completions> / <number of targets>) is displayed.



- When the Signal Check process is finished, check the results shown in the [Check] column of each signal setting tab or dialog box.

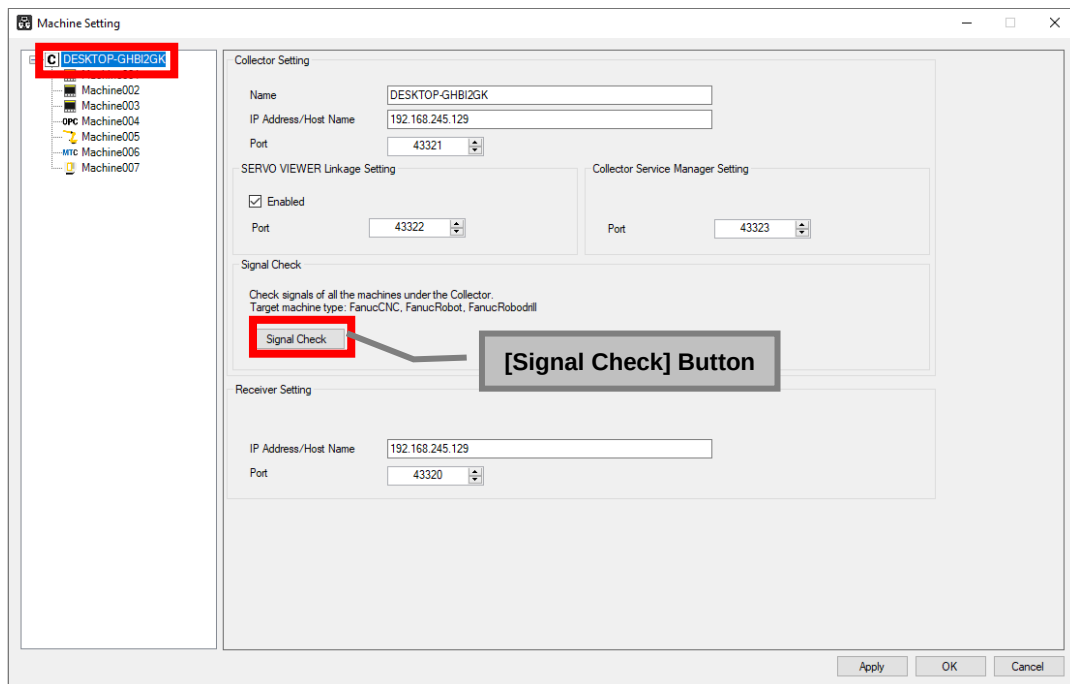
NOTE

- Signal Check per machine is executed even when the [Connect] check box is cleared in the [Machine Setting] window.
- For the signals that have the [Enable] column, the [Enable] check box is cleared when they are not targeted.
- The results of Signal Check are cleared when the [Machine Setting] window is closed.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 87/868

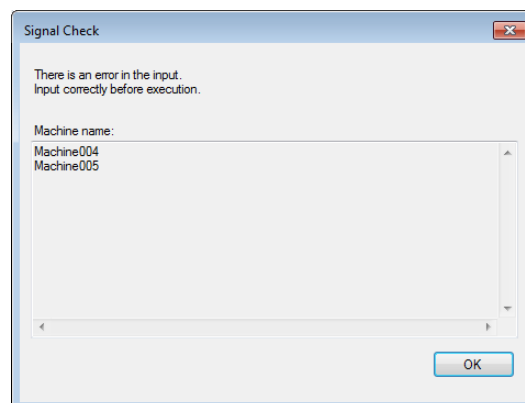
4.1.6.2 Signal Check per Collector

Connects to all the machines under the Collector selected on the Collector/Machine tree, and checks whether the signals set in each signal setting tab or dialog box can be acquired or not.

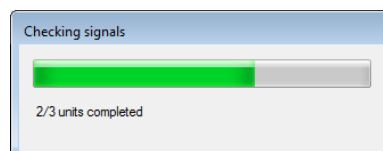


- (1) Select the targeted Collector on the Collector/Machine tree, and click the [Signal Check] button. The Signal Check per Collector process is executed.

Following message is displayed when there is one or more machines with an error in the input.

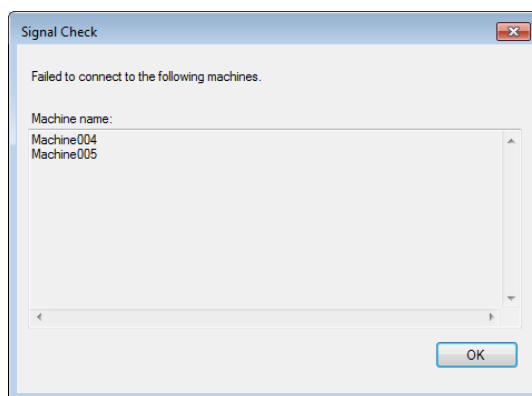


During the Signal Check process, the following progress bar is displayed which indicates the status (<number of completion> / <number of targets>).



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	88/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- (2) After the process is finished, following message is displayed when there were machines which were unable to connect. Confirm the message, and click the [OK] button.



- (3) When the Signal Check is completed, select the targeted machines on the Collector/Machine tree, and check the results shown in the [Check] column of each signal setting tab or dialog box.

NOTE

- 1 Signal Check per Collector is not executed for the machines which have the [Connect] check box cleared in the [Machine Setting] window.
- 2 For the signals that have the [Enable] column, the [Enable] check box is cleared when they are not targeted.
- 3 The results of Signal Check is cleared when the [Machine Setting] window is closed.

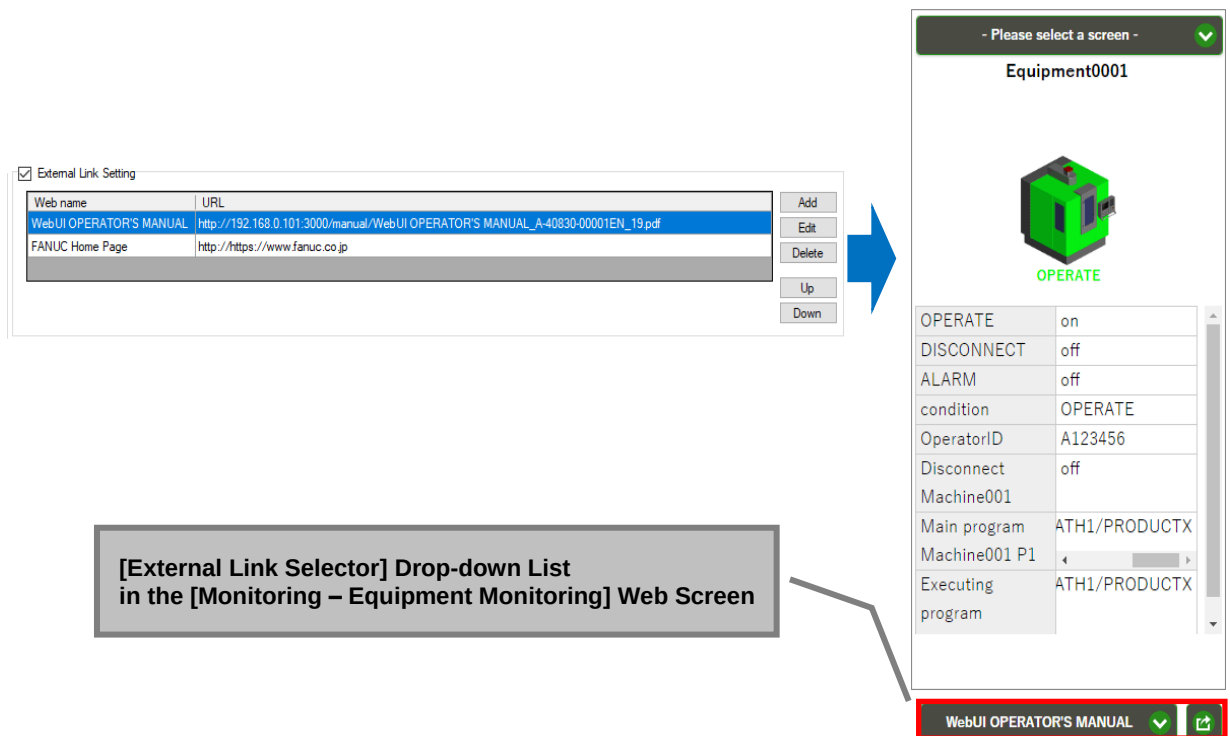
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 89/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

4.1.7 Set External Links

To view external links for a machine of a given machine type through the [Monitoring - Equipment Monitoring] Web screen (displayed by selecting an equipment icon on the [Monitoring - Overlook] screen), make settings for those external links in the [Protocol] tab of the [Machine Setting] window for the machine.

External Link function provides a way to view external links, such as Web sites, that are associated with given equipment (machines).

- When a trouble rises on machines from which MT-LINK*i* cannot gather detailed information, such as ZDT on ROBOT-LINK*i* screen and CNC screens (with Web Server function), use external links to gather them.
- Files such as operation manuals (PDFs) can be read through external links.
- Up to three external links can be registered.



- **External Link Setting**
Select this check box to enable External Link function. When it is selected, new entries can be added to the external link list. By default, it is cleared.
- **External Link list**
This list displays registered external links.
 - **Web name:** This column lists Web Names of external links.
The [External Link Selector] drop-down list and the tooltip appearing when the mouse cursor moves over the [External Link] button on the [Monitoring - Equipment Monitoring] screen show this information.
The selector lists external links in the same order as they are registered in the [Machine Setting] window of AdminTool (i.e., from top to bottom). For equipment that has multiple machines registered, the selector will sort registered external links in ascending order of machine name (machine ID) as the first key and then sort those with the same machine ID in their registration order.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 90/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

NOTE

To specify a file located in the Server PC as an external link

- 1 Store the file in the following folder:
 <MT-LINK*i* installation folder>\WebClient\public
 - 2 Set the [URL] field as follows:
 http://<IP address><Port number><File path>
 <IP address>: IP address (or hostname) of the Web Server (the Server PC)
 <Port number>: the port number for the Web Server
 <File path>: a path to the file (relative to the folder shown in step 1)
- * In Example 3, folder "manual" is created under the folder shown in step 1 and the pdf file to link to is stored in it.

- Example 4 (a screen on a robot displayed on ZDT on ROBOT-LINK*i* portal site)
 Web name: ZDT on ROBOT-LINK*i*
 URL: http://192.168.0.201/ZDTWebPortal/ec8a8021-9e86-...-8c6eb5d11101

NOTE

Follow the steps shown below to set the [URL] field to the URL of the portal site (ZDT on ROBOT-LINK*i*):

- 1) Login the portal Store and move to the screen for a robot
- 2) Copy the URL of the screen to the clipboard.
- 3) Paste the clipboard to the [URL] field in the [Add/Edit External Link] window.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION				Sheet. 92/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

4.1.8 Copy/Cut/Paste/Delete Machine Setting

(1) Copy Machine/Paste Machine

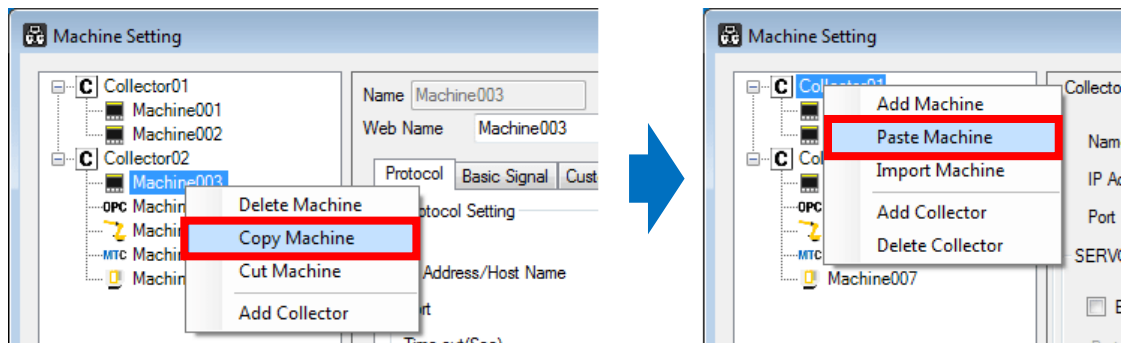
The settings of the existing (already added) machine can be copied and pasted.

Right-click the machine name of the Collector/Machine tree, and select [Copy Machine] on the drop-down menu.

* You can also copy multiple machines at the same time by selecting two or more machines.

Then right-click a Collector name, and select [Paste Machine] on the displayed drop-down menu to display the [Add Machine] dialog box.

The machine setting of the copied machine will be applied on the newly added machine. Make necessary changes for the machine setting.



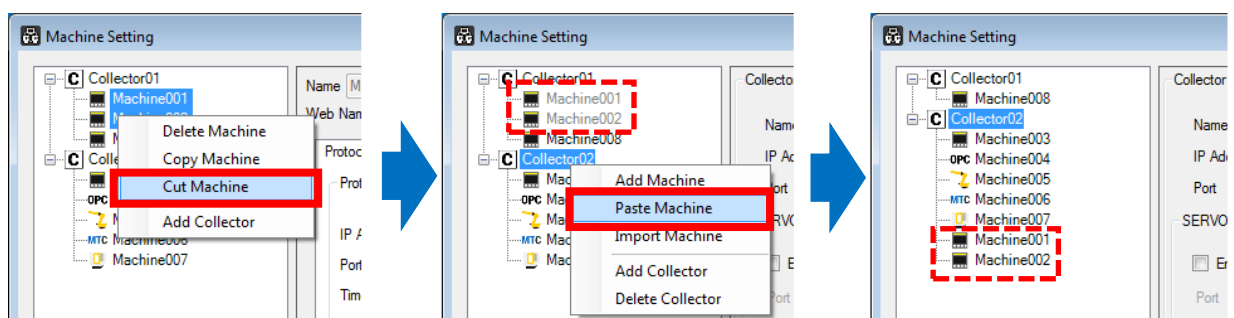
(2) Cut Machine/Paste Machine

By cutting and pasting the setting of the existing machine, the machine can be moved under a separate Collector.

Right-click the name of the machine selected on the Collector/Machine tree, and click [Cut Machine] in the menu displayed.

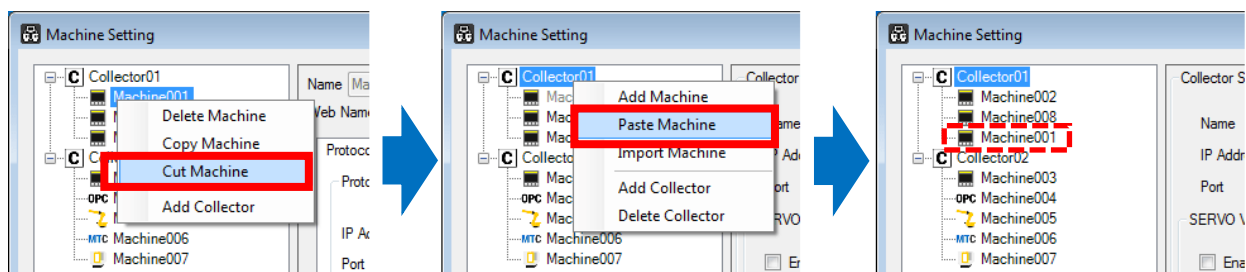
- * Two or more machines can be selected at once when cutting.
- * When the machine is cut, it will be displayed in gray. (When the [Esc] key is pressed while the machine is in a cutting state, the cutting state of the machine will be canceled.)

Then right-click a name of another Collector, and by clicking [Paste Machine] on the menu displayed, the machine being cut moves to the position of the targeted Collector. When two or more machines are selected, they are moved.



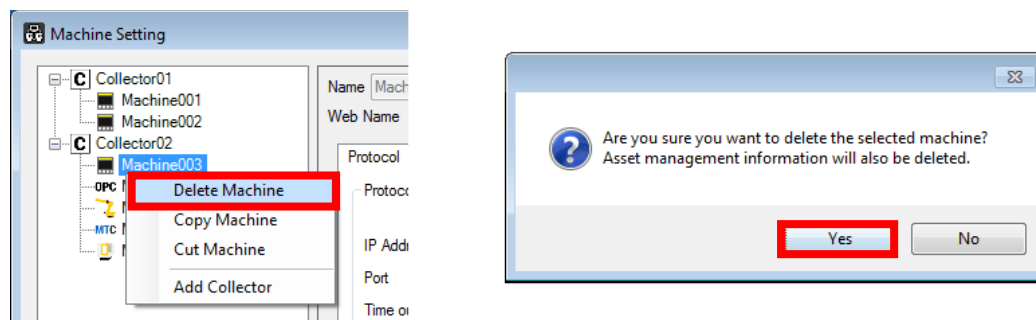
The process is the same when the machine is cut and pasted under the same Collector. In such a case, the order of the machines within the same Collector can be arranged.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	93/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					



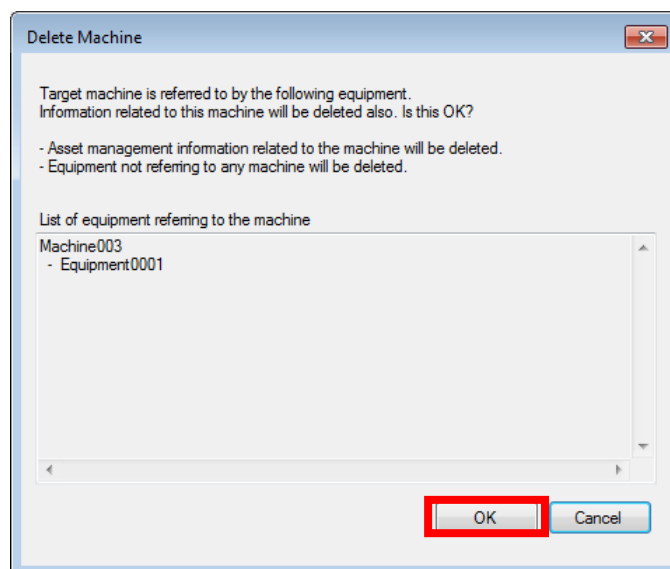
(3) Delete Machine

Right-click the machine name to be deleted on the Collector/Machine tree, and select [Delete Machine] on the displayed drop-down menu. By clicking [Yes] in the confirmation dialog box, the targeted machine for deletion will be deleted from the Collector/Machine tree.



If there are any equipment units which will have no machines to refer to by the deletion of the machine, those equipment units will also be deleted at the same time. In the case when one or more equipment units are deleted, the following confirmation message will be displayed. Check the details described in the "List of equipment referring to the machine."

By clicking the [OK] button in the confirmation message, the targeted machine will be deleted from the Collector/Machine tree.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 94/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

NOTE

In the following cases, sending setting data to the database causes an error if a machine is deleted that equipment units are referring to:

- When all or some of the machines are deleted that belong to an equipment unit (in this case, only the simple assign statuses for the machines will be deleted from the [Equipment Setting] window, and an error mark ([!]) will appear in the Multiple assign status area)
 - Click the [Automatic definition of signals] button to fix the error
- When deleting machine deletes an equipment unit assigned to group status (In/Out)
 - Correct errors in the settings made in the [Group Setting] window
- When deleting machine deletes equipment units that are subject to sending mail in delivery conditions for signal detection/alarm mail
 - Correct errors in the settings made in the [Signal Detection Mail Setting] or the [Alarm Mail Setting] window

NOTE

Until the ROBODRILL Option license is verified, no commands for FanucRobodrill type machines in the context menu are available except [Delete Machine]. The [Copy Machine] and the [Cut Machine] commands in the menu are grayed out, so that they are unclickable.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 95/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

4.1.9 Import Machine Setting

By loading the machine setting file created by the machine setting export process of the Signal Check Tool, the machine can be added under the selected Collector.

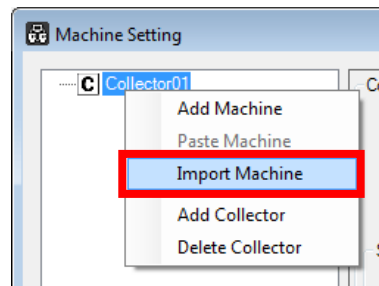
* For the details on the machine setting export process of the Signal Check Tool, refer to the "[Machine Setting Export](#)" subsection.

NOTE

- 1 Files with the same format as L0_Setting.json can be imported as the machine setting file. The files of the older versions can be imported also.
- 2 The machine setting file (L0_Setting.json) created in the [Create Collector Setting Files] dialog box is not supported.
- 3 All the mandatory signals of the imported machine setting will be enabled.
- 4 All the Web names of the imported machine setting will be set as the same names as the machine names.

Following are the steps for importing the machine setting file.

- (1) On the Collector/Machine tree, right-click the Collector you want to import the machine setting, and click [Import Machine] on the menu displayed.



- (2) [Select machine setting file] dialog box appears. Select the machine setting file to import, and click the [Open] button.

* The dialog box selects the following folder on the first session. On subsequent sessions, it selects the folder selected in the previous session.

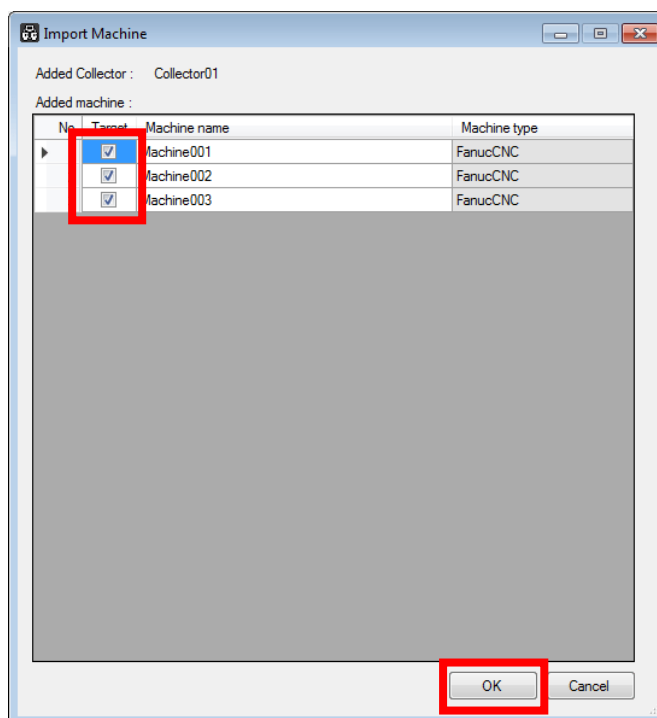
Folder name: <MT-LINK*i* installation folder>\SignalCheckTool\Export\

* Following is the name of the file created with the machine setting export process of the Signal Check Tool.

Filename: L0_Setting_<yyyyMMddHHmmss>.json
(<yyyyMMddHHmmss> is the date/time of output)

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	96/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

- (3) If the machine setting file is loaded successfully, the [Import Machine] dialog box appears.
If it fails to load the file, an error message is displayed, and the import of the machine is stopped.



- (4) Confirm the [Machine name] of the machine to import, and select the [Target] check box.
[Machine name] can be edited. If there are any errors, such as the existence of the machine with the same name, [!] will be displayed in the [Machine name] column. In such as case, make corrections on the [Machine name].
(* By moving the cursor on the [!], the detail of the error is displayed.)

NOTE

A ROBODRILL Option license is required to support FanucRobodrill machines.
If the license has not been verified yet and FanucRobodrill type machines are displayed in the [Import Machine] dialog box, clear their [Target] check boxes.

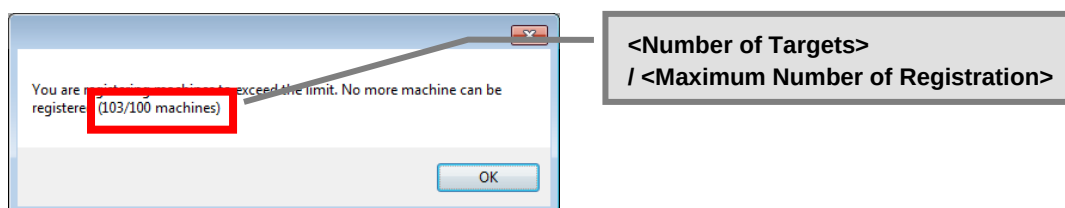
- (5) Click the [OK] button, and the import machine process starts.

NOTE

Import machine process will not start if the [Target] check box is selected for the machine with an error in the [Machine name] column. In such a case, make corrections in the [Machine name] column, or select the [Target] check box.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 97/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- (6) Following message will appear if the number of registered machines exceeds the maximum number by executing the import process. Click the [OK] button to return to the [Import Machine] dialog box.



<Number of targets>:

Total number of machines which have the [Target] check box selected or have been set in the [Machine Setting] window in advance.

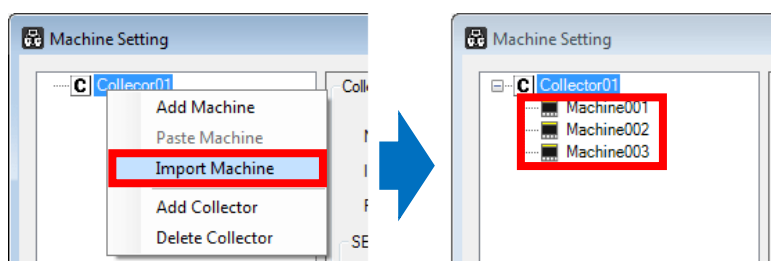
<Maximum number of registration>:

Maximum number of machines that can be registered

(Maximum number of machines that can be registered differs by license option.

For the details, refer to the "**MACHINE SETTING**" section.)

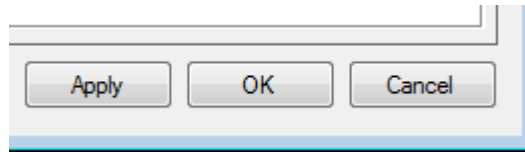
- (7) When the import machine process is completed, the target machine is added under the selected Collector.



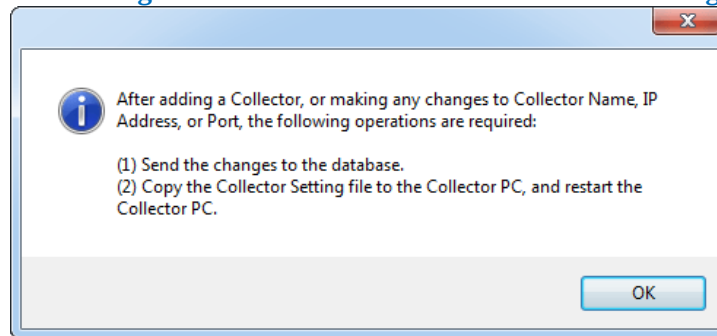
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 98/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

4.1.10 Save Machine Setting

Clicking [Apply] or [OK] saves changes in the [Machine Setting] window. If you want to discard the changes, click [Cancel].



Note that if you have made any changes to Collector settings or Receiver settings by adding a Collector, or editing them, you will receive the following message. Click [OK] to close the message and have the Collector and the database reflect the changes and restart the Collector PC as instructed in the message. For the details, refer to the "[Apply Created AdminTool Setting to Collector](#)" subsection and the "[Send Setting Data to DB](#)" subsection.



4.1.11 Apply Machine Setting

To apply changes in machine setting, select the [Send data to Collector after create] check box in the [Create Collector Setting Files] dialog box and click [Run]. For the details, refer to the "[Apply Created AdminTool Setting to Collector](#)" subsection.

NOTE

When you have added a Collector or changed settings for a Collector or a Receiver, you need to send the updated settings to the database, manually copy the Collector setting files to the Collector PC, and restart it. For the details, refer to the "[Send Setting Data to DB](#)" subsection and the "[Apply Created AdminTool Setting to Collector](#)" subsection.

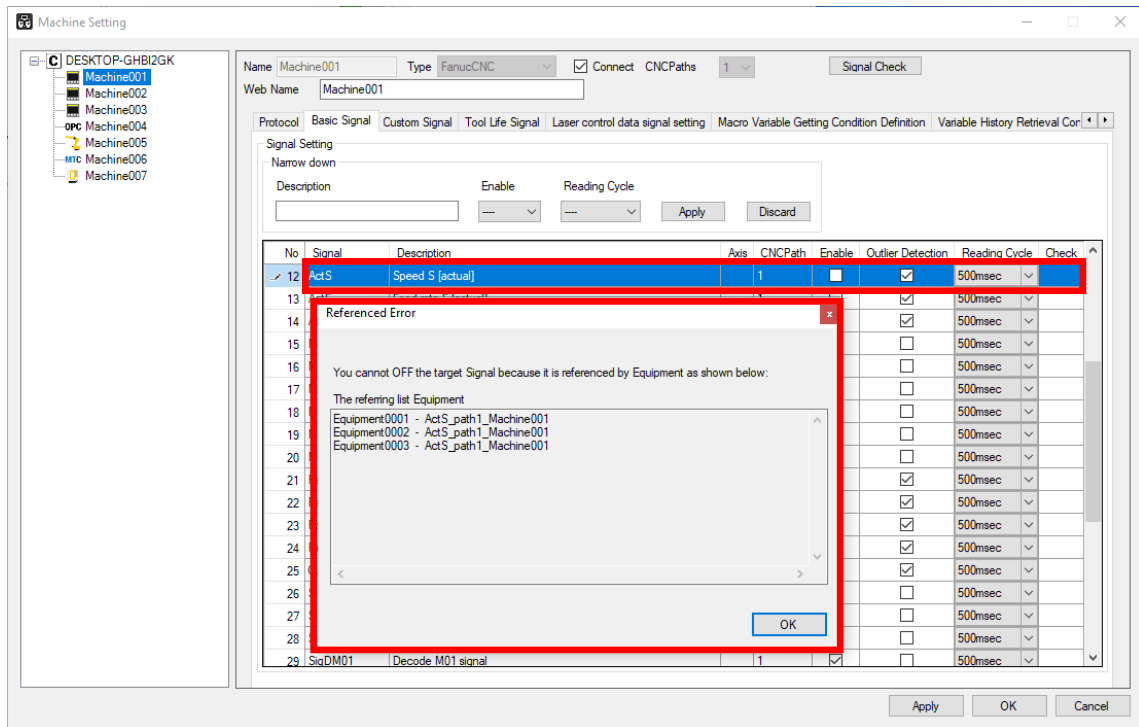
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 99/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

4.1.12 Referenced Error

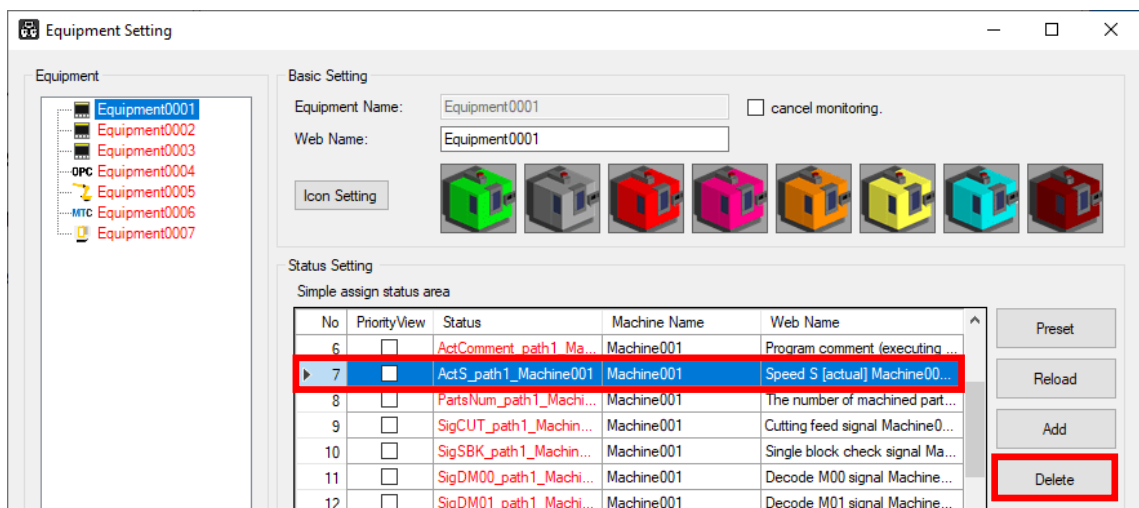
When signals are being referenced by equipment, the following operations in setting those signals will be denied with error messages:

- Clearing the [Enable] check boxes
- Renaming the signals
- Deleting the signals

Example) The [Enable] check box is cleared



In this case, check the [The referring list Equipment] group box for the referenced signals and delete them all from the [Equipment Setting] window. After deleting all the referenced signals, try to clear the [Enable] check box.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION	Sheet. 100/868		
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

4.2 MACHINE SETTING - FanucCNC

This section explains the machine setting of the machine type FanucCNC.

4.2.1 Steps for the Machine Setting of FanucCNC

Follow the steps below to set the machine.

- (1) Select [Machine Setting] in the AdminTool Menu. This displays the [Machine Setting] window.
- (2) Add Collector. For the details, refer to the "[Add Collector](#)" subsection.
- (3) Add machine type FanucCNC to each Collector. For the details, refer to the "[Add FanucCNC](#)" subsection.
- (4) Make protocol setting, connection setting, and set the information to connect to the Data Server for the machine. For the details, refer to the "[Confirm/Edit Setting of FanucCNC](#)" subsection and the "[Set Protocol Setting of FanucCNC](#)" subsection.
- (5) Perform Signal Setting for the machine. For the details, refer to the "[Set Signal of FanucCNC](#)" subsection.
- (6) Perform Custom Signal Setting for machine. For the details, refer to the "[Set Custom Signal of FanucCNC](#)" subsection.
- (7) When the control type is "L" (laser), make settings for the laser control data signal. For the details, refer to the "[Set Laser Control Data Signal of FanucCNC](#)" subsection.
- (8) When collecting tool life signals, make settings for the signals. For the details, refer to the "[Tool Life Signals of FanucCNC](#)" subsection.
- (9) Set macro variable getting conditions. For the details, refer to the "[Register Macro Variable Getting Condition Definition of FanucCNC](#)" subsection.
- (10) Set variable history retrieval conditions. For the details, refer to the "[Setting Variable History Retrieval Condition Definition of FanucCNC](#)" subsection.
- (11) Check whether the signals that are set can be acquired or not. For the details on the checking methods, refer to the "[Checking the Signals](#)" subsection.
- (12) Save all the settings to make the system reflect them. For the details, refer to the "[Save Machine Setting](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION		Sheet. 101/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

4.2.2 Add FanucCNC

(1) Add Machine

To add a machine, right-click the target Collector name in the Collector tree and point to [Add Machine].

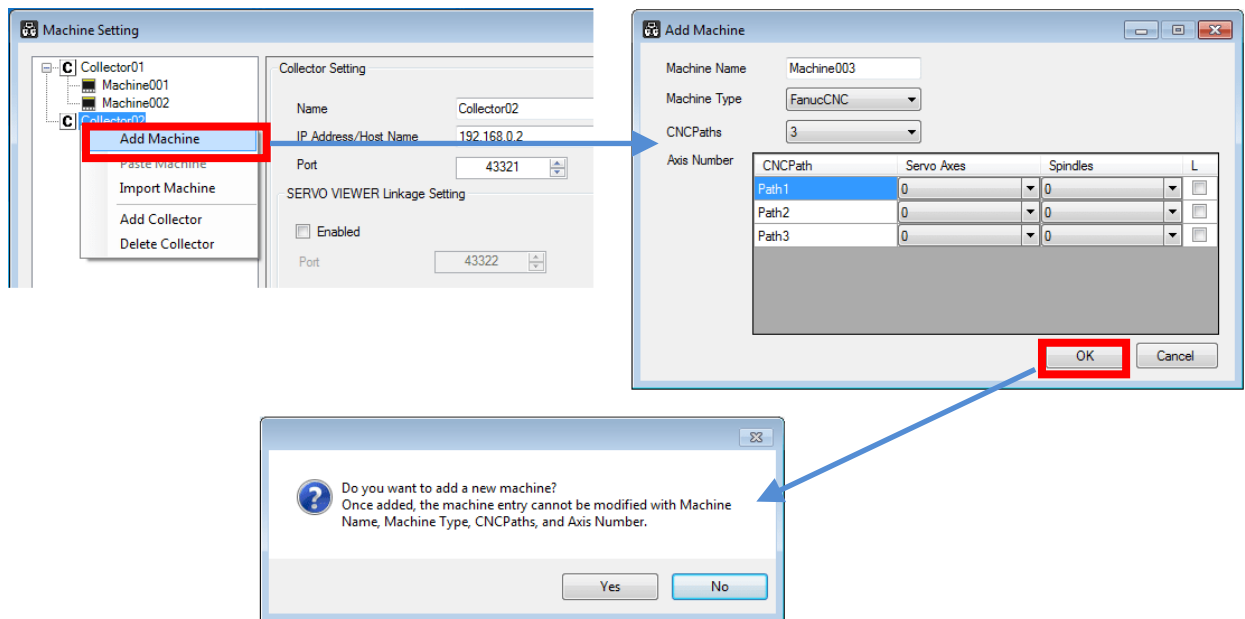
*Before adding a machine, make sure that you have registered the target Collector.

When the [Add Machine] dialog box appears, specify the settings of the target machine.

NOTE

Once set, [Machine Name], [Machine Type], [CNCPaths], and [Axis Number] ([Servo Axes], [Spindles], and [L]) will be non-editable.

If the settings of [Machine Name], [Machine Type], [CNCPaths], [Axis Number], and [L] are changed by mistake, the existing machine must be deleted. Add a new machine and make the settings for it.



■ The [Add Machine] dialog box

- **Machine Name**
Set a machine name to this field. The [Machine Name] should start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and underscores ("_"). This field can be up to 32 bytes long.
- **Machine Type**
Select [FanucCNC].
- **CNCPaths**
Set this field to the number of CNC paths for the machine.
- **Servo Axes/Spindles**
Select the numbers of the servo axes and spindles for each CNC path of the machine.
- **L**
Specify whether to collect the laser control data signal for each CNC path.
For the CNC path which collects laser control data signals, select the check box. In other cases, clear the check box. Default setting of the check box is cleared.
When one or more of these check boxes are selected, [Laser control data signal setting] tab is displayed in the [Machine Setting] window.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 102/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

- The [OK] button
Clicking this button opens a confirmation message. Click [OK] to save the settings and add the machine in the Collector tree. Or, click [No] to cancel adding the machine and close the dialog box.
- The [Cancel] button
Clicking this button discards the settings and closes the dialog box.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	103/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

4.2.3 Confirm/Edit Setting of FanucCNC

(1) Confirm and edit machine Setting

The settings of the selected Collector or machine are populated in the right pane of the [Machine Setting] window. Selecting the <Machine name> in the Collector tree will show machine setting, so that you can set and change them.

The [Web Name] is displayed on the Web screens. The value of this field should be entered in the language specified for the [Web name language] field in the [System Setting] window. The [Web Name] field is required and will be set to the default value of <Machine name> if it is not modified.

The screenshot shows the 'Machine Setting' window. On the left, a tree view shows 'Collector01' containing 'Machine001' and 'Machine002', and 'Collector02' containing 'Machine003'. 'Machine003' is selected. The right pane displays the settings for 'Machine003'. The 'Name' field is 'Machine003', 'Type' is 'FanucCNC', and the 'Connect' checkbox is checked. The 'Web Name' field is 'Machine003'. Below this, there are tabs for 'Protocol', 'Basic Signal', 'Custom Signal', 'Tool Life Signal', and 'Macro Variable G'. The 'Protocol' tab is active, showing 'Protocol Setting' with fields for 'IP Address/Host Name' (192.168.0.103), 'Port' (8193), 'Time out(Sec)' (30), and 'Access Code'. There is also a 'Time setting' section with a 'Synchronization interval' dropdown set to 'Not Synchronized'.

(2) Set Machine Connection

Set connection between machine and Collector. Select a machine in the Collector tree and select the [Connect] check box, so that the connection is available.

This screenshot is similar to the previous one, showing the 'Machine Setting' window for 'Machine003'. The main difference is that the 'Connect' checkbox in the top right of the settings pane is now checked, indicating that the connection between the machine and the collector is established.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	104/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

4.2.4 Set Protocol Setting of FanucCNC

Select a machine in the Collector tree and select the [Protocol] tab in the right pane.

■ The [Protocol] tab

Protocol Setting

- IP Address/Host Name
Specify an IP address or host name of the machine.
When entering a host name, the name can start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and periods ("."). This field can be up to 253 bytes long.
- Port
Specify the port number for connection.
- Time out
Specify the connection timeout length in second. When the specified period passes without response, the connection to the machine is regarded as having a time out issue. The default value is 30 seconds.
- Access code
For FanucCNCs on which an access code is set, select the check box and set the access code for the text field. Enter less than nine numeric characters. The entered access code will be masked with asterisks ("*").

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	105/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

NOTE

- 1 To set an access code on a FanucCNC type machine, CNC system software must be installed that supports a security function to use an access code.
- 2 If the wrong access code is set, FanucCNCs set with an access code will not be connected and thus they will be in "DISCONNECT" status.

Time Setting

- Synchronization interval
Specify the time interval for synchronizing the time of the Collector PC with the machine (FanucCNC). [Not synchronized], [12 hours] or [24 hours] can be selected. The default value is [Not synchronized].

NOTE

- When the synchronization interval is set as [12 hours] or [24 hours]
- 1 Initial time setting is executed upon connection with the machine right after the start of the Collector.
 - 2 The time synchronization after the initial time setting will be executed periodically in a synchronization interval (12 hours or 24 hours) based on the time of the initial time setting. However, time synchronization will not be executed if the machine is in operation. Time synchronization is executed when the state of the machine is other than in operation.
 - 3 For the time between 23:50 and 24:00 of the Collector PC, the synchronization process is not executed. If the time of synchronization arrives within this time, the synchronization is executed after 24:00.
 - 4 If the Collector PC and the Server PC are configured on separate PC's, the time synchronization of the Collector PC and the Server PC will not be executed in this setting. When synchronizing the time between the Collector PC and the Server PC, manage by using NTP or other means.

Data Server setting

When the Data Server functionality can be used on the connection target machine, the following functionalities can be used by setting the information to connect to the Data Server.

- The files on the Data Server can be saved and the history of the files can be managed on the machine data storing schedule, and on the [Utility – Machine Data Archive Management] screen of the Web screen.
- The files on the Data Server can be downloaded/uploaded on the [Utility – File Transfer] screen of the Web screen.
- Data Server setting
Select the check box when using the Data Server functionality. When the check box is selected, you will be able to input the information for connecting the Data Server. By default, this check box is cleared.
- IP Address/Host Name (check box)
When using an IP address or host name specific to the Data Server, select the check box. By selecting the check box, you will be able to input in the entry column of [IP Address/Host Name]. By default, this check box is cleared.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	106/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

- **IP Address/Host Name (entry column)**
Enter the IP address, or the host name of the Data Server. When entering the host name, the name can start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and periods ("."). This field can be up to 253 bytes long. In the default setting, IP address or the host name of the machine is set.
- **User name**
Enter the user name used when signing in to the Data Server. This field can be up to 31 bytes long. The default setting is blank.
- **Password**
Enter the password of the user name. This field can be up to 31 bytes long. When entering the password, "*" (asterisk) will be displayed in place of the letters to mask the password. The default setting is blank.
- **Port**
The port number of the Data Server is displayed. Fixed at "21." (cannot be changed)

NOTE

- 1 The time out period used when connecting to the Data Server is the time out value set in the protocol setting.
- 2 The connection with the Data Server is fixed to "Active mode." "Passive mode" is not supported.
- 3 The transfer mode is fixed to "ASCII (text) mode." "Binary mode" is not supported.

External Link Setting

Set fields for external links to view them through Web screens. Those links can be selected from the [Monitoring - Equipment Monitoring] screen (displayed by selecting an equipment icon on the [Monitoring - Overlook] screen).

External Link function provides a way to view external links, such as Web sites, that are associated with the selected equipment (machines).

For the details, refer to the "[Set External Links](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	107/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

4.2.5 Set Signal of FanucCNC

This setting function is provided to browse the signals of machines in MT-LINK*i*.

Select a machine in the Collector tree and select the [Basic Signal] tab in the right pane.

For each signal listed in the [Basic Signal] tab, check the [Axis] and [CNCPath] columns, set the [Reading Cycle] column, and select or clear the [Enable] check box according to whether it is to be collected.

Collector collects the signals which have the [Enable] check box selected.

The statuses will be automatically added to the [Simple assign status area] of the [Equipment Setting] window if they have been added on the [Custom Signal] tab of the [Machine Setting] window or their [Enable] check boxes have been selected on the [Basic Signal] or [Laser control data signal setting] tab of the [Machine Setting] window.

The screenshot shows the 'Machine Setting' window. On the left, a tree view shows 'Collector01' containing 'Machine001', 'Machine002', and 'Machine003'. 'Machine003' is selected. The main area shows settings for 'Machine003'. The 'Basic Signal' tab is active. Below the tabs, there's a 'Signal Setting' section with a 'Narrow down' dropdown and buttons for 'Enable', 'Reading Cycle', 'Apply', and 'Discard'. A table lists 18 signals with columns: No, Signal, Description, Axis, CNCPath, Enable, Outlier Detection, Reading Cycle, and Check. The 'Enable' column has checkboxes, and the 'Reading Cycle' column has dropdown menus.

No	Signal	Description	Axis	CNCPath	Enable	Outlier Detection	Reading Cycle	Check
1	SigOP	Automatic operation signal			☐	☐	500msec	☐
2	SigSTL	Cycle start lamp signal	1		☐	☐	500msec	☐
3	SigSPL	Feed hold lamp signal	1		☐	☐	500msec	☐
4	SigAL	Alarm signal	1		☐	☐	500msec	☐
5	EMG	Emergency stop signal	1		☐	☐	500msec	☐
6	Mode	Mode	1		☒	☐	500msec	☐
7	MainProgram	Main program	1		☒	☐	500msec	☐
8	ActProgram	Executing program	1		☒	☐	500msec	☐
9	Sequence	Sequence number	1		☐	☒	500msec	☐
10	MainComment	Program comment (main program)	1		☒	☐	500msec	☐
11	ActComment	Program comment (executing program)	1		☒	☐	500msec	☐
12	ActS	Speed S [actual]	1		☐	☒	500msec	☐
13	ActF	Feed rate F [actual]	1		☐	☒	500msec	☐
14	ActFdec	Feed rate F [actual]	1		☐	☒	500msec	☐
15	ModalS	Speed S [command]	1		☐	☐	500msec	☐
16	ModalF	Feed rate F [command]	1		☐	☐	500msec	☐
17	ModalT	T code	1		☐	☐	500msec	☐
18	ModalM	1st M code	1		☐	☐	500msec	☐

■ The [Basic Signal] tab

● Signal

This column in the signal lists the names of signals that can be set for the selected machine. To list only specific signals, narrow down the signal list. For the details on how to narrow down the list, refer to the "[Narrow down Signal List](#)" subsection.

For the details of signals, refer to the "[LIST OF FanucCNC SIGNALS](#)" section.

● Description

This column displays the Web name of each signal as its description.

● Axis

This field displays the axis of each signal. If its corresponding signal is independent of axes, this field is set to space.

● CNCPath

This field displays CNC path of signals.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	108/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- **Enable**

This is a check box specifying whether to execute data collection of signals by the Collector. If the check box is selected, the data will be collected; otherwise, the data will not be collected.

* For the following signals, this check box is fixed to selected, and may not be changed.

- Mandatory signals (MainProgram/ActProgram/CncState/CncWarning/Alarm)

NOTE

- 1 Some signals listed in the [Signal Setting] table may not be available depending on machine model. For those non-existent signals, clear the check boxes. If the check box for a non-existent signal is selected, its values will not be stored to the database, and will not be displayed on the Web screen.
- 2 When a signal is referenced by equipment, clearing the [Enable] check box for it will be denied with an error message. For the details, refer to the "**Referenced Error**" subsection.

- **Outlier Detection**

For signals with [Outlier Detection] check box selected (signals of data type Int32 or Double), outlier detection of the signal can be set by double-clicking that row. The check box column cannot be edited. For the details of the outlier detection settings, refer to the "**Set Outlier Detection of Signal**" subsection.

- **Reading Cycle**

Specify a reading interval per signal. Options are [500msec], [10sec], and [60sec].

* For the following signals, this field is fixed to 500 msec, and may not be changed.

- Mandatory signals (MainProgram/ActProgram/CncState/CncWarning/Alarm)

- Operator message/Macro message (OpMacroMsg)

- **Check**

This column displays the results of Signal Check as follows:

"OK":	The signal was retrieved
"NG" (for "not good"):	Failed to retrieve the signal
"-" (hyphen):	Not a target of Signal Check
"" (blank):	Signal Check unexecuted (initial state, machine connection failure, etc.)

For the details on Signal Check, refer to the "**Checking the Signals**" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue
FANUC CORPORATION					Sheet.	109/868	

4.2.6 Set Custom Signal of FanucCNC

This function is provided to browse the custom signals of machines in MT-LINK*i*.
Custom signals can be registered to a maximum number of 1000 per machine.
Select a <Machine name> in the Collector tree and select the [Custom Signal] tab in the right pane.

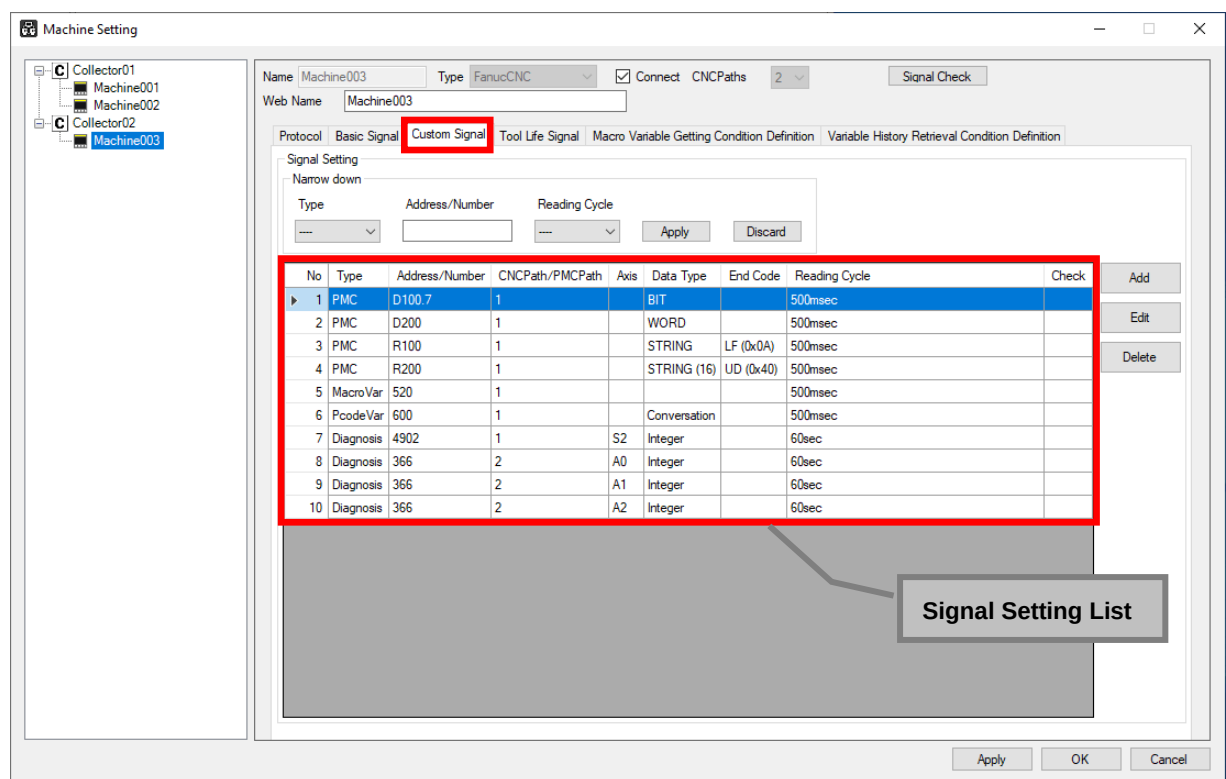
NOTE

Update Option license is required to retrieve diagnostic signals (Diagnosis).
Until Update Option license is verified, the following operations will cause an error message when a diagnostic signal has been set up:

- Clicking the [Custom Signal] tab
- Clicking the [Edit] button after selecting the row for the diagnostic signal in the signal setting list

For the details, refer to the following subsection of the "FANUC MT-LINK*i* / Integration Server Troubleshooting MANUAL":

The "AdminTool Displays License Warning" subsection in the "TROUBLES RELATED TO LICENSE" section in the "MT-LINK*i* TROUBLESHOOTING" chapter



■ The [Custom Signal] tab

This tab page lists the custom signals that are set up in the [Signal Setting] dialog box. To list only specific custom signals, narrow down the signal list. For the details on how to narrow down the list, refer to the "[Narrow down Signal List](#)" subsection.

For the details of the columns in the list, refer to [The \[Signal Setting\] dialog box](#).

• Type

This column lists the types of custom signals registered with the [Signal Setting] dialog box.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	110/868	
Ed.	Date	Design	Description									
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

- **Address/Number**
This column lists the addresses and numbers of custom signals in the following formats:
 - PMC signal (data type: BIT)
Display format: <address><number>.<bit number>
Example: D100.7 (where "D" is an address, "100" is its number, and "7" is its bit number.)
 - PMC signal (data type: other than BIT)
Display format: <address><number>
Example: D200 (where "D" is its address and "200" is its number.)
 - Macro variable/P-CODE variable/Diagnostic signal (data type: other than bit type)
Display format: <number>
Example: 501 (where "501" is its number.)
 - Diagnostic signal (data type: Bit)
Display format: <number>#<bit number>
Example: 552#0 (where "552" is its number and "0" is its bit number.)

* In the examples shown above, <addresses>, <numbers>, and <bit number> correspond respectively to the addresses, the numbers, and the bit number that must be set up in the [Signal Setting] dialog box.
- **CNCPath/PMCPATH**
This column lists PMC path numbers for PMC signals, and CNC path numbers for macro variables, P-CODE variables, and diagnostic signals.
- **Axis**
This column lists target axes of diagnostic signals in the following formats:
 - Diagnostic signal (diagnosis data type: Axis)
Display format: A <servo axis number>
Example: A1 (where "1" is its servo axis number.)
 - Diagnostic signal (diagnosis data type: Spindle)
Display format: S<spindle axis number>
Example: S0 (where "0" is its spindle axis number.)

* With the other signal types, this column lists empty values.
- **Data Type**
This column lists data types of custom signals, such as PMC signals, P-CODE variables, and diagnostic signals.
 - When the [Data Type] drop-down list is set to "STRING" and the [Data Length] text field is set to a value in the [Signal Setting] window for a PMC signal, this column populates its value as shown in the following example:
Display Format: STRING (<data length>)
Example: STRING (80) (where "80" is the length of the data.)
- **End Code**
This column lists the end codes for PMC signals only when the [Data Type] drop-down list is set to "STRING" and the [End Code] text field is set to a value in the [Signal Setting] window for those PMC signals. End codes are listed with their hexadecimal equivalent as follows:
 - NULL (0x00): when the [End Code] text field is set to "NULL"
 - ETX (0x03): when the [End Code] text field is set to "ETX"
 - LF (0x0A): when the [End Code] text field is set to "LF"
 - CR (0x0D): when the [End Code] text field is set to "CR"
 - UD (0x<xx>): when the [End Code] text field is set to "User-Designated" and the [User-designated (Hex)] text field is set to "xx" in hexadecimal

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	111/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- **Reading Cycle**
This column lists each reading interval per signal. Options are predefined as follows:
500msec
10sec
60sec
- **Check**
The results of the Signal Check, when the [Signal Check] button is clicked, are displayed.
"OK": The signal was retrieved.
"NG" (for "not good"): Failed to retrieve the signal.
"" (blank): Signal Check unexecuted (initial state, machine connection failure, etc.), or after executing Signal Check, setting was changed on the [Signal Setting] dialog box and the [OK] button was clicked.

For the details on Signal Check, refer to the "[Checking the Signals](#)" subsection.
- **The [Add] button**
Clicking this button opens the [Signal Setting] dialog box to register new custom signal.
- **The [Edit] button**
Clicking this button opens the [Signal Setting] dialog box to edit selected custom signal.
- **The [Delete] button**
Clicking this button deletes a selected custom signal. When multiple rows are selected in advance, this button can delete them at once.

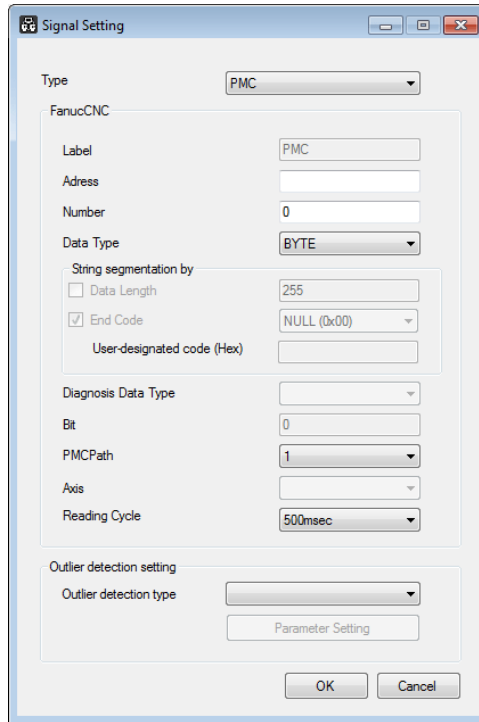
NOTE

When a signal is referenced by equipment, deleting it will be denied with an error message. For the details, refer to the "[Referenced Error](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 112/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

4.2.6.1 Add Custom Signal

Click the [Add] button. The [Signal Setting] dialog box appears for you to register a new custom signal. For the details of the outlier detection settings, refer to the "[Set Outlier Detection of Signal](#)" subsection.



The image shows the 'Signal Setting' dialog box. It has a title bar with a close button. Inside, there's a 'Type' dropdown menu set to 'PMC'. Below it, the 'FanucCNC' section contains fields for 'Label' (PMC), 'Address' (empty), 'Number' (0), 'Data Type' (BYTE), 'String segmentation by' (Data Length: 255, End Code: NULL (0x00)), and 'User-designated code (Hex)' (empty). The 'Diagnosis Data Type' is empty, 'Bit' is 0, 'PMCPATH' is 1, 'Axis' is empty, and 'Reading Cycle' is 500msec. At the bottom, there's an 'Outlier detection setting' section with 'Outlier detection type' (empty) and a 'Parameter Setting' button. 'OK' and 'Cancel' buttons are at the very bottom.

■ The [Signal Setting] dialog box

- Type
Select the type of the signal from the following:

[PMC]: PMC signal
 [MacroVar]: Macro variable
 [PcodeVar]: P-CODE variable
 [Diagnosis]: diagnostic signal

NOTE

- 1 P-CODE variables (PcodeVar) are variables used in a P-CODE macro.
- 2 Update Option license is required to retrieve diagnostic signals. Until Update Option license is verified, the [Type] drop-down list hides the [Diagnosis] type, so that the user will be unable to select it.
- 3 A new custom signal can be defined provided that there exists no custom signal with exactly the same type, address, number and data type. The only exceptions are P-CODE variables whose numbers fall in a range of 100 to 89999. Such P-CODE variables cannot coexist because those numbers are regarded as identical due to the reason that they are for the same common area.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 113/868

- Fields to set for FanucCNC type machines
Fields available to set vary depending on the type selected for the [Type] field. The following table explains what and how to set.

<For PMC signals>

Field	Values to set		Remarks
Label	No need to set a value for this field. It displays "PMC."		
Address	Set the address of the signal. Valid values are "G," "F," "Y," "X," "A," "R," "T," "K," "C," "D," "M," "N," "E," and "Z."		Example: D (in D100)
Number	Set the data number of the signal. A number in a range of 0 to 99999 is available.		Example: 100 (in D100)
Data Type	Select a data type from the drop-down list. Options are as follows: BIT (bit type) BYTE (1 byte) WORD (2 bytes) DWORD (4 bytes) REAL (4 bytes) LREAL (8 bytes) Unsigned BYTE (1 byte) Unsigned WORD (2 bytes) Unsigned DWORD (4 bytes) STRING (select to collect the PMC signal data as a string. See NOTE)		The following data types employ little-endian representation: WORD DWORD Unsigned WORD Unsigned DWORD REAL (single precision real) LREAL (double precision real)
String segmentation by		Set the [Data Length] field and/or [End Code] field when the [Data Type] field is set to "STRING."	Either or both of the [Data Length] and [End Code] fields must be set. By default, the check box for the [End Code] field is selected.
	Data Length	Select the check box and specify the data length. A number from 1 to 255 is available. When the check box is not selected, this field is regarded as set to the maximum (255 bytes).	
	End Code	Select the check box and select an end code from the drop-down list. Options are as follows: NULL (0x00): (To set NULL as the end code) ETX (0x03): (To set ETX as the end code) LF (0x0A): (To set LF as the end code) CR (0x0D): (To set CR as the end code) User-Designated: (To employ a user-designated code that should be set in hexadecimal for the [User-designated code (Hex)] field. The hexadecimal value should be within 00 to FF.)	
Bit	Set the bit position of BIT type data. Valid values are 0 to 7.		
PMCPATH	Select a PMC path from the drop-down list. Options are 1 to 5, and 9. * To get the value of a dual check safety PMC, set 9 for this field.		
Reading Cycle	Select a reading cycle from the drop-down list. Options are 500msec, 10sec, and 60 sec.		

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 114/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

NOTE**■ On encoding string type data**

- 1) String type data are encoded to ASCII character codes.
- 2) String type data can be encoded to a string of up to 255 bytes length.
- 3) When an end code is specified, the string type data will be encoded byte by byte from the top address to one-byte before the end code.
- 4) Valid ASCII character codes are 0x20 to 0x7E, and 0x0A (LF code). The other values are encoded to 0x00 (null character).
When, for example, LF code is employed as an end code, the portion from the top to one-byte before a first LF code will be segmented to a string.
- 5) To segment a string by data length (or the number of characters), data will be truncated in the specified length and converted to a string.

■ Example of encoding string type data

PMC signal data

Address of signal data	R100	R101	R102	R103	R104	R105	R106	R107	R108	R109	R110	R111	R112
Data (Binary)	53	31	32	20	2F	00	37	30	35	2F	0A	43	4E
Data (ASCII code)	P	1	2	*1	-	*2	7	0	5	/	*3	C	N

*1: space, *2: null, *3: LF (Line Feed)

Depending on values set for the [Data Length] field and the [End Code] field (or the [User-designated code (Hex)] field), portion of the data to be encoded varies. The following table shows which portion will be segmented and how they will be encoded when the [Address] field and the [Number] field are set to "R" and "100," respectively.

The mark "-" in the following table denotes that the field has no value.

Data Length	End Code	Conversion Result	Explanation
13	-	P12 - 705/ CN	The byte at R105 (NULL) is converted to a space. A line feed is performed at R110 to add a new line. As a result, data of 13 bytes from R100 to R112 will be converted.
10	-	P12 - 705/	Data of 10 bytes from R100 to R109 will be converted.
13	CR (0x0D)	P12 - 705/ CN	Data of 13 bytes from R100 to R112 will be converted because the specified end code (CR) is not found in the portion.
13	NULL (0x00)	P12 -	Data of 5 bytes from R100 to R104 will be converted because the byte at R105 has the specified end code (NULL).
-	LF (0x0A)	P12 - 705/	Data of 13 bytes from R100 to R109 will be converted because the specified end code (LF) is found at R110, which has a distance of 10 bytes (less than the maximum of 255 bytes) from the top of the data.
-	User-Designated "/" (0x2F)	P12 - 705	Data of 12 bytes from R100 to R108 will be converted because the specified end code ("/") is found at R109, which has a distance of 9 bytes (less than the maximum of 255 bytes) from the top of the data.
-	CR (0x0D)	P12 - 705/ CN....	Data of 255 bytes from R100 to R354 will be converted because the specified end code (CR) is not found in the portion. (If the specified end code exists, as with the previous example, the data will be truncated at one byte before the found end code.)

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	115/868

<For macro variables >

Field	Values to set	Remarks
Label	No need to set a value for this field. It displays "MacroVar."	
Number	Set the variable number of the signal. Valid numbers are 1 to 33, and 9 to 99999999.	
CNCPath	Select a CNC path from the drop-down list.	
Reading Cycle	Select a reading cycle from the drop-down list. Options are 500msec, 10sec, and 60 sec.	

<For P-CODE variables >

Field	Values to set	Remarks
Label	No need to set a value for this field. It displays "PcodeVar."	
Number	Set the variable number of the signal. Valid numbers are 1 to 33, and 100 to 89999.	
Data Type	Select a data type from the drop-down list. Options are as follows: Conversation Auxiliary Execution	
CNCPath	Select a CNC path from the drop-down list.	
Reading Cycle	Select a reading cycle from the drop-down list. Options are 500msec, 10sec, and 60 sec.	

<For diagnostic signals>

Field	Values to set	Remarks
Label	No need to set a value for this field. It displays "Diagnosis."	
Number	Set the diagnosis number of the signal. A number in a range of 0 to 99999 is available.	
Data Type	Select a data type from the drop-down list (The default is Integer). Options are as follows: Bit (bit type) Integer (byte, word, 2-word types) Real (real, floating point number types)	
Diagnosis Data Type	Select a type of the diagnosis data from the drop-down list (The default is Path). Options are as follows: Path Axis Spindle	NOTE 1
Bit	Set the bit position of Bit type data. Valid values are 0 to 7.	
CNCPath	Select a PMC path from the drop-down list. Options are as follows: <x>: where x is a CNC path (from 1 to the maximum CNC path number) All: all the CNC pats	

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	116/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

Field	Values to set	Remarks
Axis	Select an axis from the drop-down list. Options are as follows: <n>: where n is an axis number (from 0 to the maximum axis number minus one) All: all the axes (or the spindles)	NOTE 2 to 4
Reading Cycle	Select a reading cycle from the drop-down list. Options are 500msec, 10sec, and 60 sec.	

NOTE

- 1 When retrieving diagnosis data of spindle axes in Series 16*i* models, select [Path] for the [Diagnosis Data Type] field. If [Spindle] should be selected, no diagnosis data will be retrieved.
- 2 When the [Diagnosis Data Type] field is set to [Axis], <n> is the maximum number of the servo axes for the path set for the [CNCPath] field (however, if the number of servo axes for the path selected in the [Add Machine] dialog box is 0, the [Axis] field will be disabled).
- 3 When the [Diagnosis Data Type] field is set to [Spindle], <n> is the maximum number of the spindle axes for the path set in the [CNCPath] field (however, if the number of spindles for the path selected in the [Add Machine] dialog box is 0, the [Axis] field will be disabled).
- 4 When the [Diagnosis Data Type] field is set to [Path], the [Axis] field will be disabled).

* For Series 16*i* models, refer to the NOTE 4 in the "[Basic Signals of FanucCNC](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION		Sheet. 117/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

4.2.6.2 Edit Custom Signal

To edit a custom signal, select any signal and click [Edit]. The [Signal Setting] dialog box appears for you to edit the selected custom signal.

Note that the [Type] field is not-editable.

If outlier detection settings has been made, changing the [Data Type] field opens a confirmation message dialog box prompting that the current outlier detection settings will be initialized. Click [OK] in the message to initialize the current settings.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	118/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

■ Setting examples with the [Signal Setting] dialog box

Example 1:

A PMC signal with "D" set for the [Address] field, "100" for the [Number] field, "BIT" for the [Data Type] field, "7" for the [Bit] field, and "1" for the [PMCPATH] field:

No.	Type	Address/Number	CNCPath/PMCPATH	Axis	Data Type	End Code	Reading Cycle
1	PMC	D100.7	1		BIT		500msec

Example 2:

A PMC signal with "D" set for the [Address] field, "200" for the [Number] field, "WORD" for the [Data Type] field, and "1" for the [PMCPATH] field:

No.	Type	Address/Number	CNCPath/PMCPATH	Axis	Data Type	End Code	Reading Cycle
2	PMC	D200	1		WORD		500msec

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 119/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

A PMC signal with "R" set for the [Address] field, "100" for the [Number] field, "STRING" for the [Data Type] field, "LF (0x0A)" for the [End Code] field, and "1" for the [PMCPATH] field:

Signal Setting

Type: PMC

FanucCNC

Label: PMC

Address: R

Number: 100

Data Type: STRING

String segmentation by

☐ Data Length: 255

☒ End Code: LF (x0A)

User-designated code (Hex):

Diagnosis Data Type:

Bit: 0

PMCPath: 1

Axis:

Reading Cycle: 500msec

Outlier detection setting

Outlier detection type:

Parameter Setting

OK Cancel

No	Type	Address/Number	CNCPath/PMCPath	Axis	Data Type	End Code	Reading Cycle
1	PMC	D100.7	1		BIT		500msec
2	PMC	D200	1		WORD		500msec
3	PMC	R100	1		STRING	LF (0x0A)	500msec

A PMC signal with "R" set for the [Address] field, "200" for the [Number] field, "STRING" for the [Data Type] field, "16" for the [Data Length] field, "User-Designated" for the [End Code] field, "40" for the [User-designated code (Hex)] field, and "1" for the [PMCPATH] field:

Signal Setting

Type: PMC

FanucCNC

Label: PMC

Address: R

Number: 200

Data Type: STRING

String segmentation by

- ☒ Data Length: 16
- ☒ End Code: User-Designated

User-designated code (Hex): 40

Diagnosis Data Type:

Bit: 0

PMCPATH: 1

Axis:

Reading Cycle: 500msec

Outlier detection setting

Outlier detection type: Parameter Setting

OK Cancel

Signal Setting							
No	Type	Address/Number	CNCPath/PMCPath	Axis	Data Type	End Code	Reading Cycle
1	PMC	D100.7	1		BIT		500msec
2	PMC	D200	1		WORD		500msec
3	PMC	R100			STRING	LF (0x0A)	500msec
4	PMC	R200	1		STRING (16)	UD (0x40)	500msec

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	120/868	
Ed.	Date	Design	Description									
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.						Inoue

Example 5:

A custom macro variable with "520" set for the [Number] field, and "1" for the [CNCPath] field:

Signal Setting

Type: MacroVar

FanucCNC

Label: MacroVar

Address:

Number: 520

Data Type:

String segmentation by

☐ Data Length

☐ End Code

User-designated code (Hex):

Diagnosis Data Type:

Bit: 0

CNCPath: 1

Axis:

Reading Cycle: 500msec

Outlier detection setting

Outlier detection type:

Parameter Setting

OK Cancel

No	Type	Address/Number	CNCPath/PMCPath	Axis	Data Type	End Code	Reading Cycle
1	PMC	D100.7	1		BIT		500msec
2	PMC	D200	1		WORD		500msec
3	PMC	R100	1		STRING	LF (0x0A)	500msec
4	PMC	R200	1		STRING (16)	UD (0x40)	500msec
5	MacroVar	520	1				500msec

Example 6:

A P-CODE variable (conversation macro) with "600" set for the [Number] field, "Conversation" for the [Data Type] field, and "1" for the [CNCPath] field:

Signal Setting

Type: PcodeVar

FanucCNC

Label: PcodeVar

Address:

Number: 600

Data Type: Conversation

String segmentation by

☐ Data Length

☐ End Code

User-designated code (Hex):

Diagnosis Data Type:

Bit: 0

CNCPath: 1

Axis:

Reading Cycle: 500msec

Outlier detection setting

Outlier detection type:

Parameter Setting

OK Cancel

No	Type	Address/Number	CNCPath/PMCPath	Axis	Data Type	End Code	Reading Cycle
1	PMC	D100.7	1		BIT		500msec
2	PMC	D200	1		WORD		500msec
3	PMC	R100	1		STRING	LF (0x0A)	500msec
4	PMC	R200	1		STRING (16)	UD (0x40)	500msec
5	MacroVar	520	1				500msec
6	PcodeVar	600	1		Conversation		500msec

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	121/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

A diagnostic signal (current actual power consumption of each spindle) with "4902" set for the [Number] field, "Integer" for the [Data Type] field, "Spindle" for the [Diagnosis Data Type] field, "1" for the [CNCPath] field, and "2" for the [Axis] field:

Signal Setting

Type: Diagnosis

FanucCNC

Label: Diagnosis

Address:

Number: 4902

Data Type: Integer

String segmentation by

☐ Data Length:

☐ End Code:

User-designated code (Hex):

Diagnosis Data Type: Spindle

Bit: 0

CNCPath: 1

Axis: 2

Reading Cycle: 60sec

Outlier detection setting

Outlier detection type:

Parameter Setting

OK Cancel

Signal Setting							
No	Type	Address/Number	CNCPath/PMCPath	Axis	Data Type	End Code	Reading Cycle
1	PMC	D100.7	1		BIT		500msec
2	PMC	D200	1		WORD		500msec
3	PMC	R100	1		STRING	LF (0x0A)	500msec
4	PMC	R200	1		STRING (16)	UD (0x40)	500msec
5	MacroVar	520	1				500msec
6	PcodeVar	600	1		Conversation		500msec
7	Diagnosis	4902	1	S2	Integer		60sec

A diagnostic signal (total travel distance of axis) with number "366" set for the [Number] field, "Integer" for the [Data Type] field, "Axis" for the [Diagnosis Data Type] field, "2" for the [CNCPath] field, and "ALL" for the [Axis] field:

* Assume that the [Servo Axes] column for Path2 in the [Add Machine] dialog box is set to 3.

Signal Setting

Type: Diagnosis

FanucCNC

Label: Diagnosis

Address:

Number: 366

Data Type: Integer

String segmentation by

☐ Data Length

☐ End Code

User-designated code (Hex):

Diagnosis Data Type: Axis

Bit: 0

CNCPath: 2

Axis: ALL

Reading Cycle: 60sec

Outlier detection setting

Outlier detection type: Parameter Setting

OK Cancel

Signal Setting							
No	Type	Address/Number	CNCPath/PMCPath	Axis	Data Type	End Code	Reading Cycle
1	PMC	D100.7	1		BIT		500msec
2	PMC	D200	1		WORD		500msec
3	PMC	R100	1		STRING	LF (bx0A)	500msec
4	PMC	R200	1		STRING (16)	UD (bx0A)	500msec
5	MacroVar	520	1				500msec
6	PCodeVar	600	1		Conversation		500msec
7	Diagnosis	4902	1	S2	Integer		60sec
8	Diagnosis	366	2	A0	Integer		60sec
9	Diagnosis	366	2	A1	Integer		60sec
10	Diagnosis	366	2	A2	Integer		60sec

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	122/868	
Ed.	Date	Design	Description									
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.						Inoue

4.2.7 Set Tool Life Signal of FanucCNC

This function is provided to browse tool life signals in MT-LINK*i*.

Tool life signals can be registered up to 1039.

Select a machine in the Collector tree and click the [Tool Life Signal] tab in the right pane.

NOTE

Update Option license is required to apply this function to FanucCNC type machines. Until Update Option license is verified, the following operations will cause an error message:

- Clicking the [Tool Life Signal] tab when tool life signals have been set up
- Clicking the [Set the Number of Tool Groups] button when no tool life signals are set up

For the details on how to address such error message, refer to the following subsection of the "FANUC MT-LINK*i* / Integration Server Troubleshooting MANUAL":

The "AdminTool Displays License Warning" subsection in the "TROUBLES RELATED TO LICENSE" section in the "MT-LINK*i* TROUBLESHOOTING" chapter

The screenshot shows the 'Machine Setting' window with the 'Tool Life Signal' tab selected. The left pane shows a tree view with 'Collector01', 'Machine001', 'Machine002', 'Collector02', and 'Machine003' (selected). The right pane shows the 'Tool Life Signal' configuration for 'Machine003'. The 'Signal Setting' section has 'Narrow down' and 'Enable' buttons. Below is a table of tool life signals:

No	Signal	Description	CNCPath	Enable	Outlier Detection	Reading Cycle	Check
1	ToolGroupNum	Current tool group	1	☒	☒	500msec	☒
2	ToolLifeCount01	Tool life count 01	1	☒	☒	10sec	☒
3	ToolLifeCount02	Tool life count 02	1	☒	☒	10sec	☒
4	ToolLifeCount03	Tool life count 03	1	☒	☒	10sec	☒
5	ToolLifeCount04	Tool life count 04	1	☒	☒	10sec	☒
6	ToolLifeCount05	Tool life count 05	1	☒	☒	10sec	☒
7	ToolLifeCount06	Tool life count 06	1	☒	☒	10sec	☒
8	ToolLifeCount07	Tool life count 07	1	☒	☒	10sec	☒
9	ToolLifeCount08	Tool life count 08	1	☒	☒	10sec	☒
10	ToolLifeLimit01	Tool life limit 01	1	☒	☒	10sec	☒
11	ToolLifeLimit02	Tool life limit 02	1	☒	☒	10sec	☒
12	ToolLifeLimit03	Tool life limit 03	1	☒	☒	10sec	☒
13	ToolLifeLimit04	Tool life limit 04	1	☒	☒	10sec	☒
14	ToolLifeLimit05	Tool life limit 05	1	☒	☒	10sec	☒
15	ToolLifeLimit06	Tool life limit 06	1	☒	☒	10sec	☒
16	ToolLifeLimit07	Tool life limit 07	1	☒	☒	10sec	☒
17	ToolLifeLimit08	Tool life limit 08	1	☒	☒	10sec	☒
18	ToolLifeForenotice01	Tool life forenotice 01	1	☒	☒	10sec	☒

Buttons at the bottom: Apply, OK, Cancel. A 'Set the Number of Tool Groups' button is on the right.

■ The [Tool Life Signal] tab

- Set the Number of Tool Groups

Clicking this button opens the [Set the Number of Tool Groups] dialog box for setting the numbers of tool groups for each CNC path.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	123/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

- **Signal**
This column in the signal lists the names of signals that can be set for the selected machine. To list only specific tool life signals, narrow down the signal list. For the details on how to narrow down the list, refer to the "[Narrow down Signal List](#)" subsection.

For the detailed description of each signal, refer to the "[Tool Life Signals of FanucCNC](#)" section.
- **Description**
This column displays the Web name of each signal as its description.
- **CNCPath**
This field displays CNC path of signals.
- **Enable**
This is a check box specifying whether to execute data collection of signals by the Collector. If the check box is selected, the data will be collected; otherwise, the data will not be collected.

NOTE

- 1 Some signals listed in the [Signal Setting] table may not be available depending on machine model. For those non-existent signals, clear the check boxes. If the check box for a non-existent signal is selected, its values will not be stored to the database, and will not be displayed on the Web screen.
- 2 When a signal is referenced by equipment, clearing the [Enable] check box for it will be denied with an error message. For the details, refer to the "[Referenced Error](#)" subsection.

- **Outlier Detection**
For signals with [Outlier Detection] check box selected (signals of data type Int32), outlier detection of the signal can be set by double-clicking that row. The check box column cannot be edited.
For the details of the outlier detection settings, refer to the "[Set Outlier Detection of Signal](#)" subsection.
- **Reading Cycle**
With this field, you can specify a reading interval per signal. Options are [500msec], [10sec], and [60sec].
- **Check**
The result of the Signal Check, when the [Signal Check] button is clicked, is displayed.
 "OK" : The signal was retrieved
 "NG" : Failed to retrieve the signal
 "" (blank) : Signal Check unexecuted (initial state, machine connection failure, etc.)

For the details on Signal Check, refer to the "[Checking the Signals](#)" subsection.

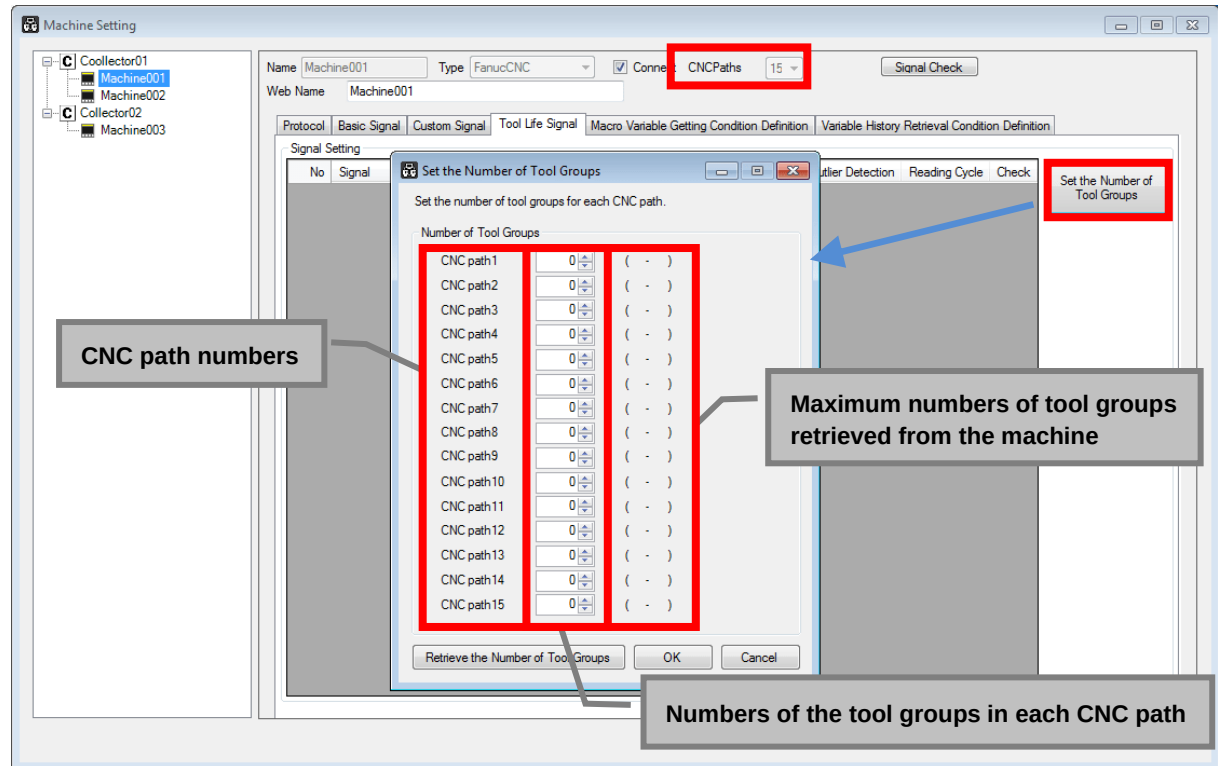
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	124/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

4.2.7.1 Set Numbers of Tool Groups

To set the numbers of tool groups, open the [Set the Number of Tool Groups] dialog box by clicking the [Set the Number of Tool Groups] button.

■ The [Set the Number of Tool Groups] dialog box

The following figure of the [Set the Number of Tool Groups] dialog box shows a setting example, where the [CNCPaths] drop-down list is set to 15 on the [Machine Setting] window.



- **CNC path numbers**
The [Number of Tool Groups] group box lists as many CNC path numbers as the number that is set for the [CNCPaths] drop-down list on the [Machine Setting] window.
- **Retrieve the Number of Tool Groups**
Clicking this button retrieves the path-wise maximum numbers of tool groups from the selected machine.
- **The [Numbers of tool groups] drop-down list**
Set each the path-wise number to a number between 0 and 256.
By default, each entry is set to 0.
Clicking the [Retrieve the Number of Tool Groups] button populates all the entries with the values retrieved from the selected machine.
- **The [Maximum numbers of tool groups] text field**
Clicking the [Retrieve the Number of Tool Groups] button populates all the entries with the values retrieved from the selected machine. Those entries are uneditable.
By default, each entry is set to "-" symbol.
- **OK**
Clicking this button saves the settings and closes the dialog box. The [Signal Setting] group box lists the tool life signals on the [Tool Life Signal] tab page of the [Machine Setting] window.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	125/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

NOTE

Decreasing the value of the [Numbers of tool groups] drop-down list may delete tool life signals of the tool groups that are being deleted by the decrease. If such a tool life signal is referenced by equipment, this decrease will be denied with an error message. For the details, refer to the **"Referenced Error"** subsection.

- Cancel
Clicking this button discards the settings and closes the dialog box.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	126/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

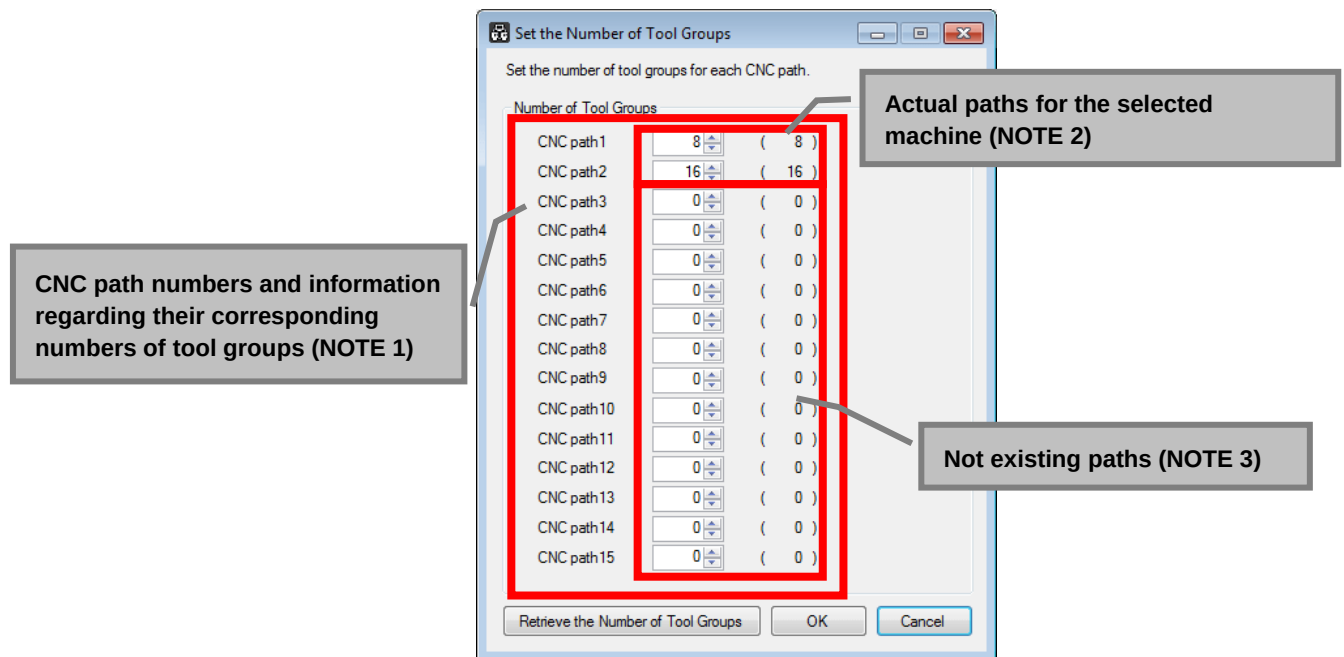
4.2.7.2 Retrieve Path-wise Numbers of Tool Groups

Click the [Retrieve the Number of Tool Groups] button to retrieve path-wise numbers of tool groups from the selected machine. A confirmation dialog box appears prompting whether to proceed. Click [Yes] to populate those retrieved numbers in the [Set the Number of Tool Groups] dialog box. Or click [No] to cancel the setting and to return to the [Machine Setting] window.

NOTE

Clicking the [Retrieve the Number of Tool Groups] button populates all the entries with the values retrieved from the selected machine.

The following figure of the [Set the Number of Tool Groups] dialog box shows an example of retrieving numbers, where the [CNCPaths] drop-down list is set to 10 on the [Machine Setting] window but actually the selected machine has two paths.



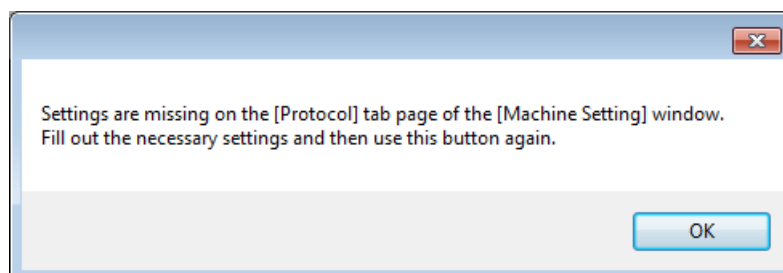
NOTE

- 1 The [Set the Number of Tool Groups] dialog box lists as many entries as the value set for the [CNCPaths] drop-down list on the [Machine Setting] window regardless of how many paths the machine actually has.
- 2 For the entries corresponding to existing paths, their drop-down lists and text fields will be set to the retrieved values.
- 3 For the entries, if any, not corresponding to existing paths, their drop-down lists and text fields will be set to 0.

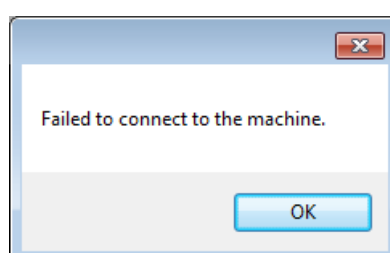
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	127/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

4.2.7.3 Messages on Failure to Retrieve Path-wise Numbers of Tool Groups

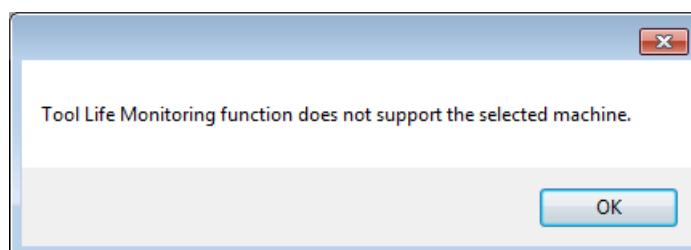
- If the [IP Address/Host Name] on the [Protocol] tab page of the [Machine Setting] window is missing, which is required for communication with the selected machine, an error message appears as below and the [Set the Number of Tool Groups] dialog box will not update the numbers.



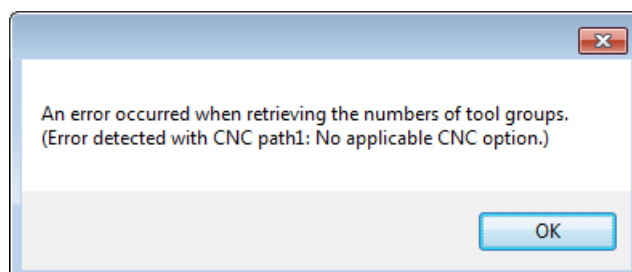
- If the [Set the Number of Tool Groups] dialog box has failed to communicate with the selected machine, an error message appears as below and the dialog box is unable to update the numbers.



- If the tool life monitoring function does not support the selected machine, an error message appears as below and the [Set the Number of Tool Groups] dialog box sets the numbers to 0.
For the supported models, refer to the "[Tool Life Signals of FanucCNC](#)" subsection.



- If an error has occurred during retrieving the numbers, an error message appears as below and the [Set the Number of Tool Groups] dialog box sets the numbers to 0.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	128/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

4.2.8 Set Laser Control Data Signal of FanucCNC

This function is provided to browse the laser control data signals of machines in MT-LINK*i*. Select a machine in the Collector tree and click the [Laser control data signal setting] tab in the right pane.

NOTE

[Laser control data signal setting] tab appears when one or more of the control type [L] check boxes are selected in the [Add Machine] dialog box.

On the [Laser control data signal setting] tab, check the [CNCPath] of each signal to select the reading cycle, and specify whether to enable the collection function or not.

When the [Enable] check box of the signal is selected, the Collector collects the signal data.

The signal which has the [Enable] check box selected on the [Laser control data signal setting] tab of the [Machine Setting] window will be added as a status on the [Simple assign status area] of the [Equipment Setting] window.

The screenshot shows the 'Machine Setting' window. On the left, a tree view shows 'DESKTOP-GHB2GK' expanded, with 'Machine001', 'Machine002', and 'Collector02' listed. 'Machine003' is selected. The main area shows the 'Laser control data signal setting' tab. It includes fields for 'Name' (Machine003), 'Type' (FanucCNC), 'Connect' (checked), 'CNCPaths' (3), and 'Web Name' (Machine003). Below these are tabs for 'Protocol', 'Basic Signal', 'Custom Signal', 'Tool Life Signal', 'Laser control data signal setting' (selected), 'Macro Variable Getting Condition Definition', and 'Variable History Retrieval Con'. The 'Signal Setting' section has a 'Narrow down' button and fields for 'Description', 'Enable', and 'Reading Cycle'. A table lists 18 signals with their respective settings.

No	Signal	Description	CNCPath	Enable	Outlier Detection	Reading Cycle	Check
1	PwrCommand	Power command	1	☒	☒	500msec	☒
2	PulseBottomCommand	Pulse bottom command	1	☐	☒	500msec	☐
3	FrequencyCommand	Frequency command	1	☐	☒	500msec	☐
4	DutyCommand	Duty command	1	☐	☒	500msec	☐
5	LaserPwr	LASER POWER	1	☐	☒	500msec	☐
6	LaserSequence	LASER SEQUENCE	1	☒	☒	500msec	☐
7	SigBeamOut	Beam output signal	1	☒	☐	500msec	☐
8	SigLaserAlarm	Laser alarm signal	1	☒	☐	500msec	☐
9	BCReflection	BC REFLECTION	1	☐	☒	500msec	☐
10	WaterFlowRate1	WATER FLOW RATE 1	1	☐	☒	500msec	☐
11	WaterFlowRate2	WATER FLOW RATE 2	1	☐	☒	500msec	☐
12	LockerTemp	LOCKER TEMPERATURE	1	☐	☒	60sec	☐
13	LockerHumidity	LOCKER HUMIDITY	1	☐	☒	60sec	☐
14	WaterTemp	WATER TEMPERATURE	1	☐	☒	60sec	☐
15	DewPoint	DEW POINT	1	☐	☒	60sec	☐
16	LC1OutPwr	LC1 OUTPUT POWER	1	☐	☒	500msec	☐
17	LC2OutPwr	LC2 OUTPUT POWER	1	☐	☒	500msec	☐
18	LC3OutPwr	LC3 OUTPUT POWER	1	☐	☒	500msec	☐

■ The [Laser control data signal setting] tab

● Signal

This column in the signal lists the names of signals that can be set for the selected machine. To list only specific laser control data signals, narrow down the signal list. For the details on how to narrow down the list, refer to the "[Narrow down Signal List](#)" subsection.

For the details on the description of each signal, refer to the "[Laser Control Data Signals of FanucCNC](#)" subsection.

● Description

This column displays the Web name of each signal as its description.

● CNCPath

CNC path of each signal is displayed.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 129/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- Enable

The check box specifies whether the Collector collects the data of the signal. When the check box is selected, the data is collected; otherwise, the data is not collected.

NOTE

- 1 Some signals listed in the [Signal Setting] table may not be available depending on machine model. For those non-existent signals, clear the check boxes. If the check box for a non-existent signal is selected, its values will not be stored to the database, and will not be displayed on the Web screen.
- 2 When a signal is referenced by equipment, clearing the [Enable] check box for it will be denied with an error message. For the details, refer to the "[Referenced Error](#)" subsection.

- Outlier Detection

As for the signals which have the check box selected in the [Outlier Detection] column (signals whose data types are Int32 or Double) the outlier detection of the signal can be set by double-clicking that row. The check box of the column cannot be edited.

For the details on the setting of the outlier detection, refer to the "[Set Outlier Detection of Signal](#)" subsection.

- Reading Cycle

With this field, you can specify a reading interval per signal.

Options are [500msec] / [10sec] / [60sec].

- Check

The result of the Signal Check, when the [Signal Check] button is clicked, is displayed.

"OK": The signal was retrieved

"NG" (for "not good"): Failed to retrieve the signal

"-" (hyphen): Not a target of Signal Check

"" (blank): Signal Check unexecuted (initial state, machine connection failure, etc.)

For the details on Signal Check, refer to the "[Checking the Signals](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue
FANUC CORPORATION					Sheet.	130/868	

4.2.9 Register Macro Variable Getting Condition Definition of FanucCNC

This setting enables the user to define how to retrieve macro variables and how to view their histories through MT-LINK*i*.

- "Macro variable getting condition" and "Variable history retrieval condition" can be registered to a maximum number of 1000 per machine.
- For one Macro Variable getting condition, the maximum number of collection is 256.

Before proceeding, read the following explanation on how MT-LINK*i* handles a trigger macro variable:

- 1 The CNC changes the value of a trigger macro variable to "1."
- 2 MT-LINK*i* keeps monitoring the variable. When the value of the variable turns to "1," MT-LINK*i* retrieves the values of the basic macro variable and as many subsequent macro variables as specified to capture.
- 3 MT-LINK*i* sets the trigger macro variable to "0."
- 4 MT-LINK*i* stores the values retrieved in step 2 into the database.



WARNING

MT-LINK*i* controls trigger macro variables using a handshaking process that waits for a CNC to turn a trigger macro variable to "1" and sets the variable to "0." If trigger macro variables reside in areas that are used on the CNC or by other applications, make sure before setting trigger macro variables that no problem should occur.

NOTE

When setting a response timeout for a trigger macro variable on CNCs, CNC operators should take into account that MT-LINK*i* may not have sufficient time to reset the value of the trigger macro variable from "1" to "0," because MT-LINK*i* is not running, or the network connection has problems.

A response timeout for a trigger macro variable should not be less than one second. The maximum value for a response timeout is the value for the connection timeout of the target CNC (*).

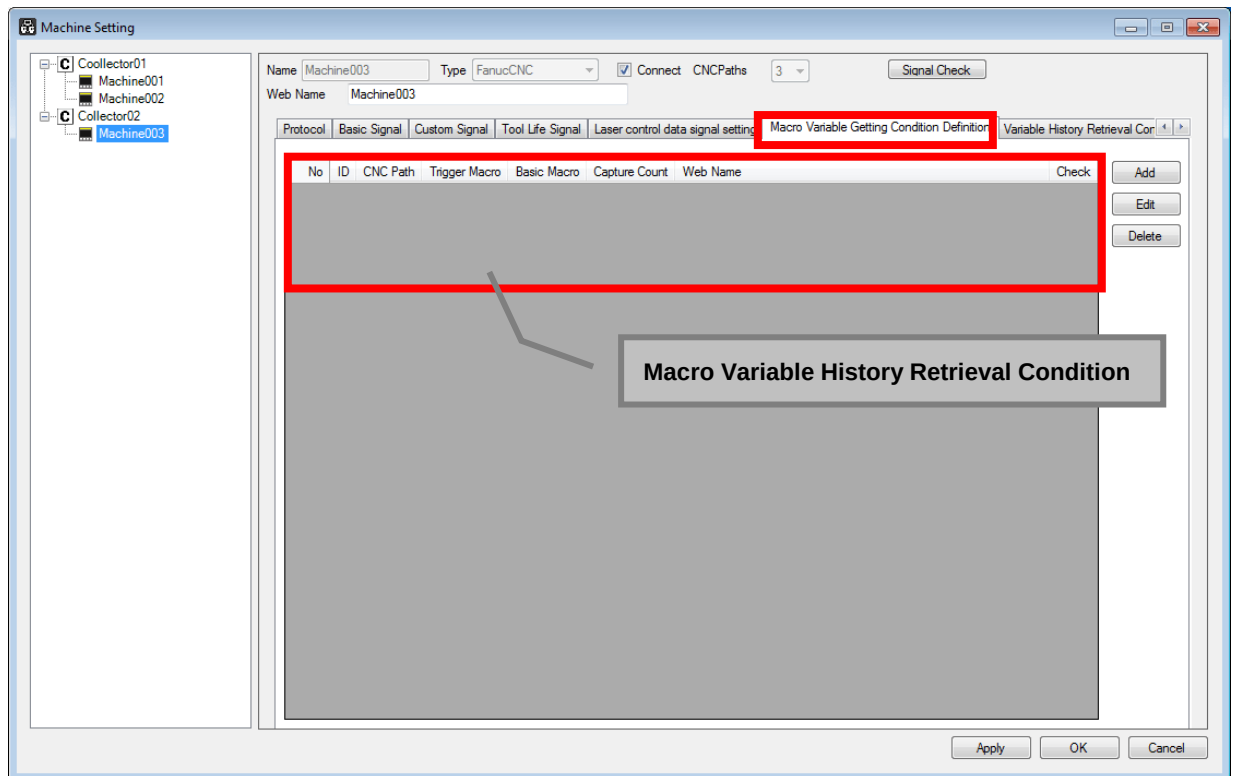
CNC operators should set an appropriate response timeout according to their production processes or environments.

* To check for the value of the connection timeout for the target CNC, select the CNC in the [Machine Setting] window, click the [Protocol] tab, and review the [Time out] for the CNC.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION		Sheet. 131/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

4.2.9.1 Setting Macro Variable Getting Condition Definition of FanucCNC

To register a condition that defines when and how a macro variable is retrieved for a machine, select the machine in the Collector tree and select the [Macro Variable Getting Condition Definition] tab on the right pane.



■ The [Macro Variable Getting Condition Definition] tab

This tab page lists the macro variable getting conditions that are set up in the [Macro Variable getting condition definition Setting] dialog box. For the details of the columns in the list, refer to **The [Macro Variable getting condition definition Setting] dialog box**.

- ID
This column lists the IDs of the registered macro variable getting conditions.
- CNC Path
This column lists the paths for the macro variables.
- Trigger Macro
This column lists the numbers of the trigger macro variables.
- Basic Macro
This column lists the numbers of basic macro variables.
- Capture Count
This column lists the numbers of the macro variables to retrieve.
- Web Name
This column lists the Web names of the macro variables to retrieve.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 132/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

- Check

The result of the Signal Check, when the [Signal Check] button is clicked, is displayed.

"OK": All of the trigger macro variable / the basic macro variable (including variables subsequent to it) were retrieved

"NG" (for "not good"): Failed to retrieve either the trigger macro variable / the basic macro variable (including variables subsequent to it)

"" (blank): Signal Check unexecuted (initial state, machine connection failure, etc.), or after executing Signal Check, setting was changed on the [Macro Variable getting condition definition Setting] dialog box and the [OK] button was clicked

For the details on Signal Check, refer to the "[Checking the Signals](#)" subsection.

- [Add] button

Click this button to add a macro variable getting condition for the selected machine.

- [Edit] button

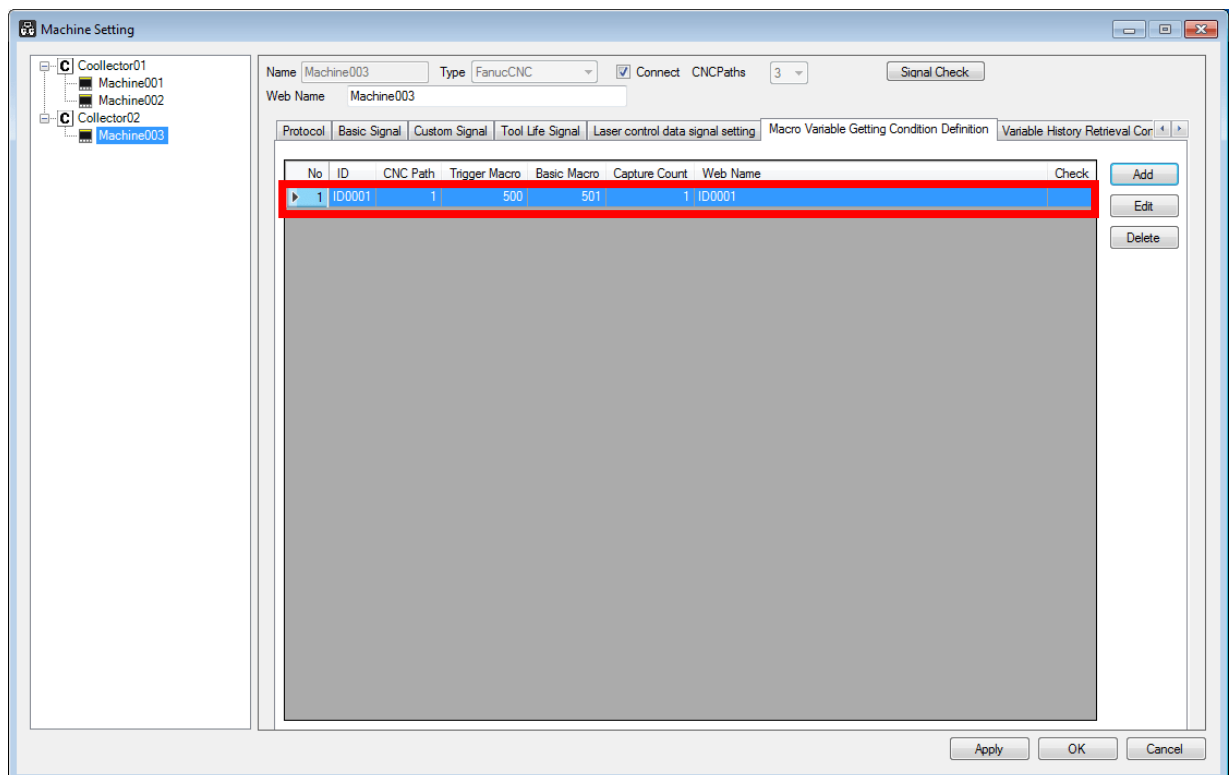
To edit a macro variable getting condition, select it in the [Macro Variable Getting Condition Definition] tab and click this button. This prompts the [Macro Variable getting condition definition Setting] dialog box for you to edit the condition.

- [Delete] button

To delete a macro variable getting condition, select it in the [Macro Variable Getting Condition Definition] tab and click this button.

(1) Adding a macro variable getting condition

Click the [Add] button. A new line for a new condition appears in the table as shown below.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 133/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

(2) Editing a macro variable getting condition

Select a row in the table and click the [Edit] button. The [Macro Variable getting condition definition Setting] dialog box appears for you to edit the selected condition. You can edit [Web Name], [CNC Path], [Trigger Macro], [Basic Macro], [Capture Count] fields and the Web Name of the Captured Macro Variable.

Once a macro variable getting condition becomes effective and when the value of the trigger macro variable turns to "1," MT-LINK*i* collects both the values of the basic macro variable and its subsequent macro variables. Then it resets the value of the trigger macro variable to "0" and stores the collected values in the database.

NOTE

If MT-LINK*i* should not get all the values of variables defined in a condition, it will get none of the values and thus no data will be displayed on the Web screen.

In the example above, when the value of the trigger macro variable #500 turns to "1," and any of the value of macro variable #501, #502, and #503 cannot be retrieved, all the values of #501, #502, and #503 are not retrieved. However, the value of the trigger macro variable is reset to "0." No data for the variables for "ID0001" will be displayed on a Web screen.

■ The [Macro Variable getting condition definition Setting] dialog box

● ID

The ID for the Macro Variable getting condition ("ID <four digit numbering>") is displayed. (Cannot be changed)

For the <four digit numbering>, an unused minimum value starting from "0001" is set automatically. (Example: ID0002)

● Web Name

Set a name for the Macro Variable getting condition. This name should be entered in the language specified for the [Web name language] in the [System Setting] window. Up to 100 bytes can be set. The default value is the same value as the [ID] field.

● CNC Path

Select CNC path number ("1" to the maximum number of CNC paths). The default value is "1."

● Trigger Macro (variable)

Set the number for trigger macro variable. Value from "100" to "999" can be set. "500" is the default value.

● Basic Macro (variable)

Set the number for basic macro variable. Value from 1 to 33 and from 100 to 99,999,999 can be used. "501" is the default value.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	134/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

- **Capture Count**
Set the number of macro variables to retrieve. The maximum number of macro variables that can be set for retrieval is "256." "1" is the default value.
 - **Check**
The result of the Signal Check, when the [Signal Check] button is clicked, is displayed.
 - "OK": The trigger macro variable was retrieved
The basic macro variable (including variables subsequent to it) was retrieved
 - "NG" (for "not good"): Failed to retrieve the trigger macro variable
Failed to retrieve the basic macro variable (including variables subsequent to it)
 - "" (blank): Signal Check unexecuted (initial state, machine connection failure, etc.), or after executing Signal Check, setting was changed on the [Macro Variable getting condition definition Setting] dialog box and the [OK] button was clicked
- For the details on Signal Check, refer to the "**Checking the Signals**" subsection.
- **[Set variable name] button**
Use this button to set the Web name of the macro variable.
 - **[OK] button**
Clicking this button saves the settings and closes the [Macro Variable getting condition definition Setting] dialog box.
 - **[Cancel] button**
Clicking this button discards the settings and closes the [Macro Variable getting condition definition Setting] dialog box.

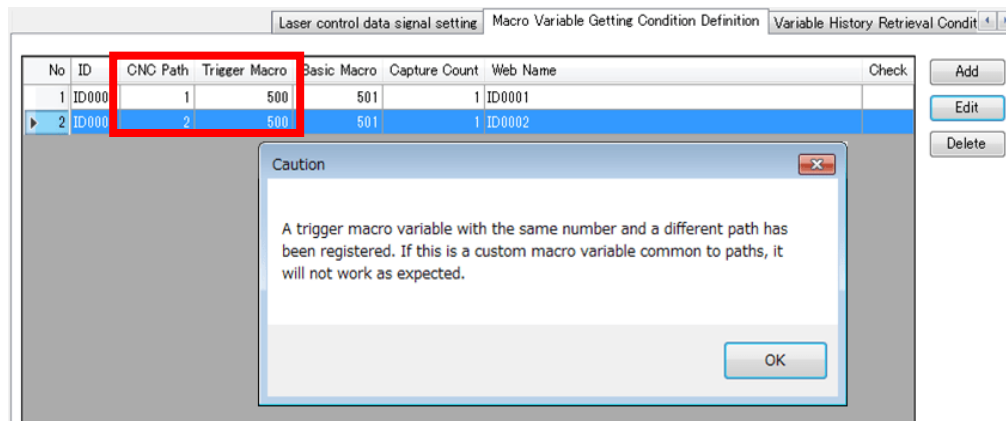
NOTE

- 1 The same trigger macro variable condition cannot be added to the same path.
 - When a same trigger macro variable is already registered for the same path in the [Macro Variable Getting Condition Definition].
 - When a same trigger macro variable is already registered for the same path in the [Variable History Retrieval Condition Definition].
 - 2 If the same trigger macro variable is registered for different paths, caution windows will appear. The detail of the caution windows are explained in the "**Caution windows displayed when the same trigger macro variable is registered for a different path**" paragraph following this NOTE.
- If the trigger macro variable is a custom macro variable common to all the paths, check the parameter of the CNC, as it will not work properly. (As for the parameter of the CNC, refer to the operator's manual of the CNC or the parameter manual on the relevant CNC parameters.)
- If it is a custom macro variable common to all the paths, then do as follows.
- Change the parameter of the custom macro variable common to all the paths.
 - Assign another macro variable as the trigger macro variable.

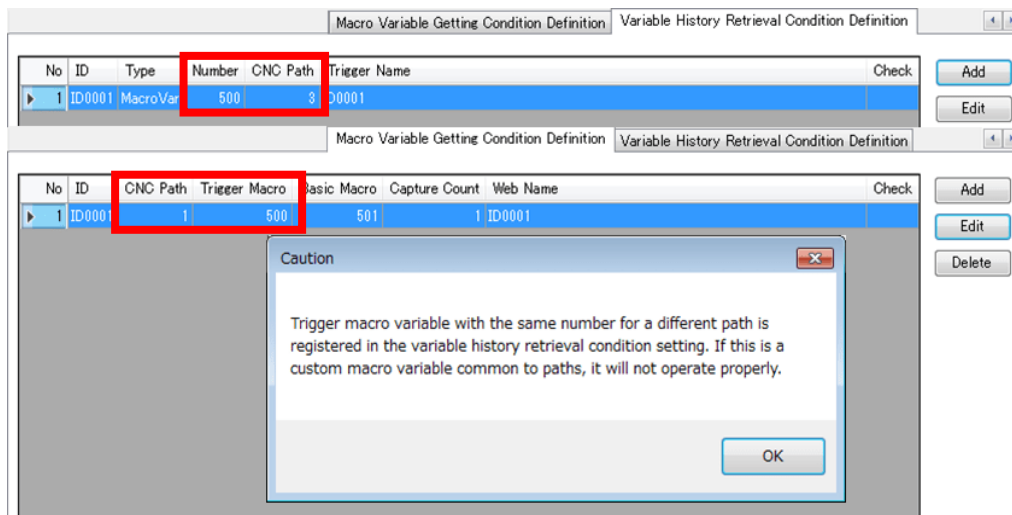
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	135/868

■ **Caution windows displayed when the same trigger macro variable is registered for a different path**

- When the same trigger macro variable is already registered for a different path in the [Macro Variable Getting Condition Definition].

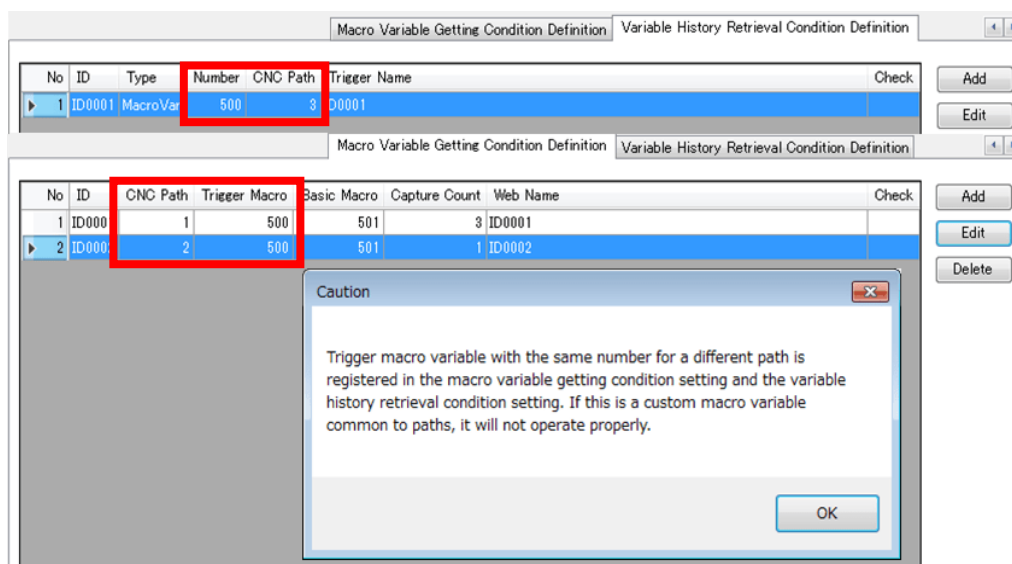


- When the same trigger variable is already registered for a different path in the [Variable History Retrieval Condition Definition].



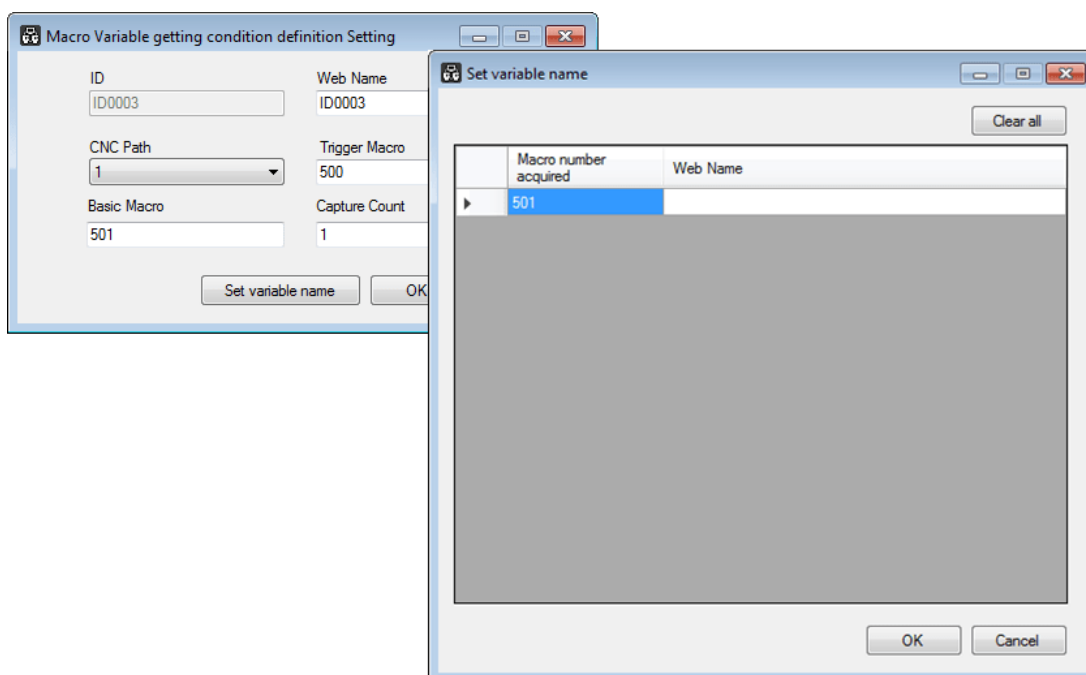
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 136/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- When the same trigger variable is already registered for different paths in the [Macro Variable Getting Condition Definition] and in the [Variable History Retrieval Condition Definition].



(3) Setting Web name of macro variable

- Click the [Set variable name] button.
- [Set variable name] dialog box appears. Enter the Web names of the Macro variable names.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	137/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

■ The [Set variable name] dialog box

- **Macro number acquired**
This number will be the basis for getting the macro variable. This number will be assigned to the collected variables starting from the basic macro variable and adding up. Up to 256 rows can be set.
- **Web Name**
Web names are set for the macro variables in the language specified for the [Web name language] in the [System Setting] window. A Web name with up to 32 bytes can be set for each variable. If the Web name is not specified, it will be displayed in the format of "value<N>" ("N" is the index number within the obtainable values) in the Web screen.
- **[Clear all] button**
Clicking this button clears all the value in the [Web name] column. When clearing all the value, click [Yes] in the confirmation dialog box.
- **[OK] button**
Clicking this button saves the edited settings, and closes the window.
- **[Cancel] button**
Clicking this button discards the edited settings, and closes the window.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	138/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

4.2.9.2 Setting File Output Function When Retrieving Macro Variable of FanucCNC

By changing the settings on the [System Setting] window or by editing the system setting file (receiver.ini) provided by MT-LINKi, CSV file can be output automatically when retrieving the macro variable.

The items that can be set on the [System Setting] window are as follows.

- Enable/disable CSV file output function (default: disable)
- Output folder of CSV file

For the details, refer to the "**Setting File Output Path of Variable History**" subsection.

The item that can be set by receiver.ini is as follows.

- Name of CSV file

For the details on the system setting file (receiver.ini), refer to the "**SYSTEM SETTING FILES**" section.

NOTE

If the receiver.ini is edited, restart the Server PC.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	139/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

4.2.9.3 Format of Output File When Retrieving Macro Variable of FanucCNC

The output folder and the file name (default) of the output file when retrieving the macro variable is shown below.

■ Output folder (default)

<MT-LINK*i* installation folder>\TriggerHistory

NOTE

- 1 In the [Macro Variable History] group box of the [System Setting] window, enable/disable setting of CSV file output and the output folder setting can be changed. For the details, refer to the "[Setting File Output Path of Macro Variable History](#)" subsection.
- 2 For the output folder, paths of disk drives such as fixed disk drives (internal disk/external disk) or UNC paths can be specified. The paths of other drives (such as optical network drives) cannot be specified.
- 3 As the size of the output data will be massive, it is recommended to specify a separate folder for the report path, such as the folder of an expanded external disk drive. Also, make sure that the disk space of the output folder does not get compressed, by deleting the output files or moving the output files to another media on a regular basis.
- 4 If the free disk space of the disk drive specified as the output destination is under 5GB, the file output process will not be executed.

■ File name (default)

Logging_<Machine Name>_<Macro ID>_<Date(yyyyMMdd)>_<Time(HH:mm:ss)>.csv

Example) Logging_Machine001_ID0001_20170420_083038.csv

NOTE

- 1 "Date (yyyyMMdd)" and "Time (HH:mm:ss)" part of the file name is the date/time of retrieval of macro variable.
- 2 By editing the system setting file (receiver.ini), the file name and the output method can be changed. For the details, refer to the "[SYSTEM SETTING FILES](#)" section.
- 3 If the output target file is being used with another program (for example, the output file organized by date, is being opened with a text editor on the day of creation), writing of data on the file will fail. In such a case, the data gotten (data retrieved) will not be rewritten on the file.
When using the output file with another program, use the file after copying it to another folder.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	140/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

■ Output layout

Order of item (NOTE 1)	Output value	Remark
ID	Macro ID is output.	
Date/time of retrieval	Date/time of retrieval of macro variable is output.	
CNC path	CNC path is output.	
Trigger macro	Number of trigger macro variable is output.	
Macro variable	Number of retrieved macro variable is output.	(NOTE 2)
Value of macro variable	Value of retrieved macro variable is output.	(NOTE 2,3)

NOTE

- 1 The row of item name will not be output.
- 2 If multiple macro variables are retrieved, the number of output for the "macro variable" and the "value of macro variable" will be the number of data retrieved.
- 3 If the value of the macro variable is "data (blank)," "NaN" will be output as the output value.

■ Character code

UTF-8 (with BOM)

■ Output sample

- When ID0002, date/time of retrieval: 2017/03/29 09:50:18, path1, Trigger macro 501, macro variable 505, number of retrieval 2

"ID0002","2017/03/29 09:50:18","1","501","505","1","506","2"

4.2.9.4 If File Output Fails When Retrieving Macro Variable of FanucCNC

If file output fails when retrieving macro variable of FanucCNC, the file with the following name will be output to the output folder.

■ Output file when the file output fails

File name	Reason for Failure
#####DISKFULL OUTPUTTING#####_yyyyMMddHHmmss	When the free disk space of the output destination disk is under 5GB

NOTE

The "yyyyMMddHHmmss" part of the file name is the output date/time of the file.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	141/868

4.2.10 Setting Variable History Retrieval Condition Definition of FanucCNC

This is a setting to retrieve PMC signal/macro variable/P-CODE variable by using a trigger variable as trigger, store the retrieved variable history data to the DB, and output them into a CSV file.

- For each machine, the maximum numbers of "Macro Variable getting conditions" and "Variable history retrieval conditions" that can be registered in total are each 1,000.
- For each variable history retrieval condition, the maximum number of basic variables that can be registered is 256.
- For each variable history retrieval condition, the maximum total number of capture count is 256.
- For each basic variable, the maximum number of capture count is 256.
- For each machine, the maximum number of entries that the [Variable History Retrieval Condition Definition] list can register is 1,000.

■ How to control trigger variables

Steps for controlling the trigger variables by CNC and MT-LINK*i* are shown below.

- 1 Change the value of a trigger variable to "1" on the CNC side.
- 2 Monitor the trigger variable on the MT-LINK*i* side. When the value is "1," retrieve the specified number of values of the basic variable.
- 3 Change the value of the trigger variable to "0" on the MT-LINK*i* side.
- 4 Record the values of the variables retrieved in step 2 to the DB, and output the CSV file.

WARNING

When using handshaking method, MT-LINK*i* waits for a CNC to turn a trigger variable to "1" and then sets the variable to "0."

If trigger variables reside in areas that are used on the CNC or by other applications, make sure before setting trigger variables that no problem should occur.

NOTE

When setting a response timeout for a trigger variable on the CNC, the CNC operators should take into account that MT-LINK*i* may not have sufficient time to reset the value of the trigger variable from "1" to "0," because MT-LINK*i* is not running, or the network connection has problems.

A response timeout set on the CNC for a trigger variable should be at least one second. The maximum value* set on MT-LINK*i* for a response timeout is the value for the connection timeout of the CNC.

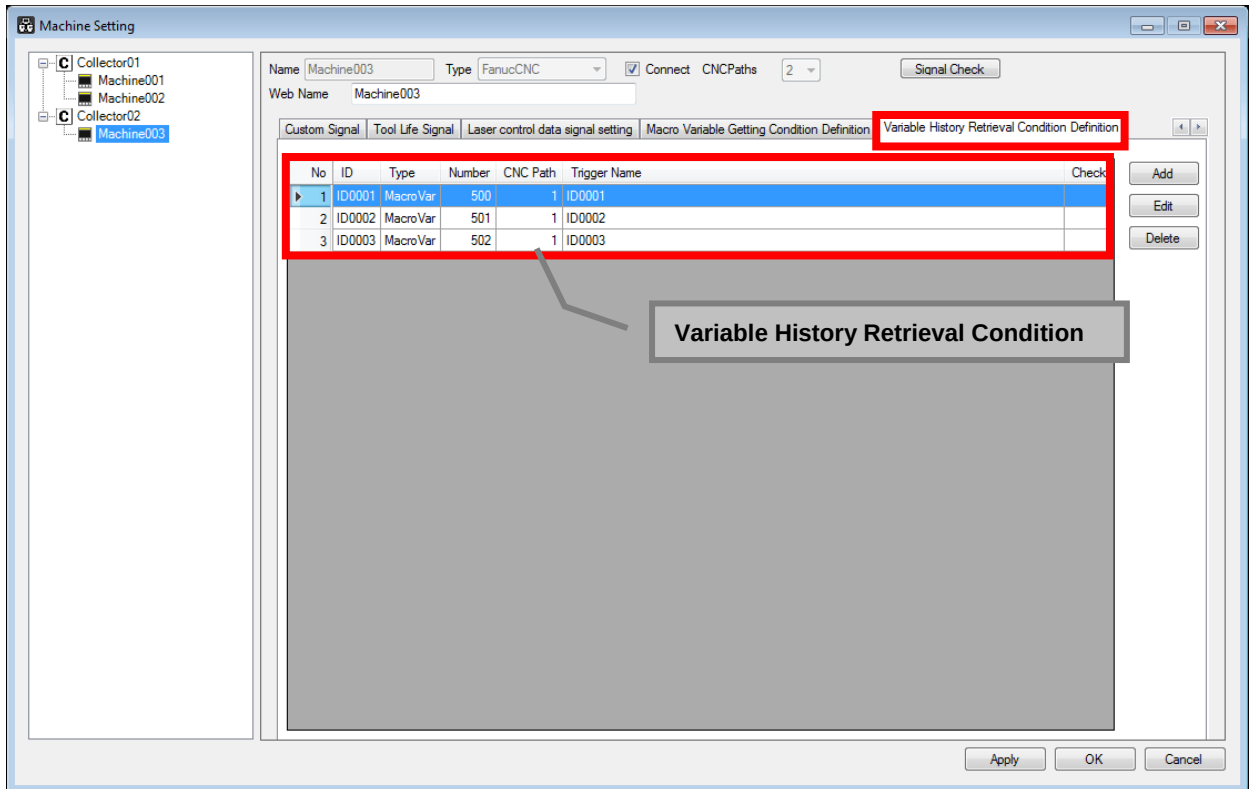
CNC operators should set an appropriate response timeout according to their production processes or environments.

* To check for the value of the connection timeout set on MT-LINK*i*, select the CNC in the [Machine Setting] window > [Protocol] tab, and review the [Time out] for the CNC.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	142/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

4.2.10.1 Setting Variable History Retrieval Condition of FanucCNC

Select a machine in the Collector tree, and select the [Variable History Retrieval Condition Definition] tab on the right pane.



■ The [Variable History Retrieval Condition Definition] tab

This tab page lists the variable history retrieval conditions that are set up in the [Variable History Retrieval Condition Setting] dialog box.

For the details of the columns in the list, refer to "The [Variable History Retrieval Condition Setting] dialog box" paragraph.

- ID
This column lists the IDs of the registered variable history retrieval conditions.
- Type
This column lists the types of trigger variables. This column is fixed to "MacroVar" (macro variable).
- Number
This column lists the numbers of trigger variables.
Format: <number>
Example: 501 (the number of the trigger variable is 501)
- CNC Path
This column lists the paths for trigger variables (macro variables).
- Trigger Name
This column lists the Web names of trigger variables (macro variables).

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	143/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

- Check

The results of the Signal Check, when the [Signal Check] button is clicked, are displayed.

"OK": All of the trigger variable / the basic variable (including variables subsequent to it) were retrieved.

"NG" (for "not good"): Failed to retrieve either the trigger variable / the basic variable (including variables subsequent to it).

"" (blank): Signal Check unexecuted (initial state, machine connection failure, etc.), or after executing Signal Check, the setting was changed on the [Variable History Retrieval Condition Setting] dialog box and the [OK] button was clicked.

For the details on Signal Check, refer to the "[Checking the Signals](#)" subsection.

- [Add] button

Click this button to add a new variable history retrieval condition.

- [Edit] button

Click this button to edit the selected variable history retrieval condition.

- [Delete] button

After a variable history retrieval condition is selected, clicking this button displays a confirmation message for the deletion.

Clicking the [Yes] button on the message deletes the selected variable history retrieval condition.

(1) Setting variable history retrieval conditions

To open the [Variable History Retrieval Condition Setting] dialog box, click the [Add] on the [Variable History Retrieval Condition Definition] tab page or select an ID and click the [Edit] button.

■ The [Variable History Retrieval Condition Setting] dialog box

- ID

This field populates the ID for the Variable history retrieval condition ("ID" + <four digit numbering>) (non-editable).

For the <four digit numbering>, an unused minimum value starting from "0001" is set automatically. (Example: ID0002)

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 144/868

- **Trigger Name**
Set a name for the trigger variable. The name should be entered in the language specified for the [Web name language] in the [System Setting] window. Up to 100 bytes can be set. The default value is the same value as [ID].

<The [Trigger Variable Setting] group box>

- **Type**
This field is fixed to "MacroVar" (macro variable).
- **Number**
Set the variable number of the trigger variable. The default value is "500."
- When [MacroVar] is selected in the [Type] drop-down list, a value from "100" to "999" can be set.
- **CNC Path**
Select the path for the trigger variable. The default value is "1."
- When [MacroVar] is selected in the [Type] drop-down list, select a CNC path ("1" to the maximum path number of the CNC).
- **Check**
The results of the Signal Check, when the [Signal Check] button is clicked, are displayed.
"OK": The trigger variable was retrieved.
The basic variable (including variables subsequent to it) was retrieved.
"NG" (for "not good"): Failed to retrieve the trigger variables.
Failed to retrieve the basic variable (including variables subsequent to it).
"" (blank): Signal Check unexecuted (initial state, machine connection failure, etc.), or after executing Signal Check, setting was changed on the [Variable History Retrieval Condition Setting] dialog box and the [OK] button was clicked.
For the details on Signal Check, refer to the "[Checking the Signals](#)" subsection.

<The [Basic Variable Setting] group box>

The following columns populate the values set with the [Basic Variable Setting] dialog box:

- **Type**
This column lists the types of basic variables as follows:
PMC (for PMC signals)
MacroVar (for macro variables)
PcodeVar (for P-CODE variables)
- **Basic Variable**
This column lists the addresses and numbers of basic variables in the following formats:
- PMC signal (data type: BIT)
Display format: <address><number>.<bit number>
Example: D100.7 (where "D" is an address, "100" is its number, and "7" is its bit number.)
- PMC signal (data type: other than BIT)
Display format: <address><number>
Example: D200 (where "D" is its address and "200" is its number.)
- Macro variable/P-CODE variable
Display format: <number>
Example: 501 (where "501" is its number.)

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	145/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

- **Data Type**
This column lists data types of basic variables.
 - When the [Data Type] drop-down list is set to "STRING" and the [Data Length] text field is set to a value in the [Basic Variable Setting] dialog box for a PMC type basic variable, this column populates its value as shown in the following example:
Display Format: STRING (<data length>)
Example: STRING (80) (where "80" is the length of the data.)
- **End Code**
This column lists the end codes for PMC type variables only when the [Data Type] drop-down list is set to "STRING" and the [End Code] text field is set to a value in the [Basic Variable Setting] dialog box for those basic variables. End codes are listed with their hexadecimal equivalent as follows:
 - NULL (0x00): when the [End Code] text field is set to "NULL"
 - ETX (0x03): when the [End Code] text field is set to "ETX"
 - LF (0x0A): when the [End Code] text field is set to "LF"
 - CR (0x0D): when the [End Code] text field is set to "CR"
 - UD (0x<xx>): when the [End Code] text field is set to "User-Designated" and the [User-designated (Hex)] text field is set to "xx" in hexadecimal
- **CNCPath/PMCPATH**
This column lists PMC path numbers for PMC type variables, and CNC path numbers for macro variables and P-CODE variables.
- **Number of Retrievals**
This column lists the numbers of variables to retrieve (including basic variables).
- **Check**
The results of the Signal Check, when the [Signal Check] button is clicked, are displayed.
 - "OK": The trigger variable was retrieved.
The basic variable (including variables subsequent to it) was retrieved.
 - "NG" (for "not good"): Failed to retrieve the trigger variable.
Failed to retrieve the basic variable (including variables subsequent to it).
 - "" (blank): Signal Check unexecuted (initial state, machine connection failure, etc.), or after executing Signal Check, setting was changed on the [Variable History Retrieval Condition Setting] dialog box and the [OK] button was clicked.

For the details on Signal Check, refer to the "[Checking the Signals](#)" subsection.
- **[Add] button**
Click this button to add a new basic variable setting.
- **[Edit] button**
Click this button to edit the selected basic variable setting.
- **[Delete] button**
After a basic variable setting is selected, clicking this button displays a confirmation message for the deletion.
Clicking the [Yes] button on the message deletes the selected basic variable setting.
- **[Variable Name Setting] button**
Click this button to set variable names in the selected basic variable setting.
- **[OK] button**
Clicking this button saves the edited settings, and closes the [Variable History Retrieval Condition Setting] dialog box. If the basic variable is not set, an error message mentioning "Basic variable is not specified" will be displayed.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	146/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- [Cancel] button
Clicking this button discards the edited settings, and closes the [Variable History Retrieval Condition Setting] dialog box.

NOTE

- 1 The variable history retrieval condition of the same trigger variable cannot be added to the same path.
 - When a same trigger variable is already registered for the same path in the [Variable History Retrieval Condition Definition].
 - When a same trigger variable is already registered for the same path in the [Macro Variable Getting Condition Definition].
- 2 If the same trigger variable is registered for different paths, caution windows will appear. The detail of the caution windows are explained in the "[Caution windows displayed when the same trigger variable is registered for a different path](#)" paragraph following this NOTE.

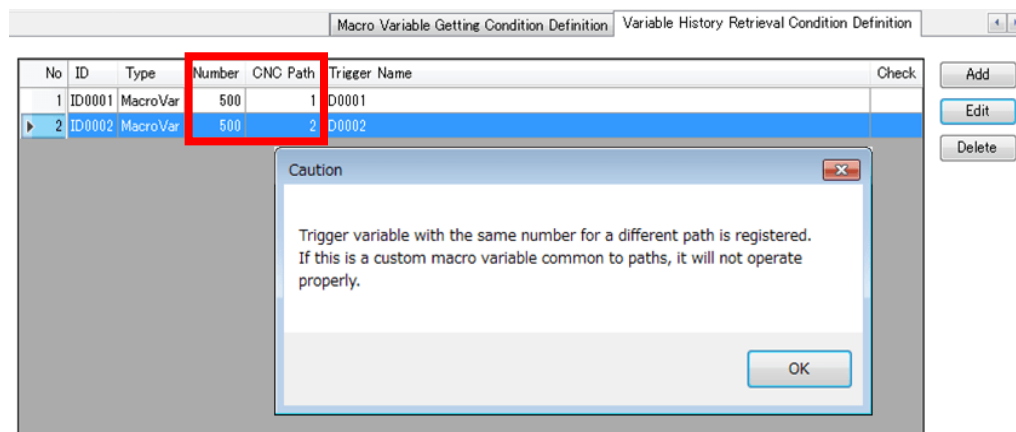
If the trigger variable is a custom macro variable common to all the paths, check the parameter of the CNC, as it will not work properly. (As for the parameter of the CNC, refer to the operator's manual of the CNC or the parameter manual on the relevant CNC parameters.)

If it is a custom macro variable common to all the paths, then perform as follows.

- Change the parameter of the custom macro variable common to all the paths.
- Assign another variable as the trigger variable.

■ Caution windows displayed when the same trigger variable is registered for a different path

- When the same trigger variable is already registered for a different path in the [Variable History Retrieval Condition Definition].



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	147/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

-
- Macro Variable Getting Condition Definition
- | No | ID | CNC Path | Trigger Macro | Basic Macro | Capture Count | Web Name | Check |
|-----|--------|----------|---------------|-------------|---------------|----------|-------|
| ▶ 1 | ID0001 | 3 | 500 | 501 | 1 | ID0001 | |
- Macro Variable Getting Condition Definition
- Variable History Retrieval Condition Definition
- | No | ID | Type | Number | CNC Path | Trigger Name | Check |
|-----|--------|---------------|--------|----------|--------------|-------|
| ▶ 1 | ID0001 | MacroVariable | 500 | 1 | 00001 | |
- Caution
- Trigger variable with the same number for a different path is registered in the macro variable getting condition setting. If this is a custom macro variable common to paths, it will not operate properly.
- OK

-
- Macro Variable Getting Condition Definition
- Variable History Retrieval Condition Definition
- | No | ID | CNC Path | Trigger Macro | Basic Macro | Capture Count | Web Name | Check |
|----|--------|----------|---------------|-------------|---------------|----------|-------|
| 1 | ID0001 | 3 | 500 | 501 | 1 | ID0001 | |
- Macro Variable Getting Condition Definition
- Variable History Retrieval Condition Definition
- | No | ID | Type | Number | CNC Path | Trigger Name | Check |
|----|--------|---------|--------|----------|--------------|-------|
| 1 | ID0001 | MacroVa | 500 | 1 | D0001 | |
| 2 | ID0002 | MacroVa | 500 | 2 | D0002 | |
- Caution**
- Trigger variable with the same number for a different path is registered in the variable history retrieval condition setting and the macro variable getting condition setting. If this is a custom macro variable common to paths, it will not operate properly.
- OK

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	148/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

(2) Setting the basic variable

When the [Add] button is clicked, or the [Edit] button is clicked after selecting a basic variable setting on the [Variable History Retrieval Condition Setting] dialog box, the [Basic Variable Setting] dialog box is displayed.

■ The [Basic Variable Setting] dialog box

● Type

Select the type of the basic variable from the following. Default variable is [MacroVar].

[PMC] (PMC signal) / [MacroVar] (Macro variable) / [PcodeVar] (P-CODE variable)

Fields available to set in this dialog box vary depending on the type selected for the [Type] field. The following table explains what and how to set.

<When [PMC] is selected >

Field	Values to set		Remarks
Address	Set the address of the variable. Valid values are "G," "F," "Y," "X," "A," "R," "T," "K," "C," "D," "M," "N," "E," and "Z."		Example: D (in D100)
Number	Set the data number of the variable. A number in a range of 0 to 99999 is available.		Example: 100 (in D100)
Data Type	Select a data type from the drop-down list. Options are as follows: BIT (bit type) BYTE (1 byte) WORD (2 bytes) DWORD (4 bytes) REAL (4 bytes) LREAL (8 bytes) Unsigned BYTE (1 byte) Unsigned WORD (2 bytes) Unsigned DWORD (4 bytes) STRING (select to retrieve the variable data as a string. See NOTE)		The following data types employ little-endian representation: WORD DWORD Unsigned WORD Unsigned DWORD REAL (single precision real) LREAL (double precision real)
String segmentation by	Set the [Data Length] field and/or [End Code] field when the [Data Type] field is set to "STRING."		Either or both of the [Data Length] and [End Code] fields must be set. By default, the check box for the [End Code] field is selected.
	Data Length	Select the check box and specify the data length. A number from 1 to 255 is available. When the check box is not selected, this field is regarded as set to the maximum (255 bytes).	

20	Nov. 22, 2023	uwano	Revised for MT-LINK ⁱ Version 4.0	Title	FANUC MT-LINK ⁱ AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK ⁱ Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK ⁱ Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK ⁱ Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK ⁱ Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK ⁱ Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK ⁱ Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 149/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

Field	Values to set		Remarks
	End Code	Select the check box and select an end code from the drop-down list. Options are as follows: NULL (0x00): (To set NULL as the end code) ETX (0x03): (To set ETX as the end code) LF (0x0A): (To set LF as the end code) CR (0x0D): (To set CR as the end code) User-Designated: (To employ a user-designated code that should be set in hexadecimal for the [User-designated code (Hex)] field. The hexadecimal value should be within 00 to FF.)	
Bit	Set the bit position of BIT type data. Valid values are 0 to 7.		
PMCPATH	Select a PMC path from the drop-down list. Options are 1 to 5, and 9. * To get the value of a dual check safety PMC, set 9 for this field.		

*For the details on encoding string type data, refer to the NOTE "■ On encoding string type data" in the "Set Custom Signal of FanucCNC" subsection.

<When [MacroVar] is selected>

Field	Values to set	Remarks
Number	Set the variable number of the variable. Valid numbers are 1 to 33, and 9 to 99999999.	
CNCPath	Select a CNC path from the drop-down list.	

<When [PcodeVar] is selected>

Field	Values to set	Remarks
Number	Set the variable number of the variable. Valid numbers are 1 to 33, and 100 to 89999.	
Data Type	Select a data type from the drop-down list. Options are as follows: Conversation Auxiliary Execution	
CNCPath	Select a CNC path from the drop-down list.	

- Number of Retrievals
Set the number of variables subsequent to the basic variable to retrieve. Up to 256 can be set. Default value is "1."
- [OK] button
Clicking this button saves the edited settings, and closes the [Basic variable setting] dialog box.
- [Cancel] button
Clicking this button discards the edited settings, and closes the [Basic variable setting] dialog box.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 150/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

NOTE

- 1 If MT-LINK*i* should not retrieve a basic variable defined in the [Basic Variable Setting] group box, it will retrieve none of the basic variables defined in the group box including variables subsequent to them.

No	Type	Basic Variable	Data Type	End Code	CNC Path/PMC Path	Number of Retrievals	Check
1	MacroVar	510			1	5	☒
2	PMC	D100	BYTE		1	1	☐
3	PMC	R100	STRING (8)		1	2	☐
4	PMC	R1000	STRING	LF (0x0A)	1	3	☐
5	PcodeVar	100	Conversation		1	1	☐
6	PcodeVar	200	Auxiliary		1	1	☐
7	PcodeVar	300	Execution		1	1	☐

In the example of the window shown above, when the value of the trigger variable turns to "1" and any of the value of the variables No.1 to No.7 cannot be retrieved, all the values of the variables (No.1 to No.7 including variables subsequent to the basic variables) cannot be retrieved. However, the value of the trigger variable will be reset to "0."

- 2 If the type of variable is [MacroVar] or [PcodeVar], specified number of variables is retrieved for each number from the basic variable.
If the type is [PMC], the range of address of the variable will change based on the data type. Examples are shown below.

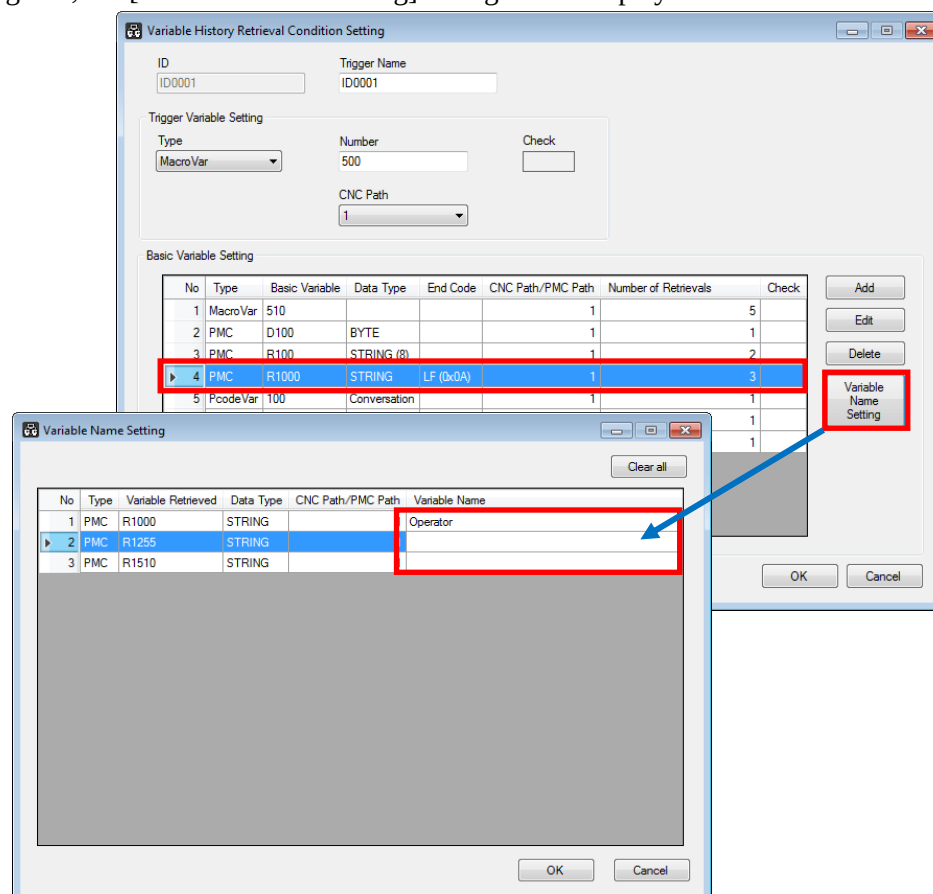
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.		Inoue
FANUC CORPORATION					Sheet.	151/868	

Type	Basic Variable	Data Type	Number of Retrieval	Variable to be Retrieved	Remarks
MacroVar	501	-	2	Variable 1: 501 Variable 2: 502	Retrieve by each number
PcodeVar	1005	-	2	Variable 1: 1005 Variable 2: 1006	Retrieve by each number
PMC	D0001	WORD	2	Variable 1: D0001 to D0002 Variable 2: D0003 to D0004	Retrieve by a word (2 bytes)
PMC	D1000.6	BIT	3	Variable 1: D1000.6 Variable 2: D1000.7 Variable 3: D1001.0	Retrieve by 1 bit
PMC	R100	STRING	2	Variable 1: R100 to R107 Variable 2: R103 to R115	When the [Data Length] field is specified to 8; the [End Code] field is not specified.
PMC	R200	STRING	3	Variable 1: R200 to R203 Variable 2: R208 to R209 Variable 3: R216 to R223	When the [Data Length] field is specified to 8; the [End Code] field is specified; the specified end code locates at R204 and R210.
PMC	R1000	STRING	3	Variable 1: R1000 to R1254 Variable 2: R1255 to R1399 Variable 3: R1510 to R1764	When the [Data Length] field is not specified (i.e., regarded as specified to 255); the [End Code] field is specified; the specified end code locates at R1400.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 152/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

(3) Setting names of variables

When the [Variable Name Setting] button is clicked on the [Variable History Retrieval Condition Setting] dialog box, the [Variable Name Setting] dialog box is displayed.



■ The [Variable Name Setting] dialog box

- Type, Data Type, CNC Path / PMC Path

For the details of those columns, refer to the explanation of the corresponding fields in the [Basic Variable Setting] group box of the [Variable History Retrieval Condition Setting] dialog box.

- Variable Retrieved

The variable to be retrieved is displayed. This column lists values in the same format as in the [Basic Variable] column in the [Basic Variable Setting] group box of the [Variable History Retrieval Condition Setting] dialog box.

- Variable Name

Set a name for the variable to be retrieved. The name should be entered in the language specified for the [Web name language] in the [System Setting] window. Up to 32bytes per variable can be set. Default is blank. If the variable name is blank, "ValueN" will be used as the item name for the output file.

* N is the index number (from "1") within the number of retrieval. (Example: Value2)

- [Clear all] button

When the [Clear all] button is clicked, confirmation message appears asking "All the variable names will be cleared. Is this OK?" If [Yes] button is clicked, all the settings in the [Variable Name] column will be cleared.

- [OK] button

Clicking this button saves the settings, and closes the [Variable Name setting] dialog box.

- [Cancel] button

Clicking this button discards the settings, and closes the [Variable Name setting] dialog box.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 153/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

4.2.10.2 Setting File Output Function When Retrieving Variable History of FanucCNC

By changing the settings on the [System Setting] window or by editing the system setting file (receiver.ini) provided by MT-LINK*i*, settings of the file output function during variable history retrieval can be set.

The items that can be set on the [System Setting] window are as follows.

- Enable/disable CSV file output function (default: enable)
- Output folder of CSV file

For the details, refer to the "[Setting File Output Path of Variable History](#)" subsection.

The item that can be set by receiver.ini is as follows.

- Name of CSV file
- A flag to indicate whether to output SinalCondition item in a CSV file (by default the item is output)
 - * This flag is available to FanucRobot type machines.

For the details on the system setting file (receiver.ini), refer to the "[SYSTEM SETTING FILES](#)" section.

NOTE

When receiver.ini is edited, restart the Server PC.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	154/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

4.2.10.3 Format of Output File When Retrieving Variable History of FanucCNC

When variable history is retrieved, the (default) output folder and the (default) file name of the output file is as follows.

- Output folder (default)
<MT-LINK*i* installation folder>\VariableHistory

NOTE

- 1 In the [Variable History] group box of the [System Setting] window, enable/disable setting of CSV file output and the output folder setting can be changed. For the details, refer to the "**Setting File Output Path of Variable History**" subsection.
- 2 For the output folder, paths of disk drives such as fixed disk drives (internal disk/external disk) or UNC paths can be specified. The paths of other drives (such as network drives) cannot be specified.
- 3 As the size of the output data will be massive, it is recommended to specify a separate folder for the report path, such as the folder of an expanded external disk drive. Also, make sure that the disk space of the output folder does not get compressed, by deleting the output files or moving the output files to another media on a regular basis.
- 4 If the free disk space of the disk drive specified as the output destination is under 5GB, the file output process will not be executed.

- File Name (default)

VariableHistory_<Machine name>_<ID>_<Machine type>_<Date (yyyyMMdd)>
_<Time (HHmmss)>.csv

Example) VariableHistory_Machine001_ID0001_FanucCNC_20170520_094049.csv

NOTE

- 1 In the file name, "Date (yyyyMMdd)" and "Time (HHmmss)" part is the date/time of variable history retrieval.
- 2 By editing the system setting file (receiver.ini), the file name can be changed. For the details, refer to the "**SYSTEM SETTING FILES**" section.
- 3 If the output target file is being used on another program (for example, in a case such as opening a daily output file using a text editor on the day the file is created), the writing process on that file will be unsuccessful. In such a case, retrieved data will not be rewritten on the file.
When using the output file on another program, use the file after copying it to another folder.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	155/868

■ Output Layout (for FanucCNC and FanucRobodrill machine types)

Item Name	Output Value	Remarks
MachineName	Machine name is output.	
<Trigger Name>	Value of the [Trigger Name] column is output.	NOTE 1
MachineType	Machine type is output. "FanucCNC" or "FanucRobodrill"	
UpdateDate	Date/time of variable history retrieval is output.	
VariableCount	Number of variables to retrieve is output.	
TriggerType	Type of trigger variable is output. * fixed to "MacroVar"	
TriggerPath	Path of trigger variable is output.	
TriggerAddress	Address of trigger variable is output. * fixed to "" (null)	
TriggerNumber	Variable number of the trigger variable is output.	
SignalCondition	Trigger condition is output that determines when to start retrieving the variable. * fixed to "" (null)	NOTE 2
Type<N>	Type of retrieved variable is output. "PMC": PMC signal "MacroVar": Macro variable "PcodeVar": P-CODE variable (common area 100 to 89999) "PcodeCon": P-CODE variable (Conversation macro 1 to 33) "PcodeAux": P-CODE variable (Auxiliary macro 1 to 33) "PcodeExe": P-CODE variable (Execution macro 1 to 33)	NOTE 3
Path<N>	Path of retrieved variable is output.	NOTE 3
Address<N>	Address of retrieved variable is output. * type of the retrieved variable: "PMC" only	NOTE 3
Number<N>	Variable number of retrieved variable is output.	NOTE 3
Bit<N>	Bit number of retrieved variable is output. * type and data type of the retrieved variable: "PMC" and "BIT" only	NOTE 3
<Variable Name> or Value<N>	Value of retrieved variable is output.	NOTE 3 to 5

NOTE

- [Trigger Name] set in the variable history retrieval condition setting is output as the item name.
- Whether to output the value of this item can be controlled by editing the system file (receiver.ini). For the details, refer to the "**SYSTEM SETTING FILES**" section.
- "N" is the index number (from 1) within the retrieved variables.
If multiple numbers are set for retrieving variables, "Type<N>" to "<Variable Name> or Value<N>" will be output for the number of variables to retrieve.
- [Variable Name] set in the variable name setting is output as the item name. If the [Variable Name] is not set (blank), "Value<N>" will be output.
- If the value of a retrieved variable is "Data (blank)," "NaN" is output as its output value.

■ Character Code

UTF-8 (with BOM)

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 156/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

■ Output Sample

Machine name: Machine001

Trigger name: RetrievalSetting1

ID: ID0001

Machine type: FanucCNC

Retrieved date/time: from 2017/07/13 13:32:08

Trigger variable:

Type: MacroVar, Path: 1, Number: 502

Basic variable 1:

Type: MacroVar, Path: 1, Number: 510, Number of variables to retrieve: 2, Variable name 1: Velocity, Variable name 2: Temperature

Basic variable 2:

Type: PMC, Path: 1, Address: D, Number: 0, Number of variables to retrieve: 1, Variable name 1: PMCD0

* In the following output example, SignalCondition item is excluded.

	A	B	C	D	E	F	G	H	I
1	MachineName	RetrievalSetting1	MachineType	UpdateDate	VariableCount	TriggerType	TriggerPath	TriggerAddress	TriggerNumber
2	Machine001	ID0001	FanucCNC	2017/07/13 13:32:08	3	MacroVar	1		502
3	Machine001	ID0001	FanucCNC	2017/07/13 13:33:22	3	MacroVar	1		502
4	Machine001	ID0001	FanucCNC	2017/07/13 13:34:36	3	MacroVar	1		502
5	Machine001	ID0001	FanucCNC	2017/07/13 13:40:17	3	MacroVar	1		502
6	Machine001	ID0001	FanucCNC	2017/07/13 13:40:26	3	MacroVar	1		502
7	Machine001	ID0001	FanucCNC	2017/07/13 13:40:50	3	MacroVar	1		502

J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA
Type1	Path1	Address1	Number1	Bit1	Velocity	Type2	Path2	Address2	Number2	Bit2	Temperature	Type3	Path3	Address3	Number3	Bit3	PMCD0
MacroVar	1		510		310	MacroVar	1		511		515	PMC	1	D		0	1
MacroVar	1		510		320	MacroVar	1		511		503	PMC	1	D		0	1
MacroVar	1		510		299	MacroVar	1		511		501	PMC	1	D		0	1
MacroVar	1		510		0	MacroVar	1		511		0	PMC	1	D		0	1
MacroVar	1		510		0	MacroVar	1		511		0	PMC	1	D		0	1
MacroVar	1		510		500	MacroVar	1		511		351	PMC	1	D		0	1

4.2.10.4 If File Output Fails When Retrieving Variable History of FanucCNC

If file output fails when retrieving variable history of FanucCNC, the file with the following name will be output to the output folder.

■ Output file when the file output fails

File name	Reason for Failure
#####DISKFULL OUTPUTTING#####_yyyyMMddHHmmss	When the free disk space of the output destination disk is under 5GB

NOTE

The "yyyyMMddHHmmss" part of the file name is the output date/time of the file.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 157/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

4.3 MACHINE SETTING - OPCUAServer

This section explains the machine setting of the machine type OPCUAServer.

4.3.1 Steps for the Machine Setting of OPCUAServer

Follow the steps below to set the OPCUAServer.

- (1) Select [Machine Setting] in the AdminTool Menu. This displays the [Machine Setting] window.
- (2) Add Collector. For the details, refer to the "[Add Collector](#)" subsection.
- (3) Add machine type OPCUAServer. For the details, refer to the "[Add OPC UA Server](#)" subsection.
- (4) Select a machine. For the details, refer to the "[Select a Machine Connected to the OPCUAServer](#)" subsection.
- (5) Make protocol and connection setting for the machine. For the details, refer to the "[Confirm/Edit Setting of OPCUAServer](#)" subsection.
- (6) Edit the endpoint when changing the connection target. For the details of the operation, refer to the "[Edit Endpoint of OPCUAServer](#)" subsection.
- (7) Make signal setting for the machine. For the details, refer to the "[Set Signal of OPCUAServer](#)" subsection.
- (8) Make alarm setting for the machine. For the details, refer to the "[Set Alarm of OPCUAServer](#)" subsection.
- (9) Save all the settings to make the system reflect them. For the details, refer to the "[Save Machine Setting](#)" subsection.

For the details of OPC UA, refer to the "[OPC UA CLIENT FUNCTIONALITY](#)" section.

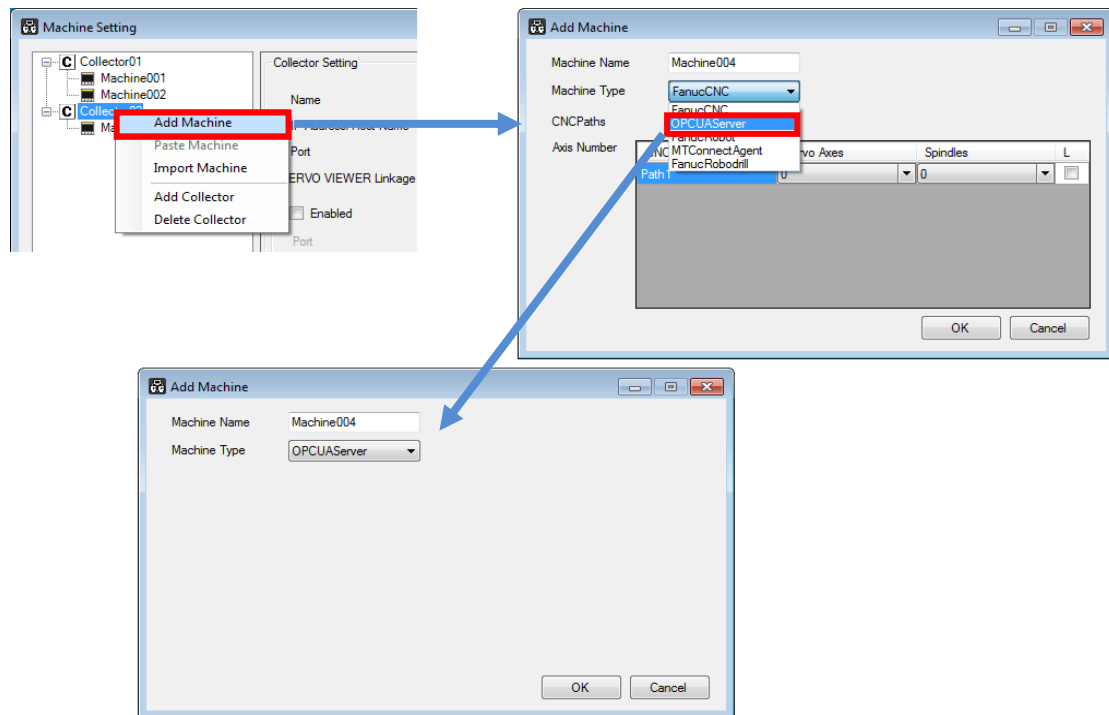
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION		Sheet. 158/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

4.3.2 Add OPC UA Server

(1) Adding a machine

In the Collector tree, right-click the target Collector under which you add a machine. A menu appears. Click [Add Machine].

* Before adding a machine, make sure that you have registered the target Collector.
When the [Add Machine] dialog box appears, set the settings of the target machine.



■ The [Add Machine] dialog box

- Machine Name
Set a machine name to this field. The [Machine Name] should start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and underscores ("_"). This field can be up to 32 bytes long. If needed, update a machine name in the [Select Machine] dialog box, which appears when clicking [OK].
- Machine Type
Select [OPCUAServer].
- [OK] button
Clicking this button opens the [Select Machine] dialog box.
- [Cancel] button
Clicking this button discards the settings being made and closes the dialog box.

NOTE

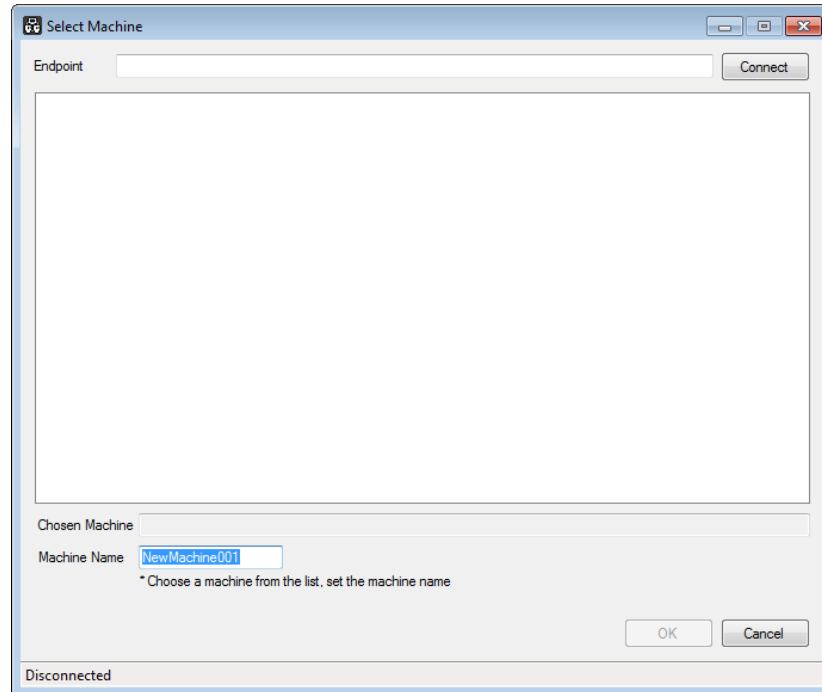
Machine type is non-editable (cannot be edited afterwards).
If the setting of the machine type is mistakenly set, you need to delete the machine already set. Add a new machine and make the settings for it.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	159/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

4.3.3 Select a Machine Connected to the OPCUAServer

(1) Selecting a machine

The [Select Machine] dialog box provides a way to assign a node managed on the OPC UA Server to a machine. This assignment makes MT-LINK*i* recognize the node as the machine populated in the [Machine Name] text field. Tags and methods for the assigned node can be associated with MT-LINK*i* signals, which will be collected through Collector and displayed on Web screens.



■ The [Select Machine] dialog box

- **Endpoint**
Set the address of the OPC UA Server to connect to. This address is used to establish a session with the OPC UA Server.
- **[Connect] button**
Clicking this button opens the [Server Configuration] dialog box. With this window, you can configure a connection with an OPC UA Server, and search OPC UA Servers.
- **Chosen Machine**
After a connection with an OPC UA Server is established, this dialog box list nodes managed on the OPC UA Server. Selecting a node from the list populates it in this field as a machine.
- **Machine Name**
Enter a name for the selected machine. This name should start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and underscores ("_"). This field can be up to 32 bytes long.
- **[OK] button**
Clicking this button saves the settings and closes the window.
- **[Cancel] button**
Clicking this button discards the settings and closes the window.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	160/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

Steps to select a machine

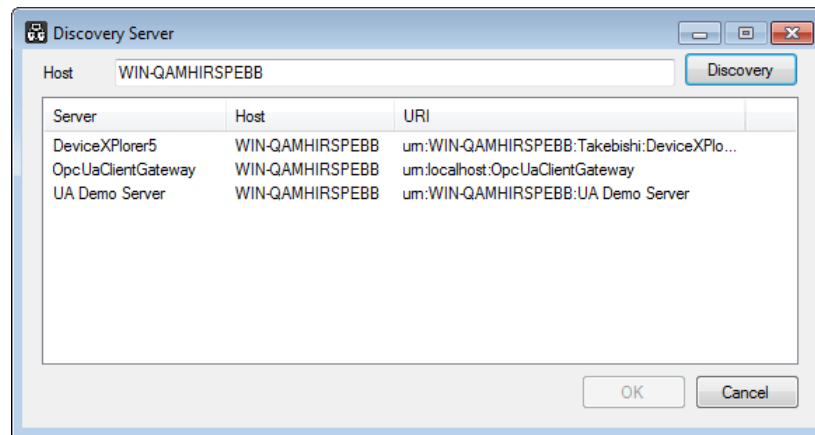
- 1 If you already have information on the endpoint of a target OPC UA Server, enter its address to [Endpoint], and click [Connect]; otherwise, leave [Endpoint] as blank and click [Connect].
- 2 The [Server Configuration] dialog box appears.
With this window, you can make settings to establish a session with an OPC UA Server.

■ The [Server Configuration] dialog box

- **Endpoint**
This field displays the endpoint to establish a session with an OPC UA Server.
- **[Discovery] button**
Clicking this button opens the [Discovery Server] dialog box. With this window, you can search for running OPC UA Servers, and retrieve server information of those servers.
- **Endpoint Information**
 - Protocol: fixed to [opc.tcp]
 - Security Mode: select the Security Mode when connected to OPC UA Server
 - Security Policy: select the Security Policy when connected to OPC UA Server
 - Message Encoding: fixed to [Binary]
- **Authentication Settings**
 - User Identity: select the User Token Type when connected to OPC UA Server
 - User: enabled when [User Name] is selected in [User Identity]
 - Password: enabled when [User Name] is selected in [User Identity]
- **[Connect] button**
Clicking this button store the settings and establishes a connection to the OPC UA Server with the settings. If a connection is established, this window closes.
- **[Validate Connection] button**
Clicking this button conducts a connection test with the OPC UA Server with the settings.
- **[Cancel] button**
Clicking this button discards the settings being made and closes the window.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 161/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- 3 If you have entered an address to the [Endpoint] on the [Select Machine] dialog box, the [Discovery] button is disabled and you can skip searching for OPC UA Servers; otherwise, click [Discovery] on the [Server Configuration] dialog box to search for OPC UA Servers. The [Discovery Server] dialog box appears.
- 4 With the [Discovery Server] dialog box, you can search for OPC UA Servers that are running, and retrieve server information of those servers



■ The [Discovery Server] dialog box

- **Host**
Specify the IP address or computer name of a PC or hardware on which an OPC UA Server is running.
- **[Discovery] button**
Clicking this button searches OPC UA Servers that are running and retrieves server information of those servers.
- **Server**
This column lists the names of the OPC UA Servers that are running on the PC specified in the [Host] text field.
- **Host**
This column lists the computer names or IP addresses of PCs or hardware on which OPC UA Servers are running.
- **URI**
This column lists the URIs of running OPC UA Servers.
- **[OK] button**
Clicking this button determines the target OPC UA Server to connect to, and closes this window.
- **[Cancel] button**
Clicking this button discards the settings being made, and closes this window.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 162/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- 5 Initially, the fields on the [Discovery Server] dialog box are populated with the information about OPC UA Server running on the local host (in this case, on the Server PC). In case no OPC UA Server is running, the window lists no entry. To search for OPC UA Servers running on other PCs or for those installed in other units, enter an IP address or a computer name in the [Host] text field and click [Discovery].
Select the target OPC UA Server listed under the [Server] column, and click [OK].

NOTE

When experiencing trouble in searching for OPC UA Servers, refer to the following subsection of the "FANUC MT-LINK*i* / Integration Server Troubleshooting MANUAL":

The "Unable to Search for OPC UA Servers" subsection in the "TROUBLES RELATED TO OPC UA SERVERS" section in the "MT-LINK*i* TROUBLESHOOTING" chapter

- 6 The [Endpoint] is populated with the endpoint information of the specified OPC UA Server. Or if you have specified the [Endpoint] on the [Select Machine] dialog box, the [Endpoint] on the [Server Configuration] is populated with the value you specified on the [Select Machine] dialog box.

- 7 If the server information of the OPC UA Server to connect to has been successfully gathered, the message "Configuration options are up to date" is displayed at the bottom of the window. In cases such as when the OPC UA Server is not running or the configuration options cannot be read properly, those setting options may not be modified.

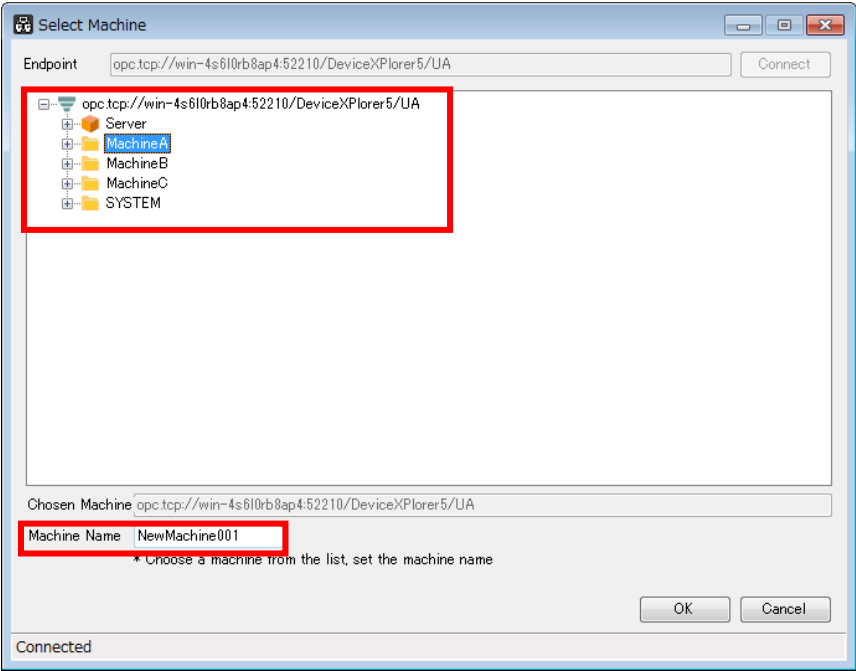
Set either option of the following according to the settings of the OPC UA Server.

- Anonymous: select this option when using an anonymous user.
- User Name: select this option when using a user identity and password.

- 8 Click [Validate Connection] to verify whether a connection with the OPC UA Server is established. If the validation is confirmed, click [Connect] to start a connection with the OPC UA Server.

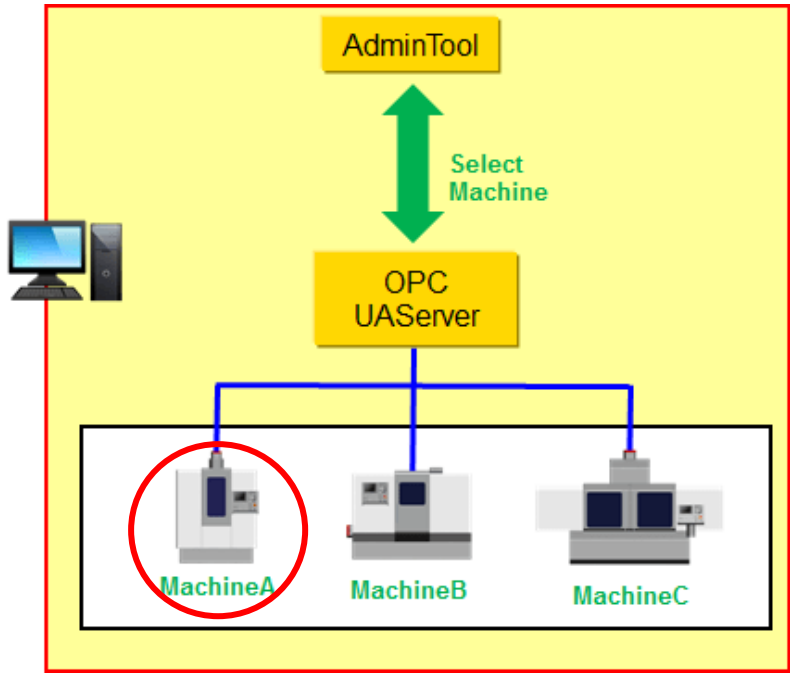
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	163/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

- 9 On the [Select Machine] dialog box, select a node from the node list in the pane. Click [OK] to close the [Select Machine] dialog box. With this, you have completed selecting a machine.



* For the icons shown in the tree, refer to "Set Signal of OPCUAServer" subsection.

- Machine Selection



NOTE

Once a machine is assigned, it turns to be non-editable.
Set a name to the assigned machine.

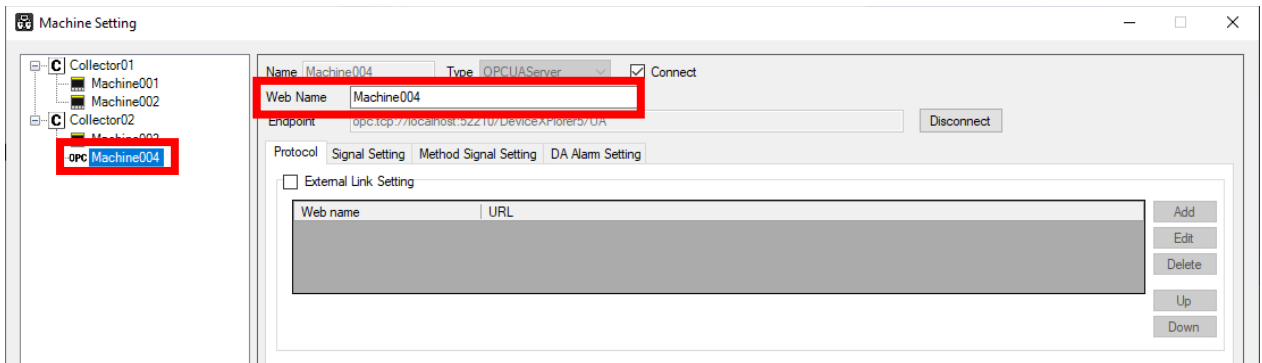
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	164/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

4.3.4 Confirm/Edit Setting of OPCUAServer

(1) Confirm and edit machine setting

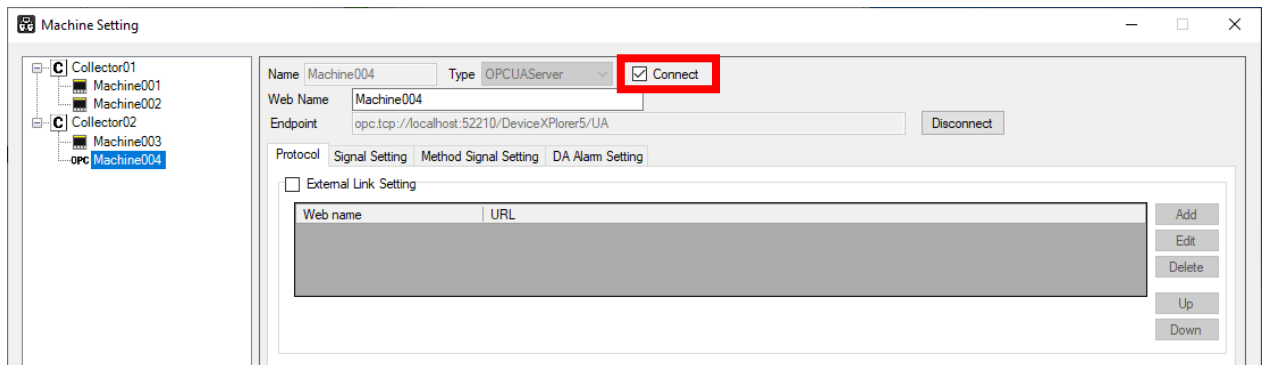
The settings of the selected Collector or machine are populated in the right pane of the [Machine Setting] window. Selecting a machine name in the Collector tree will show machine setting, so that you can set and change them.

The [Web Name] is displayed on the Web screens. The value of this field should be entered in the language specified for the [Web name language] field in the [System Setting] window. The [Web Name] field is required and will be set to the default value of <Machine name> if it is not modified.



(2) Set Machine Connection

Set connection between machine and Collector. Select a machine in the Collector tree and select the [Connect] check box, so that the connection is available.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	165/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

4.3.5 Edit Endpoint of OPCUAServer

(1) Edit endpoint in the machine setting

The connection target can be changed by editing the endpoint which was specified when the machine was selected. This applies to the following cases.

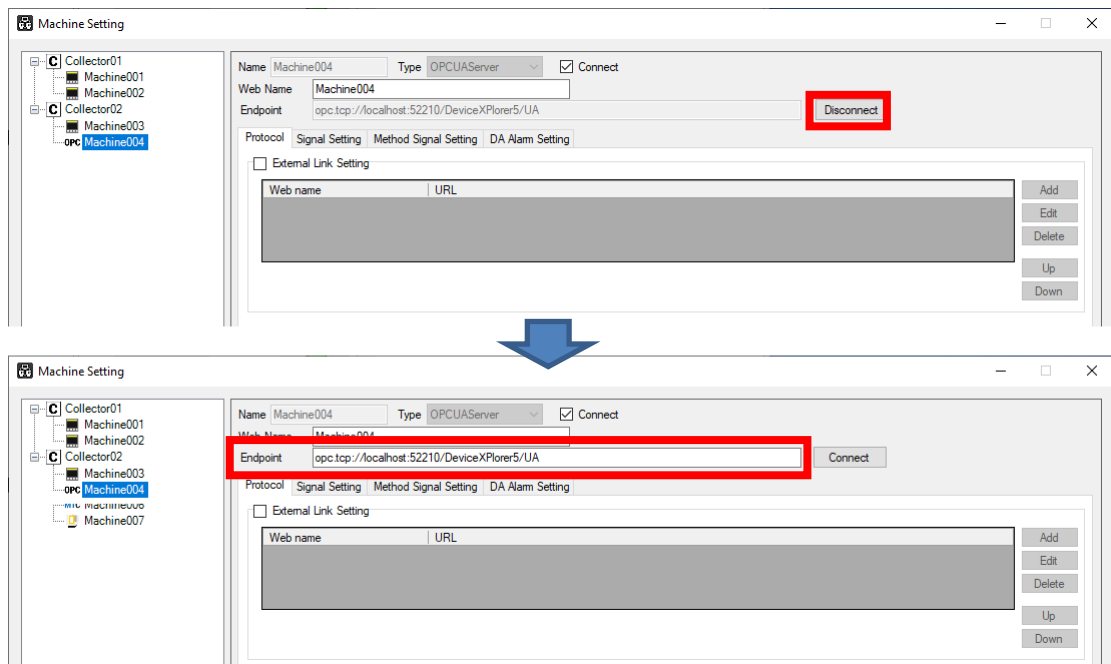
- When the IP address / host name of the OPC UA Server is changed
- When you want to utilize the existing setting to set a new OPC UA Server (see NOTE)

NOTE

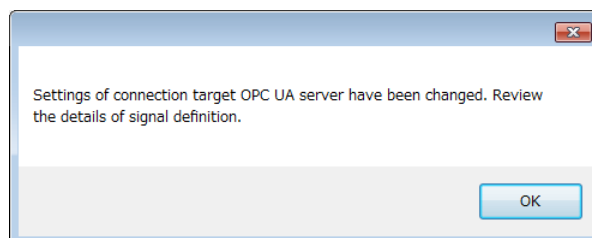
Make sure that the new connection target is the same type of machine. When the connection target is changed to a different type of machine, the configuration of the OPC UA Server may be different. In such a case, signals may not be acquired correctly.

Follow the steps below to edit the endpoint.

- 1 Select a machine on the Collector/Machine tree, and click the [Disconnect] button.



- 2 Edit the [Endpoint] entry column. Up to 32767 bytes can be entered. (Cannot be omitted)
- 3 After changing the endpoint, following message is displayed by clicking the [Connect] button (or the [Apply] button, then the [OK] button).



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	166/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

NOTE

When an endpoint is changed, the following environment must be kept unchanged before and after the change:

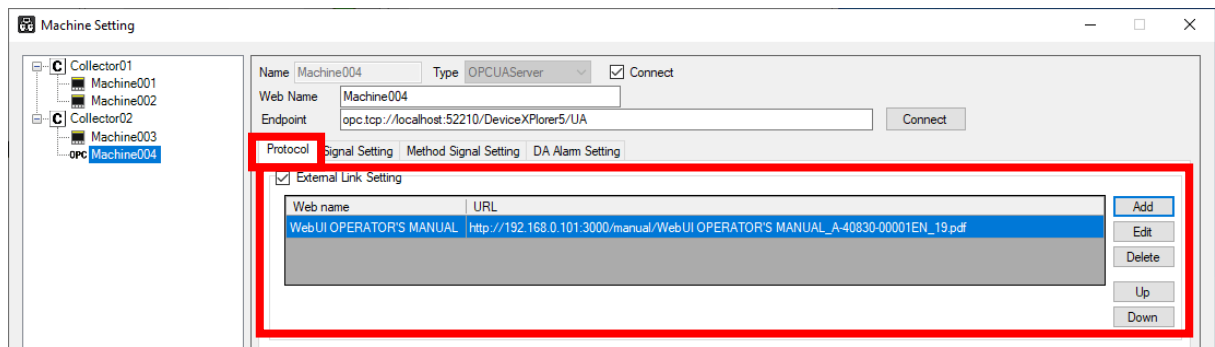
- Connection target OPC UA Server software
- Configuration of the node (containing the property), etc. for the OPC UA Server

If the environment shown above should not be identical through the change, delete the machine that has its endpoint changed, add a new machine, and make the settings for it; otherwise, the signal collection may not be executed correctly.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	167/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

4.3.6 Set Protocol Setting of OPCUAServer

Select a machine in the Collector tree and select the [Protocol] tab in the right pane.



■ The [Protocol] tab Extgernal Link Setting

Set fields for external links to view them through Web screens. Those links can be selected from the [Monitoring - Equipment Monitoring] screen (displayed by selecting an equipment icon on the [Monitoring - Overlook] screen).

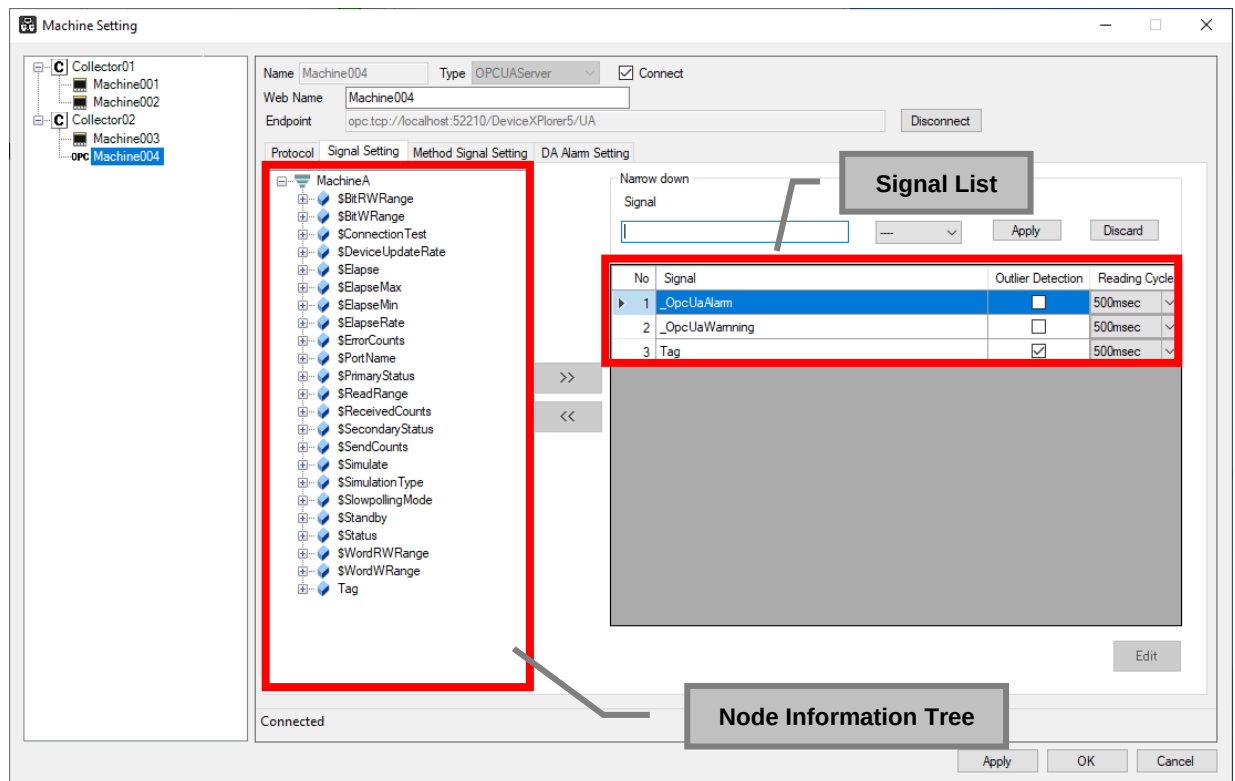
External Link function provides a way to view external links, such as Web sites, that are associated with the selected equipment (machines).

For the details, refer to the "[Set External Links](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 168/868

4.3.7 Set Signal of OPCUAServer

After a machine is selected in the [Select Machine] dialog box, a signal setting pane for the selected machine appears on the right side of the [Machine Setting] window. The signal setting pane has two tabs: one is the [Signal Setting] tab for associating a tag with a signal, and the other is the [Method Signal Setting] tab for associating a method with signals.



■ The [Machine Setting] window

- Node information tree

This tree lists nodes in a tree view. Display icons associate with them as follows:

Node type	Description	Display icon
Root	The highest node in MT-LINK <i>i</i>	
Object	A node which is an instance of Object NodeClass but does not inherit FolderType.	
Folder	A node which is an instance of Object NodeClass and inherits FolderType.	
Variable	A node which is an instance of Variable NodeClass.	
Method	A node which is an instance of Method NodeClass.	

* The other node types not listed above will be displayed with the display icon for "Object" node type.

- Signal list

The signal list of the [Signal Setting] tab page lists the basic signals and the signals that are set up with the [Signal Setting] window. To list only specific signals, narrow down the signal list. For the details on how to narrow down the list, refer to the "[Narrow down Signal List](#)" subsection.

For the basic signals, refer to the "[LIST OF OPCUAServer SIGNALS](#)" section.

The signal list of the [Method Signal Setting] tab page lists the signals that are set up with the [Method Signal Setting] window. To list only specific method signals, narrow down the signal list. For the details on how to narrow down the list, refer to the "[Narrow down Signal List](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	169/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- [Connect] check box
This is an option to enable or disable connection with the machine selected in the left pane. Selecting this check box enables the connection.
- Web Name
This is a name of the machine to display on the Web screens. The name should be entered in the language specified for the [Web name language] field in the [System Setting] window. This is a required field and will be set to the default name of <Machine name>.
- Endpoint
This field is populated with the endpoint that is used in establishing a session with the OPC UA Server.
- [Connect] / [Disconnect] button
The [Connect] button appears when the OPC UA Server is disconnected (See NOTE). Clicking this button opens the [Server Configuration] dialog box to establish connection with the OPC UA Server.
The [Disconnect] button appears when the OPC UA Server is connected. Clicking this button disconnects the OPC UA Server and enables users to edit its endpoint.

NOTE

MT-LINK*i* deems the OPC UA Server to be disconnected in the following cases:

- (1) When it has not connected with the OPC UA Server yet
- (2) When unable to retrieve any signal at an interval of the specified reading cycle
(That is when all the signal statuses are "Bad" or all the signal values is inconsistent with the settings)

However, the [Machine Setting] and the [Machine Connection Test] windows can detect only the case (1).

- [Apply] button
Clicking this button determines the settings.
- [OK] button
Clicking this button determines the settings and closes this window.
- [Cancel] button
Clicking this button discards the settings being made, and closes this window.

For the details of the [Apply], [OK], and [Cancel] buttons, refer to the "**Save Machine Setting**" subsection.

NOTE

Make sure that at least one tag signal, method signal, or DA alarm signal other than the basic signals is configured. If not, the status of the equipment, such as CNC connected to an OPC UA Server, will remain as "OPERATE" even when it is powered off.

- * For the basic signals, refer to the "**LIST OF OPCUAServer SIGNALS**" section; for DA alarm signals, refer to the "**Set OPCUAServer DA Alarms**" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	170/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

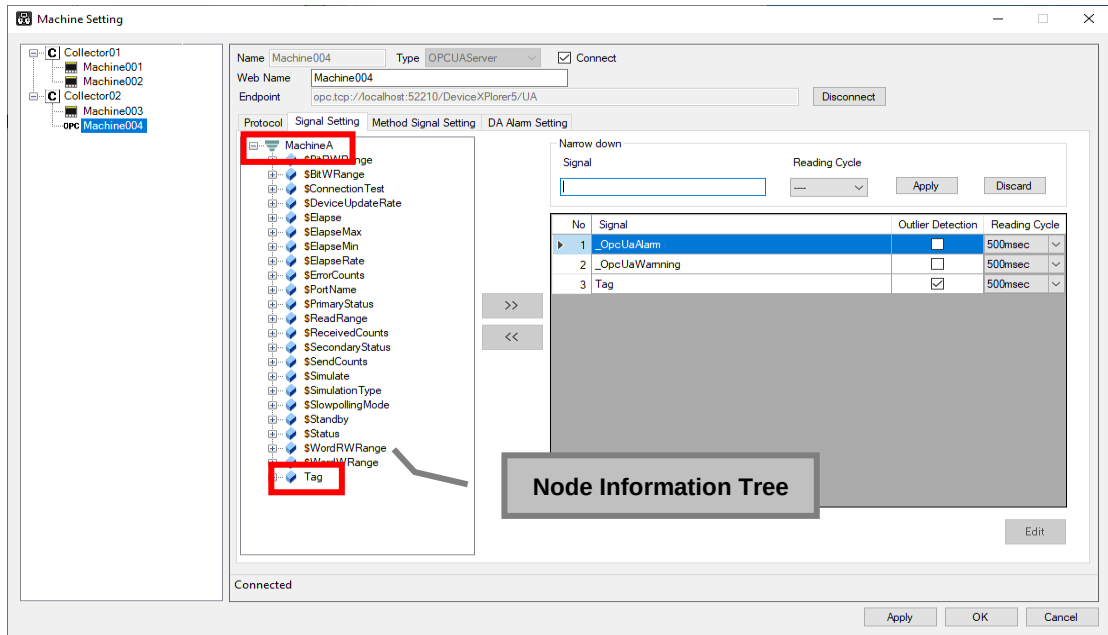
4.3.7.1 Set OPCUAServer Tags as Signals

In the [Signal Setting] tab page of the [Machine Setting] window, associate OPC UA Server tags with MT-LINK*i* signals. After a node is selected in the [Select Machine] dialog box, MT-LINK*i* recognizes the selected node as a machine and retrieves from the OPC UA Server the tags belonging to the machine. Those signals can be registered up to 1000.

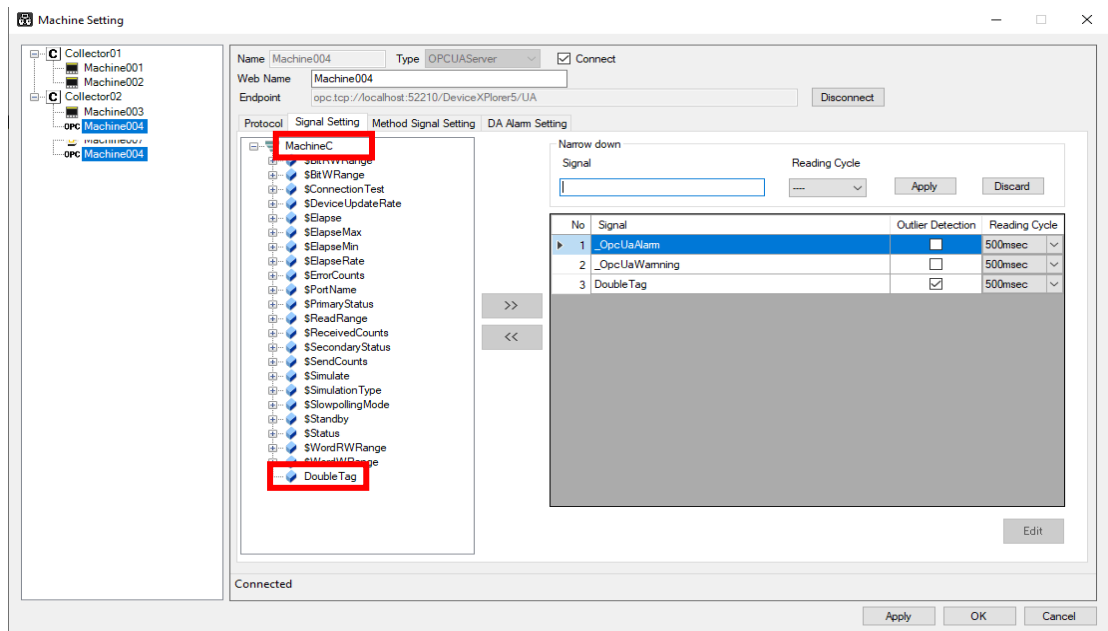
(1) Signal setting with tags

Select a tag to associate with an MT-LINK*i* signal, and click the [>>] button.

- **Example:** selecting "MachineA" in the [Select Machine] dialog box and setting the tag name "Tag" to a signal



- **Example:** selecting "MachineC" in the [Select Machine] dialog box and setting the tag name "DoubleTag" to a signal



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11							
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10							
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8							
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2							
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			
Ed.	Date	Design	Description							Sheet.
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue			

■ The [Signal Setting] tab

- Node information tree
This tree lists nodes belonging to the selected machine in a tree view except those of "Method" NodeClass.
- [>>] (Add) button
Clicking this button moves a selected tag to the signal list in the right pane, and opens the [Signal Setting] dialog box for associating the selected tag with a signal. Specify the [Signal name] and [Node ID] fields in the dialog box.
- [<<] (Delete) button
Clicking this button removes a selected tag from the signal list in the right pane.

NOTE

When a signal is referenced by equipment, deleting it will be denied with an error message. For the details, refer to the "[Referenced Error](#)" subsection.

- [Edit] button
Clicking this button opens the [Signal Setting] dialog box for editing the [Signal name] and [Node ID] fields in the dialog box. If no signal is selected, this button is disabled.
- Signal
This column holds the names of the basic signals and the signals that are set up with the [Signal Setting] window.
- Outlier Detection
For signals with [Outlier Detection] check box selected (signals of data type Int32, UInt32, or Double), outlier detection of the signal can be set by double-clicking the row. The check box column cannot be edited.
For the details of the outlier detection settings, refer to the "[Set Outlier Detection of Signal](#)" subsection.
- Reading Cycle
This drop-down list is used to select a reading cycle per signal. Its options are 500 msec, 10 sec, and 60 sec.

NOTE

If a selected tag in the Node information tree has no value, the [>>] button will be inactive and thus no signal can be added. Only tags (Variable) with a value can be configured as a signal and added to the signal list in the right pane.

If clicking the [>>] button opens the message "This tag cannot be added," it indicates the value of the selected tag cannot be retrieved and the tag cannot be configured as a signal. Verify that the machine connected to the OPC UA Server is powered on and check what options the machine has.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	172/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

(2) The [Signal Setting] dialog box

Specify names and node IDs for the selected tags, which are managed on the OPC UA Server. This setting enables MT-LINK*i* to use those names and node IDs when manipulating signals.

To open the [Signal Setting] dialog box, either select a tag from the Node information tree in the [Machine Setting] window and click the [>>] button, or select an existing signal and click the [Edit] button in the [Machine Setting] window.

■ The [Signal Setting] dialog box

• Signal name

This field is set to a name of an MT-LINK*i* signal with which to associate the selected tag. The initial value of this field is the name of the tag.

A signal name should start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and underscores ("_"). This field can be up to 64 bytes long.

NOTE

The following names are not allowed to specify for [Signal name]:

- MainProgram
- ActProgram
- Alarm
- OpMacroMsg
- CncState
- CncWarning
- RobotState
- Disconnect
- SignalLog
- VariableHistory
- AssetInfo
- WaveData
- Fault
- Warning
- FaultMessage
- WarningMessage
- AxisInfo

NOTE

When a signal is referenced by equipment, renaming it will be denied with an error message. For the details, refer to the "[Referenced Error](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	173/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

- **Node ID**
This editable field is set to an identifier which identifies the selected node in the OPC UA Server. Editing this field is prohibited when this dialog box is opened by clicking the [Edit] button and the OPC UA Server is disconnected.
- **Display name**
This field is populated with the tag name set by the OPC UA Server.
- **Data type**
This field is populated with the data type assigned by the OPC UA Server.
MT-LINK*i* determines the data type of the associated signal (NOTE 3).
This field concatenates "[]" symbol to denote the signal's data type is array.
For MT-LINK*i* data types corresponding to the supported data types, refer to the "**OPC UA Data Types**" subsection.
- **[Run Retrieving Node ID Test] button**
After updating the [Node ID] field, click this button to update the [Data type] and [Display name] fields with values retrieved from the OPCUA Server. When those fields are updated* with the successfully retrieved values, this button is disabled. (*If the retrieved values are by chance the same as the current values of the fields, they will look as if the values had not been retrieved.)
When an error occurs, this button displays an error message or "!" symbol on the row in question.
This button is disabled until the [Node ID] field is updated.
- **[OK] button**
Clicking this button saves changes, if made, to the fields, and closes the [Signal Setting] dialog box. If the [Run Retrieving Node ID Test] button is not clicked though the [Node ID] field has been modified, this button works in the same way as the [Run Retrieving Node ID Test] button.
- **[Cancel] button**
Clicking this button discards changes, if made, to the settings, and closes the [Signal Setting] dialog box.

NOTE

- 1 For the signal names used in MT-LINK*i*, half-width alphabetical character can be used for the first character, and half-width alphanumeric characters, hyphens ("–"), and underscores ("_") can be used for the subsequent characters.
No signal can be set to tags whose names have more than 64 bytes or to tags whose names are composed of unusable characters listed below.
In those cases, modify their signal name on the [Signal Setting] dialog box.
Examples of unusable tag names:
 - 1_Tag (starting with a number)
 - あいうえお (composed of non-alphanumeric characters)
 - Tag A (Space is used)
 - TAGALARM (composed of symbols other than underscores ("_") or hyphens ("-")).
- 2 In the following cases, signal setting cannot be performed:
 - The format of the node ID specified for [Node ID] is invalid.
 - The node ID specified for [Node ID] does not exist.
- 3 The node ID specified for [Node ID] already exists.
- 4 The data type of the associated signal may be ByteString or byte-array type. In order to determine which type is more proper to the signal, a prompting dialog box appears asking the user to declare.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 174/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- 5 For signal settings made with MT-LINK*i* Ver. 3.8 or earlier, values of the [Data type] field are not displayed by default but have no impact on system operation. To display such values, click the [Run Retrieving Node ID Test] button on the [Signal Setting] dialog box.
- 6 Tags whose values are of array data type cannot be configured as signals.

NOTE

Perform the following steps to add tags which are not displayed in the Node information tree.

- (1) Select any tag in the Node information tree and click the [>>] button.
- (2) Set the [Node ID] field in the [Signal Setting] dialog box to the node id of a new tag to add.
- (3) Click the [Run Retrieving Node ID Test] button.
- (4) Verify the [Signal name] and [Data type] fields and specify a new signal name for the [Signal name] field.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 175/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

4.3.7.2 Set OPCUAServer Methods as Signals

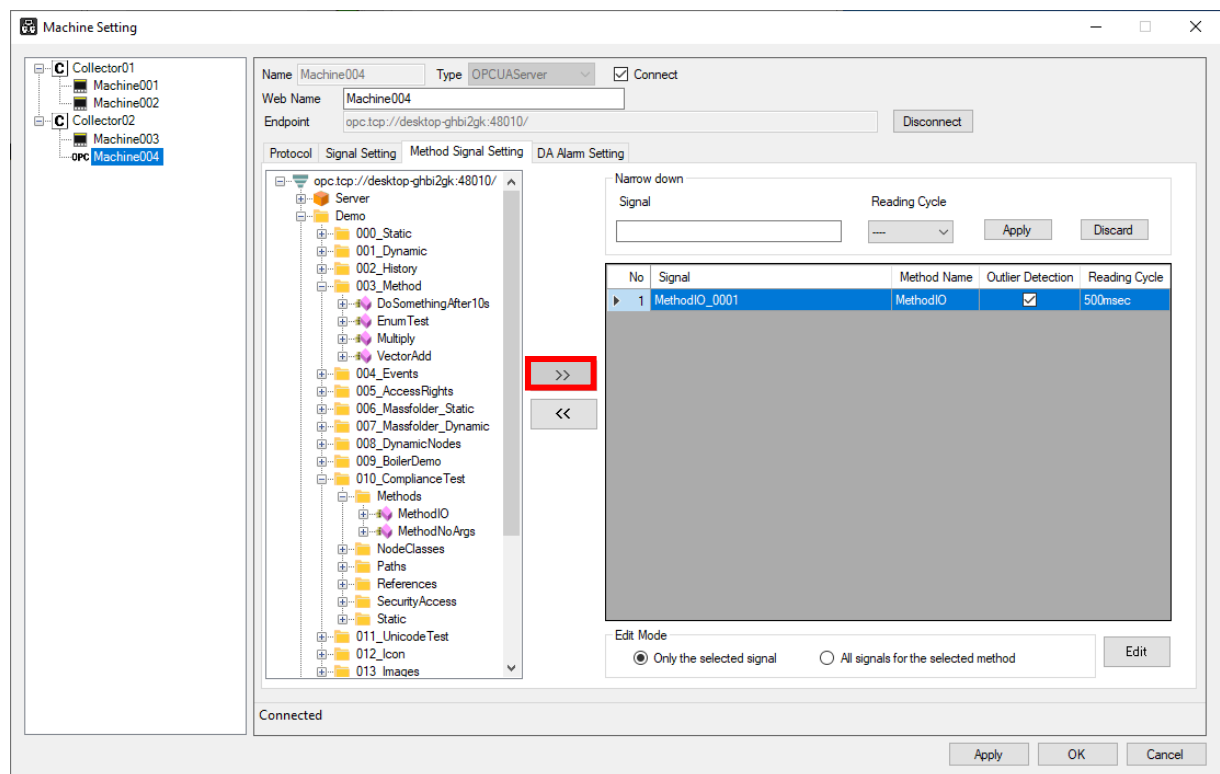
In the [Method Signal Setting] tab page of the [Machine Setting] window, associate OPC UA Server methods with MT-LINK*i* signals. Specifying input arguments and defining output arguments of a method associates it with signals and enables MT-LINK*i* to collect the signals via the output arguments. Those method signals can be registered up to 1000.

NOTE

MT-LINK*i* does not support executing methods that grant access permissions to an OPC UA Server. If needed to use such methods, execute them on another OPC UA client.

(1) Signal setting with method

Select a method to associate with MT-LINK*i* signals, and click the [>>] button.



■ The [Method Signal Setting] tab

- Node information tree
This tree lists nodes of "Object" or "Method" NodeClass belonging to the selected machine in a tree view except those of "Variable" NodeClass. The listed nodes should be of "Object" or "Method" NodeClass.
- [>>] (Add) button
Clicking this button opens the [Method Signal Setting] window to associate signals with the selected method.
The window has the [Method Name] field for the selected method, and the [Input Arguments] and [Output Arguments] group boxes to define signals.
- [<<] (Delete) button
Clicking this button removes selected entries from the signal list in the right pane.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	176/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

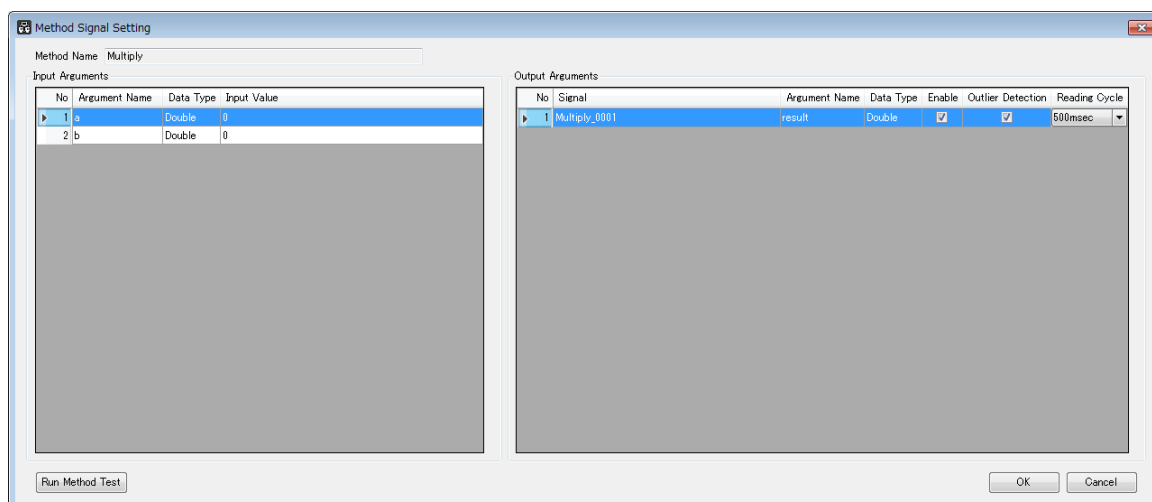
NOTE

When a signal is referenced by equipment, deleting it will be denied with an error message. For the details, refer to the "**Referenced Error**" subsection.

- **Edit Mode**
This mode has two options on how to list signals assigned to the selected method in the [Method Signal Setting] window:
 - Only the selected signal: the window lists only the selected signal.
 - All signals for the selected method: the window lists all the signals assigned to the selected method.
- **[Edit] button**
Clicking this button lists signals in the [Method Signal Setting] window as specified by the [Edit Mode] option and allows editing the [Input Arguments] and [Output Arguments] columns.
Though this button is active when the OPC UA Server is disconnected, input arguments cannot be modified.
- **Signal**
This column holds the names of method signals that are set up with the [Method Signal Setting] window.
- **Outlier Detection**
Double-clicking a row with this check box selected displays outlier detection settings of the method signal configured with the [Method Signal Setting] window. (Uneditable)
- **Reading Cycle**
This column displays reading cycles specified with the [Method Signal Setting] window.

(2) The [Method Signal Setting] window

The [Method Signal Setting] window appears as follows:

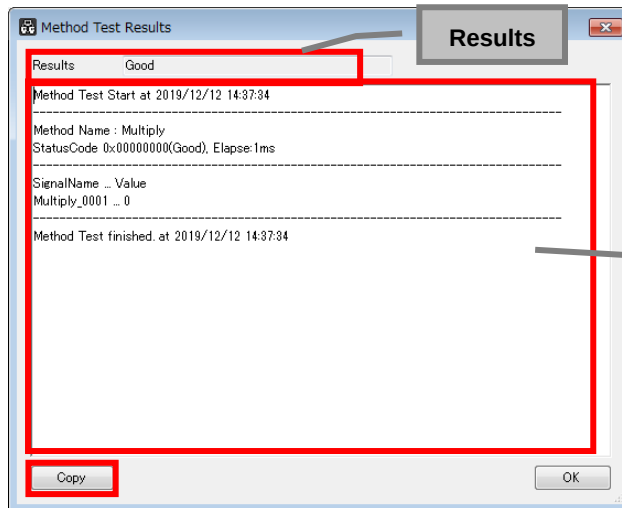


■ The [Method Signal Setting] window

- **Method Name**
This field holds the name of the selected method managed by the OPC UA Server. (Uneditable)
- **[Run Method Test] button**
Clicking this button runs a test to check if the method works as expected. The results will be shown in the [Method Test Results] window.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 177/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

■ The [Method Test Results] window



- Results
 "Good" indicates the test succeeded.
 Otherwise, text will be displayed reading what errors occurred.
 For error results other than "Good" and their countermeasures, refer to the following subsection of the "FANUC MT-LINK*i* / Integration Server Troubleshooting MANUAL":
 The "Method Test Ends with a Result Other than [Good]" subsection in the "TROUBLES RELATED TO OPC UA SERVERS" section in the "MT-LINK*i* TROUBLESHOOTING" chapter
- Detailed results
 When the test is completed in success, the method name, a StatusCode, method execution time, and the collected signal values will be displayed.
 When the test ends in failure, the method name, a StatusCode, and input/output arguments will be displayed.
- [Copy] button
 Clicking this button copies the information displayed in the Result display area to the clipboard.
- [OK] button
 Clicking this button closes the [Method Test Results] window.
- [OK] button
 Clicking this button saves changes, if made, to the settings, and closes the [Method Signal Setting] window. Signals with their [Enable] column selected will be listed on the [Method Signal Setting] tab page of the [Machine Setting] window.
- [Cancel] button
 Clicking this button discards changes, if made, to the settings, and closes the [Method Signal Setting] window.

The [Input Arguments] group box

- Argument Name
 This column holds the names of input arguments.
- Data Type
 This column holds the data types of input arguments.
- Input Value

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	178/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

This column holds the values specified for input arguments.

The following table shows supported data types for the [Input Arguments] column, and the allowed values for the [Input Value] column.

Data types defined in OPC UA standards	Size (in byte)	Input value range	Remarks
Boolean	-	FALSE / TRUE	
SByte	1	-128 to 127	
Byte	1	0 to 255	
Int16	2	-32768 to 32767	
UInt16	2	0 to 65535	
Int32	4	-2147483648 to 2147483647	
UInt32	4	0 to 4294967295	
Int64	8	-9999999999999998 to 9999999999999998	
UInt64	8	0 to 9999999999999998	
Float	4	-3.40283E+38 to 3.40283E+38	
Double	8	-1.79769313486232E+308 to 1.79769313486232E+308	
String	32767	-	
Guid	32	00000000-0000-0000-0000-000000000000 to FFFFFFFF-FFFF-FFFF-FFFF-FFFFFFFFFFFFFF	128-bit data Input arguments of this data type must be set with hexadecimal numbers in 32 bytes length.
ByteString	32766	-	Array of numbers with values from -128 to 127 Input arguments of this data type must be set with hexadecimal numbers.
NodeId	32767	-	Input arguments of this data type must be set to strings.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	179/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- Signal

- This column holds the names of signals. The first character of a name must be an alphanumeric, and the subsequent characters must be alphanumeric, hyphens ("-"), or underscores ("_"). This field can be up to 64 bytes long.

The following names are not allowed to specify for [Signal]:

- ## NOTE

- Argument Name

- Data Type

- Enable

NOTE

- Outlier Detection

For the details of the outlier detection settings, refer to the "**Set Outlier Detection of Signal**" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	180/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

- **Reading Cycle**

This drop-down list is used to select a reading cycle per signal. Its options are 500 msec, 10 sec, and 60 sec.

For MT-LINK*i* data types corresponding to the supported data types of input arguments, refer to the "**OPC UA Data Types**" subsection.

NOTE

Signals for a method cannot be defined when the following occur, in which case an error message indicating its cause appears on the left side of the [OK] button:

- Data types specified for output arguments are unsupported.
- Specified data types for output arguments are array or multidimensional array.
- The method has no output argument.
- The data types of output arguments are unsupported data type, array, or multidimensional array.
- Arguments for the method cannot be received.
- The method signal setting in MT-LINK*i* conflicts with the method setting on the OPC UA Server.

(3) Defining method signals.

Follow the steps below.

- 1 Specify the [Input Value] column in the [Input Arguments] group box.
- 2 Specify the [Signal Name] column in the [Output Arguments] group box.
- 3 Select the [Enable] check box in the [Output Arguments] group box to collect the signal; otherwise, clear it.
- 4 If needed, select the [Outlier Detection] column and perform its setting.
- 5 Select an entry from the [Reading Cycle] drop-down list.
- 6 If needed, repeat step 1 to 6. When all the necessary signals are defined, click the [Run Method Test] button.
- 7 The [Method Test Results] window appears. Review the results and click [OK].
- 8 Click [OK].

NOTE

Clicking [OK] opens an error message when the following occur:

- The number of registered signals exceeds the maximum (100 entries).
- Another signal already exists with the same settings as a specified signal is done.
- A method request fails when it is issued with specified values for the [Input Value] column of the [Input Arguments] group box.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION				Sheet. 181/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

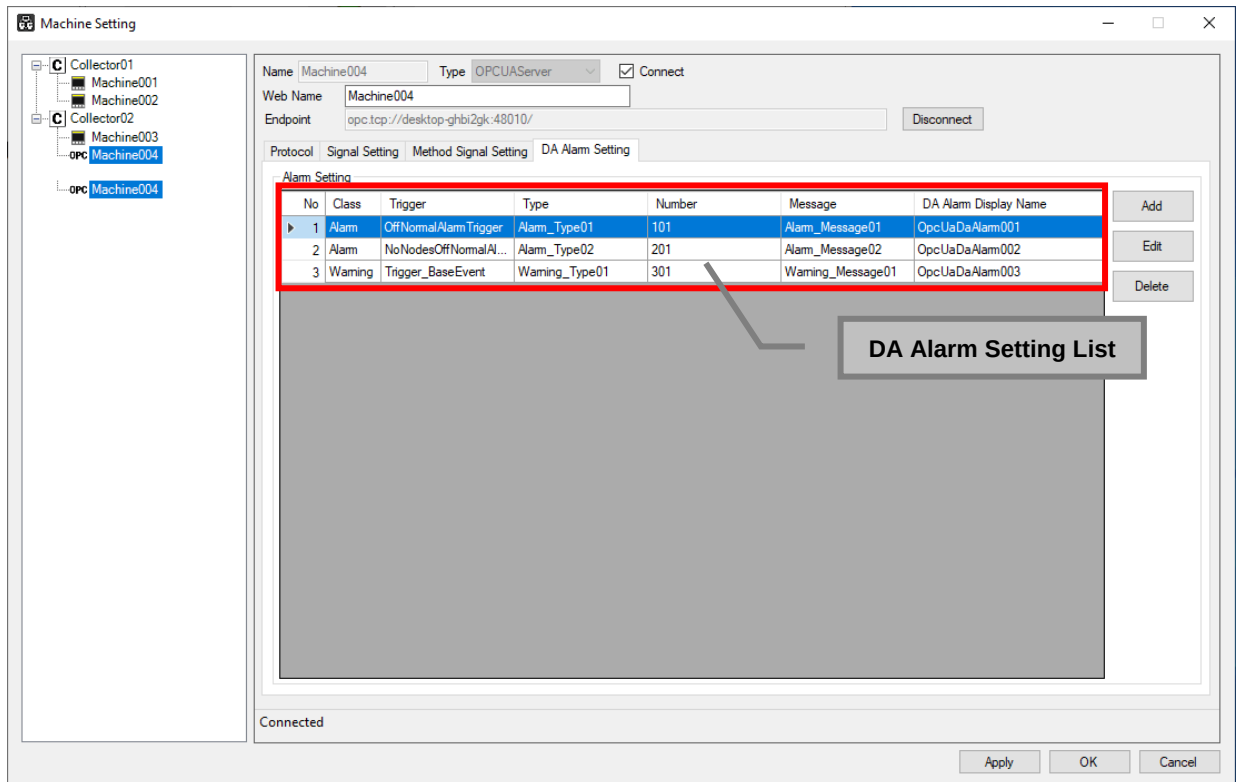
4.3.8 Set Alarm of OPCUAServer

By defining and collecting the variable data of the nodes displayed in the browse as DA alarms, the alarms occurring on the devices connected to OPC UA can be handled as alarms on the MT-LINK*i*.

4.3.8.1 Set OPCUAServer DA Alarms

In the [DA Alarm Setting] tab page of the [Machine Setting] window, associate DA alarms, collected from machine tools by OPC UA Servers, with MT-LINK*i* signals.

Those signals can be registered up to 200.



■ The [DA Alarm Setting] tab

DA alarm setting list

This list displays DA alarm settings that are made with the [DA Alarm Setting] window. For the details, refer to the "[The \[DA Alarm Setting\] window](#)" window.

- Class
The class (message class) of a DA alarm
- Type
The type of a DA alarm
- Number
The number of a DA alarm
- Message
The message of a DA alarm
- DA Alarm Display Name
The DA alarm display name of a DA alarm

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	182/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

Buttons

- [Add] button
Click this button to add a new setting of a DA alarm. This button does not work while connection to the OPC UA Server is lost.
- [Edit] button
Click this button to edit a DA alarm setting. Double-clicking on a row selected in the list should be equivalent to this button.
- [Delete] button
Click this button to delete selected DA alarm settings.

(1) Setting DA alarms

Performing the following operations with the [DA Alarm Setting] tab page of the [Machine Setting] window opens the [DA Alarm Setting] window:

- Click the [Add] button.
- Select a DA alarm setting and click the [Edit] button.
- Select a DA alarm setting and double-click on the selected row.

The screenshot shows the 'DA Alarm Setting' window. On the left is the 'Node Information Tree' listing various nodes under 'DeviceA'. A 'Selected Node' is highlighted at the bottom, showing 'Node ID: ns=2;s=DeviceA.AlarmMessage' and 'Data type: String'. The main area is divided into several sections: 'DA Alarm Display Name' (set to 'OpcUaDaAlarm0001'), 'Message Class' (radio buttons for 'Alarm' and 'Warning'), 'Trigger' (fields for 'Node Name' and 'Node ID'), 'Exclusion Setting' (checkboxes for excluding messages based on Node ID, Number, or Message), 'Type' (radio buttons for 'Text', 'Node ID', 'Number', and 'Message'), and 'Number' (radio buttons for 'Numeral' and 'Node ID'). Each section has corresponding input fields for 'Node Name' and 'Node ID'. At the bottom right are 'Test', 'OK', and 'Cancel' buttons.

■ The [DA Alarm Setting] window

- Node information tree
This tree lists the nodes for the selected machine.
- Selected node
This area populates the following information about the node selected in the tree:
 - [Node ID]: The selected node's ID defined by the OPC UA Server
 - [Data type]: The selected node's data type set on the OPC UA Server
- DA Alarm Display Name
Specify or edit the display name for the DA alarm. MT-LINKⁱ uses this display name as the signal name of the DA alarm. This field is required.

20	Nov. 22, 2023	uwano	Revised for MT-LINK ⁱ Version 4.0	Title	FANUC MT-LINK ⁱ AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK ⁱ Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK ⁱ Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK ⁱ Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK ⁱ Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK ⁱ Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK ⁱ Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	183/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

The [DA Alarm Display Name] field should start with an alphabet and the subsequent characters can be alphanumeric characters, hyphens ("-"), underscores ("_"), spaces (" "), and round brackets ("(", ")"). This field can be up to 32 bytes long and is set to "OpcUaAlarm< three digit numbering>" by default.

- **Message Class**

Select a message class for the DA alarm from the following:

- [Alarm]: An alarm message
- [Warning]: A warning

- **Trigger**

Use the following buttons to register or unregister a node as the trigger of the DA alarm:

- [>>] button: Clicking this button registers the node selected in the tree as the trigger of the alarm. This button is available to only nodes of "Boolean" data type.
When the registration of the node is set as a trigger, its node information will be populated in the text fields ([Node Name], [Node ID]) on the right side of this button.
- [<<] button: Clicking this button unregisters the node as a trigger and clears the text fields.

[Message Data]

- **Exclusion setting (alarm exclusion criteria)**

Set criteria for excluding DA alarms. Select one or more of the following check boxes then alarm messages for DA alarms will not be created if one or more of the nodes consisting of those DA alarms match the selected criteria:

- [Exclude the DA alarm if the retrieved value of the node associated with Type is null space.]
- [Exclude the DA alarm if the retrieved value of the node associated with Number is zero.]
- [Exclude the DA alarm if the retrieved value of the node associated with Message is null space.]

- **Type**

Set the alarm type of the DA alarm.

When selecting the [Text] button, specify the alarm type as follows:

Set the alarm type for the text field (not omissible).

The text should start with an alphabet and the subsequent characters can be alphanumeric characters, hyphens ("-"), underscores ("_"), spaces (" "), and round brackets ("(", ")"). This field can be up to 48 bytes long.

When selecting the [Node ID] button, use the following buttons to select the alarm type:

- [>>] button: Clicking this button associates the node selected in the tree with the alarm type of the DA alarm. This button is available to only nodes of "String" data type.
When the association of the node is completed, its node information will be populated in the text fields ([Node Name], [Node ID]) on the right side of this button.
- [<<] button: Clicking this button releases the association and clears the text fields.

- **Number**

Set the alarm number of the DA alarm.

When selecting the [Numeral] button, specify the alarm number as follows:

Set the alarm number for the text field.

The number must be "0" to "2147483647."

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	184/868

When selecting the [Node ID] button, use the following buttons to select the alarm number:

- [>>] button: Clicking this button associates the node selected in the tree with the alarm number of the alarm. This button is available to only nodes of "String," "SByte," "Byte," "Int16," "UInt16," and "Int32" data types.

When the association of the node is completed, its node information will be populated in the text fields ([Node Name], [Node ID]) on the right side of this button.

- [<<] button: Clicking this button releases the association and clears the text fields.

• Message

Specify whether to use a text or a node to compose the alarm message for the DA alarm.

When selecting the [Text] button, specify the alarm message as follows:

Set the alarm message for the text field (not omissible).

This field can be up to 100 bytes long.

When selecting the [Node ID] button, use the following buttons to select the alarm message:

- [>>] button: Clicking this button associates the node selected in the tree with the alarm message. This button is available to only nodes of "String" data type.

When the association of the node is completed, its node information will be populated in the text fields ([Node Name], [Node ID]) on the right side of this button.

- [<<] button: Clicking this button releases the association and clears the text fields.

NOTE

- 1 When the [Node ID] button is selected, messages of DA alarms are composed of values (nodes) retrieved from the OPC UA Servers except in the following cases:

- Signals cannot be collected from the OPC UA Servers
- One or more of the alarm exclusion criteria are selected and one or more of the nodes specified in the [Message] group box match the selected criteria (i.e., the value of the node specified in the [Type] or the [Message] group box is null space, or that of the node specified in the [Number] group box is 0).

- 2 If the type value of a signal collected from an OPC UA Server exceeds 45 bytes, it will be truncated to 45 bytes, concatenated with an ellipsis ("..."), and then displayed on Web screens (the [Monitoring - Alarm Monitoring] and the [Diagnosis - Alarm History] screens).

- 3 If the message value of a signal collected from an OPC UA Server exceeds 512 bytes, it will be truncated to 512 bytes, concatenated with an ellipsis ("..."), and then displayed on Web screens (the [Monitoring - Alarm Monitoring] and the [Diagnosis - Alarm History] screens). If such a long alarm message consists of multi-byte characters, the last character (at the 512nd byte position) of the truncated message may appear as garbled.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	185/868

Buttons

- [Test] button

Click this button to check the values of the DA alarm signal. A message box will appear reading the check results.

- Result: The result "OK" (success) or "Error (<error field>)" (failure)
(<error field> denotes where the error occurred.)
- Trigger: The value of the trigger (time taken in retrieving)
- Type: The value of the type (time taken in retrieving)
- Number: The value of the type (time taken in retrieving)
- Message: The value of the message (time taken in retrieving)

Example of test results (success)

Result : OK

Trigger ... True (1ms)
Type ... ExclusiveLevelAlarm (2ms)
Number ... 400 (3ms)
Message ... HighLimit exceeded. (4ms)

Example of test results (failure)

Result : Error (Type)

Trigger ... True (1ms)
Type ... 0x801F0000 (BadUserAccessDenied)
Number ... 400 (3ms)
Message ... HighLimit exceeded. (4ms)

- [OK] button

Clicking this button checks the specified values and closes the [DA Alarm Setting] window if no errors are detected.

- [Cancel] button

Clicking this button discards the specified values and closes the [DA Alarm Setting] window.

NOTE

- 1 Select a node in the Node information tree before setting a [Node ID] in the [Message Data] group box; or select a [Text] button or the [Numeral] button and then enter a text in its text field.
- 2 At least one [Node ID] text field in the [Trigger] group box and the [Message Data] group box must be set to the node ID of a node selected in the tree.
- 3 At least one of the [Type], the [Number], and the [Message] group boxes must be set.
- 4 When not specifying the [Trigger] field but selecting one or more [Node ID] buttons in the [Message Data] group box, select at least one of the alarm exclusion criteria.
- 5 Specifying a node ID in the [Trigger] group box will be regarded as a duplicate error if another DA alarm exists that is already set with that specified node ID in the [Trigger] group box.
- 6 Selecting one or more check boxes and specifying a node ID in the [Message Data] group box will be regarded as a duplicate error if another DA alarm exists that selects one or more check boxes and is already set with that specified node ID in the [Message Data] group box.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	186/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

(2) Examples of setting DA alarms

The following are DA alarm settings in typical use cases.

Setting example	Use case	Trigger of alarm	How and what to set	Remarks
1	Devices that raise only alarms of Bit type	Trigger signal (Boolean)	Use a signal to judge whether the alarm data are alarm. Users must associate Type, Number, and Message with alarm data.	Users must set Type, Number, and Message for an alarm.
2	Devices that raise alarms of String type	Type (String)	Use type in alarm data to judge whether the alarm data are alarm. Users must associate Type with alarm data.	Users need not to set Trigger for an alarm, but must select the first check box in the [Message Data] group box.
3	Devices that raise alarms of Bit type	Trigger signal (Boolean)	Use a signal to judge whether the alarm data are alarm. Users must associate Type, Number, and Message with alarm data.	If needed, users should select the second check box in the [Message Data] group box
4	Devices that raise given alarms of Int type by setting a PLC register to numeric thresholds	Number (Int32)	Use number in alarm data to judge whether the alarm data are alarm. Users must associate Number with alarm data.	Users need not to set Trigger for an alarm, but must select the third check box in the [Message Data] group box.

■ **Example 1: A device that raises only alarms of Bit type**

1) Actions of OPC UA Servers

- The OPC UA Server turns the signals of raised alarm/warning to "1 (on)."

Output signals (IO signal)

No.	Name of output signal	Data length	Description
1	ALARM001	1 bit	Internal overheating 0: Raise no alarm. 1: Inside of the amplifier is overheated.
2	ALARM002	1 bit	Insufficient torque 0: Raise no alarm. 1: Torque is insufficient. Servo motor may have its brake malfunctioned.
3	WARNING001	1 bit	Motor temperature warning 0: Raise no warning. 1: Motor temperature exceeded the predefined warning threshold.

2) Signals collected by OPC UA Servers

- Output signals (*their signal names are shown in the table in 1).

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	187/868

3) Examples of setting DA alarms

(*<n> in No.<n> in the following table corresponds to a number in the [No.] column of the table in 1))

- Select nodes for output signals in the tree and click the [>>] button to register them as trigger signals.
- Set the [Message Data] group box as shown in the table below.

Fields in the [DA Alarm Setting] window			No.1	No.2	No.3
DA Alarm Display Name			OpcUaDaAlarm0001	OpcUaDaAlarm0002	OpcUaDaWarn0001
Message Class			Alarm	Alarm	Warning
Trigger			ALARM001	ALARM002	WARNING001
Message Data	The first check box		Cleared	Cleared	Cleared
	The second check box		Cleared	Cleared	Cleared
	The third check box		Cleared	Cleared	Cleared
	Type	Radio button	Text	Text	Text
		Value	SV	DS	WARN
	Number	Radio button	Numeral	Numeral	Numeral
		Value	100	340	200
	Message	Radio button	Text	Text	Text
		Value	SERVO OVERHEAT	BRAKE TORQUE SHORTAGE	MOTOR TEMPERATURE PREWARNING

Example of the [DA Alarm Setting] window set for No.1

DA Alarm Setting

DA Alarm Display Name: OpcUaDaAlarm0001

Message Class: ☒ Alarm ☐ Warning

Trigger: >> Node Name: ALARM001 << Node ID: ns=2;s=DeviceA.Device1.ALARM001

Message Data:

- ☐ Exclude the message when the Node ID for the retrieved Type has no value in Value.
- ☐ Exclude the message when the Node ID for the retrieved Number has zero in Value.
- ☐ Exclude the message when the Node ID for the retrieved Message has no value in Value.

Type: ☒ Text SV ☐ Node ID >> Node Name << Node ID

Number: ☒ Numeral 100 ☐ Node ID >> Node Name << Node ID

Message: ☒ Text SERVO OVERHEAT ☐ Node ID >> Node Name << Node ID

Node ID: ns=2;s=DeviceA.Device1.ALARM001
Data type: Boolean

Buttons: Test, OK, Cancel

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 188/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

Example of the [DA Alarm Setting] window set for No.2

Example of the [DA Alarm Setting] window set for No.3

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				Draw	A-40829-00001EN				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	189/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

■ **Example 2: A device that raises alarms of String type**

1) Actions of OPC UA Servers

- If the OPC UA Server determines alarm data as an alarm, it collects them from the connected machine tools.

Alarm data

No.	Name of output signal	Data length	Description
1	AlarmType001	8 bytes (8 character string)	Alarm type In this example, the collected alarm type is null space: No alarm is raised.
	AlarmMessage001	256 bytes (256 character string)	Alarm message In this example, the collected alarm message is null space: No alarm is raised.
2	AlarmType002	8 bytes (8 character string)	Alarm type In this example, the collected alarm type is null space: No alarm is raised.
	AlarmMessage002	256 bytes (256 character string)	Alarm message In this example, the collected alarm message is null space: No alarm is raised.
3	AlarmType003	8 bytes (8 character string)	Alarm type In this example, the collected alarm type is null space: No alarm is raised.
	AlarmMessage003	256 bytes (256 character string)	Alarm message In this example, the collected alarm message is null space: No alarm is raised.

2) Signals collected by OPC UA Servers

- Output signals (*their signal names are shown in the table in 1).

3) Examples of setting a DA alarm

(*<n> in No.<n> in the following table corresponds to a number in the [No.] column of the table in 1))

- Make no settings in the [Trigger] group box.
- Set the [Message Data] group box as shown in the table below.
* Select the third check box for alarm exclusion to exclude DA alarms that have null space in their alarm data. (If the check box is not selected, alarm data with no message will not be regarded as an alarm.)

Fields in the [DA Alarm Setting] window		No.1	No.2	No.3
DA Alarm Display Name		OpcUaDaAlarm0001	OpcUaDaAlarm0002	OpcUaDaAlarm0003
Message Class		Alarm	Alarm	Warning
Trigger		-	-	-
Message Data	The first check box	Cleared	Cleared	Cleared
	The second check box	Cleared	Cleared	Cleared
	The third check box	Selected	Selected	Selected
	Type	Radio button	Node ID	Node ID
		Value of Node Name	AlarmType001	AlarmType002
	Number	Radio button	Numeral	Numeral
		Value	0	0
	Message	Radio button	Node ID	Node ID
		Value of Node Name	AlarmMessage001	AlarmMessage002

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	190/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

Example of the [DA Alarm Setting] window set for No.1

DA Alarm Setting

DA Alarm Display Name: OpcUsaDaAlarm001

Message Class: ☒ Alarm ☐ Warning

Trigger: >> Node Name: << Node ID: <<

Message Data

☐ Exclude the message when the Node ID for the retrieved Type has no value in Value.

☐ Exclude the message when the Node ID for the retrieved Number has zero in Value.

☒ Exclude the message when the Node ID for the retrieved Message has no value in Value.

Type: ☐ Text ☒ Node ID

Node ID: >> Node Name: AlarmType001 << Node ID: ns=2,s=DeviceA,Device2,AlarmType001

Number: ☒ Numeral: 0 ☐ Node ID

Node ID: >> Node Name: << Node ID: <<

Message: ☐ Text ☒ Node ID

Node ID: >> Node Name: AlarmMessage001 << Node ID: ns=2,s=DeviceA,Device2,AlarmMessage001

Node ID: ns=2,s=DeviceA,Device2,AlarmMessage001

Data type: String

Test OK Cancel

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	191/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

DA Alarm Setting

DA Alarm Display Name: OpcUaDaAlarm0002

Message Class: ☒ Alarm ☐ Warning

Trigger: >> Node Name
<< Node ID

Message Data: ☐ Exclude the message when the Node ID for the retrieved Type has no value in Value.
☐ Exclude the message when the Node ID for the retrieved Number has zero in Value.
☒ Exclude the message when the Node ID for the retrieved Message has no value in Value.

Type: ☐ Text
☒ Node ID >> Node Name
<< Node ID

Number: ☒ Numeral
☐ Node ID >> Node Name
<< Node ID

Message: ☐ Text
☒ Node ID >> Node Name
<< Node ID

Node ID: ns=2;s=DeviceA.Device2.AlarmMessage002
 Data type: String

Test OK Cancel

DA Alarm Setting

Node ID: ns=2;s=DeviceA.Device2.AlarmMessage003
Data type: String

DA Alarm Display Name: OpcUaDaAlarm003

Message Class: ☒ Alarm ☐ Warning

Trigger: >> Node Name:
<< Node ID:

Message Data

☐ Exclude the message when the Node ID for the retrieved Type has no value in Value.
☐ Exclude the message when the Node ID for the retrieved Number has zero in Value.
☒ Exclude the message when the Node ID for the retrieved Message has no value in Value.

Type: ☐ Text
☒ Node ID >> Node Name: AlarmType003
<< Node ID: ns=2;s=DeviceA.Device2.AlarmType003

Number: ☒ Numeral 0
☐ Node ID >> Node Name:
<< Node ID:

Message: ☐ Text
☒ Node ID >> Node Name: AlarmMessage003
<< Node ID: ns=2;s=DeviceA.Device2.AlarmMessage003

Test OK Cancel

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					FANUC CORPORATION			Sheet.	192/868
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

1) Actions of OPC UA Servers

- Input signals (IO signal)

Registered alarms (PLC register)

2) Signals collected by OPC UA Servers

- | | | | | | | | | | | | | |
|------|---------------|--------|--|-------|--|--------|-------|-------------------|---|--|--------|---------|
| 20 | Nov. 22, 2023 | uwano | Revised for MT-LINK <i>i</i> Version 4.0 | | | | | Title | FANUC MT-LINK <i>i</i>
AdminTool OPERATOR'S MANUAL | | | |
| 19 | Apr. 27, 2021 | uwano | Revised for MT-LINK <i>i</i> Version 3.11 | | | | | | | | | |
| 18 | July 28, 2020 | uwano | Revised for MT-LINK <i>i</i> Version 3.10 | | | | | | | | | |
| 17 | Jan. 30, 2020 | uwano | Revised for MT-LINK <i>i</i> Version 3.9 | | | | | Draw | A-40829-00001EN | | | |
| 16 | July 22, 2019 | uwano | Revised for MT-LINK <i>i</i> Version 3.8 | | | | | | | | | |
| 15 | Mar. 29, 2019 | ogawa | Revised for MT-LINK <i>i</i> Version 3.7.2 | | | | | | | | | |
| 14 | Jan. 29, 2019 | ogawa | Revised for MT-LINK <i>i</i> Version 3.7 | | | | | FANUC CORPORATION | | | Sheet. | 193/868 |
| Ed. | Date | Design | Description | | | | | | | | | |
| Date | Nov.30,2015 | Desig. | Ogawa | Check | | Apprv. | Inoue | | | | | |

3) Examples of setting a DA alarm

(*<n> in No.<n> in the following table corresponds to a number in the [No.] column of the table in 1))

- Select nodes for input signals in the tree and click the [>>] button to register them as trigger signals.
- Set the [Message Data] group box as shown in the table below.
* Select the third check box for alarm exclusion to exclude DA alarms that have null space in their alarm data. (If the check box is not selected, alarm data with null message will not be regarded as an alarm.)

Fields in the [DA Alarm Setting] window		No.1	No.2	No.3
DA Alarm Display Name		OpCuaDaAlarm0001	OpCuaDaWarn0001	OpCuaDaAlarm0002
Message Class		Alarm	Warning	Alarm
Trigger		Alarm001	Alarm002	Alarm003
Message Data	The first check box	Cleared	Cleared	Cleared
	The second check box	Cleared	Cleared	Cleared
	The third check box	Selected	Selected	Selected
	Type	Radio button	Text	Text
		Value	EX	OPR
	Number	Radio button	Numeral	Numeral
		Value	1000	2000
	Message	Radio button	Node ID	Node ID
		Value of Node Name	Message001	Message002
			Message002	Message003

Example of the [DA Alarm Setting] window set for No.1

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION		Sheet. 194/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

Example of the [DA Alarm Setting] window set for No.2

Example of the [DA Alarm Setting] window set for No.3

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 195/868

■ **Example 4: Device that raises given alarms of Int type by setting a PLC register to numeric thresholds**

1) Actions of OPC UA Servers

- The OPC UA Server sets the detail code of a raised Int type alarm to an output signal.

Output signals (PLC register)

No.	Name of output signal	Data length	Description
1	AlarmCode	2 bytes (value: 0 to 65535)	Alarm code 0: No alarm is raised. Otherwise: An alarm is raised (detail code).
2	WarningCode	2 bytes (value: 0 to 65535)	Warning code 0: No warning is raised. Otherwise: A warning is raised (detail code).

2) Signals collected by OPC UA Servers

- Output signals

3) Examples of setting a DA alarm

(*<n> in No.<n> in the following table corresponds to a number in the [No.] column of the table in 1))

- Make no settings in the [Trigger] group box.
- Set the [Message Data] group box as shown in the table below.
* Select the second check boxes in the [Message Data] group box. (If the check box is not selected, alarm data with alarm number "0" will not be regarded as an alarm.)

Fields on the [DA Alarm Setting] window			No.1	No.2
DA Alarm Display Name			OpcUaDaAlarm0001	OpcUaDaWarn0001
Message Class			Alarm	Warning
Trigger			-	-
Message Data	The first check box		Cleared	Cleared
	The second check box		Selected	Selected
	The third check box		Cleared	Cleared
	Type	Radio button	Text	Text
		Value	ALARM	WARN
	Number	Radio button	Node ID	Node ID
		Value of Node Name	AlarmCode	WarnCode
	Message	Radio button	Text	Text
		Value	ALARM MESSAGE	WARNING MESSAGE

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION		Sheet. 196/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

Example of the [DA Alarm Setting] window set for No.1

Example of the [DA Alarm Setting] window set for No.2

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	197/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

4.4 MACHINE SETTING - FanucRobot

This section explains the machine setting of the machine type FanucRobot.

4.4.1 Steps for the Machine Setting of FanucRobot

Follow the steps below to set the machine.

- (1) In the AdminTool Menu, select [Machine Setting] to display [Machine Setting] window.
- (2) Add a Collector. For the details of the operation, refer to the "[Add Collector](#)" subsection.
- (3) Add machine type FanucRobot. For the details, refer to the "[Add FanucRobot](#)" subsection.
- (4) Make network and connection setting. For the details, refer to the "[Confirm/Edit Setting of FanucRobot](#)" subsection and the "[Set Protocol Setting of FanucRobot](#)" subsection.
- (5) Make signal setting. For the details, refer to the "[Set Signal of FanucRobot](#)" subsection.
- (6) Make custom signal setting. For the details, refer to the "[Set Custom Signal of FanucRobot](#)" subsection.
- (7) Set variable history retrieval conditions. For the details, refer to the "[Set Variable History Retrieval Condition Definition of FanucRobot](#)" subsection.
- (8) Make position register setting. For the details, refer to the "[Set Position Register of FanucRobot](#)" subsection.
- (9) Save the machine information set, and make the system reflect them. For the details of the operation, refer to the "[Save Machine Setting](#)" subsection.

4.4.2 Add FanucRobot

- (1) Add Machine

To add a machine, right-click the target Collector name in the Collector tree and point to [Add Machine].

*Before adding a machine, make sure that you have registered the target Collector.

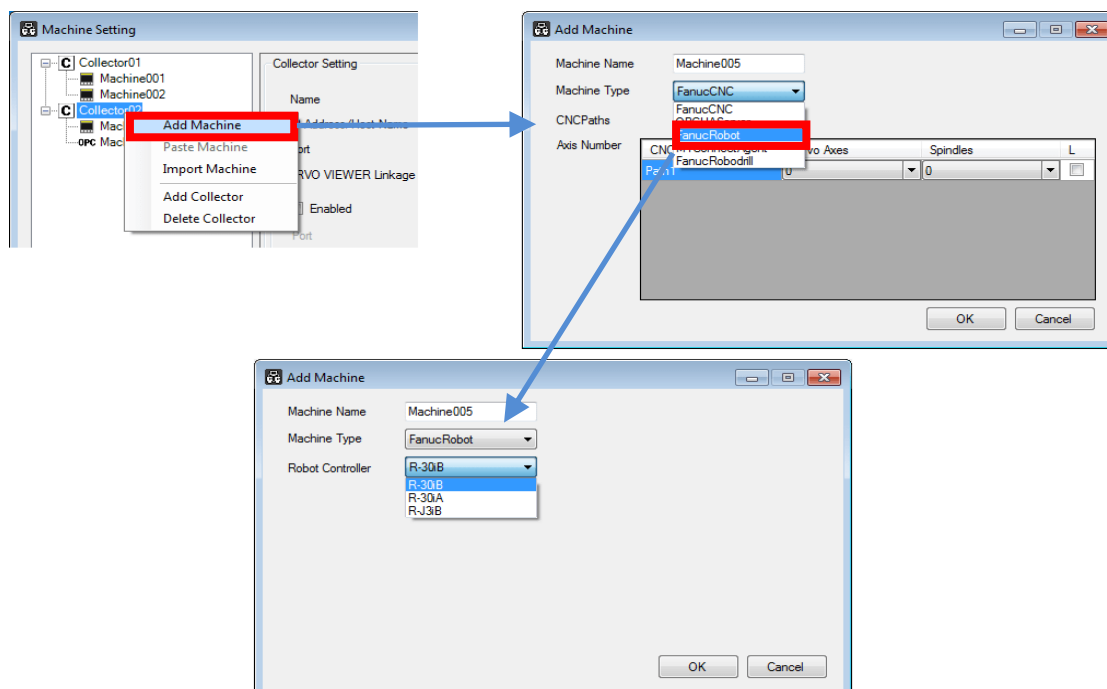
When the [Add Machine] dialog box appears, set the settings of the target machine.

NOTE

Once set, [Machine Name], [Machine Type], and [Robot Controller] will not be editable.

If the [Machine Name], [Machine Type], and [Robot Controller] are mistakenly set, the existing machine must be deleted. Add a new machine and make the settings for it.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION				Sheet. 198/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	



■ The [Add Machine] dialog box

- Machine Name
Set a machine name to this field. The [Machine Name] should start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and underscores ("_"). This field can be up to 32 bytes long.
- Machine Type
Select [FanucRobot].
- Robot Controller
Set this field to the Robot Controller model to be connected.
[R-30iB], [R-30iA], or [R-J3iB] can be selected.
However, select [R-30iB] when the Robot Controller model is R-30iB Plus / R-30iB Mate Plus.
- The [OK] button
Clicking this button adds the machine in the Collector tree.
- The [Cancel] button
Clicking this button discards the settings and closes the dialog box.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 199/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

4.4.3 Confirm/Edit Setting of FanucRobot

(1) Confirm and edit machine setting

The settings of the selected Collector or machine are populated in the right pane of the [Machine Setting] window. Selecting a machine name in the Collector tree will show machine setting, so that you can set and change them.

The [Web Name] is displayed on the Web screens. The value of this field should be entered in the language specified for the [Web name language] field in the [System Setting] window. The [Web Name] field is required and will be set to the default value of <Machine name> if it is not modified.

Machine Setting

Collector01
Machine001
Machine002
Collector02
Machine003
Machine004
Machine005

Name: Machine005 Type: FanucRobot ☒ Connect Robot Controller: R-30iB

Web Name: Machine005

Protocol: Basic Signal Custom Signal Variable History Retrieval Condition Definition Position Register

Protocol Setting

IP Address/Host Name: 192.168.0.105

Port: 60008

Time out(Sec): 30

(2) Set Machine Connection

Set connection between machine and Collector. Select a machine in the Collector tree and select the [Connect] check box, so that the connection is available.

Machine Setting

Collector01
Machine001
Machine002
Collector02
Machine003
Machine004
Machine005

Name: Machine005 Type: FanucRobot ☒ Connect Robot Controller: R-30iB

Web Name: Machine005

Protocol: Basic Signal Custom Signal Variable History Retrieval Condition Definition Position Register

Protocol Setting

IP Address/Host Name: 192.168.0.105

Port: 60008

Time out(Sec): 30

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	200/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

4.4.4 Set Protocol Setting of FanucRobot

Select a machine in the Collector tree and select the [Protocol] tab in the right pane.

■ The [Protocol] tab

Protocol Setting

- IP Address/Host Name

Specify an IP address or host name of the machine.

When entering a host name, the name can start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and periods ("."). This field can be up to 253 bytes long.



CAUTION

For FANUC Robot controllers for the following robots, if the machine setting is set by using duplicated IP address/host name (if the IP address/host name already in use is being used for a new machine setting), all the duplicated connections will be disconnected.

- R-30iA / R-30iA Mate
- R-J3iB / R-J3iB Mate

- Port

Specify the port number for connection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 201/868

NOTE

- 1 Initial value of the port number of robot controller is "60008."
- 2 When connecting RoboGuide, there are cases when the port number is changed. Check the port number and make the appropriate setting.

- Time out

Specify the connection timeout length in second. The default value is 30 seconds.

FTP Server Setting

Set fields for FTP communication with connected FanucRobot type machines. The settings made in those fields will be referred to from the following functions:

- The [Machine Data Storing Schedule] window uses them when archiving all the MDBs (memory binary devices) in machines.
- The [Utility - File Transfer] screen uses them when uploading the MDBs (memory binary devices) in machines.

- IP Address/Host Name (check box)

Select this check box when using an IP address or host name specific to the FTP Server.

When it is checked, the [IP Address/Host Name] field turns to be editable. This check box is cleared by default.

- IP Address/Host Name (not omittable text field)

Specify an IP address or host name of the FTP Server.

When entering a host name, it can start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and periods ("."). This field can be up to 253 bytes long. By default, it is set to the same value as set in the [Protocol Setting] group box.

- Port (not omittable)

Specify a connection port number for FTP communication. A number in a range of 0 to 65535 can be specified.

This field is set to 21 by default.

- User Name (optional)

Specify a user name to use when sign in to the FTP Server on a FanucRobot type machine.

Alphanumeric characters and symbols can be used for this field. This field can be up to 31 bytes long. By default, it is set to null.

- Password (optional)

Specify a password for the user name specified in the [User Name] field.

Alphanumeric characters and symbols can be used for this field. This field can be up to 31 bytes long. The entered characters will appear as asterisks (*).

By default, it is set to null.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION				Sheet. 202/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

NOTE

- 1 While connected with an FTP Server, the connection timeout length specified in the [Time out] field in the [Protocol Setting] group box will be used.
- 2 Only passive mode is available when establishing a connection with an FTP Server. Active mode is unsupported.
- 3 Only binary mode is available when transferring files. ASCII (text) mode is unsupported.

External Link Setting

Set fields for external links to view them through Web screens. Those links can be selected from the [Monitoring - Equipment Monitoring] screen (displayed by selecting an equipment icon on the [Monitoring - Overlook] screen).

External Link function provides a way to view external links, such as Web sites, that are associated with the selected equipment (machines).

For the details, refer to the "[Set External Links](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	203/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

4.4.5 Set Signal of FanucRobot

This setting function is provided to browse the signals of machines in MT-LINK*i*.

Select a machine in the Collector tree and select the [Basic Signal] tab in the right pane.

For each signal listed in the [Basic Signal] tab, select or clear the [Enable] check box according to whether it is to be collected.

Collector collects the signals which have the [Enable] check box selected.

The statuses will be automatically added to the [Simple assign status area] of the [Equipment Setting] window if they have been added on the [Custom Signal] tab of the [Machine Setting] window or their [Enable] check boxes have been selected on the [Basic Signal] tab of the [Machine Setting] window.

No	Signal	Description	Enable	Reading Cycle	Check
1	REMOTE	Remote signal	☐	500msec	▼
2	BUSY	Busy signal	☐	500msec	▼
3	HELD	Hold signal	☐	500msec	▼
4	FAULT	Alarm fault signal	☐	500msec	▼
5	BATAL	Abnormal battery alarm signal	☒	500msec	▼
6	TPENBL	Teach pendant enable signal	☐	500msec	▼
7	Mode	Mode	☒	500msec	▼
8	RobotState	Robot State	☒	500msec	▼
9	Alarm	Alarm message	☒	500msec	▼
10	ProgramName	Executing program	☒	500msec	▼
11	ProgramLineNo	Sequence number	☐	500msec	▼

■ The [Basic Signal] tab

● Signal

This column in the signal lists the names of signals that can be set for the selected machine. To list only specific signals, narrow down the signal list. For the details on how to narrow down the list, refer to the "[Narrow down Signal List](#)" subsection.

For the details of signals, refer to the "[LIST OF FanucRobot SIGNALS](#)" section.

● Description

This column displays the Web name of each signal as its description.

● Enable

This is a check box indicating whether the data is collected by the Collector or not. When the [Enable] check box is selected, the data is collected; otherwise, no collection of data.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	204/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

NOTE

- 1 The check boxes of the mandatory signals (RobotState, Alarm, and ProgramName) cannot be cleared.
- 2 When a signal is referenced by equipment, clearing the [Enable] check box for it will be denied with an error message. For the details, refer to the "[Referenced Error](#)" subsection.

- Reading Cycle

With this field, you can specify a reading interval per signal.
It is fixed at "500msec," and cannot be changed.

- Check

This column displays the results of Signal Check as follows:

"OK":	The signal was retrieved
"NG" (for "not good"):	Failed to retrieve the signal
"-" (hyphen):	Not a target of Signal Check
"" (blank):	Signal Check unexecuted (initial state, machine connection failure, etc.)

For the details on Signal Check, refer to the "[Checking the Signals](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 205/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

4.4.6 Set Custom Signal of FanucRobot

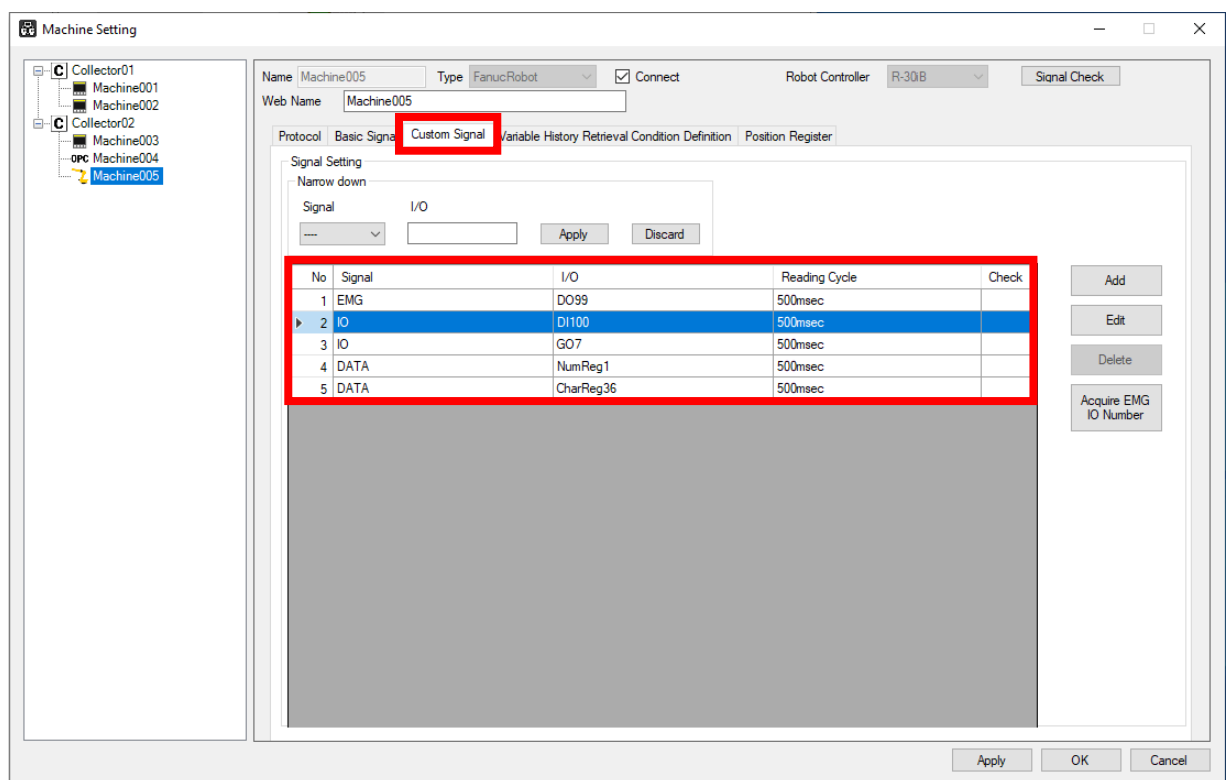
This function is provided to browse the I/O (Input/Output) signals and register data (see NOTE) of FanucRobot as custom signals in MT-LINK*i*.

Custom signals can be registered to a maximum number of 1000.

Select a machine in the Collector tree and select the [Custom Signal] tab in the right pane.

NOTE

- 1 NumReg (numeric register) and CharReg (string register) can be setup with the [Custom Signal] tab page. To setup PosReg (position register), use the [Position Register] tab page.
For the details, refer to the "[Set Position Register of FanucRobot](#)" subsection.
- 2 When setting registers in machines as signals, make sure to verify that all of those register signals can be retrieved by conducting machine connection tests.
For the details on how to conduct a machine connection test, refer to the "[Check Connection with the Machine](#)" subsection.



■ The [Custom Signal] tab

The [Signal Setting] group box lists custom signals that are set up with the [Signal Setting] dialog box. To list only specific custom signals, narrow down the signal list. For the details on how to narrow down the list, refer to the "[Narrow down Signal List](#)" subsection.

* EMG signal is pre-registered. However, its number must be set by the user.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	206/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

For the following fields, refer to "**The [Signal Setting] dialog box**" paragraph.

- **Signal**
This column lists the labels of signals:
"EMG": EMG signal
"IO": IO signal
"DATA": Register data
- **I/O**
This column lists the addresses and numbers of signals:
Example) For a signal whose address is "DO" and whose number is 99, "DO99" will be populated.
- **Reading Cycle**
This column lists each reading interval per signal (fixed to 500msec).
- **Check**
The results of the Signal Check, when the [Signal Check] button is clicked, are displayed.
"OK": The signal was retrieved.
"NG" (for "not good"): Failed to retrieve the signal.
"" (blank): Signal Check unexecuted (initial state, machine connection failure, etc.), or after executing Signal Check, setting was changed on the [Signal Setting] dialog box and the [OK] button was clicked.

For the details on Signal Check, refer to the "**Checking the Signals**" subsection.

- **The [Add] button**
Clicking this button opens the [Signal Setting] dialog box to register new custom signal.
- **The [Edit] button**
Clicking this button opens the [Signal Setting] dialog box to edit selected custom signal.
- **The [Delete] button**
Clicking this button deletes a selected custom signal.

NOTE

When a custom signal is referenced by equipment, deleting it will be denied with an error message. For the details, refer to the "**Referenced Error**" subsection.

- **The [Acquire EMG IO Number] button**
Clicking this button tries to connect with the specified FanucRobot type machine and acquire the IO (DO) number of EMG signal from it.
* For R-J3iB / R-J3iB Mate Series robots, IO (SDO) numbers will be acquired.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	207/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

NOTE

On setting the emergency stop signal on FANUC Robot controller

If the emergency stop signal of a robot is not set up on the robot controller, the emergency stop status of the robot cannot be retrieved.

Use the System configuration menu on the controller to specify a DO number (an IO number in DO)*¹ to which the emergency stop signal will be output. When an emergency stop occurs (TP external emergency stop, Operation Panel emergency stop), the emergency stop signal will be output to the DO number that is specified on the robot controller.

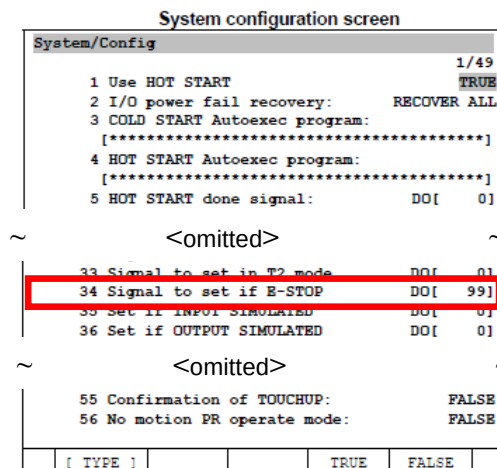
If the DO number is left as the default value ("0"), the emergency stop status cannot be retrieved from the robot.

To work this function, change the DO number to an appropriate number, power off the robot once, and power on it again.

When assigning a DO number*¹ to the emergency stop signal on a robot controller, make sure that no programs running on the FANUC Robot use it.

(*¹ For robots in R-J3iB or R-J3iB Mate Series, the emergency stop signal is output to an IO number in SDO instead)

The following figure is an example of the [System/Config] screen of a robot controller. This example screen shows the emergency stop signal is set as "DO[99]" on the robot controller.



For the details, refer to the following manuals of the respective robot controllers.

"FANUC Robot series R-30iB/ R-30iB Mate/ R-30iB Plus/ R-30iB Mate Plus CONTROLLER OPERATOR'S MANUAL (Basic Operation) (B-83284EN/05)"

"FANUC Robot series R-30iA CONTROLLER HANDLING TOOL (For series 7DA4/7DA5) OPERATOR'S MANUAL (B-83124EN-2/01)"

"FANUC Robot series R-30iA Mate CONTROLLER LR HANDLING TOOL (For series 7DA4/7DA5/7DA7) OPERATOR'S MANUAL (B-83134EN-1/01)"

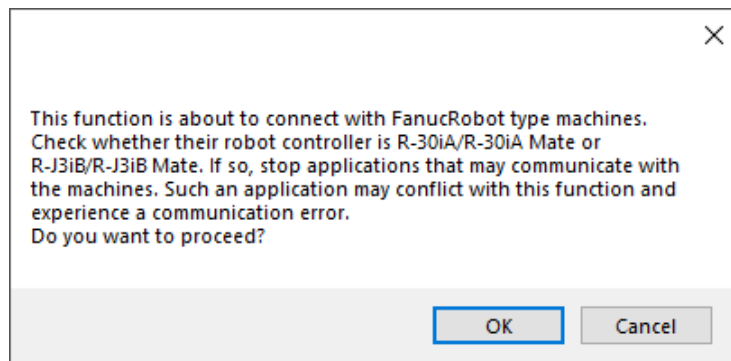
"FANUC Robot series R-J3iB CONTROLLER HANDLING TOOL OPERATOR'S MANUAL (B-81464EN-2/02)"

"FANUC Robot series R-J3iB Mate CONTROLLER LR HANDLING TOOL OPERATOR'S MANUAL (B-81524EN/02)"

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	208/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

4.4.6.1 Set EMG IO Number

When the following confirmation dialog box appears, read the note. To acknowledge it and proceed, click [OK]; otherwise, click [Cancel].

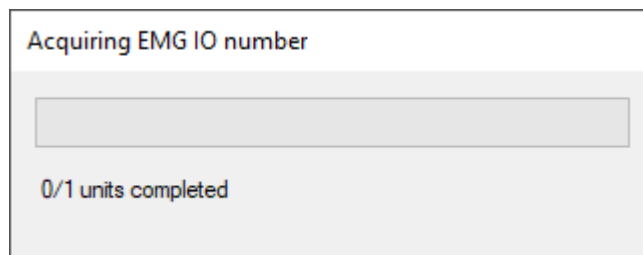


NOTE

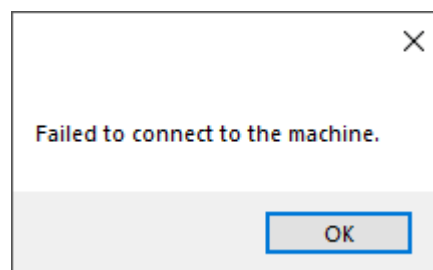
Before acquiring EMG IO numbers from the following FANUC Robot Series machines, stop applications that may communicate with those machines. If such an application is running, it may conflict with this function and experience a communication error:

- R-J3iA / R-J3iA Mate
- R-J3iB / R-J3iB Mate

While acquiring IO number, the [Acquiring EMG IO number] dialog box keeps displaying a progress bar.

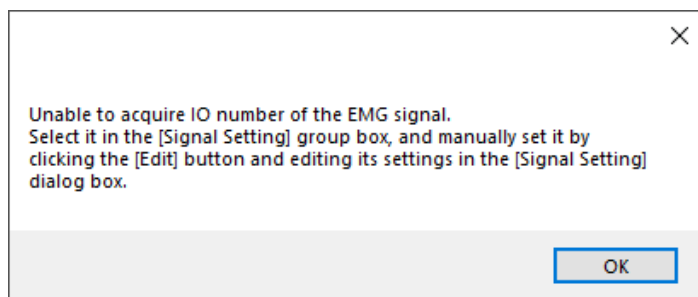


When failed to connect machines, the following error message will appears. Check whether they are connectable.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	209/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

The following message will appear if an EMG IO number is not specified or unable to be acquired.



In this case, select the row for EMG signal in the [Signal Setting] group box, click the [Edit] button, and set an EMG IO number for the [Number] field in the [Signal Setting] dialog box.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	210/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

4.4.6.2 Add Custom Signal

Click the [Add] button. The [Signal Setting] dialog box appears for you to register a new custom signal. For the details of the outlier detection settings, refer to the "[Set Outlier Detection of Signal](#)" subsection.

■ The [Signal Setting] dialog box

● Type

Select the type of the signal:

[IO]: IO signal / EMG signal

[DATA]: Registers

● The [FanucRobot] group box

Fields that can be set vary depending on signal label as follows:

<When the [Type] field is set to [EMG]>

Field	Value to set	Remarks
Label	EMG	Populated only when the [Edit] button in the [Signal Setting] group box is clicked to edit EMG signal.
Address	This field is fixed to [DO]. (For R-J3iB / R-J3iB Mate, fixed to [SDO])	-
Number	Set the EMG IO number.	Clicking the [Acquire EMG Number] button acquire an EMG IO number from the machine and automatically set one for this field. For the details, refer to the " Set EMG IO Number " subsection.
Data Type	This field is fixed to [Bool].	-
Reading Cycle	This field is fixed to [500msec].	-

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	211/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

NOTE

- 1 The EMG signal is pre-registered for retrieving the emergency stop signal set by a FANUC Robot controller.
- 2 As the EMG signal is a pre-registered signal, it is allowed to modify its settings but not to delete them.
- 3 Te [Number] field for the EMG signal must be a non-zero number. This field is set to 0 by default.
- 4 The emergency stop statuses that can be retrieved through the EMG signal are TP external emergency stop and Operation Panel emergency stop.
If assigning another signal to EMERGENCY, add a new custom signal (I/O), and change the formula of the multiple assign status in the [Equipment setting] window.

<When the [Type] field is set to [IO]>

Field	Value to set	Remarks
Label	IO	-
Address	Select the address of the IO signal. Options are as follows: [DI], [DO], [GI], [GO], [AI], [AO], [UI], [UO], [SI], [SO], [RI], [RO], [WI], [WO], [WSI]	For R-J3iB / R-J3iB Mate Series robots, read "[DI], [DO]" as "[SDI], [SDO]" and "[RI], [RO]" as "[RDI], [RDO]"
Number	Set a logical number. 1 to 5000: when the [Address] field has [DI], [DO] (or [SDI], [SDO]) 1 to 1000: when the [Address] field has [RI], [RO] (or [RDI], [RDO]), [GI], [GO], [AI], [AO] 1 to 999: when the [Address] field has [UI], [UO] 0 to 1000: when the [Address] field has [SI], [SO] 1 to 400: when the [Address] field has [WI], [WO] 1 to 600: when the [Address] field has [WSI]	-
Data Type	This field populates a data type of its basic variable in conjunction with the [Address] field. - Int32: when the [Address] field has [GI], [GO], [AI], or [AO] - Bool: when the [Address] field has another value	-
Reading Cycle	This field is fixed at 500msec.	-

NOTE

Only when [GI] / [GO] / [AI] / [AO] is selected for address, outlier detection setting can be set.
For the details of outlier detection setting, refer to the "[Set Outlier Detection of Signal](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	212/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

<When the [Type] field is set to [DATA]>

Field	Value to set	Remarks
Label	DATA	-
Address	Select the register type. [NumReg]: numeric register [CharReg]: string register	Use the [Position Register] tab to setup PosReg (position register)
Number	Set a register number. 1 to 2000: numeric register 1 to 1000: string register	-
Data Type	Select a data type for the register data. [Int32] or [Double]: when the [Address] field is set to [NumReg] [String]: when the [Address] field is set to [CharReg]	-
Reading Cycle	This field is fixed at 500msec.	-

NOTE

Only when [NumReg] (numeric register) is selected for address, outlier detection setting can be set.

For the details of outlier detection setting, refer to the "[Set Outlier Detection of Signal](#)" subsection.

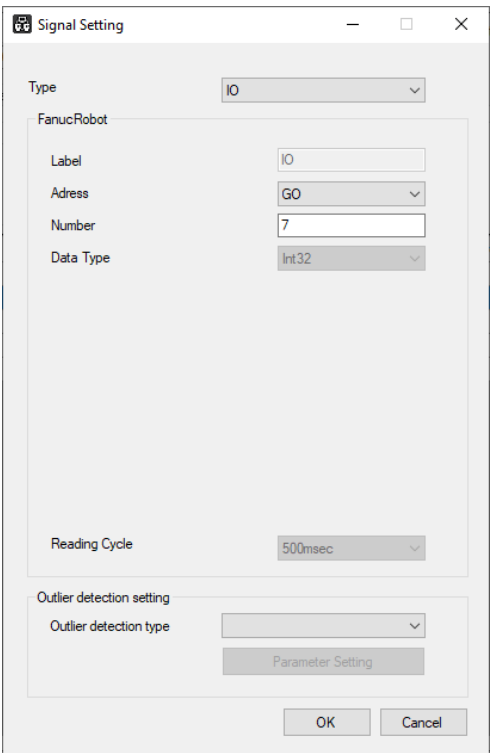
NOTE

A new custom signal can be defined provided that there exists no custom signal with exactly the same type, address, number and data type.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	213/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

4.4.6.3 Edit Custom Signal

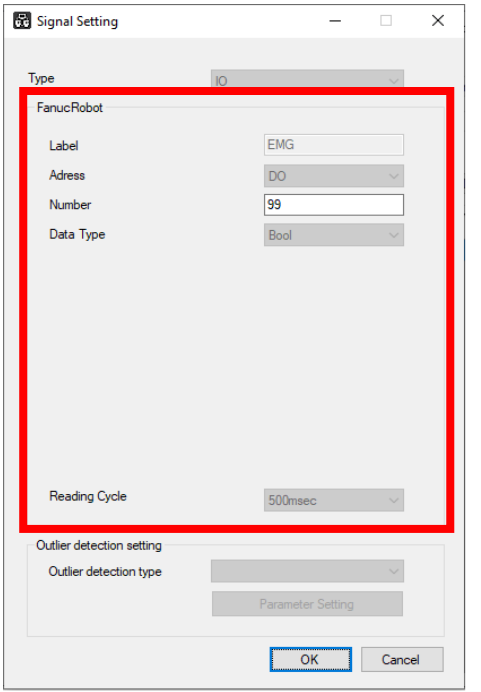
To edit a custom signal, select any signal and click [Edit]. The [Signal Setting] dialog box appears for you to edit the selected custom signal. Note that the [Type] field is not-editable.



The 'Signal Setting' dialog box is shown. It has a 'Type' dropdown set to 'IO'. Under the 'FanucRobot' section, there are fields for 'Label' (set to 'IO'), 'Address' (set to 'GO'), 'Number' (set to '7'), and 'Data Type' (set to 'Int32'). Below this is a 'Reading Cycle' dropdown set to '500msec'. At the bottom, there is an 'Outlier detection setting' section with an 'Outlier detection type' dropdown and a 'Parameter Setting' button. 'OK' and 'Cancel' buttons are at the very bottom.

The following are setting examples of custom signals.

Example 1:
To set I/O of EMG signal "DO[99]":



The 'Signal Setting' dialog box is shown with the 'Type' dropdown set to 'IO'. The 'FanucRobot' section is highlighted with a red box. The 'Label' field is set to 'EMG', 'Address' is set to 'DO', 'Number' is set to '99', and 'Data Type' is set to 'Bool'. The 'Reading Cycle' dropdown is set to '500msec'. The 'Outlier detection setting' section is visible at the bottom. 'OK' and 'Cancel' buttons are at the bottom.



No.	Signal	I/O	Reading Cycle
1	EMG	DO99	500msec

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					FANUC CORPORATION			Sheet.	214/868
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

Example 2:

To set I/O of IO signal "DO[7]":

The 'Signal Setting' dialog box is shown with the 'Type' set to 'IO'. The 'FanucRobot' section is highlighted with a red box. Inside this section, the 'Label' is 'IO', 'Address' is 'DO', 'Number' is '7', and 'Data Type' is 'Bool'. The 'Reading Cycle' is set to '500msec'. Below this section, there are 'Outlier detection setting' options, including 'Outlier detection type' and a 'Parameter Setting' button. At the bottom are 'OK' and 'Cancel' buttons.



No	Signal	I/O	Reading Cycle
1	EMG	DO99	500msec
2	IO	DO7	500msec

Example 3:

To set a numeric register "NumReg[1]" of In32 data type:

The 'Signal Setting' dialog box is shown with the 'Type' set to 'DATA'. The 'FanucRobot' section is highlighted with a red box. Inside this section, the 'Label' is 'DATA', 'Address' is 'NumReg', 'Number' is '1', and 'Data Type' is 'Int32'. The 'Reading Cycle' is set to '500msec'. Below this section, there are 'Outlier detection setting' options, including 'Outlier detection type' and a 'Parameter Setting' button. At the bottom are 'OK' and 'Cancel' buttons.



No	Signal	I/O	Reading Cycle
1	EMG	DO99	500msec
2	IO	DO7	500msec
3	DATA	NumReg1	500msec

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	215/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

4.4.7 Set Variable History Retrieval Condition Definition of FanucRobot

This is a setting to retrieve IO signal/register data by using a trigger variable as trigger, store the retrieved variable history data to the DB, and output them into a CSV file.

- For each machine, the maximum number of entries that the [Variable History Retrieval Condition Definition] list can register is 1000.

■ How to control trigger variables

Steps for controlling the trigger variables by FanucRobot type machines and MT-LINK*i* are shown below.

MT-LINK*i* supports two methods for controlling trigger variables: Change point detection method and Handshake method.

- Change point detection method

- 1 Change the value of a trigger variable to a specified value on the FanucRobot type machine side.
Example) set the trigger variable to "true" (Bool data type) or a number (Int32 data type)
- 2 Monitor the trigger variable on the MT-LINK*i* side. When it is detected to be changed to the specified value, MT-LINK*i* retrieves values of the variables as many as specified that are subsequent to the basic variable in the variable area.
- 3 MT-LINK*i* does not change the trigger variable.
- 4 MT-LINK*i* stores the values retrieved in step 2 into the database, and output them in a CSV file.

NOTE

With the change point detection method, there may occur cases where data cannot be retrieved because it has no synchronization process with a FanucRobot type machine. When use this method, make settings of MT-LINK*i* and conduct trigger control on a FanucRobot type machine taking into account the following:

When a FanucRobot type machine changes a trigger variable, it should keep it adequately longer than the reading cycle time set on MT-LINK*i*; otherwise, MT-LINK*i* may fail to detect a change and misses a chance to retrieve.

Also, MT-LINK*i* may not able to access the machine retrieve at a specified reading cycle due to the amount of MT-LINK*i* settings, network delay, or heavy workload on the machine. After making MT-LINK*i* setting, make sure MT-LINK*i* has no problem in retrieving data.

Use the [Utility - System Diagnosis] Web screen and Signal Check Tool to determine a proper reading cycle for a FanucRobot type machine.

- Handshake method

- 1 Change the value of a trigger variable to a specified value on the FanucRobot type machine side.
- 2 Monitor the trigger variable on the MT-LINK*i* side. When it is detected to be changed to the specified value, MT-LINK*i* retrieves values of the variables as many as specified that are subsequent to the basic variable in the variable area.
- 3 Change the trigger variable to "0" on the MT-LINK*i* side.
- 4 MT-LINK*i* stores the values retrieved in step 2 into the database, and output them in a CSV file.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	216/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

NOTE

When using the handshake method, which is available only when the [Address] field is set to [NumReg] (numeric register), set a FanucRobot type machine to conduct response control taking into account the following:

Set the machine to wait long enough for MT-LINK*i* to set a trigger variable to "0" after the machine set it to a specified value, taking into account time consuming cases such as when MT-LINK*i* is not running or the network is instable.

A response timeout set on the FanucRobot type machine for a trigger variable should be at least one second. The maximum value* set on MT-LINK*i* for a response timeout is the value for the connection timeout of the machine.

FanucRobot operators should set an appropriate response timeout according to their production processes or environments.

* To check for the value of the connection timeout set on MT-LINK*i*, select the FanucRobot type machine in the [Machine Setting] window > [Protocol] tab, and review the [Time out] for the machine.

 WARNING

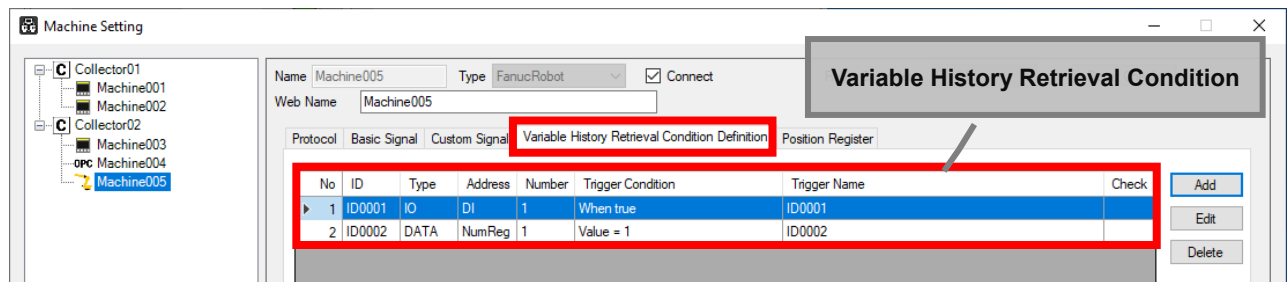
When using handshaking method, MT-LINK*i* waits for a FanucRobot type machine to turn a trigger variable to the specified value and then sets the variable to "0."

If trigger variables reside in areas that are used on the FanucRobot type machine or by other applications, make sure before setting trigger variables that no problem should occur.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	217/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

4.4.7.1 Set Variable History Retrieval Condition Definition of FanucRobot

Select a machine in the Collector tree, and select the [Variable History Retrieval Condition Definition] tab on the right pane.



■ The [Variable History Retrieval Condition Definition] tab

This tab page lists the variable history retrieval conditions that are set up in the [Variable History Retrieval Condition Setting] dialog box.

For the details of the columns in the list, refer to "The [Variable History Retrieval Condition Setting] dialog box" paragraph.

- ID
This column lists the IDs of the registered variable history retrieval conditions.
- Type
This column lists the types of trigger variables.
- Address
This column lists the addresses of trigger variables.
- Number
This column lists the numbers of trigger variables.
- Trigger Condition
This column lists the conditions for MT-LINK*i* to start retrieving data.
ON: MT-LINK*i* starts retrieving data when a trigger variable of Bool data type is set to "true."
OFF: MT-LINK*i* starts retrieving data when a trigger variable of Bool data type is set to "false."
Value = <n>: MT-LINK*i* starts retrieving data when a trigger variable of Int32 data type is set to "n" (a specified number).
- Trigger Name
This column lists the Web names of trigger variables.
- Check
The result of the Signal Check, when the [Signal Check] button is clicked, is displayed.
"OK": The trigger variable and the basic variable (including variables subsequent to it) were retrieved.
"NG" (for "not good"): Failed to retrieve either the trigger variable / the basic variable (including variables subsequent to it).
"" (blank): Signal Check unexecuted (initial state, machine connection failure, etc.), or after executing Signal Check, setting was changed on the [Variable History Retrieval Condition Setting] dialog box and the [OK] button was clicked.

For the details on Signal Check, refer to the "[Checking the Signals](#)" subsection.
- [Add] button
Click this button to add a new variable history retrieval condition.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	218/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

- [Edit] button
Click this button to edit the selected variable history retrieval condition.
- [Delete] button
After a variable history retrieval condition is selected, clicking this button displays a confirmation message for the deletion.
Clicking the [Yes] button on the message deletes the selected variable history retrieval condition.

NOTE

When setting registers in machines as trigger variables or basic variables, make sure to verify that all of those variables can be retrieved by conducting machine connection tests.

For the details on how to conduct a machine connection test, refer to the "[Check Connection with the Machine](#)" subsection.

(1) Setting variable history retrieval conditions

To open the [Variable History Retrieval Condition Setting] dialog box, click the [Add] on the [Variable History Retrieval Condition Definition] tab page or select an ID and click the [Edit] button.

No	Type	Address	Basic Variable	Data Type	Number of Retrievals	Check
1	DATA	NumReg	10	Int32	3	
2	DATA	CharReg	1	String	5	
3	DATA	PosReg	1	-	10	

■ The [Variable History Retrieval Condition Setting] dialog box

- ID
This field populates the ID for the Variable history retrieval condition ("ID" + <four digit numbering>) (non-editable).
For the <four digit numbering>, an unused minimum value starting from "0001" is set automatically. (Example: ID0002)
- Trigger Name
Set a name for the trigger variable. The name should be entered in the language specified for the [Web name language] in the [System Setting] window. Up to 100 bytes can be set. The default value is the same value as [ID].

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	219/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

<The [Trigger Variable Setting] group box>

- **Type**
Select the type of the trigger variable:
[IO]: IO signal
[DATA]: Register data
- **Address**
Select the address of the trigger variable.
 - When [IO] is selected in the [Type] drop-down list, the following options are available:
[DI], [DO], [GI], [GO], [AI], [AO], [UI], [UO], [SI], [SO], [RI], [RO], [WI], [WO], [WSI]
(For R-J3iB / R-J3iB Mate Series robots, read "[DI], [DO]" as "[SDI], [SDO]" and "[RI], [RO]" as "[RDI], [RDO]")
 - When [DATA] is selected in the [Type] drop-down list, only the following option is available:
[NumReg] (numeric register)
- **Number**
Set the variable number of the trigger variable. The default value is "0."
- **Data Type**
This field populates a data type of the basic variable in conjunction with the [Address] field.
 - [Int32]: when [Type] field is set to [IO] and the [Address] field is set to [GI], [GO], [AI], or [AO]
 - [Bool]: when [Type] field is set to [IO] and the [Address] field is set to another address
 - [Int32]: when [Type] field is set to [DATA], this is the only selectable option (fixed value)
- **Trigger Condition**
This field shows the condition for MT-LINK*i* to start retrieving data.
ON: Retrieving data starts when a trigger variable of Bool data type is set to "true."
OFF: Retrieving data starts when a trigger variable of Bool data type is set to "false."
= <n>: Retrieving data starts when a trigger variable of Int32 data type is set to "n" (a number specified in the input field next to [=]).
Use handshake: When this check box is selected, MT-LINK*i* start controlling the trigger variable with handshake method*.

* MT-LINK*i* waits until the trigger variable changes to the specified number (<n>), retrieves the basic variable and its subsequent variables as many as specified, and sets the trigger variable to "0."
- **Check**
The results of the Signal Check, when the [Signal Check] button is clicked, are displayed.
"OK": The trigger variable was retrieved.
"NG" (for "not good"): Failed to retrieve the trigger variable.
"" (blank): Signal Check unexecuted (initial state, machine connection failure, etc.), or after executing Signal Check, the setting was changed on the [Variable History Retrieval Condition Setting] dialog box and the [OK] button was clicked.
For the details on Signal Check, refer to the "[Checking the Signals](#)" subsection.

<The [Basic Variable Setting] group box>

The following columns populate the values set with the [Basic Variable Setting] dialog box:
For the details, refer to "[The \[Basic Variable Setting\] dialog box](#)" paragraph.

- **Type**
This column lists the types of basic variables.
- **Address**
This column lists the addresses of basic variables.
- **Basic Variable**
This column lists the variable numbers of basic variables.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	220/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- **Number of Retrievals**
This column lists the numbers of variables to retrieve (including basic variables).
- **Check**
The results of the Signal Check, when the [Signal Check] button is clicked, are displayed.
 "OK": All of the trigger variable / the basic variable (including variables subsequent to it) were retrieved.
 "NG" (for "not good"): Failed to retrieve either the trigger variable / the basic variable (including variables subsequent to it).
 "" (blank): Signal Check unexecuted (initial state, machine connection failure, etc.), or after executing Signal Check, the setting was changed on the [Variable History Retrieval Condition Setting] dialog box and the [OK] button was clicked.

For the details on Signal Check, refer to the "[Checking the Signals](#)" subsection.

- **[Add] button**
Click this button to add a new variable history retrieval condition.
- **[Edit] button**
Click this button to edit the selected variable history retrieval condition.
- **[Delete] button**
After a variable history retrieval condition is selected, clicking this button displays a confirmation message for the deletion.
Clicking the [Yes] button on the message deletes the selected variable history retrieval condition.
- **[Variable Name Setting] button**
Click this button to set variable names in the selected basic variable setting.
- **[OK] button**
Clicking this button saves the edited settings, and closes the [Variable History Retrieval Condition Setting] dialog box. If the basic variable is not set, an error message mentioning "Basic variable is not specified" will be displayed.
- **[Cancel] button**
Clicking this button discards the edited settings, and closes the [Variable History Retrieval Condition Setting] dialog box.

NOTE

If two variable history retrieval conditions are equivalent in that they have the same [Trigger Variable Setting] group box, one of them is not allowed to be registered if the other one has been already registered.
That is, for example, if their [Trigger Condition] fields have different values, both of them can be registered unless there exists a duplicate in registered conditions.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 221/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

(2) Setting variable history retrieval conditions

To open the [Basic Variable Setting] dialog box, click the [Add] on the [Variable History Retrieval Condition Setting] dialog box or select a basic variable setting and click the [Edit] button.

■ The [Basic Variable Setting] dialog box

• Type

Select the type of the trigger variable. By default, this field is set to [IO]:

[IO]: IO signal

[DATA]: Register data]

Fields that can be set vary depending on signal label as follows:

<When the [Type] field is set to [IO]>

Field	Value to set	Remarks
Address	Select the address of the IO signal specified as trigger variable. Options are as follows: [DI], [DO], [GI], [GO], [AI], [AO], [UI], [UO], [SI], [SO], [RI], [RO], [WI], [WO], [WSI]	For R-J3iB / R-J3iB Mate Series robots, read "[DI], [DO]" as "[SDI], [SDO]" and "[RI], [RO]" as "[RDI], [RDO]"
Number	Set the variable number of the basic variable. By default, this field is set to 0.	-
Data Type	This field populates a data type of its basic variable in conjunction with the [Address] field. - Int32: when the [Address] field has [GI], [GO], [AI], or [AO] - Bool: when the [Address] field has another value	-
Number of Retrievals	Set a number of variables to retrieve. By default, this field is set to 1.	-

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 222/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

<When the [Type] field is set to [DATA]>

Field	Value to set	Remarks
Address	Select a register type of the basic variable. [NumReg]: numeric register [CharReg]: string register [PosReg]: position register	-
Number	Set a variable number of the basic variable. By default, this field is set to 0.	-
Data Type	Select a data type of the basic variable. - [Int32] or [Double]: when the [Address] field is set to [NumReg] - [String]: when the [Address] field is set to [CharReg] - null space: when the [Address] field is set to [PosReg]	-
Action Group Number	Set an action group number of the basic variable. By default, this field is set to 1.	Available only when the [Address] field is set to [PosReg]
Number of Cartesian Coordinates	Set the number of basic variable values in Cartesian coordinate. A whole number from 6 to 9 can be set. By default, this field is set to 6.	Available only when the [Address] field is set to [PosReg]
Number of Joint Coordinates	Set the number of basic variable values in joint coordinate. A whole number from 6 to 9 can be set. By default, this field is set to 6.	Available only when the [Address] field is set to [PosReg]
Number of Retrievals	Set the number of variables to retrieve (including the basic variable). By default, this field is set to 1.	-

- [OK] button
Clicking this button saves the edited settings, and closes the [Basic Variable Setting] dialog box.
- [Cancel] button
Clicking this button discards the edited settings, and closes the [Basic Variable Setting] dialog box.

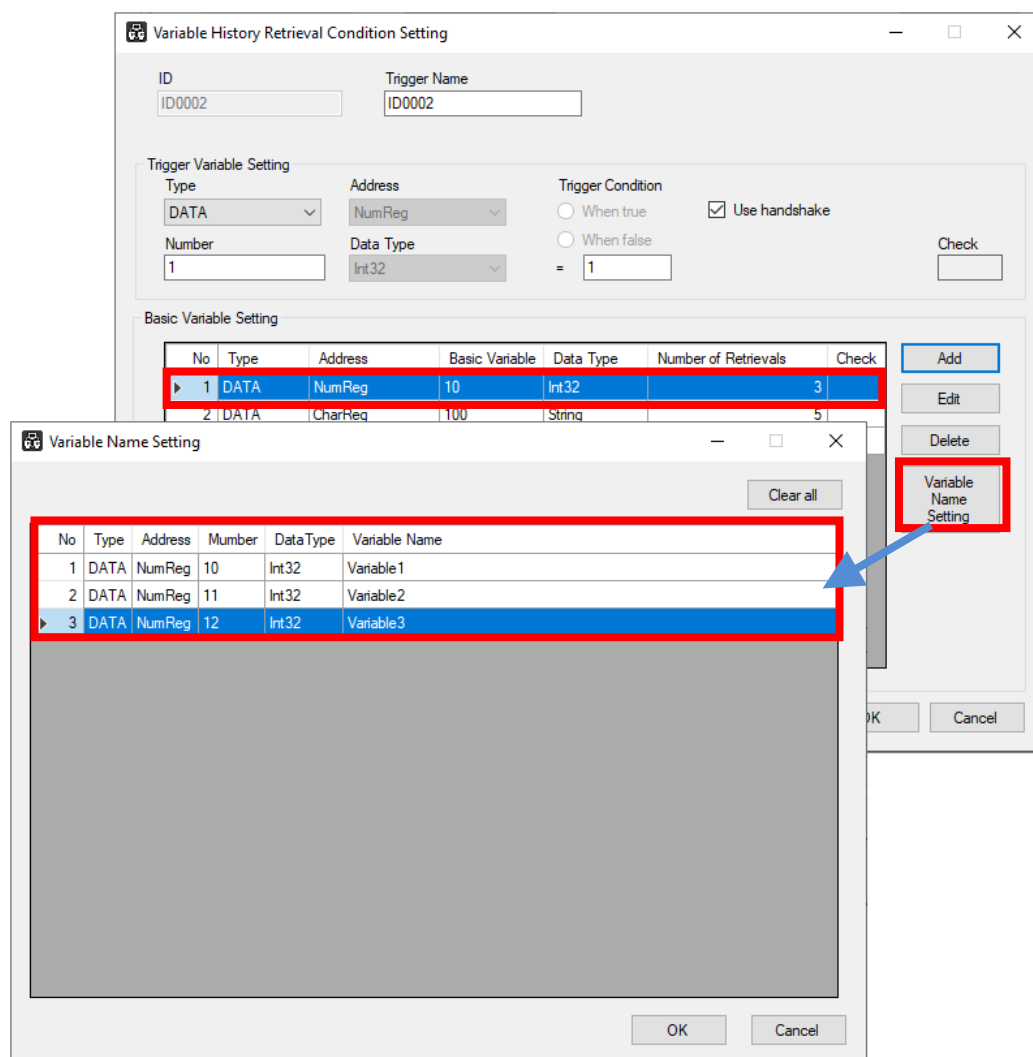
NOTE

When it is failed to retrieve a variable that is defined to be retrieved, no variable including the basic variable and variables subsequent to it will be retrieved.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	223/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

(3) Setting names of variables

To open the [Variable Name Setting] dialog box, select a basic variable setting and click the [Variable Name Setting] button on the [Variable History Retrieval Condition Setting] dialog box.



■ The [Variable Name Setting] dialog box

- Type, Address, Number, and Data Type

For the details of those columns, refer to the explanation of the corresponding fields in the [Basic Variable Setting] group box of the [Variable History Retrieval Condition Setting] dialog box.

- Variable Name

Set a name for the variable to be retrieved. The name should be entered in the language specified for the [Web name language] in the [System Setting] window. Up to 32bytes per variable can be set. Default is blank. If the variable name is blank, "ValueN" will be used as the item name for the output file.

* N is the index number (from "1") within the number of retrieval. (Example: Value2)

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	224/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

- [Clear all] button
When the [Clear all] button is clicked, confirmation message appears asking "All the variable names will be cleared. Is this OK?" If [Yes] button is clicked, all the settings in the [Variable Name] column will be cleared.
- [OK] button
Clicking this button saves the settings, and closes the [Variable Name setting] dialog box.
- [Cancel] button
Clicking this button discards the settings, and closes the [Variable Name setting] dialog box.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 225/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

4.4.7.2 On File Output When Retrieving Variable History of FanucRobot

For the details on setting file output function when retrieving variable history, refer to "[Setting File Output Function When Retrieving Variable History of FanucCNC](#)" subsection.

For the details on how to address if file output fails when retrieving variable history, refer to "[If File Output Fails When Retrieving Variable History of FanucCNC](#)" subsection.

The output file format of variable history of FanucRobot type machines is as follows:

■ Output folder (default)

<MT-LINK*i* installation folder>\VariableHistory

* For notes on the output folder, refer to NOTE in "■ Output folder (default)" paragraph in the "[Format of Output File When Retrieving Variable History of FanucCNC](#)" subsection.

■ File Name (default)

VariableHistory_<Machine name>_<ID>_<Machine type>_<Date (yyyyMMdd)>
_<Time (HHmmss)>.csv

Example) VariableHistory_Machine002_ID0002_FanucRobot_20210915_091500.csv

* For notes on an output file name, refer to NOTE in "■ File Name (default)" paragraph in the "[Format of Output File When Retrieving Variable History of FanucCNC](#)" subsection.

■ Output Layout (for FanucRobot machine type)

Item Name	Output Value	Remarks
MachineName	Machine name is output.	
<Trigger Name>	Value of the [Trigger Name] column is output.	NOTE 1
MachineType	Machine type is output. * fixed to "FanucRobot"	
UpdateDate	Date/time of variable history retrieval is output.	
VariableCount	Number of variables to retrieve is output.	
TriggerType	Type of trigger variable is output. "IO": IO signal "DATA": Register	
TriggerPath	Path of trigger variable is output. * fixed to "-" (hyphen)	
TriggerAddress	Address of trigger variable is output. - When [TriggerType] is set to "IO": [DI], [DO], [GI], [GO], [AI], [AO], [UI], [UO], [SI], [SO], [RI], [RO], [WI], [WO], [WSI] (For R-J3iB / R-J3iB Mate Series robots, read "[DI], [DO]" as "[SDI], [SDO]" and "[RI], [RO]" as "[RDI], [RDO]") - When [TriggerType] is set to "DATA": [NumReg] (numeric register)	
TriggerNumber	Variable number of the trigger variable is output.	
SignalCondition	Trigger condition is output that determines when to start retrieving the variable. "TRUE": when the trigger variable is set to "true," retrieving starts. "FALSE": when the trigger variable is set to "false," retrieving starts. <signal value>: when the trigger variable is set to <signal value>, retrieving starts.	NOTE 2

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 226/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

Item Name	Output Value	Remarks
Type<N>	Type of retrieved variable is output. "IO": IO signal "DATA": Register	NOTE 3
Path<N>	Path of retrieved variable is output. * fixed to "-" (hyphen)	NOTE 3
Address<N>	Address of trigger variable is output. - When [TriggerType] is set to "IO": [DI], [DO], [GI], [GO], [AI], [AO], [UI], [UO], [SI], [SO], [RI], [RO], [WI], [WO], [WSI] (For R-J3iB / R-J3iB Mate Series robots, read "[DI], [DO]" as "[SDI], [SDO]" and "[RI], [RO]" as "[RDI], [RDO]") - When [TriggerType] is set to "DATA": [NumReg] (numeric register) [CharReg] (string register) [PosReg] (position register)	NOTE 3
Number<N>	Variable number of retrieved variable is output.	NOTE 3
Bit<N>	Bit number of retrieved variable is output. * fixed to "-" (hyphen)	NOTE 3
<Variable Name> or Value<N>	Value of retrieved variable is output. * For NumReg, individual elements (Cartesian Coordinates, Joint Coordinates, Config, UF / UT) composing position register data each are output as a single variable.	NOTE 3, 4

NOTE

- 1 [Trigger Name] set in the variable history retrieval condition setting is output as the item name.
- 2 Whether to output the value of this item can be controlled by editing the system file (receiver.ini). For the details, refer to the "**SYSTEM SETTING FILES**" section.
- 3 "**N**" is the index number (from 1) within the retrieved variables.
If multiple numbers are set for retrieving variables, "Type<N>" to "<Variable Name> or Value<N>" will be output for the number of variables to retrieve.
- 4 [Variable Name] set in the variable name setting is output as the item name. If the [Variable Name] is not set (blank), "Value<N>" will be output.

■ Character Code

UTF-8 (with BOM)

■ Output Sample

Machine name: Machine002

Trigger name: RetrievalSetting2

ID: ID0002

Machine type: FanucRobot

Retrieved date/time: from 2021/09/15 09:15:00

Trigger variable:

Type: IO, Address: DO, Number: 100, Trigger condition: true

Basic variable 1:

Type: IO, Address: DO, Number: 5, Number of variables to retrieve: 2, Variable name 1: Velocity, Variable name 2: Temperature

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 227/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

Type: DATA, Address: PosReg, Number: 1, Action group number: 1, Number of Cartesian Coordinates: 6, Number of Joint Coordinates: 6

K	L	M	N	O	P	Q	R	S	T	U	V
Type1	Path1	Address1	Number1	Bit1	Value1	Type2	Path2	Address2	Number2	Bit2	Value2
IO	-	DO	100		FALSE	IO	-	DO	101		FALSE

W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	A
Type3	Path3	Address3	Number3	Bit3	Value3	Type4	Path4	Address4	Number4	Bit4	Value4	Type5
DATA	-	NumReg	1		0	DATA	-	NumReg	2		0	DATA

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					FANUC CORPORATION			Sheet.	228/868
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

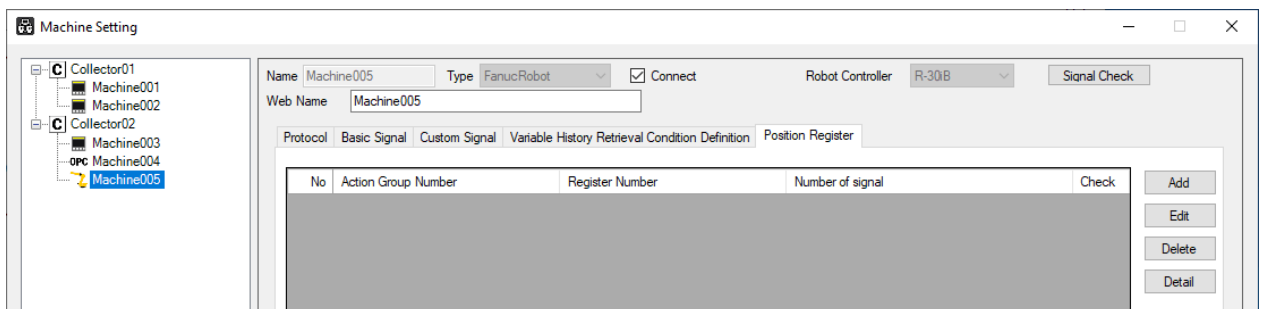
4.4.8 Set Position Register of FanucRobot

This function is provided to browse signals (data) included in position registers (see NOTE) of FanucRobot type machines in MT-LINK*i*.

Select a machine in the Collector tree and click the [Position Register] tab in the right pane.

NOTE

- 1 Only PosReg (position register) can be setup with the [Position Register] tab page. To setup NumReg (numeric register) and CharReg (string register), use the [Custom Signal] tab page. For the details, refer to the "[Set Custom Signal of FanucRobot](#)" subsection.
- 2 When setting position registers in machines as signals, make sure to verify that all of those register signals can be retrieved by conducting machine connection tests. For the details on how to conduct a machine connection test, refer to the "[Check Connection with the Machine](#)" subsection.



■ The [Position Register] tab

This tab page lists position register settings that are set up with the [Position Register Setting] dialog box.

For the following fields, refer to "[The \[Position Register Setting\] dialog box](#)" paragraph.

- Action Group Number
This column lists the action group numbers of registers.
- Register Number
This column lists the register numbers of registers.
- Number of Signals
This column lists the numbers of signals.
- Check
The results of the Signal Check, when the [Signal Check] button is clicked, are displayed.
 "OK": The signal was retrieved.
 "NG" (for "not good"): Failed to retrieve the signal.
 "" (blank): Signal Check unexecuted (initial state, machine connection failure, etc.), or after executing Signal Check, setting was changed on the [Position Register Setting] dialog box and the [OK] button was clicked.

For the details on Signal Check, refer to the "[Checking the Signals](#)" subsection.

- The [Add] button
Clicking this button opens the [Position Register Setting] dialog box to register new position register signal.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	229/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- The [Edit] button
Clicking this button opens the [Position Register Setting] dialog box to edit selected position register signal.
- The [Delete] button
Clicking this button deletes a selected position register signal.

NOTE

When a position register signal is referenced by equipment, deleting it will be denied with an error message. For the details, refer to the "**Referenced Error**" subsection.

- The [Detail] button
Clicking this button opens the [Position Register Signal Details] dialog box for the user to confirm the details of position register signals.

(1) Adding position register signals

Clicking the [Add] button opens the [Position Register Setting] dialog box to add new position register signals.

■ The [Position Register Setting] dialog box

- Action Group Number
Set an action group number for the position register to add.
A whole number from 1 to 8 can be set. By default, this field is set to 1
- Register Number
Set the register number of the position register.
A whole number from 1 to 100 can be set. By default, this field is set to 1
- Number of Cartesian Coordinates
Set the number of variable values in Cartesian coordinate to retrieve.
A whole number from 6 to 9 can be set. By default, this field is set to 6.
- Number of Joint Coordinates
Set the number of variable values in joint coordinate to retrieve.
A whole number from 6 to 9 can be set. By default, this field is set to 6.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	230/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

After setting those fields, click [OK] on the [Position Register Setting] dialog box. The list in the [Position Register] tab page adds the new position register setting.

(2) Confirming position register signals in detail

Clicking the [Detail] button on the [Position Register] tab page opens the [Position Register Signal Details] dialog box listing confirming position register signals with detailed information.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 231/868

■ The [Position Register Signal Details] dialog box

● Signal

This column lists names of signals in the following format:

Format: DATA_G<g>_PosReg<n>_<s>

<g>: Action group number (1 to 8).

<n>: Register number of the position register (1 to 100).

<s>: A signal suffix that indicates both the signal data to retrieve and its index in data array as shown in the following table.

Array index	Cartesian coordinates	Joint coordinates	Config*	Coordinate system number*
0	X	J1	conf1	UF
1	Y	J2	conf2	UT
2	Z	J3	conf3	-
3	W	J4	conf4	-
4	P	J5	TA1	-
5	R	J6	TA2	-
6	E1	JE1	TA3	-
7	E2	JE2	-	-
8	E3	JE3	-	-

* For Config and coordinate system number, the [Number of Cartesian Coordinates] and [Number of Joint Coordinates] fields do not affect how many signals to be retrieved.

Examples) when the [Number of Cartesian Coordinates] is set to "6"

the [Signal] column lists signals that have suffixes of "X" to "R" (intersections of the Cartesian coordinates column and array index rows of 0 to 5 in the table).

when the [Number of Joint Coordinates] is set to "9"

the [Signal] column lists signals that have suffixes of "J1" to "JE3" (intersections of the Joint coordinates column and array index rows of 0 to 8 in the table).

● Outlier Detection

For signals with [Outlier Detection] check box selected (signals of data type Int32 or Double), outlier detection of the signal can be set by double-clicking that row. The check box column cannot be edited.

For the details of the outlier detection settings, refer to the "[Set Outlier Detection of Signal](#)" subsection.

● Reading Cycle

This column lists reading intervals per signal. This column is fixed to "500msec."

● Check

This column displays the results of Signal Check as follows:

"OK": The signal was retrieved

"NG" (for "not good"): Failed to retrieve the signal

"" (blank): Signal Check unexecuted (initial state, machine connection failure, etc.)

For the details on Signal Check, refer to the "[Checking the Signals](#)" subsection.

(3) Editing position register signal

Clicking the [Edit] button after selecting a row (i.e., a position register setting) in the [Position Register] tab page opens the [Position Register Setting] dialog box (see NOTE 1).

The [Action Group Number], [Register Number], [Number of Cartesian Coordinates], and the [Number of Joint Coordinates] fields can be edited (see NOTE 2).

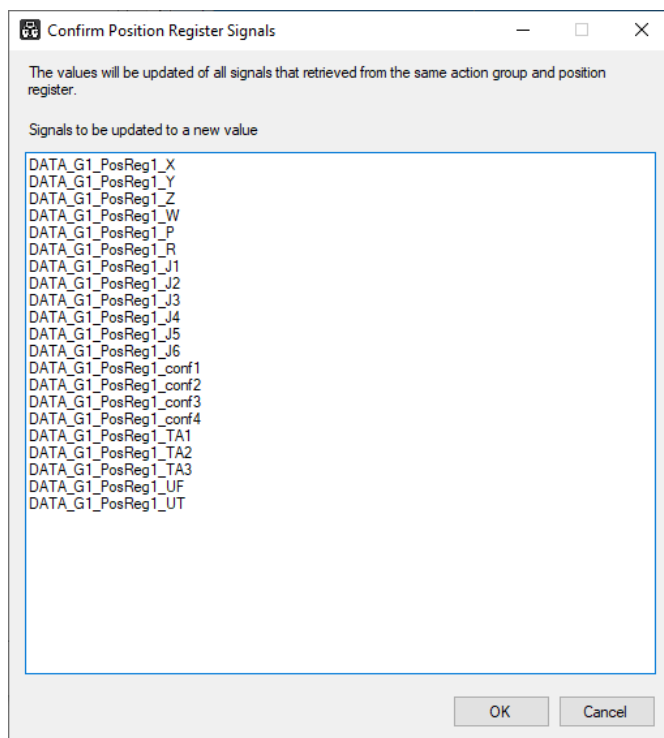
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	232/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

NOTE

- 1 When the selected signal is referenced by equipment, editing it will be denied with an error message. For the details, refer to the "[Referenced Error](#)" subsection.
- 2 If the same combination of the edited [Action Group Number] and [Register Number] fields exists in registered settings, editing the selected signal will be denied with an error message.

Clicking [OK] on the [Position Register Setting] dialog box opens the [Confirm Position Register Signals] dialog box for confirming which signals are about to be edited.

All the signals will be updated that are included in the same position register that can be identified by action group number and register number.



Clicking [OK] on the [Position Register Signal Details] dialog box updates the selected position register setting.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	233/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

To confirm the results of update, click the [Detail] button on the [Position Register] tab page. Then, the [Position Register Signal Details] dialog box appears listing position register settings with detailed information.

No	Signal	Outlier Detection	Reading Cycle	Check
1	DATA_G1_PosReg1_X	☒	500msec	
2	DATA_G1_PosReg1_Y	☒	500msec	
3	DATA_G1_PosReg1_Z	☒	500msec	
4	DATA_G1_PosReg1_W	☒	500msec	
5	DATA_G1_PosReg1_P	☒	500msec	
6	DATA_G1_PosReg1_R	☒	500msec	
7	DATA_G1_PosReg1_E1	☒	500msec	
8	DATA_G1_PosReg1_E2	☒	500msec	
9	DATA_G1_PosReg1_E3	☒	500msec	
10	DATA_G1_PosReg1_J1	☒	500msec	
11	DATA_G1_PosReg1_J2	☒	500msec	
12	DATA_G1_PosReg1_J3	☒	500msec	
13	DATA_G1_PosReg1_J4	☒	500msec	
14	DATA_G1_PosReg1_J5	☒	500msec	
15	DATA_G1_PosReg1_J6	☒	500msec	
16	DATA_G1_PosReg1_conf1	☐	500msec	
17	DATA_G1_PosReg1_conf2	☐	500msec	
18	DATA_G1_PosReg1_conf3	☐	500msec	
19	DATA_G1_PosReg1_conf4	☐	500msec	
20	DATA_G1_PosReg1_TA1	☐	500msec	
21	DATA_G1_PosReg1_TA2	☐	500msec	
22	DATA_G1_PosReg1_TA3	☐	500msec	
23	DATA_G1_PosReg1_UF	☐	500msec	
24	DATA_G1_PosReg1_UT	☐	500msec	

OK Cancel

(4) Deleting position register signal

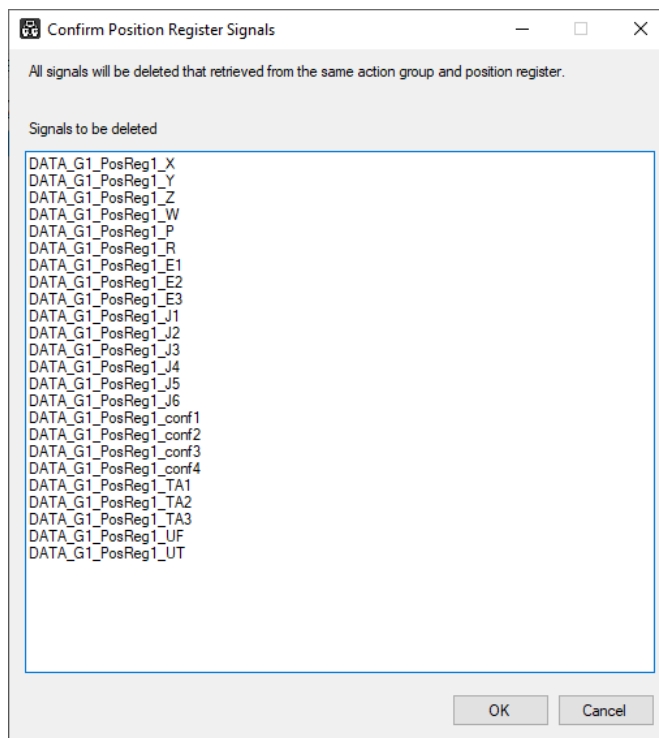
Clicking the [Delete] button after selecting a row in the [Position Register] tab page opens the [Confirm Position Register Signals] dialog box (see NOTE).

NOTE

When the selected signal is referenced by equipment, deleting it will be denied with an error message. For the details, refer to the "[Referenced Error](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	234/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

As this dialog box lists signals to be deleted, confirm your selection before deleting them.
All the signals will be deleted that are included in the same position register that can be identified by action group number and register number.



Clicking [OK] deletes the selected position register setting from the [Position Register] tab page.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION		Sheet. 235/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

4.5 MACHINE SETTING - MTConnectAgent

This section explains the machine setting of the machine type MTConnectAgent.

Unless otherwise stated, the use of MTConnect-related terms in this manual complies with MTConnect® Standard Version 1.3.0 and MTConnect® Standard Version 1.4.0.

For the details, refer to "MTConnect® Standard" documents.

* Condition signal of MTConnect Agent

The capital "C" is used for the first character in the name of the "Condition" signal in order to differentiate the signal from "condition" defined in [Multiple assign status area] of the [Equipment Setting] window:

Condition: "Condition" signal of MTConnect Agent

condition: value of "condition" status defined in [Multiple assign status area]

4.5.1 Steps for the Machine Setting of MTConnectAgent

Follow the steps below to set the MTConnectAgent.

- (1) Select [Machine Setting] in the AdminTool Menu. This displays the [Machine Setting] window.
- (2) Add Collector. For the details, refer to the "[Add Collector](#)" subsection.
- (3) Add machine type MTConnectAgent. For the details, refer to the "[Add MTConnectAgent](#)" subsection.
- (4) Select a machine. For the details, refer to the "[Select a Device Connected to the MTConnectAgent](#)" subsection.
- (5) Make protocol and connection setting for the machine. For the details, refer to the "[Confirm/Edit Setting of MTConnectAgent](#)" subsection and the "[Set Protocol Setting of MTConnectAgent](#)" subsection.
- (6) Perform Signal Setting for the machine. For the details, refer to the "[Set Signal of MTConnectAgent](#)" subsection.
- (7) Set Condition signal to be output as alarm information. For the details, refer to the "[Set Condition Signal of MTConnectAgent](#)" subsection.
- (8) Save all the settings to make the system reflect them. For the details, refer to the "[Save Machine Setting](#)" subsection.

For the details of MTConnect, refer to the "[MTConnect CLIENT FUNCTIONALITY](#)" section.

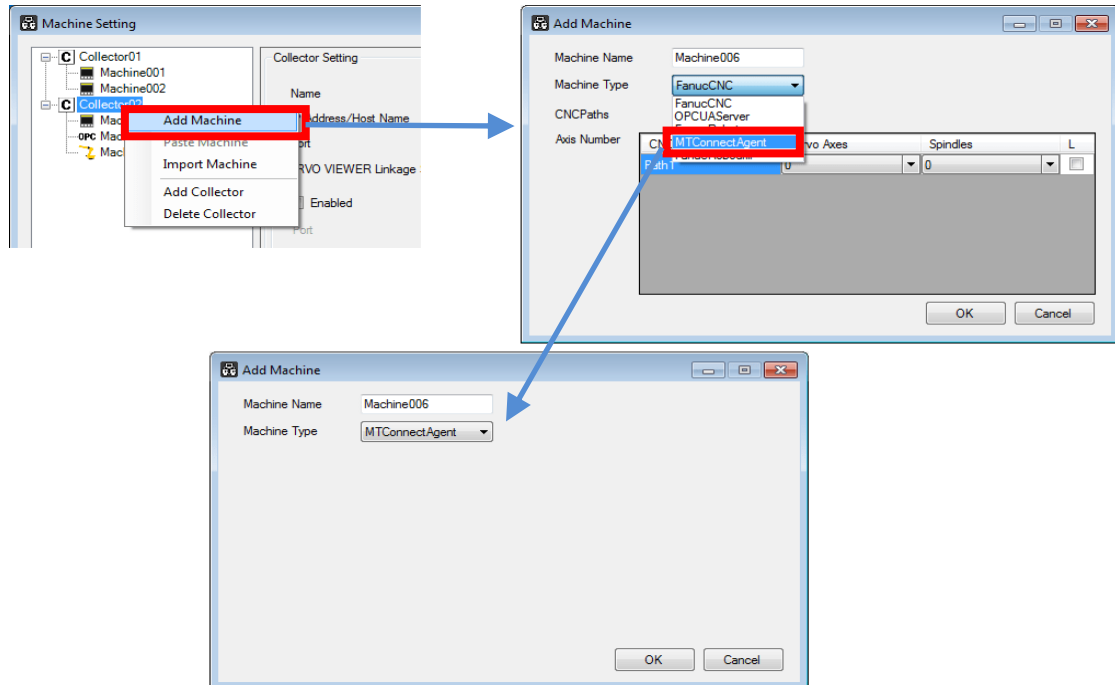
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	236/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

4.5.2 Add MTConnectAgent

(1) Adding a machine

In the Collector tree, right-click the target Collector under which you add a machine. A menu appears. Click [Add Machine].

- * Before adding a machine, make sure that you have registered the target Collector.
When the [Add Machine] dialog box appears, set the settings of the target machine.



■ The [Add Machine] dialog box

- Machine Name
Set a machine name to this field. The <Machine Name> should start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and underscores ("_"). This field can be up to 32 bytes long.
Can be set again in the [Machine Selection] dialog box which is the next window displayed.
- Machine Type
Select [MTConnectAgent].
- [OK] button
Clicking this button opens the [Machine Selection] dialog box.
- [Cancel] button
Clicking this button discards the settings being made and closes the dialog box.

NOTE

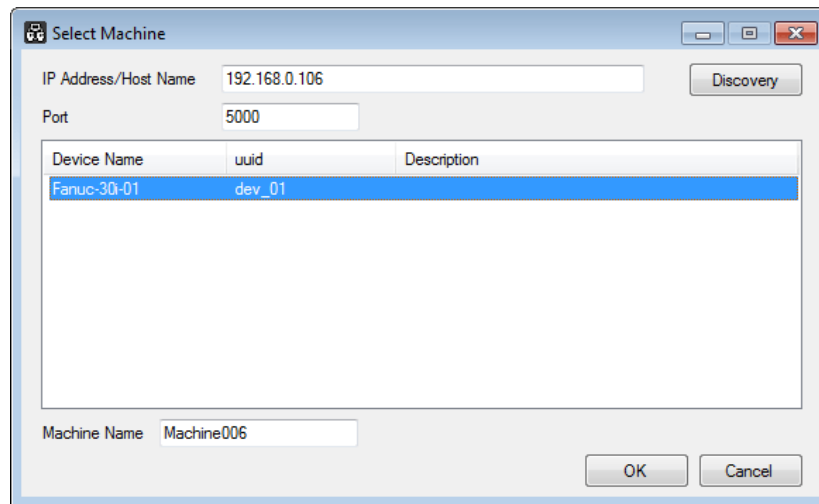
[Machine Type] is non-editable (cannot be edited afterwards).
If the setting of the [Machine Type] is mistakenly set, you need to delete the machine already set. Add a new machine and make the settings for it.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	237/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

4.5.3 Select a Device Connected to the MTConnectAgent

(1) Selecting a machine

In the [Machine Selection] dialog box, search for MTConnect Agent, and select a Device in the [Device Name] text field. MT-LINK*i* recognizes the assigned Device as a machine. After you define Device information as the signal, the Collector collects the data and you can browse them on the Web screens.



■ The [Machine Selection] dialog box

- IP Address/Host Name
Set the IP address or the host name specifying MTConnect Agent, which is the connection target. When entering the host name, the name can start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and periods ("."). This field can be up to 253 bytes long.
- Port
Set the port number to connect to MTConnect Agent. Values from 0 to 65535 can be used. The default value is set as "5000."
- Device Name
Name of the Device is displayed.
- Uuid
The ID for identifying the Device is displayed.
- Description
The description of the Device is displayed. (May not be displayed depending on the Device.)
- Machine Name
Set the machine name. The name should start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and underscores ("_"). This field can be up to 32 bytes long.
- [Discovery] button
Search for MTConnect Agent currently in operation, and the list of Devices under it are displayed. If there are no devices, a message saying "Unable to find device information." is displayed at the bottom part of the [Machine Selection] dialog box.
- [OK] button
Clicking this button confirms the selected Device, closes the [Machine Selection] dialog box, and displays the [Machine Setting] window.
- [Cancel] button
Clicking this button discards the settings being made and closes the [Machine Selection] dialog box.

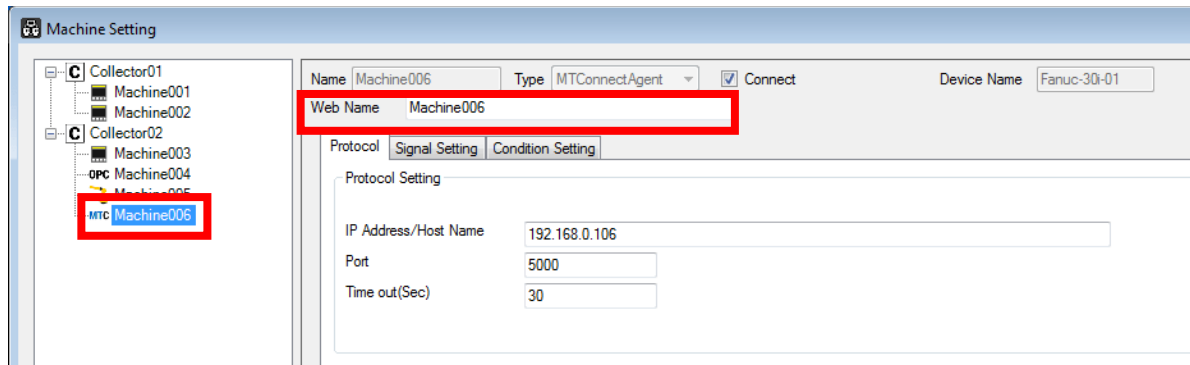
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	238/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

4.5.4 Confirm/Edit Setting of MTConnectAgent

(1) Confirm and edit machine setting

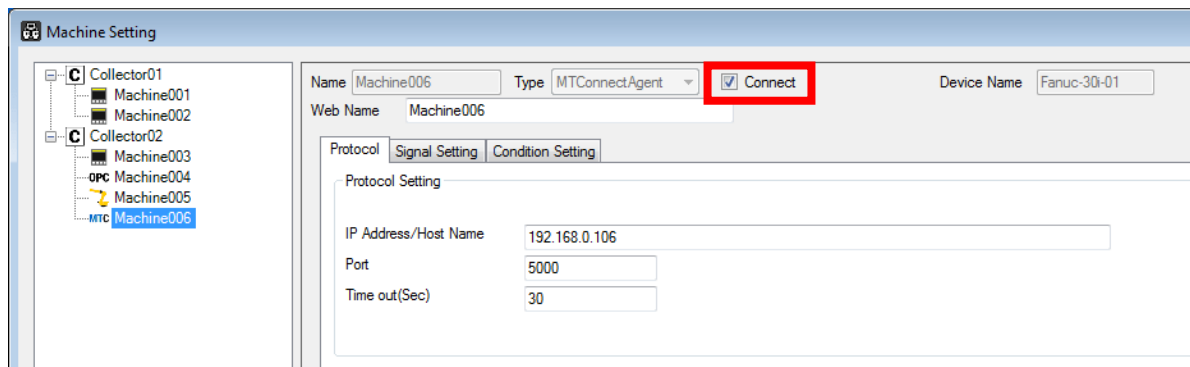
The settings of the selected Collector or machine are populated in the right pane of the [Machine Setting] window. Selecting a machine name in the Collector tree will show machine setting, so that you can set and change them.

The [Web Name] is displayed on the Web screens. The value of this field can be up to 100 bytes long and should be entered in the language specified for the [Web name language] field in the [System Setting] window. The [Web Name] field is required and will be set to the default value of <Machine name> if it is not modified.



(2) Set Machine Connection

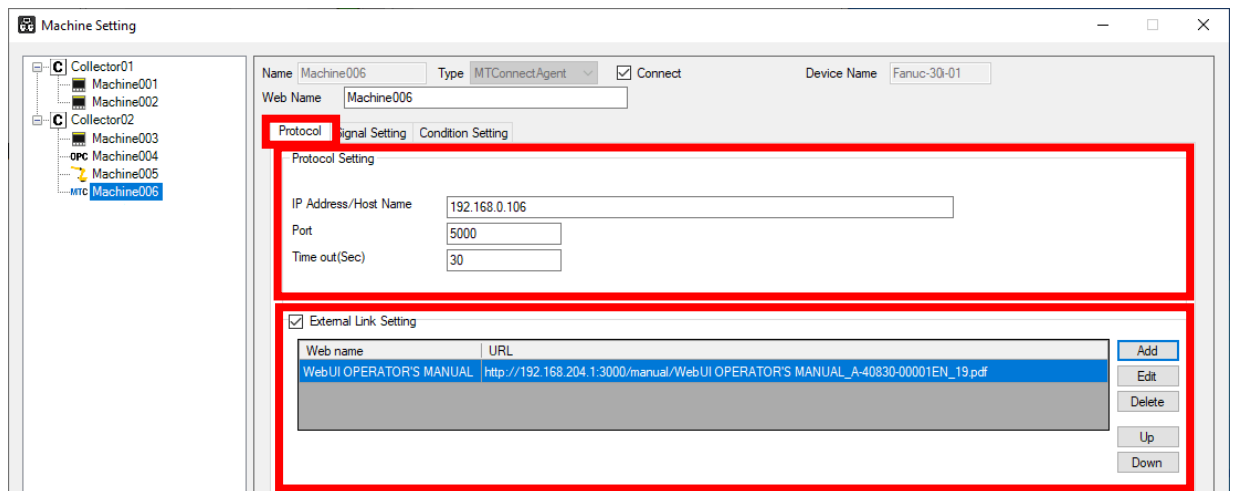
Set connection between machine and Collector. Select a machine in the Collector tree and select the [Connect] check box, so that the connection is available.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	239/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

4.5.5 Set Protocol Setting of MTConnectAgent

Select a machine in the Collector tree and select the [Protocol] tab in the right pane.



■ The [Protocol] tab

Protocol Setting

- IP Address/Host Name
The IP address or host name specified in the [Machine Setting] window is displayed. (Editable)
The first character of the host name should be a half-width alphabet character and subsequent characters can be alphanumeric characters, "-" (hyphen), and "." (dot). This field can be up to 253 bytes long.
- Port
The port number entered in the [Machine Setting] window is displayed. (Editable)
A value in a range of 0 to 65535 can be used.
- Time out
Enter the the connection timeout length in second. A value in a range of 10 to 60 can be used. The default value is 30 seconds.

External Link Setting

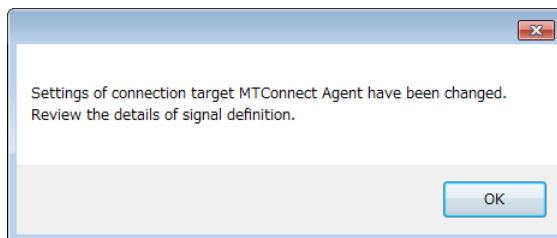
Set fields for external links to view them through Web screens. Those links can be selected from the [Monitoring - Equipment Monitoring] screen (displayed by selecting an equipment icon on the [Monitoring - Overlook] screen).

External Link function provides a way to view external links, such as Web sites, that are associated with the selected equipment (machines).

For the details, refer to the "[Set External Links](#)" subsection.

After changing the IP address / host name, following message is displayed by clicking the [Setting] button of the [Signal Setting] tab (or the [Apply] button, then the [OK] button on the [Machine Setting] window).

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	240/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					



NOTE

When the IP address / host name or port is changed, the following environment must be kept unchanged before and after the change:

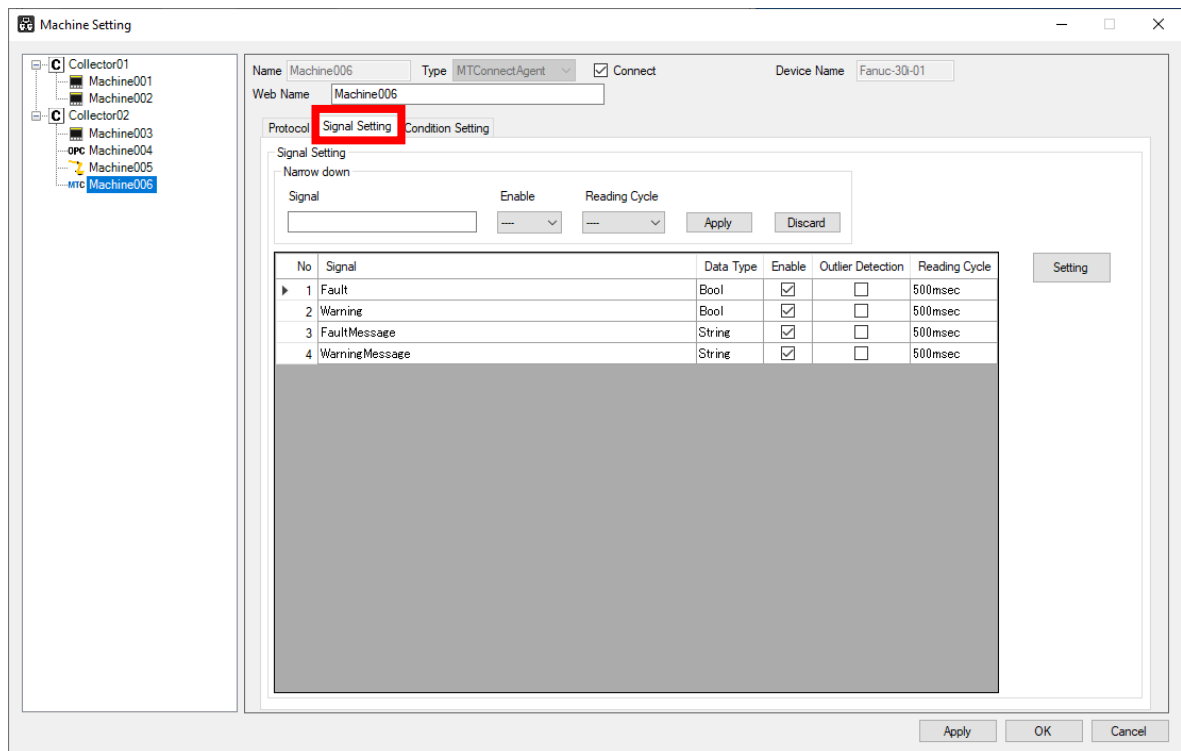
- Connection target MTConnectAgent software
- Device information containing Dataltem, Component, etc.

If the environment shown above should not be identical through the change, delete the machine that has its IP address / host name changed, add a new machine, and make the settings for it; otherwise, the signal collection may not be executed correctly.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 241/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

4.5.6 Set Signal of MTConnectAgent

Select a machine in the Collector tree and select the [Signal Setting] tab in the right pane. Basic signals and the signals set in the [Signal Setting] dialog box are displayed in a list. To list only specific signals, narrow down the signal list. For the details on how to narrow down the list, refer to the "[Narrow down Signal List](#)" subsection.



■ The [Signal Setting] Tab

- [Setting] button
Clicking this button opens the [Signal Setting] dialog box for setting machine signals to be manipulated by MT-LINK*i*.
- Signal
Signal names of the basic signals and the signals set in the [Signal Setting] dialog box are displayed. (Not-editable)
For the basic signals, refer to the "[LIST OF MTConnectAgent SIGNALS](#)" subsection.
- Data Type
Data types of the basic signals and the signals set in the [Signal Setting] dialog box are displayed. (Not-editable)
- Enable
This check box indicates whether Collector collects the corresponding basic signal or signal set in the [Signal Setting] dialog box; if it is selected, Collector collects the signal; otherwise, it does not collect.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	242/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

NOTE

- 1 The [Enable] check boxes for the mandatory basic signals (Fault, FaultMessage, and WarningMessage) cannot be cleared.
- 2 When a signal is referenced by equipment, clearing the [Enable] check box for it will be denied with an error message. For the details, refer to the "[Referenced Error](#)" subsection.

- **Outlier Detection**
Outlier detection setting of the basic signals and the signals set in the [Signal Setting] dialog box is displayed. (Not-editable)
- **Reading Cycle**
Reading cycle of the basic signals and the signals set in the [Signal Setting] dialog box is displayed. (Not-editable)

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION		Sheet. 243/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

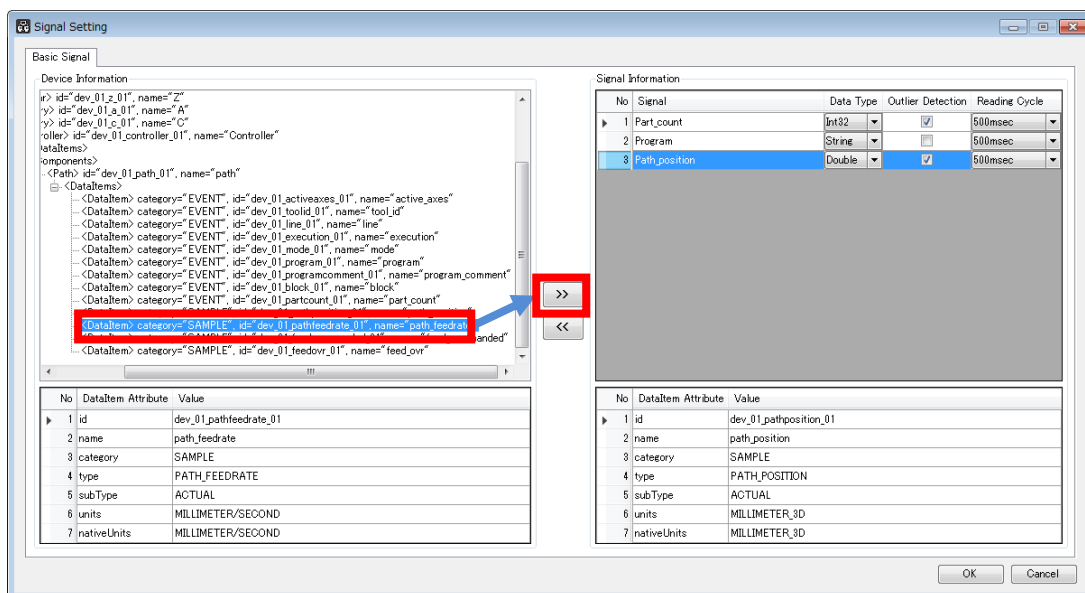
4.5.6.1 Add Signal

Clicking the [Setting] button on the [Signal Setting] tab displays the [Signal Setting] dialog box.

On the [Signal Setting] dialog box, signal of the machine is set as signal of MT-LINK*i*.

After selecting the DataItem, which will be set as signal of MT-LINK*i*, click the [>>] button to set the DataItem as signal.

Up to 1000 signals can be registered.



■ The [Signal Setting] dialog box

● Device Information

Device information (XML document) retrieved from MTConnect Agent is displayed in a tree view.

● [>>] (Add) button

Clicking this button displays the [Signal Name Entry] dialog box, which adds the DataItem selected in the Device information tree as a signal.

NOTE

Among the tags displayed in the Device information tree, DataItem tags are the only tags that can be added as signals.

However, DataItem of category="CONDITION" cannot be added as signals.

● Signal Information

Signals added from Device information tree are displayed in a list.

● [<<] (Delete) button

Clicking this button displays confirmation message for deleting the signal selected on the signal information list. When [Yes] is clicked on the message, the signal will be deleted.

NOTE

When a signal is referenced by equipment, deleting it will be denied with an error message. For the details, refer to the "[Referenced Error](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 244/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- The signal name set in the [Signal Name Entry] dialog box is displayed. (Not-editable)

- Data type set on the [Signal Name Entry] dialog box is displayed. (Editable)

For signals with [Outlier Detection] check box selected (signals of data type Int32, UInt32, or Double), outlier detection of the signal can be set by double-clicking that row. The check box column cannot be edited.

With this field, you can specify a reading interval per signal. Options are [500msec], [10sec], and [60sec].

The attribute information of the DataItem selected in the Device information tree is displayed.

- The attribute information of the DataItem selected in the Signal information list is displayed.

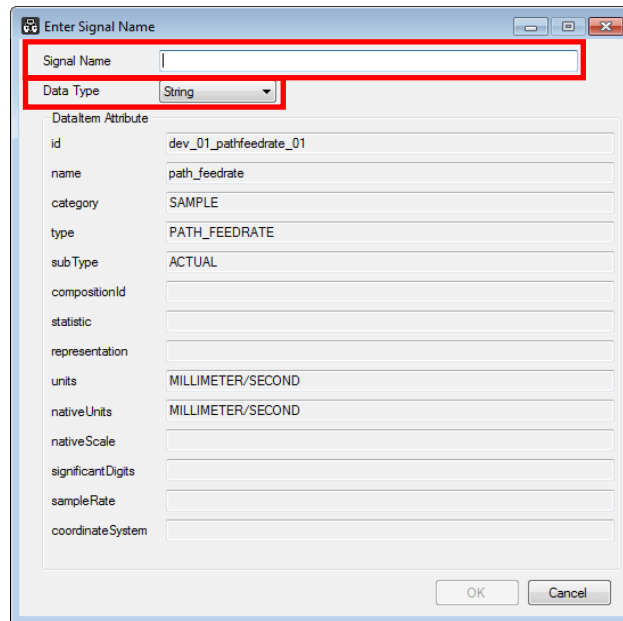
- Clicking this button confirms the settings, and closes the [Signal Setting] dialog box.

- Clicking this button discards the settings made, and closes the [Signal Setting] dialog box.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	245/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

4.5.6.2 Enter Signal Name

Clicking the [>>] button on the [Signal Setting] dialog box displays the [Signal Name Entry] dialog box. The [Signal Name] and the [Data Type] are set for the DataItem selected in the [Signal Setting] dialog box, and the signal is added.



■ The [Signal Name Entry] dialog box

- Signal Name

Set the signal name used in MT-LINK*i*. The signal name should start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and underscores ("_"). This field can be up to 64 bytes long.

NOTE

The following names are not allowed to specify for [Signal name]:

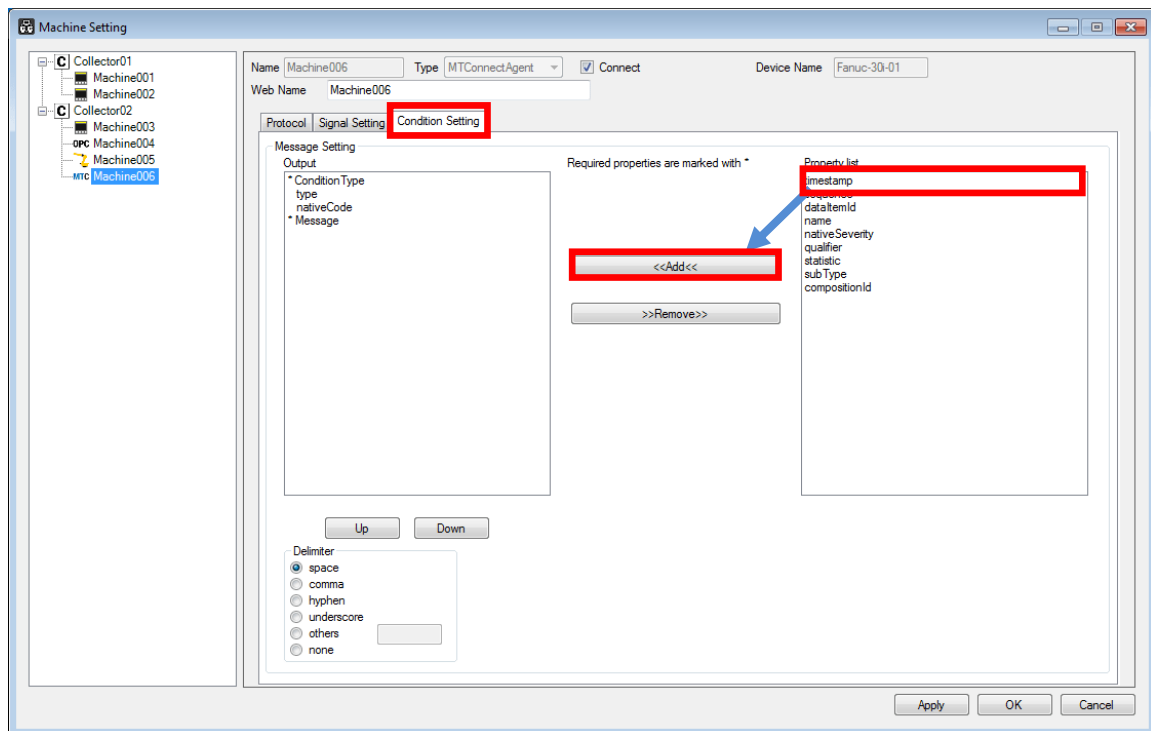
- MainProgram
- ActProgram
- Alarm
- OpMacroMsg
- CncState
- CncWarning
- RobotState
- Disconnect
- SignalLog
- VariableHistory
- AssetInfo
- WaveData
- Fault
- Warning
- FaultMessage
- WarningMessage
- AxisInfo

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	246/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

4.5.7 Set Condition Signal of MTConnectAgent

This subsection explains how to make settings to output CONDITION properties of Condition signal as an alarm message, which properties are collected from MTConnect.

- (1) Select a machine in the Collector tree and select the [Condition Setting] tab in the right pane.



■ The [Condition Setting] tab

The [Message Setting] group box

- Output

Lists properties to be output as alarm messages. Those properties can be added by selecting from the [Property list].

The properties listed in this field are output in order from top to bottom. Properties marked with an asterisk ("*") are non-removable.

- * Condition Type (required)
- type
- nativeCode
- * Message (required)

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	248/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- **Property list**
Lists property options for output as alarm messages. Those properties are as follows:
 - timestamp
 - sequence
 - type
 - dataItemId
 - name
 - nativeCode
 - nativeSeverity
 - qualifier
 - statistic
 - subType
 - compositionId
- **[<<Add<<] button**
Adds selected properties from the [Property list] to the [Output] list. This button allows multi-selection.
- **[>>Remove>>] button**
Restores selected properties from the [Output] list to the [Property list]. This button allows multi-selection. Properties marked with an asterisk ("*") are non-removable.
- **[Up] button**
Moves a selected property up by one row in the [Output] list.
This function does not work with Condition Type property.
- **[Down] button**
Moves a selected property down by one row in the [Output] list.
This function does not work with Condition Type property.
- **Delimiter**
Select a delimiter for outputs from the following options:
 - space
 - comma
 - hyphen
 - underscore
 - others
 - none

When selecting [others], enter a one-byte character as a delimiter (blank is invalid). When [none] is specified, no delimiter will be applied.

- (2) Select properties from the [Property list], and click the [<<Add<<] button to add the selected properties to the [Output] list. This button allows multi-selection.
- (3) To remove properties from the [Output] list, select properties to remove from the [Output] list and click the [>>Remove>>] button. This button allows multi-selection.
- (4) To change the order in which properties in the [Output] list are output, select a property from the [Output] list to change its output order, and click the [Up] or [Down] button.
- (5) Specify the delimiter for the properties in the [Output] list.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	249/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

- How Web screens (the [Monitoring - Alarm Monitoring] and [Diagnosis - Alarm History] screens) display
- Below is an example of the [Monitoring - Alarm Monitoring] screen displaying alarm information.

Monitoring - Alarm Monitoring						Filter
Date and time of occurrence	Equipment name	Machine name	Alarm type	Alarm number	Alarm messages	
2019/03/22 16:26:31	NewEquipment0001	NewMachine001	EX	1000	PATH01 ALARM	Alarm Information from MTConnectoAgent
2019/03/22 16:26:31	NewEquipment0001	NewMachine001	EX	1001	PATH01 ALARM	
2019/03/22 16:26:31	NewEquipment0001	NewMachine001	OPR	2000	OPE MESSAGE	
2019/03/22 16:26:31	NewEquipment0001	NewMachine001	OPR	2001	OPE MESSAGE 2	
2019/03/22 16:26:31	NewEquipment0002	NewMachine002	EX	1000	Fault 2019-03-22T07:14:46.507Z 268 SYSTEM nsystem1 n.system-1 EX1000 ALARM TEST	
2019/03/22 16:26:31	NewEquipment0002	NewMachine002	EX	1001	Fault 2019-03-22T07:14:46.507Z 268 SYSTEM nsystem1 n.system-1 EX1001 ALARM TEST 2	
2019/03/22 16:26:31	NewEquipment0002	NewMachine002		2000	Warning 2019-03-22T07:14:46.507Z 268 SYSTEM nsystem1 n.system-1 2000 OPE MESSAGE	
2019/03/22 16:26:31	NewEquipment0002	NewMachine002		2001	Warning 2019-03-22T07:14:46.507Z 268 SYSTEM nsystem1 n.system-1 2001 OPE MESSAGE 2	

- Alarm type

Alphabets at the beginning of the string value of nativeCode property are displayed as an alarm type in this column. If a string value has no alphabet at its beginning, the corresponding column is set to no string ("").

Example: When a string value of nativeCode property is "EX100," the alphabets "EX" is set for the column.
- Alarm number

The numeric characters at the end of nativeCode property value (string type) are displayed as an alarm number in this column. If a nativeCode property value has no numeric character at its end, the corresponding Alarm number column is set to zero ("0").

Example: When a nativeCode property value is "EX101," the numeric characters "101" is set for the column.
- Alarm Message

This column displays alarm messages with values of properties delimited.

- An example of XML tag for collecting condition signal by current command (NOTE)

<Fault	A: Condition Type
dataItemId = "nsystem1"	B: dataItemId
timestamp = "2019-03-22T07:14:46.507Z"	C: timestamp
name = "n.system-1"	D: name
sequence = "268"	E: sequence
nativeCode = "EX1000"	F: nativeCode
type = "SYSTEM"	G: type
>ALARM TEST</Fault>	H: Message

NOTE

Condition Type property is set to the state of a device in CONDITION category.

MT-LINK_i allows only for "FAULT" and "WARNING" device states; Condition signal from a device will be ignored when the device is in "NORMAL" or "UNAVAILABLE" state.

20	Nov. 22, 2023	uwano	Revised for MT-LINK _i Version 4.0	Title	FANUC MT-LINK _i AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK _i Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK _i Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK _i Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK _i Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK _i Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK _i Version 3.7	FANUC CORPORATION		Sheet. 250/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

- An example of an alarm message

The following example outputs values of properties in an order of "A to C to E to G to B to D to F to H" with delimiting by space when the output order and the delimiter are specified in the [Condition Setting] tab.

Fault	2019-03-22T07:14:46.507Z	268	SYSTEM	nssystem1	n.system-1	EX1000	ALARM TEST
A	C	E	G	B	D	F	H

NOTE

Alarm messages exceeding 512 bytes will be truncated to 512 bytes, concatenated with an ellipsis ("..."), and added to the listing. If such a long alarm message consists of multi-byte characters, the last character (at the 512nd byte position) of the truncated message may appear as garbled.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 251/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

4.6 MACHINE SETTING - FanucRobodrill

This section explains the machine setting of the machine type FanucRobodrill.

NOTE

ROBODRILL Option license is required to support FanucRobodrill type machines.

4.6.1 Steps for the Machine Setting of FanucRobodrill

Follow the steps below to set the machine.

- (1) Select [Machine Setting] in the AdminTool Menu. The [Machine Setting] window opens.
- (2) Add Collector. For the details, refer to the "[Add Collector](#)" subsection.
- (3) Add machine type FanucRobodrill to each Collector. For the details, refer to the "[Add FanucRobodrill](#)" subsection.
- (4) Make protocol setting and connection setting. Those settings can be performed in a similar manner as for FanucCNC type machines. For the details, refer to the "[Confirm/Edit Setting of FanucCNC](#)" subsection and the "[Set Protocol Setting of FanucCNC](#)" subsection.
- (5) Perform Signal Setting for the machine. This setting can be performed in a similar manner as for FanucCNC type machines. For the details, refer to the "[Set Signal of FanucCNC](#)" subsection.
- (6) Perform Custom Signal Setting for machine. This setting can be performed in a similar manner as for FanucCNC type machines. For the details, refer to the "[Set Custom Signal of FanucCNC](#)" subsection.
- (7) Set Robodrill signals. For the details, refer to the "[Set FanucRobodrill Specific Signal](#)" subsection.
- (8) Set macro variable getting conditions. This setting can be performed in a similar manner as for FanucCNC type machines. For the details, refer to the "[Register Macro Variable Getting Condition Definition of FanucCNC](#)" subsection.
- (9) Set variable history retrieval conditions. This setting can be performed in a similar manner as for FanucCNC type machines. For the details, refer to the "[Setting Variable History Retrieval Condition Definition of FanucCNC](#)" subsection.
- (10) Check whether the signals that are set can be acquired or not. For the details on the checking methods, refer to the "[Checking the Signals](#)" subsection.
- (11) Save all the settings to make the system reflect them. For the details, refer to the "[Save Machine Setting](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 252/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

4.6.2 Add FanucRobodrill

(1) Adding a machine

To add a machine, right-click the target Collector name in the Collector tree and point to [Add Machine].

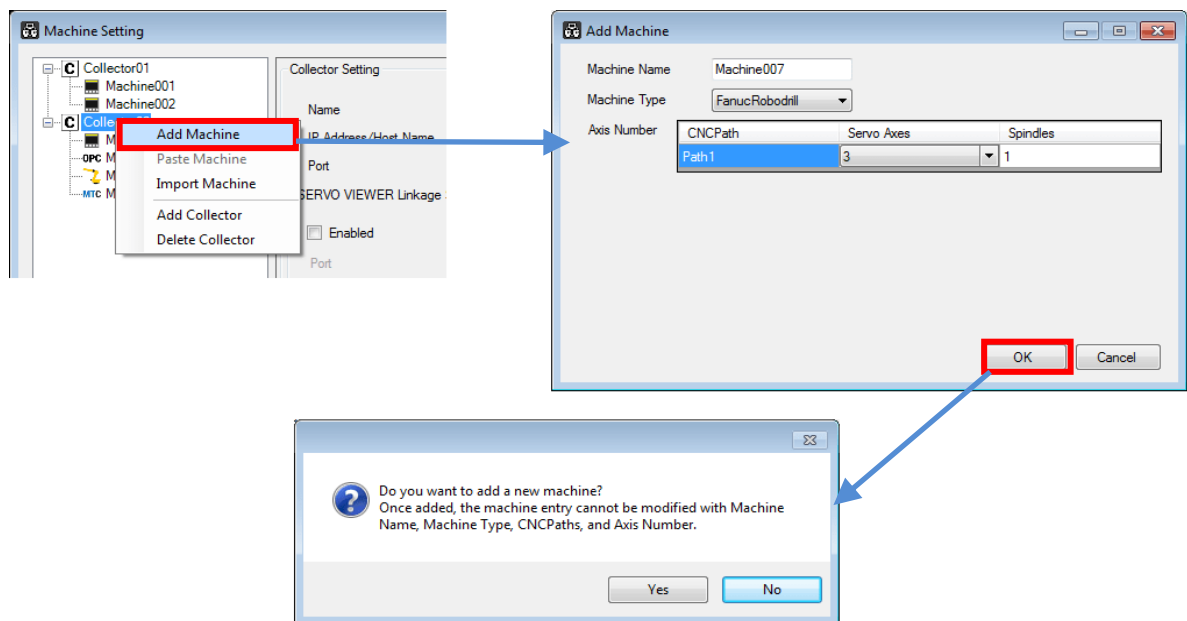
*Before adding a machine, make sure that you have registered the target Collector.

When the [Add Machine] dialog box appears, specify the settings of the target machine.

NOTE

Once set, [Machine Name], [Machine Type], [CNCPaths], and [Axis Number] ([Servo Axes] and [Spindles]) will be non-editable.

If the settings of [Machine Name], [Machine Type], [CNCPaths], [Axis Number], and [L] are changed by mistake, the existing machine must be deleted. Add a new machine and make the settings for it.



■ The [Add Machine] dialog box

- **Machine Name**
Set a machine name to this field. The [Machine Name] should start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and underscores ("_"). This field can be up to 32 bytes long.
- **Machine Type**
Select [FanucRobodrill].
- **CNCPaths**
This field is fixed to 1.
- **Servo Axes/Spindles**
Select the number of the servo axes for Path1 of the machine. The [Servo Axes] can be set to 3 to 8. It is set to 3 by default.
The [Spindles] is fixed to 1.
- **The [OK] button**
Clicking this button opens a confirmation message. Click [OK] to save the settings and add the machine in the Collector tree. Or, click [No] to cancel adding the machine and close the dialog box.
- **The [Cancel] button**
Clicking this button discards the settings and closes the dialog box.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	253/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

4.6.3 Set FanucRobodrill Specific Signal

This setting function enables Web users to browse the ROBODIRILL specific signals with the MT-LINK*i* Web Server.

Select a machine in the Collector tree and select the [Robodrill Signal Setting] tab in the right pane.

For each signal listed in the [Robodrill Signal Setting] tab, check the [Axis] and [CNCPath] columns, set the [Reading Cycle] column, and select or clear the [Enable] check box according to whether it is to be collected.

Collector collects the signals that have the [Enable] check box selected.

The statuses will be automatically added to the [Simple assign status area] of the [Equipment Setting] window if their [Enable] check boxes have been selected on the [Robodrill Signal Setting] tab of the [Machine Setting] window.

Machine Setting

Name: Machine007 Type: FanucRobodrill Connect: CNCPaths: 1 Signal Check

Web Name: Machine007

Protocol Basic Signal Custom Signal **Robodrill Signal Setting** Macro Variable Getting Condition Definition Variable History Retrieval Condition Definition

Signal Setting

Narrow down

Description Enable Reading Cycle Apply Discard

No	Signal	Description	Axis	Enable	Outlier Detection	Reading Cycle	Check
1	RobodrillToolNum	Robodrill tool number		☒	☒	500msec	☒
2	RobodrillMachiningMode	Robodrill machining mode		☐	☐	500msec	☐
3	RobodrillEsaveMode	Robodrill power save mode		☐	☐	500msec	☐
4	RobodrillRigidTapMode	Robodrill rigid tapping mode		☐	☐	500msec	☐
5	RobodrillTableLoad	Robodrill table load mode		☐	☐	60sec	☐
6	RobodrillToolLoad	Robodrill tool load mode		☐	☐	60sec	☐
7	RobodrillCounterNum1	Robodrill production counter 1		☒	☒	10sec	☐
8	RobodrillCounterTargetNum1	Robodrill production target 1		☐	☒	60sec	☐
9	RobodrillCounterPartName1	Robodrill counter part name 1		☐	☐	60sec	☐
10	RobodrillCounterNum2	Robodrill production counter 2		☒	☒	10sec	☐
11	RobodrillCounterTargetNum2	Robodrill production target 2		☐	☒	60sec	☐
12	RobodrillCounterPartName2	Robodrill counter part name 2		☐	☐	60sec	☐
13	RobodrillHMICounterNum1	Robodrill iHMI production counter 1		☐	☒	10sec	☐
14	RobodrillHMICounterTargetNum1	Robodrill iHMI production target 1		☐	☒	60sec	☐
15	RobodrillHMICounterPartName1	Robodrill iHMI counter part name 1		☐	☐	60sec	☐
16	RobodrillHMICounterNum2	Robodrill iHMI production counter 2		☐	☒	10sec	☐
17	RobodrillHMICounterTargetNum2	Robodrill iHMI production target 2		☐	☒	60sec	☐
18	RobodrillHMICounterPartName2	Robodrill iHMI counter part name 2		☐	☐	60sec	☐

Apply OK Cancel

■ The [Robodrill Signal Setting] tab

● Signal

This column in the signal lists the names of signals that can be set for the selected machine. To list only specific signals, narrow down the signal list. For the details on how to narrow down the list, refer to the "[Narrow down Signal List](#)" subsection.

For the details of signals, refer to the "[LIST OF FanucRobodrill SIGNALS](#)" section.

● Description

This column displays the Web name of each signal as its description.

● Axis

This field displays the axis of each signal. If its corresponding signal is independent of axes, this field is set to space.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	254/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- **Enable**

This is a check box specifying whether to execute data collection of signals by the Collector. If the check box is selected, the data will be collected; otherwise, the data will not be collected.

NOTE

- 1 Some signals listed in the [Signal Setting] table may not be available depending on machine model. For those non-existent signals, clear the check boxes. If the check box for a non-existent signal is selected, its values will not be stored to the database, and will not be displayed on the Web screen.
- 2 When a signal is referenced by equipment, clearing the [Enable] check box for it will be denied with an error message. For the details, refer to the "**Referenced Error**" subsection.

- **Outlier Detection**

For signals with [Outlier Detection] check box selected (signals of data type Int32 or Double), outlier detection of the signal can be set by double-clicking that row. The check box column cannot be edited. For the details of the outlier detection settings, refer to the "**Set Outlier Detection of Signal**" subsection.

- **Reading Cycle**

Specify a reading interval per signal. Options are [500msec], [10sec], and [60sec].

- **Check**

This column displays the results of Signal Check as follows:

"OK":	The signal was retrieved
"NG" (for "not good"):	Failed to retrieve the signal
"-" (hyphen):	Not a target of Signal Check
"" (blank):	Signal Check unexecuted (initial state, machine connection failure, etc.)

For the details on Signal Check, refer to the "**Checking the Signals**" subsection.

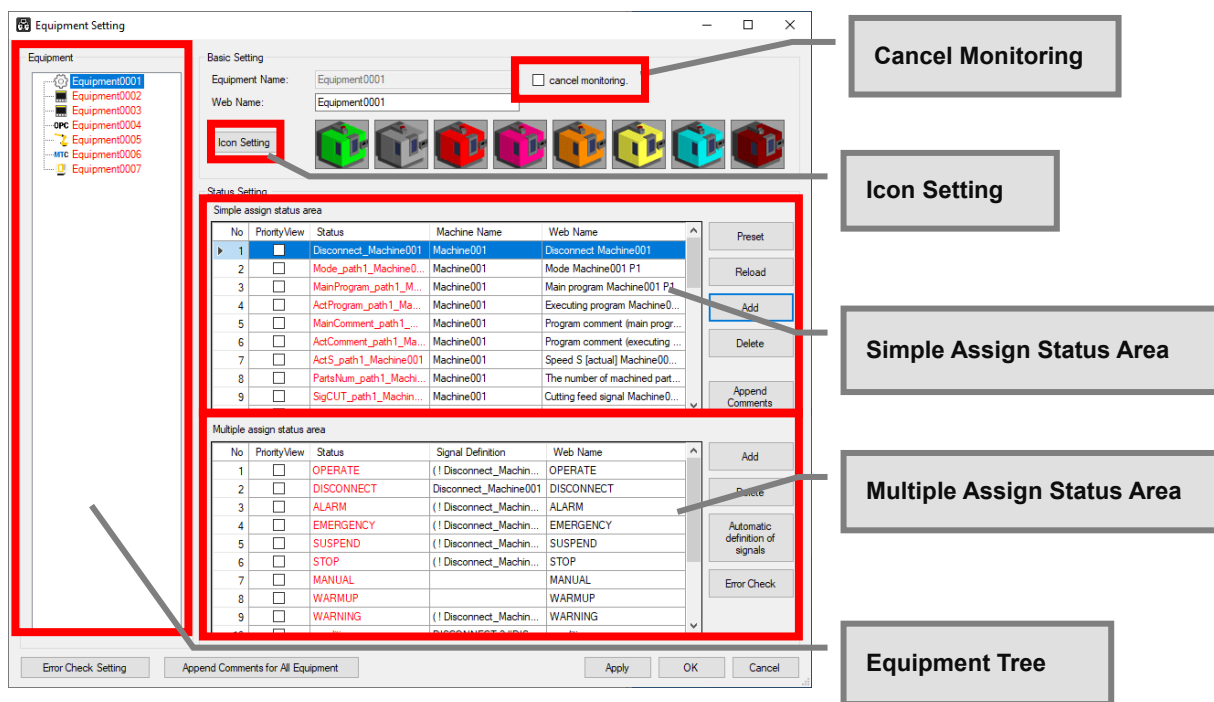
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue
FANUC CORPORATION					Sheet.	255/868	

4.7 EQUIPMENT SETTING

Use this window to set up and register equipment units. To set up an equipment unit in which to include machines that have been registered through the [Machine Setting] window, associate the equipment unit with those machines' signals with the [Equipment Setting] window.

This window allows the user to delete equipment units that have already been registered.

- Equipment units can be registered up to 100.
- Set equipment name, web name, and icons related to each status in the [Basic Setting] group box.
- In the [Status Setting] group box, apply the following settings:
 - Assign signal definition (composed of machine name, signal name and CNC path) to status in the [Simple assign status area] table.
 - Assign signal definition (composed of formula using with status name) to status in the [Multiple assign status area] table.
- Preset function facilitates setting in the [Simple assign status area] group box.
- Automatic definition of signals function automatically allows you to facilitate setting in the [Multiple assign status area] group box.
- It defines signal definitions for multi-machine and multi-path configuration and assigns them to corresponding equipment statuses.
- This window allows the user to add and delete entries both in the [Simple assign status area] and the [Multiple assign status area] tables.
- The maximum number of multiple assign statuses that can be registered to a single equipment unit is 1000.
- The maximum total number of statuses to be registered for equipment units is 70,000.
- Each of the assigned signals to a status and the assignee status will be stored in the DB.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	256/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

The Equipment tree displays equipment units with their type icons placed on their left side. Each type icon indicates that the equipment unit consists of machines of a specific machine type.

The following table lists type icons for machine types.

Type icon	Machine type	Remark
	FanucCNC	The equipment consists only of FanucCNC type machines.
OPC	OPCUAServer	The equipment consists only of OPCUAServer type machines.
	FanucRobot	The equipment consists only of FanucRobot type machines.
MTC	MTConnectAgent	The equipment consists only of MTConnectAgent type machines.
	FanucRobodrill	The equipment consists only of FanucRobodrill type machines.
	-	The equipment consists of machines of multiple machine types or consists of no machine.

* When the character string of a displayed item cannot be viewed as a whole, the end of the character string will be expressed with "..." inside the column.

By placing the mouse cursor on that item, a tool tip is displayed, and the whole character string can be confirmed.

On the [Equipment Setting] window, the following item display areas can be extended by widening the window.

- Simple assign status area: [Machine Name], [Web Name]
- Multiple assign status area: [Signal Definition], [Web Name]

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION				Sheet. 257/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

4.7.1 Steps for Equipment Setting

Follow the steps below.

- (1) Select [Equipment Setting] in the AdminTool Menu. This displays [Equipment Setting] window.
- (2) Add new equipment in the [Equipment] tree. For the details, refer to the "[Add Equipment in Equipment Tree Menu](#)" subsection.
- (3) Set the status icon (which indicates the state of the equipment unit) to the added equipment. Refer to the "[Set Equipment Status Icon](#)" subsection.
- (4) Assign monitoring status to the added equipment. Refer to the "[Set Equipment Status](#)" subsection.
- (5) When you have completed above steps, click [Apply] or [OK]. Clicking [Cancel] discards the settings you made and closes the window.

NOTE

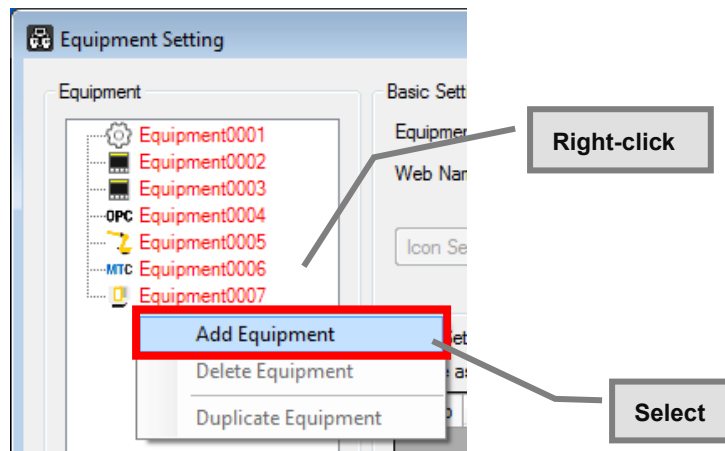
If you update from versions earlier than version 2.1, follow the steps below to update equipment icons:

- (1) Select [Equipment Setting] in the AdminTool Menu to open the [Equipment Setting] window.
- (2) Click [OK] on the [Equipment Setting] window.
- (3) Send data to the database. For the details, refer to the "[SEND TO DB](#)" section.

4.7.2 Add Equipment in Equipment Tree Menu

Follow the steps below to add equipment.

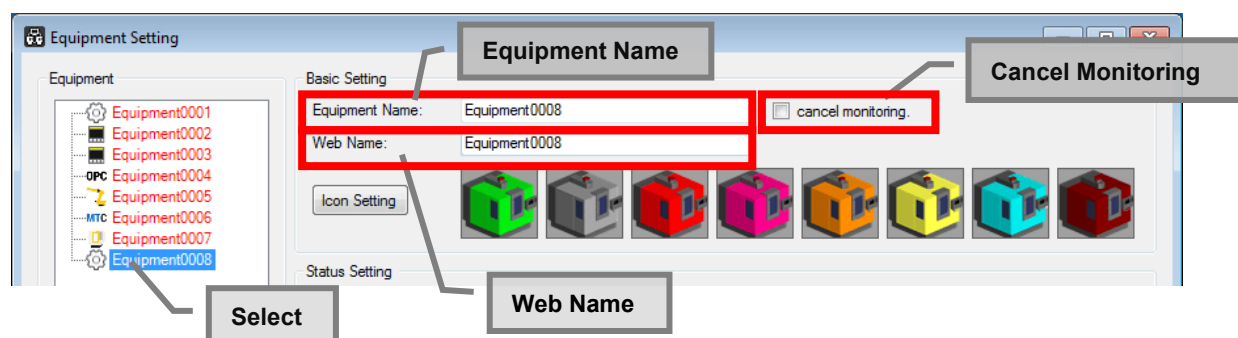
- (1) Right-click in the [Equipment] tree and select [Add Equipment].



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	258/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- (2) The new equipment is added in the [Equipment] tree. Select it and change the [Equipment Name] and the [Web Name] fields in the [Basic Setting] group box in the right pane. An equipment name should start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and underscores ("_"). This field can be up to 32 bytes long. Once the setting data is applied, this [Equipment Name] will become uneditable.

The Web name can be set in the language in which Web users view the Web screens. The [Web Name] field is required and should be entered in the language specified for the [Web name language] field in the [System Setting] window. This field can be up to 100 bytes long and will be set to the value of [Equipment Name] field by default.



- (3) Clicking [Apply] updates the equipment name in the [Equipment] tree.
- (4) The equipment, with its [cancel monitoring] check box selected, will be counted out from the monitoring target on the Web screen, and will not be able to be selected. The status of the equipment counted out from monitoring will not be stored in the database. As for the equipment counted out from monitoring, the signal detection/alarm mail will not be delivered. As for the equipment counted out from monitoring, the state of display on each Web screen will be as shown below.

Web Screen	Location	State of Display	Equipment Counted Out from Monitoring
Monitoring - Overlook	Information Display Area	Not displayed	Equipment icon will not be displayed.
	State of Equipment (Footer)	Not displayed	Will not be counted in the display for the number of equipment units (Number of equipment units by state of operation / Number of all the equipment units).
Monitoring - Group Monitoring	Filter	Displayed	No influence when selecting Group from the filter.
	Information Display Area	Displayed	No influence on the group displayed.
	State of Equipment (Footer)	Not displayed	Will not be counted in the display for the number of equipment units (Number of equipment units by state of operation / Number of all the equipment units).
Monitoring - Equipment Monitoring Monitoring - Signal Monitoring Monitoring - Alarm Monitoring Monitoring - Tool Life Monitoring Monitoring - Periodic Maintenance	Filter	Not displayed	Will not be displayed in the equipment list of filter.
	Information Display Area	Not displayed	Will not be displayed, as the equipment unit cannot be selected.
	State of Equipment (Footer)	Not displayed	Will not be counted in the display for the number of equipment units (Number of equipment units by state of operation / Number of all the equipment units).

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 259/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

Web Screen	Location	State of Display	Equipment Counted Out from Monitoring
Monitoring - Equipment Monitoring - Detail	Information Display Area	Not displayed	Will not be displayed, as the equipment unit cannot be selected.
	State of Equipment (Footer)	Not displayed	Will not be counted in the display for the number of equipment units (Number of equipment units by state of operation / Number of all the equipment units).
Results - Operational Results Results - Production Results Diagnosis - Alarm History Diagnosis - Program History Diagnosis - Signal History Diagnosis - Macro Variable History	Filter	Displayed	No influence when selecting Group from the filter.
	Information Display Area	Partially displayed	Results/history data retrieved before being counted out from monitoring will be displayed. Results/history data after being counted out from monitoring will not be displayed. (NOTE 1)
Results - Group Results Diagnosis - Report output Utility - File Transfer Utility – Simultaneous File Transmission Utility - Machine Data Archive Management	Filter	Displayed	No change in selecting group or equipment from the filter.
	Information Display Area	Displayed	No change on the information display.

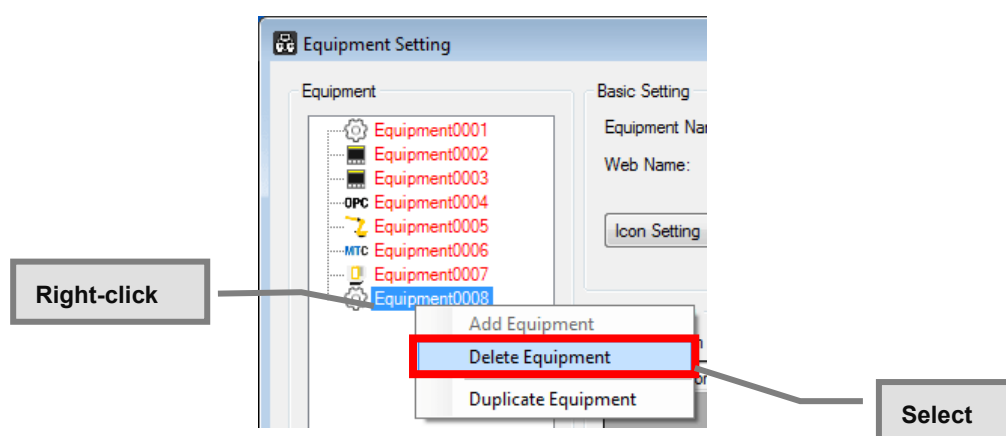
NOTE

- 1 The value of the macro variable set for retrieval in the [Macro Variable Getting Condition Definition] will be retrieved even if it is a macro variable of the equipment counted out from monitoring, and the data will be stored in the DB. It will also be displayed in the [Diagnosis - Macro Variable History] screen. When the file output function is enabled, CSV file output will be executed as well.
- 2 The value of the variable set for retrieval in the [Variable History Retrieval Condition Definition Setting] will be retrieved even if it is a variable of the equipment counted out from monitoring, and the data will be stored in the DB. When the file output function is enabled, CSV file output will be executed as well.

4.7.3 Delete Equipment in Equipment Tree Menu

Follow the steps below to delete equipment.

- (1) Right-click the target equipment name in the [Equipment] tree and select [Delete Equipment].



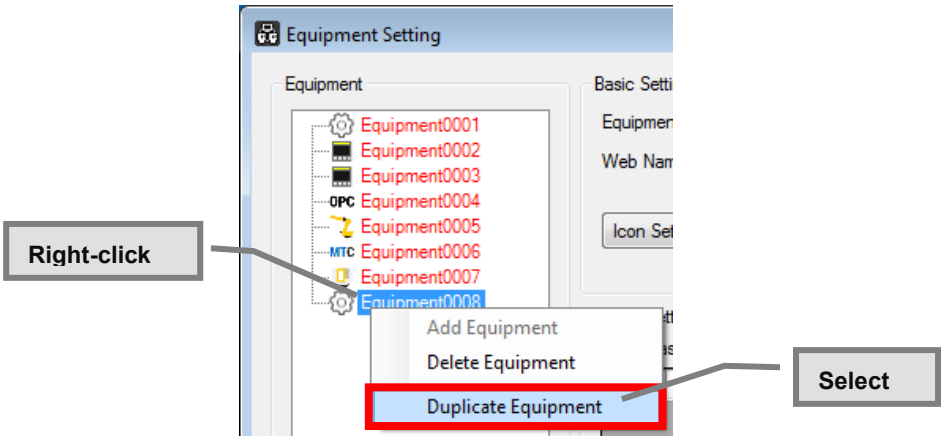
- (2) Selected icon is deleted from the [Equipment] tree.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 260/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

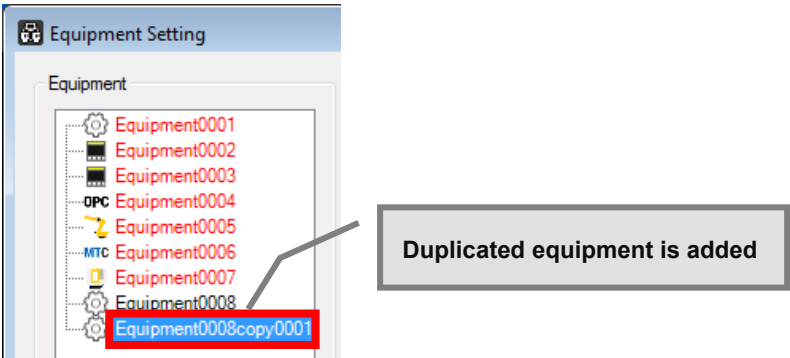
4.7.4 Duplicate Equipment in Equipment Tree Menu

Follow the steps below to duplicate equipment.

- (1) Right-click the target equipment name in the [Equipment] tree and select [Duplicate Equipment].



- (2) The duplicated equipment is added to the [Equipment] tree.

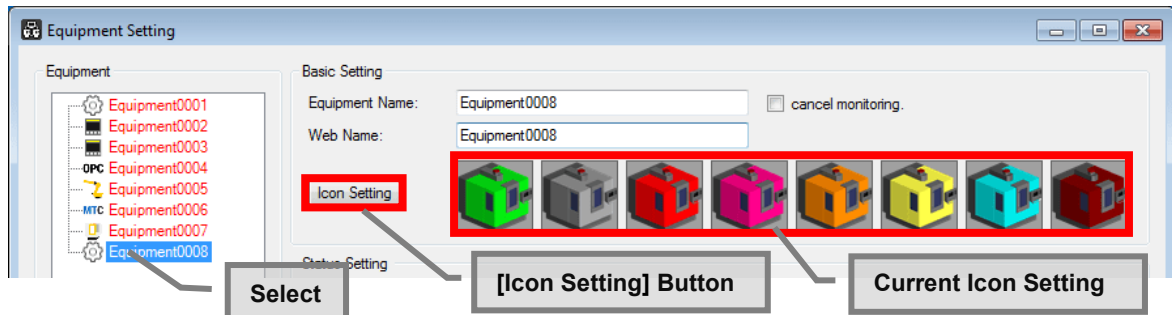


20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	261/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

4.7.5 Set Equipment Status Icon

Follow the steps below to set icon of equipment status.

- (1) Select equipment from the [Equipment] tree and verify the "Current Icon Setting" area. By default, the following icons appear. If you want to change any of those icons, click [Icon Setting].



- (2) The [Icon Setting] window appears. Click [Load Image] for the image of the state icon to replace. Then, the [Select Image File] dialog appears. Select the image file to replace with. The dialog box selects the following folder on the first session. On subsequent sessions, it selects the folder selected in the previous session.

<MT-LINK*i* installation folder>\MT-LINK*i*AdminTool\MT-LINK*i*_image\Equipment

Supported image format: png, jpg, bmp

It is required to prepare image files in advance.

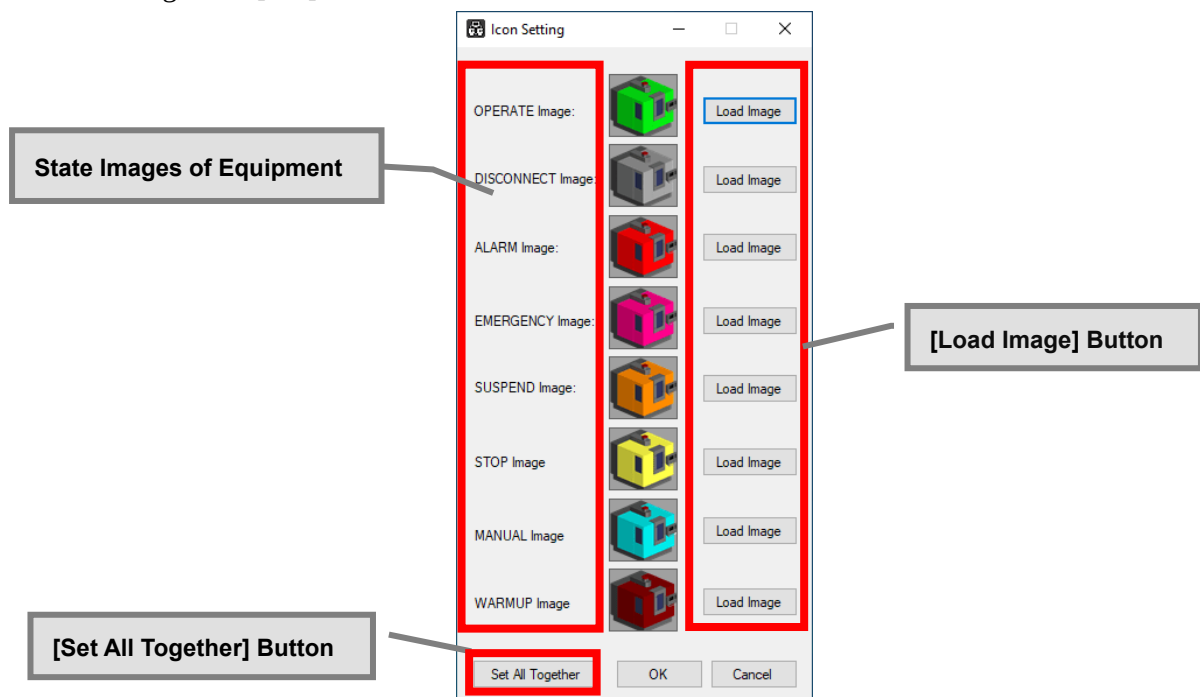
Image files must not have a resolution larger than 500 pixels wide x 500 pixels high.

NOTE

The recommended size of image for an equipment icon is 128 pixels wide x 128 pixels high (a square).

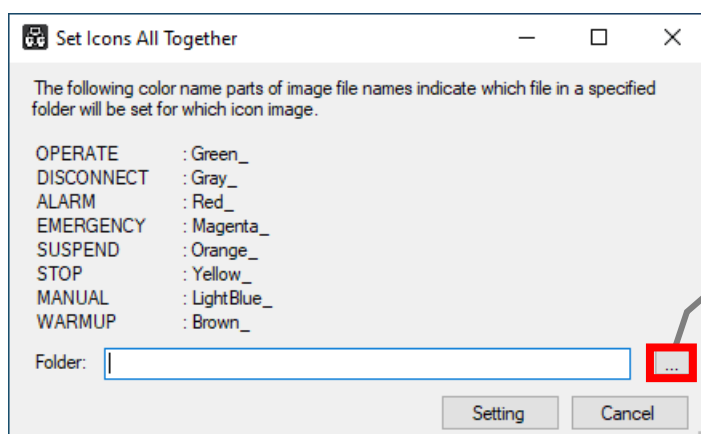
If the size of the replaced image is other than the recommended size, the image displayed on the [Monitoring - Overlook] screen of the Web screen may become corrupted.

After setting, click [OK] to close the window.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	262/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

- (3) Clicking the [Set All Together] button opens the [Set Icons All Together] window. Use this window to set image files for icon images all together.



Specify a folder which contains image files to set for icon images either by setting it for the [Folder] text field or by clicking the browse folder button ([...]) to open the [Browse For Folder] window, selecting it, and clicking [OK] to tell the [Set Icons All Together] window to populate it in the [Folder] text field.

- * The [Folder] text field populates null space.
- * On the first session, the [Browse For Folder] window selects the folder shown below. On subsequent sessions, this window selects the folder populated in the [Folder] text field:

<MT-LINK*i* installation folder>\MT-LINKiAdminTool\MT-LINKi_image\Equipment

Click the [Set] button to set image files in the specified folder for icon images indicating equipment states all together.

The second column of the following table suggests which image file will be set for which icon image:

Icon images (Equipment states)	Color name parts of image files (including the first underscores)	Remarks
Normal	Green_	<u>An error message will appear in the following cases:</u> - The specified folder cannot be found in the folder - No image file resides in the folder <u>A confirmation message will appear in the following cases:</u> - Image files have a resolution larger than 500 pixels wide x 500 pixels high - Not all image files are found in the folder The confirmation message lists icon images that were to be set to oversized or missing image files. Clicking [OK] in the message closes it and sets the other icon images.
Disconnect	Gray_	
Alarm	Red_	
Emergency	Magenta_	
Suspend	Orange_	
Stop	Yellow_	
Manual	LightBlue_	
Warmup	Brown_	

NOTE

When multiple image files are found for an icon image, the one will be selected whose name is lower than any other file name of them in alpha-numeric order. However, if the selected image file has a resolution larger than 500 pixels wide x 500 pixels high, this file selection continues.

- (4) When all icon images are set, click [OK] in the [Icon Setting] window to close it.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION				Sheet. 263/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

4.7.6 Set Equipment Status

Follow the steps below to set equipment status.

Hereafter, the term "simple assign status" indicates the status which set the signal definition composed of machine name, signal name and CNC path. The simple assign status is used as a reference when a machine signal is assigned to equipment.

In contrast, "multiple assign status" is composed of formula of simple assign statuses.

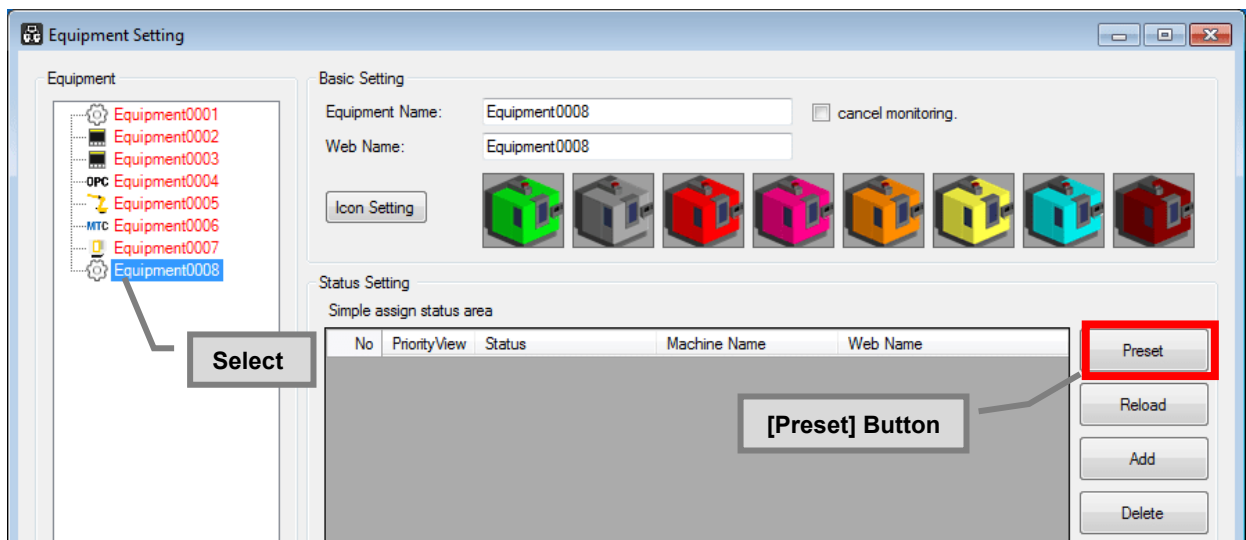
NOTE

When the existing status is preset, [Status Setting] such as [PriorityView] or [Web name] will be deleted, and return to default setting.

In a case when after presetting, [PriorityView] or [Web Name] in the [Simple assign status area] is changed, and signal to collect is added in the machine setting, use the [Reload] button instead of the [Preset] button.

For the details on the reload signal process, refer to the "[Reload Simple Assign Status](#)" subsection.

- (1) Select any equipment name and click [Preset] in the right side of the [Simple assign status area] table.

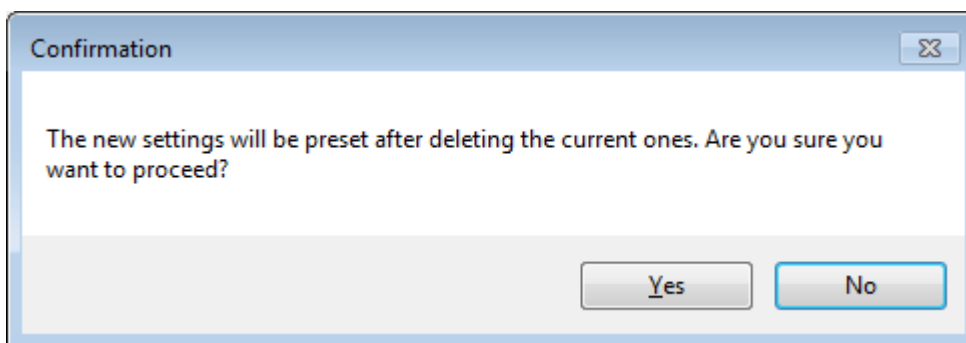


NOTE

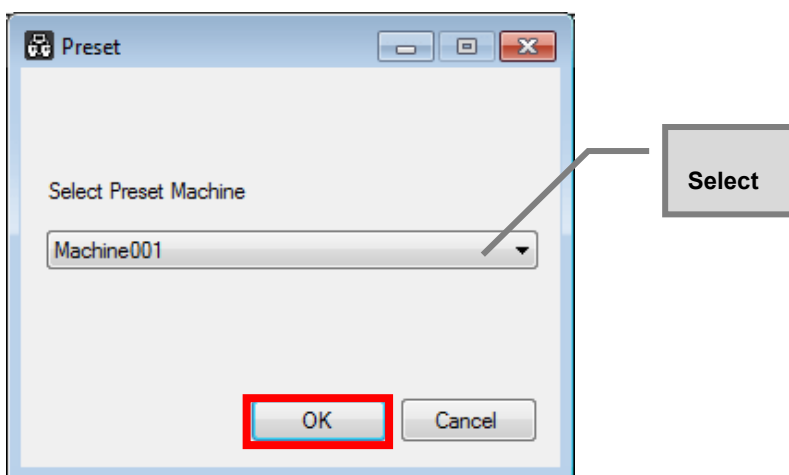
If ROBODRILL Option is neither installed yet nor recognized as installed, selecting equipment in the equipment tree disables the [Preset], the [Reload], the [Add], and the [Delete] buttons and thus no status of the equipment can be set with the [Equipment Setting] window.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	264/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

- (2) Once you performed presetting operation, clicking [Preset] shows you the following dialog box. Click [Yes] to delete the existing setting data and continue the presetting operation. Or click [No] to quit presetting.



- (3) The [Preset] dialog box appears. Select a target machine in the list box and click [OK]. If you cannot find the target machine, refer to the "[Steps for Machine Setting](#)" subsection and perform machine setting beforehand.



NOTE

If ROBODRILL Option is neither installed yet nor recognized as installed, no FanucRobodrill type machine will be listed in the [Preset] dialog box and thus will not be selected.

- (4) The following statuses will be added to the [Simple assign status area] table depending on machine type. The statuses added here are the statuses of the signals that were set in the [Machine Setting] window.

Status Setting				
Simple assign status area				
No	PriorityView	Status	Machine Name	Web Name
1	☒	Disconnect_Machine001	Machine001	Disconnect Machine001
2	☐	Mode_path1_Machine0...	Machine001	Mode Machine001 P1
3	☐	MainProgram_path1_M...	Machine001	Main program Machine001 P1
4	☐	ActProgram_path1_Ma...	Machine001	Executing program Machine0...
5	☐	MainComment_path1_...	Machine001	Program comment (main progr...
6	☐	ActComment_path1_Ma...	Machine001	Program comment (executing ...
7	☐	PartsNum_path1_Machi...	Machine001	The number of machined part...
8	☐	SigCUT_path1_Machin...	Machine001	Cutting feed signal Machine0...
9	☐	SigSBK_path1_Machin...	Machine001	Single block check signal Ma...

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 265/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

In addition, presetting operation automatically assigns signal definitions (corresponding to the selected machine) to unassigned basic statuses (such as OPERATE and DISCONNECT) in the [Multiple assign status area] table. For the basic statuses, refer to the "Add, Edit or Delete Multiple Assign Status" subsection.

Multiple assign status area

No	PriorityView	Status	Signal Definition	Web Name
1	☐	OPERATE	(! Disconnect_Machin...	OPERATE
2	☐	DISCONNECT	Disconnect_Machine001	DISCONNECT
3	☐	ALARM	(! Disconnect_Machin...	ALARM
4	☐	EMERGENCY	(! Disconnect_Machin...	EMERGENCY
5	☐	SUSPEND	(! Disconnect_Machin...	SUSPEND
6	☐	STOP	(! Disconnect_Machin...	STOP
7	☐	MANUAL		MANUAL
8	☐	WARMUP		WARMUP
9	☐	WARNING	(! Disconnect_Machin...	WARNING
10	☐	condition	DISCONNECT ? "DIS...	condition
11	☐	ProductName	MainComment_path1_...	ProductName
12	☐	ProductResultNumber	PartsNum_path1_Mac...	ProductResultNumber
13	☐	ProductSerialNumber		ProductSerialNumber
14	☐	OperatorID		OperatorID

- (5) Select the [PriorityView] check boxes for the status names which you like to browse preferentially. The values of the selected statuses in this step can be browsed in the [Monitoring - Equipment Monitoring] screen. Up to ten [PriorityView] check boxes (in the simple assign status area and the multiple assign status area combined) can be selected. The [Monitoring - Equipment Monitoring] screen shows the selected statuses in the same order as listed in the multiple assign status and the simple assign status areas.

Status Setting

Simple assign status area

No	PriorityView	Status	Machine Name	Web Name
1	☒	Disconnect_Machine002	Machine002	Disconnect Machine002
2	☒	Mode_path1_Machine0...	Machine002	Mode Machine002 P1
3	☒	MainProgram_path1_M...	Machine002	Main program Machine002 P1
4	☒	ActProgram_path1_Ma...	Machine002	Executing program Machine0...
5	☐	MainComment_path1_...	Machine002	Program comment (main prog...
6	☐	ActComment_path1_Ma...	Machine002	Program comment (executing...
7	☐	PartsNum_path1_Machi...	Machine002	The number of machined part...
8	☐	SigCUT_path1_Machin...	Machine002	Cutting feed signal Machine0...
9	☐	SigSBK_path1_Machin...	Machine002	Single block check signal Ma...

Multiple assign status area

No	PriorityView	Status	Signal Definition	Web Name
1	☒	OPERATE	(! Disconnect_Machin...	OPERATE
2	☒	DISCONNECT	Disconnect_Machine002	DISCONNECT
3	☒	ALARM	(! Disconnect_Machin...	ALARM
4	☒	EMERGENCY	(! Disconnect_Machin...	EMERGENCY
5	☒	SUSPEND	(! Disconnect_Machin...	SUSPEND
6	☒	STOP	(! Disconnect_Machin...	STOP
7	☐	MANUAL		MANUAL
8	☐	WARMUP		WARMUP
9	☐	WARNING	(! Disconnect_Machin...	WARNING

Equipment0011

Detailed

OPERATE

OPERATE	on
DISCONNECT	off
ALARM	off
EMERGENCY	off
SUSPEND	off
STOP	off
Disconnect Machine001	off
Mode Machine001 P1	MEMORY
Main program Machine001 P1	Q/PATH1/SAMPLE
Machine001 P1	

10 selected statuses are displayed

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	266/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

NOTE

If the equipment (machine) is not connected, the value of each status in the [Monitoring - Equipment Monitoring] screen will be as follows.

- Disconnect <Machine name>: on
(simple assign status)
- DISCONNECT signal: on
(multiple assign status)
- Condition (condition): DISCONNECT
- Other than above: blank

* When using the [Priority View] check box, two or more check boxes can be selected or cleared at the same time, by selecting two or more rows, right-clicking, and selecting [ON] or [OFF] on the menu displayed.

Example 1) When selecting two or more check boxes

The diagram illustrates the process of selecting multiple rows and setting a status to ON in the Simple assign status area. It consists of two screenshots of the 'Simple assign status area' table. In the first screenshot, rows 2 through 7 are selected (highlighted in blue). A callout box says 'Select rows and right-click'. A second callout box says 'Select [ON]'. In the second screenshot, the same rows are selected, and the 'Status' column for all selected rows is now 'ON'. A callout box says 'Check boxes of the selected rows are selected'.

No.	PriorityView	Status	Machine Name
1	☐	Disconnect_Machine001	Machine001
2	☒	Mode_path1_Machine001	Machine001
3	☒	MainProgram_path1_Machine001	Machine001
4	☒	ActProgram_path1_Machine001	Machine001
5	☒	MainComment_path1_Machine001	Machine001
6	☒	ActComment_path1_Machine001	Machine001
7	☒	PartsNum_path1_Machine001	Machine001
8	☐	SigCUT_path1_Machine001	Machine001
9	☐	SigSBK_path1_Machine001	Machine001

Example 2) When clearing two or more check boxes

The diagram illustrates the process of selecting multiple rows and setting a status to OFF in the Multiple assign status area. It consists of two screenshots of the 'Multiple assign status area' table. In the first screenshot, rows 2 through 8 are selected (highlighted in blue). A callout box says 'Select rows and right-click'. A second callout box says 'Select [OFF]'. In the second screenshot, the same rows are selected, and the 'Status' column for all selected rows is now 'OFF'. A callout box says 'Check boxes of the selected rows are cleared'.

No.	PriorityView	Status	Signal Definition
1	☒	OPERATE	(! Disconnect_Machine001)
2	☒	DISCONNECT	Disconnect_Machine001
3	☒	ALARM	Disconnect_Machine001
4	☒	EMERGENCY	(! Disconnect_Machine001)
5	☒	SUSPEND	(! Disconnect_Machine001)
6	☒	STOP	(! Disconnect_Machine001)
7	☒	MANUAL	(! Disconnect_Machine001)
8	☒	WARMUP	(! Disconnect_Machine001)
9	☒	WARNING	(! Disconnect_Machine001)

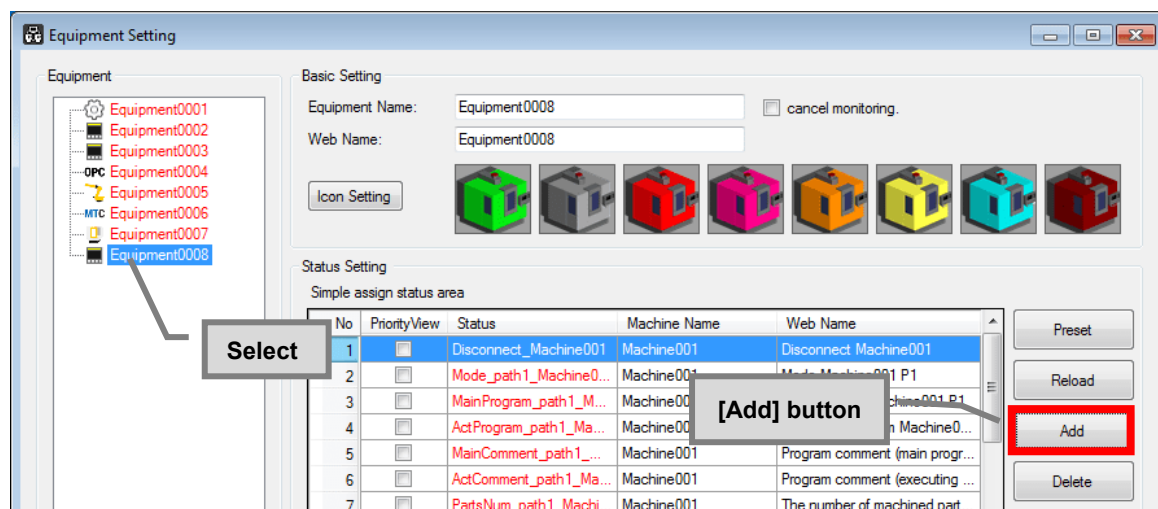
4.7.6.1 Add, Edit or Delete Simple Assign Status

Presetting operation enables you to assign all the status corresponding to machine. Refer to the "[Set Equipment Status](#)" subsection. This subsection describes how to add, edit and delete a simple assign status.

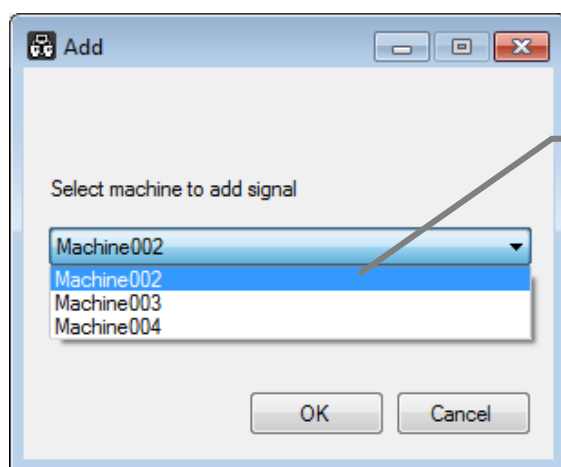
■ Add a simple assign status

- (1) Select any equipment in the [Equipment] tree and click [Add] in the right area of the [Simple assign status area] table.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION				
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				Sheet.	267/868



- (2) The [Add] dialog box appears. Select a target machine to add and click [OK]. Note that you can select from machines that have not been selected yet.



Having selected [Machine001], you can select one from [Machine002] to [Machine004] besides [Machine001].

- (3) The added status is displayed at the bottom row of the [Simple assign status area] table. This addition automatically assigns default signal definitions (corresponding to the selected machine) to unassigned basic statuses in the [Multiple assign status area] table. For the basic statuses, refer to the "[Add, Edit or Delete Multiple Assign Status](#)" subsection.

■ Edit a simple assign status

Double-click a status in the [Simple assign status area] table to open the [Signal Setting] dialog box. After editing in the window, click [OK].

- The only editable field is the [Web Name]. The value of this field should be entered in the language specified for the [Web name language] field in the [System Setting] window.
- The other fields ([Status Name], [Signal Type], [Machine Name], and [User Signal Definition]) are not-editable.

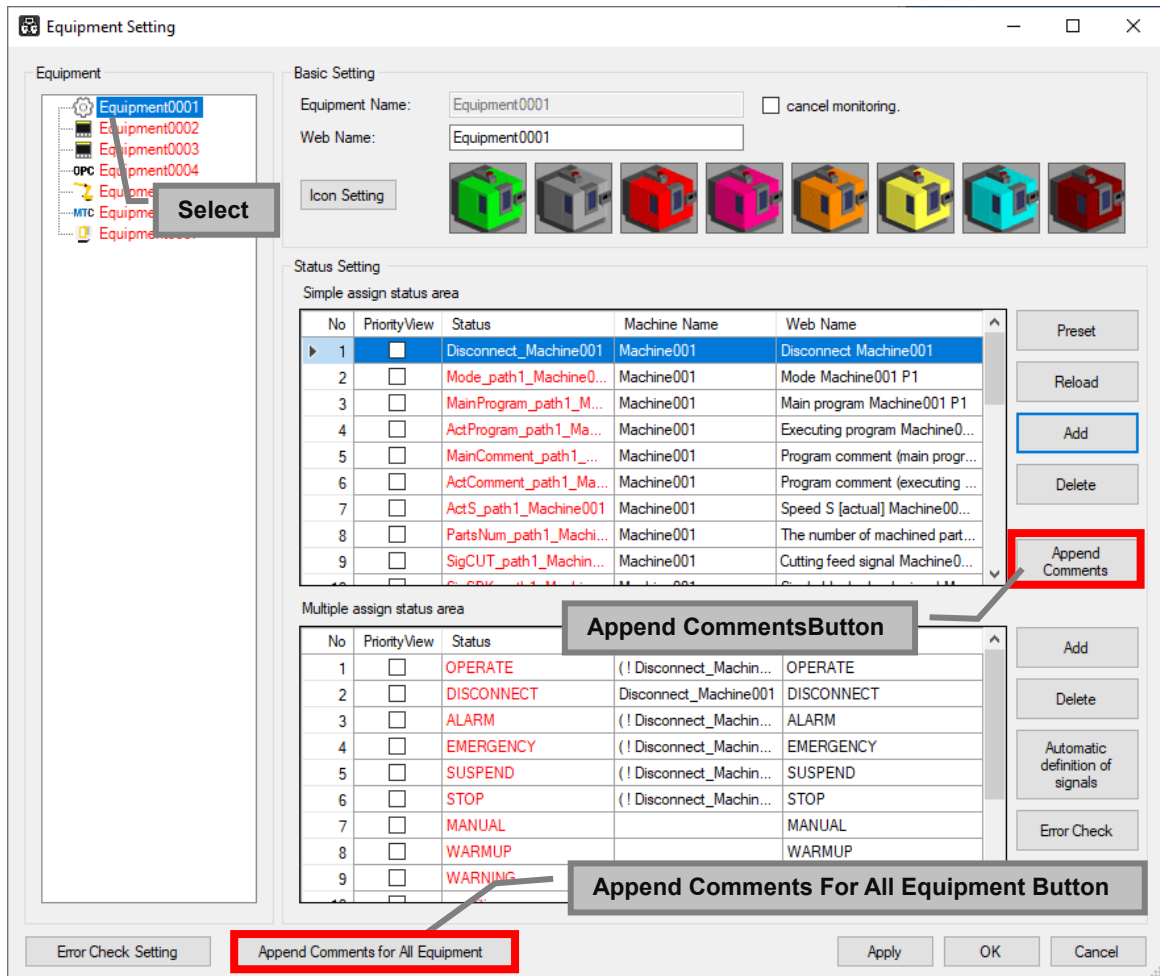
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	268/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION				Sheet.	269/868
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						

4.7.6.2 Append Comment to Web Name in Simple Assign Status Area

Comments in IO signals/register data of FanucRobot type machines can be appended to the [Web Name] column in the simple assign status area after retrieved from those machines.

The [Web Name] column appends a comment to a Web Name only when it is displayed in the language set for the [Web name language] in the [System Setting] window.



- Append Comments**
 Clicking this button retrieves comments in IO signals/register data from machines in the equipment selected in the [Equipment] tree and appends those retrieved comments to the [Web Name] column in the simple assign status area.
 If the selected equipment is not assigned with IO signals/registers of FanucRobot machine type in its simple assign status areas, this button is disabled and thus not clickable.
- Append Comments for All Equipment**
 Clicking this button retrieves comments in IO signals/register data from machines in the registered equipment and appends those retrieved comments to the [Web Name] column in their simple assign status areas.
 If none of the registered equipment is assigned with IO signal/register of FanucRobot type machines in its simple assign status area, this button is disabled and thus not clickable.

20	Nov. 22, 2023	uwano	Revised for MT-LINK ⁱ Version 4.0	Title	FANUC MT-LINK ⁱ AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK ⁱ Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK ⁱ Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK ⁱ Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK ⁱ Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK ⁱ Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK ⁱ Version 3.7	FANUC CORPORATION			Sheet. 270/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

NOTE

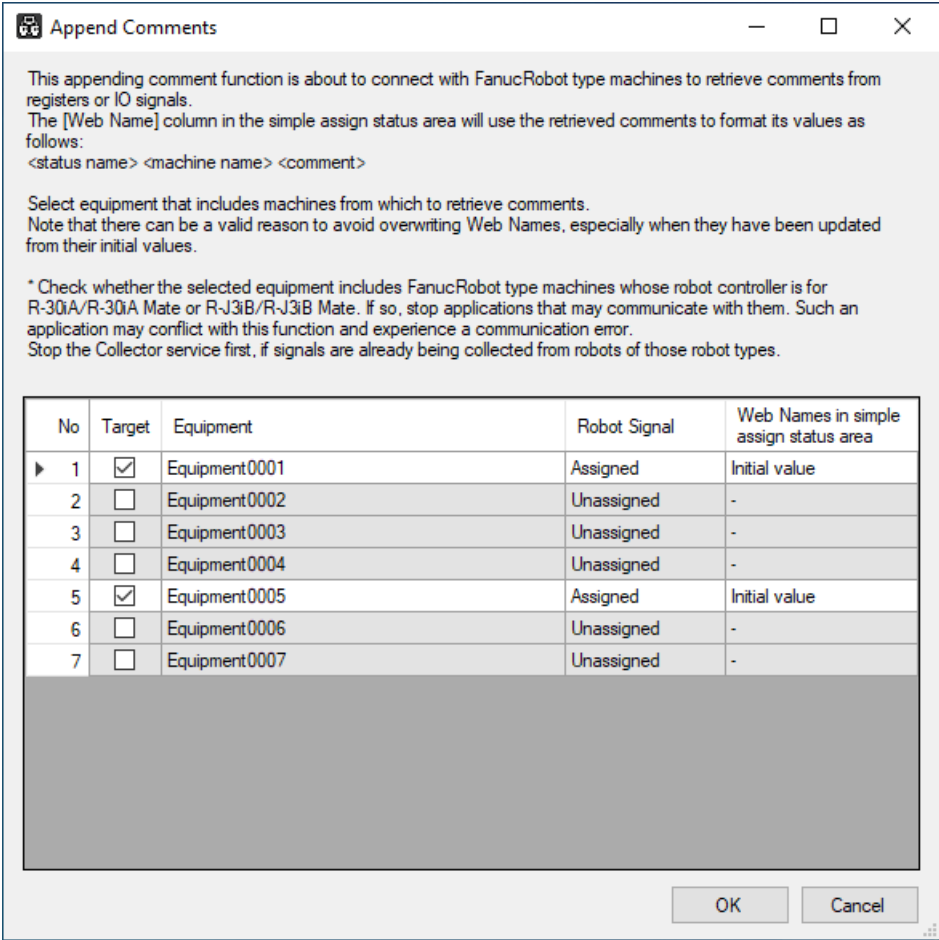
When appending comments from the following FANUC Robot Series machines, stop applications that may communicate with those machines. Such an application may conflict with this function and experience a communication error:

- R-30*i*A / R-30*i*A Mate
- R-J3*i*B / R-J3*i*B Mate

If signals are already being collected from robots of those robot types, stop the Collector service before appending comments. For the details on how to stop the service, refer to the "**START/STOP SYSTEM**" section.

■ Appending comments to Web Names in the simple assign status area

- (1) In the [Equipment] tree, select equipment that is assigned with FanucRobot signals and click the [Append Comments] button.
To append comments to all the FanucRobot signals assigned to registered equipment at once, click the [Append Comments for All Equipment] button.
- (2) When the following confirmation window (the [Append Comments] window) appears, select equipment that is assigned with FanucRobot signals to which to append comments.



The dialog box titled "Append Comments" contains the following text:

This appending comment function is about to connect with FanucRobot type machines to retrieve comments from registers or IO signals.
The [Web Name] column in the simple assign status area will use the retrieved comments to format its values as follows:
<status name> <machine name> <comment>

Select equipment that includes machines from which to retrieve comments.
Note that there can be a valid reason to avoid overwriting Web Names, especially when they have been updated from their initial values.

* Check whether the selected equipment includes FanucRobot type machines whose robot controller is for R-30*i*A/R-30*i*A Mate or R-J3*i*B/R-J3*i*B Mate. If so, stop applications that may communicate with them. Such an application may conflict with this function and experience a communication error.
Stop the Collector service first, if signals are already being collected from robots of those robot types.

No	Target	Equipment	Robot Signal	Web Names in simple assign status area
▶ 1	☒	Equipment0001	Assigned	Initial value
2	☐	Equipment0002	Unassigned	-
3	☐	Equipment0003	Unassigned	-
4	☐	Equipment0004	Unassigned	-
5	☒	Equipment0005	Assigned	Initial value
6	☐	Equipment0006	Unassigned	-
7	☐	Equipment0007	Unassigned	-

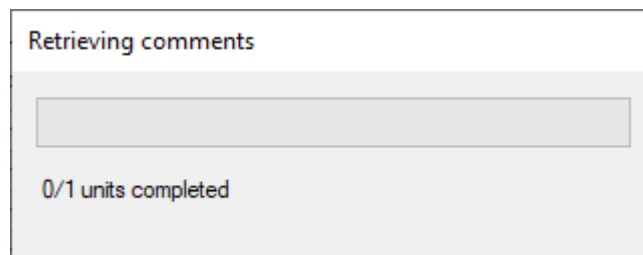
At the bottom right are "OK" and "Cancel" buttons.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 271/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

■ The [Append Comments] window

- **Target**
To append comments for machines in specific equipment, select its [Target] check box in the list shown in this window. Machines in equipment whose [Target] check box is cleared will be excluded from appending comments.
If equipment has no FanucRobot signals assigned to it, its row will be gray out and thus this check box will be unselectable.
* To select or clear multiple check boxes at once, select rows, right-click on a selected row to open a context-menu list, and select [ON] or [OFF] from it.
- **Equipment**
This column populates names of registered equipment.
- **Robot Signal**
This column shows whether the equipment is assigned with FanucRobot signals:
[Assigned]: It is assigned with one or more FanucRobot signals.
[Unassigned]: It has no FanucRobot signal assigned to it.
- **Web Names in simple assign status area**
This column shows whether Web Names are updated from their initial values:
[Initial value]: All the Web names of the IO signals/register data are left unchanged.
[Updated]: One or more of them are updated.
[-]: No FanucRobot signal is assigned to the equipment.
- **OK**
Clicking this button appends the retrieved comments to the [Web Name] columns in the simple assign status areas for the targeted equipment.
- **Cancel**
Clicking this button closes the [Append Comments] window without appending comments for the targeted equipment.

- (3) Clicking OK on the [Append Comments] window connects with FanucRobot type machines to retrieve comments from them. For checking the progress of comment retrieval, the following message box keeps displaying a progress bar until retrieving comments is completed.
Comment retrieval will be performed with all of the applicable machines even if it fails to retrieve comments from some machines.



The [Web Name] column in the simple assign status area will use successfully retrieved comments to format its values as follows:

<status name> <machine name> <comment>

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	272/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

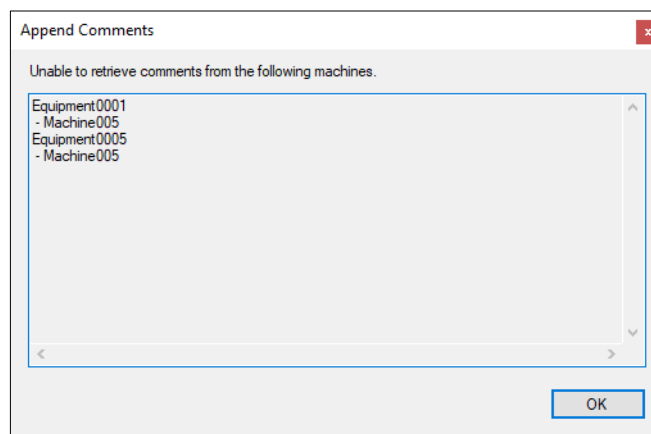
Examples of appending patterns) Web Names in the simple assign status area with retrieved comments

Result of retrieving comments	Web Names in simple assign status area* ²	
	Initial value Example: DATA_NumReg1 Machine001	Updated Example: Rotation
Successful* ¹ (a comment is retrieved)	DATA_NumReg1 Machine001 Temperature	DATA_NumReg1 Machine001 Temperature
Successful (no comment is retrieved)	DATA_NumReg1 Machine001	DATA_NumReg1 Machine001
Failed (failed to connect or retrieve)	DATA_NumReg1 Machine001	Rotation

*¹ The comment "Temperature" is already appended.

*² Updated portions from the previous Web Names are shown in blue in the table.

Even if MT-LINK*i* failed to retrieve just a single comment, the following message (the [Append Comments] dialog box) appears listing equipment and its machines from which it was unable to retrieve the comment. In such a case, check the connection settings and comment settings on the listed machines and try again to append those unretrieved comments. However, other successfully retrieved comments are appended to the [Web Name] column in the simple assign status area.



NOTE

This function excludes a comment from retrieval that is set in an IO signal with address SI or SO and the number 0.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION		Sheet. 273/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

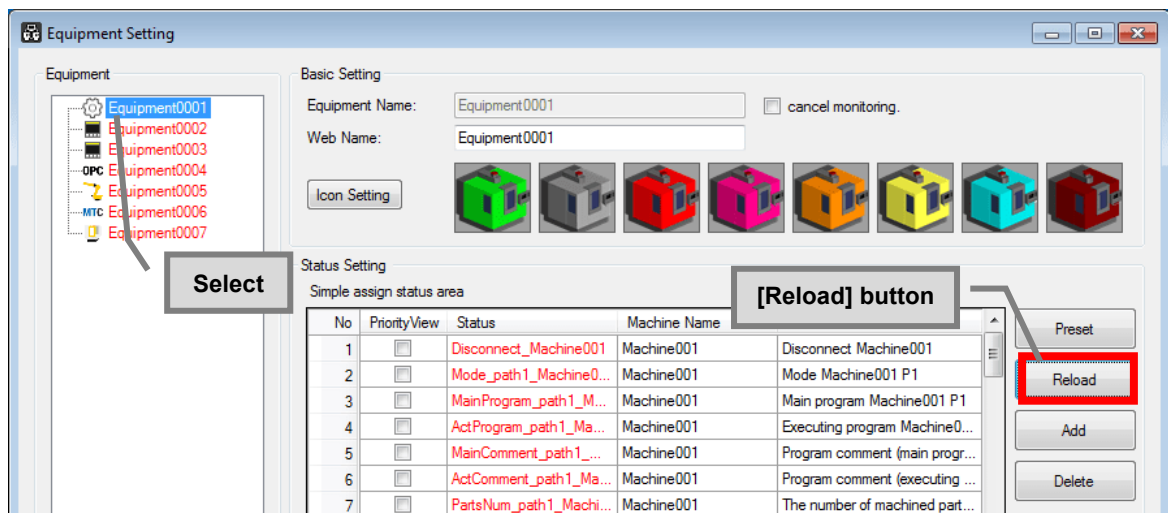
4.7.6.3 Reload Simple Assign Status

The [Preset] button enables the user to select a machine and assign all the applicable statuses to the machine at once. However, clicking this button discards the existing status settings, so that the settings such as the [PriorityView] or the [Web Name] column will be initialized. In this case, use the [Reload] button instead of the [Preset] button to update the simple assign statuses while preserving the existing status settings. This is applicable in a case as stated below.

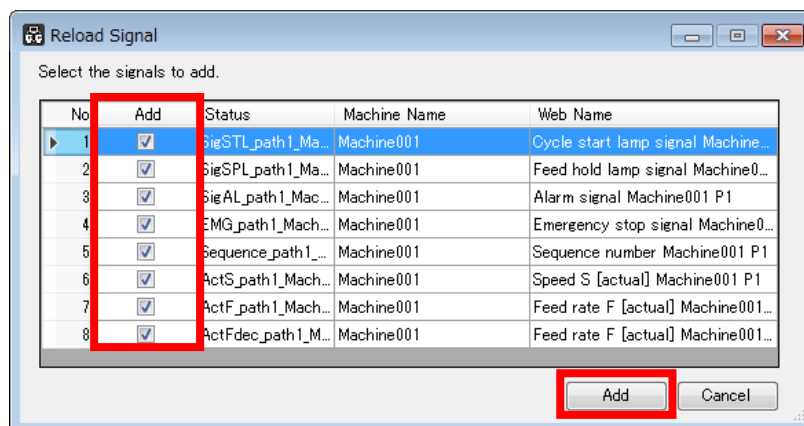
- After presetting, the user modified the [PriorityView] or the [Web Name] column in the [Simple assign status area] table and then used the [Machine Setting] window to add more signals to collect.

Follow the steps below to reload the simple assign statuses.

- (1) Select the equipment to reload the signals in the [Equipment] tree, and click the [Reload] button on the right of the [Simple assign status area].



- (2) [Reload Signal] dialog box is displayed. Select the [Add] check box for the statuses to add.



- In the default setting, all the [Add] check boxes are selected.
- Two or more [Add] check boxes can be selected or cleared simultaneously, by selecting multiple rows, right-clicking, and selecting [ON] or [OFF] from the menu displayed.
- Statuses sent to DB are displayed in red characters, and the newly added statuses are displayed in black characters.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	274/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- When there is no status registered in the [Simple assign status area], the [Reload Signal] dialog box will not be displayed,
- When there are no signals available to be added, a message saying "There are no signals that can be added." is displayed, and the [Reload Signal] window will not be displayed.

(3) When the selection of the statuses to add is completed, click the [Add] button.

The selected statuses are added to [Simple assign status area].

When the [Cancel] button is clicked, the reload signal process will not be executed, and the [Reload Signal] dialog box is closed.

NOTE

Multiple assign statuses are not updated by the reload signal function.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 275/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

4.7.6.4 Define Multiple Assign Status Automatically

Clicking the [Automatic definition of signals] button automatically assigns signal definitions to basic equipment statuses so that the statuses can support multiple machines and multiple paths. The statuses and their definitions are listed in the [Multiple assign status area] of the selected equipment.

- You need to preset statuses in the [Single assign status area] before activating the [Automatic definition of signals] button; otherwise the button is inactive (grayed out).
- This automatic functionality applies only to basic statuses such as OPERATE, DISCONNECT, and ALARM but does not to statuses you defined by clicking the [Add] button.

NOTE

If one or more of the machines consisting of one piece of equipment are disconnected, the equipment will turn to the DISCONNECT status.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 276/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

The following examples show how this functionality works.

Note that this functionality is available even when different machine types co-exist in a single equipment unit.

In the examples listed below, one piece of equipment is composed of two FanucCNC type machines.

(1) OPERATE/DISCONNECT/ALARM/EMERGENCY/SUSPEND/STOP

The signal definitions for those statuses are made with simple assign statuses concatenated by OR ("||") operator:

When you have registered a machine (Machine1) with one path and another machine (Machine2) with two paths, the signal definition will be <Machine1path1> || <Machine2path1> || <Machine2path2>.

Example) Before clicking the [Automatic definition of signals] button:

```
( ! Disconnect_Machine1 && ( CncState_path1_Machine1 == "OPERATE" ) )
```

After clicking the [Automatic definition of signals] button:

```
( ! Disconnect_Machine1 && ( CncState_path1_Machine1 == "OPERATE" ) ) ||
```

```
( ! Disconnect_Machine2 && ( CncState_path1_Machine2 == "OPERATE" ) ) ||
```

```
( ! Disconnect_Machine2 && ( CncState_path2_Machine2 == "OPERATE" ) )
```

(2) condition

The signal definition for this status is made with multiple assign statuses.

The results before clicking the [Automatic definition of signals] button will be the same as the results after clicking the [Automatic definition of signals].

Example) DISCONNECT ? "DISCONNECT"

: EMERGENCY ? "EMERGENCY"

: ALARM ? "ALARM"

: OPERATE ? "OPERATE"

: SUSPEND ? "SUSPEND"

: "STOP"

NOTE

The automatic definition set for "condition" by clicking the [Automatic definition of signals] button will be as follows depending on the machine type.

- In case the equipment includes FanucCNC or FanucRobodrill

DISCONNECT ? "DISCONNECT"

: EMERGENCY ? "EMERGENCY"

: ALARM ? "ALARM"

: OPERATE ? "OPERATE"

: SUSPEND ? "SUSPEND"

: "STOP"

- In case the equipment does not include FanucCNC or FanucRobodrill, but includes FanucRobots

DISCONNECT ? "DISCONNECT"

: EMERGENCY ? "EMERGENCY"

: ALARM ? "ALARM"

: OPERATE ? "OPERATE"

: "STOP"

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 277/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

- In case the equipment includes only OPCUAServers or MTConnectAgents
DISCONNECT ? "DISCONNECT"
: ALARM ? "ALARM"
: "OPERATE "

(3) ProductResultNumber

The signal definition for ProductResultNumber is made with simple assign statuses concatenated by ADD ("+") operator:

When you have registered a machine (Machine1) with one path (path1) and another machine (Machine2) with two paths, the signal definition will be <Machine1path1> + <Machine2path1> + <Machine2 path2>.

Example:

Before clicking the [Automatic definition of signals] button:

PartsNum_path1_Machine1

After clicking the [Automatic definition of signals] button:

PartsNum_path1_Machine1 + PartsNum_path1_Machine2 + PartsNum_path2_Machine2

(4) WARNING

The signal definition for WARNING is made with simple assign statuses concatenated by OR ("||") operator: When you have registered a machine (Machine1) with one path (path1) and another machine (Machine2) with two paths, the signal definition will be <Machine1path1> || <Machine2path1> || <Machine2path2>.

Example:

Before clicking the [Automatic definition of signals] button:

(! Disconnect_Machine1 && (CncWarning_path1_Machine1 != "NONE") && (CncWarning_path1_Machine1 != ""))

After clicking the [Automatic definition of signals] button:

(! Disconnect_Machine1 && (CncWarning_path1_Machine1 != "NONE") && (CncWarning_path1_Machine1 != "")) || (! Disconnect_Machine2 && (CncWarning_path1_Machine2 != "NONE") && (CncWarning_path1_Machine2 != "")) || (! Disconnect_Machine2 && (CncWarning_path2_Machine2 != "NONE") && (CncWarning_path2_Machine2 != ""))

(5) ProductName

The signal definition for ProductName is defined by using the signal MainComment of the path1 of the first registered machine (Machine1). The machine referred to as the first here, is specifically the first machine you selected with the [Preset] button:

MainComment_path1_Machine1.

Note that the signal definitions for ProductName before and after applying the automatic definition functionality are coincidentally identical as follows.

Example:

Before and after clicking the [Automatic definition of signals] button:

MainComment_path1_Machine001

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	278/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

4.7.6.5 Add, Edit or Delete Multiple Assign Status

The following basic statuses are pre-registered in the [Multiple assign status area] table.
Note that those basic statuses cannot be deleted.

<Equipment Operating Statuses>

Status	Type	Web Name	Remarks
OPERATE	Bool	OPERATE	The Web names of these statuses will be displayed in the following locations. [Monitoring - Overlook] screen Operation status string above the equipment icon [Monitoring - Equipment Monitoring] and [Monitoring - Equipment Monitoring - Detail] screens Operation status string under the equipment icon [Results - Operational Results] screen Operation status string on the operational results band graph
DISCONNECT	Bool	DISCONNECT	
ALARM	Bool	ALARM	
EMERGENCY	Bool	EMERGENCY	
SUSPEND	Bool	SUSPEND	
STOP	Bool	STOP	
MANUAL	Bool	MANUAL	
WARMUP	Bool	WARMUP	
WARNING	Bool	WARNING	When this status turns to true, the icon of the equipment in warning starts blinking on the [Monitoring - Overlook] screen

<Other Statuses>

Status	Type	Web Name	Description
condition	String	State	Used in the graphs to be displayed on Web screens ([Results - Group Results] and [Results - Operational Results]) Used in the operation status strings to be displayed on Web screens ([Monitoring - Overlook], [Monitoring - Equipment Monitoring], and [Monitoring - Equipment Monitoring - Detail])
ProductName	String	Product Name	Product names to be displayed on Web screens ([Results - Production Results] and [Results - Operational Results]) The maximum length of the value is 128 bytes (NOTE).
ProductResultNumber	Int32	Number of Machined Products	Used to calculate the product results to be displayed on Web screen ([Results - Production Results]) For details, refer to the " DISPLAY OF PRODUCTION RESULTS IN PRODUCTION RESULTS SCREEN " section.
ProductSerialNumber	String	Product Serial Number	Product serial numbers to be displayed on Web screens ([Results - Operational Results] and [Monitoring - Equipment Monitoring]) The maximum length of the value is 80 bytes (NOTE).
OperatorID	String	Operator ID	Operator IDs to be displayed on Web screens ([Results - Operational Results] and [Monitoring - Equipment Monitoring]) The maximum length of the value is 512 bytes (NOTE).

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 279/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

NOTE

If the length of this value exceeds the maximum length, it will be truncated and be concatenated with an ellipsis (or three dots).

In case when a truncation falls in the middle of a multi-byte character, it may be displayed corrupted.

Example) when a value of OperatorID signal is longer than 512 bytes, the value will be truncated at offset 509 bytes, and three dots will be filled in the remaining three bytes (offset 510 to 512).

The default formulae for the multiple assign status of each machine type are shown below.

<Machine type: FanucCNC (Machine001)>

Status	Type	Web Name	Signal Definition
OPERATE	Bool	Operating	(! Disconnect_Machine001 && (CncState_path1_Machine001 == "OPERATE"))
DISCONNECT	Bool	Disconnected	Disconnect_Machine001
ALARM	Bool	In Alarm Condition	(! Disconnect_Machine001 && (CncState_path1_Machine001 == "ALARM"))
EMERGENCY	Bool	Emergency Stopped	(! Disconnect_Machine001 && (CncState_path1_Machine001 == "EMERGENCY"))
SUSPEND	Bool	Suspended	(! Disconnect_Machine001 && (CncState_path1_Machine001 == "SUSPEND"))
STOP	Bool	Stopped	(! Disconnect_Machine001 && (CncState_path1_Machine001 == "STOP"))
MANUAL	Bool	Manual Operating	
WARMUP	Bool	Warming Up	
WARNING	Bool	Warning	(! Disconnect_Machine001 && (CncWarning_path1_Machine001 != "NONE") && (CncWarning_path1_Machine001 != ""))
condition	String	Condition	DISCONNECT ? "DISCONNECT" : EMERGENCY ? "EMERGENCY" : ALARM ? "ALARM" : OPERATE ? "OPERATE" : SUSPEND ? "SUSPEND" : "STOP"
ProductName	String	Product Name	MainComment_path1_Machine001
ProductResultNumber	Int32	Number of Machined Products	PartsNum_path1_Machine001
ProductSerialNumber	String	Product Serial Number	
OperatorID	String	Operator ID	

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	280/868

<Machine type: OPCUAServer (Machine002)>

Status	Type	Web Name	Signal Definition
OPERATE	Bool	Operating	
DISCONNECT	Bool	Disconnected	Disconnect_Machine002
ALARM	Bool	In Alarm Condition	(! Disconnect_Machine002 && OpcUaAlarm_Machine002)
EMERGENCY	Bool	Emergency Stopped	
SUSPEND	Bool	Suspended	
STOP	Bool	Stopped	
MANUAL	Bool	Manual Operating	
WARMUP	Bool	Warming Up	
WARNING	Bool	Warning	(! Disconnect_Machine002 && OpcUaWarning_Machine002)
condition	String	Condition	DISCONNECT ? "DISCONNECT" : ALARM ? "ALARM" : "OPERATE"
ProductName	String	Product Name	
ProductResultNumber	Int32	Number of Machined Products	
ProductSerialNumber	String	Product Serial Number	
OperatorID	String	Operator ID	

<Machine type: FanucRobot (Machine003)>

Status	Type	Web Name	Signal Definition
OPERATE	Bool	Operating	(! Disconnect_Machine003 && (RobotState_Machine003 == "OPERATE"))
DISCONNECT	Bool	Disconnected	Disconnect_Machine003
ALARM	Bool	In Alarm Condition	(! Disconnect_Machine003 && (RobotState_Machine003 == "ALARM"))
EMERGENCY	Bool	Emergency Stopped	(! Disconnect_Machine003 && (RobotState_Machine003 == "EMERGENCY"))
SUSPEND	Bool	Suspended	
STOP	Bool	Stopped	(! Disconnect_Machine003 && (RobotState_Machine003 == "EMERGENCY"))
MANUAL	Bool	Manual Operating	
WARMUP	Bool	Warming Up	
WARNING	Bool	Warning	
condition	String	Condition	DISCONNECT ? "DISCONNECT" : EMERGENCY ? "EMERGENCY" : ALARM ? "ALARM" : OPERATE ? "OPERATE" : "STOP"
ProductName	String	Product Name	
ProductResultNumber	Int32	Number of Machined Products	
ProductSerialNumber	String	Product Serial Number	
OperatorID	String	Operator ID	

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue
FANUC CORPORATION					Sheet.	281/868	

<Machine type: MTConnectAgent (Machine004)>

Status	Type	Web Name	Signal Definition
OPERATE	Bool	Operating	
DISCONNECT	Bool	Disconnected	Disconnect_Machine004
ALARM	Bool	In Alarm Condition	(! Disconnect_Machine004 && Fault_Machine004)
EMERGENCY	Bool	Emergency Stopped	
SUSPEND	Bool	Suspended	
STOP	Bool	Stopped	
MANUAL	Bool	Manual Operating	
WARMUP	Bool	Warming Up	
WARNING	Bool	Warning	(! Disconnect_Machine004 && Warning_Machine004)
condition	String	Condition	DISCONNECT ? "DISCONNECT" : ALARM ? "ALARM" : "OPERATE"
ProductName	String	Product Name	
ProductResultNumber	Int32	Number of Machined Products	
ProductSerialNumber	String	Product Serial Number	
OperatorID	String	Operator ID	

<Machine type: FanucRobodrill (Machine005)>

Status	Type	Web Name	Signal Definition
OPERATE	Bool	Operating	(! Disconnect_Machine005 && (CncState_path1_Machine005 == "OPERATE"))
DISCONNECT	Bool	Disconnected	Disconnect_Machine005
ALARM	Bool	In Alarm Condition	(! Disconnect_Machine005 && (CncState_path1_Machine005 == "ALARM"))
EMERGENCY	Bool	Emergency Stopped	(! Disconnect_Machine005 && (CncState_path1_Machine005 == "EMERGENCY"))
SUSPEND	Bool	Suspended	(! Disconnect_Machine005 && (CncState_path1_Machine005 == "SUSPEND"))
STOP	Bool	Stopped	(! Disconnect_Machine005 && (CncState_path1_Machine005 == "STOP"))
MANUAL	Bool	Manual Operating	
WARMUP	Bool	Warming Up	
WARNING	Bool	Warning	(! Disconnect_Machine005 && (CncWarning_path1_Machine005 != "NONE") && (CncWarning_path1_Machine005 != ""))
condition	String	Condition	DISCONNECT ? "DISCONNECT" : EMERGENCY ? "EMERGENCY" : ALARM ? "ALARM" : OPERATE ? "OPERATE" : SUSPEND ? "SUSPEND" : "STOP"
ProductName	String	Product Name	MainComment_path1_Machine005
ProductResultNumber	Int32	Number of Machined Products	PartsNum_path1_Machine005
ProductSerialNumber	String	Product Serial Number	
OperatorID	String	Operator ID	

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue
FANUC CORPORATION					Sheet.	282/868	

NOTE

- 1 If the signal definition status within "condition" (such as OPERATE) is undefined (blank, "null") or a definition that is unable to recognize, the condition is set to "DISCONNECT" and the status of the equipment is displayed as "Disconnected."
- 2 When editing signal definition of equipment statuses (such as DISCONNECT and OPERATE), define each status exclusively so that the state of the status and the state displayed by the value of "condition" match.
- 3 The signal definition of product serial number and operator ID are not set automatically. In order to collect product serial number and operator ID on MT-LINK i , you need to make settings to assign signal for product serial number to the status for product serial number (ProductSerialNumber), and assign signal for operator ID to the status for operator ID (OperatorID). As for the setting procedures, refer to the tables (operational expressions available for signal definition) in the pages to follow, and the steps to add, edit or delete a multiple assign status one by one stated in a NOTE later in this subsection.

The following table shows operational expressions available for signal definition.

Operator/Method	Example Expression	Example Evaluation
Arithmetic operations	Returns the operational evaluation of the first and the second operands	
Addition "+" Subtraction "-" Multiplication "*" Division "/"	Example: "equipment status name: Signal_Int, type: Int32, value: 100" Signal_Int + 1 Signal_Int - 1 Signal_Int * 2 Signal_Int / 2	101 99 200 50
Logical operations	Returns the logical evaluation of the first and the second operands	
Logical multiply "&&" Logical addition " " Logical negation "!"	Example: "equipment status name: Signal_Bool, type: Bool, value: false" When both the first and the second operands are true, the evaluation is true; otherwise, false. Signal_Bool && false Signal_Bool && true When both the first and the second operands are false, the evaluation is false, otherwise, true. Signal_Bool false Signal_Bool true When the value is false, the evaluation is true, otherwise, false. ! Signal_Bool	 false false false true true
Comparison operations	Returns the comparison evaluation of the first and the second operands	

20	Nov. 22, 2023	uwano	Revised for MT-LINK i Version 4.0	Title	FANUC MT-LINK i AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK i Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK i Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK i Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK i Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK i Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK i Version 3.7	FANUC CORPORATION		Sheet. 283/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

Operator/Method	Example Expression	Example Evaluation
Equal to "==" Not equal to "!=" Smaller than "<" Larger than ">" Equal or smaller than "<=" Equal or larger than ">="	Example: "equipment status name: Signal_Int, type: Int32, value: 100" Signal_Int == 100 Signal_Int != 100 Signal_Int < 100 Signal_Int > 100 Signal_Int <= 100 Signal_Int >= 100	true false false false true true
Ternary operation Conditional expression "?" ":"	Returns either the second or the third operand depending on the evaluation of the first operand * The type of return value should be casted to the type of the target equipment status. Example 1: "equipment status name: Signal_Bool, type: Bool, value: false" When Signal_Bool == false is true: true, When Signal_Bool == false is false: false (Signal_Bool == false) ? true : false (Signal_Bool != false) ? true : false Example 2: "equipment status name: Signal_String, type: String, value: 'on'" (Signal_String == "on") ? "on" : "off" (Signal_String != "on") ? "on" : "off"	 true false on off
Typecast (numerical value of different type) "(type name) numerical type status name" "Convert.ToInt32 (Numerical type status name)" "Convert.ToDouble (Numerical type status name)"	Casts the second operand to the specified data type at the first operand. Example 1: "equipment status name: Signal_Double, type: Double, value: 1.123" The second operand Signal_Double is casted from Double to Int, thus it is rounded to an integer. (Int32) Signal_Double Convert.ToInt32 (Signal_Double) Example 2: "equipment status name: Signal_Int, type: Int, value: 1" The second operand Signal_Int is casted from Int to Double. (Double) Signal_Int Convert.ToDouble (Signal_Int)	 1 1 1 1
Typecast (from numeric to character string)	Casts specified numeric value to character string. (Int32-->String) / (UInt32-->String) / (Double-->String)	

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 284/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

Operator/Method	Example Expression	Example Evaluation
"Convert.ToString (Numerical type status name)"	Example 1: "equipment status name: Signal Int, Type: Int32, Value: -123" Cast type of Signal_Int from Int32 to String Convert.ToString(Signal_Int) Example 2: "equipment status name: Signal UInt, Type: UInt32, Value: 123" Cast type of Signal_UInt from UInt32 to String Convert.ToString(Signal_UInt) Example 3: "equipment status name: Signal Double, Type: Double, Value: -123.45" Cast type of Signal Double from Double to String Convert.ToString(Signal_Double)	"-123" "123" "-123.45"
Typecast (from character string to numeric)	Casts specified character string to numeric value.	
"Convert.ToInt32 (String type status name)"	* In the following examples, if the equipment status name is blank, it is converted into 0. Example 1: "equipment status name: Signal_String, type: String, Value: "123"" Convert.ToInt32((Signal_String == "") ? "0" : Signal_String) Example 2: "equipment status name: Signal_String, type: String, Value: "" (blank)" Convert.ToInt32((Signal_String == "") ? "0" : Signal_String)	123 0
String split	Split character string delimited by delimiting character, and extract the character string specified by the element number.	
"String type status name .Split("delimiting character") [element number]"	Example: "equipment status name: Signal_String, type: String, Value: '0.001 0.002 0.003'" Extract character string with element numbers (0, 1, 2) from the delimited character string, and convert into Double type numerical value. * In the following examples, if the equipment status name is blank, it is converted into 0. Convert.ToDouble((Signal_String == "") ? "0" : ((String) Signal_String).Split(' ')[0]) Convert.ToDouble((Signal_String == "") ? "0" : ((String) Signal_String).Split(' ')[1]) Convert.ToDouble((Signal_String == "") ? "0" : ((String) Signal_String).Split(' ')[2])	0.001 0.002 0.003

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 285/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

Operator/Method	Example Expression	Example Evaluation
String extraction "String type status name .Substring (character position)"	Extract the character string from and after the specified character position. Example: "equipment status name: Signal_String, type: String, Value: 'A101'" Extract the numerical part from and after the character position (1), and convert into Int32 type numerical value. * In the following examples, if the equipment status name is blank, it is converted into 0. Convert.ToInt32((Signal_String == "") ? "0" : ((String) Signal_String).Substring(1))	101
String concatenation Concatenation "+"	The result of the string of first operand concatenated with the string of second operand is returned. Example 1: "name of equipment status: Signal String, Type: String, Value: ABC" Signal_String + "XYZ" Example 2: "name of equipment status: Signal String1, Type: String, Value: 123" "name of equipment status: Signal String2, Type: String, Value: 456" Signal_String1 + Signal_String2	"ABCXYZ" "123456"
Round off to negative infinity (truncation) "Math.Floor(Double type status name)"	Truncate the specified numerical value (Double) to an integer. (round off to negative infinity) * Both positive and negative numbers are rounded off to a smaller integer. Example 1: "name of equipment status: Signal Double, Type: Double, Value: 1.5" Math.Floor(Signal_Double) Example 2: "name of equipment status: Signal Double, Type: Double, Value: -1.5" Math.Floor(Signal_Double)	1 -2
Round off to 0: zero (truncation) "Math.Truncate(Double type status name)"	Truncate the specified numerical value (Double) to an integer. (round off to 0: zero) * Positive numbers are rounded off to a smaller integer. * Negative numbers are rounded off to a larger integer. Example 1: "name of equipment status: Signal Double, Type: Double, Value: 1.5" Math.Truncate(Signal_Double) Example 2: "name of equipment status: Signal Double, Type: Double, Value: -1.5" Math.Truncate(Signal_Double)	1 -1
Round off to positive infinity (upward revaluation)	Revaluate upward the specified numerical value (Double) to an integer. (round off to positive infinity) * Both positive and negative numbers are rounded off to a larger integer.	

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue
FANUC CORPORATION					Sheet.	286/868	

Operator/Method	Example Expression	Example Evaluation
"Math.Ceiling(Double type status name)"	Example 1: "name of equipment status: Signal Double, Type: Double, Value: 1.5" Math.Ceiling(Signal_Double) Example 2: "name of equipment status: Signal Double, Type: Double, Value: -1.5" Math.Ceiling(Signal_Double)	2 -1
Round off to nearest even number "Math.Round(Double type status name)" "Math.Round(Double type status name, number of decimal places)"	Round off the specified numerical value (Double) to a nearest even integer after rounding off the fractional value. * Truncate if the fractional value is less than 0.5, and reevaluate upward if 0.5 or more. However, when the fractional value is 0.5, the result will be rounded off to an even number. * The number of decimal places targeted for rounding off can be specified by the second parameter.	
	Example 1: "name of equipment status: Signal Double, Type: Double, Value: 1.5" Math.Round(Signal_Double) Example 2: "name of equipment status: Signal Double, Type: Double, Value: -1.5" Math.Round(Signal_Double) Example 3: "name of equipment status: Signal Double, Type: Double, Value: 2.5" Math.Round(Signal_Double) Example 4: "name of equipment status: Signal Double, Type: Double, Value: -2.5" Math.Round(Signal_Double) Example 5: "name of equipment status: Signal Double, Type: Double, Value: 2.345" Math.Round(Signal_Double, 3)	2 -2 2 -2 2.34
Round off "Math.Round(Double type status name, MidpointRounding.AwayFromZero)"	Round off the specified numerical value (Double) by rounding off the fractional value. * Truncate if the fractional value is less than 0.5, and reevaluate upward if 0.5 or more. * The number of decimal places targeted for rounding off can be specified by the second parameter.	

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 287/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

Operator/Method	Example Expression	Example Evaluation
"Math.Round(Double type status name, number of decimal places, MidpointRounding.AwayFromZero)"	Example 1: "name of equipment status: Signal Double, Type: Double, Value: 1.5" Math.Round(Signal_Double, MidpointRounding.AwayFromZero)	2
	Example 2: "name of equipment status: Signal Double, Type: Double, Value: -1.5" Math.Round(Signal_Double, MidpointRounding.AwayFromZero)	-2
	Example 3: "name of equipment status: Signal Double, Type: Double, Value: 2.5" Math.Round(Signal_Double, MidpointRounding.AwayFromZero)	3
	Example 4: "name of equipment status: Signal Double, Type: Double, Value: -2.5" Math.Round(Signal_Double, MidpointRounding.AwayFromZero)	-3
	Example 5: "name of equipment status: Signal Double, Type: Double, Value: 2.345" Math.Round(Signal_Double, 3, MidpointRounding.AwayFromZero)	2.35

NOTE

Following are the notes when writing the formula.

- Before and after the operator (such as "+"), a one-byte space is required.
- For the typecast, a one-byte space is required between "(type name)" and "numerical type status name."
- When writing the methods, a one-byte space is required after "(" and before ")" within the brackets.
- When delimiting the parameters with ", (comma)"s in the method, a one-byte space is required after each ", (comma)" used.
- Before and after the equipment status name, a one-byte space is required.

To add, edit or delete a multiple assign status one by one, follow the steps below:

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 288/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

NOTE

- 1 Multiple assign equipment status can be defined by using the "simple assign equipment status name" and the "multiple assign equipment status name" assigned to the same equipment.
- 2 Be careful when using multiple equipment status (example: Status B) for the definition of the multiple assign equipment status (example: Status A), as Status A might be evaluated by the previous collection value of Status B, depending on the order of definition of the two statuses in the [Multiple assign status area].
(Example)
If Status B is defined before Status A, Status A is evaluated by using the current collection value of Status B.
If Status B is defined after Status A, Status A is evaluated by using the previous collection value of Status B.
Current equipment status cannot be used for the definition of the multiple assign equipment status (self-reference).
Also, if the two equipment statuses are defined by referencing each other, value of both statuses will be undecided and will not be determined. When defining the equipment status, do not cross-refer the statuses.
- 3 The multiple assign status without the auto signal definition for default formula has no initial value.
- 4 Since OPCUAServer communicates with different types of PLC, it has no signal that expresses its unique status. Only the status of the connection (whether connected or not) is set as a default value for OPCUAServers. For OPCUAServer, set the tag for status, assign it as a signal, and enter in the [Multiple assign status area].
- 5 When the EMG signal is not correctly set for the signal setting of the machine type FanucRobot, emergency stop status is not judged correctly. If emergency stop is not judged correctly, refer to the "[Set Custom Signal of FanucRobot](#)" subsection, and check the setting.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION		Sheet. 289/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

- (1) Select any equipment name in the [Equipment] tree and click [Add] in the right area of the [Multiple assign status area] table.

Equipment Setting

Equipment

- Equipment0001
- Equipment0002
- Equipment0003
- Equipment0004
- Equipment0005
- Equipment0006
- Equipment0007

Basic Setting

Equipment Name: Equipment0001 ☐ cancel monitoring.

Web Name: Equipment0001

Icon Setting

Setting

assign status area

No	PriorityView	Status	Machine Name	Web Name
1	☐	Disconnect_Machine001	Machine001	Disconnect Machine001
2	☐	Mode_path1_Machine0...	Machine001	Mode Machine001 P1
3	☐	MainProgram_path1_M...	Machine001	Main program Machine001 P1
4	☐	ActProgram_path1_Ma...	Machine001	Executing program Machine0...
5	☐	MainComment_path1_...	Machine001	Program comment (main progr...
6	☐	ActComment_path1_Ma...	Machine001	Program comment (executing ...
7	☐	ActS_path1_Machine001	Machine001	Speed S [actual] Machine00...
8	☐	PartsNum_path1_Machi...	Machine001	The number of machined part...
9	☐	SigCUT_path1_Machin...	Machine001	Cutting feed signal Machine0...

Multiple assign status area

No	PriorityView	Status	Signal Definition	Web Name
1	☐	OPERATE	(! Disconnect_Machin...	OPERATE
2	☐	DISCONNECT	Disconnect_Machine001	DISCONNECT
3	☐	ALARM	(! Disconnect_Machin...	ALARM
4	☐	EMERGENCY	(! Disconnect_Machin...	EMERGENCY
5	☐	SUSPEND	(! Disconnect_Machin...	SUSPEND
6	☐	STOP	(! Disconnect_Machin...	STOP
7	☐	MANUAL		MANUAL
8	☐	WARMUP		WARMUP
9	☐	WARNING	(! Disconnect_Machin...	WARNING

Buttons: Preset, Reload, Add, Delete, Append Comments, Add, Delete, Automatic definition of signals, Error Check, Error Check Setting, Append Comments for All Equipment, Apply, OK, Cancel

-
- The screenshot shows the 'Signal Setting' dialog box. It has a title bar with a standard icon and window controls. The main area contains several input fields and a dropdown menu. Callouts with leader lines point to specific elements: 'Status Name' points to the 'Status Name' label and its corresponding text box; 'Web Name' points to the 'Web Name' label and its corresponding text box; 'User Signal Definition' points to the 'User Signal Definition' label and its corresponding text box; and 'Select [Signal Type]' points to the 'Signal Type' dropdown menu which currently shows 'Bool'. At the bottom right, there are 'OK' and 'Cancel' buttons, with the 'OK' button highlighted by a red rectangle.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	290/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

The value for the [Web Name] field should be entered in the language specified for the [Web name language] field in the [System Setting] window.
An entry for the [User Signal Definition] area can be defined by using the equipment status name assigned to the same equipment. Examples of setting are shown below.

Example of setting 1) When defining retrieved value of custom macro variable
Machine name: Machine001, Macro variable number: 511, Path: 1 (Equipment status name: MacroVar_511_path1_Machine001)

Signal Setting

Status Name
MacroVar_511_1

Web Name
Custom Macro Variable-511 (Path 1)

Signal Type
Double

User Signal Definition
MacroVar_511_path1_Machine001

OK Cancel

Example of setting 2) When defining the total cutting time of two machines
Machine name: Machine003, Signal name: CutTime, Path: 1 (Equipment status name: CutTime_path1_Machine003)
Machine name: Machine004, Signal name: CutTime, Path: 1 (Equipment status name: CutTime_path1_Machine004)

Signal Setting

Status Name
CutTime

Web Name
Cutting time

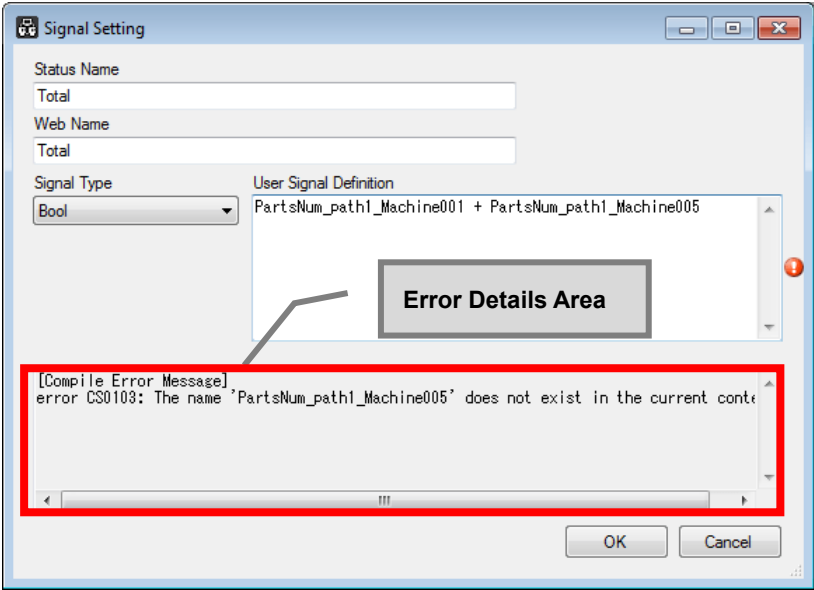
Signal Type
Double

User Signal Definition
CutTime_path1_Machine003 + CutTime_path1_Machine004

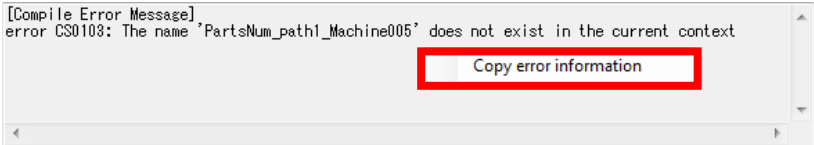
OK Cancel

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 291/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

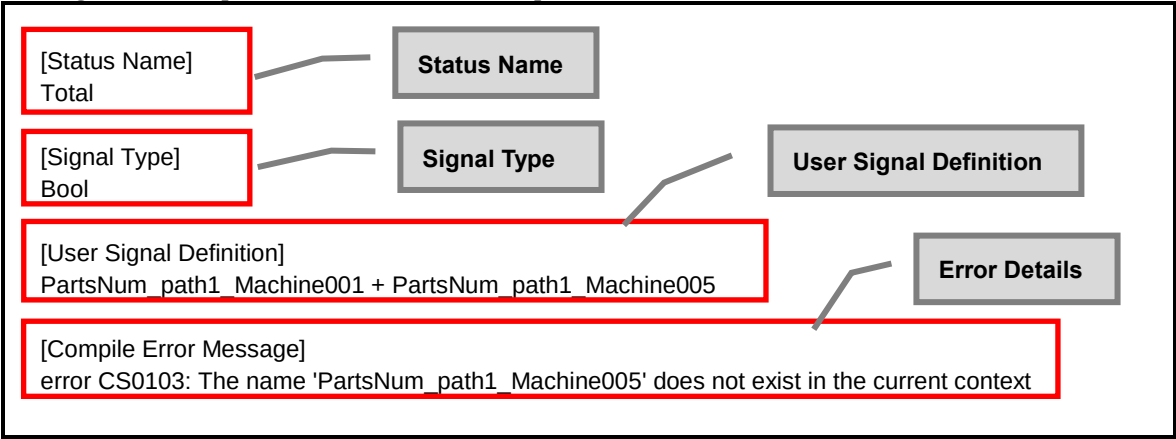
When the [Signal Setting] dialog box detects compilation errors in statuses defined in the [User Signal Definition] area, it displays an error icon ([!]) on the right side of the area and details of compilation errors in the [Error details] area.



The text in the [Error details] area, as well as the settings on the [Signal Setting] dialog box can be copied to the clipboard by right-clicking on anywhere in the area and clicking the [Copy error information] in the context menu. Save the error information in a text file for search and analysis of errors.



Following is an example of error information copied from the [Error details] area.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	292/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

(3) The added status is displayed at the bottom row of the [Multiple assign status area] table.

■ Edit a multiple assign status

To edit a status, double-click the status in the [Multiple assign status area] table. The [Signal Setting] dialog box appears. After editing the setting, click [OK].

(For basic statuses)

- [Web Name] and [User Signal Definition] are editable, and [Status Name] and [Signal Type] are not-editable. The value of the [Web Name] field should be entered in the language specified for the [Web name language] field in the [System Setting] window.

(For added statuses)

- All fields ([Status Name], [Web Name], [Signal Type], and [User Signal Definition]) are editable. The value of the [Web Name] field should be entered in the language specified for the [Web name language] field in the [System Setting] window.

Following is an example of editing a multiple assign status.

Example 1) When editing the Signal Definitions for "MANUAL" and "WARMUP"

The following statuses listed below are required to be set and edited.

Status Name	Type	Web Name	Signal Definition
MANUAL	Bool	MANUAL	Set the Signal Definition so that its value holds "true" during manual operation.
WARMUP	Bool	WARMUP	Set the Signal Definition so that its value holds "true" during warmup operation.
condition	String	condition	DISCONNECT ? "DISCONNECT" : EMERGENCY ? "EMERGENCY" : ALARM ? "ALARM" : OPERATE ? "OPERATE" : SUSPEND ? "SUSPEND" : MANUAL ? "MANUAL" : WARMUP ? "WARMUP" : "STOP"

■ Delete a multiple assign status

To delete a status, select any status other than the basic statuses in the [Multiple assign status area] table and click [Delete].

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 293/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

4.7.7 Check Equipment Status Settings

The [Equipment Setting] window automatically checks status settings when the following trigger events occur.

- (1) An equipment name is selected.
 - Clicking an equipment name in the [Equipment] tree
 - Right-clicking on an equipment name in the [Equipment] tree

- (2) A status is edited.

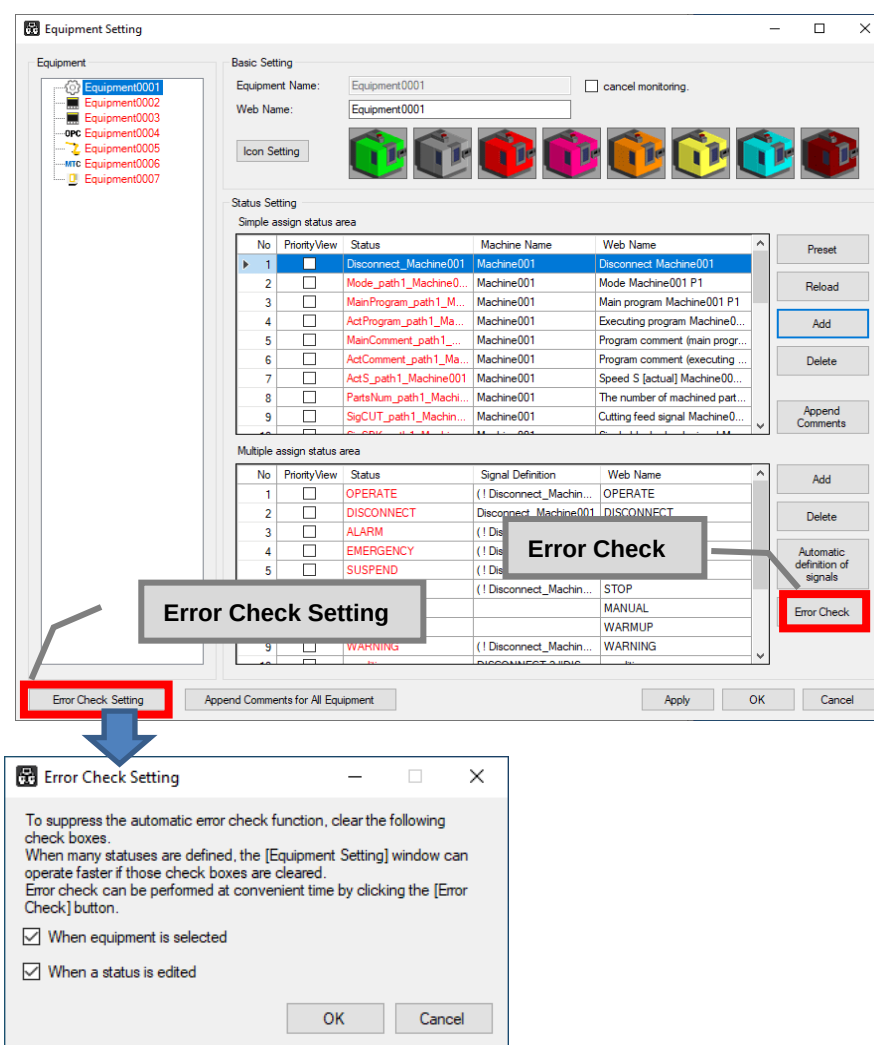
Simple assign status area

- Clicking the [Preset] button
- Clicking the [Reload] button
- Clicking the [Add] button
- Clicking the [Delete] button

Multiple assign status area

- Clicking the [Delete] button
- Double-clicking a row (with this operation, the [Signal Setting] dialog box appears)

Whether the [Equipment Setting] window automatically checks status settings is determined by the settings made in the [Error Check Setting] dialog box. Clearing both of its check boxes tells the window not to check automatically. In a case when so many equipment units and statuses are defined in the window, disabling this automatic check promotes the performance of the window.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	294/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

To open the [Error Check Setting], click the [Error Check Setting] button.

■ The [Error Check Setting] dialog box

- [When equipment is selected] check box
Clearing this check box tells the [Equipment Setting] window not to check status settings even if an equipment name is selected. By default, this check box is selected.
- [When a status is edited] check box
Clearing this check box tells the [Equipment Setting] window not to check status settings even if a status is edited. By default, this check box is selected.
- [OK] button
Clicking this button save the error check settings and closes the [Error Check Setting] dialog box.
- [Cancel] button
Clicking this button discards updates made in the [Error Check Setting] dialog box and closes it.

Even if those check boxes are cleared, clicking the [Error Check] button on the [Equipment Setting] window any time lets the window check the status settings.

NOTE

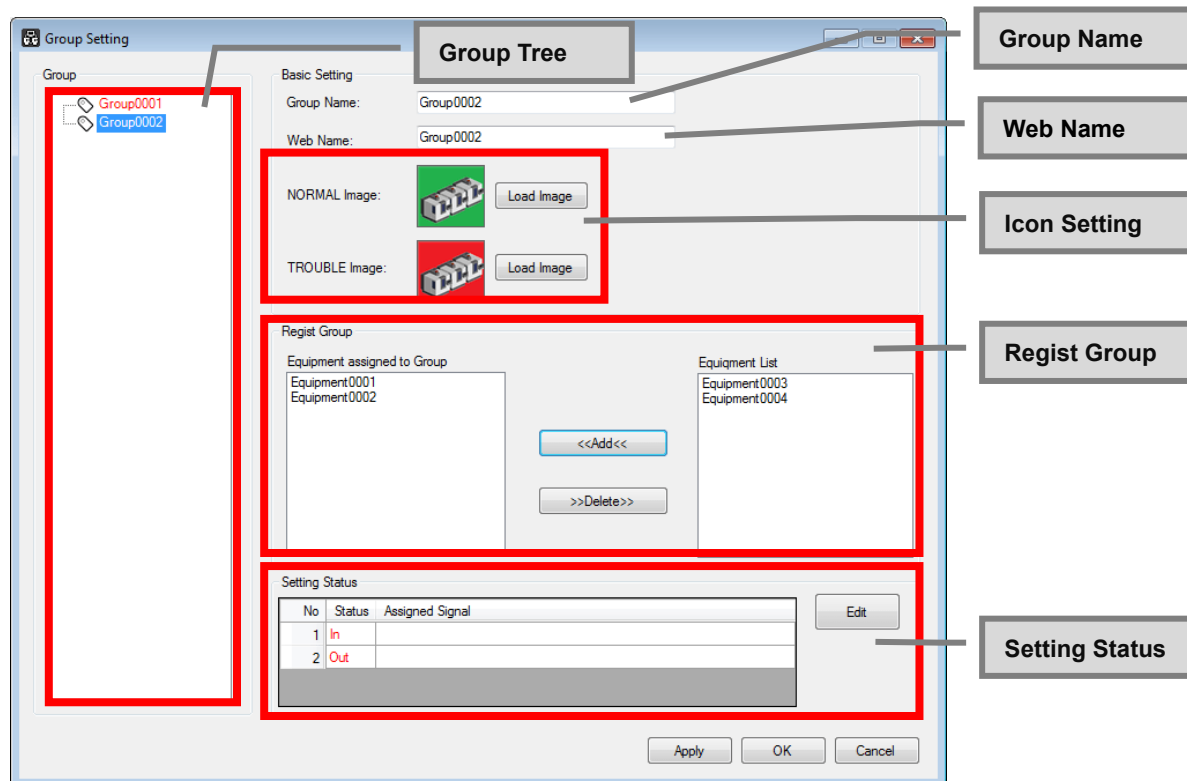
- 1 Disabling this automatic check by clearing those check boxes applies only to the [Equipment Setting] window, but not to such a child window of the [Equipment Setting] window as the [Signal Setting] dialog box.
- 2 Nevertheless, clicking [OK] or [Apply] button on the [Equipment Setting] window forces the window to check the status settings of all equipment units for errors regardless of whether those check boxes are cleared or not.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	295/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

4.8 GROUP SETTING

You can register a group in this window. Some pieces of equipment can be handled as one group. This grouping function facilitates your searching filtered by group, monitoring the status and evaluating the results by group.

- Groups can be registered up to 100.
- Set [Group Name], [Web Name], [NORMAL Image], [TROUBLE Image] for one group.
- Add equipment from [Equipment List] by group.
- Set [Status] by group.
- One piece of equipment can be assigned to several groups.



4.8.1 Steps for Group Setting

Follow the steps below to add group.

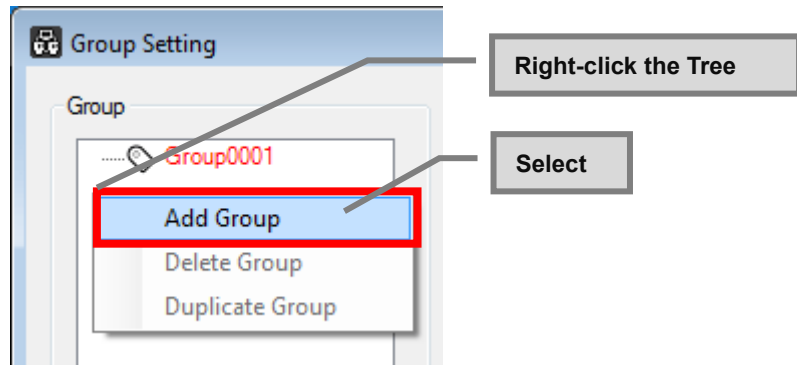
- (1) In the AdminTool Menu, selecting [Group Setting] opens the [Group Setting] window.
- (2) Add a group in the [Group] tree and set name in the [Group Name] field. For the details, refer to the "[Add Group in Group Tree Menu](#)" subsection.
- (3) Set an icon (which indicates the group status) to the added group. Refer to the "[Set Group Status Icon](#)" subsection.
- (4) Assign equipment to the added group. Refer to the "[Add Equipment to Group/Delete Equipment from Group](#)" subsection.
- (5) Assign group In/Out status to each signal. Refer to the "[Set Group Status](#)" subsection.
- (6) When you have completed above steps, click [Apply] or [OK]. Clicking [Cancel] discards the settings you made and closes the window.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	296/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

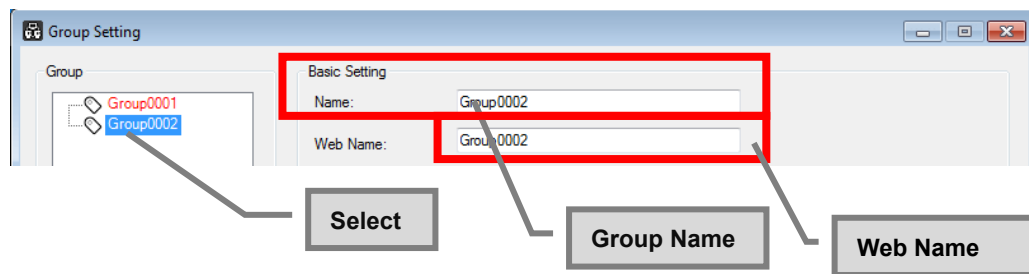
4.8.2 Add Group in Group Tree Menu

Follow the steps below to add group.

- (1) Right-click anywhere in the [Group] tree and point to [Add Group].



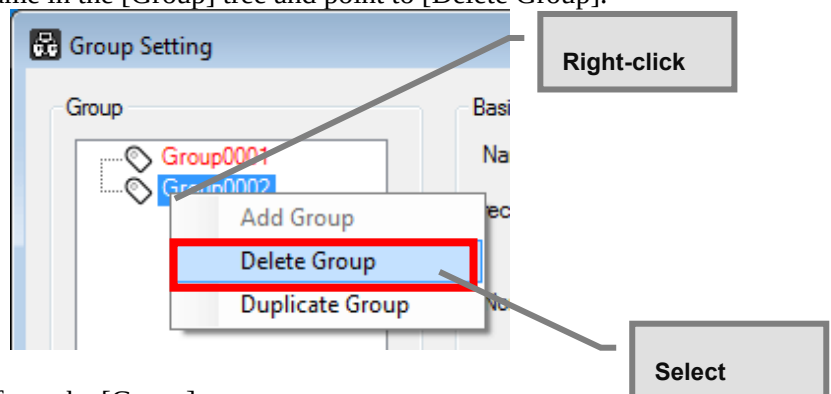
- (2) Select the new group which is added to the [Group] tree. Change the group name and Web name in the [Basic Setting] group box of the [Group Setting] window.
Group Name should start with an alphabet character and the subsequent characters can be alphanumerics, hyphens ("-") and underscores ("_"). This field can be up to 32 bytes long.
The [Web Name] is displayed on the Web screens. This field can be up to 100 bytes long. The value of this field should be entered in the language specified for the [Web name language] field in the [System Setting] window. The [Web Name] field is required and will be set to the default value of <Group name> if it is not modified.



4.8.3 Delete Group in Group Tree Menu

Follow the steps below to delete group.

- (1) Right-click the target group name in the [Group] tree and point to [Delete Group].



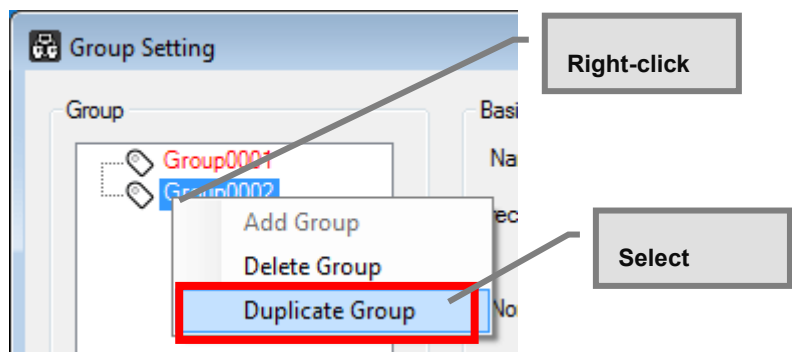
- (2) The selected group is deleted from the [Group] tree.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	297/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

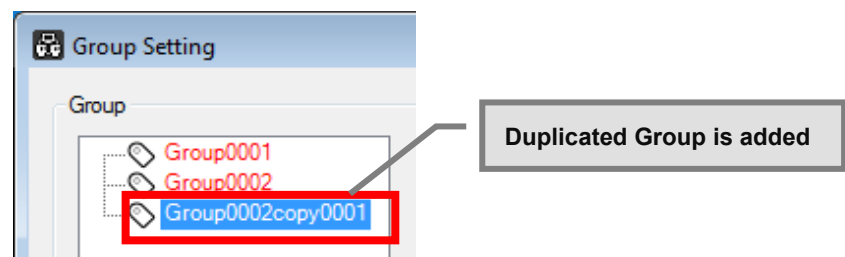
4.8.4 Duplicate Group in Group Tree Menu

Follow the steps below to duplicate group.

- (1) Right-click the target group name in the [Group] tree and point to [Duplicate Group].



- (2) The duplicated group is added to the [Group] tree.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 298/868

4.8.5 Set Group Status Icon

To set group status icon:

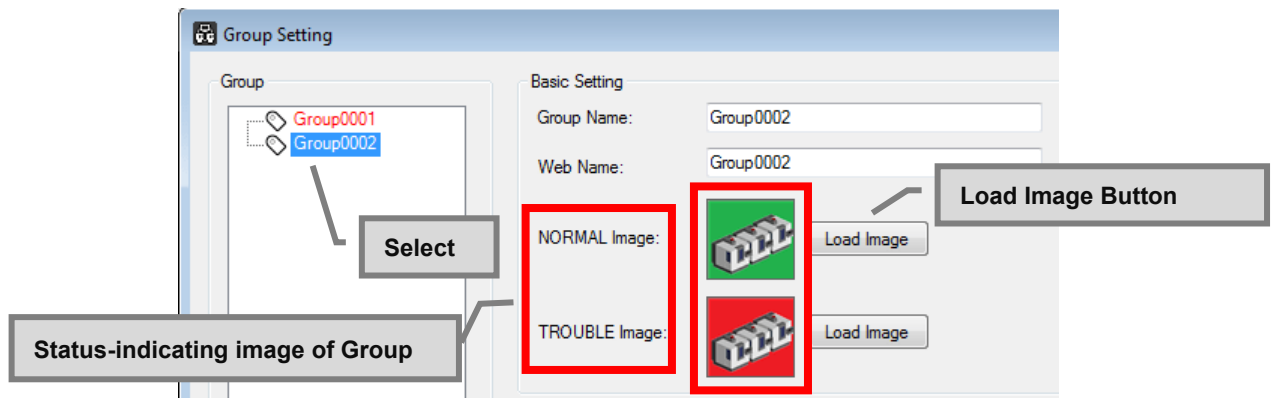
(1) To change the icon setting:

Click [Load Image] for the state icon to change. Then, the [Select Image File] dialog box appears. Select the image file to use.

The dialog box selects the following folder on the first session. This folder contains sample image files for icon images. On subsequent sessions, this dialog box selects the folder selected in the previous session.

<MT-LINK*i* installation folder>\MT-LINKiAdminTool\MT-LINKi_image\Group

- Supported image formats: png, jpg, bmp
- Prepare the image files in advance.
- Image files larger than 500 pixels wide x 500 pixels high cannot be used.



Group Operating Status	Conditions of State Transition
NORMAL	When none of the equipment units belonging to the group is in the emergency stop condition or in the alarm condition.
TROUBLE	When any of the equipment units belonging to the group is in the emergency stop condition or in the alarm condition.

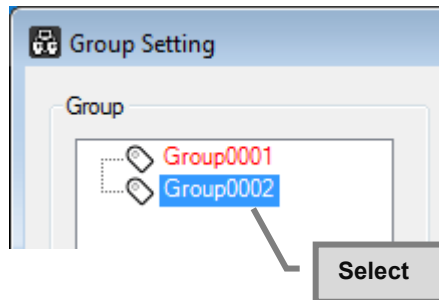
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 299/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

4.8.6 Add Equipment to Group/Delete Equipment from Group

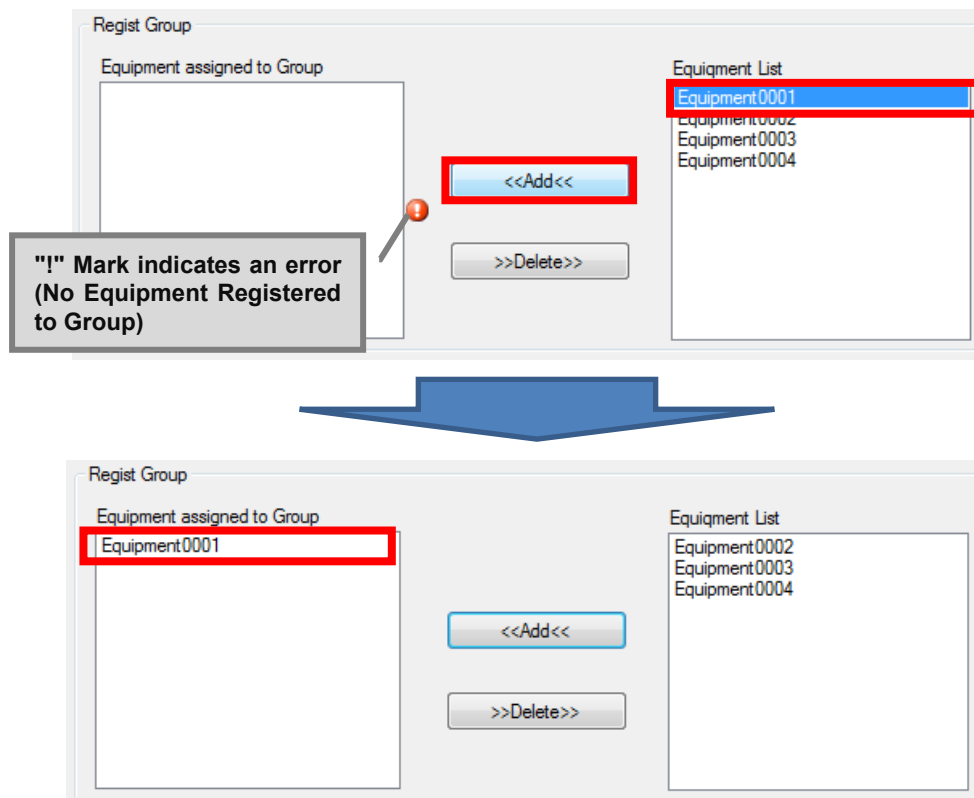
You can add/delete equipment to/from group.

■ Add equipment to group

- (1) In the [Group] tree, select the target group. The information about equipment assigned to the selected group is displayed in the [Regist Group] group box.



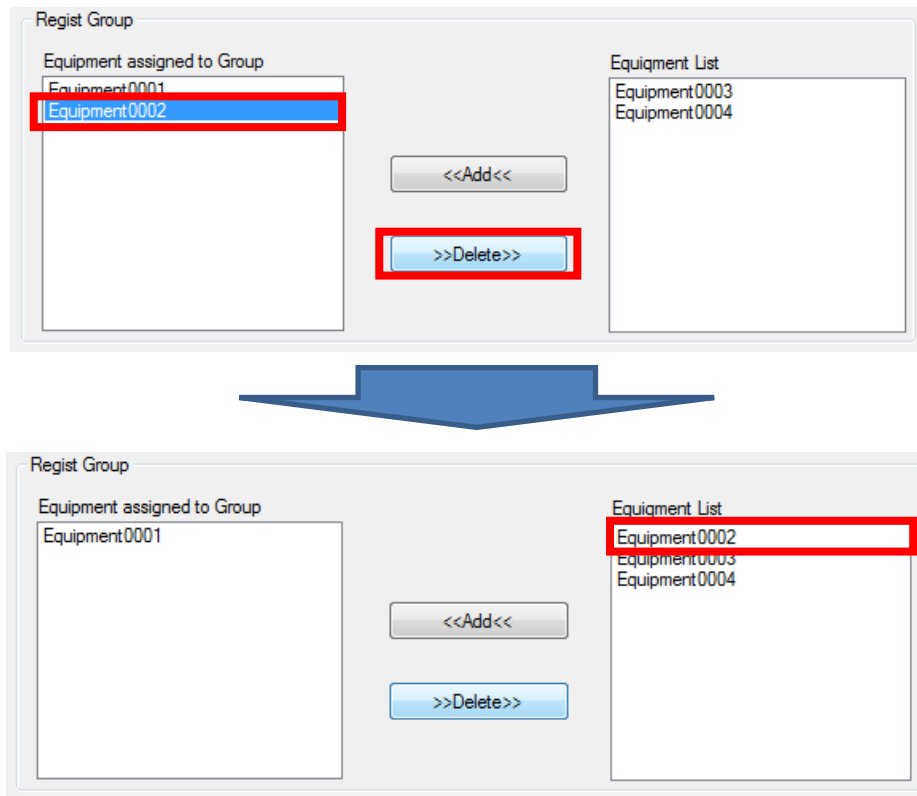
- (2) To add new equipment to this group, select the equipment in the [Equipment List] and click [<<Add<<]. If the [Equipment List] has no member, add equipment in advance. For the details, refer to the "[Steps for Equipment Setting](#)" subsection.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	300/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

■ Delete equipment from group

To delete equipment from the group, select any equipment in the [Equipment assigned to Group] list and click [>>Delete >>].

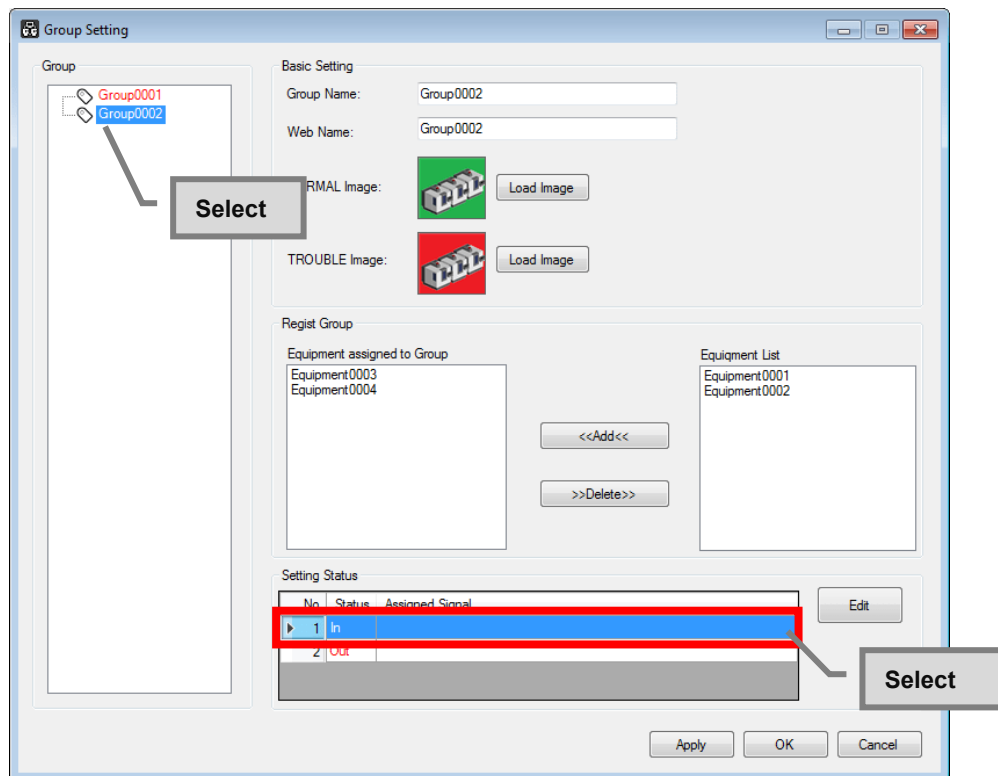


20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 301/868

4.8.7 Set Group Status

To set group status:

- (1) To change the group status, select the target group in the [Group] tree and select the target status in the [Setting Status] table. Click [Edit].

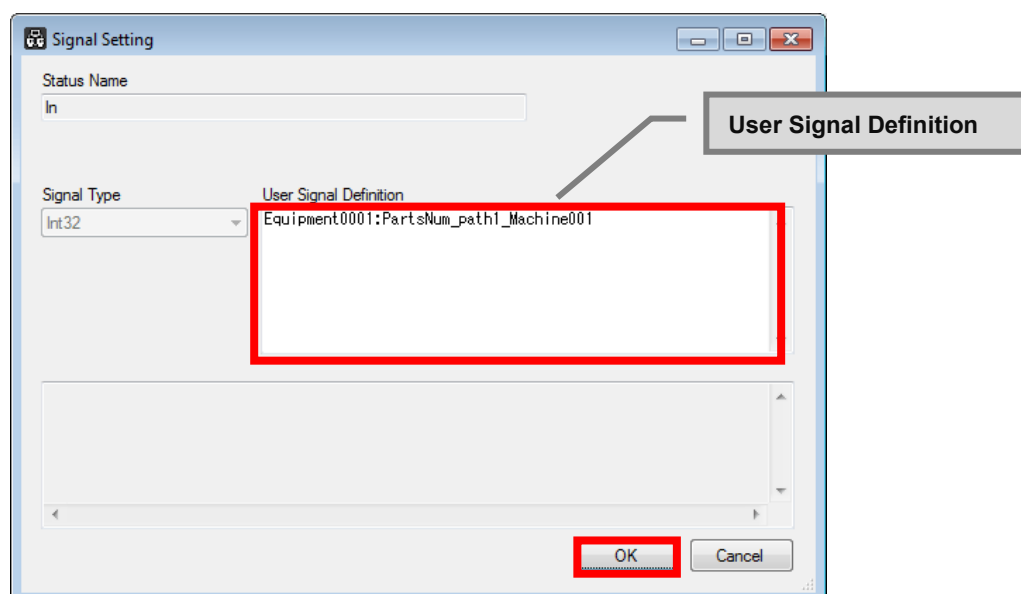


<Status of Group>

Status	Description
In	The input number displayed on the Web screens ([Results - Group Results] and [Monitoring - Group Monitoring])
Out	The output number displayed on the Web screens ([Results - Group Results] and [Monitoring - Group Monitoring])

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 302/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

(2) The [Signal Setting] dialog box appears. Set the signal definition and click [OK].

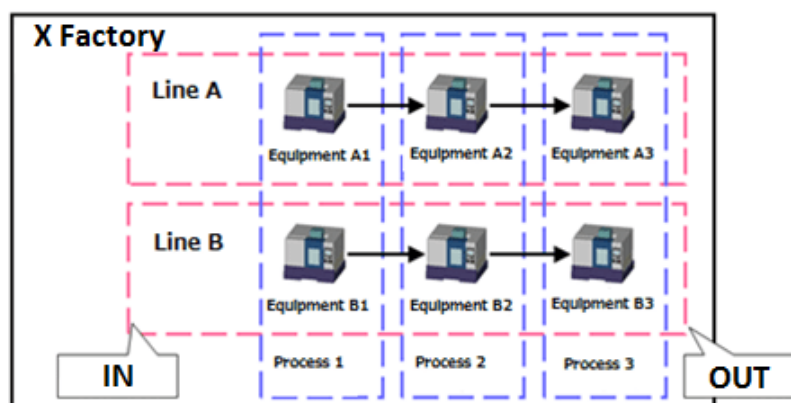


Shown below are the examples of the status setting and the signal definition.

For example, when you configure a group that is composed of a line A as shown in the figure below, the status of the group will be set as follows:

In: the work input into equipment A1

Out: the work output from equipment A3



Example 1)

Following is an example when setting the number of machined parts (PartsNum_path1_Machine001) of Equipment A1 (Equipment0001) while recognizing Line A as a group.

Signal definition:

Equipment0001:PartsNum_path1_Machine001

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	303/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

Example 2)

Following is an example of the signal definition when recognizing Process 1 as a group, and setting the value as the sum of the number of machined parts (PartsNum_path1_Machine001) of Equipment A1 (Equipment0001) and the number of machined parts (PartsNum_path1_Machine002) of Equipment B1 (Equipment0002).

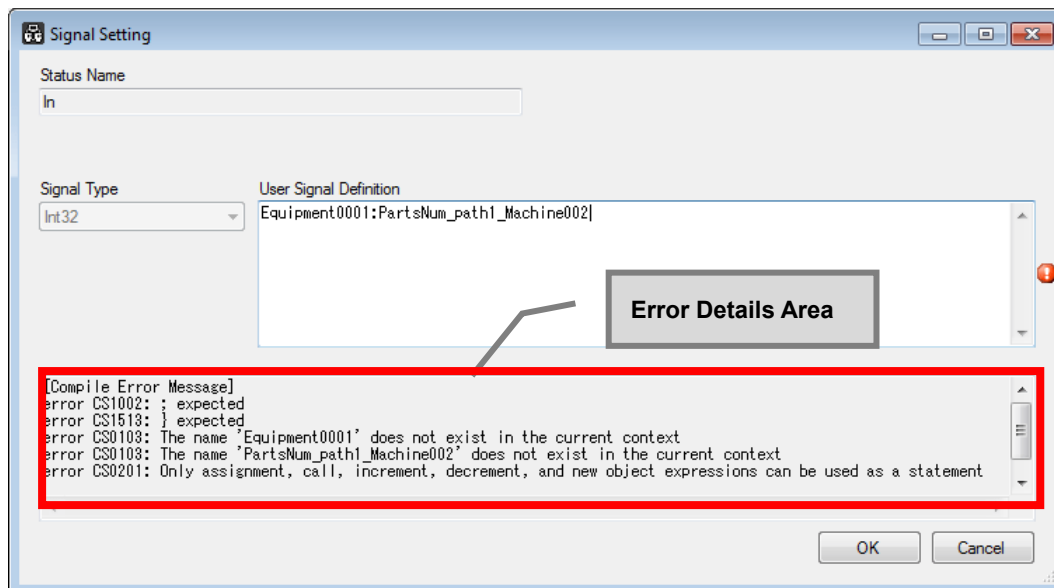
Signal definition:

Equipment0001:PartsNum_path1_Machine001 + Equipment0002:PartsNum_path1_Machine002

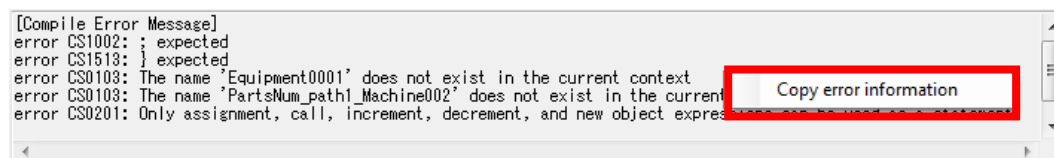
NOTE

- 3 All the simple assign equipment statuses and multiple assign equipment statuses can be used to define the statuses for In and Out.
- 4 Signal definition is given in the "<equipment name>:<status name>" format.
- 5 For the signal definition, the same formula used for the signal definition of multiple assign status can be used.

When the [Signal Setting] dialog box detects compilation errors in statuses defined in the [User Signal Definition] area, it displays an error icon (!) on the right side of the area and details of compilation errors in the [Error details] area.

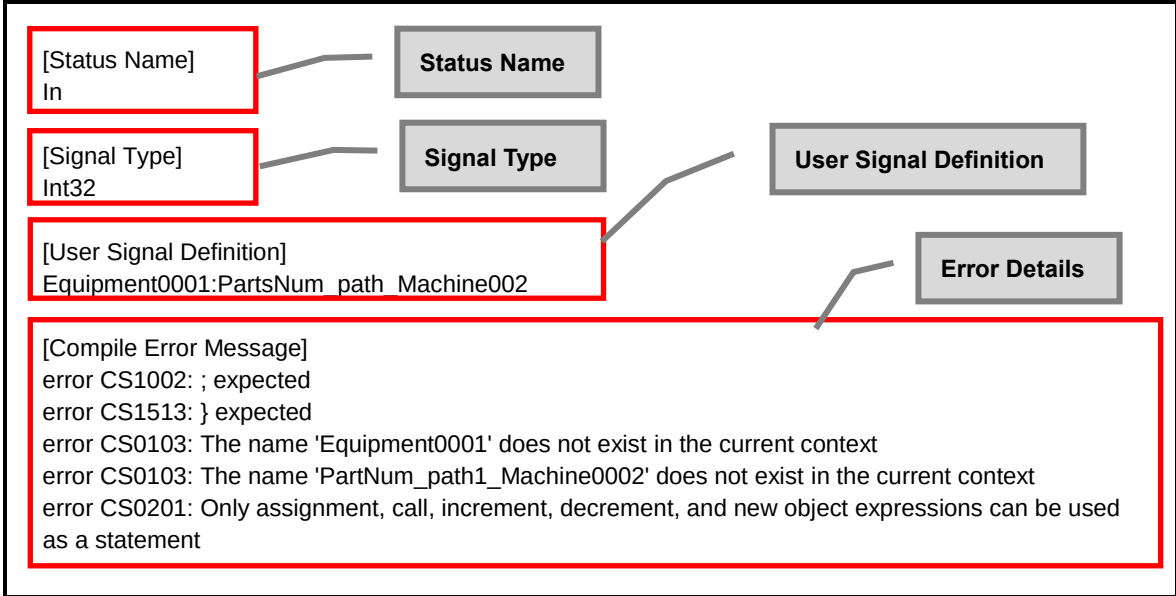


The text in the [Error details] area, as well as the settings on the [Signal Setting] dialog box can be copied to the clipboard by right-clicking on anywhere in the area and clicking the [Copy error information] in the context menu. Save the error information in a text file for search and analysis of errors.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 304/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

Following is an example of error information copied from the [Error details] area.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION				Sheet.	305/868
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						

4.9 LAYOUT SETTING

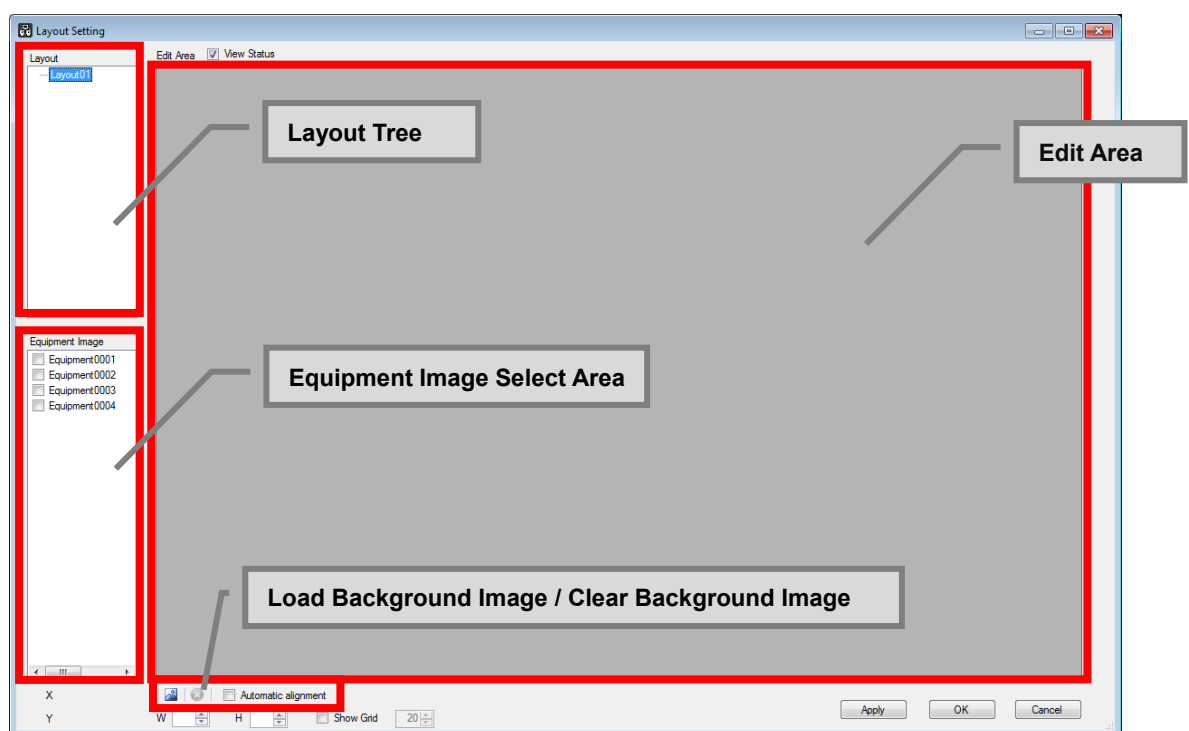
You can register a layout of the [Monitoring - Overlook] screen in this window. There are two ways for setting the layout. One way is to place the equipment icons manually, and the other way is align the equipment icons automatically.

<When editing manually>

- Multiple equipment icons can be placed on the background image.
- Equipment icons can be moved or resized.
- Layouts can be registered up to 20, including the manually edited layouts and the automatic alignment layouts.

<When using automatic alignment>

- Equipment icons (all the equipment units excluding the units counted out from monitoring) are automatically aligned on the overlook monitoring screen.
- Equipment icons cannot be moved or resized.
- Automatic alignment cannot be set on multiple layouts.



4.9.1 Steps for Layout Setting

To add a layout:

- (1) Select [Layout Setting] in the AdminTool Menu. This displays [Layout Setting] window.
- (2) Add a layout in the [Layout] tree. For the details, refer to the "[Add or Delete Layout](#)" subsection.
- (3) Set a background image to the selected layout. For the details, refer to the "[Set Background Image to Layout](#)" subsection.
- (4) Set an image of equipment to the selected layout. For the details, refer to the "[Locate Equipment to Layout](#)" subsection.
- (5) When you have completed above steps, click [Apply] or [OK]. Clicking [Cancel] discards the settings you made and closes the window.

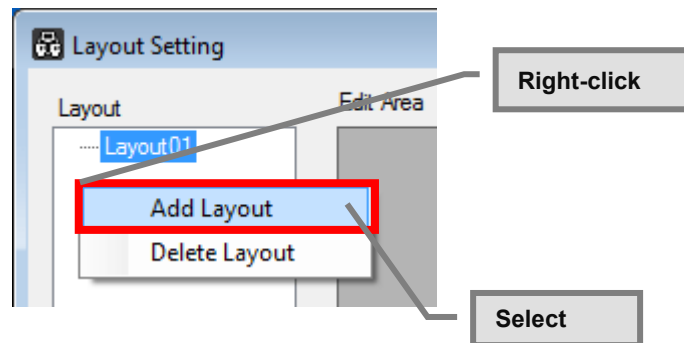
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	306/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

4.9.2 Add or Delete Layout

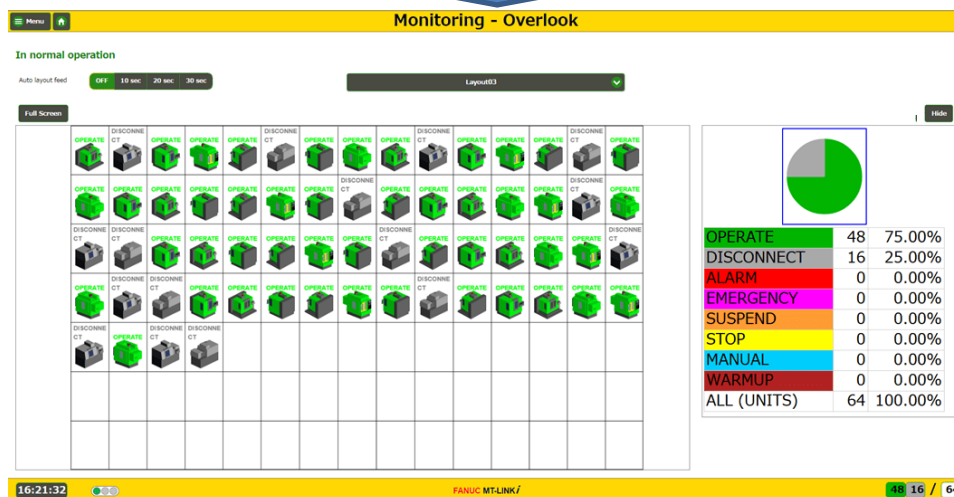
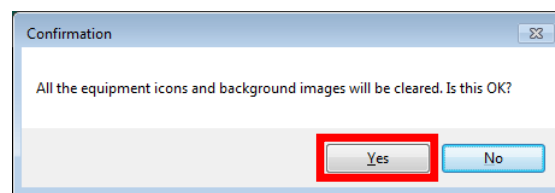
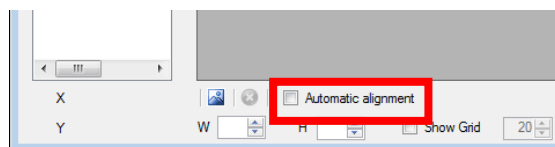
Follow the steps below.

■ Add a layout

To add a new layout, right-click anywhere in the [Layout] tree and point to [Add Layout]. A new layout with the name "Layout<NN>" (<NN>: 2digit number) is added.



When using automatic alignment option, select the [Automatic alignment] check box. The following message is shown. If you click [Yes], the editing of layout is disabled and the uncompleted layout of the equipment units and background images will be cleared.

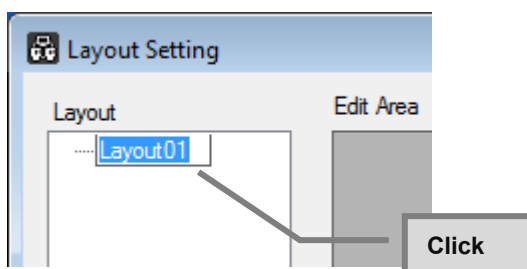


In a layout which has the [Automatic alignment] check box selected, clearing the check box restores the initial layout which has no setting, and it can be edited manually.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	307/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

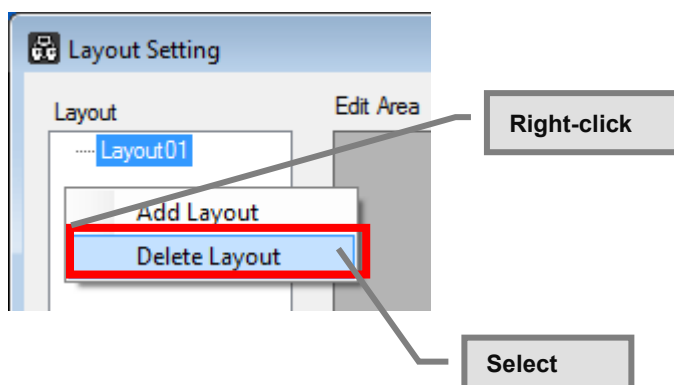
■ Change layout name

To change a layout name, click it in the [Layout] tree. Then, the name field becomes editable. The name must start with a half-width alphanumeric character and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-"), and underscores ("_"). This field can be up to 32 bytes long.



■ Delete a layout

To delete a layout, right-click it in the [Layout] tree and point to [Delete Layout].



NOTE

If the facility layout has already been set on the Integration Server, after changing the layout name or deleting the layout, resynchronize with [Facility synchronize] on the Integration Server's [Facility Layout Arrangement Setting] window to re-synchronize the layout.

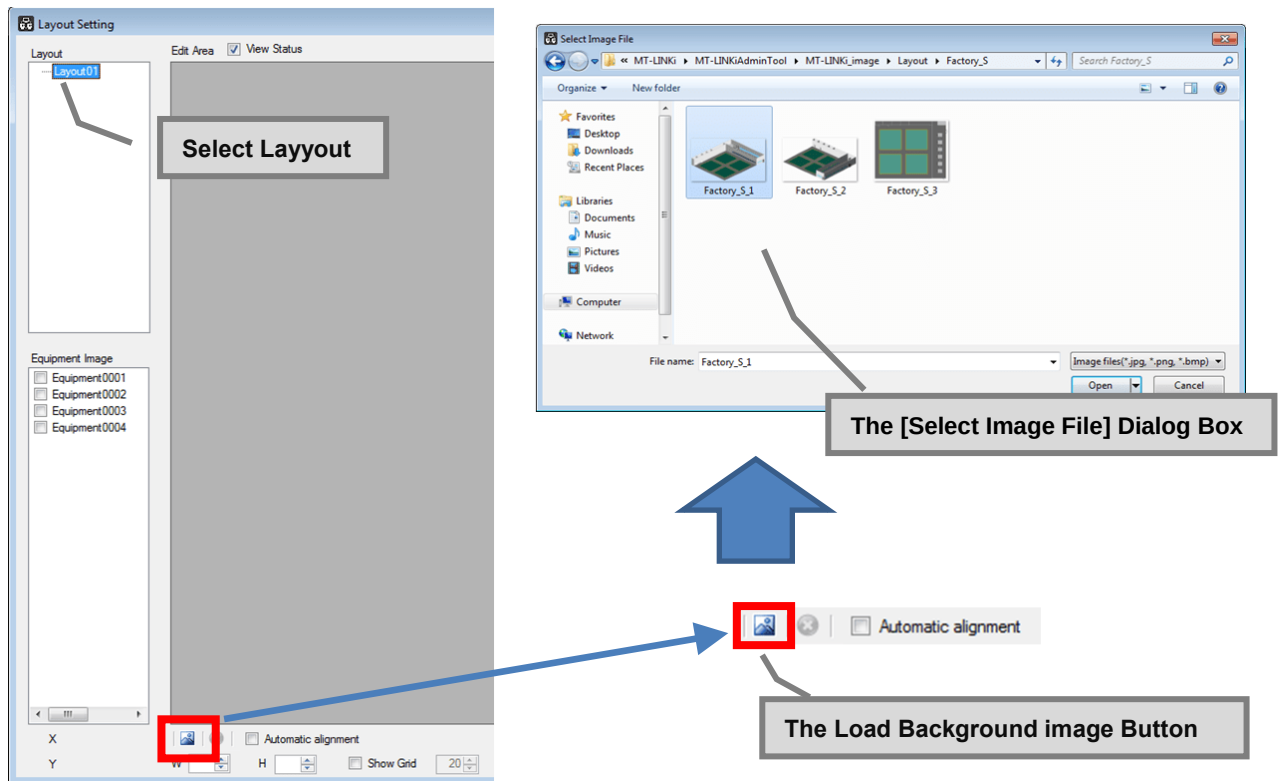
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL					
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11										
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10										
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN					
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8										
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2										
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION				Sheet.	308/868	
Ed.	Date	Design	Description										
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.							Inoue

4.9.3 Set Background Image to Layout

You can set background image per layout. When automatic alignment is not used, follow the steps below.

- (1) Select a layout in the [Layout] tree and click the Load Background Image button. The [Select Image File] dialog is displayed. Select a background file. The dialog selects the following folder on the first session. This folder contains sample background image files. On subsequent sessions, this dialog box selects the folder selected in the previous session.

<MT-LINK*i* installation folder>\MT-LINKiAdminTool\MT-LINKi_image\Layout



- (2) The image is displayed in the background.

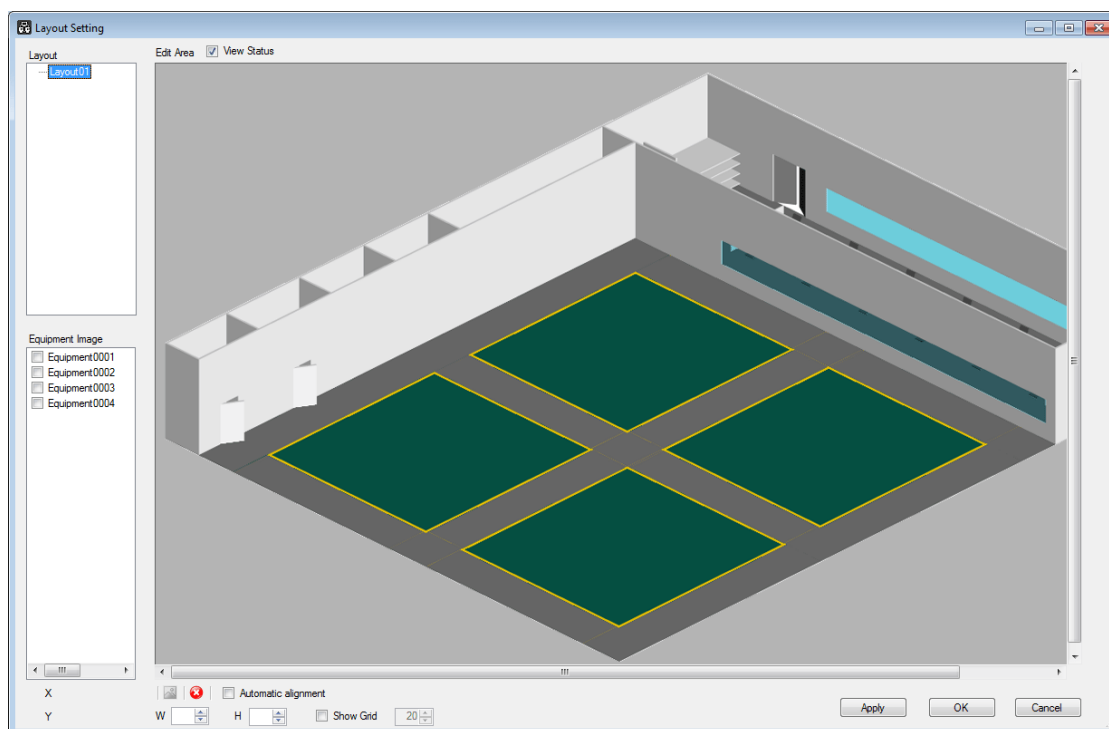
The formats that can be used for the image files are png, jpg, and bmp.

Prepare the image file in advance. You cannot edit or scale the background image file in AdminTool.

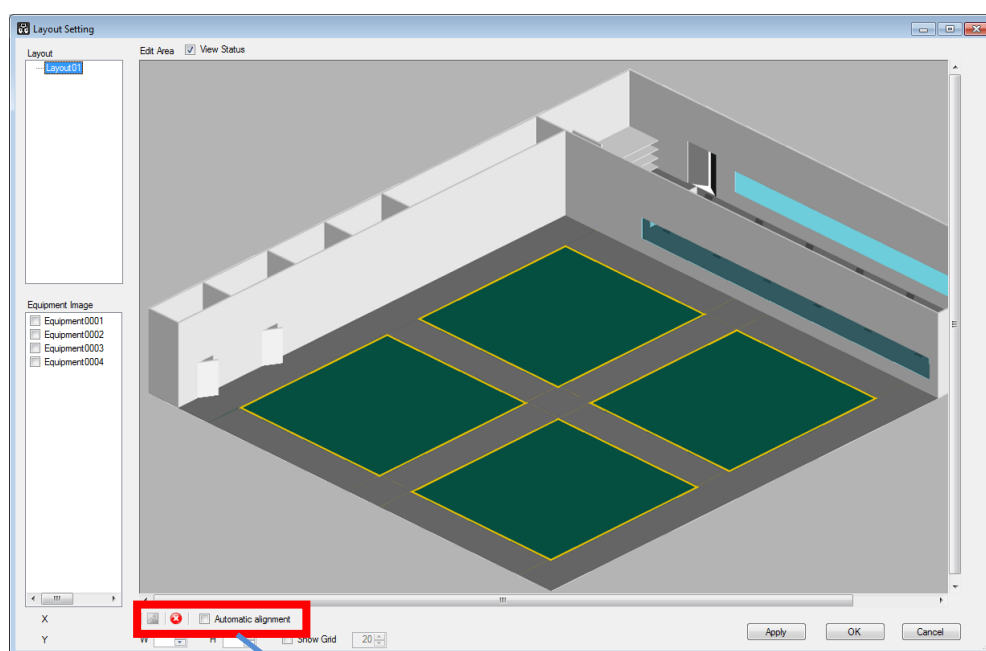
NOTE

The [Layout Setting] window will add a vertical scroll bar to the [Edit Area] when it is not large enough to display the whole of the selected background image.

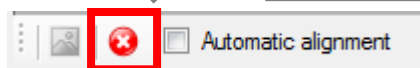
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 309/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	



(3) To delete the background, click the Clear Background Image button.



The Clear Background Image Button

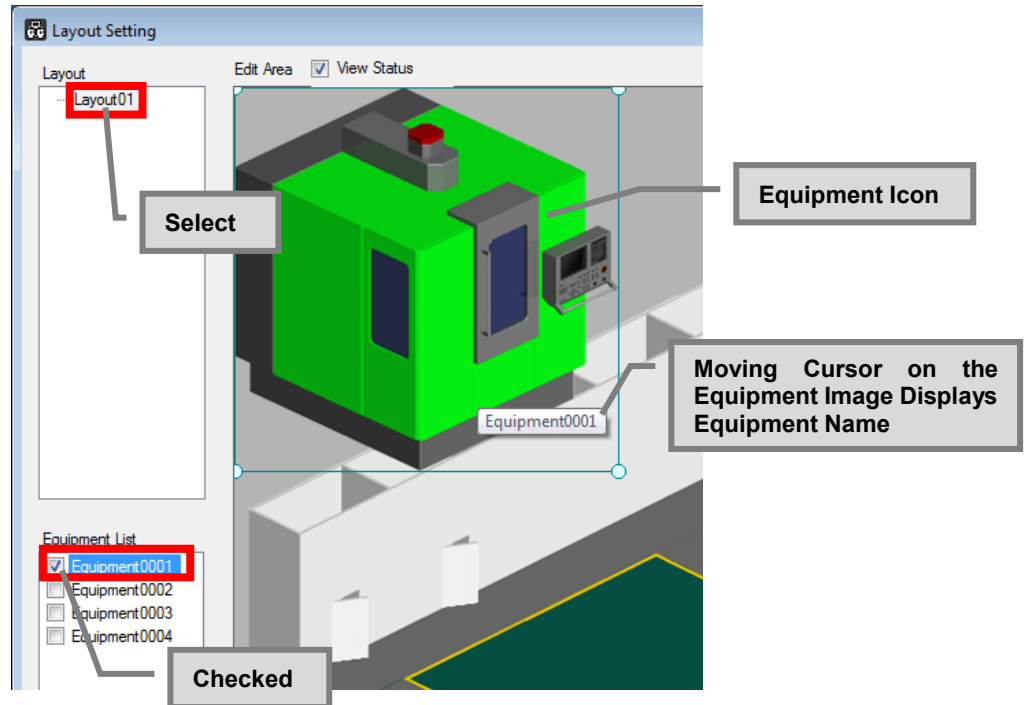


20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	310/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

4.9.4 Locate Equipment to Layout

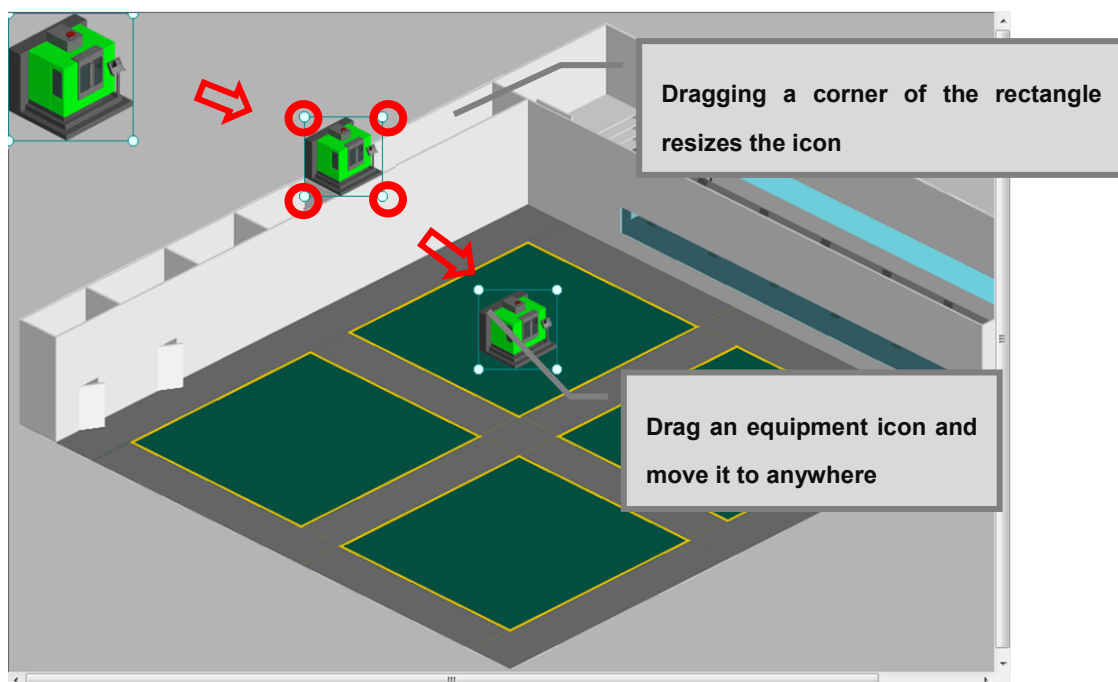
When automatic alignment is not used, equipment units can be located manually for each layout. Follow the steps below.

- (1) Select a layout from the [Layout] tree. The list of equipment check box appears in the [Equipment List] area. Selecting any equipment check box displays the corresponding equipment icon in the editing area. Moving cursor on the equipment icon shows you the equipment name.



- (2) Select an equipment icon and drag any corner of the rectangle to resize the equipment icon. The selected equipment icon can be dragged and dropped to any place in the editing area. Multiple equipment icons can be selected simultaneously by clicking the equipment icons while holding down the Shift key on the keyboard. To select all the equipment icons, hold down the Ctrl key and press A. Selected icons can be resized and moved simultaneously. Arrow keys (Up Arrow, Down Arrow, Left Arrow, and Right Arrow) can also be used to move selected equipment icons.
 - Each press of an arrow key without holding down the Ctrl key moves the icons by five pixels.
 - Each press of an arrow key while holding down the Ctrl key moves the icons by one pixel.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	FANUC CORPORATION		Sheet.	311/868

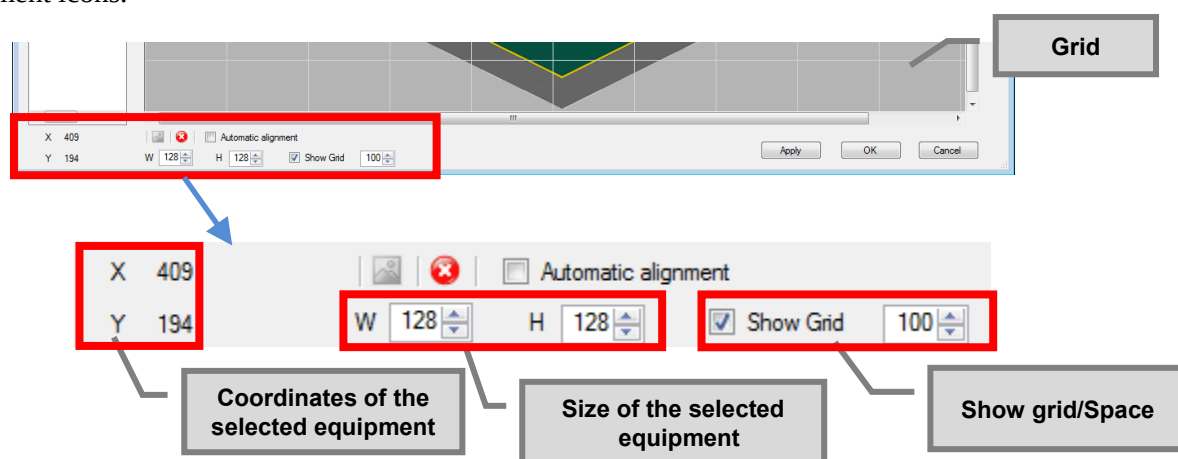


At the lower left side of the window, the coordinates [X] / [Y] (horizontal/vertical) and size [W] / [H] (width/height) of the selected equipment icon is shown. If multiple equipment icons are selected, it will not be shown. Unit of the values shown are pixels.

Numeral figures cannot be entered for [X] / [Y]. To move the equipment icon, drag it.

By entering numbers in the sizes [W] and [H], or by clicking the up/down button of the sizes [W] and [H], size of the selected equipment icon can be changed. Unit input is pixels. Input range is 15 to 1024 for [W], and 15 to 680 for [H].

Also, when the [Show grid] check box is selected, grids will be shown on the editing area. By entering numbers in [Show grid], or by clicking the up/down button of [Show grid], the space between the grids can be changed. Unit input is pixels. Use the grid as reference for adjusting the size and placement of the equipment icons.



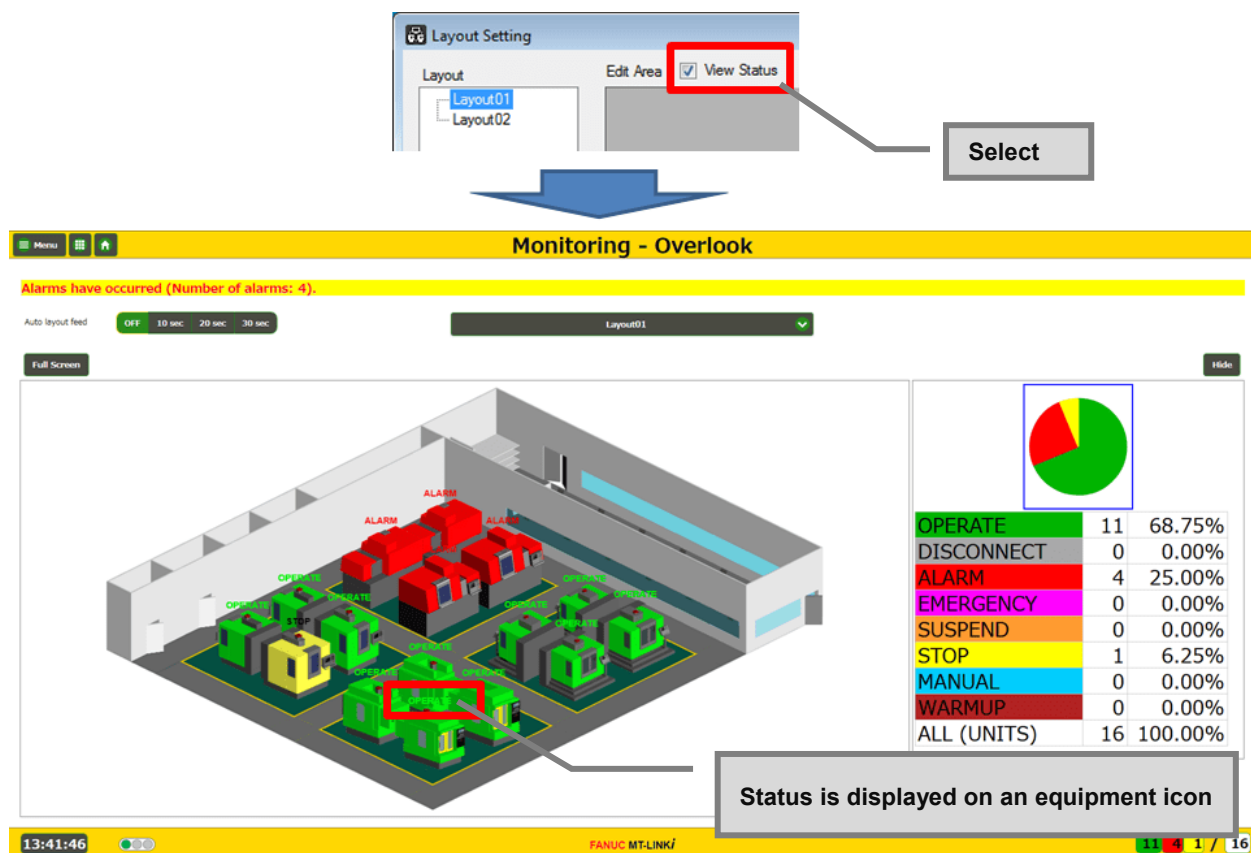
NOTE

Color of the grid can be changed. For the details, refer to the "[Customizing Layout Setting](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	312/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

4.9.5 View Status Setting

You can set whether the status string is displayed or not in the [Monitoring - Overlook] screen of MT-LINK*i*. When the [View Status] check box is selected, the status string is displayed (This can also be applied for the layout with automatic alignment).



4.9.6 Customizing Layout Setting

By editing the system setting file (AdminTool.ini) provided by MT-LINK*i*, the layout setting can be customized.

Following are the items that can be customized using AdminTool.ini.

- Layout setting grid color
Customize the color when it is difficult to view the grid display due to the background color.

NOTE

When the system setting file (AdminTool.ini) is edited, restart AdminTool.

For the details on the system setting file, refer to the "**SYSTEM SETTING FILES**" section.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION				Sheet.	313/868
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						

4.10 MANUAL DATA SETTING

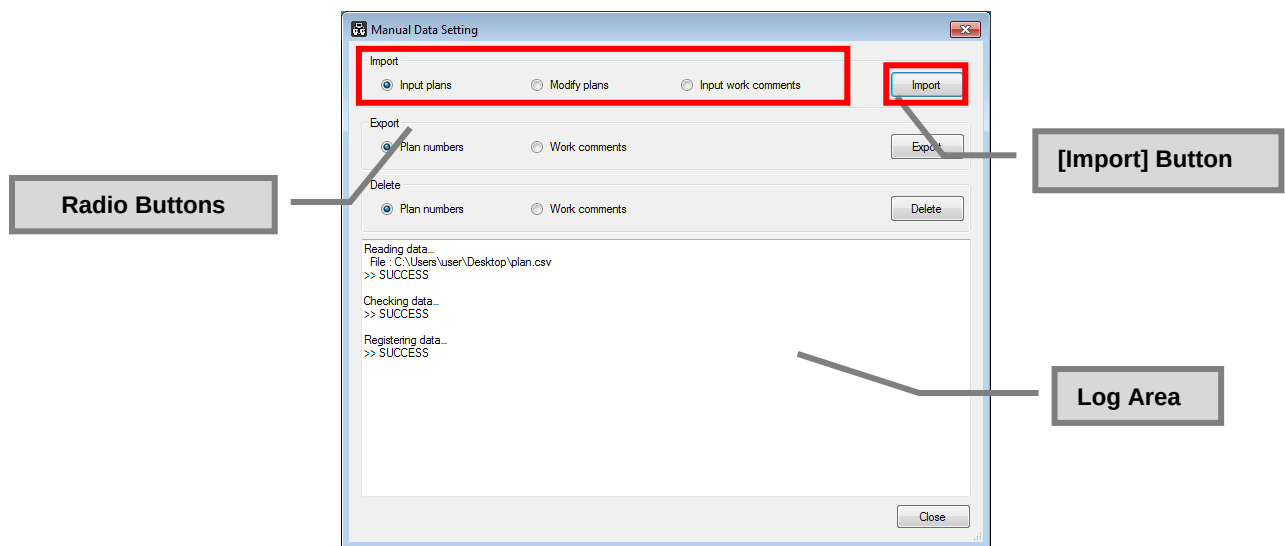
MT-LINK*i* automatically accumulates production histories of the registered equipment. Furthermore, it has Manual Data Setting functionality that enables the user to manually provide MT-LINK*i* with complementary data. MT-LINK*i* supports two kinds of complementary data; product plan data and work comments data.

The term "product plan" denotes how many of products to be machined by specific equipment in a planned duration, and "work comments" denotes explanations on what work to be machined by specific equipment, such as description of the alarm occurring and how to release it.

Hereafter, those data that are or are to be manually input are referred to as "manual data"; how many of a product is planned to be machined by MT-LINK*i* -registered equipment is referred to as "plans"; and comments on what work to have been done with specific equipment is referred to as "work comments."

The [Manual Data Setting] dialog box enables the user to do the following:

- By reading a CSV formatted product plan data, register them to the database as new entries, modify the registered product plan data, and export or delete them.
- By reading a CSV formatted work comments data, register them to the database as new entries, overwrite the registered work comments data, and export or delete them.



In the following Web screens, you can view plan data registered to the database.

Manual Data to Be Displayed on Web screen		
[Results - Production Results] screen	Transition Graph	Plans
	Summary Table	Product name
[Results - Operational Results] screen	Band Graph	Work comments

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION		Sheet. 314/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

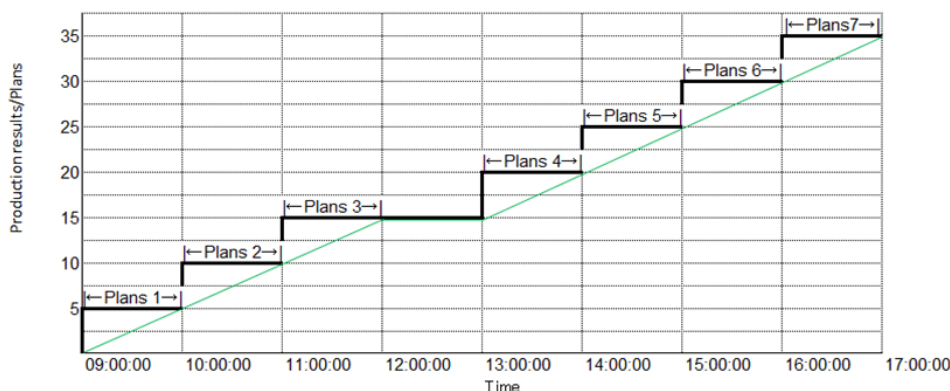
The following are example entries of product plan data in a manual data file, and their graph of their product results and plans (planned numbers) to be displayed on the [Results - Production Results] Web screen.

Example of Input Plan Data:

"Equipment"	"Start date"	"End date"	"Processed product name"	"Plans"	
"Equipment0001"	"2016/02/18 09:00:00"	"2016/02/18 10:00:00"	"MAIN COMMENT"	"5"	→ Plans 1
"Equipment0001"	"2016/02/18 10:00:00"	"2016/02/18 11:00:00"	"MAIN COMMENT"	"5"	→ Plans 2
"Equipment0001"	"2016/02/18 11:00:00"	"2016/02/18 12:00:00"	"MAIN COMMENT"	"5"	→ Plans 3
"Equipment0001"	"2016/02/18 13:00:00"	"2016/02/18 14:00:00"	"MAIN COMMENT"	"5"	→ Plans 4
"Equipment0001"	"2016/02/18 14:00:00"	"2016/02/18 15:00:00"	"MAIN COMMENT"	"5"	→ Plans 5
"Equipment0001"	"2016/02/18 15:00:00"	"2016/02/18 16:00:00"	"MAIN COMMENT"	"5"	→ Plans 6
"Equipment0001"	"2016/02/18 16:00:00"	"2016/02/18 17:00:00"	"MAIN COMMENT"	"5"	→ Plans 7



Example of Transition Graph on [Production results] screen:



* Bold black line for Plans N; green thin line for Production results

NOTE

Plans on a transition graph will not be displayed until the Start date.

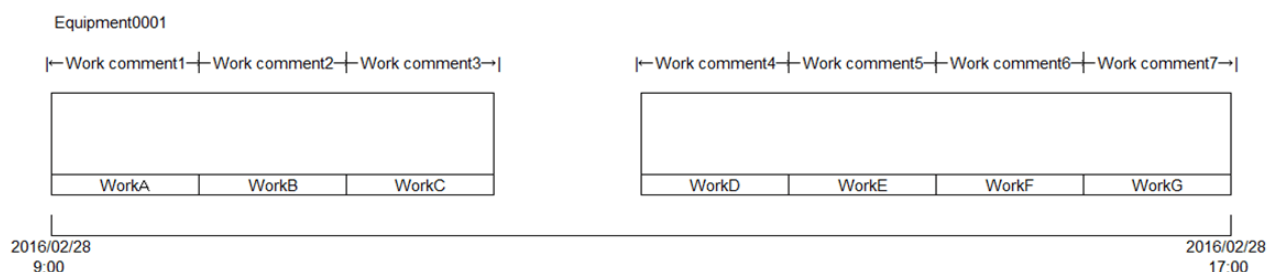
The following are examples of work comments data in a manual data file, and their band graph to be displayed on the [Results - Operational Results] Web screen.

Example of Work Comments Data

"Equipment"	"Start date"	"End date"	"Work comments"	
"Equipment0001"	"2016/02/18 09:00:00"	"2016/02/18 10:00:00"	"WorkA"	→ Work comment1
"Equipment0001"	"2016/02/18 10:00:00"	"2016/02/18 11:00:00"	"WorkB"	→ Work comment2
"Equipment0001"	"2016/02/18 11:00:00"	"2016/02/18 12:00:00"	"WorkC"	→ Work comment3
"Equipment0001"	"2016/02/18 13:00:00"	"2016/02/18 14:00:00"	"WorkD"	→ Work comment4
"Equipment0001"	"2016/02/18 14:00:00"	"2016/02/18 15:00:00"	"WorkE"	→ Work comment5
"Equipment0001"	"2016/02/18 15:00:00"	"2016/02/18 16:00:00"	"WorkF"	→ Work comment6
"Equipment0001"	"2016/02/18 16:00:00"	"2016/02/18 17:00:00"	"WorkG"	→ Work comment7



Example of Band Graph on [Operational results] screen:



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	315/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

4.10.1 Steps for Manual Data Setting

To register manual data to the database, follow the steps below:

- (1) Prepare new product plan data in a manual data file. For how to register product plan data, refer to the "[Preparing and Reading Product Plan Data](#)" subsection; for how to register work comments data, refer to the "[Preparing and Reading Work Comment](#)" subsection.
- (2) Select [Manual Data Setting] in the AdminTool Menu. The [Manual Data Setting] dialog box appears. Select the [Input plans] or the [Input work comments] button.
- (3) Load (read) the relevant manual data file. For the details, refer to the "[Reading Manual Data](#)" subsection.

4.10.2 Preparing and Reading Product Plan Data

This subsection describes on how to prepare product plan data in a manual data file and to register them to the database.

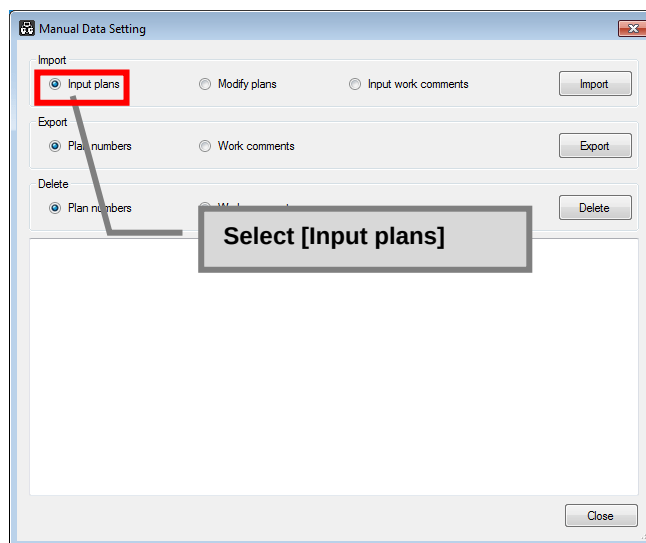
Before preparing product plan data to register, you may want to know the conditions the product plan data in a manual data file must meet.

For the conditions, refer to the "[Registration Conditions for Product Plan Data](#)" subsection.

If needed to update registered product plan data, instead of registration, follow the steps in the "[Updating Product Plan Data](#)" subsection.

To prepare product plan data and load (read) them, follow the steps below:

- (1) Prepare product plan data in a CSV text file. For the file format, refer to the "[File Format for Manual Data](#)" subsection.
- (2) Select [Manual Data Setting] in the AdminTool Menu to open the [Manual Data Setting] dialog box.
- (3) Select the [Input plans] button in the [Import] group box.



- (4) Manual data is about to be loaded. For the steps after selecting the [Input plans] button, refer to the "[Reading Manual Data](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	316/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

4.10.3 Updating Product Plan Data

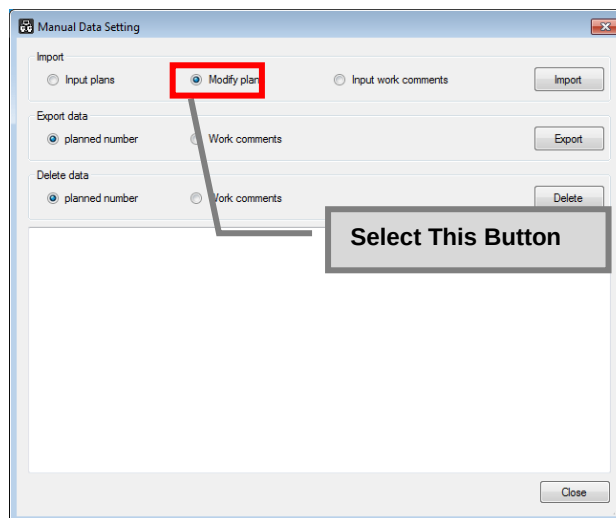
This subsection describes on how to update product plan data that have already been registered to the database.

Before updating product plan data registered in the database, you may want to know the conditions the product plan data in a manual data file must meet. For the conditions, refer to the "[Updating Conditions for Product Plan Data](#)" subsection.

When registering new product plan data, instead of updating, follow the steps in the "[Preparing and Reading Product Plan Data](#)" subsection.

To update registered product plan data, follow the steps below:

- (1) Prepare product plan data in a CSV text file. For the file format, refer to the "[File Format for Manual Data](#)" subsection. Note that plan data must meet the requirements: the CSV file format and the values of items.
- (2) Select [Manual Data Setting] in the AdminTool Menu to open the [Manual Data Setting] dialog box.
- (3) Select the [Modify plans] button in the [Import] group box.



- (4) For the steps after selecting the [Modify plans] button, refer to the "[Reading Manual Data](#)" subsection.

4.10.4 Preparing and Reading Work Comments Data

This subsection describes on how to prepare work comments data in a manual data file and to resister them to the database.

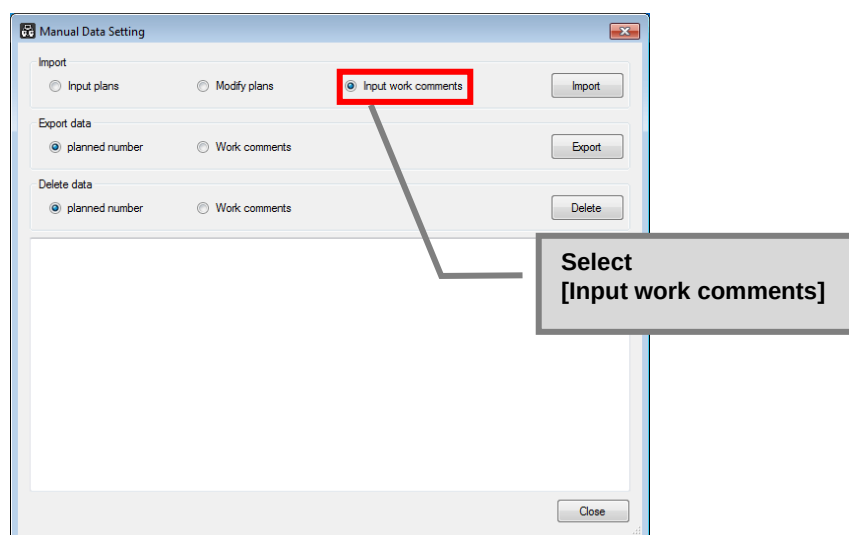
Before preparing work comments data to register, you may want to know the conditions the work comments data in a manual data file must meet. For the conditions, refer to the "[Registration Conditions for Work Comments Data](#)" subsection.

To prepare work comments data and load (read) them, follow the steps below:

- (1) Prepare work comments data in a CSV text file. For the file format, refer to the "[File Format for Manual Data](#)" subsection.
- (2) Select [Manual Data Setting] in the AdminTool Menu to open the [Manual Data Setting] dialog box.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	317/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

(3) Select the [Input work comments] button in the [Import] group box.

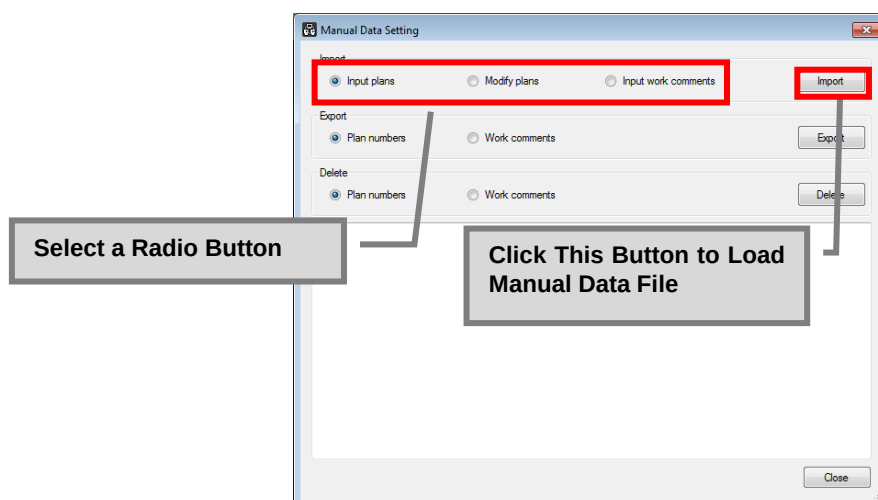


(4) Manual data is about to be loaded. For the steps after selecting the [Input work comments] button, refer to the "[Reading Manual Data](#)" subsection.

4.10.5 Reading Manual Data

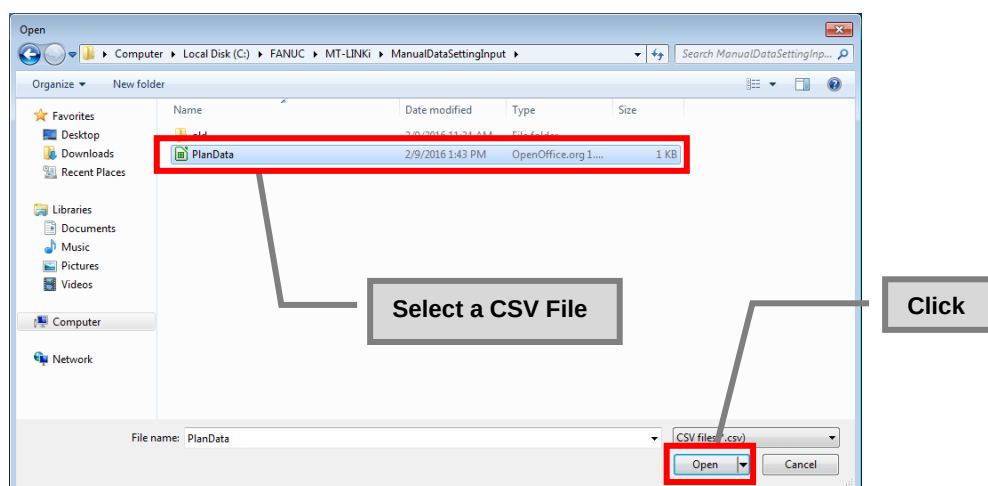
To load data from a manual data file, perform the following steps:

- (1) Select [Manual Data Setting] in the AdminTool Menu to open the [Manual Data Setting] dialog box.
- (2) Select the [Input plans], the [Modify plans], or the [Input work comments] button in the [Input] group box, and then click the [Import] button.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	318/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

(3) The [Open] dialog appears. Select the plan file you prepared and click the [Open] button.



When the [Input plans] or [Modify plans] check box is selected, this dialog box selects the following folder on the first session. On subsequent sessions, it selects the folder selected in the previous session:

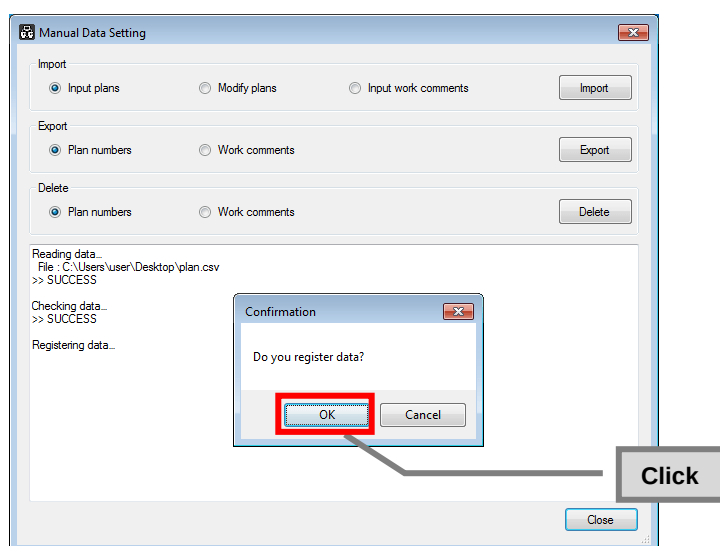
[CSV output path] in the [Product Plan Setting] group box of the [System Setting] window

When the [Input work comments] check box is selected, this dialog box selects the following folder on the first session. On subsequent sessions, it selects the folder selected in the previous session:

<MI-LINK*i* installation folder>\MT-LinkiAdminTool

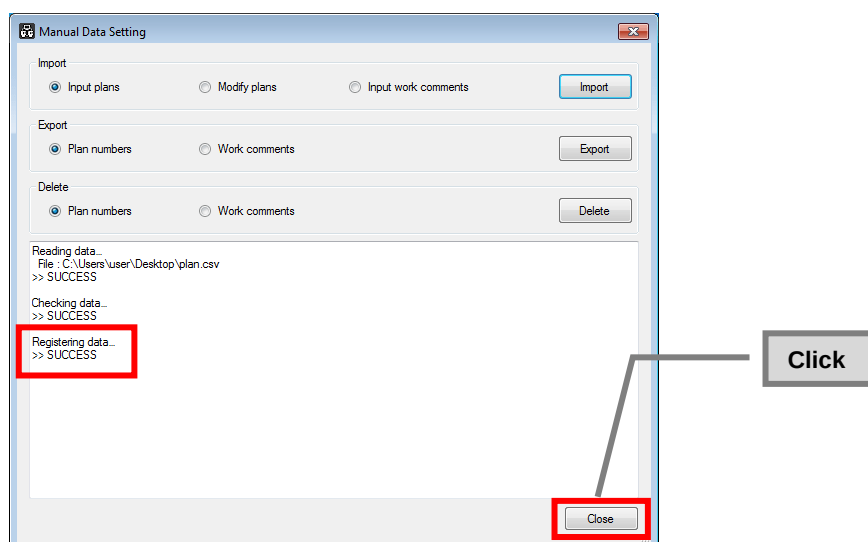
(4) When the file is loaded successfully, a message window asking you if you want to register them into the database ("Do you register data?"). Click [OK] to start to register the loaded data.

* If an error occurs during loading the file, error messages will be displayed in the log area of the [Manual Data Setting] dialog box. For those errors, refer to the "[Errors on Reading Manual Data File](#)" subsection.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	319/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

- (5) When all the data are successfully registered in the database, the messages "Data registration..." and ">> SUCCESS" appear on the log area of the [Manual Data Setting] dialog box. Click [Close].

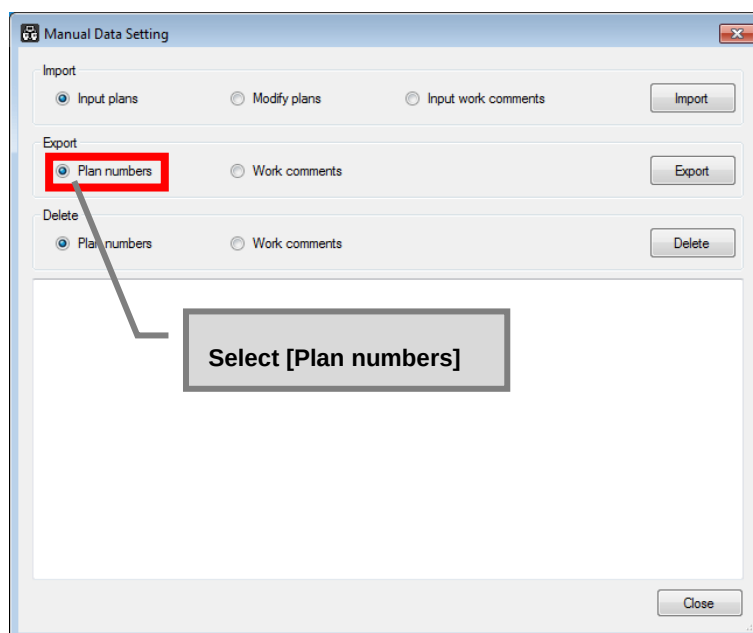


4.10.6 Exporting Product Plan Data

This subsection explains how to export product plan data that have been registered to the database.

To export product plan data (planned numbers) to a manual data file, follow the steps below.

- (1) Select [Manual Data Setting] in the AdminTool Menu to open the [Manual Data Setting] dialog box.
- (2) Select the [Plan numbers] button in the [Export] group box.



- (3) Product plan data that have been registered to the database are about to be exported to a manual data file. For the operation after selecting the [Plan numbers] button, refer to the "[Exporting Manual Data](#)" subsection.

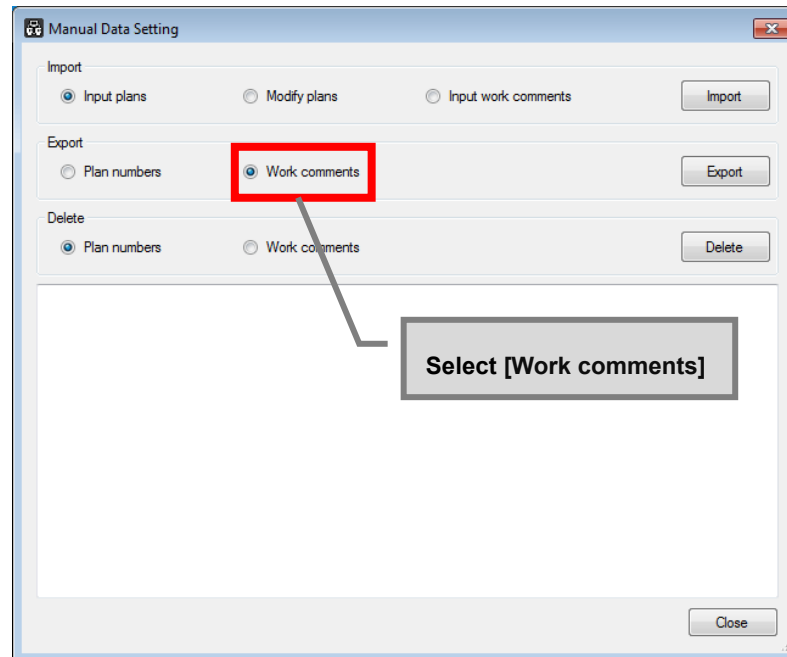
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	320/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

4.10.7 Exporting Work Comments Data

This subsection explains how to export work comments data that have been registered to the database.

To export work comments data to a manual data file, follow the steps below.

- (1) Select [Manual Data Setting] in the AdminTool Menu to open the [Manual Data Setting] dialog box.
- (2) Select the [Work comments] button in the [Export] group box.



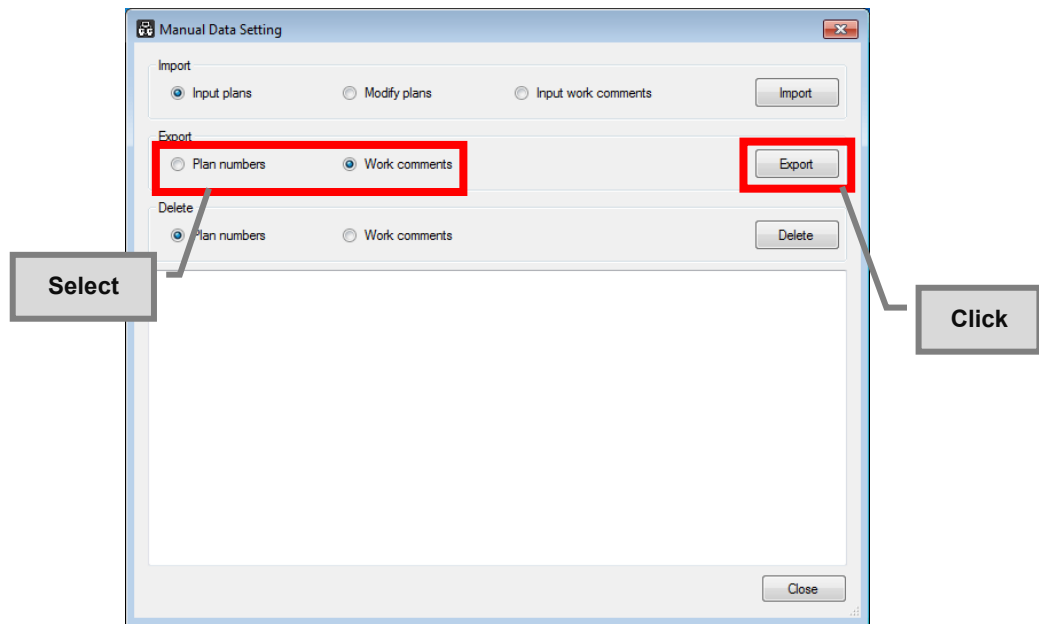
- (3) Work comments data that have been registered to the database are about to be exported to a manual data file. For the operation after selecting the [Work comments] button, refer to the "[Exporting Manual Data](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue
FANUC CORPORATION					Sheet.	321/868	

4.10.8 Exporting Manual Data

To export manual data to a manual data file from the database, follow the steps below.

- (1) Select [Manual Data Setting] in the AdminTool Menu to open the [Manual Data Setting] dialog box.
- (2) Select the [Planned number] or the [Work comments] button in the [Export] group box, and then click the [Export] button.



- (3) The [Manual Data Export] dialog box appears. Input equipment name, period (if exporting the registered manual data for the whole period select [Whole period], or if entering specific period for exporting select [Period Specified]). Either set the [Path] field to an export target folder or click the browse folder button ([...]) to open the [Browse For Folder] window, select it, and click [OK]. Click the [Export] button to start the process.

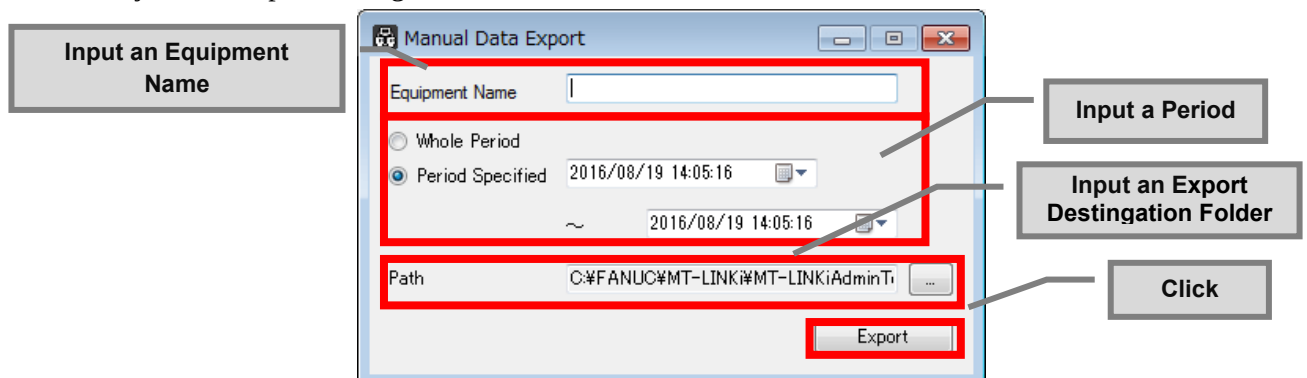
When the [Plan numbers] check box is selected, this dialog box populates the following folder on the first session. On subsequent sessions, it populates the folder specified in the previous session:

[CSV output path] in the [Product Plan Setting] group box of the [System Setting] window

When the [Work comments] check box is selected, this dialog box populates the following folder on the first session. On subsequent sessions, it populates the folder specified in the previous session:

<MT-LINK*i* installation folder>\MT-LINKiAdminTool

The [Browse For Folder] window selects the folder by default that is populated in the [Path] field in the [Manual Data Export] dialog box.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	322/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

NOTE

When the [Plan numbers] option is selected, date/time exceeding 90days from the date/time when the [Export] button is clicked cannot be specified; when the [Work comments] option is selected, date/time exceeding the date/time when the [Export] button is clicked cannot be specified.

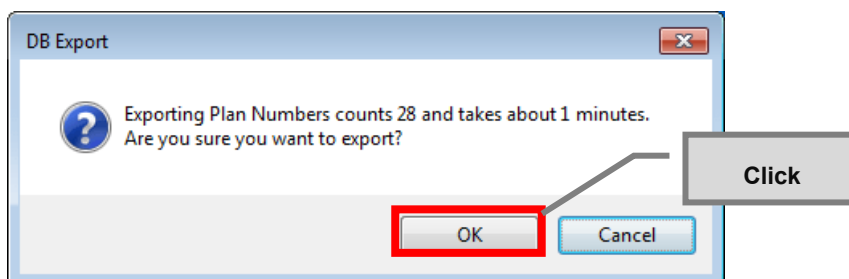
The period from the start date/time to the end date/time must not exceed one month.

(4) Following message appears. By clicking [OK], exporting procedure of manual data to the file begins.

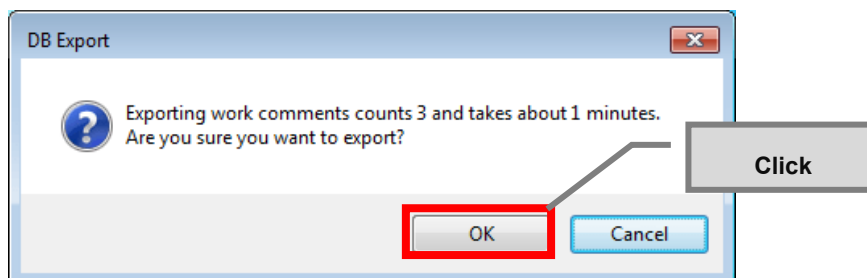
NOTE

If an error message appears while exporting the manual data, refer to the "[Errors during Exporting of Manual Data File](#)" subsection.

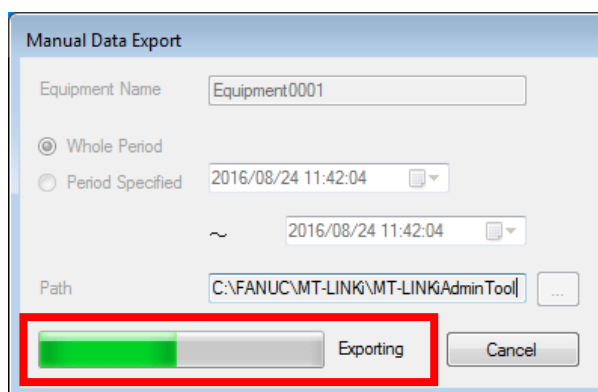
- When the [Plan numbers] button is selected



- When the [Work comments] button is selected

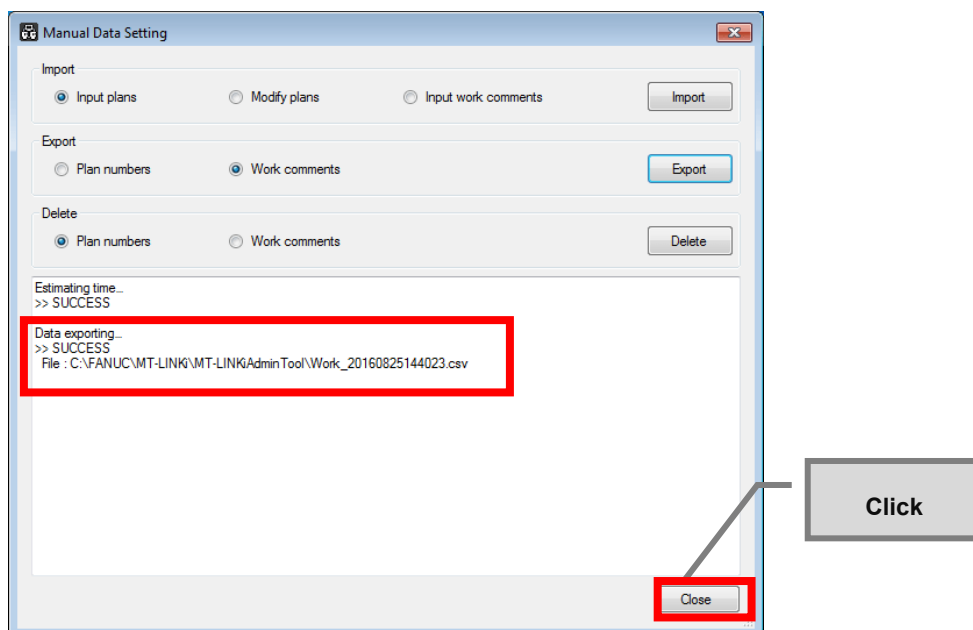


(5) As exporting procedure to the file begins, output progress bar appears on the [Manual Data Export] dialog box.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL					
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11										
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10										
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN					
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8										
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2										
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION				Sheet.	323/868	
Ed.	Date	Design	Description										
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.							Inoue

- (6) When the export procedure to the file is completed, [Manual Data Export] dialog box is closed, and [Data exporting...>> SUCCESS] will be shown in the log area of the [Manual Data Setting] dialog box. Click the [Close] button.



- (7) Manual data is saved with the file name as shown below to the folder specified as output path on [Manual Data Export] dialog box.
- When plan numbers is selected,
"ProductPlanNumber_<yyyyMMddHHmmss>.csv"
 - When work comments is selected,
"Work_<yyyyMMddHHmmss>.csv"

NOTE

The "<yyyyMMddHHmmss>" part of the file name will be the date/time just before the file is saved.
As for the format of the file, refer to the "[File Format for Manual Data](#)" subsection.

4.10.9 Deleting Product Plan Data

This subsection explains how to delete product plan data that have been registered to the database.

NOTE

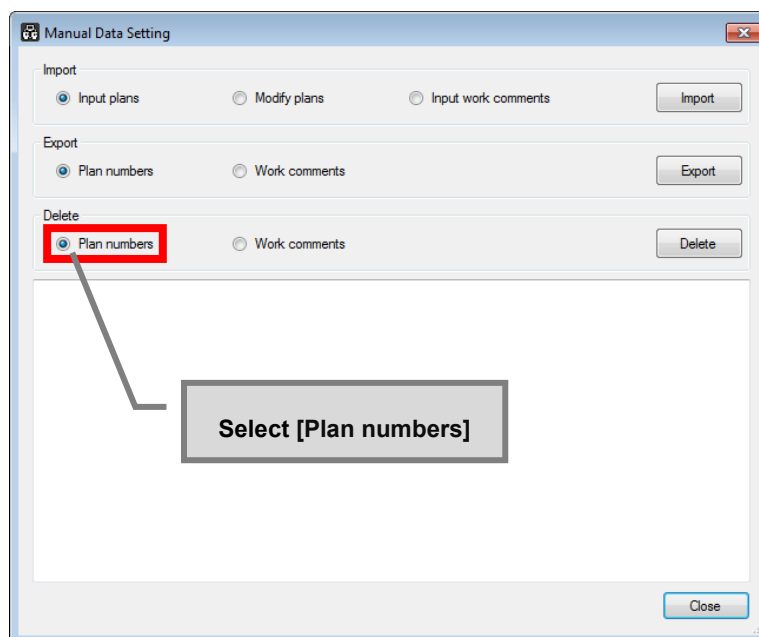
As the product plan data of the past cannot be input, if the product plan data of the past are deleted, the product plan data in the deleted period cannot be reregistered again.

To delete product plan data (planned numbers), follow the steps below.

- (1) Select [Manual Data Setting] in the AdminTool Menu to open the [Manual Data Setting] dialog box.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	324/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

(2) Select the [Plan numbers] button in the [Delete] group box.



(3) Product plan data are about to be deleted. For the operation after selecting the [Plan numbers] button, refer to the "**Deleting Manual Data**" subsection.

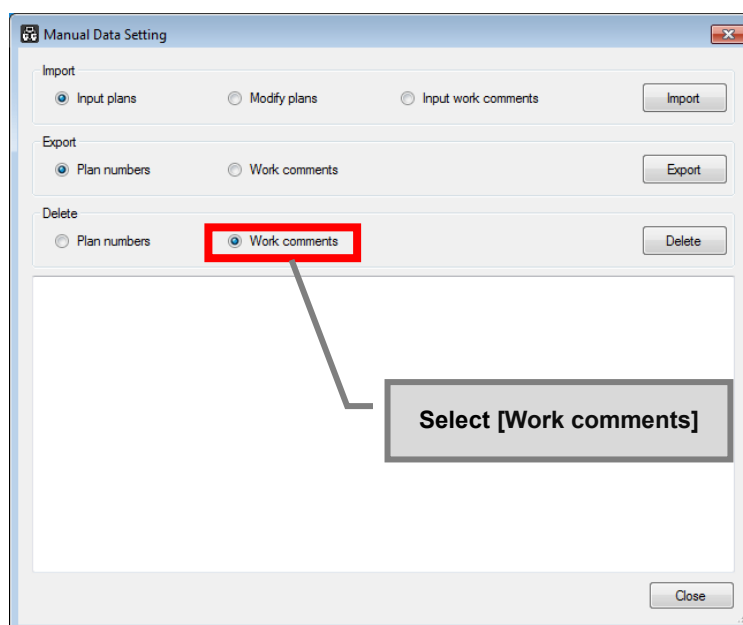
4.10.10 Deleting Work Comments Data

This subsection explains how to delete the work comments data that have been registered to the database.

To delete work comments data, follow the steps below.

(1) Select [Manual Data Setting] in the AdminTool Menu to open the [Manual Data Setting] dialog box.

(2) Select the [Work comments] button in the [Delete] group box.



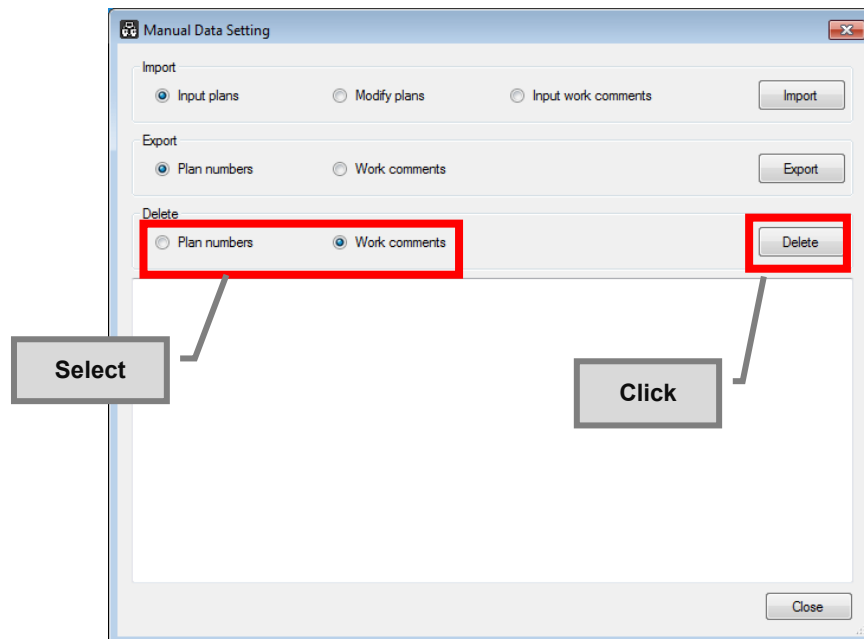
(3) Work comments data are about to be deleted. For the operation after selecting the [Work comments] button, refer to the "**Deleting Manual Data**" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 325/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

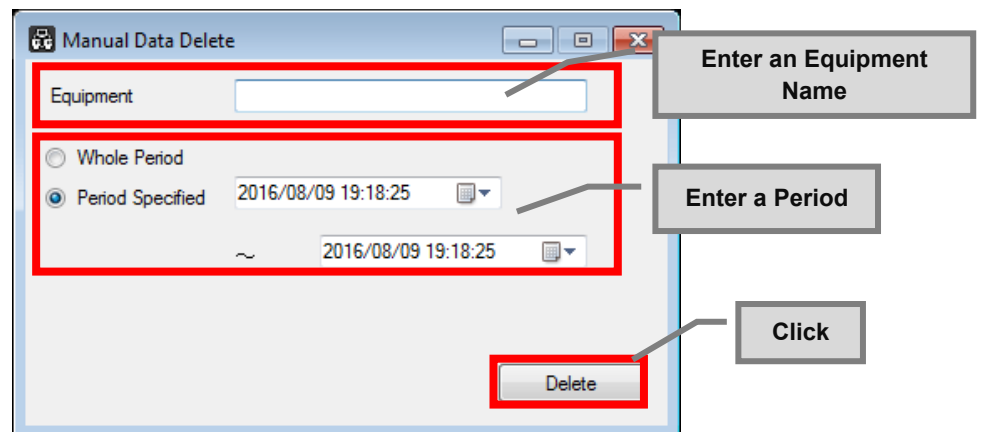
4.10.11 Deleting Manual Data

To delete registered manual data from the database, follow the steps below.

- (1) Select [Manual Data Setting] in the AdminTool Menu to open the [Manual Data Setting] dialog box.
- (2) Select the [Plan numbers] or the [Work comments] button in the [Delete] group box, and click the [Delete] button.



- (3) The [Manual Data Delete] dialog box appears. Enter an equipment name and a period (if exporting the registered manual data for the whole period, select the [Whole Period] button; or if entering specific period for exporting, select the [Period Specified] button) subject to deletion, and click the [Delete] button.

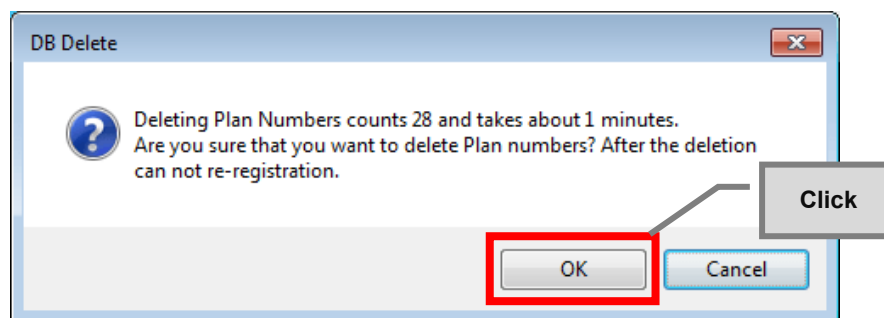


NOTE

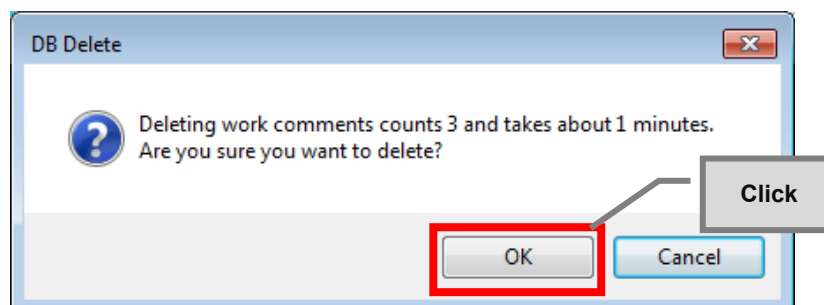
When the [Plan numbers] button is selected, date/time exceeding 90 days from the date/time when the [Delete] button is clicked cannot be specified.
When the [Work comments] button is selected, date/time later than the date/time when the [Delete] button is clicked cannot be specified.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	326/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

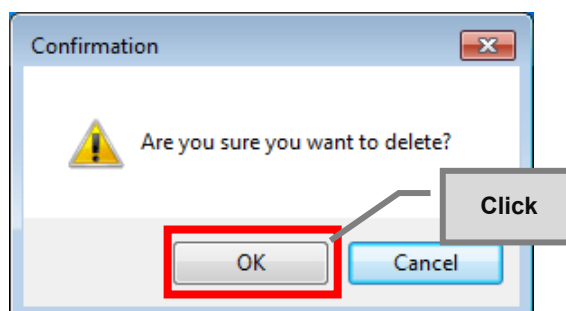
- (4) Following message is displayed. Click [OK].
- When the [Plan numbers] button is selected



- When the [Work comments] button is selected



- (5) Following message appears. When [OK] is clicked, deletion of manual data begins.



NOTE

If an error message appears while deleting the manual data, refer to the "[Errors during Deletion of Manual Data File](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	327/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

- (6) When the deletion of manual data begins, a progress bar of the deletion appears on the [Manual Data Delete] dialog box.

Manual Data Delete

Equipment

☒ Whole Period
☐ Period Specified



~ 

 Deleting

- (7) When the deletion from the database is completed, the [Manual Data Delete] dialog box is closed, and "Data deleting... >> SUCCESS" is shown in the log area of [Manual Data Setting] dialog box.

Manual Data Setting

Import

☒ Input plans ☐ Modify plans ☐ Input work comments Import

Export

☒ Plan numbers ☐ Work comments Export

Delete

☐ Plan numbers ☒ Work comments Delete

Estimating time...
>> SUCCESS

Data deleting...
>> SUCCESS

Close

Click

4.10.12 File Format for Manual Data

This subsection explains the CSV file formats for manual data.

Manual data should be prepared in a proper CSV format before performing Manual Data Setting with AdminTool. A prepared manual data file can be stored under a given folder in the Server.

A manual data file must meet the following requirements:

- It is composed of rows of data entries that have items separated by commas.
- The first row is for names of items.
- Enclose item names and setting values in double quotation marks ("").
- It must be saved in UTF-8 (with BOM).

* Item names can be either in English or Japanese.

* For the conditions for registering product plan data from a manual data file, refer to the "**Registration Conditions for Product Plan Data**" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	328/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

- * For the conditions for updating product plan data from a manual data file, refer to the "**Updating Conditions for Product Plan Data**" subsection.
- * For the conditions for registering work comments data from manual data file, refer to the "**Registration Conditions for Work Comments Data**" subsection.

NOTE

When the part that corresponds to time (displayed in the format of <HH:mm:ss>) is omitted, "00:00:00" will be displayed as the time.

■ Product plan data in a manual data file

<Requirements for product plan data in a manual data file>

Item name (English)	Item name (Japanese)	Requirement
Equipment	設備名	Name of equipment
Start date/time	開始日時	The start date and time of a production period in the format of "<yyyy/MM/dd> <HH:mm:ss>," where yyyy is 4-digit year, MM is 2-digit month, dd is 2-digit day, HH is 2-digit hour, mm is 2-digit minute, and ss is 2-digit second.
End date/time	終了日時	The end date and time of a production period in the format of "<yyyy/MM/dd> <HH:mm:ss>," where yyyy is 4-digit year, MM is 2-digit month, dd is 2-digit day, HH is 2-digit hour, mm is 2-digit minute, and ss is 2-digit second.
Product	製品名	Name of the product that the equipment produces
Plans	計画数	The planned number of the product to be produced. This number should be half-width numeric.

<Example of product plan data in a manual data file>

"Equipment","Start date","End date","Product","Plans"
 "Equipment0001","2016/02/18 09:00:00","2016/02/18 10:00:00","MAIN CONTENT","5"
 "Equipment0001","2016/02/18 10:00:00","2016/02/18 11:00:00","MAIN CONTENT","5"

■ Work comments data in a manual data file

<Requirements for work comments data in a manual data file >

Item name (English)	Item name (Japanese)	Requirement
Equipment	設備名	Name of equipment
Start date/time	開始日時	The start date and time of a production period in the format of "<yyyy/MM/dd> <HH:mm:ss>," where yyyy is 4-digit year, MM is 2-digit month, dd is 2-digit day, HH is 2-digit hour, mm is 2-digit minute, and ss is 2-digit second.
End date/time	終了日時	The end date and time of a production period in the format of "<yyyy/MM/dd> <HH:mm:ss>," where yyyy is 4-digit year, MM is 2-digit month, dd is 2-digit day, HH is 2-digit hour, mm is 2-digit minute, and ss is 2-digit second.
Work comments	作業コメント	Comments on work operated with the above equipment

<Example of work comments data in a manual data file >

"Equipment","Start date","End date","Work comments"
 "Equipment0001","2016/02/18 09:00:00","2016/02/18 10:00:00","maintenance1"
 "Equipment0001","2016/02/18 10:00:00","2016/02/18 11:00:00","maintenance2"

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 329/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

4.10.13 Registration Conditions for Product Plan Data

This subsection describes the conditions for plan data entries to be registered.

Loaded entries from a manual plan data file are checked in two ways: they are checked individually as one plan data entry, and collectively as one group of entries that have a same equipment item and a same product item in common. In each group, there must be the oldest and latest entries (we refer to the span between those entries as "the period in loaded plan data").

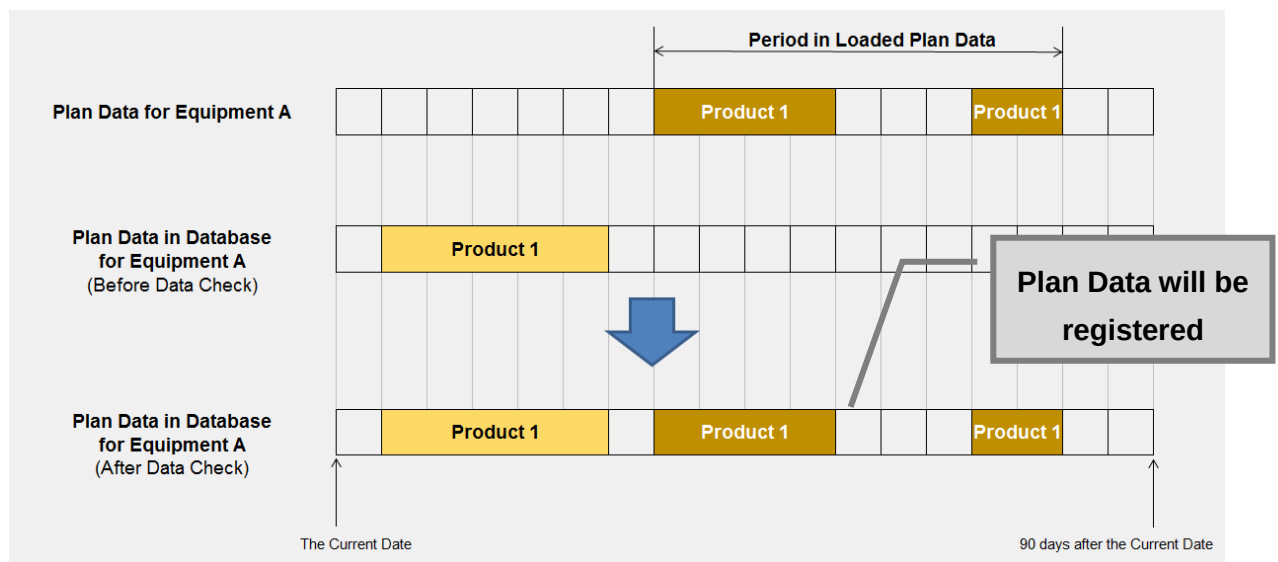
To be eligible for registration, the entries of a group must meet the following conditions:

- The period in loaded plan data does not overlap the span of the oldest and the latest of the registered plan data that have the same equipment and the same product as the group has.
- Start date of any entry is the current date or later, and End date of any entry is 90 days after the current date or earlier. (Current date / time ≤ Start date /time, and, End date / time ≤ Current date /time + 90 days)

* The period of the loaded manual plan data is the span of period between the oldest and the latest date for the data of the same equipment and the same product.

* The latest date of the loaded plan data must be later than the latest of the registered product plan data.
(For an example failing to meet this condition, see <Case B> below.)

Below is an example of manual plan data that will be registered (all the entries are eligible):



If an entry fails to meet both or either of the conditions, no plan data entry in a group will be registered; thus the database remains as is.

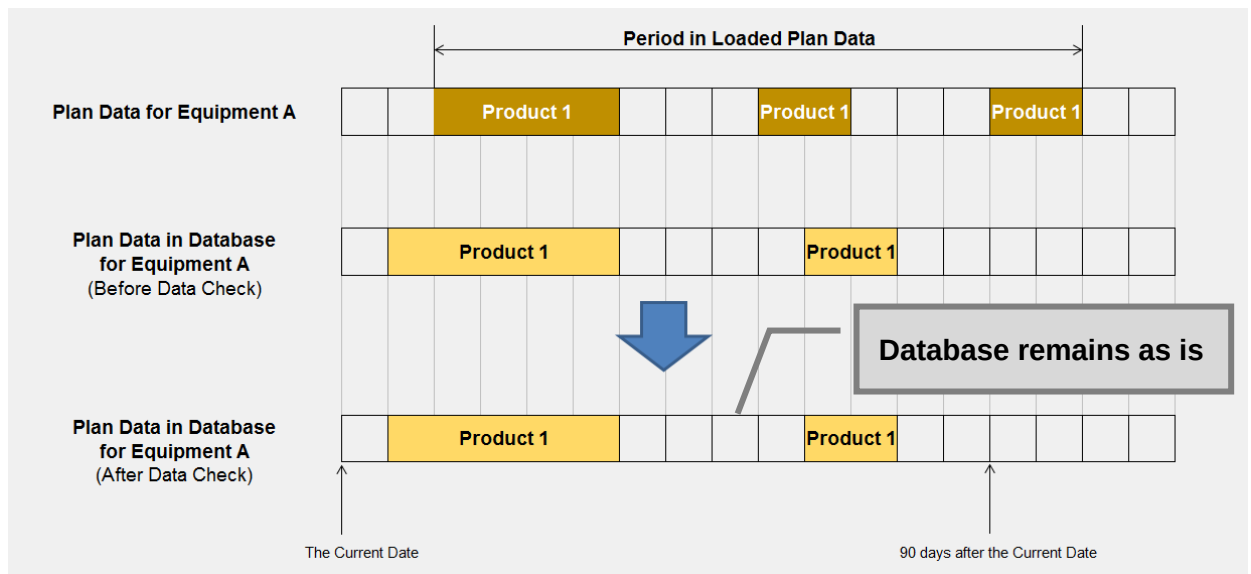
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	330/868

Below are examples of manual plan data that fail to meet the conditions:

<Case A>

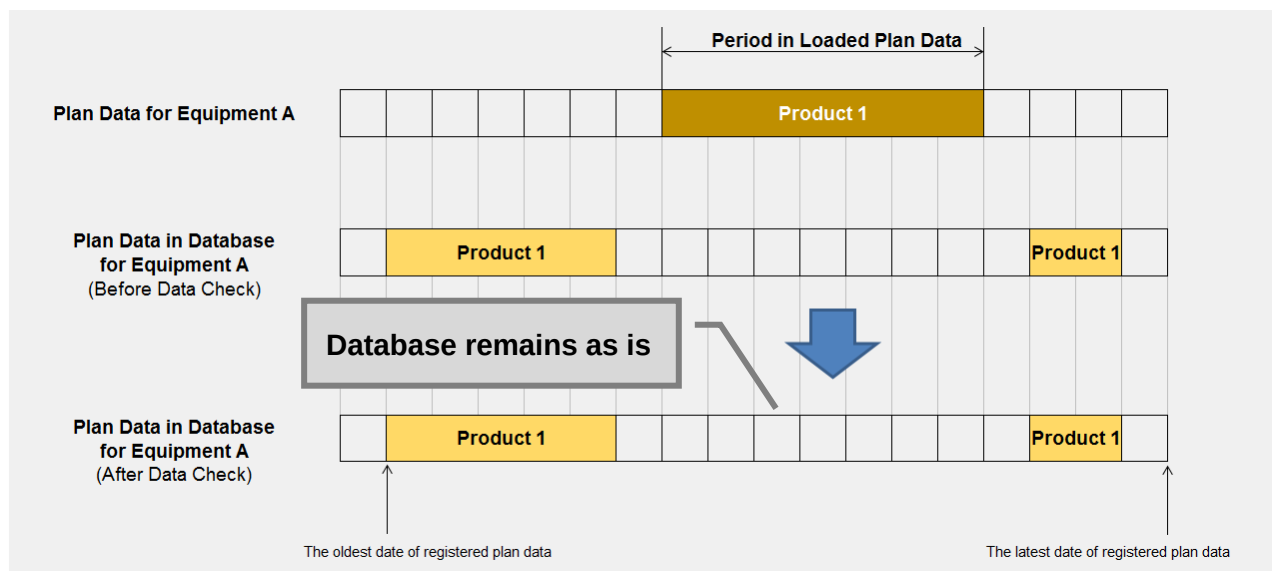
Cause of registration failure:

Overlap in the period of the loaded plan data, 90 days from current day < End date of the loaded plan data



<Case B>

Cause of registration failure: Latest date of the loaded plan data < Latest date of the registered plan data



4.10.14 Updating Conditions for Product Plan Data

This subsection describes the conditions for plan data entries to be updated with.

Loaded entries from a manual plan data file are checked in two ways: they are checked individually as one plan data entry, and collectively as one group of entries that have a same equipment item and a same product item in common. In each group, there must be the oldest and latest entries (we refer to the span between those entries as "the period in loaded plan data").

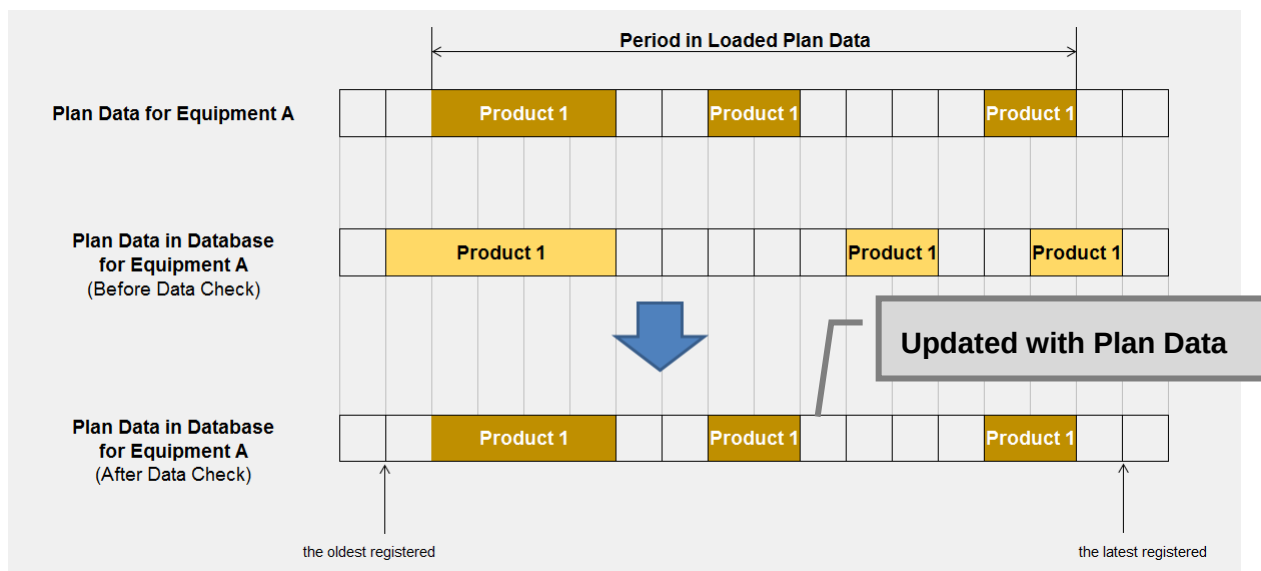
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	331/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

To be eligible for update, the entries of a group must meet the following condition:

- The period in loaded plan data falls in the span of the oldest and the latest registered product plan data that have the same equipment and the same product as the group has.
That is "the Start date of the oldest registered plan data" ≤ "Start date," and "End date" ≤ "the End date of the latest registered plan data."

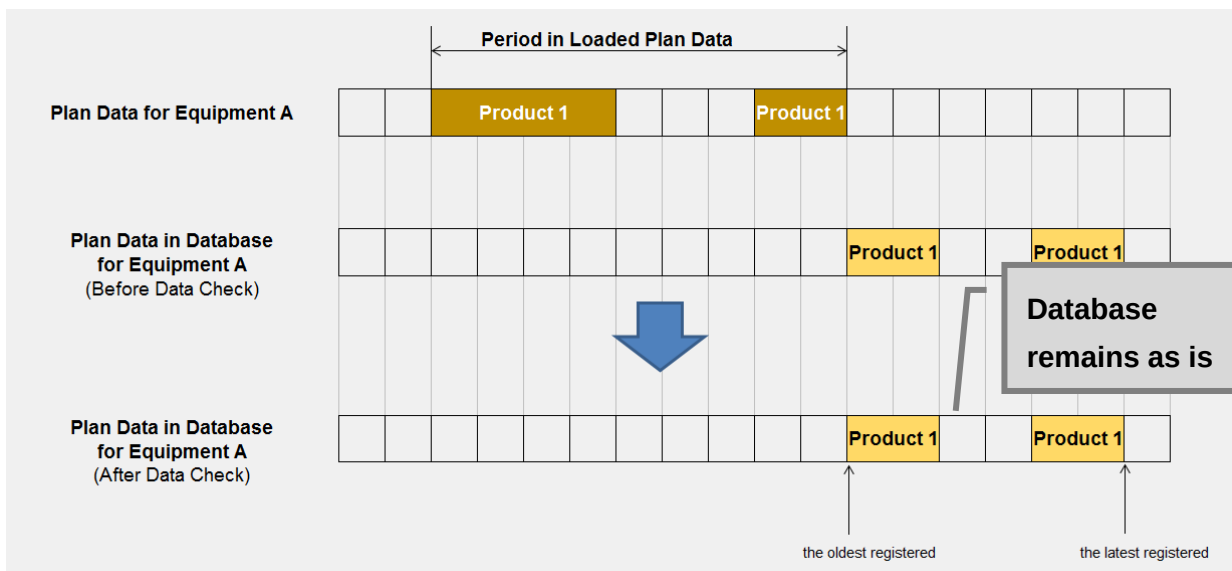
Note that when all the entries of a group are eligible, all the registered plan data within "the period in loaded plan data" will be removed and replaced with the entries.

Below is an example of manual plan data that will be registered (all the entries are eligible):



If an entry fails to meet the condition, no entry in the group will be registered; thus the database remains as is. Below is an example of manual plan data that will not be registered:

Cause of registration failure: Start/End date of the loaded plan data < Oldest date of the registered plan data



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue
FANUC CORPORATION					Sheet.	332/868	

4.10.15 Registration Conditions for Work Comments Data

This subsection describes the conditions for work comments data entries to be registered.

Loaded entries from a manual work comments data file are checked in two ways: they are checked individually as one work comments data entry, and collectively as one group of entries that have a same equipment item in common. In each group, there must be the oldest and latest entries (we refer to the span between those entries as "the period in loaded work comments data").

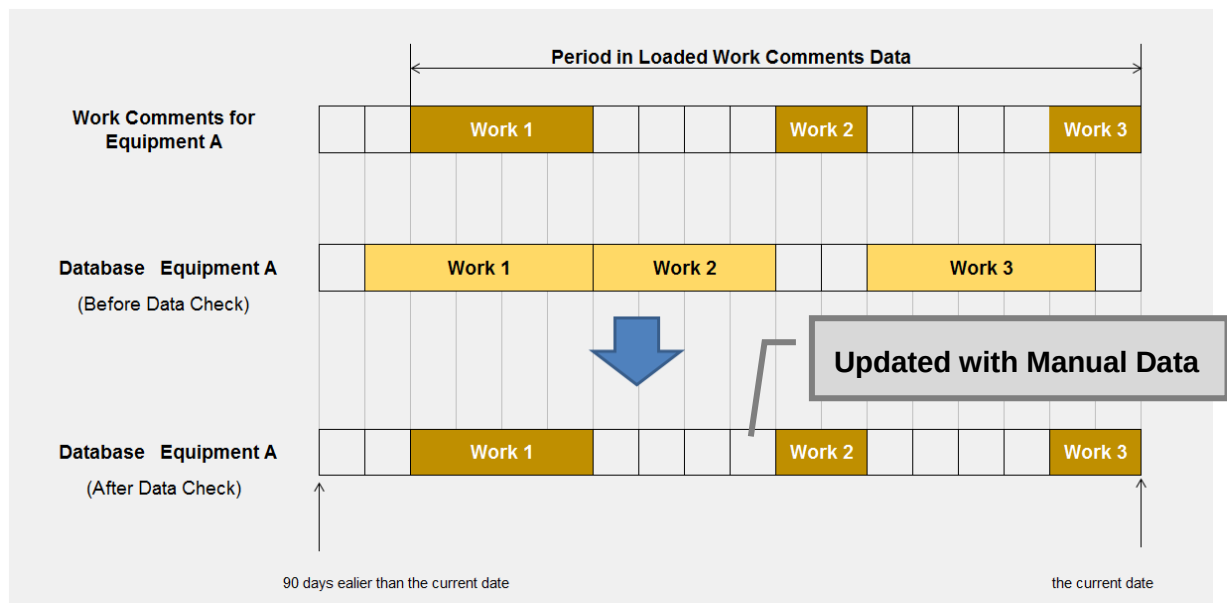
To be eligible for registration, the entries of a group must meet the following condition:

- Start date of the oldest entry is 90 days before the current date or later, and End date of the latest entry is the current date or earlier.

That is "the current date – 90 days" ≤ "Start date," and "End date" ≤ "the current date."

Note that when all the entries of a group are eligible, all registered work comments data within "the period in loaded work comments data" will be removed and replaced with the entries.

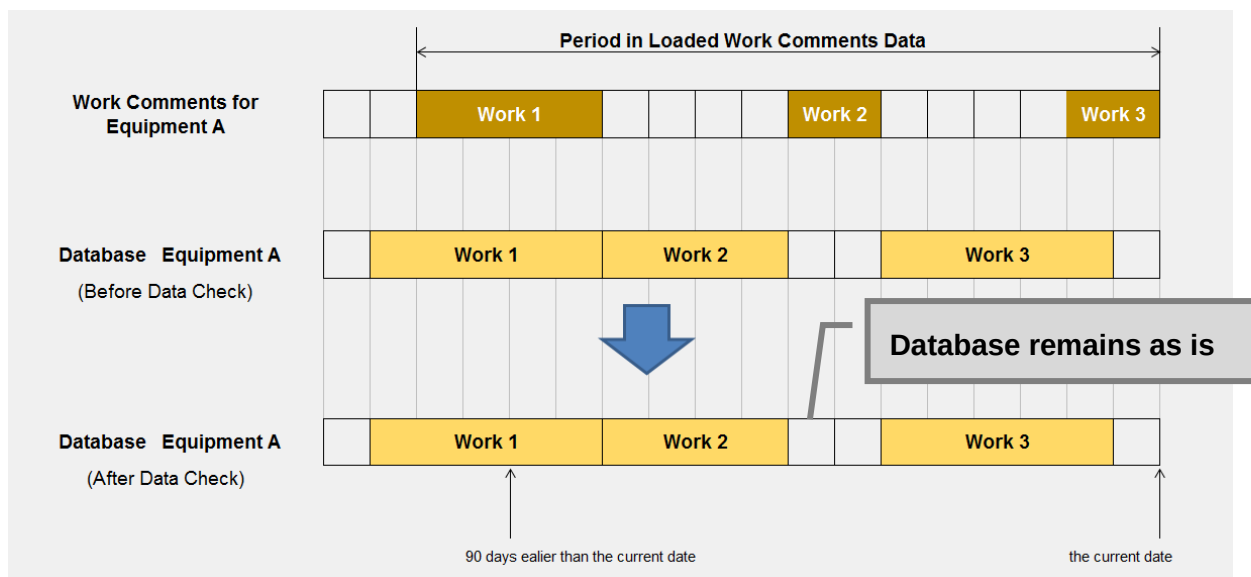
Below is an example of work comments data that will be registered (all the entries are eligible):



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 333/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

If an entry fails to meet the condition, no entry in the group will be registered; thus the database remains as is. Below is an example of manual work comments data that will not be registered:

Cause of registration failure: Start date of the loaded plan data < 90 days before current date



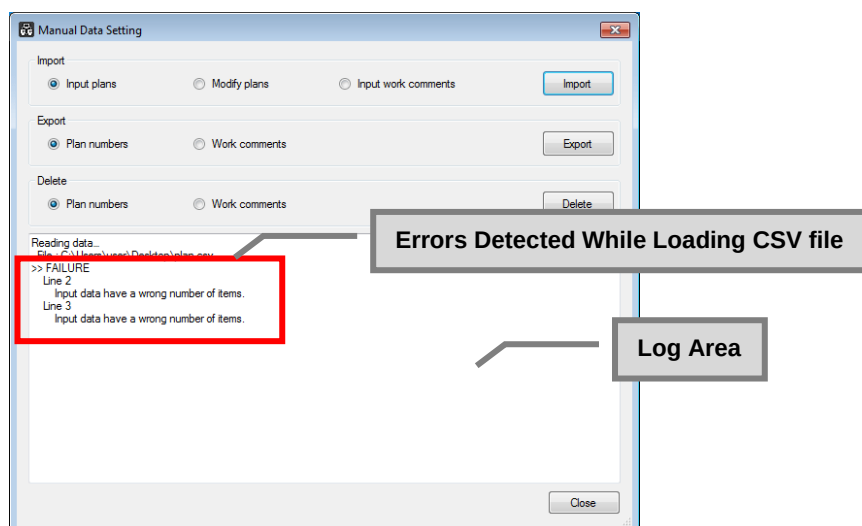
4.10.16 Errors on Reading Manual Data File

This subsection describes errors that may occur while reading a manual data file.

If an error occurs during reading a CSV file, an error message appears on the log area on the [Manual Data Setting] dialog box. At that point reading process terminates.

Correct the errors in the file by referring to the line numbers and error messages shown on the log area of the [Manual Data Setting] dialog box.

■ In case the message ">> FAILURE" appears in the next line of the message "Data reading" on the log area of the [Manual Data Setting] dialog box:

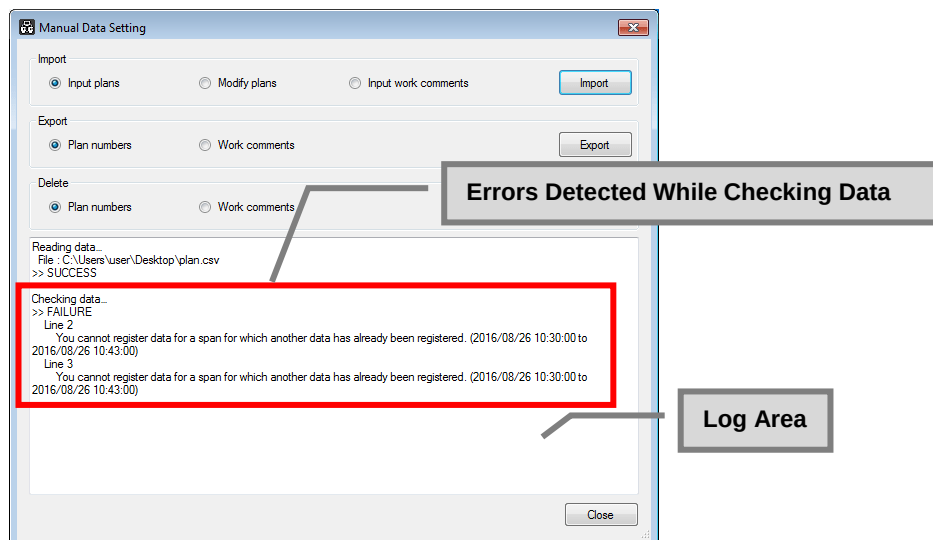


20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 334/868

The following is a possible cause of the error.

- The number of items in the CSV file is incorrect. Verify that the number of the item types is exactly the same (not less or more) as the format defines, and then correct the errors in the file by referring to the line numbers and error messages shown on the log area.

■ In case the message ">> FAILURE" appears in the next line of the message "Data checking" on the log area of the [Manual Data Setting] dialog box:



The following are possible causes of the error.

- Data loaded from the CSV file have invalid values.
- When registering product plan data, the database has product plan data whose production period is identical or overlapping with that of the loaded product plan data in error.
- When updating product plan data, the database has no product plan data whose production period is identical or overlapping with that of the loaded product plan data in error.

Correct errors in the file by referring to the requirements of plan data, and the line numbers and error messages shown on the log area.

4.10.17 Errors during Exporting of Manual Data File

If an error message appears while exporting the manual data file, the exporting process terminates. Check the description of the error message, and reconsider the entry details in the [Export Manual Data] dialog box.

Following are the possible causes of the error during export.

- Equipment name specified is invalid.
- Period specified is invalid.

4.10.18 Errors during Deletion of Manual Data File

If an error message appears while deleting the manual data file, the deleting process terminates. Check the description of the error, and reconsider the entry details in the [Delete Manual Data] dialog box.

Following are the possible causes of the error.

- Equipment name specified is invalid.
- Period specified is invalid
- Target output path specified is invalid.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 335/868

4.11 AUTHORITY SETTING

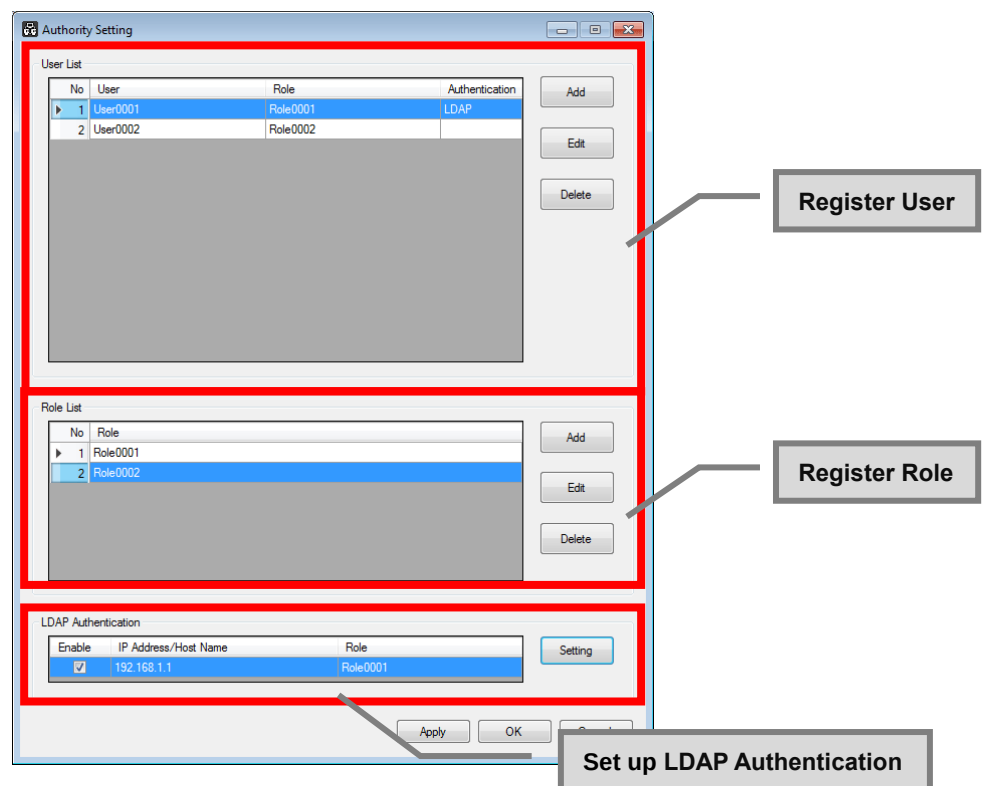
The [Authority Setting] window registers users and roles, and set up LDAP authentication functionality. Multiple roles can be assigned to a single user.

This window enables you to do the following:

- To register users (up to 1000 users)
- To register roles (up to 100 roles)
- To assign a single role to multiple users
- To set up and activate LDAP authentication functionality. However, this window allows only one entry.

NOTE

- (1) Refer to the "**LDAP AUTHENTICATION**" section for the details of LDAP authentication functionality.
- (2) In this section, an "LDAP authentication user" denotes a user whose [Use LDAP Authentication] check box is selected in the [Add/Edit User] dialog box.



This window has three group boxes as follows:

■ User List

User: The name of each registered user

Role: The user-role assigned to each registered user

Authentication: "LDAP" indicates the user is an LDAP authentication user.

(Space indicates the user is an MT-LINK*i* user.)

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	336/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

- Role List
Role: A registered user-role
- LDAP Authentication
Enable: Selecting this check box activates LDAP authentication functionality.
IP Address/Host Name: The IP address or host name of an LDAP server
Role: The default user-role of LDAP authentication users
* This list has only the header row when no entry is added.

4.11.1 Steps for Authority Setting

Follow the steps below to set authority to a user.

- (1) Select [Authority Setting] in the AdminTool Menu. The [Authority Setting] window appears.
- (2) Add a role. For the details, refer to the "[Add/Edit Role of Authority](#)" subsection.
- (3) Add a user and assign the role to it. Refer to the "[Add/Edit User](#)" subsection.
- (4) Configure LDAP authentication functionality, if needed. For the details, refer to the "[Set LDAP Authentication](#)" subsection.
- (5) Click [Apply] or [OK]. Clicking [Cancel] discards the settings you made and closes the window.
- (6) In order to activate the changes in settings, you are required to send data to DB. For the details of the operation, refer to the "[SEND TO DB](#)" section.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 337/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

4.11.2 Add/Edit Role of Authority

Follow the steps below to add role.

- (1) To add a new role, click the [Add] button in the [Role List] group box on the [Authority Setting] window. To edit an existing role, select the target role and click the [Edit] button in the [Role List] group box. The [Add/Edit Role of Authority] dialog box appears.

- (2) Specify the role name and set the types of authorities as follows.

- Setting authority by Group

Set the authority for the groups including equipment units displayed in the [Monitoring - Overlook] screen, and groups selectable with the [Filter] in the following Web screens. The default setting is "Read Only."

- Monitoring - Equipment Monitoring
- Monitoring - Group Monitoring
- Monitoring - Signal Monitoring
- Monitoring - Alarm Monitoring
- Results - Operational Results
- Results - Group Results
- Results - Production Results
- Diagnosis - Alarm History
- Diagnosis - Signal History
- Diagnosis - Program History
- Diagnosis - Macro Variable History
- Diagnosis - Report Output
- Utility - File Transfer
- Utility - Machine Data Archive Management
- Monitoring - Tool Life Monitoring
- Monitoring - Periodic Maintenance Monitoring

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	338/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- Utility - Simultaneous File Transmission
- Utility - System Setting
- Setting authority by Layout
Set the authority for the layouts selectable on the [Monitoring - Overlook] screen of the Web screen. The default setting is "Read Only."
- Setting authority by Function (WebUI screen)
Set the authority for displaying the following WebUI screens, and the authority to use those functions. The default setting is "-----" (no authority) for SERVO VIEWER linkage and System Setting, and "Read Only" for the others.
 - Monitoring - Overlook
 - Monitoring - Equipment Monitoring
 - Monitoring - Group Monitoring
 - Monitoring - Equipment Monitoring - Detail
 - Monitoring - Signal Monitoring
 - Monitoring - Alarm Monitoring
 - Results - Operational Results
 - Results - Group Results
 - Results - Production Results
 - Diagnosis - Alarm History
 - Diagnosis - Signal History
 - Diagnosis - Program History
 - Diagnosis - Macro Variable History
 - Diagnosis - Report Output
 - Utility - File Transfer
 - Utility - System Diagnosis
 - SERVO VIEWER Linkage
 - Utility - Machine Data Archive Management
 - Monitoring - Tool Life Monitoring
 - Monitoring - Periodic Maintenance Monitoring
 - Utility - Simultaneous File Transmission
 - Utility - System Setting
 - Results - Power Consumption Monitoring

- Setting authority by System
Set the authority for the selectable linkage systems on the Web screen. The default setting is "Read Only."

The authority level that can be used for the setting and its description is explained below.

Authority level	Description
----- (no authority)	Cannot browse the Web screen, select items on the Web screen, or cannot use the function. - Transition from [Monitoring - Equipment Monitoring] Web screen using the selector is disabled. - Disabled (not selectable) on the [Home] screen and the AdminTool Menu. - Equipment units included in the group set with this authority level (no authority) will not be displayed on the [Monitoring - Overlook] Web screen. - Group set with this authority level (no authority) will not be displayed on the [Filter]. - SERVO VIEWER linkage cannot be used.
Read Only	Browsing the Web screen, selecting items on the Web screen, or using the functions will be allowed. - [Download] button will be disabled in the [Utility - File transfer] Web screen. - Settings for collection of waveform data, and settings for a Measurement schedule are disabled for SERVO VIEWER linkage. - This level is unavailable for the System Setting function.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 339/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

Authority level	Description
Read/Write	Browsing the Web screen, selecting items on the Web screen, or using the function will be allowed. - All the operations can be performed in the [Utility - File transfer] and [System Setting] Web screens. - All the functions of SERVO VIEWER linkage can be used. - The [Setting authority by system] group box does not have this option. - This authority level cannot be selected for the authority setting of the linkage systems.

NOTE

- 1 Except for File Transfer, Simultaneous File Transmission, and SERVO VIEWER linkage functions, "Read only" and "Read/Write" will be regarded as the same. However, "Read Only" is unavailable for the System Setting function.
- 2 The [Setting authority by Function] column for Tool Life Monitoring is disabled until ROBODRILL Option or Update Option license is installed.
- 3 The [Setting authority by Function] columns for Periodic Maintenance Monitoring and Periodic Maintenance Monitoring are disabled until ROBODRILL Option license is installed.
- 4 The [Setting authority by Function] column for System Setting is disabled until Update Option license is installed.
- 5 The [Filter] in the [Utility - File Transfer] and the [Utility - Machine Data Archive Management] Web screens will not display equipment units or groups of equipment units if such equipment units consist only of machines that are not supported by File Transfer function (i.e., neither FanucCNC type nor FanucRobodrill type machines).
- 6 The [Filter] in the [Utility – Simultaneous File Transmission] Web screen will not display groups if they consist only of equipment units that are composed of unsupported machines by Simultaneous File Transmission function (i.e., other than FanucRobodrill type machines).
- 7 The [Results - Power Consumption Monitoring] Web screen will not display equipment units or groups if they consist only of equipment units that are composed of unsupported machines by Power Consumption Monitoring function (i.e., neither FanucCNC type nor FanucRobodrill type machines).
- 8 In order to use the SERVO VIEWER linkage function, the SERVO VIEWER and the Linkage tool for MT-LINK*i* will be required.

(3) To assign the role being edited to the users that are not assigned any role, click [Apply to unassigned users].

(4) Click [OK] to close the window.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	340/868

4.11.3 Add/Edit User

Follow the steps below to add user.

- (1) To add a new user, click the [Add] button in the [User List] group box on the [Authority Setting] window. To edit an existing user, select the target user and click the [Edit] button in the [User List] group box. The [Add/Edit User] dialog box appears.

The screenshot shows the 'Add/Edit User' dialog box with the following callouts:

- User Name**: Points to the 'User' text field containing 'User0001'.
- Use LDAP Authentication**: Points to the 'Use LDAP Authentication' checkbox.
- Role List**: Points to the 'Role List' text field containing 'Role0002'.
- Role Assigned to User**: Points to the 'Role' text field containing 'Role0001'.
- New Password**: Points to the 'New Password' text field.
- State of Password Setting**: Points to the 'Password not set' text.
- Mail Addresses**: Points to the 'E-mail 1', 'E-mail 2', and 'E-mail 3' text fields.
- Web Sign in**: Points to the 'IP Address' text field containing '192.168.137.151'.
- Calendar Period**: Points to the 'Calendar Period' table.
- Language**: Points to the 'Language' dropdown menu.
- OK**: Points to the 'OK' button.

- (2) In this window, specify [User], [Use LDAP Authentication], [Role] assigned to the user, [New Password], [Mail Address], [Web Sign in], [Calendar Period], and [Language].
The user name and password specified in this window are to be set to [User] and [New Password] in the [MT-LINKi Web Sign in] screen for signing in to the WebUI.
- (3) Select the [Use LDAP Authentication] check box to register a user as an LDAP authentication user. LDAP authentication users will be authenticated by the LDAP server at their signing in to the Web Server.
When this check box is selected, the following fields will deny modification:
 - The [Enable Password] check box (fixed to cleared)
 - The [New Password] text box
 - The [Skip] check box (fixed to cleared)

By default, this check box is cleared.
- (4) To add a role to the user, select a role from the [Role List] list and click [Add]. To delete a role from the user, select the target role from the [User Role] list and click [Remove].

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	341/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

- (5) Selecting this check box activates the [New Password] text box for typing in.
 The [New Password] text box allows a string of four to 254 bytes long. When entering the password, "*" (asterisk) will be displayed in place of the letters to mask the password. The text box denies the user to directly type in double byte characters. If needed, copy a password of double byte characters from a text editor and paste it into this text box.
 On the right side of the [New Password] column, the state of the password setting will be displayed. After clicking the [OK] button to confirm the setting, if the [Add/Edit User] dialog box for the same user is displayed again, [New Password] column will be shown as a blank column, and "Password set" will be displayed as the state of password setting in the display area.
 When the [Enable Password] check box is cleared, the user can sign in to the Web Server without entering a password. By default, [Enable Password] check box is cleared, and the state of the password setting is "Password not set."
- (6) Set the destination mail address for delivering the signal detection mail and the alarm mail.
 The signal detection mail/alarm mail of the selected target equipment unit, selected as the [Equipment] in the [Mail Delivery List] of the [Mail Function Setting] window, will be delivered to the user selected on the [User List] (the mail address set in this setting window). Up to three destination e-mail addresses can be specified. If no address is entered, it will be determined as invalid, and the mail will not be delivered.
- (7) After specifying the [User] and [New Password] (or selecting not to enter the password), you can select or clear the [Skip] check box for each IP address registered. Selecting the [Skip] check box enables the user to skip the [MT-LINKi Web Sign in] screen and to directly open the [Monitoring - Overlook] screen. When a [Skip] check box is selected, the selected IP address along with the user name and the password will be populated in a URL format in the text box in the [Web Sign in] group box. With this URL, users can open the [Monitoring - Overlook] screen directly without sign-in procedure.

NOTE

When the [Skip] check box is selected and the value of the [User] field has symbol characters other than hyphen ("-"), underscore ("_"), or dot ("."), clicking the [OK] button displays the following error message:

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 342/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

4.11.4 Set LDAP Authentication

Follow the steps below to configure LDAP authentication functionality on MT-LINK*i*.

* For the details of LDAP authentication functionality, refer to "**LDAP AUTHENTICATION**" section.

- (1) Click the [Set] button in the [LDAP Authentication] group box of the [Authority Setting] window to register or edit LDAP authentication settings.

The [LDAP Authentication Setting] window appears as follows:

- (2) This window has setting fields for a range of LDAP server information as follows:

■ LDAP Server Setting

- IP Address/Host Name (required)

This field holds the IP address or host name of an LDAP server.

When specifying a host name, place an alphabet character at its top position. The rest of the string may be alphanumeric, hyphens ("-"), or dots ("."). The specified string must be up to 253 bytes long.

- Port (required)

This field holds the number of a port to use for communication with the LDAP server.

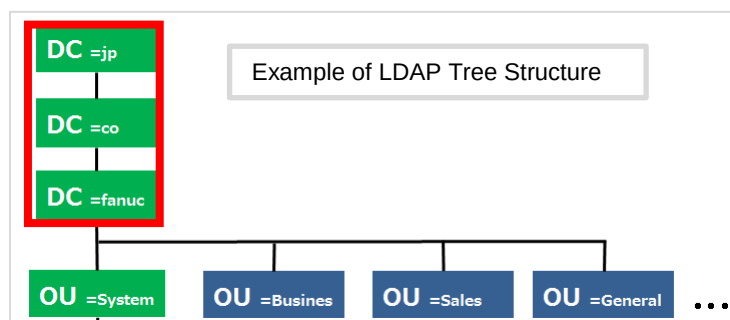
Its value must be between "0" and "65535." It has a default value of "389."

- Domain (required)

This field holds the domain's LDAP distinguished name, each of whose domain component attributes is paired with its value. These attribute-value pairs must be separated with commas. The specified LDAP distinguished name must be up to 500 bytes long.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 344/868

Example: When "fanuc.co.jp" is a domain name to pass to the LDAP server, specify the three domain component attributes as "DC=fanuc,DC=co,DC=jp" for this field.

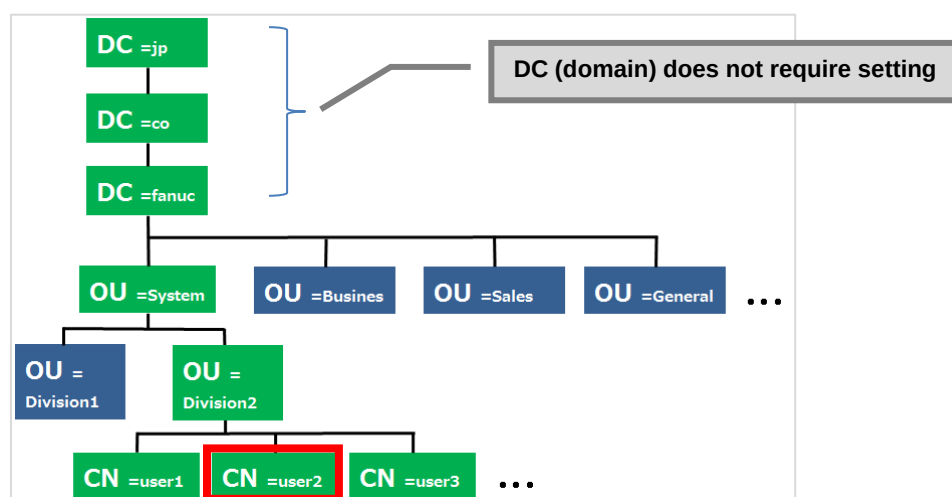


- Account (required)

This text box holds an administrative user account in LDAP distinguished name format. An LDAP distinguished name is composed of LDAP attributes, each of which is paired with its value. These attribute-value pairs must be separated with commas.

The specified LDAP distinguished name must be up to 500 bytes long.

Example: When using "user2" as the user account for the CN attribute shown in the figure below, specify "CN=user2,OU=Division2,OU=System" for this field. (In this case, DC attributes are not required.)



- Password (required)

This text box holds the password for the administrative user account specified for the [Account] text box.

It allows a string of one to 254 bytes long. It displays each of entered characters as an asterisk (*).

This text box denies the user to directly type in double byte characters. If needed, copy a password of double byte characters from a text editor and paste it into this text box.

- Attribute (User) (required)

This text box holds a user attribute for the LDAP server to use in search of users. The text box allows a string whose first character is an alphabet and is followed by alphanumerics or hyphens ("-"). A string of up to 32 bytes long can be entered in this text box.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	345/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

NOTE

Values available to the [Attribute (User)] field depend on the LDAP server (Directory service):

<Active Directory>

Either of the following values should be set.

- userPrincipalName: user name with domain name
(When signing in to the Web Server, input a user name with an atmark and domain name)
- sAMAccountName: user name without domain name
(user name for Windows 2000 or older)

<OpenLDAP>

The following value should be set.

- uid: user name without domain name

If needed to use a user attribute other than above, employ an attribute whose value is unique across the directory; otherwise, a user may be unexpectedly authenticated as the wrong user account when it has the same attribute value and thus can be retrieved from the LDAP server.

- **Attribute (E-mail address)**
This text box holds E-mail address attributes for the LDAP server to use in search of E-mail addresses. The text box allows a string whose first character is an alphabet and is followed by alphanumerics or hyphens ("-"). A string of up to 165 bytes long can be entered in this text box. When specifying multiple E-mail addresses, separate them with commas as follows.

Example: mail01, mail02, mail03

NOTE

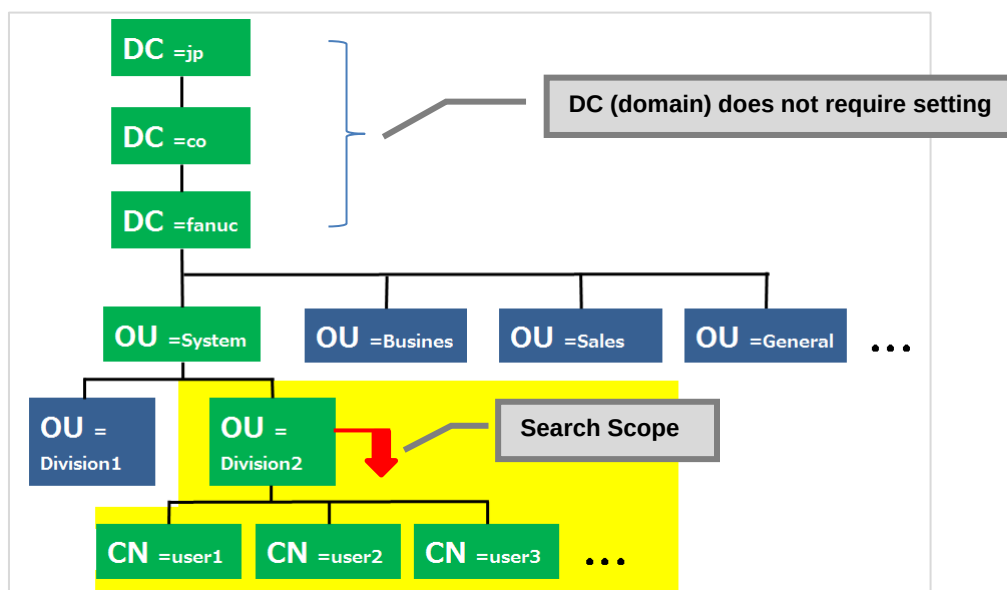
If more than three addresses are set for the [Attribute (E-mail address)] field, only the first three of the addresses retrieved from the LDAP server will be available. The retrieved addresses will be stored in the database (MongoDB). If a value of this attribute is updated on the LDAP server after automatic registration, the database (MongoDB) will not reflect the update.

For the details of automatic registration, refer to the "[How Sign-in from Web Browser Is Authenticated](#)" subsection.

- **Search Scope**
This text box holds a scope for the LDAP server to use in search of users. When this text box is left unspecified, the LDAP server searches all the entries in the domain. A scope of up to 500 bytes long can be entered in this text box.
Specifying a proper scope when searching a complex LDAP directory may help the LDAP server conduct a more effective search and thus users can sign in in a shorter time.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	346/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

Example: When specifying all the entries under "Division2" as a search scope, set "OU=Division2, OU=System" for this text box. (In this case, DC attributes are not required.)



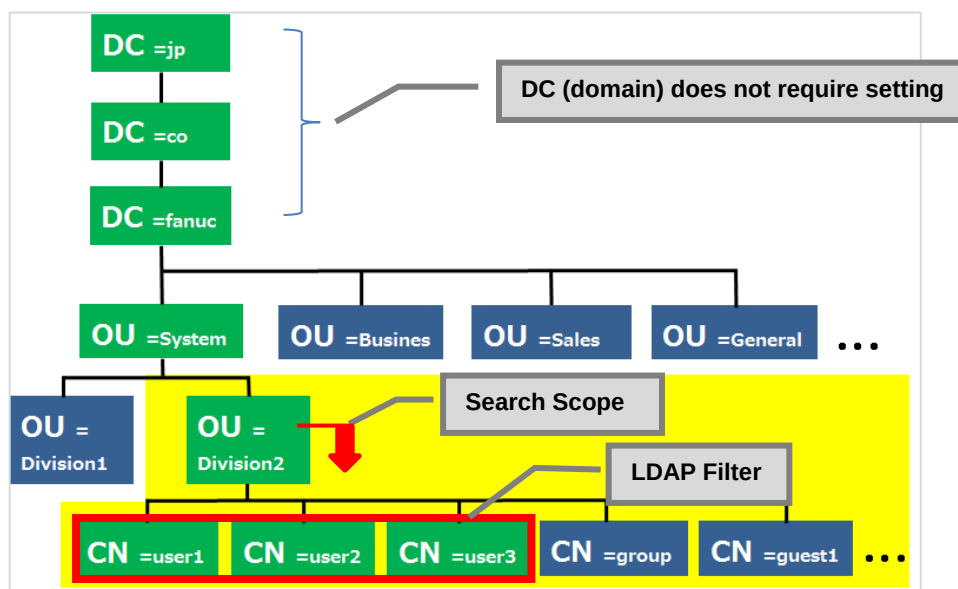
- LDAP Filter

This text box holds search criteria for the LDAP server to refine in search of users.

A value of up to 500 bytes long can be entered in this text box.

This filter can be used to refine a search query which uses comparison operators to search the specified search scope.

Example: When specifying all the entries under "Division2" as a search scope, and user name (CN) entries which start with "user" in a refined search query, set "CN=user*" for this text box. (In this case, DC attributes are not required.)



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	347/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

The following table lists the main comparison operators available for the [LDAP Filter] text box (RFC4515 compliant):

Comparison operator	Condition	Usage example	Description
=	equal to	cn =Fanuc Taro	Searches entries only whose cn is "Fanuc Taro"
>=	greater than or equal to	uidNumber >=1000	Searches entries whose uidNumber is 1000 or greater
<=	less than or equal to	uidNumber <=1000	Searches entries whose uidNumber is 1000 or less
&	logical and	(&(cn =Taro)(c =Japan))	Searches entries whose cn is "Taro" and whose c is "Japan"
	logical or	((givenName =Taro)(givenName =Jiro))	Searches entries whose givenName is "Taro" or "Jiro"
!	negation	!(cn =Fanuc Taro))	Searches entries whose cn is not "Fanuc Taro"
=*	one or more characters	cn =*	Searches all the entries whose cn has a value
		cn =Fanuc*	Searches entries whose cn has a value which starts with "Fanuc"

* The table shows LDAP attributes in blue.

- **Connection Test**
Clicking this button conducts a connection test with the LDAP server on the settings. Use this button to verify that a connection with the LDAP server can be established.
- **Result Display Area**
This area displays the results of a connection test executed by clicking the [Connection Test] button.

NOTE

Refer to the following subsections if a connection test fails or if it is uncertain how and what to set with this window:

"[Setting Example for Active Directory](#)"

"[Setting Example for OpenLDAP](#)"

For the details, contact your LDAP server administrator.

■ User Auto-registration

- **Create a user account at first sign-in**
Selecting this check box creates MT-LINK*i* user accounts for LDAP authentication users when they sign in to the Web Server for the first time. This is why those users can be authenticated at their first sign-in.

Clearing this check box disables this automatic user registration and thus users will be denied to sign in unless they have been registered as MT-LINK*i* users.

However, this check box must be cleared to address the following cases:

- The LDAP authentication users who are used in signing in to the MT-LINK*i* Web Server have been already registered as MT-LINK*i* users, and no more LDAP authentication users are to be registered.
- The number of registered users reaches the maximum and thus maintenance of user accounts is required.
- LDAP authentication users are to be registered manually.

By default, this check box is cleared.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	348/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

■ Default Authority Settings

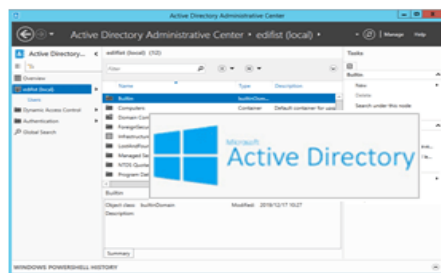
- Roll
For the details on how to assign a roll to a user, refer to step (4) in the "[Add/Edit User](#)" subsection.
- Calendar Period
For the details on how to set a calendar period, refer to step (8) in the "[Add/Edit User](#)" subsection.
- Language
For the details on how to specify a language, refer to step (9) in the "[Add/Edit User](#)" subsection.

(3) When setting is completed, click [OK] to close the window.

- OK
After verifying that the window has no error, click [OK] to save the settings made and close the window. The [LDAP Authentication] group box of the [Authority Setting] window reflects those settings. The [Enable] check box in the group box is selected by default. When an error is detected, the symbol "!" or a popup message appears to notify of an input error.
- Cancel
Clicking this button discards the settings made and closes the [LDAP Authentication Setting] window.

4.11.4.1 Setting Example for Active Directory

This subsection shows an example of setting the [LDAP Authentication Setting] window when the LDAP server is running Active Directory.

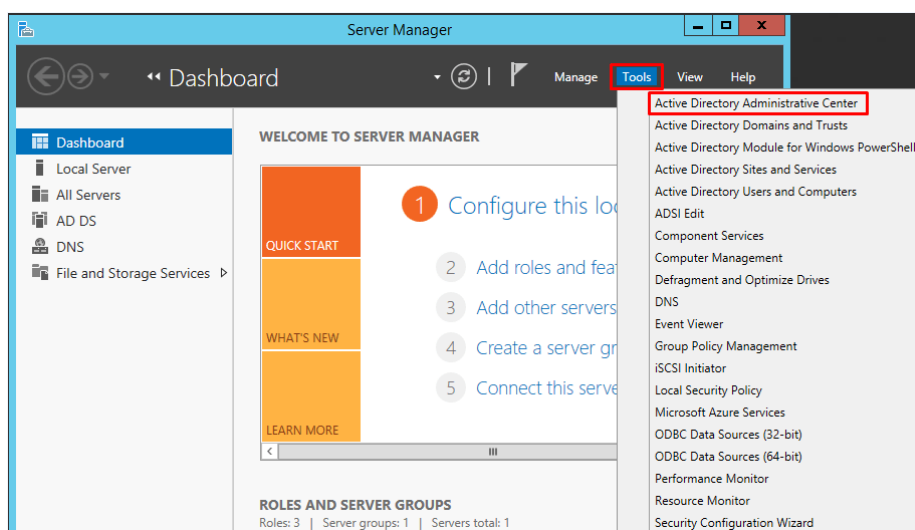


Fields to set	Example values	Refer to
IP Address/Host Name	192.168.1.1	" Set LDAP Authentication " subsection
Port	389	" Set LDAP Authentication " subsection
Domain	DC=fanuc,DC=co,DC=jp	"(1) How to verify the [Domain] field " under "The [Active Directory Administrative Center] window"
Account	CN=Fanuc Taro,CN=Users	"(2) How to verify the [Account] field " under "The [Active Directory Administrative Center] window"
Password	*****	Contact LDAP administrator when uncertain about what to set
Attribute (User)	userPrincipalName	"(3) How to verify the [Attribute (user)] field " under "The [Active Directory Administrative Center] window"
Attribute (E-mail address)	mail	"(4) How to verify the [Attribute (E-mail address)] field " under "The [Active Directory Administrative Center] window"
Search Scope	OU= Division2, OU= System	" Set LDAP Authentication " subsection
LDAP Filter	uidNumber>=1000	" Set LDAP Authentication " subsection

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 349/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

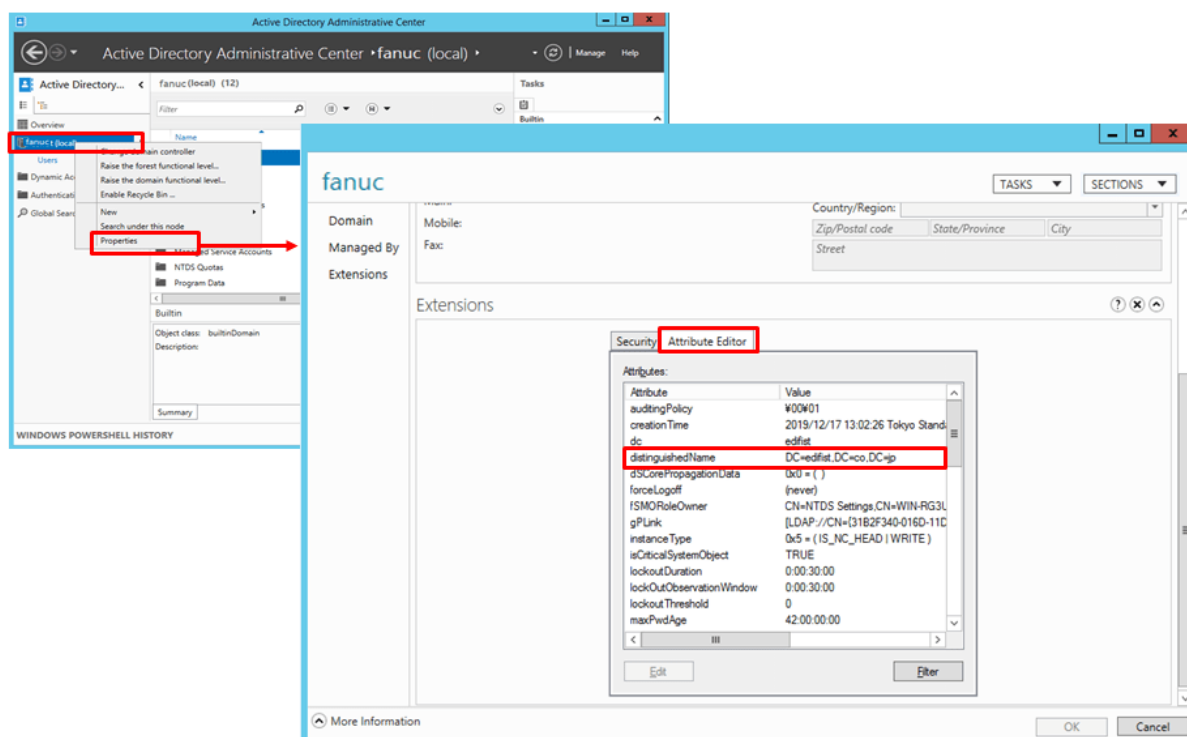
■ The [Active Directory Administrative Center] window

To open this window, click the [Start] button > [Server Manager] on the LDAP server to open the [Server Manager] window, and then select [Active Directory Administrative Center] from the context menu in the [Tool] drop-down list on the upper menu.



(1) How to verify the [Domain] field

Select a domain name and right-click it to open a context menu. Select [Properties] from the context menu to open a property window. Scroll down the window to find and select the [Attribute Editor] tab. The [Domain] field in the [LDAP Authentication Setting] window should be set to the value of the [distinguishedName] entry listed in the tab page.

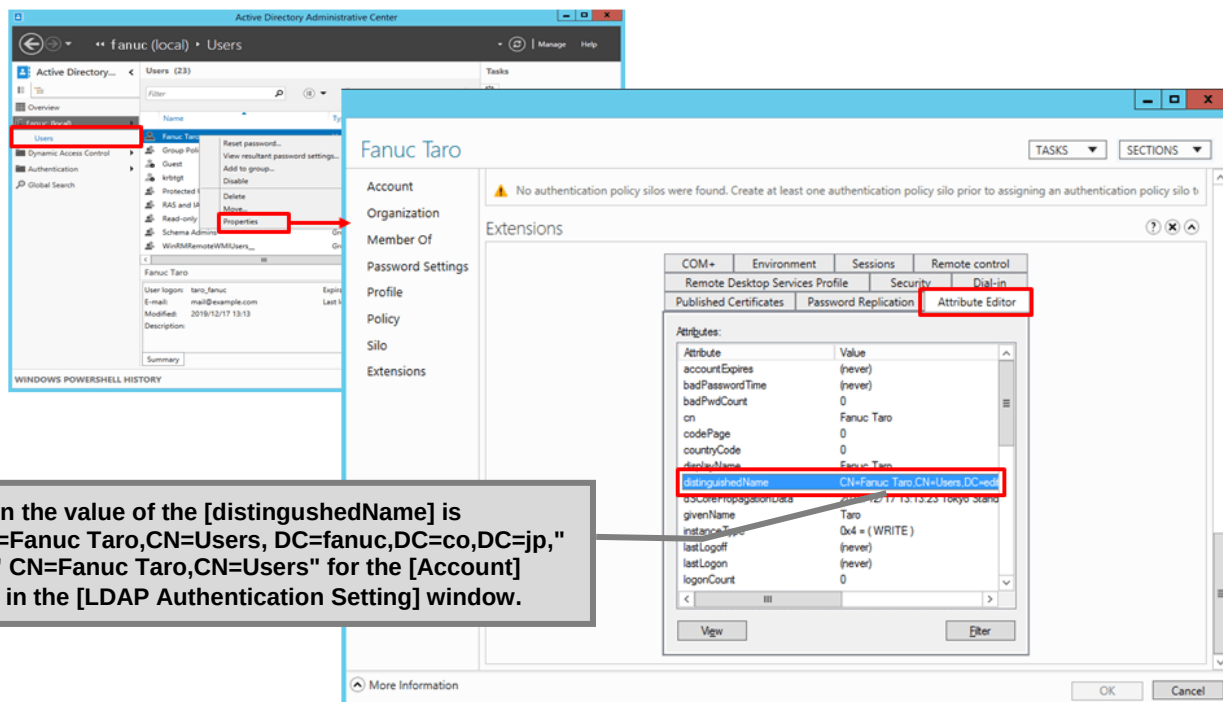


20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 350/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

(2) How to verify the [Account] field

Select [Users] to open a user list. Select an administrative user and right-click it. Select [Properties] from the context menu to open a property window. Scroll down the window to find and select the [Attribute Editor] tab.

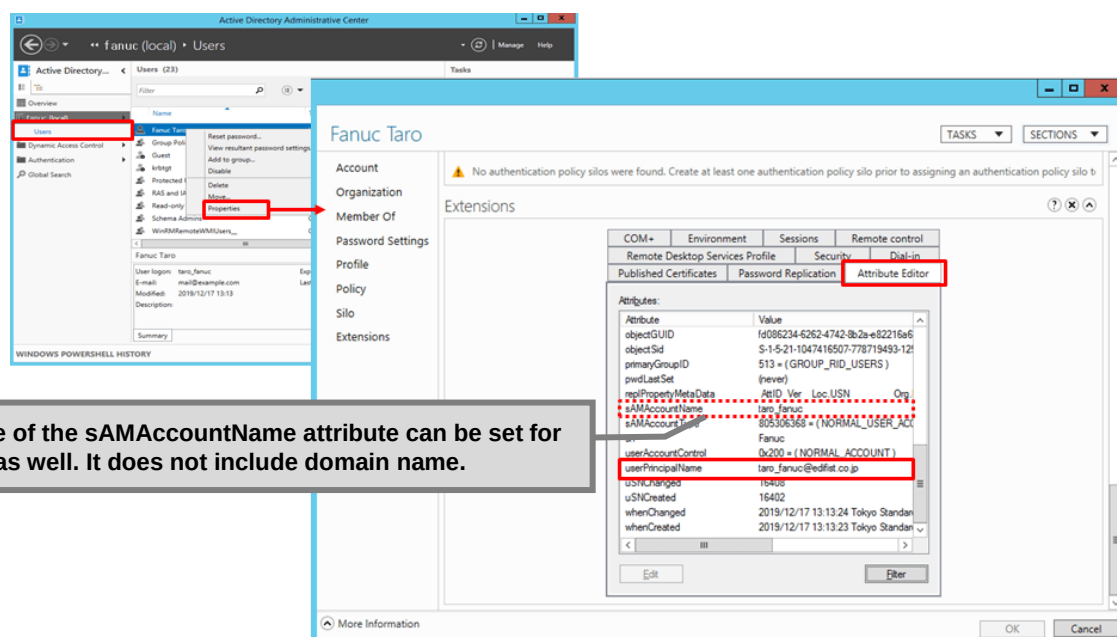
The [Account] field in the [LDAP Authentication Setting] window should be set to the value of the [distinguishedName] attribute listed in the tab page with domain components excluded from the value.



(3) How to verify the [Attribute (user)] field

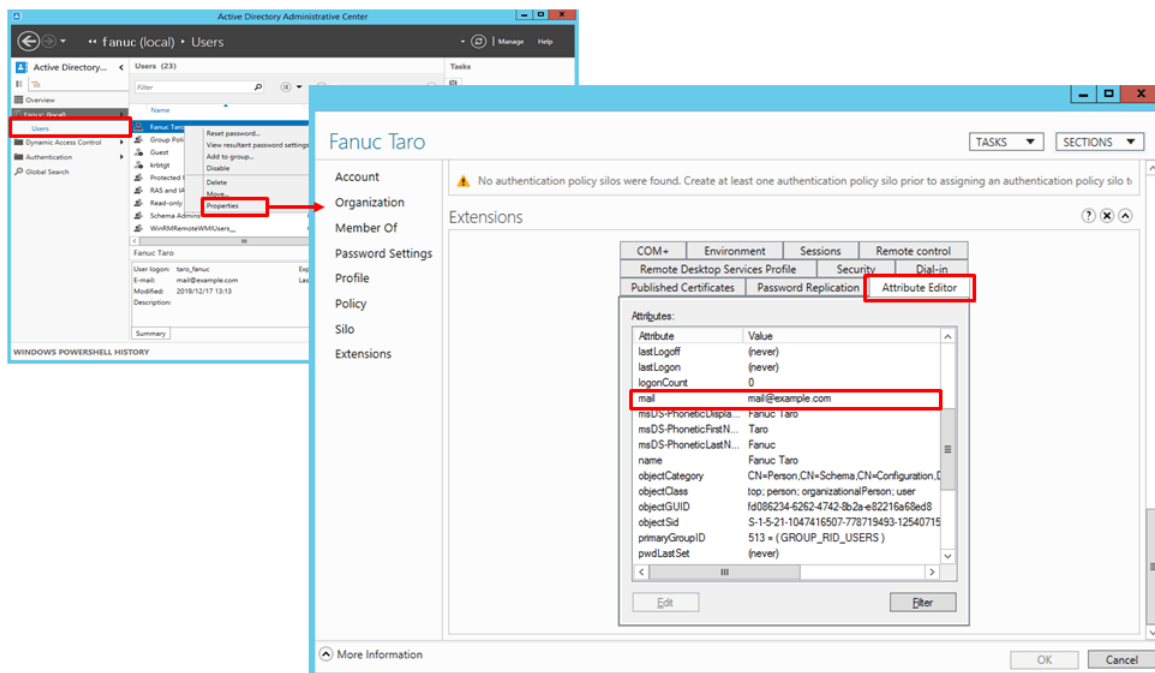
As with the [Account] field, select [Users] to open a user list. Select an administrative user and right-click it. Select [Properties] from the context menu to open a property window. Scroll down the window to find and select the [Attribute Editor] tab.

The [Attribute (User)] field in the [LDAP Authentication Setting] window should be set to the value of the [userPrincipalName] attribute listed in the tab page.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 351/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

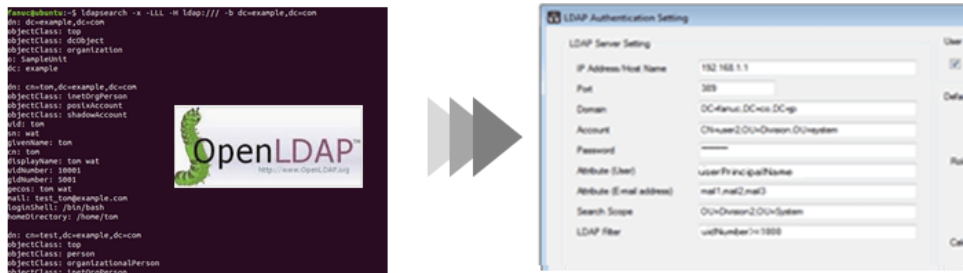
- (4) How to verify the [Attribute (E-mail address)] field
 As with the [Account] field, select [Users] to open a user list. Select an administrative user and right-click it. Select [Properties] from the context menu to open a property window. Scroll down the window to find and select the [Attribute Editor] tab.
 The [Attribute (E-mail address)] field in the [LDAP Authentication Setting] window should be set to the value of the [mail] attribute listed in the tab page.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 352/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

4.11.4.2 Setting Example for OpenLDAP

This subsection shows an example of setting the [LDAP Authentication Setting] window when the LDAP server is running OpenLDAP as directory service.



Fields to set	Example values	Refer to
IP Address/Host Name	192.168.1.1	"Set LDAP Authentication" subsection
Port	389	"Set LDAP Authentication" subsection
Domain	dc=example,dc=com	"(1) How to verify the [Domain] field" under " Command prompt window on the LDAP server (running OpenLDAP)"
Account	cn=ton	"(2) How to verify the [Account] field" under " Command prompt window on the LDAP server (running OpenLDAP)"
Password	*****	Contact LDAP administrator when uncertain about what to set
Attribute (User)	uid	"(3) How to verify the [Attribute (user)] field" under " Command prompt window on the LDAP server (running OpenLDAP)"
Attribute (E-mail address)	mail	"(4) How to verify the [Attribute (E-mail address)] field" under " Command prompt window on the LDAP server (running OpenLDAP)"
Search Scope	ou= Division2,ou= System	"Set LDAP Authentication" subsection
LDAP Filter	uidNumber>=1000	"Set LDAP Authentication" subsection

■ Command prompt window on the LDAP server (running OpenLDAP)

To display the properties and their values which are registered in OpenLDAP, execute the following command from a command prompt window on the LDAP server.

```
ldapsearch -x -b 'dc=example,dc=com'
```

(This command lists all the properties in the "example" domain.)

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	353/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

The following figure shows the execution results of the command including attribute values.

```

fanuc@ubuntu:~$ ldapsearch -x -b 'dc=example,dc=com'
# extended LDIF
#
# LDAPv3
# base <dc=example,dc=com> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#
# example.com
dn: dc=example,dc=com
objectClass: top
objectClass: dcObject
objectClass: organization
o: SampleUnit
dc: example

# tom, example.com
dn: cn=tom,dc=example,dc=com
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: tom
sn: wat
givenName: tom
cn: tom
displayName: tom wat
uidNumber: 10001
gidNumber: 5001
gecos: tom wat
mail: test_tom@example.com
loginShell: /bin/bash
homeDirectory: /home/tom

# test, example.com
dn: cn=test,dc=example,dc=com
objectClass: top
objectClass: person

```

- (1) How to verify the [Domain] field
The [Domain] field in the [LDAP Authentication Setting] window should be set to the value of the [dn] attribute displayed under "# example.com" group in the command prompt.
- (2) How to verify the [Account] field
The [Account] field in the [LDAP Authentication Setting] window should be set to the value of the [dn] attribute displayed under "# tom, example.com" group in the command prompt with domain components in the value excluded.
(the command results in this example: "cn=tom,dc=example,dc=com"; the field to be set to: "cn=tom")

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 354/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- (3) How to verify the [Attribute (user)] field
The [Attribute (User)] field in the [LDAP Authentication Setting] window should be set to the value of the [uid] attribute displayed under "# tom, example.com" group in the command prompt.
- (4) How to verify the [Attribute (E-mail address)] field
The [Attribute (E-mail address)] field in the [LDAP Authentication Setting] window should be set to the value of the [mail] attribute displayed under "# tom, example.com" group in the command prompt.

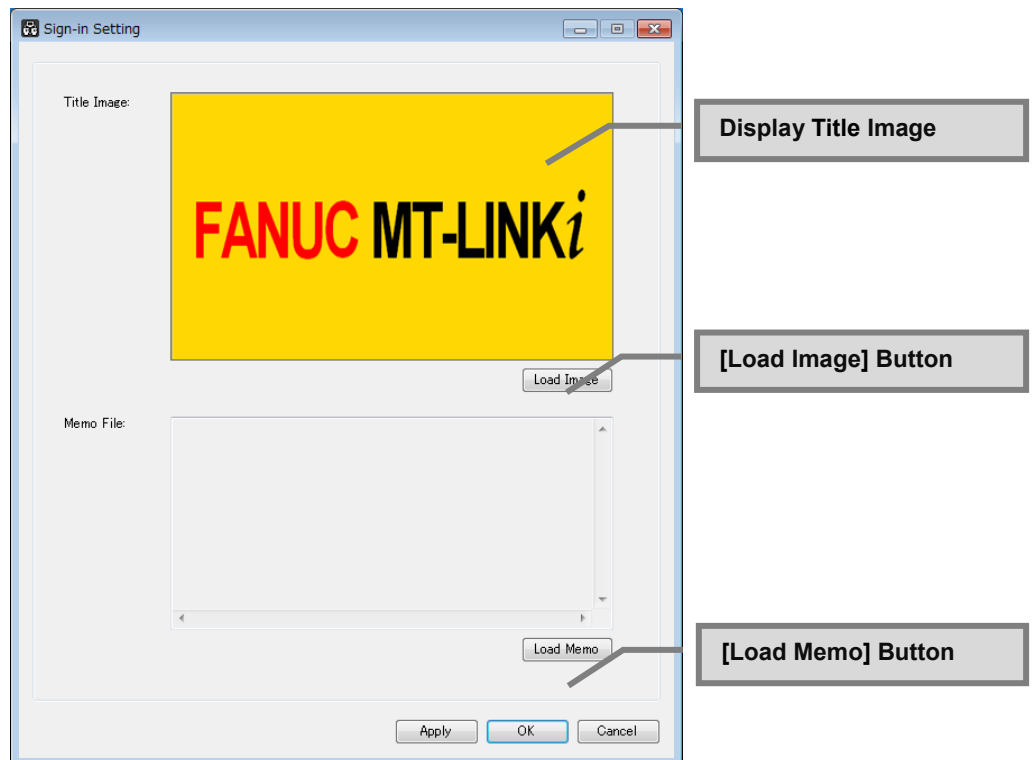
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 355/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

4.12 SIGN-IN SETTING

You can make sign-in settings and the contents to be displayed on the [MT-LINK*i* Web Sign in] screen in the [Sign-in Setting] window. Set a title image and the path to a memo file for the [MT-LINK*i* Web Sign in] screen.

The Sign-in Setting functionality enables you to do the following:

- To set the [Title Image] in the [MT-LINK*i* Web Sign in] screen.
- To set the title memo in the [MT-LINK*i* Web Sign in] screen.



4.12.1 Set Title Image

Follow the steps below to set title image.

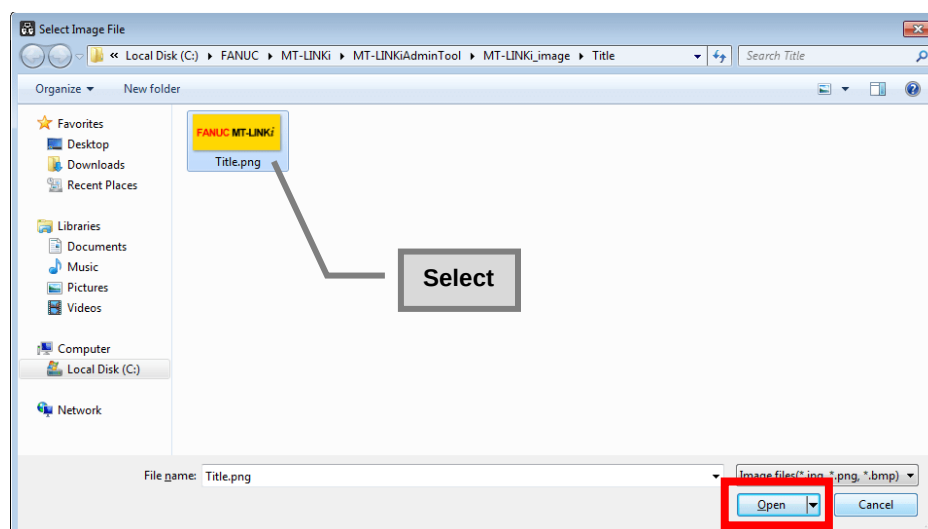
- (1) Prepare an image file in advance. Supported image formats: png, jpg, bmp
Image file larger than 1280 width x 1024 length (pixel) cannot be used.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL A-40829-00001EN		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 356/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- (2) Select [Sign-in Setting] in the AdminTool Menu. The [Sign-in Setting] window appears. Click [Load Image] to open the [Select Image File] dialog box. Select the title image file to use and click [Open].

The dialog box selects the following folder on the first session. This folder contains sample title image files. On subsequent sessions, this dialog box selects the folder selected in the previous session.

<MT-LINK*i* installation folder>\MT-LINKAdminTool\MT-LINKi_image\Title



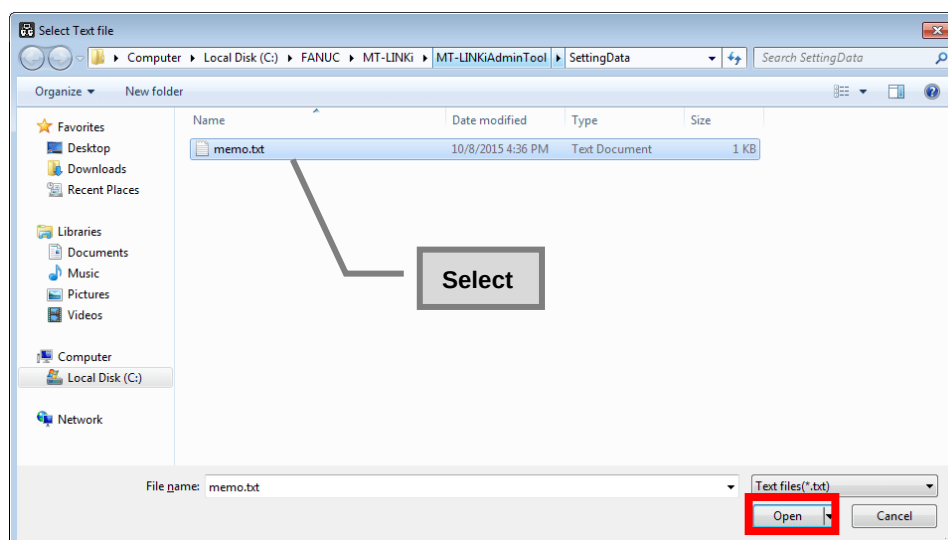
4.12.2 Set Title Memo

Follow the steps below to set title memo.

- (1) Prepare a title memo in advance.
A title memo file must be a text file (extension: .txt).
The text file should be saved in UTF-8 with BOM.
- (2) In the [Sign-in Setting] window, click [Load Memo] under the [Memo File] label. The [Select Text file] dialog box appears. Select the target file and click [Open].

The dialog box selects the following folder on the first session. On subsequent sessions, it selects the folder selected in the previous session.

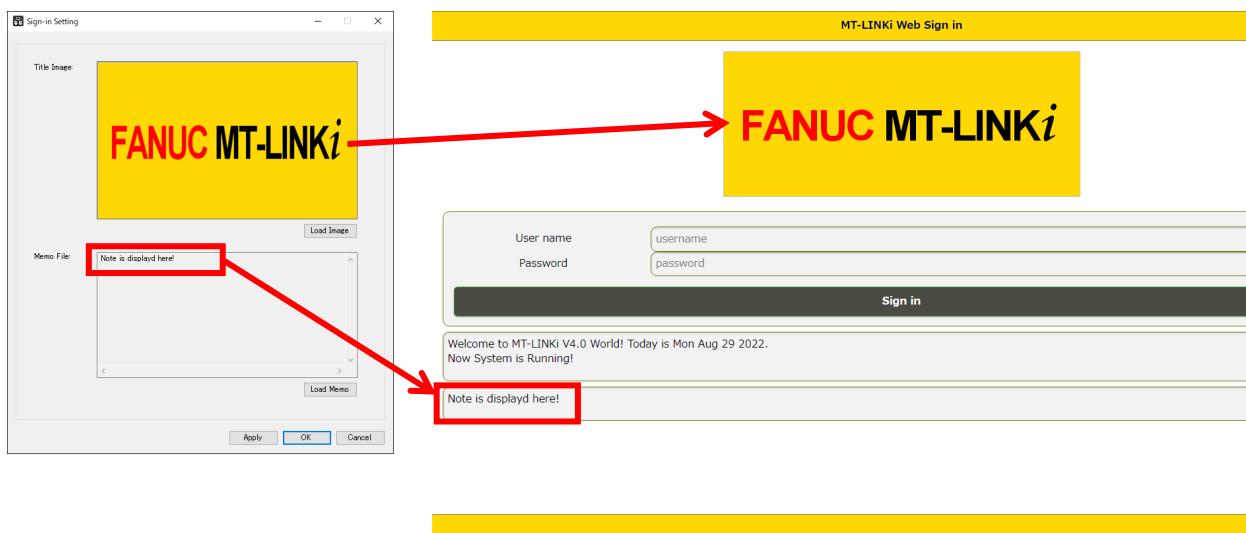
<MT-LINK*i* installation folder>



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 357/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

(3) The system loads and displays the content of the memo file.

To have the database reflect this setting and have WebUI display the memo, send setting data to the database.

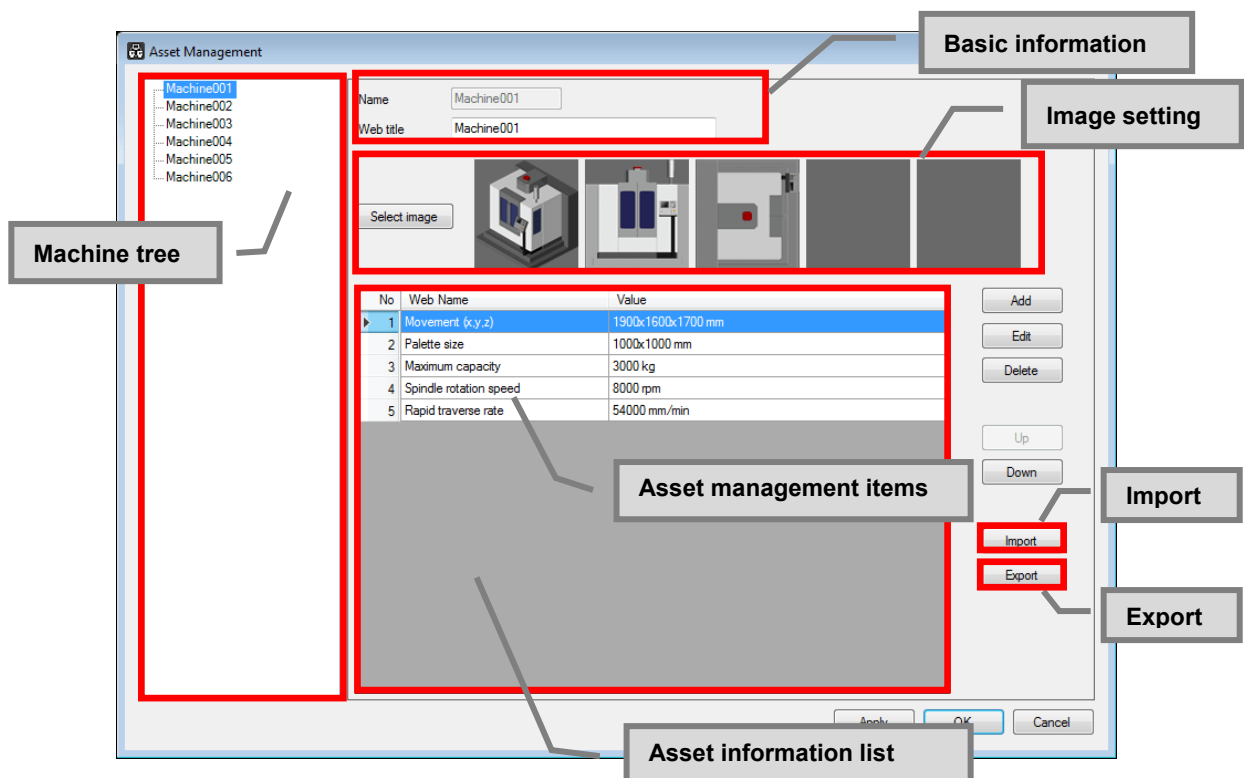


20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	358/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

4.13 ASSET MANAGEMENT

You can set the asset management settings of the machine in this window.

- In the "Machine tree," the machines registered in the machine setting are displayed.
- In the "Basic information" area, the Web title to be displayed on the Web screen is specified.
- In the "Image setting" area, the images of the machine to be displayed on the Web screen are specified. Up to five (5) images per machine can be registered.
- In the "Asset information list" area, the "Asset management items" of the machines specified are displayed in a list.
"Asset management items" can be registered up to 100 per machine.
- The "Import" function enables you to import the "Asset management items" entered in the CSV file.
- The "Export" function enables you to export the setting information of the "Asset management items" on a CSV file.



NOTE

In the machine setting, when the machine with the asset management information is deleted, the asset management information is deleted at the same time.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	359/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

The asset management function stores to the DB the specific information about the machine entered in this setting, and the online information collected directly from the machine, and displays these information on the [Monitoring - Equipment Monitoring] screen of the Web screen.

When the machine is selected by the [Asset Selection] button on the [Monitoring - Equipment Monitoring] screen (transited from equipment icon on the [Monitoring - Overlook] screen), the asset management information of the machine is displayed.

- Machine specific info.:

The specific information about the machine set in asset management is displayed.

- Online info.:

Online information (maintenance information) retrieved from the machine is displayed. (supporting only machines of FanucCNC and FanucRobodrill machine types)

- Maintenance Data Aggregation area:

Maintenance Data Aggregation functionality aggregates maintenance data for troubleshooting such as asset management information, machine data, and alarm information from machines. (supporting only machines of FanucCNC, FanucRobot, and FanucRobodrill machine types)

The screenshot shows the 'Monitoring - Equipment Monitoring' interface for 'Machine001'. It includes a sidebar with navigation options like 'OPERATE', 'DISCONNECT', 'ALARM', and 'ProductResult'. The main area is divided into several sections: 'Machine specific info.' with a table of machine specifications, 'Online info.' with a table of CNC and machine data, and 'Maintenance Data Aggregation' with buttons for 'Select Data', 'Download', and 'Detailed'. Callouts point to the 'Web title', 'Image of machine', and 'Online information' sections.

■ Online information items displayed for machine type FanucCNC

Category	Item	Remarks
CNC Information	CNC model name	-
	CNC software series	
	CNC software version	
	The maximum of controlled axes	
	CNC path	(NOTE 1)
	Ethernet board communication type	Displays the information about the Ethernet board connected to MT-LINKi. (NOTE 2)
	Data Server storage device	
Path Information	Control type Path<x>	<x>: CNC path number (1 to 15)
	The number of controlled axes Path<x>	* Number is same as the number of paths mounted. (NOTE 3)
	The number of controlled spindles Path<x>	

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 360/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

NOTE

- 1 The number of paths is displayed as follows when the Loader control path exists.
 - Series 16*i* "<Number of paths> and Loader"
(Loader control path not included in <Number of paths>)
 - Series 30*i* "< Number of paths >(Loader included)"
(Loader control path included in <Number of paths>)
- 2 "Ethernet board communication type" and "Data Server storage device" of the CNC information may not be displayed (blank) for the CNC of the earlier series number.
- 3 For the Series 16*i* models, the path information of the Loader control path is not displayed.

* For Series 30*i* and 16*i* models, refer to their descriptions respectively:

The "**Series 30i**" and "**Series 16i**" paragraphs in the "**Abbreviations of CNC models**" paragraph in the "**INTRODUCTION**" chapter.

■ Online information items displayed for machine type FanucRobodrill

Category	Item	Remarks
ROBODRILL Machine Information	ROBODRILL model name	-
	ROBODRILL serial number	
	Tool storage capacity	
	X-axis travel	
	Y-axis travel	
	Z-axis travel	
	Spindle specification	
	Spindle maximum speed	
CNC Information	CNC model name	-
	CNC software series	
	CNC software version	
	Ethernet board communication type	Displays the information about the Ethernet board connected to ROBODRILL-LINK <i>i</i> . (NOTE)
	Data Server storage device	
ROBODRILL Software Information	C-EXE software series	-
	C-EXE software version	
	LADDER software series	
	LADDER software version	
Other	Number of simultaneous controlled axes	-

NOTE

"Ethernet board communication type" and "Data Server storage device" items in CNC Information category may not be displayed (e.g., displayed as blank) for CNC software with earlier series numbers.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	361/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

4.13.1 Steps for Setting Asset Management Information

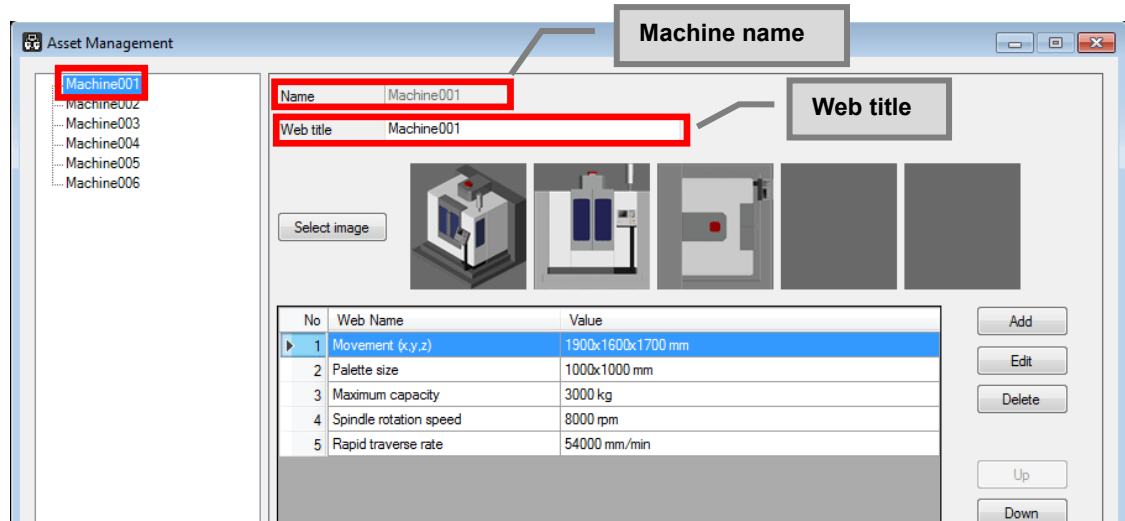
Follow the steps below to specify the asset management information.

- (1) Select [Asset Management] in the AdminTool Menu. This displays the [Asset Management] window.
- (2) Select the machine to set asset management information on the "Machine tree."
- (3) Specify the Web title in the "Basic information" area. For the details on the procedure, refer to the "[Steps for Setting Basic Information of Asset Management](#)" subsection.
- (4) Set the image of the machine in the "Image setting" area. For the details on the procedure, refer to the "[Setting Image of the Machine](#)" subsection.
- (5) Specify the asset management items of the machine in the "Asset management information list." For the details on the procedure, refer to the "[Setting Asset Management Items](#)" subsection.
- (6) When importing/exporting the asset management items, click the [Import] / [Export] button. For the details on the procedure, refer to the "[Importing / Exporting Asset Management Items](#)" subsection.
- (7) When the settings of the asset management information are completed, click [Apply] or [OK]. By clicking the [Cancel] button, the changes in the setting will be discarded.
- (8) To enable the changes in the setting, you are required to send the data to DB. For the details on the procedure, refer to the "[SEND TO DB](#)" section.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	362/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

4.13.2 Steps for Setting Basic Information of Asset Management

Follow the steps below to specify the basic information of asset management.



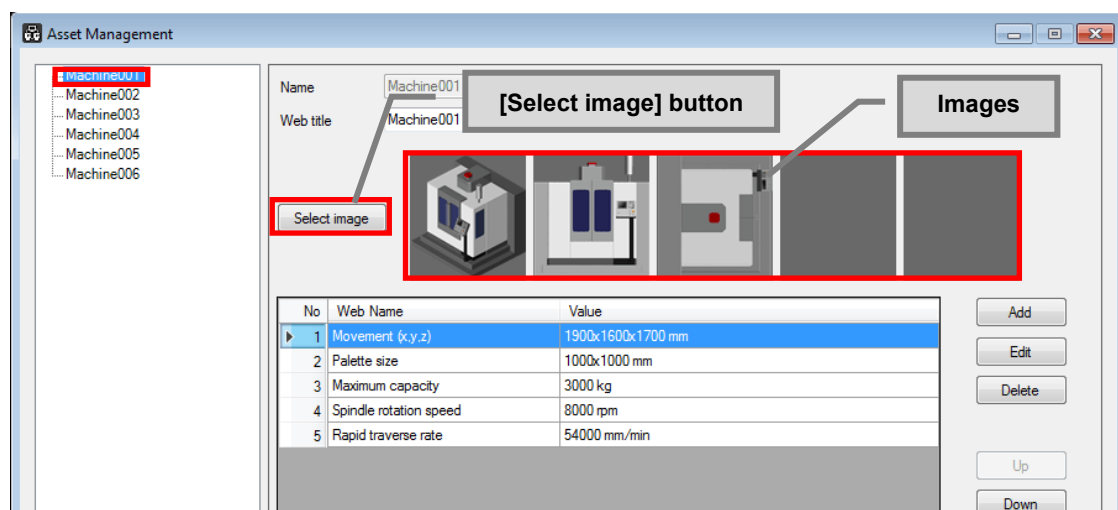
[Asset Management] window (Basic information)

- **Machine name**
The machine name being selected in the machine tree is displayed. It cannot be changed.
- **Web title**
Specifies the title to be displayed on the Web screens. The title will be displayed in the language specified for the [Web name language] field in the [System Setting] window. It cannot be omitted. This field can be up to 100 bytes long. Any character can be used. In the default setting, the machine name is specified.

4.13.3 Setting Image of the Machine

Follow the steps below to set the image for the machine.

- (1) Select a machine in the "Machine tree." There is no image specified in the default setting. If the image is not registered, image of the machine will not be displayed in the Web screen. When registering the image, click the [Select image] button.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	363/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- (2) [Select Image] dialog box is displayed. Up to five (5) images can be specified for each machine. When the [Load] button of either one of [Image1] to [Image5] is clicked, the [Select Image File] dialog box will be displayed. Select the image file to use. The dialog box selects the following folder on the first session. On subsequent sessions, it selects the folder selected in the previous session.

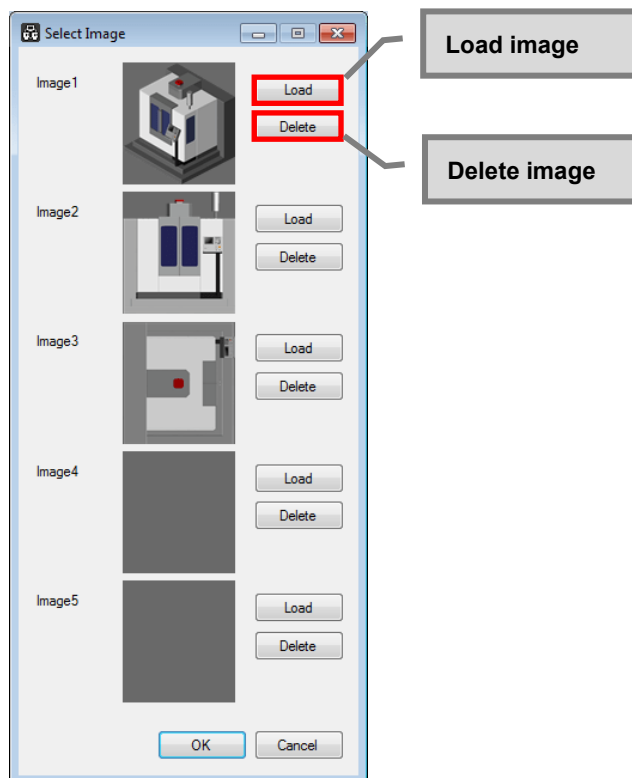
<MT-LINK*i* installation folder>\MT-LINiAdminTool\MT-LINKi_image

- Supported image formats: png, jpg, and bmp
- Prepare the image files in advance.
- Images larger than 500 pixels wide x 500 pixels high cannot be used.

When deleting the image specified, click the [Delete] button.

After the setting is completed, click the [OK] button to close the window.

If the [Cancel] button is clicked, the changes in the setting will be discarded.

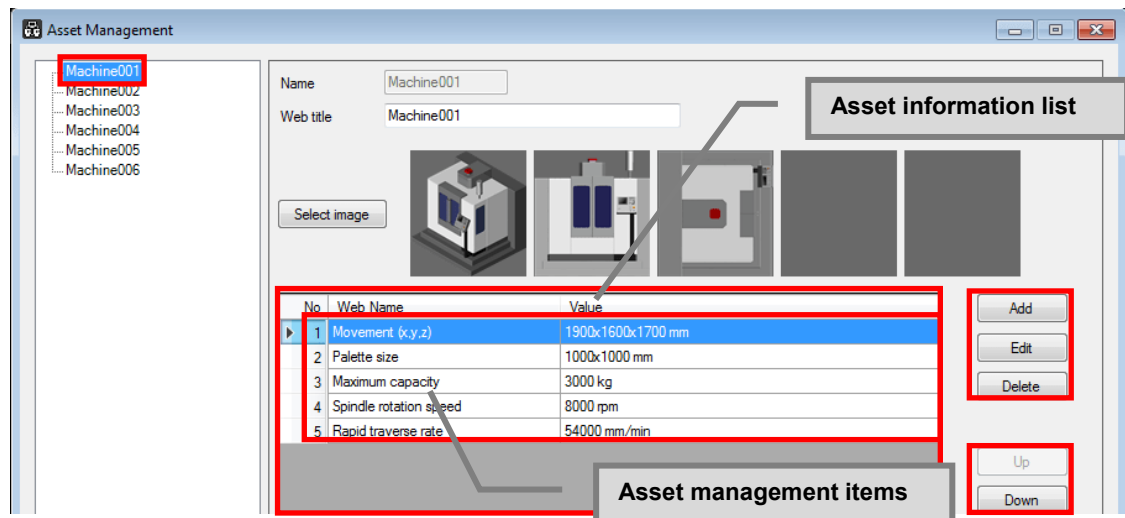


20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	364/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

4.13.4 Setting Asset Management Items

Follow the steps below to set the asset management items of the machine.

- (1) Select a machine in the "Machine tree." There are no items specified for the "Asset management items" in the default setting. Up to 100 items can be registered.



[Asset Management] window (Asset information list)

- **No.**
Index numbers of the asset management items.
On the machine specific information of the Web screen, the items are displayed in the order of the "Asset information list."
- **Web Name**
Name of asset management item. Not editable on the "Asset information list."
The name will be displayed in the language specified for the [Web name language] field in the [System Setting] window.
On the Web screen, it is displayed as an "Item" of the machine specific information.
- **Value**
Value of the asset management item. Not editable on the "Asset information list."
The value will be displayed in the language specified for the [Web name language] field in the [System Setting] window.
On the Web screen, displayed as "Specification" of the machine specific information.
- **[Add] button**
Adds an item in the "Asset information list." [Asset Management Items Setting] dialog box is displayed.
- **[Edit] button**
Edits the selected item in the "Asset information list." [Asset Management Items Setting] dialog box is displayed.
- **[Delete] button**
Deletes the selected item in the "Asset information list." Two or more items can be selected for simultaneous deletion.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	365/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- [Up] button
Moves the selected item one row upper on the "Asset information list."
- [Down] button
Moves the selected item one row lower on the "Asset information list."

- (2) Click the [Add] button when registering an asset management item.
[Asset Management Items Setting] dialog box is displayed. Set the [Web Name] and the [Value], and click the [OK] button.
If the [Cancel] button is clicked, the changes in the setting will be discarded, and the item will not be added.

The screenshot shows a dialog box titled "Asset Management Items Setting". It has two input fields: "Web Name" with the text "Feedrate" and "Value" with the text "1 mm/min + 300000 mm/min". At the bottom right, there are two buttons: "OK" and "Cancel". The "OK" button is highlighted with a red rectangular box.

[Asset Management Items Setting] dialog box

- Web Name
Specifies the name of the asset management item. The name will be displayed in the language specified for the [Web name language] field in the [System Setting] window. It cannot be omitted. Up to 100 bytes of characters can be entered. Any character can be used.
On the Web screen, displayed as an "Item" of the machine specific information.
 - Value
Specifies the value of the asset management item. The value will be displayed in the language specified for the [Web name language] field in the [System Setting] window. It cannot be omitted. Up to 100 bytes of characters can be entered. Any character can be used. Line feed (LF) can also be entered.
On the Web screen, displayed as "Specification" of the machine specific information.
- (3) When editing an asset management item, select the item to edit on the "Asset information list" and click the [Edit] button.
[Asset Management Items Setting] dialog box is displayed. Set the [Web Name] and the [Value], and click the [OK] button.
If the [Cancel] button is clicked, the changes in the setting will be discarded.
- (4) When deleting an asset management item, select the item to delete on the "Asset information list" and click the [Delete] button. The selected item will be deleted. Two or more items can be selected for simultaneous deletion.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	366/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

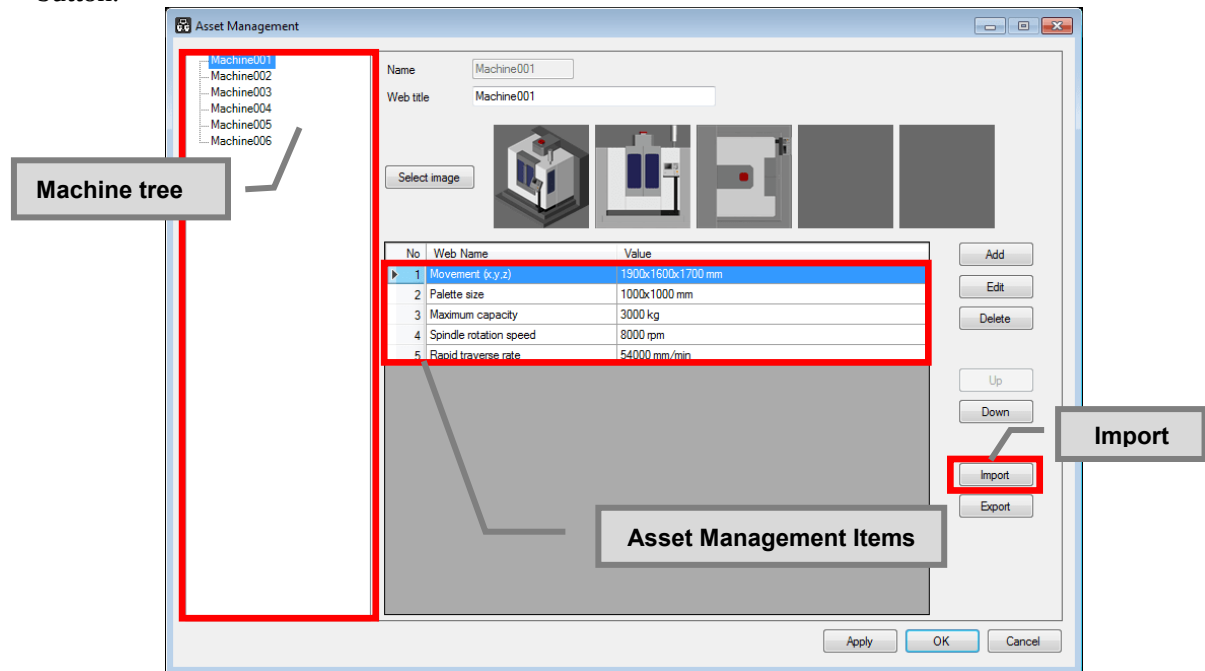
4.13.5 Importing / Exporting Asset Management Items

This subsection explains the steps to import / export the settings of the asset management items, and the format of the CSV file.

4.13.5.1 Importing Asset Management Items

When the import function is executed, all the "Asset management items" of the machine being selected in the "Machine tree" will be deleted, and the contents of the setting in the CSV file will be changed. For the details on the format of the CSV file, refer to the "[Input / Output File Format of Asset Management Items](#)" subsection. In order to import the settings of asset management items input in the CSV file, follow the steps below.

- (1) In the "Machine tree," select the machine to import the asset management items, and click the [Import] button.

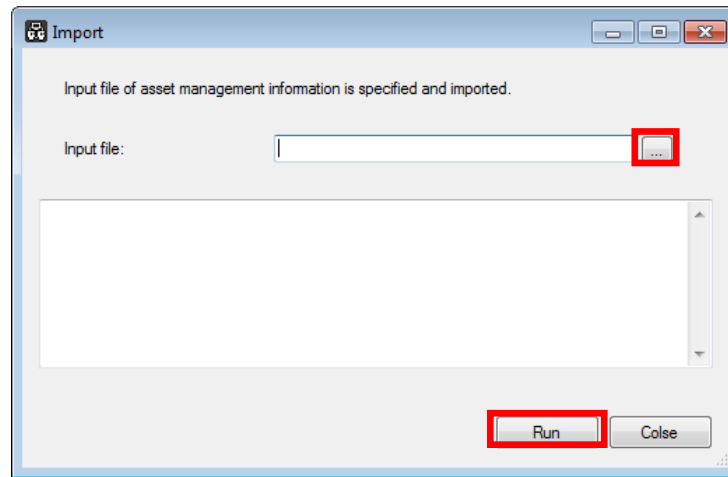


- (2) [Import] dialog box is displayed.
Either set the [Input file] field to the full-file path to a CSV file to import or click the [...] (file selection) button to open the [Open] dialog box, select it, and click the [Open] button to tell the [Input file] field in the [Import] dialog box to populate the selected file with full-path.

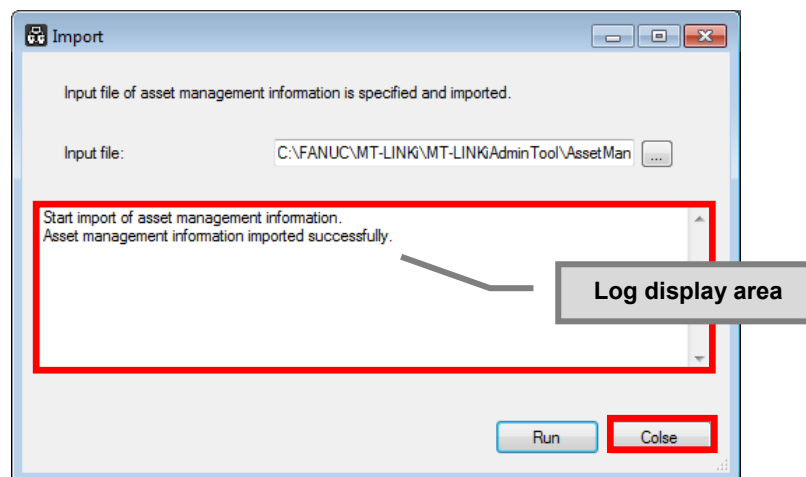
The [Open] dialog box selects the following folder on the first session. On subsequent sessions, it selects the folder selected in the previous session.

<MT-LINK*i* installation folder>\MT-LINK*i*AdminTool\Asset

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	367/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		



- (3) When the [Run] button is clicked on the [Import] dialog box, confirmation message for the deletion of the setting appears. Click the [OK] button. When the [Cancel] button is clicked, the import process will not be executed, and returns to the [Import] dialog box.
- (4) When the import process is executed successfully, "Asset management information imported successfully." is displayed in the log display area.
 When the execution of the import process fails, "Failed to import asset management information." is displayed in the log display area as an error. Execute the import process again after correcting the CSV file.
 When the import process is completed, click the [Close] button to close the dialog box.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	368/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

- (5) The contents of the "Asset management items" in the [Asset Management] window is changed to the settings of the CSV file.

The screenshot shows the 'Asset Management' window. On the left, a list of machines (Machine001 to Machine006) is shown. The main area displays the configuration for 'Machine001'. It includes fields for 'Name' (Machine001) and 'Web title' (Machine001). Below these are five image slots, with the first one containing a machine image. A 'Select image' button is present. A table with 9 rows is highlighted with a red border. The table has columns 'No', 'Web Name', and 'Value'. The rows are: 1. Movement (x,y,z) 1900x1600x1700 mm; 2. Palette Size 1000x1000 mm; 3. Maximum capacity 3000 kg; 4. Spindle rotation speed 8000 rpm; 5. Rapid traverse rate 54000 mm/min; 6. Feedrate 1 mm/min to 30000 mm/min; 7. Tool change system turret type; 8. Type of tooling JIS B 6339-2011 BT30,MAS 403-1982 P30T-1; 9. Tool storage capacity 21 tools. To the right of the table are buttons for 'Add', 'Edit', 'Delete', 'Up', 'Down', 'Import', and 'Export'. At the bottom are 'Apply', 'OK', and 'Cancel' buttons. A callout box with an arrow pointing to the table contains the text 'Imported asset management items'.

No	Web Name	Value
1	Movement (x,y,z)	1900x1600x1700 mm
2	Palette Size	1000x1000 mm
3	Maximum capacity	3000 kg
4	Spindle rotation speed	8000 rpm
5	Rapid traverse rate	54000 mm/min
6	Feedrate	1 mm/min to 30000 mm/min
7	Tool change system	turret type
8	Type of tooling	JIS B 6339-2011 BT30,MAS 403-1982 P30T-1
9	Tool storage capacity	21 tools

* When executing the import process for the other machines, repeat the above steps (1) to (4) for each machine.

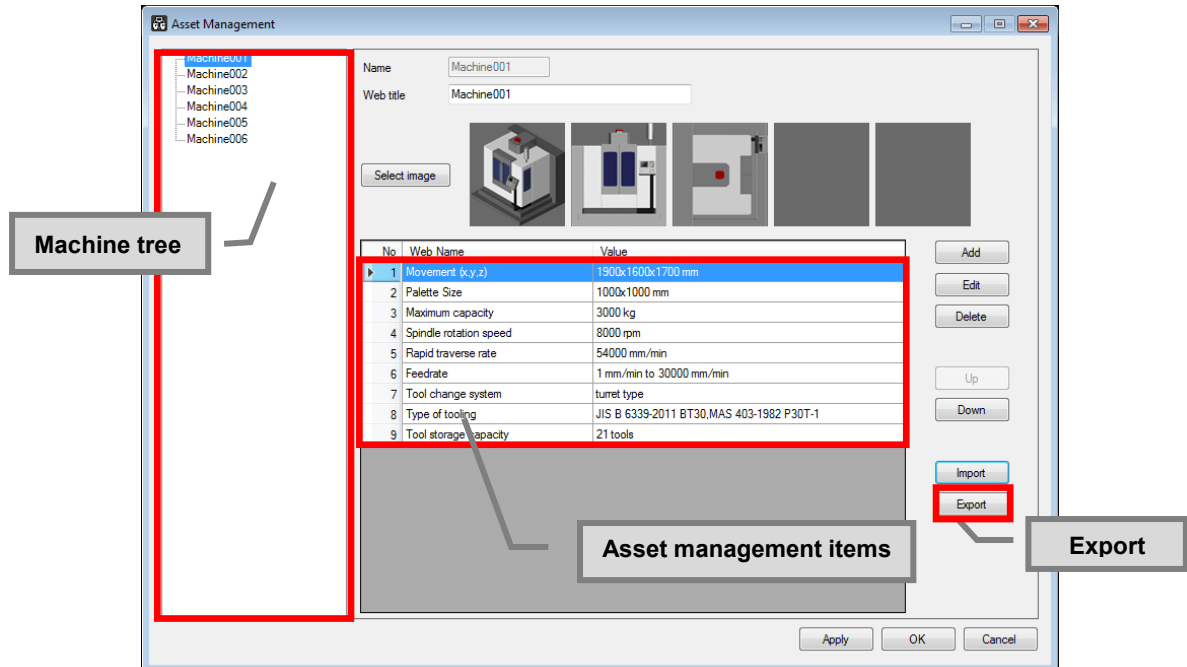
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 369/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

4.13.5.2 Exporting Asset Management Items

When the export function is executed, all the contents of the asset management items of the machine being selected in the "Machine tree" will be output. For the details on the format of the CSV file, refer to the "[Input / Output File Format of Asset Management Items](#)" subsection.

In order to export the settings of asset management items to the CSV file, follow the steps below.

- (1) In the "Machine tree," select the machine to export the asset management items, and click the [Export] button.

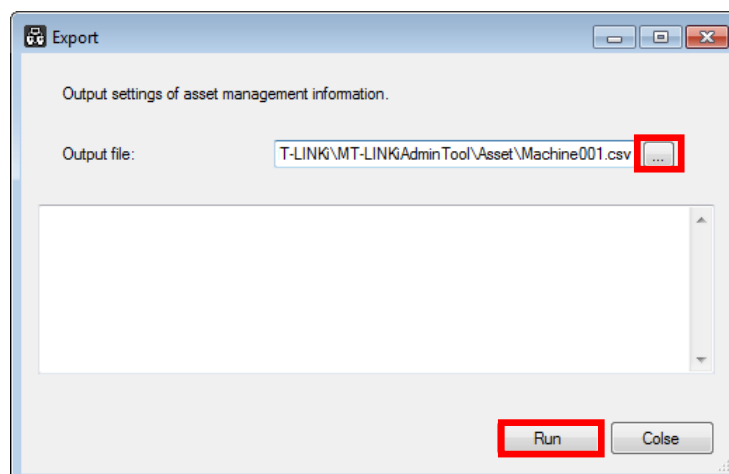


- (2) [Export] dialog box is displayed.
The [Output file] field populates the following output file name in full path:

Output file path: <MT-LINK*i* installation folder>\MT-LINKiAdminTool\Asset\<Machine name>.csv

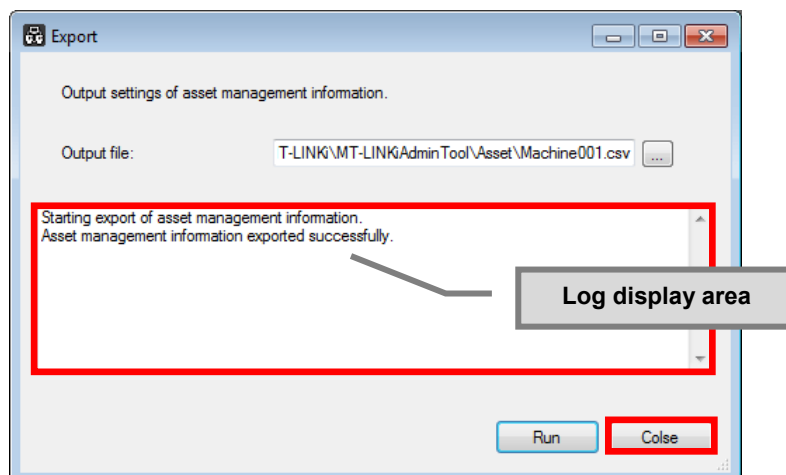
When specifying an output file other than the default file, click the [...] (file selection) button, specify it in the [Save As] dialog box, and click the [Save] button.

On subsequent sessions, the [Output file] field populates the folder selected in the previous session, and the [Save As] dialog box selects the same folder and file as populated in the [Output file] field.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 370/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- (3) Click the [Run] button in the [Export] dialog box.
When the export process is executed successfully, "Asset management information exported successfully."
is displayed in the log display area. Click the [Close] button to close the dialog box.



* When executing the export process for the other machines, repeat the above steps (1) to (3) for each machine.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 371/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

4.13.5.3 Input / Output File Format of Asset Management Items

This subsection explains the format of the input file (CSV format) used for importing the asset management items. The format of the output file used in the export process is similar to the format of the input file.

- A single data consists of asset management items (Web name/value) of each language (Japanese/English/Chinese/Local language) delimited by "," (comma), and stored by each line.
- The item name is input in the 1st line.
- The item name and the setting value are enclosed in double quotation marks ("").
- Save in character mode UTF-8 (with BOM).

■ Input / Output file format

Item Name	Description of Setting Value
WebNameJp	Web name of asset management item (Japanese)
ValueJp	Value of asset management item (Japanese)
WebNameEn	Web name of asset management item (English)
ValueEn	Value of asset management item (English)
WebNameCn	Web name of asset management item (Chinese)
ValueCn	Value of asset management item (Chinese)
WebNameLocal	Web name of asset management item (Local language)
ValueLocal	Value of asset management item (Local language)

NOTE

- 1 Create an input file for each machine when importing. (one file for one machine)
- 2 Up to 100 items can be specified on a single input file.
- 3 Up to 100 bytes can be input for the Web name and the value.
- 4 Any character can be used for the Web name and the value. However, line feed cannot be used.
- 5 The items cannot be omitted that correspond to the language selected in the [Web name language] of the [System Setting] window. The items for other languages can be omitted.
(Example) When the language selected in the [Web name language] of the [System Setting] window is "English"
[WebNameEn] and [ValueEn] cannot be omitted.
Other items can be omitted.
- 6 A local language is available when AdminTool loads the additional language resource files which may be provided separately. For the details on how to load additional language resource files in AdminTool, refer to the "[Load Additional Language Resource](#)" subsection.

■ The output folder and the file name when exporting (default)

<MT-LINK*i* installation folder>\Asset\<Machine name>.csv

(Example) C:\FANUC\MT-LINK*i*\Asset\Machine001.csv

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION				Sheet. 372/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

■ The sample of the input/output file

```
"WebNameJp","ValueJp","WebNameEn","ValueEn","WebNameCn","ValueCn","WebNameLocal","ValueLocal"
","","Distance moved (x,y,z)","1900 x 1600 x 1700 mm",","","",""
","","Palette size","1000 x 1000 mm",","","",""
","","Maximum capacity","3000 kg",","","",""
```

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11							
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10							
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8							
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2							
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			
Ed.	Date	Design	Description							Sheet.
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue			

4.14 POWER CONSUMPTION SETTING

Sets the power consumption for external device and CO2 emissions information to display on the [Results - Power Consumption Monitor] Web screen.

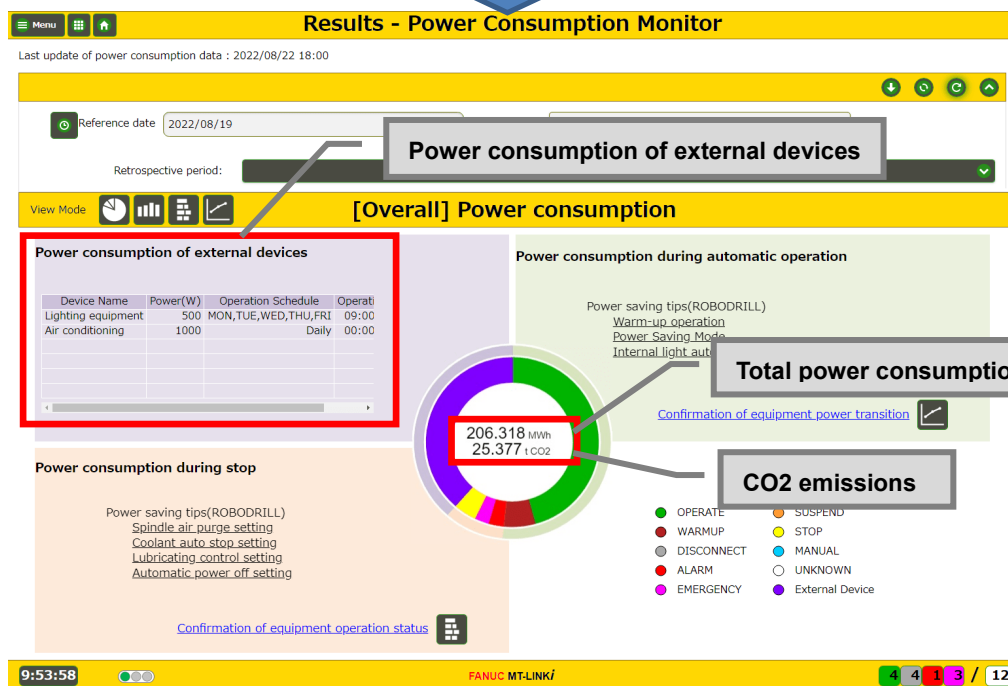
On the [Results - Power Consumption Monitor] Web screen, you can view power consumption information that aggregates power consumption signals for each group or equipment, power consumption of pre-registered external devices (lighting, air conditioning, etc.), and CO2 emissions.

The power consumption information of the target equipment and external device is aggregated every hour, saved in the DB, and reflected on the [Results - Power Consumption Monitor] Web screen.

The target equipment units are consisting of at least one FanucCNC or FanucRobodrill type machine.

The following settings can be made in the Power Consumption Settings.

- You can set display/hide CO2 emissions on the [Results - Power Consumption Monitor] Web screen and the CO2 emission factor.
- You can set the power consumption and the operation schedule/operation time for the external device.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	FANUC CORPORATION		Sheet.	374/868

For the details, refer to the "RESULTS - POWER CONSUMPTION MONITORING SCREEN" section of the "FANUC MT-LINK*i* WebUI OPERATOR'S MANUAL."

NOTE

- 1 The models of CNC and ROBODRILL that can use the power consumption monitor are as follows.

FanucCNC type machine

- Series 30*i*

FanucRobodrill type machine (* ROBODRILL Option license key is required.)

- FANUC ROBODRILL α -DiA Series
- FANUC ROBODRILL α -DiB Series
- FANUC ROBODRILL α -DiB Plus Series

- 2 In order to display the power consumption information of FanucCNC type machine on the [Result - Power Consumption Monitor] Web screen, retrieving the following signals must be enabled and assigned to the equipment status. (The FanucRobodrill type machine does not require these signals to be enabled.)

- Total current actual power consumption of all axes (PwConAllAxes)
- Accumulated power consumption of peripheral devices (AdcumPwConPeripheral)

Refer to the following section for the setting method.

Within 1 hour after the setting is completed, the setting information will be reflected on the [Result - Power Consumption Monitor] Web screen.

"MACHINE SETTING - FanucCNC" - "[Set Signal of FanucCNC](#)"

"EQUIPMENT SETTING" - "[Set Equipment Status](#)"

- * The following CNC models, "accumulated power consumption of peripheral devices" signal cannot be retrieved.

- Series 35*i*-B
- Series 30*i* /31*i* /32*i*-A
- Power Motion *i*-A

- 3 In order to display the power consumption information of FanucRobodrill type machine on the [Result - Power Consumption Monitor] Web screen, retrieving the following signals must be enabled and assigned to the equipment status.

- Robodrill electric power consumption (RobodrillElectricPowerConsumption)
- Robodrill power saving mode (RobodrillEsaveMode)

Refer to the following section for the setting method.

Within 1 hour after the setting is completed, the setting information will be reflected on the [Result - Power Consumption Monitor] Web screen.

"MACHINE SETTING - FanucRobodrill" - "[Set FanucRobodrill Specific Signal](#)"

"EQUIPMENT SETTING" - "[Set Equipment Status](#)"

- 4 For equipment that includes multiple target machines (machine type is FanucCNC or FanucRobodrill), the power consumption of each machine is added up and displayed on the [Result - Power Consumption Monitor] Web screen.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 375/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

4.14.1 Setting CO2 Emissions and Power Consumption for External Devices

Set the display/hide CO2 emissions on the [Results - Power Consumption Monitor] Web screen and the CO2 emission factor.

Also, set the power consumption and operation schedule/time of external devices (lighting, air conditioning, etc.).

Power Consumption Setting

CO2 emissions

☒ Display CO2 emissions CO2 emission factor 0.645 kg-CO2/kWh

External Device List

External Device Setting

No	Web Name	Power (W)	Operation Schedule	Operating Time
▶ 1	Lighting equipment	500	Mon,Tue,Wed,Thu,Fri	09:00 - 18:00
2	Air conditioning	1000	Daily	00:00 - 00:00

Add
Edit
Delete

Apply **OK** Cancel

[Power Consumption Setting] window

CO2 emissions

- **Display CO2 emissions**
Set the display/hide CO2 emissions on the [Results - Power Consumption Monitor] Web screen. When the check box is selected, the CO2 emissions are displayed on the Web screen. The default setting is cleared.
- **CO2 emission factor**
Specifies the CO2 emission factor required to convert power consumption to CO2 emissions. The unit is "kg-CO2/kWh".
Enter a value 0 or more and less than 1. You can enter up to three decimal places.

NOTE

For setting the CO2 emission factor, refer to the information on the Internet, etc.
CO2 emissions are calculated using the following formula.

$$\text{CO2 emissions (kg-CO2)} = \text{CO2 emission factor (kg-CO2/kWh)} \times \text{Total power consumption (kWh)}$$

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	376/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

External Device Setting

- Web Name
The Web name of the registered external device is displayed.
- Power (W)
The power consumption (W) of the registered external device is displayed.
- Operation Schedule
The operation schedule ("Daily" or day of the week "Mon"-"Sun") of the registered external device is displayed.
- Operation Time
The operation time (start time <HH:mm> - end time <HH:mm>) of the registered external device is displayed.
- [Add] button
Add the external device. [Add/Edit Device] window is displayed.
For the details, refer to the "[Add/Edit Device](#)"
The registered external device setting information is added to the "External Device List." You can register up to 100 devices.
- [Edit] button
Edit the setting information of the external device selected in the "External Device List." [Add/Edit Device] window is displayed.
For the details, refer to the "[Add/Edit Device](#)"
- [Delete] button
Delete the setting information of the external device selected in the "External Device List."

After completing the settings, click the [Apply] or [OK] button. Click [Cancel] to discard the settings.

- [Apply] button
Settings will be applied.
- [OK] button
Settings are applied and the [Power Consumption Setting] window is closed.
- [Cancel] button
Settings are discarded and the [Power Consumption Setting] window is closed.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	377/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

4.14.1.1 Add/Edit Device

On the [Power Consumption Setting] window, click the [Add] button, or select an external device registered in the "External Device List" and click the [Edit] button, [Add/Edit Device] window is displayed.
You can register a new external device and set the power consumption information, or edit the registered external device setting information.

[Add/Edit Device] window

- **Device Name**
Set the name of the external device. The value can be up to 32 bytes long. (Not omissible)
- **Power Consumption (W)**
The power consumption (W) of the external device. A number in a range of 0 to 9999999 can be specified. (Not omissible)
- **Web Name**
Set the Web name of the external device to be displayed on the [Results - Power Consumption Monitor] Web screens. The name should be entered in the language specified for the [Web name language] field in the [System Setting] window. The value can be up to 100 bytes long. (Not omissible)
- **Operation Schedule**
This check box specifies the day of the week when the external device operates.

[Daily] : When this check box is selected, all the check boxes of [Specify the day of the week] are selected, and the each check box of [Monday] to [Sunday] cannot be cleared.
When this check box is cleared, all the check boxes of [Specify the day of the week] are cleared, and the each check box of [Monday] to [Sunday] can be selected.

Specify the day of the week

[Monday]
- [Sunday] : Select the check box of the day of the week when the external device operates.
- **Operation Time**
Set the operation time (start time - end time) of the external device

[Start:] : Set the operation start time of the external device. This field is set to "09:00" by default.

[End:] : Set the operation end time of the external device. This field is set to "18:00" by default.
If "00:00" is specified for this field, it will be set to "24:00."

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	378/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- [OK] button
Settings are applied and updates the information in the "External device list" on the [Power Consumption Setting] window.
- [Cancel] button
Settings are discarded and return to the [Power Consumption Setting] window.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	379/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

[SETTING DATA] - SETTING CHECK

This chapter describes how to check the settings made in AdminTool.

The AdminTool Menu has the following menu items in the [Setting Data] - [Setting Check] group:

(1) Machine Connection Test

This menu checks whether machines set up through Machine Setting are connectable, and enables users to verify the settings.

(2) Data Validation

This menu checks settings for users to review the coherence of them before sending them to the database.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	380/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

5.1 CHECK MACHINE SETTING

Connectivity with machines can be checked on the [Machine Connection Test] window.

5.1.1 Check Connection with the Machine

This subsection describes how to check connectivity with machines.

(1) How to check

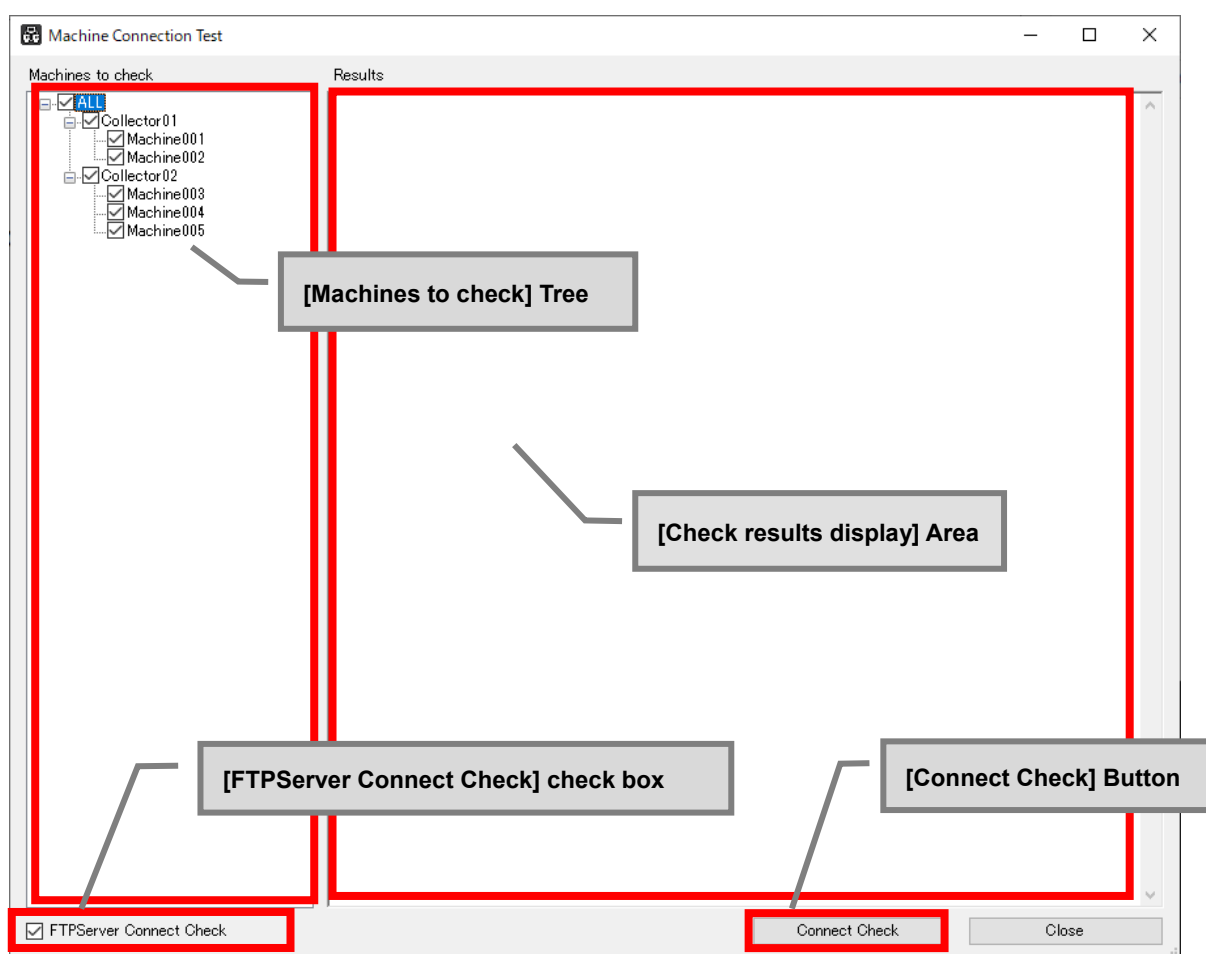
In the [Machines to check] tree, select the check boxes for machines to check.

Selecting the check box of a Collector selects the check boxes of all the machines which locate under that Collector in the tree.

Selecting the [All] check box selects all the check boxes in the tree.

When the [FTPServer Connect Check] check box is selected, the connection with the FTPServer will be checked for each selected Collector. The default setting is selected.

Clicking the [Connect Check] button conducts connection tests on Collectors and machines whose check boxes are selected.

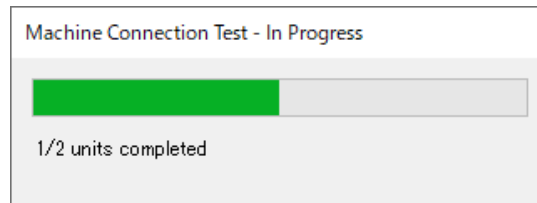


NOTE

All the machines listed in the [Machines to check] tree can be checked regardless of whether their [Connect] check box in the [Machine Setting] window may be selected or cleared.

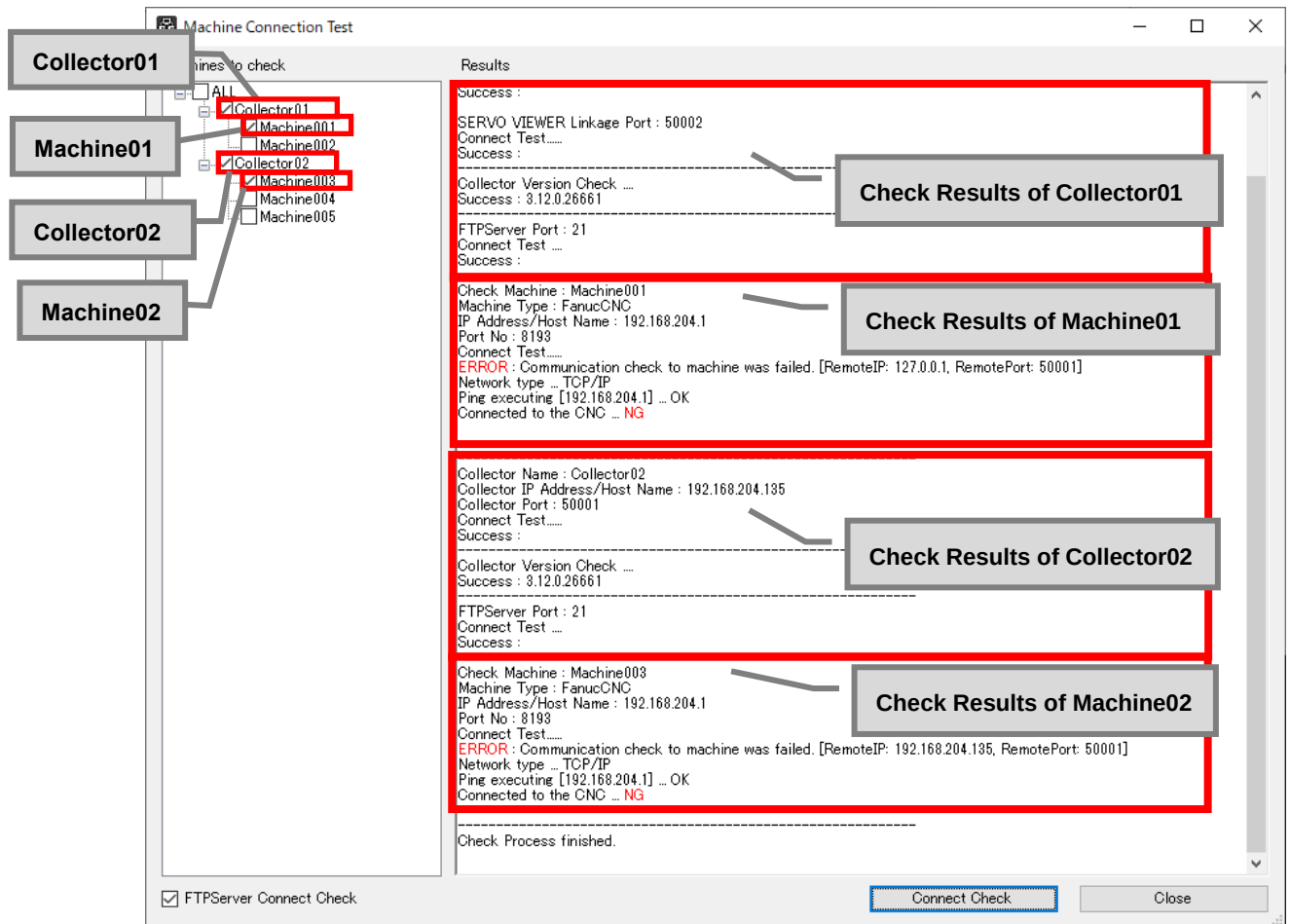
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	381/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

A progress bar appears to show test progress (the number of machines checked so far / the total number of machines to check) while connection tests are running.



(2) Displaying check results

After all the target machines are checked, the [Check results display] area displays the check items and results. When multiple machines are selected to check, their check results are listed in the same order as they are displayed in the [Machines to check] tree.



The message "Check Process finished." will be displayed in the last line.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	382/868	
Ed.	Date	Design	Description									
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.						Inoue

(3) Outputting results

Check results will be output to a text file from the [Check result display] area.

Following are where those text files will be created, how they will be named, and which character code they will be encoded in:

- Output folder
<MT-LINK*i* installation folder>\MT-LINKiAdminTool\MachineConnectionTest
- File name
(Example) Log_Result_20191911123456.txt
- Character code (encoding method)
UTF-8 (with BOM)

NOTE

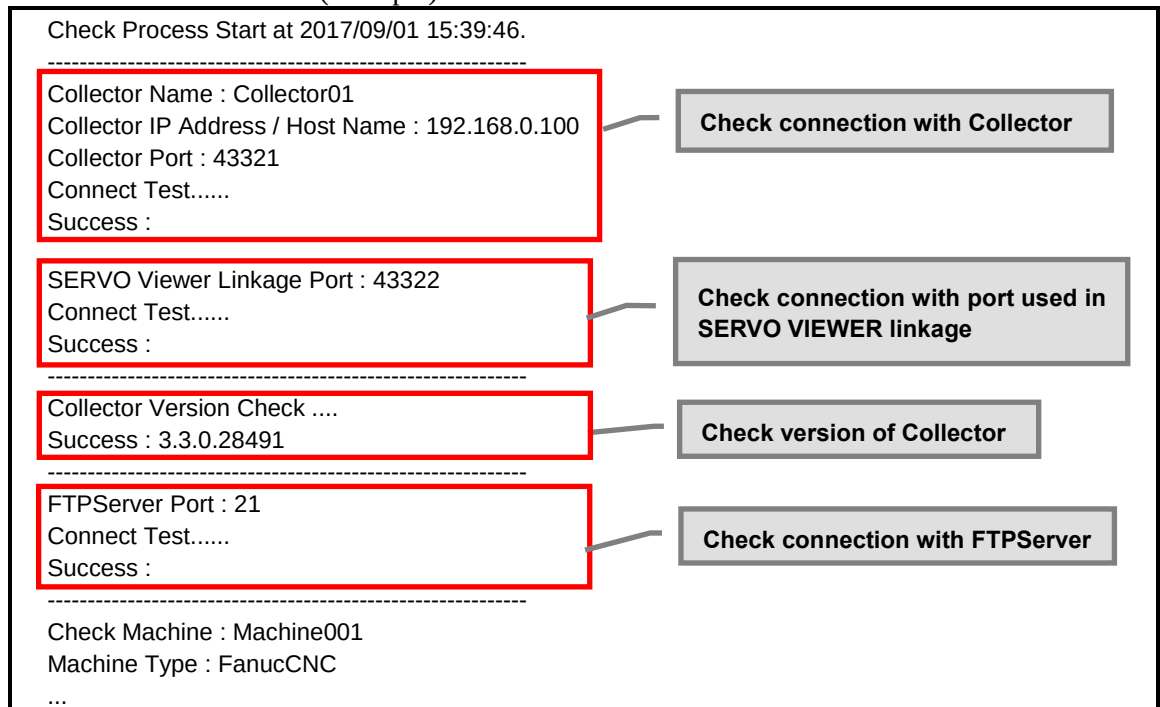
This text file will be kept for a maximum of five generations.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	383/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

5.1.2 Check Results for Collector

This subsection explains check results for Collector.

Check results for Collector (example)



■ Checking connection with Collector

The first check item is whether connection with the Collector can be established.

When the connection is successful, "Success" will be displayed after "Connect Test....." If the connection is unsuccessful, "ERROR" and the detail of the error will be displayed.

NOTE

If an "ERROR" is displayed, the Collector PC may not be connected.

Perform the following:

- (1) Check the protocol setting of the machine.
- (2) Create a Collector setting file and have the Collector PC reflect the setting.
For the details, refer to the "[Apply Created AdminTool Setting to Collector](#)" subsection.
- (3) Check if MT-LINK*i* Collector service is running. For the details, refer to the following subsection of the "FANUC MT-LINK*i* / Integration Server Troubleshooting MANUAL":

The "ERROR" When Checking Connection with Collector" subsection in the "TROUBLES RELATED TO CHECKING MACHINE SETTING" section in the "MT-LINK*i* TROUBLESHOOTING" chapter

■ Checking connection with the port used in SERVO VIEWER linkage

If the SERVO VIEWER linkage function is enabled, the connection with the port used in SERVO VIEWER linkage is checked.

When the connection is successful, "Success" will be displayed after "Connect Test....." If the connection is unsuccessful, "ERROR" and the detail of the error will be displayed.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	384/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

- Checking connection with the FTPServer
If the [FTPServer Connect Check] check box is selected, the connection with the FTPServer is checked. When the connection is successful, "Success" will be displayed after "Connect Test....." If the connection is unsuccessful, "ERROR" and the detail of the error will be displayed.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	385/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

5.1.3 Check Results for FanucCNC Type

This subsection explains check results for FanucCNC type machines.

Check results for FanucCNC type machine (example)

Check Machine : Machine001 Machine Type : FanucCNC IP Address / Host Name : 192.168.0.101 Port No : 8193 Connect Test..... Success : Network type ... TCP/IP Ping executing [192.168.0.101] ... OK Connected to the CNC ... OK	Check connection with machine
< Machine Information > -- CNC Model = Series30iB -- Max Path = 1 -- CNC Type = M -- Axis = 3 -- Spindle = 1	Retrieve machine information
Data Server Setting Check IP Address / Host Name : 192.168.0.201 User Name : user Port No : 21 Connect Test..... Success :	Check connection with Data Server
< Signal Setting > ...	Verify signal setting
< Signal Log Setting > ...	Verify macro variable getting setting
< Variable History Setting > ...	Verify variable history retrieval setting
----- Check Machine : Machine002 ... ----- Check Process finished.	

■ Checking connection with machine

If the connection with the Collector is successful, then the connection with the machine is checked. When the connection is successful, "Success" will be displayed after "Connect Test....." If the connection is unsuccessful, "ERROR" and the detail of the error will be displayed.

If an "ERROR" is displayed, solve the problem considering the location and the detail of the error. (Example) If an error occurs in Ping Executing, check the IP address/host name settings of the machine or the network environment, taking in account the description of the error displayed.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	386/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

NOTE

When checking the machine setting, the time out for communication is set at 30 seconds per machine. If an "ERROR" (such as "Communication protocol error ...") occurs, there may be a communication time out occurring between the machine and the Collector. In such a case, check the following items.

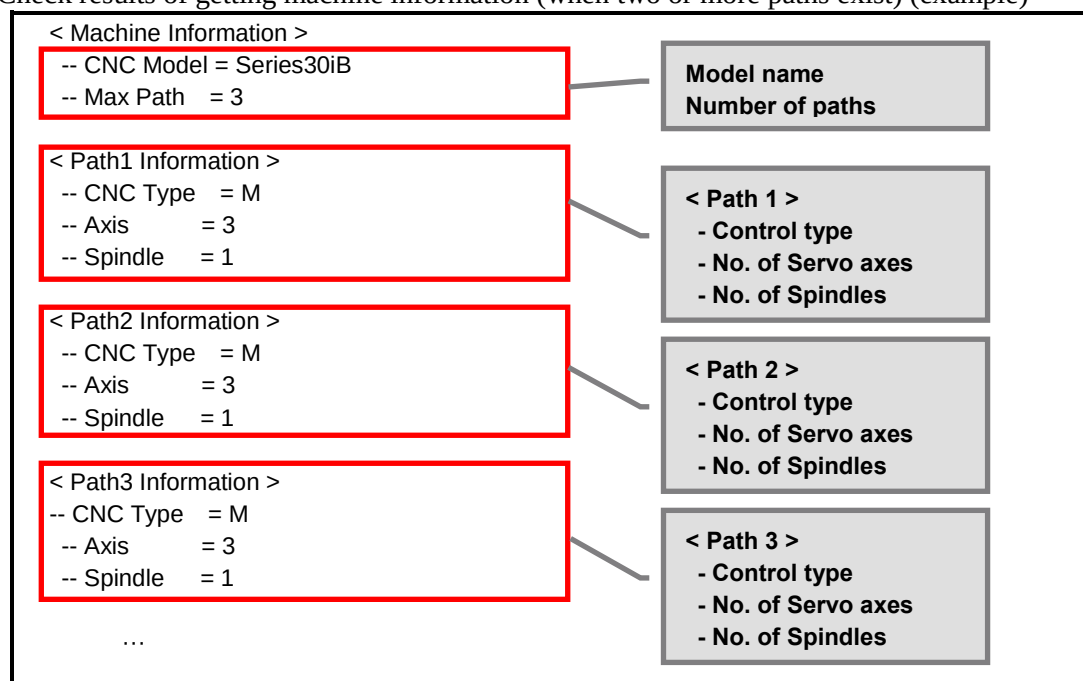
- (1) There is a possibility that the network is slow, or there is a high workload on the network. Check the network connection between the machine and the Collector.
- (2) The low performance of the network communication card or the NC may be the cause. Check the hardware configuration.

■ Retrieving machine information

After succeeding in the check of connection with the machine, the information retrieved from the machine is displayed in the lines following "< Machine Information >."

If there are two or more paths, the control type and the number of axes for each path are displayed in the lines following "< Path<x> Information >" (<x>: path number).

Check results of getting machine information (when two or more paths exist) (example)



NOTE

The path information of the Loader control path is not displayed for Series 16i models.

* For Series 16i models, refer to the NOTE 4 in the "Basic Signals of FanucCNC" subsection.

■ Checking connection with Data Server

Connection with the Data Server is checked when the information to connect to the Data Server is set in the machine setting.

The term "Success" will be displayed after "Connect Test....." when the connection is established successfully. "ERROR" and the details of the error will be displayed when the connection is not established.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 387/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

■ Verifying signal setting

Verify whether each signal set in the signal setting is being loaded correctly.

The results will appear under "< Signal Setting >."

If loading succeeded, "OK" appears; otherwise, "NG (meaning Not Good)" appears.

- When you find "NG" for a basic signal, select the [Basic Signal] tab in the right pane of the [Machine Setting] window and clear the [Enable] check box.
- When you find "NG" for a custom signal, select the [Custom Signal] tab of the [Machine Setting] window, and check the details of the signal setting.
- When you find "NG" for a laser control data signal, check the type of laser oscillator connected and the signals that are retrievable, and then select the [Laser control data signal setting] tab of the [Machine Setting] window and clear the [Enable] check box.
- When you find "NG" for a tool life signal, determine the tool life signals to collect per CNC path, and then select the [Tool Life Signal] tab of the [Machine Setting] window and clear the [Enable] check box.

Check results of signal setting (example)

< Signal Setting >

-- SigOP_path1 ... OK

-- Mode_path1 ... OK

-- MainProgram_path1 ... OK

-- ActProgram_path1 ... OK

-- MainComment_path1 ... OK

-- CncState_path1 ... OK

...

-- PMC_D200_W_path1 ... OK

-- MacroVar_510_path1 ... OK

-- PcodeCon_1_path1 ... OK

-- PcodeAux_2_path1 ... OK

-- PcodeExe_3_path1 ... OK

-- PcodeVar_100_path1 ... OK

-- Diagnosis_365_1_2_path1 ... OK

...

-- ToolGroupNum_path1 ... OK

-- ToolLifeCount01_path1 ... OK

-- ToolLifeCount02_path1 ... OK

-- ToolLifeCount03_path1 ... OK

-- ToolLifeCount04_path1 ... OK

...

-- PwrCommand_path1 ... OK

-- LaserPwr_path1 ... OK

-- BCReflection_path1 ... NG

...

Check results of basic signal setting

Check results of custom signal setting

Check results of tool life signal setting

Check results of laser control data signal setting

NOTE

The signal check results of the non-existent path and the Loader control path will all be NG for Series 16i models.

* For Series 16i models, refer to the NOTE 4 in the "Basic Signals of FanucCNC" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	388/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

NOTE

Update Option license is required to retrieve diagnostic signals (Diagnosis) and tool life signals.

- Verifying Macro Variable getting setting
Verify the macro variable getting setting.
The results are displayed under "< Signal Log Setting >" for each ID.
If retrieval is successful, "OK" appears; otherwise, "NG (meaning Not Good)" appears.
For the macro variable with "NG," select the [Macro Variable Getting Condition Definition] tab of the [Machine Setting], and check the details of the setting.

Check results of macro variable getting setting (example)

< Signal Log Setting >
-- Log ID : ID0001
-- Trigger Signal : #501 ... OK
-- Capture Signal(1) : #510 ... OK
-- Capture Signal(2) : #511 ... OK
-- Capture Signal(3) : #522 ... OK
-- Capture Signal(4) : #533 ... OK

-- Log ID : ID0002
...

Check results of Trigger macro variable

Check results of retrieved macro variable

- Verifying variable history retrieval setting
Verify the variable history retrieval setting.
The results are displayed under "< Variable History Setting >" for each ID.
If retrieval is successful, "OK" appears; otherwise, "NG (meaning Not Good)" appears.
For the variable with "NG," select the [Variable History Retrieval Condition Definition] tab of the [Machine Setting], and check the details of the setting.

Check results of variable history retrieval setting (example)

< Variable History Setting >
-- ID : ID0001
-- Trigger : MacroVar_500_path1 ... OK
-- Variable(1) : MacroVar_199_path1 ... OK
-- Variable(2) : MacroVar_200_path1 ... NG
-- Variable(3) : PcodeVar_100_path1 ... OK
-- Variable(4) : PcodeVar_101_path1 ... OK
-- Variable(5) : PcodeCon_30_path1 ... OK
-- Variable(6) : PMC_D100_0_path1 ... OK
-- Variable(7) : PMC_D100_D_path1 ... OK

-- ID : ID0002
...

Check results of Trigger variable

Check results of retrieved variable

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 389/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

The formats of the signal names displayed in the machine setting check (check result of signal setting/variable history retrieval setting) are as follows.

Signal Type		Display Format	Display Example	Remarks
Basic Signal				
Spindle axis		<signal name>_<m>_path<x>	SpindleLoad_3_path1	<m>: spindle axis number (0 to 3) <n>: servo axis number (0 to 23) <x>: path number (1 to 15)
Servo axis		<signal name>_<n>_path<x>	ServoSpeed_5_path1	
Others		<signal name>_path<x>	CncState_path1	
Custom Signal				
Custom macro variable		MacroVar_<n>_path<x>	MacroVar_500_path1	<n>: custom macro variable number <x>: CNC path number (1 to 15)
PMC Signal				
BIT		PMC_<a>_<t>_path<x>	PMC_D200_0_path1	<a>: PMC signal address*1 <address>+<number> <t>: bit position (0 to 7) <x>: PMC path number*2 (1 to 5, 9)
BYTE		PMC_<a>_path<x>	PMC_D200_path1	
WORD		PMC_<a>_W_path<x>	PMC_D200_W_path1	
DWORD		PMC_<a>_D_path<x>	PMC_D200_D_path1	
REAL		PMC_<a>_R_path<x>	PMC_D200_R_path1	
LREAL		PMC_<a>_L_path<x>	PMC_D200_L_path1	
Unsigned BYTE		PMC_<a>_UB_path<x>	PMC_D200_UB_path1	
Unsigned WORD		PMC_<a>_UW_path<x>	PMC_D200_UW_path1	
Unsigned DWORD		PMC_<a>_UD_path<x>	PMC_D200_UD_path1	
STRING		PMC_<a>_S_path<x>	PMC_D200_S_path1	
P-CODE				
Conversation macro		PcodeCon_<n>_path<x>	PcodeCon_1_path1	<n>: variable number (1 to 33) <x>: CNC path number (1 to 15)
Auxiliary macro		PcodeAux_<n>_path<x>	PcodeAux_2_path1	
Execution macro		PcodeExe_<n>_path<x>	PcodeExe_3_path1	
Common area		PcodeVar_<n>_path<x>	PcodeVar_600_path1	<n>: variable number (100 to 89999) <x>: CNC path number (1 to 15)
Diagnostic Signal				
Bit	Path	Diagnosis_<d>_<t>_path<x>	Diagnosis_100_0_path1	<d>: diagnosis number (0 to 99999) <t>: bit position (0 to 7) <x>: CNC path number (1 to 15) <y>: servo axis number (0 to <g>*3) <z>: spindle axis number (0 to <h>*4)
	Servo axis	Diagnosis_<d>_<t>_path<x>_axis<y>	Diagnosis_101_1_path1_axis2	
	Spindle axis	Diagnosis_<d>_<t>_path<x>_spindle<z>	Diagnosis_102_1_path1_spindle3	
Other Than Bit	Path	Diagnosis_<d>_path<x>	Diagnosis_103_path1	
	Servo axis	Diagnosis_<d>_path<x>_axis<y>	Diagnosis_104_path1_axis1	
	Spidle axis	Diagnosis_<d>_path<x>_spindle<z>	Diagnosis_105_path1_spindle2	

*1 <address>: "G" / "F" / "Y" / "X" / "A" / "R" / "T" / "K" / "C" / "D" / "M" / "N" / "E" / "Z"

*2 PMC path number when retrieving value of dual check safety PMC: "9"

*3 <g>: The maximum servo axis number for the path set for the [CNCPath] box (the [Signal Setting] dialog box)

*4 <h>: The maximum spindle axis number for the path set for the [CNCPath] box (the [Signal Setting] dialog box)

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 390/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

NOTE

Update Option license is required to retrieve diagnostic signals.

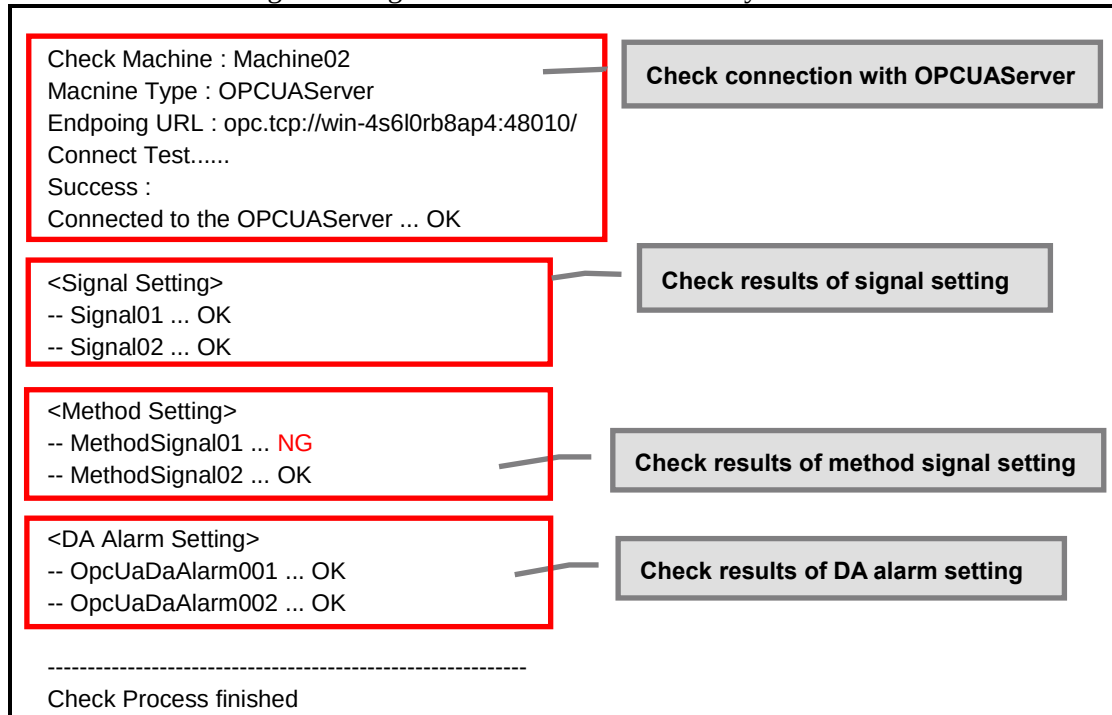
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	391/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

5.1.4 Check Results for OPCUAServer Type

This subsection explains check results for OPCUAServer type machines.

Check results for OPCUAServer type machine (example)

The term "NG" standing for "not good" is used here as an antonym of "OK."



■ Check connection with OPCUAServer

When a connection with the OPC UA Server is established successfully, "Success" will be displayed under the text "Connect Test..."; otherwise, the text "ERROR" and description of the error will be displayed.

In a case when the text "ERROR" is displayed, refer to the following subsection of the "FANUC MT-LINK*i* / Integration Server Troubleshooting MANUAL":

The "Connection with OPC UA Server Fails" subsection in the "TROUBLES RELATED TO OPC UA SERVERS" section in the "MT-LINK*i* TROUBLESHOOTING"

■ Check results of signal setting

These results show whether the signals that are set up can be retrieved. They are displayed under the text "<Signal Setting >."

When a signal is retrieved successfully, the text "OK" will be displayed; otherwise, the text "NG" (standing for "Not Good") will be displayed.

In a case when the text "NG" is displayed, check the status of the OPC UA Server and verify the settings made in the [Signal Setting] tab page of the [Machine Setting] window for the machine.

■ Check results of method signal setting

When finding methods with "NGs" in <Method Setting> section, conduct a detailed check for the methods by clicking the [Run Method Test] button in the [Method Signal Setting] window.

For the details, refer to the "[Set OPCUAServer Methods as Signals](#)" subsection.

■ Check results of DA alarm setting

When finding methods with "NGs" in <DA Alarm Setting> section, conduct a detailed check for DA alarm settings by clicking the [Test] button in the [DA Alarm Setting] window.

For the details, refer to the "[Set OPCUAServer DA Alarms](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	392/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

5.1.5 Check Results for FanucRobot Type

This subsection explains check results for FanucRobot type machines.

Check results for FanucRobot type machine (example)

Check Machine : Machine003
 Machine Type : FanucRobot
 RobotController : R-30iB
 IP Address / Host Name : 192.168.0.103
 Port No : 60008
 Connect Test.....
 Success :
 Network type ... TCP/IP
 Ping executing [192.168.0.103] ... OK
 Connected to the ROBOT ... OK

< Machine Information >
 -- Connected Robot = R-30iB
 -- Connected Robot Version = V8.30180 8/18/2016

Robot Version Check
 Success :

< Signal Setting >
 -- BATAL ... OK
 -- Mode ... OK
 -- RobotState ... OK
 -- EMG ... OK
 -- IO_DO100 ... OK
 -- DATA_NumReg5 ... OK
 -- DATA_G2_PosReg1_X ... OK

< Variable History Setting >
 -- ID : ID0001

-- Trigger : IO_SO1 ... OK

-- Variable(1) : IO_SO8 ... OK

-- Variable(2) : DATA_NumReg1 ... NG

-- Variable(3) : DATA_G1_PosReg2_X ... OK

FTP Server Setting Check ...
 IP Address / Host Name : 192. 168.0.202
 User Name : user
 Port No : 21
 Connect Test.....
 Success :

.....

Check Process finished.

Check connection with
Robot controller

Model name
Version information

Check results of
Robot controller version

Check results of
basic / custom signals

Check results of
variable history setting

Check results of
trigger variable

Check results of
retrieved variables

Check connection with
FTP Server

If the model of the robot controller connected is different from the model of the robot controller set in the machine setting, text "Error" will be displayed in the "Robot Version Check..." In such a case, check the model setting of the robot controller in the [Machine Setting] window.

When connection to the FTP Server fails to establish, text "Error" will be displayed in the "FTP Server Setting Check..." In such a case, check the FTP Server setting in the [Machine Setting] window.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	393/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

NOTE

- 1 When connecting to R-30*i*B / R-30*i*B Mate / R-30*i*B Plus / R-30*i*B Mate Plus / R-30*i*A / R-30*i*A Mate, if R650 North America dedicated setting is applied for the FANUC Robot controller, R553 touch panel communication function option is required.
- 2 FANUC Robot controllers for the following robots cannot check the Machine Information and the Signal Setting.
 - R-30*i*A / R-30*i*A Mate
 - R-J3*i*B / R-J3*i*B Mate

Signal type		Format	Example	Remarks
Basic		<Signal Name>	RobotState	-
Custom Signal	EMG (Emergency stop signal)	EMG	EMG	-
	IO (input / output signal)	IO_<a>	IO_DO100	<a>: address of IO signal* ¹ <address>+<number>
	DATA (register)			
	numeric register	DATA_NumReg<n>	DATA_NumReg10	<n>: register number (1 to 2000)
	string register	DATA_CharReg<n>	DATA_CharReg3	<n>: register number (1 to 1000)
	position register	DATA_G<g>_PosReg<n>_<s>	DATA_G1_PosReg2_X	<g>: action group number (1 to 8) <n>: number (1 to 100) <s>: signal array suffix* ²

*¹ <address>: DI, DO, GI, GO, AI, AO, UI, UO, SI, SO, RI, RO, WI, WO, or WSI

(For R-J3*i*B / R-J3*i*B Mate Series robots, read DI and DO as SDI and SDO; and RI and RO as RDI and RDO)

<number>: one of the following numbers determined in accordance with Address.

- DI (SDI) or DO (SDO): 1 to 5000
- RI (RDI), RO (RDO), GI, GO, AI, or AO: 1 to 1000
- UI or UO: 1 to 999
- SI or SO: 0 to 1000
- WI or WO: 1 to 400
- WSI: 1 to 600

*² <signal suffix>: a signal suffix that indicates both the signal data to retrieve and its index in data array as shown in the following table.

Array index	Cartesian coordinates	Joint coordinates	Config*	Coordinate system number*
0	X	J1	conf1	UF
1	Y	J2	conf2	UT
2	Z	J3	conf3	-
3	W	J4	conf4	-
4	P	J5	TA1	-
5	R	J6	TA2	-
6	E1	JE1	TA3	-
7	E2	JE2	-	-
8	E3	JE3	-	-

* For Config and coordinate system number, there is no way to specify how many signals to retrieve.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	394/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

Examples) when the number of Cartesian coordinates is 6

the signals with suffixes of "X" to "R" (intersections of the Cartesian coordinates column and array index rows of 0 to 5 in the table) will be retrieved.

when the number of joint coordinates is 9

the signals with suffixes of "J1" to "JE3" (intersections of the Joint coordinates column and array index rows of 0 to 8 in the table) will be retrieved.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	395/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

5.1.6 Check Results for MTConnectAgent Type

This subsection explains check results for MTConnectAgent type machines.

Check results for MTConnectAgent type machine (example)

Check Machine : Machine004
 Machine Type : MTConnectAgent
 Device Name : FANUC_CNC
 Device uuid : 000
 IP Address / Host Name : 192.168.0.104
 Port No : 5000
 Connect Test.....
 Success :
 Network type ... TCP/IP
 Ping executing [192.168.0.104] ... OK
 Connected to the MTConnectAgent ... OK

<Information>
 -- Version ... 1.3.0.4

< Signal Setting >
 -- AVAILABILITY ... OK
 -- EmergencyStop ... OK
 -- Execution ... OK
 -- Program ... OK
 -- PartsCount ... **NG** [Value is UNAVAILABLE]

 Check Process finished.

Check result of connection with MTConnect Agent

Version info of MTConnect

Checkresults of signal retrieval

If the check result of connection with MTConnect Agent, or the confirmation result of signal retrieval is "NG," then the reason of failure will be displayed in the format of "NG [<message>]." In such a case, check the state of MTConnect Agent / Device, or the settings in the [Signal Setting] of [Machine Setting].

Examples of <message> and countermeasures when check result is "NG" (meaning Not Good)

Check Item	Message	Description and Countermeasures
Connection with MTConnect Agent	Value of Availability is UNAVAILABLE	State of MTConnect Adapter / Device is unusable. There is a possibility of failure in connection with any one of MT-LINK <i>i</i> , MTConnect Agent, MTConnect Adapter, and Device. Check the following. - Error in setting of connection target (IP address / host name, port) - MTConnect Agent is not started - MTConnect Adapter is not started - Device is not started - No connection between MTConnect Agent and Device
	Value of Availability is UNKNOWN	Availability of MTConnect Agent has an invalid value. Check the settings of the MTConnect Agent.
	Version of MTConnect is not supported	Version of MTConnect is not supported by MT-LINK <i>i</i> . Check version of connection target MTConnect. - Version of MTConnect supported by MT-LINK <i>i</i> is "1.<x>.<x>" (<x>: an optional number)

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 396/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

Check Item	Message	Description and Countermeasures
	Device does not exist	Unable to find Device information. Check the setting of the Device information at the connection target MTConnect Agent. - 1 or more Device information exist - There are no overlap of uuid's among multiple Device information
	Failed to get device	Failed to retrieve Device information. Check the state of connection target MTConnect Agent.
Confirmation of Signal Retrieval	ID of signal is not found	Failed to retrieve signal value. Signal (ID) corresponding to the Device information does not exist. Check Device information of connection target.
	Value is UNAVAILABLE	State of signal value is unusable. Failed to retrieve signal setting corresponding to the Device information. Check Device information of connection target.
	Data type is invalid	Data type is invalid. Data type is different for the signal corresponding to the Device information. Check Device information of connection target.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 397/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

5.1.7 Check Results for FanucRobodrill Type

This subsection explains check results for FanucRobodrill type machines.

Check results for FanucRobodrill type machine (example)

Check Machine : Machine004
 Machine Type : FanucRobodrill
 IP Address / Host Name : 192.0.51.60
 Port No : 8193
 Connect Test.....
 Success :
 Network type ... TCP/IP
 Ping executing [192.0.51.60] ... OK
 Connected to the CNC ... OK

Check results of connection with FanucRobodrill

< Machine Information >
 -- CNC Model = Series31iB
 -- Max Path = 1

-- Machine Model = TiF Series

ROBODRILL model name

<Signal Setting>

-- SigOP_path1 ... OK
 -- Mode_path1 ... OK
 ...

-- PMC_D200_W_path1 ... OK
 -- MacroVar_510_path1 ... OK
 ...

-- RobodrillToolNum_path1 ... OK
 -- RobodrillCounterNum1_path1 ... OK
 ...

Check results of basic signal setting

Check results of custom signal setting

Check results of Robodrill signal setting

Check Process finished.

When checking connection with the Data Server detects errors, "ERROR : <error details>" will be displayed in the "Connect Test..." line.

If "NGs" (standing for "not good") are found in the check results of Robodrill signal setting section, verify on a model by model basis that those error marked signals are irretrievable and clear their [Enable] check boxes in the [Robodrill Signal Setting] tab of the [Machine Setting] window.

Examples of <message> and countermeasures when check result is "NG" (meaning Not Good)

Check Item	Message	Description and Countermeasures
Connection with FanucRobodrill machines	ROBODRILL license option is required.	A ROBODRILL Option license is required to support FanucRobodrill machines. - Check if the license is verified if it is installed. - Delete FanucRobodrill type machines if no license is installed.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 398/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

5.2 DATA VALIDATION

This function checks the setting data integrity.

When a lack or an error is present in the setting files, an error may occur just before transmitting in the [Send to DB] window.

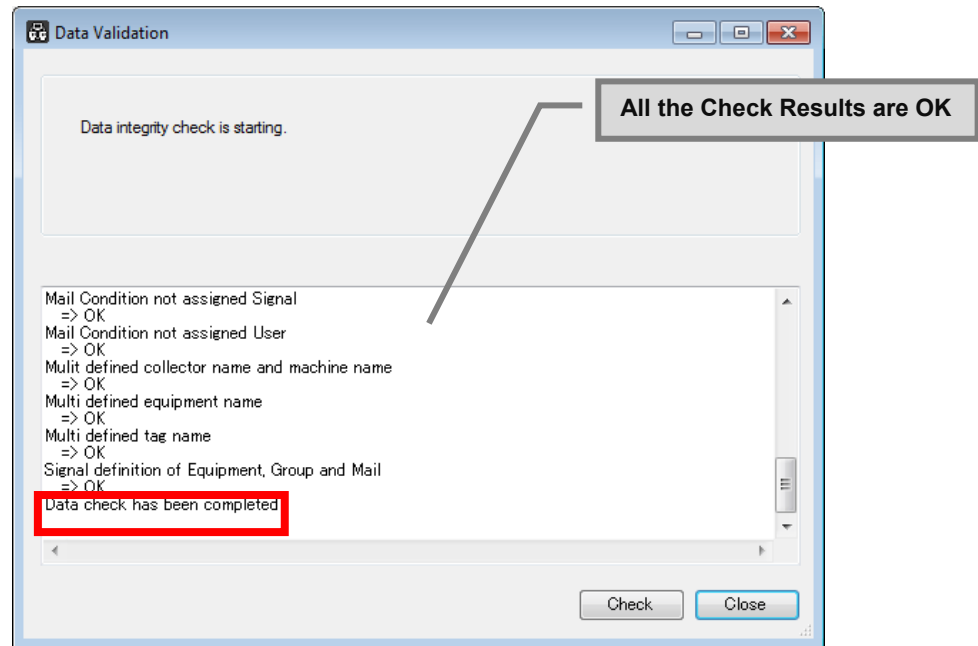
You can avoid this error previously by checking consistency in the [Data Validation] window.

Selecting [Data Validation] in the AdminTool Menu opens the following [Data Validation] window.

Clicking [Check] executes data validation and displays the execution log.

■ When the result is OK:

Verify that all the check results are "OK" and the "Data check has been completed" message is displayed at the bottom row. This indicates that the data validation was executed successfully.

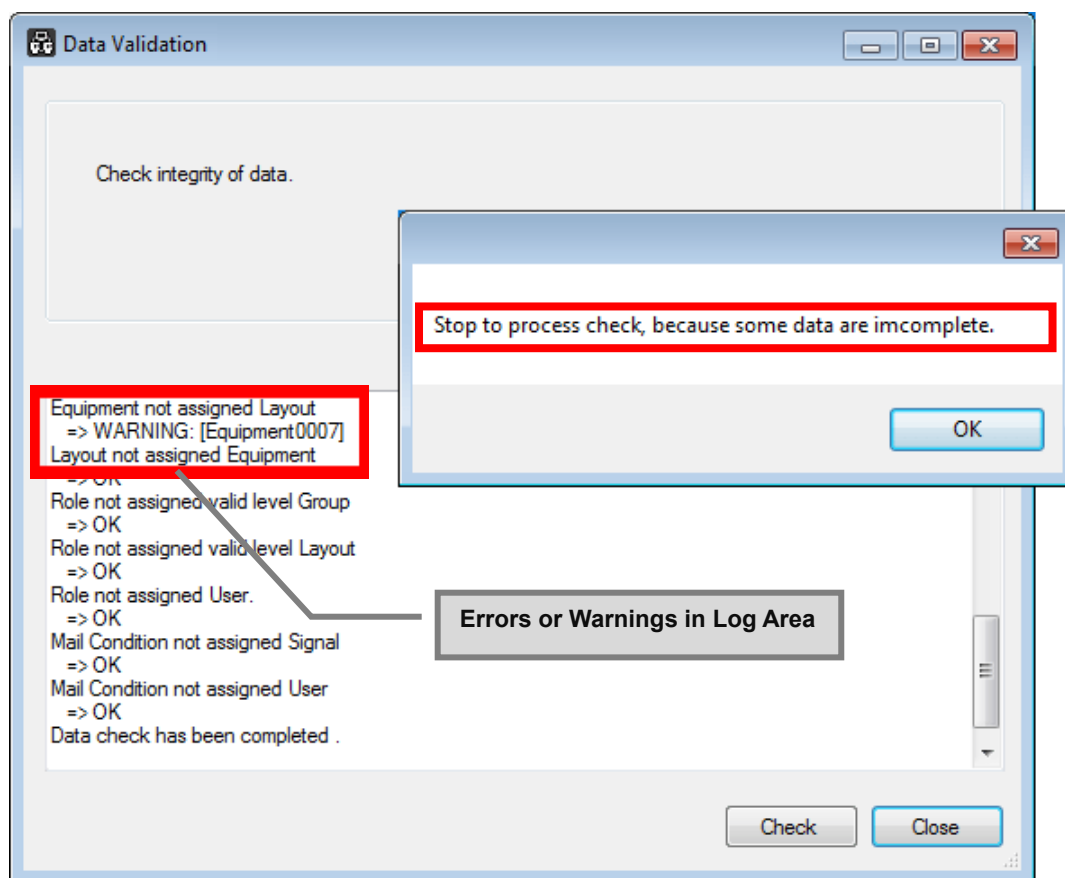


20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	399/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

■ **When the result is NG:**

When data validation fails, a message appears reading that the data validation has aborted, and the errors or warnings that caused the data validation to abort are listed in the log area.

Correct the settings by referring to those errors and warnings.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 400/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

6

[APPLYING SETTINGS] - APPLICATION OF SETTINGS

This chapter describes how to apply the settings made in AdminTool.

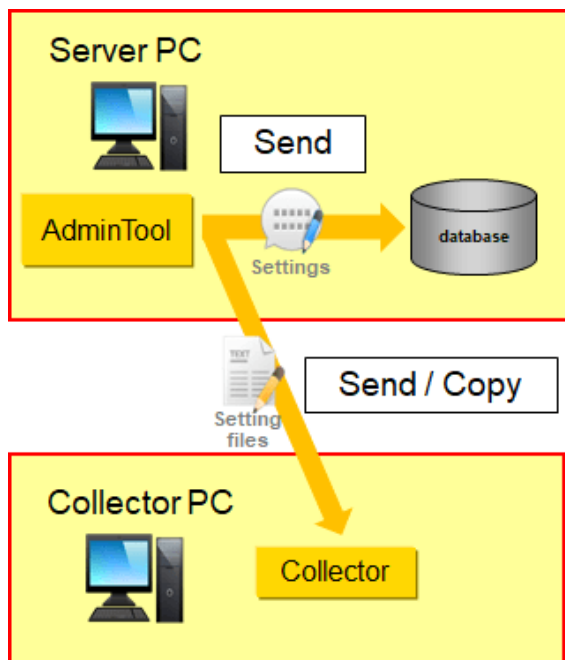
The AdminTool Menu has the following menu items in the [Applying Settings] - [Application of Settings] group:

(1) Send to DB

This menu sends setting data to the database and applies them to the system.

(2) Create Collector Setting Files

This menu creates setting files for Collectors and applies them to the Collectors.



- AdminTool transfers setting data to the database (Mongo DB).
- AdminTool generates setting files and send setting files to Collector.
- When you cannot send setting file, copy setting file to setting folder of Collector manually.

WARNING

When updating machine settings with the [Machine Setting] window of AdminTool, the following steps must be performed:

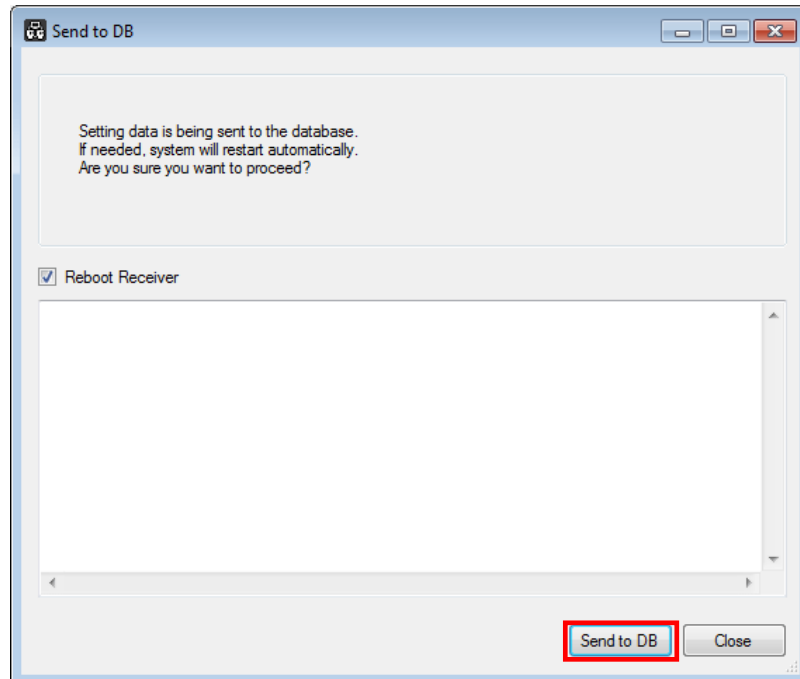
- (1) Send updated setting data to the database.
- (2) Create Collector setting files and send them to Collector.

If updated machine settings are not sent to the database and Collector and thus they cannot reflect the updates, such as deletion or addition of signals, unexpected troubles may occur because the database and Collector will continue to refer to the older settings.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	401/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

6.1.1 Send Setting Data to DB

- (1) Select [Send to DB] in the AdminTool Menu. The following [Send to DB] window appears.

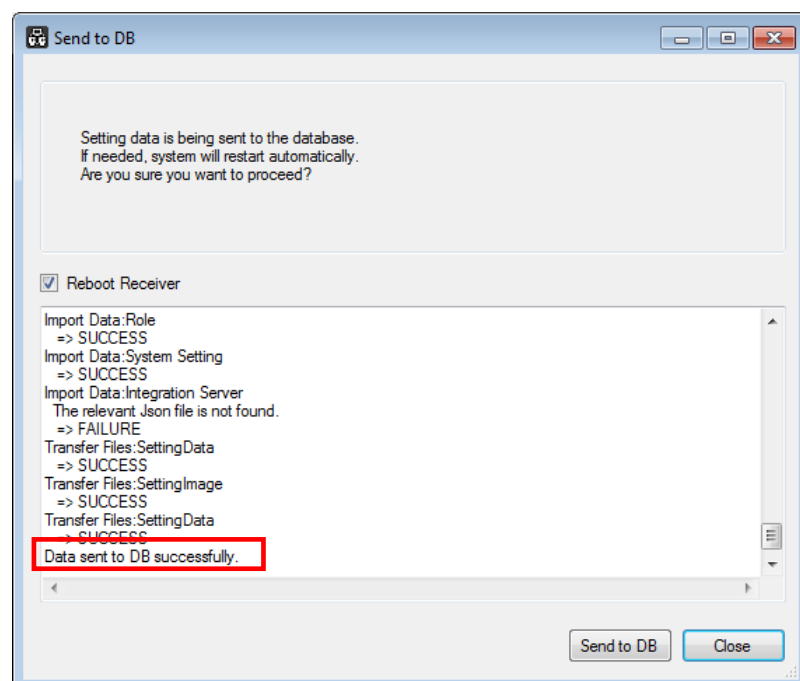


- (2) Click the [Send to DB] button to check the setting data and transfer to the Database. If the [Reboot Receiver] check box is selected, the system restarts the Receiver after data transfer. During this restart process, the Receiver reflects the transferred settings.

■ Example of normal case:

If data is sent to DB normally, the "Data sent to DB successfully." message is displayed.

If the setting data for SERVO VIEWER linkage is sent to the Collector, "Data sent to Collector successfully." message is displayed.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	403/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

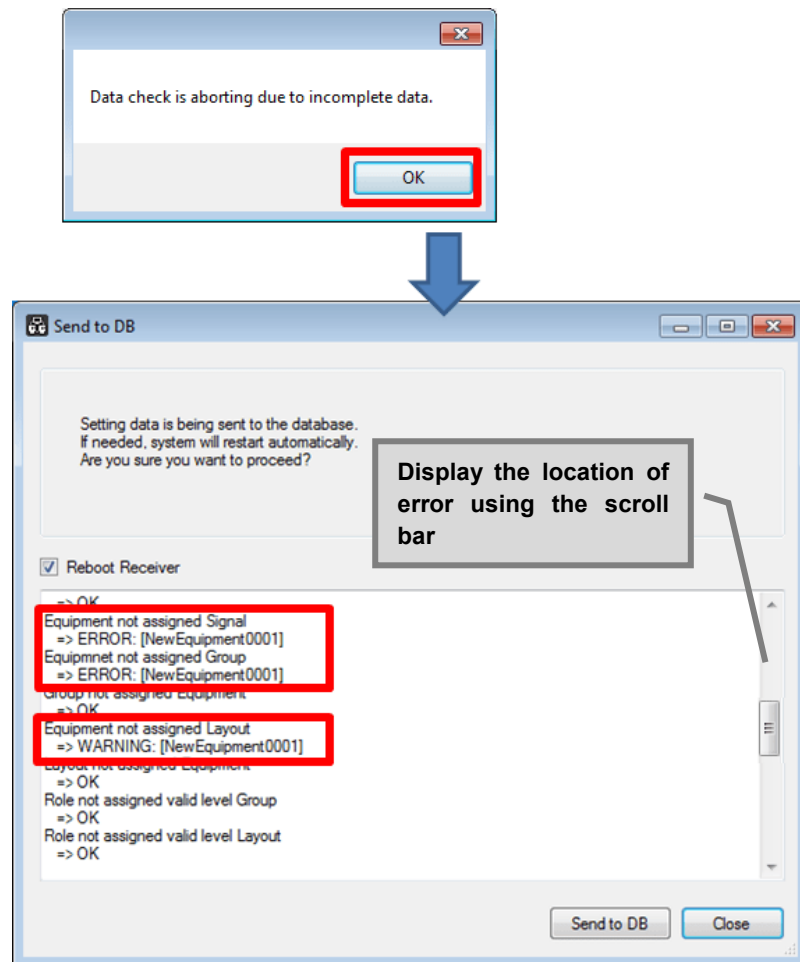
■ Example of error case:

Following is an error case in checking the setting data.

A window is displayed informing you that the data checking process is stopped.

Click the [OK] button, and check the detailed information of the error using the scroll bar on the [Send to DB] window.

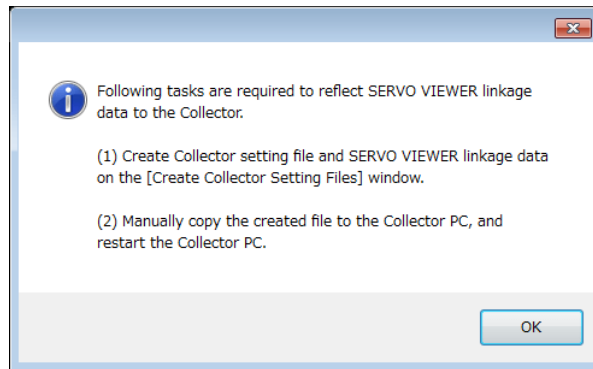
Correct the settings, referring to the error message.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 404/868

NOTE

When sending data to DB, the following message may be displayed.



When the above message is displayed, select the [Create SERVO VIEWER linkage data.] check box in the [Create Collector Setting Files] dialog box and click the [Run] button. After the Collector setting file and the setting file for the SERVO VIEWER linkage is created, copy these files to the Collector PC manually and restart the Collector PC.

For the details, refer to the "■ Steps for manual update:" part of the "[Apply Created AdminTool Setting to Collector](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 405/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

6.1.2 Recognize Transferred Item

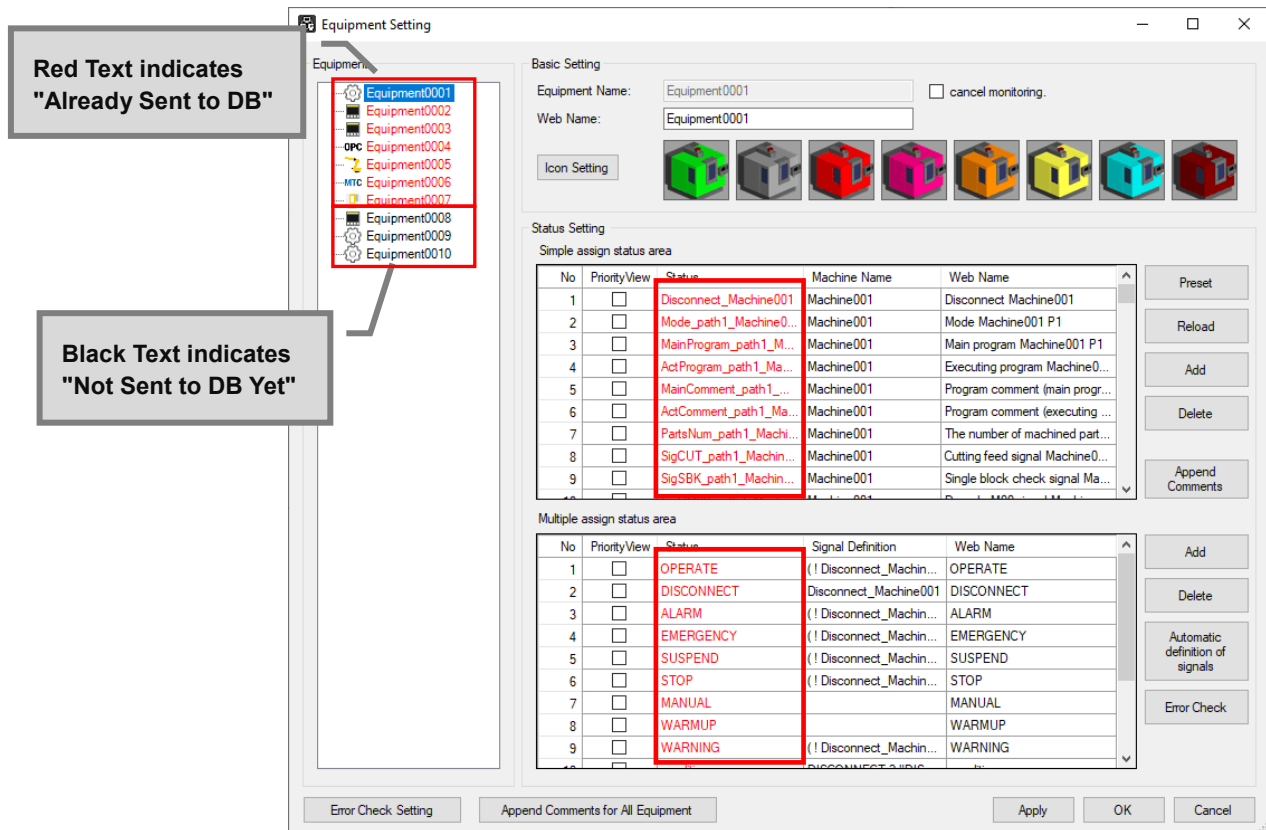
After the transmission, the following names are turned RED:

- Equipment name/signal name in the [Equipment Setting] window
- Group name in the [Group Setting] window

In contrast, names of non-transferred item stay BLACK. So you can recognize which item has transferred or not by its color.

Example 1:

In the [Equipment Setting] window, transferred equipment names are RED and new equipment names are BLACK.



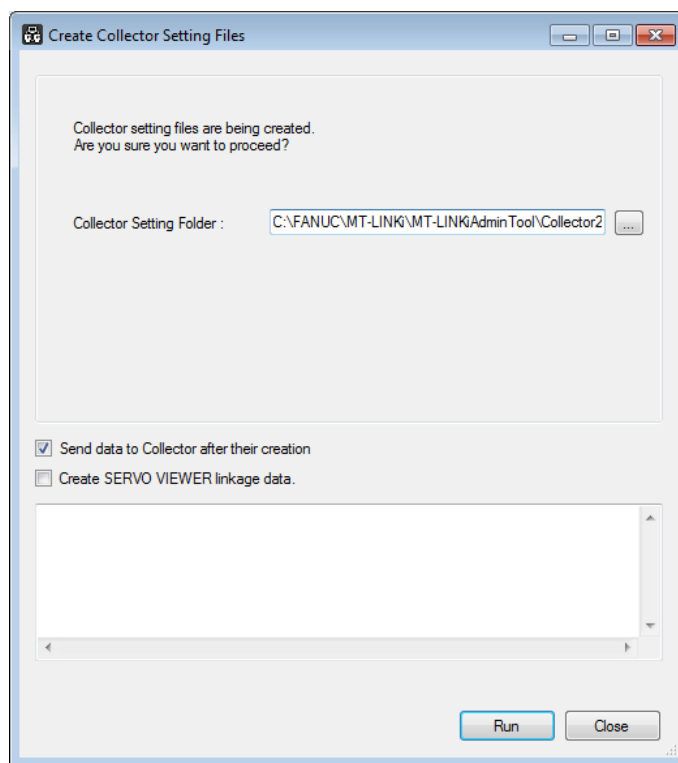
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	406/868

6.2 CREATE COLLECTOR SETTING FILES

When you execute the new setting or setting change in the [Machine Setting] window, you should create Collector setting files and save them to the destination folder. When you change any setting in the [Machine Setting] window, execute this operation.

6.2.1 Apply Created AdminTool Setting to Collector

Select [Create Collector Setting Files] in the AdminTool Menu. The following [Create Collector Setting Files] dialog box appears.



Each time Collector setting files are created, a new Collector setting folder is generated to store the setting files. The following is the default folder for a Collector setting file:

<MT-LINK*i* installation folder>\MT-LINKiAdminTool\Collector<yyyyMMddHHmmss>

If creating Collector setting files for the first time after installation, perform the "**■ Steps for manual update**" procedure to pass Collector setting information to the Collector manually; otherwise, if Collector is running already, perform the "**■ Steps for automatic update**" procedure.

If the Servo Viewer Linkage Collector is not running when the Servo Viewer Linkage setting is enabled, perform "**■ Steps for manual update**"; otherwise, if the Servo Viewer Linkage Collector is running, data for Servo Viewer Linkage will be transferred when "Send to DB" is executed. [Create SERVO VIEWER linkage data.] checkbox to create data for linking with the servo viewer.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	407/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

■ Steps for manual update:

If the Collector is not started (Collector setting file has never been created), or if failing to restart the Collector, execute the following steps.

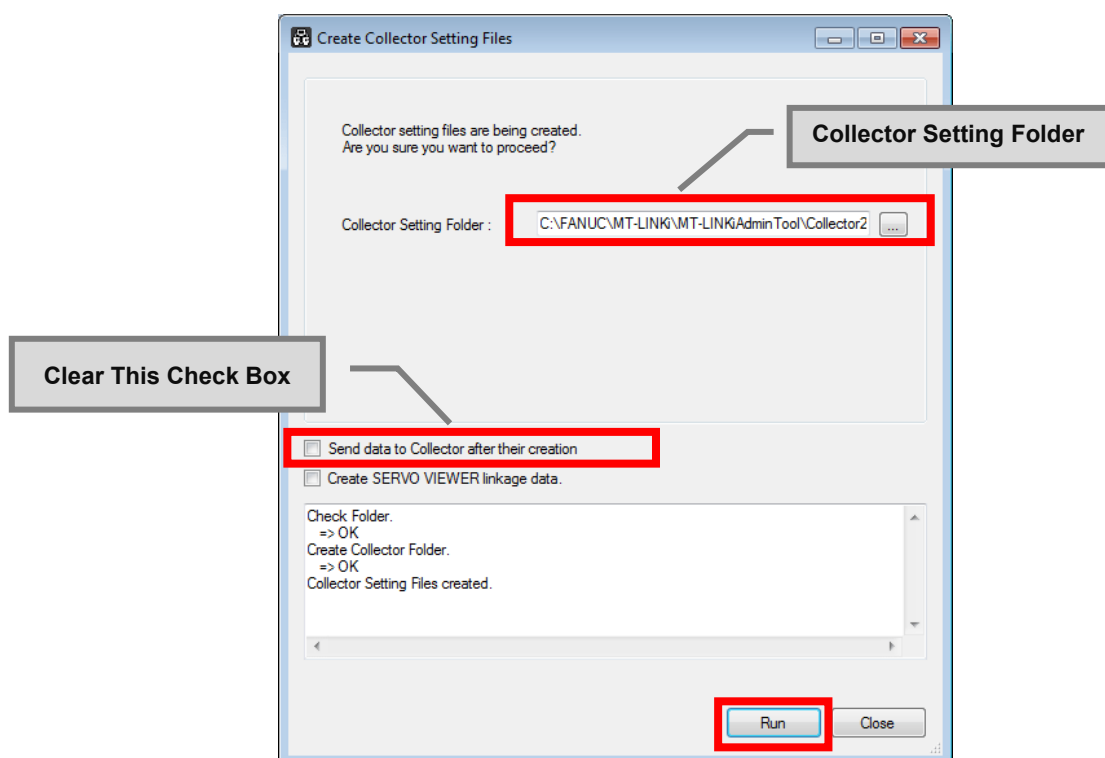
- (1) Clear the [Send data to Collector after their creation] check box.

When creating the setting file for the SERVO VIEWER linkage function at the same time, select the [Create SERVO VIEWER linkage data.] check box.

NOTE

The [Create SERVO VIEWER linkage data.] check box is displayed when the setting of SERVO VIEWER Linkage function is enabled in the [Machine Setting] window.

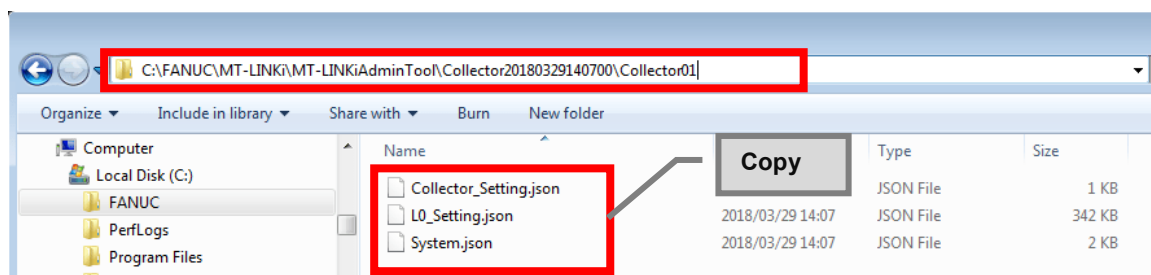
Click the [Run] button to create the Collector setting file.



- (2) On the Server PC, start Windows Explorer and specify the Collector setting folder path (created in the step 1). You should find the folder given the same name as Collector. When you open this folder, you should find the following three files. In a case when two or more Collectors are registered, same number of folders with the names of the Collectors will be created, and the following three files can be found in each folder.

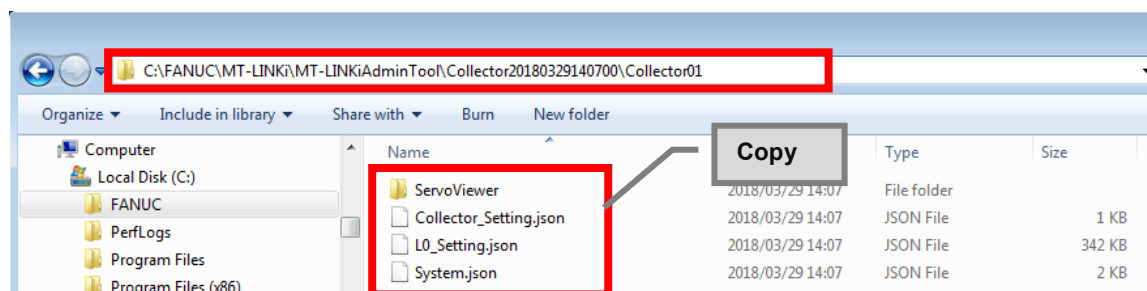
- Collector_Setting.json
- L0_Setting.json
- System.json

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 408/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	



When the [Create SERVO VIEWER linkage data.] check box is selected, the following setting files for SERVO VIEWER linkage are created.

- ServoViewer\L0_SamplingSetting.json
- File under ServoViewer\Profile\ Measure profile (*.svwraw)
- File under ServoViewer\Schedule\ Measurement schedule (*.svwsch)

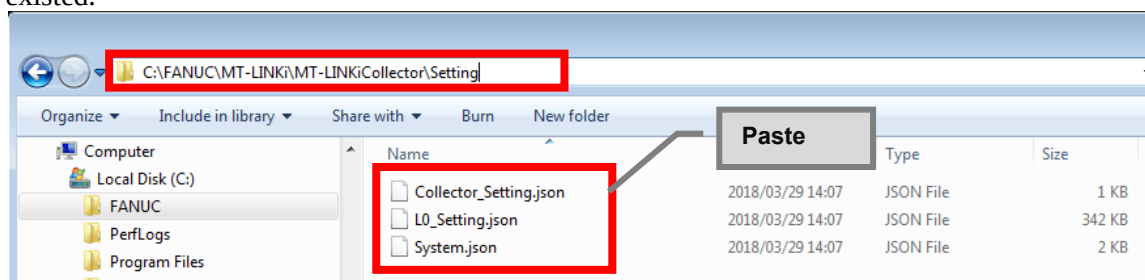


* All the contents under the folder with the name of the Collector should be copied.
(If there is a sub folder created under this folder, the sub folder and the files contained there should be copied altogether)

- (3) On the Collector PC with the name of the targeted Collector, start Windows Explorer and open the folder storing the Collector setting files.

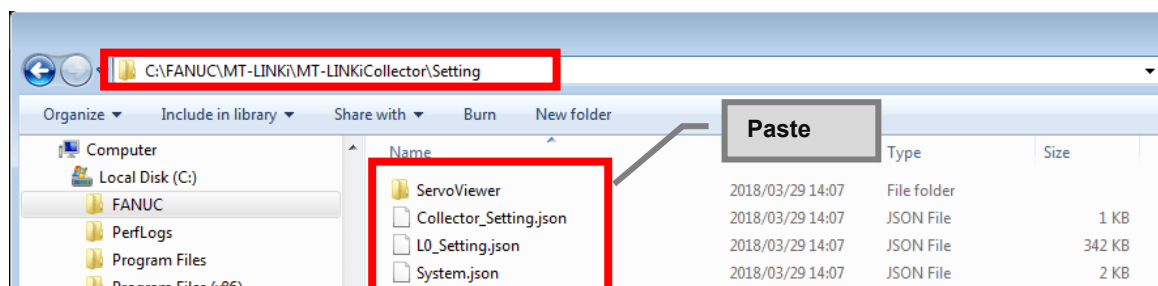
<MT-LINKi Collector installation folder>\Setting

Paste the three files (copied in the step 2) to the Setting folder. Overwrite them when the same name files are existed.



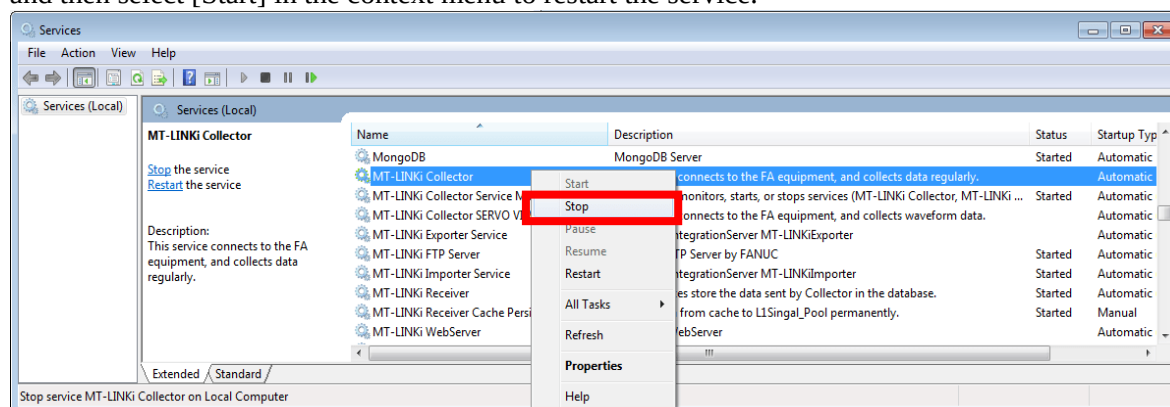
When the [Create SERVO VIEWER linkage data.] check box is selected, delete the folder for SERVO VIEWER linkage below, and then paste the folder for SERVO VIEWER linkage along with the three files.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 409/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	



(4) To apply the setting of the Collector setting files, you need to restart the MT-LINKi Collector service or the PC.
If the [Create SERVO VIEWER linkage data.] check box is selected, restart the Collector PC.

(5) To restart the MT-LINKi Collector service, on the PC in which MT-LINKi Collector is installed, click the [Start] button > [Control Panel] > [Administrative Tools] > [Services]. Point to [MT-LINKi Collector], right-click it to open a context menu. Select [Stop] in the context menu to stop MT-LINKi Collector once and then select [Start] in the context menu to restart the service.



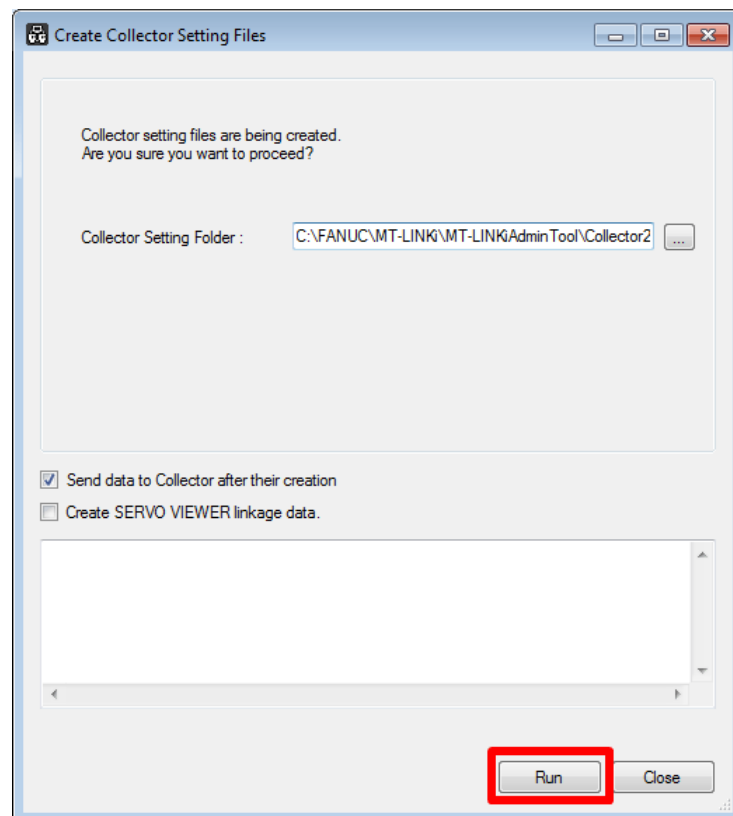
* In a case when two or more Collectors are registered, perform steps (2) to (5) for all the Collector PCs.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 410/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

■ Steps for automatic update:

If the Collector is started (Collector setting file has been created), execute the following steps.

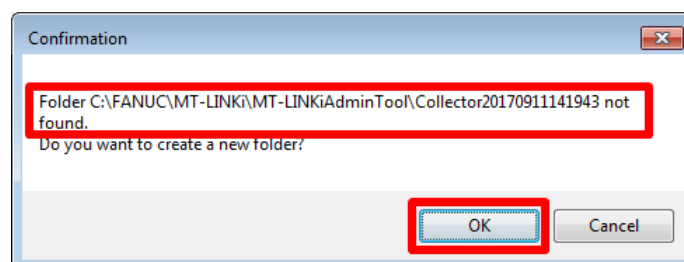
- (1) Select the [Send data to Collector after their creation] check box in the [Create Collector Setting Files] dialog box and click [Run].



NOTE

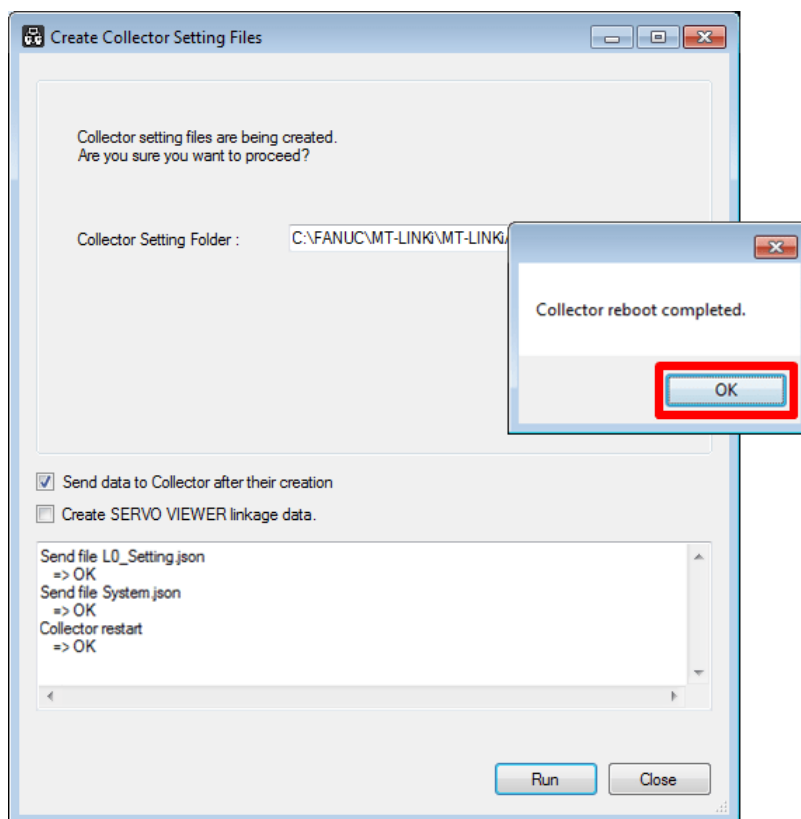
Selecting the [Create SERVO VIEWER linkage data.] check box disables the [Send data to Collector after their creation] check box. In this case, automatic update does not work.

- (2) Click [OK] in the confirm dialog.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	411/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- (3) The following "Collector reboot completed." message indicates successful termination. Click [OK] and close the window.



If "Failed to reboot the Collector" message dialog box appears, execute the [Steps for manual update] explained above.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	412/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

7

[CONTROLLING SYSTEM] - SYSTEM CONTROL

This chapter describes how to control MT-LINK*i* system.

The AdminTool Menu has the following menu items in the [Controlling System] - [System Control] group:

(1) Start/Stop System

This menu starts and stop services, such as Receiver and Collector, that compose MT-LINK*i* system.

The targeted services when the system is started or stopped will be as follows.

For Server PC

- MongoDB (not targeted to be stopped)
- MT-LINK*i* Exporter Service
- MT-LINK*i* Receiver
- MT-LINK*i* WebServer

For Collector PC

- MT-LINK*i* Collector
- MT-LINK*i* Collector SERVO VIEWER Linkage
- MT-LINK*i* FTP Server

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 413/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

7.1 START/STOP SYSTEM

The services composing MT-LINK*i* system can be started and stopped by using the [Start/Stop System] window of AdminTool.

Also, the system can be started and stopped by starting and stopping those services manually.

7.1.1 Start/Stop System

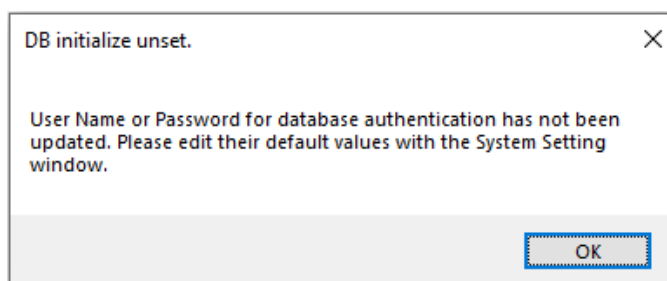
To start or stop the system, follow the steps below.

- (1) Select [Start/Stop System] in the AdminTool Menu.

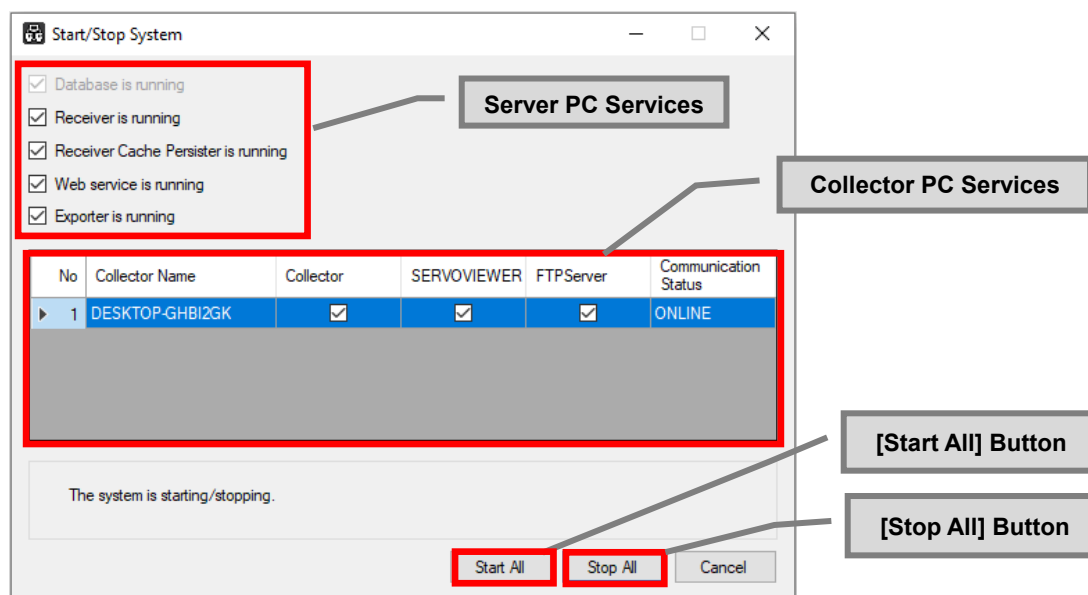
NOTE

Right after the installation, initial values of user and password are both set as "admin."

If user and password are not changed from initial values, following message is displayed. Click [OK] and specify new username and password on the [System Setting] window, and then click [Start/Stop System] again.



- (2) The following [Start/Stop System] window is displayed.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	414/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

■ [Start/Stop System] window

Server PC services

- Database is running
The state of MongoDB service is displayed.
When DB is being stopped for some reason, the check box will be cleared. In such a case, DB can be started by selecting the check box. However, the check box cannot be cleared (stop DB).
- Receiver is running
The state of MT-LINK*i* Receiver service is displayed. (NOTE 1, 2)
- Receiver Cache Persister is running
The state of MT-LINK*i* Receiver Cache Persister service is displayed. (NOTE 1, 2)
- web service is running
The state of MT-LINK*i* WebServer service is displayed. (NOTE 1, 2)
- Exporter is running
The state of MT-LINK*i* Exporter Service service is displayed. (NOTE 1, 2)

Collector PC services

- Collector Name
The name of the Collector running on the Collector PC is displayed. (Cannot be changed)
- Collector
The state of MT-LINK*i* Collector service is displayed. (NOTE 1, 2, 3)
- SERVVIEWER
The state of MT-LINK*i* Collector SERVO VIEWER Linkage service is displayed. (NOTE 1, 2, 3, 4)
- FTPServer
The state of MT-LINK*i* FTP Server service is displayed. (NOTE 1, 2, 3)
- Communication Status
The current status of communication with the Collector Service Manager running on a Collector.
ONLINE: indicates the manager is available for communication.
OFFLINE: indicates the manager is unavailable for communication. Rows with this status will be grayed out. (NOTE 3, 5)

NOTE

- 1 If the check box is selected, the service is running. The service stops when the check box is cleared.
- 2 If the check box is cleared, the service is being stopped. The service starts when the check box is selected.
- 3 Check boxes will be pained in gray when corresponding rows have "OFFLINE" in the [Communication Status] column. In such a case, the states of the services are unknown, and selecting or clearing the check boxes cannot restart/stop those services.
- 4 If SERVO VIEWER Linkage function is set disabled for a Collector on the [Machine Setting] window, "Disabled" is displayed in the [SERVVIEWER] column for the Collector.
- 5 When, immediately after the Collector PC started, the [Communication Status] column for the Collector on the [Restart System] window may be set to "OFFLINE" because Collector Service Manager has not started at that point in time. In such a case, wait for seconds and check the column again.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 415/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

* After selecting a row, right-click anywhere in the row to open a popup menu, and select [ON] or [OFF] from the menu to activate or deactivate all the services on the selected Collector PC. When multiple rows are selected, all the services on the selected Collector PCs are to be activated or deactivated.

Command buttons

- [Start All] button
Clicking this button starts all the services that are being stopped (check box is cleared).
- [Stop All] button
Clicking this button stops all the services that are running (check box is selected).
- [Cancel] button
Clickin this button closes the [Start/Stop System] window without restarting the system.

- (3) To start the system, click the [Start All] button; to stop the system, click the [Stop All] button. When closing the [Start/Stop System] window without starting or stopping the system, click the [Cancel] button.

NOTE

It may take some time to start and stop the Receiver.



WARNING

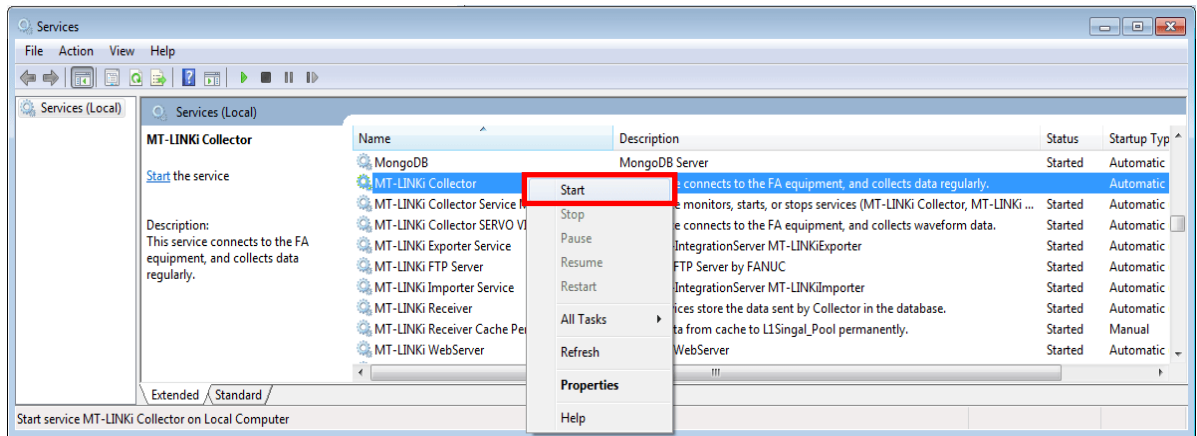
Before turning off the power of the Server PC, make sure that the system is shut down.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 416/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

7.1.2 Start Services Manually

Following are the steps to start services of MT-LINK*i* manually, and to restart the system.

- (1) For the Server PC or the Collector PC, click the [Start] button > [Control Panel] > [Administrative Tools] > [Services] to open the [Services] window.
- (2) Select the targeted service on the [Services] window, right-click it, and select [Start] in the context menu to start the service.



The targeted services will be as follows.

For Server PC

- MongoDB
- MT-LINK*i* Exporter Service
- MT-LINK*i* Receiver
- MT-LINK*i* WebServer
- MT-LINK*i* Receiver Cache Persister

For Collector PC

- MT-LINK*i* Collector
- MT-LINK*i* Collector Service Manager
- MT-LINK*i* Collector SERVO VIEWER Linkage
- MT-LINK*i* FTP Server

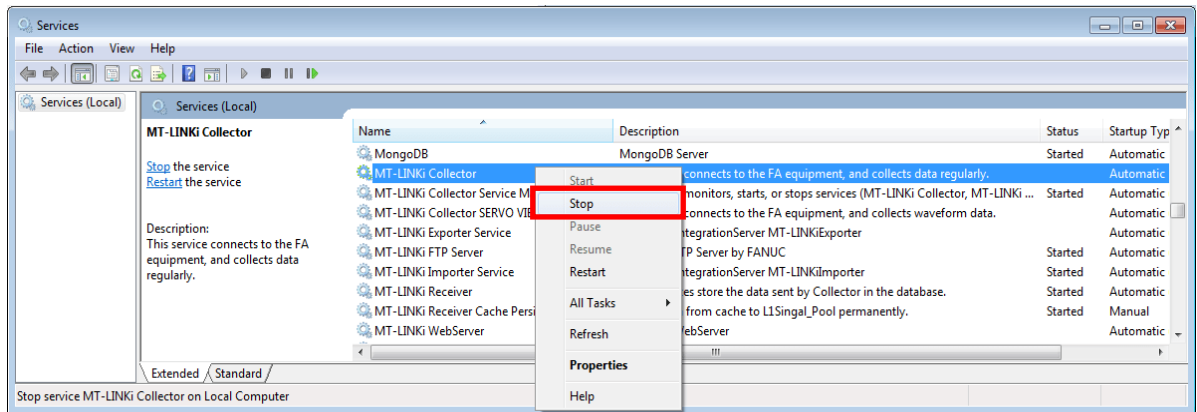
* When two or more Collectors are registered, follow the steps above for all the Collector PCs that are targeted.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 417/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

7.1.3 Stop Services Manually

Following are the steps to stop services of MT-LINK*i* manually, and to stop the system.

- (1) For the Server PC or the Collector PC, click the [Start] button > [Control Panel] > [Administrative Tools] > [Services] to open the [Services] window.
- (2) Select the targeted service on the [Services] window, right-click it, and select [Stop] in the context menu to stop the service.



The targeted services will be as follows.

For Server PC

- MongoDB
- MT-LINK*i* Exporter Service
- MT-LINK*i* Receiver
- MT-LINK*i* WebServer
- MT-LINK*i* Receiver Cache Persister

For Collector PC

- MT-LINK*i* Collector
- MT-LINK*i* Collector Service Manager
- MT-LINK*i* Collector SERVO VIEWER Linkage
- MT-LINK*i* FTP Server

* When two or more Collectors are registered, follow the steps above for all the Collector PCs that are targeted.

NOTE

When the Server PC or the Collector PC is restarted, the services will start automatically. If you need to stop any service when the PC is restarted, stop the service each time the PC is restarted.

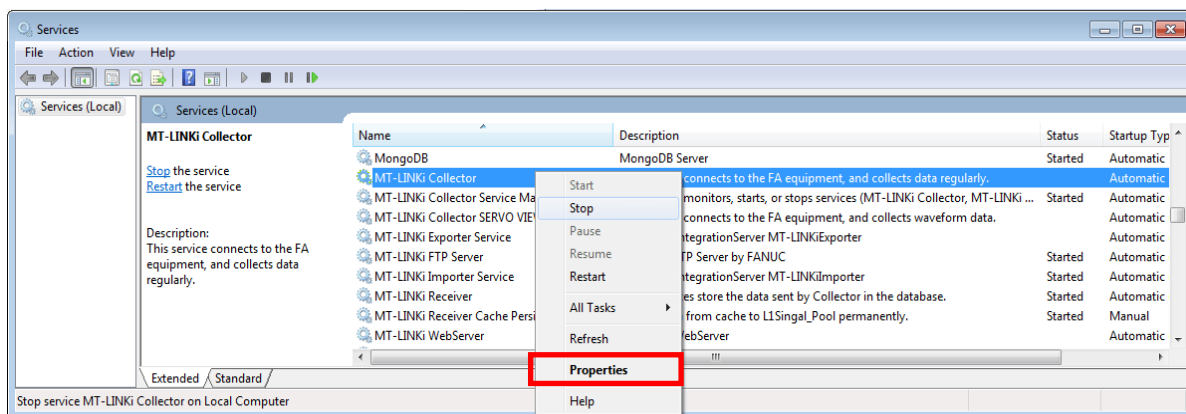
For the steps to prevent the services to start automatically when the PC is restarted, refer to the "[Disable Automatic Start of Services](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 418/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

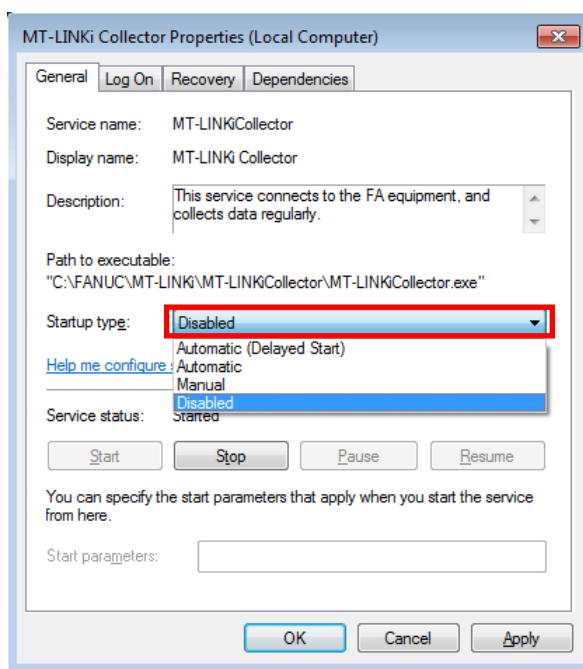
7.1.4 Disable Automatic Start of Services

Following are the steps to make settings to prevent the services of MT-LINK*i* to start automatically when the Server PC or the Collector PC is restarted.

- (1) For the Server PC or the Collector PC, click the [Start] button > [Control Panel] > [Administrative Tools] > [Services] to open the [Services] window.
- (2) Select the targeted service in the [Name] column of the [Services] window, right-click it, and select [Properties] in the context menu.



- (3) Set [Startup type] to "Disabled" and click the [OK] button.



The targeted services and the Startup type (default) of each service will be as follows.

For Server PC

- MongoDB: Automatic
- MT-LINK*i* Exporter Service: Automatic (Delayed Start)
- MT-LINK*i* Receiver: Automatic (Delayed Start)
- MT-LINK*i* WebServer: Automatic (Delayed Start)
- MT-LINK*i* Receiver Cache Persister: Manual

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	419/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

For Collector PC

- MT-LINK*i* Collector: Automatic (Delayed Start)
- MT-LINK*i* Collector Service Manager: Automatic (Delayed Start)
- MT-LINK*i* Collector SERVO VIEWER Linkage: Automatic (Delayed Start)
- MT-LINK*i* FTP Server: Automatic (Delayed Start)

* When two or more Collectors are registered, follow the steps above for all the Collector PCs that are targeted.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	420/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

8

[SETTING SYSTEM FUNCTIONS] - SYSTEM SETTING

This chapter describes how to make system settings of MT-LINK*i*.

The AdminTool Menu has the following menu items in the [Setting System Functions] - [System Setting] group:

(1) System Setting

This menu makes settings of various systems composing MT-LINK*i*.

8.1 SYSTEM SETTING

Set the basic items of the system, such as language, database settings, path settings, Web Server settings, and Web screen display settings.

Administrators can perform the following:

- choose language used in AdminTool,
- change the language used to display and set Web names on a variety of setting windows,
- set the database connection string and execute a connection test,
- set the data storage path of the database (DB path),
- set the data retention period for the database,
- clean up the database,
- optimize the database,
- set the port number of the Web Server,
- set the show/hide setting of the footer text and the background color of the header/footer,
- set the linkage with other systems. Up to 10 systems can be registered,
- set the report path,
- set the file output path of macro variable history,
- set the file output path of variable history,
- set the file output path of product plan setting,
- set the display order in the operational results information display area of the Web screen,
- open Web screens in an added language by loading additional language resource files which may be provided separately.

Selecting [System Setting] in the AdminTool Menu opens the following [System Setting] window.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	421/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

■ [Setting 1] tab

Language Setting for AdminTool

Language: English (If you change language of AdminTool, you need to restart it.)

DB Connection User name/Password

Web name language: English

DB Connecting Test

DB Setting

DB Connecting String: mongod://localhost/MT-LINKi Connecting Test

User: admin1 Password: ***** Apply

Web Name Language

DB Path Edit

Report Report path: C:\FANUC\MT-LINKi\Report

Data Retention Period Edit

Report Path

Macro Variable History

Output path: C:\FANUC\MT-LINKi\TriggerHistory

Output Path of Macro Variable History

Variable History

Output path: C:\FANUC\MT-LINKi\VariableHistory

Output Path of Variable History

Product Plan Setting

CSV output path: C:\FANUC\MT-LINKi\ProductPlan

Output Path of Product Plan Setting

DB Cleanup (Last executed: -) Clean Up

DB Optimization (Last executed: -) Optimize

DB Cleanup

DB Optimization

Web URL: http://192.168.146.152:3000/login.html Port: 3000

Color of header/footer #ff0000 Color setting Show footer characters

Root path of Web Server, Port Number

Linkage with other systems

Web name URL

Linkage with Other Systems

Header/Footer Color Setting / Show/Hide Footer Text

Reset all values to default OK Cancel

Reset all values to default

■ [Setting 2] tab

System Setting

Language: English Web name language: English (If you change language of AdminTool, you need to restart it.)

Setting 1 Setting 2

Display Order Setting of Operational Results Information Display Area

No	Display	Item
1	☒	Product Name
2	☒	Work Comments
3	☐	Product Serial Number
4	☐	Operator ID

Up Down

Load Additional Language Resource

Additional Language Resource

Local Language: Language Code: CodePage: Version: UpdateDate: Load

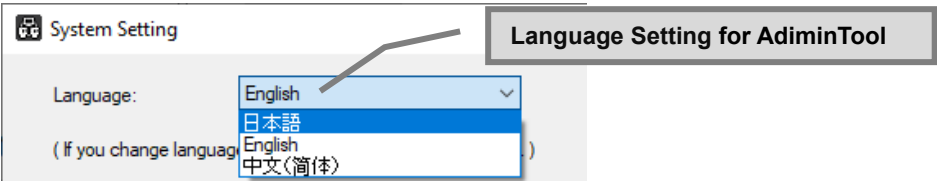
Reset all values to default OK Cancel

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	422/868

8.1.1 Change Language Used in MT-LINKi

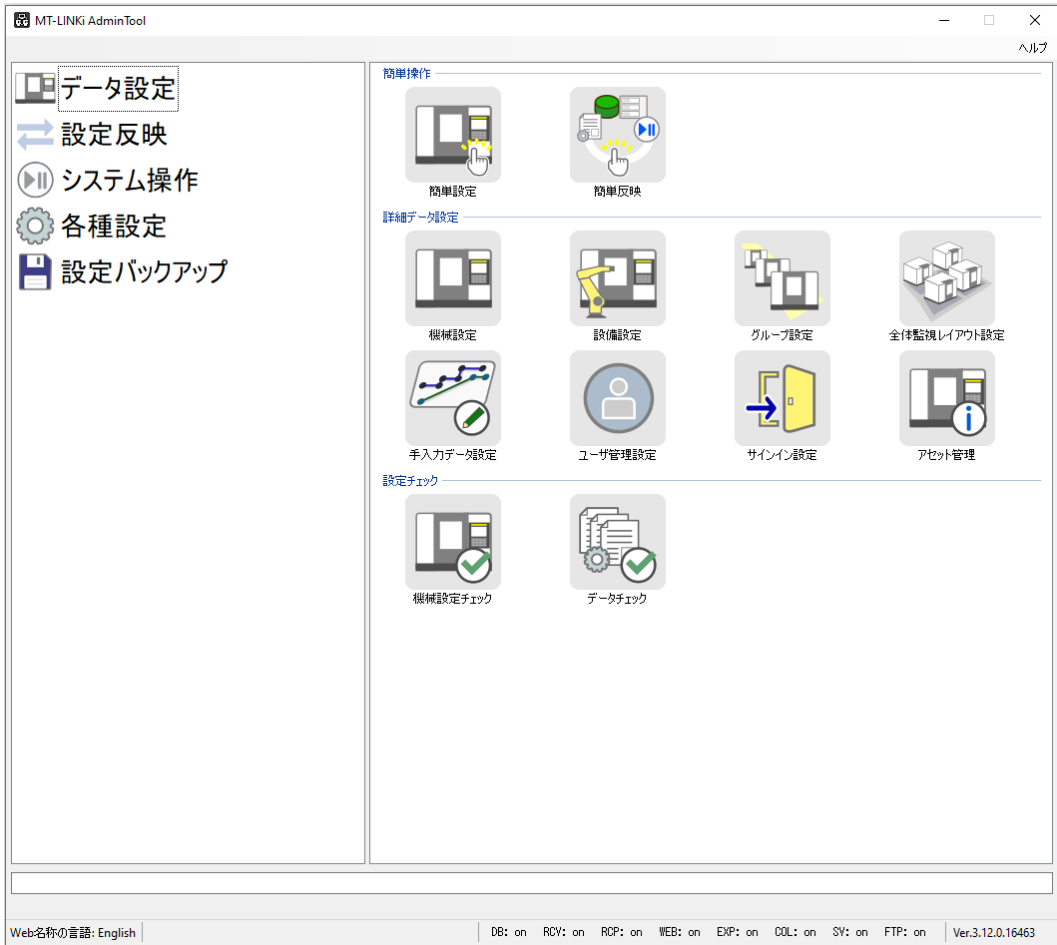
To change the language used in AdminTool of MT-LINKi:

(1) Choose your language in the [Language] drop-down list box.



(2) Clicking [OK] saves the settings. Restart AdminTool.

(3) Once restarted, the characters in the AdminTool are changed to the language you chose.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					FANUC CORPORATION			Sheet.	423/868
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

8.1.2 Change Language for Web Names

This subsection explains how to specify the language in which to display and set Web names on a variety of setting windows.

Web names can be set per language by selecting it from the [Web name language] drop-down list on the [System Setting] window. Web screens display Web names in the language that is specified for the current Web user in the [Add/Edit User] dialog box. For example, when Web names are set in English through setting windows after selecting [English] in the [Web name language] drop-down list, a Web user whose language is set to [English] in the [Add/Edit User] dialog box can view the Web names in English on Web screens.

After loading additional language resource files, AdminTool adds a local language to language options to display Web names. For the details on how AdminTool loads additional language resource files, refer to the "[Load Additional Language Resource](#)" subsection.

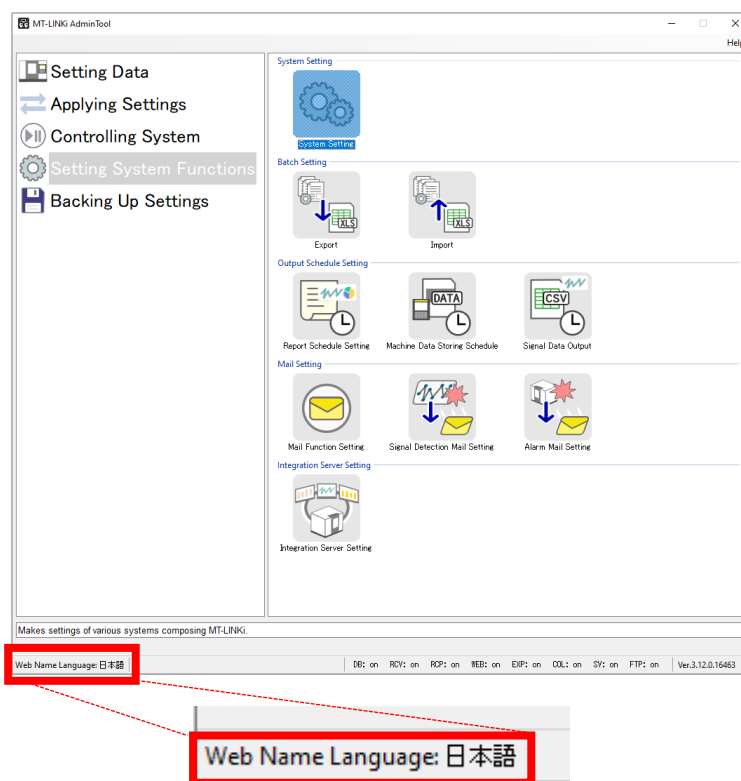
Follow the steps below to make settings for a language for Web names.

- (1) Select a language from the [Web name language] drop-down list box.



- (2) Click the [OK] button on the [System Setting] window for AdminTool to apply the selection.

- (3) The selected language is displayed in the [Web Name Language] field on the AdminTool Menu.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.		Inoue
FANUC CORPORATION					Sheet.	424/868	

NOTE

- 1 Click the [OK] button on the [System Setting] window of AdminTool to apply a change in the [Web name language]. AdminTool does not require restarting to reflect the change.
- 2 When no additional language resource file is loaded, the [Web name language] drop-down list box displays no local language.

- Fields on the setting windows (or function names) to be displayed in the language selected in the [Web name language]

Window	Field	Explanation
Machine Setting	Web Name	The Web name for a machine
Machine Setting ([Macro Variable Getting Condition Definition] tab page)	Web Name	The Web name for a Macro Variable getting condition (not editable)
Machine Setting ([Macro Variable getting condition definition Setting])	Web Name	The Web name for a Macro Variable getting condition
Machine Setting ([Macro Variable getting condition definition Setting] – [Set variable name])	Web Name	The Web name for a macro variable to be retrieved
Machine Setting ([Variable History Retrieval Condition Definition] tab page)	Trigger Name	The Web name for a trigger variable (not editable)
Machine Setting ([Variable History Retrieval Condition Setting])	Trigger Name	The Web name for a trigger variable
Machine Setting ([Variable History Retrieval Condition Setting] – [Variable Name Setting])	Variable Name	The Web name for a variable to be retrieved
Equipment Setting ([Basic Setting] group box)	Web Name	The Web name for equipment
Equipment Setting ([Simple assign status area])	Web Name	The Web name for a simple assign status
Equipment Setting ([Multiple assign status area])	Web Name	The Web name for a multiple assign status
Equipment Setting ([Signal Setting])	Web Name	The Web name for a simple or multiple assign status
Group Setting	Web Name	The Web name for a group
Signal Detection Mail Setting	Text	Strings to be displayed in the [Text] group box
Alarm Mail Setting	Text	Strings to be displayed in the [Text] group box
Asset Management	Web title	The title for a machine to be displayed on Web screens (not editable)
Asset Management	Web Name	The Web name for an asset item
Asset Management	Value	The value of an asset item
Asset Management ([Asset Management Items Setting])	Web Name	The Web name for an asset item
Asset Management ([Asset Management Items Setting])	Value	The value of an asset item
System Setting ([Linkage with other systems])	Web name	The Web name for a linkage with other system (not editable)
System Setting ([Linkage with other systems] – [Linkage with other systems Add/Edit])	Web name	The Web name for a linkage with other system
Integration Server Setting	Web Name	The Web name for a facility
Manual Data Setting (Input plans/Modify plans)	Product	Product name in manual data
Manual Data Setting (Input work comments)	Work comments	Work comments in manual data
Asset Management (CSV file for Import)	WebName...	Language for required asset items
Asset Management (CSV file for Import)	Value...	Language for required values

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 425/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

Pattern setting file in Batch Setting (Asset Management)	Web name	Language for required Web names
Pattern setting file in Batch Setting (Asset Management)	Value	Language for required values

- Fields on the setting windows (or function names) to be displayed in the language selected in the [Language]

Window	Field	Explanation
System Setting	Language	Language in which to display AdminTool windows
Authority Setting ([Add/Edit User])	Language	Default language for a new user
Authority Setting ([Add/Edit Role of Authority])	Name of system	The name of other system (not editable)
Authority Setting (LDAP Authentication Setting)	Language	Default language for a new user
System Setting ([Report] group box)	User	The user name when specified in UNC path
System Setting ([Macro Variable History] group box)	User	The user name when specified in UNC path
System Setting ([Variable History] group box)	User	The user name when specified in UNC path
Signal Data Output ([Output Setting] group box)	User ID	The user name when specified in UNC path
Sign-in Setting (title memo file)	(text in the file)	Text to be displayed in Web sign-in screen
Variable History (CSV file)	Trigger Name	Trigger name in the CSV header (Web name)
Variable History (CSV file)	Variable Name	Variable name in the CSV header (Web name)

NOTE

- 1 Report output does not depend on the language selection in either the [Web name language] or [Language] because it uses template files for the language the user prefers to use.
- 2 Web names (Machine name, Equipment name, Status name) will be written in the language selected in [Web name language] for their corresponding Web names in mail body of a mail which is sent through signal detection mail delivery/alarm mail delivery.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 426/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

8.1.3 Database Setting

This subsection describes how to perform the following for the database:

- Set [DB connecting string], [User], and [Password] required for connecting to the database; clicking the [Connecting Test] button verifies the effectiveness of the connection settings.
- Set the data storage path (DB path); setting this field changes the data storing location for the database.
- Set the data retention period of the database; the data meeting the period criteria will be deleted automatically.
- Perform DB cleanup to delete unnecessary data residing in the database.
- Optimize the database; unused areas in the database will be released.

The screenshot shows the 'DB Setting' dialog box with the following components and callouts:

- DB Connecting String:** mongodb://localhost/MTLINKi
- User:** admin1
- Password:** *****
- Connecting Test:** Button to verify connection settings.
- Apply:** Button to save connection settings.
- DB Data Storing Setting:**
 - Database path:** C:\FANUC\MT-LINKi\MongoDB
 - Edit:** Button to change the database path.
- DB data retention period setting:**
 - Deletion time:** -
 - Deletion interval (day[s]):** -
 - Retention period (day[s]):** -
 - Edit:** Button to change retention settings.
- DB Cleanup:** (Last executed: -) with a **Clean Up** button.
- DB Optimization:** (Last executed: -) with an **Optimize** button.

Callout boxes on the right side of the dialog include: 'DB Connecting String', 'Password', 'DB Connecting Test', 'User', 'Apply', 'Edit DB path', 'Edit DB Data Retention Period', 'DB Cleanup', and 'Optimize'.

8.1.3.1 Settings Required for Connecting to the Database

To set the settings for database connection, follow the steps below.

- (1) Specify the [User] and the [Password]. Changing those fields activates the [Apply] button. Click [Apply].

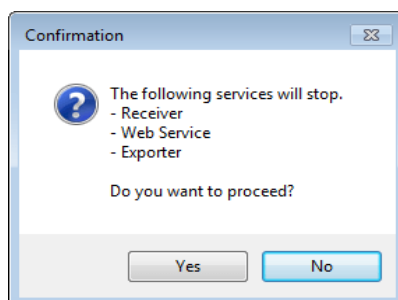
This close-up shows the 'DB Connecting String' field set to 'mongodb://localhost/MTLINKi'. The 'User' field is 'dbuser' and the 'Password' field is '*****'. The 'Connecting Test' button is visible, and the 'Apply' button is highlighted with a red rectangular box.

NOTE

For the setting of [User] and [Password], "admin" cannot be used.

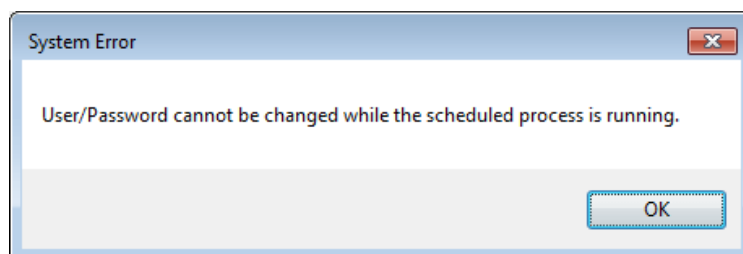
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
13	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.6				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	427/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

- MT-LINK*i* Receiver
- MT-LINK*i* WebServer
- MT-LINK*i* Exporter Service

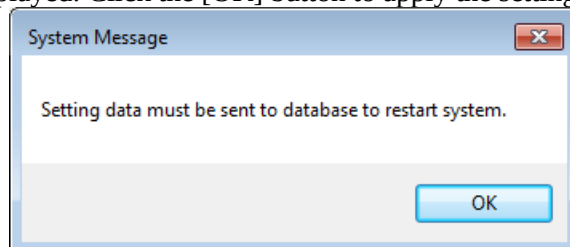


NOTE

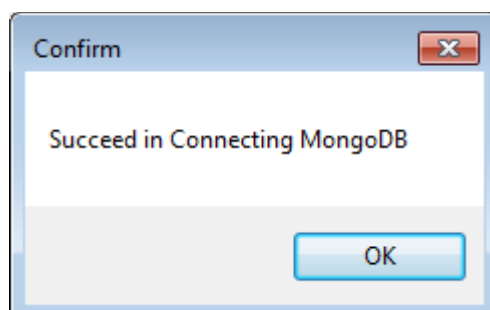
If the following message is displayed, [User] and [Password] cannot be set.



- (3) The following message is displayed. Click the [OK] button to apply the setting.



- (4) To test if a connection to the database can be established, click [Connecting Test]. When the connection test succeeds, the following dialog box appears. Click [OK].



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	428/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

Otherwise, the error icon appears as shown below. Start the database and try another connection test. If the test still fails, restart the Server PC and retry the connection test.

- (5) Click the [OK] button on the [System Setting] window.

If the error icon still appears, this is probably because the database is not running. Start the database and retry the connection test. If the connection test still fails even when the database is running, restart the Server PC and retry the connection test.

NOTE

If the [OK] button is clicked without clicking the [Apply] button when changing the [User] and the [Password], you will need to execute steps (2) and (3).

- (6) Send data to DB.

For the details, refer to the "[Send Setting Data to DB](#)" subsection.

- (7) Start the system.

For the details, refer to the "[Start/Stop System](#)" subsection.

8.1.3.2 Data Storing Setting of Database

The storage path of the database can be changed.

- When the available free space of the current drive of database storage path is low, the data storage destination of the database can be switched to another fixed disk or to another folder.

NOTE

When the free disk space of the database storage path is under 5GB, the following services will be stopped.

- MT-LINK*i* Receiver service
- MT-LINK*i* Receiver Cache Persister service

- If a newly created folder is specified, new database is created in that folder and connected to that database.
- If a folder storing an existing database is specified, the database in that folder is connected.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	429/868

NOTE

- 1 The specification (change) of the data storage location of the database (DB) alone will not carry over the data before the change to the database after the change. In order to carry over the data before the change, follow the steps below.
 - (1) Stop the system. For the details, refer to the "[Start/Stop System](#)" subsection.
 - (2) Stop the AdminTool. For the details, refer to the "[How to Start/End](#)" subsection.
 - (3) Stop the service of MongoDB. To stop, click [Control Panel] > [Administrative Tools] > [Services], and check the status of the service of "Mongo DB." If the status is "Started," right-click and select [Stop] on the drop-down menu.
 - (4) Copy the whole folder of the existing database (DB) data (mtl_data) to the folder after the change.
For example, copy the mtl_data folder under [C:\FANUC\MT-LINKi\MongoDB] (folder before the change) to under [D:\MongoDB] (folder after the change).
 - (5) Start the [MongoDB] service. To start, click [Control Panel] > [Administrative Tools] > [Services], and check the status of the service of [Mongo DB]. The status should be blank, so right-click it and select [Start] in the context menu.
 - (6) Start the [AdminTool]. For the details, refer to the "[How to Start/End](#)" subsection.
 - (7) In the DB data storing setting, specify the folder where the files were copied to in step 4 to change the data storing location.
 - (8) Transfer AdminTool setting data to DB. For the details, refer to the "[SEND TO DB](#)" section.
 - (9) Start the system. For the details, refer to the "[Start/Stop System](#)" subsection.
 - (10) Delete the whole folder of the database data that was copied from. For example, delete "C:\FANUC\MT-LINKi\MongoDB\mtl_data" (source of copy).

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 430/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

- 2 When carrying over the DB data, the DB data that can be carried over will be the DB data of MT-LINK*i* with the same major version. If the major version is different, DB data cannot be carried over.
If the major version is the same, the DB data can be carried over even if the minor version is different.

Cases when the DB data can be carried over:

- MT-LINK*i* in operation: Version 3.11,
DB data stored in the folder after the change: Version 3.6
- MT-LINK*i* in operation: Version 2.1,
DB data stored in the folder after the change: Version 2.2

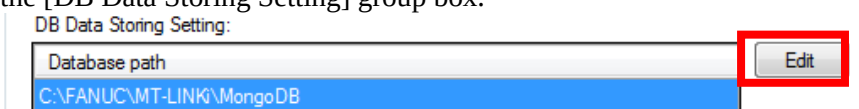
Cases when the DB data cannot be carried over:

- MT-LINK*i* in operation: Version 3.0,
DB data stored in the folder after the change: Version 2.2
- MT-LINK*i* in operation: Version 2.1,
DB data stored in the folder after the change: Version 3.0

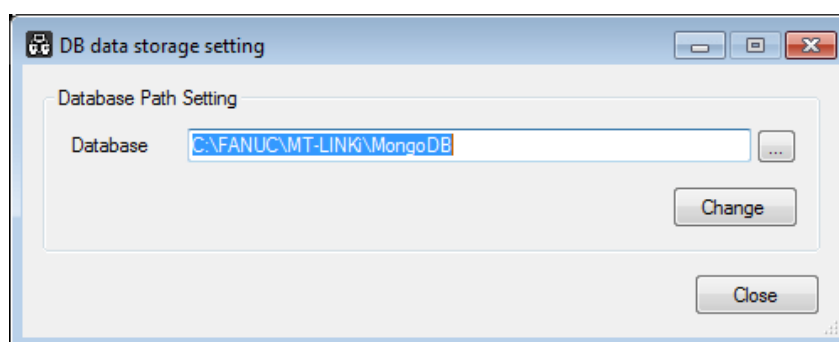
- 3 When carrying over DB data from an older version of MT-LINK*i* through changing the data storage location, updating the DB may occur. Since this update process may require hours depending on the data amount, it is recommended to plan a maintenance schedule so that changing the data storing location will not interfere with the system operation.

To specify the storage path of the database, follow the steps below:

- (1) Click [Edit] in the [DB Data Storing Setting] group box.



- (2) The [DB data storage setting] dialog box appears. The [Database] field populates the current database path. To change the path, either set the database storage location for the [Database] field or click the browse folder button ([...]) to open the [Browse For Folder] window, select it, and click [OK]. The [Browse For Folder] window selects the folder currently populated in the [Database] field. After changing the [Database] field, click the [Change] button in the [Database Path Setting] group box.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 431/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- 1 With the [Change] button clicked, the database path setting is enabled.
- 2 For the database path, specify the path of the fixed disk drive (internal/external disk drive). The paths of other drives (such as network drives) or UNC paths cannot be specified.

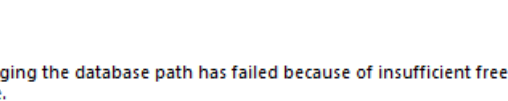
A screenshot of a Windows-style confirmation dialog box. The title bar is light blue and contains the word "Confirmation" on the left and a red close button with a white 'X' on the right. The main area has a white background and contains the text: "Once the location changing process starts, you will be unable to access Web pages or to save data until it is completed. Are you sure you want to proceed?". At the bottom, there is a light gray bar containing two buttons: "OK" (highlighted with a blue border) and "Cancel" (disabled, with a gray border).

When the location changing process starts, the system ceases to operate. Web screens cannot be displayed, and data cannot be stored in the database until the system starts again. Also, the database cannot be accessed from an external application during the process.

After completing this location changing process, send the data to the database, and start the system.

Before changing the database location, plan a maintenance schedule in advance such that location changing process will not interfere with the system operation.

Fix the problem by referring to the following subsection of the "FANUC MT-LINK*i* / Integration Server Troubleshooting MANUAL":

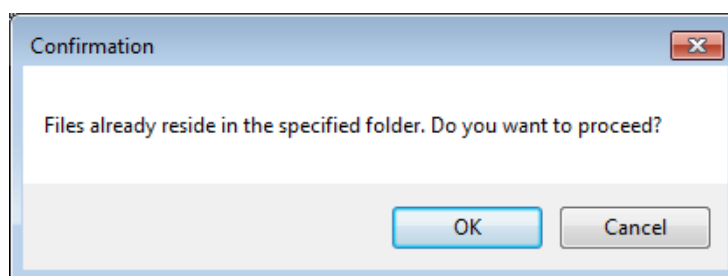


Changing the database path has failed because of insufficient free disk space.
A free disk space of at least 10 GB is required.
Free enough disk space by deleting or moving unnecessary files.
For the details, refer to the troubleshooting manual.

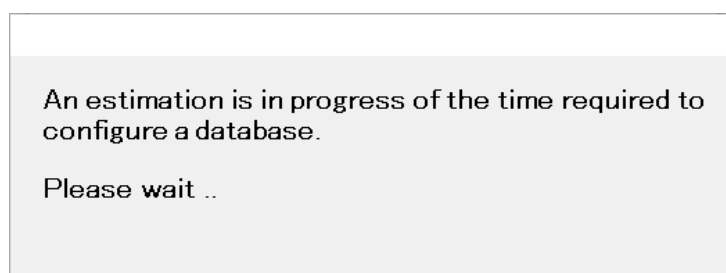
OK

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	432/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

(5) If the folder specified in the [Database] field has a database file, the following message appears. Click [OK].



(a) The following message appears and starts estimating the time required to configure a database.



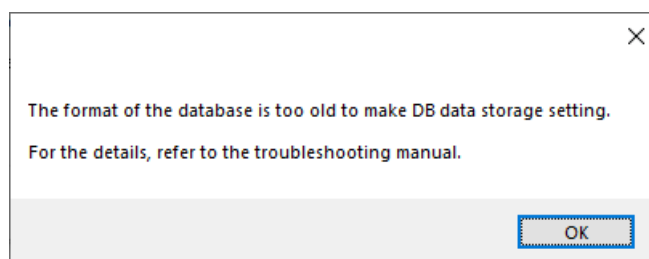
(b) If this estimation is not completed in five minutes, a confirmation message will appear reading "The estimation is still in progress, taking longer than expected. Do you want to proceed?"

Clicking [Yes] returns you to the previous step (a) and continues estimating; clicking [NO] returns you to the [DB data storage setting] dialog box.

(c) If the database in the specified folder has too old a format, the following message appear notifying that database data storage setting is prohibited.

Fix the problem by referring to the following subsection of the "FANUC MT-LINK*i* / Integration Server Troubleshooting MANUAL":

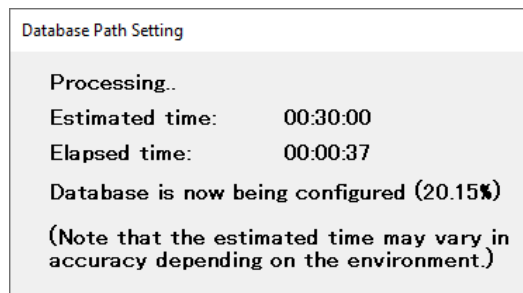
The "Unable to Make DB Data Storage Setting Due to Difference in Database Format" subsection in the "TROUBLES RELATED TO DATABASE (MongoDB)" section in the "MT-LINK*i* TROUBLESHOOTING" chapter



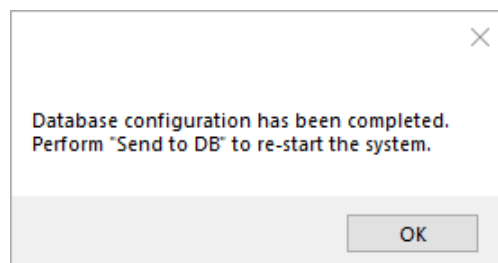
(d) When the estimation is completed, the following message appear displaying an approximate time required to configure the database:

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	433/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	434/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					



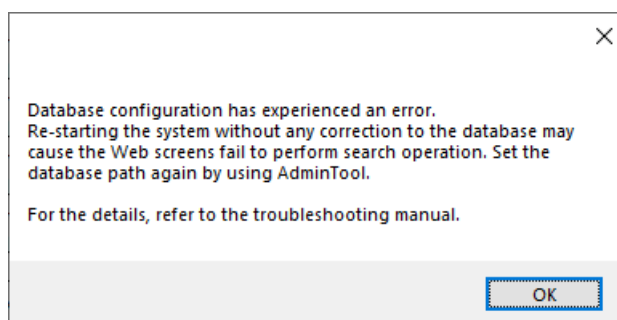
(b) When database configuration is completed, the following message will appear.



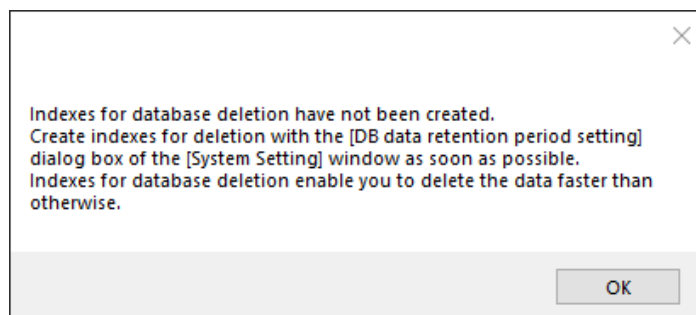
(c) If database configuration fails, the following message will appear:

Fix the problem by referring to the following subsection of the "FANUC MT-LINK*i* / Integration Server Troubleshooting MANUAL":

Case (1) in the "Configuration of Database (MongoDB) Terminates with an Error" subsection in the "TROUBLES RELATED TO DATABASE (MongoDB)" section in the "MT-LINK*i* TROUBLESHOOTING" chapter



(8) If indexes for database deletion have not been created, the following message will appear:
For the details, refer to the **"DB Data Retention Period Setting"** subsection.



(9) Click the [Close] button in the [DB data storage setting] dialog box.

(10) Click [OK] on the [System Setting] window.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	435/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

- (11) Send data to the database.

For the operational procedures, refer to the **"Send Setting Data to DB"** subsection.

- (12) Start the system.

For the details on how to start the system, refer to the "[Start/Stop System](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	436/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

8.1.3.3 DB Data Retention Period Setting

This subsection describes how to set a data retention period of the database and its deletion settings.

The [DB data retention period setting] window enables users to do the following:

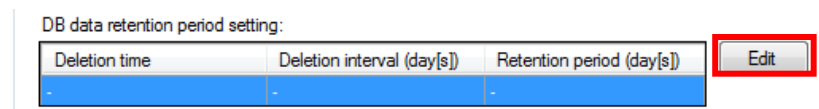
- Registering a schedule to delete data in the database
- Manually deleting data from the database
- Creating indexes for faster deletion when manually deleting data from the database

NOTE

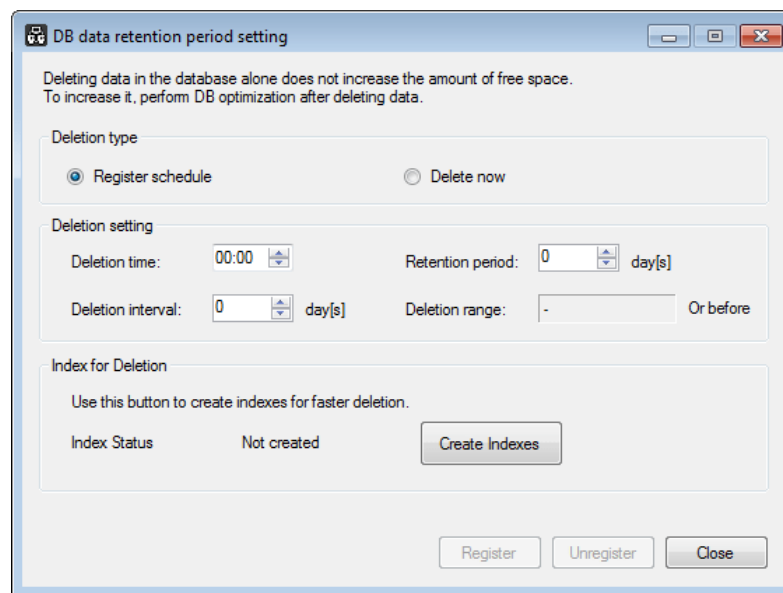
- 1 Setting operation with AdminTool is not allowed while DB data is being deleted manually.
- 2 Deleting data in the database alone does not increase the amount of free space on the disk.
To increase the amount of free space in the database, perform DB optimization after deleting data. For the details, refer to the "[DB Optimization](#)" subsection.

To set the data retention period of the database, follow the steps below.

- (1) Click the [Edit] button in [DB data retention period setting] group box.



- (2) The [DB data retention period setting] dialog box appears.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 437/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION				Sheet.	438/868
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						

NOTE

- 1 As the database gets larger, so much the longer time will be taken to create indexes. It is recommended to once create indexes for database deletion immediately after a start of the system.
- 2 Because indexes are created in a background process, the system can keep operating while the process is running.

(iii) With indexes for database deletion created, the [Index Status] field changes to "Created."

NOTE

- 1 Once creation of indexes has started, do not stop MongoDB service, shutdown the Server PC, or optimize the database until it is completed.
- 2 The [Index Status] field shows whether indexes for database deletion has been created.

■ When registering data deletion schedule of DB

- (i) Select the [Register schedule] button in the [Deletion type] group box.
- (ii) Set the time of data deletion for the [Deletion time] field, the interval days of data deletion for the [Deletion interval] field, and retention period in days for keeping the data for the [Retention period] field. In the [Deletion range] field, the first day of data deletion (the first day exceeding the retention period) is displayed.
 - * For the [Deletion time] drop-down list, time from "0:00" to "23:59" can be selected.
 - * For the [Deletion interval] drop-down list, day(s) from "1" to "365" can be selected.
 - * For the [Retention period] field, day(s) from "1" to "99999" can be specified.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
13	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.6			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 439/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

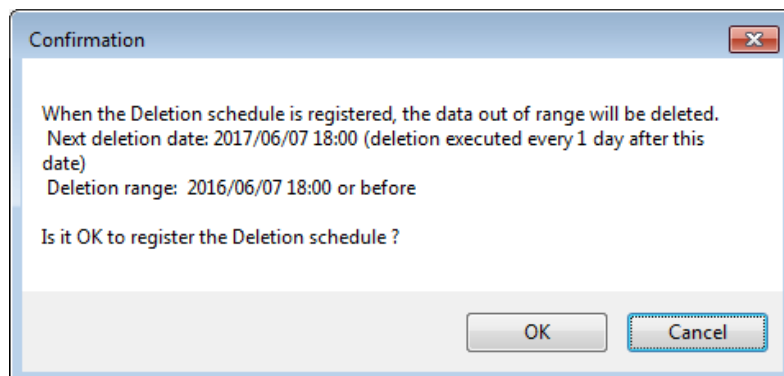
(iii) Clicking the [Register] button displays confirmation message for the registration of data deletion schedule.

Clicking the [OK] button completes the registration of schedule for data deletion.

If you are not registering the deletion schedule, click the [Cancel] button.

NOTE

If schedule is already registered, [Register] button cannot be clicked.
Register the new schedule after unregistering the currently registered schedule.



(iv) Clicking the [Unregister] button deletes the currently registered schedule.

NOTE

- 1 If DB data deletion schedule is executed while operating the Web screen, the response may become slow. Make sure the time of deletion (schedule setting) does not overlap with the operating hours of the machine.
For example, if the Web screen will be operated from 9:00 to 17:00, set the schedule so that the deletion of DB data starts after 18:00.
- 2 There is no need to start AdminTool in preparation for the date/time of deletion schedule.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue
FANUC CORPORATION					Sheet.	440/868	

- When deleting DB data manually
 - (i) Select the [Delete now] button in the [Deletion type] group box.
 - (ii) Select a retention period for keeping the data in days from the [Retention period] drop-down list
In the [Deletion range] field, the first day of data deletion (the first day exceeding the retention period) is displayed.
 - * For [Retention period], day(s) from "1" to "99999" can be specified.

DB data retention period setting

Deleting data in the database alone does not increase the amount of free space.
To increase it, perform DB optimization after deleting data.

Deletion type

☐ Register schedule ☒ Delete now

Deletion setting

Retention period: 365 day[s]

Deletion range: 2019/06/16 Or before

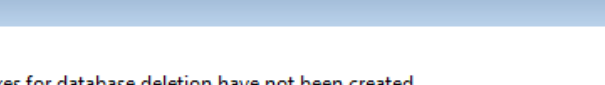
Index for Deletion

Use this button to create indexes for faster deletion.

Index Status	Create Indexes
Not created	<button>Create Indexes</button>

Execute Close

- (iii) If indexes for database deletion have not been created, clicking the [Execute] button displays the following confirmation message.



Indexes for database deletion have not been created.

It may take a longer time to delete the data without the indexes for database depending on the amount of them.

Are you sure you want to delete them without indexes for deletion?

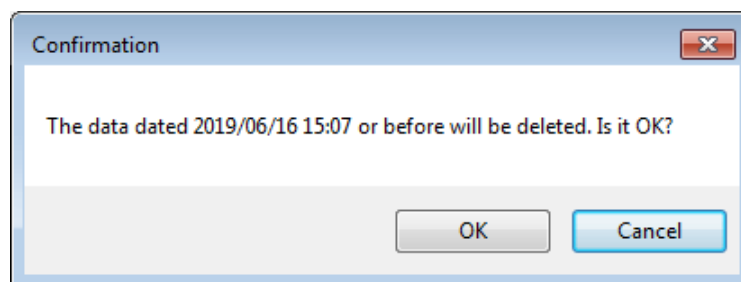
OK Cancel

NOTE

If a database deletion is executed without indexes for database deletion, it takes much longer time than done with the indexes. It is highly recommended to create indexes for database deletion before a database deletion is executed.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION				Sheet.	441/868
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						

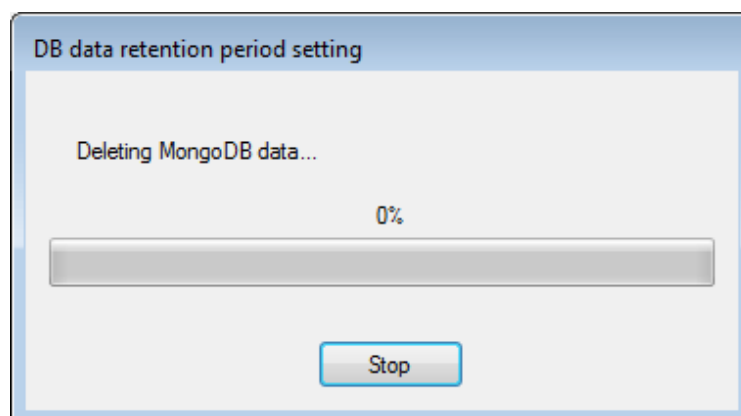
Clicking the [OK] button opens the following confirmation message for database deletion; otherwise clicking the [Cancel] button returns you to the [DB data retention period setting] dialog box.



To initiate data deletion, click the [OK] button; otherwise, click the [Cancel] button.

(iv) DB data deletion starts.

To stop the process, click the [Stop] button.



NOTE

- 1 During a deletion process, it is unable to execute setting operations on the AdminTool windows.
- 2 A deletion process may take some time, especially when the deletion range contains a very large amount of data.
- 3 When the [Delete now] button is selected, a deletion process divides the data contained in the deletion range into blocks, and then deletes the blocks one by one. Even if the [Stop] button is clicked, the deletion process will keep running until it deletes the currently handling block. This time lag may be shortened by setting DELETE_TERM in the system setting file (AdminTool.ini) to a better suited value.

For the details, refer to the "[AdminTool.ini](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue
FANUC CORPORATION					Sheet.	442/868	

- (3) Click the [Close] button on the [DB data retention period setting] dialog box to complete DB data retention period setting procedures.

NOTE

Every time a signal changes its value, the following data will be stored in the database as a single record.

- (a) Value of the signal (hereafter referred to as Value (a))
- (b) Start date/time: date/time the signal changed its value to Value (a)
- (c) End date/time: date/time the signal changed its value from Value (a) to another

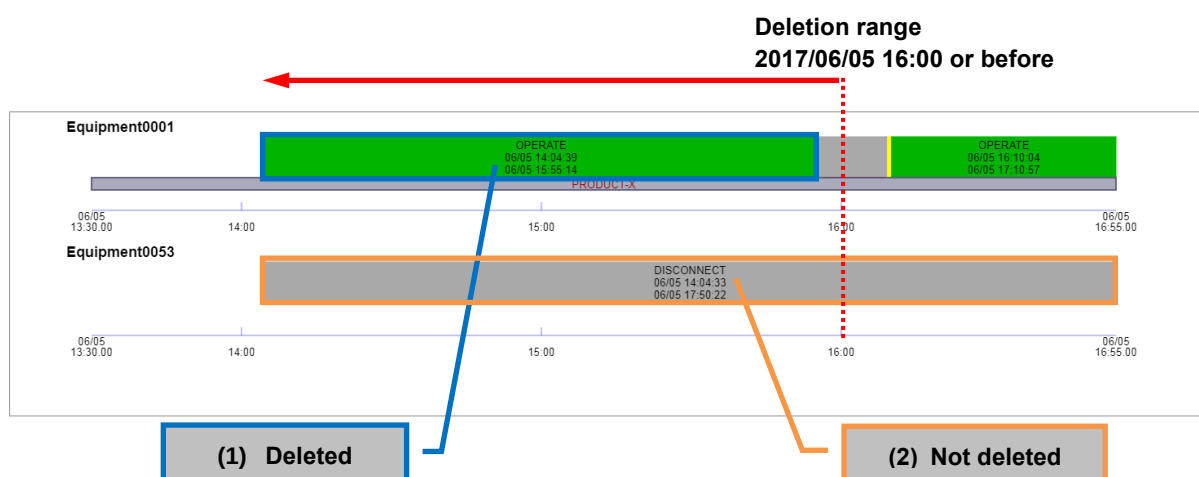
All the records whose End date/time is within the deletion range will be deleted; any records whose End date/time is out of the range will be retained.

(Example)

Assume that the deletion range is "2017/06/05 16:00 or before" and the operating statuses of the example equipment are as follows;

- Operating state of Equipment0001:
OPERATE (2017/06/05 14:04:39 to 2017/06/05 15:55:14).....Data (1)
- Operating state of Equipment0053:
DISCONNECT (2017/06/05 14:04:33 to 2017/06/05 16:37:02)....Data (2)

--> Data (1) will be deleted and Data (2) will be retained.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	443/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

8.1.3.4 DB Cleanup

When SERVO VIEWER Linkage function requests Linkage tool for MT-LINK*i* to delete waveform data from the database, the data will remain in the database but never be utilized again.

In such a case, DB cleanup will remove those unnecessary data residing in the database.

Make sure to execute DB cleanup after using Linkage tool for MT-LINK*i* to delete waveform data.

NOTE

- 1 Do not shutdown Server PC nor stop the database while a DB cleanup process is running.
- 2 Executing DB cleanup does not necessarily increase free space of the disk. To realize that purpose, execute DB cleanup first and then DB optimization. For the details, refer to the "[DB Optimization](#)" subsection.
- 3 Since this process may take a long time depending on the amount of data stored in the database, it is recommended to plan a maintenance schedule so that cleaning up the database will not interfere with the system operation.

DB Cleanup: (Last executed: 2021/01/26 17:05:53)

Clean Up

■ DB cleanup

1. Last executed

This area indicates the last date and time when the database is successfully cleaned up.

Format: <yyyy/MM/dd HH:mm:ss> (Example: 2020/12/24 12:34:56)

In case of no DB cleanup has been executed, "-" will be displayed in this area.

This field will not be updated unless a DB cleanup process is completed.

2. [Clean Up] button

Clicking this button starts cleaning up the database.

To clean up the database, perform the steps below.

(1) Click the [Clean Up] button.

When a confirmation message appears, click the [OK] button.

(2) While a DB cleanup process is running, the [DB Cleanup] window displays the progress bar indicating the progress (in percentage) of the process.

Clicking the [Stop] button on the window terminates the process and return the user to the [System Setting] window.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
13	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	444/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

8.1.3.5 DB Optimization

Though DB data retention period setting and DB cleanup delete data in the database, the deleted data are actually only marked as "unused."

DB optimization increases the amount of free space on the disk by releasing unused areas of the database. After deleting data from the database by using the following functions, perform DB optimization to increase the amount of free space on the disk:

- DB data retention period setting
- DB cleanup

NOTE

- 1 Do not shut down the Server PC during the process, or the database may be damaged.
- 2 The process cannot be suspended.
Since this process may require hours and, in an extreme case, could take more than 24 hours, depending on the data amount, it is recommended to plan a maintenance schedule so that optimizing the database will not interfere with the system operation.
- 3 If neither DB data retention period setting nor DB cleanup is executed before DB optimization, it will not increase free space in the disk but will take longer time to complete. Execute DB optimization after executing one or both of the following:
 - DB data retention period setting
 - DB cleanup
- 4 The following will be suspended during the process:
 - Receiver service
 - Receiver Cache Persister service
 - Web Server service
 - Data transfer process to the Integration Server using Exporter service (only in an environment using the Integration Server)
 - Report output schedule
 - Signal data output schedule
 - Machine data storing schedule
 - DB data deletion schedule set by the [DB data retention period setting]

Also, the database cannot be accessed from an external application during the process (since the database is locked during DB optimization, external applications must wait until the process is completed before accessing the database).

- 5 Optimizing the database cannot be performed during the following processes are in progress. Wait until these processes are completed, and then optimize the database:
 - Execution of DB data deletion schedule set by DB data retention period setting
 - Data transfer to the Integration Server using Exporter service (only in an environment using the Integration Server)
 - Execution of creating indexes for database deletion

DB Optimization: (Last executed: 2019/06/13 17:26:05)

Optimize

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	445/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

- DB Optimization

3. Last executed

This area indicates the last date and time when the database is successfully optimized.

Format: <yyyy/MM/dd HH:mm:ss> (Example: 2019/05/10 12:34:56)

In case of no optimization has been executed, "-" will be displayed in this area.

4. [Optimize] button

Clicking this button starts optimizing the database.

To optimize the database, perform the steps below.

(1) Click the [Optimize] button.

When a confirmation message appears, click the [OK] button.

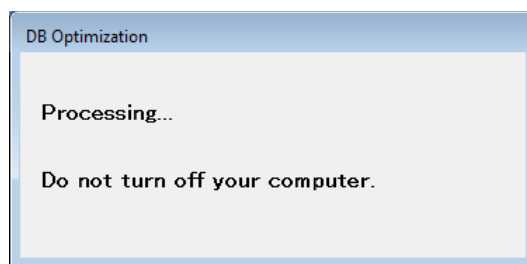
When another confirmation message appears, click the [OK] button.

(2) A check is performed on whether any conflicting process is running.

- If a DB data deletion schedule set by DB data retention period setting is running, the "DB deletion is in process" message appears.
- If the Exporter service is transferring data to the Integration Server, the "Exporter service is in process" message appears.

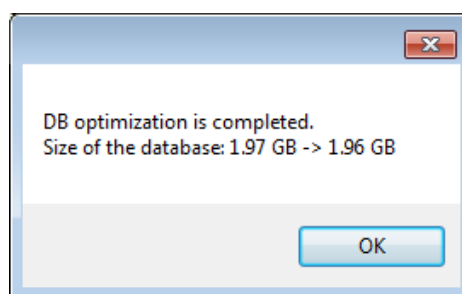
In either case, clicking the [OK] button returns to the [System Setting] window.

(3) The following message remains during the optimization process.



(4) When the process is completed, the following message appears:

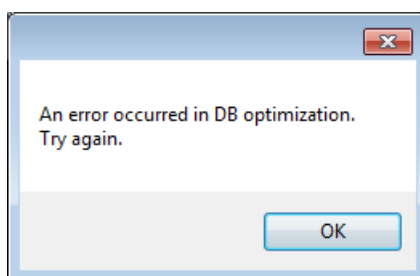
- When the process succeeded;



The [Size of the database] displays a change in the size of the database between before and after optimization. The size values are displayed in GB (rounded up to the hundredths decimal place).

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					FANUC CORPORATION			Sheet.	446/868
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

- When the process failed;

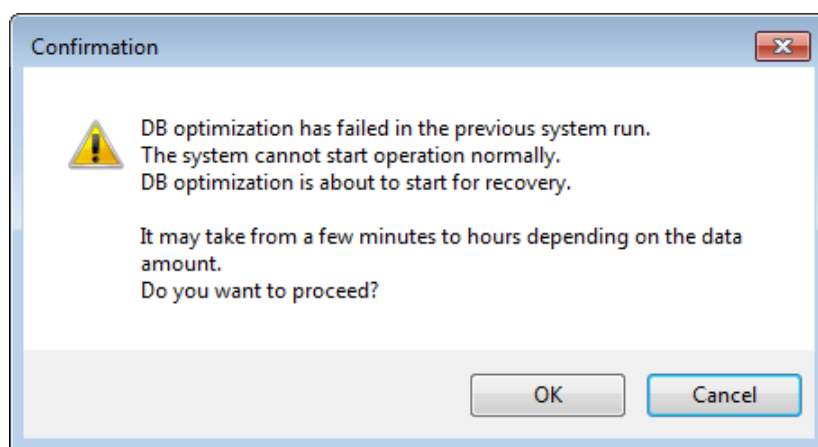


NOTE

- Stopping MongoDB service
- Shutdown the Server PC
- Creating indexes for database deletion

If such a case should take place by accident, restart the AdminTool.

Click the [OK] button to start optimizing the database.



The "Database (MongoDB) Optimization Fails" subsection in TROUBLESHOOTING RELATED TO DATABASE (MongoDB)" section in the "MT-LINKi TROUBLESHOOTING" chapter

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	447/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

8.1.4 Set Port Number of Web Server

When you need to change the port number of the Web Server, perform the following steps.

- The [Web URL] combo box holds the IP addresses of the Web Server. You can use this information to view Web screens and to provide your Web users with a URL of the Web Server.
- In the following figure, "192.168.200.128" is an IP address of the Web Server.
- The [Web URL] combo box shows as many URLs of the Web Server as its network adapters.



Web URL: Port:

NOTE

- 1 The list entries of the [Web URL] combo box are merely informational and thus modifying or deleting those entries does not affect the Web Server's URLs. As an administrator, you can select a URL from the list to copy and pass it to Web users.
- 2 When the list has multiple entries, any URL in the list is available to copy even when it does not currently appear at the top of the list.

- (1) When you change the [Port] field, the corresponding URL will appear in the [Web URL] field.
- (2) Click [OK].
- (3) Transfer the updated settings to DB. For the details, refer to the "[Send Setting Data to DB](#)" subsection.
- (4) Select [Start/Stop System] in the AdminTool Menu to stop the Web Server once, and start it again. For the details, refer to the "[Start/Stop System](#)" subsection.

Specifying this URL enables Web users to sign in to the Web Server.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	448/868

8.1.5 Show Footer Text and Set Background Color of Header/Footer

Setting to show/hide the footer text and the background color of the header/footer can be set on the Web screen.

The background color setting of the header/footer will be reflected in the area shown below.



■ To show/hide the footer text, follow the steps below.

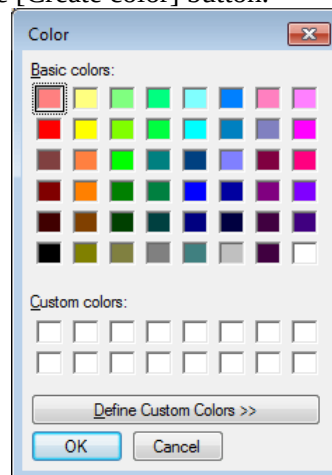
- (1) When adding or showing the footer text, select the [Show footer characters] check box. When hiding the footer text, clear the check box.

Color of header/footer #ffd702 Color setting ☒ Show footer characters

- (2) Click the [OK] button on the [System Setting] window.

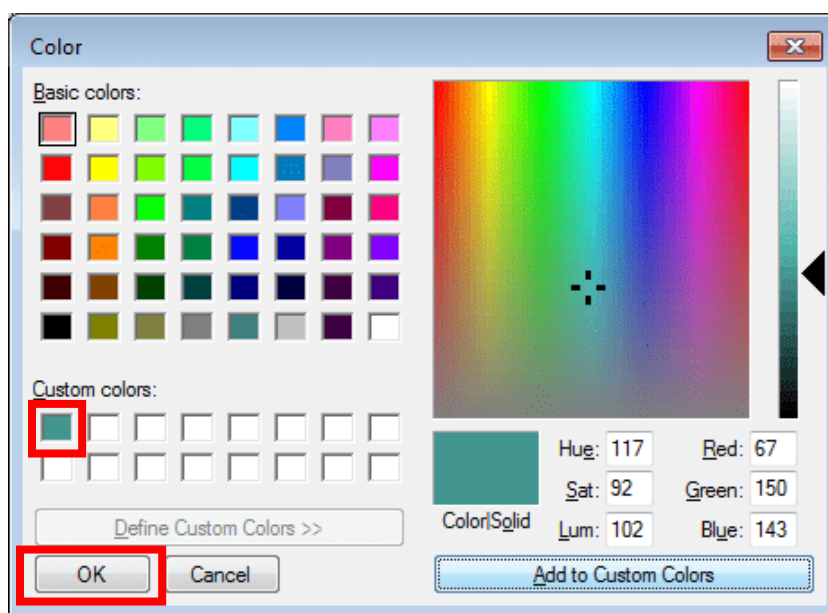
■ For the setting of the background color of the header/footer, follow the steps below.

- (1) Click the [Color setting] button.
- (2) [Set color] dialog box appears. Click the [Create color] button.

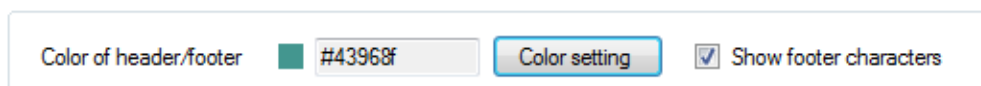


20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	449/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

(3) Select the appropriate color, and click [OK].



(4) Click the [OK] button of the [System Setting] window to apply the setting.

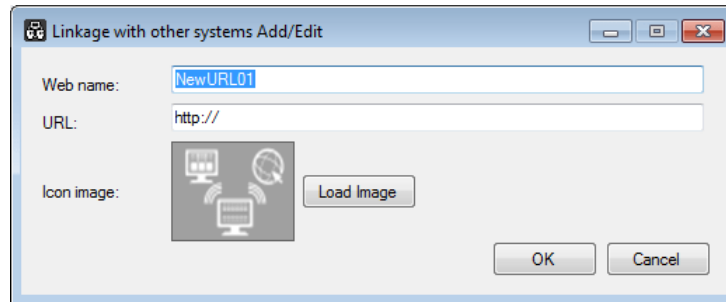


20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	450/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

8.1.6 Set Linkage with Other Systems

Follow the steps below to set linkage with other systems.

- (1) When adding or editing setting information for the linkage with other systems, click the [Add] or the [Edit] button. The [Linkage with other systems Add/Edit] dialog box shown below appears.



- (2) Set each item on the [Linkage with other systems Add/Edit] dialog box.

- Web name

The name specified in this window will be displayed in the Menu of the Web screen. It should be entered in the language specified for the [Web name language] in the [System Setting] window. Up to 20 bytes each can be entered for the names. The initial value when added is "NewURL<##>" (<##> is the 2 digit number assigned starting from 01 and adding up).

- URL

Set the URL of the system to be linked. Up to 2083 bytes can be entered. When added, "http://" is the initial value.

- Icon image

The icon image specified here will be displayed on the Web screen ([Home] screen). Click the [Load image] button, and specify the icon image from the [Select image file] dialog box. The dialog box selects the following folder on the first session. This folder contains default icon image files. On subsequent sessions, it selects the folder selected in the previous session.

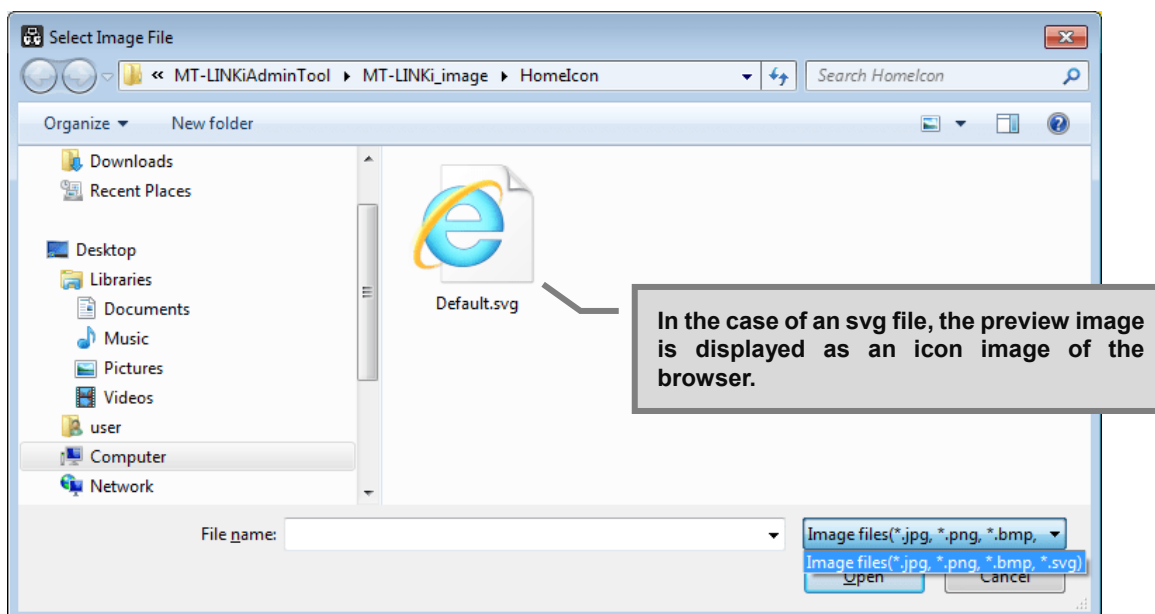
<MT-LINK*i* installation folder>\MT-LINK*i*AdminTool\MT-LINK*i*_image\HomeIcon

Supported file formats are png, jpg, bmp, and svg.

Image files larger than 500 pixels wide x 500 pixels high cannot be used.

* If an icon image of 65 pixels wide x 500 pixels high (with an aspect ratio of 13:9) is used, the display on the Web screen will be best displayed.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 451/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	



- (3) When finalizing the contents added or edited, click the [OK] button on the [Linkage with other systems Add/Edit] dialog box. When deleting the setting contents, click the [Cancel] button. [Linkage with other systems Add/Edit] dialog box will be closed.
- (4) The current setting of linkage with other systems will be displayed in the linkage list with other systems in the [System setting] window. As the systems added or edited in the [System setting] window will be automatically reflected on the authority setting list per system of the [Add/Edit Role of Authority] dialog box, set the role of the authority appropriately. For the details of adding and editing the role, refer to the "[Add/Edit Role of Authority](#)" subsection.
- (5) Click the [OK] button.
- (6) Select [Send to DB] in the AdminTool Menu to send the data to database. For the details, refer to the "[SEND TO DB](#)" section.
- (7) Select [Start/Stop System] in the AdminTool Menu to stop the Web Server once, and start it again. For the details, refer to the "[Start/Stop System](#)" section.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 452/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

8.1.7 Report Path Setting

This subsection describes how to set the [Report path].

Make sure to send the setting data to DB (execute [Send to DB]) after changing the report path.

Folder Selection Button

■ Report

◆ [Report path]

Specify the path to store reports. This field can be up to 260 bytes long.

This field populates the following folder by default. On subsequent sessions after changed once, it populates the last modified value:

<MT-LINK*i* installation folder>\Report

◆ [...] (Folder selection) button

Clicking this browse folder button opens the [Browse For Folder] window to either select an existing folder or create a new folder. By default, the [Browse For Folder] window selects the folder populated in the [Report path] field.

◆ [UNC Permitted] check box

When specifying a UNC path for the report path, select the check box.

When the [UNC Permitted] check box is selected, [User] and [Password] can be entered.

◆ [User]

Set the user required for authentication of UNC path. This field can be up to 255 bytes long.

◆ [Password]

Set the password required for authentication of UNC path. Up to 127 half-width alphanumeric and symbol characters (space included) can be input.

NOTE

- 1 For the report path, paths of disk drives such as fixed disk drives (internal disk/external disk) or UNC paths can be specified. The paths of other drives (such as network drives) cannot be specified.
- 2 As the size of the output data will be massive, it is recommended to specify a separate folder for the report path, such as the folder of an expanded external disk drive. Also, make sure that the disk space of the output folder does not get compressed, by deleting the output files or moving the output files to another media on a regular basis.
- 3 If the free disk space of the disk drive specified as the storage destination of the reports is under 300 MB, the execution of the report output process will be suspended.
- 4 For the details on the confirmation of operation by OS combination when UNC path is specified as output path, refer to the "[Confirmation of Operation when UNC Path Is Specified as Output Path](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 453/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

8.1.8 Setting File Output Path of Macro Variable History

Set the output setting and output destination path for the automatic output of CSV file, when the macro variable set in the [Macro Variable Getting Condition Definition] is being retrieved.

Make sure to send the setting data to DB (click the [Send to DB] button) and restart Receiver after changing the path.

For the details, refer to the "[Send Setting Data to DB](#)" subsection.

■ Macro Variable History

● [File output] check box

When enabling CSV file output during retrieval of macro variable, select the check box. The default setting is "disable" (Check box is cleared).

● [Output path]

Specify the folder to output CSV files. This field can be up to 260 bytes long.

This field populates the following folder by default. On subsequent sessions after changed once, it populates the last modified value:

<MT-LINK*i* installation folder>\TriggerHistory

● [...] (Folder selection) button

Clicking this browse folder button opens the [Browse For Folder] window to either select an existing folder or create a new folder. By default, the [Browse For Folder] window selects the folder populated in the [Output path] field.

● [UNC Permitted] check box

When specifying a UNC path for the output path, select the check box.

When the [UNC Permitted] check box is selected, [User] and [Password] can be entered.

● [User]

Set the user required for authentication of UNC path. This field can be up to 255 bytes long.

● [Password]

Set the password required for authentication of UNC path. Up to 127 half-width alphanumeric and symbol characters (space included) can be input.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	454/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

NOTE

- 1 For the output path, paths of disk drives such as fixed disk drives (internal disk/external disk) or UNC paths can be specified. The paths of other drives (such as network drives) cannot be specified.
- 2 As the size of the output data will be massive, it is recommended to specify a separate folder for the output path, such as the folder of an expanded external disk drive. Also, make sure that the disk space of the output folder does not get compressed, by deleting the output files or moving the output files to another media on a regular basis.
- 3 If the free disk space of the disk drive specified as the storage destination is under 5GB, the file output process will not be executed.
- 4 For the details on the confirmation of operation by OS combination when UNC path is specified as output path, refer to the "**Confirmation of Operation when UNC Path Is Specified as Output Path**" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	455/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

8.1.9 Setting File Output Path of Variable History

Set the output setting and output path for the output of CSV file, when the variable set in the [Variable History Retrieval Condition Setting] is being retrieved.

Make sure to send the setting data to DB (click the [Send to DB] button) and restart Receiver after changing the path.

For the details, refer to the "[Send Setting Data to DB](#)" subsection.

■ Variable History

● [File output] check box

When enabling CSV file output during retrieval of variable history, select the check box. The default setting is "enable" (This check box is selected).

● [Output path]

Specify the folder to output CSV files. This field can be up to 260 bytes long.

This field populates the following folder by default. On subsequent sessions after changed once, it populates the last modified value:

<MT-LINK*i* installation folder>\VariableHistory

● [...] (Folder selection) button

Clicking this browse folder button opens the [Browse For Folder] window to either select an existing folder or create a new folder. By default, the [Browse For Folder] window selects the folder populated in the [Output path] field.

● [UNC Permitted] check box

When specifying a UNC path for the output path, select the check box.

When the [UNC Permitted] check box is selected, [User] and [Password] can be entered.

● [User]

Set the user required for authentication of UNC path. This field can be up to 255 bytes long.

● [Password]

Set the password required for authentication of UNC path. Up to 127 half-width alphanumeric and symbol characters (space included) can be input.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 456/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

NOTE

- 1 For the output path, paths of disk drives such as fixed disk drives (internal disk/external disk) or UNC paths can be specified. The paths of other drives (such as network drives) cannot be specified.
- 2 As the size of the output data will be massive, it is recommended to specify a separate folder for the output path, such as the folder of an expanded external disk drive. Also, make sure that the disk space of the output folder does not get compressed, by deleting the output files or moving the output files to another media on a regular basis.
- 3 If the free disk space of the disk drive specified as the storage destination is under 5GB, the file output process will not be executed.
- 4 For the details on the confirmation of operation by OS combination when UNC path is specified as output path, refer to the "[Confirmation of Operation when UNC Path Is Specified as Output Path](#)" subsection.

8.1.10 Setting File Output Path of Product Plan Setting

Set the output path for a product plan data file (CSV file) to be used by the [Input Product Plan Data] web screen. The [Input Product Plan Data] tab page of the [Utility – System Setting] screen uses this path as the default path when writing and reading CSV files.

Make sure to send the updated settings to the database after changing the path.

For the details on how to send settings to the database, refer to the "[Send Setting Data to DB](#)" subsection.

The screenshot shows a web interface titled "Product Plan Setting". It contains a label "CSV output path:" followed by a text input field containing the path "C:\FANUC\MT-LINKi\ProductPlan". To the right of the input field is a small red square button with three dots "...". Below the input field is a grey button labeled "Folder Selection Button". A line connects the red square button to the "Folder Selection Button".

■ Product Plan Setting

◆ [CSV output path]

Specify the folder to output CSV files. This field can be up to 260 bytes long.

This field populates the following folder by default. On subsequent sessions after changed once, it populates the last modified value:

<MT-LINK*i* installation folder>\ProductPlan

◆ [...] (Folder selection) button

Clicking this browse folder button opens the [Browse For Folder] window to either select an existing folder or create a new folder. By default, the [Browse For Folder] window selects the folder populated in the [CSV output path] field.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	457/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

NOTE

- 1 The [CSV output path] field allows only a path to a fixed disk drive (an internal or external disk drive); No other path, such as a UNC path or one to a network drive, is available.
- 2 The data to be stored in the folder of this path will increase to a relatively large amount. It is recommended that this field is set to the path to a folder in a newly added external disk drive. In addition to that, ensure that the folder has sufficient space by periodically deleting stored files or moving them to another media.

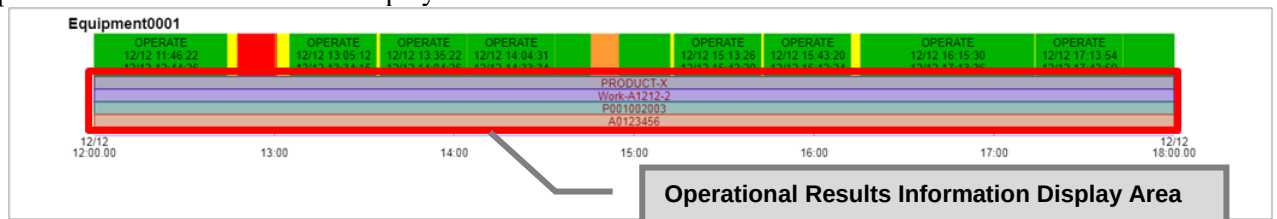
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	458/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

8.1.11 Display Order Setting of Operational Results Information Display Area

For the following items displayed in the operational results information display area ([Results - Operational Results] screen and [Monitoring - Equipment Monitoring] screen) in the Web screen, specify the display/hide status and the order of display.

- Product name
- Work comments
- Product serial number
- Operator ID

■ Operational results information display area of the Web screen



■ Display order setting of operational results information display area

Display Order Setting for Operational Results Info. Display Area

No	Display	Item
1	☒	Product Name
2	☒	Work Comments
3	☐	Product Serial Number
4	☐	Operator ID

Up Down

- [No]
The order of display in the operational results information display area.
- [Display] check box
Set the display/hide status of the display items on the operational results information display area.
[Item] which has the check box selected will be displayed in the operational results information display area in the order shown in [No].
[Item] which has the check box cleared will not be displayed in the operational results information display area.
- [Item]
The items that will be displayed in the operational results information display area.
- [Up] button
[Display] / [Item] of the selected row move up one row exchanging places with the contents in that row.
- [Down] button
[Display] / [Item] of the selected row move down one row exchanging places with the contents in that row.

The default settings are as follows.

No	Display (Check Box)	Item
1	Display (ON)	Product name
2	Display (ON)	Work comments
3	Hide (OFF)	Product serial number
4	Hide (OFF)	Operator ID

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue
FANUC CORPORATION					Sheet.	459/868	

8.1.12 Load Additional Language Resource

Additional language resources are the data files that define settings and texts for a local language to be added to MT-LINK*i* system.

AdminTool adds a local language to the language options after loading the local language's additional language resource files which may be provided separately. Web users that the local language is set for can view Web screens in the local language.

■ Additional Language Resource

- Local Language
Displays the name of a language whose resource files are loaded.
- Language Code
Displays the language code of a language whose resource files are loaded.
- CodePage
Displays the codepage of a language whose resource files are loaded.
- Version
Displays the version of the loaded resource files.
- Updated Date/Time
Displays the date and time when the resource files have been updated.

NOTE

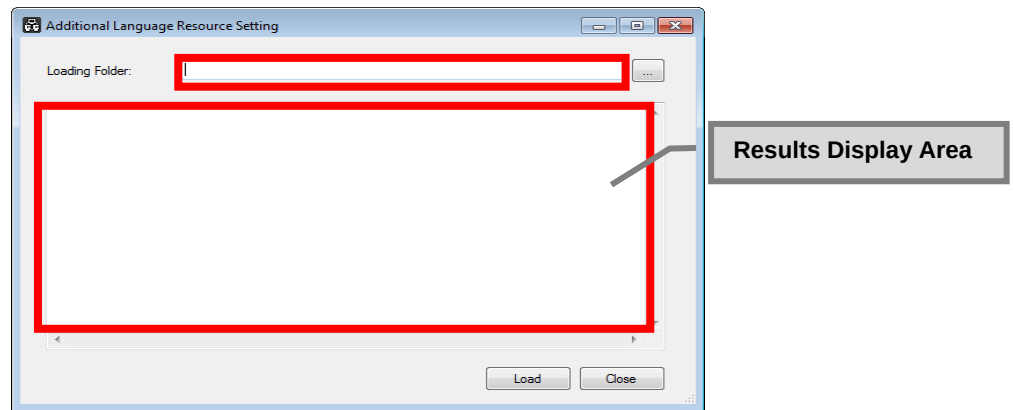
Load the same version of additional language resource files as that of MT-LINK*i*. Difference in version between them may cause the following:

- When the version of the resource files is higher than that of MT-LINK*i*, an error message appears and loading them will be prohibited.
- When the version of the resource files is lower than that of MT-LINK*i*, they can be loaded. However, if a higher version of MT-LINK*i* than the loaded resource files has added texts that they do not support, Web screens will display such texts in English.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	460/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

Follow the steps shown below to load additional language resource files.

- (1) Click the [Load] button to open the [Additional Language Resource Setting] window.



- (2) Specify the [Loading Folder] with the name of the folder in which additional language resource files are saved. Clicking this browse folder button opens the [Browse For Folder] window to either select an existing folder or create a new folder. The [Loading Folder] field populates the folder path selected in the [Browse For Folder] window.

The [Browse For Folder] window selects the following folder on the first session. On subsequent sessions, it selects the folder specified in the previous session:

<MT-LINK*i* installation folder>\MT-LINK*i*AdminTool

- (3) Click the [Load] button to load the additional language resource files and to conduct a validity check on the loaded files. The check results are displayed in the [Results display area].

NOTE

When an error is detected, AdminTool stops loading additional language resource files.

■ Additional language resource files

File Name	Description
webserver.yml	Additional language resource file for Web Server (Defines texts displayed on Web screens)
admintool.yml	Additional language resource file for AdminTool (Defines Web names used in AdminTool)
config.yml	Setting file for AdminTool (Sets values for a local language)
daily-report-template.xls	Report template (Daily)
weekly-report-template.xls	Report template (Weekly)
monthly-report-template.xls	Report template (Monthly)

■ Example of success results

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	461/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

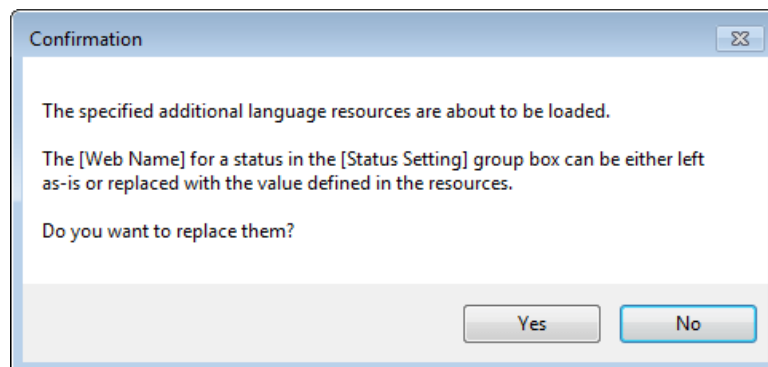
Start Loading Additional Language Resource Files
 Additional Language Resource File for Web Server (Web screen texts)
 => OK
 Setting File for AdminTool
 => OK
 Additional Language Resource File for AdminTool (Web names)
 => OK
 Report template (Daily)
 => OK
 Report template (Weekly)
 => OK
 Report template (Monthly)
 => OK
 Loading Additional Language Resource Completed
 Loaded files: 6
 Failed to load: 0

■ Example of error results

Start Loading Additional Language Resource Files
 Additional Language Resource File for Web Server (Web screen texts)
 => Failed to load the file.
 => NG
 Setting File for AdminTool
 => LanguageCode: XYZ is invalid.
 => NG
 Additional Language Resource File for AdminTool (Web names)
 => RadFan2SrvComPwStatus: The length of the Web Name exceeds 58 bytes.
 => RadFan1SpindleAmpSpeed: The length of the Web Name exceeds 58 bytes.
 => NG
 Report template (Daily)
 => No file is found.
 => NG
 Report template (Weekly)
 => No file is found.
 => NG
 Report template (Monthly)
 => No file is found.
 => NG
 Failed to load additional language resources.
 Loaded files: 0
 Failed to load: 6

(4) Click the [Close] button to return to the [System Setting] window.

(5) Click the [OK] button to open the following confirmation dialog box.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	462/868

(6) Select [Send to DB] in the AdminTool Menu to send the loaded additional language resources to the DB. For the details on how to send data to the DB, refer to the "**SEND TO DB**" section.

When using report template files for a local language, copy them from the Sample folder (Basic) to the Master folder after additional resource files are loaded. For the details, refer to the "**PREPARE REPORT TEMPLATE**" section.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	463/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

8.1.13 Reset All Values to Default

Clicking the [Reset all values to default] button in the [System Setting] window resets the following values to the default values. All of the following values are reset to default values regardless of the tab opened.

- [DB Connecting String]
- [Port]
- [Color of header/footer]
- [Show footer characters]
- [Report]
- [Macro Variable History]
- [Variable History]
- [Product Plan Setting]
- [Display Order Setting of Operational Results Info. Display Area]
- [Additional Language Resource]

By clicking the [OK] button, the setting is applied.

After all the values are set to default, send the setting changes to the database and reboot Receiver.

For the details on how to send the settings and reboot Receiver, refer to the "[Send Setting Data to DB](#)" subsection.

■ [Setting 1] tab

System Setting

Language: English Web name language: English

(If you change language of AdminTool, you need to restart it.)

Setting 1 Setting 2

DB Setting

DB Connecting String: Connecting Test

User: admin1 Password: ***** Apply

DB Data Storing Setting:

Database path: Edit

DB data retention period setting:

Deletion time: Deletion interval (day[s]): Retention period (day[s]): Edit

DB Cleanup: (Last executed: -) Clean Up

DB Optimization: (Last executed: -) Optimize

Web URL: Port:

Color of header/footer: Color setting

☒ Show footer characters

Linkage with other systems:

Web name: URL

Add Edit Delete Up Down

Report

Report path: ...

☐ UNC Permitted User: Password:

Macro Variable History

Output path: ...

☐ UNC Permitted User: Password:

☐ File output

Variable History

Output path: ...

☐ UNC Permitted User: Password:

☒ File output

Product Plan Setting

CSV output path: ...

Reset all values to default

OK Cancel

NOTE

Note that the [User] and [Password] for the database connection will not be reset to the default. This is why we recommend the users should use other values for those fields than the default values and this Default Setting functionality helps the users to change them.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	464/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

■ [Setting 2] tab

System Setting

Language: Web name language:

(If you change language of AdminTool, you need to restart it.)

Setting 1 Setting 2

Display Order Setting for Operational Results Info. Display Area

No	Display	Item
1	☒	Product Name
2	☒	Work Comments
3	☐	Product Serial Number
4	☐	Operator ID

Up Down

Additional Language Resource

Local Language:

Language Code:

CodePage:

Version:

UpdateDate: Read

Reset all values to default

OK Cancel

NOTE

After additional language resource files are loaded, clicking the [Reset all values to default] button and the [OK] button clears all the values set by loading as follows:

- [Web name language] is set to the value selected in the [Language] of the [System Setting] window.
- [Language] in the [Add/Edit User] dialog box is set to the value selected in the [Language] of the [System Setting] window.
- All the Web names (in a local language) are reset to their default values.
- Report templates (samples) are deleted.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	465/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

[SETTING SYSTEM FUNCTIONS] - BATCH SETTING

This chapter describes how to utilize Batch Setting.

The AdminTool Menu has the following menu items in the [Setting System Functions] - [Batch Setting] group:

(1) Export

This menu exports current setting data in batch setting files.

(2) Import

This menu imports setting data from batch setting files.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	466/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

9.1 EXPORT

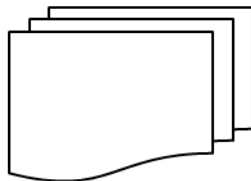
Export function outputs machine setting, equipment setting, group setting, layout setting, authority setting, and sign-in setting of AdminTool as input file, pattern setting files or external files. The exported file can be imported using the import function.

■ Export

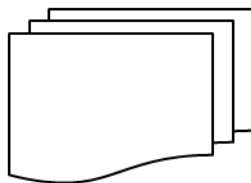
Input File (CollectiveSettingInput.xls)



Pattern Setting Files



External Files (not editable)



Export

Export AdminTool settings.

Export target folder:

NOTE

<Settings unsupported by Export function>

System Setting (except DB Data Retention Period Setting)

Correlation Check, Exporting

Run

Close

NOTE

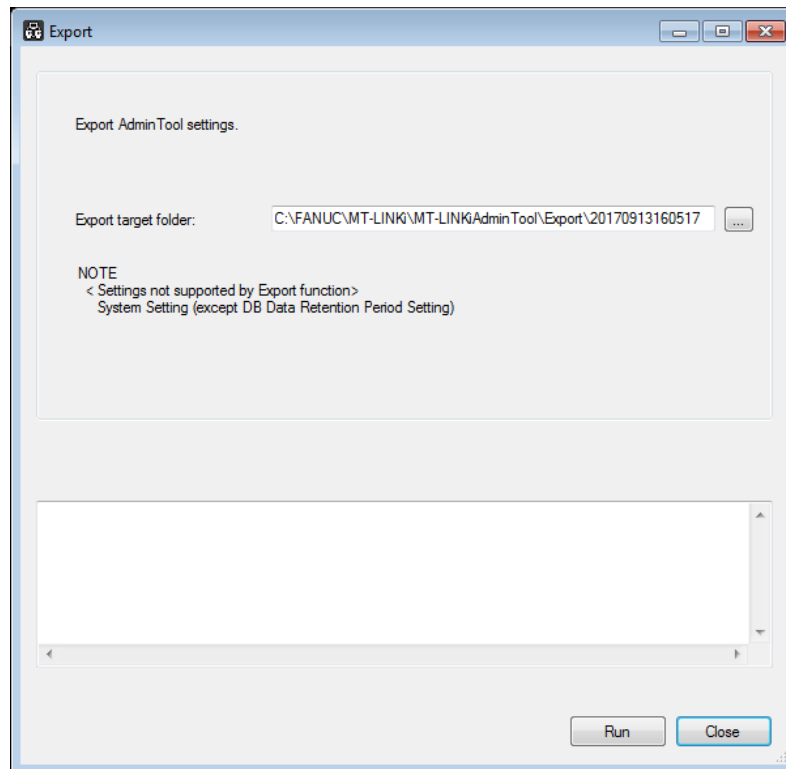
Export function does not support setting data made through System Setting (except for DB Data Retention Period Setting).

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	467/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

9.1.1 Exporting Machine Setting Information

The machine setting information is exported by following the steps below.

- (1) Select [Export] in the AdminTool Menu. The [Export] window appears.



- (2) The [Export target folder] field of the [Export] window populates the following default export target folder:

<MT-LINK*i* installation folder>\MT-LINKiAdminTool\Export\<yyyyMMddHHmmss>

* If the specified folder is not empty, an error message ("For the storage target folder, specify an empty folder or a folder with a name that does not exist.") will appear.

- (3) Click the [Run] button.
- (4) The execution log appears on the log display area.
- (5) If an error occurs, correct the AdminTool settings by referring to the contents of the error. In case of a warning, check if there is any problem, and make corrections if necessary. After correcting the settings, save the settings and repeat step (2) once again.
- (6) When the error is resolved, and it is determined that there is no problem related to the warning, click the [Close] button on the [Export] window to close the window.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	468/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

9.1.2 Export File of Export Function

CollectiveSettingInput.xls, pattern setting files, and the external files exported using the export function can be used as the input file, the pattern setting files, or the external files for the import function. However, editable files are the input file and the pattern setting files.

The input file, the pattern setting files, and the external files are stored in the following paths.

File Type	Storing Folder	File Name
Input file	<User specified folder>	CollectiveSettingInput.xls
Pattern setting file	<User specified folder>\DefineSetting	SignalSetting.xls
		CustomSignalSetting.xls
		LaserSignalSetting.xls
		ToolLifeSignalSetting.xls
		RobodrillSignalSetting.xls
		MacroSetting.xls
		VariableHistorySetting.xls
		EquipmentStatusSetting.xls
		AssetSetting.xls
External file	<User specified folder>\InputDisabled	L0_Setting_Ex.json
		Layout_Setting.json
		User.json
		Role.json
		Ldap.json
		FilterSetting.xls
		StatusSignal.xls
		AlarmMailSetting.json
		E MailServer Setting.json
		SignalMailSetting.json
		ReportSchedule.json
		TimeSeriesCsvSchedule.json
		TimeSeriesCsvManual.json
		FileUploadSchedule.json
		IntegrationServer_Setting.json
		DBDataDelete.json
		DataServer.json
		AccessCode.json
		PowerConsumptionSetting.json
	<User specified folder>\SettingData	Login.txt
	<User specified folder>\SettingImage	(Images, etc.)

This subsection explains the format and the setting values of the editable input file and pattern setting files. When editing, refer to the "**Configuring Input File**" subsection and the "**Editing Pattern Setting File**" subsection, where procedures for manual entry are explained.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	469/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

■ CollectiveSettingInput.xls

(1) This sheet is created for machine setting, equipment setting, group setting, and sign-in setting.

(a) CollectorSetting sheet (Machine setting)

Machine Setting (Collector Setting)									
Necessary		Necessary		Optional		Optional*		Necessary	
Receiver Setting				Collector Setting		Collector Service Manager		Connected Machine	
IP Address/Host Name		Port		SERVO VIEWER Linkage		Port		Machine 1, Machine 2, Machine 100	
example	192.168.1.10		50000						
No	Name	IP Address/Host Name	Port	Use/Not Use	Port				
example	NewCollector01	192.168.1.1	50001	0			50003	NewMachine001	
1									
2									
3									
4									
5									
6									
7									
8									
9									
10									
11									
12									
13									
14									
15									
16									
17									
18									
19									
20									
21									
22									
23									
24									
25									
26									
27									
28									
29									
30									
31									
32									
33									
34									
35									
36									
37									
38									
39									
40									
41									
42									
43									
44									
45									
46									
47									
48									
49									
50									
51									
52									
53									
54									
55									
56									
57									
58									
59									
60									
61									
62									
63									
64									
65									
66									
67									
68									
69									
70									
71									
72									
73									
74									
75									
76									
77									
78									
79									
80									
81									
82									
83									
84									
85									
86									
87									
88									
89									
90									
91									
92									
93									
94									
95									
96									
97									
98									
99									
100									

Setting Entries

- (i) Receiver Setting
The IP Address / Host Name and Port number for the Server PC.
- (ii) CollectorSetting (Read Signal)
The Name, IP Address / Host Name, and Port number for the Collector PC.
- (iii) CollectorSetting (SERVO VIEWER Linkage)
The state of usage for SERVO VIEWER linkage and the port number.
- (iv) CollectorSetting (Collector Service Manager)
The Port number used by Collector Service Manager running on the Collector PC.
- (v) Connected Machine
The machine connected to the Collector PC.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 470/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

(b) MachineSetting sheet (Machine setting)

CollectiveSettingInput_en.xls - LibreOffice Calc

File Edit View Insert Format Styles Sheet Data Tools Window Help

1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	RD	BE	BF	BD		
2	Machine Setting(Machine Register)				Web language types (not settable)																		
3					(i)		Type:0:FanucCNC/1:OPCUAServer/2:FanucRobot/3:MTConnectClient/4:Synchronization interval(0:Not set. Mandatory for FanucCNC, unnecessary for others)																
4	Machine No.		Machine Name		Optional		Necessary		Necessary*		Necessary*		Necessary*		Necessary*		Necessary*		Necessary to match the CNC path numbers*				
5	No.	Machine Name	Web Name				Type	Connection	IP Address/Host Name		Port	Timeout(Sec)	Synchronization interval (0)	CNC Path	Path1			Path15					
6			Japanese	English	Chinese	Local language									Servo axis number	Spindle axis number	L	Servo axis number	Spindle axis number	L			
7	1	NewMachine001	001	Machine001	机床001	Machine001	0	1	192.168.1.11		8193	30	0	2	1	1	0						
8	2						0	1			8193	30	0										
9	3						0	1			8193	30	0										
10	4						0	1			8193	30	0										
11	5						0	1			8193	30	0										
12	6						0	1			8193	30	0										
13	7						0	1			8193	30	0										
14	8						0	1			8193	30	0										
15	9						0	1			8193	30	0										
16	10						0	1			8193	30	0										
17	11						0	1			8193	30	0										
18	12						0	1			8193	30	0										
19	13						0	1			8193	30	0										
20	14						0	1			8193	30	0										
21	15						0	1			8193	30	0										
22	16						0	1			8193	30	0										
23	17						0	1			8193	30	0										
24	18						0	1			8193	30	0										
25	19						0	1			8193	30	0										
26	20						0	1			8193	30	0										
27	21						0	1			8193	30	0										
28	22						0	1			8193	30	0										
29	23						0	1			8193	30	0										
30	24						0	1			8193	30	0										
31	25						0	1			8193	30	0										
32	26						0	1			8193	30	0										
33	27						0	1			8193	30	0										
34	28						0	1			8193	30	0										
35	29						0	1			8193	30	0										
36	30						0	1			8193	30	0										
37	31						0	1			8193	30	0										
38	32						0	1			8193	30	0										
39	33						0	1			8193	30	0										
40	34						0	1			8193	30	0										
41	35						0	1			8193	30	0										
42	36						0	1			8193	30	0										
43	37						0	1			8193	30	0										
44	38						0	1			8193	30	0										
45	39						0	1			8193	30	0										
46	40						0	1			8193	30	0										
47	41						0	1			8193	30	0										
48	42						0	1			8193	30	0										
49	43						0	1			8193	30	0										
50	44						0	1			8193	30	0										
51	45						0	1			8193	30	0										
52	46						0	1			8193	30	0										
53	47						0	1			8193	30	0										
54	48						0	1			8193	30	0										
55	49						0	1			8193	30	0										
56	50						0	1			8193	30	0										
57	51						0	1			8193	30	0										
58	52						0	1			8193	30	0										
59	53						0	1			8193	30	0										
60	54						0	1			8193	30	0										
61	55						0	1			8193	30	0										
62	56						0	1			8193	30	0										
63	57						0	1			8193	30	0										
64	58						0	1			8193	30	0										
65	59						0	1			8193	30	0										
66	60						0	1			8193	30	0										
67	61						0	1			8193	30	0										
68	62						0	1			8193	30	0										
69	63						0	1			8193	30	0										
70	64						0	1			8193	30	0										
71	65						0	1			8193	30	0										
72	66						0	1			8193	30	0										
73	67						0	1			8193	30	0										
74	68						0	1			8193	30	0										
75	69						0	1			8193	30	0										
76	70						0	1			8193	30	0										
77	71						0	1			8193	30	0										
78	72						0	1			8193	30	0										
79	73						0	1			8193	30	0										
80	74						0	1			8193	30	0										
81	75						0	1			8193	30	0										
82	76						0	1			8193	30	0										
83	77						0	1			8193	30	0										
84	78						0	1			8193	30	0										
85	79						0	1			8193	30	0										
86	80						0	1			8193	30	0										
87	81						0	1			8193	30	0										
88	82						0	1			8193	30	0										
89	83						0	1			8193	30	0										
90	84						0	1			8193	30	0										
91	85						0	1			8193	30	0										
92	86						0	1			8193	30	0										
93	87						0	1			8193	30	0										
94	88						0	1			8193	30	0										
95	89						0	1			8193	30	0										
96	90						0	1			8193	30	0										
97	91						0	1			8193	30	0										
98	92						0	1			8193	30	0										
99	93						0	1			8193	30	0										
100	94						0	1			8193	30	0										
101	95						0	1			8193	30	0										
102	96						0	1			8193	30	0										
103	97						0	1			8193	30	0										
104	98						0	1			8193	30	0										
105	99						0	1			8193	30	0										
106	100						0	1			8193	30	0										
107	101						0	1			8193	30	0										
108	102						0	1			8193	30	0										
109	103						0	1			8193	30	0										
110	104						0	1			8193	30	0										
111	105						0	1			8193	30	0										
112	106						0	1			8193	30	0										
113	107						0	1			8193	30	0										
114	108						0	1			8193	30	0										
115	109						0	1			8193	30	0										
116	110																						

CollectiveSettingInput_en.xls - LibreOffice Calc																							
File	Edit	View	Insert	Format	Styles	Sheet	Data	Tools	Window	Help													
	BI	BJ	BK	BL	BM	BN	BO	BP	BQ	BR	BS	BT	BU	BV	BW	BX	BY	BZ	CA	CB	CC	CD	CE
1																							
2																							
3																							
4																							
5																Optional	Optional	Optional	Optional	Optional	Optional		
6	External Link Validity	External Link 1					External Link 2					External Link 3					Use of outlier detection setting file	Signal Pattern	Laser Signal Pattern	ROBOCELL Signal Pattern	Asset management setting pattern	Tool Life Signal Pattern	Note
7		Japanese	English	Chinese	Local language	URL		Japanese	English	Chinese	Local language	URL		Japanese	English	Chinese	Local language	URL					
8	1	リンク1	LINK1	关联1	LINK1	http://cnc								0	1	1	1	1	1	1			
9	0																						
10	0																						
11	0																						
12	0																						
13	0																						
14	0																						
15	0																						
16	0																						
17	0																						
18	0																						
19	0																						
20	0																						
21																							
22																							
23																							
24																							
25																							
26																							
27																							
28																							
29																							
30																							
31																							
32																							
33																							
34																							
35																							
36																							
37																							
38																							
39																							
40																							
41																							
42																							
43																							
44																							
45																							
46																							
47																							
48																							
49																							
50																							
51																							
52																							
53																							
54																							
55																							
56																							
57																							
58																							
59																							
60																							
61																							
62																							
63																							
64																							
65																							
66																							
67																							
68																							
69																							
70																							
71																							
72																							
73																							
74																							
75																							
76																							
77																							
78																							
79																							
80																							
81																							
82																							
83																							
84																							
85																							
86																							
87																							
88																							

Setting Entries

- (i) Web language types
This is an entry that notifies how many languages are supported for (iii) Web name. The value on the template file is set.
- (ii) Machine Name
The name identifies each machine that is included in the equipment.
- (iii) Web Name (Japanese/English/Chinese/Local language)
The Web names of the equipment units.
- (iv) Type
The type of the machine is set.
- (v) Connection (NOTE 1, 2)
This is an entry that sets the connection status of the machine.
- "1" is set for this column when connection check box is selected.
- (vi) Network (NOTE 1, 2)
This is an entry that sets IP address/host name, port, time out (sec), and synchronization interval (H) of the machine.
- (vii) CNC Path (NOTE 1, 2)
CNC Path can be selected from the drop-down menu.
- (viii) Servo axis Number / Spindle axis Number / L (NOTE 1, 2)

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 471/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

This is an entry that sets the number of servo axes and spindle axes of the path, and whether the control type is L or not.

- (ix) External Link Validity
The validity (1: valid or 0: invalid) of the external link is set.
- (x) External Link 1 / External Link 2 / External Link 3
Web names (in English, Japanese, Chinese, and a local language) and URLs of the external links are set.
- (xi) Use of outlier detection setting (NOTE 1, 2)
If there are outlier detection settings, "1" (Use) is set, and if not "0" (Not Use) is set.
- (xii) Signal Pattern (NOTE 1, 2)
The values of this column are sequentially numbered starting with "101" in the order in which machines are registered to AdminTool.
- (xiii) Laser Signal Pattern (NOTE 1)
For the machines whose [L] columns are set to "1," this column will be set to the same number as the [Signal Pattern] column.
- (xiv) ROBODRILL Signal Pattern (NOTE 2)
The values of this column are sequentially numbered starting with "101" in the order in which machines are registered to AdminTool.
For the machines of other than FanucRobodrill machine types, this column will be set to blank.
- (xv) Asset Setting Pattern
The values of this column are sequentially numbered starting with "101" in the order in which machines are registered to AdminTool.
The asset setting pattern of the machine that has no asset management information registered will be set to blank.
- (xvi) Tool Life Signal Pattern (NOTE 1, 2)
The values of this column are sequentially numbered starting with "101" in the order in which machines are registered to AdminTool.

NOTE

- 1 Applicable when the [Type] column is set to "0" (FanucCNC).
- 2 Applicable when the [Type] column is set to "4" (FanucRobodrill).

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	472/868

CollectiveSettingInput_en.xls - LibreOffice Calc															
File Edit View Insert Format Sheet Data Tools Window Help															
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Equipment Setting					Web language types (not-editable)	4	(i)							
2	Necessary	Necessary		Optional				Necessary	not-editable	Optional	Optional	Optional	Optional	Optional	Optional
3	No	Equipment Name	Web Name				Cancel monitor	Record Cycle	Equipment Status Pattern	Use of simple assign file	Normal Image	Disconnect Image	Alarm Image		
4			Japanese	English	Chinese	Local language									
5	example	NewEquipment0001	設備0001	Equipment0001	设备0001	Equipment0001	0	1	1	0	C:\FANUC\WMT-LINK\WMT-LINK\AdminTool\WMT-LINK\Image\Equipment\MachiningCenter_SWRightGreen_MachiningCenter_S.png	C:\FANUC\WMT-LINK\WMT-LINK\AdminTool\WMT-LINK\Image\Equipment\MachiningCenter_SWRightGreen_MachiningCenter_S.png	C:\FANUC\WMT-LINK\WMT-LINK\AdminTool\WMT-LINK\Image\Equipment\MachiningCenter_SWRightRed_MachiningCenter_S.png		
6	1						0	1							
7	2						0	1							
8	3						0	1							
9	4						0	1							
10	5						0	1							
11	6						0	1							
12	7						0	1							
13	8						0	1							
14	(ii)			(iii)			(iv)	(v)	(vi)	(vii)					
15	90						0	1							
16	91						0	1							
17	92						0	1							
18	93						0	1							
19	94						0	1							
20	95						0	1							
21	96						0	1							
22	97						0	1							
23	98						0	1							
24	99						0	1							
25	100						0	1							
26	101						0	1							
27	102						0	1							
28	103						0	1							
29	104						0	1							
30	105						0	1							
31	106						0	1							
32	107						0	1							
33	108						0	1							
34	109						0	1							
35	110						0	1							
36	111						0	1							

[illegible]

(i) Web language types

(ii) Equipment Name

(iii) Web Name (Japanese/English/Chinese/Local language)

- (iv) Cancel monitoring

(v) Record Cycle

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					FANUC CORPORATION			Sheet.	473/868
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

- (vi) Equipment Status Pattern
The values of this column are sequentially numbered starting with "101" in the order in which equipment units are registered to AdminTool.
- (vii) Use of Simple Assign File
"1" (Use) is set.
- (viii) Icon Setting
This is an entry that sets the (8) icons of the equipment. It is a relative path from the export target (export folder of CollectiveSettingInput.xls).
- (ix) Machine Selection
Machine name of the machine is set.

(d) GroupSetting sheet (Group setting)

Setting Entries

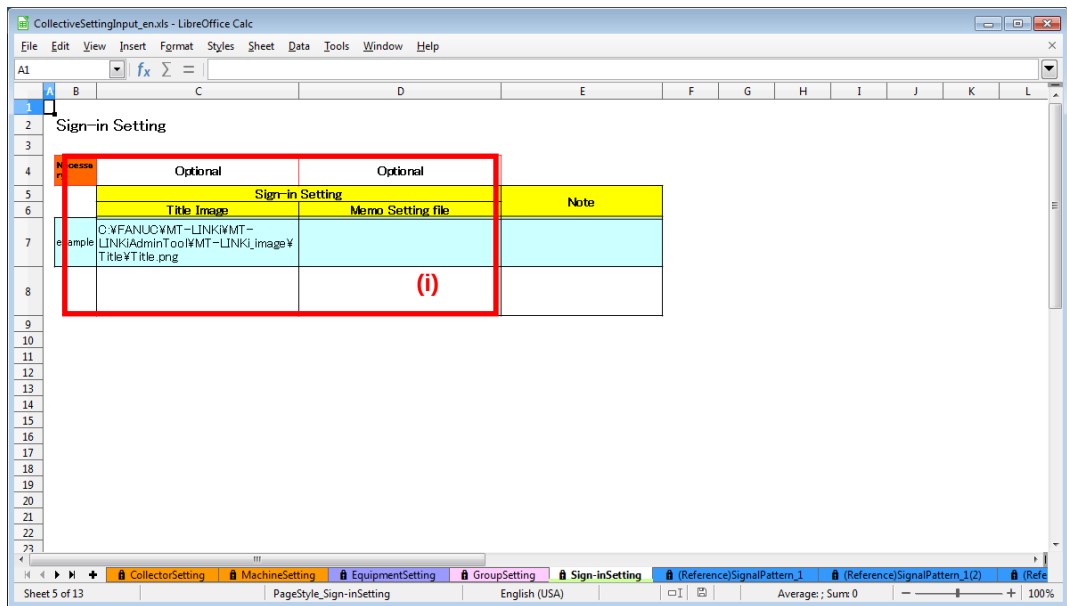
- (i) Web language types
This is an entry that notifies how many languages are supported for (iii) Web name. The value on the template is set.
- (ii) Group Name
The name of the group is set.
- (iii) Web Name (Japanese/English/Chinese/Local language)
The Web name of the group is set.
- (iv) Record Cycle
"1" is set.
- (v) Read Image
This entry is the path to the icon that indicates the status of the group (NORMAL, TROUBLE). It is a relative path from the export target (export folder of CollectiveSettingInput.xls).
- (vi) Equipment Selection
Equipment units of the group are set.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 474/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

(vii) Status Setting

Enter the settings of assigned signals of the group for "In" and "Out."

(e) Sign-inSetting sheet (Sign-in setting)



Setting Entries

(i) Sign-in Setting

- The path to the title image file of the [MT-LINKi Web Sign in] screen is set as the title image.

- The path to the memo setting file is set as the memo setting file.

* Both of above are a relative path from the export target (export folder of CollectiveSettingInput.xls).

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
13	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	475/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

■ Pattern Setting File

Each pattern setting file is created by copying the pattern setting template file for export.

When the specified machine type is limited to only OPCUAServer or to only FANUC Robot, or if exported with the incomplete machine setting, a dummy file of the pattern setting file will be exported.

(1) Signal Setting (SignalSetting.xls)

- This sheet is created for each signal pattern.

Signal	Null	0 axis	1 axis	2 axis	3 axis	4 axis	5 axis	6 axis	7 axis	8 axis	9 axis	10 axis	11 axis	12 axis	13 axis	14 axis	15 axis	16 axis	17 axis	18 axis	19 axis	20 axis	21 axis	22 axis	23 axis	Note
SigOP	0																									Automatic operation signal
SigSTL	0																									Cycle start lamp signal
SigSPL	0																									Feed hold lamp signal
SigL	0																									Alarm signal
EMG	0																									Emergency stop signal
Mode	500																									Mode
MainProgram	500																									Main program
ActProgram	500																									Executing program
Sequence	0																									Sequence number
MainComment	500																									Program comment (main program)
ActComment	500																									Program comment (executing program)
ActS	0																									Speed S (actual)
ActF	0																									Feed rate F (actual)
ActFdec	0																									Feed rate F (actual)
ModalS	0																									Speed S (command)
ModalF	0																									Feed rate F (command)
ModalT	0																									T code
ModalM	0																									1st M code
ModalM2	0																									2nd M code
ModalM3	0																									3rd M code
PartsNum	500																									The number of machined parts
PartsNumAll	0																									The total number of machined parts
PowOnTime	0																									Power-on period
RunTime	0																									Operation time
CutTime	0																									Cutting time
SigCUT	500																									Cutting feed signal
SigSBK	500																									Single block check signal
SigM00	500																									Decode M00 signal
SigM01	500																									Decode M01 signal

Setting Entries

(a) Signal Pattern

The values of this column are sequentially numbered starting with "101" in the order in which machines are registered to AdminTool.

When the pattern setting file is a dummy file, "999" will be set.

(b) CNC Path

CNC path of the machine is set.

(c) Axis

The reading cycle of each axis of signal is set. When there is no axis, the [Null] column is used.

- If the signal is enabled, the reading cycle set in the AdminTool is set.

- If the signal is disabled, "0" will be set.

- If the pattern setting file is a dummy file, "0" will be set.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 476/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

(2) Custom Signal Setting (CustomSignalSetting.xls)

- This sheet is created for each signal pattern.

CustomSignalSetting.xls - LibreOffice Calc

W1041

No	Type	CNC Path/PMC Path	Address*1	Number*2	Data Type	Data Length	End Code	Diagnosis Data Type	Bit	Axis	Reading Cycle	Note
1	PMC	Necessary*3	Necessary	Necessary	Necessary	Necessary*10	Necessary*10+11	Null	Necessary*4	Null	Necessary	*3 Enter the PMC path *4 Necessary if Data type is "BIT" *10 Necessary if Data type is "STRING" To disable this property, set no value for it
2	MacroVar	Necessary*5	Null	Necessary	Null	Null	Null	Null	Null	Null	Necessary	*5 Enter the CNC path
3	PcodeVar	Necessary*5	Null	Necessary	Necessary*6	Null	Null	Null	Null	Null	Necessary	*5 Enter the CNC path *6 Data types are as following: Conversation: Conversation macro Auxiliary: Auxiliary macro Execution: Execution macro
4	Diagnosis	Necessary*5	Null	Necessary	Necessary*7	Null	Null	Necessary*8	Necessary*4	Necessary*9	Necessary	*4 Necessary if Data type is "BIT" *5 Enter the CNC path *7 Data types are as following: BIT: Bit DWORD: Integer REAL: Real *8 Diagnosis Data Types are as following: Path: Path Axis: Axis Spindle: Spindle *9 Necessary if Diagnosis Data Type is "Axis" or "Spindle"

Signal Pattern Setting

Signal Pattern Number is assigned as following table

- *1 ~ 1000 : For FANUC
- *1001 ~ 2000 : For others

Signal Pattern

No	CNC Path/PMC Path	Type	Label	Address	Number	Data Type	Data Length	End Code	Diagnosis Data Type	Bit	Axis	Reading Cycle (msec)	Note
1	PMC	PMC	A	1234	BIT					0		500	
2	PMC	PMC	B	5678	WORD							60000	
3	PMC	PMC	B	0	WORD	255		ETX (0x03)				500	
4	PMC	PMC	B	255	WORD	0		JF				500	
5	MacroVar	MacroVar		123								10000	
6	PcodeVar	PcodeVar		33	Conversation							500	
7	Diagnosis	Diagnosis		1234	BIT				Path	0		60000	
8	Diagnosis	Diagnosis		5678	DWORD				Axis		0	60000	
9	PMC	PMC	D	100	WORD							500	

101 (a)

(b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Setting Entries

(a) Signal Pattern

The values of this column are sequentially numbered starting with "101" in the order in which machines are registered to AdminTool.

If the pattern setting file is a dummy file, "999" will be set.

(b) CNC Path/PMC Path

CNC path or the PMC path of the custom signal is set.

If the pattern setting file is a dummy file, there will be no setting.

(c) Type

The type of the custom signal is set.

- Options are "PMC," "MacroVar," "PcodeVar," and "Diagnosis."

- If the pattern setting file is a dummy file, there will be no setting.

(d) Address

- If the [Type] column is set to "PMC," this column is set to the value that is set in the AdminTool.

- If the [Type] column is set to other than "PMC," there will be no setting.

(e) Number

The number of a custom signal

If the pattern setting file is a dummy file, there will be no setting.

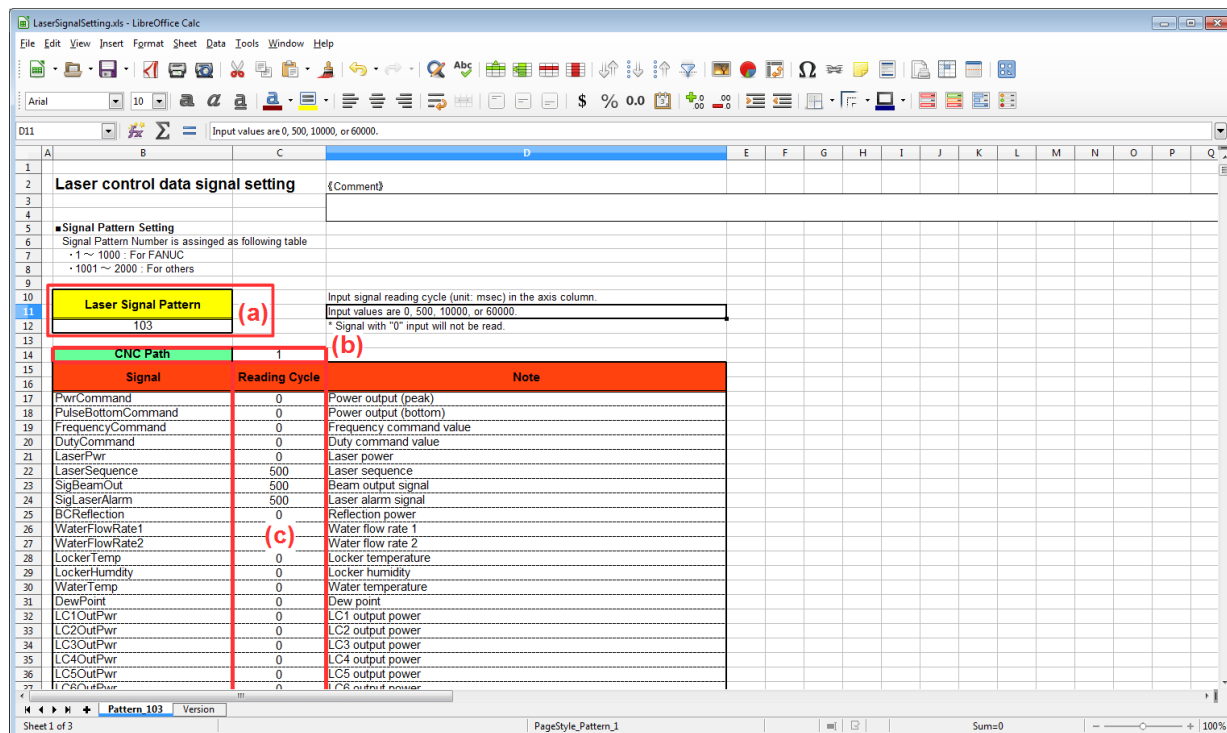
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	477/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

- (f) Data Type
- If the [Type] column is set to "PMC," "PcodeVar," or "Diagnosis," this column is set to the value that is set in the AdminTool.
 - If the [Type] column is set to "MacroVar," there will be no setting.
 - If the pattern setting file is a dummy file, there will be no setting.
- (g) Data Length
- This column is set to the length of string data in bytes.
- If the [Type] column and the [Data Type] column are set to "PMC" and "STRING," respectively, this column is set to the value that is set in the AdminTool.
 - Otherwise, no value is set for this column.
 - If the pattern setting file is a dummy file, there will be no setting.
- (h) End Code
- If the [Type] column and the [Data Type] column are set to "PMC" and "STRING," respectively, this column is set to the value that is set in the AdminTool.
 - Otherwise, no value is set for this column.
 - If the pattern setting file is a dummy file, there will be no setting.
- (i) Diagnosis Data Type
- If the [Type] column is set to "Diagnosis," this column is set to the value that is set in the AdminTool.
 - If the [Type] column is set to other type than "Diagnosis," there will be no setting.
 - If the pattern setting file is a dummy file, there will be no setting.
- (j) Bit
- If the [Type] column is set to "PMC" or "Diagnosis," this column is set to the value that is set in the AdminTool.
 - If the [Type] column is set to "MacroVar" or "PcodeVar," there will be no setting.
 - If the pattern setting file is a dummy file, there will be no setting.
- (k) Axis
- If the [Type] column is set to "Diagnosis," this column is set to the value that is set in the AdminTool.
 - If the [Type] column is set to other type than "Diagnosis," there will be no setting.
 - If the pattern setting file is a dummy file, there will be no setting.
- (l) Reading Cycle
- The cycle time (msec) of reading a custom signal
- If the pattern setting file is a dummy file, there will be no setting.
- (3) Tool Life Signal setting (ToolLifeSignalSetting.xls)
- This sheet is created for each tool life signal pattern.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	478/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

ToolLifeSignalSetting.xls - LibreOffice Calc										
File Edit View Insert Format Styles Sheet Data Tools Window Help										
1	A	B	C	D	E	F	G	H	I	J
2		Tool Life Signal Setting		{Comment}						
3										
4										
5		■ Signal Pattern Setting								
6		Signal Pattern Number is assigned as following table								
7		• 1 ~ 1000 : For FANUC								
8		• 1001 ~ 2000 : For others								
9										
10		Signal Pattern (a)		Input signal reading cycle (unit: msec) in the axis column.						
11		999		Input values are 0, 500, 10000, or 60000.						
12				with "0" input will not be read.						
13										
14		CNC Path	1							
15		Tool Groups	0							
16										
17		Signal	Reading Cycle	Note						
18		ToolGroupNum	0	Current tool group						
19		ToolLifeCount01	0	Tool life count 01						
20		ToolLifeCount02	0	Tool life count 02						
21		ToolLifeCount03	0	Tool life count 03						
22		ToolLifeCount04	0	Tool life count 04						
23		ToolLifeCount05	0	Tool life count 05						
24		ToolLifeCount06	0	Tool life count 06						
25		ToolLifeCount07	0	Tool life count 07						
26		ToolLifeCount08	0	Tool life count 08						
1040		ToolLifeCountType254	0	Tool life count type 254						
1041		ToolLifeCountType255	0	Tool life count type 255						
1042		ToolLifeCountType256	0	Tool life count type 256						
1043										
1044		CNC Path	2							
1045		Tool Groups	0	Set a value between 0 and 256 to the [Tool Groups] column. (Example: when 3 is set for the column, signals for tool group num						
1046										
1047		Signal	Reading Cycle	Note						
1048		ToolGroupNum	0	Current tool group						
1049		ToolLifeCount01	0	Tool life count 01						
1050		ToolLifeCount02	0	Tool life count 02						
1051		ToolLifeCount03	0	Tool life count 03						
1052		ToolLifeCount04	0	Tool life count 04						
1053		ToolLifeCount05	0	Tool life count 05						
1054		ToolLifeCount06	0	Tool life count 06						
1055		ToolLifeCount07	0	Tool life count 07						
1056		ToolLifeCount08	0	Tool life count 08						
1057										
1058										
1059										
1060										
1061										
1062										
1063										
1064										
1065										
1066										
1067										
1068										
1069										
1070										
1071										
1072										
1073										
1074										
1075										
1076										
1077										
1078										
1079										
1080										
1081										
1082										
1083										
1084										
1085										
1086										
1087										
1088										
1089										
1090										
1091										
1092										
1093										
1094										
1095										
1096										
1097										
1098										
1099										
1100										
1101										
1102										
1103										
1104										
1105										
1106										
1107										
1108										
1109										
1110										
1111										
1112										
1113										
1114										
1115										
1116										
1117										
1118										
1119										
1120										
1121										
1122										
1123										
1124										
1125										
1126										
1127										
1128										
1129										
1130										
1131										
1132										
1133										
1134										
1135										
1136										
1137										
1138										
1139										
1140										
1141										
1142										
1143										
1144										
1145										
1146										
1147										
1148										
1149										
1150										
1151										
1152										
1153										
1154										
1155										
1156										
1157										
1158										
1159										
1160										
1161										
1162										
1163										
1164										
1165										
1166										
1167										
1168										
1169										
1170										
1171										
1172										
1173										
1174										
1175										
1176										
1177										
1178										
1179										
1180										
1181										
1182										
1183										
1184										
1185										
1186										
1187										
1188										
1189										
1190										
1191										
1192										
1193										
1194										
1195										
1196										
1197										
1198										
1199										
1200										
1201										
1202										
1203										
1204										
1205										
1206										
1207										
1208										
1209										
1210										
1211										
1212										
1213										
1214										
1215										
1216										
1217										
1218										
1219										
1220										
1221										
1222										
1223										
1224										
1225										
1226										

- (4) Laser control data signal setting (LaserSignalSetting.xls)
 - This sheet is created for each laser signal pattern.



Setting Entries

- (a) Laser Signal Pattern
 The same number is set as signal pattern of the signal setting for the machine including a CNC path with "1" set in the L.
- (b) CNC Path
 CNC path of the machine is set.
- (c) Reading Cycle
 The reading cycle (msec) of the laser control data signal is set.
 - When signal is enabled, the reading cycle set on AdminTool is set.
 - When signal is disabled, "0" is set.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	480/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

(5) ROBODRILL Signal setting (RobodrillSignalSetting.xls)

- This sheet is created for each Robodrill signal pattern. This is available for FanucRobodrill type machines.

Signal	Null	X axis	Y axis	Z axis	B axis	A axis	U axis	V axis	W axis	Note
RobodrillToolNum	500									Robodrill tool number
RobodrillMachiningMode	0									Robodrill machining mode
RobodrillPowerSaveMode	0									Robodrill power save mode
RobodrillRigidTappingMode	0									Robodrill rigid tapping mode
RobodrillTableLoad	0									Robodrill table load mode
RobodrillToolLoad	0									Robodrill tool load mode
RobodrillCounterNum1	10000									Robodrill production counter 1
RobodrillCounterTargetNum1	0									Robodrill production target 1
RobodrillCounterPartName1	0									Robodrill counter part name 1
RobodrillCounterNum2	10000									Robodrill production counter 2
RobodrillCounterTargetNum2	0									Robodrill production target 2
RobodrillCounterPartName2	0									Robodrill counter part name 2
RobodrillIHMICounterNum1	0									Robodrill IHMI production counter 1
RobodrillIHMICounterTargetNum1	0									Robodrill IHMI production target 1
RobodrillIHMICounterPartName1	0									Robodrill IHMI counter part name 1
RobodrillIHMICounterNum2	0									Robodrill IHMI production counter 2
RobodrillIHMICounterTargetNum2	0									Robodrill IHMI production target 2
RobodrillIHMICounterPartName2	0									Robodrill IHMI counter part name 2
RobodrillIHMICounterNum3	0									Robodrill IHMI production counter 3
RobodrillIHMICounterTargetNum3	0									Robodrill IHMI production target 3
RobodrillIHMICounterPartName3	0									Robodrill IHMI counter part name 3
RobodrillProgramProtection	0									Robodrill program protection
RobodrillOperationLevel	0									Robodrill operation level
RobodrillElectricPowerConsumption	0									Robodrill electric power consumption
ThermalCompEnableX	0									X-axis thermal displacement compensation enable
ThermalCompEnableY	0									Y-axis thermal displacement compensation enable
ThermalCompEnableZ	0									Z-axis thermal displacement compensation enable
ThermalCompAdjValX	0									X-axis thermal displacement compensation adjust value
ThermalCompAdjValY	0									Y-axis thermal displacement compensation adjust value

Setting Entries

(a) Signal Pattern

The values of this column are sequentially numbered starting with "101" in the order in which machines are registered to AdminTool.

If the pattern setting file is a dummy file, "999" will be set.

(b) CNC Path

This cell is fixed to "1."

(c) Axis

The reading cycle of the signal is set for the [Null] column.

- When signal is enabled, the reading cycle set on AdminTool is set.
- When signal is disabled, "0" is set.
- When the pattern setting file is a dummy file, "0" is set.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 481/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

(6) Macro Variable Getting Condition Setting (MacroSetting.xls)

- This sheet is created for each signal pattern.

MacroSetting.xls - LibreOffice Calc

File Edit View Insert Format Sheet Data Tools Window Help

A B C D E F G H I J K L M N O P HL HP

1

Macro variable getting condition definition

Web language types (not-editable)

4

(a)

(Comment)

4

■Signal Pattern Setting

Signal Pattern Number is assigned as following table

・1 ~ 1000 : For FANUC

・1001 ~ 2000 : For others

9

Signal Pattern

101

(b)

14

No	ID	Web Name				Trigger Macro Variable	CNC Path	Standard Macro	Collecting number	Web Name1/53/105/157/209					Name52/104/156
		Japanese	English	Chinese	Local language					Japanese	English	Chinese	Local language	Local language	
example	ID001	マクロ1	Macro1	宏1	Macro1	500	8	501	1	1~	マクロ1	Macro1	宏1	Macro1	Macro52
										53~	マクロ53	Macro53	宏53	Macro53	Macro104
										105~	マクロ105	Macro105	宏105	Macro105	Macro156
										157~	マクロ157	Macro157	宏157	Macro157	Macro208
										209~	マクロ209	Macro209	宏209	Macro209	
1~	温度	Temperature	温度												
1	ID0001	マクロID0001	ID0001	ID0001	ID0001	500	1	501	5	53~					
										105~					
										157~					
										209~					
										1~					
										53~					
										105~					
										157~					
										209~					
										1~					

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

31

Pattern 101

Pattern 102

Pattern 103

Pattern 104

Pattern 105

Pattern 106

Pattern 107

Version

Setting Entries

(a) Web language types

This is an entry that notifies how many languages are supported for (d) Web name. The value on the pattern setting template file is set.

If the pattern setting file is a dummy file, there will be no setting.

(b) Signal Pattern

The values of this column are sequentially numbered starting with "101" in the order in which machines are registered to AdminTool.

If the pattern setting file is a dummy file, "999" will be set.

(c) ID

The ID of the Macro Variable getting condition definition

If the pattern setting file is a dummy file, there will be no setting.

(d) Web Name (Japanese/English/Chinese/Local language)

Web name of the macro variable getting condition.

(e) Trigger Macro Variable

Trigger macro variable of the macro variable getting condition

If the pattern setting file is a dummy file, there will be no setting.

(f) CNC Path

The CNC path to be used for the macro variable getting condition definition

If the pattern setting file is a dummy file, there will be no setting.

(g) Standard Macro

This macro variable is set as the base variable for the Macro Variable getting condition definition.

If the pattern setting file is a dummy file, there will be no setting.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 482/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

(h) Collecting Number

The number of macro variables to be collected

If the pattern setting file is a dummy file, there will be no setting.

(i) Web Name of Macro Variable (Japanese/English/Chinese/Local language)

Web name of the macro variable displayed on the Web browser

(7) Variable history retrieval condition definition (VariableHistorySetting.xls)

- This sheet is created for each signal pattern.

VariableHistorySetting_en.xls - LibreOffice Calc

HX77

1	Variable History Retrieval Condition Definition																			
2	[Comment]																			
3	[Comment]																			
4	[Comment]																			
5	[Comment]																			
6	Variable Items Requiring Entry																			
7	No.	Type	Path	Address#1	Number#2	Data type	Data Length	End Code	BIT	Note										
8	1	PMC#7	Necessary#3	Necessary	Necessary	Necessary	Necessary#9	Necessary#10	Necessary#4	P3 Enter the PMC path. P4 Necessary if Data Type is "BIT". P5 Necessary if Data Type is "STRING". To disable this property, set no value for P.										
9	2	MacroVar	Necessary#5	N/A	Necessary	N/A	N/A	N/A	N/A	P3 Enter the CHD path. P4 Data Types are as follows: Conversion : Conversion macro Auxiliary : Auxiliary macro Exception : Exception macro P4 Address can be 3 digit.										
10	3	ProbeVar#7	Necessary#5	N/A	Necessary	Necessary#9	N/A	N/A	N/A	P3 Enter the CHD path. P4 Data Types are as follows: Conversion : Conversion macro Auxiliary : Auxiliary macro Exception : Exception macro P4 Address can be 3 digit (F or MacroVar 8 digit). P7 Cannot set as Tripper variable. P8 To describe us and code not listed in the drop-down list, enter it in hexadecimal (2 to FF).										
11																				
12																				
13																				
14																				
15																				
16	Signal Pattern Setting																			
17	Signal Pattern Number is assigned as following table.																			
18	*1 ~ 1000 : For FANUC																			
19	*1001 ~ 2000 : For others																			
20	Signal Pattern																			
21	10																			
22																				
23																				
24																				
25																				
26																				
27																				
28																				
29																				
30																				
31																				
32																				
33																				
34																				
35																				
36																				
37																				
38																				
39																				
40																				
41																				
42																				
43																				
44																				
45																				
46																				
47																				
48																				
49																				
50																				
51																				
52																				
53																				
54																				
55																				
56																				
57																				
58																				
59																				
60																				
61																				
62																				
63																				
64																				
65																				
66																				
67																				
68																				
69																				
70																				
71																				
72																				
73																				
74																				
75																				
76																				
77																				
78																				
79																				
80																				
81																				
82																				
83																				
84																				
85																				
86																				
87																				
88																				
89																				
90																				
91																				
92																				
93																				
94																				
95																				
96																				
97																				
98																				
99																				
100																				
101																				
102																				
103																				
104																				
105																				
106																				
107																				
108																				
109																				
110																				
111																				
112																				
113																				
114																				
115																				
116																				
117																				
118																				
119																				
120																				
121																				
122																				
123																				
124																				
125																				
126																				
127																				
128																				
129																				
130																				
131																				
132																				
133																				
134																				
135																				
136																				
137																				
138																				
139																				
140																				
141																				
142																				
143																				
144																				
145																				
146																				
147																				
148																				
149																				
150																				
151																				
152																				
153																				
154																				
155																				
156																				
157																				
158																				
159																				
160																				
161																				
162																				
163																				
164																				
165																				
166																				
167																				
168																				
169																				
170																				
171																				
172																				
173																				
174																				
175																				
176																				
177																				
178																				
179																				
180																				
181																				
182																				
183																				
184																				
185																				
186																				
187																				
188																				
189																				
190																				
191																				
192																				
193																				
194																				
195																				
196																				
197																				
198																				
199																				
200																				
201																				
202																				
203																				
204																				
205																				
206																				
207																				
208																				
209																				
210																				
211																				
212																				
213																				
214																				
215																				
216																				
217																				
218																				
219																				
220																				
221																				
222																				
223																				
224																				
225																				
226																				
227																				
228																				
229																				
230																				
231																				
232																				
233																				
234																				
235																				
236																				
237																				
238																				
239																				
240																				
241																				
242																				
243																				
244																				
245																				
246																				
247																				
248																				
249																				
250																				
251																				
252																				
253																				
254																				
255																				
256																				
257																				
258																				
259																				
260																				
261																				
262																				
263																				
264																				
265																				
266																				
267																				
268																				
269																				
270																				
271																				
272																				
273																				
274																				
275																				
276																				
277																				
278																				
279																				
280																				
281																				
282																				
283																				
284																				
285																				
286																				
287																				
288																				
289																				
290																				
291																				
292																				
293																				
294																				
295																				
296																				
297																				
298																				
299																				
300																				
301																				
302																				
303																				
304																				
305																				
306																				
307																				
308																				
309																				
310																				
311																				
312																				
313																				
314																				
315																				
316																				
317																				
318																				
319																				
320																				
321																				
322																				
323																				
324																				
325																				
326																				
327																				
328																				
329																				
330																				
331																				
332																				
333																				
334																				
335																				
336																				
337																				
338																				
339																				
340																				
341																				
342																				
343																				
344																				
345																				
346																				
347																				
348																				
349																				
350																				
351																				
352																				
353																				
354																				
355																				
356																				
357																				
358																				
359																				
360																				
361																				
362																				
363																				
364																				
365																				
366																				
367																				
368																				
369																				
370																				
371																				
372																				
373																				
374																				
375																				
376																				
377																				
378																				
379																				
380																				
381																				
382																				
383																				
384																				
385																				
386																				
387																				
388																				
389																				
390																				
391																				
392																				
393																				
394																				
395																				
396																				
397																				
398																				
399																				
400																				
401																				
402																				
403																				
404																				
405																				
406																				
407																				
408																				
409																				
410																				
411																				
412																				
413																				
414																				
415																				
416																				
417																				
418																				
419																				
420																				
421																				
422																				
423																				
424																				
425																				
426																				
427																				
428																				
429																				
430																				
431																				
432																				
433																				
434																				
435																				
436																				
437																				
438																				
439																				
440																				
441																				
442																				
443																				
444																				
445																				
446																				
447																				
448																				
449																				
450																				
451																				
452																				
453																				
454																				
455																				
456																				
457																				
458																				
459																				
460																				
461																				
462																				
463																				
464																				
465																				
466																				
467																				
468																				
469																				
470																				
471																				
472																				
473																				
474																				
475																				
476																				
477																				
478																				
479																				
480																				
481																				
482																				
483																				
484																				
485																				
486																				
487																				
488																				
489																				
490																				
491																				
492																				
493																				
494																				
495																				
496																				
497																				
498																				
499																				
500																				
501																				
502																				
503																				
504																				
505																				
506																				
507																				
508																				
509																				
510																				
511																				
512																				
513																				
514																				
515																				
516																				
517																				
518																				
519																				
520																				
521																				
522																				
523																				
524																				
525																				
526																				
527																				
528																				
529																				
530																				
531																				
532																				
533																				
534																				
535																				
536																				
537																				
538																				
539																				
540																				
541																				
542																				
543																				
544																				
545																				
546																				
547																				
548																				
549																				
550																				
551																				
552																				
553																				
554																				
555																				
556																				
557																				
558																				
559																				

- During the export process, when there are multiple "Basic variable setting" for the same "ID," the "Trigger Name" and the "Variable history retrieval condition" will be output only for the setting of the first basic variable will be output. From the second variable of the same "ID," the "Trigger Name" and the "Variable history retrieval condition" will not be output.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	484/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

- (8) Equipment Status Setting (EquipmentStatusSetting.xls)
- This sheet for each equipment status pattern is created.

No	Priority View	Status	Japanese	English	Chinese	Local language	Type	Signal Definition
example	0	Status001	ステータス001	status001	状态001	status001	Int32	PartsNum.path1.Machine01
example	1	OPERATE	運転中	Operate	运行中	OPERATE	Bool	! Disconnect.Machine01 && (CncState.path1.Machine01 == "OPERATE")
example	0	STOP	停止中	Stop	停止中	STOP	Bool	! Disconnect.Machine01 && (CncState.path1.Machine01 == "STOP")
example	1	DISCONNECT	切断中	DISCONNECT	断开中	DISCONNECT	Bool	! Disconnect.Machine01
example	1	ALARM	アラーム中	ALARM	报警中	ALARM	Bool	! Disconnect.Machine01 && (CncState.path1.Machine01 == "ALARM")
example	0	EMERGENCY	非常停止中	EMERGENCY	紧急停止中	EMERGENCY	Bool	! Disconnect.Machine01 && (CncState.path1.Machine01 == "EMERGENCY")
example	0	SUSPEND	一時停止中	SUSPEND	暂停停止中	SUSPEND	Bool	! Disconnect.Machine01 && (CncState.path1.Machine01 == "SUSPEND")
example	0	STOP	停止中	Stop	停止中	STOP	Bool	! Disconnect.Machine01 && (CncState.path1.Machine01 == "STOP")
example	0	MANUAL	手動運転中	MANUAL	手动运转中	MANUAL	Bool	! Disconnect.Machine01 && (MacroVar.S30.path1.Machine01 == 1)
example	0	WARNING	警告	WARNING	警告	WARNING	Bool	! Disconnect.Machine01 && (CncWarning.path1.Machine01 == "NONE") && (CncWarning.path1.Machine01 == "WARNING")
example	0	condition	状态	condition	状态	condition	String	! Disconnect.Machine01 && (CncWarning.path1.Machine01 == "WARNING")
example	0	Product Name	加工部品名	Product Name	加工零件名	Product Name	String	MainComment.path1.Machine01
example	0	Product Result Number	加工部品数	Product Result Num	加工零件数	Product Result Num	Int32	PartsNum.path1.Machine01
example	0	Product Serial Number	製造番号	Product Serial Num	制造编号	Product Serial Num	String	"P001002009"
example	0	Operator ID	操作者ID	Operator ID	操作者ID	Operator ID	String	"A123456"

Setting Entries

- (a) Web language types
This is an entry that notifies how many languages are supported for (d) Web name. The value on the pattern setting template file is set.
If the pattern setting file is a dummy file, there will be no setting.
- (b) Equipment Status Pattern
The values of this column are sequentially numbered starting with "101" in the order in which equipment units are registered to AdminTool.
If the pattern setting file is a dummy file, "999" will be set.
- (c) Priority View
The viewing priority for each equipment status of the equipment unit is set.
If the pattern setting file is a dummy file, there will be no setting.
- (d) Status
The status of the equipment unit shown in the [Multiple assign status area]
If the pattern setting file is a dummy file, there will be no setting.
- (e) Web Name (Japanese/English/Chinese/Local language)
Web name of the equipment status
If the pattern setting file is a dummy file, there will be no setting.
- (f) Type
The type of the equipment status
- There are three types of equipment status: "Bool" / "Int32" / "UInt32" / "Double" / "String"
- If the pattern setting file is a dummy file, there will be no setting.
- (g) Signal Definition
The signal definition of the equipment status
If the pattern setting file is a dummy file, there will be no setting.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 485/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- This sheet is created for each setting pattern.



- (a) Web language types
Number of languages that support (c) Web title, (e) Web name, and (f) Value. The values of the pattern setting template file are set. In the case of dummy pattern setting file, the values are not set.
- (b) Setting Pattern
The values of this column are sequentially numbered starting with "101" in the order in which machines are registered to AdminTool (only for the machines with the asset management information registered). In the case of dummy pattern setting file, "999" will be set.
- (c) Web Title (Japanese/English/Chinese/Local language)
The titles displayed on a Web browser.
- (d) Image Setting (Images 1 to 5)
The name of the image file for the machine.
The relative path from the output destination of export process (output folder of CollectiveSettingInput.xls)
- (e) Web Name (Japanese/English/Chinese/Local language)
The names of the asset management items.
- (f) Value (Japanese/English/Chinese/Local language)
The values of the asset management items.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	486/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

Files that have been exported can be used as an input file and pattern setting files.

Also, if external files exist, this function imports them and executes relevant settings as shown in the following table.

File Type	Description of Setting	Editable
Input file	- Machine setting (FanucCNC / FanucRobodrill) - Equipment setting - Group setting - Sign-in setting	YES
Pattern setting file	- Signal setting - Custom signal - Tool life signal - Laser control data signal setting - Robodrill signal setting - Macro variable getting condition definition - Variable history retrieval condition definition - Equipment status - Asset setting	YES
External file	- Machine setting (OPCUAServer, FanucRobot, MTConnectAgent) - Overview layout - Authority setting - LDAP authentication setting - Simple assign status - Mail function setting - Signal detection mail delivery setting - Alarm mail delivery setting - Report schedule - Signal data output - Machine data storing schedule - Integration Server setting - DB data retention period setting - Data server setting - Access code setting - Power consumption setting	NO

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 488/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

NOTE

- 1 It is prohibited to export and import files between different versions of MT-LINK*i*. Once MT-LINK*i* is updated, files exported from the earlier version will be invalid. Make sure to follow the steps below when importing files.
 - (1) Update MT-LINK*i* to the latest version.
 - (2) Execute the export procedure.
 - (3) Execute the import procedure using the file exported in step (2).
- 2 The current data set by any setting, be it Machine Setting, Equipment Setting, Group Setting, Layout Setting, Authority Setting, or Sign-in Setting will be set with the values you specified in the input file.
- 3 The import function prohibits using an input file to add a new entry of OPCUAServer, FanucRobot, or MTConnectAgent machine type to the [Machine Setting] window. The only way for this function to import entries of those machine types is to import entries that have already been exported from the window.
- 4 When using exported files to run the import function, make sure to edit the input file and pattern files in the same file structure as when exported. Also, keep in mind that it is prohibited to edit the external files. For the details on the exported file, refer to the "[Export File of Export Function](#)" subsection.

Follow the steps below to import.

- (1) Specify values in the setting entries for Machine Setting, Equipment Setting, Group Setting, and Sign-in Setting in your input file (*.xls). For the details, refer to the "[Configuring Input File](#)" subsection.
- (2) Select [Import] in the AdminTool Menu. The [Import] window appears.
- (3) Import the input file from the [Import] window. For the details, refer to the "[Importing Machine Setting Information](#)" subsection.
- (4) After importing the input file, perform Layout Setting, Manual Data Setting, Authority Setting, Data Validation, and Send to DB operations. For the details, refer to the "[Settings after Importing Input File](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	489/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

9.2.1 Configuring Input File

Use the freeware LibreOffice Calc for configuring the input file.

Follow the steps below to complete specifying entries in the input file. For the input file, the exported file can be used.

(1) Select an input file in the following folder.

- When using the default input file:

<MT-LINK*i* installation folder>\CollectiveSettingInput\

Languages	Name of File
English	CollectiveSettingInput_en.xls
Japanese	CollectiveSettingInput_jp.xls
Chinese	CollectiveSettingInput_cn.xls

- When using the exported file:

<User specified export folder>\CollectiveSettingInput.xls

- When using the sample of the export file for FanucCNC machine type:

<MT-LINK*i* installation folder>\CollectiveSettingInput\SampleData\

Languages	Sub folder/Name of File
English	en\CollectiveSettingInput.xls
Japanese	jp\CollectiveSettingInput.xls
Chinese	cn\CollectiveSettingInput.xls

* This is an exported file with the settings shown as an example for FanucCNC machine type explained in the "FANUC MT-LINK*i* Startup MANUAL."

- When using the sample of the export file for FanucRobodrill machine type:

<MT-LINK*i* installation folder>\CollectiveSettingInput\SampleData_RD\

Languages	Sub folder/Name of File
English	en\CollectiveSettingInput.xls
Japanese	jp\CollectiveSettingInput.xls
Chinese	cn\CollectiveSettingInput.xls

* This is an exported file with the settings shown as an example for FanucRobodrill machine type explained in the "FANUC MT-LINK*i* Startup MANUAL."

NOTE

ROBODRILL Option license is required to use the sample export file for FanucRobodrill machine type.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION				Sheet. 490/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

(2) Specify setting entries for Machine Setting, Equipment Setting, Group Setting, and Sign-in Setting. Hereafter, we will explain how to specify the entries sheet by sheet.

(a) CollectorSetting sheet: Machine Setting

- This sheet has setting entries for Receiver and Collector.

Setting Entries

(i) Receiver Setting

IP Address/Host Name and Port number for the Server PC

- Host Name should start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-") and periods ("."). Host Name must not exceed 253 bytes.
- A value between "1024" and "65535" can be used as a Port number.

(ii) Collector Setting (Read Signal)

For each row, specify Name, IP Address/Host Name, and Port number for a Collector PC.

- You can specify up to 100 Collector PCs.
- A name in the [Name] column should be identical to the computer name of the corresponding Collector PC. For how to identify computer name, refer to the "[Add Collector](#)" subsection.
- A name in the [Name] column can be any combination of half-width alphanumeric characters, hyphens ("-"), and underscores ("_"), provided that its first character should be a half-width alphabet. Name must not exceed 32 bytes.
- If the [Name] column is omitted, the values in the same row will be ignored and thus will not be imported.
- Host Name should start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-"), and periods ("."). Host Name must not exceed 253 bytes.
- A value between "1024" and "65535" can be used as a Port number.

(iii) Collector Setting (SERVO VIEWER Linkage)

Specify the use of SERVO VIEWER linkage function and the port number.

- Select the use of function from "0" (Not Use) / "1" (Use) on the drop-down menu.
- A value between 1024 and 65535 can be used as a Port number.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 491/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

(iv) Collector Setting (Collector Service Manager)

Enter the Port number used by Collector Service Manager running on the Collector PC.

- A value between "1024" and "65535" can be specified as a Port number.

(v) Connected Machine

Specify the machine to be connected to the Collector PC.

- We recommend you to complete the MachineSetting sheet before specifying this entry. Since a drop-down list shows up all the machines you specified in the MachineSetting sheet, you can set the machines easily.

(b) MachineSetting sheet: Machine Setting

- This sheet has setting entries for machines.

The top screenshot shows the 'Machine Setting' sheet with columns for Machine Name, Web Name, Type, Connection, IP Address/Host Name, Port, TimeoutSec, Synchronization Interval (s), GNC Path, and various GNC Path parameters. Red boxes and labels (i) through (viii) highlight specific areas.

The bottom screenshot shows the 'External Link' sheet with columns for External Link, External Link 1, External Link 2, External Link 3, and various External Link parameters. Red boxes and labels (ix) through (xvi) highlight specific areas.

Setting Entries

(i) Web language types (not-editable)

This is a non-editable entry that only notifies how many languages support Web displaying.

(ii) Machine Name

For each row, specify the name of a machine to connect to a Collector that should be defined in the CollectorSetting sheet.

- Up to 100 entries can be specified (the maximum number of connectable machines varies depending on the license option).
- This name can be any combination of half-width alphanumeric characters, hyphens ("-"), and underscores ("_"), provided that its first character should be a half-width alphabet. Machine Name must not exceed 32 bytes.
- Make sure that you have specified correct machine names. You cannot edit machine names on AdminTool after Batch Setting is completed.

20	Nov. 22, 2023	uwano	Revised for MT-LINK ⁱ Version 4.0	Title	FANUC MT-LINK ⁱ AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK ⁱ Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK ⁱ Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK ⁱ Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK ⁱ Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK ⁱ Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK ⁱ Version 3.7	FANUC CORPORATION		Sheet. 492/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

- If this entry is omitted, the values in the same row will be ignored and thus will not be imported.

(iii) Web Name (Japanese/English/Chinese/Local language)

Names for the equipment to be displayed in the three languages on a Web browser

- If this entry is omitted, (ii) Machine Name will be set automatically during import of data.

(iv) Type

Select machine type from the drop-down menu:

"0" (FanucCNC) / "1" (OPCUAServer) / "2" (FanucRobot) / "3" (MTConnectAgent)
/ "4" (FanucRobodrill)

If "1," "2," or "3" is selected, the external file must be exported in advance.

(v) Connection (NOTE 1, 2)

Select "0" or "1" from the drop-down menu, where "0" indicates not to connect, "1" to connect.

- Default value: "1" (to connect)

(vi) Network (NOTE 1, 2)

For each row, specify the IP Address/Host Name, Port number, and Time out (in sec) for the machine defined in the same row.

- Host Name should start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-"), and periods ("."). Host Name must not exceed 253 bytes.
- Default Port number: "8193"
- Default Time out: "30" (Seconds)

Select the synchronization interval (H) from the drop-down menu 0 (Not Synchronized) / 12 (Synchronize (12 hours)) / 24 (Synchronize (24 hours)).

- Default synchronization interval: 0 (Not Synchronized)

(vii) CNC Path (NOTE 1, 2)

For each row, specify the number of CNC paths for the machine in the same row. The drop-down menu provides the options to specify:

- CNC path number should be in a range of "1" to "15" when the [Type] column is set to "0" (FanucCNC).
- CNC path number should be "1" (fixed) when the [Type] column is set to "4" (FanucRobodrill).
- Make sure that the specified numbers are correct. They turn non-editable on AdminTool after Batch Setting is once completed.

(viii) Servo axis Number / Spindle axis Number / L (NOTE 1, 2)

For each path, select the number of servo axes, the number of spindle axes, and the number corresponding to the status of L from the drop-down list.

- "Servo axis number" should be in a range of "0" to "24," and " Spindle axis Number" in a range of "0" to "4" when the [Type] column is set to "0" (FanucCNC).
- "Servo axis number" should be in a range of "3" to "8," and " Spindle axis Number" is fixed to "1" when the [Type] column is set to "4" (FanucRobodrill).
- L can be selected from "0" (no collection of laser control data signals) or "1" (collection of laser control data signals) when the [Type] column is set to "0" (FanucCNC).
- Specify as many the [Path<n>] columns as the value (<n>) set for the [CNC Path] column.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	493/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

- (ix) External Link Validity
Select the validity of the external link from the drop-down list.
0: invalid, 1: valid
- (x) External Link 1 / External Link 2 / External Link 3
Specify a Web name and URL for up to three external links
- English / Japanese / Chinese / a local language
Specify the Web name of each external link in each language.
 - URL
Specify the URL for each external link. The maximum length of this item is 4200 bytes.
- (xi) Use of outlier detection setting file (NOTE 1, 2)
Select from the drop-down menu whether to use or not the outlier detection setting file (FilterSetting.xls), which is an external file, when importing: "0" (Not use), "1" (Use).
- (xii) Signal Pattern (NOTE 1, 2)
Set the pattern number of the signal to use.
"Signal Pattern" can be one of signal patterns that are pre-defined in the pattern setting files; SignalSetting.xls, CustomSignalSetting.xls, MacroSetting.xls, or VariableHistorySetting.xls, or ToolLifeSignalSetting.xls. For the details, refer to the "[Editing Pattern Setting File](#)" subsection.
If you omit this entry, preset values will be set.
- All of the signal pattern definitions mentioned above will be applied.
- (xiii) Laser Signal Pattern (NOTE 1)
Set the pattern number of the laser control data signal to use.
"Laser Signal Pattern" can be one of the signal patterns that are pre-defined in the pattern setting file; [LaserSignalSetting.xls]. For the details, refer to the "[Editing Pattern Setting File](#)" subsection.
If you omit this entry, preset values will be set.
- (xiv) ROBODRILL Signal Pattern (NOTE 2)
Set the pattern number of the ROBODRILL specific signals to use.
"ROBODRILL Signal Pattern" can be one of the signal patterns that are pre-defined in the pattern setting file; [RobodrillSignalSetting.xls]. For the details, refer to the "[Editing Pattern Setting File](#)" subsection.
If you omit this entry, preset values will be set.
- (xv) Asset Setting Pattern
Set the pattern number of the asset management information to use.
"Asset Setting Pattern" can be one of setting patterns that are pre-defined in the asset management information pattern setting file; [AssetSetting.xls]. For the details, refer to the "[Editing Pattern Setting File](#)" subsection.
In the case of the machine that does not register the asset management information, omit this entry (leave it as "blank").
- (xvi) Tool Life Signal Pattern (NOTE 1, 2)
Set the pattern number of the tool life signals to be used.
"Tool Life Signal Pattern" can be one of the signal patterns that are pre-defined in the pattern setting file; [ToolLifeSignalSetting.xls]. For the details, refer to the "[Editing Pattern Setting File](#)" subsection.
If you omit this entry, preset values will be set.

NOTE

- 1 Applicable when the [Type] column is set to "0" (FanucCNC).
- 2 Applicable when the [Type] column is set to "4" (FanucRobodrill).

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	494/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

(c) EquipmentSetting sheet: Equipment Setting

- This sheet has entries for icons to indicate statuses of the equipment, and for machines to include in the equipment.

CollectiveSettingInput_encls - LibreOffice Calc															
File Edit View Insert Format Sheet Data Tools Window Help															
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Equipment Setting														
2	Web Image type (not-editable) 4 (i)														
3															
4	Necessary	Necessary	Optional				Necessary	not-editable	Optional	Optional	Optional	Optional	Optional	Optional	
5	No	Equipment Name	Web Name				Cancel monitor	Record Cycle	Equipment Status Pattern	Use of simple assign file	Normal Image	Disconnect Image	Alarm Image		
6			Japanese	English	Chinese	Local language									
7	example	NewEquipment0001	設備0001	Equipment0001	设备0001	Equipment0001	0	1	1	0	C:\FANUC\MT-LINKi\MT-LINKiAdminTool\MT-LINKiImage\Equipment\MachiningCenter\SVRight\Green_MachiningCenter.S.png	C:\FANUC\MT-LINKi\MT-LINKiAdminTool\MT-LINKiImage\Equipment\MachiningCenter\SVRight\Green_MachiningCenter.S.png	C:\FANUC\MT-LINKi\MT-LINKiAdminTool\MT-LINKiImage\Equipment\MachiningCenter\SVRight\Red_MachiningCenter.S.png		
8	1						0	1							
9	2						0	1							
10	3						0	1							
11	4						0	1							
12	5						0	1							
13	6						0	1							
14	7						0	1							
15	8						0	1							
16		(ii)	(iii)				(iv)	(v)	(vi)	(vii)					
17							0	1							
97	90						0	1							
98	91						0	1							
99	92						0	1							
100	93						0	1							
101	94						0	1							
102	95						0	1							
103	96						0	1							
104	97						0	1							
105	98						0	1							
106	99						0	1							
107	100						0	1							
108							0	1							
CollectiveSetting MachineSetting EquipmentSetting GroupSetting LoginSetting (Reference)SignalPattern_1 (Reference)SignalPattern_1(2) (Reference)SignalPattern_1(5) (Reference)SignalPattern_RDL (Reference)EquipPattern_1 Version															

CollectiveSettingInput_encls - LibreOffice Calc											
File Edit View Insert Format Sheet Data Tools Window Help											
	O	P	Q	R	S	T	U	DO	DP	DQ	DR
1											
2											
3											
4	Optional	Optional	Optional	Optional	Optional	Necessary					
5	Icon Setting					Machine Selection			Note		
6	Emergency Image	Suspend Image	Stop Image	Manual Image	Warmup Image	Machine 1	Machine 2	Machine 100			
7	C:\FANUC\MT-LINKi\MT-LINKiAdminTool\MT-LINKiImage\Equipment\MachiningCenter\SVRight\Green_MachiningCenter.S.png	C:\FANUC\MT-LINKi\MT-LINKiAdminTool\MT-LINKiImage\Equipment\MachiningCenter\SVRight\Orange_MachiningCenter.S.png	C:\FANUC\MT-LINKi\MT-LINKiAdminTool\MT-LINKiImage\Equipment\MachiningCenter\SVRight\Yellow_MachiningCenter.S.png	C:\FANUC\MT-LINKi\MT-LINKiAdminTool\MT-LINKiImage\Equipment\MachiningCenter\SVRight\LightBlue_MachiningCenter.S.png	C:\FANUC\MT-LINKi\MT-LINKiAdminTool\MT-LINKiImage\Equipment\MachiningCenter\SVRight\Brown_MachiningCenter.S.png	NewMachine001					
8											
9											
10											
11											
12											
13											
14											
15											
16											
17											
18											
19											
20											
21											
22											
23											
24											
25											
26											
27											
28											
29											
30											
31											
32											
33											
34											
35											
36											
37											
38											
39											
40											
41											
42											
43											
44											
45											
46											
47											
48											
49											
50											
51											
52											
53											
54											
55											
56											
57											
58											
59											
60											
61											
62											
63											
64											
65											
66											
67											
68											
69											
70											
71											
72											
73											
74											
75											
76											
77											
78											
79											
80											
81											
82											
83											
84											
85											
86											
87											
88											
89											
90											
91											
92											
93											
94											
95											
96											
97											
98											
99											
100											
101											
102											
103											
104											
105											
106											
107											
108											
109											
110											
111											
112											
113											
114											
115											
116											
117											
118											
119											
120											
121											
122											
123											
124											
125											
126											
127											
128											
129											
130											
131											
132											
133											
134											
135											
136											
137											
138											
139											
140											
141											
142											
143											
144											
145											
146											
147											
148											
149											
150											
151											
152											
153											
154											
155											
156											
157											
158											
159											
160											
161											
162											
163											
164											
165											
166											
167											
168											
169											
170											
171											
172											
173											
174											
175											
176											
177											
178											
179											
180											
181											
182											
183											
184											
185											
186											
187											
188											
189											
190											
191											
192											
193											
194											
195											
196											
197											
198											
199											
200											
201											
202											
203											
204											
205											
206											
207											
208											
209											
210											
211											
212											
213											
214											
215											
216											
217											
218											
219											
220											
221											
222											
223											
224											
225											
226											
227											
228											
229											
230											
231											
232											
233											
234											
235											
236											
237											
238											

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL					
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11										
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10										
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN					
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8										
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2										
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION				Sheet.	495/868	
Ed.	Date	Design	Description										
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue						

Setting Entries

- (i) Web language types (not-editable)
This is a non-editable entry that only notifies how many languages support Web displaying.
- (ii) Equipment Name
For each row, specify a name of the equipment.
 - You can specify up to 100 entries.
 - This name can be any combination of half-width alphanumeric characters, hyphens ("-"), and underscores ("_"), provided that its first character should be a half-width alphabet. Equipment Name must not exceed 32 bytes.
 - If this entry is omitted, the values in the same row will be ignored and thus will not be imported.
- (iii) Web Name (Japanese/English/Chinese/Local language)
Names for the equipment to be displayed in the three languages on a Web browser
If omitted, these entries will be regarded as the name specified in the [Equipment Name] column.
- (iv) Cancel monitoring
Either "0" or "1", where "0" indicates the equipment to be monitored, and "1" to exclude it from monitoring targets.
 - Default value: "0" (To be monitored)
 - * Equipment counted out from monitoring will not be included in the total number of the states of equipment display on the Web screen.
For the details on the states of equipment display, refer to "FANUC MT-LINK*i* WebUI OPERATOR'S MANUAL."
- (v) Record Cycle
This entry is non-editable and fixed to one ("1").
- (vi) Equipment Status Pattern
The number of an equipment status pattern prepared in EquipmentStatusSetting.xls.
For the details, refer to the "**Editing Pattern Setting File**" subsection.
A status pattern number according to the statuses in the multiple assign status area will be created for the equipment.
 - If you omit this entry, AdminTool behaves as if you clicked the [Automatic definition of signals] button with the equipment.
 - If you specify a status pattern number, AdminTool creates statuses in the multiple assign status area for the equipment, based on the pattern number. Note that the newly created statuses overwrite and are added to the area.
- (vii) Use of simple assign file
Select from the drop-down menu, whether to use or not the simple assign status file, which is an external file, when importing: "0" (Not use), "1" (Use). On the imported simple assign file, the settings for the Web names of the simple assign status and the display priority is recorded.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	496/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

(viii) Icon Setting

Icons to be used for indicating the 8 (eight) states of equipment units: Normal, Disconnect, Alarm, Emergency, Suspend, Stop, Manual, and Warmup

- For each status, specify the absolute path or the relative path from the input file that includes the image file.

If omitted, these eight entries will be set with the paths to the corresponding default images.

Sample images are available stored in the following folder:

- <MT-LINK*i* installation folder>\MT-LINKiAdminTool\MT-LINKi_image\Equipment
- Supported image formats: png, jpg, and bmp

(ix) Machine Selection

For each row, specify machines to include in the equipment defined in the same row.

If you have completed the MachineSetting sheet before specifying this sheet, you can select machines in a drop-down list.

(d) GroupSetting sheet: Group Setting

- This sheet has entries for the equipment that compose the group, and for the icons that are to be used for the group.

CollectiveSettingInput_uts - LibreOffice Calc

File Edit View Insert Format Data Tools Window Help

A B C D E F G H I J K L DF DG DH DI

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

37

38

39

40

41

42

43

44

45

46

47

48

49

50

51

52

53

54

55

56

57

58

59

60

61

62

63

64

65

66

67

68

69

70

71

72

73

74

75

76

77

78

79

80

81

82

83

84

85

86

87

88

89

90

91

92

93

94

95

96

97

98

99

100

Group Setting

Web language types (not-editable)4(i)

Necessary No	Necessary Group Name	Optional Web Name				not-editable Record Cycle	Optional Read Image		Optional TROUBLE Image		Necessary Equipment Selection			Optional Status setting		Note
		Japanese	English	Chinese	Local language		NORMAL Image				Equipment 1	Equipment 2	Equipment 100	In	Out	
7	example NewGroup0001	グループ0001	Group0001	组别0001	Group0001	1	C:\FANUC\WMT-LINK\WMT-LINK\AdminTool\IWM-TLINK_image\Group\MachiningCenter\Green_MachiningCenter.png		C:\FANUC\WMT-LINK\WMT-LINK\AdminTool\IWM-TLINK_image\Group\MachiningCenter\Red_MachiningCenter.png		New Equipment 0001			New Equipment 001 ProductResultNumber	New Equipment 002 ProductResultNumber	
8	1					1										
9	2					1										
10	3					1										
11	4					1										
12	5					1										
13	6					1										
14	7					1										
15	8					1										
16	9					1										
17	(ii)		(iii)		(iv)			(v)			(vi)			(vii)		
91						1										
92						1										
93						1										
94						1										
95						1										
96						1										
97						1										
98						1										
99						1										
100	100					1										

CollectorSettingMachineSettingEquipmentSettingGroupSettingLoginSetting(Reference)SignalPattern_1(Reference)SignalPattern_1(2)(Reference)SignalPattern_1(5)(Reference)SignalPattern_RD1(Reference)EquipPattem_1Version

Setting Entries

(i) Web language types (not-editable)

This is a non-editable entry that only notifies how many languages support Web displaying.

(ii) Group Name

For each row, specify a name of the group.

- You can specify up to 100 entries.
- Group Name should start with a half-width alphabet and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-"), and periods ("."). Group Name must not exceed 32 bytes.
- If this entry is omitted, the values in the same row will be ignored and thus will not be imported.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	497/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

(iii) Web Name (Japanese/English/Chinese/Local language)
Names for the equipment to be displayed in the three languages on a Web browser
If omitted, these entries will be regarded as the name specified in the [Group Name] column.

(iv) Record Cycle
This entry is non-editable and fixed to one ("1").

(v) Load Image
Icons you use to indicate the normal and alarm statuses

- For each status, you can specify the absolute path to an image file, including the file itself, or the relative path from the input file.

Sample images of the icons are stored in the following folder:

- <MT-LINK*i* installation folder>\MT-LINK*i*AdminTool\MT-LINK*i*_image\Group
- Supported image formats: png, jpg, and bmp

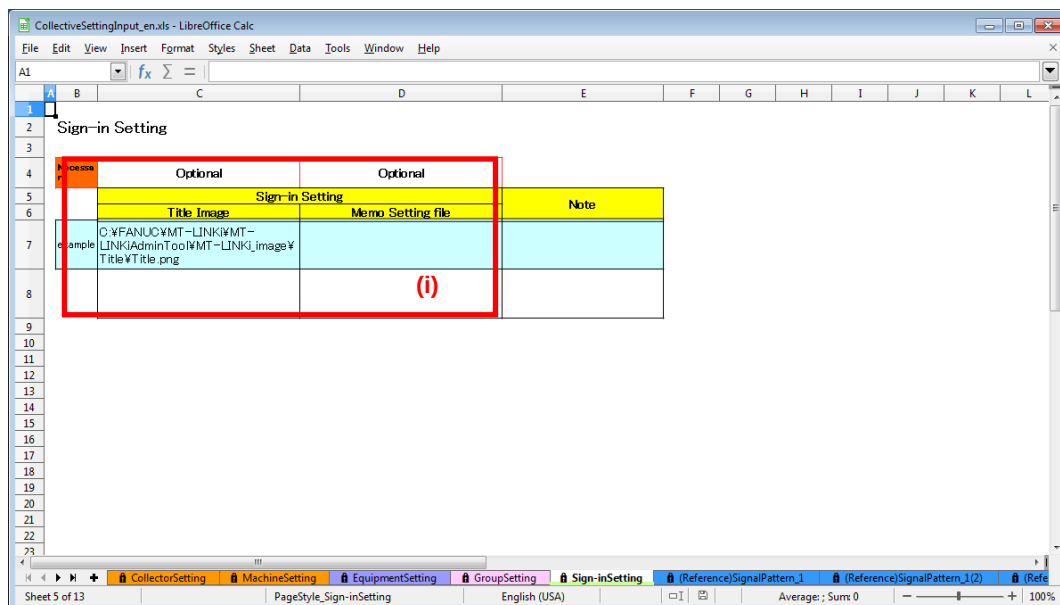
If omitted, these two entries will be set with the paths to the corresponding default images.

(vi) Equipment Selection
For each row, specify the pieces of equipment which the group in the same row consists of.
If you have completed the "EquipmentSetting" sheet before specifying entries in this sheet, you can select equipment in a drop-down list.

(vii) Status Setting
Enter the settings of assigned signals for "In" and "Out."

(e) Sign-inSetting sheet: Sign-in Setting

- This sheet has entries for a title image for the [MT-LINK*i* Web Sign in] screen, and for a text file for displaying information at sign-in.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	FANUC CORPORATION			Sheet.	498/868

Setting Entries

(i) Sign-in Setting

Specify the [Title Image] column with a path to a title image for the [MT-LINK*i* Web Sign in] screen.

- You can specify the absolute path to an image file, including the file itself, or the relative path from the input file.
- Supported image formats are png, jpg, and bmp.
- If omitted, the entry will be set with the path to the default image.

Specify a Memo Setting file with a path to a text file for displaying information at sign-in.

- You can specify the absolute path to a text file, including the file itself, or the relative path from the input file.

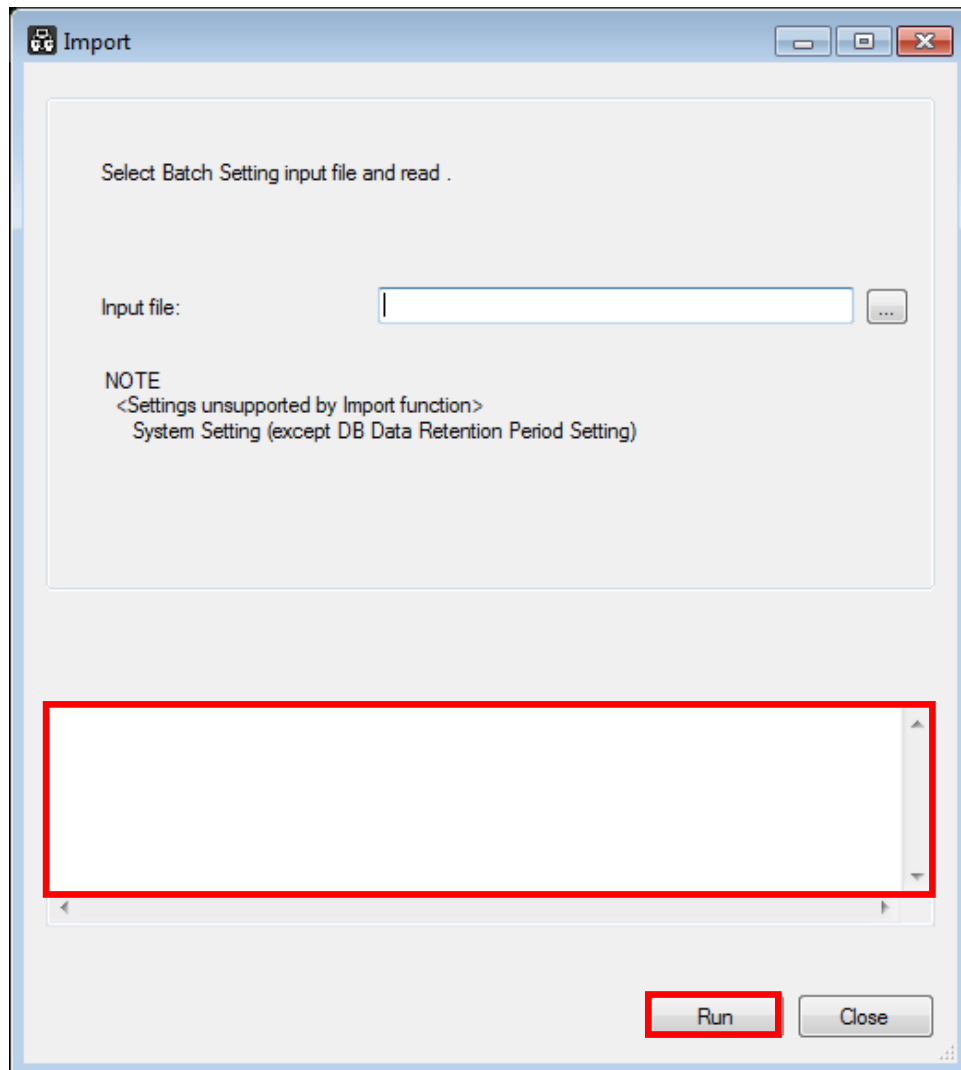
(3) Overwrite save and close the input file.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION				Sheet. 499/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

9.2.2 Importing Machine Setting Information

Perform the following steps to import your input file to AdminTool.

- (1) Select [Import] in the AdminTool Menu.



- (2) The [Import] window appears. Either set the [Input file] field to an input file (.xls) for Batch Setting or click the browse folder button ([...]) to open the [Select XLS File] dialog box, select it, and click the [Open] button to tell the [Input file] field to populate the file path to the selected file. If the specified file is not an xls file, an error message will appear.

The [Select XLS File] dialog box selects the following folder on the first session. On subsequent sessions, it selects the folder selected in the previous session:

<MT-LINK*i* installation folder>\CollectiveSettingInput

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 500/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

(3) Click the [Run] button.

NOTE

- 1 Current setting data, if any, for Machine Setting, Equipment Setting, Group Setting, Layout Setting, Authority Setting, or Sign-in Setting will be overwritten with the new setting values in the input file.
- 2 If schedules are registered using the following functions, after running the [Import] process, a message appears which asks for the confirmation on the registering/unregistering of the schedules.
 - DB data retention period setting (System Setting)
 - Report schedule (daily / weekly / monthly)
 - Signal data output schedule
 - Machine data storing schedule

When [Yes] is selected, the process for registering/unregistering schedule will be executed.

If there is a schedule running at that time, that schedule will be completed without interruption.

When [No] is selected, the process for registering/unregistering schedule will not be executed, and the currently registered schedules will remain as they are.

- (4) The execution log appears in the log area of the [Batch Setting] window.
With the three types of messages, AdminTool informs you whether it can import the input file:
- OK: no errors
 - Warning: minor errors (Data can be imported.)
 - ERROR: critical errors (Until no error is present, AdminTool will stop processing.)

Warnings are shown in the following format:

" => Warning: [<row number> <name>: <column title>: <warning message>]"

Example:

Machine Setting (Collector Setting)

" => Warning: [No.1 Collector01: Machine name: at least one Collector is required for one machine setting] "

Errors are shown in the following format:

" => ERROR: [<row number> <name>: <column title>: <error message>]"

Example:

Machine Setting (Collector Setting)

" => ERROR: [No.1 Machine001: IP address/host name: Invalid IP Address] "

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	501/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

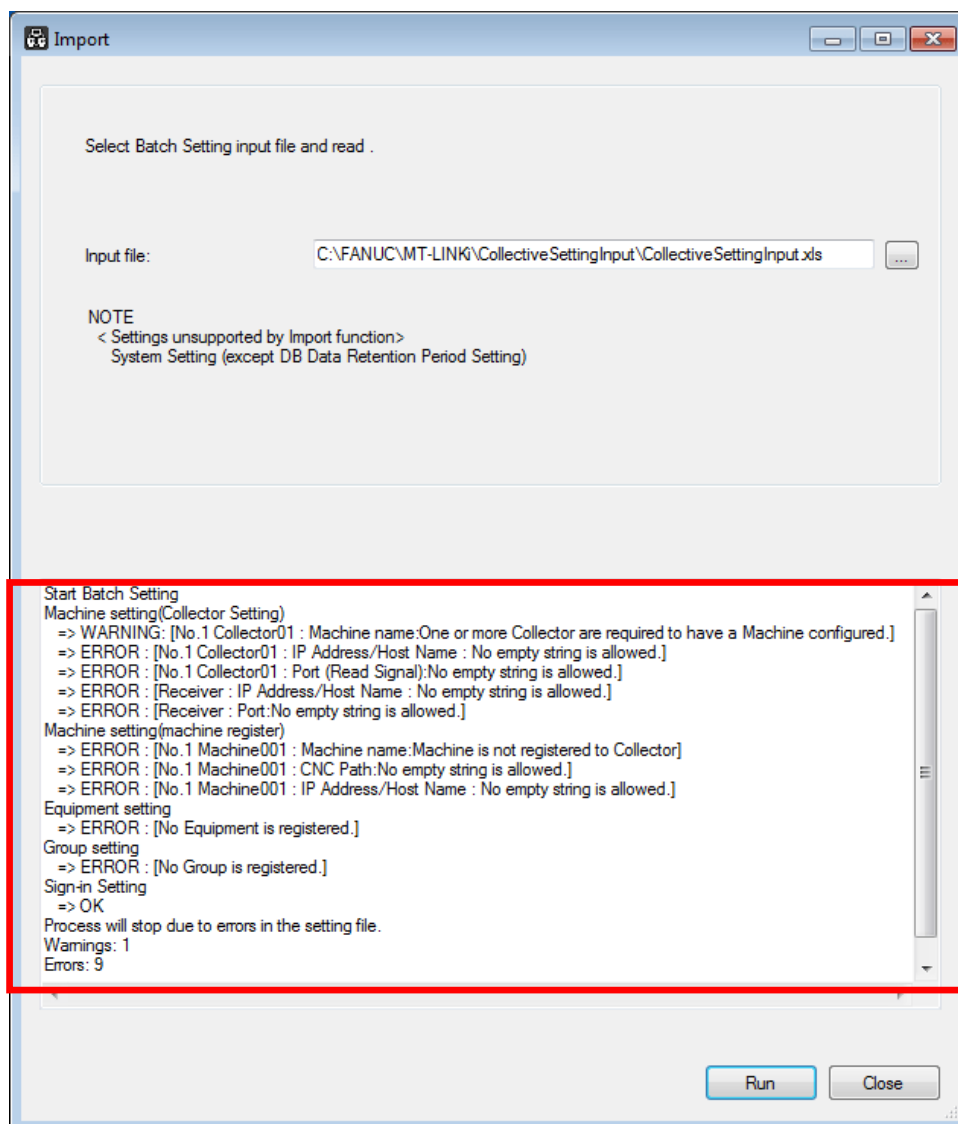
If all the required values, such as a machine name, an equipment name or a group name, are omitted, only an error message will show up.

" => ERROR: [<message>] "

Example:

Equipment setting

" => ERROR: [No Item registered to Equipment.] "



- (5) If errors are present, correct all the defective entries in the input file by carefully reviewing the error messages. If you have warning messages, you may have specified wrong data. Verify again the entries you specified referring to the warning messages.
After correcting the entries, save and close the input file. Repeat step (2) through (5).
- (6) When Batch Setting is completed successfully, the message "Batch Setting completed." appears. Click the [Close] button to close the [Import] window, and perform the "[Settings after Importing Input File.](#)"

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	502/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

9.2.3 Settings after Importing Input File

After importing the input file for the Batch Setting, follow the steps below to perform other setting operations and to perform the [Send to DB] operation.

- If the input file is the exported file (CollectiveSettingInput.xls), there is no need to perform the settings already done in the previous export operation.
- In other cases, it is required to perform the settings.

- (1) Perform layout setting. For the details, refer to the "**LAYOUT SETTING**" section.
- (2) Select [Manual Data Setting] in the AdminTool Menu. For the details, refer to the "**MANUAL DATA SETTING**" section.
- (3) Select [Authority Setting] in the AdminTool Menu to perform Authority Setting. For the details, refer to the "**AUTHORITY SETTING**" section.
- (4) Select [Data Validation] in the AdminTool Menu to check the setting data. For the details, refer to the "**DATA VALIDATION**" section.

NOTE

As explained regarding "(c) EquipmentSetting sheet" in subsection 8.1.1, if you have specified an **Equipment Status Pattern**(vi) in the input file, AdminTool may prompt you with the message "Cannot compile the scripts."

This means that statuses shown in the Multiple assign status area have errors. The "**Add, Edit or Delete Multiple Assign Status**" subsection may help you correct the errors.

- (5) Select [Send to DB] in the AdminTool Menu to send the current setting data to the Database. For the details, refer to the "**SEND TO DB**" section.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	503/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

9.2.4 Editing Pattern Setting File

The pattern setting file is referred to when importing the input file using the import function.

There are two types of pattern setting files, one is created during installation, and the other is created during export process.

The language of the pattern setting file created during installation is dependent on the language setting of the installer. Also, the language of the pattern setting file created during export process is dependent on the language setting of AdminTool.

When importing, you can use either the installed pattern setting file, or the pattern setting file that was automatically created during export. Both pattern setting files are editable.

For the details on the format and setting values of the exported file, refer to the "[Export File of Export Function](#)" subsection.

■ Pattern setting files

No.	Setting Types	File Name
1	Signal	SignalSetting.xls
2	Custom signal	CustomSignalSetting.xls
3	Tool life signal	ToolLifeSignalSetting.xls
4	Laser control data signal	LaserSignalSetting.xls
5	Robodrill signal setting	RobodrillSignalSetting.xls
6	Macro variable getting condition definition	MacroSetting.xls
7	Variable history retrieval condition definition	VariableHistorySetting.xls
8	Equipment status	EquipmentStatusSetting.xls
9	Asset management	AssetSetting.xls

■ Pattern setting file templates (the table lists optional languages and file names)

No.	Language	Setting Types	File Name
1	English	Signal	SignalSetting_en.xls
		Custom signal	CustomSignalSetting_en.xls
		Tool life signal	ToolLifeSignalSetting_en.xls
		Laser control data signal setting	LaserSignalSetting_en.xls
		Robodrill signal setting	RobodrillSignalSetting_en.xls
		Macro variable getting condition definition	MacroSetting_en.xls
		Variable history retrieval condition definition	VariableHistorySetting_en.xls
		Equipment status	EquipmentStatusSetting_en.xls
		Asset management	AssetSetting_en.xls
2	Japanese	Signal	SignalSetting_jp.xls
		Custom signal	CustomSignalSetting_jp.xls
		Tool life signal	ToolLifeSignalSetting_jp.xls
		Laser control data signal setting	LaserSignalSetting_jp.xls
		Robodrill signal setting	RobodrillSignalSetting_jp.xls
		Macro variable getting condition definition	MacroSetting_jp.xls
		Variable history retrieval condition definition	VariableHistorySetting_jp.xls
		Equipment status	EquipmentStatusSetting_jp.xls
		Asset management	AssetSetting_jp.xls
3	Chinese	Signal	SignalSetting_cn.xls
		Custom signal	CustomSignalSetting_cn.xls

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 504/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

No.	Language	Setting Types	File Name
		Tool life signal	ToolLifeSignalSetting_cn.xls
		Laser control data signal setting	LaserSignalSetting_cn.xls
		Robodrill signal setting	RobodrillSignalSetting_cn.xls
		Macro variable getting condition definition	MacroSetting_cn.xls
		Variable history retrieval condition definition	VariableHistorySetting_cn.xls
		Equipment status	EquipmentStatusSetting_cn.xls
		Asset management	AssetSetting_cn.xls

■ Folders for the pattern setting files and the pattern setting file templates

* The pattern setting files are the files used for editing. (Note that templates are prepared in case a pattern setting file is collapsed.)

Upon installation, the following folders are used to store those files.

Pattern setting files:

<MT-LINK*i* installation folder>\MT-LINK*i*AdminTool\DefineSetting

Pattern setting file templates:

<MT-LINK*i* installation folder>\MT-LINK*i*AdminTool\DefineSetting\DefineTemplate

* Pattern setting file templates are provided for use as backup of the pattern setting files. In case when one of the pattern setting files corrupts, those template files can be helpful to restore it.

Upon exporting those files, the following folders are used to store them.

Pattern setting files:

<User specified export folder>\DefineSetting

Pattern setting file templates:

<MT-LINK*i* installation folder>\MT-LINK*i*AdminTool\DefineSetting\ExportTemplate.

* MT-LINK*i* uses the pattern setting file templates in exporting files.

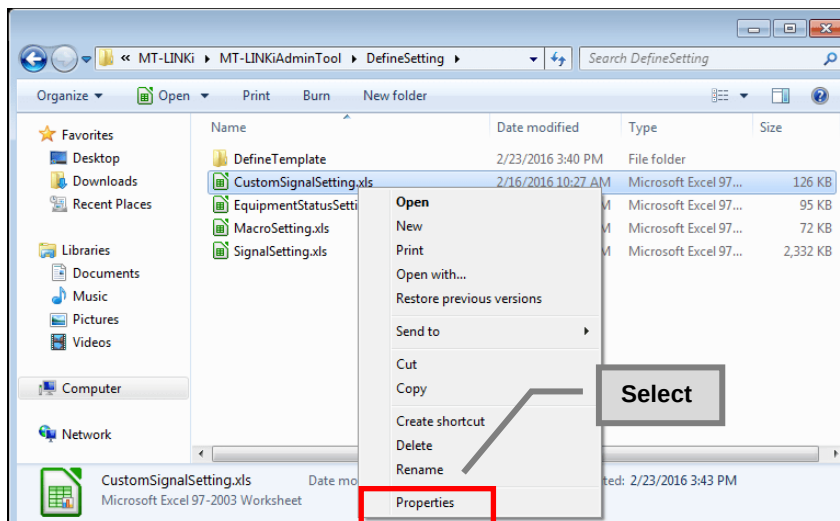
Do not edit the contents of those templates nor overwrite them with pattern setting files because they are not identical to each other.

Note that the pattern setting files have read-only property and are protected except for the areas where input is required. When you edit a pattern file, change the read-only property by following the steps below.

NOTE

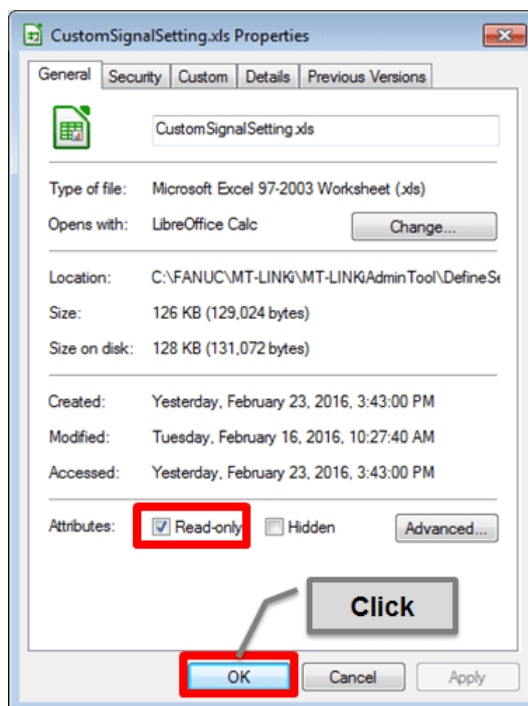
Do not unprotect areas that are not required to input.

(1) Right-click the target file. A pop-up menu appears. Select [Properties]



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	505/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

(2) Clear the [Read-only] check box and click [OK].



■ Entering settings of the pattern setting file

Follow the steps below to enter settings in the pattern setting files. Follow the steps below also when editing the exported pattern setting file. As for the setting values of the pattern setting file, refer to the "**Export File of Export Function**" subsection.

(1) Signal Setting (SignalSetting.xls)

- This file defines signals per machine.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	506/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

Setting Entries

(a) Signal Pattern

Set the pattern number that identifies the signal setting pattern to use

- Default value: "1"
- If you add a signal pattern, add a new sheet named "Pattern_<x>" by copying the "Pattern_1" sheet. (Note that <x> should be a numeric character; otherwise, Batch Setting cannot load the file. Set a signal pattern for <x>.) * In case that this file has duplicate sheet names, Batch Setting will load only the first sheet.

(b) Axis

Set the reading cycle of the signal for each axis column. If there are no axes, use the [Null] column. For the axis which the signal will be collected, select the reading cycle from "500" / "10000" / "60000" (unit: msec).

Set "0" for the axis of which the signal will not be collected.

- Grey-marked cells in the sheet are non-editable because such combinations of a signal and an axis never occur.
- For the following signals, the reading cycle will be fixed at "500" regardless of the specified value. (Not editable)
 - Mandatory signals (MainProgram / ActProgram / CncState / CncWarning / Alarm)
 - Operator message/Macro message (OpMacroMsg)

(2) Custom Signal (CustomSignalSetting.xls)

- This file defines custom signals per machine.
- Custom signals can be registered to a maximum number of 1000.
- A row whose entries have no value will not be loaded.
- Even if you edit "not Necessary" entries, the edited value will not be loaded.

CustomSignalSetting_en.xls - LibreOffice Calc

No	Type	CNC Path/PMC Path	Address#1	Number#2	Data Type	Data Length	End Code	Diagnosis Data Type	Bit	Axis	Reading Cycle	Note
1	PMC	Necessary*3	Necessary	Necessary	Necessary	Necessary*10	Necessary*10*11	Null	Necessary*4	Null	Necessary	*3 Enter the PMC path *4 Necessary if Data type is "BIT" *10 Necessary if Data type is "STRNG" To disable this property, set no value for it
2	MacroVar	Necessary*5	Null	Necessary	Null	Null	Null	Null	Null	Null	Necessary	*5 Enter the CNC path
3	ProbeVar	Necessary*5	Null	Necessary	Necessary*6	Null	Null	Null	Null	Null	Necessary	*5 Enter the CNC path *6 Data types are as following: Conversation: Conversation macro Auxiliary: Auxiliary macro Execution: Execution macro
4	Diagnosis	Necessary*5	Null	Necessary	Necessary*7	Null	Null	Necessary*8	Necessary*4	Necessary*9	Necessary	*4 Necessary if Data type is "BIT" *5 Enter the CNC path *7 Data types are as following: BIT: Bit DWORD: Integer REAL: Real *8 Diagnosis Data Types are as following: Path: Path Axis: Axis Spindle: Spindle *9 Necessary if Diagnosis Data Type is "Axis" or "Spindle"

*1 Address can be 3 digit
*2 Number can be 5 digit (For MacroVar: 6 digit)
*11 To designate an end code not listed in the drop-down list, enter it in hexadecimal (01 to FF)

No	CNC Path / PMC Path	Type	Label	Address	Number	Data Type	Data Length	End Code	Diagnosis Data Type	Bit	Axis	Reading Cycle (msec)	Note
example 1	PMC	PMC	PMC	A	1234	BIT	8			0		500	
example 2	PMC	PMC	PMC	B	5678	WORD	8					60000	
example 3	PMC	PMC	PMC	B	8	WORD	255	ETX (0x03)				500	
example 4	PMC	PMC	PMC	B	255	WORD	8	7F				500	
example 5	MacroVar	MacroVar	MacroVar		123	Conversation						10000	
example 6	ProbeVar	ProbeVar	ProbeVar		1234	BIT						500	
example 7	Diagnosis	Diagnosis	Diagnosis		1234	BIT				0		60000	
example 8	Diagnosis	Diagnosis	Diagnosis		5678	DWORD					0	60000	

(b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (j)

Setting Entries

(a) Signal Pattern

Set the pattern number that identifies the custom signal setting pattern to use

- Default value: "1"

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	507/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

- If you add a signal pattern, add a new sheet named "Pattern_<x>" by copying the "Pattern_1" sheet. (Note that <x> should be a numeric character; otherwise, Batch Setting cannot load the file. Set a signal pattern for <x>.) * In case that this file has duplicate sheet names, Batch Setting will load only the first sheet.
- (b) CNC Path / PMC Path
Select the CNC Path or PMC Path from the drop-down list to use for a custom signal
- When setting PMC to the [Type] column of the same row, select a number from "1" to "5" or set number nine ("9"). * To get the value of a dual check safety PMC, specify number nine ("9").
 - When setting MacroVar, PcodeVar, or Diagnosis to the [Type] column of the same row, set a CNC path.
- (c) Type
Select the type of custom signal from the drop-down list
- Options are "PMC," "MacroVar," "PcodeVar," and "Diagnosis."
 - The [Label] column is set with the value of the [Type] column. There is no need to edit the [Label] column.
- (d) Address
Enter the address of a custom signal
- If you set PMC to the [Type] column of the same row, you must specify an appropriate value.
 - Address should be set to one of "G" / "F" / "Y" / "X" / "A" / "R" / "T" / "K" / "C" / "D" / "M" / "N" / "E" / "Z".
- (e) Number
Enter the number of a custom signal
- (f) Data Type
Select the data type of a custom signal (PMC and PcodeVar) from the drop-down list
- Necessary to set a value if "PMC," "PcodeVar," or "Diagnosis" is selected for (c) Type.
 - PMC: Select from the following
"BIT" / "BYTE" / "WORD" / "DWORD" / "REAL" (single precision real) / "LREAL" (double precision real) / "UBYTE" (unsigned byte) / "UWORD" (unsigned word) / "UDWORD" (unsigned double word) / "STRING" (string)
 - PcodeVar: Select from the following
"Conversation" (macro) / "Auxiliary" (macro) / "Execution" (macro)
 - Diagnosis: Select from the following
"BIT" (Bit) / "DWORD" (Integer) / "REAL" (Real)
- (g) Data Length
Set the length of string data in bytes.
- If the [Type] column and the [Data Type] column are set to "PMC" and "STRING," respectively, this column must be set a value.
 - Valid values are numbers in a range of 1 to 255.
- (h) End Code
Select an end code from the drop-down list, or designate an end code.
- If the [Type] column and the [Data Type] column are set to "PMC" and "STRING," respectively, this column must be set a value.
 - Selectable options in the drop-down list are "NULL (0x00)," "ETX (0x03)," "LF (0x0A)," and "CR (0x0D)."

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	508/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

- Set a hexadecimal value in a range of 00 to FF when designating an end code other than pre-defined options.
- (i) **Diagnosis Data Type**
Select a diagnosis data type from the drop-down list.
- If the [Type] column of the same row is set to "Diagnosis," the [Diagnosis Data Type] column must be set to "Path," "Axis," or "Spindle."
- (j) **Bit**
Select a bit position from the drop-down list.
- If the [Type] column of the same row is set to "PMC" or "Diagnosis" and the [Data Type] column is set to "BIT," the [Bit] column must be set to a value from "0" to "7."
- (k) **Axis**
Select an axis number from the drop-down list.
- Necessary to set a value if "Diagnosis" is selected for (c) Type.
 - Axis: 0 to <servo axis number>, where <servo axis number> is a servo axis number for the CNC path that is set in the CNC Path / PMC Path ((b) shown above), and its value is calculated by subtracting 1 from the value of its [Servo axis number] column of the [MachineSetting] sheet in the input file (CollectiveSettingInput.xls).
 - Spindle: 0 to <spindle axis number>, where <spindle axis number> is a spindle axis number for the CNC path that is set in the CNC Path / PMC Path ((b) shown above), and its value is calculated by subtracting 1 from the value of its [Spindle axis number] column of the [MachineSetting] sheet in the input file (CollectiveSettingInput.xls).
- (l) **Reading Cycle**
Select the cycle time of reading a signal from the drop-down list.
- Options are "500" / "10000" / "60000" in unit of milliseconds.

■ Required (reqd.) or not required (not reqd.) items for Type

Type	CNC Path/ PMC Path	Address	Number	Data Type	Data Length	End Code	Diagnosis Data Type	Bit	Axis	Reading Cycle
PMC	reqd. (NOTE 1)	reqd.	reqd.	reqd.	reqd. (NOTE 1)	reqd. (NOTE 1)	not reqd.	reqd. (NOTE 3)	not reqd.	reqd.
MacroVar	reqd. (NOTE 4)	not reqd.	reqd.	not reqd.	not reqd.	not reqd.	not reqd.	not reqd.	not reqd.	reqd.
PcodeVar	reqd. (NOTE 4)	not reqd.	reqd.	reqd.	not reqd.	not reqd.	not reqd.	not reqd.	not reqd.	reqd.
Diagnosis	reqd. (NOTE 4)	not reqd.	reqd.	reqd.	not reqd.	not reqd.	reqd. (NOTE 5)	reqd. (NOTE 3)	reqd. (NOTE 6)	reqd.

NOTE

- 1 A PMC path number is required.
- 2 When the [Data Type] column is set to "STRING," either or both of the [Data Length] and the [End Code] columns must be set to a value.
- 3 Required when the [Data Type] column is set to "BIT."
- 4 A CNC path number is required.
- 5 When retrieving diagnosis data of spindle axes from Series 16*i* models, specify "Path" for the [Diagnosis Data Type] column.
- 6 Required when the [Diagnosis Data Type] column is set to "Axis" or "Spindle."

For Series 16*i* models, refer to the NOTE 4 in the "**Basic Signals of FanucCNC**" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	509/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

(3) Tool Life Signal Setting (ToolLifeSignalSetting.xls)

- This file defines the retrieval condition of tool life signals per machine.

Signal	Reading Cycle	Note
ToolGroupNum	500	Current tool group
ToolLifeCount01	10000	Tool life count 01
ToolLifeCount02	10000	Tool life count 02
ToolLifeCount03	10000	Tool life count 03
ToolLifeCount04	10000	Tool life count 04
ToolLifeCount05	10000	Tool life count 05
ToolLifeCount06	10000	Tool life count 06
ToolLifeCount07	10000	Tool life count 07
ToolLifeCount08	10000	Tool life count 08
ToolLifeCountType254	10000	Tool life count type 254
ToolLifeCountType255	10000	Tool life count type 255
ToolLifeCountType256	10000	Tool life count type 256

Setting Entries

(a) Signal Pattern

Enter the pattern number that identifies the tool life signal pattern to use

- the default value: "1"
- If you add a tool life signal pattern, add a new sheet named "Pattern_<x>" by copying the "Pattern_1" sheet. (Note that <x> should be a numeric character; otherwise, Batch Setting cannot load the file. Set a tool life signal pattern for <x>.)

* In case that this file has duplicate sheet names, Batch Setting will load only the first sheet.

(b) Tool Groups

Enter numbers of the tool groups for each CNC path; the path number of those CNC paths ranges from 1 to 15.

- the default value: "0"
- the value can be in a range of "0" to "256."

NOTE

The total number of tool groups across all the CNC paths must not exceed 256.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 510/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

(c) Reading Cycle

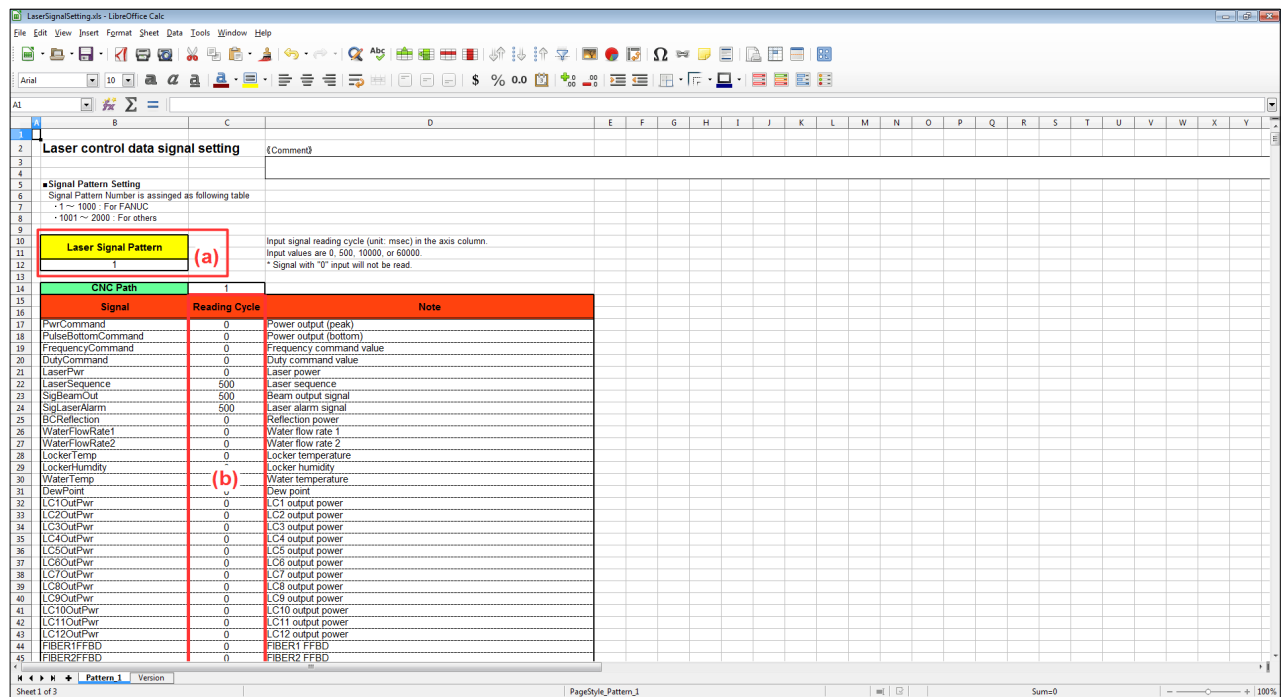
Enter the reading cycle of the tool life signal.

When retrieving the signal, options are "500" / "10000" / "60000" (unit: msec).

Enter "0" when not retrieving the signal.

(4) Laser control data signal setting (LaserSignalSetting.xls)

- This file defines the retrieval condition of laser control data signals per machine.



Setting Entries

(a) Laser Signal Pattern

Enter the pattern number that identifies the laser signal setting to use

- the default value: "1"

- If you add a laser signal pattern, add a new sheet named "Pattern_<x>" by copying the "Pattern_1" sheet. (Note that <x> should be a numeric character; otherwise, Batch Setting cannot load the file. Set a laser signal pattern for <x>.)

* In case that this file has duplicate sheet names, Batch Setting will load only the first sheet.

(b) Reading Cycle

Enter the reading cycle of the laser control data signal.

When retrieving the signal, options are "500" / "10000" / "60000" (unit: msec).

Enter "0" when not retrieving the signal.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 511/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

(5) ROBODRILL Signal Setting (RobodrillSignalSetting.xls)

- This file defines the retrieval condition of ROBODRILL specific signals per machine and is available for FanucRobodrill type machines.

Signal Pattern		CNC Path		Axis		Note	
Signal	Null	X axis	Y axis	Z axis	B axis	A axis	U axis
Robodrill tool number	500						
Robodrill machining mode	0						
Robodrill power save mode	0						
Robodrill rigid tapping mode	0						
Robodrill table load mode	0						
Robodrill tool load mode	0						
Robodrill production counter 1	10000						
Robodrill production target 1	0						
Robodrill counter part name 1	0						
Robodrill production counter 2	10000						
Robodrill production target 2	0						
Robodrill counter part name 2	0						
Robodrill iHMI production counter 1	0						
Robodrill iHMI production target 1	0						
Robodrill iHMI counter part name 1	0						
Robodrill iHMI production counter 2	0						
Robodrill iHMI production target 2	0						
Robodrill iHMI counter part name 2	0						
Robodrill iHMI production counter 3	0						
Robodrill iHMI production target 3	0						
Robodrill iHMI counter part name 3	0						
Robodrill program protection	0						
Robodrill operation level	0						
Robodrill electric power consumption	0						
X-axis thermal displacement compensation enable	0						
Y-axis thermal displacement compensation enable	0						
Z-axis thermal displacement compensation enable	0						
X-axis thermal displacement compensation adjust value	0						
Y-axis thermal displacement compensation adjust value	0						

Setting Entries

(a) Signal Pattern

Enter the pattern number that identifies the Robodrill signal pattern to use

- the default value: "1"
- If you add a tool life signal pattern, add a new sheet named "Pattern_<x>" by copying the "Pattern_1" sheet. (Note that <x> should be a numeric character; otherwise, Batch Setting cannot load the file. Set a Robodrill signal pattern for <x>.)

* In case that this file has duplicate sheet names, Batch Setting will load only the first sheet.

(b) Axis

Set the reading cycle of the signal to the [Null] column. Selectable options are "500" / "10000" / "60000" (unit: msec). Set "0" when the signal will not be collected.

- Grey-marked cells in the sheet are non-editable because such combinations of a signal and an axis never occur.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 512/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

(6) Macro variable getting condition definition (MacroSetting.xls)

- This file defines the retrieval condition of macro variables per machine.
- "Macro Variable getting condition" and "Variable history retrieval condition" can be registered to a maximum number of 1000 in total.

Setting Entries

- (a) Web language types (not-editable)
This is a non-editable entry that only notifies how many languages support displaying Web names. See entry (d).
- (b) Signal Pattern
Set the pattern number that identifies the pattern setting of the Macro Variable getting condition definition to use
- Default value: "1"
 - If you add a signal pattern, add a new sheet named "Pattern_<x>" by copying the "Pattern_1" sheet. (Note that <x> should be a numeric character; otherwise, Batch Setting cannot load the file. Use signal pattern for <x>.) * In case that this file has duplicate sheet names, Batch Setting will load only the first sheet.
- (c) ID
Enter "ID" at the top in the field, and then a four digit number.
The four digit number should be the unused minimum number that is equal to "0001" or greater (for example, ID0001).
- (d) Web Name (Japanese/English/Chinese/Local language)
Enter the name to display on a Web browser. Web Name must not exceed 100 bytes.
If this entry is omitted, the [ID] column will be set automatically during import of data.
- (e) Trigger Macro Variable
Enter the trigger macro variable to be used for the Macro Variable getting condition definition
- Default: 500
 - Available macro variables: 100 to 999

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	513/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

- (f) CNC Path
Select the CNC Path from the drop-down list to use for the Macro Variable getting condition definition
- Available value: 1 to 15
- (g) Standard Macro
Enter the basic variable for the Macro Variable getting condition definition
- Default: 501
 - Available macro variables: 1 to 33, and 100 to 99999999
- (h) Collecting Number
Set the number of variables to retrieve
- Default: 1
 - Maximum number: 256
- (i) Web Name of Macro Variable (Japanese/English/Chinese/Local language)
Enter the name of the macro variable to display on a Web browser. Up to 256 Web names can be set for each language.
- Maximum length of the character string: 32 bytes.
- (7) Variable History Retrieval Condition Definition (VariableHistorySetting.xls)
- This file defines the retrieval condition of variable histories per machine.
 - "Macro Variable getting condition" and "Variable history retrieval condition" can be registered to a maximum number of 1000 in total.

Setting Entries

(a) Web language types (not editable)

This is a non-editable entry that only notifies how many languages support displaying Trigger names.
See entry (d).

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 514/868

- (b) Signal Pattern
Enter the pattern number that identifies the pattern setting of the Variable history retrieval condition definition to use
- Default: 1
 - If you add a signal pattern, add a new sheet named "Pattern_<x>" by copying the "Pattern_1" sheet. (Note that <x> should be a numeric character; otherwise, Batch Setting cannot load the file. Use signal pattern for <x>.) * In case that this file has duplicate sheet names, Batch Setting will load only the first sheet.
- (c) ID
- Enter "ID" at the top in the field, and then a four digit number.
 - The four digit number should be the unused minimum number that is equal to "0001" or greater (for example, ID0001).
 - "Macro Variable getting condition" and "Variable history retrieval condition" can be registered to a maximum number of 1000 in total. (Duplicate ID's are counted)
- (d) Trigger Name (Japanese/English/Chinese/Local language)
Enter the name to display on a Web browser.
- If this entry is omitted, the [ID] column will be set automatically during import of data.
 - Maximum length of the character string: 32 bytes.
- (e) Trigger Variable Setting
- (i) Type
Select the type of the trigger variable from the drop-down menu.
- Only "MacroVar" can be selected.
- (ii) Number
Enter the number of the trigger variable.
- Default: 500
 - When Type is set to "MacroVar," a number in the range of 100 to 999 is available.
- (iii) Path
Select a CNC path.
- A number in the range of 1 to 15 is available.
- (f) Basic Variable Setting
- (i) Type
From the drop-down menu, select the type of the variable to retrieve.
- Options are "PMC," "MacroVar," and "PcodeVar."
- (ii) Address
Enter the address of the variable to retrieve.
- Required only when Type is set to "PMC."
- (iii) Number
Enter the number of the variable to retrieve.
- Default: 0
 - When Type is set to "PMC," enter the address of the PMC signal. Numbers in the range of 100 to 99999 are available.
 - When Type is set to "MacroVar," numbers in the ranges of 1 to 33 and 100 to 99999999 are available.
 - When Type is set to "PcodeVar," numbers in the ranges of 1 to 33 and 100 to 89999 are available.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	515/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

(iv) Data Type

Select the data type of the variable from the drop-down menu.

- When Type is set to "MacroVar," this column does not need to be set.
- When Type is set to "PMC" or "PcodeVar," select one of the following:
"BIT" / "BYTE" / "WORD" / "DWORD" / "REAL" / "LREAL" / "UBYTE" / "UWORD" /
"UDWORD" / "STRING" (string) / "Conversation" / "Auxiliary" / "Execution"
- * Consistency of the type and the data type is checked during import process.

(v) Data Length

Set the length of string data in bytes.

- If the [Type] column and the [Data Type] column are set to "PMC" and "STRING," respectively, this column must be set a value.
- Valid values are numbers in a range of 1 to 255.

(vi) End Code

Select an end code from the drop-down list, or designate an end code.

- If the [Type] column and the [Data Type] column are set to "PMC" and "STRING," respectively, this column must be set a value.
- Selectable options in the drop-down list are "NULL (0x00)," "ETX (0x03)," "LF (0x0A)," and "CR (0x0D)."
- Set a hexadecimal value in a range of 00 to FF when designating an end code other than pre-defined options.

(vii) Bit

Select the bit number of the variable to retrieve from the drop-down menu.

- This column is available only when the trigger variable is of PMC type and BIT data type.
- A value in the range of 0 to 7 is available.

(viii) Path

Select a CNC/PMC path.

- When Type is set to "MacroVar" or "PcodeVar," select a CNC path.
- When Type is set to "PMC," select a PMC path.
- A number in the range of 1 to 15 is available.

(ix) Collecting number

Enter the number of variables to retrieve

- Default: 1
- Maximum number: 256

(g) Variable Name (Japanese/English/Chinese/Local language)

Enter the name of the variable to retrieve

- Up to 256 names can be set for each language.
- Maximum length of the character string is 32 bytes.

NOTE

During the import process, when there are multiple "Basic Variable Setting" for the same "ID," the "Trigger Name" and the "Variable history retrieval condition" written in the setting of the first basic variable will be valid. From the second variable of the same "ID," the "Trigger Name" and the "Variable history retrieval condition" can be omitted as it will not be used.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	516/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

(8) Equipment Status (EquipmentStatusSetting.xls)

- This file defines equipment statuses per equipment to be assigned as multiple assign statuses.

Equipment Status Setting.xls (read-only) - LibreOffice Calc

File Edit View Insert Format Sheet Data Tools Window Help

A B C D E F G H I J

1

2 Equipment Status Web language types (not-editable) 4 (a) (Comment)

3

4 ■Equipment Status Pattern Setting

5 Equipment Status Pattern Number is assigned as following table

6 ·1 ~ 1000 : For FANUC

7 ·1001 ~ 2000 : For others

8

9 Equipment Status Pattern (b)

10 1

11

12

13

14 No Priority View Status Web Name Type Signal Definition

15 Japanese English Chinese Local language

16 example 0 Status001 ステータス001 status001 status001 Int32 ParaNum_path1 [MACHINE]

17 example 1 OPERATE 運転中 Operate Bool (! Disconnect [MACHINE]) && (OnCState_path1 [MACHINE]) == "OPERATE")) || (! Disconnect [MACHINE]) && (OnCState_path1 [MACHINE]) == "OPERATE"))

18 example 0 STOP 停止中 Bool

19 2 Bool

20 3 Bool

21 4 Bool

22 (c) (d) (e) (f) (g)

1011

1012 998 0 Bool

1013 999 0 Bool

1014 999 0 Bool

1015 999 0 Bool

1016 1000 0 Bool

1017

Pattern 1 Version

Setting Entries

- (a) Web language types (not-editable)
This is a non-editable entry that only notifies how many languages support displaying Web names.
See entry (e).
- (b) Equipment Status Pattern
Enter the pattern number that identifies the equipment status setting pattern to use
 - If you add a status pattern, add a new sheet named "Pattern_<x>" by copying the "Pattern_1" sheet. (Note that <x> should be a numeric character; otherwise, Batch Setting cannot load the file. Use signal pattern for <x>.)
 - * In case that this file has duplicate sheet names, Batch Setting will load only the first sheet.
- (c) Priority View
Select either "0" or "1" from the drop-down list, where "0" indicates as being not selected in the [PriorityView] drop-down list on the [Equipment Setting] window of AdminTool, and "1" as being selected.
- (d) Status
Enter the status to add to the equipment in the [Multiple assign status] table on the [Equipment Setting] window of AdminTool
 - If you specify a name of pre-set statuses, the pre-set status will be replaced with the new status.
- (e) Web Name (Japanese/English/Chinese/Local language)
Enter the name for the equipment status to display in the three languages on a Web browser
If this entry is omitted, the [Status] column will be set automatically during import of data.
- (f) Type
Select type ("Bool" / "Int32" / "UInt32" / "Double" / "String") from the drop-down list.
- (g) Signal Definition
Enter the definition of the signal
 - At the point when configuring this file, there is no way to know the names of the machines to compose the equipment. We employ the following rule to map a machine name in this file to the name of an actual machine. Note that if the equipment does not have a machine specified in this file, data checking after loading pattern files will alarm because the mapping rule cannot apply to this case:

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION				Sheet.	517/868
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						

■ Rule for mapping machine name

(Example)

[MACHINE1] => the first machine of an equipment unit

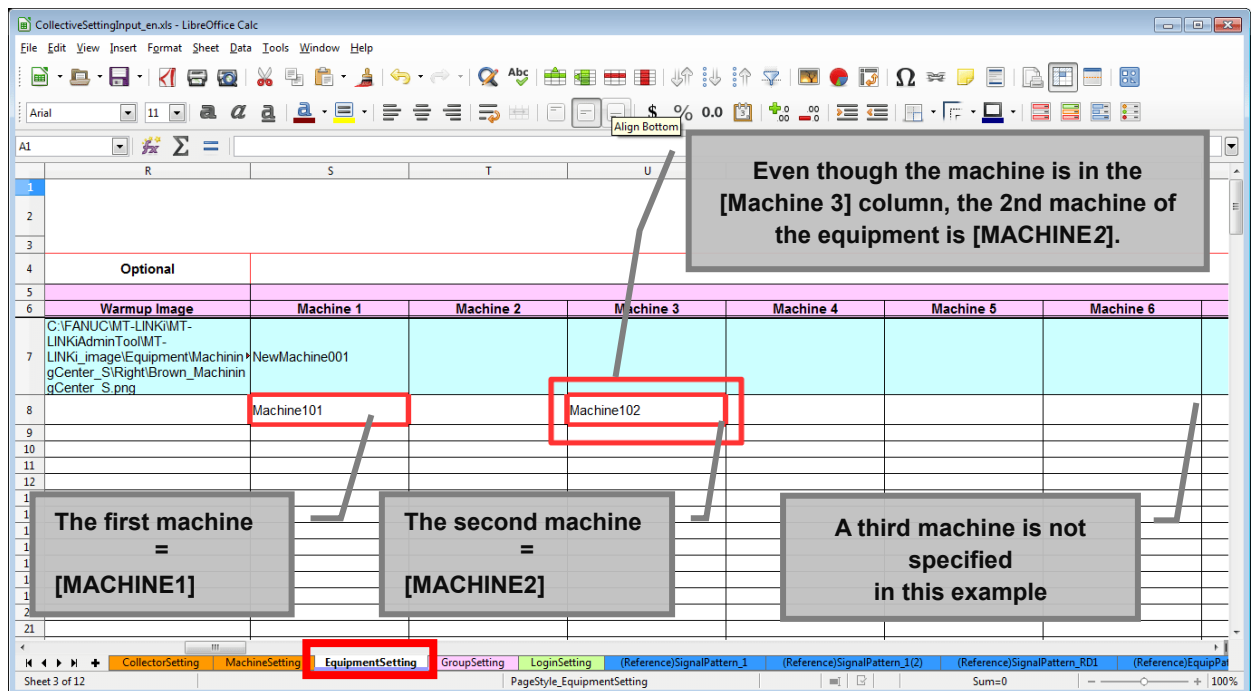
(this is referred to as Machine101 in the example below)

[MACHINE2] => the second machine of the equipment unit

(this is referred to as Machine102 in the example below)

[MACHINE3] => the equipment has no third machine

(this will be detected as error on loading pattern setting files)



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 518/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

(9) Asset management setting (AssetSetting.xls)

- This file defines the asset management information set for each machine.
- Setting items (Web name / value) can be registered to a maximum number of 100.

Setting Entries

- (a) Web language types (not-editable)
Number of languages that support (c) Web title, (e) Web name, and (f) Value
- (b) Setting Pattern
Enter the asset setting pattern to be used
- Default: 1
 - If you add a signal pattern, add a new sheet named "Pattern_<x>" by copying the "Pattern_1" sheet. (Note that <x> should be a numeric character; otherwise, Batch Setting cannot load the file. Use signal pattern for <x>.)
- * In case that this file has duplicate sheet names, Batch Setting will load only the first sheet.
- (c) Web Title (Japanese/English/Chinese/Local language) (NOTE)
Enter the title to display on a Web browser
- Web Title must not be omitted.
 - Web Title must not exceed 100 bytes.
 - Any character can be used.
- (d) Image Setting (Images 1 to 5)
Enter the name of the image file for a machine. The image file will be used to display the machine on the Web browser
- Write the absolute path of the path including the name of the image file, or the relative path from the input file.
 - The format of the image files that can be used are png, jpg, and bmp.
 - The image of the machine will not be displayed if omitted.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	519/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

(e) Web name (Japanese/English/Chinese/Local language) (NOTE)

Enter the name of an asset management item

- Web name must not exceed 100 bytes.
- Any character can be used.

(f) Value (Japanese/English/Chinese/Local language) (NOTE)

Enter the value of an asset management item

- Value must not exceed 100 bytes.
- Any character can be used. Line feed (LF) can also be entered.

NOTE

The items cannot be omitted that correspond to the language selected in the [Web name language] of the [System Setting] window. The items for other languages can be omitted.

(Example) When the language selected in the [Web name language] of the [System Setting] window is "English"
[Web Name] and [Value] cannot be omitted. [Web Name] and [Value] for other languages can be omitted.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	520/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

[SETTING SYSTEM FUNCTIONS] - OUTPUT SCHEDULE SETTING

This chapter describes how to make settings of output schedules.

The AdminTool Menu has the following menu items in the [Setting System Functions] - [Output Schedule Setting] group:

(1) Report Schedule Setting

This menu sets schedules to create reports (daily/weekly/monthly) of equipment operation statuses.

(2) Machine Data Storing Schedule

This menu sets schedules to periodically store manage machine programs and data for managing change history of them.

(3) Signal Data Output

This menu sets schedules to chronologically output signal data in CSV files.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION				Sheet.	521/868
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						

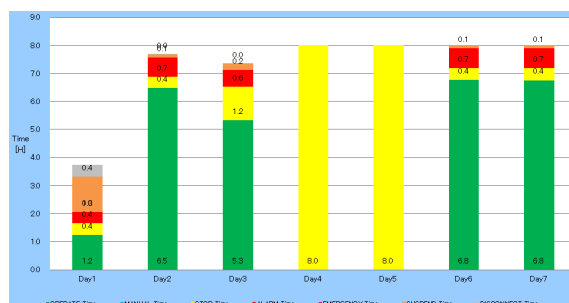
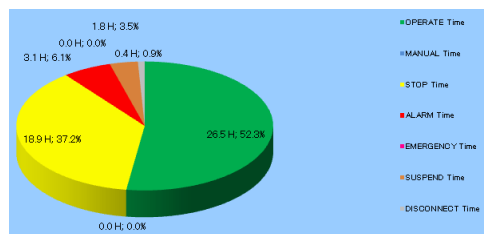
10.1 REPORT SCHEDULE SETTING

With the report schedule setting, the schedule for creating the reports to output can be set on the Web screen.

AdminTool

Web screen

Report file (xls)



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	522/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

NOTE

- 1 If scheduled output of report is executed while operating the Web screen, the Receiver may stop.
Make sure that the report output time and the operation on the Web screen (operation time of the machine) do not overlap (consider setting of Schedule Output).
For instance, when using the Web screen while operating the machine from 9:00 to 17:00, set the schedule for report output at 18:00 or later.
- 2 Report templates are used for the output of report.
For details of report template, refer to the "**PREPARE REPORT TEMPLATE**" section.
- 3 There is no need to start AdminTool in advance on the date/time of the execution of the schedule.
- 4 Report schedule function can be customized. For the details on the items that can be customized, refer to the "**Customizing the Report Schedule Function**" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	523/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

10.1.1 PREPARE REPORT TEMPLATE

In order to use reporting function, you must prepare a report template beforehand.

MT-LINK*i* (Version 3.8 or later) provides default report templates in the following folders.

For the details on how to use sample templates, refer to the "[Use Default Report Template](#)" subsection.

Template files in the [Sample] folder are provided as samples. For details on how to use the template samples, refer to the "[About the Template Samples](#)" subsection.

Also, you can customize the report template file. For the details, refer to the "[Customize the Report Template](#)" subsection.

For file names for report templates and folders to store them, refer to "[Template Filename and Destination Folder](#)" subsection.

To edit the default template, use LibreOffice Calc.

NOTE

The template files edited by LibreOffice Calc should be saved in "Microsoft Excel 97 / 2000 / XP / 2003" format.

■ Folder structure for the default report templates:

Location	File Name of Template		Explanation
<u>Master folder</u> <MT-LINK <i>i</i> installation folder> └ Template └ Report └ Master	daily-report-template.xls daily-report-template_[equipment].xls weekly-report-template.xls weekly-report-template_[equipment].xls monthly-report-template.xls monthly-report-template_[equipment].xls		Template files for report output (NOTE 3 to 6)
<u>Sample folder (Basic)</u> <MT-LINK <i>i</i> installation folder> └ Template └ Report └ Sample └ Basic	\ja \en \zh-Hans \<xx> (NOTE 1, 2)	daily-report-template.xls daily-report-template_[equipment].xls weekly-report-template.xls weekly-report-template_[equipment].xls monthly-report-template.xls monthly-report-template_[equipment].xls	Each sample template file for a language resides in a folder corresponding to the language.
<u>Sample folder (Extended)</u> <MT-LINK <i>i</i> installation folder> └ Template └ Report └ Sample └ Extended	\ja \en \zh-Hans (NOTE 1)	daily-report-template.xls daily-report-template_[equipment].xls weekly-report-template.xls weekly-report-template_[equipment].xls monthly-report-template.xls monthly-report-template_[equipment].xls	Those sample template files are extended from the basic sample templates. As with them, each extended sample template for a language resides in a folder corresponding to the language.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION		Sheet. 524/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

- Following are the details of each template file.

File Name	Details of Each Template
daily-report-template.xls	Daily Report (Graph for analysis on 2nd sheet), common to equipment
daily-report-template_[equipment].xls	Daily Report (Graph for analysis on 2nd sheet), exclusive to equipment ([equipment])
weekly-report-template.xls	Weekly Report (Graph for analysis on 2nd sheet), common to equipment
weekly-report-template_[equipment].xls	Weekly Report (Graph for analysis on 2nd sheet), exclusive to equipment ([equipment])
monthly-report-template.xls	Monthly Report (Graph for analysis on 2nd sheet), common to equipment
monthly-report-template_[equipment].xls	Monthly Report (Graph for analysis on 2nd sheet), exclusive to equipment ([equipment])

NOTE

- 1 Report templates for English are provided in the "\en" folder; for Japanese, in the "ja" folder; for Chinese, in the "zh-Hans" folder.
- 2 The "\<xx>" folder is for a local language, where <xx> is a language code of the language. (Example: When a local language is French, "\fr" will be the folder path of the sample templates for French.)
When additional language resource files are loaded, folders are created and report template files for the local language are stored in the folders. When using the report template files for the local language, copy necessary report template files to the Master folder.

For the details on how to load additional language resource files, refer to the "[Load Additional Language Resource](#)" subsection.

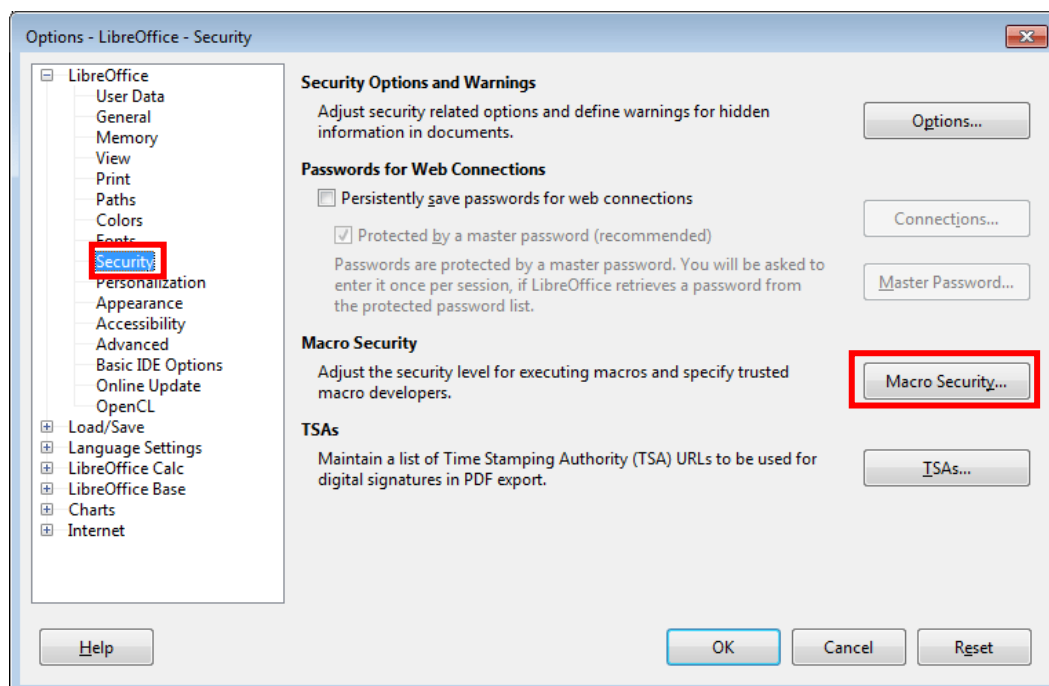
- 3 In case of clean installing the version of MT-LINK*i* is 3.8 or later, when AdminTool starts for the first time since its installation, it copies to the Master folder the report template files installed in Sample (Basic) folder for (*) the same language as does the Windows operating system to the Master folder. When using report template files for another language, manually copy necessary report template files for the language in the Sample folder to the Master folder.

* When the operating system is respectively English, Japanese, or Chinese, the language will be English, Japanese, or Chinese; otherwise, it will be English.

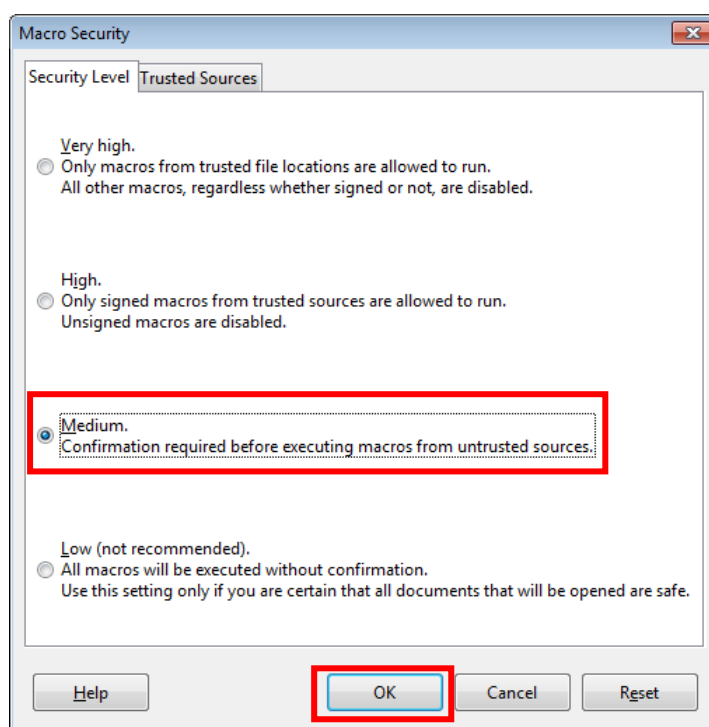
- 4 When updated to version 3.8 or later, MT-LINK*i* continues using report template files stored in the Master folder. If additional language resource files are loaded, it also continues using the report template files for the local language stored in the Sample (Basic) folder.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION		Sheet. 525/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

In using LibreOffice Calc, you need to set [Macro Security].
 Start [LibreOffice Calc] and click [Tools (T)]-[Options (O)] to display the [Options - LibreOffice - Security] window. Select [LibreOffice]-[Security] on the tree in the left pane and click [Macro Security...].

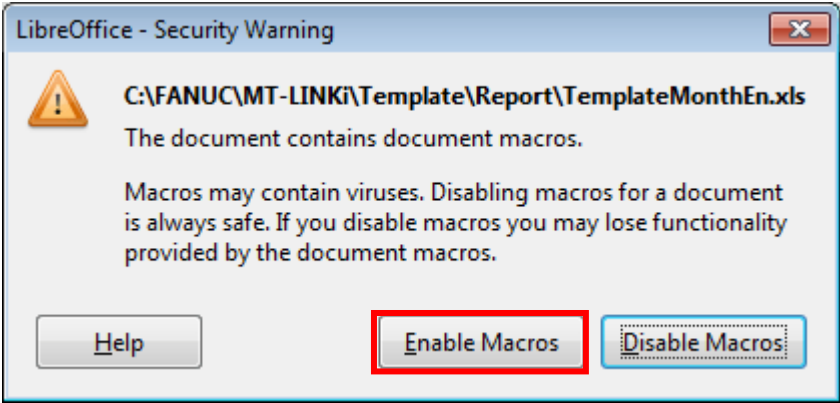


Set to [Medium] and click [OK] on the [Macro Security] window.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	527/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

When you open the template file, the following window appears.
Click the [Enable Macros] button.



■ **Example:** In the case of the Monthly report (English)
Monthly report filename (English): "monthly-report-template.xls"

Equipment Name to output, Collection Period, and Processing Time

Create Keyword

[Create Keyword] Button

Suffix

Totaling Period

Prefix

Keyword

	A	B	C	D	E	F	G	H	I
1			Equipment Name :	%EQUIPMENT_NAME%					
2			Collection Period :	%START_TIME%~%END_TIME%					
3			Processing Time :	%START_AVAILABLE_TIME%~%END_AVAILABLE_TIME%					
7									
9			OPERATE Time	STOP Time	ALARM Time	EMERGENCY Time	MANUAL Time	SUSPEND Time	DISCONNECT Time
10		Prefix / Suffix	OPERATE.statustime	STOP.statustime	ALARM.statustime	EMERGENCY.statustime	MANUAL.statustime	SUSPEND.statustime	DISCONNECT.statustime
11	Past 1 month	MONTH	%MONTH.OPERATE.statustime%	%MONTH.STOP.statustime%	%MONTH.ALARM.statustime%	%MONTH.EMERGENCY.statustime%	%MONTH.MANUAL.statustime%	%MONTH.SUSPEND.statustime%	%MONTH.DISCONNECT.statustime%
12	1Weeks	1WEEKS	%1WEEKS.OPERATE.statustime%	%1WEEKS.STOP.statustime%	%1WEEKS.ALARM.statustime%	%1WEEKS.EMERGENCY.statustime%	%1WEEKS.MANUAL.statustime%	%1WEEKS.SUSPEND.statustime%	%1WEEKS.DISCONNECT.statustime%
13	2Weeks	2WEEKS	%2WEEKS.OPERATE.statustime%	%2WEEKS.STOP.statustime%	%2WEEKS.ALARM.statustime%	%2WEEKS.EMERGENCY.statustime%	%2WEEKS.MANUAL.statustime%	%2WEEKS.SUSPEND.statustime%	%2WEEKS.DISCONNECT.statustime%
14	3Weeks	3WEEKS	%3WEEKS.OPERATE.statustime%	%3WEEKS.STOP.statustime%	%3WEEKS.ALARM.statustime%	%3WEEKS.EMERGENCY.statustime%	%3WEEKS.MANUAL.statustime%	%3WEEKS.SUSPEND.statustime%	%3WEEKS.DISCONNECT.statustime%
15	4Weeks	4WEEKS	%4WEEKS.OPERATE.statustime%	%4WEEKS.STOP.statustime%	%4WEEKS.ALARM.statustime%	%4WEEKS.EMERGENCY.statustime%	%4WEEKS.MANUAL.statustime%	%4WEEKS.SUSPEND.statustime%	%4WEEKS.DISCONNECT.statustime%
16	5Weeks	5WEEKS	%5WEEKS.OPERATE.statustime%	%5WEEKS.STOP.statustime%	%5WEEKS.ALARM.statustime%	%5WEEKS.EMERGENCY.statustime%	%5WEEKS.MANUAL.statustime%	%5WEEKS.SUSPEND.statustime%	%5WEEKS.DISCONNECT.statustime%
17	Monday	MON	%MON.OPERATE.statustime%	%MON.STOP.statustime%	%MON.ALARM.statustime%	%MON.EMERGENCY.statustime%	%MON.MANUAL.statustime%	%MON.SUSPEND.statustime%	%MON.DISCONNECT.statustime%
18	Tuesday	TUE	%TUE.OPERATE.statustime%	%TUE.STOP.statustime%	%TUE.ALARM.statustime%	%TUE.EMERGENCY.statustime%	%TUE.MANUAL.statustime%	%TUE.SUSPEND.statustime%	%TUE.DISCONNECT.statustime%
19	Wednesday	WED	%WED.OPERATE.statustime%	%WED.STOP.statustime%	%WED.ALARM.statustime%	%WED.EMERGENCY.statustime%	%WED.MANUAL.statustime%	%WED.SUSPEND.statustime%	%WED.DISCONNECT.statustime%
20	Thursday	THU	%THU.OPERATE.statustime%	%THU.STOP.statustime%	%THU.ALARM.statustime%	%THU.EMERGENCY.statustime%	%THU.MANUAL.statustime%	%THU.SUSPEND.statustime%	%THU.DISCONNECT.statustime%
21	Friday	FRI	%FRI.OPERATE.statustime%	%FRI.STOP.statustime%	%FRI.ALARM.statustime%	%FRI.EMERGENCY.statustime%	%FRI.MANUAL.statustime%	%FRI.SUSPEND.statustime%	%FRI.DISCONNECT.statustime%
22	Saturday	SAT	%SAT.OPERATE.statustime%	%SAT.STOP.statustime%	%SAT.ALARM.statustime%	%SAT.EMERGENCY.statustime%	%SAT.MANUAL.statustime%	%SAT.SUSPEND.statustime%	%SAT.DISCONNECT.statustime%
23	Sunday	SUN	%SUN.OPERATE.statustime%	%SUN.STOP.statustime%	%SUN.ALARM.statustime%	%SUN.EMERGENCY.statustime%	%SUN.MANUAL.statustime%	%SUN.SUSPEND.statustime%	%SUN.DISCONNECT.statustime%
24	0 TO 1 O'clock	HR_OT01	%HR_OT01.OPERATE.statustime%	%HR_OT01.STOP.statustime%	%HR_OT01.ALARM.statustime%	%HR_OT01.EMERGENCY.statustime%	%HR_OT01.MANUAL.statustime%	%HR_OT01.SUSPEND.statustime%	%HR_OT01.DISCONNECT.statustime%

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11										
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10										
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9										
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2										
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7										
Ed.	Date	Design	Description						FANUC CORPORATION			Sheet.	528/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue						

- (3) Input the suffix corresponding to the summary value to output on the report for each suffix cell. Although the default value exists it can be edited. For detailed information about suffix, refer to the "[About the Suffixes](#)" subsection.

		Processing Time:	%START_AVAILABLE_TIME%~%END_AVAILABLE_TIME%		
Create Keyword		OPERATE Time	STOP Time	ALARM Time	EMERGENCY Time
Prefix / Suffix		OPERATE.statustime	STOP.statustime	ALARM.statustime	EMERGENCY.statustime

- Status Name: STOP
- Obtained Code: statustime
- Content: Total time of stoppage

- Status Name: OPERATE
- Obtained Code: statustime
- Content: Total operation time

- (4) After you have completed the suffix cells, click the [Create Keyword] button. Report keywords are automatically generated, corresponding to each totaling period.

Create Keyword					
		[Create Keyword] button			
	Prefix / Suffix	OPERATE.statustime	STOP.statustime	ALARM.statustime	EMERGENCY.statustime
Past 1 month	MONTH	%MONTH.OPERAT E.statustime%	%MONTH.STOP.st atustime%	%MONTH.ALARM.s tatustime%	%MONTH.EMERGE NCY.statustime%
1Weeks	1WEEKS	%1 WEEKS.OPERAT E.statustime%	%1 WEEKS.STOP.st atustime%	%1 WEEKS.ALARM.s tatustime%	%1 WEEKS.EMERGE NCY.statustime%
2Weeks	2WEEKS	%2 WEEKS.OPERAT E.statustime%	%2 WEEKS.STOP.st atustime%	%2 WEEKS.ALARM.s tatustime%	%2 WEEKS.EMERGE NCY.statustime%
3Weeks	3WEEKS	%3 WEEKS.OPERAT E.statustime%	%3 WEEKS.STOP.st atustime%	%3 WEEKS.ALARM.s tatustime%	%3 WEEKS.EMERGE NCY.statustime%
4Weeks	4WEEKS	%4 WEEKS.OPERAT E.statustime%	%4 WEEKS.STOP.st atustime%	%4 WEEKS.ALARM.s tatustime%	%4 WEEKS.EMERGE NCY.statustime%

Report
Keywords

NOTE

There are following limitations with the ranges (columns) applied to the keyword when automatically creating keyword with the [Create Keyword] button.

- When the default template file is used: automatically created to column N
- When the sample template file is used: automatically created to column U

For the details on how to expand the range to apply to the automatically created keywords, refer to the following subsection of the "FANUC MT-LINK_i / Integration Server Troubleshooting MANUAL":

The "Keyword of the Report Template Is not Created Automatically" subsection in the "TROUBLES RELATED TO REPORT OUTPUT OR SIGNAL DATA OUTPUT" section in the "MT-LINK_i TROUBLESHOOTING" chapter

- (5) Save this workbook. As for the filename and destination, refer to the "[Template Filename and Destination Folder](#)" subsection.
- (6) Now you are ready for using the default template.

20	Nov. 22, 2023	uwano	Revised for MT-LINK _i Version 4.0	Title	FANUC MT-LINK _i AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK _i Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK _i Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK _i Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK _i Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK _i Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK _i Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	529/868

10.1.1.2 Customize the Report Template

Default report templates are provided from FANUC when MT-LINK*i* is installed. However, you can customize the template files.

- (1) Create new workbook with LibreOffice Calc.
- (2) Input the report keywords you want on the LibreOffice Calc workbook after referring to the "[About Report Keyword](#)" subsection.
- (3) Save the LibreOffice Calc workbook. As for filename and destination, refer to the "[Template Filename and Destination Folder](#)" subsection.
- (4) Now you are ready for using your report template.

NOTE

When creating a file with LibreOffice Calc, save the file in "Microsoft Excel 97 / 2000 / XP / 2003" format.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	530/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

10.1.1.3 Template Filename and Destination Folder

You should save the report template in the specified filename to the destination below:

■ Filename

Type of Report	Filename	Remarks
Daily	daily-report-template.xls	Common to equipment
	daily-report-template_[equipment].xls Example: when equipment name is Equipment0001 "daily-report-template_Equipment0001.xls"	Exclusive to equipment
Weekly	weekly-report-template.xls	Common to equipment
	weekly-report-template_[equipment].xls Example: when equipment name is Equipment0002 "weekly-report-template_Equipment0002.xls"	Exclusive to equipment
Monthly	monthly-report-template.xls	Common to equipment
	monthly-report-template_[equipment].xls Example: when equipment name is Equipment0003 "monthly-report-template_Equipment0003.xls"	Exclusive to equipment

Template files common to equipment are used for the report output of all the targeted equipment units. Template files exclusive to equipment have a file name with an equipment name added to it, and are used as the template files only for that equipment.

■ Destination folder

<MT-LINK*i* installation folder>\Template\Report\Master

NOTE

- 1 Reports are output in the language which their template files saved in the folder shown above are prepared for.
- 2 If a report template file exclusive to equipment is stored in the above destination folder, report is output using the report template file corresponding to the output target equipment. If the corresponding report template file is not stored, report is output using the common report template file for equipment units.
- 3 Even when a report template file exclusive to equipment is saved, report template files common to equipment are necessary. Make sure that those template files common to equipment are saved.
- 4 To output report Simple assign status, a template file dedicated to that equipment unit is required.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION				Sheet. 531/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

10.1.1.4 About Report Keyword

While the report template holds report keyword in each cell, each summary value is calculated in each report. This is how reporting system works in MT-LINK*i*. The following is the list of report keyword.

- Equipment Name: %EQUIPMENT_NAME%
Equipment name to output
Example) Equipment0003
- Collection Period: %START_TIME%~ %END_TIME%
Output range (period)
Example) 2020/0102 - 2020/02/01
- Processing Time: %START_AVAILABLE_TIME%~ %END_AVAILABLE_TIME%
Output range (time)
Example) 09:00 - 17:00
- Prefix / Suffix: %<prefix>.<suffix>%

As for the detail of prefix, refer to the "[About the Prefixes](#)" subsection.
As for the detail of suffix, refer to the "[About the Suffixes](#)" subsection.

Example 1:

If during operation hours on the 1st day of summary
%DAY_1.OPERATE.statustime%

Example 2:

If average value of feed rate (path 1, machine 002) between 15:00 and 16:00
%HR_15TO16.ActF_path1_Machine002.avg%

Example 3:

If average value of override (path 1, machine 002) on the 1st week of summary
%1WEEKS.Override_path1_Machine002.avg%

Example 4:

If maximum value of SigCUT time (path 1, machine 002) on Monday
%MON.SigCUT_path1_Machine002.max%

Example 5:

If minimum value of spindle load (spindle 0, path 1, machine 002) in the past 1 month
%MONTH.SpindleLoad_0_path1_Machine002.min%

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	532/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

10.1.1.5 About the Prefixes

Following are the prefixes. ("OK" denotes that the report type supports the period.)

Prefix	Content	Monthly	Weekly	Daily
MONTH	Summary for the last one month	OK	--	--
[N]WEEKS	Summary for <N> (1-5) weeks	OK	OK (<N>=1)	--
HR_[N]TO[M]	Summary for <N> (0-23):00 - <M> (1-24):00	OK	OK	OK
DAY_[N]	Summary for <N> (1-31) days	OK	OK (<N>=1-7)	OK (<N>=1)
MON, TUE, WED, THU, FRI, SAT, SUN	Summary for day of week (Monday to Sunday)	OK	OK	--

By deleting specific rows of the report template, cases such as the following can be managed.

- When you want to display data of a certain time zone on the daily report
- When you want to hide data of a certain day of the week on the weekly or the monthly report

NOTE

Although the rows are deleted, data of the deleted row are included in the summarized (accumulated) value for each prefix.

- Example: When displaying only the data from 12 to 17 o'clock on the daily report

(1) On the report template (daily report), delete the rows from "0 TO 1 O'clock" to "11 TO 12 O'clock" and from "17 TO 18 O'clock" to "23 TO 24 O'clock."

The screenshot shows a spreadsheet with columns A through I. Rows 12 to 17 and 23 to 24 are highlighted in red, indicating they are to be deleted. A grey box with the text "Delete Rows Other than 12 to 17" is overlaid on the spreadsheet, pointing to the rows to be deleted. The spreadsheet contains various data points, including equipment names, collection periods, processing times, and status information for different time intervals.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	Sheet.	533/868

(2) On the output of the daily report, only the data from 12 to 17 o'clock is displayed.

The screenshot shows a LibreOffice Calc spreadsheet with the following data table:

	A	B	C	D	E							
1			Equipment Name:	Equipment0001								
2			Collection Period:	2018/10/29~2018/10/29								
3			Processing Time:	09:00~17:00								
7												
9	Create Keyword		OPERATE Time	STOP Time	ALARM Time	EMERGENCY Time	MANUAL Time	SUSPEND Time	DISCONNECT Time			
10		Prefix / Suffix	OPERATE status	STOP status	ALARM status	EMERGENCY status	MANUAL status	SUSPEND status	DISCONNECT status			
11	Aggregate 1 day	DAY_1	2162.5	10058.5	0	0	0	0	1	1906		
12	12 TO 13 O'clock	HR_12TO13	0	0	0	0	0	0	0	0		
13	13 TO 14 O'clock	HR_13TO14	0	0	0	0	0	0	0	0		
14	14 TO 15 O'clock	HR_14TO15	688.5	1446	0	0	0	0	0	1261.5		
15	15 TO 16 O'clock	HR_15TO16	0	3600	0	0	0	0	0	0		
16	16 TO 17 O'clock	HR_16TO17	700	2899	0	0	0	0	1	0		
17												

A red box highlights the row for 'Aggregate 1 day' (row 11). A callout box states: 'Data of the Deleted Rows are Included the Summarized (Accumulated) Value'.

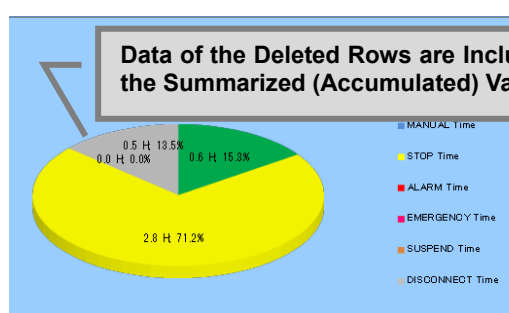


Fig1.1 Today's Status Occurrence Time and Ratio

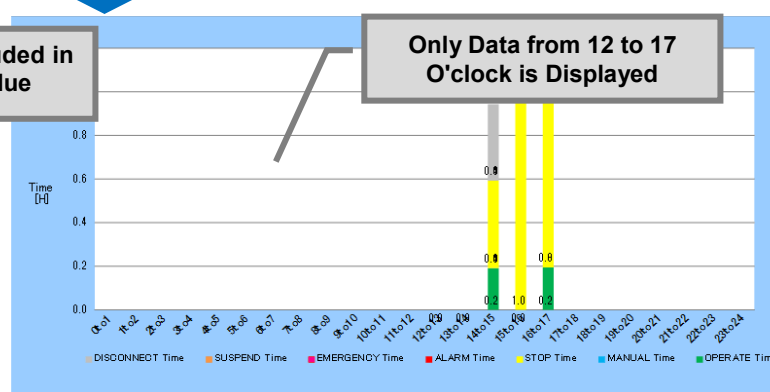


Fig1.2 Hourly Status Occurrence Time

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	534/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

10.1.1.6 About the Suffixes

Following are the suffixes. Monthly, weekly and daily reports are common.

Suffix	Content
<Equipment status name>.statustime	Time period while <value of condition for multiple assign status> = <equipment status name>
<Equipment status name>.time	Time period while <bool type status value> = true
<Equipment status name>.count	Count for <bool type status value> = true (NOTE 1)
<Equipment status name>.countRise	Count for change to <bool type signal value> = true (NOTE 1)
<Equipment status name>.errortime	Time period while <Outlier detection decision result> = error (NOTE 2)
<Equipment status name>.errorcount	Number of occurrence of <Outlier detection decision result> = error (NOTE 1) (NOTE 2)
<Equipment status name>.errorcountRise	Count for change to <Outlier detection decision result> = error (NOTE 1) (NOTE 2)
<Equipment status name>.avg	Average of <integer type status value> (NOTE 3)
<Equipment status name>.max	Maximum value of <integer type status value> (NOTE 3)
<Equipment status name>.min	Minimum value of <integer type status value> (NOTE 3)
.FULL_TIME	Total time of summary period (NOTE 4)

NOTE

- For the details of the count obtained, refer to the "[About the Counts Obtained](#)" subsection.
- To output report for <Outlier detection decision result> = error, outlier detection setting for the target status is required.
Set outlier detection referring to the "[Set Outlier Detection of Signal](#)" subsection.
- When the status is Bool type or String type, the value will be "0" (zero).
- For details on FULL_TIME, refer to the "[About FULL_TIME](#)" subsection.
- In cases when data collection is not executed, such as power of the machine is off, the period will not be included in the summary result.

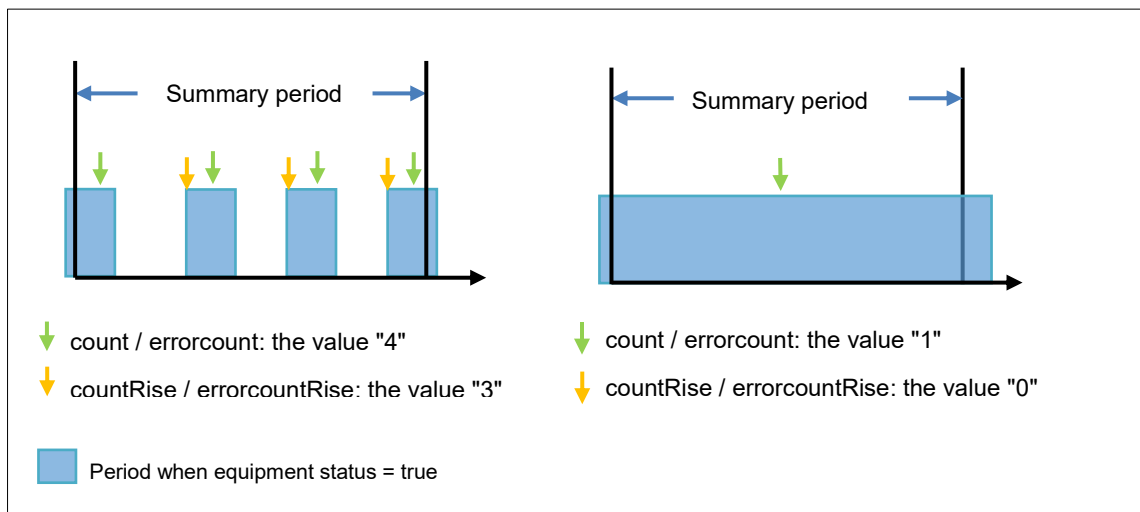
You can use <Equipment status name> that is set in the target equipment on the [Equipment Setting] window. To confirm the status of each piece of equipment, select [AdminTool - Equipment Setting] and point the target equipment, see the status value in the [simple assign status list] or the [multiple assign status list]. As for the equipment status, refer to the following subsections: "[Set Equipment Status](#)," "[Add, Edit or Delete Simple Assign Status](#)," and "[Add, Edit or Delete Multiple Assign Status](#)."

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	535/868

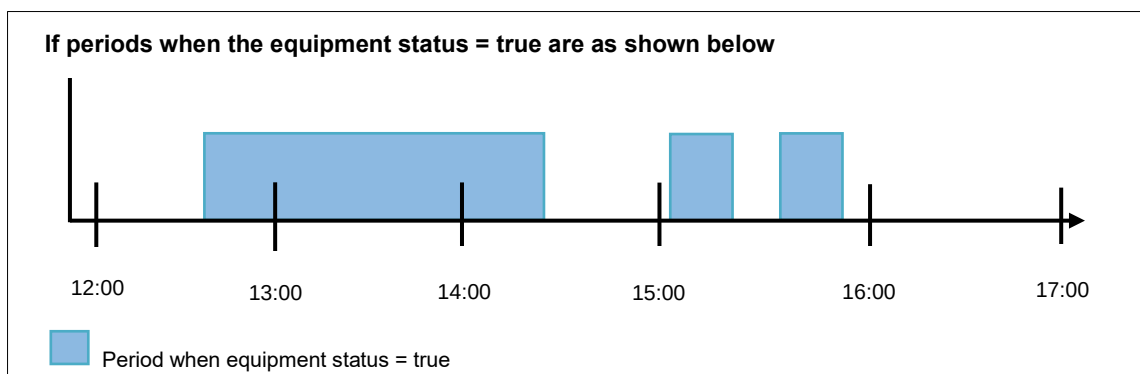
10.1.1.7 About the Counts Obtained

Values for counter suffixes (i.e., count, countRise, errorcount, and errorcountRise) are counted as follows:

- For count and errorcount suffixes, how many times the equipment status is "true" during the summary period.
- For countRise and errorcountRise suffixes, how many times the equipment status turns from "false" to "true" during the summary period.



In the following example, the values for count and countRise suffixes are shown in the table below.



The table below shows examples of obtained values by summary period

Summary period	Value (count)	
	count suffix	countRise suffix
12:00 to 13:00	1	1
13:00 to 14:00	1	0
14:00 to 15:00	1	0
15:00 to 16:00	2	2
16:00 to 17:00	0	0
12:00 to 17:00	3	3

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	536/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

10.1.1.8 About FULL_TIME

If the suffix FULL_TIME is used, total time within the summary period can be obtained in seconds. FULL_TIME can be used as a denominator for calculating ratios.

Report output keywords for FULL_TIME are as follows.

%<prefix>..FULL_TIME%

Examples of retrieved values by summary period

(Where the output period of Weekly report is 8 hours from 9:00)

Summary period	Report output keyword	Retrieved values when using FULL_TIME (seconds)
Day 1 total	%DAY_1..FULL_TIME%	28800 (8 hours x 60 minutes x 60 seconds)
Day 2 total	%DAY_2..FULL_TIME%	28800 (8 hours x 60 minutes x 60 seconds)
.		
.		
.		
Day 7 total	%DAY_7..FULL_TIME%	28800 (8 hours x 60 minutes x 60 seconds)
0:00 to 1:00	%HR_0TO1..FULL_TIME%	0 (Out of time range specified)
1:00 to 2:00	%HR_1TO2..FULL_TIME%	0 (Out of time range specified)
.		
.		
.		
8:00 to 9:00	%HR_8TO9..FULL_TIME%	0 (Out of time range specified)
9:00 to 10:00	%HR_9TO10..FULL_TIME%	25200 (7 days x 1 hour x 60 minutes x 60 seconds)
10:00 to 11:00	%HR_10TO11..FULL_TIME%	25200 (7 days x 1 hour x 60 minutes x 60 seconds)
.		
.		
.		
16:00 to 17:00	%HR_16TO17..FULL_TIME%	25200 (7 days x 1 hour x 60 minutes x 60 seconds)
17:00 to 18:00	%HR_17TO18..FULL_TIME%	0 (Out of time range specified)
.		
.		
.		
23:00 to 24:00	%HR_23TO24..FULL_TIME%	0 (Out of time range specified)

Also, FULL_TIME can be used solely as a report output keyword in a cell as follows. In this case, the cell will hold the length of the period in seconds:

%FULL_TIME%

Examples of retrieved values in Daily report

Start date	Start time	Period	Retrieved values when using FULL_TIME (seconds)
2020/01/01	09:00	1 hour	3600 (1 hour x 60 minutes x 60 seconds)
2020/01/01	09:00	8 hours	28800 (8 hours x 60 minutes x 60 seconds)

Examples of retrieved values in Weekly report

Start date	Start time	Period	Retrieved values when using FULL_TIME (seconds)
2020/01/01	09:00	1 hour	25200 (7 days x 1 hour x 60 minutes x 60 seconds)
2020/01/01	09:00	8 hours	201600 (7 days x 8 hours x 60 minutes x 60 seconds)

Examples of retrieved values in Monthly report

Start date	Start time	Period	Retrieved values when using FULL_TIME (seconds)
2020/01/01	09:00	1 hour	90000 (25 days x 1 hour x 60 minutes x 60 seconds)
2020/01/01	09:00	8 hours	806400 (28 days x 8 hours x 60 minutes x 60 seconds)

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue
FANUC CORPORATION					Sheet.	537/868	

10.1.1.9 About the Template Samples

To use the template samples, edit the sample template files stored in the folder shown below, and store them in a standard folders. As for the file names and the standard folder to save edited template files, refer to the "**Template Filename and Destination Folder**" subsection.

Folder storing sample template files: <MT-LINK*i* installation folder>\Template\Report\Sample

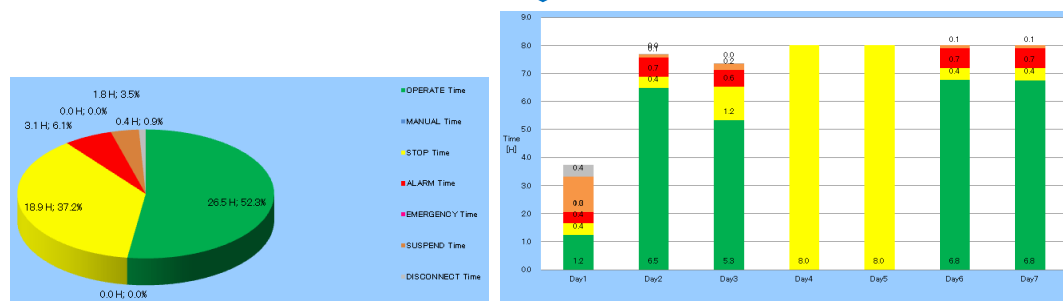
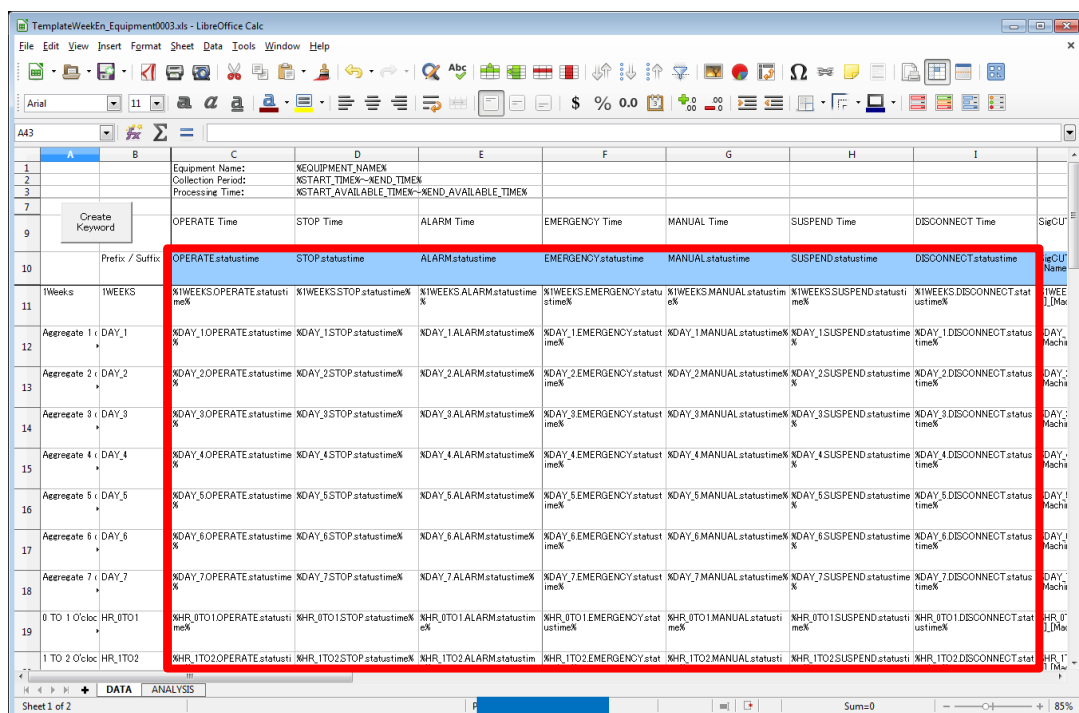
Open the template files prepared for each equipment unit, and edit the suffix of the datasheet as follows.

- When using SigCUT time with multiple paths (path 1, machine 002)
"SigCUT_path<path>_<machine name>.time" --> "SigCUT_path1_Machine002.time"
- When using Spindle Load (average) with multiple paths and multiple axes (axis 0, path 1, machine 002)
"SpindleLoad_<axis>_path<path>_<machine name>.avg" --> "SpindleLoad_0_path1_Machine002.avg"

NOTE

The output target signal must be activated on the signal setting of the AdminTool [Machine Setting] window.

When the edited report template is output, the report is output as shown below.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	538/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

NOTE

As shown in the following example, the status occurrence time table of each report type will set retrieved values for its index columns after converting each of them from seconds to hours and rounding off it to the tenths place. This is why the value in a total index column, usually the second-leftmost column as is the [week] index column in the example, may not be equal to the total of the other index columns in the same row.

Report type	The total of index columns other than the total index column	Total time in hours (The value of the total index column)
Daily	Total of 24 hourly values (The total of values in the [0to1] index column to the [23to24] index column)	Total occurrence time for one day (The value of the [1day] index column)
Weekly	Total of 7 daily values (The total of values in the [Day1] index column to the [Day7] index column)	Total occurrence time for one week (The value of the [week] index column)
Monthly	Total of 5 weekly values (The total of values in the [1stweek] index column to the [5thweek] index column)	Total occurrence time for one month (The value of the [1Month] index column)

Example) "This Week's Daily Status Occurrence Time" table

This Week's State of Operation

Equipment Name: Equipment0001
 Collection Period: 2020/06/24~2020/06/30
 Processing Time: 09:00~17:00

The total of the [Day1] to the [Day7] index columns = 7.2 H

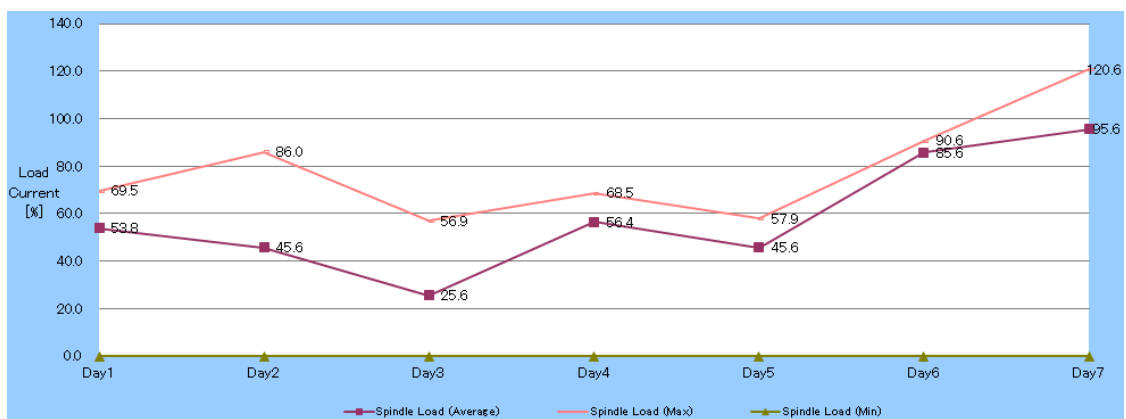
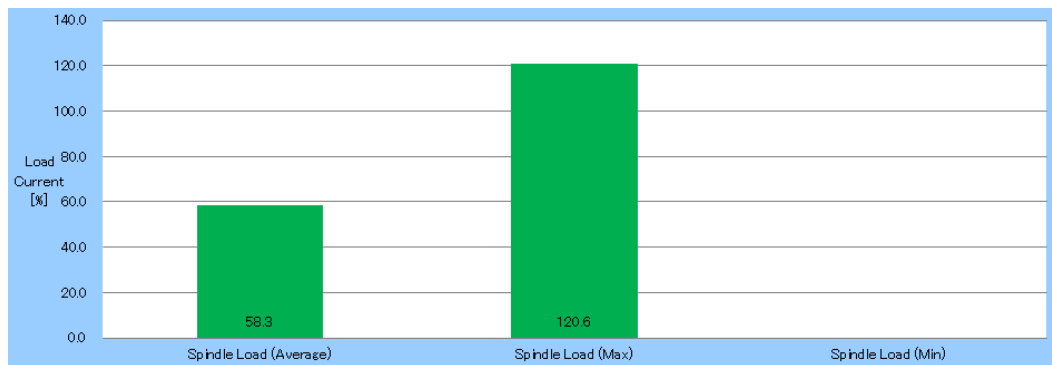
Index	week	Day1	Day2	Day3	Day4	Day5	Day6	Day7	Unit [H]
OPERATE Time	7.1	1.4	4.2	1.6	0.0	0.0	0.0	0.0	
MANUAL Time	1.4	0.4	1.0	0.0	0.0	0.0	0.0	0.0	
STOP Time							0.0	0.0	
ALARM Time							0.0	0.0	
EMERGENCY Time	0.9	0.0	0.9	0.0	0.0	0.0	0.0	0.0	
SUSPEND Time	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
DISCONNECT Time	7.1	0.1	0.4	1.8	0.0	0.0	3.8	1.1	

The value in the [week] index column = 7.1 H

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	539/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	540/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

Template/WeekEn_Equipment0003.xls - LibreOffice Calc									
File Edit View Insert Format Sheet Data Tools Window Help									
Arial 11									
A43									
1	A	B	J	K	L	M	N	O	P
2									
3									
7									
9	Create Keyword	SigOUT Time	SigSBK Time	SpindleLoad (Average)	SpindleLoad (Max)	SpindleLoad (Min)	M00 Stop Time	M01 Stop Time	
10	Prefix / Suffix	SigOUT_path[Path]_MachineNameTime	SigSBK_path[Path]_MachineNameTime	SpindleLoad_Axis[Path]_MachineNameTime	SpindleLoad_Axis[Path]_MachineNameTime	SpindleLoad_Axis[Path]_MachineNameTime	SigDM00_path[Path]_MachineNameTime	SigDM01_path[Path]_MachineNameTime	
11	1Weeks	1WEEKS	1WEEKS	1WEEKS	1WEEKS	1WEEKS	1WEEKS	1WEEKS	
12	Aggregate 1 : DAY_1	XDAY_1SigOUT_path[Path]_MachineNameTime	XDAY_1SigSBK_path[Path]_MachineNameTime	XDAY_1SpindleLoad_Axis[Path]_MachineNameTime	XDAY_1SpindleLoad_Axis[Path]_MachineNameTime	XDAY_1SpindleLoad_Axis[Path]_MachineNameTime	XDAY_1SigDM00_path[Path]_MachineNameTime	XDAY_1SigDM01_path[Path]_MachineNameTime	
13	Aggregate 2 : DAY_2	XDAY_2SigOUT_path[Path]_MachineNameTime	XDAY_2SigSBK_path[Path]_MachineNameTime	XDAY_2SpindleLoad_Axis[Path]_MachineNameTime	XDAY_2SpindleLoad_Axis[Path]_MachineNameTime	XDAY_2SpindleLoad_Axis[Path]_MachineNameTime	XDAY_2SigDM00_path[Path]_MachineNameTime	XDAY_2SigDM01_path[Path]_MachineNameTime	
14	Aggregate 3 : DAY_3	XDAY_3SigOUT_path[Path]_MachineNameTime	XDAY_3SigSBK_path[Path]_MachineNameTime	XDAY_3SpindleLoad_Axis[Path]_MachineNameTime	XDAY_3SpindleLoad_Axis[Path]_MachineNameTime	XDAY_3SpindleLoad_Axis[Path]_MachineNameTime	XDAY_3SigDM00_path[Path]_MachineNameTime	XDAY_3SigDM01_path[Path]_MachineNameTime	
15	Aggregate 4 : DAY_4	XDAY_4SigOUT_path[Path]_MachineNameTime	XDAY_4SigSBK_path[Path]_MachineNameTime	XDAY_4SpindleLoad_Axis[Path]_MachineNameTime	XDAY_4SpindleLoad_Axis[Path]_MachineNameTime	XDAY_4SpindleLoad_Axis[Path]_MachineNameTime	XDAY_4SigDM00_path[Path]_MachineNameTime	XDAY_4SigDM01_path[Path]_MachineNameTime	
16	Aggregate 5 : DAY_5	XDAY_5SigOUT_path[Path]_MachineNameTime	XDAY_5SigSBK_path[Path]_MachineNameTime	XDAY_5SpindleLoad_Axis[Path]_MachineNameTime	XDAY_5SpindleLoad_Axis[Path]_MachineNameTime	XDAY_5SpindleLoad_Axis[Path]_MachineNameTime	XDAY_5SigDM00_path[Path]_MachineNameTime	XDAY_5SigDM01_path[Path]_MachineNameTime	
17	Aggregate 6 : DAY_6	XDAY_6SigOUT_path[Path]_MachineNameTime	XDAY_6SigSBK_path[Path]_MachineNameTime	XDAY_6SpindleLoad_Axis[Path]_MachineNameTime	XDAY_6SpindleLoad_Axis[Path]_MachineNameTime	XDAY_6SpindleLoad_Axis[Path]_MachineNameTime	XDAY_6SigDM00_path[Path]_MachineNameTime	XDAY_6SigDM01_path[Path]_MachineNameTime	
18	Aggregate 7 : DAY_7	XDAY_7SigOUT_path[Path]_MachineNameTime	XDAY_7SigSBK_path[Path]_MachineNameTime	XDAY_7SpindleLoad_Axis[Path]_MachineNameTime	XDAY_7SpindleLoad_Axis[Path]_MachineNameTime	XDAY_7SpindleLoad_Axis[Path]_MachineNameTime	XDAY_7SigDM00_path[Path]_MachineNameTime	XDAY_7SigDM01_path[Path]_MachineNameTime	
19	0 TO 1 O'clock	XHR_0TO1SigOUT_path[Path]_MachineNameTime	XHR_0TO1SigSBK_path[Path]_MachineNameTime	XHR_0TO1SpindleLoad_Axis[Path]_MachineNameTime	XHR_0TO1SpindleLoad_Axis[Path]_MachineNameTime	XHR_0TO1SpindleLoad_Axis[Path]_MachineNameTime	XHR_0TO1SigDM00_path[Path]_MachineNameTime	XHR_0TO1SigDM01_path[Path]_MachineNameTime	
20	1 TO 2 O'clock	XHR_1TO2SigOUT_path[Path]_MachineNameTime	XHR_1TO2SigSBK_path[Path]_MachineNameTime	XHR_1TO2SpindleLoad_Axis[Path]_MachineNameTime	XHR_1TO2SpindleLoad_Axis[Path]_MachineNameTime	XHR_1TO2SpindleLoad_Axis[Path]_MachineNameTime	XHR_1TO2SigDM00_path[Path]_MachineNameTime	XHR_1TO2SigDM01_path[Path]_MachineNameTime	



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	Sheet.	541/868
FANUC CORPORATION								

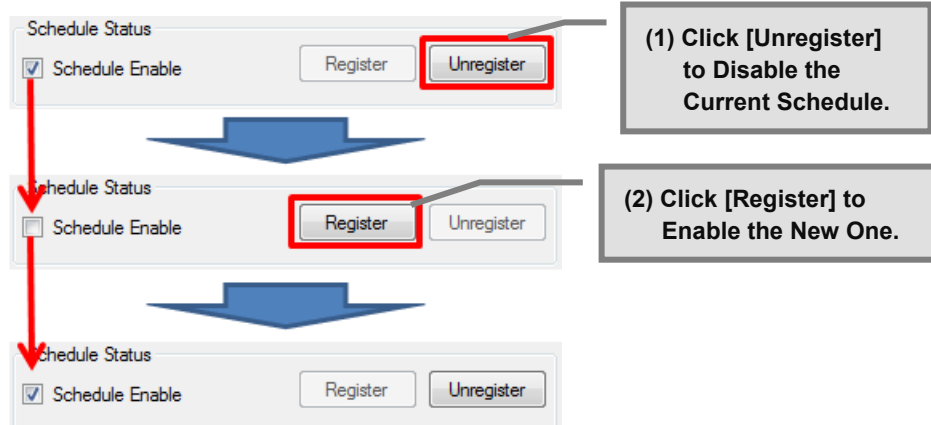
10.1.2 Steps for Report Schedule Setting

Follow the steps below to set report schedule.

- (1) In the [Equipment] list, select the check boxes next to all of the equipment to be reported about.
- (2) Select [Daily], [Weekly] or [Monthly] in the [Report Type] group box.
- (3) Set the [Start time] and the [Period] lists.
 Generally, the [Start time] and [Period] lists are set to business hours and are used as a denominator of production ratio.
 The [Period] field allows a value equal to or less than "24:00" (<hour>:<minute>).
 The latest moment specified in the [Period] lists will be excluded from the output range.
 Example:
 In the following case:
 The [Start time] is set to "13:00:00" and [Period] is set to "01: 00" (end time will be "14:00:00").
 The exact range is from 13:00:00.000 to 13:59:59.999.
 Note that data at 14:00:00 is excluded from the output range.
- (4) Follow the message in the window and set the schedule.
 After setting schedule, click [Unregister] to disable the current schedule. Then, click [Register] to enable the new one.

NOTE

Output schedule can be registered for each report type (daily/weekly/monthly).



*After you modify the reporting schedule, never fail to do this disable/enable action.

- (5) Click [Close].
- (6) When the specified start date/time comes, report output process is executed. At the completion of the output process, a file is output to the report output folder. Output process set by report schedule cannot be canceled during the output process.
 For the details on output file, refer to the "[Report Output File](#)" subsection. Also, when output is failing, refer to the "[When Report Output Fails](#)" subsection.

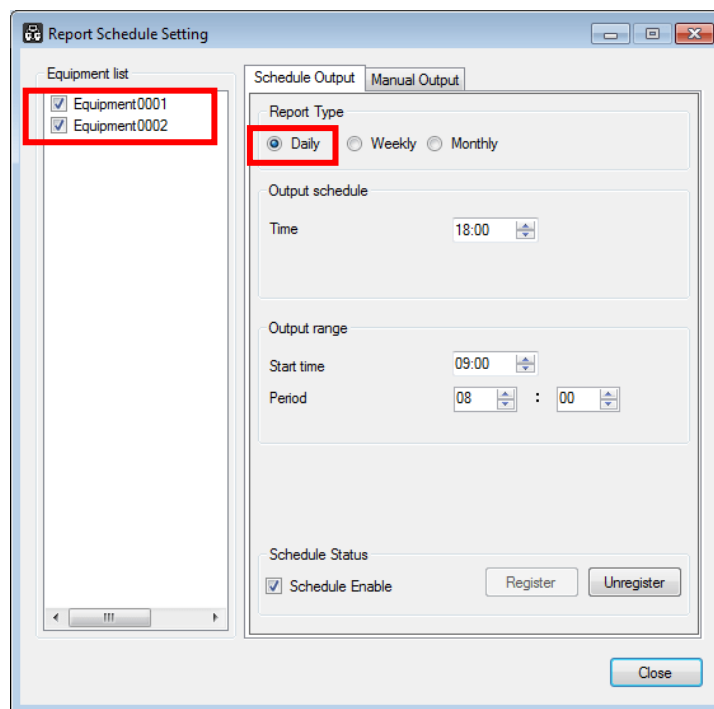
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	542/868
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue					

NOTE

- 1 When the PC is shut down during schedule output, some of the report may not be output, as the report output is terminated before the completion of the process. The reports that have not been output will not be output in the succeeding schedule output. If you need the reports that have not been output, use manual output function to output the report separately.
- 2 When the time for the next schedule output arrives while schedule output is in process, the next schedule output will be executed after the current schedule output process is completed.
- 3 When the schedule output of the daily, weekly, and monthly reports are registered on the same time, and when the schedule output of these reports are executed on the same date/time, the schedule output will be executed in the order of daily report, weekly report, and monthly report, with 20 seconds interval in between. If the output process takes more than 20 seconds, the next schedule output is executed after the current schedule output.

10.1.3 Scheduling Daily Report

Follow the steps below to set daily report schedule.



- (1) In the [Equipment] list, select all the check boxes with the equipment to be reported.
- (2) Select [Daily] in the [Report Type] group box.
- (3) Set the output time of the daily report in the [Output schedule] group box.
- (4) Set the start time and duration of the daily report in the [Output range] group box.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 543/868

Example 1:

In the following case:

Time	18:00
Start time	09:00
Period	08:00
Date/Time when the setting contents are set	2015/5/20 15:00

Next output:

- Time/Date: at 18:00 on 2015/5/20 (the same date when the setting contents are set)
- Range: from 09:00 to 17:00 (2015/5/20)

Example 2:

In this example the ending time of the time range is later than the output time.

In the following case:

Time	16:00
Start time	09:00
Period	08:00
Date/Time when the setting contents are set	2015/5/20 15:00

Next output:

- Time/Date: at 16:00 on 2015/5/20 (the same date when the setting contents are set)
- Range: from 09:00 to 17:00 (2015/5/19)

*Note that the data of the previous day will be output.

Example 3:

In this example, the ending time of the time range falls in the next day (exceeding hour 24).

In the following case:

Time	03:00
Start time	23:00
Period	05:00
Date/Time when the setting contents are set	2015/5/20 15:00

Next output:

- Time/Date: at 03:00 on 2015/5/21 (the same date when the setting contents are set)
- Range: from 23:00 on 2015/5/19 to 04:00 2015/5/20

*Note that the data of the previous day will be output.

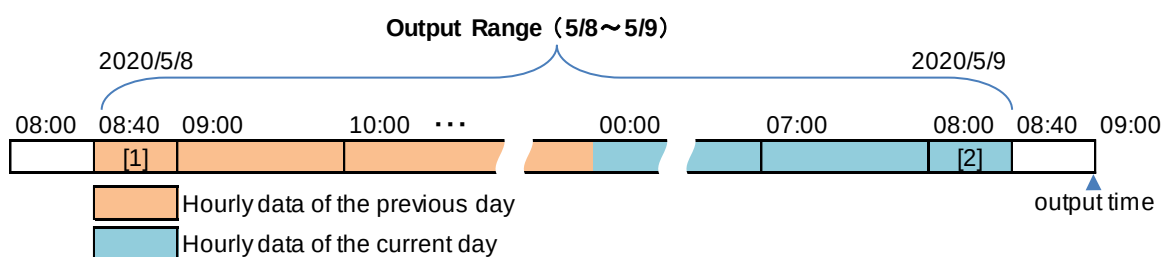
20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 544/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

NOTE

Daily report outputs 24-hour summary data in at most 24 rows of hourly spans. However, when the [Start time] is set to a time whose minutes part is not "00" and the [Period] is set to "24:00" as in the following example, the row whose span includes the [Start time] will be used for not only the data of the current day but the data of the previous day as well.

Example)

Output time: 09:00
Start time: 08:40
Period: 24:00
Current date/time: 2020/5/9 09:00



Output Results of Report (Daily)

hour 0 to 1	
hour 1 to 2	
...	...
hour 6 to 7	
hour 7 to 8	
hour 8 to 9	[2] [1]
hour 9 to 10	
hour 10 to 11	
...	...
hour 22 to 23	
hour 23 to 24	

This row is used to sum [1] and [2]

[1] = data of from 8:40 to 9:00 of the previous day
[2] = data of from 8:00 to 8:40 of the current day

To avoid such a sum over two days in a single hourly row, perform a schedule output and a manual output.

As for the example shown above, perform two outputs with the following settings:

Settings for a schedule output)

Output time: 09:00
Start time: 08:40
Period: 23:20

Settings for a manual output)

Start time: 08:00
Period: 00:40

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	FANUC CORPORATION		Sheet.	545/868
				Apprv.				
				Inoue				

10.1.4 Scheduling Weekly Report

Follow the steps below to set weekly report schedule.

- (1) In the [Equipment] list, select all the check box with the equipment to be reported.
- (2) Select [Weekly] in the [Report Type] group box.
- (3) In the [Output schedule] group box, set the day of the week and time to output the weekly report.
- (4) In the [Output range] group box, set the [Start time], the [Period], and the [Closing day] (weekly closing day in the weekly report).

Example 1:

Here is the example to output the results from Wednesday to Tuesday on Wednesday.

In the following case:

[Every week] in [Output schedule] (output date of the week)	Wednesday
[Closing day] in [Output range]	Tuesday
The current date/time when the setting is made	Friday, May 15

Next output:

- Date: 5/20 Wed.
- Range: from 5/13 Wed. to 5/19 Tue.

May, 2015						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
26	27	28	29	30	1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31	1	2	3	4	5	6

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	546/868

Example 2:

In contrast to above, here is the example that the ending time of the summarize time span is earlier than the output time.

[Every week] in [Output schedule] (output date of the week)	Tuesday
[Closing day] in [Output range]	Wednesday
The current date/time when the setting is made	Friday, May 15

Next output:

- Date: 5/19 Tue.
- Range: from 5/7 (Thu.) to 5/13 (Wed.)

May, 2015						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
26	27	28	29	30	1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31	1	2	3	4	5	6

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	547/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue				

10.1.5 Scheduling Monthly Report

Follow the steps below to set monthly report schedule.

- (1) In the [Equipment] list, select all the check boxes of the equipment to be reported.
 - (2) Select [Monthly] in the [Report Type] group box.
 - (3) In the [Output schedule] group box, set the day of the month and time to output the monthly report.
 - (4) In the [Output range] group box, set the [Start time], the [Period], and the [Closing day] (monthly closing day in the monthly report).
- * The [Every month] in the [Output schedule] group box provides options of monthly output day between 1 and 28, as well as an option of [End of month]. When you specify the [End of month] option, monthly reports will be issued on the last day of each month.
 - * The [Closing day] in the [Output range] group box provides options of monthly closing day also between 1 and 28, as well as an option of [End of month]. When you specify the [End of month] option, the [Closing day] for a month will be set to the last day of each month.

As for monthly report, the beginning date of the output span is the next day of the closing date of the last month.
Example: when the closing date is 25, the range will be from 26 of last month to 25 of this month.

Example 1:

Here is the example that the output date is later than the closing date.
In the following case:

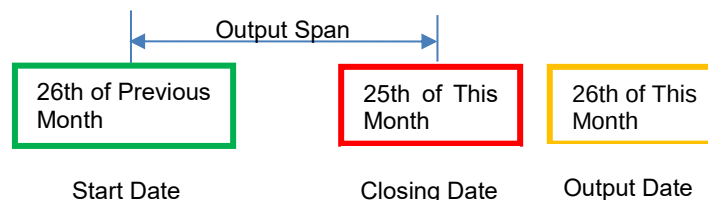
[Every month] in [Output schedule] (output date of the month)	26 th
[Closing day] in [Output range]	25 th
The current date/time when the setting is made	2015/7/15

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	548/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

Next output:

- Date: 2015/7/26
- Range: days within the frame

June, 2015							July, 2015						
Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat
31	1	2	3	4	5	6	28	29	30	1	2	3	4
7	8	9	10	11	12	13	5	6	7	8	9	10	11
14	15	16	17	18	19	20	12	13	14	15	16	17	18
21	22	23	24	25	26	27	19	20	21	22	23	24	25
28	29	30	1	2	3	4	26	27	28	29	30	31	1
5	6	7	8	9	10	11	2	3	4	5	6	7	8



Example 2:

In this example the output date is earlier than the closing date.

In the following case:

[Every month] in [Output schedule] (output date of the month)	5 th
[Closing day] in [Output range]	25 th
The current date/time when the setting is made	2015/7/15

Next output:

- Date: 2015/8/5
- Range: days within the following frame

June, 2015							July, 2015							August, 2015						
Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat
31	1	2	3	4	5	6	28	29	30	1	2	3	4	26	27	28	29	30	31	1
7	8	9	10	11	12	13	5	6	7	8	9	10	11	2	3	4	5	6	7	8
14	15	16	17	18	19	20	12	13	14	15	16	17	18	9	10	11	12	13	14	15
21	22	23	24	25	26	27	19	20	21	22	23	24	25	16	17	18	19	20	21	22
28	29	30	1	2	3	4	26	27	28	29	30	31	1	23	24	25	26	27	28	29
5	6	7	8	9	10	11	2	3	4	5	6	7	8	30	31	1	2	3	4	5



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	549/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

10.1.6 Steps for Manual Output

Manual Output functionality enables you to get the report output immediately, without waiting for report output schedule.

- (1) Select the [Manual Output] tab.
- (2) In the [Equipment] list, select all the check boxes for the equipment to be reported.
- (3) Select [Daily], [Weekly] or [Monthly] in the [Report Type] group box.

Set the start date, end date and time range.

The maximum value of the [Period] is 24:00 and the latest moment specified in the [Period] lists will be excluded from the output range.

Example:

In the following case:

The [Start time] is set to 13:00 and [Period] is set to 01: 00 (end time will be 14:00).

The exact range is from 13:00:00.000 to 13:59:59.999.

Note that data at 14:00:00 is excluded from the range.

NOTE

If a future date/time is included in the output range, it will be an error. Specify the output range so that it will not include date/time of the future.

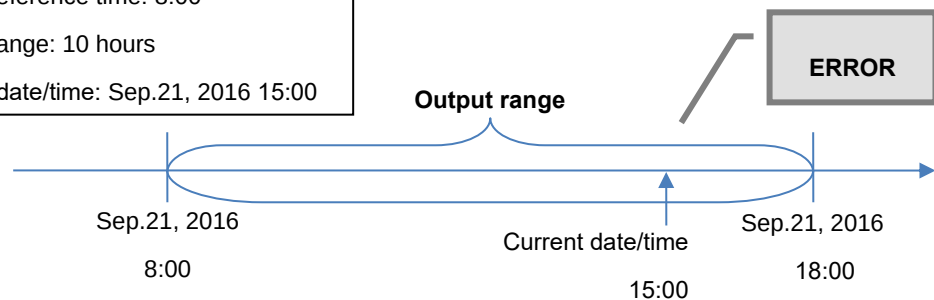
(Example)

Output reference date: Sep.21, 2016

Output reference time: 8:00

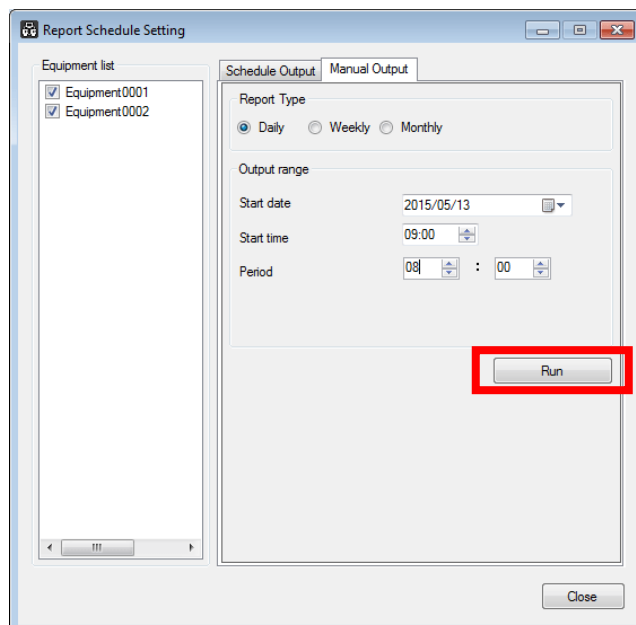
Output range: 10 hours

Current date/time: Sep.21, 2016 15:00



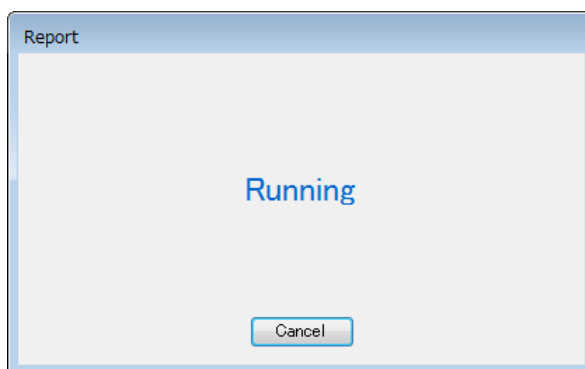
20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	550/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

(4) Click [Run].

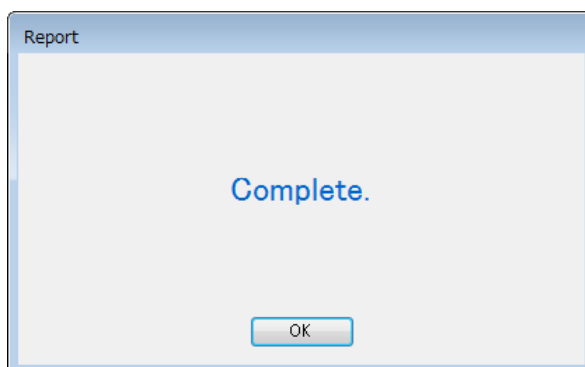


(5) The following dialog box appears while the output is being processed.

* If wishing to cancel the output, click [Cancel].



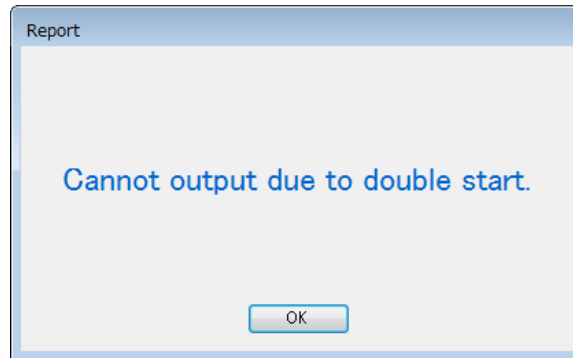
(6) When the output is completed, the following dialog box appears. Click [OK].



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION				Sheet.	551/868
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						

NOTE

- 1 Output process may take some time to complete. Especially weekly and monthly reports require much time to calculate their large amount of data.
- 2 During the output process, it is unable to execute setting operations on the AdminTool windows.
- 3 If the output fails, refer to the "**When Report Output Fails**" subsection.
- 4 If you request a manual output while a report schedule output or a signal data schedule output is in process, your request is not accepted and the following error message appears. Try the manual output after a while.

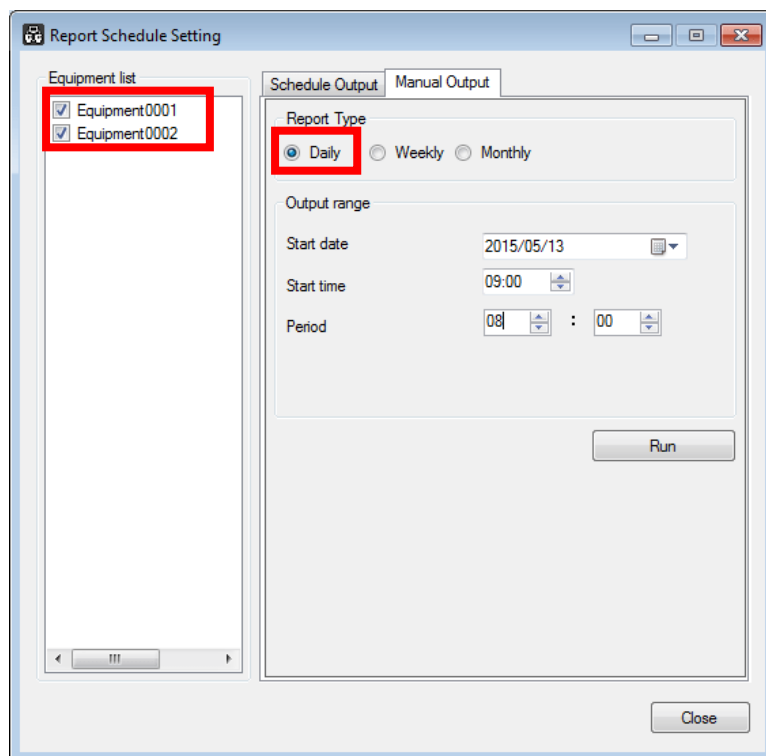


- 5 If your manual output is being processed when a scheduled output is about to start on schedule, the scheduled output is postponed and will start after the manual output is completed.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet.	552/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

10.1.7 Manual Output Daily Report

To output daily report manually, follow the steps below.



- (1) Select the [Manual Output] tab.
- (2) In the [Equipment] list, select all the check boxes for the equipment to be reported.
- (3) Select [Daily] in the [Report Type] group box.
- (4) Specify the [Start date], the [Start time], and the [Period] in the [Output range] group box.

Example 1:

With the following settings, the output range will be from 2015/5/13 09:00 to 2015/5/13 17:00.

Start date	2015/5/13
Start time	09:00
Period	08:00

Example 2:

With the following settings, the output range will be from 2015/5/13 23:00 to 2015/5/14 04:00 (exceeding hour 24).

Start date	2015/5/13
Start time	23:00
Period	05:00

- (5) Click the [Run] button.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	FANUC CORPORATION		Sheet.	553/868

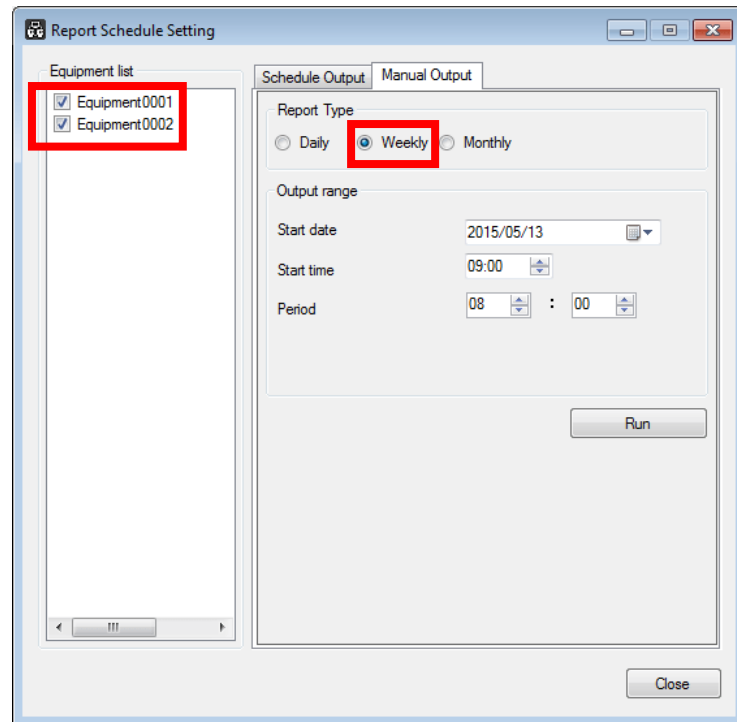
NOTE

Daily report outputs 24-hour summary data in at most 24 rows of hourly spans. However, when the [Start time] is set to a time whose minutes part is not "00" and the [Period] is set to "24:00," the row whose span includes the [Start time] will be used for not only the data of the current day but the data of the previous day as well.

For the details, refer to **NOTE** in the "Scheduling Daily Report" subsection.

10.1.8 Manual Output Weekly Report

To output weekly report manually, follow the steps below.



- (1) Select the [Manual Output] tab.
- (2) In the [Equipment] list, select all the check boxes for the equipment to be reported.
- (3) Select [Weekly] in the [Report Type] group box.
- (4) Specify the [Start date], the [Start time], and the [Period] in the [Output range] group box.

Example 1:

With the following settings, data of seven days during 2015/5/13 to 2015/5/19 from 9:00 to 17:00 will be output.

Start date	2015/5/13
Start time	09:00
Period	08:00

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 554/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

May, 2015						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
26	27	28	29	30	1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31	1	2	3	4	5	6

(5) Click the [Run] button.

10.1.9 Manual Output Monthly Report

To output monthly report manually, follow the steps below.

- (1) Select the [Manual Output] tab.
- (2) In the [Equipment] list, select all the check boxes for the equipment to be reported.
- (3) Select [Monthly] in the [Report Type] group box.
- (4) Specify the [Start date], the [Start time], the [Period], and the [Closing day] in the [Output range] group box.

* The [Closing day] in the [Output range] group box provides options of monthly closing day between 1 and 28, as well as an option of [End of month]. When you specify the [End of month] option, the [Closing day] is set to the last day of each month.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	555/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

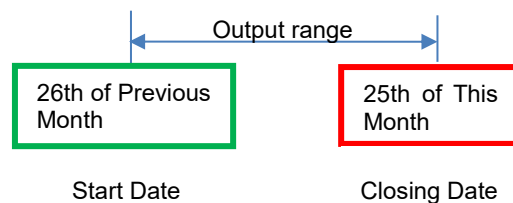
Example 1:

In this example the start date is later than the closing day.

With the following settings, data of 30 days during 2015/6/26 to 2015/7/25 from 9:00 to 17:00 will be output as shown.

Start date	2015/6/26
Start time	09:00
Period	08:00
Closing day	25

June, 2015							July, 2015						
Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat
31	1	2	3	4	5	6	28	29	30	1	2	3	4
7	8	9	10	11	12	13	5	6	7	8	9	10	11
14	15	16	17	18	19	20	12	13	14	15	16	17	18
21	22	23	24	25	26	27	19	20	21	22	23	24	25
28	29	30	1	2	3	4	26	27	28	29	30	31	1
5	6	7	8	9	10	11	2	3	4	5	6	7	8

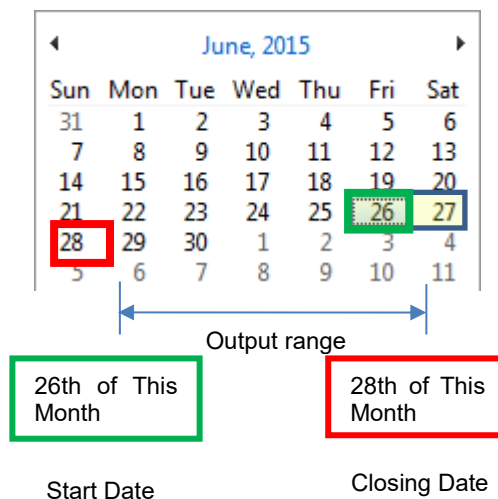


Example 2:

In this example the start date is the closing day or earlier.

With the following settings, data of everyday during 2015/6/26 to 2015/6/30 from 9:00 to 17:00 will be output.

Start date	2015/6/26
Start time	09:00
Period	08:00
Closing day	28



(5) Click the [Run] button.

Note that if you set non-existent date to the [Closing day] and click the [Run] button, you will receive an error message.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	556/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

10.1.10 Report Output File

The report output file is stored in the folder specified at the report path in the [System Setting] window.

NOTE

If a report with the same file name is output two or more times, the contents of the previous output file are overwritten by the contents of the succeeding output file.

The file name will be as follows:

report_output_<Report Type>_<Equipment Name>_<Date (yyyyMMdd)>.xls

■ Report Type

[Day]: daily report

[Week]: weekly report

[Month]: monthly report

■ Equipment Name

This is the name of the target equipment.

■ Date (yyyyMMdd)

- As for a daily report, the date is the day the output is performed. When the output range exceeds one day, the date is the second day in the output range.
- As for a weekly report, the date is the closing day of the week when it is a scheduled output; the date is the last day of the output range when it is a manual output.
- As for a monthly report, the date is the closing day of the month.

NOTE

If the %END_TIME% key word for Collection Period is deleted from a report template, the "Date (yyyyMMdd)" portion in the name of a report output file that uses the customized report template will be set to the date of output, regardless of report type.

For the key words available in report templates, refer to the "[About Report Keyword](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	557/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

10.1.11 When Report Output Fails

When the report output fails, a file with the following name is output to the output target folder.

- Output file when report output fails

File Name	Cause of Failure
#####CANCEL OUTPUTTING#####_<yyyyMMddHHmmss>	When output process is canceled
#####TIMEOUT OUTPUTTING#####_<yyyyMMddHHmmss>	When output process timed out
#####ITEMERROR OUTPUTTING#####_<yyyyMMddHHmmss>	When there is a file with output error (NOTE 3)
#####DISKFULL OUTPUTTING#####_<yyyyMMddHHmmss>	When the free disk space of disk drive specified as the storage destination of the reports is under 5GB

NOTE

- 1 <yyyyMMddHHmmss> part of the file name is the output date/time of the file.
- 2 When the output fails by time out, there is a possibility that the cause of failure is due to the excessive amount of equipment units and the excessive time expensed for processing the data. In such a case, narrow down the equipment units targeted for output on the equipment list, and output the report again.
- 3 The equipment name of the file with the output error will be stored in the file.
(Example) Equipment0001

10.1.12 Customizing the Report Schedule Function

The report output schedule function can be customized by editing the system setting file (AdminTool.ini) provided by MT-LINK*i*.

Following are the items that can be customized by AdminTool.ini.

- Time out period of report schedule output
Customize for cases when time out occurs during the report output schedule process.
- Time out period of report manual output
Customize for cases when time out occurs during the report manual output process.

NOTE

If the system setting file (AdminTool.ini) is edited, restart AdminTool.
If the report schedule output is being registered, in order to enable the settings after editing, [Unregister] the schedule and [Register] the schedule again.

For the details of the system setting files, refer to the "**SYSTEM SETTING FILES**" section.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	558/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

10.2 MACHINE DATA STORING SCHEDULE

By using Machine Data Storing Schedule, schedule can be set for storing the program files and the data files on the machine periodically.

Also, archive management settings for displaying archives of the machine data and the description of changes can be set on the [Utility - Machine Data Archive Management] screen.

Machine Data Storing Schedule provides three output options as follows:

- Schedule Output:
To automatically archive machine data on a regular basis as scheduled
(The archived data can be reviewed with the [Utility - Machine Data Archive Management] screen only if the [Enabled Archive Management] check box is selected before archiving.)
- Manual Output:
To manually archive machine data
(The archived data cannot be reviewed with the [Utility - Machine Data Archive Management] screen.)
- History Manual Output:
To manually archive machine data
(The archived data can be reviewed with the [Utility - Machine Data Archive Management] screen only if the [Enabled Archive Management] check box is selected before archiving.)

The following are machine data to be stored through this function.

(1) CNC data (in FanuCNC and FanucRobodrill type machines) are stored in the following folders:

CNCDATA folder:

- Parameter
- Offset data
- Workpiece coordinate system data
- Pitch error compensation data
- Custom macro common variable
- Operation history

CNCPROG folder:

CNC_MEM folder:

- Data in the program memory area of the CNC

DATA_SV folder:

- Data under the sign-in folder of the Data Server

PMCDATA folder:

- PMC parameter

FROM folder:

- FROM data

SRAM folder:

- SRAM data

(2) Robot data (in FanucRobot type machines) are stored in the ROBOTDATA folder.

ROBOTDATA folder:

- All data in MDBs (memory binary devices) of the robot

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION				Sheet. 559/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

NOTE**On names of files in robot data**

If a file name includes characters that are undisplayable due to the regional settings made on the operating system, the file name may be garbled on platforms with those regional settings. Especially when the garbled file name includes such invalid characters as a question mark ("?"), Machine Data Storing function may fail.

Example) when a robot's regional settings are set to Japanese and the Collector PC's regional settings are set to another language

If the robot uses half-width kanas to name a file, those characters will be garbled on the [Utility - File Transfer] screen.

In such a case, use an operating system of the same language on both sides to fix this problem.

The following are general notes on Machine Data Storing Schedule function.

NOTE

- 1 When the storing schedule is executed while using File Transfer function on the Web screen, the machine data which is being used for File Transfer function may not be stored. Try not to use File Transfer function while the storing schedule is in process. Also, output manually the machine data that were not stored.
- 2 If File Transfer function is used on the Web screen while the storing schedule is in process, file transfer may not be successful. Try not to use File Transfer function while the storing schedule is in process.
- 3 There is no need to have AdminTool running on the date/time the output schedule will be executed.
- 4 Machine Data Storing function supports FanucCNC, FanucRobot, and FanucRobodrill type machines; Machine Data Archive Management function supports FanucCNC and FanucRobodrill machines.
- 5 Only the description of changes on the text files can be verified on the [Utility - Machine Data Archive Management] screen.
In the case of binary files, the description of changes cannot be verified because they are treated as text files and will become unreadable.
- 6 In the difference display area of the [Machine Data Archive Management] window, the line feed code will not be the target for comparing the difference.
- 7 After registering a schedule, it is required to execute the "Send to DB" process.
- 8 When storing schedule is executed on the machine that is collecting signal data, the collection process will take more time. Use the storing schedule so that it will not affect the collection process.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	560/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

- 9 After a schedule is registered, if the machine for the schedule is deleted or any of the following settings of that machine is modified in the [Machine Setting] window, re-register the schedule by clicking the [Unregister] button and then the [Register] button on the [Machine Data Storing Schedule] Window.

Window Name		Setting Field Name	
Machine Setting	(When the Collector is selected to connect to the target machine)	Collector Setting	IP Address/Host Name
	(When the target machine is selected) The [Protocol] tab	-	Web Name
		Protocol Setting	IP Address/Host Name
			Port
			Time out (Sec)
		Data server setting ^{*1}	Data server setting
			IP Address/Host Name
			User Name
			Password
			Port
		FTP Server Setting ^{*2}	IP Address/Host Name
			User Name
			Password
			Port

^{*1}: available for FanucCNC and FanucRobodrill type machines

^{*2}: available for FanucRobot type machines

This note also applies to machines for which a registered schedule has been set if they are deleted or any of their fields shown above is modified through the following functions:

Batch Setting: Importing is performed after deleting machines from the exported files or modifying the exported files on any setting fields that correspond to those shown above.

Restore Settings: Restoring backed-up settings may delete machines or modify any setting fields shown above.

Receive from DB: Receiving settings from the database may delete machines or modify any setting fields shown above.

Easy Setting: This function may delete machines or modify any setting fields shown above.

- 10 CNC models and data types supported by Machine Data Storing and Machine Data Archive Management functions, and robot data types supported by Machine Data Storing function are the same as those supported by File Transfer function.

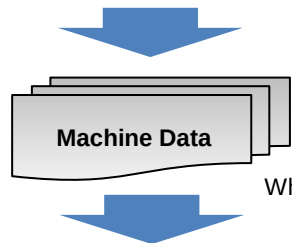
For the details, refer to the "UTILITY- FILE TRANSFER SCREEN" section of the "FANUC MT-LINK*i* WebUI OPERATOR'S MANUAL."

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION		Sheet. 561/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

■ [Schedule Output] tab page

The screenshot shows the 'Machine Data Storing Schedule' dialog box with the following components and annotations:

- Target Data:** A red box highlights the 'Target Data' tab and the 'Equipment list' and 'Machine list' sections. The 'Equipment list' contains a list of equipment units (0001 to 0012). The 'Machine list' contains a list of machines (001 to 003) with sub-items like CNCDATA, CNCPROG, PMCDATA, FROM, and SRAM.
- Enable Archive Management:** A red box highlights the 'Enable Archive Management' section, which includes a checked checkbox, a 'Maximum Number of Archives' field set to 100, and a list of files excluded from archive management (FROM, SRAM).
- Storing Folder:** A red box highlights the 'Storing Folder' field, which is set to 'C:\FANUC\MT-LINK\#FileTransfer\Backup'.
- Schedule Setting:** A red box highlights the 'Schedule Setting' section, which includes a 'Storing Interval' dropdown set to 'Daily' and a 'Start Time' field set to '18:00'.
- Schedule Status:** A red box highlights the 'Schedule Status' section, which includes a 'Schedule Enable' checkbox and 'Register' and 'Unregister' buttons.
- Settings for Archive Management:** A red box highlights the 'Settings for Archive Management' section, which includes a 'Machine Name' dropdown set to 'Machine001' and a 'File' dropdown set to 'CNCDATA\CNC-PARA.TXT'.



Schedule Output

When the [Enable Archive Management] check box is selected

After output, history and changes of machine data can be reviewed on the [Utility – Machine Data Archive Management] screen

The screenshot shows the 'Utility - Machine Data Archive Management' screen with the following components:

- Machine001:** A dropdown menu showing the selected machine.
- Search For All Data:** A button to search for all data.
- Display Difference:** A button to display the difference between the source and destination.
- Display File:** A button to display the file.
- Start date of search:** A date field set to '2020/10/25'.
- Show Folder Tree:** A button to show the folder tree.
- Folder Tree:** A tree view showing the folder structure of the machine data.
- Comparison source:** A text field showing the source of the comparison, set to '2018/09/20 09:44:01'.
- Comparison destination:** A text field showing the destination of the comparison, set to '2018/11/16 09:50:02'.
- Display mode:** A dropdown menu showing the display mode, set to 'Split'.
- Display file:** A dropdown menu showing the display file, set to 'Comparison source'.
- Machine Data:** A table showing the machine data, including columns for Folder, File, Last Update, Type, and Archive.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	FANUC CORPORATION			Sheet.	562/868

- Use the Schedule Output option in the following case (with the [Enable Archive Management] check box cleared):
 - It is necessary to archive machine data periodically on a regular basis so that those archived data can be reviewed.
- Use the Schedule Output option in the following case (with the [Enable Archive Management] check box selected):
 - It is necessary to archive machine data periodically on a regular basis so that changes and change history of them can be reviewed with a Web screen (the [Utility – Machine Data Archive Management] screen).

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	563/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

■ [Manual Output] tab page

Machine Data Storing Schedule

Schedule Output **Manual Output**

Target Data

Equipment list

- ☐ All equipment units
- ☒ Equipment0001
- ☒ Equipment0002
- ☒ Equipment0003
- ☒ Equipment0004
- ☒ Equipment0005
- ☒ Equipment0006
- ☒ Equipment0007
- ☒ Equipment0008
- ☒ Equipment0009
- ☒ Equipment0010
- ☒ Equipment0011
- ☒ Equipment0012

Machine list

- ☒ All machines
- ☒ Machine002
- ☒ CNCDATA
- ☒ CNCPROG
- ☒ PMCDATA
- ☒ FROM
- ☒ SRAM

Manual Output

Storing Folder: C:\FANUC\MT-LINK\File Transfer\BackupManual

Run

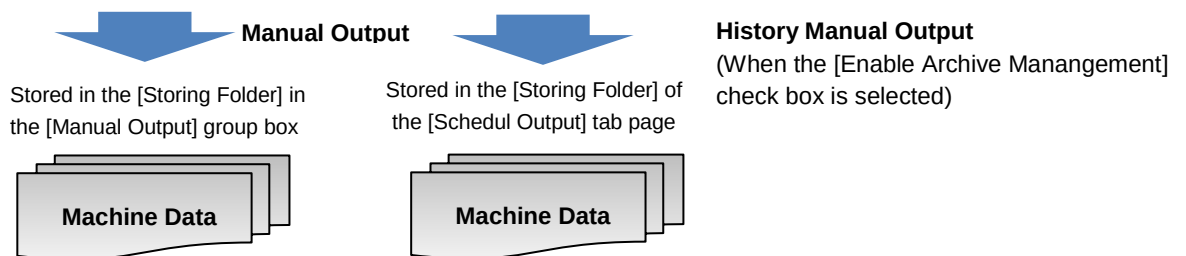
History Manual Output

Run

Copy

Result display area

Close



After output, history and changes of machine data can be reviewed on the [Utility – Machine Data Archive Management] screen

Utility - Machine Data Archive Management

Machine002

Start date of search: 2020/10/25

Show Folder Tree

Folder	File	Last Update	Type
/CNCDATA	CNC-PARA.TXT	2020/10/21 13:38:39	Change
/CNCDATA	EXT_WKZ.TXT	2017/11/27 16:01:09	Add
/CNCDATA	MACRO.TXT	2020/10/16 17:11:05	Change
/CNCDATA	OPRT_HIS.TXT	2020/10/21 13:38:39	Change
/CNCDATA	TOOLOFST.TXT	2017/11/27 16:01:09	Add
/CNCPROG/CNC_MEM/USER/PATH1	01	2020/10/16 17:11:05	Delete
/CNCPROG/CNC_MEM/USER/PATH1	02	2020/10/16 17:11:05	Delete
/CNCPROG/CNC_MEM/USER/PATH1	0400	2020/10/16 17:11:05	Delete
/CNCPROG/CNC_MEM/USER/PATH1	050	2020/10/16 17:11:05	Delete

Comparison source : 2019/12/19 13:39:03 | Comparison destination : 2019/12/19 13:42:47

Display mode: Split Unified

Display file: Comparison source Comparison destination

Archive	Type
2020/10/21 13:38:39	Change
2020/10/16 17:11:05	Change
2019/12/19 13:42:47	Change
2019/12/19 13:39:03	Change
2017/12/04 16:01:08	Change
2017/11/28 16:01:06	Change
2017/11/27 16:01:09	Add

10:47:52 AM FANUC MT-LINK/ 11 4 1 / 16

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	FANUC CORPORATION		Sheet.	564/868

- | | | | | | | | | | | | | |
|------|---------------|--------|--|-------|--|--------|-------|-------------------|---|--|--------|---------|
| 20 | Nov. 22, 2023 | uwano | Revised for MT-LINK <i>i</i> Version 4.0 | | | | | Title | FANUC MT-LINK <i>i</i>
AdminTool OPERATOR'S MANUAL | | | |
| 19 | Apr. 27, 2021 | uwano | Revised for MT-LINK <i>i</i> Version 3.11 | | | | | | | | | |
| 18 | July 28, 2020 | uwano | Revised for MT-LINK <i>i</i> Version 3.10 | | | | | | | | | |
| 17 | Jan. 30, 2020 | uwano | Revised for MT-LINK <i>i</i> Version 3.9 | | | | | Draw | A-40829-00001EN | | | |
| 16 | July 22, 2019 | uwano | Revised for MT-LINK <i>i</i> Version 3.8 | | | | | | | | | |
| 15 | Mar. 29, 2019 | ogawa | Revised for MT-LINK <i>i</i> Version 3.7.2 | | | | | | | | | |
| 14 | Jan. 29, 2019 | ogawa | Revised for MT-LINK <i>i</i> Version 3.7 | | | | | FANUC CORPORATION | | | Sheet. | 565/868 |
| Ed. | Date | Design | Description | | | | | | | | | |
| Date | Nov.30,2015 | Desig. | Ogawa | Check | | Apprv. | Inoue | | | | | |

10.2.1 Steps for Registering the Machine Data Storing Schedule

To set the machine data storing schedule and the machine data archive management settings, follow the steps below.

(1) Select the [Schedule Output] tab.

The screenshot shows the 'Machine Data Storing Schedule' dialog box with the 'Schedule Output' tab selected. The 'Equipment list' on the left has 'All equipment units' checked, and the 'Machine list' on the right has 'All machines' checked. The 'Storing Folder' is set to 'C:\FANUC\MT-LINK\#FileTransfer\#Backup'. Under 'Schedule Setting', the 'Storing Interval' is set to 'Daily' with a 'Start Time' of '18:00'. The 'Enable Archive Management' checkbox is checked, and the 'Machine Name' is set to 'Machine001'. The 'File' dropdown is set to 'CNCDATA\CNC-PARA.TXT'. The 'Schedule Status' section shows 'Schedule Enable' is unchecked. There are 'Register', 'Unregister', and 'Close' buttons at the bottom right.

(2) Select the check box of the equipment units on the [Equipment list] which include the machine to be backed up.

When the check box for [All equipment units] is selected, all the check boxes of the equipment units included in the [Equipment list] are selected. The machines included in the equipment with the check box selected are displayed on the [Machine list].

This screenshot shows the same dialog box after selecting 'All equipment units' in the 'Equipment list'. All individual equipment units from 'Equipment0001' to 'Equipment0011' are now checked. The 'Machine list' on the right now shows 'Machine001' and 'Machine002' selected, with their respective data files (CNCDATA, CNCPROG, PMCDATA, FROM, SRAM) listed under them.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION	Sheet.	566/868	
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

NOTE

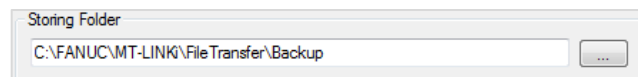
- 1 On the [Equipment list], equipment units are displayed that include FanucCNC, FanucRobot or FanucRobodrill type machines.
- 2 On the [Machine list], FanucCNC, FanucRobot or FanucRobodrill type machines are displayed.
- 3 Until the ROBODRILL Option license is verified, FanucRobodrill type machines are grayed out on the [Machine list] and their check boxes are unclickable.

- (3) Select the check boxes of the folders of the machines to be backed up on the [Machine list].
 When the check box of a machine is selected, all the check boxes of the folders placed under this folder are selected.
 When the [All machines] check box is selected, check boxes of all the machines in the [Machine list] and the check boxes of the folders placed under these machines are selected.

NOTE

For the machines which has information set for connecting to the Data Server in the machine setting, the data within the Data Server is saved in the "DATA_SV" folder under the "CNCPROG" folder.

- (4) Specify the folder to store the machine data as the [Storing Folder].
 Clicking the browse folder button ([...]) opens the [Browse For Folder] window to either select an existing folder or create a new folder. The [Browse For Folder] window selects the folder populated in the [Storing Folder] field.



In case of a scheduled output, the [Storing Folder] field populates the following folder by default. On subsequent sessions, it populates the folder specified in the previous session:

<MT-LINK*i* installation folder>\FileTransfer\Backup

NOTE

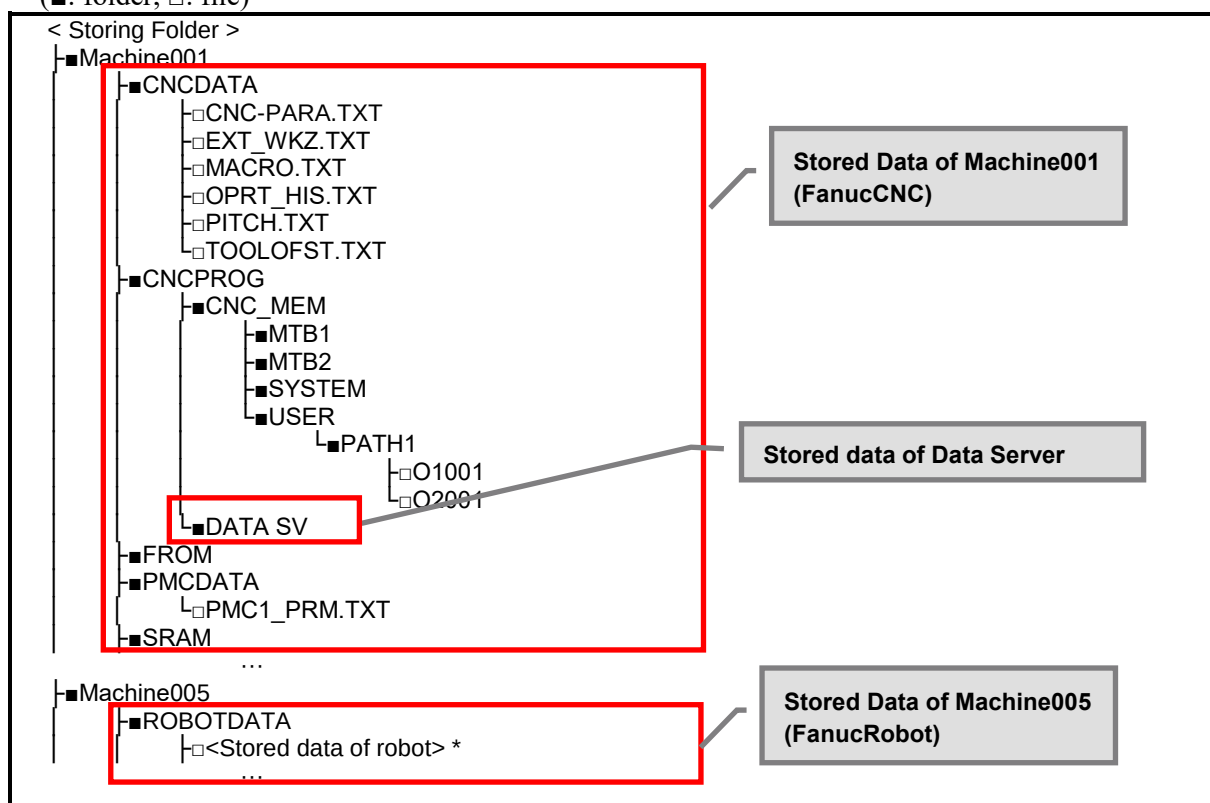
- 1 For the storing folder, paths of disk drives such as fixed disk drives (internal disk/external disk) can be specified. The paths of other drives (such as optical disk drives or network drives) or UNC paths cannot be specified.
- 2 As the size of the storing data will be massive, it is recommended to specify a separate folder for the report path, such as the folder of an expanded external disk drive. Also, make sure that the disk space of the output folder does not get compressed, by deleting the output files or moving the output files to another media on a regular basis.
- 3 If the free disk space of the disk drive specified as the storage destination is under 5GB, the execution of the file storing process will not start.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	567/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

In the storing folder, the folders specified on the [Machine list] of each machine, and files under the folders are stored.

Shown below is an example for storing when the [Enable Archive Management] check box (mentioned in item (6) in the succeeding procedures) is cleared (function is disabled).

(■: folder, □: file)



* Stored data of robot: All the data in MDBs (memory binary devices) on a robot

When the [Enable Archive Management] check box (mentioned in item (6) in the succeeding procedures) is selected (function is enabled), a folder for time stamps is created under the folder with the name of the machine, and within that folder, the machine metadata file (MachineMetaData.json) for managing the machine data and the archive will be stored.

The name of the folder for the time stamp will be in the format of "yyyyMMddTHH:mm:ssZ" in UTC (Coordinated Universal Time).

("yyyyMMdd" and "HH:mm:ss" are the storing date/time (UTC+0) of the machine data.)

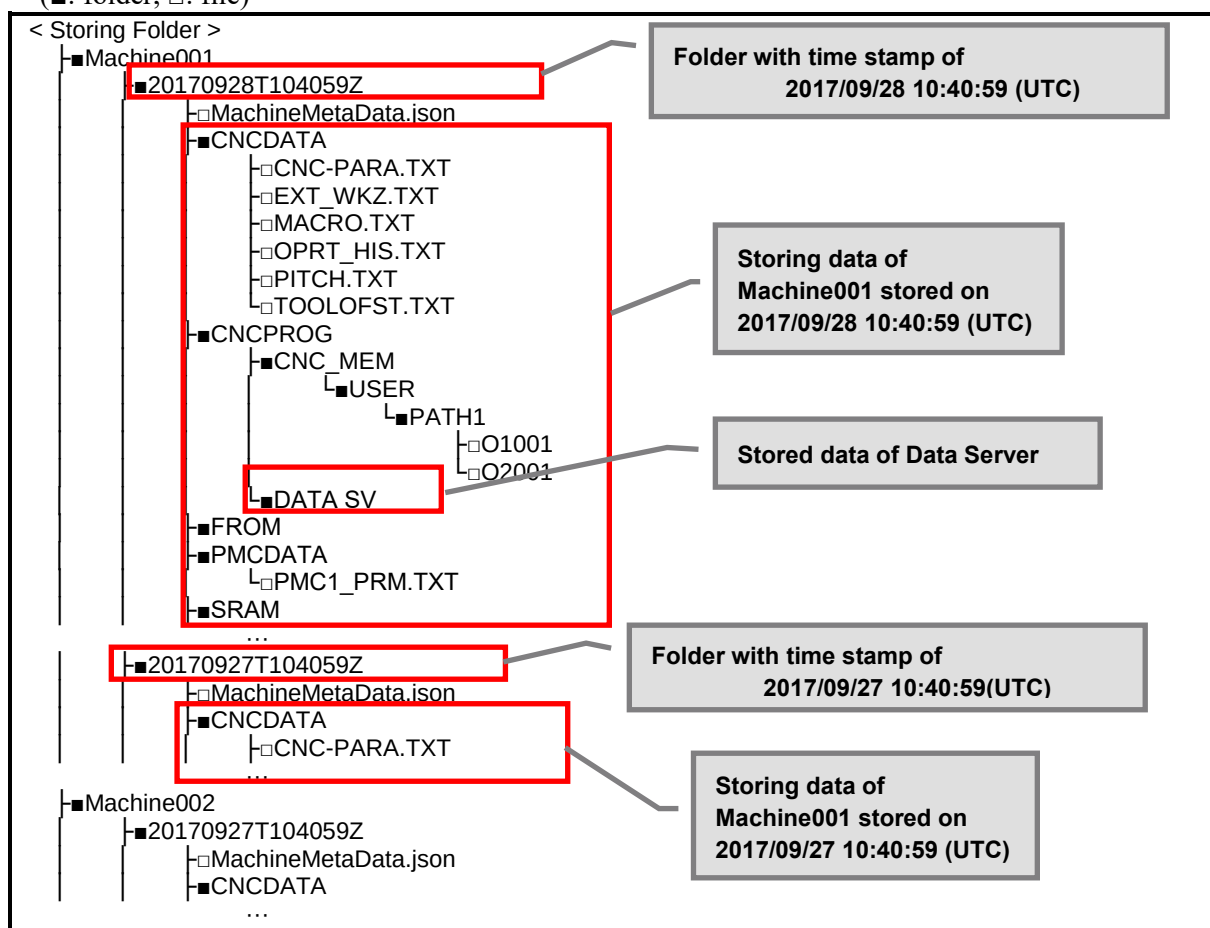
NOTE

The machine metadata file (MachineMetaData.json) contains the information needed for archive management (such as folder structure and check sum of files). Do not modify or delete the files or folders in the storing folder including this file. If the files or folders are modified or deleted, archive information on the [Utility - Machine Data Archive Management] screen of the Web screen will not be displayed properly.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION		Sheet. 568/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

Shown below is an example for storing when the [Enable Archive Management] check box (mentioned in item (6) in the succeeding procedures) is selected (function is enabled).

(■: folder, □: file)



NOTE

When the [Enable Archive Management] check box is selected (function is enabled), the existing archive data cannot be browsed on the [Utility - Machine Data Archive Management] screen of the Web screen, if the storing folder is changed. However, by changing the storing folder back to the original folder before changing, the archive data stored before the change can be browsed. If you want to carry over the archive data, copy the archive data to the newly created storing folder.

(Example)

Steps for carrying over the archive data when changing the storing folder

* When "C:\FANUC\MT-LINKi\FileTransfer\Backup" is the current storing folder, and "D:\Backup" is the new storing folder.

- 1) Copy all the folders under "C:\FANUC\MT-LINKi\FileTransfer\Backup" to under "D:\Backup."
- 2) Change the storing folder to "D:\Backup," and register the schedule.
- 3) Execute [Send to DB] procedure.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	569/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

(5) Specify the [Storing Interval] and the [Time]. For the [Storing Interval], [Daily], [Weekly] or [Monthly] can be selected.

- If [Daily] is selected, [Start Time] is specified.

For the [Start Time], the time to start storing the machine data is specified. Default setting is "18:00."

The screenshot shows the 'Schedule Setting' dialog box. Under 'Storing Interval', the 'Daily' radio button is selected. In the 'Time' section, the 'Start Time' is set to 18:00.

- If [Weekly] is selected, [Start time] and [Every week] are specified.

For the [Start Time], the time to start storing the machine data is specified. Default setting is "18:00."

For the [Every week], the day of the week (Sunday to Saturday) to store the machine data is selected. Default setting is "Saturday."

The screenshot shows the 'Schedule Setting' dialog box with 'Weekly' selected. The 'Every week' dropdown menu is open, displaying a list of days from Sunday to Saturday, with 'Saturday' highlighted at the bottom.

- If [Monthly] is selected, [Start Time] and [Every month] are specified.

For the [Start Time], the time to start storing the machine data is specified. Default setting is "18:00."

For the [Every month], the day of the month ("1 day" to "28 day", "End of month") to store the machine data is selected. Default setting is "1 day."

The screenshot shows the 'Schedule Setting' dialog box with 'Monthly' selected. The 'Every month' dropdown menu is open, displaying a list of intervals from '1 day' to '28 day', plus 'End of month'. '1 day' is highlighted at the top of the list.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 570/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- (6) Set the [Enable Archive Management] check box.

When enabling the archive management function, select the check box. By default, this check box is cleared.

* If the [Enable Archive Management] check box is cleared (disabled), the settings in the following steps (7) to (9) are not required.

- (7) Set the [Maximum Number of Archives].

Set the maximum number of data archives for each equipment unit. The value of the maximum number can be set from 1 to 3650. When a value that is out of range is specified, either the upper limit value or the lower limit value which is the closest to the specified value will be set. The default value is "100."

If the total number of data archives exceeds the maximum number, data archive will be deleted from the oldest archive when the archive data is being stored. The archive data to be deleted are deleted recursively from the files and folders that are stored in the folder with the timestamp.

- (8) Set the [Automatic Deletion of Archive with No Difference] check box.

When the check box is selected, the latest archive data will be deleted automatically during the storing of archive data, if there is no difference between the latest archive data and the previous archive data of each machine (NOTE 1). By default, this check box is cleared.

The archive data to be deleted are deleted recursively from the files and folders that are stored in the folder with the timestamp.

NOTE

- 1 For each machine, the differences in all the files and folders stored in the folders of the latest timestamp and the previous timestamp. If there are no differences, the folder with the latest timestamp will be deleted. However, the files registered as the [Files Excluded from Archive Management] as in step (9) will be excluded from the files to determine the differences.
- 2 It is recommended to select the [Automatic Deletion of Archive with No Difference] check box during operation.
When the number of stored archives becomes large, the response of the [Utility - Machine Data Archive Management] screen of the Web screen may become slow.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 571/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

(9) Set the [Files Excluded from Archive Management].

The files registered as the files excluded from archive management are omitted from the files used in determining the difference between the archive data.

Up to 1000 files can be registered.

In the default setting, the files registered are all the files under the following FROM/SRAM folder.

- "*"FROM*"
- "*"SRAM*"

NOTE

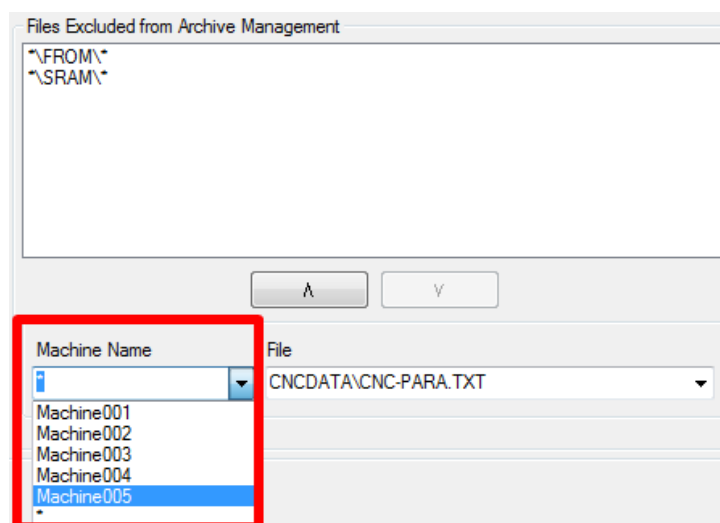
- 1 In the case of binary files, the description of changes cannot be verified because the characters will become unreadable.
- 2 Do not delete the default setting of the files excluded from archive management. Since binary data is included in the files under the FROM/SRAM folder, the changes in these files cannot be verified.

The files excluded from archive management are registered by the following steps.

1) Select a [Machine Name] from the drop-down menu.

FanucCNC or FanucRobodrill type machines can be selected that have been registered on the [Machine Setting] window.

[Machine Name] is editable also. The first character of a machine name should be an alphabet character and the subsequent characters can be alphanumeric characters, hyphens ("-"), and underscores ("_"). Up to 32 bytes can be entered.



NOTE

- 1 Until the ROBODRILL Option license is verified, no names of FanucRobodrill type machines are listed in the [Machine Name] drop-down list and thus they are not selectable.
- 2 The [Machine Name] drop-down list does not list FanucRobot type machines and thus cannot be selected.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 572/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- 2) Select a [File] from the drop-down menu.
Data files in CNCDATA or program files in CNCPROG can be selected.
Also, a selected file can be edited. The path to the file must be up to 259 full-width characters long.
(", " / ":" / "/" / """" / "<" / ">" / "|" and control codes from 0x0000 to 0x001F cannot be used.)

Files Excluded from Archive Management

~\FROM~
~\SRAM~

Machine Name: Machine001

File: CNCDATA\CNC-PARA.TXT

Schedule Status: ☒ Schedule Enable

Files Excluded from Archive Management List:

- CNCDATA\CNC-PARA.TXT
- CNCDATA\CNCPARAM.DAT
- CNCDATA\TOOLOFST.TXT
- CNCDATA\TOOLOFST.DAT
- CNCDATA\TOOL-OFS.TXT
- CNCDATA\EXT_WKZ.TXT
- CNCDATA\WORK-G54.DAT
- CNCDATA\EXT-WKZ.TXT
- CNCDATA\PITCH.TXT
- CNCDATA\PITCHERR.DAT
- CNCDATA\MACRO.TXT
- CNCDATA\MACROVAR.DAT
- CNCDATA\OPRT_HIS.TXT
- CNCDATA\OPT-HIST.DAT
- CNCDATA*
- CNCPROG*
- PMCDATA\PMC1_PRM.txt
- PMCDATA*
- FROM~
- SRAM~

- 3) Click the [^] (add) button.
A character string of the path ("value selected as [Machine Name]" + "\" + "value selected as [File]") for the file excluded from archive management is added to the list box.

Files Excluded from Archive Management

~\FROM~
~\SRAM~

Machine Name: Machine005

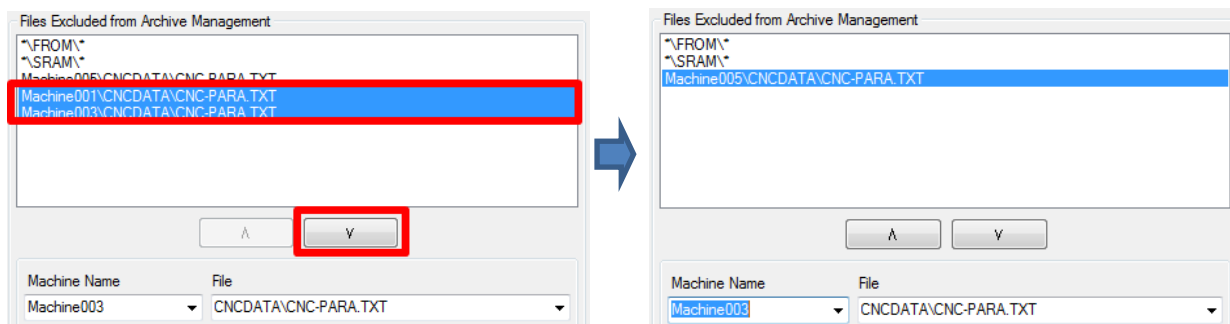
File: CNCDATA\CNC-PARA.TXT

Files Excluded from Archive Management List:

- Machine005\CNCDATA\CNC-PARA.TXT

By selecting a file excluded from archive management in the list box, and clicking the [V] (delete) button, the selected row will be deleted from the list box. Two or more rows can be selected and deleted also.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	573/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue



* When specifying the [Machine Name] and the [File], an asterisk ("*") or a question mark ("?") can be used as a wildcard character. An asterisk represents a character string of an optional length, and a question mark represents a single optional character.

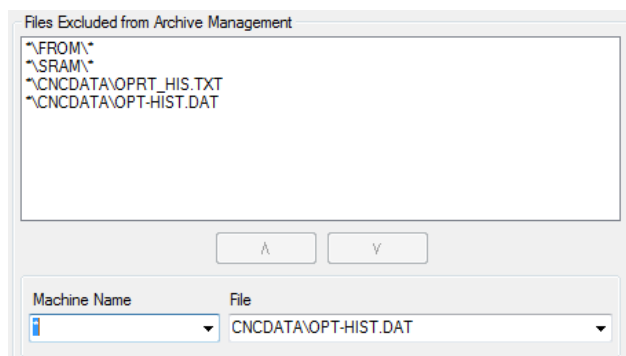
<Example for setting files excluded from archive management>

Example 1: If excluding operation history files for all the machines

The following two files are registered for operation history, as there are 2 types of files by model.

Machine Name: "*"; File: "CNCDATA\OPRT_HIS.TXT"

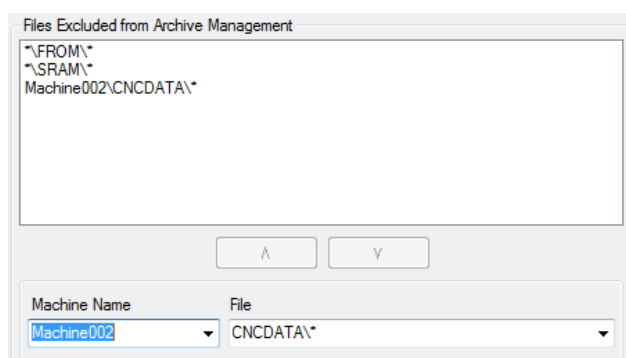
Machine Name: "*"; File: "CNCDATA\OPT-HIST.DAT"



Example 2: When setting all the data files in the CNCDATA folder as files excluded from archive management for a specific machine "Machine002"

Register as follows.

Machine Name: "Machine002"; File: "CNCDATA*"



20	Nov. 22, 2023	uwano	Revised for MT-LINK ⁱ Version 4.0	Title	FANUC MT-LINK ⁱ AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK ⁱ Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK ⁱ Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK ⁱ Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK ⁱ Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK ⁱ Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK ⁱ Version 3.7	FANUC CORPORATION			Sheet. 574/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

Example 3: When excluding programs "PATH1\O1001" to "O1099" for the machine group with the machine name starting with "B"

Register as follows.

Machine Name: "B*"; File: "CNCPROG\CNC_MEM\USER\PATH1\O10??"

NOTE

- 1 If the schedule is registered after changing the setting of [Files Excluded from Archive Management], the changes will not be reflected on the files of the past. Even if the setting for the [Automatic Deletion of Archive with No Difference] is enabled, archive data of the past will not be deleted retrospectively.
- 2 The files excluded from archive management cannot be browsed on the [Utility - Machine Data Archive Management] screen of the Web screen.

(10) Click the [Register] button in the [Schedule Status] group box to register the schedule and the archive management settings.

When the schedule is registered, [Schedule Enable] check box is selected, [Register] button is disabled, and [Unregister] button is enabled.

When unregistering a registered schedule and the archive management setting, click the [Unregister] button. When the [Close] button is clicked, and the [Utility - Machine Data Archive Management] screen is opened once again, the display of the details of the settings excluding the [Storing Folder] and the [Files Excluded from Archive Management] are reset to default setting.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	575/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue		

NOTE

- 1 In order to change the schedule when a schedule is already registered, click the [Unregister] button to unregister the schedule, and register the new schedule. The changes entered when the schedule is already registered cannot be registered, and will be cancelled when unregistering.
- 2 The schedule that can be registered is either one of Daily, Weekly, or Monthly.

NOTE

If the ROBODRILL Option license has not been verified yet and FanucRobodrill type machines are selected in the [Machine list] group box, clicking the [Register] button opens an error message dialog box.
In this case, delete the FanucRobodrill type machines using the [Machine Setting] window and then register a schedule.

- (11) Click the [Close] button.

If the [Register] button is not clicked, the settings will be canceled.

- (12) When the settings are set or changed in the archive management, execute [Send to DB] procedure after registering the schedule.

- (13) The storing process starts when the specified start date/time for storing the machine data arrives. When the storing process is completed, data is stored in the storage folder. The storing process that is registered as a schedule cannot be canceled while the process is in action. Also, when the storing process was unsuccessful, refer to the "**When Failing to Store the Machine Data**" subsection.

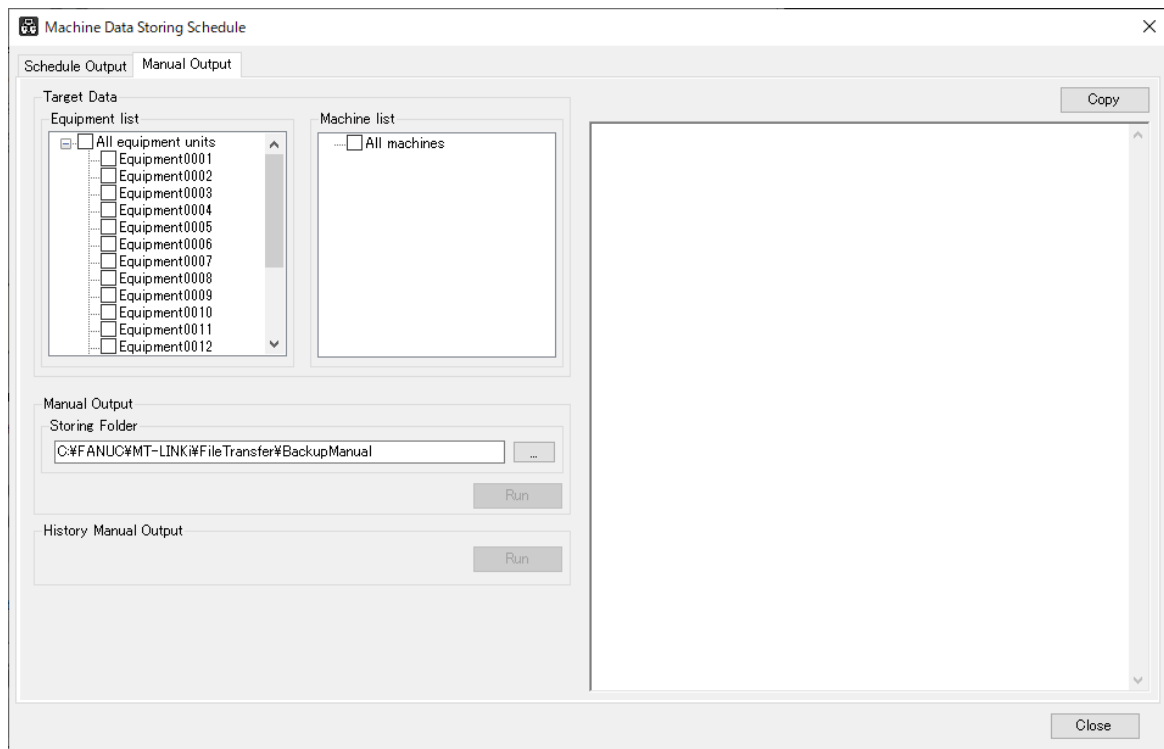
NOTE

- 1 When the PC is shut down while storing the schedule, a part of the data may not be stored, as the storing process of the machine data has not been completed. For the data that had not been stored, store the data separately by outputting manually.
- 2 When the time of the next storing schedule arrives while a storing schedule process is in progress, the next storing schedule is executed after the current storing schedule process is completed.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 576/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

10.2.2 Steps for Storing Machine Data Using Manual Output

The machine data can be stored by using manual output. When the [Enable Archive Management] check box is selected (enabled) on the [Schedule Output] tab, machine data archive can be output manually.



Follow the steps below to output manually.

- (1) Select the [Manual Output] tab.
- (2) On the [Equipment list], select the check box of the equipment including the machine to be backed up.
* For the details, refer to step (2) of the "Steps for Registering the Machine Data Storing Schedule" subsection.
- (3) On the [Machine list], select the check box of the folder of the machine to be backed up.
* For the details, refer to step (3) of the "Steps for Registering the Machine Data Storing Schedule" subsection.
- (4) For the [Storing Folder], specify the folder to store the machine data.
Clicking the browse folder button ([...]) opens the [Browse For Folder] window to either select an existing folder or create a new folder. By default, the [Browse For Folder] window selects the folder populated in the [Storing Folder] field.

In case of a manual output, the [Storing Folder] field populates the following folder by default. On subsequent sessions, it populates the folder specified in the previous session:

<MT-LINK*i* installation folder>\FileTransfer\BackupManual

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	577/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

NOTE

- 1 For the storing folder, paths of disk drives such as fixed disk drives (internal disk / external disk) can be specified. The paths of other drives (such as optical disk drives or network drives) or UNC paths cannot be specified.
- 2 As the size of the storing data will be massive, it is recommended to specify a separate folder for the report path, such as the folder of an expanded external disk drive. Also, make sure that the disk space of the output folder does not get compressed, by deleting the output files or moving the output files to another media on a regular basis.
- 3 If the free disk space of the disk drive specified as the storage destination is under 5GB, the execution of the file storing process will not start.

(5) Click the [Run] button in the [Manual Output] group box to store the machine data in the storing folder.

When the [Enable Archive Management] check box is selected (enabled) on the [Schedule Output] tab

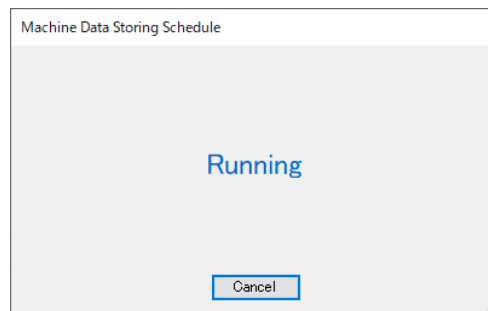
By clicking the [Run] button in the [History Manual Output] group box, the data archive of the machine selected on the [Machine list] of the [Manual Output] tab is stored in the [Storing Folder] of the [Schedule Output] tab. The archive data that was stored manually will be subject to archive management on the [Utility - Machine Data Archive Management] screen.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7								
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue	FANUC CORPORATION		Sheet.	578/868

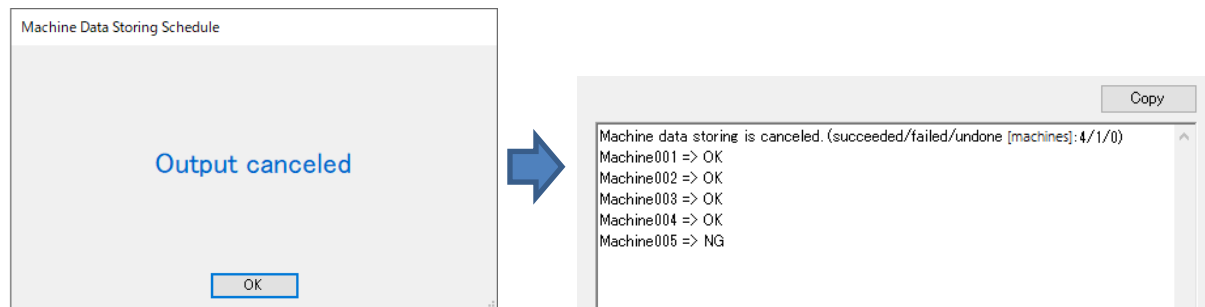
NOTE

In order to browse the manually stored archive data on the [Utility - Machine Data Archive Management] screen of the Web screen, you need to specify the settings for the archive management on the [Schedule Output] tab in advance, and execute the [Send to DB] procedure after registering the schedule.

- (6) The following dialog box will be shown while manual output is in progress.

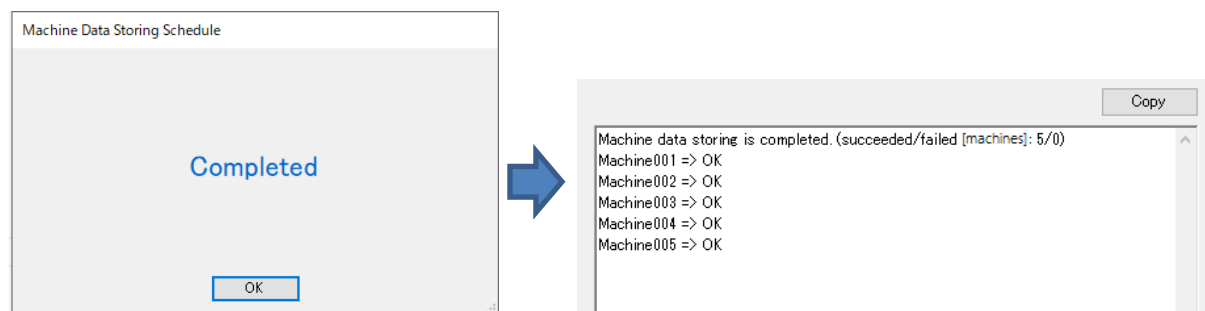


When the [Cancel] button is clicked, the following dialog box will be shown and output process can be canceled. Click the [OK] button to display the output results for each machine in the result display area.

**NOTE**

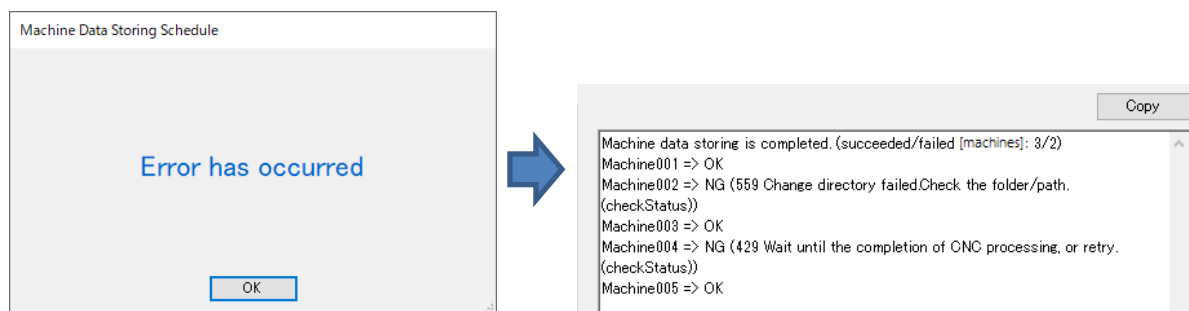
When the [Cancel] button is clicked, the file that has finished the storing process will remain, but the file that is in the storing process at that moment will be deleted.

- (7) When the output process is completed, following dialog box appears. Click the [OK] button to display the output results for each machine in the result display area.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 579/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

When error occurs in the output process, following dialog box appears.
Click the [OK] button to display the output results and error details for each machine in the result display area.

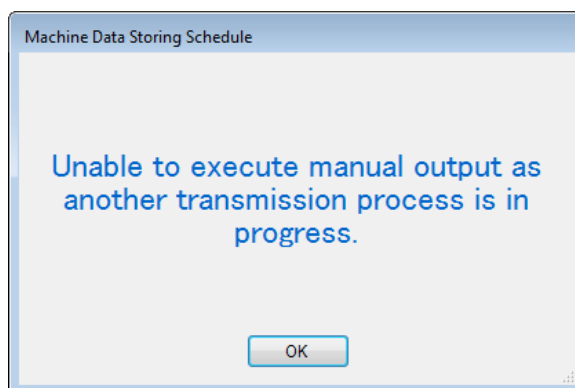


* When the [Copy] button is clicked, the information displayed in the Result display area is copied to the clipboard.

(8) Click the [Close] button.

NOTE

- 1 The output process may require considerable amount of time. Be careful not to specify too many equipment units or machines to store data.
- 2 During the output process, it is unable to execute setting operations on the AdminTool windows.
- 3 When the output process is unsuccessful, refer to the "**When Failing to Store the Machine Data**" subsection.
- 4 If manual output is executed while scheduled storing process is in progress, the following error message is displayed, and the manual output will not be executed. Try to execute manual output again after a while.



- 5 When the time of the storing schedule arrives while manual output process is in progress, the storing schedule is executed after the manual output process is completed.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	580/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

10.2.3 When Failing to Store the Machine Data

When the machine data storing process fails, the files with the names shown below are output to the storage folder.

■File output when the storing process of machine data fails

File Name	Cause of Failure
#####CANCEL OUTPUTTING#####_<yyyyMMddHHmmss>_<ID>	Output canceled by the user
#####ERROR OUTPUTTING#####_<yyyyMMddHHmmss>_<ID>	Output process ended with error
#####ITEMERROR OUTPUTTING#####_<yyyyMMddHHmmss>_<ID>	File with output error exists (NOTE 3, 4)
#####DISKFULL OUTPUTTING#####_<yyyyMMddHHmmss>_<ID>	Free space of storage disk is under 5GB

NOTE

- 1 <yyyyMMddHHmmss> part of the file name is the output date/time of the file.
- 2 <ID> part of the file name is any 4-digit ID that uniquely identifies the file name.
- 3 The error code, the machine name and the detailed message (status string returned from the FTP Client) are stored in the file.
(Example) connect_error@Machine001 <Detailed message>

Refer to the "List of Errors for File Transfer Functionality" subsection in the "TROUBLES RELATED TO FILE TRANSFER FUNCTIONALITY" section in the "MT-LINK*i* TROUBLESHOOTING" chapter of the "FANUC MT-LINK*i* / Integration Server Troubleshooting MANUAL" to identify the description of the error that may occur from the items shown below, and remove the cause of the error.
 - Upload/download→Upload
 - CNC model →Model of the connection target CNC
 - Type of data →Type of data of the manipulation target file
- 4 In case of the following errors, the storing process will not be suspended.
 - The machine has no option (no_option_error)
 - Error of parameter 3202#0, #4 being "1" when program of O8000's/O9000's is being stored (protect_error)
- 5 If the storing process of the machine data is suspended by an error occurring during the process, that machine data will be excluded from archive management.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION				Sheet.	581/868	

10.3 SIGNAL DATA OUTPUT

By using the signal data output function, up to 24 hours (unit: 0.5 second) of signal history information data for the equipment can be output as a CSV file in time-series.

The CSV file can output 24 hours of signal data, by splitting the data into up to four parts.

NOTE

- 1 If schedule output is executed while operating the Web screen, the response may become bad.
Operate the schedule output so that the time of signal data output and the time of machine operation do not overlap. For instance, if the operation of the Web screen is from 9:00 to 17:00, set the schedule output so that signal data will be output from 18:00 or later.
- 2 There is no need to have AdminTool running on the date/time the output schedule will be executed.
- 3 Signal data output function can be customized. For the details on the items that can be customized, refer to the "[Customizing the Signal Data Output Function](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	582/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

10.3.1 Creating and Registering the Output Target Data of Signal Data Output

This subsection explains how to create the output target data for the signal data output. The output target data is a CSV file setting the equipment name of the output target, the enable/disable status of the output function, and the name of the equipment status.

The output target is created based on the following conditions.

- Item of the output target equipment unit is counted as one unit of data, each stored in one line separated by commas.
- Item name is entered in the first line.
- Enclose item names and setting values in double quotation marks ("").
- UTF-8 is the character code used for storing the data.

The output target data is created by following the steps mentioned later.

The format of the output target data is shown in the following.

Name of item	Description of setting value
Equipment	Enter the equipment name of the registered equipment.
TargetEquip	"+": Output target equipment, "-": Non output target equipment
TargetData1	Enter equipment status name 1 which is the output target.
.	.
.	.
.	.
TargetData<N>	Enter equipment status name <N> which is the output target.

NOTE

The name of the items from TargetData1 to TargetData<N> must be created with sequential numbering starting from 1. There is no upper limit for the number of <N>. If the numbering is not sequential, output target data cannot be registered.

Sample of output target data

```
"Equipment","TargetEquip","TargetData1","TargetData2","TargetData3"
"Equipment0001","+","SigOP_path1_Machine001","EMG_path1_Machine001",
"SigCUT_path1_Machine001"
"Equipment0002","-","PartsNum_path1_Machine002",
"AbsPos_0_path1_Machine002","MacroVar_101_path1_Machine002"
```

The meaning of the above sample is as follows.

Equipment name (Equipment): Equipment0001
Output target (TargetEquip): Enabled
Equipment status 1 (TargetData1): SigOP_path1_Machine001,
Equipment status 2 (TargetData2): EMG_path1_Machine001,
Equipment status 3 (TargetData3): SigCUT_path1_Machine001

Equipment name (Equipment): Equipment0002
Output target (TargetEquip): Disabled
Equipment status 1 (TargetData1): PartsNum_path1_Machine002,
Equipment status 2 (TargetData2): AbsPos_0_path1_Machine002,
Equipment status 3 (TargetData3): MacroVar_101_path1_Machine002

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue
FANUC CORPORATION					Sheet.	583/868

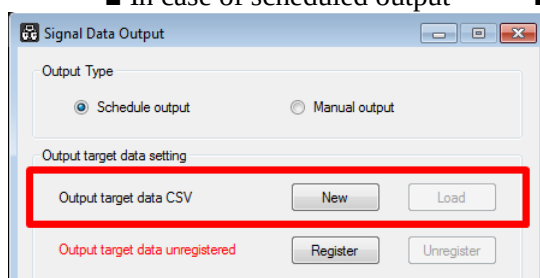
Output target data is created and registered by the following steps.

- (1) The output target data can be newly created by clicking the [New] button on the [Output target data CSV] line in the [Output target data setting] area. All the equipment units and all the equipment statuses registered in the database will be output on the newly created output target data file.

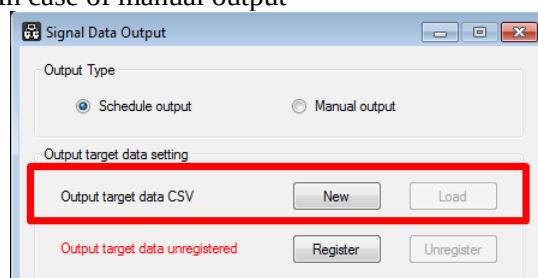
NOTE

- 1 If there are equipment units and equipment statuses not sent to the database, output target data cannot be created. Create the data after executing [Send to DB].
- 2 The output target data created by the [New] button in [Schedule output] and in [Manual output] are the same.

■ In case of scheduled output



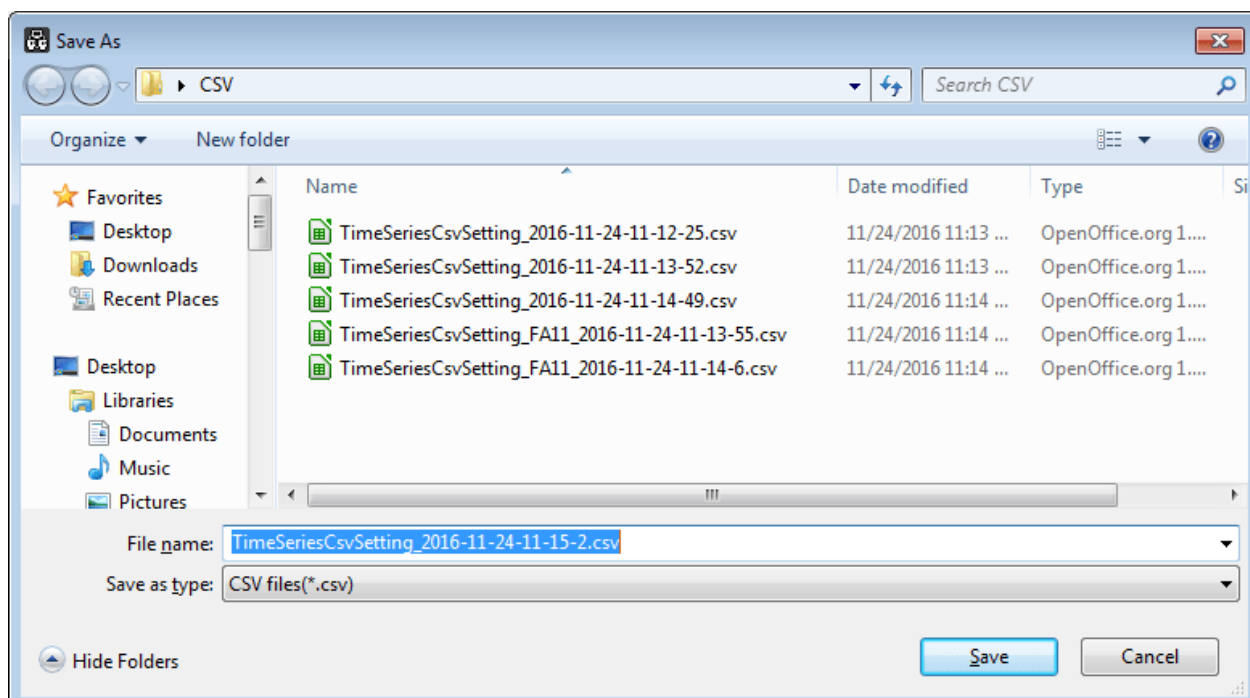
■ In case of manual output



When the [New] button is clicked, [Save As] dialog box appears. Specify an adequate location and click [Save] button.

The [Save As] dialog box selects the following folder on the first session. On subsequent sessions, it selects the folder selected in the previous session:

<MT-LINK*i* installation folder>\MT-LINK*i*AdminTool



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION			Sheet. 584/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- (2) Open the saved output target data file with LibreOffice Calc.

A1						
	A	B	C	D	E	F
1	Equipment	TargetEquip	TargetData1	TargetData2	TargetData3	TargetData4
2	Equipment0001	-	Disconnect_Machine001	SigOP_path1_Machine001	SigSTL_path1_Machine001	SigSPL_path1_Mi
3	Equipment0002	-	Disconnect_Machine002	SigOP_path1_Machine002	SigSTL_path1_Machine002	SigSPL_path1_Mi
4	Equipment0003	-	Disconnect_Machine003	SigOP_path1_Machine003	SigSTL_path1_Machine003	SigSPL_path1_Mi
5	Equipment0004	-	Disconnect_Machine001	SigOP_path1_Machine001	SigSTL_path1_Machine001	SigSPL_path1_Mi
6	Equipment0005	-	Disconnect_Machine001	SigOP_path1_Machine001	SigSTL_path1_Machine001	SigSPL_path1_Mi
7	Equipment0006	-	Disconnect_Machine002	SigOP_path1_Machine002	SigSTL_path1_Machine002	SigSPL_path1_Mi

NOTE

When the number of equipment statuses targeted for output is 1023 or more, the CSV file of the output target cannot be opened normally by LibreOffice Calc, due to the limitation of the maximum number of columns in LibreOffice Calc. In such a case, edit the output target data by using applications such as text editors.

- (3) For the [TargetEquip] column of the row with the equipment name of the output target, set "+" as the value.

Example) When Equipment0001 is the output target...

A2						
	A	B	C	D	E	F
1	Equipment	TargetEquip	TargetData1	TargetData2	TargetData3	TargetData4
2	Equipment0001	+	Disconnect_Machine001	SigOP_path1_Machine001	SigSTL_path1_Machine001	SigSPL_path1_Machin
3	Equipment0002	-	Disconnect_Machine002	SigOP_path1_Machine002	SigSTL_path1_Machine002	SigSPL_path1_Machin
4	Equipment0003	-	Disconnect_Machine003	SigOP_path1_Machine003	SigSTL_path1_Machine003	SigSPL_path1_Machin
5	Equipment0004	-	Disconnect_Machine001	SigOP_path1_Machine001	SigSTL_path1_Machine001	SigSPL_path1_Machin
6	Equipment0005	-	Disconnect_Machine001	SigOP_path1_Machine001	SigSTL_path1_Machine001	SigSPL_path1_Machin
7	Equipment0006	-	Disconnect_Machine002	SigOP_path1_Machine002	SigSTL_path1_Machine002	SigSPL_path1_Machin

- (4) Find the equipment statuses which are the output targets, and delete the cells with the equipment status names that are not output targets. However, if all the equipment statuses are output targets, this step is not required.

F11										
	A	B	C	D	E	F	G	H	I	J
1	Equipment	TargetEquip	TargetData1	TargetData2	TargetData3	TargetData4	TargetData5	TargetData6	TargetData7	TargetData8
2	Equipment0001	+		SigOP_path1_Machine001						
3	Equipment0002	+						EMG_path1_Machine002		
4	Equipment0003	+							Mode_path1_Machine003	
5	Equipment0004	-								
6	Equipment0005	-								
7	Equipment0006	-								
8	Equipment0007	-								
9	Equipment0008	-								

- (5) When deleting the cells that are not output targets, it can be done by deleting the columns with equipment statuses that are not output targets, instead of deleting the cells. However, if all the equipment statuses are output targets, this step is not required.

C1						
	A	B	C	D	E	F
1	Equipment	TargetEquip	TargetData2	TargetData6	TargetData16	
2	Equipment0001	+	SigOP_path1_Machine001	EMG_path1_Machine001	ModalS_path1_Machine001	
3	Equipment0002	-	SigOP_path1_Machine002	EMG_path1_Machine002	ModalS_path1_Machine002	
4	Equipment0003	-	SigOP_path1_Machine003	EMG_path1_Machine003	ModalS_path1_Machine003	
5	Equipment0004	-	SigOP_path1_Machine001	EMG_path1_Machine001	ModalS_path1_Machine001	
6	Equipment0005	-	SigOP_path1_Machine001	EMG_path1_Machine001	ModalS_path1_Machine001	
7	Equipment0006	-	SigOP_path1_Machine002	EMG_path1_Machine002	ModalS_path1_Machine002	

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	585/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

NOTE

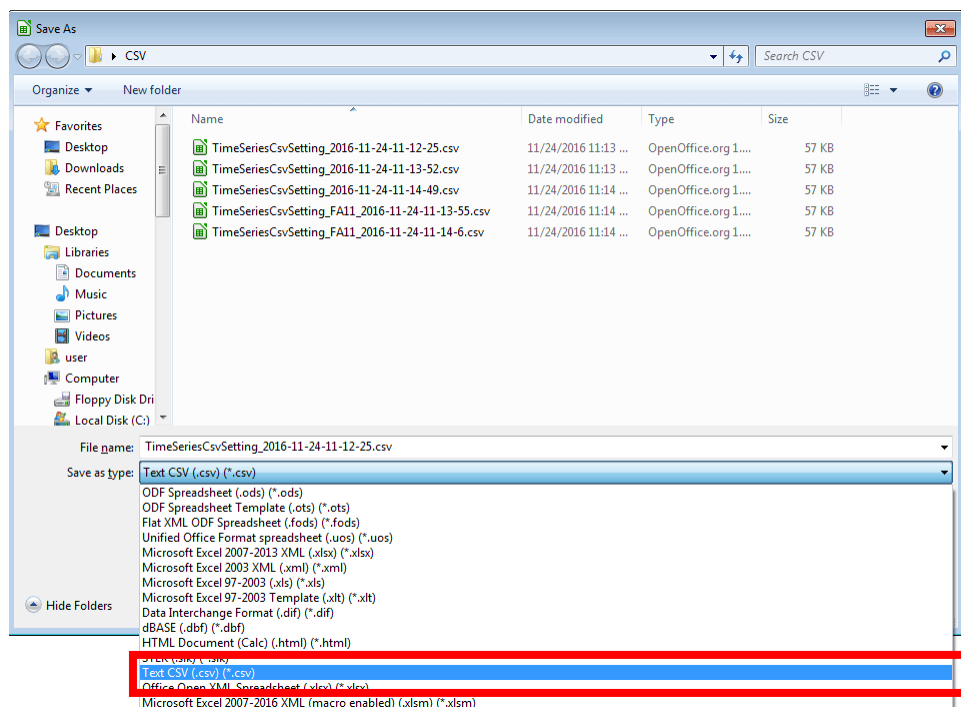
The same equipment status does not always belong in the same column.
When deleting by columns, be careful with the equipment statuses to be deleted.

- (6) Change the numbering of the item name of TargetData<N> to sequential numbering starting from 1.
However, if all the equipment statuses are output targets, this step is not required.

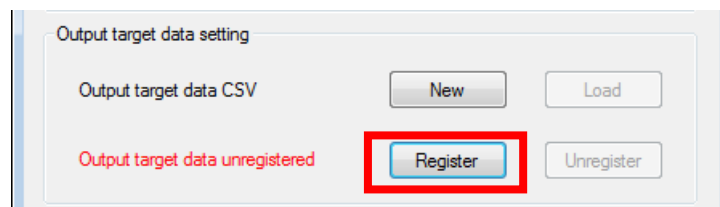
TargetData3					
	A	B	C	D	E
1	Equipment	TargetEquip	TargetData1	TargetData2	TargetData3
2	Equipment0001	+	SigOP_path1_Machine001	EMG_path1_Machine001	ModalS_path1_Machine001
3	Equipment0002	-	SigOP_path1_Machine002	EMG_path1_Machine002	ModalS_path1_Machine002
4	Equipment0003	-	SigOP_path1_Machine003	EMG_path1_Machine003	ModalS_path1_Machine003
5	Equipment0004	-	SigOP_path1_Machine001	EMG_path1_Machine001	ModalS_path1_Machine001
6	Equipment0005	-	SigOP_path1_Machine001	EMG_path1_Machine001	ModalS_path1_Machine001
7	Equipment0006	-	SigOP_path1_Machine002	EMG_path1_Machine002	ModalS_path1_Machine002

- (7) Save the corrected file.

Select [Save As] and after selecting [Text CSV (csv) (*.csv)] save the file to an adequate location.



- (8) Click the [Register] button in the [Output target data setting] group box.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet.	586/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

NOTE

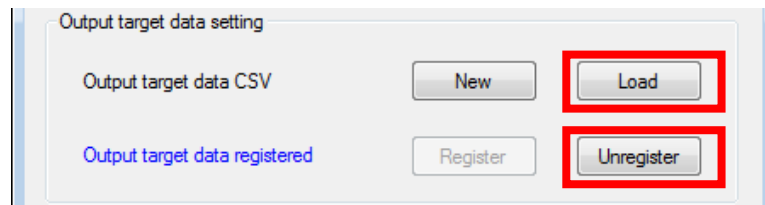
If the [Register] button is disabled (unable to click), the output target data has been registered already.

To check the registered output target data by outputting the CSV file, click the [Load] button, select any folder in the [Save As] dialog box, and click the [Save] button. The dialog box selects the following folder on the first session. On subsequent sessions, it selects the folder selected in the previous session.

<MT-LINK*i* installation folder>\MT-LINKiAdminTool

When changing the output target data, click the [Unregister] button to cancel the output target data, and then register the new data.

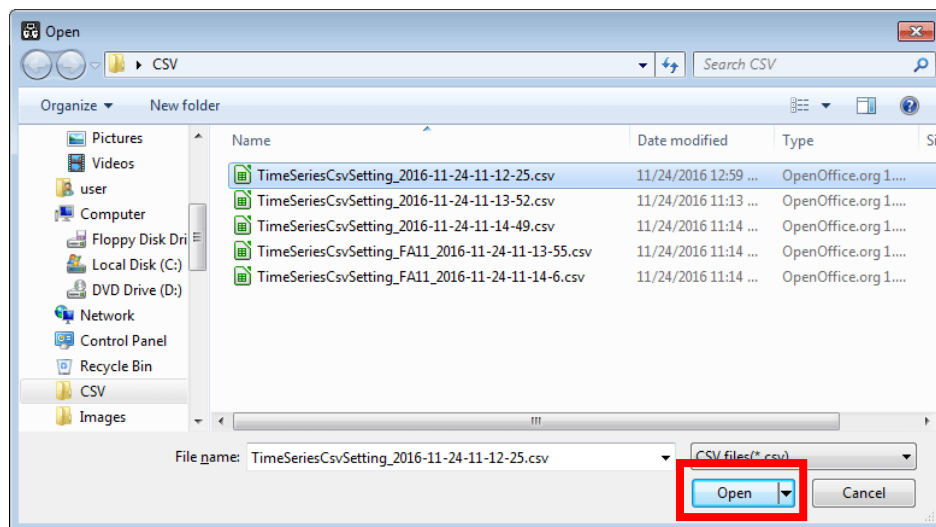
For the details of how to unregister the output target data, refer to the "[Unregistering the Registered Output Target Data of Signal Data Output](#)" subsection.



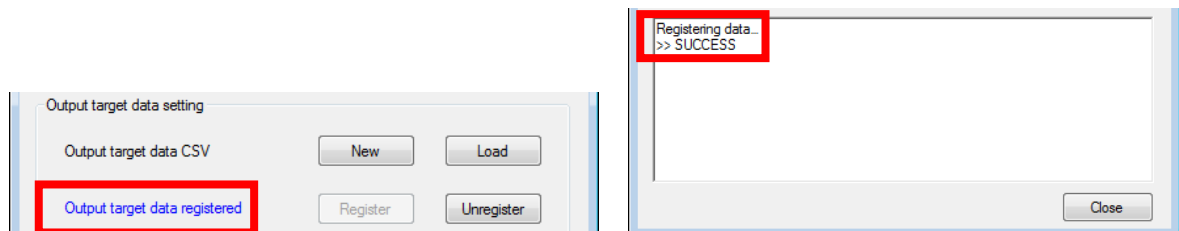
- (9) Select the output target data, and click [Open].

The [Open] window selects the following folder on the first session. On subsequent sessions, it selects the folder selected in the previous session.

<MT-LINK*i* installation folder>\MT-LINKiAdminTool



- (10) If the registration is successful, [Output target data registered] will be displayed on the left of the [Register] button. Also, in the message area at the bottom, "Registering data... >> SUCCESS" will be displayed.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	587/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

NOTE

If the registration is unsuccessful, an error message is displayed. In this case, the contents or the file format of the CSV file containing the output target data may be corrupted. Correct or recreate the output target data file, and register it.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	588/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

10.3.2 Unregistering the Registered Output Target Data of Signal Data Output

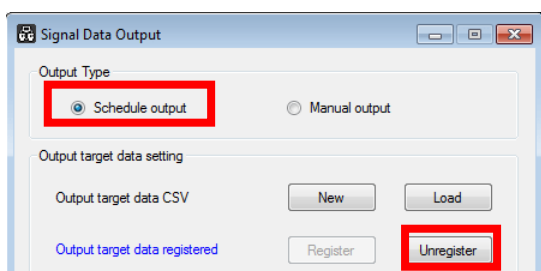
Once the output target data is registered by scheduled output or manual output, the output target data is retained until unregistered.

The output target data is saved for scheduled output and manual output respectively.

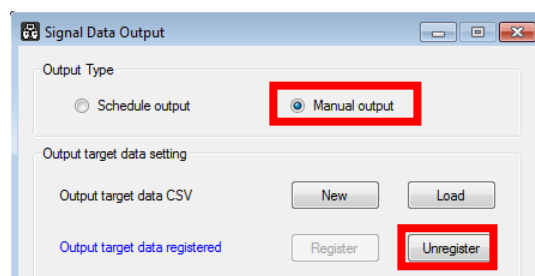
To unregister the output target data, follow the steps below.

- (1) When unregistering the output target data for scheduled output, select [Schedule output] for the output type. Also, when unregistering the output target data for manual output, select [Manual output] for the output type.
- (2) Click the [Unregister] button in the [Output target data setting] group box.

In case of scheduled output

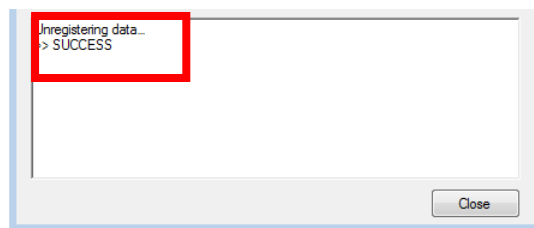
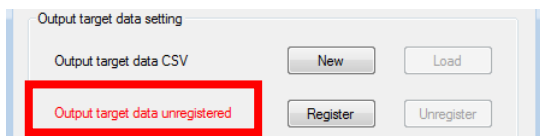


In case of manual output



- (3) If the unregistering is successful, [Output target data unregistered] is displayed on the left of the [Register] button.

Also, in the message area at the bottom, "Unregistering data... >> SUCCESS" will be displayed.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	589/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

10.3.3 Registering the Output Schedule of Signal Data Output

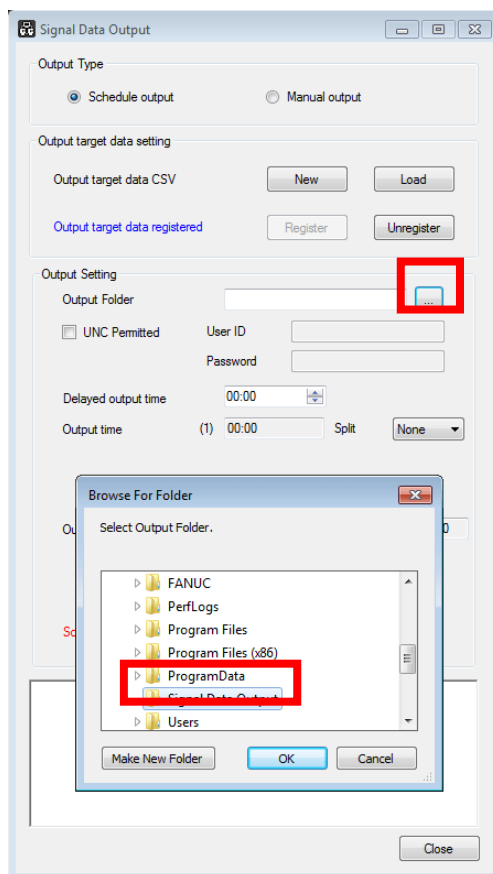
Signal data output can be executed in a periodical basis by setting the output schedule.

With the scheduled signal data output, output process starts at an output time which is specified in advance.

The signal data that will be output can be output on CSV file by splitting the 24 hours of signal data into up to four parts.

The output target data for the schedule output must be created and registered in advance. For the details, refer to the "[Creating and Registering the Output Target Data of Signal Data Output](#)" subsection.

(1) Select [Schedule output] for the [Output Type], and specify the output folder for the output setting.



- [Output Folder]

Specify the folder to output signal data. This field can be up to 260 bytes long. It populates "" (null space) on the first session. On subsequent sessions after changed once, it populates the last modified value.

- [...] (Folder selection) button

Clicking this browse folder button opens the [Browse For Folder] window to either select an existing folder or create a new folder. The [Output Folder] field populates the folder path selected in the [Browse For Folder] window.

The [Browse For Folder] window selects the following folder on the first session. On subsequent sessions, it selects the folder selected in the previous session:

<MT-LINK*i* installation folder>

- [UNC Permitted] check box

When specifying a UNC path for the output path, select the check box.

When the [UNC Permitted] check box is selected, the [User ID] and the [Password] can be entered.

- [User ID]

Set the user required for authentication of UNC path. This field can be up to 255 bytes long.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	590/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

•[Password]

Set the password required for authentication of UNC path. Up to 127 half-width alphanumeric and symbol characters (space included) can be input.

NOTE

- 1 For the output folder, paths of disk drives such as fixed disk drives (internal disk/external disk) or UNC paths can be specified. The paths of other drives (such as network drives) cannot be specified.
- 2 As the size of the output data becomes massive, it is recommended to specify for the output target folder, folders such as the folder of the expanded external disk drive. Also, make sure that the disk space of the output folder does not get compressed, by deleting the output files or moving the output files to another media on a regular basis.
- 3 If the free disk space of the disk drive specified as the output target is under 5GB, the execution of the output process will be suspended.
- 4 For the details on the confirmation of operation by OS combination when UNC path is specified as output path, refer to the "[Confirmation of Operation when UNC Path Is Specified as Output Path](#)" subsection.

- (2) For the [Split] setting, select how to split the output of CSV file from the drop-down menu (No split and split into 2 to 4 parts: [None] / [2] / [3] / [4]). [None] is set as the default value.

The screenshot shows the 'Signal Data Output' dialog box. The 'Output Type' section has 'Schedule output' selected. The 'Output target data setting' section has 'Output target data CSV' and 'Output target data registered' buttons. The 'Output Setting' section includes 'Output Folder' (C:\Signal Data Output), 'UNC Permitted' checkbox, 'User ID' and 'Password' fields, 'Delayed output time' (00:00), 'Output time' (1) (00:00), and 'Output target range' (1) (y'day 00:00:00.000 -> y'day 23:59:59.500). The 'Split' dropdown menu is highlighted with a red box, showing options: None, 2, 3, and 4. The 'None' option is selected. The 'Schedule Unregistered' status is shown at the bottom, with 'Register' and 'Unregister' buttons.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	591/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

(3) Set the [Delayed output time]. Enter the delayed time from the Output time (0:00 A.M.).

- The [Delayed output time] that can be entered will be as follows, depending on the number of splits for the output of CSV file.

Split	Range of Delayed Output Time that can be Entered
None	00:00 to 23:59
2 (Split into two parts)	00:00 to 11:59
3 (Split into three parts)	00:00 to 07:59
4 (Split into four parts)	00:00 to 05:59

- The [Output time] and [Output target range] will be as follows, based on the number of splits for the output of CSV file. When the [Delayed output time] and [Split] is changed, the display of [Output time] and [Output target range] is updated instantly.

Splits	Output Number	Output Time	Output Target Range
None	(1)	Delayed output time	y'day 00:00:00 --> y'day 23:59:59
2 (two)	(1)	Delayed output time	y'day 12:00:00 --> y'day 23:59:59
	(2)	Delayed output time + 12 hours	00:00:00 --> 11:59:59
3 (three)	(1)	Delayed output time	y'day 16:00:00 --> y'day 23:59:59
	(2)	Delayed output time + 8 hours	00:00:00 --> 07:59:59
	(3)	Delayed output time + 16 hours	08:00:00 --> 15:59:59
4 (four)	(1)	Delayed output time	y'day 18:00:00 --> y'day 23:59:59
	(2)	Delayed output time + 6 hours	00:00:00 --> 05:59:59
	(3)	Delayed output time + 12 hours	06:00:00 --> 11:59:59
	(4)	Delayed output time + 18 hours	12:00:00 --> 17:59:59

NOTE

The "y'day" displayed in the [Output target range] column stands for yesterday.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 592/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

(4) To register the schedule, click the [Register] button in the [Output Setting] group box.

NOTE

If the [Register] button is disabled, schedule is already registered. To change the schedule, click the [Unregister] button to unregister the schedule, and then register the new schedule. For the details of how to unregister the schedule, refer to the "[Unregistering the Registered Schedule of Signal Data Output](#)" subsection.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 593/868

NOTE

- 1 When the PC is shut down during schedule output, some of the signal data may not be output, as the report output is terminated before the completion of the process. The signal data that have not been output will not be output in the succeeding schedule output. If you need the signal data that have not been output, use manual output function to output the signal data separately.
- 2 When the time for the next schedule output arrives while schedule output is in process, the next schedule output will be executed after the current schedule output process is completed.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 595/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

10.3.4 Unregistering the Registered Schedule of Signal Data Output

To unregister the registered schedule, follow the steps below.

- (1) Select [Schedule output] for the [Output Type], and click the [Unregister] button in the [Output Setting] area.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 596/868

(2) If the unregistering of schedule is successful, [Schedule unregistered] will be displayed on the left of [Register] button.

Also, in the message area at the bottom, "Unregistering schedule... >> SUCCESS" will be displayed.

The screenshot shows the 'Output Setting' dialog box. The 'Output Folder' is set to 'C:\Signal Data Output'. The 'UNC Permitted' checkbox is checked. The 'User ID' and 'Password' fields are empty. The 'Delayed output time' is set to '05:00'. The 'Output time' section has four entries: (1) 05:00, (2) 11:00, (3) 17:00, and (4) 23:00. The 'Split' dropdown is set to '4'. The 'Output target range' section has four entries: (1) y'day 18:00:00.000 --> y'day 23:59:59.500, (2) 00:00:00.000 --> 05:59:59.500, (3) 06:00:00.000 --> 11:59:59.500, and (4) 12:00:00.000 --> 17:59:59.500. The 'Schedule Unregistered' label is highlighted in red on the left of the 'Register' button. The message area at the bottom shows 'Unregistering data... >> SUCCESS'.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 597/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

10.3.5 Manual Output of Signal Data

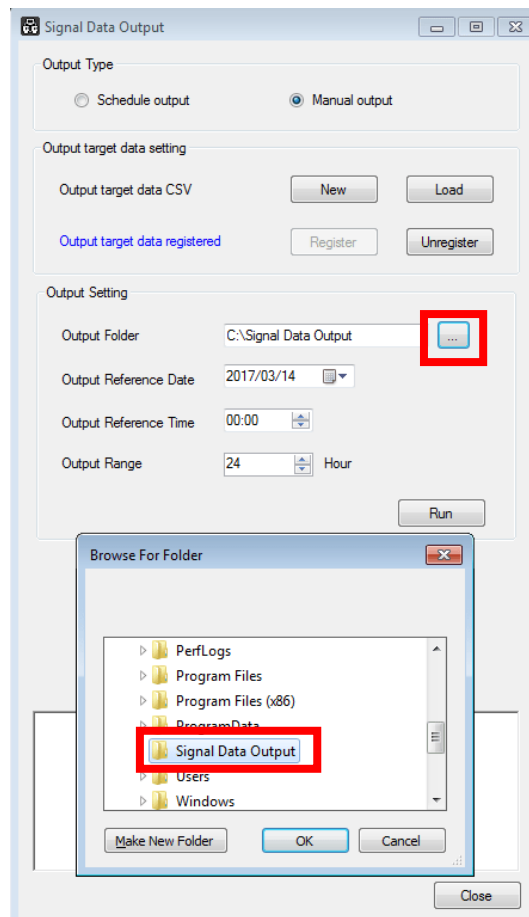
Signal data can be output manually.

In the manual output, up to 24 hours of data starting from the specified output reference date/time can be output. The output process starts right after the [Run] button is clicked. Manual output can be unregistered.

When the output is failing, refer to the "[When Report Output Fails](#)" subsection.

The output target data for the manual output must be created and registered in advance. For the details, refer to the "[Creating and Registering the Output Target Data of Signal Data Output](#)" subsection.

- (1) Select [Manual output] for the output type, and specify the output target folder in the [Output Setting] area.



- [Output Folder]

Specify the folder to output signal data. This field can be up to 260 bytes long. It populates "" (null space) on the first session. On subsequent sessions after changed once, it populates the last modified value.

- [...] (Folder selection) button

Clicking this browse folder button opens the [Browse For Folder] window to either select an existing folder or create a new folder. The [Output Folder] field populates the folder path selected in the [Browse For Folder] window.

The [Browse For Folder] window selects the following folder on the first session. On subsequent sessions, it selects the folder selected in the previous session:

<MT-LINK*i* installation folder>

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	598/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

NOTE

- 1 For the output folder, paths of disk drives such as fixed disk drives (internal disk/external disk) or UNC paths can be specified. The paths of other drives (such as network drives) cannot be specified.
- 2 As the size of the output data becomes massive, it is recommended to specify for the output target folder, folders such as the folder of the expanded external disk drive. Also, make sure that the disk space of the output folder does not get compressed, by deleting the output files or moving the output files to another media on a regular basis.
- 3 If the free disk space of the disk drive specified as the output target is under 5GB, an error message will appear, and the execution of the manual output process will be suspended.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	599/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

- (2) Set the [Output Reference Date], [Output Reference Time], and the [Output Range]. Enter the target date of output for the [Output Reference Date], the start time of the targeted output period for the [Output Reference Time], and the time span of targeted output period for the [Output Range].

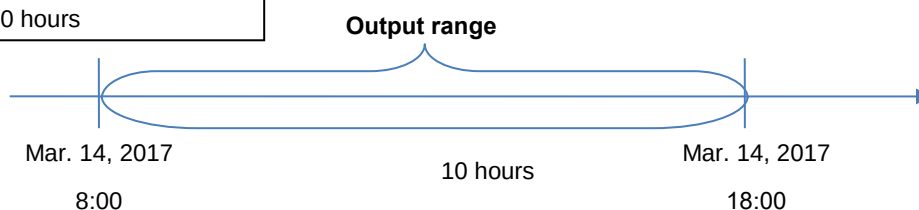
In the following example, the targeted output period will be from Mar. 14, 2017 8:00 to Mar. 14, 2017 18:00.

(Example)

Output reference date: Mar. 14, 2017

Output reference time: 8:00

Output range: 10 hours



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	600/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

NOTE

If a future date/time is included in the output range, it will be an error. Specify the output range so that it will not include date/time of the future.

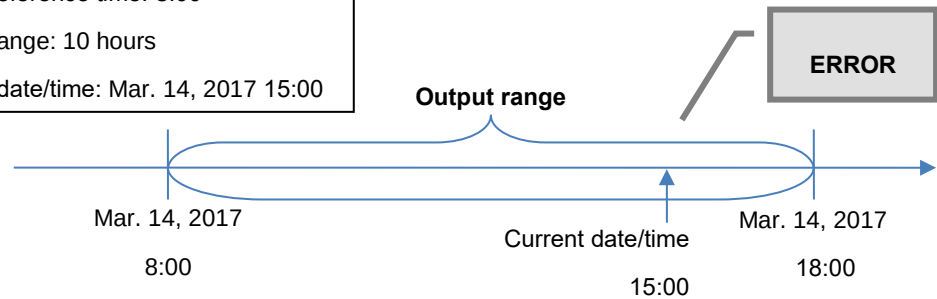
(Example)

Output reference date: Mar. 14, 2017

Output reference time: 8:00

Output range: 10 hours

Current date/time: Mar. 14, 2017 15:00



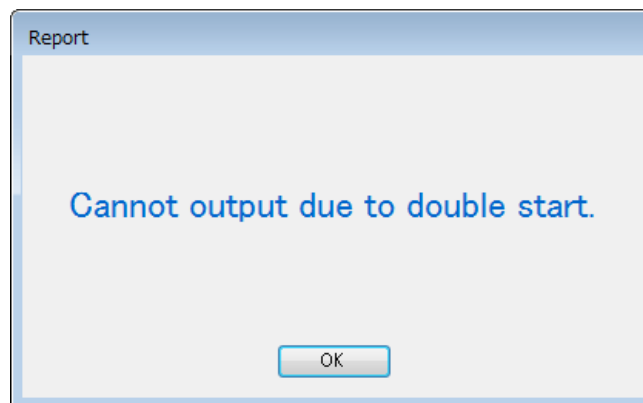
(3) Click the [Run] button.

The 'Signal Data Output' dialog box is shown. It has three main sections: 'Output Type' with 'Schedule output' and 'Manual output' (selected); 'Output target data setting' with buttons for 'New', 'Load', 'Register', and 'Unregister'; and 'Output Setting' with fields for 'Output Folder' (C:\Signal Data Output), 'Output Reference Date' (2017/03/14), 'Output Reference Time' (08:00), and 'Output Range' (10 Hour). The 'Run' button is highlighted with a red rectangle.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	FANUC CORPORATION		Sheet.	601/868

NOTE

- 1 If the output target is not specified or if a future date/time is included in the output range, an error message will be displayed. In such a case, output the file after correcting the setting based on the contents of the message.
- 2 It may take some time to output. Be careful, as it will take time if there are too many equipment statuses specified as output target.
- 3 During the output process, it is unable to execute setting operations on the AdminTool windows.
- 4 If the output is unsuccessful, refer to the "**When Signal Data Output was Unsuccessful**" subsection.
- 5 If you execute manual output while a signal data schedule output or a report schedule output is in process, the following message is displayed, and the output will not be executed. Execute manual output after waiting for a while.



- 6 If the time for the signal data schedule output arrives while executing manual output, the schedule output will be executed after the manual output is completed.

10.3.6 Output File Format of Signal Data Output

(1) The format of the file output by signal data output will be as follows.

■ File name format

timeseries_output@<Equipment name>@<Equipment status name>
@<yyyy-MM-dd-HH-mm-ss>.csv

NOTE

- 1 <yyyy-MM-dd-HH-mm-ss> part of the file name is the starting date/time of signal data output period.
- 2 If a report with the same file name is output two or more times, the contents of the previous output file are overwritten by the contents of the succeeding output file.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9								
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7								
Ed.	Date	Design	Description					FANUC CORPORATION		Sheet.	602/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

■ Output layout

Name of item	Output value	Remark
date	Date/time information. Output by unit of 500ms.	NOTE 1
<Equipment name>@<Equipment status name>@Value	"Raw value of signal" for the specified equipment by date/time is output.	
<Equipment name>@<Equipment status name>@Score	"Converted value of signal" for the specified equipment by date/time is output.	NOTE 2
<Equipment name>@<Equipment status name>@Judge	"Judgment result for the converted value of signal (Score) against threshold" for the specified equipment by date/time is output. 0: Normal 1: Warning (Converted value of signal exceeds the upper limit warning value, or did not reach the lower limit warning value) 2: Error (Converted value of signal exceeds the upper limit error value, or did not reach the lower limit error value)	NOTE 2

NOTE

- 1 By editing the system setting file (AdminTool.ini), the time interval of signal data output can be changed. For the details, refer to the "[Customizing the Signal Data Output Function](#)" subsection.
- 2 Output when data type of signal data is set to "1" in the system setting file (AdminTool.ini). For the details, refer to the "[Customizing the Signal Data Output Function](#)" subsection.

■ Output details of each item (example)

Item name (NOTE1)	Output Details							
	In Normal Operation				Receiver Stopped/ System Stopped (No DB recording)	Machine Disconnected / Collector Stopped (DB value = null)	Custom Macro / P-CODE Variables Is "data (blank)" (DB value = NaN)	
	String type	Numeric type (NOTE3)	Numeric type (NOTE3)	Bool type (NOTE5)			Numeric type (NOTE3)	Numeric type (NOTE4)
Value	String	Numerical value	Numerical value	0 / 1	-	null	NaN	NaN
Score	null	Numerical value	null	null	-	null	NaN	null
Judge	null	0 / 1 / 2	null	null	-	null	null	null

NOTE

- 1 For the item names, the expression "<Equipment name>@<Equipment status name>@" is omitted.
- 2 "-" in the table means that nothing will be output.
- 3 Refers to the case when upper limit / lower limit value is set in the outlier detection setting.
- 4 Refers to the case when upper limit / lower limit value is not set in the outlier detection setting.
- 5 Refers to the case of Bool type signal (false: 0 / true: 1).

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	603/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

- Character code
UTF-8 (with BOM)

- Output sample
 - When the data type of signal data is "0" (default)

```
"date","Equipment0001@SigOP_path1_Machine001@Value"
2017/03/21 13:32:44.000,1
2017/03/21 13:32:44.500,1
2017/03/21 13:32:45.000,1
2017/03/21 13:32:45.500,1
2017/03/21 13:32:46.000,1
```

- When the data type of signal data is "1"

```
"date","Equipment0001@SigOP_path1_Machine001@Value","Equipment0001@SigOP_path1_Machine001@Score","Equipment0001@SigOP_path1_Machine001@Judge"
2017/03/21 13:32:44.000,1,null,null
2017/03/21 13:32:44.500,1,null,null
2017/03/21 13:32:45.000,1,null,null
2017/03/21 13:32:45.500,1,null,null
2017/03/21 13:32:46.000,1,null,null
```

NOTE

- 1 In the initial setting, compression function of the output file is disabled. To enable the compression function, refer to the "[Customizing the Signal Data Output Function](#)" subsection.
- 2 The files targeted for compression are all the files with CSV extension which are stored in the specified output target folder. Be careful as the files stored in previous occasions will also be compressed.
- 3 If the compression function is enabled, all the files will be compressed altogether after all the output files, which are not compressed yet, have been output. Be careful as there will be considerable workload on the disk space of the output target folder while the output files are being compressed.

10.3.7 When Signal Data Output was Unsuccessful

When the signal data output was unsuccessful, file with the following name is output to the output folder.

The output file when signal data output was unsuccessful

Name of file	Reason of failure
#####CANCEL OUTPUTTING#####_<yyyyMMddHHmmss>	Output is canceled by the user
#####TIMEOUT OUTPUTTING#####_<yyyyMMddHHmmss>	Output process timed out
#####ERROR OUTPUTTING#####_<yyyyMMddHHmmss>	Output process ended abnormally
#####ITEMERROR OUTPUTTING#####_<yyyyMMddHHmmss>	When there is a file with output error (NOTE 3)
#####DISKFULL OUTPUTTING#####_<yyyyMMddHHmmss>	When the free disk space of the disk drive specified as the output target is under 5GB

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 604/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

NOTE

- 1 <yyyyMMddHHmmss> part of the file name is the output date/time of the file.
- 2 If time out is the cause of failure, there is a possibility that the amount of output target data was excessive, and the time expended for processing was too much. In such a case, narrow down the equipment units and equipment statuses of the output target data, and execute the output process again.
- 3 The equipment name of the file with the output error will be stored in the file.
(Example) Equipment0001@ActS_path1_Machine001

10.3.8 Customizing the Signal Data Output Function

The signal data output function can be customized by editing the system setting file (AdminTool.ini) provided by MT-LINK*i*.

Following items can be customized using AdminTool.ini.

- Maximum number of status output
Customize when limiting the number of statuses of the signal data to be output.
- Time interval of signal data output
Customize when reducing the size of the signal data output file to be output.
- Enable/disable compression of signal data
Customize when reducing the size of the signal data output file to be output.
- Data type of signal data
Customize when switching the data type of signal data to be output.
- Number of rows split for signal data output
Customize when outputting the split signal data output file.
- Time out period of signal data schedule output
Customize when time out occurs during schedule output process on signal data.
- Time out period of signal data manual output
Customize when time out occurs during manual output process on signal data.

NOTE

When the system setting file (AdminTool.ini) is edited, restart AdminTool.
If the signal data schedule output is being registered, in order to enable the settings after editing, [Unregister] the schedule and [Register] the schedule again.

For the details on the system setting files, refer to the "**SYSTEM SETTING FILES**" section.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION				Sheet. 605/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

11

[SETTING SYSTEM FUNCTIONS] - MAIL SETTING

This chapter describes how to make settings of mail delivery.

The AdminTool Menu has the following menu items in the [Setting System Functions] - [Mail Setting] group:

(1) Mail Function Setting

This menu makes primary settings for signal detection and alarm mail delivery, such as settings of mail server and mail delivery destination users.

(2) Signal Detection Mail Setting

This menu makes settings of delivery conditions and target equipment to send mail when signals are detected that meet the conditions.

(3) Alarm Mail Setting

This menu makes settings of delivery conditions and target equipment to send mail when alarms occur on them.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION				Sheet. 606/868
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue	

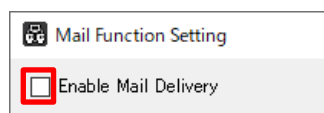
11.1 MAIL FUNCTION SETTING

This section explains the settings of the mail functions for delivering alarm mail and signal detection mail. Select [Mail Function Setting] in the AdminTool Menu to display the [Mail Function Setting] window.

■ [Mail Function Setting]

• [Enable Mail Delivery] check box

Use this check box to enable/disable the mail delivery function. By default, this check box is cleared. Selecting it enables users to use the mail delivery function and to specify the setting fields, and also activates the [Signal Detection Mail Setting] and [Alarm Mail Setting] menus in the AdminTool Menu. Clearing it disables the mail delivery function, and deactivates all the setting made for the function and the [Signal Detection Mail Setting] and [Alarm Mail Setting] menus as well.



Select the [Enable Mail Delivery] check box to enable the mail delivery function

The image shows the full 'Mail Function Setting' window. The 'Enable Mail Delivery' checkbox is now checked. The window is divided into several sections:

- Delivery Setting:** Includes 'Administrator Email Address' (mail_user001@mail_test.com) and a checkbox for 'Send by BCC to delivery users'.
- Mail Server Setting:**
 - Server Info:** 'Server (SMTP)' (127.0.0.1), 'Port' (587), and 'Time out' (30 sec).
 - User Info:** 'Host domain' (mail_test.com), 'User' (mail_user), and 'Password' (****). Below these are checkboxes for 'SMTP-AUTH Required' and 'Use SSL/TLS'. A dropdown menu next to 'Use SSL/TLS' is set to 'SMTP over SSL(SMTPS)', which is highlighted with a red box and a callout labeled 'SSL/TLS Method'.
- Mail Flow Control:** A slider set to 100 Requests / 10 sec.
- Mail Delivery List:** A list of equipment (Equipment0001 to Equipment0111) and a 'User List' with checkboxes for User0001, User0002, User0003, and User0004.

 At the bottom are buttons for 'Test', 'Apply', 'OK', and 'Cancel'.

■ [Delivery Setting]

• [Administrator Email Address] (entry required)

Set the Administrator email address. The address will be used as the delivery destination (To:) and the sender (From:).

This field can be up to 256 bytes long.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	607/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

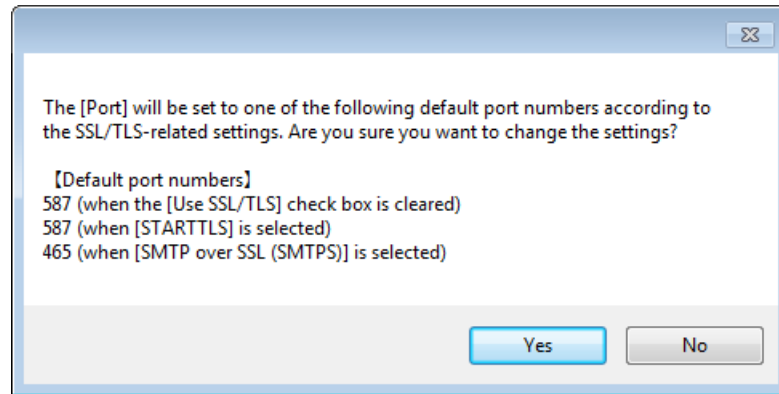
- [Send by BCC to delivery users] check box

If you do not wish the mail addresses of the users selected on the [User List] to be displayed to other recipients, select the check box. In this case, only the Administrator Email Address will be set on the delivery destination (To:) of the mail header, the mail addresses of the delivery destination users will be set as Bcc:. The default setting is selected.

■ [Mail Server Setting]

NOTE

When the user tries to modify the [Use SSL/TLS] check box or to change the selection from its drop-down list, the following dialog box appears.



Clicking [Yes] will set a value for the [Port] field of the [Mail Function Setting] window as follows:

Use SSL/TLS or not	Selection of [Use SSL/TLS] (SSL/TLS method)	Port number to be set to
Not use SSL/TLS (the check box is cleared)	N.A	587
Use SSL/TLS (the check box is selected)	SMTP over SSL (SMTPS)	465
	STARTTLS	587

If having second thoughts, click [No] to leave the settings including the [Port] field as they are.

- [Server Info] group box

- [Server (SMTP)] (entry required)

Set the server to deliver mail by the IP address or the host name. The following mail servers are supported. In the case of the host name, a half-width alphabet can be used as the first character, and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-"), and periods ("."). This field can be up to 256 bytes long.

Supported mail servers are shown below:

- SMTP server (without authentication)
- SMTP server (with authentication) without SSL
- SMTP-AUTH

- [Port]

Set the port number used for the communication with the mail server. The default setting is 587.

- [Time out]

Set the communication time out period (seconds) with the mail server. The default setting is 30 (seconds).

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 608/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

- [User Info] group box

- [Host domain]
Host domain is not used.
- [SMTP-AUTH Required] check box
If authentication is required on the SMTP server, select the [SMTP-AUTH Required] check box, and set the [User] and [Password] text boxes. By default, this check box is cleared (no authentication).
- [Use SSL/TLS] check box
When using an SSL/TLS method in mail delivery, select this check box and select the method from the drop-down list. By default, this check box is cleared.
- SSL/TLS method drop-down list (next to the [Use SSL/TLS] check box)
When using an SSL/TLS method, select a method from this list box. By default, this list box is set to [SMTP over SSL (SMTPS)].
- [User]
Set the user for the authentication of the mail server. This field can be up to 256 bytes long.
- [Password]
Set the password for the authentication of the mail server. This field can be up to 256 bytes long.

■ [Mail Flow Control]

Set the upper limit value for the number of total mail delivery requests per unit time (seconds). (NOTE 1)
If the number of requests per unit time exceeds the upper limit value, the requested mail pieces will be delivered up to the request limit but the rest will not. The count of the requests will be reset after every unit time. (NOTE 2)
For the number of requests, 0 to 10000 can be entered. For the unit time, 1 to 1000 (seconds) can be entered. The default setting is 100 (requests) / 10 seconds (unit time).

NOTE

- 1 The number of requests is counted by the number of mail bodies to be delivered. Mail bodies are created per language for which users registered in the delivery destination user list.
(Example) If there are Japanese and English users registered in the delivery destination user list, this single mail delivery is counted as 2 requests.
- 2 The information about the mail undelivered due to flow control will be recorded in the log file. Refer to the following log file in case you need to check the undelivered mail.
 - Storage location of the log file
<MT-LINK*i* installation folder>\MT-LINK*i*Receiver\mailflowcontrol.log
 - The contents recorded in the log file are as follows.
 - Time stamp
 - Classification of event (Signal Detection / Clearance, or Alarm / Clearance)
 - Equipment name
 - Destination address
 - Contents of messages

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	609/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

■ [Mail Delivery List]

Associates the target equipment unit with the users to deliver mail to.

[Equipment => User] Tab

For each target equipment unit, select the users to deliver mail to.

• [Equipment]

The equipment units will be shown on a list, select the target equipment unit. The alarms occurring and the signals detected in the selected equipment unit will become the target of mail delivery.

• [User List]

Users registered in the authority setting will be shown on a list. For each target equipment unit, select the check box of the users to deliver mail to. The mail of the target equipment unit will be delivered to the mail addresses (up to 3 addresses) that are registered for the user with the check box selected.

[User => Equipment] Tab

For each user to deliver mail to, select the target equipment units.

• [User]

Users registered in the authority setting will be shown on a list, select the target user. The mail of the target equipment unit will be delivered to the mail addresses (up to 3 addresses) that are registered for the selected user.

• [Equipment List]

The equipment units will be shown on a list. For each user, select the check box of the target equipment units. The alarms occurring and the signals detected in the selected equipment unit will become the target of mail delivery.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	610/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

- * When using the check box of [User List] or [Equipment List], two or more check boxes can be selected or cleared at the same time, by selecting two or more rows.
 - If you select two or more rows, and select the check boxes, the check boxes of the selected rows will be selected.
 - If you select two or more rows, and clear the check boxes, the check boxes of the selected rows will be cleared.
- * When the [OK] / [Apply] button is clicked, the [Mail Delivery List] settings are checked. When a check error occurs, the display color of the equipment units in the [Equipment] and [Equipment List] will be shown in red. The display color of the equipment units will be as follows depending on the setting status of the [Signal Detection Mail Setting] / [Alarm Mail Setting].
 - Black : For the equipment units without the Signal Detection / Alarm Mail Settings.
 - Blue : For the equipment units with the Signal Detection / Alarm Mail Settings and with the settings of the users to deliver mail to.
 - Red : For the equipment units with the Signal Detection / Alarm Mail Settings and without the settings of the users to deliver mail to.

NOTE

Settings of the delivery destination mail addresses for each user is done in the [Add/Edit User] dialog box of the authority setting.

For the details, refer to the "[Add/Edit Role of Authority](#)" subsection.

■ Buttons

- [Test] button
Execute a test to determine if mail is delivered with the settings of the mail server information.
A test mail is sent to the representative mail address.
- [Apply] button
Settings will be applied.
- [OK] button
Settings are applied and the window is closed.
- [Cancel] button
Settings are discarded and the window is closed.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 611/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

11.2 SIGNAL DETECTION MAIL DELIVERY SETTING

This window sets the mail delivery settings for signal detection.

The settings for the mail server information and the selection of mail delivery destination users are done in the [Mail Function Setting] window. For the details, refer to the "**MAIL FUNCTION SETTING**" section.

In the signal detection mail delivery, mail is delivered when a signal is detected that matches the delivery condition set for each equipment unit.

Also, if the delivery condition is cleared, detection clearance notice is delivered by mail.

NOTE

The judgement of the delivery condition is done in terms of each equipment unit.

- As for the signal detection mail, delivery condition targeting multiple (2 or more) equipment units cannot be set.

(Example: Mail is delivered when delivery condition is established on both equipment 1 and equipment 2)

- Signal detection clearance notice is delivered by mail when all the established delivery conditions of an equipment unit are cleared.

- Delivery conditions (CONDITION) can be registered up to 30 in total of the signal detection mail delivery setting and the alarm mail delivery setting.
- Details for delivery condition (DETAIL) can be registered up to 5 per delivery condition (CONDITION).

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	612/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- * When the character string of a displayed item cannot be viewed as a whole, the end of the character string will be expressed with "..." inside the column.
By placing the mouse cursor on that item, a tool tip is displayed, and the whole character string can be confirmed.
On the [Signal Detection Mail Setting] window, the following item display areas can be extended by widening the window.
- [Detail Setting] group box: [Equipment], [Status]

NOTE

Simple assign status and multiple assign status of the equipment status can be used for the condition setting of the signal detection mail delivery. (Data types Bool, Int32, Double, and String can be used.)
Among these, simple assign status assigned with machine signal set with outlier detection can use the judging value for surpassing threshold (Judge) and the converted signal value (Score).

11.2.1 Steps for Setting Signal Detection Mail Delivery

Follow the steps below to set the signal detection mail delivery.

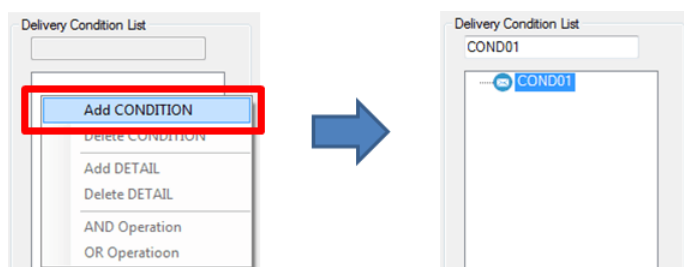
- (1) Select [Signal Detection Mail Setting] in the AdminTool Menu to open the [Signal Detection Mail Setting] window.
- (2) On the [Delivery Condition List], add a new CONDITION (delivery condition).
- (3) Select the equipment unit associated to CONDITION.
- (4) For the CONDITION, enter the text character string to display on the signal detection mail body.
- (5) For the CONDITION, define DETAIL (detail of delivery condition).
- (6) After completing the settings, click the [Apply] button or the [OK] button. If the [Cancel] button is clicked, the settings will be discarded.

For the details, refer to the "[Adding Delivery Condition for Signal Detection Mail Delivery](#)" subsection.

11.2.2 Adding Delivery Condition for Signal Detection Mail Delivery

Follow the steps below to add CONDITION (delivery condition) and DETAIL (detail of delivery condition) for the signal detection mail delivery, and define the delivery condition.

- (1) Right-click in the tree area of the [Delivery Condition List], and select [Add CONDITION] from the menu displayed.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	613/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- (2) CONDITION is added to the [Delivery Condition List].

The default name of CONDITION is "COND<NN>." (<NN> is the index number starting from "01" and the smallest unused value will be used.)

Also, the name of the **CONDITION** can be changed in the text box above. Only alphabetic letters can be used for the first character, and alphanumeric characters and "_" (underscore) can also be used for the remaining characters of the name. This name can be up to 32 bytes long. The name of **CONDITION** will be displayed in the "CONDITION" part of the signal detection mail text.



- (3) Select a **CONDITION**, and on the **[Equipment]** list, select the check box of the equipment unit that will be the target equipment unit for signal detection mail delivery. At least one equipment unit must be selected. The equipment units displayed on the **[Equipment]** list are the equipment units registered in the **[Equipment Setting]** window.

Delivery Condition List

Detail Setting
Value Type

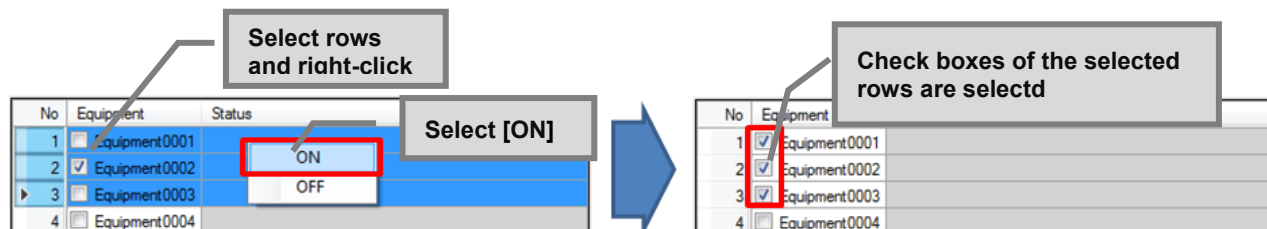
Text
☒ Enable

.....COND01

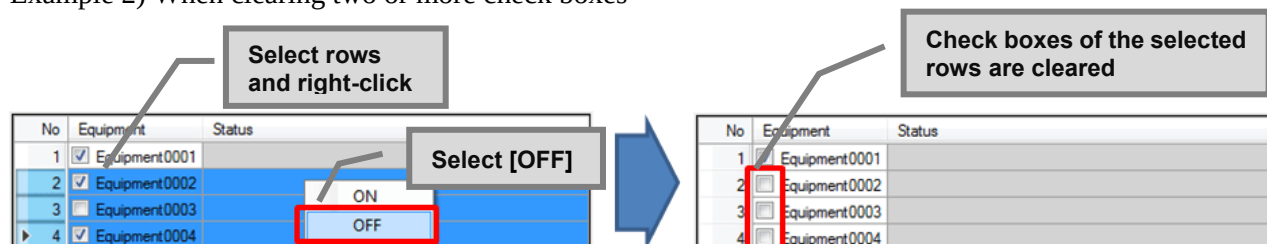
No	Equipment	Status
1	☒ Equipment0001	
2	☐ Equipment0002	
3	☒ Equipment0003	
4	☐ Equipment0004	

* When using the equipment check box, two or more check boxes can be selected or cleared at the same time, by selecting two or more rows, right-clicking, and selecting [ON] or [OFF] on the menu displayed.

Example 1) When selecting two or more check boxes



Example 2) When clearing two or more check boxes



- (4) When the [Enable] check box is selected in the [Text] group box, characters to be set in the [TEXT] part of the signal detection mail body can be entered. Up to 1000 bytes can be entered.
The string for this field should be entered in the language specified for the [Web name language] field in the [System Setting] window.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION				Sheet.	614/868
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						

If there are no characters set in the text, the state of [TEXT] will be empty. If the check box is cleared, [TEXT] will not be displayed.

In the default setting, check box is selected, and the character string of text is empty.

- (5) Select and right-click the added CONDITION, and select [Add DETAIL] on the menu displayed. At least one DETAIL must be set.

The name of DETAIL is "DETAIL<N>." (<N> is the index number starting from "1" and the smallest unused value will be used.)

The name of DETAIL cannot be changed.



- (6) Select the added DETAIL, select judgement condition from the drop-down menu for each item in the [Detail Setting] and enter the value of the item.

In the default setting, there are no settings for all the items. The signal and the delivery condition must be set for all the selected equipment units.

- Equipment

The check box is selected for the target equipment unit selected when the CONDITION is added. This cannot be changed.

- Status

Select the signal assigned to the equipment unit from the drop-down menu.

- Value Type

Select the type of the status. The statuses of the selected type will be listed in the drop-down menu of the [Status] column.

For the type, [bool], [int32], [uint32], [double], [string], [judge], or [score] can be selected.

[bool] is selected in the default setting.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	615/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

- [Search] button

If a part of the status name is entered in the text box and the [Search] button is clicked, statuses with the partial match among the statuses of the selected [Value Type] will be listed in the drop-down menu. At this moment, the status at the top of the list will be set as the default value.

NOTE

- 1 The [Search] button is useful for collective setting in which the same status and comparing condition is applied to multiple equipment units.
- 2 It is possible to set different statuses for multiple equipment units. However, in such a case, only one comparison operator and one value can be selected. Take note that the comparing condition will be the same for the different statuses.

- Operator

Select the operator for comparing the status value of the selected [Status].

- Value

Select the value to compare with the status value of the selected [Status].

■ Setting values of [Operator] and [Value] for each [Value Type] are as follows.

Value Type	Operator	Value	Remarks
bool	== (Equal to "Value") != (Not equal to "Value")	Select from true / false	-
int uint32 double	== (Equal to "Value") != (Not equal to "Value") < (Less than "Value") > (More than "Value") <= ("Value" or less) >= ("Value" or more)	Enter numerical value	-
string	== (Equal to "Value") != (Not equal to "Value") partial match	Enter character string	Up to 1024 bytes can be entered.
judge	== (Equal to "Value") != (Not equal to "Value") < (Less than "Value") > (More than "Value") <= ("Value" or less) >= ("Value" or more)	Select from 0 / 1 / 2	Signal that outputs the judging result of the threshold when [Outlier detection] is set in the [Signal Setting] of the [Machine Setting]. (0: normal value / 1: alert value / 2:error value) * ".FR" added at the end of the status name

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 616/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

Value Type	Operator	Value	Remarks
score	== (Equal to "Value") != (Not equal to "Value") < (Less than "Value") > (More than "Value") <= ("Value" or less) >= ("Value" or more)	Enter numerical value	Converted signal used for judging the threshold when [Outlier detection] is set in the [Signal Setting] of the [Machine Setting]. * ".FV" added at the end of the status name

NOTE

If "judge" for type, and custom macro variable / P-CODE variable for status is set in the detailed setting, the judgement of the delivery condition cannot be done when the value of variable on the NC side is "data (blank)."

Example of setting)

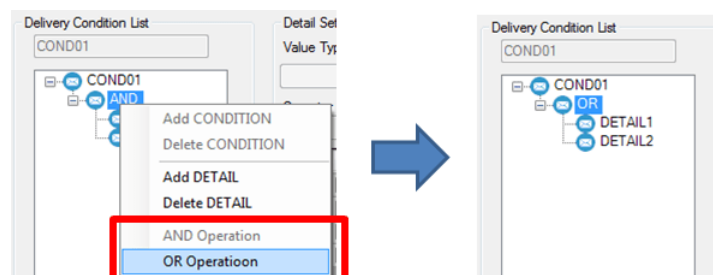
Type: judge
 Status: MacroVar_510_path1_Machine001.FR
 Operator: !=
 Value: 0

In the example above, delivery condition will be established when the value is other than "0" (normal value), and the signal detection mail will be delivered. However, if the value of the status variable is "data (blank)," delivery condition will not be established because the judgement if the delivery condition cannot be made, and the signal detection mail will not be delivered.

(7) If you want to combine multiple DETAILS, repeat steps (5) and (6).

When multiple DETAILS are registered, the associating condition (default setting is "AND") of the DETAILS is inserted in the 2nd layer of the [Delivery Condition List]. In order to change the associating condition, select and right-click the associating condition, and select [AND Operation] or [OR Operation] from the menu displayed.

- [AND Operation]: Associates the judgement conditions of each DETAIL with AND.
The condition is established for CONDITION when the judgement conditions of all the DETAILS are established.
- [OR Operation]: Associates the judgement conditions of each DETAIL with OR.
The condition is established for CONDITION when at least one of the judgement conditions of the DETAILS is established.



(8) In order to add another CONDITION, repeat steps (1) to (7) for each CONDITION added.

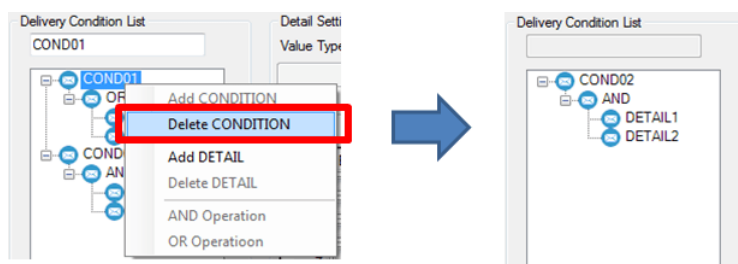
When all the settings are completed, click the [OK] button or the [Apply] button to apply the settings. If the [Cancel] button is clicked, the settings made will be discarded.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	617/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

11.2.3 Deleting Delivery Condition of Signal Detection Mail Delivery

- Follow the steps below to delete CONDITION (delivery condition) of the signal detection mail delivery.

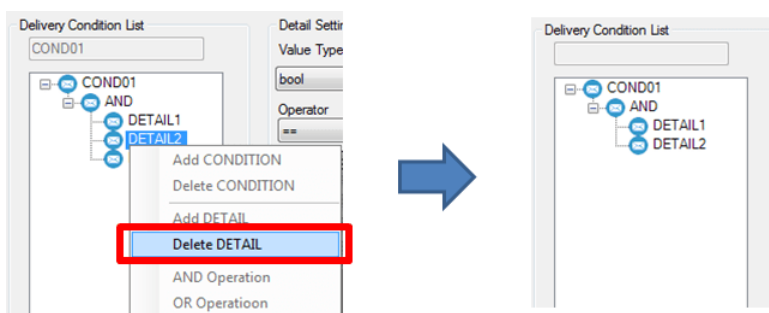
- (1) Select and right-click the CONDITION to be deleted in the tree area of the [Delivery Condition List], and select [Delete CONDITION] from the menu displayed. Two or more CONDITIONS can be selected.



- (2) The CONDITION selected in the tree area of the [Delivery Condition List] is deleted. If two or more CONDITIONS have been selected, all the selected CONDITIONS will be deleted. If there are DETAILS set for the CONDITIONS, the DETAILS will be deleted together with the CONDITIONS.

- Follow the steps below to delete the DETAIL (detail of delivery condition) of the signal detection mail delivery.

- (1) Select and right-click the DETAIL to be deleted in the tree area of the [Delivery Condition List], and select [Delete DETAIL] from the menu displayed. Two or more DETAILS can be selected.



- (2) The DETAIL selected in the tree area of the [Delivery Condition List] is deleted. If two or more DETAILS have been selected, all the selected DETAILS will be deleted. The associating condition (AND / OR) will be deleted automatically when the number of DETAILS become one or less after the deletion. The index number (<N>) for the name of DETAIL (DETAIL<N>) will be renumbered automatically. (Example: When the registered details are DETAIL1, DETAIL2, and DETAIL3, and if "DETAIL2" is deleted, the name of "DETAIL3" will become "DETAIL2.")

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	618/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

11.2.4 Examples of Delivery Condition Settings for Signal Detection Mail Delivery

Following are the examples of the settings made for the delivery condition of the signal detection mail delivery.

(Example 1)

This setting is a delivery condition for Equipment0001 and Equipment0002. Mail is delivered when multiple assign status "condition" is "OPERATE," and when the signal value of "ServoTemp_0_path1" (the temperature of servo motor) exceeds the threshold (error value).

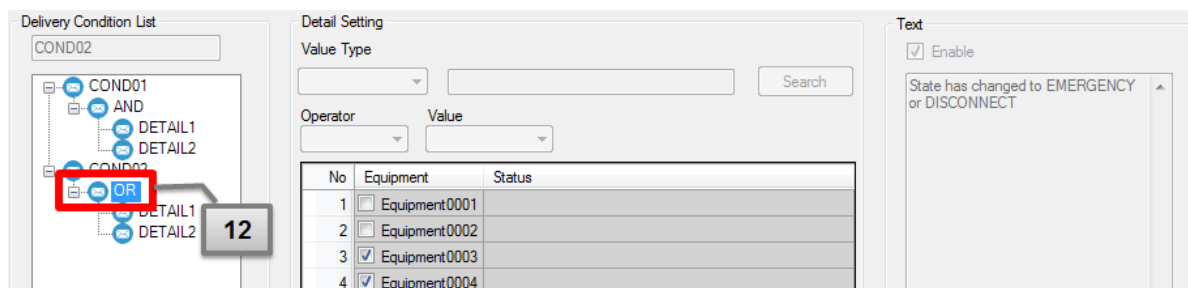
- 1) Add a CONDITION (COND01).
- 2) Select the target equipment units (Equipment0001 and Equipment0002).
- 3) Enter character string in the [Text] area after selecting the [Enable] check box.

- 4) Add DETAIL (DETAIL1).
- 5) Select [string] for [Value Type], enter "cond" in the text box, and click the [Search] button.
- 6) [condition] is set in the [Status] column.
- 7) Select [==] for [Operator], and enter "OPERATE" in the [Value] column.

- 8) Add [DETAIL2].
- 9) Select [judge] for [Value Type], enter "ServoTemp" in the text box, and click the [Search] button.
- 10) [ServoTemp_0_path1_<Machine name>.FR] is set in the [Status] column.
- 11) Select [==] for [Operator], and select [2] (error value) in the [Value] column.
- 12) Check that the associating condition for DETAILS is "AND," and the setting is completed.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	619/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

12) Change the associating condition of the DETAILS from "AND" to "OR," and the setting is completed.



11.2.5 Mail Format for Signal Detection Mail Delivery

The format of the signal detection mail is shown below.

(1) The structure of the signal detection mail

- Basic structure: Use multipart/alternative and divide mail text into text mail part and html mail part.
- Character code: utf-8
- Encode: base64

Mail header

From: < Administrator Email Address >
 To: < Delivery destination mail address >
 Date: < Date and time of delivery >
 Subject: < Title >
 Content-Type: multipart/alternative;

Mail body

Text mail part

html mail part

(2) Signal detection mail header

- From: <Administrator Email Address>
 The [Administrator Email Address] set in the [Mail Function Setting] is set.
- To: <Delivery destination mail address>
 - If the [Send by BCC] check box in the [Mail Function Setting] window is selected (enable).
 Only the [Administrator Email Address] will be set.
 - If the [Send by BCC] check box in the [Mail Function Setting] window is cleared (disable).
 The [Administrator Email Address] and the mail address of the user selected on the [Mail Delivery List] will be set.
- Subject: <Title>
 - Signal Detection Notice
 "MT-LINKi SIGNAL Detection Mail"
 - Signal Detection Clearance Notice
 "MT-LINKi SIGNAL Clear Mail"

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	621/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

(3) Signal detection mail body

- The structure of the text mail body
 - Signal detection notice (example when delivery conditions for 2 statuses exist)

Signal condition is satisfied on [<Equipment name>].

[CONDITION], <Delivery condition (CONDITION name)>

[MACHINE], <Machine name>

[<Status name>], <Value>

[MACHINE], <Machine name>

[<Status name>], <Value>

- Signal detection clearance notice

Signal condition has not been established on [<Equipment name>].

[CONDITION], <Delivery condition (CONDITION name)>

- The structure of the html mail body

- Signal detection notice (example when 3 statuses in delivery condition 1, and 1 status in delivery condition 2 exist)

Signal condition is satisfied on [Equipment name].

CONDITION	<Delivery condition 1 (CONDITION name)>
TEXT	<Text character string> * The character string set in the [Text] of [Signal Detection Mail Delivery Setting] is displayed. (NOTE 3)
MACHINE	<Machine name>
< Status name>	<Value>
< Status name>	<Value>
MACHINE	<Machine name>
< Status name>	<Value>

CONDITION	<Delivery condition 2 (CONDITION name)>
TEXT	<Text character string> * The character string set in the [Text] of [Signal Detection Mail Delivery Setting] is displayed. (NOTE 3)
MACHINE	<Machine name>
< Status name>	<Value>

- Signal detection notice (example when text character string of the delivery condition is set as disabled)

Signal condition is satisfied on [**<Equipment name>**].

CONDITION	<Delivery condition (CONDITION name)>
MACHINE	<Machine name>
< Status name>	<Value>
< Status name>	<Value>

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	622/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

- Signal detection clearance notice (example when both conditions are not established at the same time)

Signal condition has not been established on [<Equipment name>].

CONDITION	<Delivery condition 1 (CONDITION name)>
CONDITION	<Delivery condition 2 (CONDITION name)>

- * If the value type of status is [judge], ".FR" is attached at the end of <Status name>.
(Example) ServoTemp_0_path1_Machine001.FR
- * If the value type of status is [score], ".FV" is attached at the end of <Status name>.
(Example) ActF_path1_Machine002.FV

NOTE

- 1 Web name is displayed for <Equipment name>, <Machine name>, and <Status name>.
- 2 When <Status name> is a multiple assign status, "-" will be displayed for the <Machine name>.
- 3 In the <Text character string>, the character string set as the [Text] in the [Signal Detection Mail Setting] window, will be displayed in the same language selected as the language of the user of the destination address (the [Language] in the [Add/Edit User] dialog box).

(Example) If the language of the user selected as the destination is "English"
The string set in the [Text] in the [Signal Detection Mail Setting] window, with the [Web name language] in the [System Setting] window set to "English," will be displayed in the [TEXT] area.
- 4 If there are two or more delivery conditions (CONDITIONs) set for the same equipment unit, the delivery conditions established at the same time will be notified in a single mail.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 623/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

11.3 ALARM MAIL DELIVERY SETTING

On this window, the settings related to the mail delivery at the occurrence of alarm can be set. The settings for the mail server information and the selection of mail delivery destination users are done in the [Mail Function Setting] window. For the details, refer to the "**MAIL FUNCTION SETTING**" section.

In the alarm mail delivery, mail is delivered when an alarm occurs that matches the delivery condition set for each equipment unit.

Also, if all the alarms matching the delivery conditions are cleared, alarm clearance notice is delivered by mail.

NOTE

The judgement of the delivery condition is done in terms of each equipment unit.

- As for the alarm mail, delivery condition targeting multiple (2 or more) equipment units cannot be set.
(Example: Mail is delivered when delivery condition is established on both equipment 1 and equipment 2)
- Alarm clearance notice is delivered by mail when all the established delivery conditions of an equipment unit are cleared.

- Delivery conditions (CONDITION) can be registered up to 30 in total of the signal detection mail delivery setting and the alarm mail delivery setting.
- Details for delivery condition (DETAIL) can be registered up to 5 per delivery condition (CONDITION).

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	624/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

11.3.1 Steps for Setting Alarm Mail Delivery

Follow the steps below to set the alarm mail delivery.

- (1) Select [Alarm Mail Setting] in the AdminTool Menu to open the [Alarm Mail Setting] window.
- (2) On the [Delivery Condition List], add a new CONDITION (delivery condition).
- (3) Select the equipment unit to be corresponded to the delivery condition.
- (4) For the delivery condition, enter the text character string to display on the alarm mail body.
- (5) For the delivery condition, define the detail of delivery condition.
- (6) After completing the settings, click the [Apply] button or the [OK] button. If the [Cancel] button is clicked, the settings will be discarded.

For the details, refer to the "[Adding Delivery Condition for Alarm Mail Delivery](#)" subsection.

11.3.2 Adding Delivery Condition for Alarm Mail Delivery

Follow the steps below to add CONDITION (delivery condition) and DETAIL (detail of delivery condition) for the alarm mail delivery, and define the delivery condition.

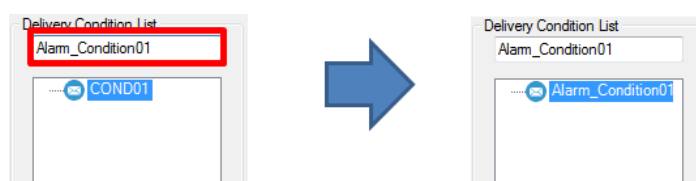
- (1) Right-click in the tree area of the [Delivery Condition List], and select [Add CONDITION] from the menu displayed.



- (2) CONDITION is added to the [Delivery Condition List].

The default name of CONDITION is "COND<NN>." ("<NN>" is the index number starting from "01" and the smallest unused value will be used.)

The name of the CONDITION can be changed in the text box above. Only an alphabet character can be used for the top character, and alphanumeric characters and underscores ("_") can also be used for the remaining characters of the name. This field can be up to 32 bytes long. The name of CONDITION will be displayed in the "CONDITION" part of the alarm mail text.



20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 625/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- (3) Select a CONDITION, and on the [Equipment List], select the check box of the equipment unit that will be the target equipment unit for alarm mail delivery. At least one equipment unit must be selected. The equipment units displayed on the [Equipment List] are the equipment units registered in the [Equipment Setting] window.

- * When using the check box of [Equipment List], two or more check boxes can be selected or cleared at the same time, by selecting two or more rows
- If you select two or more rows, and select the check boxes, the check boxes of the selected rows will be selected.
 - If you select two or more rows, and clear the check boxes, the check boxes of the selected rows will be cleared.

- (4) When the [Enable] check box is selected in the [Text] group box, characters to be set in the [TEXT] part of the alarm mail body can be entered. Up to 1000 bytes can be entered. The string for this field should be entered in the language specified for the [Web name language] field in the [System Setting] window. If there are no characters set in the text, the state of [TEXT] will be empty. If the check box is cleared, [TEXT] will not be displayed.

In the default setting, check box is selected, and the character string of text is empty.

- (5) Select and right-click the added CONDITION, and select [Add DETAIL] on the menu displayed. At least one DETAIL must be set.

The name of DETAIL is "DETAIL<N>." (<N> is the index number starting from "1" and the smallest unused value will be used.)

The name of DETAIL cannot be changed.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	626/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

- (6) Select the added DETAIL, select judgement condition from the drop-down menu for each item in the [Detail Setting] and enter the value of the item.

In the default setting, there are no settings for all the items. At least one item must be set.

The image shows two screenshots of the 'Detail Setting' interface. The top screenshot shows the default state with empty fields for Machine, Type, and Number, and a dropdown set to 'Equal to'. The bottom screenshot shows the fields populated: Machine is 'Machine001', Type is 'EX', and Number is '5'. The dropdown menu is open, showing options: 'Equal to', 'Not equal to', 'Or more', and 'Or less'. A blue arrow points from the top screenshot to the bottom one.

- **Machine**
Set the judgement condition for the machine name.
Select the judgement condition ([Equal to] / [Not equal to] / [Partial Match]), and enter the name of the target machine. Up to 32 bytes can be entered for the machine name.
If the setting for the machine name is omitted, it will be unconditional (all the machines will be targeted).
- **Type**
Set the judgement condition for the alarm type.
Select the judgement condition ([Equal to] / [Not equal to] / [Partial Match]), and enter the alarm type. Up to 32 bytes can be entered for the alarm type.
If the setting for the alarm type is omitted, it will be unconditional (all the alarm types will be targeted).
- **Number**
Set the judgement condition for the alarm number.
Select the judgement condition ([Equal to] / [Not equal to] / [Or more] / [Or less]) for the 1st criteria (in the upper column), and enter the alarm number.
If combining the conditions of two numbers, select either of the associating condition (AND / OR), and set the judgement condition and the alarm number of the 2nd criteria (in the lower column).
If the setting for the alarm number is omitted, it will be unconditional (all the alarm numbers will be targeted).

NOTE

All the items of the [Detail Setting] are associated with the AND condition. The delivery condition of DETAIL is established when the judgement conditions of [Machine], [Type], and [Number] are all established.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 627/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- When multiple DETAILS are registered, the associating condition (default setting is "AND") of the DETAILS is inserted in the 2nd layer of the [Delivery Condition List]. In order to change the associating condition, select and right-click the associating condition, and select [AND Operation] or [OR Operation] from the menu displayed.

-
- The diagram illustrates the process of changing a condition operation. On the left, a 'Delivery Condition List' shows 'COND01' with an 'AND' operation. A context menu is open with 'OR Operation' highlighted. A large blue arrow points to the right, where the same list now shows 'COND01' with an 'OR' operation and two sub-conditions, 'DETAIL1' and 'DETAIL2'.

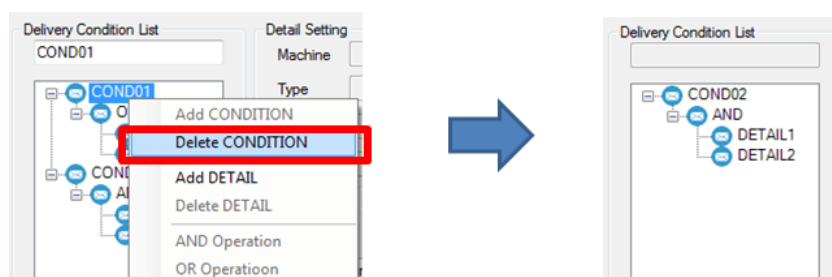
- When all the settings are completed, click the [OK] button or the [Apply] button to apply the settings. If the [Cancel] button is clicked, the settings made will be discarded.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0					Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9									
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8					Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7									
Ed.	Date	Design	Description					FANUC CORPORATION			Sheet.	628/868
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue					

11.3.3 Deleting Delivery Condition of Alarm Mail Delivery

- Follow the steps below to delete the CONDITION (delivery condition) of the alarm mail delivery.

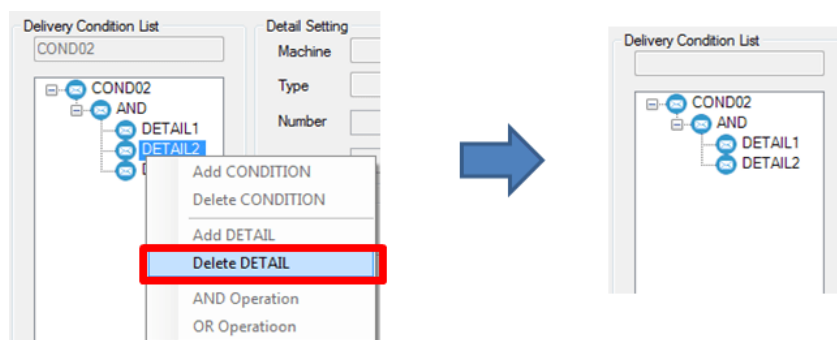
- (1) Select and right-click the CONDITION to be deleted in the tree area of the [Delivery Condition List], and select [Delete CONDITION] from the menu displayed. Two or more CONDITIONS can be selected.



- (2) The CONDITION selected in the tree area of the [Delivery Condition List] is deleted. If two or more CONDITIONS have been selected, all the selected CONDITIONS will be deleted.
If there are DETAILS set for the CONDITIONS, the DETAILS will be deleted together with the CONDITIONS.

- Follow the steps below to delete the DETAIL (detail of delivery condition) of the alarm mail delivery.

- (1) Select and right-click the DETAIL to be deleted in the tree area of the [Delivery Condition List], and select [Delete DETAIL] from the menu displayed. Two or more DETAILS can be selected.



- (2) The DETAIL selected in the tree area of the [Delivery Condition List] is deleted. If two or more DETAILS have been selected, all the selected DETAILS will be deleted. The associating condition (AND / OR) will be deleted automatically when the number of DETAILS become one or less after the deletion. The index number (<N>) for the name of DETAIL (DETAIL<N>) will be renumbered automatically.
(Example: When the registered details are DETAIL1, DETAIL2, and DETAIL3, and if "DETAIL2" is deleted, the name of "DETAIL3" will become "DETAIL2.")

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	629/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

11.3.4 Examples of Delivery Condition Settings for Alarm Mail Delivery

Following are the examples of the settings made for the delivery condition of the alarm mail delivery.

(Example 1)

This is an example of the settings for Equipment0001. Mail is delivered when either of the following two conditions is established.

In this example, Machine001 is assigned to Equipment0001.

- DETAIL1 (detail of delivery condition)
 - [Machine] (Name): [Equal to] "Machine001"
 - [Type]: [Equal to] "IE"
 - [Number]: "1" [Or more] "8" [Or less]
- DETAIL2 (detail of delivery condition)
 - [Machine] (Name): [Equal to] "Machine001"
 - [Type]: [Equal to] "OT"
 - [Number]: [Equal to] "500"

- 1) Add CONDITION (COND01).
- 2) Select "Equipment0001" on the [Equipment List] (Select the check box).
- 3) Enter character strings in the [Text] area after selecting the [Enable] check box.

The screenshot shows the 'Delivery Condition List' with 'COND01' selected. The 'Detail Setting' screen is open, showing 'Machine' set to 'Machine001', 'Type' set to 'IE', and 'Number' set to '1' [Or more] '8' [Or less]. The 'Equipment List' shows 'Equipment0001' selected. The 'Text' area shows 'Alarm is occurring on the equipment.'

- 4) Add DETAIL1.
- 5) For [Machine], select [Equal to] and enter "Machine001" in the text box.
- 6) For [Kind / Type], select [Equal to] and enter "IE" in the text box.
- 7) For [Number] (upper row), select [Or more] and enter "1" as the number.
- 8) For [Number] (lower row), select [Or less] and enter "8" as the number.
- 9) Select [AND] as the associating condition for the two numbers.

The screenshot shows the 'Delivery Condition List' with 'COND01' selected. The 'Detail Setting' screen is open, showing 'Machine' set to 'Machine001', 'Type' set to 'IE', and 'Number' set to '1' [Or more] '8' [Or less]. The 'Equipment List' shows 'Equipment0001' selected. The 'Text' area shows 'Alarm is occurring on the equipment.'

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 630/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

- 10) Add DETAIL2.
- 11) For [Machine], select [Equal to] and enter "Machine0001" in the text box.
- 12) For [Type], select [Equal to] and enter "OT" in the text box.
- 13) For [Number] (upper column), select [Equal to] and enter "500" as the number.

- 14) Change the associating condition of DETAILS from AND to OR. Click [OK] and the setting is completed.

(Example 2)

This is an example of the settings for Equipment0002. Mail is delivered when either of the following two conditions is established.

In this example, Machine002 and Machine003 are assigned to Equipment0002.

- DETAIL1 (detail of delivery condition)
 - [Machine] (Name): [Equal to] "Machine002"
 - [Type]: [Equal to] "SV"
 - [Number]: [Equal to] "10"
- DETAIL2 (detail of delivery condition)
 - [Machine] (Name): [Equal to] "Machine003"
 - [Type]: [Equal to] "SV"
 - [Number]: [Equal to] "10"

- 1) Add CONDITION (COND02).
- 2) Select "Equipment0002" on the [Equipment List].
- 3) Enter character strings in the [Text] area after selecting the [Enable] check box.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	631/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.					

- 4) Add DETAIL1.
- 5) For [Machine], select [Equal to] and enter "Machine0002" in the text box.
- 6) For [Type], select [Equal to] and enter "SV" in the text box.
- 7) For [Number] (upper column), select [Equal to] and enter "10" as the number.

- 8) Add DETAIL2.
- 9) For [Machine], select [Equal to] and enter "Machine0003" in the text box.
- 10) For [Type], select [Equal to] and enter "SV" in the text box.
- 11) For [Number] (upper column), select [Equal to] and enter "10" as the number.
- 12) Change the associating condition of DETAILS from AND to OR. Click [OK] and the setting is completed.

NOTE

The delivery condition of the alarm mail is determined by each alarm message. Therefore, alarm mail is not delivered when DETAIL1 and DETAIL2 is associated with AND as in the example below. (There is no alarm message with the type being "SW" and "PW" at the same time.)

- Associating condition of DETAIL1 and DETAIL2: AND
- DETAIL1 (detail of delivery condition)
 - [Machine] (Name): [Equal to] "Machine001"
 - [Type]: [Equal to] "SW"
- DETAIL2 (detail of delivery condition)
 - [Machine] (Name): [Equal to] "Machine003"
 - [Type]: [Equal to] "PW"

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9					
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7					
Ed.	Date	Design	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	FANUC CORPORATION	Sheet. 632/868

11.3.5 Mail Format for Alarm Mail Delivery

The format of the alarm mail is shown below.

(1) The structure of the alarm mail

- Basic structure: Use multipart/alternative and separate mail text into text mail part and html mail part.
- Character code: utf-8
- Encode: base64

Mail header

From: <Administrator Email Address>
 To: <Delivery destination mail address>
 Date: <Date and time of delivery>
 Subject: <Title>
 Content-Type: multipart/alternative;

Mail body

Text mail part

html mail part

(2) Alarm mail header

- From: < Administrator Email Address >
 The [Administrator Email Address] set in the [Mail Function Setting] is set.
- To: < Delivery destination mail address >
 - If the [Send by BCC] check box in the [Mail Function Setting] window is selected (enable).
 Only the [Administrator Email Address] will be set.
 - If the [Send by BCC] check box in the [Mail Function Setting] window is cleared (disable).
 The [Administrator Email Address] and the mail address of the user selected on the [Mail Delivery List] will be set.
- Subject: <Title>
 - Alarm occurrence notice
 "MT-LINKi ALARM Detection Mail"
 - Alarm clearance notice
 "MT-LINKi ALARM Clear Mail"

(3) Alarm mail body

- The structure of the text mail body
 - Alarm occurrence notice (example when delivery conditions for 2 alarms exist)

Alarm occurred on [<Equipment name>].

[CONDITION], <Delivery condition (CONDITION name)>
 [MACHINE], <Machine name>
 [KIND], <Alarm type>
 [NUMBER], <Alarm number>
 [MESSAGE], <Alarm message>
 [MACHINE], <Machine name>
 [KIND], <Alarm type>
 [NUMBER], <Alarm number>
 [MESSAGE], <Alarm message>

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11					
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10					
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8					
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2					
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet.	633/868
Ed.	Date	Desig.	Description					
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue		

- Alarm cleared on [<Equipment name>].

- Alarm occurrence notice (example when 2 alarms exists in delivery condition 1, and 1 alarm exist in delivery condition 2)

Alarm occurred on [<Equipment name>].

CONDITION	<Delivery condition 1 (CONDITION name)>
TEXT	<Text character string> * The character string set in the [Text] of [Alarm Mail Setting] is displayed. (NOTE 2)
MACHINE	<Machine name>
KIND	<Alarm type>
NUMBER	<Alarm number>
MESSAGE	<Alarm message>
MACHINE	<Machine name>
KIND	<Alarm type >
NUMBER	<Alarm number>
MESSAGE	<Alarm message>

CONDITION	<Delivery condition 2 (CONDITION name)>
TEXT	<Text character string> * The character string set in the [Text] of [Alarm Mail Setting] is displayed. (NOTE 2)
MACHINE	<Machine name>
KIND	<Alarm type>
NUMBER	<Alarm number>
MESSAGE	<Alarm message>

- Alarm occurred on [<Equipment name>].

CONDITION	<Delivery condition (CONDITION name)>
MACHINE	<Machine name>
KIND	<Alarm type>
NUMBER	<Alarm number>
MESSAGE	<Alarm message>

- Alarm cleared on [<Equipment name>].

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	634/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

- 2 Web name is displayed for <Equipment name> and <Machine name>.
- 3 The character string set as the [Text] of [Alarm Mail Setting] window, written in the same language selected as the [Language] in the [Add/Edit User] dialog box, will be displayed in the [TEXT] area above.

(Example) If the language of the user selected as the destination is "English"
The string set in the [Text] in the [Alarm Mail Setting] window, with the [Web name language] in the [System Setting] window set to "English," will be displayed in the [TEXT] area.
- 4 When there are two or more delivery conditions (CONDITION) set for the same equipment unit, the delivery conditions established at the same time will be notified in a single mail.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	635/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

12

[SETTING SYSTEM FUNCTIONS] - INTEGRATION SERVER SETTING

This chapter describes how to make settings of MT-LINK*i* Integration Server.

The AdminTool Menu has the following menu items in the [Setting System Functions] - [Integration Server Setting] group:

(1) Integration Server Setting

This menu sets data about the MT-LINK*i* system for Integration Servers to manage it.

Hereafter, an MT-LINK*i* Server might be referred to as a "facility" when it is emphasized as an exporter of operational results to Integration Servers.

12.1 INTEGRATION SERVER SETTING

With the [Integration Server Setting] window, you can configure a facility to be integrated with Integration Servers, which consolidate operational results imported from integrated MT-LINK*i* Servers or facilities.

Before Integration Server works with MT-LINK*i*, you need to configure Integration Server with Integration Server AdminTool, as well as with AdminTool. For the details of setting Integration Server, refer to the "FANUC MT-LINK*i* Integration Server AdminTool OPERATOR'S MANUAL."

The [Integration Server Setting] window enables you to do the following:

- To specify the name and the Web name of the facility and the Web Server URL of the facility, along with such Integration Server data as the name and the IP address/host name, the start date to import operational results, and the interval to subsequent results acquisitions.
- To view the registered Integration Servers in the [Integration Server List] table.
- To register a maximum number of three (3) Integration Servers per MT-LINK*i*.
- To check the connection with the Integration Server specified.
- To register the facility information of the Integration Server after confirming the settings.

To open the [Integration Server Setting] window, select [Integration Server Setting] in the AdminTool Menu.

The screenshot shows the 'Integration Server Setting' window. It has several sections and controls:

- Facility Information:**
 - Facility Name:** A text field containing 'Facility01'.
 - Web Name:** A text field containing 'Facility01'.
 - URL of the facility Web server:** A dropdown menu showing 'http://192.168.137.135:3000/'.
- Integration Server List:** A table with columns: No, Server Name, IP Address/Host Name, Acquisition Date, and Acquisition Interval. It contains one entry: '1 server 127.0.0.1 2018/06/18 1'.
- Buttons:** 'Add', 'Edit', 'Delete', 'Connection Test', 'Apply', 'OK', and 'Cancel'.
- Result Display Area:** A large empty rectangular area at the bottom for displaying results.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	636/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

[Integration Server Setting] window

- Facility information
The information about the facility (Facility name, Web name, Facility Web Server URL) is specified.
- Integration Server list
Add, edit or delete the information about the Integration Server. The information about the Integration Servers (Server name, IP address / host name, acquisition date, acquisition interval) is displayed in a list.
- Result display area
When the [Connection Test] button is clicked, the confirmation of the connection result is displayed. The registration result of the facility information is displayed when the [OK] button on the registration confirmation message of the facility information which is displayed after clicking the [Apply] or the [OK] button.

12.1.1 Steps for Integration Server Setting

This subsection instructs how to register Integration Servers for a facility.

- (1) In the AdminTool Menu, select [Integration Server Setting] to open the [Integration Server Setting] window.
- (2) In this window, you can add, edit, or delete Integration Servers. For the details, refer to the "[Add, Edit, or Delete Integration Server](#)" subsection.
- (3) Click the [Connection Test] button, and check the connection with the added Integration Server. For the details on the operation, refer to the "[Check Connection with the Integration Server](#)" subsection.
- (4) After adding the Integration Server and specifying the settings, click [Apply] or [OK] to save the settings. To discard the settings of the Integration Server just being made, click [Cancel].
- (5) Click [OK] in confirmation message of the registration of the facility information which is displayed after saving the settings in step (4), to register the facility information on the Integration Server. For the details on the operation, refer to the "[Register Facility Information to the Integration Server](#)" subsection.
- (6) After completing the settings of the Integration Server, you need to perform "Send to DB" procedure. For the details, refer to the "[SEND TO DB](#)" section.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN	
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8			
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION		Sheet. 637/868
Ed.	Date	Design	Description			
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

12.1.2 Add, Edit, or Delete Integration Server

This subsection instructs how to add, edit, and delete Integration Server settings in the [Integration Server Setting] window.

■ Adding an Integration Server

- (1) In the [Integration Server Setting] window, specify the [Facility Name], [Web Name], and the [Facility Web Server URL] if they are not populated. Click the [Add] button next to the [Integration Server List].

[Integration Server Setting] Window (Add, edit, or delete Integration Server)

Facility Information

- Facility name
Set the name of the facility. This field allows a half-width alphanumeric string. Note that all the groups of equipment registered to an MT-LINK*i* can be regarded collectively as a single facility.
- Web Name
Set the Web name of the facility to be displayed on Web screens. The name should be entered in the language specified for the [Web name language] field in the [System Setting] window.
- Facility Web Server URL
Select the Web Server URL of the facility from the drop-down list. The URL displayed in the [Web URL] field of the [System Setting] window ([Setting 1] tab) can be selected. Used to connect to the individual facility screen displayed in the [Facility Monitoring] screen of the Integration Server.

Integration Server List

- Server name
Name of the connection target Integration Server is displayed.
- IP address / host name
IP address or the host name of the connection target Integration Server is displayed.
- Acquisition date
The start date/time of the acquisition of operational results data is displayed.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	638/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

- Acquisition interval
The interval period of the acquisition of operational results data is displayed.
- [Add] button
Information about an Integration Server is added to the Integration Server list.
- [Edit] button
Information about an Integration Server selected on the Integration Server list is edited.
- [Delete] button
Information about an Integration Server selected on the Integration Server list is deleted.

(2) The [Add/Edit Integration Server] dialog box appears. Specify the information about the Integration Server.

[Add/Edit Integration Server] dialog box

- [Server Name]
Enter the name of the Integration Server used in the authentication for connection of the Integration Server to be connected. Up to 48 bytes of alphanumeric characters can be input.
- [IP Address/Host Name]
This field holds the IP address or the host name of the Integration Server specified in the [Server Name]. In the case of the host name, a half-width alphabet can be used as the first character, and the subsequent characters can be any combination of half-width alphanumeric characters, hyphens ("-"), and periods ("."). This field can be up to 253 bytes long.

NOTE

[Server Name] and [IP Address/Host Name] can be used as the name of the storing folder for the operational results data file, as well as the file name. When the length of the path of this file becomes 260 bytes or more due to the settings of [IP Address/Host Name], a message dialog box saying "Parameter setting error" appears by clicking the [OK] button. In such a case, check the file path specified in the message, and either change [IP Address/Host Name] so that the length of the path becomes 259 bytes or less, or set the [IP address/Host Name] with the IP address.

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0				Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL				
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11									
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10									
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9				Draw	A-40829-00001EN				
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8									
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2									
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7				FANUC CORPORATION			Sheet.	639/868	
Ed.	Date	Design	Description									
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.						Inoue

- [Password]
Enter the password used in the authentication for connection of the Integration Server to be connected.
- [Acquisition Date]
This field holds the start date for the Integration Server to import (or acquire) operational results from the facility. The hour part of the [Acquisition Date] allows "0" to "23." Note that a date of the past or a future date more than one month later cannot be specified in this field.
Acquiring production results starts at the specified start time on the day specified in the [Acquisition Date]. The first acquisition imports the data dating prior to the start time by the specified interval hours or less. The subsequent acquisitions will start and import the corresponding data every time the specified interval hours passes.
- [Acquisition Interval]
This field holds the interval period between adjacent results acquisitions. The [Acquisition Interval] allows "1" to "24" hours.
- [OK] button
Clicking the [OK] button saves the settings being made, and closes the [Add/Edit Integration Server] dialog box. The Integration Servers that are successfully added will be listed in the [Integration Server List] table.
- [Cancel] button
Clicking the [Cancel] button discards the settings being made, and closes the [Add/Edit Integration Server] dialog box.

The following example shows which production results are imported when the acquisition date and interval are specified as follows:

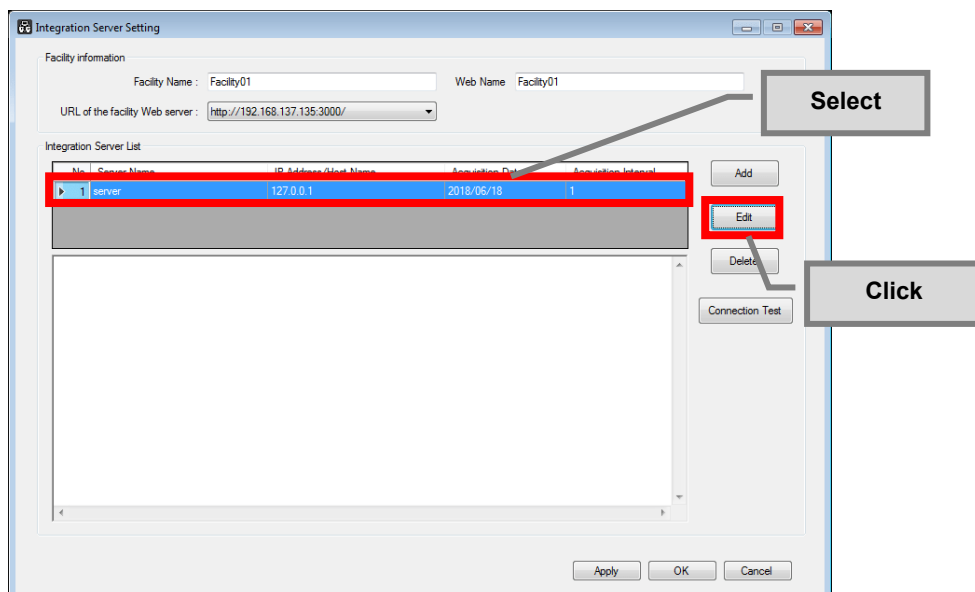
Acquisition Date	at 14:00 on September 13th, 2017
Acquisition Interval	24 hours
acquired data range	from 14:00 on September 13th, 2017 to 24 hours later

The next acquisition will start at 14:00 on September 13th, 2017 to import results (dating from 14:00:00.0 on September 13th, 2017 to 13:59:59.0 on September 14th, 2017).

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7	FANUC CORPORATION			Sheet. 640/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.	Inoue

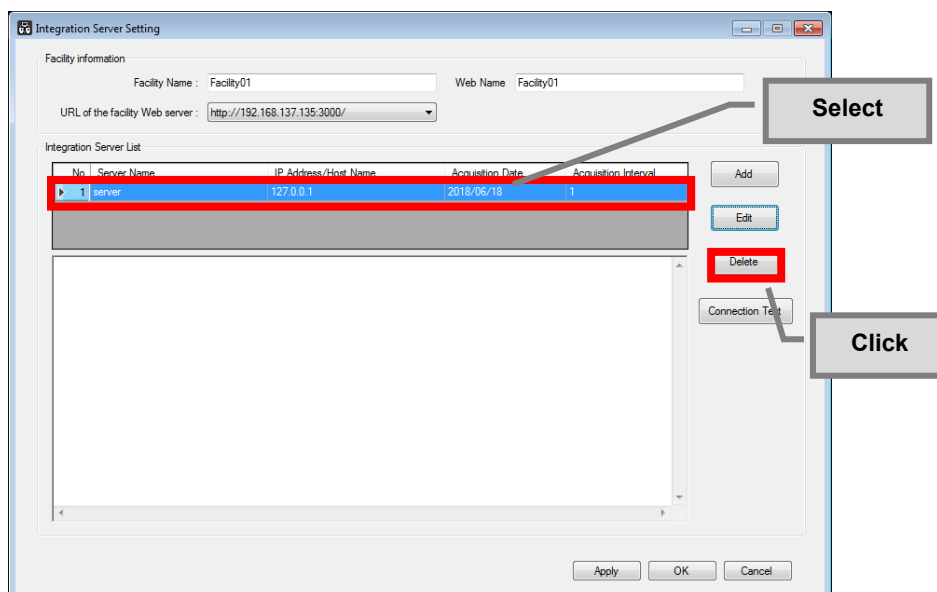
■ Editing a registered Integration Server

- (1) Select an Integration Server to edit in the [Integration Server List] table.
- (2) Click the [Edit] button next to the [Integration Server List] table.
- (3) The [Add/Edit Integration Server] dialog box appears. Edit the fields need modifying.
- (4) Click [OK] or [Cancel].



■ Deleting a registered Integration Server

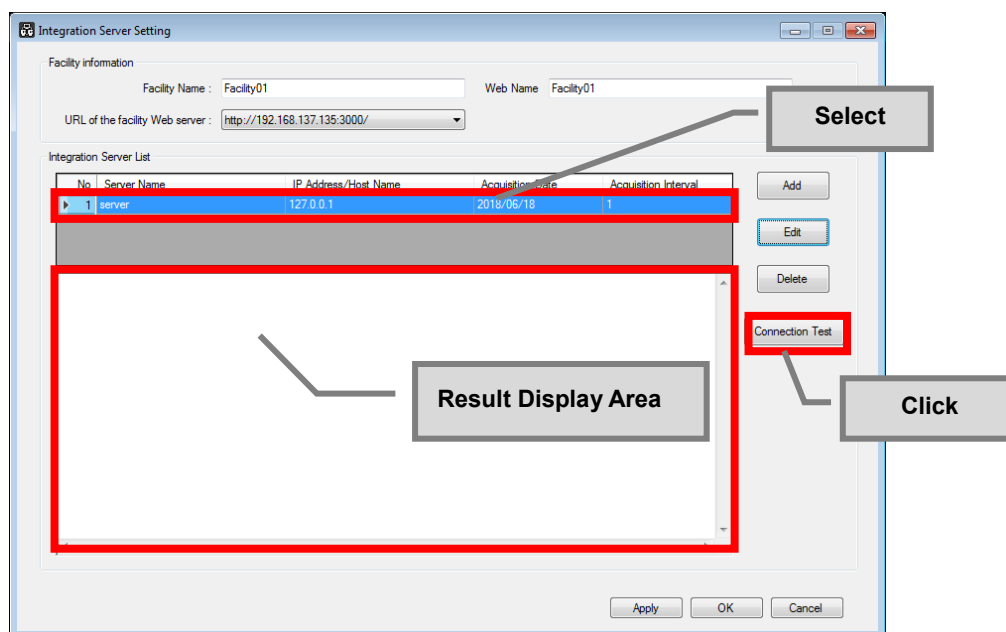
- (1) Select an Integration Server to delete in the [Integration Server List] table.
- (2) Click the [Delete] button next to the [Integration Server List] table.



20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	641/868
Ed.	Date	Design	Description								
Date	Nov.30,2015	Desig.	Ogawa	Check		Apprv.					

12.1.3 Check Connection with the Integration Server

This subsection explains how to check the connection with the Integration Server.



[Integration Server Setting] window (Check connection with Integration Server)

Integration Server List

- [Connection Test] button
Perform connection test with the Integration Server selected on the [Integration Server List].
- Result Display Area
Result of the connection test is displayed when the [Connection Test] button is clicked.

■ Follow the steps below to perform the connection test.

- (1) Select an Integration Server to check the connection from the [Integration Server List] in the [Integration Server Setting] window.
- (2) Click the [Connection Test] button.
- (3) The result of the connection check is displayed in the result display area.
If the connection check is successful, "Succeeded in a connection test to the Integration Server." will be displayed in the final row. If the test fails, the message will be displayed in red characters.

■ Following are the details of the items confirmed with the connection test.

- (1) Communication establishment check
Establishment of the communication with the Integration Server is checked.
- (2) Version check
Versions of MT-LINK*i* and Integration Server is checked.
Version check will be unsuccessful if the version of Integration Server is older than MT-LINK*i*.
- (3) Authentication check
Checks if the Integration Server name / Integration Server password match with the setting of the Integration Server.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9	Draw	A-40829-00001EN		
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8				
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7	FANUC CORPORATION			Sheet. 642/868
Ed.	Date	Design	Description				
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue	

(4) Number of registered facilities check

Checks if the MAC address is registered in the Integration Server.

If not registered, checks if the number of registered facilities is not exceeding the upper limit (20 facilities).

NOTE

Connection test is a function to check if there is communication with the Integration Server. Therefore, facility information is not registered to the Integration Server.

- Following are the confirmation results (examples) of the connection test displayed in the result display area, and the action to take in case the test was unsuccessful.

- Example of display when the test was successful

Server Name: server
IP Address/Host Name: 192.168.0.1
=> Facility information registration status Unregistered
Succeeded in a connection test to the Integration Server.

- Example of display when the communication establishment check was unsuccessful

Server Name: server
IP Address/Host Name: 192.168.0.1
=> Communication establishment check FAILURE
The connection target cannot be found.

Action:

- 1) Check if the IP address / host name of the Integration Server specified in the [Add/Edit Integration Server] dialog box is correct.
- 2) Check if the Importer service of the Integration Server is started.
Click the [Restart System] button of the AdminTool Menu of the Integration Server to display the [Restart System] window. Select the [Importer service is running] check box, if it is cleared.
- 3) Check if there is no problem with the network connection between the MT-LINKi Server PC and the Integration Server PC.

- Example of display when the communication establishment check was unsuccessful
(communication time out occurred)

Server Name: server
IP Address/Host Name: 192.168.0.1
=> Communication establishment check FAILURE
Communication timeout has occurred in connecting to the Integration Server.

Action: Check if there is no problem in the network connection between the MT-LINKi Server PC and the Integration Server PC.

- Example of display when the version check was unsuccessful

Server Name: server
IP Address/Host Name: 192.168.0.1
=> Version check FAILURE
The version of the Integration Server is older than the oldest compatible version with the current version of MT-LINKi.

Action: Update the Integration Server.

For the details, refer to the "FANUC MT-LINKi Integration Server Install MANUAL."

20	Nov. 22, 2023	uwano	Revised for MT-LINKi Version 4.0	Title	FANUC MT-LINKi AdminTool OPERATOR'S MANUAL	
19	Apr. 27, 2021	uwano	Revised for MT-LINKi Version 3.11			
18	July 28, 2020	uwano	Revised for MT-LINKi Version 3.10			
17	Jan. 30, 2020	uwano	Revised for MT-LINKi Version 3.9			
16	July 22, 2019	uwano	Revised for MT-LINKi Version 3.8	Draw	A-40829-00001EN	
15	Mar. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7.2			
14	Jan. 29, 2019	ogawa	Revised for MT-LINKi Version 3.7			
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet. 643/868
Date	Nov.30,2015	Desig.	Ogawa	Check	Apprv.	Inoue

- Example of display when the authentication check was unsuccessful

Server Name: server
 IP Address/Host Name: 192.168.0.1
 => Authentication check FAILURE
 Either or both of Server Name and Password differ from those recognized by the target Integration Server.

Action: Check if the Server name and the password of Integration Server specified in the [Add/Edit Integration Server] dialog box are correct.

- Example of display when the number of registered facilities check was unsuccessful

Server Name: server
 IP Address/Host Name: 192.168.0.1
 => Facility registration limit check FAILURE
 The number of registered facilities has reached the maximum. No more facility can be registered (maximum: 20 facilities).

Action: Register the facility to another Integration Server which has not reached the upper limit for registering the facilities, or register the facility after setting up a new Integration Server.

- Example of display when an internal error has occurred (when using a proxy server)

Server Name: server
 IP Address/Host Name: 192.168.0.1
 Internal error has occurred.[The remote server returned an unexpected response: (407) Proxy Authorization Required.]

Action: Connection with Integration Server will fail if the setting is to use a proxy server.
 Do not use a proxy server to connect the MT-LINKi Server PC and the Integration Server PC.

- Example of display when an internal error has occurred (when an exception occurred in AdminTool)

Server Name: server
 IP Address/Host Name: 192.168.0.1
 Internal error has occurred.[The remote server returned an unexpected response: (417) Expectation failed.]

Action: Contact FANUC.

- Example of display when an internal error has occurred (when an exception occurred in the Importer)

Server Name: server
 IP Address/Host Name: 192.168.0.1
 Internal error has occurred. [Importer Error.]

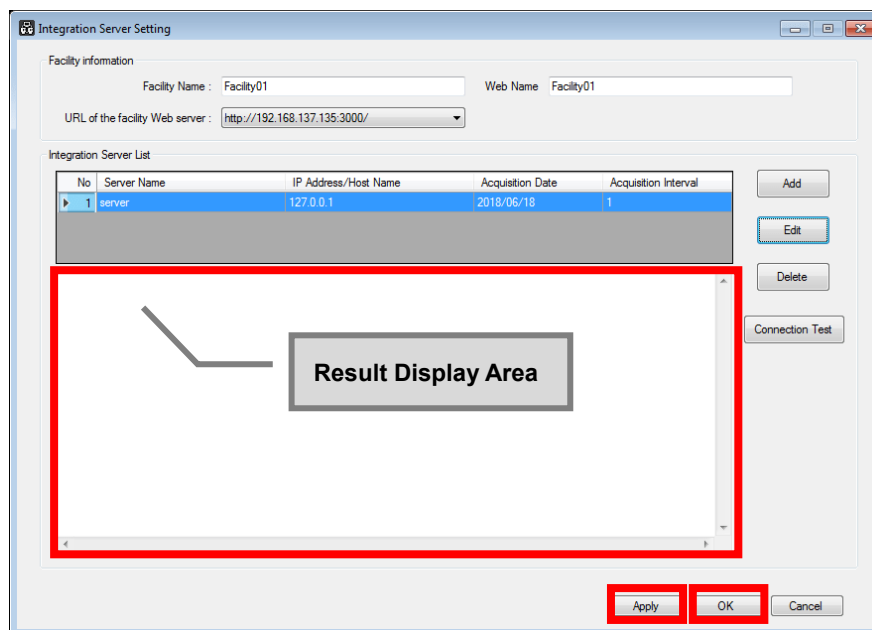
Action: Contact FANUC.

* For the details on how to contact FANUC, refer to the "HOW TO CONTACT US FOR TROUBLESHOOTING" chapter of the "FANUC MT-LINKi / Integration Server Troubleshooting MANUAL."

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0				Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL			
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11								
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10								
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				Draw	A-40829-00001EN			
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8								
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2								
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				FANUC CORPORATION			Sheet.	644/868
Ed.	Date	Design	Description								
Date	Nov.30.2015	Desig.	Ogawa	Check		Apprv.	Inoue				

12.1.4 Register Facility Information to the Integration Server

This subsection explains how to register the facility information after confirming the setting information of the Integration Server.



[Integration Server Setting] window
(Confirmation of Integration Server setting information and registration of facility information)

Integration Server List

- Result display area
When the [OK] button is clicked in the registration confirmation message of the facility information, registration result of the facility information is displayed.
- [OK] button
The setting details are confirmed by displaying the setting completion message of the Integration Server setting.
Next, the registration confirmation message of the facility information is displayed.
After finishing the operation of the registration confirmation message, close the [Integration Server Setting] window.
- [Apply] button
The setting details are confirmed by displaying the setting completion message of the Integration Server setting.
Next, the registration confirmation message of the facility information is displayed.
After finishing the operation of the registration confirmation message, return to the [Integration Server Setting] window.
- [Cancel] button
Discard the setting details, and close the [Integration Server Setting] window.

20	Nov. 22, 2023	uwano	Revised for MT-LINK <i>i</i> Version 4.0	Title	FANUC MT-LINK <i>i</i> AdminTool OPERATOR'S MANUAL		
19	Apr. 27, 2021	uwano	Revised for MT-LINK <i>i</i> Version 3.11				
18	July 28, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.10				
17	Jan. 30, 2020	uwano	Revised for MT-LINK <i>i</i> Version 3.9				
16	July 22, 2019	uwano	Revised for MT-LINK <i>i</i> Version 3.8	Draw	A-40829-00001EN		
15	Mar. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7.2				
14	Jan. 29, 2019	ogawa	Revised for MT-LINK <i>i</i> Version 3.7				
Ed.	Date	Design	Description	FANUC CORPORATION		Sheet.	645/868
Date	Nov.30,2015	Desig.	Ogawa				
		Check		Apprv.	Inoue		

- Follow the steps below to confirm the Integration Server setting information, and register the facility information to the Integration Server.
 - (1) After completing to add and set the Integration Server on the [Integration Server Setting] window, click the [OK] or the [Apply] button.
 - (2) The following setting completion message for the Integration Server setting appears. Click the [OK] button.

- (3) The registration confirmation message of the facility information will be displayed as shown below.

- (4) When the [OK] button is clicked, the result of the facility information registration is displayed in the result display area.
If the facility information registration is successful, "=> Registered successfully" will be displayed in the final row (in the case of multiple registration, the final row of each Integration Server). If the registration fails, message in red is displayed.
If the [Cancel] button is clicked, the facility information will not be registered.
- (5) When the setting is confirmed by clicking the [OK] button in the [Integration Server Setting] window, the [Integration Server Setting] window closes.
When the setting is confirmed by clicking the [Apply] button in the [Integration Server Setting] window, it returns to the [Integration Server Setting] window.
However, when the facility information registration is unsuccessful, it will return to the [Integration Server Setting] window.

- | Data Item | Description |
|---------------------------|---------------------------------------|
| MAC Address | MAC address of facility |
| Facility Name | Name of facility |
| Web Name (English) | Web name of facility (English) |
| Web Name (Japanese) | Web name of facility (Japanese) |
| Web Name (Chinese) | Web name of facility (Chinese) |
| Web Name (Local language) | Web name of facility (Local language) |
| Facility Web Server URL | Web Server URL of facility |

- Example of display when the registration was successful (when two Integration Servers are registered)

```
Server Name: server2
IP Address/Host Name: 192.168.0.2
=> Registered successfully
```

- ```
Server Name: server1
IP Address/Host Name: 192.168.0.1
=> Registered successfully

Server Name: server2
IP Address/Host Name: 192.168.0.2
=> Facility information transmission FAILURE
Failed to send facility registration information.
```

Action: Check if the usage rate of the CPU or the memory of MT-LINK*i* Server PC is in a high state.  
Terminate any application or resident programs that are not related to MT-LINK*i*.  
Restart the Server PC if the error occurs continuously.

|      |               |        |                                            |       |  |        |                   |                                                       |  |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  |  | Sheet. | 647/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |  |        |         |

# 13

## [BACKING UP SETTINGS] - SETTINGS BACKUP

This chapter describes how to back up and restore the settings made with AdminTool.

The AdminTool Menu has the following menu items in the [Backing Up Settings] - [Settings Backup] group:

(1) Backup Settings

This menu backs up current settings of AdminTool as backup data.

(2) Restore Settings

This menu restores setting data of AdminTool that were backed up through Backup Settings.

(3) Receive from DB

This menu retrieves stored setting data from the database and overwrites the current settings of AdminTool.

### NOTE

The following setting files for the SERVO VIEWER Linkage function, which are set by the Linkage tool for MT-LINK*i*, are not supported by Backup Settings/Restore Settings function.

- Measure profile
- Measurement schedule

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 648/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         |

## 13.1 BACKUP SETTINGS

Use this function to make backup copies of the settings at any time; for example, when you have completed the AdminTool settings once or before you update the settings.

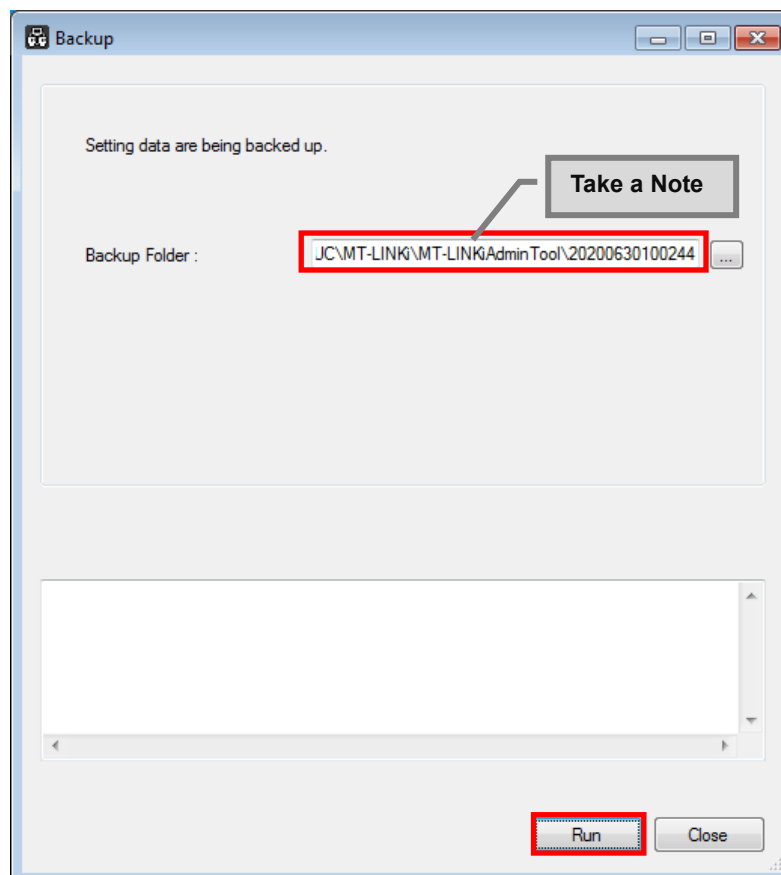
- Back up the settings of MT-LINK*i* at a certain point.

### 13.1.1 Backup Setting Data

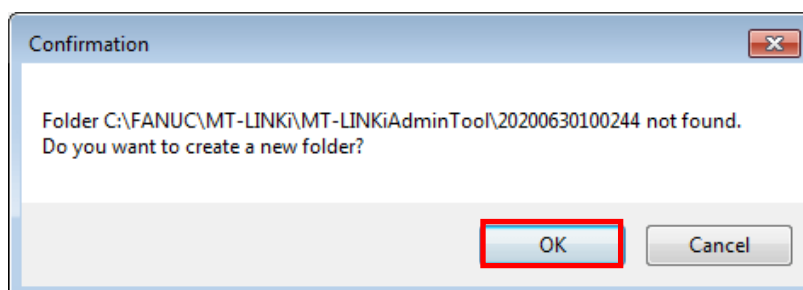
- (1) Selecting [Backup Settings] in the AdminTool Menu opens the [Backup Settings] window as shown below. Take a note of the path populated in the [Backup Folder] field.

The [Backup Folder] field populates the following folder by default:

<MT-LINK*i* installation folder>\MT-LINKiAdminTool\<yyyyMMddHHmmss>

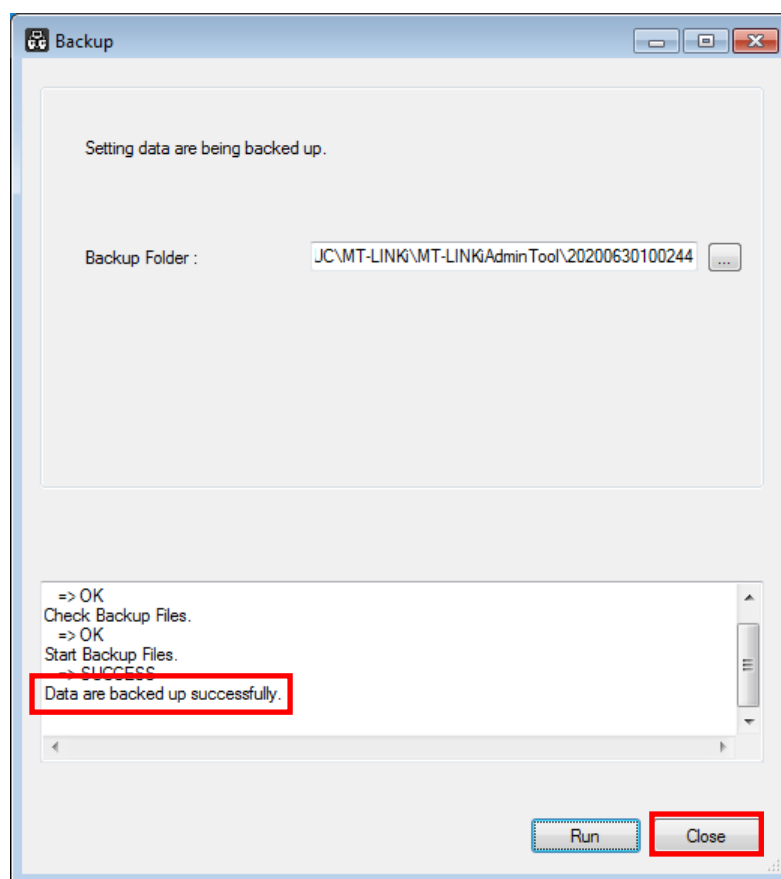


- (2) Clicking [Run] displays the [Confirmation] dialog box. Click [OK].

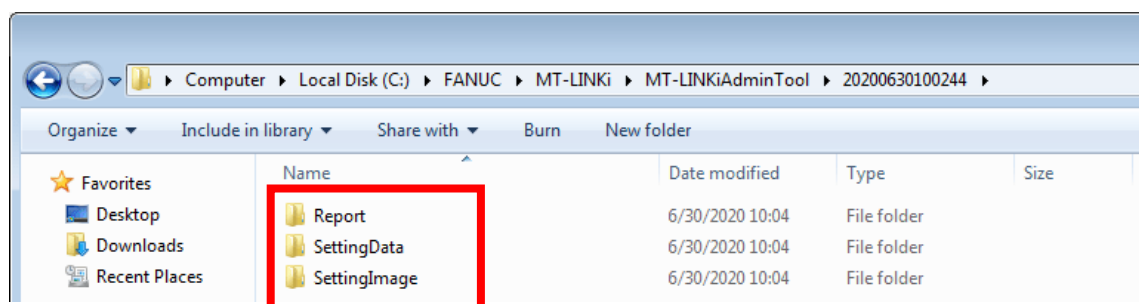


|      |               |        |                                            |                          |                                                       |       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |       | Sheet. 649/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue |                |

- (3) Check whether "Data are backed up successfully" has appeared on the window. The text means backup copies of the settings are successfully created. Click the [Close] button to close the window.



- (4) The created backup copies will be stored in the [Backup Folder] as shown in the following figure.



- The [Report] folder stores report template files.
- The [SettingData] folder stores setting files such as AdminTool.ini.
- The [SettingImage] folder stores images used in AdminTool.

|      |               |        |                                    |       |  |        |                   |                                               |  |        |         |
|------|---------------|--------|------------------------------------|-------|--|--------|-------------------|-----------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |                   |                                               |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |                   |                                               |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        | Draw              | A-40829-00001EN                               |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        |                   |                                               |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |                   |                                               |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        | FANUC CORPORATION |                                               |  | Sheet. | 650/868 |
| Ed.  | Date          | Design | Description                        |       |  |        |                   |                                               |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check |  | Apprv. |                   |                                               |  |        |         |

## 13.2 RESTORE SETTINGS

This restore function allows you to load the saved setting files with Backup Settings function and to restore the settings.

- This function retrieves all the AdminTool settings from the backup files.

## NOTE

1. Be careful when executing Restore Settings, as the existing settings will be overwritten.
2. If schedules are registered using the following functions, after running the [Restore Settings] process, a message appears which asks for the confirmation on the registering/unregistering of the schedules.
  - DB data retention period setting (System Setting)
  - Report schedule (daily/weekly/monthly)
  - Signal data output schedule
  - Machine data storing schedule

When [Yes] is selected, the process for registering/unregistering schedule will be executed.

If there is a schedule running at that time, that schedule will be completed without interruption.

When [No] is selected, the process for registering/unregistering schedule will not be executed, and the currently registered schedules will remain as they are.

|      |               |        |                                            |       |  |        |       |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        |       | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                   |                                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |       | Draw              | A-40829-00001EN                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                   |                                                       |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       | FANUC CORPORATION |                                                       |  | Sheet. | 651/868 |
| Date | Nov.30.2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                   |                                                       |  |        |         |

## 13.2.1 Restore Backup Setting

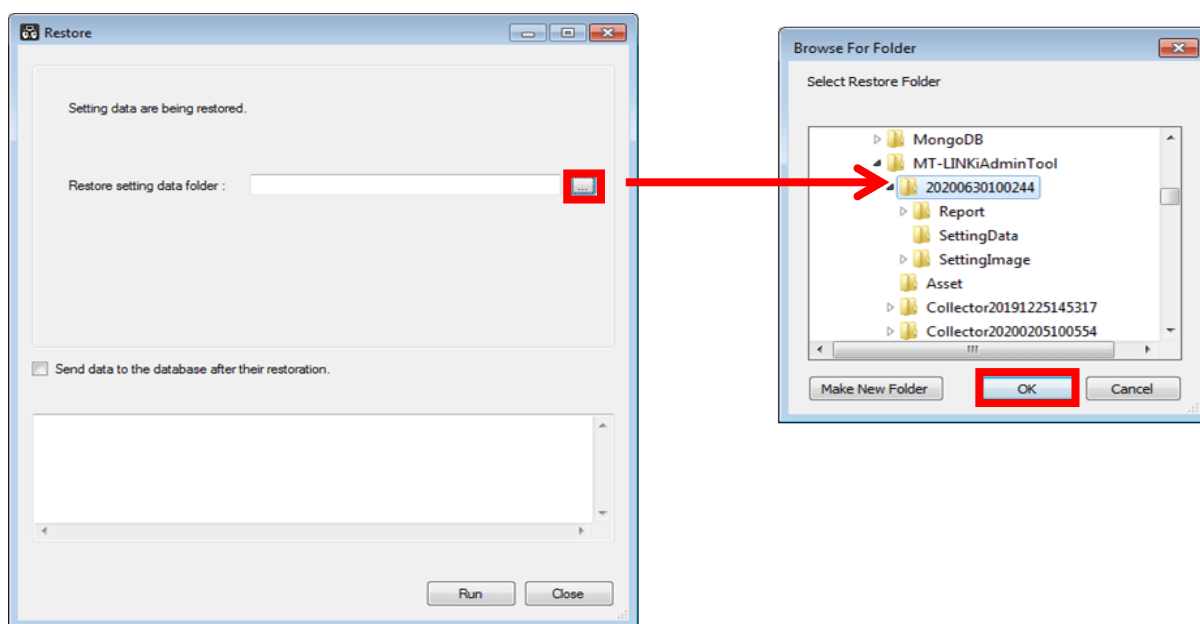
- (1) Selecting [Restore Settings] opens the following [Restore Settings] window.  
Click the upload file button attached at the [Restore setting data folder] field. Select the folder you set in the backup function.

Either set the [Restore setting data folder] field to the folder specified in the [Backup Folder] field of the [Backup Settings] window or click the browse folder button ([...]) to open the [Browse For Folder] window, select it, and click [OK] to tell the [Restore setting data folder] field in the [Restore Settings] dialog box to populate the selected folder.

The [Browse For Folder] window selects the following folder on the first session.

<MT-LINK*i* installation folder>\MT-LINKiAdminTool

On subsequent sessions, it selects the folder specified in the [Restore setting data folder] field in the last session.

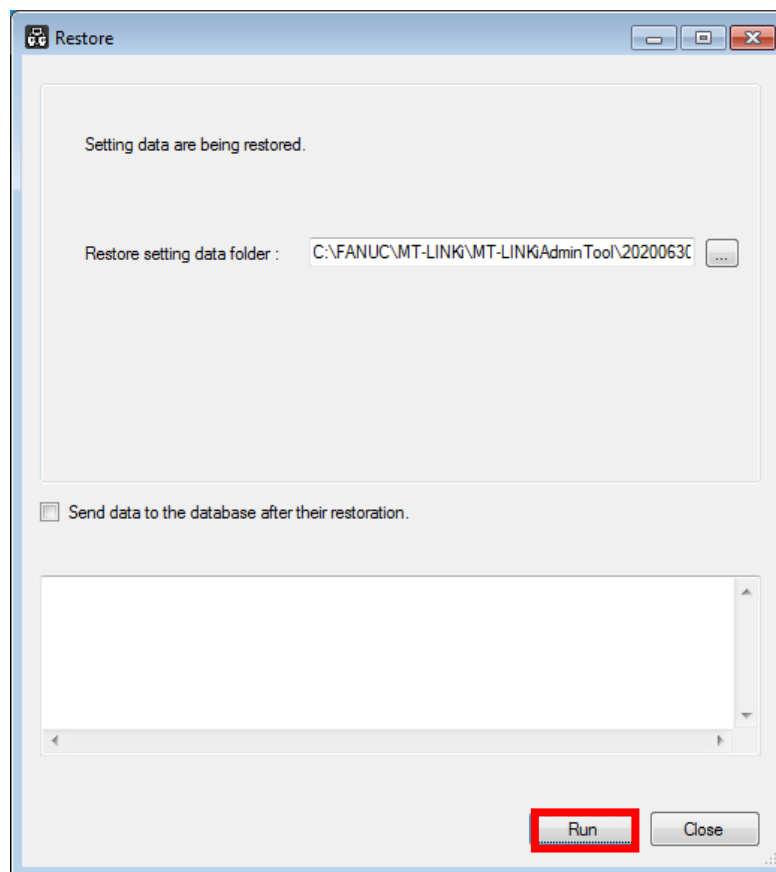


- (2) Set the check box for [Send data to the database after their restoration.].  
If the check box for [Send data to the database after their restoration.] is selected, the restored setting data is sent to DB.  
If the checkbox for [Send data to the database after their restoration.] is cleared, the restored setting data is reflected only inside AdminTool.

|      |               |        |                                            |       |  |        |       |                                                       |  |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------------------------------------------|--|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                                                       |  |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                                                       |  |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                                                       |  |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        | Draw  | A-40829-00001EN                                       |  |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                                                       |  |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                                                       |  |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       | FANUC CORPORATION                                     |  |  | Sheet. | 652/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                                                       |  |  |        |         |

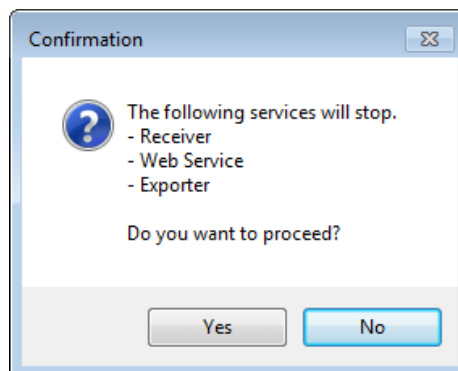


(3) Click the [Run] button.



If any of the following services is running, it will be stopped. The confirmation message shown below appears. Click the [Yes] button:

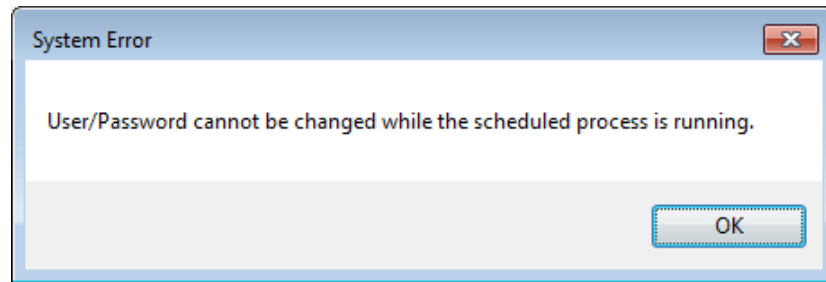
- MT-LINK*i* Receiver
- MT-LINK*i* WebServer
- MT-LINK*i* Exporter Service



|      |               |        |                                            |                          |                                                       |       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |       | Sheet. 653/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue |                |

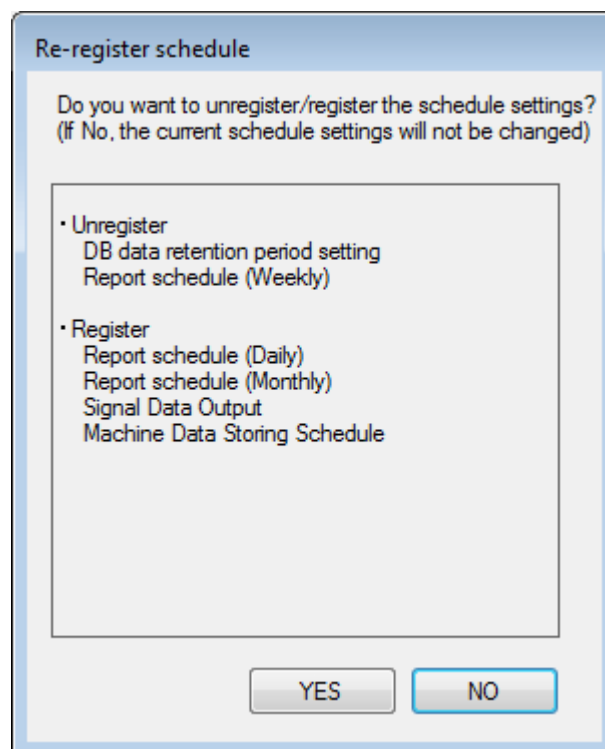
**NOTE**

If the following message is displayed, Restore Settings cannot run.



- (4) The following dialog box appears.

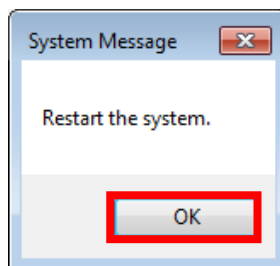
Clicking [NO] avoids re-registering schedules and leaves the currently registered schedules as-is; clicking [Yes] unregisters those currently registered but re-registers the schedules that had been or were registered at the point in time of the backup.



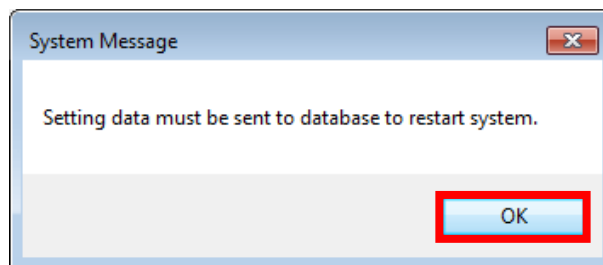
|      |               |        |                                    |                          |                                               |       |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |       | Sheet. 654/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue |                |

(5) The following message appears. Clicking [OK] starts to restore the setting data to AdminTool.

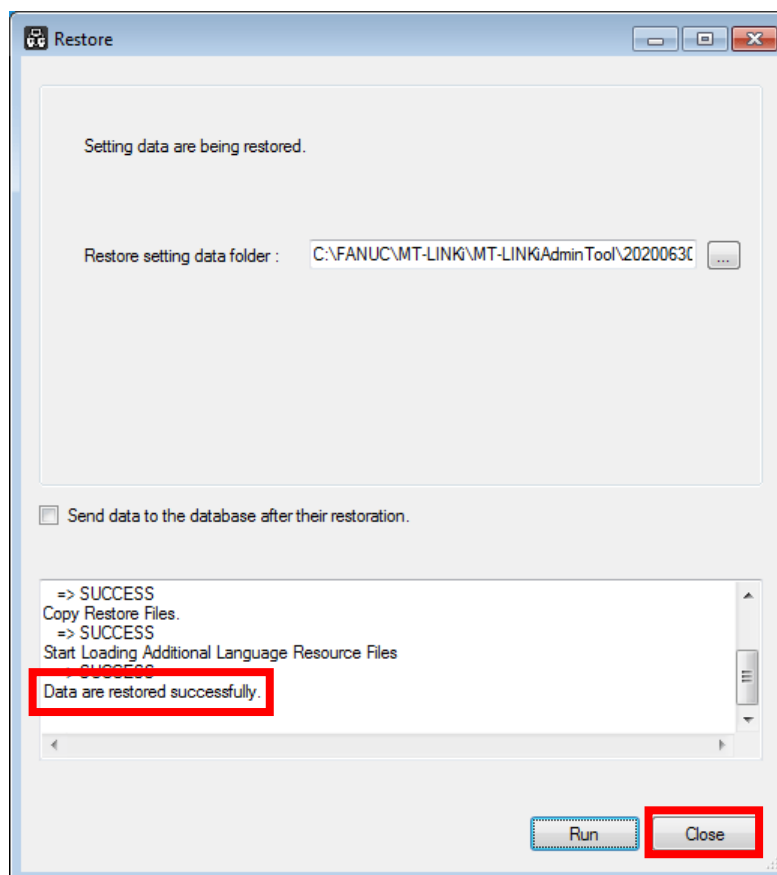
- If the [Send data to the database after their restoration.] check box is selected  
When the following message appears, click [OK].



- If the [Send data to the database after their restoration.] check box is cleared  
When the following message appears, click [OK].



(6) Check whether "Data are restored successfully" has appeared on the window. The text means backed up copies are successfully restored. Click the [Close] button to close the window.  
Proceed to step (8) if "Restart the system" was displayed in step (5).



|      |               |        |                                    |       |  |        |       |                                               |  |  |        |         |
|------|---------------|--------|------------------------------------|-------|--|--------|-------|-----------------------------------------------|--|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |       |                                               |  |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |       |                                               |  |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        |       |                                               |  |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        | Draw  | A-40829-00001EN                               |  |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |       |                                               |  |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        |       |                                               |  |  |        |         |
| Ed.  | Date          | Design | Description                        |       |  |        |       | FANUC CORPORATION                             |  |  | Sheet. | 655/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check |  | Apprv. | Inoue |                                               |  |  |        |         |



## 13.3 RECEIVE FROM DB

The [Receive from DB] window retrieves the stored settings of AdminTool from the database and overwrites them. Selecting [Receive from DB] in the AdminTool Menu opens the window.

This window has the following functions:

- Retrieving the setting data from the database and overwriting AdminTool settings with them
- Checking the connection to the database and notifying users of an error event if unable to connect to the database

### NOTE

- 1 Beware that clicking the [Receive from DB] button will overwrite the settings of AdminTool, even when they are being edited.
- 2 It is not allowed to overwrite the settings of AdminTool by copying setting data from an older database to the current database and then clicking the [Receive from DB] button.
- 3 When the following schedules are registered, after running the [Receive from DB] process, a message appears which asks for the confirmation on the registering/unregistering of the schedules.
  - DB data retention period setting (System setting)
  - Report schedule (daily/weekly/monthly)
  - Signal data output schedule
  - Machine data storing schedule

When [Yes] is selected, the process for registering/unregistering schedule will be executed.

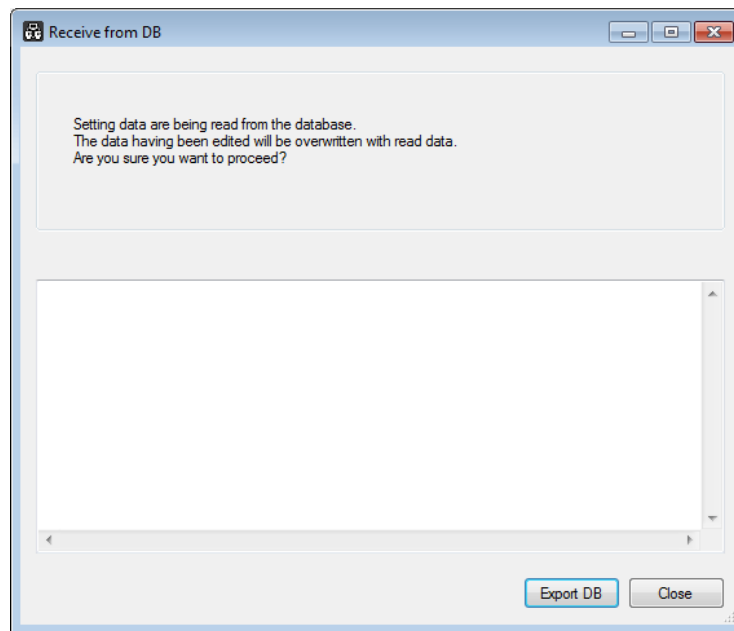
If there is a schedule running at that time, that schedule will be completed without interruption.

When [No] is selected, the process for registering/unregistering schedule will not be executed, and the currently registered schedules will remain as they are.

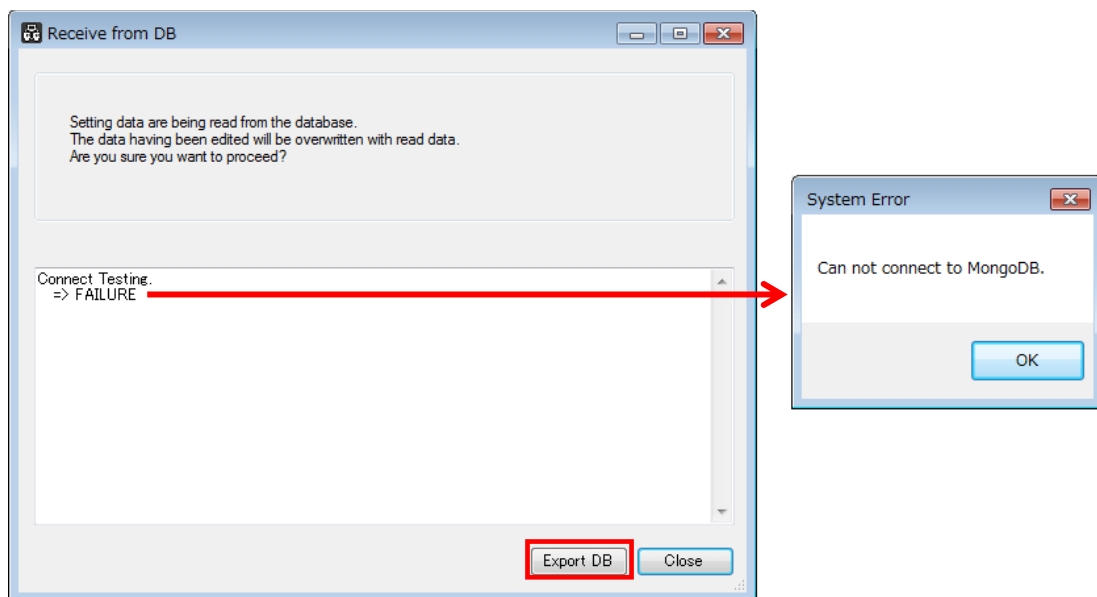
|      |               |        |                                    |       |  |        |                   |                                               |  |        |         |
|------|---------------|--------|------------------------------------|-------|--|--------|-------------------|-----------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |                   |                                               |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |                   |                                               |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        | Draw              | A-40829-00001EN                               |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        |                   |                                               |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |                   |                                               |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        | FANUC CORPORATION |                                               |  | Sheet. | 657/868 |
| Ed.  | Date          | Design | Description                        |       |  |        |                   |                                               |  |        |         |
| Date | Nov.30.2015   | Desig. | Ogawa                              | Check |  | Apprv. |                   |                                               |  |        |         |

## 13.3.1 Receive Setting Data from DB

(1) Selecting [Receive from DB] in the AdminTool Menu opens the [Receive from DB] window as below.

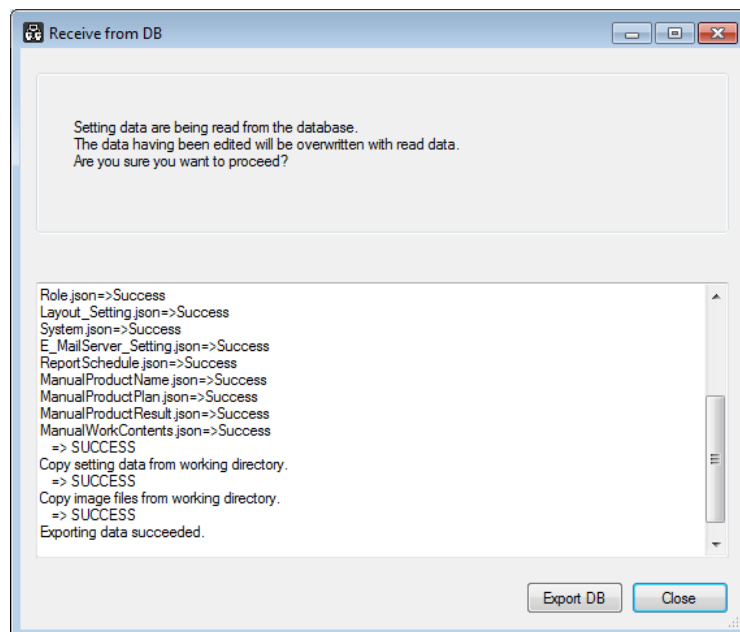


(2) Clicking the [Export DB] button checks the connection to DB. If connection fails, an error is displayed.



|      |               |        |                                    |                          |                                               |        |        |         |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |        | Sheet. | 658/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    |                                               | Apprv. | Inoue  |         |

- (3) When connection succeeds, get the setting data from DB. This step automatically overwrites the AdminTool settings by the received data.



|      |               |        |                                            |                          |                                                       |       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |       | Sheet. 659/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue |                |

# 14 OTHERS

## 14.1 LIST OF FanucCNC SIGNALS

### 14.1.1 Basic Signals of FanucCNC

The table below shows basic signals retrievable from FanucCNC.

<m>, <n>, and <p> in the table denote as follows:

<m>: signal definition that includes the <m>-th axis of the spindle axes.

<n>: signal definition that includes the <n>-th axis of the servo axes.

<p>: the thousands digit of the PMC address. The path number is calculated by the formula <p> + 1  
(Example: PMC F1000.7, the path number will be 2 (1 + 1)).

For Series 30*i* and 16*i* models, refer to their descriptions respectively:

The "Series 30*i*" and "Series 16*i*" paragraphs in the "Abbreviations of CNC models" paragraph in the "INTRODUCTION" chapter.

For the signals that can be retrieved for each CNC, refer to Table 2 and Table 3.

**Table 1. Basic Signals**

| #  | Signal Type | Description                                     |                    | Data Type | Reading Cycle<br>(NOTE 1)<br>[Default] | Enabled<br>(NOTE 1)<br>[Default] | <m>/<br><n>/<br><p> |
|----|-------------|-------------------------------------------------|--------------------|-----------|----------------------------------------|----------------------------------|---------------------|
|    |             | Series 30 <i>i</i><br>Series 16 <i>i</i>        | Series 15 <i>i</i> |           |                                        |                                  |                     |
| 01 | SigOP       | Automatic operation signal                      |                    | Bool      | 500 msec                               | NO                               | <p>                 |
|    |             | PMC F<p>000.7                                   | PMC F<p>000.5      |           |                                        |                                  |                     |
| 02 | SigSTL      | Cycle start lamp signal                         |                    | Bool      | 500 msec                               | NO                               | <p>                 |
|    |             | PMC F<p>000.5                                   | PMC F<p>000.4      |           |                                        |                                  |                     |
| 03 | SigSPL      | Feed hold lamp signal                           |                    | Bool      | 500 msec                               | NO                               | <p>                 |
|    |             | PMC F<p>000.4                                   | PMC F<p>000.3      |           |                                        |                                  |                     |
| 04 | SigAL       | Alarm signal                                    |                    | Bool      | 500 msec                               | NO                               | <p>                 |
|    |             | PMC F<p>001.0                                   | PMC F<p>000.1      |           |                                        |                                  |                     |
| 05 | EMG         | Emergency stop signal                           |                    | Bool      | 500 msec                               | NO                               |                     |
| 06 | Mode        | Mode                                            |                    | String    | 500 msec                               | YES                              |                     |
| 07 | MainProgram | Main program (NOTE 2)                           |                    | String    | 500 msec                               | YES                              |                     |
| 08 | ActProgram  | Executing program (NOTE 2)                      |                    | String    | 500 msec                               | YES                              |                     |
| 09 | Sequence    | Sequence number                                 |                    | Int32     | 500 msec                               | NO                               |                     |
| 10 | MainComment | Program comment (main program)<br>(NOTE 3)      |                    | String    | 500 msec                               | YES                              |                     |
| 11 | ActComment  | Program comment (executing program)<br>(NOTE 3) |                    | String    | 500 msec                               | YES                              |                     |
| 12 | ActS        | Speed S (actual value) (rpm) (NOTE 5)           |                    | Int32     | 500 msec                               | NO                               |                     |

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 660/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue             |                                                       |  |        |         |



| #  | Signal Type | Description                                                                                          |                          | Data Type | Reading Cycle (NOTE 1) [Default] | Enabled (NOTE 1) [Default] | <m>/<n>/<p> |
|----|-------------|------------------------------------------------------------------------------------------------------|--------------------------|-----------|----------------------------------|----------------------------|-------------|
|    |             | Series 30i<br>Series 16i                                                                             | Series 15i               |           |                                  |                            |             |
| 13 | ActF        | Feed rate F (actual value) (mm/min or inch/min) (NOTE 5, 7)<br>* Rounded down to the nearest integer |                          | Int32     | 500 msec                         | NO                         |             |
| 14 | ActFdec     | Feed rate F (actual value) (mm/min or inch/min) (NOTE 5, 6, 7)                                       |                          | Double    | 500 msec                         | NO                         |             |
| 15 | ModalS      | Speed S (command value) (rpm)                                                                        |                          | String    | 500 msec                         | NO                         |             |
| 16 | ModalF      | Feed rate F (command value) (mm/min or inch/min)                                                     |                          | String    | 500 msec                         | NO                         |             |
| 17 | ModalT      | T code                                                                                               |                          | String    | 500 msec                         | NO                         |             |
| 18 | ModalM      | 1 <sup>st</sup> M code                                                                               |                          | String    | 500 msec                         | NO                         |             |
| 19 | ModalM2     | 2 <sup>nd</sup> M code                                                                               |                          | String    | 500 msec                         | NO                         |             |
| 20 | ModalM3     | 3 <sup>rd</sup> M code                                                                               |                          | String    | 500 msec                         | NO                         |             |
| 21 | PartsNum    | The number of machined parts                                                                         |                          | Int32     | 500 msec                         | YES                        |             |
|    |             | parameter (No.6711)                                                                                  | parameter (No.107)       |           |                                  |                            |             |
| 22 | PartsNumAll | The total number of machined parts                                                                   |                          | Int32     | 500 msec                         | NO                         |             |
|    |             | parameter (No.6712)                                                                                  | parameter (No.108)       |           |                                  |                            |             |
| 23 | PowOnTime   | Power-on period (sec)                                                                                |                          | Double    | 500 msec                         | NO                         |             |
|    |             | parameter (No.6750)                                                                                  | parameter (No.100)       |           |                                  |                            |             |
| 24 | RunTime     | Operation time (sec)                                                                                 |                          | Double    | 500 msec                         | NO                         |             |
|    |             | parameter (No.6751 - 6752)                                                                           | parameter (No.101 - 102) |           |                                  |                            |             |
| 25 | CutTime     | Cutting time (sec)                                                                                   |                          | Double    | 500 msec                         | NO                         |             |
|    |             | parameter (No.6753 - 6754)                                                                           | parameter (No.103 - 104) |           |                                  |                            |             |
| 26 | SigCUT      | Cutting feed signal<br>PMC F<p>002.6                                                                 |                          | Bool      | 500 msec                         | YES                        | <p>         |
| 27 | SigSBK      | Single block check signal<br>PMC F<p>004.3                                                           |                          | Bool      | 500 msec                         | YES                        | <p>         |
| 28 | SigDM00     | Decode M00 signal                                                                                    |                          | Bool      | 500 msec                         | YES                        | <p>         |
|    |             | PMC F<p>009.7                                                                                        | PMC Fp009.4              |           |                                  |                            |             |
| 29 | SigDM01     | Decode M01 signal                                                                                    |                          | Bool      | 500 msec                         | YES                        | <p>         |
|    |             | PMC F<p>009.6                                                                                        | PMC Fp009.5              |           |                                  |                            |             |

|      |               |        |                                            |       |  |        |                   |                                                       |  |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  |  | Sheet. | 661/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |  |        |         |

| #  | Signal Type | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                          | Data Type | Reading Cycle<br>(NOTE 1)<br>[Default] | Enabled<br>(NOTE 1)<br>[Default] | <m>/<br><n>/<br><p> |
|----|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------|----------------------------------|---------------------|
|    |             | Series 30 <i>i</i><br>Series 16 <i>i</i>                                                                                                                                                                                                                                                                                                                                                                                                                               | Series 15 <i>i</i>                                                                                                                                                                       |           |                                        |                                  |                     |
| 30 | SigMDRN     | Dry run check signal                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                          | Bool      | 500 msec                               | YES                              | <p>                 |
|    |             | PMC F<p>002.7                                                                                                                                                                                                                                                                                                                                                                                                                                                          | PMC Fp004.1                                                                                                                                                                              |           |                                        |                                  |                     |
| 31 | Override    | Cutting feed speed override (%)                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                          | Int32     | 500 msec                               | NO                               | <p>                 |
|    |             | When PMC G<p>006.4 (cancel signal) is enabled ("1"): override is set to100<br>When PMC G<p>006.4 (cancel signal) is disabled ("0"): override is equal to PMC G<p>012 (federate override)                                                                                                                                                                                                                                                                               | When PMC G<p>004.4 (cancel signal) is enabled ("1"): override is set to100<br>When PMC G<p>004.4 (cancel signal) is disabled ("0"): override is equal to PMC G<p>012 (federate override) |           |                                        |                                  |                     |
| 32 | SigINP      | In-position signal                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                          | Bool      | 500 msec                               | NO                               | <n>/<br><p>         |
|    |             | PMC F<p>104.<n><br><br>Series 30 <i>i</i><br>- Maximum number of controlled axes is less than 72<br><p>: tens digit of parameter (No.3021)<br><n>: ones digit of parameter (No.3021)<br>- Maximum number of controlled axes is 72 or more<br><p>: tens digit of parameter (No.3025)<br><n>: ones digit of parameter (No.3025)<br>* When parameter has no setting (all "zero"), same as Series 16 <i>i</i> .<br><br>Series 16 <i>i</i><br><p>: path<br><n>: axis number | PMC F<p>064.6<br>- PMC F<p>124.6 (axis 1 – 16)<br>PMC F<p>288.6<br>- PMC F<p>316.6 (axis 17 – 24)<br>* per 4 addresses each                                                              |           |                                        |                                  |                     |

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |       |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|-------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |       |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |       |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |       |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |       |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |       |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |       |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 662/868 |       |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |       |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         | Inoue |

| #  | Signal Type  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                    | Data Type | Reading Cycle<br>(NOTE 1)<br>[Default] | Enabled<br>(NOTE 1)<br>[Default] | <m>/<br><n>/<br><p> |
|----|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------|----------------------------------------|----------------------------------|---------------------|
|    |              | Series 30 <i>i</i><br>Series 16 <i>i</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Series 15 <i>i</i> |           |                                        |                                  |                     |
| 33 | AbsPos       | Absolute position (mm) or (inch)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                    | Double    | 500 msec                               | NO                               | <n>                 |
| 34 | RelPos       | Relative position (mm) or (inch)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                    | Double    | 500 msec                               | NO                               | <n>                 |
| 35 | McPos        | Machine position (mm) or (inch)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                    | Double    | 500 msec                               | NO                               | <n>                 |
| 36 | SpindleLoad  | Spindle motor load (%)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                    | Double    | 500 msec                               | NO                               | <m>                 |
| 37 | SigENB       | Spindle axis operating signal<br>Series 30 <i>i</i><br>PMC F<p>001.4 (<Bit position>=1)<br>PMC F<p>038.2 (<Bit position>=2)<br>PMC F<p>038.3 (<Bit position>=3)<br>PMC F<p>039.1 (<Bit position>=4)<br>- Maximum number of controlled axes is less than 72<br><p>: tens digit of parameter (No.3022)<br><Bit position>: ones digit of parameter (No.3022)<br>- Maximum number of controlled axes is 72 or more<br><p>: tens digit of parameter (No.3025)<br><Bit position>: ones digit of parameter (No.3025)<br>* When parameter has no setting (all "zero"), same as Series 16 <i>i</i> .<br><br>Series 16 <i>i</i><br>PMC F<p>001.4 (first spindle)<br>PMC F<p>038.2 (second spindle)<br>PMC F<p>038.3 (third spindle)<br>PMC F<p>039.1 (fourth spindle) |                    | Bool      | 500 msec                               | NO                               | <m>/<br><p>         |
| 38 | SpindleSpeed | Speed of spindle motor (rpm)<br>Speed of spindle axis without gear ratio;<br>- Diagnostic data (No.411)<br>or<br>- Value of speed of spindle motor when:<br>bit 6 (OPS) of parameter No.3111 is set to 0<br>of the operating monitor display when:<br>bit 5 (OPM) of parameter No.3111 is set to 1<br>in the POS screen                                                                                                                                                                                                                                                                                                                                                                                                                                     |                    | Double    | 500 msec                               | NO                               | <m>                 |

|      |               |        |                                            |                          |                                                       |        |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|--------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |        |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |        |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |        |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |        |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |        | Sheet. 663/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |        |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    |                                                       | Apprv. | Inoue          |

| #  | Signal Type | Description                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                             | Data Type | Reading Cycle<br>(NOTE 1)<br>[Default] | Enabled<br>(NOTE 1)<br>[Default] | <m>/<br><n>/<br><p> |
|----|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------|----------------------------------|---------------------|
|    |             | Series 30i<br>Series 16i                                                                                                                                                                                                                                                                          | Series 15i                                                                                                                                                                                                  |           |                                        |                                  |                     |
| 39 | SpindleTemp | Temperature of spindle motor (deg C)                                                                                                                                                                                                                                                              |                                                                                                                                                                                                             | Double    | 60<br>sec                              | YES                              | <m>                 |
|    |             | For Series 30i:<br>Equal to diagnostic data (No.403)<br><br>For Series 16i:<br>Equal to diagnostic data (No.403) (first spindle)<br>Equal to diagnostic data (No.404) (second spindle)<br>Equal to diagnostic data (No.433) (third spindle)<br>Equal to diagnostic data (No.434) (fourth spindle) | For Series 15i-A:<br>Equal to diagnostic data (No.1506)<br>For Series 15i-B:<br>Equal to diagnostic data (No.1563)                                                                                          |           |                                        |                                  |                     |
| 40 | ServoSpeed  | Speed of servo motor (rpm)<br>NOT composite speed of all axis;<br>Speed of each axis, that is;<br>actual speed RPM of the servo tuning screen when:<br>bit 0 (SVS) of parameter No.3111 is set to 1                                                                                               |                                                                                                                                                                                                             | Double    | 500<br>msec                            | NO                               | <n>                 |
| 41 | ServoTemp   | Temperature of servo motor winding (deg C)                                                                                                                                                                                                                                                        |                                                                                                                                                                                                             | Double    | 60<br>sec                              | YES                              | <n>                 |
|    |             | Equal to diagnostic data (No.308)<br>0 to 255 (deg C) (140 (deg C) causes overheat alarm)                                                                                                                                                                                                         | Equal to diagnostic data (No.3520)<br>0 to 255 (deg C) (140 (deg C) causes overheat alarm)                                                                                                                  |           |                                        |                                  |                     |
| 42 | ServoError  | Deviation value of servo position (detection unit)                                                                                                                                                                                                                                                |                                                                                                                                                                                                             | Double    | 500<br>msec                            | NO                               | <n>                 |
|    |             | Position deviation value between command pulse and feedback pulse (detection unit)<br>-> If you want to specify a different unit, you need to calculate (diagnostic data (No.300) or servo tuning screen).                                                                                        | Position deviation value between command pulse and feedback pulse (detection unit)<br>-> If you want to specify a different unit, you need to calculate (diagnostic data (No.3000) or servo tuning screen). |           |                                        |                                  |                     |

|      |               |        |                                    |       |  |                   |                                               |  |        |         |
|------|---------------|--------|------------------------------------|-------|--|-------------------|-----------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |                   |                                               |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |                   |                                               |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |                   |                                               |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  | Draw              | A-40829-00001EN                               |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |                   |                                               |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |                   |                                               |  |        |         |
| Ed.  | Date          | Design | Description                        |       |  | FANUC CORPORATION |                                               |  | Sheet. | 664/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check |  |                   |                                               |  |        |         |

| #  | Signal Type     | Description                                                                                                                                                                       |                                                                                                       | Data Type | Reading Cycle (NOTE 1) [Default] | Enabled (NOTE 1) [Default] | <m>/<n>/<p> |
|----|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-----------|----------------------------------|----------------------------|-------------|
|    |                 | Series 30i<br>Series 16i                                                                                                                                                          | Series 15i                                                                                            |           |                                  |                            |             |
| 43 | ServoLoad       | Servo load (%)<br>Percentage of a rated current<br>- Servo load meter of the operating monitor display when:<br>bit 5 (OPM) of parameter No.3111 is set to 1<br>in the POS screen |                                                                                                       | Double    | 500 msec                         | NO                         | <n>         |
| 44 | ServoCurrent    | Servo load current (A)                                                                                                                                                            |                                                                                                       | Double    | 500 msec                         | NO                         | <n>         |
|    |                 | Actual current value (A) in servo tuning screen when:<br>bit 0 (SVS) of parameter No.3111 is set to 1                                                                             | Actual current value (A) in servo tuning screen when:<br>bit 0 (SVS) of parameter No.0015 is set to 1 |           |                                  |                            |             |
| 45 | ServoCurrentPer | Servo load current (%)                                                                                                                                                            |                                                                                                       | Double    | 500 msec                         | NO                         | <n>         |
|    |                 | Actual current value (%) in servo tuning screen when:<br>bit 0 (SVS) of parameter No.3111 is set to 1                                                                             | Actual current value (%) in servo tuning screen when:<br>bit 0 (SVS) of parameter No.0015 is set to 1 |           |                                  |                            |             |
| 46 | PulseCoderTemp  | Temperature of servo pulse coder (deg C)                                                                                                                                          |                                                                                                       | Double    | 60 sec                           | YES                        | <n>         |
|    |                 | Equal to diagnostic data (No.309)<br>0 to 255 (deg C) (100 (deg C) causes overheat alarm)                                                                                         | Equal to diagnostic data (No.3521)<br>0 to 255 (deg C) (100 (deg C) causes overheat alarm)            |           |                                  |                            |             |
| 47 | Alarm           | Alarm message (NOTE 4)<br>- Alarms<br>- External alarm message (EX alarm)<br>- Macro alarm message of #3000 (MC alarm)<br>- CNC warning information (NOTE 8)                      |                                                                                                       | -         | 500 msec                         | YES                        |             |
| 48 | OpMacroMsg      | Operator message/Macro message (NOTE 4)<br>- External operator message<br>- Macro message of custom macro variable #3006                                                          |                                                                                                       | -         | 500 msec                         | YES                        |             |

|      |               |        |                                    |                   |                                               |                |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |                   |                                               |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw              | A-40829-00001EN                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |                   |                                               |                |
| Ed.  | Date          | Design | Description                        | FANUC CORPORATION |                                               | Sheet. 665/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             | Apprv.                                        |                |

| #  | Signal Type           | Description                                                                                                                                                                                                                                                                                                                               |            | Data Type | Reading Cycle<br>(NOTE 1)<br>[Default] | Enabled<br>(NOTE 1)<br>[Default] | <m>/<br><n>/<br><p> |
|----|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|----------------------------------------|----------------------------------|---------------------|
|    |                       | Series 30i<br>Series 16i                                                                                                                                                                                                                                                                                                                  | Series 15i |           |                                        |                                  |                     |
| 49 | CncState              | State of CNC:<br>"EMERGENCY,"<br>"ALARM,"<br>"OPERATE,"<br>"SUSPEND,"<br>"STOP,"<br>"UNKNOWN"                                                                                                                                                                                                                                             |            | String    | 500 msec                               | YES                              |                     |
| 50 | CncWarning            | CNC warning information<br>(NOTE 8)<br>Without warning: "NONE"<br>With a following warning<br>Battery warning:<br>"BATTERY LOW,"<br>Fan warning: "FAN,"<br>Power warning: "PS,"<br>FSSB warning: "FSSB,"<br>Insulation warning:<br>"LEAKAGE,"<br>Encoder warning<br>"ENCODER,"<br>PMC warning: "PMC<br>ALARM,"<br>Other warnings: "OTHER" |            | String    | 500 msec                               | YES                              |                     |
| 51 | CncBatLow             | CNC battery low                                                                                                                                                                                                                                                                                                                           |            | Bool      | 60 sec                                 | YES                              | <n>                 |
| 52 | ApcBatZero            | Battery voltage zero<br>Equal to diagnostic data (bit 2 of No.202)                                                                                                                                                                                                                                                                        |            | Bool      | 60 sec                                 | YES                              | <n>                 |
| 53 | ApcBatLow             | Battery voltage low<br>Equal to diagnostic data (bit 5 of No.202)                                                                                                                                                                                                                                                                         |            | Bool      | 60 sec                                 | YES                              | <n>                 |
| 54 | IndBatZero            | Inductosyn battery voltage zero<br>Equal to diagnostic data (bit 2 of No.310)                                                                                                                                                                                                                                                             |            | Bool      | 60 sec                                 | YES                              | <n>                 |
| 55 | SpdBatZero            | Separate detector battery voltage zero<br>Equal to diagnostic data (bit 3 of No.310)                                                                                                                                                                                                                                                      |            | Bool      | 60 sec                                 | YES                              | <n>                 |
| 56 | SSpdBatZero           | Serial separate detector battery voltage zero<br>Equal to diagnostic data (bit 5 of No.311)                                                                                                                                                                                                                                               |            | Bool      | 60 sec                                 | YES                              | <n>                 |
| 57 | ServoLeakResistData   | Servo motor insulation data<br>Equal to diagnostic data (No.1701)<br>(to be retrieved when bit 1 of diagnostic data<br>No.1700 is 1 (1: not supported))<br>(written as "NaN" in the database)                                                                                                                                             |            | Double    | 60 sec                                 | YES                              | <n>                 |
| 58 | SpindleLeakResistData | Spindle motor insulation data<br>Equal to diagnostic data (No.1703)<br>(to be retrieved when bit 1 of diagnostic data<br>No.1702 is 1 (1: not supported))<br>(written as "NaN" in the database)                                                                                                                                           |            | Double    | 60 sec                                 | YES                              | <m>                 |
| 59 | CncFan1Speed          | CNCFAN1 speed<br>Equal to diagnostic data (No.1002)                                                                                                                                                                                                                                                                                       |            | Double    | 60 sec                                 | NO                               |                     |
| 60 | CncFan2Speed          | CNCFAN2 speed<br>Equal to diagnostic data (No.1003)                                                                                                                                                                                                                                                                                       |            | Double    | 60 sec                                 | NO                               |                     |

|      |               |        |                                    |                   |                                               |        |                |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|--------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |        |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |        |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw              | A-40829-00001EN                               |        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                   |                                               |        |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |        |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | FANUC CORPORATION |                                               |        | Sheet. 666/868 |
| Ed.  | Date          | Design | Description                        |                   |                                               |        |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             |                                               | Apprv. | Inoue          |

| #  | Signal Type   | Description                                                                                                                                                                                                                                                                                                                                                                                                            |            | Data Type | Reading Cycle<br>(NOTE 1)<br>[Default] | Enabled<br>(NOTE 1)<br>[Default] | <m>/<br><n>/<br><p> |
|----|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|----------------------------------------|----------------------------------|---------------------|
|    |               | Series 30i<br>Series 16i                                                                                                                                                                                                                                                                                                                                                                                               | Series 15i |           |                                        |                                  |                     |
| 61 | CncFan3Speed  | CNC FAN3 speed<br>Equal to diagnostic data (No.1490)                                                                                                                                                                                                                                                                                                                                                                   |            | Double    | 60<br>sec                              | NO                               |                     |
| 62 | CncFan4Speed  | CNC FAN4 speed<br>Equal to diagnostic data (No.1491)                                                                                                                                                                                                                                                                                                                                                                   |            | Double    | 60<br>sec                              | NO                               |                     |
| 63 | CncFan1Status | CNC FAN1 exchange information<br>Diagnostic data (bit 5 of No.1495)=1 (no FAN1): "NON"<br>Diagnostic data (bits 7 to 6 of No.1495)=0 (normal): "NORMAL"<br>Diagnostic data (bits 7 to 6 of No.1495)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 7 to 6 of No.1495)=2 (startup delay due to FAN1 fixed): "FIXED"<br>Diagnostic data (bits 7 to 6 of No.1495)=3 (speed low and fixed): "SPEED LOW AND FIXED" |            | String    | 60<br>sec                              | YES                              |                     |
| 64 | CncFan2Status | CNC FAN2 exchange information<br>Diagnostic data (bit 1 of No.1495)=1 (no FAN2): "NON"<br>Diagnostic data (bits 3 to 2 of No.1495)=0 (normal): "NORMAL"<br>Diagnostic data (bits 3 to 2 of No.1495)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 3 to 2 of No.1495)=2 (startup delay due to FAN2 fixed): "FIXED"<br>Diagnostic data (bits 3 to 2 of No.1495)=3 (speed low and fixed): "SPEED LOW AND FIXED" |            | String    | 60<br>sec                              | YES                              |                     |
| 65 | CncFan3Status | CNC FAN3 exchange information<br>Diagnostic data (bit 5 of No.1496)=1 (no FAN3): "NON"<br>Diagnostic data (bits 7 to 6 of No.1496)=0 (normal): "NORMAL"<br>Diagnostic data (bits 7 to 6 of No.1496)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 7 to 6 of No.1496)=other value than 0 or 1: "UNKNOWN"                                                                                                      |            | String    | 60<br>sec                              | YES                              |                     |
| 66 | CncFan4Status | CNC FAN4 exchange information<br>Diagnostic data (bit 1 of No.1496)=1 (no FAN4): "NON"<br>Diagnostic data (bits 3 to 2 of No.1496)=0 (normal): "NORMAL"<br>Diagnostic data (bits 3 to 2 of No.1496)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 3 to 2 of No.1496)=other value than 0 or 1: "UNKNOWN"                                                                                                      |            | String    | 60<br>sec                              | YES                              |                     |
| 67 | PwConAllAxes  | Total current actual power consumption of all axes (unit :W)<br>Equal to diagnostic data (No.4900)                                                                                                                                                                                                                                                                                                                     |            | Double    | 500<br>msec                            | NO                               |                     |

|      |               |        |                                    |                   |                                               |        |                |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|--------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |        |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |        |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw              | A-40829-00001EN                               |        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                   |                                               |        |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |        |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | FANUC CORPORATION |                                               |        | Sheet. 667/868 |
| Ed.  | Date          | Design | Description                        |                   |                                               |        |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             |                                               | Apprv. | Inoue          |

| #  | Signal Type                | Description                                                                                                                                                                                                                                                                                                                                    |            | Data Type | Reading Cycle<br>(NOTE 1)<br>[Default] | Enabled<br>(NOTE 1)<br>[Default] | <m>/<br><n>/<br><p> |
|----|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|----------------------------------------|----------------------------------|---------------------|
|    |                            | Series 30i<br>Series 16i                                                                                                                                                                                                                                                                                                                       | Series 15i |           |                                        |                                  |                     |
| 68 | AdcumPwConPeriphera<br>l   | Accumulated power consumption of peripheral devices (unit: 0.001 kW)<br>Equal to diagnostic data (No. 4943)                                                                                                                                                                                                                                    |            | Double    | 10<br>sec                              | NO                               |                     |
| 69 | SpindleTotalRev1           | Spindle total rotation1<br>Equal to diagnostic data (No.1520)                                                                                                                                                                                                                                                                                  |            | Double    | 60<br>sec                              | YES                              | <m>                 |
| 70 | SpindleTotalRev2           | Spindle total rotation2<br>Equal to diagnostic data (No.1521)                                                                                                                                                                                                                                                                                  |            | Double    | 60<br>sec                              | YES                              | <m>                 |
| 71 | InFan1SrvAmpSpeed          | Servo amplifier inside cooling FAN1 speed<br>Equal to diagnostic data (No.1711)                                                                                                                                                                                                                                                                |            | Double    | 60<br>sec                              | NO                               | <n>                 |
| 72 | InFan2SrvAmpSpeed          | Servo amplifier inside cooling FAN2 speed<br>Equal to diagnostic data (No.1712)                                                                                                                                                                                                                                                                |            | Double    | 60<br>sec                              | NO                               | <n>                 |
| 73 | InFan1SrvAmpStatus         | Servo amplifier inside cooling FAN1 exchange information<br>Diagnostic data (bit 5 of No.1710)=1 (no FAN1): "NON"<br>Diagnostic data (bits 7 to 6 of No.1710)=0 (normal): "NORMAL"<br>Diagnostic data (bits 7 to 6 of No.1710)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 7 to 6 of No.1710)=other value than 0 or 1: "UNKNOWN"   |            | String    | 60<br>sec                              | YES                              | <n>                 |
| 74 | InFan2SrvAmpStatus         | Servo amplifier inside cooling FAN2 exchange information<br>Diagnostic data (bit 1 of No.1710)=1 (no FAN2): "NON"<br>Diagnostic data (bits 3 to 2 of No.1710)=0 (normal): "NORMAL"<br>Diagnostic data (bits 3 to 2 of No.1710)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 3 to 2 of No.1710)=other value than 0 or 1: "UNKNOWN"   |            | String    | 60<br>sec                              | YES                              | <n>                 |
| 75 | RadCoolFan1SrvAmpS<br>peed | Servo amplifier radiator cooling FAN1 speed<br>Equal to diagnostic data (No.1714)                                                                                                                                                                                                                                                              |            | Double    | 60<br>sec                              | NO                               | <n>                 |
| 76 | RadCoolFan2SrvAmpS<br>peed | Servo amplifier radiator cooling FAN2 speed<br>Equal to diagnostic data (No.1715)                                                                                                                                                                                                                                                              |            | Double    | 60<br>sec                              | NO                               | <n>                 |
| 77 | RadFan1SrvAmpStatus        | Servo amplifier radiator cooling FAN1 exchange information<br>Diagnostic data (bit 5 of No.1713)=1 (no FAN1): "NON"<br>Diagnostic data (bits 7 to 6 of No.1713)=0 (normal): "NORMAL"<br>Diagnostic data (bits 7 to 6 of No.1713)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 7 to 6 of No.1713)=other value than 0 or 1: "UNKNOWN" |            | String    | 60<br>sec                              | YES                              | <n>                 |

|      |               |        |                                    |                          |                                               |        |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|--------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |        |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |        |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |        |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |        |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |        | Sheet. 668/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |        |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    |                                               | Apprv. | Inoue          |



| #  | Signal Type           | Description                                                                                                                                                                                                                                                                                                                                             |            | Data Type | Reading Cycle<br>(NOTE 1)<br>[Default] | Enabled<br>(NOTE 1)<br>[Default] | <m>/<br><n>/<br><p> |
|----|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|----------------------------------------|----------------------------------|---------------------|
|    |                       | Series 30i<br>Series 16i                                                                                                                                                                                                                                                                                                                                | Series 15i |           |                                        |                                  |                     |
| 78 | RadFan2SrvAmpStatus   | Servo amplifier radiator cooling FAN2 exchange information<br>Diagnostic data (bit 1 of No.1713)=1 (no FAN2): "NON"<br>Diagnostic data (bits 3 to 2 of No.1713)=0 (normal): "NORMAL"<br>Diagnostic data (bits 3 to 2 of No.1713)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 3 to 2 of No.1713)=other value than 0 or 1: "UNKNOWN"          |            | String    | 60 sec                                 | YES                              | <n>                 |
| 79 | InFan1SrvComPwSpeed   | Servo motor common power inside cooling FAN1 speed<br>Equal to diagnostic data (No.1717)                                                                                                                                                                                                                                                                |            | Double    | 60 sec                                 | NO                               | <n>                 |
| 80 | InFan2SrvComPwSpeed   | Servo motor common power inside cooling FAN2 speed<br>Equal to diagnostic data (No.1718)                                                                                                                                                                                                                                                                |            | Double    | 60 sec                                 | NO                               | <n>                 |
| 81 | InFan1SpdlComPwSpeed  | Spindle motor common power inside cooling FAN1 speed<br>Equal to diagnostic data (No.1729)                                                                                                                                                                                                                                                              |            | Double    | 60 sec                                 | NO                               | <m>                 |
| 82 | InFan2SpdlComPwSpeed  | Spindle motor common power inside cooling FAN2 speed<br>Equal to diagnostic data (No.1730)                                                                                                                                                                                                                                                              |            | Double    | 60 sec                                 | NO                               | <m>                 |
| 83 | InFan1SrvComPwStatus  | Servo motor common power inside cooling FAN1 exchange information<br>Diagnostic data (bit 5 of No.1716)=1 (no FAN1): "NON"<br>Diagnostic data (bits 7 to 6 of No.1716)=0 (normal): "NORMAL"<br>Diagnostic data (bits 7 to 6 of No.1716)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 7 to 6 of No.1716)=other value than 0 or 1: "UNKNOWN"   |            | String    | 60 sec                                 | YES                              | <n>                 |
| 84 | InFan1SpdlComPwStatus | Spindle motor common power inside cooling FAN1 exchange information<br>Diagnostic data (bit 5 of No.1728)=1 (no FAN1): "NON"<br>Diagnostic data (bits 7 to 6 of No.1728)=0 (normal): "NORMAL"<br>Diagnostic data (bits 7 to 6 of No.1728)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 7 to 6 of No.1728)=other value than 0 or 1: "UNKNOWN" |            | String    | 60 sec                                 | YES                              | <m>                 |

|      |               |        |                                    |                          |                                               |       |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |       | Sheet. 669/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue |                |

| #  | Signal Type           | Description                                                                                                                                                                                                                                                                                                                                             |            | Data Type | Reading Cycle (NOTE 1) [Default] | Enabled (NOTE 1) [Default] | <m>/<n>/<p> |
|----|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|----------------------------------|----------------------------|-------------|
|    |                       | Series 30i<br>Series 16i                                                                                                                                                                                                                                                                                                                                | Series 15i |           |                                  |                            |             |
| 85 | InFan2SrvComPwStatus  | Servo motor common power inside cooling FAN2 exchange information<br>Diagnostic data (bit 1 of No.1716)=1 (no FAN2): "NON"<br>Diagnostic data (bits 3 to 2 of No.1716)=0 (normal): "NORMAL"<br>Diagnostic data (bits 3 to 2 of No.1716)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 3 to 2 of No.1716)=other value than 0 or 1: "UNKNOWN"   |            | String    | 60 sec                           | YES                        | <n>         |
| 86 | InFan2SpdlComPwStatus | Spindle motor common power inside cooling FAN2 exchange information<br>Diagnostic data (bit 1 of No.1728)=1 (no FAN2): "NON"<br>Diagnostic data (bits 3 to 2 of No.1728)=0 (normal): "NORMAL"<br>Diagnostic data (bits 3 to 2 of No.1728)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 3 to 2 of No.1728)=other value than 0 or 1: "UNKNOWN" |            | String    | 60 sec                           | YES                        | <m>         |
| 87 | RadFan1SrvComPwSpeed  | Servo motor common power radiator cooling FAN1 speed<br>Equal to diagnostic data (No.1720)                                                                                                                                                                                                                                                              |            | Double    | 60 sec                           | NO                         | <n>         |
| 88 | RadFan2SrvComPwSpeed  | Servo motor common power radiator cooling FAN2 speed<br>Equal to diagnostic data (No.1721)                                                                                                                                                                                                                                                              |            | Double    | 60 sec                           | NO                         | <n>         |
| 89 | RadFan1SpdlComPwSpeed | Spindle motor common power radiator cooling FAN1 speed<br>Equal to diagnostic data (No.1732)                                                                                                                                                                                                                                                            |            | Double    | 60 sec                           | NO                         | <m>         |
| 90 | RadFan2SpdlComPwSpeed | Spindle motor common power radiator cooling FAN2 speed<br>Equal to diagnostic data (No.1733)                                                                                                                                                                                                                                                            |            | Double    | 60 sec                           | NO                         | <m>         |
| 91 | RadFan1SrvComPwStatus | Servo motor common power radiator cooling FAN1 exchange information<br>Diagnostic data (bit 5 of No.1719)=1 (no FAN1): "NON"<br>Diagnostic data (bits 7 to 6 of No.1719)=0 (normal): "NORMAL"<br>Diagnostic data (bits 7 to 6 of No.1719)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 7 to 6 of No.1719)=other value than 0 or 1: "UNKNOWN" |            | String    | 60 sec                           | YES                        | <n>         |

| #  | Signal Type            | Description                                                                                                                                                                                                                                                                                                                                               |            | Data Type | Reading Cycle<br>(NOTE 1)<br>[Default] | Enabled<br>(NOTE 1)<br>[Default] | <m>/<br><n>/<br><p> |
|----|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|----------------------------------------|----------------------------------|---------------------|
|    |                        | Series 30i<br>Series 16i                                                                                                                                                                                                                                                                                                                                  | Series 15i |           |                                        |                                  |                     |
| 92 | RadFan1SpdlComPwStatus | Spindle motor common power radiator cooling FAN1 exchange information<br>Diagnostic data (bit 5 of No.1731)=1 (no FAN1): "NON"<br>Diagnostic data (bits 7 to 6 of No.1731)=0 (normal): "NORMAL"<br>Diagnostic data (bits 7 to 6 of No.1731)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 7 to 6 of No.1731)=other value than 0 or 1: "UNKNOWN" |            | String    | 60 sec                                 | YES                              | <m>                 |
| 93 | RadFan2SrvComPwStatus  | Servo motor common power radiator cooling FAN2 exchange information<br>Diagnostic data (bit 1 of No.1719)=1 (no FAN2): "NON"<br>Diagnostic data (bits 3 to 2 of No.1719)=0 (normal): "NORMAL"<br>Diagnostic data (bits 3 to 2 of No.1719)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 3 to 2 of No.1719)=other value than 0 or 1: "UNKNOWN"   |            | String    | 60 sec                                 | YES                              | <n>                 |
| 94 | RadFan2SpdlComPwStatus | Spindle motor common power radiator cooling FAN2 exchange information<br>Diagnostic data (bit 1 of No.1731)=1 (no FAN2): "NON"<br>Diagnostic data (bits 3 to 2 of No.1731)=0 (normal): "NORMAL"<br>Diagnostic data (bits 3 to 2 of No.1731)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 3 to 2 of No.1731)=other value than 0 or 1: "UNKNOWN" |            | String    | 60 sec                                 | YES                              | <m>                 |
| 95 | InFan1SpindleAmpSpeed  | Spindle amplifier inside cooling FAN1 speed<br>Equal to diagnostic data (No.1723)                                                                                                                                                                                                                                                                         |            | Double    | 60 sec                                 | NO                               | <m>                 |
| 96 | InFan2SpindleAmpSpeed  | Spindle amplifier inside cooling FAN2 speed<br>Equal to diagnostic data (No.1724)                                                                                                                                                                                                                                                                         |            | Double    | 60 sec                                 | NO                               | <m>                 |
| 97 | InFan1SpindleAmpStatus | Spindle amplifier inside cooling FAN1 exchange information<br>Diagnostic data (bit 5 of No.1722)=1 (no FAN1): "NON"<br>Diagnostic data (bits 7 to 6 of No.1722)=0 (normal): "NORMAL"<br>Diagnostic data (bits 7 to 6 of No.1722)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 7 to 6 of No.1722)=other value than 0 or 1: "UNKNOWN"            |            | String    | 60 sec                                 | YES                              | <m>                 |

|      |               |        |                                    |                   |                                               |        |                |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|--------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |        |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |        |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw              | A-40829-00001EN                               |        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                   |                                               |        |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |        |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | FANUC CORPORATION |                                               |        | Sheet. 671/868 |
| Ed.  | Date          | Design | Description                        |                   |                                               |        |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             |                                               | Apprv. | Inoue          |

| #   | Signal Type             | Description                                                                                                                                                                                                                                                                                                                                      |            | Data Type | Reading Cycle<br>(NOTE 1)<br>[Default] | Enabled<br>(NOTE 1)<br>[Default] | <m>/<br><n>/<br><p> |
|-----|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|----------------------------------------|----------------------------------|---------------------|
|     |                         | Series 30i<br>Series 16i                                                                                                                                                                                                                                                                                                                         | Series 15i |           |                                        |                                  |                     |
| 98  | InFan2SpindleAmpStatus  | Spindle amplifier inside cooling FAN2 exchange information<br>Diagnostic data (bit 1 of No.1722)=1 (no FAN2): "NON"<br>Diagnostic data (bits 3 to 2 of No.1722)=0 (normal): "NORMAL"<br>Diagnostic data (bits 3 to 2 of No.1722)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 3 to 2 of No.1722)=other value than 0 or 1: "UNKNOWN"   |            | String    | 60 sec                                 | YES                              | <m>                 |
| 99  | RadFan1SpindleAmpSpeed  | Spindle amplifier radiator cooling FAN1 speed Equal to diagnostic data (No.1726)                                                                                                                                                                                                                                                                 |            | Double    | 60 sec                                 | NO                               | <m>                 |
| 100 | RadFan2SpindleAmpSpeed  | Spindle amplifier radiator cooling FAN2 speed Equal to diagnostic data (No.1727)                                                                                                                                                                                                                                                                 |            | Double    | 60 sec                                 | NO                               | <m>                 |
| 101 | RadFan1SpindleAmpStatus | Spindle amplifier radiator cooling FAN1 exchange information<br>Diagnostic data (bit 5 of No.1725)=1 (no FAN1): "NON"<br>Diagnostic data (bits 7 to 6 of No.1725)=0 (normal): "NORMAL"<br>Diagnostic data (bits 7 to 6 of No.1725)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 7 to 6 of No.1725)=other value than 0 or 1: "UNKNOWN" |            | String    | 60 sec                                 | YES                              | <m>                 |
| 102 | RadFan2SpindleAmpStatus | Spindle amplifier radiator cooling FAN2 exchange information<br>Diagnostic data (bit 1 of No.1725)=1 (no FAN2): "NON"<br>Diagnostic data (bits 3 to 2 of No.1725)=0 (normal): "NORMAL"<br>Diagnostic data (bits 3 to 2 of No.1725)=1 (fan speed low): "SPEED LOW"<br>Diagnostic data (bits 3 to 2 of No.1725)=other value than 0 or 1: "UNKNOWN" |            | String    | 60 sec                                 | YES                              | <m>                 |

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               | Sheet. 672/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |

## NOTE

- 1 For each signal, whether to retrieve it or not (enabled/disabled) can be specified. Also, the reading cycle can be selected for each signal (500msec / 10sec / 60sec). However, the OpMacroMsg signal is fixed to "500msec."
  - \* For the following mandatory signals, retrieval of the signal cannot be disabled (the [Enable] check box in the Signal Setting of [Basic Signal] tab in the [Machine Setting] window cannot be cleared), and the reading cycle cannot be changed either.

| Mandatory Signals | Description             |
|-------------------|-------------------------|
| MainProgram       | Main program            |
| ActProgram        | Executing program       |
| CncState          | State of CNC            |
| CncWarning        | CNC warning information |
| Alarm             | Alarm message           |

- 2 The [Diagnosis – Program History] web screen supports only FanucCNC and FanucRobodrill type machines.  
Program history information (executed by Main programs or Executing programs) is stored to the database when the CncState signal has "OPERATE" (operating) status and can be viewed through the [Diagnosis - Program History] web screen.
- 3 In order to lessen workloads on MT-LINK*i* as well as on CNCs, it sets a restriction on reading program comments from specific CNC models by putting an upper limit (threshold) to the number of files in the program folder. For CNCs running system software with the following versions, if the number of the files residing in the program folder of a CNC exceeds its upper limit, the program comment signals will not be retrieved from the CNC and will be handled as though they had space values. The file upper limit is defined in the system setting file (collector.ini) per the target device (i.e., CNC\_MEM, Data Server, or a program storage memory card) and can be modified.

For the details on the file upper limit, refer to the "**collector.ini**" subsection of the "SYSTEM SETTING FILES" section.

A program comment will be retrieved that is written on the same line in a program as the program number. However, if multiple program comments are on the line, only the program comment immediately following the program number will be retrieved.

|      |               |        |                                            |                          |                                                       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       | Sheet. 673/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue          |

| CNC Model                   | System Software                  | Version           |
|-----------------------------|----------------------------------|-------------------|
| 31i-WB                      | G4A1                             | Earlier than 15.0 |
| 31i-WA /31i-WA5             | all                              | all versions      |
| 30i-LB                      | G3C1 / G3D1                      | Earlier than 26.0 |
|                             | G3C2 / G3D2                      | Earlier than 02.0 |
| 31i-LB                      | G4C1                             | Earlier than 26.0 |
|                             | G4C2                             | Earlier than 02.0 |
| 0i-PF                       | D8G1                             | Earlier than 03.0 |
| 30i-PB                      | G3E1 / G3F1                      | Earlier than 25.0 |
| 31i-PB                      | G4E1                             | Earlier than 25.0 |
| 30i-A /31i-A /31i-A5 /32i-A | all                              | all versions      |
| 30i-B                       | G301 / G311 / G321 / G331 / G351 | Earlier than 66.0 |
| 31i-B                       | G401 / G411                      | Earlier than 66.0 |
| 31i-B5                      | G421 / G431                      | Earlier than 66.0 |
| 32i-B                       | G501 / G511                      | Earlier than 66.0 |
| 0i-F                        | D4G1 / D6G1                      | Earlier than 5.0  |

- 4 The "Alarm" and "OpeMacroMsg" are signals for the alarm mail delivery function and the [Alarm Monitoring] / [Alarm History] web screen, and are not used by the signal collection function. These signals cannot be used with the following functions because they cannot be assigned to the simple assign status of the equipment status.
  - Signal detection mail delivery
  - Reporting
  - Signal data output
  - [Signal Monitoring] / [Signal History] web screen
- 5 The "ActS", "ActF" and "ActFdec" are retrieved the values displayed on the CNC position display screen.
- 6 The "ActFdec" depends on parameters for the number of digits after the decimal point. The applicable parameter numbers are as follows for each CNC model. Please refer to the PARAMETER MANUAL for details.
  - Series 30i  
Parameter No.3135
  - Series 16i  
No applicable parameter
  - Series 15i  
Parameter No.2223
- 7 The "ActF" and "ActFdec" are retrieved as values in mm/min (feed per minute) even in mm/rev (feed per revolution).
- 8 Series 30i /Series 16i CNC warning information cannot be obtained while the following alarm messages are occurring.
  - Alarms
  - External alarm message (EX alarm)
  - Macro alarm message of #3000 (MC alarm)

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               | Sheet. 674/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |

### ■ Details of Signal CncState

Signal CncState determines the state of CNC based on the cnc\_statinfo of the FOCAS2 function as shown in the following table.

| Priority | Value of cnc_statinfo         |                                       | State of CNC | Remarks            |
|----------|-------------------------------|---------------------------------------|--------------|--------------------|
|          | Series 30 <i>i</i> (NOTE 3)   | Series 15 <i>i</i> (NOTE 4)           |              |                    |
| 1        | emergency = 1                 |                                       | EMERGENCY    | Emergency stopped  |
| 2        | alarm = 1                     |                                       | ALARM        | In alarm condition |
| 3        | run = 3                       | run = 2 or 13                         | OPERATE      | Operating          |
| 4        | run = 1 or 2 or 4             | run = 0 or 1 or 3 or 4 or 5 or 6 or 7 | SUSPEND      | Suspended          |
| 5        | run = 0                       | run = 8                               | STOP         | Stopped            |
| 6        | run = other than above values |                                       | UNKNOWN      | Unknown            |

\* Highest priority is 1.

Following are the details of the value of cnc\_statinfo. (For the details of other values, refer to the specification document of FOCAS2 function.)

| Item      | Details of the cnc_statinfo value                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | Series 30 <i>i</i> (NOTE 3)                                                                                                                                                                        | Series 15 <i>i</i> (NOTE 4)                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| emergency | Status of emergency<br>0: (Not emergency)<br>1: EMerGency<br>2: ReSET<br>3: WAIT(FS35 <i>i</i> only)                                                                                               | Status of emergency<br>0: (Not emergency)<br>1: EMerGency                                                                                                                                                                                                                                                                                                                                                                                                                            |
| alarm     | Status of alarm<br>0: *** (Others)<br>1: ALarM<br>2: BATtery low<br>3: FAN(NC or Servo amplifier)<br>4: PS Warning<br>5: FSsB warning<br>6: INSulate warning<br>7: ENCoder warning<br>8: PMC alarm | Status of alarm<br>0: (No alarm)<br>1: ALarM                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| run       | Status of automatic operation<br>0: ****(reset)<br>1: STOP<br>2: HOLD<br>3: STaRT<br>4: MSTR(during retraction and re-positioning of tool retraction and recovery, and operation of JOG MDI)       | Status of automatic operation<br>0: STOP<br>1: HOLD<br>2: STaRT<br>3: MSTR(jog mdi)<br>4: ReStArT(not blinking)<br>Except under manual mode and under cutter radius compensation outside corner<br>5: PRSR(program restart)<br>6: NSRC(sequence number search)<br>7: ReStArT(blinking)<br>Under manual mode or under cutter radius compensation outside corner<br>8: ReSET<br>9: (Not used)<br>10: (Not used)<br>11: (Not used)<br>12: (Not used)<br>13: HPCC(during RISC operation) |

|                   |               |        |                                            |       |                                                       |         |
|-------------------|---------------|--------|--------------------------------------------|-------|-------------------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |                                                       |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |                                                       |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |                                                       |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   | Draw  | <b>A-40829-00001EN</b>                                |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |                                                       |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |                                                       |         |
| Ed.               | Date          | Design | Description                                |       |                                                       |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                                      | Check | Apprv.                                                | Inoue   |
| FANUC CORPORATION |               |        |                                            |       | Sheet.                                                | 675/868 |

The following are the status and signal definition rules when you assign a signal definition to the status in the [Equipment Setting] window.

#### ■ Spindle axis

Status: <Signal name>\_<m>\_path<x>\_<Machine name>

Signal definition: <Machine name>\_<Signal name>\_<m>\_path<x>

<m>: the <m>-th axis of the spindle axes (0 to 3)

<x>: CNC path (1 to 15)

#### Example

Given the following:

Machine name: Machine001, signal name: SpindleLoad, m: 3, CNC path: 1

Status: SpindleLoad\_3\_path1\_Machine001

Signal definition: Machine001\_SpindleLoad\_3\_path1

#### ■ Servo axis

Status: <Signal name>\_<n>\_path<x>\_<Machine name>

Signal definition: <Machine name>\_<Signal name>\_<n>\_path<x>

<n>: the <n>-th axis of the servo axes (0 to 23)

<x>: CNC path (1 to 15)

#### Example

Given the following:

Machine name: Machine001, signal name: ServoSpeed, servo axis number: 5, CNC path: 1

Status: ServoSpeed\_5\_path1\_Machine001

Signal definition: Machine001\_ServoSpeed\_5\_path1

|      |               |        |                                            |                          |                                                       |        |        |         |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |        | Sheet. | 676/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    |                                                       | Apprv. | Inoue  |         |



The following tables list retrievable signals from CNC models.

**Table 2. Retrievable Signals Listed By CNC Model**  
(Signals retrievable from the model are marked with "YES")

| #  | Signal Type  | 30i/31i/32i-B,<br>30i/31i/32i-B Plus,<br>31i-B5, 31i-B5 Plus,<br>35i-B, 0i-F, 0i-F Plus | PMi-A | 30i/31i/<br>32i-A<br>31i-A5 | 0i-D | 16i/18i/<br>21i-A/B,<br>0i-B/C | PMi-D | PMi-H |
|----|--------------|-----------------------------------------------------------------------------------------|-------|-----------------------------|------|--------------------------------|-------|-------|
| 01 | SigOP        | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 02 | SigSTL       | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 03 | SigSPL       | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 04 | SigAL        | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 05 | EMG          | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 06 | Mode         | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 07 | MainProgram  | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 08 | ActProgram   | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 09 | Sequence     | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 10 | MainComment  | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 11 | ActComment   | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 12 | ActS         | YES                                                                                     |       | YES                         | YES  | YES                            | YES   | YES   |
| 13 | ActF         | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 14 | ActFdec      | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 15 | ModalS       | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 16 | ModalF       | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 17 | ModalT       | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 18 | ModalM       | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 19 | ModalM2      | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 20 | ModalM3      | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 21 | PartsNum     | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 22 | PartsNumAll  | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 23 | PowOnTime    | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 24 | RunTime      | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 25 | CutTime      | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 26 | SigCUT       | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 27 | SigSBK       | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 28 | SigDM00      | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 29 | SigDM01      | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 30 | SigMDRN      | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 31 | Override     | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 32 | SigINP       | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 33 | AbsPos       | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 34 | RelPos       | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 35 | McnPos       | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 36 | SpindleLoad  | YES                                                                                     |       | YES                         | YES  | YES                            | YES   |       |
| 37 | SigENB       | YES                                                                                     |       | YES                         | YES  | YES                            | YES   | YES   |
| 38 | SpindleSpeed | YES                                                                                     |       | YES                         | YES  | YES                            | YES   |       |
| 39 | SpindleTemp  | YES                                                                                     |       | YES                         | YES  | YES                            |       |       |
| 40 | ServoSpeed   | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 41 | ServoTemp    | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |

|      |               |        |                                    |       |  |        |                   |                                               |  |        |         |
|------|---------------|--------|------------------------------------|-------|--|--------|-------------------|-----------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |                   |                                               |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |                   |                                               |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        | Draw              | A-40829-00001EN                               |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        |                   |                                               |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |                   |                                               |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        | FANUC CORPORATION |                                               |  | Sheet. | 677/868 |
| Ed.  | Date          | Design | Description                        |       |  |        |                   |                                               |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check |  | Apprv. |                   |                                               |  |        |         |

| #  | Signal Type           | 30i/31i/32i-B,<br>30i/31i/32i-B Plus,<br>31i-B5, 31i-B5 Plus,<br>35i-B, 0i-F, 0i-F Plus | PMi-A | 30i/31i/<br>32i-A<br>31i-A5 | 0i-D | 16i/18i/<br>21i-A/B,<br>0i-B/C | PMi-D | PMi-H |
|----|-----------------------|-----------------------------------------------------------------------------------------|-------|-----------------------------|------|--------------------------------|-------|-------|
| 42 | ServoError            | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 43 | ServoLoad             | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 44 | ServoCurrent          | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 45 | ServoCurrentPer       | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 46 | PulseCoderTemp        | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 47 | Alarm                 | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 48 | OpMacroMsg            | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 49 | CncState              | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 50 | CncWarning            | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 51 | CncBatLow             | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 52 | ApcBatZero            | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 53 | ApcBatLow             | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 54 | IndBatZero            | YES                                                                                     | YES   | YES                         | YES  | YES                            |       |       |
| 55 | SpdBatZero            | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 56 | SSpdBatZero           | YES                                                                                     | YES   | YES                         | YES  | YES                            | YES   | YES   |
| 57 | ServoLeakResistData   | YES                                                                                     | YES   |                             |      |                                |       |       |
| 58 | SpindleLeakResistData | YES                                                                                     |       |                             |      |                                |       |       |
| 59 | CncFan1Speed          | YES                                                                                     | YES   | YES                         |      |                                |       |       |
| 60 | CncFan2Speed          | YES                                                                                     | YES   | YES                         |      |                                |       |       |
| 61 | CncFan3Speed          | YES                                                                                     | YES   | YES                         |      |                                |       |       |
| 62 | CncFan4Speed          | YES                                                                                     | YES   | YES                         |      |                                |       |       |
| 63 | CncFan1Status         | YES                                                                                     | YES   |                             |      |                                |       |       |
| 64 | CncFan2Status         | YES                                                                                     | YES   |                             |      |                                |       |       |
| 65 | CncFan3Status         | YES                                                                                     | YES   |                             |      |                                |       |       |
| 66 | CncFan4Status         | YES                                                                                     | YES   |                             |      |                                |       |       |
| 67 | PwConAllAxes          | YES                                                                                     | YES   | YES                         | YES  |                                |       |       |
| 68 | AdcumPwConPeripheral  | YES (*)                                                                                 |       |                             | YES  |                                |       |       |
| 69 | SpindleTotalRev1      | YES                                                                                     |       | YES                         | YES  |                                |       |       |
| 70 | SpindleTotalRev2      | YES                                                                                     |       | YES                         | YES  |                                |       |       |
| 71 | InFan1SrvAmpSpeed     | YES                                                                                     | YES   |                             |      |                                |       |       |
| 72 | InFan2SrvAmpSpeed     | YES                                                                                     | YES   |                             |      |                                |       |       |
| 73 | InFan1SrvAmpStatus    | YES                                                                                     | YES   |                             |      |                                |       |       |
| 74 | InFan2SrvAmpStatus    | YES                                                                                     | YES   |                             |      |                                |       |       |
| 75 | RadFan1SrvAmpSpeed    | YES                                                                                     | YES   |                             |      |                                |       |       |
| 76 | RadFan2SrvAmpSpeed    | YES                                                                                     | YES   |                             |      |                                |       |       |
| 77 | RadFan1SrvAmpStatus   | YES                                                                                     | YES   |                             |      |                                |       |       |
| 78 | RadFan2SrvAmpStatus   | YES                                                                                     | YES   |                             |      |                                |       |       |
| 79 | InFan1SrvComPwSpeed   | YES                                                                                     | YES   |                             |      |                                |       |       |
| 80 | InFan2SrvComPwSpeed   | YES                                                                                     | YES   |                             |      |                                |       |       |
| 81 | InFan1SpdlComPwSpeed  | YES                                                                                     |       |                             |      |                                |       |       |
| 82 | InFan2SpdlComPwSpeed  | YES                                                                                     |       |                             |      |                                |       |       |
| 83 | InFan1SrvComPwStatus  | YES                                                                                     | YES   |                             |      |                                |       |       |
| 84 | InFan1SpdlComPwStatus | YES                                                                                     |       |                             |      |                                |       |       |
| 85 | InFan2SrvComPwStatus  | YES                                                                                     | YES   |                             |      |                                |       |       |

|      |               |        |                                    |       |  |        |       |                   |                                               |  |        |         |
|------|---------------|--------|------------------------------------|-------|--|--------|-------|-------------------|-----------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        |       | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |       |                   |                                               |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |       |                   |                                               |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        |       | Draw              | A-40829-00001EN                               |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        |       |                   |                                               |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |       |                   |                                               |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        |       |                   |                                               |  |        |         |
| Ed.  | Date          | Design | Description                        |       |  |        |       | FANUC CORPORATION |                                               |  | Sheet. | 678/868 |
| Date | Nov.30.2015   | Desig. | Ogawa                              | Check |  | Apprv. | Inoue |                   |                                               |  |        |         |



| #  | Signal Type           | 15i-A/B | 30i/31i-PB,<br>30i/31i-LB,<br>0i-LF Plus,<br>0i-PF | 31i-WB | 31i-WA,<br>31i-WA5 | 0i-PD | 16i/18i-PA,<br>16i-LA/LPA | 16i/18i-PB/LB,<br>16i-LPB,<br>0i-PB/PC |
|----|-----------------------|---------|----------------------------------------------------|--------|--------------------|-------|---------------------------|----------------------------------------|
| 21 | PartsNum              | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 22 | PartsNumAll           | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 23 | PowOnTime             | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 24 | RunTime               | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 25 | CutTime               | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 26 | SigCUT                |         | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 27 | SigSBK                | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 28 | SigDM00               | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 29 | SigDM01               | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 30 | SigMDRN               | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 31 | Override              | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 32 | SigINP                | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 33 | AbsPos                | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 34 | RelPos                | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 35 | McnPos                | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 36 | SpindleLoad           | YES     |                                                    |        |                    |       |                           |                                        |
| 37 | SigENB                |         |                                                    |        |                    |       |                           |                                        |
| 38 | SpindleSpeed          | YES     |                                                    |        |                    |       |                           |                                        |
| 39 | SpindleTemp           | YES     |                                                    |        |                    |       |                           |                                        |
| 40 | ServoSpeed            | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 41 | ServoTemp             | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 42 | ServoError            | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 43 | ServoLoad             | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 44 | ServoCurrent          | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 45 | ServoCurrentPer       | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 46 | PulseCoderTemp        | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 47 | Alarm                 | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 48 | OpMacroMsg            | YES     | YES                                                | YES    | YES                | YES   | YES (*)                   | YES                                    |
| 49 | CncState              | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 50 | CncWarning            | YES     | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 51 | CncBatLow             |         | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 52 | ApcBatZero            |         | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 53 | ApcBatLow             |         | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 54 | IndBatZero            |         | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 55 | SpdBatZero            |         | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 56 | SSpdBatZero           |         | YES                                                | YES    | YES                | YES   | YES                       | YES                                    |
| 57 | ServoLeakResistData   |         | YES                                                | YES    |                    |       |                           |                                        |
| 58 | SpindleLeakResistData |         |                                                    |        |                    |       |                           |                                        |
| 59 | CncFan1Speed          |         | YES                                                | YES    | YES                |       |                           |                                        |
| 60 | CncFan2Speed          |         | YES                                                | YES    | YES                |       |                           |                                        |
| 61 | CncFan3Speed          |         | YES                                                | YES    | YES                |       |                           |                                        |
| 62 | CncFan4Speed          |         | YES                                                | YES    | YES                |       |                           |                                        |
| 63 | CncFan1Status         |         | YES                                                | YES    |                    |       |                           |                                        |
| 64 | CncFan2Status         |         | YES                                                | YES    |                    |       |                           |                                        |

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 680/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue             |                                                       |  |        |         |

| #   | Signal Type             | 15i-A/B | 30i/31i-PB,<br>30i/31i-LB,<br>0i-LF Plus,<br>0i-PF | 31i-WB | 31i-WA,<br>31i-WA5 | 0i-PD | 16i/18i-PA,<br>16i-LA/LPA | 16i/18i-PB/LB,<br>16i-LPB,<br>0i-PB/PC |
|-----|-------------------------|---------|----------------------------------------------------|--------|--------------------|-------|---------------------------|----------------------------------------|
| 65  | CncFan3Status           |         | YES                                                | YES    |                    |       |                           |                                        |
| 66  | CncFan4Status           |         | YES                                                | YES    |                    |       |                           |                                        |
| 67  | PwConAllAxes            |         | YES                                                | YES    | YES                | YES   |                           |                                        |
| 68  | AdcumPwConPeripheral    |         | YES                                                | YES    |                    | YES   |                           |                                        |
| 69  | SpindleTotalRev1        |         |                                                    |        |                    |       |                           |                                        |
| 70  | SpindleTotalRev2        |         |                                                    |        |                    |       |                           |                                        |
| 71  | InFan1SrvAmpSpeed       |         | YES                                                | YES    |                    |       |                           |                                        |
| 72  | InFan2SrvAmpSpeed       |         | YES                                                | YES    |                    |       |                           |                                        |
| 73  | InFan1SrvAmpStatus      |         | YES                                                | YES    |                    |       |                           |                                        |
| 74  | InFan2SrvAmpStatus      |         | YES                                                | YES    |                    |       |                           |                                        |
| 75  | RadFan1SrvAmpSpeed      |         | YES                                                | YES    |                    |       |                           |                                        |
| 76  | RadFan2SrvAmpSpeed      |         | YES                                                | YES    |                    |       |                           |                                        |
| 77  | RadFan1SrvAmpStatus     |         | YES                                                | YES    |                    |       |                           |                                        |
| 78  | RadFan2SrvAmpStatus     |         | YES                                                | YES    |                    |       |                           |                                        |
| 79  | InFan1SrvComPwSpeed     |         | YES                                                | YES    |                    |       |                           |                                        |
| 80  | InFan2SrvComPwSpeed     |         | YES                                                | YES    |                    |       |                           |                                        |
| 81  | InFan1SpdlComPwSpeed    |         |                                                    |        |                    |       |                           |                                        |
| 82  | InFan2SpdlComPwSpeed    |         |                                                    |        |                    |       |                           |                                        |
| 83  | InFan1SrvComPwStatus    |         | YES                                                | YES    |                    |       |                           |                                        |
| 84  | InFan1SpdlComPwStatus   |         |                                                    |        |                    |       |                           |                                        |
| 85  | InFan2SrvComPwStatus    |         | YES                                                | YES    |                    |       |                           |                                        |
| 86  | InFan2SpdlComPwStatus   |         |                                                    |        |                    |       |                           |                                        |
| 87  | RadFan1SrvComPwSpeed    |         | YES                                                | YES    |                    |       |                           |                                        |
| 88  | RadFan2SrvComPwSpeed    |         | YES                                                | YES    |                    |       |                           |                                        |
| 89  | RadFan1SpdlComPwSpeed   |         |                                                    |        |                    |       |                           |                                        |
| 90  | RadFan2SpdlComPwSpeed   |         |                                                    |        |                    |       |                           |                                        |
| 91  | RadFan1SrvComPwStatus   |         | YES                                                | YES    |                    |       |                           |                                        |
| 92  | RadFan1SpdlComPwStatus  |         |                                                    |        |                    |       |                           |                                        |
| 93  | RadFan2SrvComPwStatus   |         | YES                                                | YES    |                    |       |                           |                                        |
| 94  | RadFan2SpdlComPwStatus  |         |                                                    |        |                    |       |                           |                                        |
| 95  | InFan1SpindleAmpSpeed   |         |                                                    |        |                    |       |                           |                                        |
| 96  | InFan2SpindleAmpSpeed   |         |                                                    |        |                    |       |                           |                                        |
| 97  | InFan1SpindleAmpStatus  |         |                                                    |        |                    |       |                           |                                        |
| 98  | InFan2SpindleAmpStatus  |         |                                                    |        |                    |       |                           |                                        |
| 99  | RadFan1SpindleAmpSpeed  |         |                                                    |        |                    |       |                           |                                        |
| 100 | RadFan2SpindleAmpSpeed  |         |                                                    |        |                    |       |                           |                                        |
| 101 | RadFan1SpindleAmpStatus |         |                                                    |        |                    |       |                           |                                        |
| 102 | RadFan2SpindleAmpStatus |         |                                                    |        |                    |       |                           |                                        |

|      |               |        |                                    |       |  |        |                   |                                               |  |        |         |
|------|---------------|--------|------------------------------------|-------|--|--------|-------------------|-----------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |                   |                                               |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |                   |                                               |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        | Draw              | A-40829-00001EN                               |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        |                   |                                               |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |                   |                                               |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        | FANUC CORPORATION |                                               |  | Sheet. | 681/868 |
| Ed.  | Date          | Design | Description                        |       |  |        |                   |                                               |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check |  | Apprv. |                   |                                               |  |        |         |

## 14.1.2 Custom Signals of FanucCNC

The following table lists custom signals that can be retrieved from FanucCNC type machines.

**Table 4. Custom Signals**

| Signal Type | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Data Type                      |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| PMC         | PMC signal <ul style="list-style-type: none"> <li>PMC type: the first to the fifth path PMC, PMC for dual check safety (DCS)</li> <li>Address: G / F / Y / X / A / R / T / K / C / D / M / N / E / Z</li> <li>Data Type: BIT / BYTE / UBYTE / WORD / Unsigned WORD / DWORD / Unsigned DWORD / REAL (single precision real) / LREAL (double precision real) / STRING (WORD / DWORD / Unsigned WORD / Unsigned DWORD / REAL / LREAL support only little endian mode.)</li> </ul> | Bool / Int32 / Double / String |
| MacroVar    | Custom Macro Common Variables <ul style="list-style-type: none"> <li>Local variable, common variable, system variable</li> </ul>                                                                                                                                                                                                                                                                                                                                               | Double                         |
| PcodeVar    | P-CODE variable <ul style="list-style-type: none"> <li>Conversation macro</li> <li>Auxiliary macro</li> <li>Execution macro</li> </ul>                                                                                                                                                                                                                                                                                                                                         | Double                         |
| Diagnosis   | Diagnostic signal (NOTE 2)                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Bool / Int32 / Double          |

### NOTE

- 1 Reading cycle can be specified for each signal (500msec, 10sec, or 60sec).
- 2 Update Option license is required to retrieve diagnostic signals.

The following are the status and signal definition rules when you assign a signal definition to the status in the [Equipment Setting] window.

#### ■ PMC signal

Status: PMC\_<a>\_<t>\_path<x>\_<Machine name>

Signal definition: <MachineName>\_PMC\_<a>\_<t>\_path<x>

<a>: PMC signal address (address + number)

\* Address should be one of "G" / "F" / "Y" / "X" / "A" / "R" / "T" / "K" / "C" / "D" / "M" / "N" / "E" / "Z."

\* Number should be a value in the range of 0 to 99999.

<t>: bit position (0 to 7) or data type

\* BIT: bit position, BYTE: omitted, WORD: "W," DWORD: "D," REAL: "R," LREAL: "L," UBYTE: "UB," UWORD: "UW," UDWORD: "UD," STRING: "S"

<x>: PMC path number (1 to 5, or 9)

\* PMC path number when retrieving value of dual check safety PMC: "9"

### Examples

Given the following:

Machine name: Machine002, Address: D, Data type: BIT, bit position: 0, PMC path number: 1

Status: PMC\_D200\_0\_path1\_Machine002

Signal definition: Machine002\_PMC\_D200\_0\_path1

|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | <b>A-40829-00001EN</b>                        |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 682/868 |

Given the following:

Machine name: Machine002, Address: D, Data type: BYTE, PMC path number: 2

Status: PMC\_D200\_path2\_Machine002

Signal definition: Machine002\_PMC\_D200\_path2

Given the following:

Machine name: Machine002, Address: D, Data type: WORD, PMC path number: 3

Status: PMC\_D200\_W\_path3\_Machine002

Signal definition: Machine002\_PMC\_D200\_W\_path3

Given the following:

Machine name: Machine002, Address: D, Data type: DWORD, PMC path number: 4

Status: PMC\_D200\_D\_path4\_Machine002

Signal definition: Machine002\_PMC\_D200\_D\_path4

Given the following:

Machine name: Machine002, Address: D, Data type: REAL, PMC path number: 5

Status: PMC\_D200\_R\_path5\_Machine002

Signal definition: Machine002\_PMC\_D200\_R\_path5

Given the following:

Machine name: Machine002, Address: D, Data type: LREAL, PMC path number: 9

Status: PMC\_D200\_L\_path9\_Machine002

Signal definition: Machine002\_PMC\_D200\_L\_path9

Given the following:

Machine name: Machine002, Address: D, Number: 200, Data type: STRING, PMC path number: 1

Status: PMC\_D200\_S\_path1\_Machine002

Signal definition: Machine002\_PMC\_D200\_S\_path1

#### ■ Custom macro variable

Status: MacroVar\_<n>\_path<x>\_<Machine name>

Signal definition: <Machine name>\_MacroVar\_<n>\_path<x>

<n>: custom macro variable number (1 to 33, or 100 to 99999999)

<x>: CNC path number (1 to 15)

#### Example

Given the following:

Machine name: Machine003, custom macro variable number: 500, CNC path number: 1

Status: MacroVar\_500\_path1\_Machine003

Signal definition: Machine003\_MacroVar\_500\_path1

#### ■ P-CODE variable

Data type: Conversation macro

Status: PcodeCon\_<n>\_path<x>\_<Machine name>

Signal definition: <Machine name>\_PcodeCon\_<n>\_path<x>

<n>: P-CODE variable number (1 to 33)

<x>: CNC path number (1 to 15)

#### Example

Given the following:

Machine name: Machine004, P-CODE variable number: 1, CNC path number: 1

Status: PcodeCon\_1\_path1\_Machine004

Signal definition: Machine004\_PcodeCon\_1\_path1

|      |               |        |                                            |       |  |        |       |                                                       |  |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------------------------------------------|--|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                                                       |  |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                                                       |  |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                                                       |  |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        | Draw  | A-40829-00001EN                                       |  |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                                                       |  |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                                                       |  |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       | FANUC CORPORATION                                     |  |  | Sheet. | 683/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                                                       |  |  |        |         |

Data Type: Auxiliary macro

Status: PcodeAux\_<n>\_path<x>\_<Machine name>

Signal definition: <Machine name>\_PcodeAux\_<n>\_path<x>

<n>: P-CODE variable number (1 to 33)

<x>: CNC path number (1 to 15)

### Example

Given the following:

Machine name: Machine004, P-CODE variable number: 1, CNC path number: 1

Status: PcodeAux\_1\_path1\_Machine004

Signal definition: Machine004\_PcodeAux\_1\_path1

Data type: Execution macro

Status: PcodeExe\_<n>\_path<x>\_<Machine name>

Signal definition: <Machine name>\_PcodeExe\_<n>\_path<x>

<n>: P-CODE variable number (1 to 33)

<x>: CNC path number (1 to 15)

### Example

Given the following:

Machine name: Machine004, P-CODE variable number: 1, CNC path number: 1

Status: PcodeExe\_1\_path1\_Machine004

Signal definition: Machine004\_PcodeExe\_1\_path1

Data Type: Conversation, Auxiliary, Execution macro (in common)

Status: PcodeVar\_<n>\_path<x>\_<Machine name>

Signal definition: <Machine name>\_PcodeVar\_<n>\_path<x>

<n>: P-CODE variable number (100 to 89999)

<x>: CNC path number (1 to 15)

### Example

Given the following:

Machine name: Machine004, P-CODE variable number: 600, CNC path number: 1

Status: PcodeVar\_600\_path1\_Machine004

Signal definition: Machine004\_PcodeVar\_600\_path1

### NOTE

For the machines below, if P-CODE variable number is specified between 1 and 33, it will be retrieved in the predefined data type regardless of the specification of data type.

- Series 15*i*-A/B: "Conversation macro"
- Series 16*i* /18*i* /21*i*-A/B, Series 0*i*-B/C, Power Mate *i*-D/H: "Execution macro"

### ■Diagnostic signal

Data type: Bit, Diagnosis Data Type: Path

Status: Diagnosis\_<n>\_<t>\_path<x>\_<Machine name>

Signal definition: <Machine name>\_Diagnosis\_<n>\_<t>\_path<x>

<n>: diagnosis number

<t>: bit position (0 to 7)

<x>: CNC path number (1 to 15)

|      |               |        |                                            |                          |                                                       |        |        |         |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |        | Sheet. | 684/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    |                                                       | Apprv. | Inoue  |         |



### Example

Given the following:

Machine name: Machine005, diagnosis number: 110, bit position: 0, CNC path number: 1

Status: Diagnosis\_110\_0\_path1\_Machine005

Signal definition: Machine005\_Diagnosis\_110\_0\_path1

Data type: other than Bit, Diagnosis Data Type: Axis

Status: Diagnosis\_<n>\_path<x>\_axis<y>\_<Machine name>

Signal definition: <Machine name>\_Diagnosis\_<n>\_path<x>\_axis<y>

<n>: diagnosis number

<x>: CNC path number (1 to 15)

<y>: servo axis number (0 to the maximum servo axis number for the CNC path whose path number is specified with <x>)

### Example

Given the following:

Machine name: Machine005, diagnosis number: 114, servo axis number: 4, CNC path number: 5

Status: Diagnosis\_114\_path5\_axis4\_Machine005

Signal definition: Machine005\_Diagnosis\_114\_path5\_axis4

Data type: other than Bit, Diagnosis Data Type: Spindle

Status: Diagnosis\_<n>\_path<x>\_spindle<z>\_<Machine name>

Signal definition: <Machine name>\_Diagnosis\_<n>\_path<x>spindle<z>

<n>: diagnosis number

<x>: CNC path number (1 to 15)

<z>: spindle axis number (0 to the maximum spindle axis number for the CNC path whose path number is specified with <x>)

### Example

Given the following:

Machine name: Machine005, diagnosis number: 115, spindle axis number: 5, CNC path number: 6

Status: Diagnosis\_115\_path6\_spindle5\_Machine005

Signal definition: Machine005\_Diagnosis\_115\_path6\_spindle5

|      |               |        |                                    |                          |                                               |        |        |         |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |        | Sheet. | 685/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    |                                               | Apprv. | Inoue  |         |

## 14.1.3 Tool Life Signals of FanucCNC

The following table lists the tool life signals that can be retrieved from FanucCNC type machines.

**Table 5. Tool Life Signals**

| Signal Type           | Description                             | Data Type | Reading Cycle (Default) (NOTE 1) | Enabled (Default) (NOTE 2) | Remarks                                     |
|-----------------------|-----------------------------------------|-----------|----------------------------------|----------------------------|---------------------------------------------|
| ToolGroupNum          | Tool group number in use                | Int32     | 500 ms                           | NO                         | -                                           |
| ToolLifeCount<n>      | Tool life count for tool group <n>      | Int32     | 10 sec                           | NO                         | <n>: tool group number (01 to 256) (NOTE 3) |
| ToolLifeLimit<n>      | Tool life value for tool group <n>      | Int32     | 10 sec                           | NO                         |                                             |
| ToolLifeForenotice<n> | Tool life forenotice for tool group <n> | Int32     | 10 sec                           | NO                         |                                             |
| ToolLifeCountType<n>  | Tool life count type for tool group <n> | String    | 10 sec                           | NO                         |                                             |

The following table lists the tool life signals retrievable from CNC models.

**Table 6. Retrievable Signals Listed By CNC Model**  
(Signals retrievable from the model are marked with "YES")

| Signal Type           | 30i/31i/32i-B, 30i/31i/32i-B Plus, 31i-B5, 31i-B5 Plus, 35i-B, 30i/31i/32i-A, 16i/18i-B, 16i/18i-A | 0i-F, 0i-F Plus, 0i-D, 21i-B, 21i-A, 0i-B, 0i-C, 15i-A, 15i-B | Remarks                                     |
|-----------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------|
| ToolGroupNum          | YES                                                                                                | YES                                                           | -                                           |
| ToolLifeCount<n>      | YES                                                                                                | YES                                                           | <n>: tool group number (01 to 256) (NOTE 3) |
| ToolLifeLimit<n>      | YES                                                                                                | YES                                                           |                                             |
| ToolLifeForenotice<n> | YES<br>Retrievable from M (machining models)                                                       |                                                               |                                             |
| ToolLifeCountType<n>  | YES                                                                                                | YES                                                           |                                             |

### NOTE

- The reading cycle can be selected for each signal (selectable cycle: 500msec / 10sec / 60sec).
- For each signal, whether to retrieve it or not (enabled/disabled) can be specified.
- When the tool group number is fewer than 10, express it as a two-digit number with 0 in the tens place.  
Example) When the tool group number is three, the tool life count signal for the group should be ToolLifeCount03.
- CNC models that support tool life signals are as follows:  
30i /31i /32i-B, 30i /31i /32i-B Plus, 31i-B5, 31i-B5 Plus, 35i-B, 0i-F, 0i-F Plus, 30i /31i /32i-A, 0i-D, 16i /18i /21i-B, 16i /18i /21i-A, 0i-B, 0i-C, 15i-A, 15i-B
- Update Option license is required to retrieve tool life signals from FanucCNC type machines.

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               | Sheet. 686/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |

When assigning a signal to equipment by defining a status for the signal on the [Equipment Setting] window, conform to the signal definitions shown below:

#### ■ ToolGroupNum

Status: ToolGroupNum\_path<x>\_<machine name>

Signal definition: <machine name>\_ToolGroupNum\_path<x>

<x>: CNC path number (1 to 15)

#### Example

Given the following:

Machine name: Machine002, CNC path number: 1

Status: ToolGroupNum\_path1\_Machine002

Signal definition: Machine002\_ToolGroupNum\_path1

#### ■ ToolLifeCount

Status: ToolLifeCount<n>\_path<x>\_<machine name>

Signal definition: <machine name>\_ToolLifeCount<n>\_path<x>

<n>: tool group number (01 to 256)

<x>: CNC path number (1 to 15)

#### Example

Given the following:

Machine name: Machine003, tool group number: 01, CNC path number: 2

Status: ToolLifeCount01\_path2\_Machine003

Signal definition: Machine003\_ToolLifeCount01\_path2

#### ■ ToolLifeLimit

Status: ToolLifeLimit<n>\_path<x>\_<machine name>

Signal definition: <machine name>\_ToolLifeLimit<n>\_path<x>

<n>: tool group number (01 to 256)

<x>: CNC path number (1 to 15)

#### Example

Given the following:

Machine name: Machine004, tool group number: 09, CNC path number: 3

Status: ToolLifeLimit09\_path3\_Machine004

Signal definition: Machine004\_ToolLifeLimit09\_path3

#### ■ ToolLifeForenotice

Status: ToolLifeForenotice<n>\_path<x>\_<machine name>

Signal definition: <machine name>\_ToolLifeForenotice<n>\_path<x>

<n>: tool group number (01 to 256)

<x>: CNC path number (1 to 15)

|      |               |        |                                    |                          |                                               |        |       |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|--------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |        |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |        |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |        |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |        |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |        |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |        |       | Sheet. 687/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |        |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    |                                               | Apprv. | Inoue |                |

## Example

Given the following:

Machine name: Machine005, tool group number: 99, CNC path number: 4

Status: ToolLifeForenotice99\_path4\_Machine005

Signal definition: Machine005\_ToolLifeLimit99\_path4

### ■ ToolLifeCountType

Status: ToolLifeCountType<n>\_path<x>\_<machine name>

Signal definition: <machine name>\_ToolLifeCountType<n>\_path<x>

<n>: tool group number (01 to 256)

<x>: CNC path number (1 to 15)

## Example

Given the following:

Machine name: Machine006, tool group number: 100, CNC path number: 5

Status: ToolLifeCountType100\_path5\_Machine006

Signal definition: Machine006\_ToolLifeCountType100\_path5

|      |               |        |                                            |                          |                                                       |        |        |         |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |        | Sheet. | 688/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    |                                                       | Apprv. | Inoue  |         |

#### 14.1.4 Laser Control Data Signals of FanucCNC

The following table lists the laser control data signals that can be retrieved from FanucCNC type machines.

**<p>** in the table denotes as follows:

**<p>**: value of thousands place digit of a PMC address (the path number = <p>+1)

(Example: PMC F1221.7, as  $\langle p \rangle = 1$ , the path number is 2.)

For the signals that can be retrieved by laser oscillator produced by FANUC, refer to [Table 8](#);

for the signals that can be retrieved by laser oscillator produced by other manufacturers, refer to [Table 9](#);

and for the relationship between the type of FanucCNC supporting laser control data signals and the laser oscillators, refer to the "FANUC MT-LINK*i* Startup MANUAL."

### Table 7. Laser Control Data Signals

| #  | Type of Signal     | Description of Signal                                                                                                                                                                                                                                               | Data Type | Reading Cycle (Default) | Enabled (Default) | <p> |
|----|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------|-------------------|-----|
| 01 | PwrCommand         | Value of laser power output (peak) (W)                                                                                                                                                                                                                              | Int32     | 500 msec                | NO                |     |
| 02 | PulseBottomCommand | Value of laser power output (bottom) (W)                                                                                                                                                                                                                            | Int32     | 500 msec                | NO                |     |
| 03 | FrequencyCommand   | Pulse frequency command value (Hz)                                                                                                                                                                                                                                  | Int32     | 500 msec                | NO                |     |
| 04 | DutyCommand        | Pulse duty command value (%)                                                                                                                                                                                                                                        | Int32     | 500 msec                | NO                |     |
| 05 | LaserPwr           | Actual output value detected by power sensor (W)                                                                                                                                                                                                                    | Int32     | 500 msec                | NO                |     |
| 06 | LaserSequence      | Laser oscillator start state<br>Same data as diagnostic data (No.908)                                                                                                                                                                                               | Int32     | 500 msec                | YES               |     |
| 07 | SigBeamOut         | Beam output signal (notifies that the beam is output)<br>In case PMC BEAM F<p>221.7 is enabled ("1"):<br>State of beam is "on"<br>In case PMC BEAM F<p>221.7 is disabled ("0"):<br>State of beam is "off"                                                           | Bool      | 500 msec                | YES               | <p> |
| 08 | SigLaserAlarm      | Laser alarm<br>In case the alarm related to laser oscillator and laser control occurs, PMC LARM F<p>221.0 becomes 1.                                                                                                                                                | Bool      | 500 msec                | YES               | <p> |
| 09 | BCReflection       | Reflection power from workpiece (W)<br>Same data as diagnostic data (No.8505)                                                                                                                                                                                       | Int32     | 500 msec                | NO                |     |
| 10 | WaterFlowRate1     | Flow rate of cooling water of laser oscillator (0.1L/min)<br>Same data as diagnostic data (No.2500)                                                                                                                                                                 | Int32     | 500 msec                | NO                |     |
| 11 | WaterFlowRate2     | If laser oscillator is "FANUC FIBER LASER series":<br>Flow rate of cooling water of BC unit (0.1L/min)<br>If laser oscillator is "FANUC LASER C series":<br>Flow rate of cooling water of laser oscillator (0.1L/min)<br><br>Same data as diagnostic data (No.2501) | Int32     | 500 msec                | NO                |     |

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 689/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         |

| #  | Type of Signal | Description of Signal                                                                                                                                   | Data Type | Reading Cycle (Default) | Enabled (Default) | <p> |
|----|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------|-------------------|-----|
| 12 | LockerTemp     | Temperature in the locker of laser oscillator (0.1°C)<br>Same data as diagnostic data (No.8500)                                                         | Int32     | 60 sec                  | NO                |     |
| 13 | LockerHumidity | Relative humidity in the locker of laser oscillator (%)<br>Same data as diagnostic data (No.8501)                                                       | Int32     | 60 sec                  | NO                |     |
| 14 | WaterTemp      | Temperature of cooling water of laser oscillator (0.1°C)<br>Same data as diagnostic data (No.8502)                                                      | Int32     | 60 sec                  | NO                |     |
| 15 | DewPoint       | Dew point (0.1°C)<br>Dew point of the case that the air in the locker is exposed to the cooling water is displayed<br>Same as diagnostic data (No.8503) | Int32     | 60 sec                  | NO                |     |
| 16 | LC1OutPwr      | Output power of LC unit 1 (W)<br>Same data as diagnostic data (No.8511)                                                                                 | Int32     | 500 msec                | NO                |     |
| 17 | LC2OutPwr      | Output power of LC unit 2 (W)<br>Same data as diagnostic data (No.8512)                                                                                 | Int32     | 500 msec                | NO                |     |
| 18 | LC3OutPwr      | Output power of LC unit 13 (W)<br>Same data as diagnostic data (No.8513)                                                                                | Int32     | 500 msec                | NO                |     |
| 19 | LC4OutPwr      | Output power of LC unit 4 (W)<br>Same data as diagnostic data (No.8514)                                                                                 | Int32     | 500 msec                | NO                |     |
| 20 | LC5OutPwr      | Output power of LC unit 5 (W)<br>Same data as diagnostic data (No.8515)                                                                                 | Int32     | 500 msec                | NO                |     |
| 21 | LC6OutPwr      | Output power of LC unit 6 (W)<br>Same data as diagnostic data (No.8516)                                                                                 | Int32     | 500 msec                | NO                |     |
| 22 | LC7OutPwr      | Output power of LC unit 7 (W)<br>Same data as diagnostic data (No.8517)                                                                                 | Int32     | 500 msec                | NO                |     |
| 23 | LC8OutPwr      | Output power of LC unit 8 (W)<br>Same data as diagnostic data (No.8518)                                                                                 | Int32     | 500 msec                | NO                |     |
| 24 | LC9OutPwr      | Output power of LC unit 9 (W)<br>Same data as diagnostic data (No.8519)                                                                                 | Int32     | 500 msec                | NO                |     |
| 25 | LC10OutPwr     | Output power of LC unit 10 (W)<br>Same data as diagnostic data (No.8520)                                                                                | Int32     | 500 msec                | NO                |     |
| 26 | LC11OutPwr     | Output power of LC unit 11 (W)<br>Same data as diagnostic data (No.8521)                                                                                | Int32     | 500 msec                | NO                |     |
| 27 | LC12OutPwr     | Output power of LC unit 12 (W)<br>Same data as diagnostic data (No.8522)                                                                                | Int32     | 500 msec                | NO                |     |
| 28 | FIBER1FFBD     | Output of FFBD of FIBER1 (mW)<br>Same data as diagnostic data (No.8675)                                                                                 | Int32     | 500 msec                | NO                |     |
| 29 | FIBER2FFBD     | Output of FFBD of FIBER2 (mW)<br>Same data as diagnostic data (No.8676)                                                                                 | Int32     | 500 msec                | NO                |     |
| 30 | FIBER3FFBD     | Output of FFBD of FIBER3 (mW)<br>Same data as diagnostic data (No.8677)                                                                                 | Int32     | 500 msec                | NO                |     |
| 31 | FIBER4FFBD     | Output of FFBD of FIBER4 (mW)<br>Same data as diagnostic data (No.8678)                                                                                 | Int32     | 500 msec                | NO                |     |
| 32 | FIBER5FFBD     | Output of FFBD of FIBER5 (mW)<br>Same data as diagnostic data (No.8679)                                                                                 | Int32     | 500 msec                | NO                |     |
| 33 | FIBER6FFBD     | Output of FFBD of FIBER6 (mW)<br>Same data as diagnostic data (No.8680)                                                                                 | Int32     | 500 msec                | NO                |     |
| 34 | PSU1-1Current  | Current of LD PSU (PSU1-1) (10mA)<br>Same data as diagnostic data (No.8615)                                                                             | Int32     | 500 msec                | NO                |     |

|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | <b>A-40829-00001EN</b>                        |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 690/868 |

| #  | Type of Signal | Description of Signal                                                        | Data Type | Reading Cycle (Default) | Enabled (Default) | <p> |
|----|----------------|------------------------------------------------------------------------------|-----------|-------------------------|-------------------|-----|
| 35 | PSU1-2Current  | Current of LD PSU (PSU1-2) (10mA)<br>Same data as diagnostic data (No.8616)  | Int32     | 500 msec                | NO                |     |
| 36 | PSU2-1Current  | Current of LD PSU (PSU2-1) (10mA)<br>Same data as diagnostic data (No.8617)  | Int32     | 500 msec                | NO                |     |
| 37 | PSU2-2Current  | Current of LD PSU (PSU2-2) (10mA)<br>Same data as diagnostic data (No.8618)  | Int32     | 500 msec                | NO                |     |
| 38 | PSU3-1Current  | Current of LD PSU (PSU3-1) (10mA)<br>Same data as diagnostic data (No.8619)  | Int32     | 500 msec                | NO                |     |
| 39 | PSU3-2Current  | Current of LD PSU (PSU3-2) (10mA)<br>Same data as diagnostic data (No.8620)  | Int32     | 500 msec                | NO                |     |
| 40 | PSU4-1Current  | Current of LD PSU (PSU4-1) (10mA)<br>Same data as diagnostic data (No.8621)  | Int32     | 500 msec                | NO                |     |
| 41 | PSU4-2Current  | Current of LD PSU (PSU4-2) (10mA)<br>Same data as diagnostic data (No.8622)  | Int32     | 500 msec                | NO                |     |
| 42 | PSU5-1Current  | Current of LD PSU (PSU5-1) (10mA)<br>Same data as diagnostic data (No.8623)  | Int32     | 500 msec                | NO                |     |
| 43 | PSU5-2Current  | Current of LD PSU (PSU5-2) (10mA)<br>Same data as diagnostic data (No.8624)  | Int32     | 500 msec                | NO                |     |
| 44 | PSU6-1Current  | Current of LD PSU (PSU6-1) (10mA)<br>Same data as diagnostic data (No.8625)  | Int32     | 500 msec                | NO                |     |
| 45 | PSU6-2Current  | Current of LD PSU (PSU6-2) (10mA)<br>Same data as diagnostic data (No.8626)  | Int32     | 500 msec                | NO                |     |
| 46 | PSU7-1Current  | Current of LD PSU (PSU7-1) (10mA)<br>Same data as diagnostic data (No.8627)  | Int32     | 500 msec                | NO                |     |
| 47 | PSU7-2Current  | Current of LD PSU (PSU7-2) (10mA)<br>Same data as diagnostic data (No.8628)  | Int32     | 500 msec                | NO                |     |
| 48 | PSU8-1Current  | Current of LD PSU (PSU8-1) (10mA)<br>Same data as diagnostic data (No.8629)  | Int32     | 500 msec                | NO                |     |
| 49 | PSU8-2Current  | Current of LD PSU (PSU8-2) (10mA)<br>Same data as diagnostic data (No.8630)  | Int32     | 500 msec                | NO                |     |
| 50 | PSU9-1Current  | Current of LD PSU (PSU9-1) (10mA)<br>Same data as diagnostic data (No.8631)  | Int32     | 500 msec                | NO                |     |
| 51 | PSU9-2Current  | Current of LD PSU (PSU9-2) (10mA)<br>Same data as diagnostic data (No.8632)  | Int32     | 500 msec                | NO                |     |
| 52 | PSU10-1Current | Current of LD PSU (PSU10-1) (10mA)<br>Same data as diagnostic data (No.8633) | Int32     | 500 msec                | NO                |     |
| 53 | PSU10-2Current | Current of LD PSU (PSU10-2) (10mA)<br>Same data as diagnostic data (No.8634) | Int32     | 500 msec                | NO                |     |
| 54 | PSU11-1Current | Current of LD PSU (PSU11-1) (10mA)<br>Same data as diagnostic data (No.8635) | Int32     | 500 msec                | NO                |     |
| 55 | PSU11-2Current | Current of LD PSU (PSU11-2) (10mA)<br>Same data as diagnostic data (No.8636) | Int32     | 500 msec                | NO                |     |
| 56 | PSU12-1Current | Current of LD PSU (PSU12-1) (10mA)<br>Same data as diagnostic data (No.8637) | Int32     | 500 msec                | NO                |     |
| 57 | PSU12-2Current | Current of LD PSU (PSU12-2) (10mA)<br>Same data as diagnostic data (No.8638) | Int32     | 500 msec                | NO                |     |
| 58 | PSU1-1Voltage  | Voltage of LD PSU (PSU1-1) (0.1V)<br>Same data as diagnostic data (No.8639)  | Int32     | 500 msec                | NO                |     |

|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | <b>A-40829-00001EN</b>                        |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 691/868 |

| #  | Type of Signal | Description of Signal                                                        | Data Type | Reading Cycle (Default) | Enabled (Default) | <p> |
|----|----------------|------------------------------------------------------------------------------|-----------|-------------------------|-------------------|-----|
| 59 | PSU1-2Voltage  | Voltage of LD PSU (PSU1-2) (0.1V)<br>Same data as diagnostic data (No.8640)  | Int32     | 500 msec                | NO                |     |
| 60 | PSU2-1Voltage  | Voltage of LD PSU (PSU2-1) (0.1V)<br>Same data as diagnostic data (No.8641)  | Int32     | 500 msec                | NO                |     |
| 61 | PSU2-2Voltage  | Voltage of LD PSU (PSU2-2) (0.1V)<br>Same data as diagnostic data (No.8642)  | Int32     | 500 msec                | NO                |     |
| 62 | PSU3-1Voltage  | Voltage of LD PSU (PSU3-1) (0.1V)<br>Same data as diagnostic data (No.8643)  | Int32     | 500 msec                | NO                |     |
| 63 | PSU3-2Voltage  | Voltage of LD PSU (PSU3-2) (0.1V)<br>Same data as diagnostic data (No.8644)  | Int32     | 500 msec                | NO                |     |
| 64 | PSU4-1Voltage  | Voltage of LD PSU (PSU4-1) (0.1V)<br>Same data as diagnostic data (No.8645)  | Int32     | 500 msec                | NO                |     |
| 65 | PSU4-2Voltage  | Voltage of LD PSU (PSU4-2) (0.1V)<br>Same data as diagnostic data (No.8646)  | Int32     | 500 msec                | NO                |     |
| 66 | PSU5-1Voltage  | Voltage of LD PSU (PSU5-1) (0.1V)<br>Same data as diagnostic data (No.8647)  | Int32     | 500 msec                | NO                |     |
| 67 | PSU5-2Voltage  | Voltage of LD PSU (PSU5-2) (0.1V)<br>Same data as diagnostic data (No.8648)  | Int32     | 500 msec                | NO                |     |
| 68 | PSU6-1Voltage  | Voltage of LD PSU (PSU6-1) (0.1V)<br>Same data as diagnostic data (No.8649)  | Int32     | 500 msec                | NO                |     |
| 69 | PSU6-2Voltage  | Voltage of LD PSU (PSU6-2) (0.1V)<br>Same data as diagnostic data (No.8650)  | Int32     | 500 msec                | NO                |     |
| 70 | PSU7-1Voltage  | Voltage of LD PSU (PSU7-1) (0.1V)<br>Same data as diagnostic data (No.8651)  | Int32     | 500 msec                | NO                |     |
| 71 | PSU7-2Voltage  | Voltage of LD PSU (PSU7-2) (0.1V)<br>Same data as diagnostic data (No.8652)  | Int32     | 500 msec                | NO                |     |
| 72 | PSU8-1Voltage  | Voltage of LD PSU (PSU8-1) (0.1V)<br>Same data as diagnostic data (No.8653)  | Int32     | 500 msec                | NO                |     |
| 73 | PSU8-2Voltage  | Voltage of LD PSU (PSU8-2) (0.1V)<br>Same data as diagnostic data (No.8654)  | Int32     | 500 msec                | NO                |     |
| 74 | PSU9-1Voltage  | Voltage of LD PSU (PSU9-1) (0.1V)<br>Same data as diagnostic data (No.8655)  | Int32     | 500 msec                | NO                |     |
| 75 | PSU9-2Voltage  | Voltage of LD PSU (PSU9-2) (0.1V)<br>Same data as diagnostic data (No.8656)  | Int32     | 500 msec                | NO                |     |
| 76 | PSU10-1Voltage | Voltage of LD PSU (PSU10-1) (0.1V)<br>Same data as diagnostic data (No.8657) | Int32     | 500 msec                | NO                |     |
| 77 | PSU10-2Voltage | Voltage of LD PSU (PSU10-2) (0.1V)<br>Same data as diagnostic data (No.8658) | Int32     | 500 msec                | NO                |     |
| 78 | PSU11-1Voltage | Voltage of LD PSU (PSU11-1) (0.1V)<br>Same data as diagnostic data (No.8659) | Int32     | 500 msec                | NO                |     |
| 79 | PSU11-2Voltage | Voltage of LD PSU (PSU11-2) (0.1V)<br>Same data as diagnostic data (No.8660) | Int32     | 500 msec                | NO                |     |
| 80 | PSU12-1Voltage | Voltage of LD PSU (PSU12-1) (0.1V)<br>Same data as diagnostic data (No.8661) | Int32     | 500 msec                | NO                |     |
| 81 | PSU12-2Voltage | Voltage of LD PSU (PSU12-2) (0.1V)<br>Same data as diagnostic data (No.8662) | Int32     | 500 msec                | NO                |     |

|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | <b>A-40829-00001EN</b>                        |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 692/868 |



| #  | Type of Signal    | Description of Signal                                                                           | Data Type | Reading Cycle (Default) | Enabled (Default) | <p> |
|----|-------------------|-------------------------------------------------------------------------------------------------|-----------|-------------------------|-------------------|-----|
| 82 | PSU1HeatSinkTemp  | Heat sink temperature of LD PSU (PSU1)<br>(0.1°C)<br>Same data as diagnostic data (No.8663)     | Int32     | 500 msec                | NO                |     |
| 83 | PSU2HeatSinkTemp  | Heat sink temperature of LD PSU (PSU2)<br>(0.1°C)<br>Same data as diagnostic data (No.8664)     | Int32     | 500 msec                | NO                |     |
| 84 | PSU3HeatSinkTemp  | Heat sink temperature of LD PSU (PSU3)<br>(0.1°C)<br>Same data as diagnostic data (No.8665)     | Int32     | 500 msec                | NO                |     |
| 85 | PSU4HeatSinkTemp  | Heat sink temperature of LD PSU (PSU4)<br>(0.1°C)<br>Same data as diagnostic data (No.8666)     | Int32     | 500 msec                | NO                |     |
| 86 | PSU5HeatSinkTemp  | Heat sink temperature of LD PSU (PSU5)<br>(0.1°C)<br>Same data as diagnostic data (No.8667)     | Int32     | 500 msec                | NO                |     |
| 87 | PSU6HeatSinkTemp  | Heat sink temperature of LD PSU (PSU6)<br>(0.1°C)<br>Same data as diagnostic data (No.8668)     | Int32     | 500 msec                | NO                |     |
| 88 | PSU7HeatSinkTemp  | Heat sink temperature of LD PSU (PSU7)<br>(0.1°C)<br>Same data as diagnostic data (No.8669)     | Int32     | 500 msec                | NO                |     |
| 89 | PSU8HeatSinkTemp  | Heat sink temperature of LD PSU (PSU8)<br>(0.1°C)<br>Same data as diagnostic data (No.8670)     | Int32     | 500 msec                | NO                |     |
| 90 | PSU9HeatSinkTemp  | Heat sink temperature of LD PSU (PSU9)<br>(0.1°C)<br>Same data as diagnostic data (No.8671)     | Int32     | 500 msec                | NO                |     |
| 91 | PSU10HeatSinkTemp | Heat sink temperature of LD PSU (PSU10)<br>(0.1°C)<br>Same data as diagnostic data (No.8672)    | Int32     | 500 msec                | NO                |     |
| 92 | PSU11HeatSinkTemp | Heat sink temperature of LD PSU (PSU11)<br>(0.1°C)<br>Same data as diagnostic data (No.8673)    | Int32     | 500 msec                | NO                |     |
| 93 | PSU12HeatSinkTemp | Heat sink temperature of LD PSU (PSU12)<br>(0.1°C)<br>Same data as diagnostic data (No.8674)    | Int32     | 500 msec                | NO                |     |
| 94 | LaserRunTime      | Total run time of laser oscillator (0.1hour)<br>Same data as diagnostic data (No.8719)          | Int32     | 60 sec                  | NO                |     |
| 95 | BCBeamOnTime      | Integrated value of BC unit beam on time<br>(0.1hour)<br>Same data as diagnostic data (No.8720) | Int32     | 60 sec                  | NO                |     |
| 96 | CuttingIndex1     | Cutting index 1<br>Same data as diagnostic data (No.8721)                                       | Int32     | 500 msec                | NO                |     |
| 97 | CuttingIndex2     | Cutting index 2<br>Same data as diagnostic data (No.8722)                                       | Int32     | 500 msec                | NO                |     |
| 98 | CuttingIndex3     | Cutting index 3<br>Same data as diagnostic data (No.8723)                                       | Int32     | 500 msec                | NO                |     |
| 99 | RetryTimes        | Retry times<br>Same data as diagnostic data (No.8724)                                           | Int32     | 500 msec                | NO                |     |

|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | <b>A-40829-00001EN</b>                        |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 693/868 |



| #   | Type of Signal | Description of Signal                                                                     | Data Type | Reading Cycle (Default) | Enabled (Default) | <p> |
|-----|----------------|-------------------------------------------------------------------------------------------|-----------|-------------------------|-------------------|-----|
| 114 | RFVoltage8     | RF voltage impressed to discharge tube No.8 (V)<br>Same data as diagnostic data (No.923)  | Int32     | 500 msec                | NO                |     |
| 115 | RFCurrent1     | RF current impressed to discharge tube No.1 (mA)<br>Same data as diagnostic data (No.910) | Int32     | 500 msec                | NO                |     |
| 116 | RFCurrent2     | RF current impressed to discharge tube No.2 (mA)<br>Same data as diagnostic data (No.912) | Int32     | 500 msec                | NO                |     |
| 117 | RFCurrent3     | RF current impressed to discharge tube No.3 (mA)<br>Same data as diagnostic data (No.914) | Int32     | 500 msec                | NO                |     |
| 118 | RFCurrent4     | RF current impressed to discharge tube No.4 (mA)<br>Same data as diagnostic data (No.916) | Int32     | 500 msec                | NO                |     |
| 119 | RFCurrent5     | RF current impressed to discharge tube No.5 (mA)<br>Same data as diagnostic data (No.918) | Int32     | 500 msec                | NO                |     |
| 120 | RFCurrent6     | RF current impressed to discharge tube No.6 (mA)<br>Same data as diagnostic data (No.920) | Int32     | 500 msec                | NO                |     |
| 121 | RFCurrent7     | RF current impressed to discharge tube No.7 (mA)<br>Same data as diagnostic data (No.922) | Int32     | 500 msec                | NO                |     |
| 122 | RFCurrent8     | RF current impressed to discharge tube No.8 (mA)<br>Same data as diagnostic data (No.924) | Int32     | 500 msec                | NO                |     |
| 123 | DCVoltage1     | DC voltage of power supply No.1 (0.1V)<br>Same data as diagnostic data (No.925)           | Int32     | 500 msec                | NO                |     |
| 124 | DCVoltage2     | DC voltage of power supply No.2 (0.1V)<br>Same data as diagnostic data (No.927)           | Int32     | 500 msec                | NO                |     |
| 125 | DCVoltage3     | DC voltage of power supply No.3 (0.1V)<br>Same data as diagnostic data (No.929)           | Int32     | 500 msec                | NO                |     |
| 126 | DCVoltage4     | DC voltage of power supply No.4 (0.1V)<br>Same data as diagnostic data (No.931)           | Int32     | 500 msec                | NO                |     |
| 127 | DCVoltage5     | DC voltage of power supply No.5 (0.1V)<br>Same data as diagnostic data (No.933)           | Int32     | 500 msec                | NO                |     |
| 128 | DCVoltage6     | DC voltage of power supply No.6 (0.1V)<br>Same data as diagnostic data (No.935)           | Int32     | 500 msec                | NO                |     |
| 129 | DCVoltage7     | DC voltage of power supply No.7 (0.1V)<br>Same data as diagnostic data (No.937)           | Int32     | 500 msec                | NO                |     |
| 130 | DCVoltage8     | DC voltage of power supply No.8 (0.1V)<br>Same data as diagnostic data (No.939)           | Int32     | 500 msec                | NO                |     |
| 131 | DCCurrent1     | DC current of power supply No.1 (0.1A)<br>Same data as diagnostic data (No.926)           | Int32     | 500 msec                | NO                |     |
| 132 | DCCurrent2     | DC current of power supply No.2 (0.1A)<br>Same data as diagnostic data (No.928)           | Int32     | 500 msec                | NO                |     |
| 133 | DCCurrent3     | DC current of power supply No.3 (0.1A)<br>Same data as diagnostic data (No.930)           | Int32     | 500 msec                | NO                |     |

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |                          |                                               |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |                          |                                               |                |
| Ed.  | Date          | Design | Description                        | <b>FANUC CORPORATION</b> |                                               | Sheet. 695/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |

| #   | Type of Signal | Description of Signal                                                               | Data Type | Reading Cycle (Default) | Enabled (Default) | <p> |
|-----|----------------|-------------------------------------------------------------------------------------|-----------|-------------------------|-------------------|-----|
| 134 | DCCurrent4     | DC current of power supply No.4 (0.1A)<br>Same data as diagnostic data (No.932)     | Int32     | 500 msec                | NO                |     |
| 135 | DCCurrent5     | DC current of power supply No.5 (0.1A)<br>Same data as diagnostic data (No.934)     | Int32     | 500 msec                | NO                |     |
| 136 | DCCurrent6     | DC current of power supply No.6 (0.1A)<br>Same data as diagnostic data (No.936)     | Int32     | 500 msec                | NO                |     |
| 137 | DCCurrent7     | DC current of power supply No.7 (0.1A)<br>Same data as diagnostic data (No.938)     | Int32     | 500 msec                | NO                |     |
| 138 | DCCurrent8     | DC current of power supply No.8 (0.1A)<br>Same data as diagnostic data (No.940)     | Int32     | 500 msec                | NO                |     |
| 139 | DCW1           | DC electric power of power supply No.1 (W)<br>Same data as diagnostic data (No.941) | Int32     | 500 msec                | NO                |     |
| 140 | DCW2           | DC electric power of power supply No.2 (W)<br>Same data as diagnostic data (No.942) | Int32     | 500 msec                | NO                |     |
| 141 | DCW3           | DC electric power of power supply No.3 (W)<br>Same data as diagnostic data (No.943) | Int32     | 500 msec                | NO                |     |
| 142 | DCW4           | DC electric power of power supply No.4 (W)<br>Same data as diagnostic data (No.944) | Int32     | 500 msec                | NO                |     |
| 143 | DCW5           | DC electric power of power supply No.5 (W)<br>Same data as diagnostic data (No.945) | Int32     | 500 msec                | NO                |     |
| 144 | DCW6           | DC electric power of power supply No.6 (W)<br>Same data as diagnostic data (No.946) | Int32     | 500 msec                | NO                |     |
| 145 | DCW7           | DC electric power of power supply No.7 (W)<br>Same data as diagnostic data (No.947) | Int32     | 500 msec                | NO                |     |
| 146 | DCW8           | DC electric power of power supply No.8 (W)<br>Same data as diagnostic data (No.948) | Int32     | 500 msec                | NO                |     |

|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | <b>A-40829-00001EN</b>                        |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 696/868 |

The following table lists the signals retrievable from laser oscillators produced by FANUC.

**Table 8. Retrievable Signals Listed By Laser Oscillator Series  
(YES: the signal is retrievable from the model)**

| #  | Type of Signal     | FANUC FIBER LASER series |                |            | FANUC LASER C series |                                       |
|----|--------------------|--------------------------|----------------|------------|----------------------|---------------------------------------|
|    |                    | 30i /31i-LB              | 30i /31i/32i-B | 0i-LF Plus | 30i /31i-LB          | 16i-LA, 16i /18i-LB, 16i-LPA, 16i-LPB |
| 01 | PwrCommand         | YES                      | YES (NOTE 2)   | YES        | YES                  | YES (16i-LPA cannot retrieve)         |
| 02 | PulseBottomCommand | YES                      | YES (NOTE 2)   | YES        |                      |                                       |
| 03 | FrequencyCommand   | YES                      | YES (NOTE 2)   | YES        | YES                  | YES (16i-LPA cannot retrieve)         |
| 04 | DutyCommand        | YES                      | YES (NOTE 2)   | YES        | YES                  | YES (16i-LPA cannot retrieve)         |
| 05 | LaserPwr           | YES                      | YES (NOTE 2)   | YES        | YES                  | YES (16i-LPA cannot retrieve)         |
| 06 | LaserSequence      | YES                      | YES            | YES        | YES                  | YES                                   |
| 07 | SigBeamOut         | YES                      | YES            | YES        | YES                  | YES                                   |
| 08 | SigLaserAlarm      | YES                      | YES            | YES        | YES                  | YES                                   |
| 09 | BCReflection       | YES                      | YES            | YES        |                      |                                       |
| 10 | WaterFlowRate1     | YES                      | YES            | YES        | YES                  |                                       |
| 11 | WaterFlowRate2     | YES                      | YES            | YES        | YES                  |                                       |
| 12 | LockerTemp         | YES                      | YES            | YES        |                      |                                       |
| 13 | LockerHumdity      | YES                      | YES            | YES        |                      |                                       |
| 14 | WaterTemp          | YES                      | YES            | YES        |                      |                                       |
| 15 | DewPoint           | YES                      | YES            | YES        |                      |                                       |
| 16 | LC1OutPwr          | YES                      | YES            | YES        |                      |                                       |
| 17 | LC2OutPwr          | YES                      | YES            | YES        |                      |                                       |
| 18 | LC3OutPwr          | YES                      | YES            | YES        |                      |                                       |
| 19 | LC4OutPwr          | YES                      | YES            | YES        |                      |                                       |
| 20 | LC5OutPwr          | YES                      | YES            | YES        |                      |                                       |
| 21 | LC6OutPwr          | YES                      | YES            | YES        |                      |                                       |
| 22 | LC7OutPwr          | YES                      | YES            | YES        |                      |                                       |
| 23 | LC8OutPwr          | YES                      | YES            | YES        |                      |                                       |
| 24 | LC9OutPwr          | YES                      | YES            | YES        |                      |                                       |
| 25 | LC10OutPwr         | YES                      | YES            | YES        |                      |                                       |
| 26 | LC11OutPwr         | YES                      | YES            | YES        |                      |                                       |
| 27 | LC12OutPwr         | YES                      | YES            | YES        |                      |                                       |
| 28 | FIBER1FFBD         | YES                      | YES            | YES        |                      |                                       |
| 29 | FIBER2FFBD         | YES                      | YES            | YES        |                      |                                       |
| 30 | FIBER3FFBD         | YES                      | YES            | YES        |                      |                                       |
| 31 | FIBER4FFBD         | YES                      | YES            | YES        |                      |                                       |
| 32 | FIBER5FFBD         | YES                      | YES            | YES        |                      |                                       |
| 33 | FIBER6FFBD         | YES                      | YES            | YES        |                      |                                       |
| 34 | PSU1-1Current      | YES                      | YES            | YES        |                      |                                       |
| 35 | PSU1-2Current      | YES                      | YES            | YES        |                      |                                       |
| 36 | PSU2-1Current      | YES                      | YES            | YES        |                      |                                       |
| 37 | PSU2-2Current      | YES                      | YES            | YES        |                      |                                       |
| 38 | PSU3-1Current      | YES                      | YES            | YES        |                      |                                       |
| 39 | PSU3-2Current      | YES                      | YES            | YES        |                      |                                       |
| 40 | PSU4-1Current      | YES                      | YES            | YES        |                      |                                       |
| 41 | PSU4-2Current      | YES                      | YES            | YES        |                      |                                       |

|      |               |        |                                    |                          |                                               |        |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|--------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |        |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |        |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |        |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |        |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |        | Sheet. 697/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |        |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    |                                               | Apprv. | Inoue          |

| #  | Type of Signal   | FANUC FIBER LASER series |                |            | FANUC LASER C series |                                       |
|----|------------------|--------------------------|----------------|------------|----------------------|---------------------------------------|
|    |                  | 30i /31i-LB              | 30i /31i/32i-B | 0i-LF Plus | 30i /31i-LB          | 16i-LA, 16i /18i-LB, 16i-LPA, 16i-LPB |
| 42 | PSU5-1Current    | YES                      | YES            | YES        |                      |                                       |
| 43 | PSU5-2Current    | YES                      | YES            | YES        |                      |                                       |
| 44 | PSU6-1Current    | YES                      | YES            | YES        |                      |                                       |
| 45 | PSU6-2Current    | YES                      | YES            | YES        |                      |                                       |
| 46 | PSU7-1Current    | YES                      | YES            | YES        |                      |                                       |
| 47 | PSU7-2Current    | YES                      | YES            | YES        |                      |                                       |
| 48 | PSU8-1Current    | YES                      | YES            | YES        |                      |                                       |
| 49 | PSU8-2Current    | YES                      | YES            | YES        |                      |                                       |
| 50 | PSU9-1Current    | YES                      | YES            | YES        |                      |                                       |
| 51 | PSU9-2Current    | YES                      | YES            | YES        |                      |                                       |
| 52 | PSU10-1Current   | YES                      | YES            | YES        |                      |                                       |
| 53 | PSU10-2Current   | YES                      | YES            | YES        |                      |                                       |
| 54 | PSU11-1Current   | YES                      | YES            | YES        |                      |                                       |
| 55 | PSU11-2Current   | YES                      | YES            | YES        |                      |                                       |
| 56 | PSU12-1Current   | YES                      | YES            | YES        |                      |                                       |
| 57 | PSU12-2Current   | YES                      | YES            | YES        |                      |                                       |
| 58 | PSU1-1Voltage    | YES                      | YES            | YES        |                      |                                       |
| 59 | PSU1-2Voltage    | YES                      | YES            | YES        |                      |                                       |
| 60 | PSU2-1Voltage    | YES                      | YES            | YES        |                      |                                       |
| 61 | PSU2-2Voltage    | YES                      | YES            | YES        |                      |                                       |
| 62 | PSU3-1Voltage    | YES                      | YES            | YES        |                      |                                       |
| 63 | PSU3-2Voltage    | YES                      | YES            | YES        |                      |                                       |
| 64 | PSU4-1Voltage    | YES                      | YES            | YES        |                      |                                       |
| 65 | PSU4-2Voltage    | YES                      | YES            | YES        |                      |                                       |
| 66 | PSU5-1Voltage    | YES                      | YES            | YES        |                      |                                       |
| 67 | PSU5-2Voltage    | YES                      | YES            | YES        |                      |                                       |
| 68 | PSU6-1Voltage    | YES                      | YES            | YES        |                      |                                       |
| 69 | PSU6-2Voltage    | YES                      | YES            | YES        |                      |                                       |
| 70 | PSU7-1Voltage    | YES                      | YES            | YES        |                      |                                       |
| 71 | PSU7-2Voltage    | YES                      | YES            | YES        |                      |                                       |
| 72 | PSU8-1Voltage    | YES                      | YES            | YES        |                      |                                       |
| 73 | PSU8-2Voltage    | YES                      | YES            | YES        |                      |                                       |
| 74 | PSU9-1Voltage    | YES                      | YES            | YES        |                      |                                       |
| 75 | PSU9-2Voltage    | YES                      | YES            | YES        |                      |                                       |
| 76 | PSU10-1Voltage   | YES                      | YES            | YES        |                      |                                       |
| 77 | PSU10-2Voltage   | YES                      | YES            | YES        |                      |                                       |
| 78 | PSU11-1Voltage   | YES                      | YES            | YES        |                      |                                       |
| 79 | PSU11-2Voltage   | YES                      | YES            | YES        |                      |                                       |
| 80 | PSU12-1Voltage   | YES                      | YES            | YES        |                      |                                       |
| 81 | PSU12-2Voltage   | YES                      | YES            | YES        |                      |                                       |
| 82 | PSU1HeatSinkTemp | YES                      | YES            | YES        |                      |                                       |
| 83 | PSU2HeatSinkTemp | YES                      | YES            | YES        |                      |                                       |
| 84 | PSU3HeatSinkTemp | YES                      | YES            | YES        |                      |                                       |
| 85 | PSU4HeatSinkTemp | YES                      | YES            | YES        |                      |                                       |
| 86 | PSU5HeatSinkTemp | YES                      | YES            | YES        |                      |                                       |
| 87 | PSU6HeatSinkTemp | YES                      | YES            | YES        |                      |                                       |
| 88 | PSU7HeatSinkTemp | YES                      | YES            | YES        |                      |                                       |
| 89 | PSU8HeatSinkTemp | YES                      | YES            | YES        |                      |                                       |
| 90 | PSU9HeatSinkTemp | YES                      | YES            | YES        |                      |                                       |

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               | Sheet. 698/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |

| #   | Type of Signal              | FANUC FIBER LASER series |                |            | FANUC LASER C series |                                       |
|-----|-----------------------------|--------------------------|----------------|------------|----------------------|---------------------------------------|
|     |                             | 30i /31i-LB              | 30i /31i/32i-B | 0i-LF Plus | 30i /31i-LB          | 16i-LA, 16i /18i-LB, 16i-LPA, 16i-LPB |
| 91  | PSU10HeatSinkTemp           | YES                      | YES            | YES        |                      |                                       |
| 92  | PSU11HeatSinkTemp           | YES                      | YES            | YES        |                      |                                       |
| 93  | PSU12HeatSinkTemp           | YES                      | YES            | YES        |                      |                                       |
| 94  | LaserRunTime                | YES<br>(NOTE 2)          | YES            | YES        |                      |                                       |
| 95  | BCBeamOnTime                | YES<br>(NOTE 2)          | YES            | YES        |                      |                                       |
| 96  | CuttingIndex1               | YES<br>(NOTE 2)          | YES            | YES        |                      |                                       |
| 97  | CuttingIndex2               | YES<br>(NOTE 2)          | YES            | YES        |                      |                                       |
| 98  | CuttingIndex3               | YES<br>(NOTE 2)          | YES            | YES        |                      |                                       |
| 99  | RetryTimes                  | YES<br>(NOTE 2)          | YES            | YES        |                      |                                       |
| 100 | PwrCalibrationCoefficient   |                          |                |            | YES                  | YES                                   |
| 101 | LaserTubePres               |                          |                |            | YES                  | YES                                   |
| 102 | LaserGasSupplyPres          |                          |                |            | YES                  |                                       |
| 103 | AddTimeAfterStartUp         |                          |                |            | YES                  | YES                                   |
| 104 | AddTimeForOpExPump          |                          |                |            | YES                  | YES                                   |
| 105 | LaserElectricPwrConsumption |                          |                |            | YES<br>(NOTE 3)      |                                       |
| 106 | IntPwrConsumption           |                          |                |            | YES<br>(NOTE 3)      |                                       |
| 107 | RFVoltage1                  |                          |                |            | YES                  | YES                                   |
| 108 | RFVoltage2                  |                          |                |            | YES                  | YES                                   |
| 109 | RFVoltage3                  |                          |                |            | YES                  | YES                                   |
| 110 | RFVoltage4                  |                          |                |            | YES                  | YES                                   |
| 111 | RFVoltage5                  |                          |                |            | YES                  | YES                                   |
| 112 | RFVoltage6                  |                          |                |            | YES                  | YES                                   |
| 113 | RFVoltage7                  |                          |                |            | YES                  | YES                                   |
| 114 | RFVoltage8                  |                          |                |            | YES                  | YES                                   |
| 115 | RFCurrent1                  |                          |                |            | YES                  | YES                                   |
| 116 | RFCurrent2                  |                          |                |            | YES                  | YES                                   |
| 117 | RFCurrent3                  |                          |                |            | YES                  | YES                                   |
| 118 | RFCurrent4                  |                          |                |            | YES                  | YES                                   |
| 119 | RFCurrent5                  |                          |                |            | YES                  | YES                                   |
| 120 | RFCurrent6                  |                          |                |            | YES                  | YES                                   |
| 121 | RFCurrent7                  |                          |                |            | YES                  | YES                                   |
| 122 | RFCurrent8                  |                          |                |            | YES                  | YES                                   |
| 123 | DCVoltage1                  |                          |                |            | YES                  | YES                                   |
| 124 | DCVoltage2                  |                          |                |            | YES                  | YES                                   |
| 125 | DCVoltage3                  |                          |                |            | YES                  | YES                                   |
| 126 | DCVoltage4                  |                          |                |            | YES                  | YES                                   |
| 127 | DCVoltage5                  |                          |                |            | YES                  | YES                                   |
| 128 | DCVoltage6                  |                          |                |            | YES                  | YES                                   |
| 129 | DCVoltage7                  |                          |                |            | YES                  | YES                                   |
| 130 | DCVoltage8                  |                          |                |            | YES                  | YES                                   |
| 131 | DCCurrent1                  |                          |                |            | YES                  | YES                                   |

|      |               |        |                                    |                          |                                               |        |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|--------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |        |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |        |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |        |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |        |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |        | Sheet. 699/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |        |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    |                                               | Apprv. | Inoue          |

| #   | Type of Signal | FANUC FIBER LASER series |                |            | FANUC LASER C series |                                       |
|-----|----------------|--------------------------|----------------|------------|----------------------|---------------------------------------|
|     |                | 30i /31i-LB              | 30i /31i/32i-B | 0i-LF Plus | 30i /31i-LB          | 16i-LA, 16i /18i-LB, 16i-LPA, 16i-LPB |
| 132 | DCCurrent2     |                          |                |            | YES                  | YES                                   |
| 133 | DCCurrent3     |                          |                |            | YES                  | YES                                   |
| 134 | DCCurrent4     |                          |                |            | YES                  | YES                                   |
| 135 | DCCurrent5     |                          |                |            | YES                  | YES                                   |
| 136 | DCCurrent6     |                          |                |            | YES                  | YES                                   |
| 137 | DCCurrent7     |                          |                |            | YES                  | YES                                   |
| 138 | DCCurrent8     |                          |                |            | YES                  | YES                                   |
| 139 | DCW1           |                          |                |            | YES                  | YES                                   |
| 140 | DCW2           |                          |                |            | YES                  | YES                                   |
| 141 | DCW3           |                          |                |            | YES                  | YES                                   |
| 142 | DCW4           |                          |                |            | YES                  | YES                                   |
| 143 | DCW5           |                          |                |            | YES                  | YES                                   |
| 144 | DCW6           |                          |                |            | YES                  | YES                                   |
| 145 | DCW7           |                          |                |            | YES                  | YES                                   |
| 146 | DCW8           |                          |                |            | YES                  | YES                                   |

The following table lists the signals retrievable from laser oscillators that are produced by other manufacturers.

**Table 9. Retrievable Signals Listed By Laser Oscillator Produced By Other Manufacturers  
(YES: the signal is retrievable from the model)**

| #  | Type of Signal     | Laser Oscillator Produced by Other Manufacturers |                  |            |                         |
|----|--------------------|--------------------------------------------------|------------------|------------|-------------------------|
|    |                    | 30i /31i-LB                                      | 30i /31i /32i -B | 0i-LF Plus | 16i-LA, 16i-LB, 16i-LPB |
| 01 | PwrCommand         | YES (NOTE 2, 3)                                  | YES (NOTE 2)     | YES        | YES                     |
| 02 | PulseBottomCommand | YES (NOTE 2)                                     | YES (NOTE 2)     | YES        |                         |
| 03 | FrequencyCommand   | YES (NOTE 2, 3)                                  | YES (NOTE 2)     | YES        | YES                     |
| 04 | DutyCommand        | YES (NOTE 2, 3)                                  | YES (NOTE 2)     | YES        | YES                     |
| 05 | LaserPwr           | YES (NOTE 2)                                     | YES (NOTE 2)     | YES        |                         |
| 06 | LaserSequence      |                                                  |                  |            |                         |
| 07 | SigBeamOut         | YES (NOTE 2, 3)                                  | YES              | YES        | YES                     |
| 08 | SigLaserAlarm      | YES (NOTE 2, 3)                                  | YES              | YES        | YES                     |
| 09 | BCReflection       | YES (NOTE 2)                                     | YES              | YES        |                         |

\* The signals numbered 10 or more on [Table 7](#) cannot be retrieved.

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               | Sheet. 700/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |



**NOTE**

- 1 In the case of 30*i* /31*i*-LB series, 30*i* /31*i* /32*i*-B series, each laser oscillator series supports the FanucCNC series shown below.

| Laser Oscillator         | Supported Series                                                                                |
|--------------------------|-------------------------------------------------------------------------------------------------|
| FANUC FIBER LASER series | G3C3 / G3D3 / G4C3 / G3C5 / G3D5 / G4B5 / G4C5 / G5L3 / G4L3 / G4M3 / G4N3 / G4P3 / G3L3 / G3M3 |
| FANUC LASER C series     | G3C2 / G3D2 / G4C2 / G3C1 / G3D1 / G4C1 / G3C4 / G3D4 / G4B4 / G4C4                             |

- 2 For the series supported by FANUC FIBER LASER series, signals are retrievable on the following versions.

| Series                                         | Versions    |
|------------------------------------------------|-------------|
| G3C3 / G3D3 / G4C3                             | 07 or later |
| G5L3 / G4L3 / G4M3 / G4N3 / G4P3 / G3L3 / G3M3 | 05 or later |
| G3C5 / G3D5 / G4B5 / G4C5                      | 01 or later |

- 3 For the series supported by FANUC LASER C series, signals are retrievable on the following versions.

| Series                                  | Versions    |
|-----------------------------------------|-------------|
| G3C2 / G3D2 / G4C2 / G3C1 / G3D1 / G4C1 | 02 or later |
| G3C4 / G3D4 / G4B4 / G4C4               | 01 or later |

- 4 Signals with "YES" (without any NOTE) indicate that signals are retrieval by all versions of the FanucCNC series supported by each laser oscillator series.

|      |               |        |                                            |                          |                                                       |        |       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|--------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |        |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |        |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |        |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |        |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |        |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |        |       | Sheet. 701/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |        |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    |                                                       | Apprv. | Inoue |                |

## 14.2 LIST OF OPCUAServer SIGNALS

### 14.2.1 Basic Signals of OPCUAServer

The table below lists basic signals retrievable from OPCUAServer type machines.

### Table 10. Basic Signals

| Signal Type   | Description                                                                                                                                            | Data Type | Reading Cycle (NOTE 1) (Default) | Enabled (Default) (NOTE 2) |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------|----------------------------|
| _OpcUaAlarm   | Alarm state of OPC UA Server<br>true: in alarm state (when OPC UA alarms are occurring)<br>false: no alarm (when no OPC UA alarms are occurring)       | Bool      | 500 msec                         | YES                        |
| _OpcUaWarning | Warning state of MTConnect<br>true: in warning state (when OPC UA warnings are occurring)<br>false: no warning (when no OPC UA warnings are occurring) | Bool      | 500 msec                         | YES                        |

## NOTE

- 1 Reading cycle cannot be changed for each signal. (Selectable cycle: 500 msec)
- 2 The following mandatory basic signals cannot be disabled.

| Mandatory Signal | Description      |
|------------------|------------------|
| _OpcUaAlarm      | State of alarm   |
| _OpcUaWarning    | State of warning |

The signals can be added that are associated with tags and methods for OPCUAServer type machines. For the details, refer to the **"Set Signal of OPCUAServer"** subsection.

In addition to above, the signals can be added that are associated with DA alarms to be collected from OPCUAServer type machines.

For the details, refer to the "**Set Alarm of OPCUAServer**" subsection.

|      |               |        |                                            |       |  |        |                   |                                                       |  |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  |  | Sheet. | 702/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |  |        |         |

# 14.3 LIST OF FanucRobot SIGNALS

## 14.3.1 Basic Signals of FanucRobot

The table below lists the basic signals retrievable from FanucRobot type machines.

MT-LINK*i* can retrieve signals from the following robots:

- R-30*i*B / R-30*i*B Mate / R-30*i*B Plus / R-30*i*B Mate Plus
- R-30*i*A / R-30*i*A Mate
- R-J3*i*B / R-J3*i*B Mate

**Table 11. Basic Signals**

| Signal Type   | Description                                                               | Data Type | Reading Cycle<br>(NOTE 1)<br><Default> | Enabled<br>(NOTE 2)<br><Default> |
|---------------|---------------------------------------------------------------------------|-----------|----------------------------------------|----------------------------------|
| REMOTE        | Remote signal                                                             | Bool      | 500 msec                               | NO                               |
| BUSY          | Operating signal                                                          | Bool      | 500 msec                               | NO                               |
| HELD          | Hold signal                                                               | Bool      | 500 msec                               | NO                               |
| FAULT         | Alarm signal                                                              | Bool      | 500 msec                               | NO                               |
| BATAL         | Battery trouble signal                                                    | Bool      | 500 msec                               | YES                              |
| TPENBL        | Teach pendant active signal                                               | Bool      | 500 msec                               | NO                               |
| Mode          | Mode                                                                      | String    | 500 msec                               | YES                              |
| RobotState    | States of Robot<br>"EMERGENCY"<br>"ALARM"<br>"OPERATE"<br>"STOP"          | String    | 500 msec                               | YES                              |
| Alarm         | Alarm message                                                             | String    | 500 msec                               | YES                              |
| ProgramName   | The name of the currently running program                                 | String    | 500 msec                               | YES                              |
| ProgramLineNo | The line number of a program the machine is currently processing (NOTE 3) | Int32     | 500 msec                               | NO                               |

### NOTE

- 1 The [Reading Cycle] column is non-editable (fixed to 500ms).
- 2 Whether to retrieve or not (enabled/disabled) can be specified for each signal. However, this retrieval setting cannot be disabled for the following mandatory signals.

| Mandatory Signal | Description                           |
|------------------|---------------------------------------|
| RobotState       | State of Robot                        |
| Alarm            | Alarm message                         |
| ProgramName      | The name of currently running program |

- 3 When a FanucRobot type machine processes the program within 500ms or less, this signal may not be retrieved.

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |       |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|-------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |       |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |       |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |       |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |       |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |       |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |       |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 703/868 |       |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |       |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         | Inoue |
|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |       |

## 14.3.2 Custom Signals of FanucRobot

The table below lists the custom signals retrievable from FanucRobot type machines.

**Table 12. Custom Signals**

| Signal Type            | Description                                                                                                                                                                                                               | Data Type       |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| I/O                    | EMG (Emergency stop signal) ( <b>NOTE 1, 2</b> )                                                                                                                                                                          | Bool            |
|                        | I/O (Input/Output signal)<br>General I/O<br>- Digital I/O DI / DO (*1)<br>Special I/O<br>- Peripheral (UOP) I/O UI / UO<br>- Operation Panel (SOP) I/O SI / SO<br>- Robot I/O RI / RO (*1)<br>- Welding I/O WI / WO / WSI | Bool            |
|                        | I/O (Input/Output signal)<br>General I/O<br>- Group I/O GI / GO<br>- Analog I/O AI / AO                                                                                                                                   | Int32           |
| DATA ( <b>NOTE 4</b> ) | NumReg (numeric register)                                                                                                                                                                                                 | Int32<br>Double |
|                        | CharReg (string register)                                                                                                                                                                                                 | String          |
|                        | PosReg (position register)<br>- Cartesian coordinates: X, Y, Z, W, P, R, E1, E2, E3<br>- Joint coordinates: J1, J2, J3, J4, J5, J6, JE1, JE2, JE3                                                                         | Double          |
|                        | PosReg (position register)<br>- Config: X, Y, Z, W, P, R, E1, E2, E3<br>- Coordinate system number: J1, J2, J3, J4, J5, J6, JE1, JE2, JE3                                                                                 | Int32           |

### NOTE

- 1 It is mandatory to set the EMC signal.
- 2 Assign a DO number (\*1) to the emergency stop signal on a robot controller so as not to conflict with the numbers that are used in FanucRobot programs.
- 3 Reading cycle cannot be changed for each signal. (Selectable cycle: 500 msec)
- 4 Numreg (numeric register) and CharReg (string register) can be added with the [Custom Singnal] tab page.  
PosReg (position register) can be added with the [Position Register] tab page.

\*1 For R-J3iB and R-J3iB Mate, read DI/DO as SDI/SDO, and RI/RO as RDI/RDO.

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 704/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue             |                                                       |  |        |         |

The following are the rules with which the [Equipment Setting] window assigns a signal definition to a status.

#### ■ IO

Status: IO\_<a>\_<Machine name>

Signal definition: <Machine name>\_IO\_<a>

<a>: the address of the IO signal (Address and Number)

\* Address: the name of an IO signal listed in table "Custom Signals" shown above.

\* Number: one of the following numbers determined in accordance with Address.

- DI (SDI) or DO (SDO): 1 to 5000
- RI (RDI), RO (RDO), GI, GO, AI, or AO: 1 to 1000
- UI or UO: 1 to 999
- SI or SO: 0 to 1000
- WI or WO: 1 to 400
- WSI: 1 to 600

Example) Machine name: Machine001; Address: SO; Number: 9

Status: IO\_SO9\_Machine001

Signal definition: Machine001\_IO\_SO9

#### ■ DATA (numeric register)

Status: DATA\_NumReg<n>\_<s>\_<Machine name>

Signal definition: <Machine name>\_DATA\_NumReg<n>

<n>: Register number (1 to 2000)

Example) Machine name: Machine002; Register number: 10

Status: DATA\_NumReg10\_Machine002

Signal definition: Machine002\_DATA\_NumReg10

#### ■ DATA (string register)

Status: DATA\_CharReg<n>\_<s>\_<Machine name>

Signal definition: <Machine name>\_DATA\_CharReg<n>

<n>: Register number (1 to 1000)

Example) Machine name: Machine003; Register number: 3

Status: DATA\_CharReg3\_Machine003

Signal definition: Machine003\_DATA\_CharReg3

#### ■ DATA (position register)

Status: DATA\_G<g>\_PosReg<n>\_<s>\_<Machine name>

Signal definition: <Machine name>\_DATA\_G<g>\_PosReg<n>\_<s>

<g>: Action group number (1 to 8)

<n>: Register number (1 to 100)

<s>: A signal suffix that indicates both the signal data to retrieve and its index in data array as shown in the following table.

| Array index | Cartesian coordinates | Joint coordinates | Config*1 | Coordinate system number*1 |
|-------------|-----------------------|-------------------|----------|----------------------------|
| 0           | X                     | J1                | conf1    | UF                         |
| 1           | Y                     | J2                | conf2    | UT                         |
| 2           | Z                     | J3                | conf3    | -                          |
| 3           | W                     | J4                | conf4    | -                          |
| 4           | P                     | J5                | TA1      | -                          |
| 5           | R                     | J6                | TA2      | -                          |
| 6           | E1                    | JE1               | TA3      | -                          |
| 7           | E2                    | JE2               | -        | -                          |
| 8           | E3                    | JE3               | -        | -                          |

|      |               |        |                                            |                          |                                                       |        |        |         |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |        | Sheet. | 705/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    |                                                       | Apprv. | Inoue  |         |

\*<sup>1</sup> For Config and coordinate system number, all signal data in a position register will be retrieved because there is no way to specify how many and which signal data to retrieve.

Example) Machine name: Machine004; Action group number: 1; Register number: 2; Number of Cartesian coordinates: 6; Number of Joint coordinates: 9

- Among six sets\*<sup>2</sup> of Cartesian coordinates, one with array index 0 will be defined as follows:

Status: DATA\_G1\_PosReg2\_X\_Machine004

Signal definition: Machine004\_DATA\_G1\_PosReg2\_X

\*<sup>2</sup> Six sets of Cartesian coordinates will be retrieved and defined each with a signal suffix from "X" to "R."

- Among nine sets\*<sup>3</sup> of Joint coordinates, one with array index 8 will be defined as follows:

Status: DATA\_G1\_PosReg2\_JE3\_Machine004

Signal definition: Machine004\_DATA\_G1\_PosReg2\_JE3

\*<sup>3</sup> Nine sets of Joint coordinates will be retrieved and defined each with a signal suffix from "J1" to "JE3."

- The fifth Config data will be defined as follows:

Status: DATA\_G1\_PosReg2\_TA1\_Machine004

Signal definition: Machine004\_DATA\_G1\_PosReg2\_TA1

- The first Coordinate system number data will be defined as follows:

Status: DATA\_G1\_PosReg2\_UF\_Machine004

Signal definition: Machine004\_DATA\_G1\_PosReg2\_UF

|      |               |        |                                    |                          |                                               |        |        |         |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |        | Sheet. | 706/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    |                                               | Apprv. | Inoue  |         |

|      |               |        |                                    |       |  |        |                   |                                               |  |        |         |
|------|---------------|--------|------------------------------------|-------|--|--------|-------------------|-----------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |                   |                                               |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |                   |                                               |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        | Draw              | A-40829-00001EN                               |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        |                   |                                               |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |                   |                                               |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        | FANUC CORPORATION |                                               |  | Sheet. | 707/868 |
| Ed.  | Date          | Design | Description                        |       |  |        |                   |                                               |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check |  | Apprv. |                   |                                               |  |        |         |

## 14.5 LIST OF FanucRobodrill SIGNALS

The following table lists the ROBODRILL specific signals that are retrievable from FanucRobodrill type machines.

| ROBODRILL Models                          |
|-------------------------------------------|
| FANUC ROBODRILL $\alpha$ -TiA Series      |
| FANUC ROBODRILL $\alpha$ -TiB Series      |
| FANUC ROBODRILL $\alpha$ -TiC Series      |
| FANUC ROBODRILL $\alpha$ -TiD Series      |
| FANUC ROBODRILL $\alpha$ -TiE Series      |
| FANUC ROBODRILL $\alpha$ -TiF Series      |
| FANUC ROBODRILL $\alpha$ -DiA Series      |
| FANUC ROBODRILL $\alpha$ -DiB Series      |
| FANUC ROBODRILL $\alpha$ -DiB Plus Series |

## NOTE

ROBODRILL Option license is required to retrieve signals from FanucRobodrill type machines.

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 708/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         |



### 14.5.1 Basic Signals of FanucRobodrill

For the basic signals that can be retrieved from FanucRobodrill type machines, refer to the column for Series 30i of "**Table 1. Basic Signals**" in the "**Basic Signals of FanucCNC**" section.

The following table lists, on a model-by-model basis, the basic signals retrievable from ROBODRILLs.

**Table 14. Basic Signals Retrievable From ROBODRILLs**  
(Signals retrievable from the model are marked with "YES")

| #  | Signal Type  | DiA Series<br>DiB Series<br>DiB Plus Series | TiE Series<br>TiF Series | TiA Series<br>TiB Series<br>TiC Series<br>TiD Series |
|----|--------------|---------------------------------------------|--------------------------|------------------------------------------------------|
| 01 | SigOP        | YES                                         | YES                      | YES                                                  |
| 02 | SigSTL       | YES                                         | YES                      | YES                                                  |
| 03 | SigSPL       | YES                                         | YES                      | YES                                                  |
| 04 | SigAL        | YES                                         | YES                      | YES                                                  |
| 05 | EMG          | YES                                         | YES                      | YES                                                  |
| 06 | Mode         | YES                                         | YES                      | YES                                                  |
| 07 | MainProgram  | YES                                         | YES                      | YES                                                  |
| 08 | ActProgram   | YES                                         | YES                      | YES                                                  |
| 09 | Sequence     | YES                                         | YES                      | YES                                                  |
| 10 | MainComment  | YES                                         | YES                      | YES                                                  |
| 11 | ActComment   | YES                                         | YES                      | YES                                                  |
| 12 | ActS         | YES                                         | YES                      | YES                                                  |
| 13 | ActF         | YES                                         | YES                      | YES                                                  |
| 14 | ActFdec      | YES                                         | YES                      | YES                                                  |
| 15 | ModalS       | YES                                         | YES                      | YES                                                  |
| 16 | ModalF       | YES                                         | YES                      | YES                                                  |
| 17 | ModalT       | YES                                         | YES                      | YES                                                  |
| 18 | ModalM       | YES                                         | YES                      | YES                                                  |
| 19 | ModalM2      | YES                                         | YES                      | YES                                                  |
| 20 | ModalM3      | YES                                         | YES                      | YES                                                  |
| 21 | PartsNum     | YES                                         | YES                      | YES                                                  |
| 22 | PartsNumAll  | YES                                         | YES                      | YES                                                  |
| 23 | PowOnTime    | YES                                         | YES                      | YES                                                  |
| 24 | RunTime      | YES                                         | YES                      | YES                                                  |
| 25 | CutTime      | YES                                         | YES                      | YES                                                  |
| 26 | SigCUT       | YES                                         | YES                      | YES                                                  |
| 27 | SigSBK       | YES                                         | YES                      | YES                                                  |
| 28 | SigDM00      | YES                                         | YES                      | YES                                                  |
| 29 | SigDM01      | YES                                         | YES                      | YES                                                  |
| 30 | SigMDRN      | YES                                         | YES                      | YES                                                  |
| 31 | Override     | YES                                         | YES                      | YES                                                  |
| 32 | SigINP       | YES                                         | YES                      | YES                                                  |
| 33 | AbsPos       | YES                                         | YES                      | YES                                                  |
| 34 | RelPos       | YES                                         | YES                      | YES                                                  |
| 35 | McnPos       | YES                                         | YES                      | YES                                                  |
| 36 | SpindleLoad  | YES                                         | YES                      | YES                                                  |
| 37 | SigENB       | YES                                         | YES                      | YES                                                  |
| 38 | SpindleSpeed | YES                                         | YES                      | YES                                                  |
| 39 | SpindleTemp  | YES                                         | YES                      | YES                                                  |
| 40 | ServoSpeed   | YES                                         | YES                      | YES                                                  |
| 41 | ServoTemp    | YES                                         | YES                      | YES                                                  |

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 709/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         |

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |       |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|-------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |       |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |       |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |       |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |       |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |       |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |       |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 710/868 |       |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |       |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         | Inoue |



## 14.5.3 ROBODRILL Specific Signal

### 14.5.3.1 Robodrill Signals of FanucRobodrill

The table below lists the Robodrill signals that can be retrieved from FanucRobodrill type machines. Refer to [Table 16](#) for retrievable signals on a model-by-model basis of ROBODRILL.

**Table 15. Robodrill Signals**

| #  | Signal Type                       | Description                                                 | Data Type | Reading Cycle (NOTE 1) | Enabled (NOTE 2) |
|----|-----------------------------------|-------------------------------------------------------------|-----------|------------------------|------------------|
| 01 | RobodrillToolNum                  | Robodrill tool number                                       | Int32     | 500 msec               | YES              |
| 02 | RobodrillMachiningMode            | Robodrill machining mode                                    | String    | 500 msec               | NO               |
| 03 | RobodrillEsaveMode                | Robodrill power saving mode                                 | Bool      | 500 msec               | NO               |
| 04 | RobodrillRigidTapMode             | Robodrill rigid tapping mode                                | String    | 500 msec               | NO               |
| 05 | RobodrillTableLoad                | Robodrill table load mode                                   | String    | 60 sec                 | NO               |
| 06 | RobodrillToolLoad                 | Robodrill tool load mode                                    | String    | 60 sec                 | NO               |
| 07 | RobodrillCounterNum1              | Robodrill production counter 1                              | Int32     | 10 sec                 | YES              |
| 08 | RobodrillCounterTargetNum1        | Robodrill production target 1                               | Int32     | 60 sec                 | NO               |
| 09 | RobodrillCounterPartName1         | Robodrill counter part name 1                               | String    | 60 sec                 | NO               |
| 10 | RobodrillCounterNum2              | Robodrill production counter 2                              | Int32     | 10 sec                 | YES              |
| 11 | RobodrillCounterTargetNum2        | Robodrill production target 2                               | Int32     | 60 sec                 | NO               |
| 12 | RobodrillCounterPartName2         | Robodrill counter part name 2                               | String    | 60 sec                 | NO               |
| 13 | RobodrilliHMICounterNum1          | Robodrill iHMI production counter 1                         | Int32     | 10 sec                 | NO               |
| 14 | RobodrilliHMICounterTargetNum1    | Robodrill iHMI production target 1                          | Int32     | 60 sec                 | NO               |
| 15 | RobodrilliHMICounterPartName1     | Robodrill iHMI counter part name 1                          | String    | 60 sec                 | NO               |
| 16 | RobodrilliHMICounterNum2          | Robodrill iHMI production counter 2                         | Int32     | 10 sec                 | NO               |
| 17 | RobodrilliHMICounterTargetNum2    | Robodrill iHMI production target 2                          | Int32     | 60 sec                 | NO               |
| 18 | RobodrilliHMICounterPartName2     | Robodrill iHMI counter part name 2                          | String    | 60 sec                 | NO               |
| 19 | RobodrilliHMICounterNum3          | Robodrill iHMI production counter 3                         | Int32     | 10 sec                 | NO               |
| 20 | RobodrilliHMICounterTargetNum3    | Robodrill iHMI production target 3                          | Int32     | 60 sec                 | NO               |
| 21 | RobodrilliHMICounterPartName3     | Robodrill iHMI counter part name 3                          | String    | 60 sec                 | NO               |
| 22 | RobodrillProgramProtection        | Robodrill program protection                                | Bool      | 10 sec                 | NO               |
| 23 | RobodrillOperationLevel           | Robodrill operation level                                   | Int32     | 10 sec                 | NO               |
| 24 | RobodrillElectricPowerConsumption | Robodrill electric power consumption -99.999 to 99.999 (kW) | Double    | 500 msec (NOTE 1)      | NO               |
| 25 | ThermalCompEnableX                | X-axis thermal displacement compensation enable             | Bool      | 60 sec                 | NO               |
| 26 | ThermalCompEnableY                | Y-axis thermal displacement compensation enable             | Bool      | 60 sec                 | NO               |
| 27 | ThermalCompEnableZ                | Z-axis thermal displacement compensation enable             | Bool      | 60 sec                 | NO               |
| 28 | ThermalCompAdjValX                | X-axis thermal displacement compensation adjust value       | Int32     | 60 sec                 | NO               |
| 29 | ThermalCompAdjValY                | Y-axis thermal displacement compensation adjust value       | Int32     | 60 sec                 | NO               |

|      |               |        |                                    |                   |                                               |                |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |                   |                                               |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw              | <b>A-40829-00001EN</b>                        |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |                   |                                               |                |
| Ed.  | Date          | Design | Description                        | FANUC CORPORATION |                                               | Sheet. 712/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             | Apprv.                                        | Inoue          |

| #  | Signal Type         | Description                                             | Data Type | Reading Cycle (NOTE 1) | Enabled (NOTE 2) |
|----|---------------------|---------------------------------------------------------|-----------|------------------------|------------------|
| 30 | ThermalCompAdjValZ  | Z-axis thermal displacement compensation adjust value   | Int32     | 60 sec                 | NO               |
| 31 | ThermalCompSavePosX | X-axis thermal displacement compensation save position  | Int32     | 60 sec                 | NO               |
| 32 | ThermalCompSavePosY | Y-axis thermal displacement compensation save position  | Int32     | 60 sec                 | NO               |
| 33 | ThermalCompSavePosZ | Z-axis thermal displacement compensation save position  | Int32     | 60 sec                 | NO               |
| 34 | ThermalCompValX     | X-axis thermal displacement compensation value (NOTE 4) | Int32     | 60 sec                 | NO               |
| 35 | ThermalCompValY     | Y-axis thermal displacement compensation value (NOTE 4) | Int32     | 60 sec                 | NO               |
| 36 | ThermalCompValZ     | Z-axis thermal displacement compensation value (NOTE 4) | Int32     | 60 sec                 | NO               |
| 37 | CoolantDensity      | Coolant density                                         | Double    | 60 sec                 | NO               |
| 38 | CoolantDensityType  | Coolant density type                                    | String    | 60 sec                 | NO               |
| 39 | CoolantPH           | Coolant pH                                              | Double    | 60 sec                 | NO               |
| 40 | CoolantTemp         | Coolant temperature                                     | Double    | 60 sec                 | NO               |

#### NOTE

- 1 Reading cycle cannot be changed for each signal. (Selectable cycles: 500 msec, 10 sec, and 60 sec)
- 2 Whether to retrieve or not (enabled/disabled) can be specified for each signal.
- 3 Retrieving the data regarding Robodrill electric power consumption from  $\alpha$ -DiA Series models may take longer time than from other models.
- 4 Estimated thermal displacement of the machine can be retrieved only when recording history option is enabled on the CNC operation screen.

The following table lists the signals retrievable from ROBODRILL models.

**Table 16. Retrievable Signals Listed By ROBODRILL Model**  
(Signals retrievable from the model are marked with "YES")

| #  | Signal Type                    | DiB Plus Series | DiB Series | DiA Series | TiF Series | TiE Series | TiD Series | TiC Series | TiB Series | TiA Series |
|----|--------------------------------|-----------------|------------|------------|------------|------------|------------|------------|------------|------------|
| 01 | RobodrillToolNum               | YES             | YES        | YES        | YES        | YES        | YES        | YES        | YES        | YES        |
| 02 | RobodrillMachiningMode         | YES             | YES        | YES        | YES        |            |            |            |            |            |
| 03 | RobodrillEsaveMode             | YES             | YES        | YES        |            |            |            |            |            |            |
| 04 | RobodrillRigidTapMode          | YES             | YES        | YES        | YES        |            |            |            |            |            |
| 05 | RobodrillTableLoad             | YES             | YES        | YES        | YES        | YES        |            |            |            |            |
| 06 | RobodrillToolLoad              | YES             | YES        | YES        | YES        | YES        |            |            |            |            |
| 07 | RobodrillCounterNum1           | YES             | YES        | YES        | YES        | YES        | YES        | YES        | YES        | YES        |
| 08 | RobodrillCounterTargetNum1     | YES             | YES        | YES        | YES        | YES        | YES        | YES        | YES        | YES        |
| 09 | RobodrillCounterPartName1      | YES             | YES        | YES        | YES        | YES        |            |            |            |            |
| 10 | RobodrillCounterNum2           | YES             | YES        | YES        | YES        | YES        | YES        | YES        | YES        | YES        |
| 11 | RobodrillCounterTargetNum2     | YES             | YES        | YES        | YES        | YES        | YES        | YES        | YES        | YES        |
| 12 | RobodrillCounterPartName2      | YES             | YES        | YES        | YES        | YES        |            |            |            |            |
| 13 | RobodrilliHMICounterNum1       | YES             | YES        |            |            |            |            |            |            |            |
| 14 | RobodrilliHMICounterTargetNum1 | YES             | YES        |            |            |            |            |            |            |            |

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 713/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         |

| #  | Signal Type                       | DiB Plus Series | DiB Series | DiA Series | TiF Series | TiE Series | TiD Series | TiC Series | TiB Series | TiA Series |
|----|-----------------------------------|-----------------|------------|------------|------------|------------|------------|------------|------------|------------|
| 15 | RobodrilliHMICounterPartName1     | YES             | YES        |            |            |            |            |            |            |            |
| 16 | RobodrilliHMICounterNum2          | YES             | YES        |            |            |            |            |            |            |            |
| 17 | RobodrilliHMICounterTargetNum2    | YES             | YES        |            |            |            |            |            |            |            |
| 18 | RobodrilliHMICounterPartName2     | YES             | YES        |            |            |            |            |            |            |            |
| 19 | RobodrilliHMICounterNum3          | YES             | YES        |            |            |            |            |            |            |            |
| 20 | RobodrilliHMICounterTargetNum3    | YES             | YES        |            |            |            |            |            |            |            |
| 21 | RobodrilliHMICounterPartName3     | YES             | YES        |            |            |            |            |            |            |            |
| 22 | RobodrillProgramProtection        |                 |            | YES        | YES        | YES        | YES        | YES        | YES        | YES        |
| 23 | RobodrillOperationLevel           | YES             | YES        |            |            |            |            |            |            |            |
| 24 | RobodrillElectricPowerConsumption | YES             | YES        | YES        |            |            |            |            |            |            |
| 25 | ThermalCompEnableX                | YES             | YES        | YES        | YES        | YES        | YES*       |            |            |            |
| 26 | ThermalCompEnableY                | YES             | YES        | YES        | YES        | YES        | YES*       |            |            |            |
| 27 | ThermalCompEnableZ                | YES             | YES        | YES        | YES        | YES        | YES        | YES        | YES        |            |
| 28 | ThermalCompAdjValX                | YES             | YES        | YES        | YES*       | YES*       |            |            |            |            |
| 29 | ThermalCompAdjValY                | YES             | YES        | YES        | YES*       | YES*       |            |            |            |            |
| 30 | ThermalCompAdjValZ                | YES             | YES        | YES        | YES*       | YES*       |            |            |            |            |
| 31 | ThermalCompSavePosX               | YES             | YES        | YES        | YES        | YES        | YES*       |            |            |            |
| 32 | ThermalCompSavePosY               | YES             | YES        | YES        | YES        | YES        | YES*       |            |            |            |
| 33 | ThermalCompSavePosZ               | YES             | YES        | YES        | YES        | YES        | YES        | YES        | YES        |            |
| 34 | ThermalCompValX                   | YES             | YES        | YES        | YES*       | YES*       |            |            |            |            |
| 35 | ThermalCompValY                   | YES             | YES        | YES        | YES*       | YES*       |            |            |            |            |
| 36 | ThermalCompValZ                   | YES             | YES        | YES        | YES*       | YES*       |            |            |            |            |
| 37 | CoolantDensity                    | YES*            | YES*       |            |            |            |            |            |            |            |
| 38 | CoolantDensityType                | YES*            | YES*       |            |            |            |            |            |            |            |
| 39 | CoolantPH                         | YES*            | YES*       |            |            |            |            |            |            |            |
| 40 | CoolantTemp                       | YES*            | YES*       |            |            |            |            |            |            |            |

\* Signals may not be retrieved depending on the software version.

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 714/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue             |                                                       |  |        |         |

### 14.5.3.2 Tool Life Signals of FanucRobodrill

The following table lists the Robodrill tool life signals that can be retrieved from FanucRobodrill type machines. Refer to [Table 18](#) for retrievable signals listed on a model-by-model basis.

**Table 17. Tool Life Signals**

| Signal Type           | Description                            | Data Type | Reading Cycle (NOTE 1) [Default] | Enabled (NOTE 2) [Default] | Remarks                               |
|-----------------------|----------------------------------------|-----------|----------------------------------|----------------------------|---------------------------------------|
| ToolGroupNum          | Current tool group                     | Int32     | 500 msec                         | YES                        | -                                     |
| ToolLifeCount<m>      | Tool life count of tool group <m>      | Int32     | 10 sec                           | YES                        | <m>:<br>tool group num.<br>(01 to 16) |
| ToolLifeCount<n>      | Tool life count of tool group <n>      | Int32     | 10 sec                           | NO                         | <n>:<br>tool group num.<br>(17 to 32) |
| ToolLifeLimit<m>      | Tool life limit of tool group <m>      | Int32     | 10 sec                           | YES                        | <m>:<br>tool group num.<br>(01 to 16) |
| ToolLifeLimit<n>      | Tool life limit of tool group <n>      | Int32     | 10 sec                           | NO                         | <n>:<br>tool group num.<br>(17 to 32) |
| ToolLifeForenotice<m> | Tool life forenotice of tool group <m> | Int32     | 10 sec                           | YES                        | <m>:<br>tool group num.<br>(01 to 16) |
| ToolLifeForenotice<n> | Tool life forenotice of tool group <n> | Int32     | 10 sec                           | NO                         | <n>:<br>tool group num.<br>(17 to 32) |
| ToolLifeCountType<m>  | Count type of tool group <m>           | String    | 10 sec                           | YES                        | <m>:<br>tool group num.<br>(01 to 16) |
| ToolLifeCountType<n>  | Count type of tool group <n>           | String    | 10 sec                           | NO                         | <n>:<br>tool group num.<br>(17 to 32) |

#### NOTE

- 1 Reading cycle cannot be changed for each signal. (Selectable cycles: 500 msec, 10 sec, and 60 sec)
- 2 Whether to retrieve or not (enabled/disabled) can be specified for each signal.
- 3 The values of "Tool life count," "Tool life limit," and "Tool life forenotice" are displayed without a decimal point.
- 4 While tool life information managed by Tool Life Management function can be retrieved, those managed by Tool Management function cannot be retrieved.

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               | Sheet. 715/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |

The following table lists the tool life signals retrievable from ROBODRILL models.

**Table 18. Retrievable Tool Life Signals Listed By ROBODRILL Model**  
(Signals retrievable from the model are marked with "YES")

| Signal Type           | DiB Plus Series | DiB Series | DiA Series | TiF Series | TiE Series | TiD Series | TiC Series | TiB Series | TiA Series | Remarks                               |
|-----------------------|-----------------|------------|------------|------------|------------|------------|------------|------------|------------|---------------------------------------|
| ToolGroupNum          | YES             | YES        | YES        | YES        | YES        | YES        | YES        | YES        | YES        | —                                     |
| ToolLifeCount<m>      | YES             | YES        | YES        | YES        | YES        | YES        | YES        | YES        | YES        | <m>:<br>tool group num.<br>(01 to 14) |
| ToolLifeCount<n>      | YES             | YES        | YES        | YES        | YES        | YES        | YES        |            |            | <n>:<br>tool group num.<br>(15 to 32) |
| ToolLifeLimit<m>      | YES             | YES        | YES        | YES        | YES        | YES        | YES        | YES        | YES        | <m>:<br>tool group num.<br>(01 to 14) |
| ToolLifeLimit<n>      | YES             | YES        | YES        | YES        | YES        | YES        | YES        |            |            | <n>:<br>tool group num.<br>(15 to 32) |
| ToolLifeForenotice<m> | YES             | YES        | YES        | YES        | YES        | YES        | YES        | YES        | YES        | <m>:<br>tool group num.<br>(01 to 14) |
| ToolLifeForenotice<n> | YES             | YES        | YES        | YES        | YES        | YES        | YES        |            |            | <n>:<br>tool group num.<br>(15 to 32) |
| ToolLifeCountType<m>  | YES             | YES        | YES        | YES        | YES        | YES        | YES        | YES        | YES        | <m>:<br>tool group num.<br>(01 to 14) |
| ToolLifeCountType<n>  | YES             | YES        | YES        | YES        | YES        | YES        | YES        |            |            | <n>:<br>tool group num.<br>(15 to 32) |

|      |               |        |                                    |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 716/868 |
| Ed.  | Date          | Design | Description                        |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check |  | Apprv. |                   |                                                       |  |        |         |



### 14.5.3.3 Periodic Maintenance Signals of FanucRobodrill

The following table lists the Robodrill periodic maintenance signals that can be retrieved from FanucRobodrill type machines.

Refer to [Table 20](#) for retrievable signals listed on a model-by-model basis.

**Table 19. Periodic Maintenance Signals**

| #  | Signal Type                 | Description                                    | Data Type | Reading Cycle (NOTE 1) | Enabled (NOTD 2) |
|----|-----------------------------|------------------------------------------------|-----------|------------------------|------------------|
| 01 | SpindleGearLub              | Spindle and Turret gear lubrication            | Int32     | 60 sec                 | NO               |
| 02 | SpindleGearLubInterval      | Spindle and Turret gear lubrication interval   | Int32     | 60 sec                 | NO               |
| 03 | SpindleGearLubForenotice    | Spindle and Turret gear lubrication forenotice | Int32     | 60 sec                 | NO               |
| 04 | TurretCamLub                | Turret and UC cam lubrication                  | Int32     | 60 sec                 | NO               |
| 05 | TurretCamLubInterval        | Turret and UC cam lubrication interval         | Int32     | 60 sec                 | NO               |
| 06 | TurretCamLubForenotice      | Turret and UC cam lubrication forenotice       | Int32     | 60 sec                 | NO               |
| 07 | LMGuideLub                  | LM guide lubrication                           | Int32     | 60 sec                 | NO               |
| 08 | LMGuideLubInterval          | LM guide lubrication interval                  | Int32     | 60 sec                 | NO               |
| 09 | LMGuideLubForenotice        | LM guide lubrication forenotice                | Int32     | 60 sec                 | NO               |
| 10 | FlexibleHoseChk             | Flexible hose check                            | Int32     | 60 sec                 | NO               |
| 11 | FlexibleHoseChkInterval     | Flexible hose check interval                   | Int32     | 60 sec                 | NO               |
| 12 | FlexibleHoseChkForenotice   | Flexible hose check forenotice                 | Int32     | 60 sec                 | NO               |
| 13 | FireExConstChk              | Fire-extinguisher visual check                 | Int32     | 60 sec                 | NO               |
| 14 | FireExConstChkInterval      | Fire-extinguisher visual check interval        | Int32     | 60 sec                 | NO               |
| 15 | FireExConstChkForenotice    | Fire-extinguisher visual check forenotice      | Int32     | 60 sec                 | NO               |
| 16 | FireExFuncChk               | Fire-extinguisher functional check             | Int32     | 60 sec                 | NO               |
| 17 | FireExFuncChkInterval       | Fire-extinguisher functional check interval    | Int32     | 60 sec                 | NO               |
| 18 | FireExFuncChkForenotice     | Fire-extinguisher functional check forenotice  | Int32     | 60 sec                 | NO               |
| 19 | ConUnitFanChk               | Fan motor check                                | Int32     | 60 sec                 | NO               |
| 20 | ConUnitFanChkInterval       | Fan motor check interval                       | Int32     | 60 sec                 | NO               |
| 21 | ConUnitFanChkForenotice     | Fan motor check forenotice                     | Int32     | 60 sec                 | NO               |
| 22 | ACSpindleMotorChk           | AC spindle motor check                         | Int32     | 60 sec                 | NO               |
| 23 | ACSpindleMotorChkInterval   | AC spindle motor check interval                | Int32     | 60 sec                 | NO               |
| 24 | ACSpindleMotorChkForenotice | AC spindle motor check forenotice              | Int32     | 60 sec                 | NO               |
| 25 | AirFilterChk                | Air filter and regulator check                 | Int32     | 60 sec                 | NO               |
| 26 | AirFilterChkInterval        | Air filter and regulator check interval        | Int32     | 60 sec                 | NO               |
| 27 | AirFilterChkForenotice      | Air filter and regulator check forenotice      | Int32     | 60 sec                 | NO               |
| 28 | CTCPartChk                  | Center-through part check                      | Int32     | 60 sec                 | NO               |
| 29 | CTCPartChkInterval          | Center-through part check interval             | Int32     | 60 sec                 | NO               |
| 30 | CTCPartChkForenotice        | Center-through part check forenotice           | Int32     | 60 sec                 | NO               |
| 31 | DoorCylinderChk             | Door cylinder check                            | Int32     | 60 sec                 | NO               |
| 32 | DoorCylinderChkInterval     | Door cylinder check interval                   | Int32     | 60 sec                 | NO               |
| 33 | DoorCylinderChkForenotice   | Door cylinder check forenotice                 | Int32     | 60 sec                 | NO               |
| 34 | FrontDoorPadChk             | Front door rubber and roller check             | Int32     | 60 sec                 | NO               |
| 35 | FrontDoorPadChkInterval     | Front door rubber and roller check interval    | Int32     | 60 sec                 | NO               |
| 36 | FrontDoorPadChkForenotice   | Front door rubber and roller check             | Int32     | 60 sec                 | NO               |
| 37 | DryerChk                    | Dryer check                                    | Int32     | 60 sec                 | NO               |
| 38 | DryerChkInterval            | Dryer check interval                           | Int32     | 60 sec                 | NO               |
| 39 | DryerChkForenotice          | Dryer check forenotice                         | Int32     | 60 sec                 | NO               |
| 40 | AbsPulseCodaBtrReplace      | Pulse coder battery replacement                | Int32     | 60 sec                 | NO               |

|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | <b>A-40829-00001EN</b>                        |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 717/868 |

| #  | Signal Type                                | Description                                         | Data Type | Reading Cycle (NOTE 1) | Enabled (NOTD 2) |
|----|--------------------------------------------|-----------------------------------------------------|-----------|------------------------|------------------|
| 41 | AbsPulseCodaBtrReplaceInterval             | Pulse coder battery replacement interval            | Int32     | 60 sec                 | NO               |
| 42 | AbsPulseCodaBtrReplaceForenotice           | Pulse coder battery replacement forenotice          | Int32     | 60 sec                 | NO               |
| 43 | CNCmemBtrReplace                           | CNC memory backup battery replacement               | Int32     | 60 sec                 | NO               |
| 44 | CNCmemBtrReplaceInterval                   | CNC memory backup battery replacement interval      | Int32     | 60 sec                 | NO               |
| 45 | CNCmemBtrReplaceForenotice                 | CNC memory backup battery replacement forenotice    | Int32     | 60 sec                 | NO               |
| 46 | InternalLightReplace                       | Internal light replacement                          | Int32     | 60 sec                 | NO               |
| 47 | InternalLightReplaceInterval               | Internal light replacement interval                 | Int32     | 60 sec                 | NO               |
| 48 | InternalLightReplaceForenotice             | Internal light replacement forenotice               | Int32     | 60 sec                 | NO               |
| 49 | XaxisCoverReplace                          | X-cover cushion rubber replacement                  | Int32     | 60 sec                 | NO               |
| 50 | XaxisCoverReplaceInterval                  | X-cover cushion rubber replacement interval         | Int32     | 60 sec                 | NO               |
| 51 | XaxisCoverReplaceForenotice                | X-cover cushion rubber replacement forenotice       | Int32     | 60 sec                 | NO               |
| 52 | YaxisCoverReplace                          | Y-cover cushion rubber replacement                  | Int32     | 60 sec                 | NO               |
| 53 | YaxisCoverReplaceInterval                  | Y-cover cushion rubber replacement interval         | Int32     | 60 sec                 | NO               |
| 54 | YaxisCoverReplaceForenotice                | Y-cover cushion rubber replacement forenotice       | Int32     | 60 sec                 | NO               |
| 55 | DrawBarUnitReplace                         | Draw bar unit replacement                           | Int32     | 60 sec                 | NO               |
| 56 | DrawBarUnitReplaceInterval                 | Draw bar unit replacement interval                  | Int32     | 60 sec                 | NO               |
| 57 | DrawBarUnitReplaceForenotice               | Draw bar unit replacement forenotice                | Int32     | 60 sec                 | NO               |
| 58 | FrontDoorPadReplace                        | Front door rubber and roller replacement            | Int32     | 60 sec                 | NO               |
| 59 | FrontDoorPadReplaceInterval                | Front door rubber and roller replacement interval   | Int32     | 60 sec                 | NO               |
| 60 | FrontDoorPadReplaceForenotice              | Front door rubber and roller replacement forenotice | Int32     | 60 sec                 | NO               |
| 61 | CTCPartReplace                             | Center-through part replacement                     | Int32     | 60 sec                 | NO               |
| 62 | CTCPartReplaceInterval                     | Center-through part replacement interval            | Int32     | 60 sec                 | NO               |
| 63 | CTCPartReplaceForenotice                   | Center-through part replacement forenotice          | Int32     | 60 sec                 | NO               |
| 64 | DoorCylinderReplace                        | Door cylinder replacement                           | Int32     | 60 sec                 | NO               |
| 65 | DoorCylinderReplaceInterval                | Door cylinder replacement interval                  | Int32     | 60 sec                 | NO               |
| 66 | DoorCylinderReplaceForenotice              | Door cylinder replacement forenotice                | Int32     | 60 sec                 | NO               |
| 67 | DDR4thAxisRotaryJointSeal                  | DDR 4th axis rotary joint seal                      | Int32     | 60 sec                 | NO               |
| 68 | DDR4thAxisRotaryJointSealInterval          | DDR 4th axis rotary joint seal interval             | Int32     | 60 sec                 | NO               |
| 69 | DDR4thAxisRotaryJointSealForenotice        | DDR 4th axis rotary joint seal forenotice           | Int32     | 60 sec                 | NO               |
| 70 | DDR4thAxisSupportRotaryJointSeal           | DDR 4th axis support rotary joint seal              | Int32     | 60 sec                 | NO               |
| 71 | DDR4thAxisSupportRotaryJointSealInterval   | DDR 4th axis support rotary joint seal interval     | Int32     | 60 sec                 | NO               |
| 72 | DDR4thAxisSupportRotaryJointSealForenotice | DDR 4th axis support rotary joint seal forenotice   | Int32     | 60 sec                 | NO               |
| 73 | DDR5thAxisRotaryJointSeal                  | DDR 5th axis rotary joint seal                      | Int32     | 60 sec                 | NO               |
| 74 | DDR5thAxisRotaryJointSealInterval          | DDR 5th axis rotary joint seal interval             | Int32     | 60 sec                 | NO               |
| 75 | DDR5thAxisRotaryJointSealForenotice        | DDR 5th axis rotary joint seal forenotice           | Int32     | 60 sec                 | NO               |

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               | Sheet. 718/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |

| #   | Signal Type                                | Description                                       | Data Type | Reading Cycle (NOTE 1) | Enabled (NOTD 2) |
|-----|--------------------------------------------|---------------------------------------------------|-----------|------------------------|------------------|
| 76  | DDR5thAxisSupportRotaryJointSeal           | DDR 5th axis support rotary joint seal            | Int32     | 60 sec                 | NO               |
| 77  | DDR5thAxisSupportRotaryJointSealInterval   | DDR 5th axis support rotary joint seal interval   | Int32     | 60 sec                 | NO               |
| 78  | DDR5thAxisSupportRotaryJointSealForenotice | DDR 5th axis support rotary joint seal forenotice | Int32     | 60 sec                 | NO               |
| 79  | CustomMaint1Name                           | Custom periodic maintenance name 1                | String    | 60 sec                 | NO               |
| 80  | CustomMaint1                               | Custom periodic maintenance 1                     | Int32     | 60 sec                 | NO               |
| 81  | CustomMaint1Interval                       | Custom periodic maintenance interval 1            | Int32     | 60 sec                 | NO               |
| 82  | CustomMaint1Forenotice                     | Custom periodic maintenance forenotice 1          | Int32     | 60 sec                 | NO               |
| 83  | CustomMaint1CountType                      | Custom periodic maintenance count type 1          | String    | 60 sec                 | NO               |
| 84  | CustomMaint2Name                           | Custom periodic maintenance name 2                | String    | 60 sec                 | NO               |
| 85  | CustomMaint2                               | Custom periodic maintenance 2                     | Int32     | 60 sec                 | NO               |
| 86  | CustomMaint2Interval                       | Custom periodic maintenance interval 2            | Int32     | 60 sec                 | NO               |
| 87  | CustomMaint2Forenotice                     | Custom periodic maintenance forenotice 2          | Int32     | 60 sec                 | NO               |
| 88  | CustomMaint2CountType                      | Custom periodic maintenance count type 2          | String    | 60 sec                 | NO               |
| 89  | CustomMaint3Name                           | Custom periodic maintenance name 3                | String    | 60 sec                 | NO               |
| 90  | CustomMaint3                               | Custom periodic maintenance 3                     | Int32     | 60 sec                 | NO               |
| 91  | CustomMaint3Interval                       | Custom periodic maintenance interval 3            | Int32     | 60 sec                 | NO               |
| 92  | CustomMaint3Forenotice                     | Custom periodic maintenance forenotice 3          | Int32     | 60 sec                 | NO               |
| 93  | CustomMaint3CountType                      | Custom periodic maintenance count type 3          | String    | 60 sec                 | NO               |
| 94  | CustomMaint4Name                           | Custom periodic maintenance name 4                | String    | 60 sec                 | NO               |
| 95  | CustomMaint4                               | Custom periodic maintenance 4                     | Int32     | 60 sec                 | NO               |
| 96  | CustomMaint4Interval                       | Custom periodic maintenance interval 4            | Int32     | 60 sec                 | NO               |
| 97  | CustomMaint4Forenotice                     | Custom periodic maintenance forenotice 4          | Int32     | 60 sec                 | NO               |
| 98  | CustomMaint4CountType                      | Custom periodic maintenance count type 4          | String    | 60 sec                 | NO               |
| 99  | CustomMaint5Name                           | Custom periodic maintenance name 5                | String    | 60 sec                 | NO               |
| 100 | CustomMaint5                               | Custom periodic maintenance 5                     | Int32     | 60 sec                 | NO               |
| 101 | CustomMaint5Interval                       | Custom periodic maintenance interval 5            | Int32     | 60 sec                 | NO               |
| 102 | CustomMaint5Forenotice                     | Custom periodic maintenance forenotice 5          | Int32     | 60 sec                 | NO               |
| 103 | CustomMaint5CountType                      | Custom periodic maintenance count type 5          | String    | 60 sec                 | NO               |
| 104 | CustomMaint6Name                           | Custom periodic maintenance name 6                | String    | 60 sec                 | NO               |
| 105 | CustomMaint6                               | Custom periodic maintenance 6                     | Int32     | 60 sec                 | NO               |
| 106 | CustomMaint6Interval                       | Custom periodic maintenance interval 6            | Int32     | 60 sec                 | NO               |
| 107 | CustomMaint6Forenotice                     | Custom periodic maintenance forenotice 6          | Int32     | 60 sec                 | NO               |
| 108 | CustomMaint6CountType                      | Custom periodic maintenance count type 6          | String    | 60 sec                 | NO               |
| 109 | CustomMaint7Name                           | Custom periodic maintenance name 7                | String    | 60 sec                 | NO               |
| 110 | CustomMaint7                               | Custom periodic maintenance 7                     | Int32     | 60 sec                 | NO               |
| 111 | CustomMaint7Interval                       | Custom periodic maintenance interval 7            | Int32     | 60 sec                 | NO               |
| 112 | CustomMaint7Forenotice                     | Custom periodic maintenance forenotice 7          | Int32     | 60 sec                 | NO               |
| 113 | CustomMaint7CountType                      | Custom periodic maintenance count type 7          | String    | 60 sec                 | NO               |
| 114 | CustomMaint8Name                           | Custom periodic maintenance name 8                | String    | 60 sec                 | NO               |
| 115 | CustomMaint8                               | Custom periodic maintenance 8                     | Int32     | 60 sec                 | NO               |
| 116 | CustomMaint8Interval                       | Custom periodic maintenance interval 8            | Int32     | 60 sec                 | NO               |
| 117 | CustomMaint8Forenotice                     | Custom periodic maintenance forenotice 8          | Int32     | 60 sec                 | NO               |
| 118 | CustomMaint8CountType                      | Custom periodic maintenance count type 8          | String    | 60 sec                 | NO               |

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               | Sheet. 719/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |

**NOTE**

- 1 Reading cycle cannot be changed for each signal. (Selectable cycles: 500 msec, 10 sec, and 60 sec)
- 2 Whether to retrieve or not (enabled/disabled) can be specified for each signal.

**Table 20. Retrievable Periodic Maintenance Signals Listed By ROBODRILL Model**  
**(Signals retrievable from the model are marked with "YES")**

| #  | Signal Type                 | DiB Plus Series | DiB Series | DiA Series | TiF Series | TiE Series | TiD Series | TiC Series | TiB Series | TiA Series |
|----|-----------------------------|-----------------|------------|------------|------------|------------|------------|------------|------------|------------|
| 01 | SpindleGearLub              | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 02 | SpindleGearLubInterval      | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 03 | SpindleGearLubForenotice    | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 04 | TurretCamLub                | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 05 | TurretCamLubInterval        | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 06 | TurretCamLubForenotice      | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 07 | LMGuideLub                  | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 08 | LMGuideLubInterval          | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 09 | LMGuideLubForenotice        | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 10 | FlexibleHoseChk             | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 11 | FlexibleHoseChkInterval     | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 12 | FlexibleHoseChkForenotice   | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 13 | FireExConstChk              | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 14 | FireExConstChkInterval      | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 15 | FireExConstChkForenotice    | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 16 | FireExFuncChk               | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 17 | FireExFuncChkInterval       | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 18 | FireExFuncChkForenotice     | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 19 | ConUnitFanChk               | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 20 | ConUnitFanChkInterval       | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 21 | ConUnitFanChkForenotice     | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 22 | ACSpindleMotorChk           | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 23 | ACSpindleMotorChkInterval   | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 24 | ACSpindleMotorChkForenotice | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 25 | AirFilterChk                | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 26 | AirFilterChkInterval        | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 27 | AirFilterChkForenotice      | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 28 | CTCPartChk                  | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 29 | CTCPartChkInterval          | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 30 | CTCPartChkForenotice        | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 31 | DoorCylinderChk             | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 32 | DoorCylinderChkInterval     | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 33 | DoorCylinderChkForenotice   | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 34 | FrontDoorPadChk             | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 35 | FrontDoorPadChkInterval     | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 36 | FrontDoorPadChkForenotice   | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 37 | DryerChk                    | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 38 | DryerChkInterval            | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 39 | DryerChkForenotice          | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 40 | AbsPalseCodaBtrReplace      |                 |            | YES        | YES        |            |            |            |            |            |

|      |               |        |                                            |       |  |        |       |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        | Draw  | A-40829-00001EN                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                                                       |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue | FANUC CORPORATION                                     |  | Sheet. | 720/868 |

| #  | Signal Type                                | DiB Plus Series | DiB Series | DiA Series | TiF Series | TiE Series | TiD Series | TiC Series | TiB Series | TiA Series |
|----|--------------------------------------------|-----------------|------------|------------|------------|------------|------------|------------|------------|------------|
| 41 | AbsPulseCodaBtrReplaceInterval             |                 |            | YES        | YES        |            |            |            |            |            |
| 42 | AbsPulseCodaBtrReplaceForenotice           |                 |            | YES        | YES        |            |            |            |            |            |
| 43 | CNCmemBtrReplace                           | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 44 | CNCmemBtrReplaceInterval                   | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 45 | CNCmemBtrReplaceForenotice                 | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 46 | InternalLightReplace                       |                 |            | YES        | YES        |            |            |            |            |            |
| 47 | InternalLightReplaceInterval               |                 |            | YES        | YES        |            |            |            |            |            |
| 48 | InternalLightReplaceForenotice             |                 |            | YES        | YES        |            |            |            |            |            |
| 49 | XaxisCoverReplace                          | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 50 | XaxisCoverReplaceInterval                  | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 51 | XaxisCoverReplaceForenotice                | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 52 | YaxisCoverReplace                          | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 53 | YaxisCoverReplaceInterval                  | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 54 | YaxisCoverReplaceForenotice                | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 55 | DrawBarUnitReplace                         | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 56 | DrawBarUnitReplaceInterval                 | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 57 | DrawBarUnitReplaceForenotice               | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 58 | FrontDoorPadReplace                        | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 59 | FrontDoorPadReplaceInterval                | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 60 | FrontDoorPadReplaceForenotice              | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 61 | CTCPartReplace                             | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 62 | CTCPartReplaceInterval                     | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 63 | CTCPartReplaceForenotice                   | YES*2           | YES        | YES        | YES        |            |            |            |            |            |
| 64 | DoorCylinderReplace                        | YES*2           | YES*1      |            |            |            |            |            |            |            |
| 65 | DoorCylinderReplaceInterval                | YES*2           | YES*1      |            |            |            |            |            |            |            |
| 66 | DoorCylinderReplaceForenotice              | YES*2           | YES*1      |            |            |            |            |            |            |            |
| 67 | DDR4thAxisRotaryJointSeal                  | YES*2           | YES*1      |            |            |            |            |            |            |            |
| 68 | DDR4thAxisRotaryJointSealInterval          | YES*2           | YES*1      |            |            |            |            |            |            |            |
| 69 | DDR4thAxisRotaryJointSealForenotice        | YES*2           | YES*1      |            |            |            |            |            |            |            |
| 70 | DDR4thAxisSupportRotaryJointSeal           | YES*2           | YES*1      |            |            |            |            |            |            |            |
| 71 | DDR4thAxisSupportRotaryJointSealInterval   | YES*2           | YES*1      |            |            |            |            |            |            |            |
| 72 | DDR4thAxisSupportRotaryJointSealForenotice | YES*2           | YES*1      |            |            |            |            |            |            |            |
| 73 | DDR5thAxisRotaryJointSeal                  | YES*2           | YES*1      |            |            |            |            |            |            |            |
| 74 | DDR5thAxisRotaryJointSealInterval          | YES*2           | YES*1      |            |            |            |            |            |            |            |
| 75 | DDR5thAxisRotaryJointSealForenotice        | YES*2           | YES*1      |            |            |            |            |            |            |            |
| 76 | DDR5thAxisSupportRotaryJointSeal           | YES*2           | YES*1      |            |            |            |            |            |            |            |
| 77 | DDR5thAxisSupportRotaryJointSealInterval   | YES*2           | YES*1      |            |            |            |            |            |            |            |
| 78 | DDR5thAxisSupportRotaryJointSealForenotice | YES*2           | YES*1      |            |            |            |            |            |            |            |

|      |               |        |                                            |                   |                                                       |        |       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------------------|-------------------------------------------------------|--------|-------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |       |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                   |                                                       |        |       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                   |                                                       |        |       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw              | A-40829-00001EN                                       |        |       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                   |                                                       |        |       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                   |                                                       |        |       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | FANUC CORPORATION |                                                       |        |       |  | Sheet. | 721/868 |
| Ed.  | Date          | Design | Description                                |                   |                                                       |        |       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check             |                                                       | Apprv. | Inoue |  |        |         |

| #   | Signal Type            | DiB Plus Series | DiB Series | DiA Series | TiF Series | TiE Series | TiD Series | TiC Series | TiB Series | TiA Series |
|-----|------------------------|-----------------|------------|------------|------------|------------|------------|------------|------------|------------|
| 79  | CustomMaint1Name       | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 80  | CustomMaint1           | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 81  | CustomMaint1Interval   | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 82  | CustomMaint1Forenotice | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 83  | CustomMaint1CountType  | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 84  | CustomMaint2Name       | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 85  | CustomMaint2           | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 86  | CustomMaint2Interval   | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 87  | CustomMaint2Forenotice | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 88  | CustomMaint2CountType  | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 89  | CustomMaint3Name       | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 90  | CustomMaint3           | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 91  | CustomMaint3Interval   | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 92  | CustomMaint3Forenotice | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 93  | CustomMaint3CountType  | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 94  | CustomMaint4Name       | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 95  | CustomMaint4           | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 96  | CustomMaint4Interval   | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 97  | CustomMaint4Forenotice | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 98  | CustomMaint4CountType  | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 99  | CustomMaint5Name       | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 100 | CustomMaint5           | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 101 | CustomMaint5Interval   | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 102 | CustomMaint5Forenotice | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 103 | CustomMaint5CountType  | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 104 | CustomMaint6Name       | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 105 | CustomMaint6           | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 106 | CustomMaint6Interval   | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 107 | CustomMaint6Forenotice | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 108 | CustomMaint6CountType  | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 109 | CustomMaint7Name       | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 110 | CustomMaint7           | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 111 | CustomMaint7Interval   | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 112 | CustomMaint7Forenotice | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 113 | CustomMaint7CountType  | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 114 | CustomMaint8Name       | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 115 | CustomMaint8           | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 116 | CustomMaint8Interval   | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 117 | CustomMaint8Forenotice | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |
| 118 | CustomMaint8CountType  | YES*2           | YES        | YES        | YES*1      |            |            |            |            |            |

\*1 Signals may not be retrieved depending on the software version.

\*2 Signals may not be retrieved depending on the specifications of FanucRobodril.

|      |               |        |                                    |       |                                               |        |       |                   |         |  |
|------|---------------|--------|------------------------------------|-------|-----------------------------------------------|--------|-------|-------------------|---------|--|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |       |                   |         |  |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |        |       |                   |         |  |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |        |       |                   |         |  |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |        |       |                   |         |  |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | <b>A-40829-00001EN</b>                        |        |       |                   |         |  |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |        |       |                   |         |  |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |        |       |                   |         |  |
| Ed.  | Date          | Design | Description                        |       |                                               |        |       |                   |         |  |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check |                                               | Apprv. | Inoue | FANUC CORPORATION |         |  |
|      |               |        |                                    |       |                                               |        |       | Sheet.            | 722/868 |  |

## 14.6 OPC UA CLIENT FUNCTIONALITY

### 14.6.1 OPC UA Summary

OPC UA is a new standard for manufacturing software, enabling communication with a variety of units such as PLCs, and providing high functionalities including those that OPC Classic has provided.

Without using communication environment based on DCOM technology underlying OPC Classic, OPC UA Clients communicate with OPC UA Servers, and retrieve information from the OPC UA Servers.

### 14.6.2 System Configuration with OPC UA Client Functionality

Following are the examples of the main system configuration patterns incorporating the OPC UA Client functionality.

- Pattern 1: OPC UA Server software and MT-LINK*i* Collector is installed in the same PC
- Pattern 2: OPC UA Server software and MT-LINK*i* Collector is installed in a separate PC
- Pattern 3: OPC UA Server built-in device used by MT-LINK*i*
- Pattern 4: Ethernet I/O Converter used by MT-LINK*i*

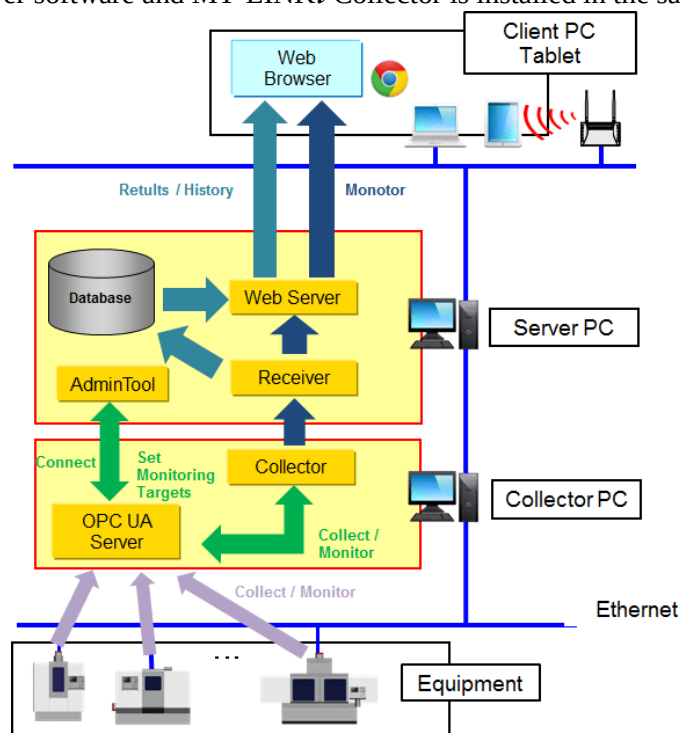
Following are the descriptions of the system configuration examples of each pattern.

#### NOTE

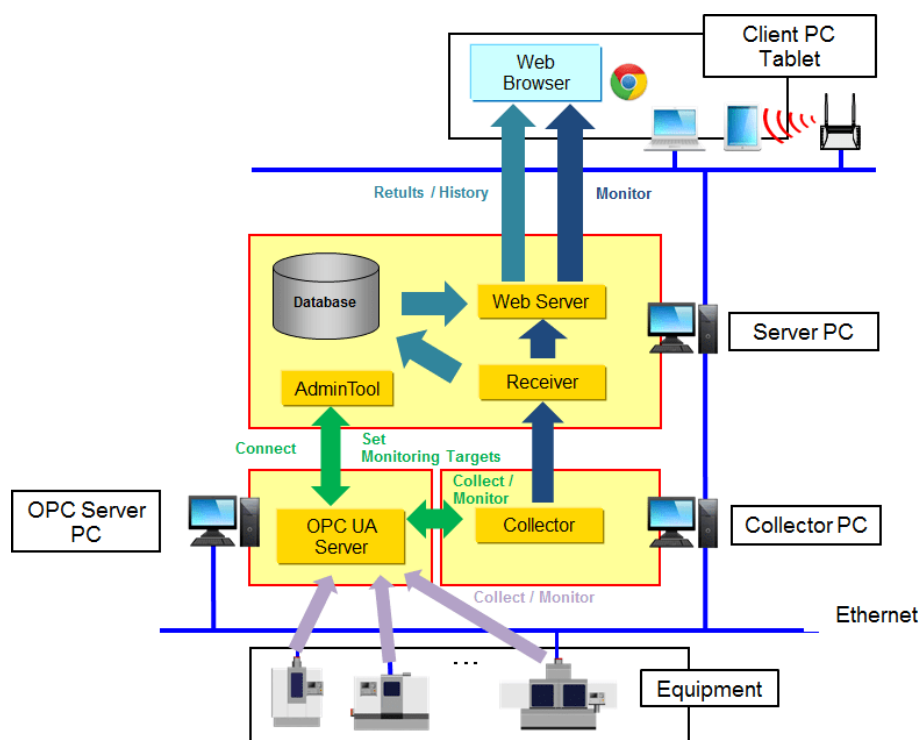
You should prepare other equipment except operation management software MT-LINK*i*.

|      |               |        |                                            |                          |                                                       |        |        |         |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |        | Sheet. | 723/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    |                                                       | Apprv. | Inoue  |         |

- Pattern 1: OPC UA Server software and MT-LINK*i* Collector is installed in the same PC



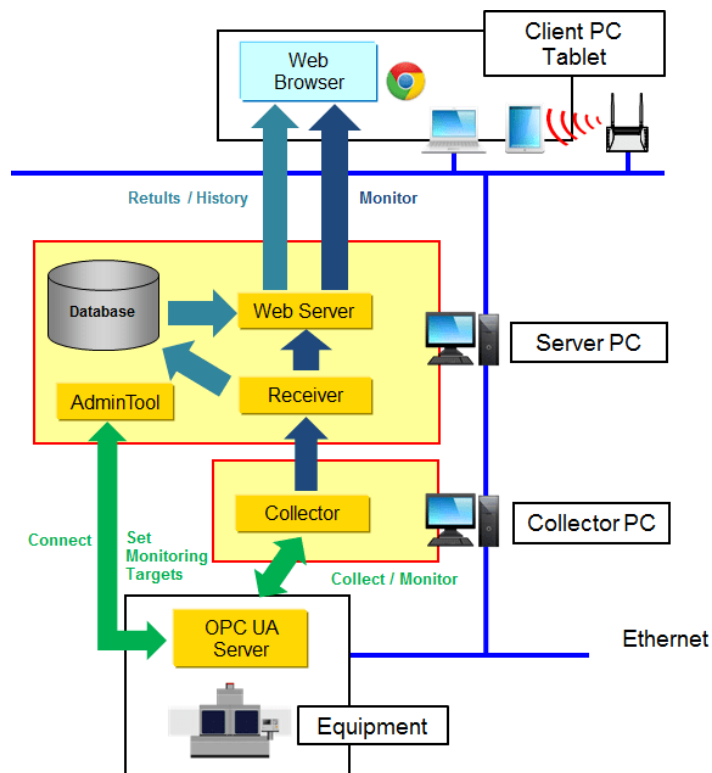
- Pattern 2: OPC UA Server software and MT-LINK*i* Collector is installed in a separate PC



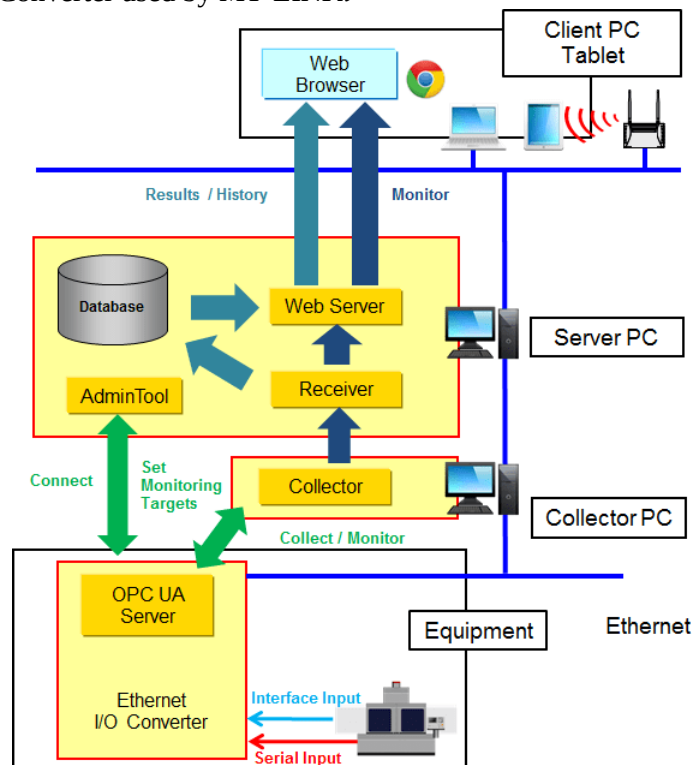
|      |               |        |                                            |                          |                                                       |       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |                          |                                                       |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   | Draw                     | <b>A-40829-00001EN</b>                                |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |                          |                                                       |       |                |
| Ed.  | Date          | Design | Description                                | <b>FANUC CORPORATION</b> |                                                       |       | Sheet. 724/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue |                |



- Pattern 3: OPC UA Server built-in device used by MT-LINK*i*



- Pattern 4: Ethernet I/O Converter used by MT-LINK*i*



Ethernet I/O Converter is a hardware unit with a built-in OPC UA Server functionality.  
For the details, refer to the "[Ethernet I/O Converter](#)" subsection.

|      |               |        |                                            |                          |                                                       |       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |                          |                                                       |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   | Draw                     | <b>A-40829-00001EN</b>                                |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |                          |                                                       |       |                |
| Ed.  | Date          | Design | Description                                | <b>FANUC CORPORATION</b> |                                                       |       | Sheet. 725/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue |                |

## 14.6.3 OPC UA Data Types

The following table shows OPC UA data types and their corresponding MT-LINK*i* data types.

| OPC UA Data Type | MT-LINK <i>i</i> Data Type |
|------------------|----------------------------|
| Boolean          | Bool                       |
| SByte            | Int32                      |
| Byte             |                            |
| Int16            |                            |
| UInt16           |                            |
| Int32            |                            |
| UInt32           | UInt32                     |
| Float            | Double                     |
| Double           |                            |
| Int64            | String                     |
| UInt64           |                            |
| String           |                            |
| DateTime         |                            |
| Guid             |                            |
| ByteString       |                            |
| XmlElement       |                            |
| NodeId           |                            |
| ExpandedNodeId   |                            |
| StatusCode       |                            |
| QualifiedName    |                            |
| LocalizedText    |                            |
| ExtensionObject  |                            |
| DataValue        |                            |

### NOTE

- 1 MT-LINK*i* supports only the data types shown in the above table. The tag of the data types not mentioned in the above table cannot be used as signals.
- 2 MT-LINK*i* can handle up to 10240 bytes of String type data. Data exceeding 10240 bytes will be cut off to 10237 bytes. (When the data is cut off, string data "... " will be added in the 10238<sup>th</sup> to 10240<sup>th</sup> bytes.)
- 3 Data types for method output arguments correspond to MT-LINK*i* data types in the same way as OPC UA data types do.

|      |               |        |                                            |                          |                                                       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       | Sheet. 726/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue          |

## 14.6.4 OPC UA Supported Information Models

This table shows information models stipulated in OPC UA.

| Information Model         | Supported |
|---------------------------|-----------|
| Data Access (DA)          | YES       |
| Alarm and Conditions (AC) | NO        |
| Historical Access (HA)    | NO        |
| Programs (Prog)           | NO        |

This table shows OPC UA Read API supported by MT-LINK*i*.

| Read API                           | Supported |
|------------------------------------|-----------|
| Synchronized Read                  | YES       |
| Asynchronous Read                  | NO        |
| Subscription (Change Notification) | NO        |

## 14.6.5 OPC UA Supported Security Models

This table shows OPC UA security modes and whether they are supported by MT-LINK*i*.

| Security Mode  | Description                                   | Supported |
|----------------|-----------------------------------------------|-----------|
| None           | Does not apply security process on messages.  | YES       |
| Sign           | Adds signature to messages.                   | YES       |
| SignAndEncrypt | Adds signature to messages and encrypts them. | YES       |

This table shows OPC UA security policies and whether they are supported by MT-LINK*i*.

| Security Policy | Description                                                              | Supported |
|-----------------|--------------------------------------------------------------------------|-----------|
| None            | Uses the procedure configured for the lowest level requirement.          | YES       |
| Basic128Rsa15   | Uses the procedure configured for the middle level requirement.          | YES       |
| Basic256        | Uses the procedure configured for the middle and high level requirement. | YES       |
| Basic256Sha256  | Uses the procedure configured for the high level requirement.            | NO        |

This table shows OPC UA user token types and whether they are supported by MT-LINK*i*.

| User Token Type    | Description                                              | Supported |
|--------------------|----------------------------------------------------------|-----------|
| User Name Password | Identifies by the combination of user name and password. | YES       |
| X509               | Identifies by certificate based on X50.                  | NO        |
| Kerberos           | Identifies by Kerberos token.                            | NO        |
| Anonymous          | accepts anonymous users                                  | YES       |

### NOTE

When using other than "None" for security mode, security policy, application certificate located under "C:\ProgramData\FANUC\MT-LINK*i*\pki\own\certs" (MT-LINK*i*\_OpcUaClient <xxxxx>.der) must be uploaded to the connection target OPC UA Server.

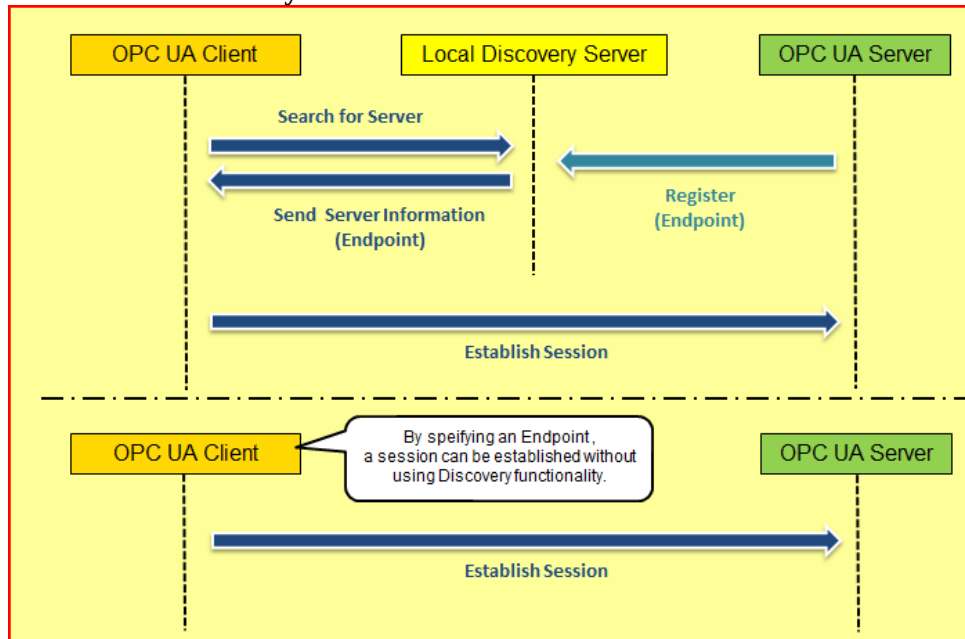
As the uploading procedure of the application certificate varies depending on the OPC UA Server, follow the procedures for the respective OPC UA Server.

|      |               |        |                                            |       |  |        |       |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        | Draw  | A-40829-00001EN                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                                                       |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       | FANUC CORPORATION                                     |  | Sheet. | 727/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                                                       |  |        |         |

## 14.6.6 Local Discovery Server

Local Discovery Server, provided by the OPC Foundation, is software for search for running OPC UA Servers. Local Discovery Server responses to an inquiry for OPC UA Servers from an OPC UA Client with information about the OPCUA Server running on the client PC.

### - Conceptual view of Local Discovery Server



- Upper part from the chain line: when you use Local Discovery Server.  
 The OPC UA Server registers its information to Local Discovery Server.  
 The OPC UA Client requests Local Discovery Server for searching OPC UA Servers.  
 Local Discovery Server sends the server information to the client.

- Lower part from the chain line: when you specify an endpoint.  
 By specifying an endpoint an OPC UA Server if you have its information, you can establish a session with the OPC UA Server.  
 MT-LINK*i* supports this direct way of establishing session with OPC UA Servers.

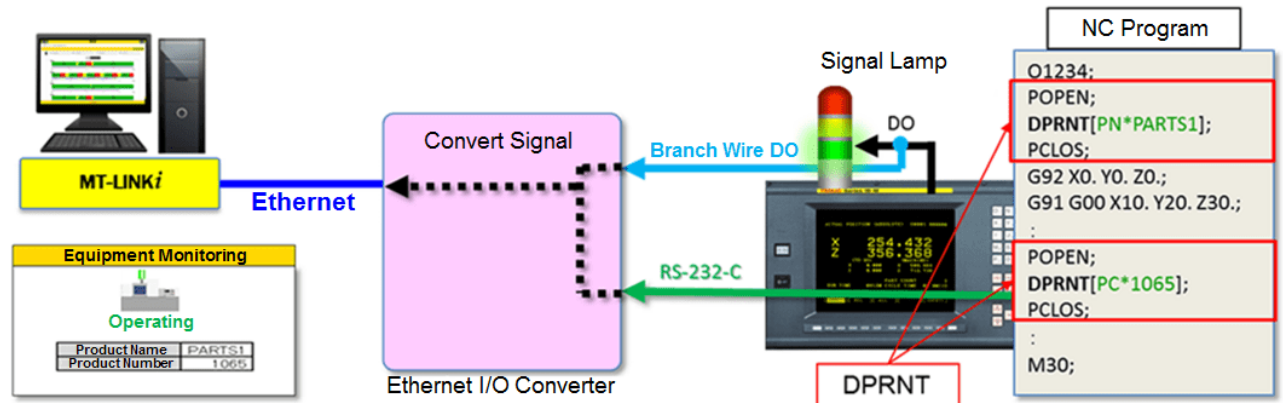
#### NOTE

OPC UA Server and Local Discovery Server must be installed on a same PC.  
 If Local Discovery Server is not installed, or the connected OPC UA Server does not support Local Discovery Server, the Search for Server function cannot be used.  
 In such a case, directly input the endpoint of the connected OPC UA Server.

|      |               |        |                                            |                   |                                                       |        |         |
|------|---------------|--------|--------------------------------------------|-------------------|-------------------------------------------------------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                   |                                                       |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                   |                                                       |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |                   |                                                       |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   | Draw              | <b>A-40829-00001EN</b>                                |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                   |                                                       |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |                   |                                                       |        |         |
| Ed.  | Date          | Design | Description                                | FANUC CORPORATION |                                                       | Sheet. | 728/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check             | Apprv.                                                | Inoue  |         |

### 14.6.7 Ethernet I/O Converter

Ethernet I/O Converter is a hardware unit with the built-in OPC UA server function. In addition to the existing OPC UA server function, it receives information from the machine via 2 ways, (1) via interface input using branch wire from the signal lamp, or (2) via serial input using RS-232C. It converts it into signal that can be used on MT-LINK*i*, and stores it to a specific node on the OPC UA Server. MT-LINK*i* uses the OPC UA client function to connect to the Ethernet I/O Converter.



## NOTE

- 1 The maximum number of OPC UA clients that can connect to Ethernet I/O Converter is 2, including MT-LINK*i*. (When MT-LINK*i* is in operation, since the Collector service is constantly in connection with the Ethernet I/O Converter, the connectable number of clients becomes 1.) If the number of clients exceeds the maximum connectable number, an error ("BadTooManySessions") is notified.
- 2 The contents of the PrintOutput node on the Ethernet I/O Converter will be cleared after the client reads it. Other client must not Read the node while MT-LINK*i* is in operation, as it may lead to unintended consequences such as loss of data.

To use the Ethernet I/O Converter with MT-LINK*i*, it is necessary to perform the following steps.

- Setting of NC parameters relating to connection of RS-232C
- Addition of description to the command for external output to NC program
- Setting of MT-LINK*i* (AdminTool)

|      |               |        |                                            |       |  |        |       |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        |       | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                   |                                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |       | Draw              | A-40829-00001EN                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                   |                                                       |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       | FANUC CORPORATION |                                                       |  | Sheet. | 729/868 |
| Date | Nov.30.2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                   |                                                       |  |        |         |

The details of each step are explained in the following.

■ Setting of NC parameters relating to connection of RS-232C

Set the parameters related to the connection of RS-232C, in order to connect NC to the Ethernet I/O Converter. The recommended setting values are shown below.

(1) In the case of FANUC Series 16/18/21

| # | Item                            | Setting                                                                                              | Parameter                                                |
|---|---------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| 1 | I/O Channel Separation Function | Separate the channel used for data input from the channel used for data output (Recommended setting) | No.110#0=1                                               |
| 2 | I/O Channel                     | Channel 2 for data output (Recommended setting only when DPRNT is in use)                            | No.21=2 (If No.110#0=1)<br>OR<br>No.20=2 (If No.110#0=0) |
| 3 | Line feed code for DPRNT        | LF(0Fh) + CR(0Dh)                                                                                    | No.6001#4=1                                              |
|   |                                 | LF(0Fh)<br>(Recommended setting)                                                                     | No.6001#4=0                                              |
| 4 | Leading zero for DPRNT          | Do not output space<br>(Recommended setting)                                                         | No.6001#1=1                                              |
|   |                                 | Output space                                                                                         | No.6001#1=0                                              |
| 5 | Output code                     | ISO code<br>(Recommended setting)                                                                    | No.000#1=1                                               |

Recommended setting values of the parameters for serial port channel 2

| # | Item                                     | Setting                                          | Parameter  |
|---|------------------------------------------|--------------------------------------------------|------------|
| 1 | Stop bit                                 | 2                                                | No.121#0=1 |
| 2 | Output code                              | ISO code                                         | No.121#3=0 |
| 3 | Line feed code before and after the data | None                                             | No.121#7=1 |
| 4 | Specification of connected device        | General RS232-C device<br>(Use DC1 to DC4 codes) | No.122=0   |
| 5 | Baud rate                                | 4800bps                                          | No.123=10  |

(2) In case of FANUC Series 15

| # | Item                     | Setting                                                                      | Parameter                  |
|---|--------------------------|------------------------------------------------------------------------------|----------------------------|
| 1 | I/O channel              | Channel 2 for data output<br>(Recommended setting only when DPRNT is in use) | No.21=2                    |
| 2 | Line feed code for DPRNT | LF(0Fh) + CR(0Dh) + CR(0Dh)                                                  | No.0000#3=0                |
|   |                          | LF(0Fh)<br>(Recommended setting)                                             | No.0000#3=1                |
| 3 | Leading zero for DPRNT   | Do not output space<br>(Recommended setting)                                 | No.7000#7=1                |
|   |                          | Output space                                                                 | No.7000#7=0                |
| 4 | Output code              | ISO code without parity bit<br>(Recommended setting)                         | No.0000#2=1<br>No.0000#4=0 |

Recommended setting values of the parameters for serial port channel 2

| # | Item                                     | Setting                                          | Parameter              |
|---|------------------------------------------|--------------------------------------------------|------------------------|
| 1 | Stop bit                                 | 2                                                | No.5161=2              |
| 2 | Line feed code before and after the data | None                                             | Setting unnecessary    |
| 3 | Specification of connected device        | General RS232-C device<br>(Use DC1 to DC4 codes) | No.5002=6<br>No.5160=3 |
| 4 | Baud rate                                | 4800bps                                          | No.5162=10             |

|      |               |        |                                    |                   |                                               |        |        |         |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw              | A-40829-00001EN                               |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                   |                                               |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | FANUC CORPORATION |                                               |        | Sheet. | 730/868 |
| Ed.  | Date          | Design | Description                        |                   |                                               |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             |                                               | Apprv. | Inoue  |         |

■ Addition of description to the command for external output to NC program

Macro command (external output command) is used to transmit data from the NC program to the Ethernet I/O Converter.

The following macro commands can be used.

- DPRNT
- POPEN
- PCLOS

The details of each macro command are dependent on the NC device in use.

(1) Open command POPEN

POPEN command is a command to establish connection with the Ethernet I/O Converter. It is commanded before a set of data output commands. The string "POPEN," is added to PrintOutput node, which is the node on the Ethernet I/O Converter.

(2) Data output command DPRNT

DPRNT[a b #c [de] b<sub>1</sub> #c<sub>1</sub> [d<sub>1</sub>e<sub>1</sub>] ...]

a: Identifier (output destination node)

b: String to be output

c: Number of parameter storing the data to be output

d: Number of significant figures before decimal point for the parameter

e: Number of significant figures after decimal point for the parameter

The DPRNT command outputs the string and the contents of the parameter to the Ethernet I/O Converter. The output data is output additionally to the PrintOutput node on the Ethernet I/O Converter separated with commas (","). To output to a specific node other than PrintOutput node, use the identifier to specify the output destination. If the identifier is omitted, data output is executed only for PrintOutput node. However, strings reserved for use as identifiers cannot be used as "strings for output."

List of Usable Identifiers

| #  | Identifier | Output destination node of Ethernet I/O Converter |                                       |        |
|----|------------|---------------------------------------------------|---------------------------------------|--------|
|    |            | Name of Node Displayed                            | Meaning                               | Type   |
| 1  | (none)     | PrintOutput                                       | Output string                         | String |
| 2  | PN*        | ProductName                                       | Product name                          | String |
| 3  | PC*        | ProductResultNumber                               | Total number of products (PartsCount) | Int32  |
| 4  | V01*       | value01                                           | General-purpose numerical type 1      | Double |
| 5  | V02*       | value02                                           | General-purpose numerical type 2      | Double |
| 6  | V03*       | value03                                           | General-purpose numerical type 3      | Double |
| 7  | V04*       | value04                                           | General-purpose numerical type 4      | Double |
| 8  | V05*       | value05                                           | General-purpose numerical type 5      | Double |
| 9  | V06*       | value06                                           | General-purpose numerical type 6      | Double |
| 10 | V07*       | value07                                           | General-purpose numerical type 7      | Double |
| 11 | V08*       | value08                                           | General-purpose numerical type 8      | Double |
| 12 | V09*       | value09                                           | General-purpose numerical type 9      | Double |
| 13 | V10*       | value10                                           | General-purpose numerical type 10     | Double |
| 14 | SR01*      | string01                                          | General-purpose string type 1         | String |

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               | Sheet. 731/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |

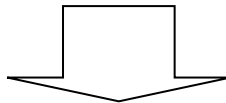
| #  | Identifier | Output destination node of Ethernet I/O Converter |                                |        |
|----|------------|---------------------------------------------------|--------------------------------|--------|
|    |            | Name of Node Displayed                            | Meaning                        | Type   |
| 15 | SR 02*     | string02                                          | General-purpose string type 2  | String |
| 16 | SR 03*     | string03                                          | General-purpose string type 3  | String |
| 17 | SR 04*     | string04                                          | General-purpose string type 4  | String |
| 18 | SR 05*     | string05                                          | General-purpose string type 5  | String |
| 19 | SR 06*     | string06                                          | General-purpose string type 6  | String |
| 20 | SR 07*     | string07                                          | General-purpose string type 7  | String |
| 21 | SR 08*     | string08                                          | General-purpose string type 8  | String |
| 22 | SR 09*     | string09                                          | General-purpose string type 9  | String |
| 23 | SR 10*     | string10                                          | General-purpose string type 10 | String |

### (3) Close command PCLOS

The PCLOS command is the command to release the connection with Ethernet I/O Converter. It is commanded after all the data output commands finish. The string "PCLOS" is added to PrintOutput node on the Ethernet I/O Converter.

#### Example of program description

|                      |                                                                     |
|----------------------|---------------------------------------------------------------------|
| O1234 ;              |                                                                     |
| #100 =999 ;          | Assign value "999" in common parameter #100                         |
| POPEN ;              | Declare start of output to node, add string "POPEN" to PrintOutput  |
| DPRNT[ABC] ;         | Add string "ABC" + ", " to PrintOutput                              |
| DPRNT[PN*PARTS01] ;  | Set "PARTS01" for ProductName                                       |
| DPRNT[PC*#100[30]] ; | Set content of common parameter #100 ("999") to ProductResultNumber |
| DPRNT[] ;            | Add ", " to PrintOutput                                             |
| PCLOS ;              | Declare end of output to node, add string "PCLOS" to PrintOutput    |
| M30 ;                |                                                                     |
| %                    |                                                                     |



#### Contents of node after running the program

| # | Name of Node Displayed | Value                                                                                               |
|---|------------------------|-----------------------------------------------------------------------------------------------------|
| 1 | PrintOutput            | POPEN,ABC,PN PARTS01,PC 999,,PCLOS,<br>* The values will be cleared after PrintOutput node is read. |
| 2 | MainProgram            | PARTS01                                                                                             |
| 3 | ProductResultNumber    | 999                                                                                                 |

### NOTE

- 1 Always write DPRNT command in the range between POPEN command and PCLOS command. If DPRNT command is written outside of the range between POPEN command and PCLOS command, the output contents to the node will not be guaranteed.
- 2 As long as it is within the range between POPEN command and PCLOS command, DPRNT command can be written in the subprogram, etc. by writing POPEN command and PCLOS command in the caller program. The details such as the extent of applicable overlapping of programs caused by the calling of the programs are dependent on the macro commands of each NC device.

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               | Sheet. 732/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |



■ Setting of MT-LINK*i* (AdminTool)

Perform setting for the connection to the OPC UA Server on AdminTool. Select Ethernet I/O Converter for the connection target OPC UA Server.

<Machine Setting>

Select machine type OPCUAServer, and select Ethernet I/O Converter for connection target OPC UA Server. When using Ethernet I/O Converter, the nodes described in the List of Usable Identifiers can be used. The signals are added from each node based on its necessity.

<Equipment Setting>

The following signal definition can be set to the multiple assign status, if necessary.

| # | Name (Status)       | Web Name           | Input Contents of Signal Definition                 | Input Example of Signal Definition |
|---|---------------------|--------------------|-----------------------------------------------------|------------------------------------|
| 1 | ProductName         | Product name       | Name of signal assigned to ProductName node         | ProductName_NewMachine001          |
| 2 | ProductResultNumber | Number of products | Name of signal assigned to ProductResultNumber node | ProductResultNumber_NewMachine001  |

|      |               |        |                                            |                          |                                                       |        |       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|--------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |        |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |        |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |        |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |        |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |        |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |        |       | Sheet. 733/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |        |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    |                                                       | Apprv. | Inoue |                |

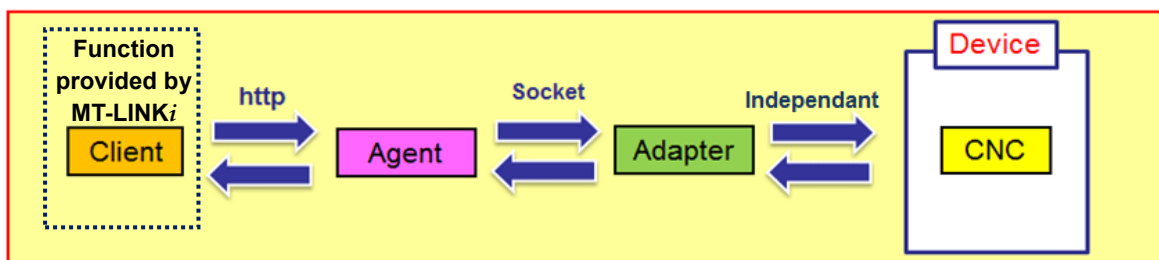
## 14.7 MTConnect CLIENT FUNCTIONALITY

### 14.7.1 MTConnect Summary

MTConnect is a standard based on an open protocol targeted for data integration, and it is a standard for retrieving information from devices such as machine tools.

### 14.7.2 System Configuration Using MTConnect Client Functionality

An example of a system configuration of MTConnect is shown below.  
It provides Client functionality for MT-LINK*i*.



| Function Name | Description                                            |
|---------------|--------------------------------------------------------|
| Client        | Retrieve information from Agent via http communication |
| Agent         | Create XML based on the data received from Adapter     |
| Adapter       | Collect information from Device, and send to Agent     |
| Device        | Various devices such as CNC                            |

Following are the examples of the main system configuration patterns incorporating the MTConnect Client functionality.

- Pattern 1: MTConnect Agent / MTConnect Adapter and MT-LINK*i* Collector is installed in the same PC
- Pattern 2: MTConnect Agent / MTConnect Adapter and MT-LINK*i* Collector is installed in separate PCs
- Pattern 3: MTConnect Agent / MTConnect Adapter built-in device used by MT-LINK*i*

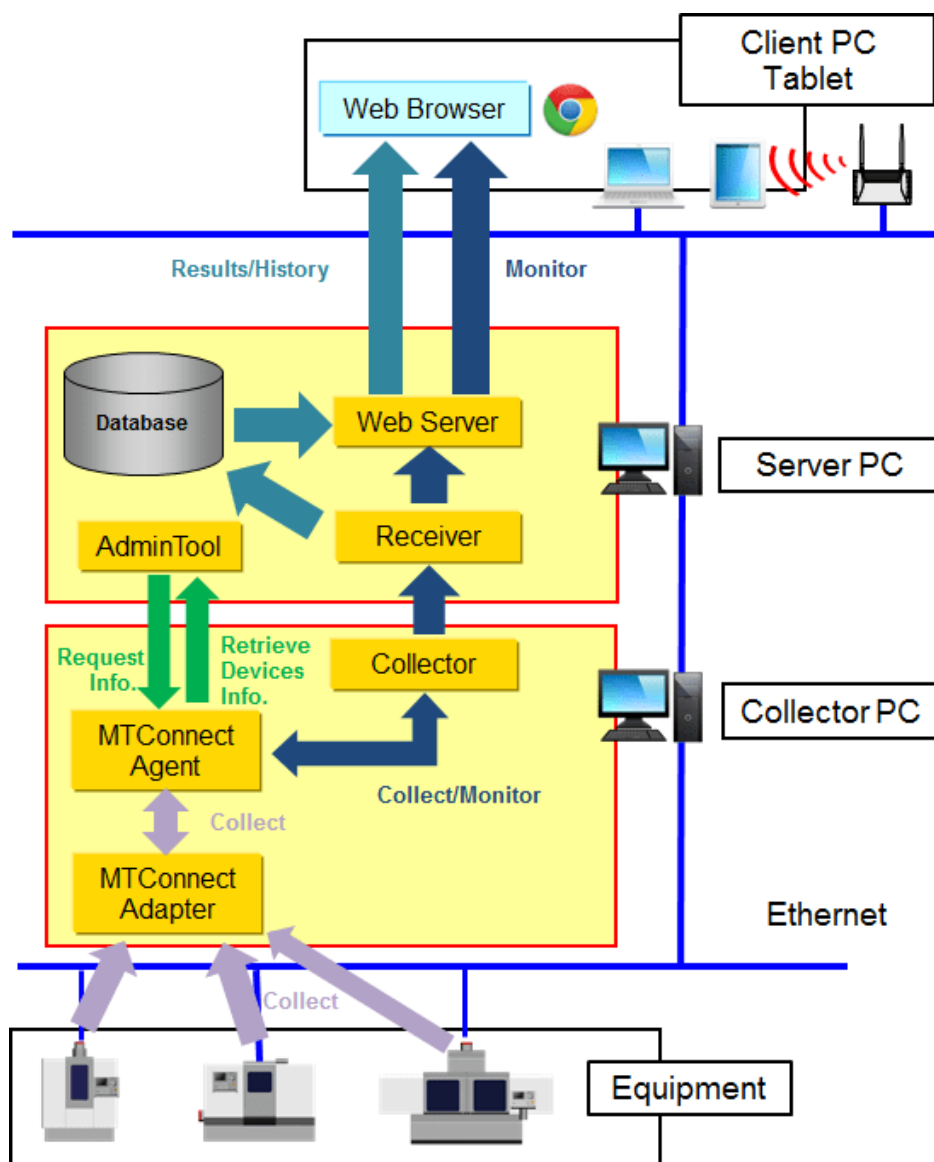
The system configuration examples of each pattern are described in the following.

#### NOTE

Other functionalities besides the operation management software MT-LINK*i* must be prepared by the user.

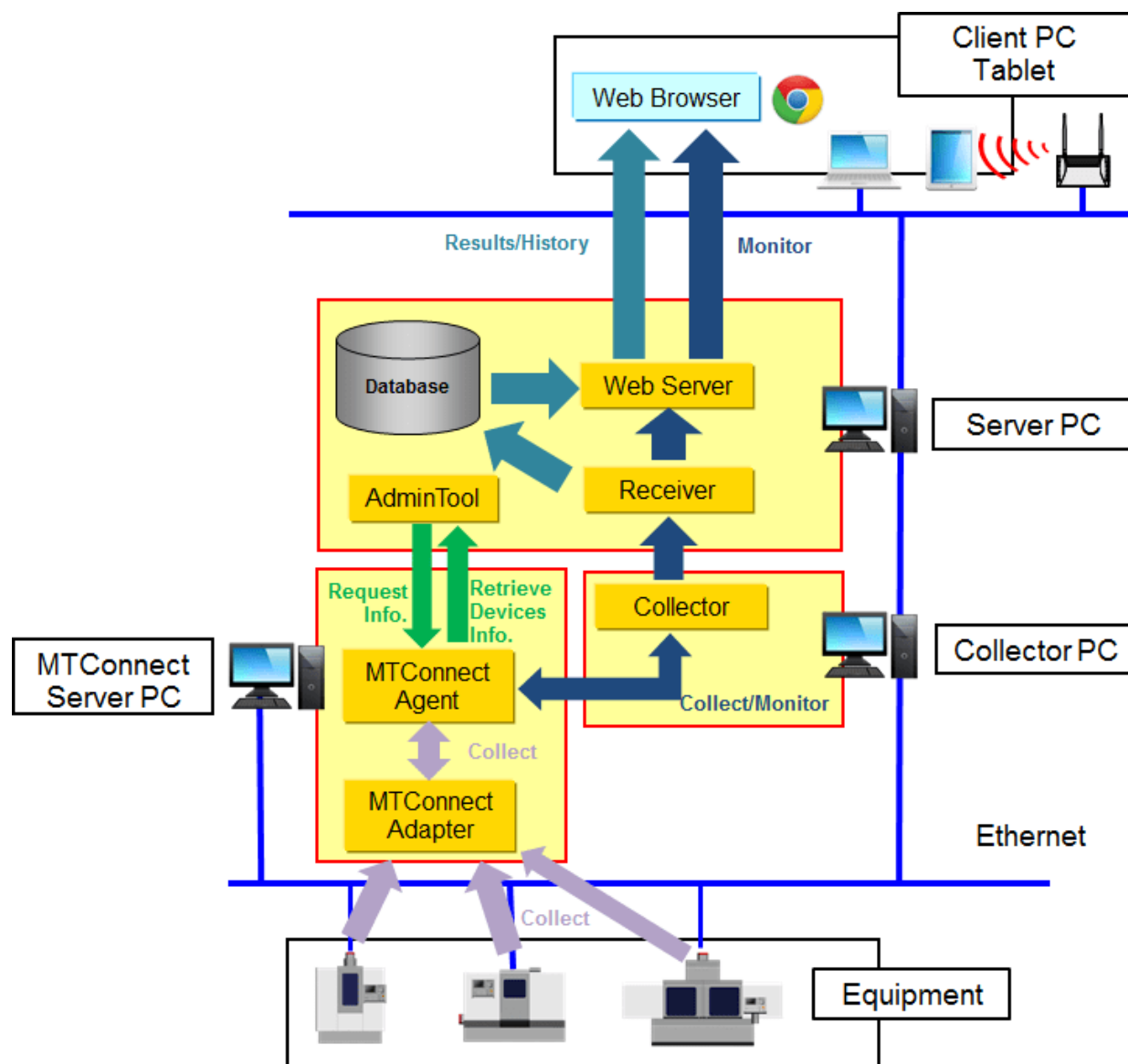
|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 734/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         |

- Pattern 1: MTConnect Agent / MTConnect Adapter and MT-LINK*i* Collector is installed in the same PC



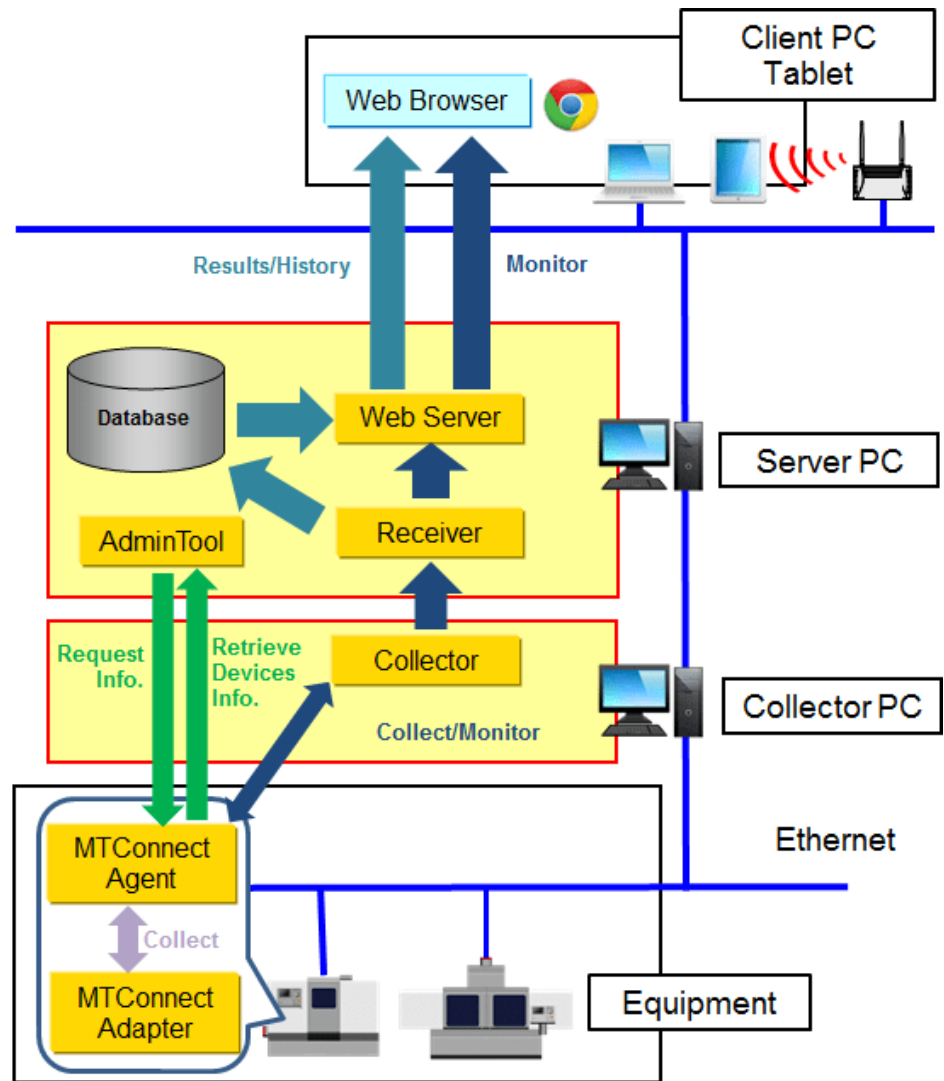
|      |               |        |                                            |                          |                                                       |        |         |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |                          |                                                       |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   | Draw                     | <b>A-40829-00001EN</b>                                |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |                          |                                                       |        |         |
| Ed.  | Date          | Design | Description                                | <b>FANUC CORPORATION</b> |                                                       | Sheet. | 735/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue  |         |

- Pattern 2: MTConnect Agent / MTConnect Adapter and MT-LINK*i* Collector is installed in separate PCs



|      |               |        |                                            |                          |                                                       |       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |                          |                                                       |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   | Draw                     | <b>A-40829-00001EN</b>                                |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |                          |                                                       |       |                |
| Ed.  | Date          | Design | Description                                | <b>FANUC CORPORATION</b> |                                                       |       | Sheet. 736/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue |                |

- Pattern 3: MTConnect Agent / MTConnect Adapter built-in device used by MT-LINK*i*



\* This example is a case when MTConnect Agent and MTConnect Adapter are built-in in the CNC.

|      |               |        |                                            |       |  |        |       |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        |       | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                   |                                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |       | Draw              | A-40829-00001EN                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                   |                                                       |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       | FANUC CORPORATION |                                                       |  | Sheet. | 737/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                   |                                                       |  |        |         |
|      |               |        |                                            |       |  |        |       |                   |                                                       |  |        |         |

## 14.8 SERVO VIEWER LINKAGE

This section describes the overview of SERVO VIEWER Linkage function, the measure sequence, and the settings required for the use of the function.

 **CAUTION**

For the collection of servo data from a single unit of machine (CNC), only one method of collection can be executed at the same time.

When MT-LINK<sub>i</sub> is collecting servo data from a single unit of machine (CNC), other methods to collect servo data cannot be used. If you attempt to collect data by using two or more methods at the same time, the measurement commands of the methods interfere with each other, causing failure in the normal retrieval of the servo data.

Therefore, the following applications cannot be used simultaneously with the collection of servo data by MT-LINK*i*. When executing automatic collection of servo data by MT-LINK*i*, such applications as shown below should not be running at the same time.

- SERVO VIEWER (PC version)
- SERVO GUIDE (PC version)
- SERVO VIEWER (iHMI version)
- SERVO GUIDE MATE (CNC screen)
- Application for C language executor using high-speed sampling function for servo/spindle data
- MTB application using high-speed sampling FOCAS function for servo/spindle data
- On-machine measurement system using continuous position data output function (CNC option A02B-xxxx-J989)  
(Example: Renishaw plc, SPRINT™ system)

When using the above applications, make sure that the MT-LINK*i* is stopped, or either disable or temporarily stop the SERVO VIEWER linkage functionality.

For the details on how to stop the MT-LINK<sub>i</sub> system, refer to the "Start/Stop System" subsection.

For details on how to stop SERVO VIEWER linkage service (MT-LINK<sup>i</sup> Collector SERVO VIEWER Linkage), refer to the "[Start/Stop System](#)" subsection.

For details on how to disable the SERVO VIEWER linkage functionality, refer to the "Deleting Measurement Data and Disabling SERVO VIEWER Linkage Function" subsection of the "FANUC MT-LINK*i* Startup MANUAL."

For details on how to temporarily stop the SERVO VIEWER linkage functionality, refer to the "Stop Using SERVO VIEWER Linkage Function Temporarily" subsection of the "FANUC MT-LINK*i* Startup MANUAL."

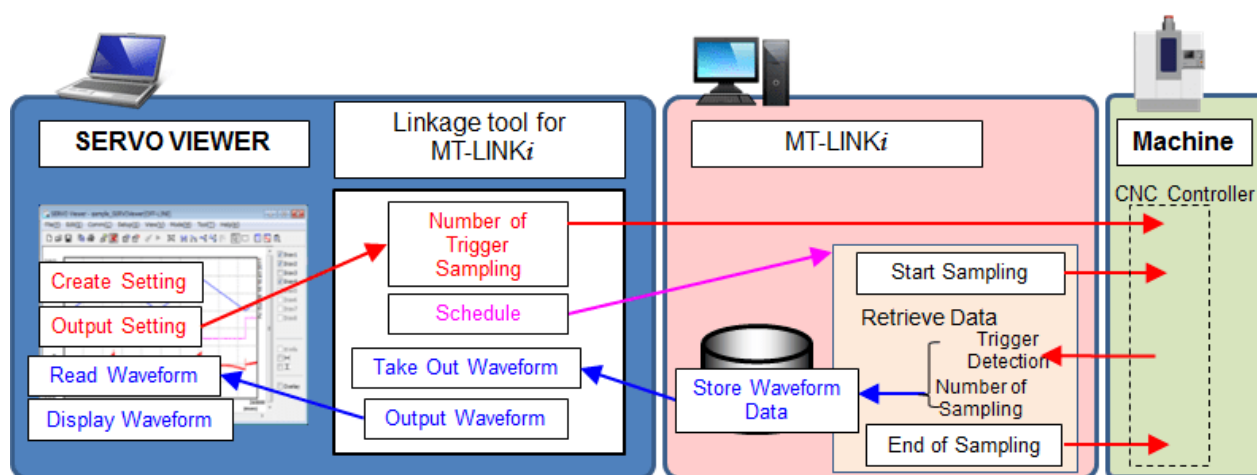
|      |               |        |                                    |       |  |        |                   |                                               |  |        |         |
|------|---------------|--------|------------------------------------|-------|--|--------|-------------------|-----------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |                   |                                               |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |                   |                                               |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        | Draw              | A-40829-00001EN                               |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        |                   |                                               |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |                   |                                               |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        | FANUC CORPORATION |                                               |  | Sheet. | 738/868 |
| Ed.  | Date          | Design | Description                        |       |  |        |                   |                                               |  |        |         |
| Date | Nov.30.2015   | Desig. | Ogawa                              | Check |  | Apprv. |                   |                                               |  |        |         |

## 14.8.1 SERVO VIEWER Linkage Summary

The SERVO VIEWER Linkage function creates files with the waveform data retrieved from the machine, and stores these files to database (DB) and manages them.

The retrieval process and the display process of the waveform data is executed by the following steps.

1. On the SERVO VIEWER, create setting data (Measure profile) for the retrieval of waveform data.
2. Use the Linkage tool for MT-LINK*i* to set the Measure profile and Measurement schedule on MT-LINK*i*.
3. On MT-LINK*i*, the retrieval of waveform data and storing to the database (DB) is executed based on the setting information.
4. Linkage tool for MT-LINK*i* takes out the waveform data retrieved.
5. SERVO VIEWER reads the waveform data and displays it.



For the details on the settings for retrieving waveform data and how to set the Measurement schedule, refer to the following documents.

"FANUC SERVO VIEWER OPERATOR'S MANUAL (B-65444EN)"

"Linkage tool for MT-LINK*i* OPERATOR'S MANUAL (A-95363EN-024)"

"How to observe servo data through linkage with MT-LINK*i* (A-95363EN-023)"

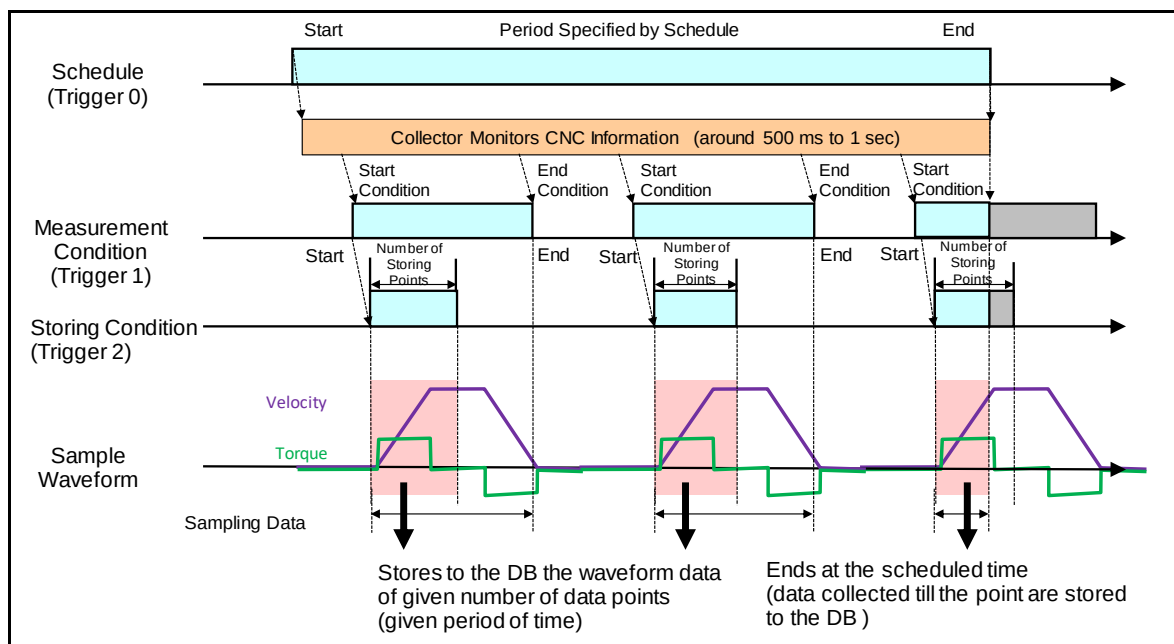
### NOTE

- 1 If the IP address of the machine targeted for retrieval is changed on the AdminTool, data is retrieved by the newly changed IP address. Be cautious when there is a change in the type of CNC model before and after the change of IP address, and there may be a difference between the setting of retrieval (Measure profile) and the CNC models of retrieval target.
- 2 Try to restrict the maximum number of units to be connected with the SERVO VIEWER linkage function within 20 units. If too many units are connected at the same time, the operating speed of the system might become slow.
- 3 Multiple SERVO VIEWER linkage functions should not be applied to a single unit of CNC simultaneously. If done so, the requests may interfere with each other, causing failure in the normal execution of the data collection process.

|      |               |        |                                            |                   |                                                       |        |         |
|------|---------------|--------|--------------------------------------------|-------------------|-------------------------------------------------------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                   |                                                       |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                   |                                                       |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |                   |                                                       |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   | Draw              | <b>A-40829-00001EN</b>                                |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                   |                                                       |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |                   |                                                       |        |         |
| Ed.  | Date          | Design | Description                                | FANUC CORPORATION |                                                       | Sheet. | 739/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      |                   |                                                       |        |         |
|      |               | Check  |                                            | Apprv.            | Inoue                                                 |        |         |

## 14.8.2 Measure Sequence of SERVO VIEWER Linkage

The measure sequence of SERVO VIEWER linkage is shown below.



- Within the period specified by the Measurement schedule, from the start condition specified by the Measure profile, the data points up to the number of points specified by the Measure profile are stored to the DB.
- If the end condition is established, the data collection will not be continued thereafter although it is still within the schedule period.
- If the schedule period ends during data collection process, the process will be suspended.
- When the data collection is suspended before the number of data points collected reaches the specified number, the data collected till the suspended point in time will be stored to the DB.
- If multiple schedules are specified at the same time, the schedule starting later will be prioritized. The data collection in process will be stopped instantly.
- There are a few seconds of time lag between the start time of the schedule and the time when the sampling of data actually starts.

### (1) Schedule (Trigger 0)

Specify the start date/time and the end date/time.

Multiple schedules can be registered for a single machine. However, overlapping time ranges cannot be specified.

The schedules are specified from the Measurement schedule of the Linkage tool for MT-LINK*i*.

|      |               |        |                                            |                          |                                                       |       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |                          |                                                       |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   | Draw                     | <b>A-40829-00001EN</b>                                |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |                          |                                                       |       |                |
| Ed.  | Date          | Design | Description                                | <b>FANUC CORPORATION</b> |                                                       |       | Sheet. 740/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue |                |



(2) Measurement condition (Trigger 1)

Specify the conditions to start or end the measurement. Following can be used as measurement conditions.

| Measurement Condition | Number of Conditions that can be Specified | Conditions that can be Specified                                                                                              |
|-----------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Start Condition       | Maximum 3 (NOTE 1)                         | Program name (NOTE 2)                                                                                                         |
|                       |                                            | PMC signal (NOTE 3)                                                                                                           |
|                       |                                            | Macro variable (NOTE 3)                                                                                                       |
| End Condition         | 1 (NOTE 4)                                 | PMC signal (NOTE 3)                                                                                                           |
|                       |                                            | Macro variable (NOTE 3)                                                                                                       |
| Repeat Condition      | 1 (NOTE 4)                                 | Specify the repeat count of the measurement. (NOTE 5)<br>(Example: If 5 is specified, ends when 5 measurements are completed) |

Data collection starts when all of the following conditions are established.

- If matching with start condition
- If not matching with end condition
- If within the range of the repeat condition

Measurement conditions are specified from the Measurement schedule of the Linkage tool for MT-LINK*i*.

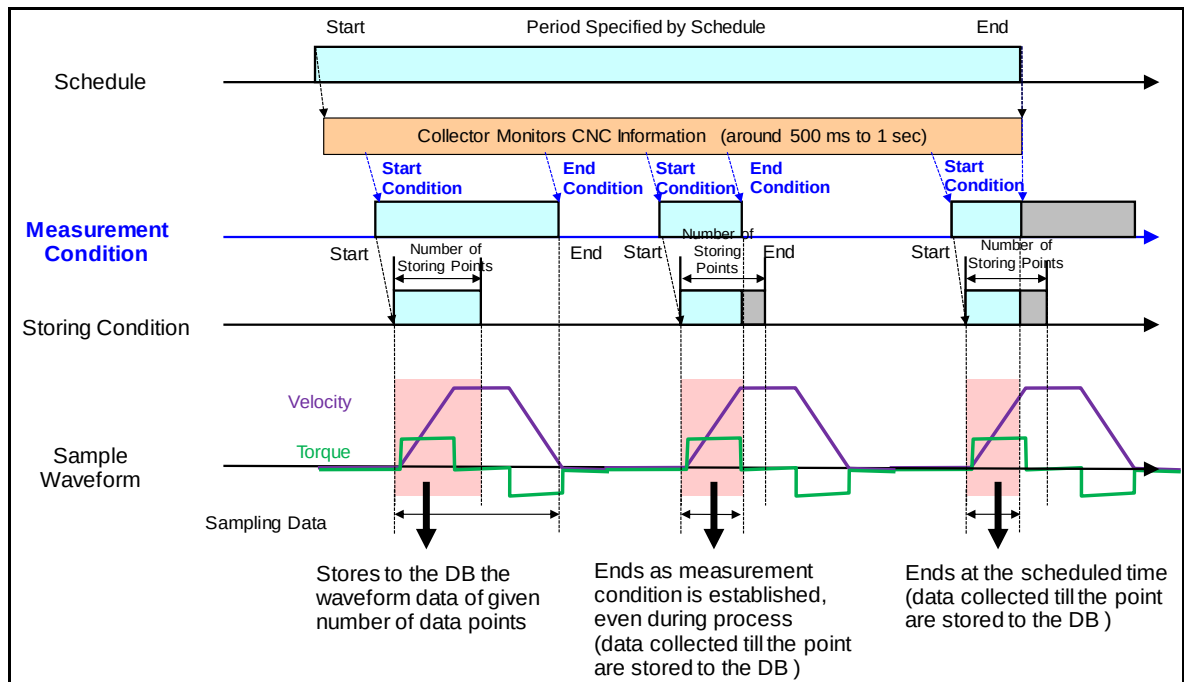
**NOTE**

- 1 The condition for starting is established when all the conditions specified are established.
- 2 Only one program name can be specified for a single start condition.
- 3 The number of signals (PMC signals and macro variables) that can be monitored is five in maximum, including all the start and end conditions of the measurement conditions.
- 4 If specifying the measurement condition, at least one of either the end condition or the repeat condition must be specified. The condition to end is established when either the end condition is established or the repeat condition is fulfilled.
- 5 If the measurement fails, it will not be counted in the repeat count.

|      |               |        |                                            |                          |                                                       |        |       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|--------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |        |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |        |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |        |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |        |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |        |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |        |       | Sheet. 741/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |        |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    |                                                       | Apprv. | Inoue |                |

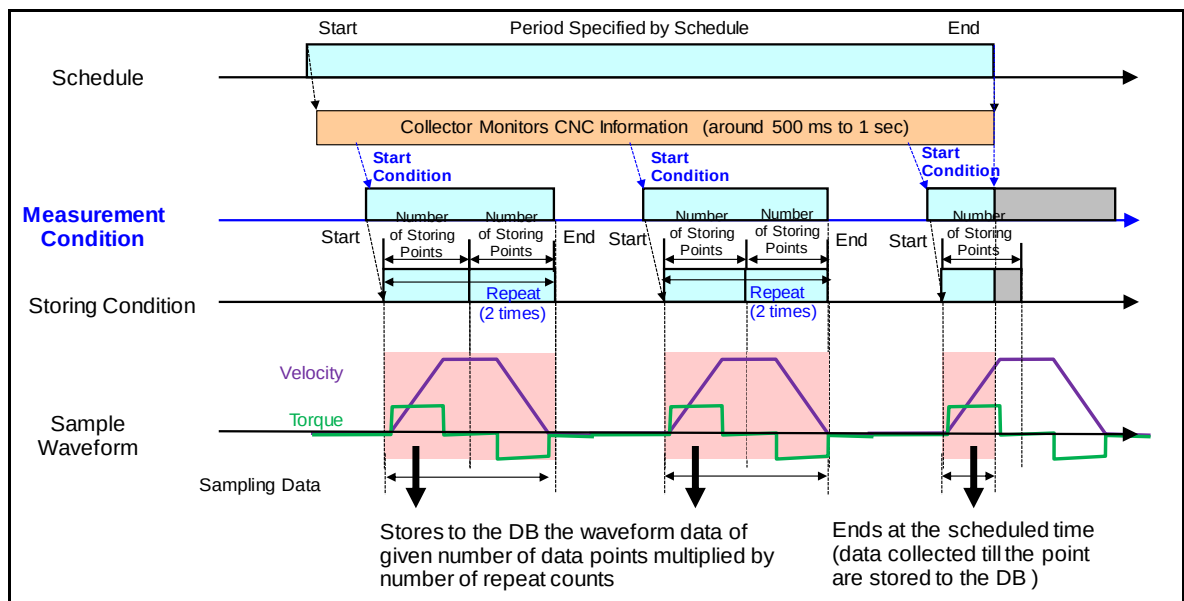
\* The patterns (examples) of combination for the conditions of measurement condition are shown below.

■ Pattern 1: When specifying "start condition" and "end condition"



- After the start condition is established, stores to the DB the storing points up to the number of points specified as waveform data.
- When the end condition is established, stops data collection and waits for the next start condition to be established.

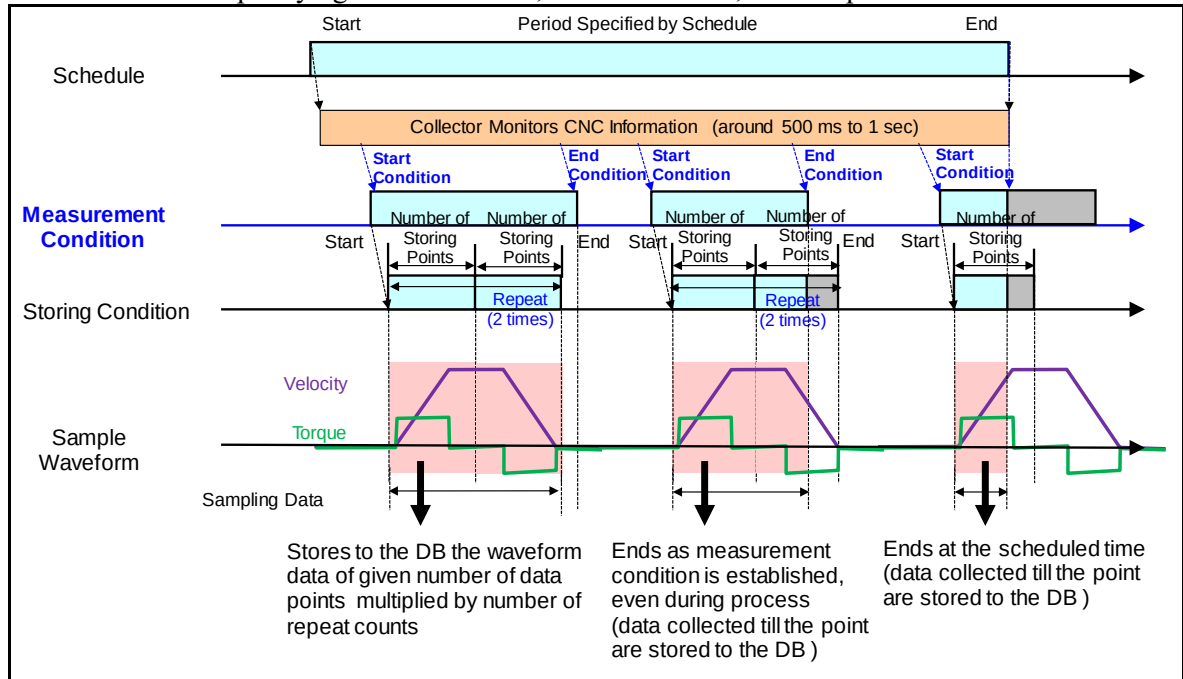
■ Pattern 2: When specifying "start condition" and "repeat condition"



- After the start condition is established, stores to the DB the storing points up to the number of points specified multiplied by the number of repeat counts as waveform data.
- When the repeat condition is finished, waits for the next start condition to be established.

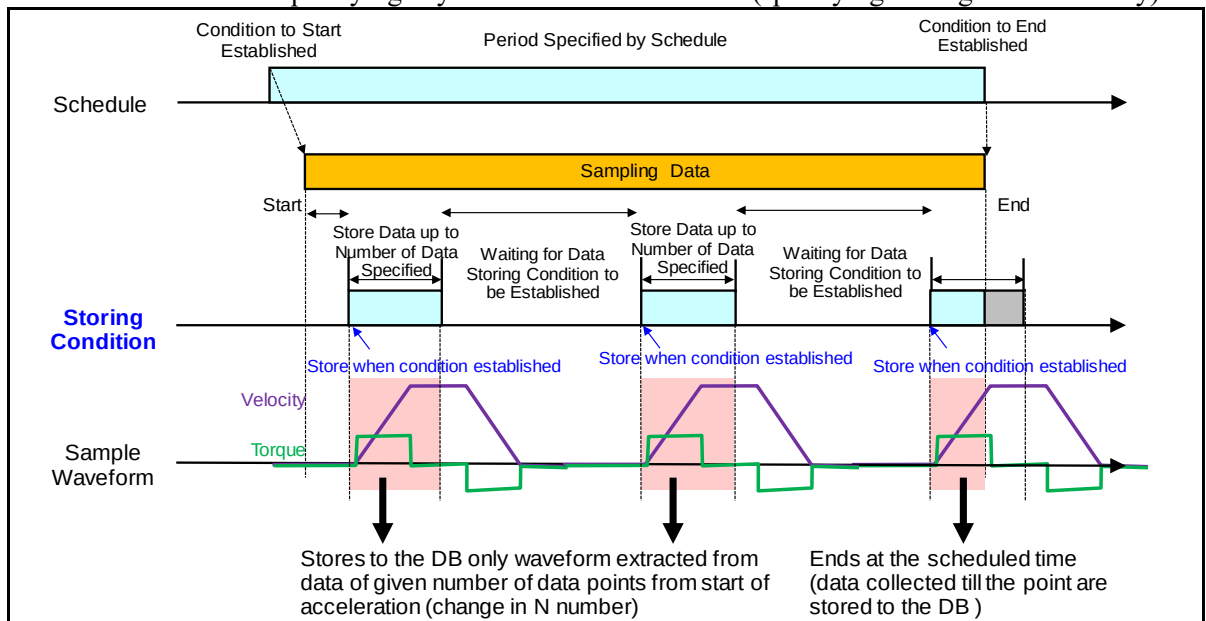
|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | A-40829-00001EN                               |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 742/868 |

### ■ Pattern 3: When specifying "start condition," "end condition," and "repeat condition"



- After the start condition is established, stores to the DB the storing points up to the number of points specified multiplied by the number of repeat counts as waveform data.
- When the repeat condition is fulfilled, waits for the next start condition to be established.
- When the end condition is established, stops data collection and waits for the next start condition to be established.

### ■ Pattern 4: When not specifying any measurement conditions (specifying storing condition only)



- Stores to the DB the storing points up to the number of points specified as waveform data from the point when the storing condition (trigger 2) is established.
- Waits for the next storing condition (trigger 2) to be established.

|      |               |        |                                            |       |  |        |       |                                                       |  |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------------------------------------------|--|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                                                       |  |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                                                       |  |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                                                       |  |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        | Draw  | A-40829-00001EN                                       |  |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                                                       |  |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                                                       |  |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       | FANUC CORPORATION                                     |  |  | Sheet. | 743/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                                                       |  |  |        |         |

(3) Storing condition (Trigger 2)

Specify the start condition for storing the collected data.

The storing condition is used as a trigger for the machine, and the measurement points up to the specified number of points are stored as the collected data.

Storing conditions are specified by the Measure profile of the Linkage tool for MT-LINK*i*.

## 14.8.3 MT-LINK*i* Settings Required for SERVO VIEWER Linkage

Settings required on MT-LINK*i* when using the SERVO VIEWER linkage function are explained below.

(1) SERVO VIEWER Linkage Setting

Settings to enable the use of the SERVO VIEWER linkage function.

Enable the SERVO VIEWER linkage function, and specify the port number to use in communicating with the Collector PC. For the details refer to the "[Add Collector](#)" subsection.

(2) Authority Setting

The user's authority and password is set.

When signing in to MT-LINK*i* from Linkage tool for MT-LINK*i*, use the sign-in user of MT-LINK*i*. If the sign-in user of MT-LINK*i* wishes to use the SERVO VIEWER linkage function, the adequate authority level (Read Only, Read/Write) must be set for the role of the user. For the details, refer to the "[Steps for Authority Setting](#)" subsection.

### NOTE

- 1 By editing the system setting file (ServoViewer.ini) provided by MT-LINK*i*, you can customize the setting when retrieving the waveform data by using the SERVO VIEWER linkage function.

Following are the items that can be specified by ServoViewer.ini.

- Rate of reading interval when retrieving waveform data

For the details on the system setting file (ServoViewer.ini), refer to the "[SYSTEM SETTING FILES](#)" section.

- 2 LDAP authentication users can sign in to the Web Server via the Linkage tool for MT-LINK*i* provided that they are registered as MT-LINK*i* users.  
Ask your administrator for an MT-LINK*i* user account, or get your LDAP authentication user account registered as an MT-LINK*i* user by performing as follows:
  - Sign in to the MT-LINK*i* Web Server using an LDAP authentication user account.  
MT-LINK*i* automatically creates an MT-LINK*i* user account for the user to sign in to the Web Server via the Linkage tool for MT-LINK*i*.

|      |               |        |                                            |                          |                                                       |        |        |         |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |        | Sheet. | 744/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    |                                                       | Apprv. | Inoue  |         |

## 14.9 SYSTEM DIAGNOSIS

This section describes the functions and the operating procedures of the system diagnosis function of the Web screen.

For details on how to display the [Utility - System Diagnosis] screen, refer to the "FANUC MT-LINK*i* WebUI OPERATOR'S MANUAL."

### 14.9.1 System Diagnosis Summary

The system diagnosis function retrieves the state of connection and workload of the Receiver and the Collector, stores the retrieved information to the DB, and displays the diagnosis information on the [Utility - System Diagnosis] screen of the Web screen. For the details on the [Utility - System Diagnosis] screen, refer to the "[Utility - System Diagnosis Screen](#)" subsection.

Also, the retrieved diagnosis information is output as a CSV file. For the details on the CSV file output of system diagnosis function, refer to the "[CSV File Output of System Diagnosis Information](#)" subsection.

The items displayed on the [Utility - System Diagnosis] screen are shown below.

- (1) Last update date/time
- (2) System diagnosis information of the Receiver
  - CPU usage rate of the Server PC
  - Memory usage rate of the Server PC
  - State of accumulated queue for the Receiver
- (3) Cache information of the Receiver
  - End date/time of the last signal data change point (\*1) stored to the DB
  - Usage rate and the state of cache area
- (4) Signal received by the Receiver / State of data stored to DB
  - Number of signal data change points received by the Receiver (average value per one second)
  - Increase in the number of signal data change points stored to the DB compared to the previous value (average value per one second)
- (5) Disk information
  - The free disk space of the disk storing the DB, and its state.
- (6) System diagnosis information of the Collector
  - CPU usage rate of the Collector PC
  - Memory usage rate of the Collector PC
  - State of accumulated queue for the Collector
  - Cycle time and the state of the machine connected to the Collector

\*1: When the value of each signal data changes, the following information is stored to DB as a single record.

- (a) Value of signal data
  - (b) Start date/time: Date/time when the value of the signal data changed to the value of (a)
  - (c) End date/time: Date/time when the value of the signal data changed to another value from (a)
- The items (b) and (c) are called the change points of the signal data.

#### NOTE

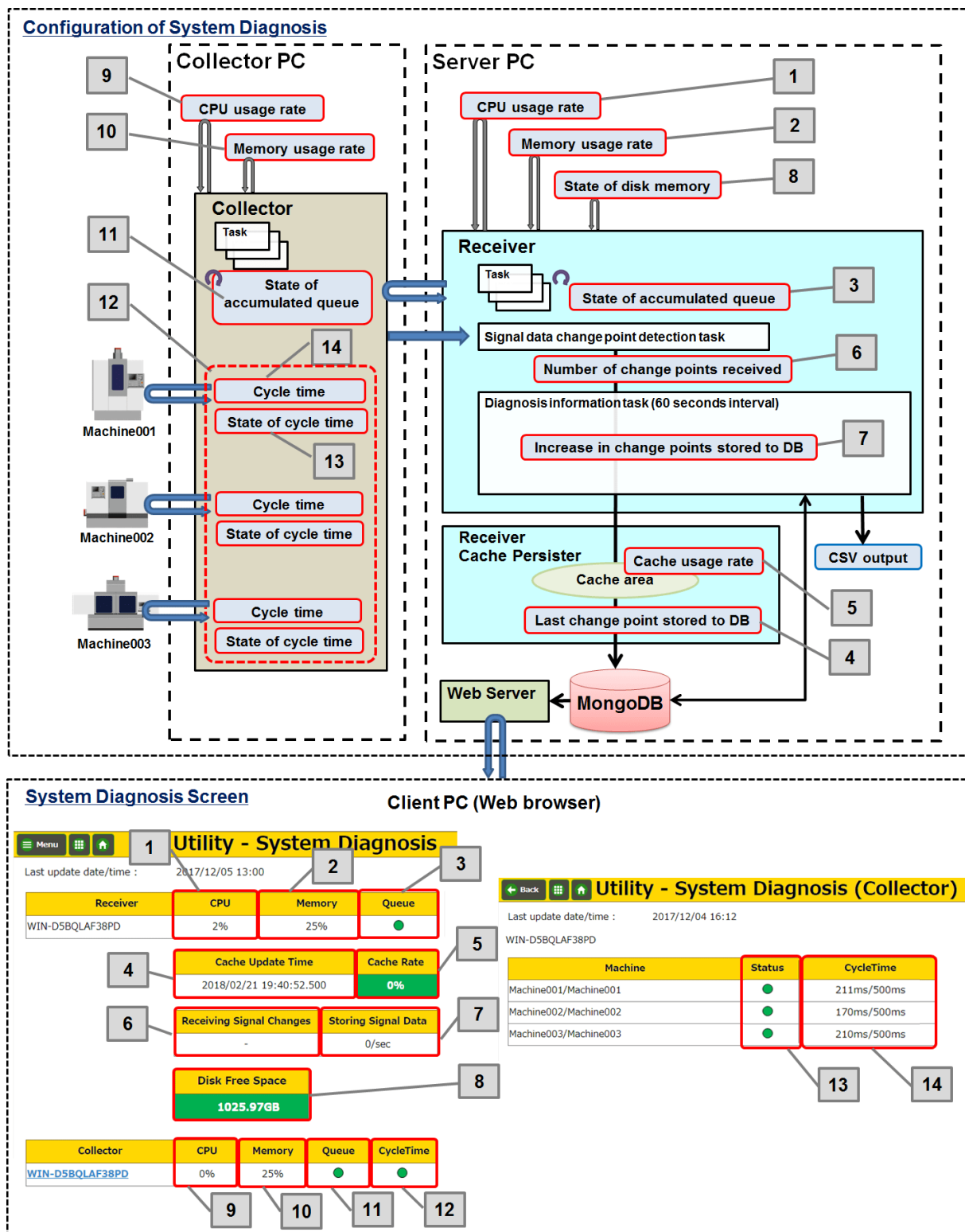
The signal data that are subject to the cache information and the state of signal received by Receiver / state of data stored to the DB are the simple assign statuses / the multiple assign statuses (excluding condition) specified on the [Equipment Setting] window.

|      |               |        |                                            |                          |                                                       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       | Sheet. 745/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue          |

### 14.9.2 Configuration of System Diagnosis

A conceptual diagram for the configuration of system diagnosis is shown below.

The numbers in the pop-up in "Configuration of System Diagnosis" correspond to the numbers in the pop-up of each item in "System Diagnosis Screen."



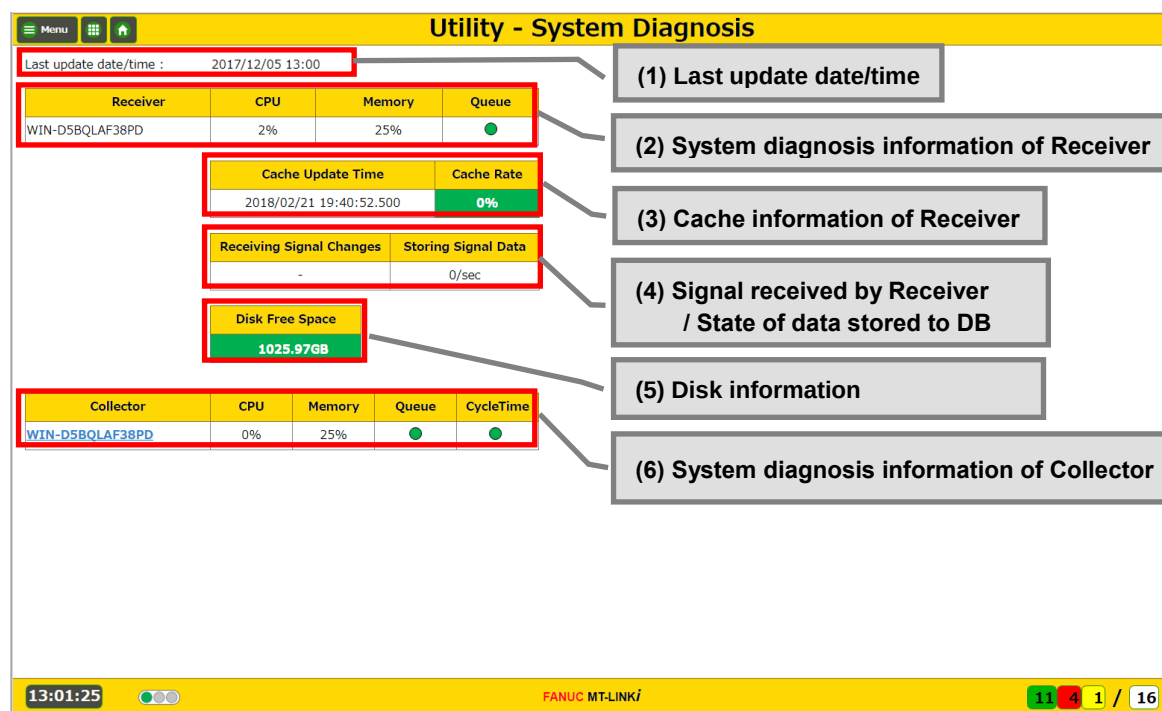
|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 746/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         |

## 14.9.3 Utility - System Diagnosis Screen

Displays on the Web screen, the state of connection and workload of Receiver and Collector.

### 14.9.3.1 Utility - System Diagnosis Screen (System Diagnosis Information)

Displayed when System Diagnosis is selected on the Web screen.



#### ■ [Utility - System Diagnosis] screen (System diagnosis information)

##### (1) Last update date/time

The last date/time the diagnosis information was collected from the Receiver and the Collector. The information is updated in an interval of 60 seconds.

##### (2) System diagnosis information of the Receiver

The system diagnosis information of the Receiver is displayed. Also, when the diagnosis information cannot be collected, such as when the Receiver is stopped, the message "Failed to get system diagnosis information of Receiver." will be displayed.

- Receiver  
Name of the Server PC which has the Receiver installed
- CPU  
CPU usage rate (%) of the Server PC which has the Receiver installed
- Memory  
Memory usage rate (%) of the Server PC which has the Receiver installed

|      |               |        |                                    |                   |                                               |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw              | A-40829-00001EN                               |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                   |                                               |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | FANUC CORPORATION | Sheet. 747/868                                |
| Ed.  | Date          | Design | Description                        |                   |                                               |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             |                                               |
|      |               |        |                                    | Apprv.            | Inoue                                         |



- Queue

The processing load status of the Receiver is displayed with the following colored icons according to the accumulated number of queues.

| Processing Load Status                                                            |              | Description                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Good         | The Receiver is operating normally.                                                                                                                                                                                               |
|  | Warning      | The Receiver is operating normally, but the processing load is becoming high.<br>There is no problem if it is occurring on a short term basis, but if this status continues on a long term basis, contact with the administrator. |
|  | Alarm        | The Receiver is not operating normally because of high processing load.<br>If the alarm status is occurring, settings must be reviewed. Contact the administrator.                                                                |
|  | Disconnected | The Receiver is not connected normally.                                                                                                                                                                                           |

**NOTE**

The judging criteria for the alarm status of the processing load (duration time of alarm status) can be customized by changing the system setting file (receiver.ini). For the details, refer to the "**SYSTEM SETTING FILES**" section.

(3) Cache information of the Receiver

- Cache Update Time

The latest change point (end date/time) of the signal data that was stored to the DB from the cache area of the Receiver.

- Cache Rate

Based on the usage rate of the cache area, the usage rate (%) and the status is displayed using the following icons.

| Status of Cache                                                                     |         | Judging Criteria (Default)                       |
|-------------------------------------------------------------------------------------|---------|--------------------------------------------------|
|  | Good    | Less than 10%                                    |
|  | Warning | 10% or more and less than 50%                    |
|  | Alarm   | 50 to 100%                                       |
| -                                                                                   | Unknown | When the status of the cache cannot be retrieved |

**NOTE**

The judging criteria for the status of the cache can be customized by changing the system setting file (receiver.ini). For the details, refer to the "**SYSTEM SETTING FILES**" section.

|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | <b>A-40829-00001EN</b>                        |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 748/868 |



(4) Signal received by the Receiver / State of data stored to DB

- Receiving Signal Changes  
Displays the number of change points of the signal data that the Receiver receives (average value per one second).  
If the Receiver has just been started or there are no values to display, the character "-" will be displayed instead.
- Storing Signal Data  
Displays the increase in the number of change points of the signal data (average value per one second) that are stored to the DB compared with the previous number of the stored data.  
If the Receiver has just been started and there is no previous value, the character "-" will be displayed instead.

**NOTE**

When DB deletion process set by [DB Data Retention Period Setting] (System Setting) is in progress (whether by deletion schedule, or by manual deletion), "-" will be displayed in the [Storing Signal Data] column.

(5) Disk information

- Disk Free Space  
Following state and free disk space is displayed according to the free disk space ratio (free disk space / total disk space) of the disk storing the DB. Unit of free disk space is GB, and it is rounded to the second decimal place (third decimal place is rounded up). The state of the disk is judged by the free disk space ratio, or by the threshold of free disk space (fixed to 10GB). If the disk space information cannot be acquired, "-" is displayed as the state being abnormal.

| Status of Disk |          | Judging Criteria (Default)                                           |
|----------------|----------|----------------------------------------------------------------------|
| 1025GB         | Good     | Free disk space ratio is 10% or more                                 |
| 25GB           | Warning  | Free disk space ratio is 5% or more and less than 10%                |
| 10GB           | Alarm    | Free disk space ratio is less than 5%, or free space is 10GB or less |
| -              | Abnormal | Disk space information cannot be acquired                            |

**NOTE**

The judging criteria for the status (Good/Warning) of the disk can be customized by changing the system setting file (receiver.ini). For the details, refer to the "**SYSTEM SETTING FILES**" section.

(6) System diagnosis information of the Collector

The diagnosis information of the Collector is displayed. Also, when the diagnosis information cannot be collected, such as when the Receiver is stopped, the message "Failed to get system diagnosis information of Collector." will be displayed.

Up to 100 rows of Collector system diagnosis information are displayed.

- Collector  
Name of the Collector PC which has the Collector installed is displayed.  
If you click the name of the Collector PC, [Utility - System Diagnosis (Collector)] is displayed. For the details, refer to the "**Utility - System Diagnosis (Collector) Screen**" subsection.
- CPU  
CPU usage rate (%) of the Collector PC which has the Collector installed is displayed.
- Memory  
Memory usage rate (%) of the Collector PC which has the Collector installed is displayed.

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               | Sheet. 749/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |

- Queue

The processing load status of the Collector is displayed with the following colored icons according to the accumulated number of queues.

| Processing Load Status                                                            |              | Description                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Good         | The Collector and the machines connected are operating normally.                                                                                                                                                                   |
|  | Warning      | The Collector is operating normally, but the processing load is becoming high.<br>There is no problem if it is occurring on a short term basis, but if this status continues on a long term basis, contact with the administrator. |
|  | Alarm        | The Collector is not operating normally because of high processing load.<br>If the alarm status is occurring, review of the setting is required. Contact the administrator.                                                        |
|  | Disconnected | The Collector or the machines are not connected normally                                                                                                                                                                           |

- CycleTime

This is the cycle time status of the Collector. The status of the cycle time of the machine connected is judged comprehensively based on the judging criteria of the following table.

| Status of Cycle Time                                                                |              | Judging Criteria                                                                                   |
|-------------------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------|
|    | Good         | Status is Good for all the machines connected. There are no machines with Warning or Alarm status. |
|    | Warning      | Status is Warning for some of the machines connected. There are no machines with Alarm status.     |
|   | Alarm        | Status is Alarm for some of the machines connected.                                                |
|  | Disconnected | Status is Disconnected for all of the machines targeted for connection.                            |

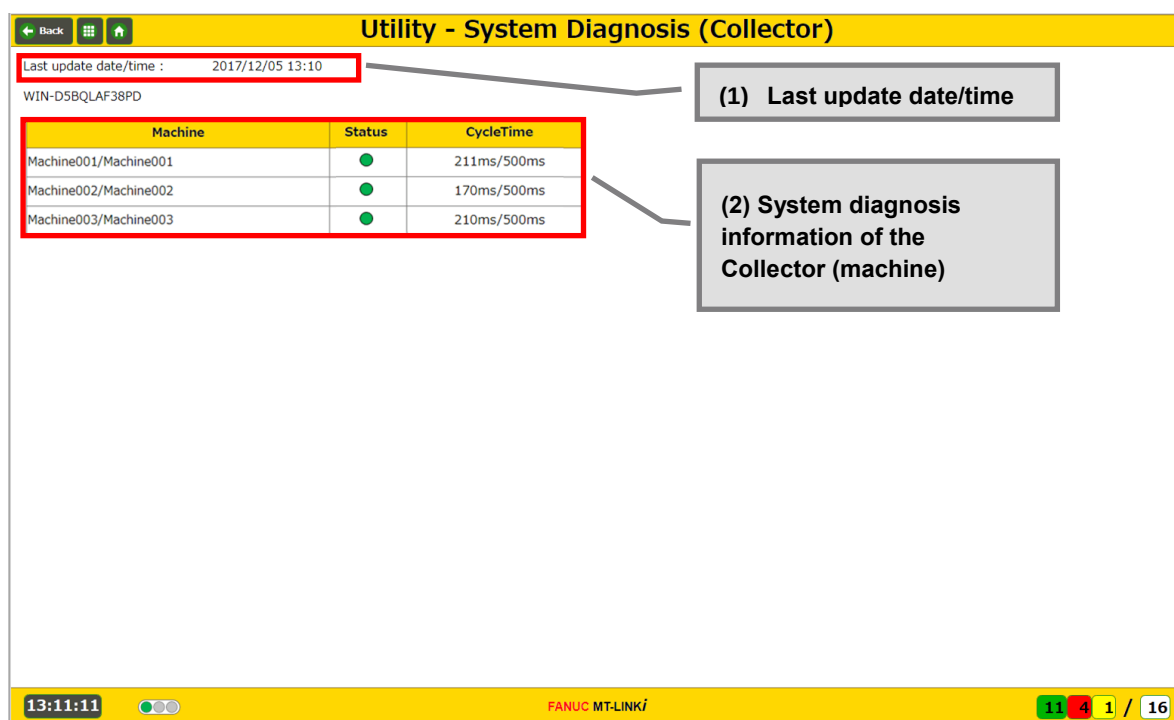
**NOTE**

- 1 For the details on the "Status of the machines," refer to the "**Status**" part of the "**Utility - System Diagnosis (Collector) Screen**" subsection.
- 2 Even though the Receiver and the Collector is running on the same PC, the displayed values may differ, as the time for the collection of usage rate of the CPU or the memory may be different.

|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | <b>A-40829-00001EN</b>                        |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 750/868 |

## 14.9.3.2 Utility - System Diagnosis (Collector) Screen

Displays the diagnosis information of the machines connected to the Collector, which was selected on the [Utility - System Diagnosis] screen (System diagnosis information).



### ■ [Utility - System Diagnosis (Collector)] screen

#### (1) Last update date/time

The last date/time the diagnosis information was collected from the Collector. The information is updated in an interval of 60 seconds.

#### (2) System diagnosis information of the Collector (machine)

The system diagnosis information of the machine connected to the Collector is displayed. Also, when the system diagnosis information cannot be collected, such as when the Receiver is stopped, the message "Failed to get system diagnosis information of Collector." will be displayed.

##### - Machine

Name of the machine is displayed in the form of "<Machine ID>/<Machine name>." Machine ID is a name that uniquely identifies the equipment defined with AdminTool.

##### - Status

The status of the connected machine is displayed with the following icons based on the cycle time.

| Status                                                                              |              | Range of Cycle Time (Default) |
|-------------------------------------------------------------------------------------|--------------|-------------------------------|
|  | Good         | 0 - 500ms                     |
|  | Warning      | 501 - 2000ms                  |
|  | Alarm        | 2001ms -                      |
|  | Disconnected | -                             |

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |                          |                                               |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |                          |                                               |                |
| Ed.  | Date          | Design | Description                        | <b>FANUC CORPORATION</b> |                                               | Sheet. 751/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |

## NOTE

The range of the cycle time can be customized by changing the system setting file (collector.ini). For the details, refer to the "**SYSTEM SETTING FILES**" section.

- CycleTime

The cycle time of the machine is displayed in the form of "<Cycle time>/500ms." However, when the Collector is disconnected, "-/-" will be displayed, and when the Collector is connected but the machine is disconnected, "0ms/500ms" will be displayed.

|      |               |        |                                            |       |  |        |       |                   |                                                       |  |        |         |  |  |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------|-------------------------------------------------------|--|--------|---------|--|--|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        |       | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |  |  |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                   |                                                       |  |        |         |  |  |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                   |                                                       |  |        |         |  |  |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       | Draw              | A-40829-00001EN                                       |  |        |         |  |  |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |       |                   |                                                       |  |        |         |  |  |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                   |                                                       |  |        |         |  |  |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       | FANUC CORPORATION |                                                       |  | Sheet. | 752/868 |  |  |
| Ed.  | Date          | Design | Description                                |       |  |        |       |                   |                                                       |  |        |         |  |  |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                   |                                                       |  |        |         |  |  |

## 14.9.4 CSV File Output of System Diagnosis Information

The retrieved diagnosis information is stored to the DB on an interval of 60seconds, and output as a CSV file at the same time.

By showing graphically the process load status in time series using the CSV file, the trend of the process load on the system can be analyzed.

The output destination folder, the file name, and the output format are shown below.

■ Output destination folder

<MT-LINK*i* installation folder>\MT-LINKiReceiver\DiagnoseLog

**NOTE**

If the free space of the output destination disk drive is less than 5GB, the file output process will not be executed.

■ File name (default)

diagnoselog\_<yyyyMMddHHmmss>.csv

Example) diagnoselog\_20171127111427.csv

**NOTE**

- 1 The "<yyyyMMddHHmmss>" part of the file name indicates the date/time of the file creation.
- 2 The file is created when the Receiver is started. When the date changes, a new file is created at the change of date.
- 3 The output files are retained for a retention period (45 days by default). Files which are older than the retention period will be deleted automatically when their new files are created.  
A retention period for CSV files can be customized by modifying the system setting file (receiver.ini).  
For the details, refer to the "**SYSTEM SETTING FILES**" section.
- 4 If the output target file is used with another program (for example, the output file is opened by a Text Editor on the day of creation), data writing process on that file will fail. In such a case, writing process of the retrieved data will not be attempted again.  
When using the output file with another program, use the file after making a copy of that file in another folder.

■ Output layout

| Item Name                         | Output Value                                                                           | Remarks   |
|-----------------------------------|----------------------------------------------------------------------------------------|-----------|
| Time                              | Output date/time of diagnosis information                                              |           |
| R_RECEIVER_CPU                    | CPU usage rate (%) of the Server PC                                                    | NOTE 1    |
| R_RECEIVER_MEMORY                 | Memory usage rate (%) of the Server PC                                                 | NOTE 1    |
| R_RECEIVER_QUEUE                  | Accumulation status of the Queue for the Receiver                                      | NOTE 2    |
| R_RECEIVER_RECEIVINGSIGNALCHANGES | Number of signal data change points received by the Receiver (number of points/second) | NOTE 1    |
| R_RECEIVER_STORINGSIGNALDATA      | Number of signal data change points stored to the DB (number of point/second)          | NOTE 1, 3 |
| S_STORAGE_DISKFREE                | Free disk space of the disk storing DB (GB)                                            |           |

|      |               |        |                                            |                   |                                                       |        |         |
|------|---------------|--------|--------------------------------------------|-------------------|-------------------------------------------------------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                   |                                                       |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                   |                                                       |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |                   |                                                       |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   | Draw              | <b>A-40829-00001EN</b>                                |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                   |                                                       |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |                   |                                                       |        |         |
| Ed.  | Date          | Design | Description                                | FANUC CORPORATION |                                                       | Sheet. | 753/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check             | Apprv.                                                | Inoue  |         |

| Item Name                    | Output Value                                                         | Remarks   |
|------------------------------|----------------------------------------------------------------------|-----------|
| S_STORAGE_DISKSTATUS         | Status of free disk space ratio for the disk storing DB              | NOTE 2    |
| P_PERSISTER_UPDATETIME       | Last change point (end date/time) stored to the DB                   |           |
| P_PERSISTER_RATE             | Usage rate of the cache area                                         | NOTE 1    |
| P_PERSISTER_STATUS           | Usage status of the cache area                                       | NOTE 2    |
| C_<Collector name>_CPU       | CPU usage rate (%) of the Collector PC                               | NOTE 1, 4 |
| C_<Collector name>_MEMORY    | Memory usage rate (%) of the Collector PC                            | NOTE 1, 4 |
| C_<Collector name>_QUEUE     | Accumulation status of the Queue for the Collector                   | NOTE 2, 4 |
| C_<Collector name>_CYCLETIME | Status of cycle time for all the machines connected to the Collector | NOTE 2, 4 |
| M_<Machine name>_STATUS      | Status of cycle time for each machine                                | NOTE 2, 5 |
| M_<Machine name>_CYCLETIME   | Cycle time of each machine                                           | NOTE 1, 5 |

## NOTE

- 1 If there is no value, due to reasons such as the Receiver had just been started, the character "-" will be output.
- 2 Following values are output according to each status.
 

|                                                                                   |                       |   |
|-----------------------------------------------------------------------------------|-----------------------|---|
|  | Unknown/Disconnected: | 0 |
|  | Good:                 | 1 |
|  | Warning:              | 2 |
|  | Alarm:                | 3 |
- 3 When DB deletion process set by [DB Data Retention Period Setting] (System Setting) is in progress, "-" will be output.
- 4 If two or more Collectors are registered, the number of outputs of "C\_<Collector name>\_CPU" to "M\_<Machine name>\_CYCLETIME" will be equal to the number of Collectors. The order of output for the Collectors is same as the order of display (ascending order) in the [Collector] column of the [Utility – System Diagnosis] screen.
- 5 If two or more machines are connected to the Collector, number of outputs of "M\_<Machine name>\_STATUS" to "M\_<Machine name>\_CYCLETIME" will be equal to the number of machines. The order of output for the machines is same as the order of display (ascending order) in the [Machine] column of the [Utility – System Diagnosis (Collector)] screen.

- Character code  
UTF-8 (with BOM)

|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | <b>A-40829-00001EN</b>                        |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 754/868 |

■ Output sample

Output date/time: 2017/11/27 11:14:27 to 11:22:26

Collector name: Collector01

Machine name: Machine001, Machine002, Machine003

|    | A                   | B              | C                 | D                | E                                 | F                            |
|----|---------------------|----------------|-------------------|------------------|-----------------------------------|------------------------------|
| 1  | Time                | R_RECEIVER_CPU | R_RECEIVER_MEMORY | R_RECEIVER_QUEUE | R_RECEIVER_RECEIVINGSIGNALCHANGES | R_RECEIVER_STORINGSIGNALDATA |
| 2  | 2018/06/06 15:46:13 | 10             | 30                | 1                | 19                                | -                            |
| 3  | 2018/06/06 15:47:12 | 11             | 31                | 1                | 0                                 | 0                            |
| 4  | 2018/06/06 15:48:12 | 10             | 30                | 1                | 29                                | 29                           |
| 5  | 2018/06/06 15:49:12 | 12             | 32                | 1                | 55                                | 54                           |
| 6  | 2018/06/06 15:50:12 | 11             | 31                | 1                | 40                                | 40                           |
| 7  | 2018/06/06 15:51:12 | 11             | 31                | 1                | 57                                | 58                           |
| 8  | 2018/06/06 15:52:12 | 9              | 29                | 1                | 53                                | 53                           |
| 9  | 2018/06/06 15:53:12 | 12             | 32                | 1                | 46                                | 46                           |
| 10 | 2018/06/06 15:54:12 | 9              | 29                | 1                | 59                                | 59                           |

|    | G                       | H                | I                  | J                  | K                    |
|----|-------------------------|------------------|--------------------|--------------------|----------------------|
| 1  | P_PERSISTER_UPDATETIME  | P_PERSISTER_RATE | P_PERSISTER_STATUS | S_STORAGE_DISKFREE | S_STORAGE_DISKSTATUS |
| 2  | 2018/06/06 15:45:50.000 | 0                | 1                  | 194.14             | 1                    |
| 3  | 2018/06/06 15:46:50.500 | 0                | 1                  | 194.14             | 1                    |
| 4  | 2018/06/06 15:48:11.000 | 0                | 1                  | 194.14             | 1                    |
| 5  | 2018/06/06 15:49:10.000 | 0                | 1                  | 194.14             | 1                    |
| 6  | 2018/06/06 15:50:10.500 | 0                | 1                  | 194.14             | 1                    |
| 7  | 2018/06/06 15:51:10.500 | 0                | 1                  | 194.14             | 1                    |
| 8  | 2018/06/06 15:52:11.000 | 0                | 1                  | 194.14             | 1                    |
| 9  | 2018/06/06 15:53:11.000 | 0                | 1                  | 194.14             | 1                    |
| 10 | 2018/06/06 15:54:11.000 | 0                | 1                  | 194.14             | 1                    |

|    | L                 | M                    | N                   | O                       |
|----|-------------------|----------------------|---------------------|-------------------------|
| 1  | C_Collector01_CPU | C_Collector01_MEMORY | C_Collector01_QUEUE | C_Collector01_CYCLETIME |
| 2  | 9                 | 29                   | 1                   | 1                       |
| 3  | 12                | 32                   | 1                   | 1                       |
| 4  | 9                 | 29                   | 1                   | 1                       |
| 5  | 10                | 30                   | 1                   | 1                       |
| 6  | 10                | 30                   | 1                   | 1                       |
| 7  | 11                | 31                   | 1                   | 1                       |
| 8  | 14                | 34                   | 1                   | 1                       |
| 9  | 11                | 31                   | 1                   | 1                       |
| 10 | 9                 | 29                   | 1                   | 1                       |

|    | P                   | Q                      | R                   | S                      | T                   | U                      |
|----|---------------------|------------------------|---------------------|------------------------|---------------------|------------------------|
| 1  | M_Machine001_STATUS | M_Machine001_CYCLETIME | M_Machine002_STATUS | M_Machine002_CYCLETIME | M_Machine003_STATUS | M_Machine003_CYCLETIME |
| 2  | 1                   | 351                    | 1                   | 289                    | 1                   | 315                    |
| 3  | 1                   | 353                    | 1                   | 291                    | 1                   | 314                    |
| 4  | 1                   | 357                    | 1                   | 292                    | 1                   | 318                    |
| 5  | 1                   | 362                    | 1                   | 293                    | 1                   | 322                    |
| 6  | 1                   | 357                    | 1                   | 293                    | 1                   | 319                    |
| 7  | 1                   | 370                    | 1                   | 300                    | 1                   | 331                    |
| 8  | 1                   | 371                    | 1                   | 307                    | 1                   | 333                    |
| 9  | 1                   | 380                    | 1                   | 311                    | 1                   | 340                    |
| 10 | 1                   | 371                    | 1                   | 303                    | 1                   | 333                    |

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |       |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|-------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |       |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |       |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |       |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |       |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |       |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |       |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 755/868 |       |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |       |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         | Inoue |
|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |       |

#### 14.9.5 Operation of System Diagnosis and Countermeasures against System Anomaly

Just after starting the operation of MT-LINK*i*, or if any of the settings is changed (addition of machine or equipment, addition of signal to retrieve, etc.), display the [Utility - System Diagnosis] screen to check if there are no problems in the operation of the system.

Also, try to check the [Utility - System Diagnosis] screen periodically during the operation.

If the diagnosis signal lamp is lighting in a color other than green (indicating "Good") or is not lighting on a Web screen, or if items on the [Utility - System Diagnosis] screen are indicating anomaly states, take appropriate countermeasures by referring to the following subsection of the "FANUC MT-LINK*i* / Integration Server Troubleshooting MANUAL":

The "Operation of System Diagnosis and Countermeasures against Anomaly" subsection in the "TROUBLES RELATED TO SYSTEM DIAGNOSIS" section in the "MT-LINK*i* TROUBLESHOOTING" chapter

|      |               |        |                                            |       |  |        |       |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        |       | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                   |                                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |       | Draw              | A-40829-00001EN                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                   |                                                       |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       | FANUC CORPORATION |                                                       |  | Sheet. | 756/868 |
| Date | Nov.30.2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                   |                                                       |  |        |         |



## 14.10 CUSTOMIZING THE WEB SCREEN

By editing the system setting files (Setting.js, collector.ini, receiver.ini) provided by MT-LINK*i*, items such as the colors displayed on the Web screen or the threshold values of the system diagnosis function can be customized.

- (1) The following items can be set with Setting.js.
- Colors of the scale on the [Results - Operational Results] screen, etc. (CSS attribute value)  
Set this setting to customize the color of the scale of the graph on the [Results - Operational Results] screen, etc.
  - Colors of the state of equipment text (CSS attribute value)  
Set this setting to customize the colors of the state of equipment text on the monitoring screens.
  - Colors of the state of equipment shown on the graph of [Results - Operational Results] screen, etc. (CSS attribute value)  
Set this setting to customize the colors of the statuses of equipment shown on the table or graph of [Results - Operational Results] screen, etc.
  - Legend name of equipment operation status proportion table (CSS attribute value)  
Set this setting to customize the legend name of equipment operation status proportion table displayed for each operation status of NC.
  - Update interval of the screens  
Set this item to customize intervals for the monitoring screens to refresh the states of equipment units and the information about signals, tool life signals, and periodic maintenance signals.
  - Previous value retention setting for the [Monitoring - Equipment Monitoring [equipment]] screen, to which users can move from the [Overlook] screen  
Set this setting to customize whether to retain the previous values for the screens to which users can move from the [Monitoring - Equipment Monitoring [equipment]] screen using the screen selector.
  - Show/hide [Home] screen  
Set this setting to customize whether to show/hide the [Home] screen when signing in to the Web screen.
  - Auto refresh interval on results screens  
Set this setting to customize the auto refresh interval for results screens (the [Results - Group Results], [Results - Operational Results], and [Results - Production Results] screens).
  - Alarm history exclusion setting  
Set this setting to narrow down the search results to be listed in the [Alarm] list of the [Diagnosis - Alarm History] screen by excluding the alarm history data which match the specified exclusion conditions.
  - Display format of real numbers for the [Diagnosis - Macro Variable History] screen  
Set this setting to display values (real numbers with fractional part) of macro variables on the screen in the display format specified with this setting. The screen will refer to this setting in the first session.

### NOTE

When Setting.js is edited, restart the Server PC.

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 757/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         |

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |       |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|-------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |       |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |       |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |       |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |       |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |       |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |       |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 758/868 |       |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |       |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         | Inoue |

## 14.11 BACKING UP AND RESTORING DATABASE

It is recommended to periodically back up MT-LINK*i* database in case of trouble.  
This section describes how to back up and restore the database.

### 14.11.1 Backing up Database

This subsection describes how to back up MT-LINK*i* database.

#### NOTE

Prior to backing up the database, make sure to stop MongoDB service; otherwise, copying the database may fail.

- (1) Start MT-LINK*i* AdminTool. Select [Start/Stop System] in the AdminTool Menu.
- (2) The [Start/Stop System] window appears. Click the [Stop All] button to stop MT-LINK*i*.
- (3) Close the AdminTool Menu.
- (4) Perform the following to stop MongoDB service.  
Click [Control Panel] > [Administrative Tool] > [Services]. Right-click the [MongoDB] service and select [Stop] in the context menu.
- (5) Copy the MT-LINK*i* database folder (mtl\_data folder), including its subfolders and files, to a back-up folder.  
  
MT-LINK*i* database folder: <MT-LINK*i* installation folder>\MongoDB\mtl\_data  
\* Note that the path to the database may have been changed from its default through DB data storage setting feature.
- (6) Perform the following to start MongoDB service.  
Click [Control Panel] > [Administrative Tool] > [Services]. Right-click the [MongoDB] service and select [Start] in the context menu.
- (7) Start MT-LINK*i* AdminTool. Select [Start/Stop System] in the AdminTool Menu.
- (8) The [Start/Stop System] window appears. Click the [Start All] button to start MT-LINK*i*.

|      |               |        |                                            |                   |                                                       |        |                |
|------|---------------|--------|--------------------------------------------|-------------------|-------------------------------------------------------|--------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                   |                                                       |        |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                   |                                                       |        |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw              | A-40829-00001EN                                       |        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                   |                                                       |        |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                   |                                                       |        |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | FANUC CORPORATION |                                                       |        | Sheet. 759/868 |
| Ed.  | Date          | Design | Description                                |                   |                                                       |        |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check             |                                                       | Apprv. | Inoue          |

## 14.11.2 Restoring Database

This subsection describes how to restore a backed-up MT-LINK*i* database.

### 14.11.2.1 Restoring It to the Same Server PC as It Was Backed up from

When restoring a backed-up MT-LINK*i* database data to the same Server PC as it was backed up from, perform the procedure written in **NOTE**, subsequent to the following paragraph in the "**Data Storing Setting of Database**" subsection:

"If a folder storing an existing database is specified, the database in that folder is connected"

### 14.11.2.2 Restoring It to a Different Server PC

When restoring a backed-up MT-LINK*i* database data to the new Server PC (other than the old Server PC on which the backup was made), perform the procedure written in the "STEPS TO MIGRATE MT-LINK*i* (SERVER PC)" section in the "STEPS FOR MIGRATING SERVER PC / COLLECTOR PC WITH DATABASE" chapter in the "FANUC MT-LINK*i* Install MANUAL."

#### NOTE

After step (5) in the "Migration steps to perform on the new Server PC" procedure in the "STEPS TO MIGRATE MT-LINK*i* (SERVER PC)" section is performed, MT-LINK*i* database is migrated to the folder mtl\_data on the new Server PC. At this point, restore the backed-up database on the new Server PC. The folder path to mtl\_data is as follows unless the storage path of the database has been changed from the default:

<MT-LINK*i* installation folder>\MongoDB\

|      |               |        |                                            |                          |                                                       |        |        |         |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |        | Sheet. | 760/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    |                                                       | Apprv. | Inoue  |         |

# 14.12 LDAP AUTHENTICATION

## 14.12.1 Overview on LDAP Authentication

LDAP stands for Lightweight Directory Access Protocol, a client-server protocol for accessing a directory service. MT-LINK*i* users can sign in to the Web Server by using LDAP authentication functionality of MT-LINK*i*. Alternatively, when LDAP authentication users sign in to the MT-LINK*i* Web Server for the first time, their MT-LINK*i* user accounts can be automatically registered and then their authorities can be modified after the registration.

MT-LINK*i* supports the following directory services:

- Active Directory (developed by Microsoft)
- OpenLDAP (open source implementation)

### ■ LDAP basics

#### (1) LDAP distinguished name (DN)

An LDAP distinguished name is comprised of one or more LDAP attributes separated by commas.

Example: DC=fanuc,DC=co,DC=jp

DNs can be specified for the [Domain], [Account], [Search Scope], and [LDAP Filter] fields of the [LDAP Authentication Setting] window.

The following table lists main LDAP attributes.

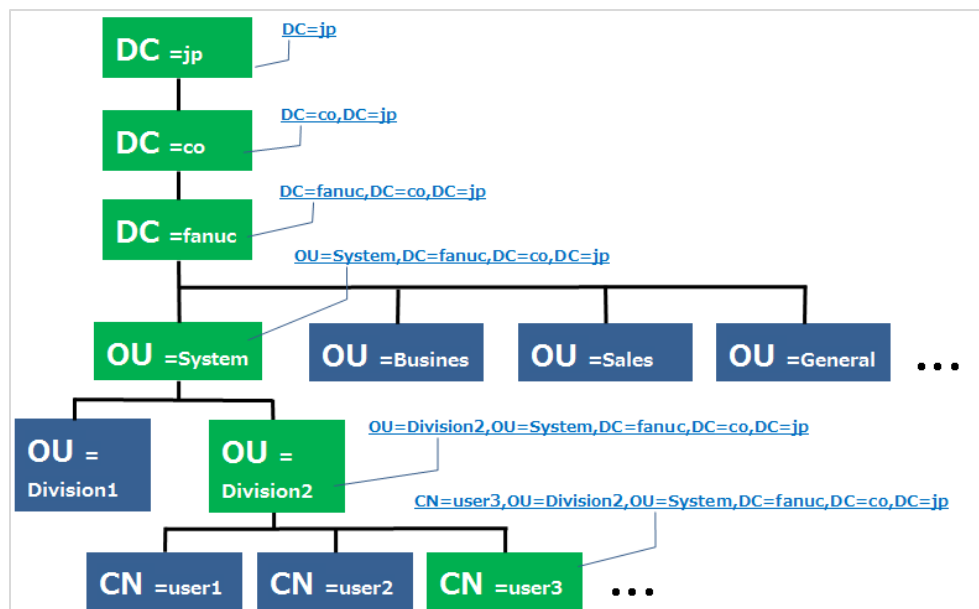
| LDAP attribute | Standing for        | Description                                               |
|----------------|---------------------|-----------------------------------------------------------|
| DC             | Domain Component    | A domain name                                             |
| OU             | Organizational Unit | An organization unit                                      |
| CN             | Common Name         | A user name, group name, computer name, or container name |

\* LDAP attributes are defined as case-insensitive.

\* DN (Distinguished Name) denotes a distinguished name comprised of comma-separated LDAP attributes.

#### (2) LDAP tree structure

The following figure shows an example of LDAP tree structure.



\* An LDAP distinguished name (DN) is comprised of comma-separated LDAP attributes ascending the tree from left to right.

|      |               |        |                                            |                   |                                                       |        |         |
|------|---------------|--------|--------------------------------------------|-------------------|-------------------------------------------------------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                   |                                                       |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                   |                                                       |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |                   |                                                       |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   | Draw              | A-40829-00001EN                                       |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                   |                                                       |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |                   |                                                       |        |         |
| Ed.  | Date          | Design | Description                                | FANUC CORPORATION |                                                       | Sheet. | 761/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      |                   |                                                       |        |         |
|      |               | Check  |                                            | Apprv.            | Inoue                                                 |        |         |

## 14.12.2 Settings for MT-LINK*i* to Activate LDAP Authentication

This subsection describes how to activate LDAP authentication functionality of MT-LINK*i*.

(1) LDAP Authentication Setting

Use this window to set up LDAP authentication functionality on MT-LINK*i*.

This window holds such information as the IP address, host name, and port number of an LDAP server. For the details, refer to "[Set LDAP Authentication](#)" subsection.

To activate LDAP authentication functionality, select the [Enable] check box in the [LDAP Authentication] group box of the [Authority Setting] window.

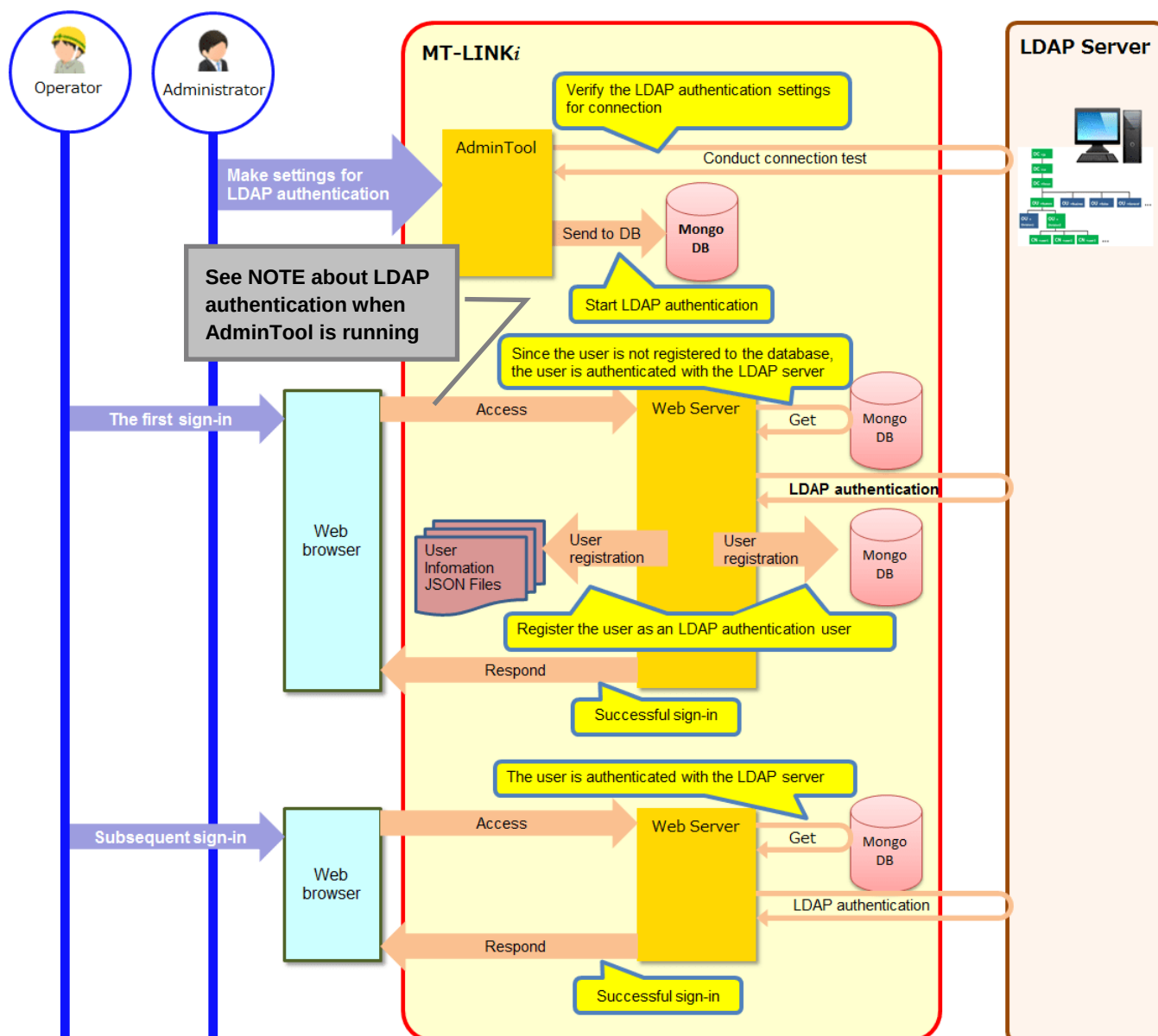
(2) Authority Setting

Use this window to make authority settings for LDAP authentication users to access functionalities of MT-LINK*i*. To activate LDAP authentication functionality for a user, select the [Use LDAP Authentication] check box in the [Add/Edit User] dialog box for the user. For the details, refer to "[AUTHORITY SETTING](#)" section.

|      |               |        |                                            |                          |                                                       |       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |       | Sheet. 762/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue |                |

## 14.12.3 Operational Flow of Using LDAP Authentication Functionality

The following figure shows a basic operational flow of using LDAP authentication functionality.



|      |               |        |                                            |       |  |        |                   |                                                       |  |  |        |         |       |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--|--------|---------|-------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |  |        |         |       |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |  |        |         |       |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |  |        |         |       |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |  |        |         |       |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |  |        |         |       |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |  |        |         |       |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  |  | Sheet. | 763/868 |       |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |  |        |         |       |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |  |        |         | Inoue |
|      |               |        |                                            |       |  |        |                   |                                                       |  |  |        |         |       |

**NOTE**

- LDAP authentication when AdminTool is running  
When an LDAP authentication user signs in to Web Server for the first time while the administrator is using specific setting windows of AdminTool to modify user settings, the user is denied to sign in.

Those specific setting windows are listed below. As soon as completing user setting, the administrator should close AdminTool:

| Functionality           | Setting window    |
|-------------------------|-------------------|
| Easy Operation          | Easy Setting      |
| Specific Data Setting   | Authority Setting |
| Setting Check           | Data Validation   |
| Application of Settings | Send to DB        |
| System Control          | Start/Stop System |
| Batch Setting           | Export            |
| Settings Backup         | Backup Settings   |
|                         | Restore Settings  |
|                         | Receive from DB   |
|                         | Import            |

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               | Sheet. 764/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |



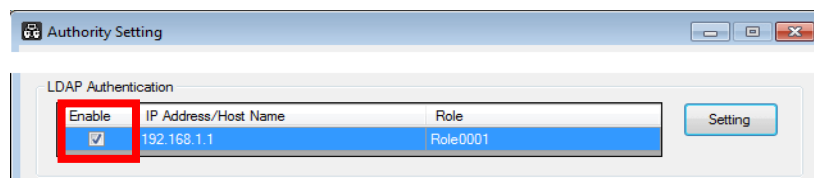
## 14.12.4 How Sign-in from Web Browser Is Authenticated

When a user tries to sign in to the Web Server from a Web browser, MT-LINK*i* authenticates the user either with the LDAP server or with itself against the password registered to AdminTool. By referring to the selection of the following check boxes, MT-LINK*i* determines per user which server to authenticate with:

(A) LDAP authentication functionality setting (applicable to all the users)

The [Enable] check box in the [LDAP Authentication] group box of the [Authority Setting] window

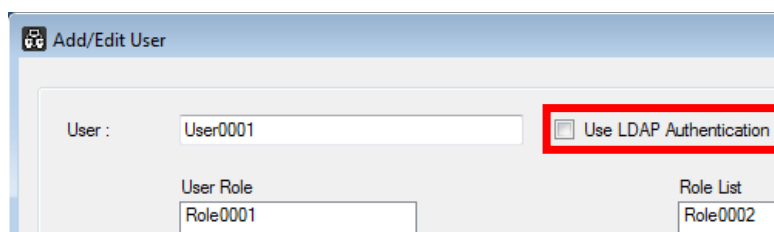
Selecting this check box activates LDAP authentication functionality of MT-LINK*i*.



(B) LDAP authentication user setting (applicable per user)

The [Use LDAP Authentication] check box in the [Add/Edit User] dialog box

Selecting this check box enables a user to authenticate with the LDAP server when signing in from a Web browser.



### NOTE

To register a user as an LDAP authentication user, select both check boxes described above.

The following table shows how a user is authenticated when signing in from a Web browser.

| Check box selection |                                   | User credentials are registered in the database<br>(Yes: registered, No: unregistered) |                                     | Authentication result                                                                 |
|---------------------|-----------------------------------|----------------------------------------------------------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------|
| (A)<br>Enable       | (B)<br>Use LDAP<br>Authentication | Registered to<br>AdminTool                                                             | Registered in<br>the LDAP<br>server |                                                                                       |
| Selected            | (Unregistered)                    | No                                                                                     | Yes                                 | Authenticated by the LDAP server and automatically registered to AdminTool (see NOTE) |
|                     | Selected                          | Yes                                                                                    | Yes                                 | Authenticated by the LDAP server                                                      |
|                     |                                   | Yes                                                                                    | No                                  | Sign-in fails: LDAP authentication error                                              |
|                     | Cleared                           | Yes                                                                                    | Yes                                 | Authenticated against the password registered to AdminTool                            |
|                     |                                   | Yes                                                                                    | No                                  |                                                                                       |
| Cleared             | (Unregistered)                    | No                                                                                     | Yes                                 | Sign-in fails: LDAP authentication error due to not being registered to AdminTool     |
|                     | Selected                          | Yes                                                                                    | Yes                                 | Sign-in fails: (see * below)                                                          |
|                     |                                   | Yes                                                                                    | No                                  |                                                                                       |

|      |               |        |                                            |                          |                                                       |        |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|--------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |        |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |        |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |        |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |        |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |        | Sheet. 765/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |        |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    |                                                       | Apprv. | Inoue          |

|  |         |     |     |                                                            |
|--|---------|-----|-----|------------------------------------------------------------|
|  | Cleared | Yes | Yes | Authenticated against the password registered to AdminTool |
|  |         | Yes | No  |                                                            |

\* When the [Enable] check box is cleared and the [Use LDAP Authentication] check box is selected, the user will be denied to sign in.

#### NOTE

■ Automatic registration of LDAP authentication users as MT-LINK*i* users:

(1) If the [Create a user account at first sign-in] check box is selected in the [LDAP Authentication Setting] window, an LDAP authentication user will be assigned an MT-LINK*i* user account. MT-LINK*i* retrieves the following information about the user from the LDAP server:

- User name
- E-mail address

However, since MT-LINK*i* will not preserve the passwords of the LDAP authentication users, they must be managed by the LDAP server.

(2) In the following cases, however, automatic registration will not work:

- AdminTool displays specific windows (for the details, refer to "[Operational Flow of Using LDAP Authentication Functionality](#)" subsection).
- The [Create a user account at first sign-in] check box is cleared in the [LDAP Authentication Setting] window.
- The number of MT-LINK*i* users has reached the limit (1000 users).

Also automatic registration will not work in the following cases when signing in is performed not via the [MT-LINK*i* Web Sign in] screen:

- LDAP authentication users try to sign in via Web APIs for basic authentication.
- LDAP authentication users try to sign in via the Linkage tool for MT-LINK*i* to use SERVO VIEWER linkage functionality of MT-LINK*i*.

|      |               |        |                                            |                          |                                                       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       | Sheet. 766/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue          |

## 14.13 COLLECT PRODUCTION RESULTS/PRODUCT NAME AUTOMATICALLY

Hereafter, the term "production results" denotes "production results" displayed on the [Results – Production Results] Web screen, and the term "a product name" denotes "a product name" displayed on both the [Results – Production Results] and the [Results - Operational Results] screens.

To collect production results and product name automatically, you need to assign a signal to the production results status (ProductResultNumber), and a signal to the product name status (ProductName).

ProductResultNumber and ProductName are a basic status that are preregistered in the [Multiple assign status area] table in the [Equipment Setting] window. By clicking the [Preset] button, CNC parameter PartsNum is assigned to ProductResultNumber as a signal for the number of machined parts (or of machined products), and the program comment signal MainComment is assigned to ProductName as a signal for the name of the product automatically.

Usually you can use this automatic assignment without modification in most of machine configuration scenarios. For the details, refer to the "[Assigning CNC Parameter and Program Comment Signal to Basic Signals](#)" subsection.

However, you may not collect data from a machine because of the following reasons:

For the details, contact your distributor.

- (1) "Run hour" and "parts count display" option is set as disabled in the machine.
- (2) Your distributor has customized the machine to output the number of machined parts to a custom macro variable, a PMC signal, or a P-CODE variable (not to a CNC parameter).

In the case (2) above, you can assign and set a custom signal (a custom macro variable, a PMC signal, or a P-CODE variable) that outputs the number of machined parts to ProductResultNumber. For the details, refer to the "[Assigning FanucCNC / FanucRobodrig Custom Signal to ProductResultNumber](#)" subsection.

Also, you can assign another signal, such as MainProgram, other than MainComment, to ProductName. For the details, refer to the "[Assigning FanucCNC / FanucRobodrig Signal to ProductName](#)" subsection.

Note that if ProductName is not collected, "NO PRODUCT" will be displayed in Product name column on the [Results - Production Results] and the [Results - Operational Results] screens.

<ProductResultNumber>

| Data Source                                  | Signal for Production Results    |
|----------------------------------------------|----------------------------------|
| CNC parameter (the number of machined parts) | PartsNum_path<x> (<x>: CNC path) |
| Custom macro variable                        | Custom signal                    |
| PMC signal                                   | Custom signal                    |
| P-CODE variable                              | Custom signal                    |

<ProductName>

| Data Source     | Signal for Product Name |
|-----------------|-------------------------|
| Program comment | MainComment_path1       |
| Main program    | MainProgram_path1       |

|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | <b>A-40829-00001EN</b>                        |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 767/868 |

## NOTE

A program comment will be retrieved that is written on the same line in a program as the program number. However, if multiple comments are on the line, only the program comment immediately following the program number will be retrieved.

The following examples of program comments show how to write a program comment in a program to be retrieved:

Example 1) A program comment is retrieved that is written on the same line as the program number.

```
O0001 (TEST_PROGRAM COMMENT);
M08 (COOLANT ON);
```

In this case, "TEST\_PROGRAM COMMENT" will be retrieved as a program comment. The program comment on the second line will be discarded.

Example 2) No program comment is written on the same line as the program number.

```
O0001;
(TEST_PROGRAM COMMENT);
M08 (COOLANT ON);
```

In this case, no program comment will be retrieved.

Example 3) Multiple program comments are on the same line as the program number. The program comment immediately after the program number is retrieved.

```
O0001 (COMMENT) (TEST_PROGRAM COMMENT);
M08 (COOLANT ON);
```

In this case, "COMMENT" will be retrieved as a program comment.

|      |               |        |                                    |                          |                                               |        |        |         |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |        | Sheet. | 768/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    |                                               | Apprv. | Inoue  |         |

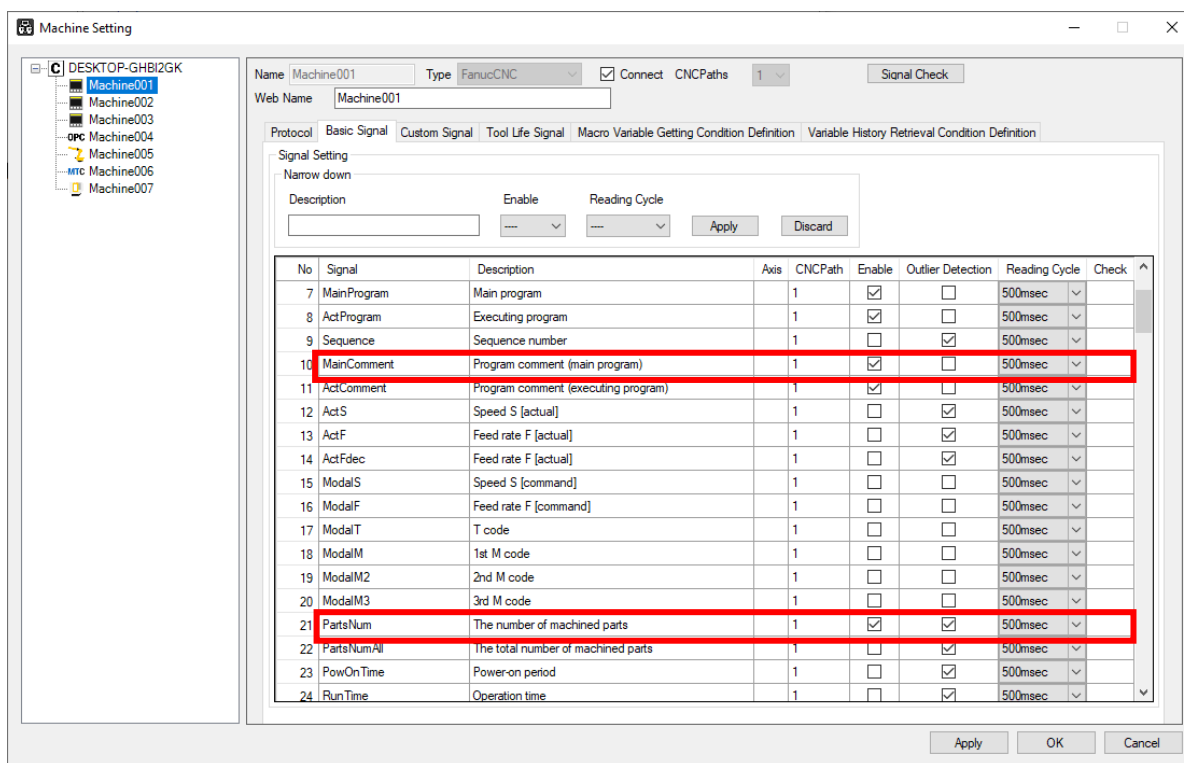
## 14.13.1 Assigning CNC Parameter and Program Comment Signal to Basic Signals

This subsection explains how to assign FanucCNC parameter PartsNum and a program comment signal to basic statuses.

In most scenarios, all you have to do with signal settings is to perform the presetting steps below. In case, however, you need to assign a specific signal other than the preset signals, refer to the "[Assigning FanucCNC / FanucRobodrill Custom Signal to ProductResultNumber](#)" subsection and the "[Assigning FanucCNC / FanucRobodrill Signal to ProductName](#)" subsection.

Perform the steps below:

- (1) Confirm that MainComment and PartsNum signals are enabled in [Signal setting] tab of the [Machine Setting] window.



|      |               |        |                                    |                   |                                               |        |         |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |                   |                                               |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw              | A-40829-00001EN                               |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |                   |                                               |        |         |
| Ed.  | Date          | Design | Description                        | FANUC CORPORATION |                                               | Sheet. | 769/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             | Apprv.                                        | Inoue  |         |

- 
- Equipment Setting
- Equipment
- Equipment0001
  - Equipment0002
  - Equipment0003
  - opc Equipment0004
  - mtc Equipment0005
  - Equipment0006
  - Equipment0007
  - Equipment0008
- Basic Setting
- Equipment Name:  ☐ cancel monitoring.
- Web Name:
- Icon Setting
- Status Setting
- Simple assign status area
- | No. | Priority/View | Status | Machine Name | Web Name |
|-----|---------------|--------|--------------|----------|
|     |               |        |              |          |
- Preset
- Reload
- Add
- Delete
- Append Comments
- Multiple assign status area
- | No. | Priority/View            | Status                | Signal Definition | Web Name              |
|-----|--------------------------|-----------------------|-------------------|-----------------------|
| 6   | <input type="checkbox"/> | STOP                  |                   | STOP                  |
| 7   | <input type="checkbox"/> | MANUAL                |                   | MANUAL                |
| 8   | <input type="checkbox"/> | WARMUP                |                   | WARMUP                |
| 9   | <input type="checkbox"/> | WARNING               |                   | WARNING               |
| 10  | <input type="checkbox"/> | condition             |                   | condition             |
| 11  | <input type="checkbox"/> | Product Name          |                   | Product Name          |
| 12  | <input type="checkbox"/> | Product Result Number |                   | Product Result Number |
| 13  | <input type="checkbox"/> | Product Serial Number |                   | Product Serial Number |
| 14  | <input type="checkbox"/> | Operator ID           |                   | Operator ID           |
- Add
- Delete
- Automatic definition of signals
- Error Check
- Error Check Setting
- Append Comments for All Equipment
- Apply
- OK
- Cancel
- Click
- Product Name  
Product Result Number  
Product Serial Number  
Operator ID are red.

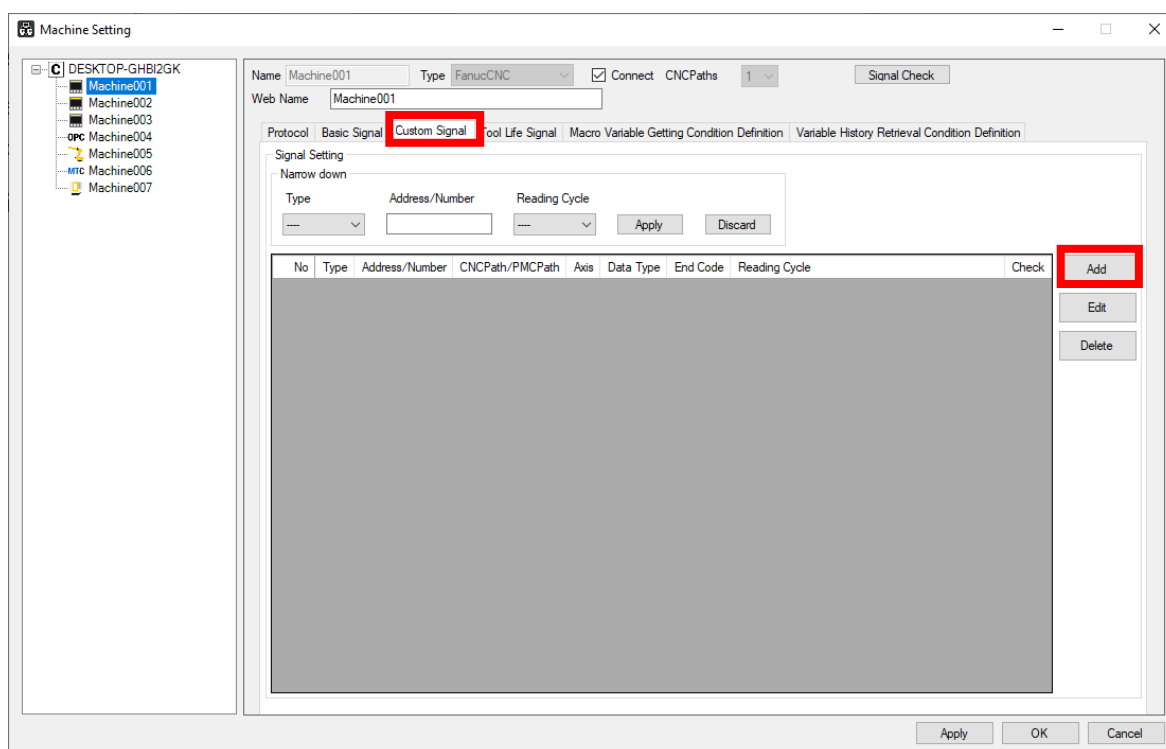
- 
- Equipment Setting**
- Equipment
- Equipment001
  - Equipment002
  - Equipment003
  - Equipment004
  - Equipment005
  - Equipment006
  - Equipment007
  - Equipment008**
- Basic Setting
- Equipment Name:  ☐ cancel monitoring.
- Web Name:
- Icon Setting
- Status Setting
- Simple assign status area
- | No | Priority                            | Status                  | Machine Name | Web Name                        |
|----|-------------------------------------|-------------------------|--------------|---------------------------------|
| 1  | <input checked="" type="checkbox"/> | Disconnect_Machine001   | Machine001   | Disconnect_Machine001           |
| 2  | <input type="checkbox"/>            | Mode_path1_Machine0...  | Machine001   | Mode Machine001 P1              |
| 3  | <input type="checkbox"/>            | MainProgram_path1_M...  | Machine001   | Main program Machine001 P1      |
| 4  | <input type="checkbox"/>            | ActProgram_path1_Ma...  | Machine001   | Executing program Machine0...   |
| 5  | <input type="checkbox"/>            | MainComment_path1_...   | Machine001   | Program comment (main progr...  |
| 6  | <input type="checkbox"/>            | ActComment_path1_Ma...  | Machine001   | Program comment (executing ...  |
| 7  | <input type="checkbox"/>            | PartsNum_path1_Machi... | Machine001   | The number of machined part...  |
|    | <input type="checkbox"/>            | SigCUT_path1_Machin...  | Machine001   | Cutting feed signal Machine0... |
|    | <input type="checkbox"/>            | SigSBK_path1_Machin...  | Machine001   | Single block check signal Ma... |
- Signal Definition
- | Priority | Status                                       | Signal Definition       | Web Name            |
|----------|----------------------------------------------|-------------------------|---------------------|
|          | <input type="checkbox"/> STOP                | (1 Disconnect_Machin... | STOP                |
|          | <input type="checkbox"/> MANUAL              |                         | MANUAL              |
|          | <input type="checkbox"/> WARMUP              |                         | WARMUP              |
| 9        | <input type="checkbox"/> WARNING             | (1 Disconnect_Machin... | WARNING             |
| 10       | <input type="checkbox"/> condition           | DISCONNECT 2 DISC...    | condition           |
| 11       | <input type="checkbox"/> ProductName         | MainComment_path1_...   | productName         |
| 12       | <input type="checkbox"/> ProductResultNumber | PartsNum_path1_Mac...   | ProductResultNumber |
| 13       | <input type="checkbox"/> ProductSerialNumber |                         | ProductSerialNumber |
| 14       | <input type="checkbox"/> OperatorID          |                         | OperatorID          |
- MainComment is assigned to ResultNumber, and Comment is assigned to Name
- Buttons: Preset, Reload, Add, Delete, Append Comments, Add, Delete, Automatic definition of signals, Error Check, Apply, OK, Cancel

|      |               |        |                                    |       |  |        |                   |                                               |  |        |         |
|------|---------------|--------|------------------------------------|-------|--|--------|-------------------|-----------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |                   |                                               |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |                   |                                               |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        | Draw              | A-40829-00001EN                               |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        |                   |                                               |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |                   |                                               |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        | FANUC CORPORATION |                                               |  | Sheet. | 770/868 |
| Ed.  | Date          | Design | Description                        |       |  |        |                   |                                               |  |        |         |
| Date | Nov.30.2015   | Desig. | Ogawa                              | Check |  | Apprv. |                   |                                               |  |        |         |

## 14.13.2 Assigning FanucCNC / FanucRobodrill Custom Signal to ProductResultNumber

In case that a machine distributor has customized signals so that the machine outputs the production results to a PMC signal, a macro variable, or a P-CODE variable, you can collect production results automatically by assigning the customized signals to ProductResultNumber status.

- (1) Before assigning, get information about the signal, such as its type and address.
- (2) Based on the gathered information, configure the signal as a custom signal for a machine. To do that, select [Machine Setting] in the AdminTool Menu to open the [Machine Setting] window, select a target machine, and select the [Custom Signal] tab.



|      |               |        |                                            |                   |                                                       |        |         |
|------|---------------|--------|--------------------------------------------|-------------------|-------------------------------------------------------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                   |                                                       |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                   |                                                       |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |                   |                                                       |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   | Draw              | <b>A-40829-00001EN</b>                                |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                   |                                                       |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |                   |                                                       |        |         |
| Ed.  | Date          | Design | Description                                | FANUC CORPORATION |                                                       | Sheet. | 771/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check             | Apprv.                                                | Inoue  |         |

(3) Click the [Add] button to open the [Signal Setting] dialog box.

(4) Specify the required fields with the gathered information.

(5) Double-click the [ProductResultNumber] row in the [Multiple assign status area] table.

|      |               |        |                                            |       |  |        |       |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        | Draw  | A-40829-00001EN                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                                                       |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue | FANUC CORPORATION                                     |  | Sheet. | 772/868 |



- (6) The [Signal Setting] dialog box appears. Set the [Web Name] field to a signal name that is to be displayed on Web screens, and the [User Signal Definition] field to a signal definition of the signal. Click [OK].

| Type of Custom Signal |                    | Value of Signal Definition                                               | Remark                                                                                                                                |
|-----------------------|--------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Custom macro variable |                    | MacroVar_<n>_path<x>_<Machine name><br>Ex) MacroVar_500_path1_Machine001 | <n>: custom macro variable number<br><x>: CNC path number (1 to 15)                                                                   |
| PMC                   | BIT                | PMC_<a>_<t>_path<x>_<Machine name><br>Ex) PMC_D200_0_path1_Machine001    | <a>: PMC signal address<br><address>+<number>(* 1)<br><t>: bit position (* 2)<br>(0 to 7)<br><x>: PMC path number(* 3)<br>(1 to 5, 9) |
|                       | BYTE               | PMC_<a>_path<x>_<Machine name><br>Ex) PMC_D200_path1_Machine001          |                                                                                                                                       |
|                       | WORD               | PMC_<a>_W_path<x>_<Machine name><br>Ex) PMC_D200_W_path1_Machine001      |                                                                                                                                       |
|                       | DWORD              | PMC_<a>_D_path<x>_<Machine name><br>Ex) PMC_D200_D_path1_Machine001      |                                                                                                                                       |
|                       | REAL               | PMC_<a>_R_path<x>_<Machine name><br>Ex) PMC_D200_R_path1_Machine001      |                                                                                                                                       |
|                       | LREAL              | PMC_<a>_L_path<x>_<Machine name><br>Ex) PMC_D200_L_path1_Machine001      |                                                                                                                                       |
|                       | Unsigned BYTE      | PMC_<a>_UB_path<x>_<Machine name><br>Ex) PMC_D200_UB_path1_Machine001    |                                                                                                                                       |
|                       | Unsigned WORD      | PMC_<a>_UW_path<x>_<Machine name><br>Ex) PMC_D200_UW_path1_Machine001    |                                                                                                                                       |
|                       | Unsigned DWORD     | PMC_<a>_UD_path<x>_<Machine name><br>Ex) PMC_D200_UD_path1_Machine001    |                                                                                                                                       |
|                       | STRING             | PMC_<a>_S_path<x>_<Machine name><br>Ex) PMC_D200_S_path1_Machine001      |                                                                                                                                       |
| P-CODE                | Conversation macro | PcodeCon_<n>_path<x>_<Machine name><br>Ex) PcodeCon_1_path1_Machine001   | <n>: variable number (1 to 33)<br><x>: CNC path number (1 to 15)                                                                      |
|                       | Auxiliary macro    | PcodeAux_<n>_path<x>_<Machine name><br>Ex) PcodeAux_2_path1_Machine001   |                                                                                                                                       |
|                       | Execution macro    | PcodeExe_<n>_path<x>_<Machine name><br>Ex) PcodeExe_3_path1_Machine001   |                                                                                                                                       |
|                       | Common area        | PcodeVar_<n>_path<x>_<Machine name><br>Ex) PcodeVar_600_path1_Machine001 | <n>: variable number (100 to 89999)<br><x>: CNC path number (1 to 15)                                                                 |

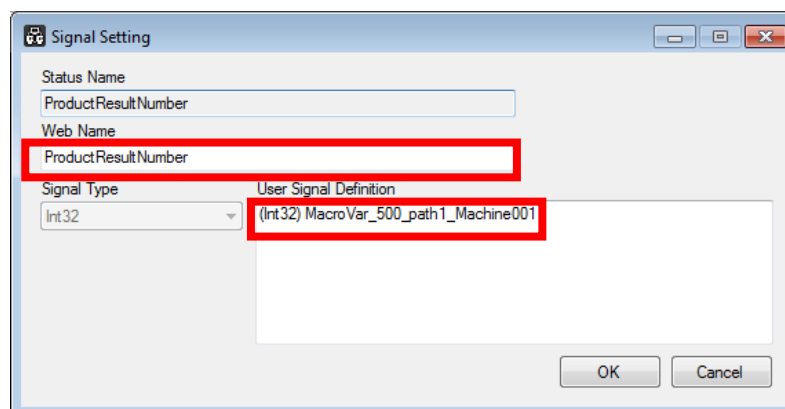
\* 1 <address>: One of the character among "G" / "F" / "Y" / "X" / "A" / "R" / "T" / "K" / "C" / "D" / "M" / "N" / "E" / "Z"

\* 2 BIT type cannot be used to specify production results.

\* 3 PMC path number when retrieving value of dual check safety PMC: "9"

### Example of configuration 1:

The case of custom macro variable number 500 (CNC path: 1):



In the [User Signal Definition], the expression converts the value of the custom macro variable to Int32 type. A half-width space is required between "(Int32)" and "MacroVar\_500\_path1\_Machine001".

|      |               |        |                                    |       |  |        |                   |                                               |  |        |         |
|------|---------------|--------|------------------------------------|-------|--|--------|-------------------|-----------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |                   |                                               |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |                   |                                               |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        | Draw              | A-40829-00001EN                               |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        |                   |                                               |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |                   |                                               |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        | FANUC CORPORATION |                                               |  | Sheet. | 773/868 |
| Ed.  | Date          | Design | Description                        |       |  |        |                   |                                               |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check |  | Apprv. |                   |                                               |  |        |         |

## Example of configuration 2:

The case of PMC address D100 and WORD type (PMC Path: 1):

The dialog box 'Signal Setting' shows the following configuration:

- Status Name: ProductResultNumber
- Web Name: ProductResultNumber
- Signal Type: Int32
- User Signal Definition: PMC\_D100\_W\_path1\_Machine00

Red boxes highlight the 'Web Name' field and the 'User Signal Definition' text.

## Example of configuration 3:

The case of P-CODE variable number 600 (CNC path: 1):

The dialog box 'Signal Setting' shows the following configuration:

- Status Name: ProductResultNumber
- Web Name: ProductResultNumber
- Signal Type: Int32
- User Signal Definition: (Int32) PcodeVar\_600\_path1\_Machine001

Red boxes highlight the 'Web Name' field and the 'User Signal Definition' text.

In the [User Signal Definition], the expression converts the value of the P-CODE variable to Int32 type. A blank is required between "(Int32)" and "PcodeVar\_600\_path1\_Machine001."

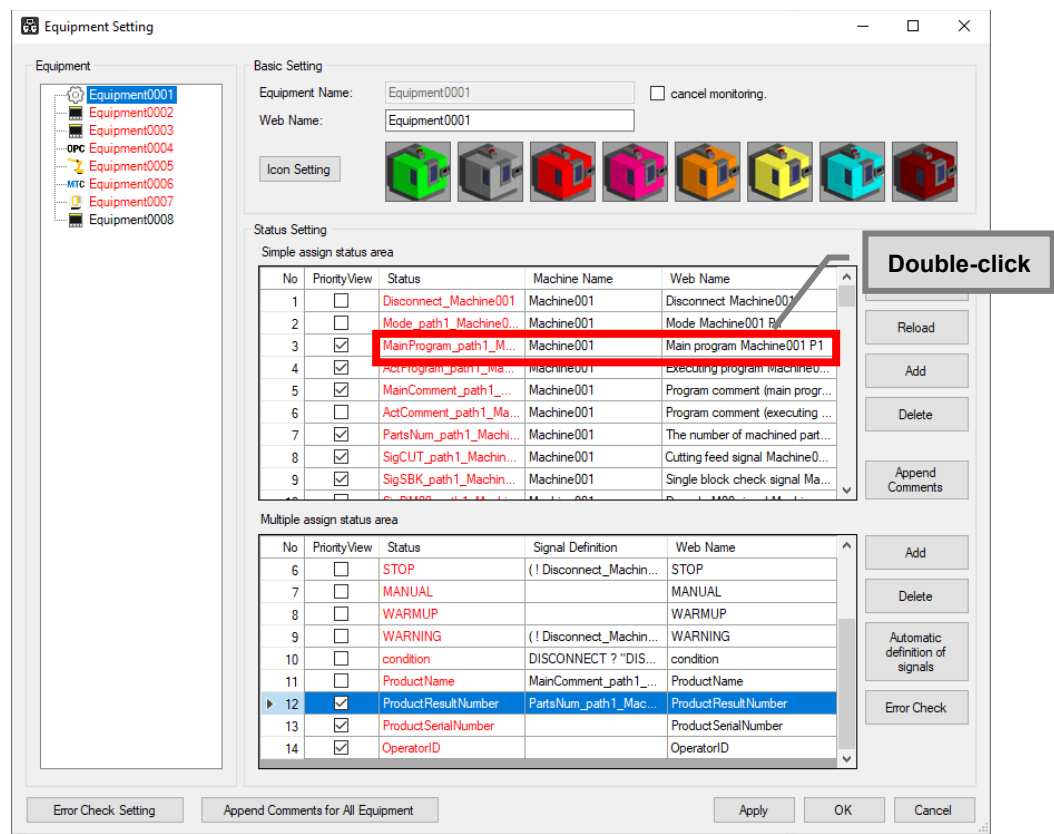
- (7) The changes to the status are applied to the [ProductResultNumber] row of the [Multiple assign status area] table. Click [OK] on the [Equipment Setting] window to close the window.

|      |               |        |                                    |       |  |        |                   |                                               |  |        |         |
|------|---------------|--------|------------------------------------|-------|--|--------|-------------------|-----------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |                   |                                               |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |                   |                                               |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        | Draw              | A-40829-00001EN                               |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        |                   |                                               |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |                   |                                               |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        | FANUC CORPORATION |                                               |  | Sheet. | 774/868 |
| Ed.  | Date          | Design | Description                        |       |  |        |                   |                                               |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check |  | Apprv. |                   |                                               |  |        |         |

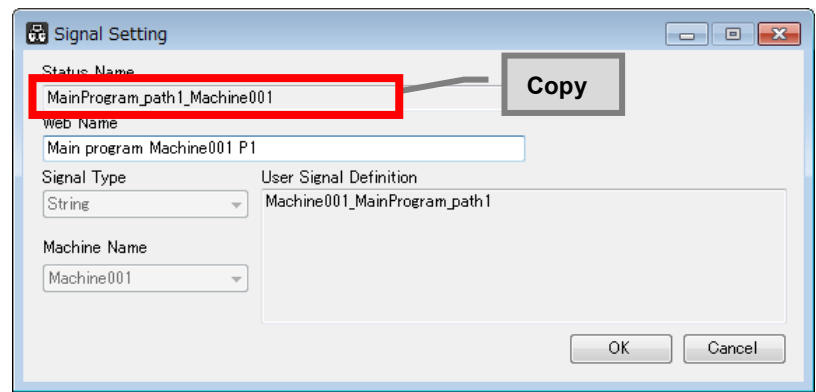
# 14.13.3 Assigning FanucCNC / FanucRobodrill Signal to ProductName

In an environment that a signal other than MainComment is used for product names, you also can assign the signal to ProductName status and configure an automatic data collection. Any signal will work but we recommend using a signal of string type. This subsection explains how to assign a signal to ProductName through an example of using MainProgram signal.

- (1) Double-click the entry of "MainProgram\_path1\_<machine name>" in the [Simple assign status area] table as shown below.

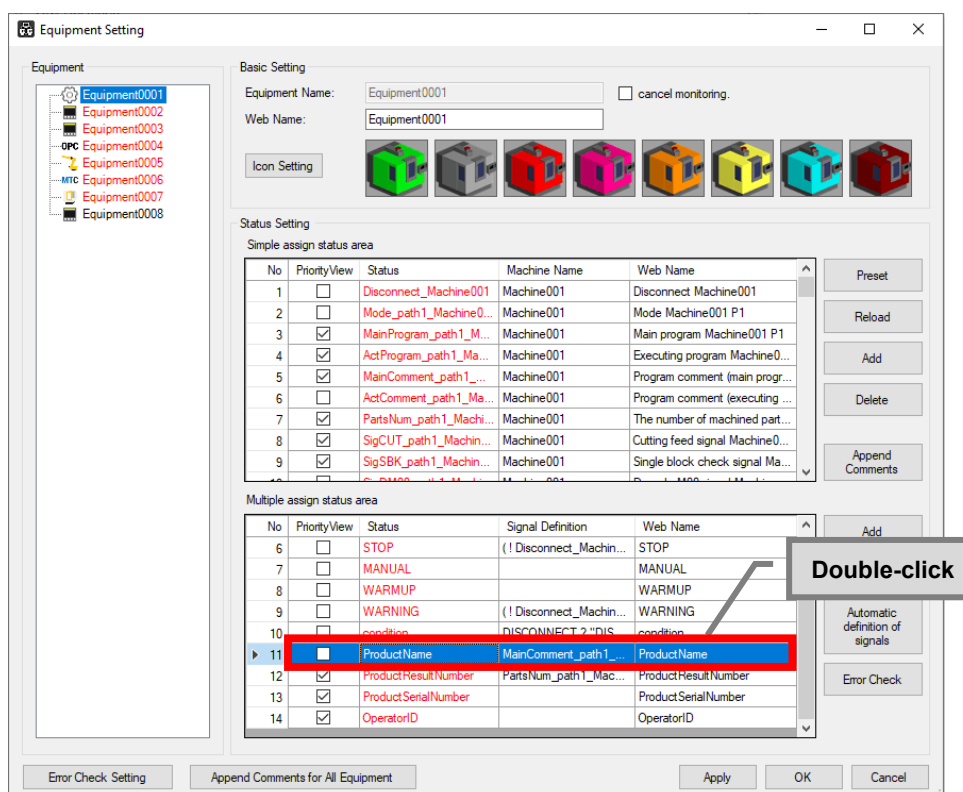


- (2) The [Signal Setting] dialog box appears. Copy the [Status Name] text field and click the [Cancel] button.

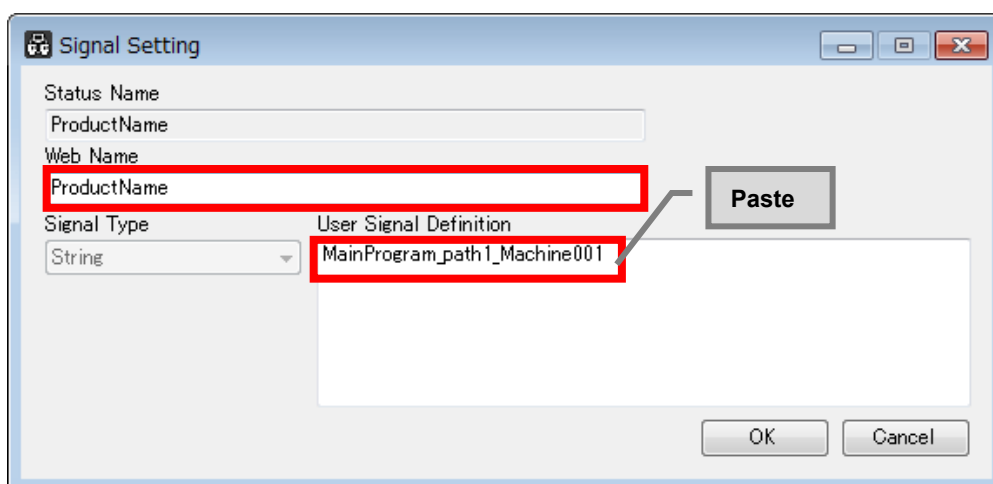


|      |               |        |                                    |                   |                                               |                |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw              | A-40829-00001EN                               |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                   |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | FANUC CORPORATION |                                               | Sheet. 775/868 |
| Ed.  | Date          | Design | Description                        |                   |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             | Apprv.                                        | Inoue          |

(3) Double-click the entry for "ProductName" in the [Multiple assign status area] table as shown below.



(4) The [Signal Setting] dialog box appears. Specify a Web name for ProductName in the [Web Name] field and paste the copied status name to the [User Signal Definition] field. Click [OK]. Below is an example of assigning MainProgram signal to ProductName status.



(5) The entry for ProductName in the [Multiple assign status area] table reflects the change. Click [OK] to close the [Signal Setting] dialog box.

|      |               |        |                                    |       |  |        |                   |                                               |  |        |         |
|------|---------------|--------|------------------------------------|-------|--|--------|-------------------|-----------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |                   |                                               |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |                   |                                               |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        | Draw              | A-40829-00001EN                               |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        |                   |                                               |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |                   |                                               |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        | FANUC CORPORATION |                                               |  | Sheet. | 776/868 |
| Ed.  | Date          | Design | Description                        |       |  |        |                   |                                               |  |        |         |
| Date | Nov.30.2015   | Desig. | Ogawa                              | Check |  | Apprv. |                   |                                               |  |        |         |

**Results - Production Results**

\* Calendar has the upper limit for time span (Upper limit [day]: 7).

Auto Refresh: OFF | Calendar View Mode: Shift

Start date: 2017/11/28 | Start time: 09:00:00 | End date: 2017/11/28 | End time: 18:00:00

Eval. Kind: Equipment/Product

Equipment name: Equipment0001 | Product name: PRODUCT-X

**Summary Table**

|                                  | Equipment     | Product   | Production result | Plans | Rate    |
|----------------------------------|---------------|-----------|-------------------|-------|---------|
| <input checked="" type="radio"/> | Equipment0001 | PRODUCT-X | 65                | 65    | 100.00% |

**Transition Graph**

Production result/Plans

Time series

2017/11/28 14:30:00  
Production result: 33  
Plans: 55

11:04:18 | FANUC MT-LINKi | 11 / 4 / 1 / 16

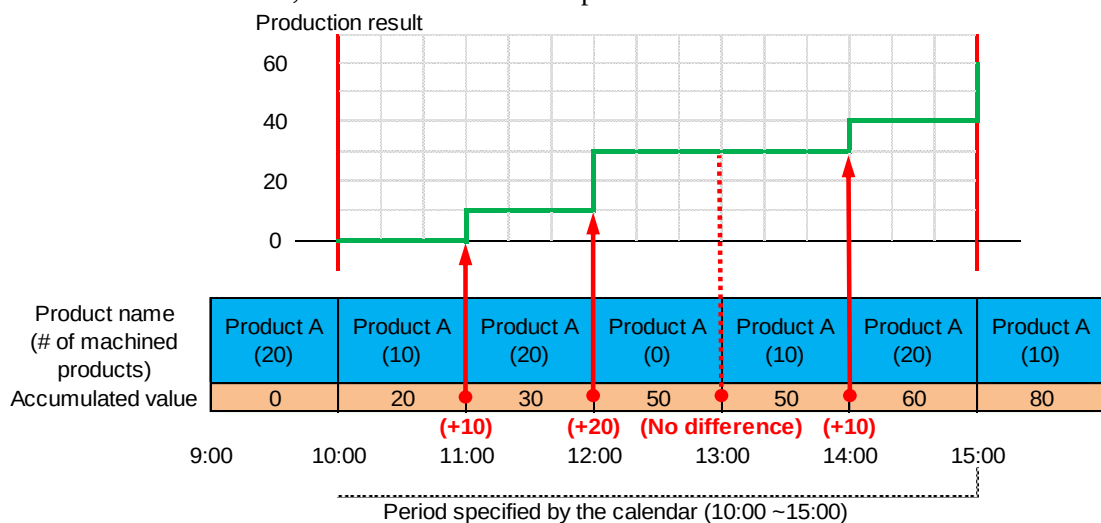
### 14.14.1 Calculation of Production Results

For the production results on the [Results – Production Results] screen, the incremental difference of "ProductResultNumber" (number of machined products assigned to multiple assign status) within the period specified by the calendar are displayed.

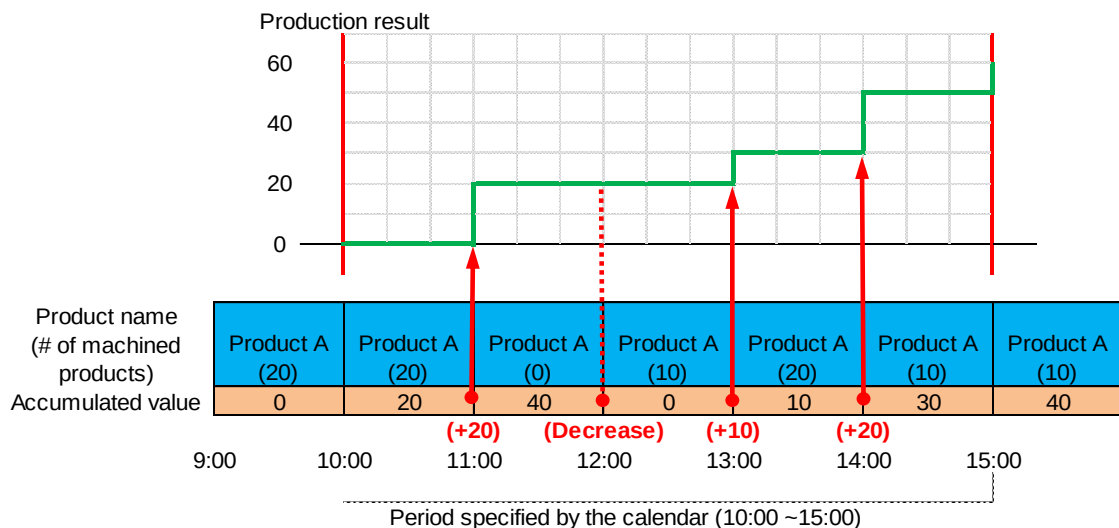
The "Product name (# of machined products)" and the "Accumulated value" used in the explanation below are as follows.

- Product name (# of machined products):  
Indicates the product name and the number of machined products in the specified period.
- Accumulated value:  
Indicates the signal value (accumulated value of machined products) assigned to "ProductResultNumber."

- (1) When the difference in the number of machined products increases, it is calculated as a production result.  
When there is no difference, it is not calculated as a production result.



- (2) When the difference in the number of machined products decreases, it is judged as there is no incremental difference, and it not calculated as a production result.



|      |               |        |                                            |       |  |        |       |                   |                                                       |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------|-------------------------------------------------------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        |       | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                   |                                                       |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                   |                                                       |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                   |                                                       |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |       | Draw              | A-40829-00001EN                                       |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                   |                                                       |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                   |                                                       |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       | FANUC CORPORATION |                                                       | Sheet. | 778/868 |
| Date | Nov.30.2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                   |                                                       |        |         |

#### 14.14.2 Display Example of Production Results

The display of production results change by controlling the signal value (accumulated value) assigned to "ProductResultNumber."

This subsection explains the following cases

- (1) When the product was changed during operation  
(Cases depending on whether the accumulated value was cleared to 0 or not)
- (2) When the product was not changed while communication was disconnected  
(Cases depending on whether there was increase/decrease in the accumulated value or not)
- (3) When the product was changed while communication was disconnected  
(Cases depending on whether there was increase/decrease in the accumulated value or not)

\* A disconnected communication is a state in which data cannot be recorded to the DB due to stoppage of the Receiver.

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 779/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         |

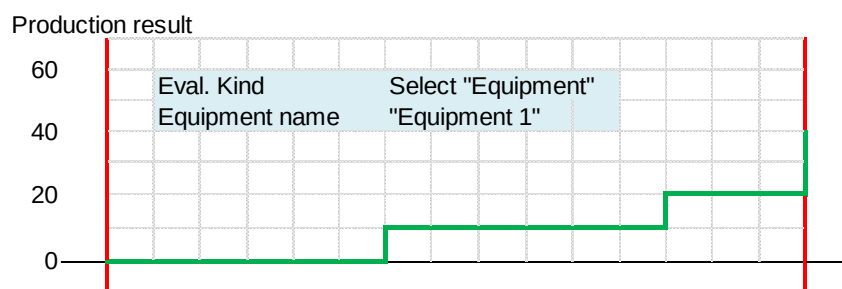
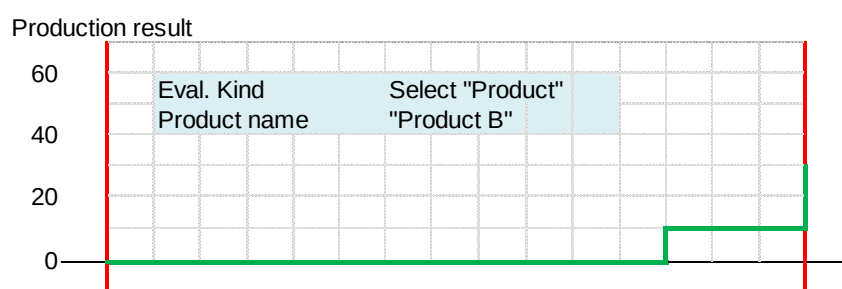
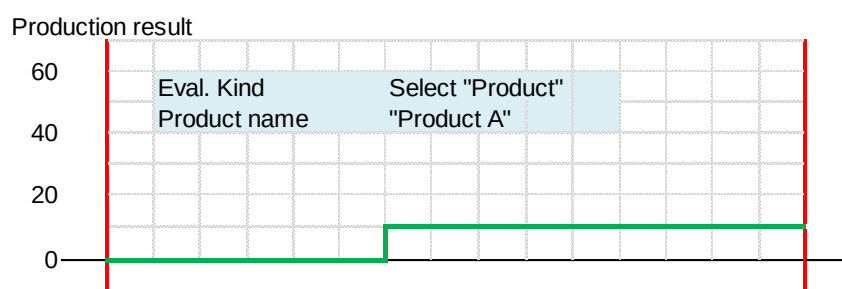
## 14.14.2.1 When Product was Changed during Operation

This subsection explains the cases when the product was changed during operation, and depending on whether the accumulated value was cleared to 0 or not.

Following are the display examples of production results when "Product A" / "Product B" / "Equipment 1" are selected in the [Results – Production Results] screen.

- (1) When "Product A" is changed to "Product B" on "Equipment 1," and when the accumulated value is not cleared to 0 before or after the change

After changing to "Product B," the record of "Product B (increase in number of machined products)" is recorded at the timing when the number of machined parts increases, and the value is calculated into production result.



↓ Change to "Product B"

|                                          |                   |                   |                   |                   |                   |       |       |
|------------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------|-------|
| Product name<br>(# of machined products) | Product A<br>(20) | Product A<br>(10) | Product B<br>(10) | Product B<br>(20) | Product B<br>(10) |       |       |
| Accumulated value                        | 0                 | 20                | 30                | 40                | 60                |       |       |
|                                          | 9:00              | 10:00             | 11:00             | 12:00             | 13:00             | 14:00 | 15:00 |

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |                          |                                               |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |                          |                                               |                |
| Ed.  | Date          | Design | Description                        | <b>FANUC CORPORATION</b> |                                               | Sheet. 780/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |

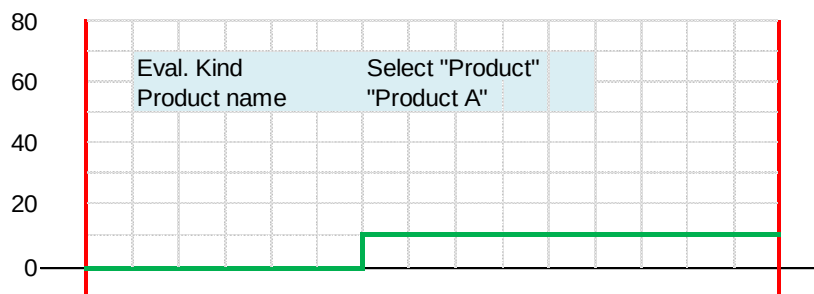


- (2) When "Product A" is changed to "Product B" on "Equipment 1," and when the accumulated value is cleared to 0 after the change

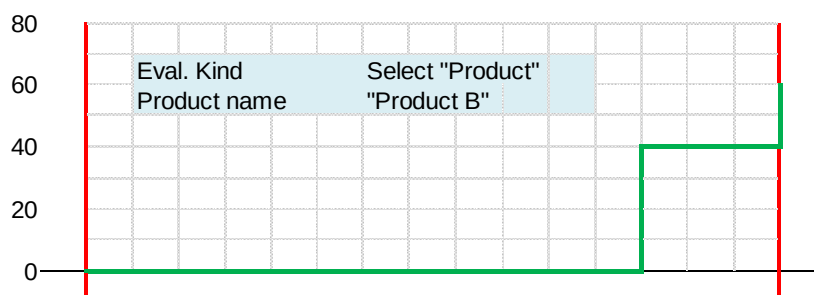
Right after the change to "Product B," the record of "Product B (number of machined products = 0)" is recorded.

After this, the record of "Product B (increase in machined products)" is recorded at the timing when the number of machined parts increases, and the value is calculated into the production result.

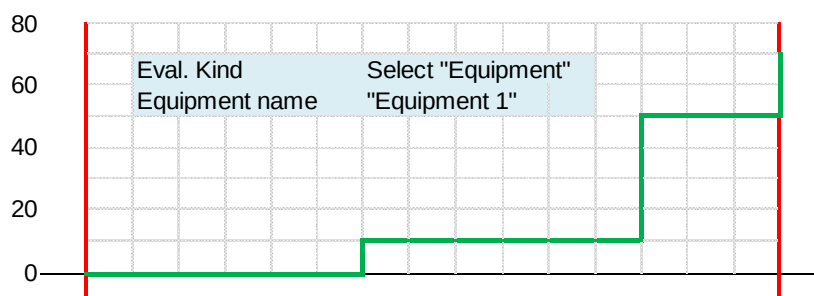
Production result



Production result



Production result



↓ Change to "Product B"

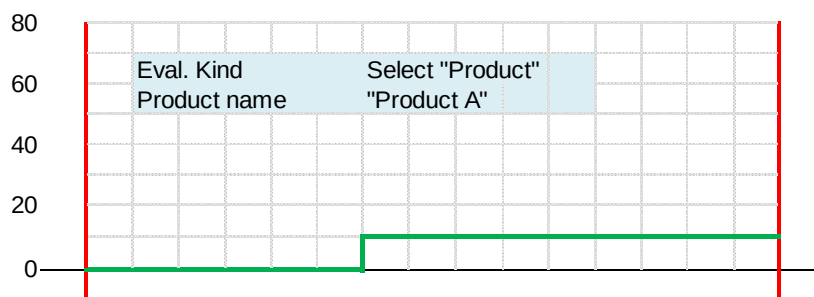
|                                             |                   |                   |                  |                   |                   |                   |
|---------------------------------------------|-------------------|-------------------|------------------|-------------------|-------------------|-------------------|
| Product name<br>(# of machined<br>products) | Product A<br>(20) | Product A<br>(10) | Product B<br>(0) | Product B<br>(40) | Product B<br>(20) | Product B<br>(10) |
| Accumulated value                           | 0                 | 20                | 30               | 0                 | 40                | 60                |
|                                             | 9:00              | 10:00             | 11:00            | 12:00             | 13:00             | 14:00             |

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |                          |                                               |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |                          |                                               |                |
| Ed.  | Date          | Design | Description                        | <b>FANUC CORPORATION</b> |                                               | Sheet. 781/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |

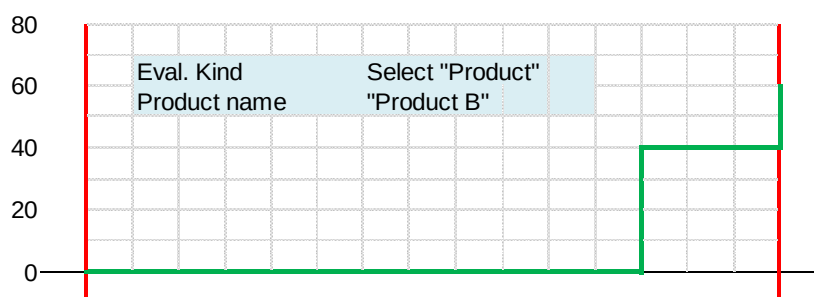
- (3) When "Product A" is changed to "Product B" on "Equipment 1," and when the accumulated value is cleared to 0 before the change

Before the change to "Product B," the record of "Product A (number of machined products = 0)" is recorded. After this, the record of "Product B (increase in machined products)" is recorded at the timing when the number of machined parts increases, and the value is calculated into the production result.

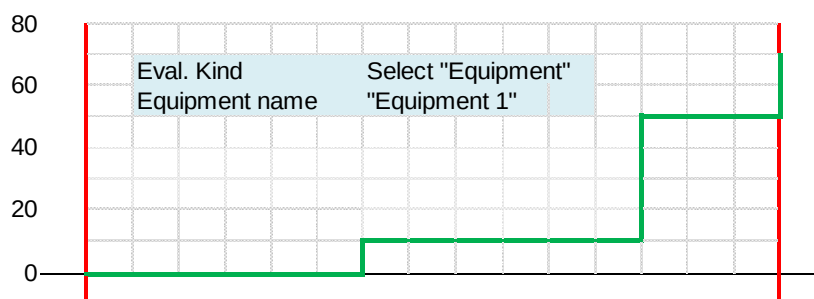
Production result



Production result



Production result



↓ Change to "Product B"

|                                             |                   |                   |                  |                   |                   |                   |
|---------------------------------------------|-------------------|-------------------|------------------|-------------------|-------------------|-------------------|
| Product name<br>(# of machined<br>products) | Product A<br>(20) | Product A<br>(10) | Product A<br>(0) | Product B<br>(40) | Product B<br>(20) | Product B<br>(10) |
| Accumulated value                           | 0                 | 20                | 30               | 0                 | 40                | 60                |
|                                             | 9:00              | 10:00             | 11:00            | 12:00             | 13:00             | 14:00             |

|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | <b>A-40829-00001EN</b>                        |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 782/868 |

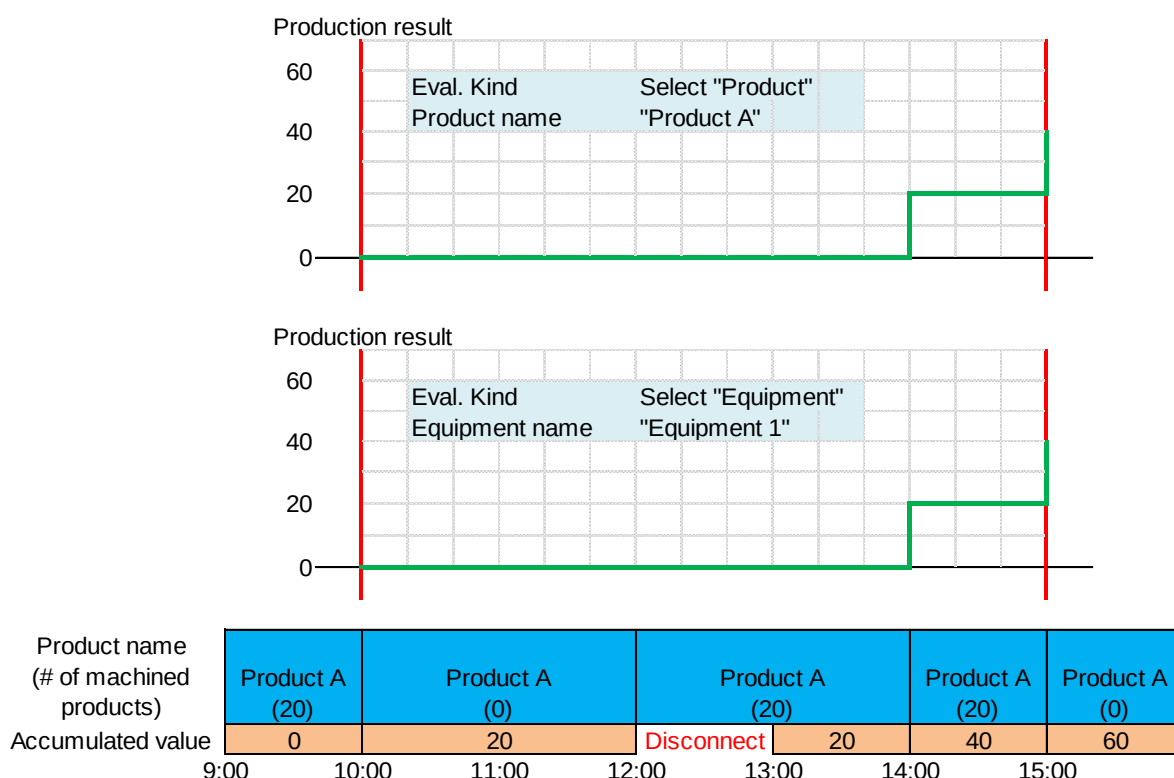
## 14.14.2.2 When Product Was Not Changed While Communication Was Disconnected

This subsection explains the cases when the product is not changed after recovering from a disconnected communication caused by the stoppage of the Receiver, and depending on whether there was increase/decrease in the accumulated value or not after recovery.

Following are the display examples of production results when "Product A" / "Equipment 1" are selected in the [Results – Production Results] screen.

- (1) When there is no increase/decrease in the accumulated value for "Product A" on "Equipment 1" after recovering from a disconnected communication caused by the stoppage of the Receiver

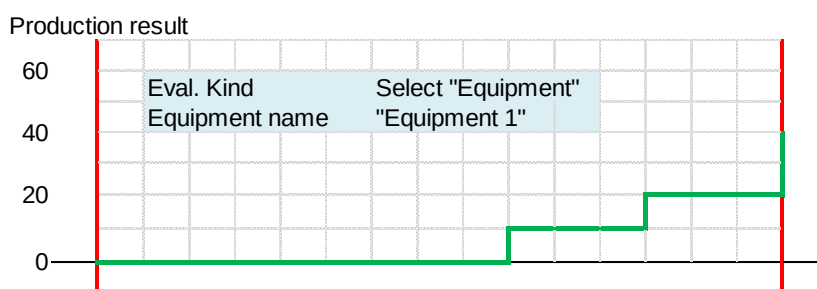
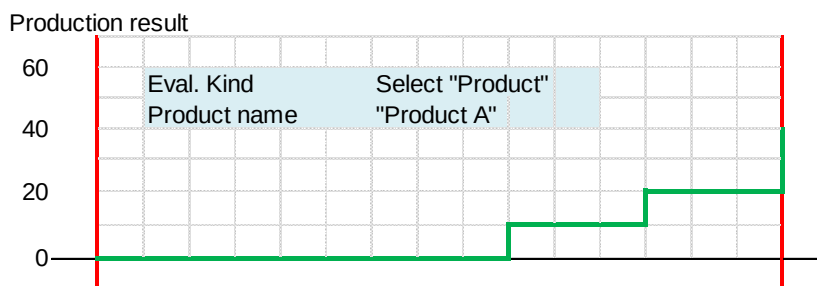
After recovering from a disconnected communication, the record of "Product A (increase in number of machined products)" is recorded at the timing when the accumulated value increases, and the value is calculated into production result.



|      |               |        |                                    |                   |                                               |       |                |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |                   |                                               |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw              | A-40829-00001EN                               |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |                   |                                               |       |                |
| Ed.  | Date          | Design | Description                        | FANUC CORPORATION |                                               |       | Sheet. 783/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             | Apprv.                                        | Inoue |                |

- (2) When there is increase in the accumulated value for "Product A" on "Equipment 1" after recovering from a disconnected communication caused by the stoppage of the Receiver

Just after recovering from a disconnected communication, the record of "Product A (increase in number of machined products)" is recorded, and the value is calculated into production result.



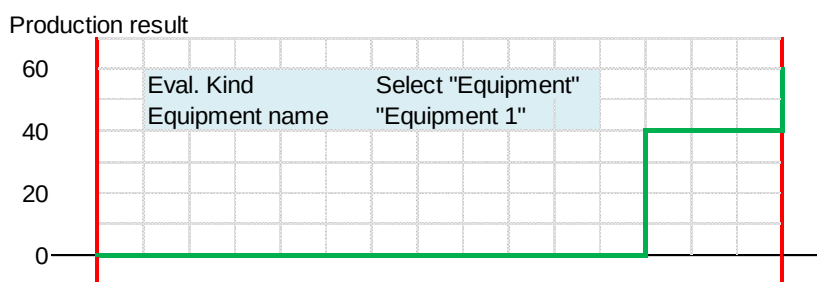
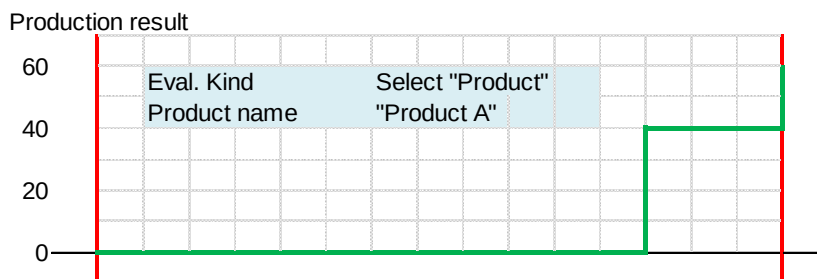
|                                             |                   |       |                  |       |                   |                   |                   |                   |
|---------------------------------------------|-------------------|-------|------------------|-------|-------------------|-------------------|-------------------|-------------------|
| Product name<br>(# of machined<br>products) | Product A<br>(20) |       | Product A<br>(0) |       | Product A<br>(10) | Product A<br>(10) | Product A<br>(20) | Product A<br>(10) |
| Accumulated value                           | 0                 |       | 20               |       | Disconnect        | 30                | 40                | 60                |
|                                             | 9:00              | 10:00 | 11:00            | 12:00 | 13:00             | 14:00             | 15:00             |                   |

|      |               |        |                                    |                          |                                               |       |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |       | Sheet. 784/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue |                |

- (3) When there is decrease in the accumulated value for "Product A" on "Equipment 1" after recovering from a disconnected communication caused by the stoppage of the Receiver

Just after recovering from a disconnected communication, record of "Product A (number of machined products = 0)" is recorded.

After this, the record of "Product A (increase in number of machined products)" is recorded at the timing when the accumulated value increases, and the value is calculated into production result.



|                                             |                   |                  |                  |                   |                   |                   |       |
|---------------------------------------------|-------------------|------------------|------------------|-------------------|-------------------|-------------------|-------|
| Product name<br>(# of machined<br>products) | Product A<br>(20) | Product A<br>(0) | Product A<br>(0) | Product A<br>(40) | Product A<br>(20) | Product A<br>(10) |       |
| Accumulated value                           | 0                 | 20               | Disconnect       | 0                 | 40                | 60                |       |
|                                             | 9:00              | 10:00            | 11:00            | 12:00             | 13:00             | 14:00             | 15:00 |

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               | Sheet. 785/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |

### 14.14.2.3 When Product Was Changed while Communication Was Disconnected

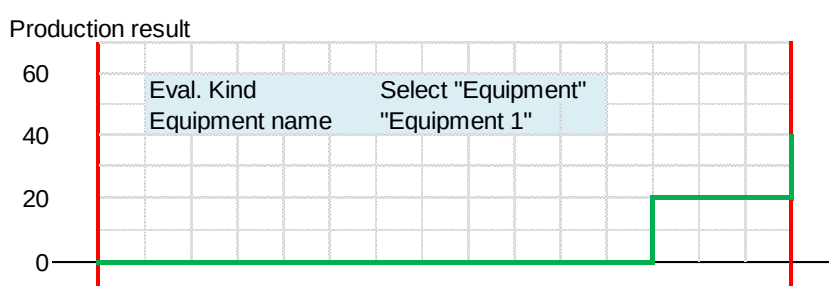
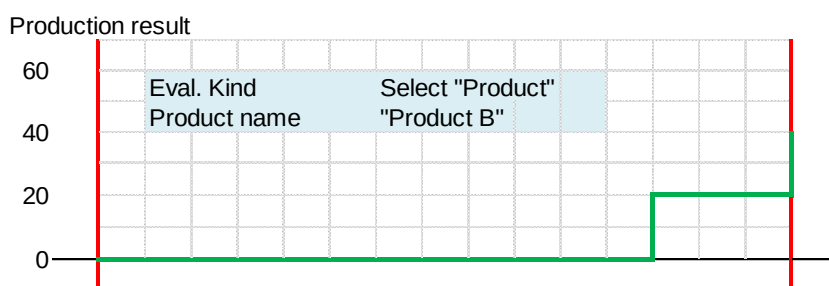
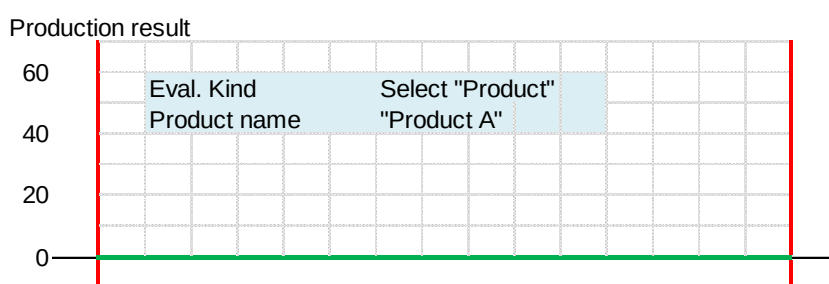
This subsection explains the cases when the product is changed after recovering from a disconnected communication caused by the stoppage of the Receiver, and depending on whether there was increase/decrease in the accumulated value or not after recovery.

Following are the display examples of production results when "Product A" / "Product B" / "Equipment 1" are selected on the [Results – Production Results] screen.

- (1) When "Product A" is changed to "Product B" on "Equipment 1" after recovering from a disconnected communication caused by the stoppage of the Receiver, and when there is no increase/decrease in the accumulated value

Just after recovering from a disconnected communication, record of "Product B (number of machined products = 0)" is recorded.

After this, the record of "Product B (increase in number of machined products)" is recorded at the timing when the accumulated value increases, and the value is calculated into production result.



↓ Change to "Product B"

|                                          |                   |                  |                  |                   |                   |                   |       |
|------------------------------------------|-------------------|------------------|------------------|-------------------|-------------------|-------------------|-------|
| Product name<br>(# of machined products) | Product A<br>(20) | Product A<br>(0) | Product B<br>(0) | Product B<br>(20) | Product B<br>(20) | Product B<br>(10) |       |
| Accumulated value                        | 0                 | 20               | Disconnect       | 20                | 40                | 60                |       |
|                                          | 9:00              | 10:00            | 11:00            | 12:00             | 13:00             | 14:00             | 15:00 |

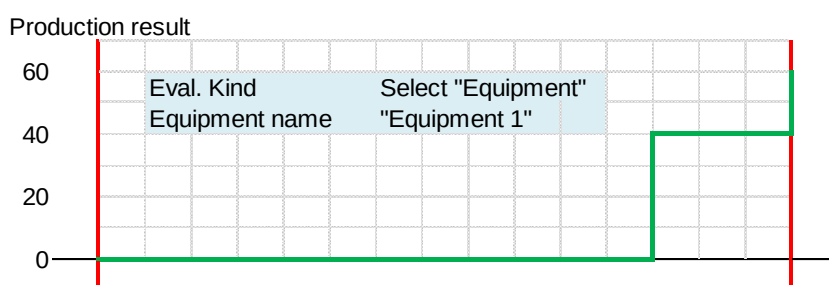
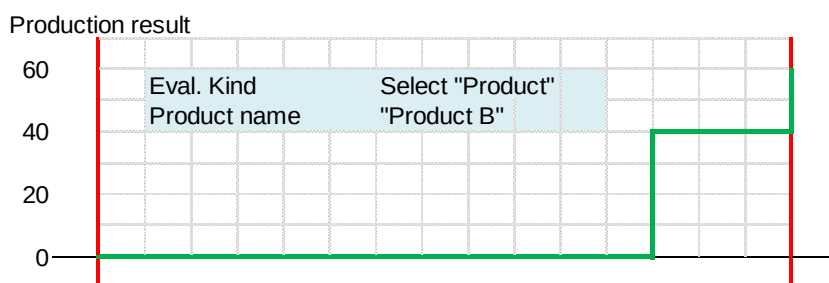
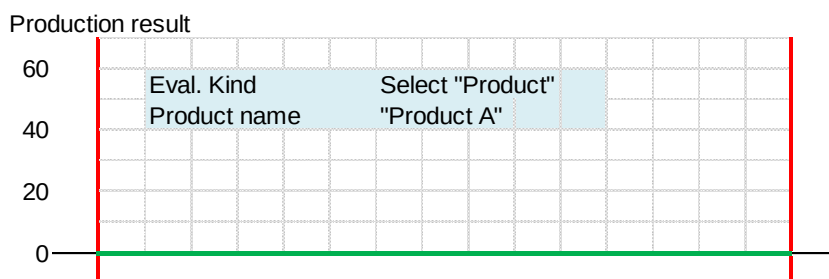
|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |                          |                                               |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |                          |                                               |                |
| Ed.  | Date          | Design | Description                        | <b>FANUC CORPORATION</b> |                                               | Sheet. 786/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |



- (3) When "Product A" is changed to "Product B" on "Equipment 1" after recovering from a disconnected communication caused by the stoppage of the Receiver, and when there is decrease in the accumulated value

Just after recovering from a disconnected communication, record of "Product B (number of machined products = 0)" is recorded.

After this, the record of "Product B (increase in number of machined products)" is recorded at the timing when the accumulated value increases, and the value is calculated into production result.



↓ Change to "Product B"

|                                             |           |       |            |       |       |       |       |
|---------------------------------------------|-----------|-------|------------|-------|-------|-------|-------|
| Product name<br>(# of machined<br>products) | Product A |       | Product B  |       |       |       |       |
|                                             | (20)      | (0)   | (0)        | (40)  | (20)  | (10)  |       |
| Accumulated value                           | 0         | 20    | Disconnect | 0     | 40    | 60    |       |
|                                             | 9:00      | 10:00 | 11:00      | 12:00 | 13:00 | 14:00 | 15:00 |

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |                          |                                               |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |                          |                                               |                |
| Ed.  | Date          | Design | Description                        | <b>FANUC CORPORATION</b> |                                               | Sheet. 788/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |





## 15.1.1 AdminTool.ini

The setting items of AdminTool.ini are listed in the following.

("YES" in the [Initial] column indicates that the item is set upon installation, "YES" in the [Edit] column indicates that the item is editable, and "NO" in the [Edit] column indicates that the item should not be edited.)

| Section Name                                             | Item Name                       | Description                                                                                              | Initial | Edit | Remarks |
|----------------------------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------|---------|------|---------|
| MTLINKi<br>_ADMIN TOOL                                   | User                            | DB User                                                                                                  | YES     | NO   |         |
|                                                          | Password                        | DB password                                                                                              | YES     | NO   |         |
|                                                          | MONGOKEY                        | DB connection string                                                                                     | YES     | NO   |         |
|                                                          | ADMINLANGUAGE                   | Language of AdminTool                                                                                    | NO      | NO   |         |
|                                                          | GRIDCOLOR                       | (1) Layout setting grid color<br>(Layout setting function)                                               | NO      | YES  | *1      |
|                                                          | SHOWLAYOUTGRID                  | Display grid in the overview layout                                                                      | NO      | NO   | *2      |
|                                                          | LAYOUTGRIDPITCH                 | Grid space in the overview layout                                                                        | NO      | NO   | *2      |
|                                                          | TIMESERIES_DATA_LIMIT           | (2) Maximum number of status output<br>(Signal data output function)                                     | NO      | YES  | *3      |
|                                                          | TIMESERIES_INTERVAL             | (3) Time interval of signal data output<br>(Signal data output function)                                 | NO      | YES  | *3      |
|                                                          | TIMESERIES_COMPRESSION          | (4) Enable/disable compression of signal<br>data<br>(Signal data output function)                        | NO      | YES  | *3      |
|                                                          | TIMESERIES_KIND                 | (5) Data type of signal data<br>(Signal data output function)                                            | NO      | YES  | *3      |
|                                                          | TIMESERIES_SPLIT_RECORD         | (6) Number of rows split for signal data<br>output<br>(Signal data output function)                      | NO      | YES  | *3      |
|                                                          | TIMESERIES_SCHEDULE_TIME<br>OUT | (7) Time out period of signal data<br>schedule output<br>(Signal data output function)                   | NO      | YES  | *3      |
|                                                          | TIMESERIES_MANUAL_TIMEO<br>UT   | (8) Time out period of signal data manual<br>output<br>(Signal data output function)                     | NO      | YES  | *3      |
|                                                          | REPORTTOOL_SCHEDULE_TI<br>MEOUT | (9) Time out period of report schedule<br>output<br>(Report schedule function)                           | NO      | YES  | *4      |
|                                                          | REPORTTOOL_MANUAL_TIME<br>OUT   | (10) Time out period of report manual<br>output<br>(Report schedule function)                            | NO      | YES  | *4      |
|                                                          | COLMGR_INTERVAL                 | (11) Interval for inquiring Collector Service<br>Manager<br>(Start/Stop System function)                 | NO      | YES  | *5      |
|                                                          | COLMGR_COM_TIMEOUT              | (12) Time out period for communication<br>with Collector Service Manager<br>(Start/Stop System function) | NO      | YES  | *5      |
| DBDATASAVE<br>(Data Storing<br>Setting)                  | COUNT                           | DB path setting count                                                                                    | NO      | YES  | *6      |
|                                                          | DBPATH1                         | DB path (1st setting)                                                                                    | NO      | NO   | *6      |
|                                                          | DBSIZE_ENABLE1                  | DB enable size (1st setting)                                                                             | NO      | NO   | *6      |
|                                                          | DBSIZE1                         | DB size (1st setting)                                                                                    | NO      | NO   | *6      |
| DBDATADELETE<br>(DB Data<br>Retention Period<br>Setting) | DELETE_TERM                     | (13) Time unit in dividing DB data for block-<br>wise deletion                                           | NO      | YES  | *5      |

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 790/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         |

| Section Name             | Item Name          | Description                                               | Initial | Edit | Remarks |
|--------------------------|--------------------|-----------------------------------------------------------|---------|------|---------|
| OPCUA<br>(OPC UA Server) | KEEPALIVE_INTERVAL | (14) Interval for communicating with OPC UA Server        | NO      | YES  | *5      |
|                          | KEEPALIVE_TIMEOUT  | (15) Time out period for communicating with OPC UA Server | NO      | YES  | *5      |

#### NOTE

- When the [Layout Setting] window is displayed, the items will be reflected on the system setting file.
- When the [Show Grid] check box is selected in the [Layout Setting] window, the items will be reflected on the system file setting file.
- When the [Signal Data Output] window is displayed, the items will be reflected on the system setting file.
- When the [Report Schedule Setting] window is displayed, the items will be reflected on the system setting file.
- In order to change the setting, add the item.
- When storage settings on the [System Setting] window are made, the items will be reflected on the system setting file.
- In order to apply the setting of the system setting file (AdminTool.ini), restart AdminTool.
- When the signal data schedule output is being registered, in order to enable the edited settings of the items (2) to (7), [Unregister] the schedule and [Register] the schedule again.
- When the report schedule output is being registered, in order to enable the edited setting of item (9), [Unregister] the schedule and [Register] the schedule again.

The following part explains each item that is editable (with "YES" in the [Edit] column).

#### (1) Layout setting grid color

Set the grid color displayed on the editing area of the [Layout Setting] window.

Change the color when there is difficulty in seeing the grid display due to the background color.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                                                                               |
|----------------------|-------------------------------------------------------------------------------|
| <b>Name of item</b>  | GRIDCOLOR                                                                     |
| <b>Parameter</b>     | #000000 to #FFFFFF<br>(write HTML color code in 6 digit hex character string) |
| <b>Default value</b> | #D3D3D3 (LightGrey)                                                           |

#### NOTE

When setting the HTML color name, refer to the information on the Internet, etc.

#### (2) Maximum number of status output

Set number of equipment statuses that can be registered as the output target data of signal data output.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                       |
|----------------------|-----------------------|
| <b>Name of item</b>  | TIMESERIES_DATA_LIMIT |
| <b>Parameter</b>     | 1 to 5000 (items)     |
| <b>Default value</b> | 3000                  |

|             |               |               |                                                |                          |                                                           |                |
|-------------|---------------|---------------|------------------------------------------------|--------------------------|-----------------------------------------------------------|----------------|
| 20          | Nov. 22, 2023 | uwano         | Revised for MT-LINK <sup>i</sup> Version 4.0   | Title                    | FANUC MT-LINK <sup>i</sup><br>AdminTool OPERATOR'S MANUAL |                |
| 19          | Apr. 27, 2021 | uwano         | Revised for MT-LINK <sup>i</sup> Version 3.11  |                          |                                                           |                |
| 18          | July 28, 2020 | uwano         | Revised for MT-LINK <sup>i</sup> Version 3.10  |                          |                                                           |                |
| 17          | Jan. 30, 2020 | uwano         | Revised for MT-LINK <sup>i</sup> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                    |                |
| 16          | July 22, 2019 | uwano         | Revised for MT-LINK <sup>i</sup> Version 3.8   |                          |                                                           |                |
| 15          | Mar. 29, 2019 | ogawa         | Revised for MT-LINK <sup>i</sup> Version 3.7.2 |                          |                                                           |                |
| 14          | Jan. 29, 2019 | ogawa         | Revised for MT-LINK <sup>i</sup> Version 3.7   | <b>FANUC CORPORATION</b> |                                                           | Sheet. 791/868 |
| <b>Ed.</b>  | <b>Date</b>   | <b>Design</b> | <b>Description</b>                             |                          |                                                           |                |
| <b>Date</b> | Nov.30,2015   | <b>Desig.</b> | Ogawa                                          | <b>Check</b>             | <b>Apprv.</b>                                             | Inoue          |

(3) Time interval of signal data output

Set the time interval for signal data output.

For example, when signal data for 24 hours is output with 500msec interval, 172,800 rows of signal data will be output for each output file. If you wish to reduce the output volume of signal data, the setting value should be large.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                                                                      |
|----------------------|----------------------------------------------------------------------|
| <b>Name of item</b>  | TIMESERIES_INTERVAL                                                  |
| <b>Parameter</b>     | 500 to 3600000 (msec)<br>* Can be specified by multiple of 500 msec. |
| <b>Default value</b> | 500                                                                  |

(4) Enable/disable compression of signal data

Set whether to compress the signal data output file (CSV) to be output in ZIP format.

When this item is set to 1, not only the signal data files that will be output but also all the signal data files that have been already stored in the output folder.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                                    |
|----------------------|------------------------------------|
| <b>Name of item</b>  | TIMESERIES_COMPRESSION             |
| <b>Parameter</b>     | 0: No compression<br>1: Compressed |
| <b>Default value</b> | 0                                  |

(5) Data type of signal data

Set the data type of the signal data to be output.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                                                                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name of item</b>  | TIMESERIES_KIND                                                                                                                                                                            |
| <b>Parameter</b>     | 0: Output only value of Value (raw value of the signal)<br>1: Output values of<br>Value (raw value of the signal),<br>Score (converted value of the signal), and<br>Judge (judging result) |
| <b>Default value</b> | 0                                                                                                                                                                                          |

(6) Number of rows split for signal data output

Split the record of the signal data to be output in specified number of rows.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                                             |
|----------------------|---------------------------------------------|
| <b>Name of item</b>  | TIMESERIES_SPLIT_RECORD                     |
| <b>Parameter</b>     | 0: No split of rows<br>174 to 172799 (Rows) |
| <b>Default value</b> | 0                                           |

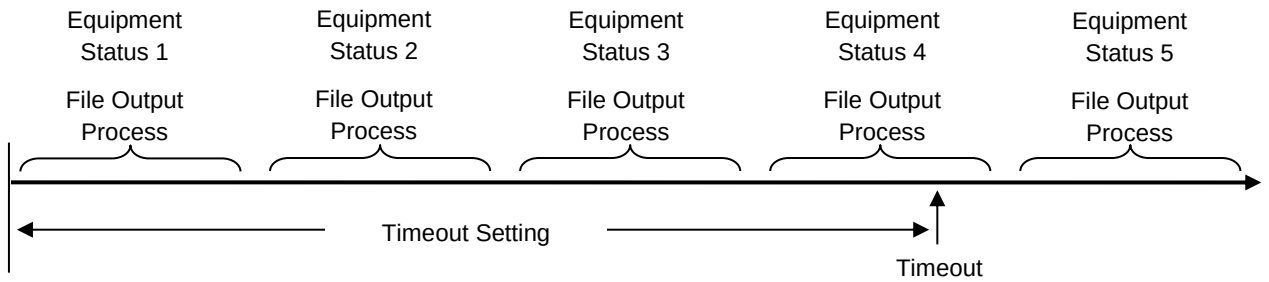
|      |               |        |                                            |       |  |        |                   |                                                       |  |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  |  | Sheet. | 792/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |  |        |         |

(7) Time out period of signal data schedule output

Set time out period of signal data schedule output.

File is output per equipment status, and the part after the timeout setting period will not be output.

In the following case, the signal data output files of equipment statuses 1, 2, and 3 will be output.



If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                             |
|----------------------|-----------------------------|
| <b>Name of item</b>  | TIMESERIES_SCHEDULE_TIMEOUT |
| <b>Parameter</b>     | 1 to 1200 (minutes)         |
| <b>Default value</b> | 720                         |

(8) Time out period of signal data manual output

Set time out period of signal data manual output.

File is output per equipment status, and the part after the timeout setting period will not be output.

The behavior after the timeout setting period is the same as shown in the above item (8).

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

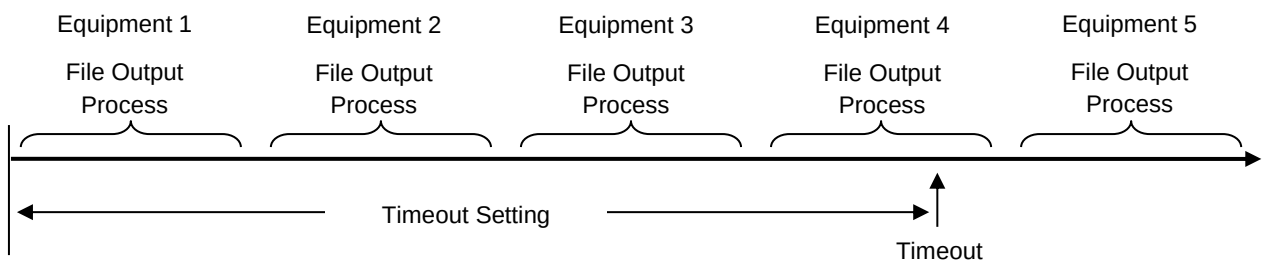
|                      |                           |
|----------------------|---------------------------|
| <b>Name of item</b>  | TIMESERIES_MANUAL_TIMEOUT |
| <b>Parameter</b>     | 1 to 1200 (minutes)       |
| <b>Default value</b> | 720                       |

(9) Time out period of report schedule output

Set time out period of report schedule output.

File is output per equipment unit, and the part after the timeout setting period will not be output.

In the following case, the signal data output files of equipment units 1, 2, and 3 will be output.



If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                             |
|----------------------|-----------------------------|
| <b>Name of item</b>  | REPORTTOOL_SCHEDULE_TIMEOUT |
| <b>Parameter</b>     | 1 to 1200 (minutes)         |
| <b>Default value</b> | 720                         |

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |  |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|--|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |  |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |  |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |  |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |  |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |  |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |  |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 793/868 |  |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |  |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue             |                                                       |  |        |         |  |

(10) Time out period of report manual output

Set time out period of report manual output.

File is output per equipment unit, and the part after the timeout setting period will not be output.

The behavior after the timeout setting period is the same as shown in the above item (10).

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                           |
|----------------------|---------------------------|
| <b>Name of item</b>  | REPORTTOOL_MANUAL_TIMEOUT |
| <b>Parameter</b>     | 1 to 1200 (minutes)       |
| <b>Default value</b> | 720                       |

(11) Interval for inquiring Collector Service Manager

Set an interval at which to inquire from the Collector PC to Collector Service Manager.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                   |
|----------------------|-------------------|
| <b>Name of item</b>  | COLMGR_INTERVAL   |
| <b>Parameter</b>     | 5 to 60 (seconds) |
| <b>Default value</b> | 5                 |

(12) Time out period for communication with Collector Service Manager

Set a time out period for communication with Collector Service Manager.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                     |
|----------------------|---------------------|
| <b>Name of item</b>  | COLMGR_COM_TIMEOUT  |
| <b>Parameter</b>     | 1 to 1000 (seconds) |
| <b>Default value</b> | 60                  |

(13) Time unit in dividing DB data for block-wise deletion

Set a time unit in minutes for this item. A manual deletion process determines the data that fall in the deletion range and divides them into blocks such that each of them contains data of the specified minutes or less. Then it proceeds to block-wise deletion and continues it until all the blocks are deleted.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                     |
|----------------------|---------------------|
| <b>Name of item</b>  | DELETE_TERM         |
| <b>Parameter</b>     | 1 to 1440 (minutes) |
| <b>Default value</b> | 60                  |

(14) Interval for communicating with OPC UA Server

Set an interval at which to periodically communicate with OPC UA Server.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                      |
|----------------------|----------------------|
| <b>Name of item</b>  | KEEPALIVE_INTERVAL   |
| <b>Parameter</b>     | 5000 to 60000 (msec) |
| <b>Default value</b> | 5000                 |

(15) Time out period for communicating with OPCUA Server

Set a time out period for communication with OPCUA Server.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                      |
|----------------------|----------------------|
| <b>Name of item</b>  | KEEPALIVE_TIMEOUT    |
| <b>Parameter</b>     | 5000 to 60000 (msec) |
| <b>Default value</b> | 10000                |

|             |               |               |                                            |                          |                                                       |               |              |                |
|-------------|---------------|---------------|--------------------------------------------|--------------------------|-------------------------------------------------------|---------------|--------------|----------------|
| 20          | Nov. 22, 2023 | uwano         | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |               |              |                |
| 19          | Apr. 27, 2021 | uwano         | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |               |              |                |
| 18          | July 28, 2020 | uwano         | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |               |              |                |
| 17          | Jan. 30, 2020 | uwano         | Revised for MT-LINK <i>i</i> Version 3.9   |                          |                                                       |               |              |                |
| 16          | July 22, 2019 | uwano         | Revised for MT-LINK <i>i</i> Version 3.8   | Draw                     | <b>A-40829-00001EN</b>                                |               |              |                |
| 15          | Mar. 29, 2019 | ogawa         | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |               |              |                |
| 14          | Jan. 29, 2019 | ogawa         | Revised for MT-LINK <i>i</i> Version 3.7   |                          |                                                       |               |              |                |
| 13          | Jan. 29, 2019 | ogawa         | Revised for MT-LINK <i>i</i> Version 3.7   |                          |                                                       |               |              |                |
| <b>Ed.</b>  | <b>Date</b>   | <b>Design</b> | <b>Description</b>                         | <b>FANUC CORPORATION</b> |                                                       |               |              | Sheet. 794/868 |
| <b>Date</b> | Nov.30,2015   | <b>Desig.</b> | <b>Ogawa</b>                               | <b>Check</b>             |                                                       | <b>Apprv.</b> | <b>Inoue</b> |                |

Following is a sample when the default values are defined for AdminTool.ini.

```
[MTLINKi_ADMINTOOL]
User=dbuser
Password= dbuser
MONGOKEY=mongodb://localhost/MTLINKi
ADMINLANGUAGE=Japanese
GRIDCOLOR=#D3D3D3
TIMESERIES_DATA_LIMIT=3000
TIMESERIES_FREE_SPACE=20
TIMESERIES_INTERVAL=500
TIMESERIES_COMPRESSION=0
TIMESERIES_KIND=0
TIMESERIES_SPLIT_RECORD=0
TIMESERIES_SCHEDULE_TIMEOUT=720
TIMESERIES_MANUAL_TIMEOUT=720
REPORTTOOL_SCHEDULE_TIMEOUT=720
REPORTTOOL_MANUAL_TIMEOUT=720
COLMGR_INTERVAL=5
COLMGR_COM_TIMEOUT=60
```

```
[DBDATASAVE]
COUNT=1
DBPATH1=C:\FANUC\MT-LINK\iMongoDB
DBSIZE_ENABLE1=false
DBSIZE1=0
```

```
[DBDATADELETE]
DELETE_TERM=60
```

```
[OPCUA]
KEEPALIVE_INTERVAL=5000
KEEPALIVE_TIMEOUT=10000
```

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |       |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|-------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |       |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |       |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |       |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |       |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |       |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |       |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 795/868 |       |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |       |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         | Inoue |

## 15.1.2 receiver.ini

The setting items of receiver.ini are listed in the following.

("YES" in the [Initial] column indicates that the item is set upon installation, "YES" in the [Edit] column indicates that the item is editable, and "NO" in the [Edit] column indicates that the item should not be edited.)

| Section Name                                                                | Item Name                                                          | Description                                                                                                                                                                                             | Initial | Edit |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------|
| SystemDiagnosis<br>(System Diagnosis)                                       | RECEIVER_MONITORING_CYCLE_SEC                                      | Receiver information collection cycle                                                                                                                                                                   | YES     | NO   |
|                                                                             | COLLECTOR_MONITORING_CYCLE_SEC                                     | Collector information collection cycle                                                                                                                                                                  | YES     | NO   |
|                                                                             | UPDATE_MONITORING_CYCLE_SEC                                        | Information collection record cycle                                                                                                                                                                     | YES     | NO   |
|                                                                             | RECEIVER_ALERT_THRESHOLD                                           | Receiver queue alert threshold                                                                                                                                                                          | YES     | NO   |
|                                                                             | RECEIVER_ALERT_DELAY                                               | (1) Duration period of overflow in number of Receiver queue                                                                                                                                             | YES     | YES  |
|                                                                             | CACHE_ALERT_THRESHOLD                                              | (2) Threshold of DB cache usage (Status: Alarm)                                                                                                                                                         | YES     | YES  |
|                                                                             | CACHE_WARN_THRESHOLD                                               | (3) Threshold of DB cache usage (Status: Warning)                                                                                                                                                       | YES     | YES  |
|                                                                             | FREE_DISK_SPACE_THRESHOLD_GREEN                                    | (4) Threshold of free disk space ratio (Status: Good)                                                                                                                                                   | YES     | YES  |
|                                                                             | FREE_DISK_SPACE_THRESHOLD_YELLOW                                   | (5) Threshold of free disk space ratio (Status: Warning)                                                                                                                                                | YES     | YES  |
| LicenseKey                                                                  | CheckIntervalTime                                                  | License check cycle (in minutes)                                                                                                                                                                        | YES     | NO   |
| LogInfo                                                                     | Collector<br>Receiver<br>Mapper<br>Keeper<br>IPC<br>Alarm<br>Macro | Collector log output enabled<br>Receiver log output enabled<br>Mapper log output enabled<br>Keeper log output enabled<br>IPC log output enabled<br>Alarm log output enabled<br>Macro log output enabled | YES     | NO   |
| TRIGGER_LOG<br>(Output in CSV file when macro variable is retrieved)        | FILE                                                               | (6) Name of CSV file when macro variable is retrieved                                                                                                                                                   | YES     | YES  |
| VARIABLE_HISTORY<br>(Output in CSV file when variable history is retrieved) | FILE                                                               | (7) Name of CSV file when variable history is retrieved                                                                                                                                                 | YES     | YES  |
|                                                                             | OUTPUT_SIGNAL_CONDITION                                            | (8) A flag to indicate whether to output SignalCondition item                                                                                                                                           | YES     | YES  |
| DiagnoseLog<br>(Output in CSV file of system diagnosis information)         | RETENTION_PERIOD                                                   | (9) The Retention period for system diagnosis information log (a CSV file)                                                                                                                              | YES     | YES  |

The following part explains each item that is editable (with "YES" in the [Edit] column).

### (1) Duration period of overflow in number of Receiver queue

Set the duration period after number of Receiver queue has exceeded the "Receiver queue alert threshold."

When the status exceeding the "Receiver queue alert threshold" continued for the period of the setting, it is judged as Alarm condition, and the  will be displayed on the system diagnosis (Queue of Receiver) and the diagnosis signal lamp of the Web screen.

When the status exceeding the "Receiver queue alert threshold" did not continue for the period of the setting, it is judged as Good condition, and the  will be displayed on the system diagnosis (Queue of Receiver) and the diagnosis signal lamp of the Web screen, and the duration period will be cleared.

In an environment which a high workload on the Receiver occurs intermittently (while machine is in operation), it may become difficult to detect the Alarm status if the setting of the duration period is too long.

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               | Sheet. 796/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |



Set the value as 1800 seconds or less, for the Receiver may stop before judging the Alarm status if the setting value is too large.

If this item is unspecified, the default value will be set.

|                      |                      |
|----------------------|----------------------|
| <b>Name of item</b>  | RECEIVER_ALERT_DELAY |
| <b>Parameter</b>     | 1 to 1800 (seconds)  |
| <b>Default value</b> | 300                  |

(2) Threshold of DB cache usage (Status: Alarm)

Set the threshold value of DB cache usage for the "Alarm" status.

If the amount of DB cache usage is equal or more than the setting value, the status is judged as "Alarm" and the background color of [Cache Rate] column in the [Utility - System Diagnosis] screen will be red.

If this item is unspecified, or an invalid parameter is specified, the default value will be set.

|                      |                       |
|----------------------|-----------------------|
| <b>Name of item</b>  | CACHE_ALERT_THRESHOLD |
| <b>Parameter</b>     | 0 to 100 (%)          |
| <b>Default value</b> | 50                    |

**NOTE**

As for increase in the [Cache Rate] column on the [Utility - System Diagnosis] screen, refer to the following subsection of the "FANUC MT-LINK*i* / Integration Server Troubleshooting MANUAL":

The "Operation of System Diagnosis and Countermeasures against Anomaly" subsection in the "TROUBLES RELATED TO SYSTEM DIAGNOSIS" section in the "MT-LINK*i* TROUBLESHOOTING" chapter

(3) Threshold of DB cache usage (Status: Warning)

Set the threshold value of DB cache usage for the "Warning" status.

If the amount of DB cache usage is less than the setting value of (2) above, and equal or more than the setting value, the status is judged as "Warning" and the background color of [Cache Rate] column in the [Utility - System Diagnosis] screen will be yellow. If the amount of DB cache usage is less than the setting value, the status is judged as "Good" and the background color of [Cache Rate] column in the [Utility - System Diagnosis] screen will be green.

If this item is unspecified, or an invalid parameter is specified, the default value will be set.

|                      |                      |
|----------------------|----------------------|
| <b>Name of item</b>  | CACHE_WARN_THRESHOLD |
| <b>Parameter</b>     | 0 to 100 (%)         |
| <b>Default value</b> | 10                   |

**NOTE**

As for increase in the [Cache Rate] column on the [Utility - System Diagnosis] screen, refer to the following subsection of the "FANUC MT-LINK*i* / Integration Server Troubleshooting MANUAL":

The "Operation of System Diagnosis and Countermeasures against Anomaly" subsection in the "TROUBLES RELATED TO SYSTEM DIAGNOSIS" section in the "MT-LINK*i* TROUBLESHOOTING" chapter

|             |               |               |                                            |                          |                                                       |               |       |                |
|-------------|---------------|---------------|--------------------------------------------|--------------------------|-------------------------------------------------------|---------------|-------|----------------|
| 20          | Nov. 22, 2023 | uwano         | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |               |       |                |
| 19          | Apr. 27, 2021 | uwano         | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |               |       |                |
| 18          | July 28, 2020 | uwano         | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |               |       |                |
| 17          | Jan. 30, 2020 | uwano         | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |               |       |                |
| 16          | July 22, 2019 | uwano         | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |               |       |                |
| 15          | Mar. 29, 2019 | ogawa         | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |               |       |                |
| 14          | Jan. 29, 2019 | ogawa         | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |               |       | Sheet. 797/868 |
| <b>Ed.</b>  | <b>Date</b>   | <b>Design</b> | <b>Description</b>                         |                          |                                                       |               |       |                |
| <b>Date</b> | Nov.30,2015   | <b>Desig.</b> | Ogawa                                      | <b>Check</b>             |                                                       | <b>Apprv.</b> | Inoue |                |

(4) Threshold of free disk space ratio (Status: Good)

Set the threshold value of free disk space ratio for the "Good" status.

If the free disk space ratio is equal or more than the setting value, the status is judged as "Good" and the background color will be green for the system diagnosis (Disk Free Space), and the  is displayed on the diagnosis signal lamp of the Web screen.

If this item is unspecified, the default value will be set.

|                      |                                 |
|----------------------|---------------------------------|
| <b>Name of item</b>  | FREE_DISK_SPACE_THRESHOLD_GREEN |
| <b>Parameter</b>     | 0 or more (%)                   |
| <b>Default value</b> | 10                              |

(5) Threshold of free disk space ratio (Status: Warning)

Set the threshold value of free disk ratio space for the "Warning" status.

If the free disk space ratio is smaller than the setting value of item (4) above, and equal or more than the setting value, the status is judged as "Warning" and the background color will be yellow for the system diagnosis (Disk Free Space), and the  is displayed on the diagnosis signal lamp of the Web screen. If the free disk space is smaller than the setting value, or the free disk space is under 10GB, the status is judged as "Alert" and the background color will be red for the system diagnosis (Disk Free Space), and the  is displayed on the diagnosis signal lamp of the Web screen.

If this item is unspecified, the default value will be set.

|                      |                                  |
|----------------------|----------------------------------|
| <b>Name of item</b>  | FREE_DISK_SPACE_THRESHOLD_YELLOW |
| <b>Parameter</b>     | 0 or more (%)                    |
| <b>Default value</b> | 5                                |

(6) Name of CSV file when macro variable is retrieved

Set the output file name for the CSV file output when getting the macro variable.

If this item is unspecified, or an invalid parameter is specified, the default value will be set.

For the output file name, reserved words are prepared, which can be replaced to a machine name.

- Machine name (%M)  
If Machine name = Machine01, %M is replaced by "Machine01."
- Macro ID (%L)  
If Macro ID = ID0001, %L is replaced by "ID0001."
- Date (%D)  
If the date of getting (data retrieval) is "2017/01/01" (Jan. 1st, 2017), %D is replaced by "20170101."  
The files can be stored separately by each date.
- Time (%T)  
Time (%T) should be used in a set with date (%D).  
If the time of getting (data retrieval) is "12:34:56" %T is replaced by "123456." The time is set in 24 hours format (00 to 23).  
The files are created based on each date/time, enabling the files to be stored based on each event occurrence of macro variable getting condition.

|                      |                         |
|----------------------|-------------------------|
| <b>Name of item</b>  | FILE                    |
| <b>Parameter</b>     | Name of CSV file        |
| <b>Default value</b> | Logging_%M_%L_%D_%T.csv |

|      |               |        |                                    |       |  |        |                   |                                               |  |        |         |       |
|------|---------------|--------|------------------------------------|-------|--|--------|-------------------|-----------------------------------------------|--|--------|---------|-------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |        |         |       |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |                   |                                               |  |        |         |       |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |                   |                                               |  |        |         |       |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        | Draw              | A-40829-00001EN                               |  |        |         |       |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        |                   |                                               |  |        |         |       |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |                   |                                               |  |        |         |       |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        | FANUC CORPORATION |                                               |  | Sheet. | 798/868 |       |
| Ed.  | Date          | Design | Description                        |       |  |        |                   |                                               |  |        |         |       |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check |  | Apprv. |                   |                                               |  |        |         | Inoue |

## NOTE

For the name of the file, make sure that the three reserved words, the machine name (%M), the macro ID (%L) and the date (%D) are always specified.

(Example of setting)

Logging\_%M\_%L\_%D.csv

Logging\_%M\_%L\_%D\_%T.csv

(7) Name of CSV file when variable history is retrieved

Set the output file name for the CSV file output when variable history is retrieved.

If this item is unspecified, or an invalid parameter is specified, the default value will be set.

For the output file name, reserved words are prepared, which can be replaced to a machine name.

- Machine name (%M)  
If Machine name = Machine01, %M is replaced by "Machine01."
- Macro ID (%I)  
If ID = ID0001, %I is replaced by "ID0001."
- Machine type (%K)  
If Machine type = FanucCNC, %K is replaced by "FanucCNC."
- Date (%D)  
If the date of getting (data retrieval) is "2017/01/01" (Jan. 1st, 2017), %D is replaced by "20170101."  
The files can be stored separately by each date.
- Time (%T)  
Time (%T) should be used in a set with date (%D).  
If the time of getting (data retrieval) is "12:34:56" %T is replaced by "123456." The time is set in 24 hours format (00 to 23).  
The files are created based on each date/time, enabling the files to be stored based on each event occurrence of macro variable getting condition.

|                      |                                 |
|----------------------|---------------------------------|
| <b>Name of item</b>  | FILE                            |
| <b>Parameter</b>     | Name of CSV file                |
| <b>Default value</b> | VariableHistory_%M_%I_%D_%T.csv |

## NOTE

For the name of the file, make sure that the four reserved words, the machine name (%M), the macro ID (%I), the machine type (%K) and the date (%D) are always specified.

(Example of setting)

VariableHistory\_%M\_%I\_%K\_%D.csv

VariableHistory %M %I %K %D %T.csv

(8) Flag to indicate whether to output SignalCondition item

Specify whether to output SignalCondition item into a CSV file of variable histories.

SignalCondition item is available only to FanucRobot type machines.

Set this flag to 0 (do not output) if the item is not needed in variable history CSV files.

If this item is unspecified, or an invalid parameter is specified, the default value will be set.

|                      |                               |
|----------------------|-------------------------------|
| <b>Name of item</b>  | OUTPUT_SIGNAL_CONDITION       |
| <b>Parameter</b>     | 0: do not output<br>1: output |
| <b>Default value</b> | 1                             |

|      |               |        |                                            |       |  |        |       |                   |                                                       |  |        |         |  |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------|-------------------------------------------------------|--|--------|---------|--|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        |       | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |  |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                   |                                                       |  |        |         |  |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                   |                                                       |  |        |         |  |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       | Draw              | A-40829-00001EN                                       |  |        |         |  |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |       |                   |                                                       |  |        |         |  |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                   |                                                       |  |        |         |  |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       | FANUC CORPORATION |                                                       |  | Sheet. | 799/868 |  |
| Ed.  | Date          | Design | Description                                |       |  |        |       |                   |                                                       |  |        |         |  |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                   |                                                       |  |        |         |  |

(9) Retention period for system diagnosis information log (a CSV file)

Set a retention period for system diagnosis information log output by "CSV file Output of System Diagnosis Information" function.

If this item is unspecified, or an invalid parameter is specified, the default value will be set.

|                      |                  |
|----------------------|------------------|
| <b>Name of item</b>  | RETENTION_PERIOD |
| <b>Parameter</b>     | 1 to 90 days     |
| <b>Default value</b> | 45               |

Following is a sample when the default values are defined for receiver.ini.

```
[SystemDiagnosis]
RECEIVER_MONITORING_CYCLE_SEC = 5
COLLECTOR_MONITORING_CYCLE_SEC = 15
UPDATE_MONITORING_CYCLE_SEC = 60
RECEIVER_ALERT_THRESHOLD = 30
RECEIVER_ALERT_DELAY = 300
CACHE_ALERT_THRESHOLD = 50
CACHE_WARN_THRESHOLD = 10
FREE_DISK_SPACE_THRESHOLD_GREEN = 10
FREE_DISK_SPACE_THRESHOLD_YELLOW = 5
```

```
[LicenseKey]
CheckIntervalTime = 60
```

```
[LogInfo]
Collector = false
Receiver = false
Mapper = false
Keeper = false
IPC = false
Alarm = false
Macro = false
```

```
[TRIGGER_LOG]
FILE=Logging_%M_%L_%D_%T.csv
```

```
[VARIABLE_HISTORY]
FILE=VariableHistory_%M_%l_%K_%D_%T.csv
OUTPUT_SIGNAL_CONDITION = 1
```

```
[DiagnoseLog]
RETENTION_PERIOD = 45
```

|             |               |               |                                    |                          |                                               |               |              |         |
|-------------|---------------|---------------|------------------------------------|--------------------------|-----------------------------------------------|---------------|--------------|---------|
| 20          | Nov. 22, 2023 | uwano         | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |               |              |         |
| 19          | Apr. 27, 2021 | uwano         | Revised for MT-LINKi Version 3.11  |                          |                                               |               |              |         |
| 18          | July 28, 2020 | uwano         | Revised for MT-LINKi Version 3.10  |                          |                                               |               |              |         |
| 17          | Jan. 30, 2020 | uwano         | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |               |              |         |
| 16          | July 22, 2019 | uwano         | Revised for MT-LINKi Version 3.8   |                          |                                               |               |              |         |
| 15          | Mar. 29, 2019 | ogawa         | Revised for MT-LINKi Version 3.7.2 |                          |                                               |               |              |         |
| 14          | Jan. 29, 2019 | ogawa         | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |               | Sheet.       | 800/868 |
| <b>Ed.</b>  | <b>Date</b>   | <b>Design</b> | <b>Description</b>                 |                          |                                               |               |              |         |
| <b>Date</b> | Nov.30,2015   | <b>Desig.</b> | <b>Ogawa</b>                       | <b>Check</b>             |                                               | <b>Apprv.</b> | <b>Inoue</b> |         |

## 15.1.3 collector.ini

The setting items of collector.ini are listed in the following.

("YES" in the [Initial] column indicates that the item is set upon installation, "YES" in the [Edit] column indicates that the item is editable, and "NO" in the [Edit] column indicates that the item should not be edited.)

| Section Name                                    | Item Name                       | Description                                                                                                                                          | Initial | Edit | Remarks |
|-------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------|---------|
| SystemDiagnosis<br>(System<br>Diagnosis)        | CYCLE_TIMEOVER_THRESHOLD_GREEN  | (1) Threshold of cycle time over<br>(Status: Good)                                                                                                   | YES     | YES  |         |
|                                                 | CYCLE_TIMEOVER_THRESHOLD_YELLOW | (2) Threshold of cycle time over<br>(Status: Warning)                                                                                                | YES     | YES  |         |
| Asset<br>(Asset<br>Management)                  | ASSET_GET                       | Enable/disable asset function                                                                                                                        | YES     | NO   |         |
|                                                 | SEND_CYCLE_SEC                  | Transmission interval of maintenance<br>information from Receiver (sec)                                                                              | YES     | NO   |         |
| ReadComment<br>(Reading<br>program<br>comments) | READ_ITEM_COUNTER_CNC_MEM       | (3) A max. number of pieces of the<br>folder information to read at a<br>time when it is read in multiple<br>times. (CNC MEM)                        | YES     | YES  | NOTE 1  |
|                                                 | READ_INTERVAL_CNC_MEM           | (4) An interval at which folder<br>information is read when it is<br>read in multiple times.<br>(CNC MEM)                                            | YES     | YES  | NOTE 1  |
|                                                 | READ_ITEM_COUNTER_DATA_SV       | (5) A max. number of pieces of the<br>folder information to read at a<br>time when it is read in multiple<br>times. (Data Server)                    | YES     | YES  | NOTE 1  |
|                                                 | READ_INTERVAL_DATA_SV           | (6) An interval at which folder<br>information is read when it is<br>read in multiple times.. (Data<br>Server)                                       | YES     | YES  | NOTE 1  |
|                                                 | READ_ITEM_COUNTER_MEMCARD       | (7) A max. number of pieces of the<br>folder information to read at a<br>time when it is read in multiple<br>times. (program storage memory<br>card) | YES     | YES  | NOTE 1  |
|                                                 | READ_INTERVAL_MEMCARD           | (8) An interval at which folder<br>information is read when it is<br>read in multiple times. (program<br>storage memory card)                        | YES     | YES  | NOTE 1  |
|                                                 | SKIP_READ_COMMENT_CNC_MEM       | (9) A flag to skip or not to skip<br>reading program comments.<br>(CNC MEM)                                                                          | YES     | YES  | NOTE 1  |
|                                                 | LIMIT_FILE_NUM_CNC_MEM          | (10) An upper limit (threshold) to skip<br>reading program comments.<br>(CNC MEM)                                                                    | YES     | YES  | NOTE 1  |
|                                                 | SKIP_READ_COMMENT_DATA_SV       | (11) A flag to skip or not to skip<br>reading program comments.<br>(Data Server)                                                                     | YES     | YES  | NOTE 1  |
|                                                 | LIMIT_FILE_NUM_DATA_SV          | (12) An upper limit (threshold) to skip<br>reading program comments.<br>(Data Server)                                                                | YES     | YES  | NOTE 1  |

|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | <b>A-40829-00001EN</b>                        |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 801/868 |

| Section Name             | Item Name                 | Description                                                                                     | Initial | Edit | Remarks |
|--------------------------|---------------------------|-------------------------------------------------------------------------------------------------|---------|------|---------|
|                          | SKIP_READ_COMMENT_MEMCARD | (13) A flag to skip or not to skip reading program comments. (program storage memory card)      | YES     | YES  | NOTE 1  |
|                          | LIMIT_FILE_NUM_MEMCARD    | (14) An upper limit (threshold) to skip reading program comments. (program storage memory card) | YES     | YES  | NOTE 1  |
| OPCUA<br>(OPC UA Server) | KEEPALIVE_INTERVAL        | (15) Interval for communicating with the OPC UA Server                                          | NO      | YES  | NOTE 2  |
|                          | KEEPALIVE_TIMEOUT         | (16) Time out period for communicating with the OPC UA Server                                   | NO      | YES  | NOTE 2  |
|                          | TAG_BULK_REQUEST          | (17) The number of bulk tag signal requests to the OPC UA Server                                | NO      | YES  | NOTE 2  |
|                          | METHOD_BULK_REQUEST       | (18) The number of bulk method call requests to the OPC UA Server                               | NO      | YES  | NOTE 2  |
| AxisConfiguration        | AXIS_CONG_GET             | Enable / disable to collect axis information                                                    | YES     | NO   |         |
|                          | READ_CYCLE_MIN            | Interval at which to collect axis information                                                   | YES     | NO   |         |

#### NOTE

- For the CNC models that are subject to ReadComment section, refer to the seventh note in NOTE of the "**Basic Signals of FanucCNC**" subsection.
- In order to change the setting, the user is required to add the item.

The following part explains each item that is editable (with "YES" in the [Edit] column).

#### (1) Threshold of cycle time over (Status: Good)

Set the threshold value of cycle time over for the "Good" status.

If the time taken for getting signal from the machine is equal or less than the setting value, the status is judged as "Good" and the  is displayed on the system diagnosis (Cycle Time of Collector) and the diagnosis signal lamp of the Web screen.

|               |                                |
|---------------|--------------------------------|
| Name of item  | CYCLE_TIMEOVER_THRESHOLD_GREEN |
| Parameter     | 0 or more (msec)               |
| Default value | 500                            |

#### (2) Threshold of cycle time over (Status: Warning)

Set the threshold value of cycle time over for the "Warning" status.

If the time taken for getting signal from the machine is more than the setting value of item (1) above, and equal or less than the setting value, the status is judged as "Warning" and the  is displayed on the system diagnosis (Cycle Time of Collector) and the diagnosis signal lamp of the Web screen. If the time taken for getting signal from the machine exceeds the setting value, the status is judged as "Alert" and the  is displayed on the system diagnosis (Cycle Time of Collector) and the diagnosis signal lamp of the Web screen.

|               |                                 |
|---------------|---------------------------------|
| Name of item  | CYCLE_TIMEOVER_THRESHOLD_YELLOW |
| Parameter     | 0 or more (msec)                |
| Default value | 2000                            |

MT-LINK*i* offers a customization in reading program comments from specific CNC models.

It can lessen the workload on a CNC by reading the information about the program folder in multiple times.

\* For the CNC models that this subsection applies to, refer to the seventh note in NOTE of the "**Basic Signals of FanucCNC**" subsection.

|      |               |        |                                            |                          |                                                       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       | Sheet. 802/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue          |

As for the following items (3) to (8), set both the maximum number of pieces of folder information to read at a time and an interval for reading the folder information.

## NOTE

Modifying the following settings to higher values may cause workloads on CNCs and affect their processing. Make sure beforehand that the modification should not interfere with CNCs.

- (3) A maximum number of pieces of folder information to read at a time when it is read in multiple times (CNC\_MEM)

This item should be set to a maximum number of program comments to be read at a time when they are read in multiple times and the target device type is CNC\_MEM.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                           |
|----------------------|---------------------------|
| <b>Name of item</b>  | READ_ITEM_COUNTER_CNC_MEM |
| <b>Parameter</b>     | 1 to 20000                |
| <b>Default value</b> | 200                       |

- (4) An interval of reading folder information when it is read in multiple times (CNC\_MEM)

This item should be set to an interval of reading program comments when they are read in multiple times and the target device type is CNC\_MEM.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                       |
|----------------------|-----------------------|
| <b>Name of item</b>  | READ_INTERVAL_CNC_MEM |
| <b>Parameter</b>     | 1 to 500 (msec)       |
| <b>Default value</b> | 100                   |

- (5) A maximum number of pieces of folder information to read at a time when the folder information is read in multiple times (Data Server)

This item should be set to a maximum number of program comments to be read at a time when they are read in multiple times and the target device type is Data Server.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                           |
|----------------------|---------------------------|
| <b>Name of item</b>  | READ_ITEM_COUNTER_DATA_SV |
| <b>Parameter</b>     | 1 to 20000                |
| <b>Default value</b> | 50                        |

- (6) An interval of reading folder information when it is read in multiple times (Data Server)

This item should be set to an interval of reading program comments when they are read in multiple times and the target device type is Data Server.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                       |
|----------------------|-----------------------|
| <b>Name of item</b>  | READ_INTERVAL_DATA_SV |
| <b>Parameter</b>     | 1 to 500 (msec)       |
| <b>Default value</b> | 100                   |

- (7) A maximum number of pieces of folder information to read at a time when the folder information is read in multiple times (program storage memory card)

This item should be set to a maximum number of program comments to be read at a time when they are read in multiple times and the target device type is program storage memory card.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                           |
|----------------------|---------------------------|
| <b>Name of item</b>  | READ_ITEM_COUNTER_MEMCARD |
| <b>Parameter</b>     | 1 to 20000                |
| <b>Default value</b> | 200                       |

|      |               |        |                                            |       |  |        |       |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        |       | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                   |                                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |       | Draw              | A-40829-00001EN                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                   |                                                       |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       | FANUC CORPORATION |                                                       |  | Sheet. | 803/868 |
| Date | Nov.30.2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                   |                                                       |  |        |         |

- (8) An interval of reading folder information when it is read in multiple times (program storage memory card)  
This item should be set to an interval of reading program comments when they are read in multiple times and the target device type is program storage memory card.  
If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                       |
|----------------------|-----------------------|
| <b>Name of item</b>  | READ_INTERVAL_MEMCARD |
| <b>Parameter</b>     | 1 to 500 (msec)       |
| <b>Default value</b> | 100                   |

MT-LINK*i* sets a restriction on reading program comments from specific CNC models by using an upper limit (threshold).

During reading program comments in the program folder of a CNC, if MT-LINK*i* detects the number of files in the folder exceeds the file upper limit (threshold) for it, it skips reading program comments and they will be handled as space.

\* For the CNC models that this subsection applies to, refer to the seventh note in NOTE of the "**Basic Signals of FanucCNC**" subsection.

As for the following items (9) to (14), set both the flag indicating to skip or not to skip and an upper limit (threshold).

## NOTE

Modifying the following settings to higher values may cause workloads on CNCs and affect their processing. Make sure beforehand that the modification should not interfere with CNCs.

- (9) The flag to skip or not to skip retrieving program comments (CNC\_MEM)  
This item should be set whether reading program comments is to be skipped when the target device type is CNC\_MEM.  
If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Name of item</b>  | SKIP_READ_COMMENT_CNC_MEM                                                                                 |
| <b>Parameter</b>     | 0: not skip<br>1: skip when the number of files in the program folder exceeds the upper limit (threshold) |
| <b>Default value</b> | 1                                                                                                         |

- (10) The upper limit (threshold) to be used to skip or not to skip reading program comments (CNC\_MEM)  
This item should be set to an upper limit for the files in the program folder when the target device type is CNC\_MEM.  
If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                        |
|----------------------|------------------------|
| <b>Name of item</b>  | LIMIT_FILE_NUM_CNC_MEM |
| <b>Parameter</b>     | 1 to 20000             |
| <b>Default value</b> | 1000                   |

- (11) The flag to skip or not to skip retrieving program comments (Data Server)  
This item should be set whether reading program comments is to be skipped when the target device type is Data Server.  
If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Name of item</b>  | SKIP_READ_COMMENT_DATA_SV                                                                                 |
| <b>Parameter</b>     | 0: not skip<br>1: skip when the number of files in the program folder exceeds the upper limit (threshold) |
| <b>Default value</b> | 1                                                                                                         |

|      |               |        |                                            |       |  |        |       |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        |       | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                   |                                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |       | Draw              | A-40829-00001EN                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                   |                                                       |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       | FANUC CORPORATION |                                                       |  | Sheet. | 804/868 |
| Date | Nov.30.2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                   |                                                       |  |        |         |



- |                      |                        |
|----------------------|------------------------|
| <b>Name of item</b>  | LIMIT_FILE_NUM_DATA_SV |
| <b>Parameter</b>     | 1 to 20000             |
| <b>Default value</b> | 100                    |

- |                      |                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Name of item</b>  | SKIP_READ_COMMENT_MEMCARD                                                                                 |
| <b>Parameter</b>     | 0: not skip<br>1: skip when the number of files in the program folder exceeds the upper limit (threshold) |
| <b>Default value</b> | 1                                                                                                         |

- |                      |                        |
|----------------------|------------------------|
| <b>Name of item</b>  | LIMIT_FILE_NUM_MEMCARD |
| <b>Parameter</b>     | 1 to 20000             |
| <b>Default value</b> | 1000                   |

- |                      |                      |
|----------------------|----------------------|
| <b>Name of item</b>  | KEEPALIVE_INTERVAL   |
| <b>Parameter</b>     | 5000 to 60000 (msec) |
| <b>Default value</b> | 5000                 |

- |                      |                      |
|----------------------|----------------------|
| <b>Name of item</b>  | KEEPALIVE_TIMEOUT    |
| <b>Parameter</b>     | 5000 to 60000 (msec) |
| <b>Default value</b> | 10000                |

- |                      |                  |
|----------------------|------------------|
| <b>Name of item</b>  | TAG_BULK_REQUEST |
| <b>Parameter</b>     | 1 to 1000        |
| <b>Default value</b> | 100              |

|      |               |        |                                    |       |  |        |                   |                                               |  |        |         |
|------|---------------|--------|------------------------------------|-------|--|--------|-------------------|-----------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |                   |                                               |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |                   |                                               |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        | Draw              | A-40829-00001EN                               |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        |                   |                                               |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |                   |                                               |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        | FANUC CORPORATION |                                               |  | Sheet. | 805/868 |
| Ed.  | Date          | Design | Description                        |       |  |        |                   |                                               |  |        |         |
| Date | Nov.30.2015   | Desig. | Ogawa                              | Check |  | Apprv. |                   |                                               |  |        |         |

- (18) The number of bulk method call requests to the OPC UA Server

Set the number of tag signal requests to be sent in bulk to the OPCUA Server.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                       |
|----------------------|-----------------------|
| <b>Name of item</b>  | METHOD_BULK_REQUEST   |
| <b>Parameter</b>     | 1 to 100 method calls |
| <b>Default value</b> | 100                   |

Following is a sample when the default values are defined for collector.ini.

```
[SystemDiagnosis]
CYCLE_TIMEOVER_THRESHOLD_GREEN = 500
CYCLE_TIMEOVER_THRESHOLD_YELLOW = 2000
```

```
[Asset]
ASSET_GET = 0
SEND_CYCLE_SEC = 3600
```

```
[ReadComment]
READ_ITEM_COUNTER_CNC_MEM = 200
READ_INTERVAL_CNC_MEM = 100
READ_ITEM_COUNTER_DATA_SV = 50
READ_INTERVAL_DATA_SV = 100
READ_ITEM_COUNTER_MEMCARD = 200
READ_INTERVAL_MEMCARD = 100
SKIP_READ_COMMENT_CNC_MEM = 1
LIMIT_FILE_NUM_CNC_MEM = 1000
SKIP_READ_COMMENT_DATA_SV = 1
LIMIT_FILE_NUM_DATA_SV = 100
SKIP_READ_COMMENT_MEMCARD = 1
LIMIT_FILE_NUM_MEMCARD = 1000
```

```
[OPCUA]
KEEPALIVE_INTERVAL = 5000
KEEPALIVE_TIMEOUT = 10000
TAG_BULK_REQUEST = 100
METHOD_BULK_REQUEST = 100
```

```
[AxisConfiguration]
 AXIS_CONG_GET = 0
 READ_CYCLE_MIN = 0
```

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 806/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         |

## 15.1.4 ServoViewer.ini

The setting items of ServoViewer.ini (SERVO VIEWER linkage setting file) are listed in the following. ("YES" in the [Initial] column indicates that the item is set upon installation, "YES" in the [Edit] column indicates that the item is editable, and "NO" in the [Edit] column indicates that the item should not be edited.)

| Section Name                                               | Item Name               | Description                                                                               | Initial | Edit |
|------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------|---------|------|
| DataReadParameter<br>(Waveform data retrieval)             | READ_BLOCK_NUM          | Number of waveform data blocks read                                                       | YES     | NO   |
|                                                            | READ_INTERVAL           | Interval of reading waveform data                                                         | YES     | NO   |
|                                                            | INTERVAL_RATE           | (1) Rate of reading interval                                                              | YES     | YES  |
|                                                            | NC_LOCK_INTERVAL        | A length of time for which to occupy sampling function                                    | YES     | NO   |
|                                                            | SAVE_START_TIMING_OLD   | (2) A flag to start storing data at the same timing as is employed in an older method     | YES     | YES  |
|                                                            | AVE_COUNTER             | Output interval of waveform data retrieval log                                            | YES     | NO   |
|                                                            | StopWatchLog            | Switch output of waveform data retrieval log                                              | YES     | NO   |
| ProfileTrigger<br>(Condition monitoring)                   | StopWatchTriggerLog     | Switch output of condition monitoring log                                                 | YES     | NO   |
|                                                            | TRIGGER1_CHECK_INTERVAL | Condition monitoring interval                                                             | YES     | NO   |
| ThreadPoolParameter<br>(ThreadPool optimization parameter) | EnableParamChange       | (3)Necessity to optimize threads                                                          | YES     | YES  |
|                                                            | minThreadForOneL0       | (4)The minimum threads per machine                                                        | YES     | YES  |
| ExclusiveRelease<br>(Exclusive forced release)             | REJECT_Release30i       | (5)Necessity of forcible cancellation of sampling function occupancy for Series30i series | YES     | YES  |
|                                                            | REJECT_Count30i         | Number of Forced Release Rejects for Series 30i                                           | YES     | NO   |
|                                                            | REJECT_Release16i       | (6)Necessity of forcible cancellation of sampling function occupancy for Series16i series | YES     | YES  |
|                                                            | REJECT_Count16i         | Number of Forced Release Rejects for Series 16i                                           | YES     | NO   |
|                                                            | OcpEnd_BusyRetryCnt     | Waveform data acquisition end Retry count at busy error                                   | YES     | NO   |
|                                                            | OcpEnd_RetrySleepTime   | Waveform data acquisition end retry interval                                              | YES     | NO   |
| Timeapi<br>(Timer Precision API)                           | TimeBeginPeriod         | (7)Necessity of executing timer precision change                                          | YES     | YES  |
|                                                            | PeriodSetTime           | (8)Value to set for TimeBeginPeriod                                                       | YES     | YES  |
|                                                            | SetProcessInformation   | Necessity of executing QoS level change on Windows 11                                     | YES     | NO   |

The following part explains each item that is editable (with "YES" in the [Edit] column).

### (1) Rate of reading interval

Specifies the rate (%) of the time interval, from the end of the previous reading of waveform data to the start of the next reading of waveform data, against the time required to read the data (msec). If this item is unspecified, or an invalid parameter is specified, the default value will be set.

The changed value will be reflected from the next reading of waveform data.

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |  |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|--|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |  |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |  |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |  |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |  |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |  |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |  |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 807/868 |  |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |  |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue             |                                                       |  |        |         |  |



(5) Release the right of possession forcibly or not for Series 30i

If disconnect during sampling servo data in SERVO VIEWER Linkage function, the right of possession for servo data will stay.

In the case of the situation, the response of CNC value return reject error (error code 13 EW\_REJECT) in next sampling and this will loop.

If release the right of possession forcibly, set value "1".

|                      |                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------|
| <b>Name of item</b>  | REJECT_Release30i                                                                                 |
| <b>Parameter</b>     | 0: Not to release the right of possession forcibly<br>1: Release the right of possession forcibly |
| <b>Default value</b> | 1                                                                                                 |

**NOTE:**

In case of this configuration setting "1", please make sure thoroughly that there is no application collecting servo data except MT-LINKi because of releasing the right of possession forcibly.

The details of the reject error (error code 13 EW\_REJECT) can be confirmed in the following log.

Log file name:

<MT-LINKi Collector installation folder>\LogSERVOVIEWERLinkage.txt

(6) Release the right of possession forcibly or not for Series 16i

If disconnect during sampling servo data in SERVO VIEWER Linkage function, the right of possession for servo data will stay.

In the case of the situation, the response of CNC value return reject error (error code 13 EW\_REJECT) in next sampling and this will loop.

If release the right of possession forcibly, set value "1".

|                      |                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------|
| <b>Name of item</b>  | REJECT_Release16i                                                                                 |
| <b>Parameter</b>     | 0: Not to release the right of possession forcibly<br>1: Release the right of possession forcibly |
| <b>Default value</b> | 1                                                                                                 |

**NOTE:**

In case of this configuration setting "1", please make sure thoroughly that there is no application collecting servo data except MT-LINKi because of releasing the right of possession forcibly.

The details of the reject error (error code 13 EW\_REJECT) can be confirmed in the following log.

Log file name:

<MT-LINKi Collector installation folder>\LogSERVOVIEWERLinkage.txt

(7) Necessity of executing timer precision change

Set enable/disable for Operating System timer resolution.

|                      |                                                           |
|----------------------|-----------------------------------------------------------|
| <b>Name of item</b>  | TimeBeginPeriod                                           |
| <b>Parameter</b>     | 0: Not to set timer resolution<br>1: Set timer resolution |
| <b>Default value</b> | 1                                                         |

**NOTE:**

|      |               |        |                                    |                          |                                               |        |       |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|--------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |        |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |        |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |        |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |        |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |        |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |        |       | Sheet. 809/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |        |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    |                                               | Apprv. | Inoue |                |

Prior to Windows 10, version 2004, this setting affects a global Windows setting. For all processes Windows uses the lowest value (that is, highest resolution) requested by this setting.  
If it cause something problem, disable this setting to zero.

(8) Value to set for TimeBeginPeriod

Set timer resolution value for operating system. (Unit: msec)

|                      |                |
|----------------------|----------------|
| <b>Name of item</b>  | PeriodSetTime  |
| <b>Parameter</b>     | 1 to 12 (msec) |
| <b>Default value</b> | 1              |

Following is a sample when the default values are defined for ServoViewer.ini.

```
[DataReadParameter]
READ_BLOCK_NUM = 5
READ_INTERVAL = -1
INTERVAL_RATE = 30
NC_LOCK_INTERVAL = 1
SAVE_START_TIMING_OLD = 0
AVE_COUNTER = 100
StopWatchLog = 0

[ProfileTrigger]
StopWatchTriggerLog = 0
TRIGGER1_CHECK_INTERVAL = 500

[ThreadPoolParameter]
EnableParamChange = 1
minThreadForOneL0 = 2

[ExclusiveRelease]
REJECT_Release30i = 0
REJECT_Count30i = 4
REJECT_Release16i = 0
REJECT_Count16i = 4
OcpEnd_BusyRetryCnt = 3
OcpEnd_RetrySleepTime = 500

[Timeapi]
TimeBeginPeriod = 1
PeriodSetTime = 1
SetProcessInformation = 1
```

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |       |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|-------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |       |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |       |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |       |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |       |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |       |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |       |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 810/868 |       |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |       |
| Date | Nov.30.2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         | Inoue |

## 15.1.5 FTPServer.ini

The setting items of FTPServer.ini (FTP Server setting file) are listed in the following.  
("YES" in the [Initial] column indicates that the item is set upon installation, "YES" in the [Edit] column indicates that the item is editable, and "NO" in the [Edit] column indicates that the item should not be edited.)

| Section Name                           | Item Name                   | Description                                                                                                          | Initial | Edit |
|----------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------|---------|------|
| ServicePort                            | FTPPort                     | Port used by FTP Server                                                                                              | YES     | NO   |
| ReadProgramlist<br>(Reading file list) | ReadItemCount_ExternalBoard | (1) The maximum number of pieces of folder information per reading when folder information is read in multiple times | YES     | YES  |
|                                        | ReadInterval_ExternalBoard  | (2) The reading interval when set to read folder information in multiple times                                       | YES     | YES  |

The following part explains each item that is editable (with "YES" in the [Edit] column).

File Transfer, Machine Data Storing Schedule, and Maintenance Data Aggregation functionalities of MT-LINK*i* offer a customization in reading file information from specific CNC models. It can lessen the workload on a CNC by reading pieces of the folder information at a time and repeating it at a specified interval until the entire information is read.

The following table lists the CNC models to which ReadProgramList setting (Reading file directory) applies:

| CNC model                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Ethernet device installed                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Series 30 <i>i</i> /31 <i>i</i> /32 <i>i</i> /35 <i>i</i> -B<br>Series 31 <i>i</i> -B5<br>Series 30 <i>i</i> /31 <i>i</i> /32 <i>i</i> -A<br>Series 31 <i>i</i> -A5<br>Power Motion <i>i</i> -A<br>Series 0 <i>i</i> -F<br>Series 0 <i>i</i> -D<br>Series 30 <i>i</i> /31 <i>i</i> -PB<br>Series 30 <i>i</i> /31 <i>i</i> -LB<br>Series 31 <i>i</i> -WB<br>Series 31 <i>i</i> -WA<br>Series 31 <i>i</i> -WA5<br>Series 0 <i>i</i> -PF<br>Series 0 <i>i</i> -PD | Fast Ethernet board<br>Fast Data Server board<br>FL-net board |

### NOTE

Modifying the following settings to higher values may cause workloads on CNCs and affect their processing. Make sure beforehand that the modification should not interfere with CNCs.

- (1) A maximum number of pieces of folder information to read at a time when it is read in multiple times  
This item should be set to a number to which the folder information can be read at a time when it is read in multiple times.  
If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|               |                             |
|---------------|-----------------------------|
| Name of item  | ReadItemCount_ExternalBoard |
| Parameter     | 1 to 20000                  |
| Default value | 100                         |

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 811/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30.2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue             |                                                       |  |        |         |

- (2) An interval of reading folder information when it is read in multiple times

This item should be set to an interval of reading folder information when the folder information is read in multiple times.

If this item is unspecified, or the value of the parameter is out of range, the default value will be set.

|                      |                            |
|----------------------|----------------------------|
| <b>Name of item</b>  | ReadInterval_ExternalBoard |
| <b>Parameter</b>     | 1 to 1000 (msec)           |
| <b>Default value</b> | 500                        |

Following is a sample when the default values are defined for FTPServer.ini.

[ServicePort]

FTPPort = 21

[ReadProgramList]

ReadItemCount\_ExternalBoard = 100

ReadInterval\_ExternalBoard = 500

|      |               |        |                                            |       |  |        |       |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        |       | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |       |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       | FANUC CORPORATION |                                                       |  | Sheet. | 812/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |       |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                   |                                                       |  |        |         |



## 15.1.6 Setting.js

The following table lists the setting items of Setting.js.

("YES" in the [Initial] column indicates that the item is set upon installation, "YES" in the [Edit] column indicates that the item is editable, and "NO" in the [Edit] column indicates that the item should not be edited.)

| Variable Name                | Attribute Name                                                                                       | Description                                                                                                                                                                                                          | Initial | Edit |
|------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------|
| colorGrpMap                  | axis                                                                                                 | (1) Color of scale<br>(Color mapping of graph)                                                                                                                                                                       | YES     | YES  |
| colorStatusMap               | EMERGENCY<br>ALARM<br>OPERATE<br>MANUAL<br>WARMUP<br>SUSPEND<br>STOP<br>DISCONNECT<br>UNKNOWN        | (2) Color of status text of equipment icon (monitoring screen)<br>Emergency Stopped<br>In Alarm Condition<br>Operating<br>Manual Operating<br>Warming Up<br>Suspended<br>Stopped<br>Disconnected<br>Unknown          | YES     | YES  |
| colorMap                     | EMERGENCY<br>ALARM<br>OPERATE<br>MANUAL<br>WARMUP<br>SUSPEND<br>STOP<br>DISCONNECT<br>UNKNOWN<br>ALL | (3) Color of equipment status for table/graph (results screen)<br>Emergency Stopped<br>In Alarm Condition<br>Operating<br>Manual Operating<br>Warming Up<br>Suspended<br>Stopped<br>Disconnected<br>Unknown<br>All   | YES     | YES  |
| UserSettingStr.ja.StatusName | Emergency<br>Alarm<br>Operate<br>Manual<br>Warmup<br>Suspend<br>Stop<br>Disconnect                   | (4) Legend name of the equipment operation status proportion table (Japanese user)<br>Emergency Stopped<br>In Alarm Condition<br>Operating<br>Manual Operating<br>Warming Up<br>Suspended<br>Stopped<br>Disconnected | YES     | YES  |
| UserSettingStr.en.StatusName | Emergency<br>Alarm<br>Operate<br>Manual<br>Warmup<br>Suspend<br>Stop<br>Disconnect                   | (5) Legend name of the equipment operation status proportion table (English user)<br>Emergency Stopped<br>In Alarm Condition<br>Operating<br>Manual Operating<br>Warming Up<br>Suspended<br>Stopped<br>Disconnected  | YES     | YES  |

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               | Sheet. 813/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |

| Variable Name                                 | Attribute Name                                                                     | Description                                                                                                                                                                                                         | Initial | Edit |
|-----------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------|
| UserSettingStr.zh-Hans.StatusName             | Emergency<br>Alarm<br>Operate<br>Manual<br>Warmup<br>Suspend<br>Stop<br>Disconnect | (6) Legend name of the equipment operation status proportion table (Chinese user)<br>Emergency Stopped<br>In Alarm Condition<br>Operating<br>Manual Operating<br>Warming Up<br>Suspended<br>Stopped<br>Disconnected | YES     | YES  |
| UserSettingStr.local.StatusName               | Emergency<br>Alarm<br>Operate<br>Manual<br>Warmup<br>Suspend<br>Stop<br>Disconnect | (7) Legend name of the equipment operation status proportion table (Local language user)<br>* By default, this variable is commented out.                                                                           | YES     | YES  |
| SET_STATUS_UPDATE_INTERVAL_TIME               | -                                                                                  | (8) Update interval of equipment condition                                                                                                                                                                          | YES     | YES  |
| SET_SIGNAL_UPDATE_INTERVAL_TIME               | -                                                                                  | (9) Update interval of signal information                                                                                                                                                                           | YES     | YES  |
| SET_TOOL_LIFE_UPDATE_INTERVAL_TIME            | -                                                                                  | (10) Update interval of tool life information                                                                                                                                                                       | YES     | YES  |
| SET_PERIODIC_MAINTENANCE_UPDATE_INTERVAL_TIME | -                                                                                  | (11) Update interval of periodic maintenance information                                                                                                                                                            | YES     | YES  |
| SET_TOOL_LIFE_DISP_COUNT                      | -                                                                                  | (12) The maximum number of rows of tool life information to display                                                                                                                                                 | YES     | YES  |
| SET_PERIODIC_MAINTENANCE_DISP_COUNT           | -                                                                                  | (13) The maximum number of rows of periodic maintenance information to display                                                                                                                                      | YES     | YES  |
| SET_EQUIPMENT_LOCALSTRAG E                    | -                                                                                  | (14) Previous value retention setting for the [Monitoring - Equipment Monitoring [equipment]] screen                                                                                                                | YES     | YES  |
| SET_IHMI_DISPLAY                              | -                                                                                  | (15) Show/hide [Home] screen                                                                                                                                                                                        | YES     | YES  |
| SET_AUTO_REFRESH_A                            | -                                                                                  | (16) Auto refresh interval on results screens                                                                                                                                                                       | YES     | YES  |
| SET_AUTO_REFRESH_B                            |                                                                                    |                                                                                                                                                                                                                     |         |      |
| SET_AUTO_REFRESH_C                            |                                                                                    |                                                                                                                                                                                                                     |         |      |
| SET_EXCLUSION_ALARM                           | -                                                                                  | (17) Alarm history exclusion setting                                                                                                                                                                                | YES     | YES  |
| SET_MACROVALUE_DIGITS_AFT ER_DECIMAL_POINT    | -                                                                                  | (18) Display format of real numbers for the [Diagnosis - Macro Variable History] screen                                                                                                                             | YES     | YES  |

|      |               |        |                                    |                          |                                               |        |        |         |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |        | Sheet. | 814/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    |                                               | Apprv. | Inoue  |         |

| Variable Name    | Attribute Name                                                                                                                                                    | Description                                                                                                                                                    | Initial | Edit |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------|
| powerConColorMap | EXTERNAL<br>OTHERS<br>E_SAVE_ON<br>E_SAVE_OFF<br>CATEGORY<br>OPERATE<br>STOP<br>EXTERNAL<br>PIE_MODE<br>TABLE_HEADER<br>BACKGROUND<br>EXTERNAL<br>OPERATE<br>STOP | (19) Display color for [Results - Power Consumption Monitor] screen.<br><br>* Display color of each equipment status will be specified in item (3) "colorMap." | YES     | YES  |

The following part explains each item that is editable (with "YES" in the [Edit] column).

|      |               |        |                                            |                          |                                                       |        |        |         |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |        | Sheet. | 815/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    |                                                       | Apprv. | Inoue  |         |

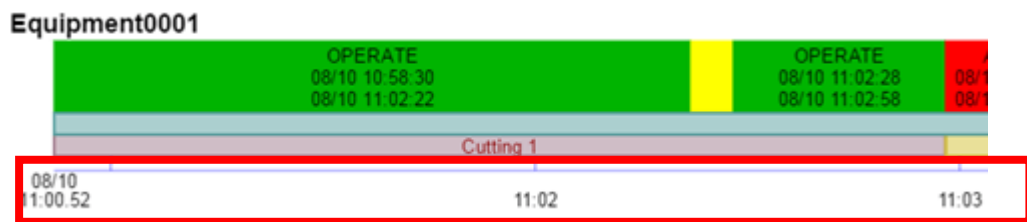
(1) Color of scale

The color of scale in the graph displayed in [Results - Operational Results] screen and [Diagnosis - Program history] screen can be customized.

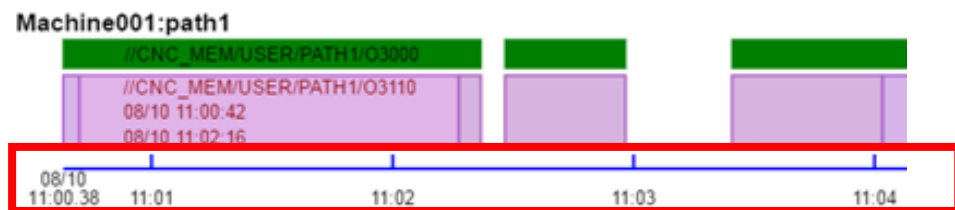
If this item is unspecified, the default value will be set.

If a parameter out of range is specified for this item, the value will not be set.

■ Color of the time scale for the operational results band graph



■ Color of the time scale for the program history band graph



|                      |                                                                |
|----------------------|----------------------------------------------------------------|
| <b>Name of item</b>  | colorGrpMap (axis)                                             |
| <b>Parameter</b>     | "black" to "white" (write HTML color name in character string) |
| <b>Default value</b> | blue                                                           |

**NOTE**

When setting the HTML color name, refer to the information on the Internet, etc.

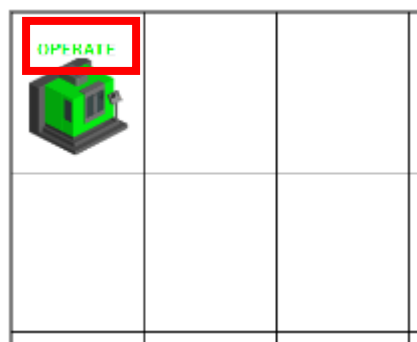
(2) Color of status text of equipment icon (monitoring screen)

The color of the text for each operating status of the equipment icon on the [Monitoring - Overview], [Monitoring - Equipment Monitoring], and [Monitoring - Equipment Monitoring -Detail] screens can be customized.

If this item is unspecified, the default value will be set.

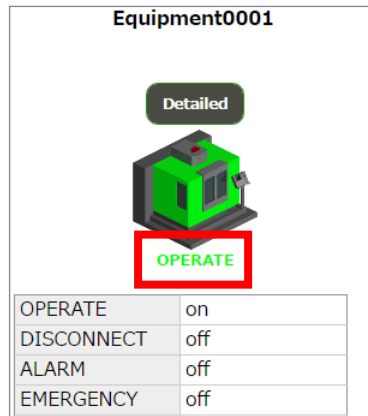
If a parameter out of range is specified for this item, the value will not be set.

■ Color of the text for each operating status of the equipment icon on the [Monitoring - Overview] screen

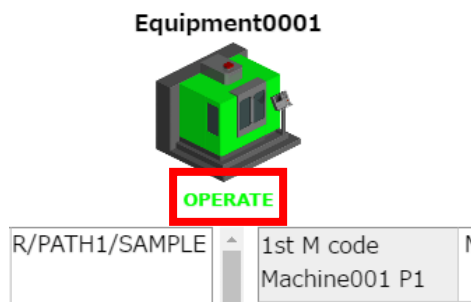


|             |               |               |                                            |                          |                                                       |               |                |
|-------------|---------------|---------------|--------------------------------------------|--------------------------|-------------------------------------------------------|---------------|----------------|
| 20          | Nov. 22, 2023 | uwano         | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |               |                |
| 19          | Apr. 27, 2021 | uwano         | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |               |                |
| 18          | July 28, 2020 | uwano         | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |               |                |
| 17          | Jan. 30, 2020 | uwano         | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |               |                |
| 16          | July 22, 2019 | uwano         | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |               |                |
| 15          | Mar. 29, 2019 | ogawa         | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |               |                |
| 14          | Jan. 29, 2019 | ogawa         | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |               | Sheet. 816/868 |
| <b>Ed.</b>  | <b>Date</b>   | <b>Design</b> | <b>Description</b>                         |                          |                                                       |               |                |
| <b>Date</b> | Nov.30,2015   | <b>Desig.</b> | Ogawa                                      | <b>Check</b>             |                                                       | <b>Apprv.</b> | Inoue          |

- Color of the text for each operating status of the equipment icon on the [Monitoring - Equipment Monitoring] screen



- Color of the text for each operating status of the equipment icon on the [Monitoring - Equipment Monitoring -Detail] screen



|                      |                                                                                                                                                                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name of item</b>  | colorStatusMap<br>(EMERGENCY, ALARM, OPERATE, MANUAL, WARMUP, SUSPEND, STOP, DISCONNECT, UNKNOWN)                                                                                                                                                 |
| <b>Parameter</b>     | "#000000" to "#FFFFFF" (write HTML color code in 6 digit hex character string)                                                                                                                                                                    |
| <b>Default value</b> | EMERGENCY: #FF00FF (pink)<br>ALARM: #FF0000 (red)<br>OPERATE: #00FF00 (green)<br>MANUAL: #000000 (black)<br>WARMUP: #000000 (black)<br>SUSPEND: #000000 (black)<br>STOP: #000000 (black)<br>DISCONNECT: #808080 (grey)<br>UNKNOWN: #808080 (grey) |

## NOTE

When setting the HTML color name, refer to the information on the Internet, etc.

|      |               |        |                                            |       |  |        |       |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        |       | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                   |                                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |       | Draw              | A-40829-00001EN                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                   |                                                       |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       | FANUC CORPORATION |                                                       |  | Sheet. | 817/868 |
| Date | Nov.30.2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                   |                                                       |  |        |         |

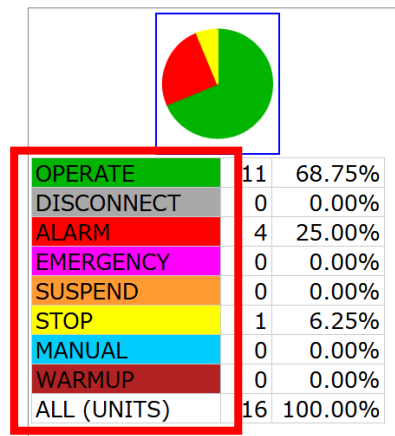
(3) Color of equipment status for table/graph (results screen)

The color of the table/graph for each operating status of the equipment on the [Monitoring - Overlook], [Results - Group Results], and [Results - Production Results] screens can be customized.

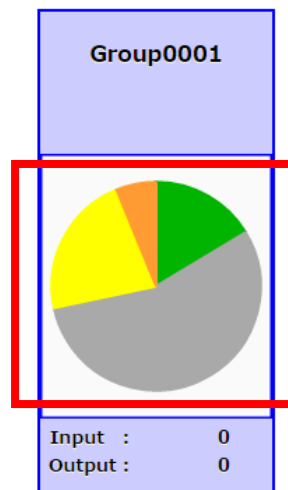
If this item is unspecified, the default value will be set.

If a parameter out of range is specified for this item, the value will not be set.

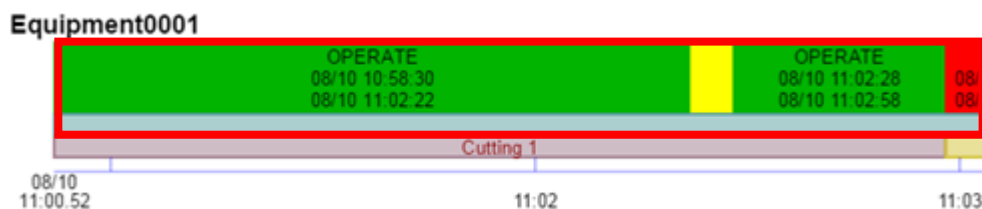
■ Background color for each operating status on the equipment operation status proportion table



■ Color for each operating status on the operational results pie graph



■ Color for each operating status on the operational results band graph



|      |               |        |                                            |       |  |        |       |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        | Draw  | A-40829-00001EN                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                                                       |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       | FANUC CORPORATION                                     |  | Sheet. | 818/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                                                       |  |        |         |

|                      |                                                                                                                                                                                                                                                                             |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name of item</b>  | colorStatusMap<br>(EMERGENCY, ALARM, OPERATE, MANUAL, WARMUP, SUSPEND, STOP, DISCONNECT, UNKNOWN)                                                                                                                                                                           |
| <b>Parameter</b>     | "#000000" to "#FFFFFF" (write HTML color code in 6 digit hex character string)                                                                                                                                                                                              |
| <b>Default value</b> | EMERGENCY: #FF00FF (pink)<br>ALARM: #FF0000 (red)<br>OPERATE: #00B400 (green)<br>MANUAL: #02CDDC (cyan)<br>WARMUP: #B22222 (brown)<br>SUSPEND: #FF9B32 (orange)<br>STOP: #FFFF00 (yellow)<br>DISCONNECT: #A9A9A9 (grey)<br>UNKNOWN: #FFFFFF (white)<br>ALL: #FFFFFF (white) |

#### NOTE

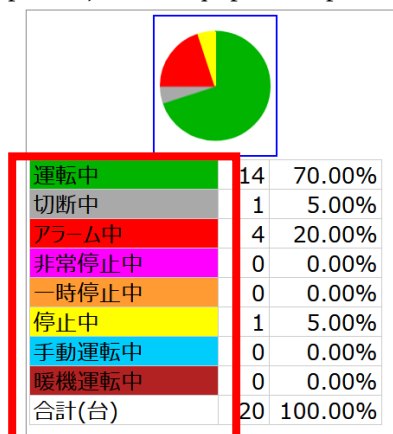
When setting the HTML color name, refer to the information on the Internet, etc.

#### (4) Legend name of the equipment operation status proportion table (Japanese user)

The Japanese user signing in to the Web screen can customize the legend names of each operation status for the equipment operation status proportion table.

If this item is unspecified, the default value will be set.

##### ■ Legend names (Japanese) of the equipment operation status proportion table



|                      |                                                                                                                                      |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name of item</b>  | UserSettingStr.ja.StatusName<br>(Emergency, Alarm, Operate, Manual, Warmup, Suspend, Stop, Disconnect)                               |
| <b>Parameter</b>     | Legend character string (Japanese) for equipment operation status proportion table                                                   |
| <b>Default value</b> | Emergency: 非常停止中<br>Alarm: アラーム中<br>Operate: 運転中<br>Manual: 手動運転中<br>Warmup: 暖機運転中<br>Suspend: 一時停止中<br>Stop: 停止中<br>Disconnect: 切断中 |

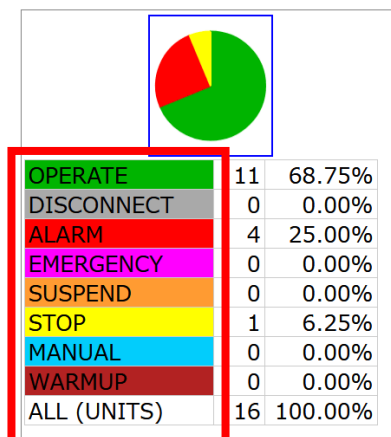
|             |               |               |                                    |                          |                                               |              |         |  |
|-------------|---------------|---------------|------------------------------------|--------------------------|-----------------------------------------------|--------------|---------|--|
| 20          | Nov. 22, 2023 | uwano         | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |              |         |  |
| 19          | Apr. 27, 2021 | uwano         | Revised for MT-LINKi Version 3.11  |                          |                                               |              |         |  |
| 18          | July 28, 2020 | uwano         | Revised for MT-LINKi Version 3.10  |                          |                                               |              |         |  |
| 17          | Jan. 30, 2020 | uwano         | Revised for MT-LINKi Version 3.9   |                          |                                               |              |         |  |
| 16          | July 22, 2019 | uwano         | Revised for MT-LINKi Version 3.8   | Draw                     | A-40829-00001EN                               |              |         |  |
| 15          | Mar. 29, 2019 | ogawa         | Revised for MT-LINKi Version 3.7.2 |                          |                                               |              |         |  |
| 14          | Jan. 29, 2019 | ogawa         | Revised for MT-LINKi Version 3.7   |                          |                                               |              |         |  |
| <b>Ed.</b>  | <b>Date</b>   | <b>Design</b> | <b>Description</b>                 | <b>FANUC CORPORATION</b> |                                               | Sheet.       | 819/868 |  |
| <b>Date</b> | Nov.30,2015   | <b>Desig.</b> | <b>Ogawa</b>                       | <b>Check</b>             | <b>Apprv.</b>                                 | <b>Inoue</b> |         |  |

(5) Legend name of the equipment operation status proportion table (English user)

The English user signing in to the Web screen can customize the legend names of each operation status for the equipment operation status proportion table.

If this item is unspecified, the default value will be set.

■ Legend names (English) for of the equipment operation status proportion table



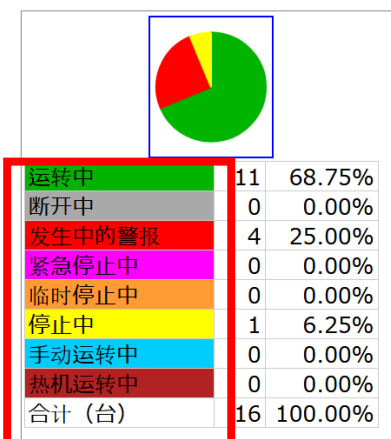
| Name of item  | UserSettingStr.en.StatusName<br>(Emergency, Alarm, Operate, Manual, Warmup, Suspend, Stop, Disconnect)                                                   |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameter     | Legend character string (in English) for NC operation status                                                                                             |
| Default value | Emergency: EMERGENCY<br>Alarm: ALARM<br>Operate: OPERATE<br>Manual: MANUAL<br>Warmup: WARMUP<br>Suspend: SUSPEND<br>Stop: STOP<br>Disconnect: DISCONNECT |

(6) Legend name of the equipment operation status proportion table (Chinese user)

The Chinese user signing in to the Web screen can customize the legend names of each operation status for the equipment operation status proportion table.

If this item is unspecified, the default value will be set.

■ Legend names (Chinese) of the of equipment operation status proportion table



|      |               |        |                                    |                   |                                               |       |                |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw              | A-40829-00001EN                               |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                   |                                               |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | FANUC CORPORATION |                                               |       | Sheet. 820/868 |
| Ed.  | Date          | Design | Description                        |                   |                                               |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             | Apprv.                                        | Inoue |                |



|                      |                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name of item</b>  | UserSettingStr.zh-Hans.StatusName<br>(Emergency, Alarm, Operate, Manual, Warmup, Suspend, Stop, Disconnect)                           |
| <b>Parameter</b>     | Legend character string (in Chinese) for NC operation status                                                                          |
| <b>Default value</b> | Emergency: 紧急停止中<br>Alarm: 发生中的警报<br>Operate: 运转中<br>Manual: 手动运转中<br>Warmup: 热机运转中<br>Suspend: 临时停止中<br>Stop: 停止中<br>Disconnect: 断开中 |

(7) Legend name of the equipment operation status proportion table (Local language user)

The Local language user signing in to the Web screen can customize the legend names of each operation status for the equipment operation status proportion table.

If this item is unspecified, the default value (values specified in additional language resource files) will be set. By default, the parameters are commented out but will be set to the same values as specified for "English." To modify those parameters for the local language, clear the comment-outs and specify appropriate values for them.

|                      |                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Name of item</b>  | UserSettingStr.local.StatusName<br>(Emergency, Alarm, Operate, Manual, Warmup, Suspend, Stop, Disconnect) |
| <b>Parameter</b>     | Legend character string (in Local language) for NC operation status                                       |
| <b>Default value</b> | Values specified in additional language resource files are used for this purpose.                         |

(8) Update interval of equipment condition

The interval for updating the equipment status can be customized for the following monitoring screens.

- Monitoring - Overlook
- Monitoring - Group Monitoring
- Monitoring - Equipment Monitoring
- Monitoring - Equipment Monitoring -Detail
- Monitoring - Alarm Monitoring
- Monitoring – Tool Life Monitoring
- Monitoring – Periodic Maintenance Monitoring

If this item is unspecified, the default value will be set.

If a parameter out of range is specified for this item, the value will not be set.

|                      |                                 |
|----------------------|---------------------------------|
| <b>Name of item</b>  | SET_STATUS_UPDATE_INTERVAL_TIME |
| <b>Parameter</b>     | 1000 to 60000 (msec)            |
| <b>Default value</b> | 1000                            |

**NOTE**

- 1 Update interval of current time is not applicable (cannot be changed).
- 2 Signal data display area of the [Monitoring - Signal Monitoring] screen is not applicable. It can be customized in the setting of the next item, "Update interval of signal information."

|             |               |               |                                                |                          |                                                           |              |                |
|-------------|---------------|---------------|------------------------------------------------|--------------------------|-----------------------------------------------------------|--------------|----------------|
| 20          | Nov. 22, 2023 | uwano         | Revised for MT-LINK <sup>i</sup> Version 4.0   | Title                    | FANUC MT-LINK <sup>i</sup><br>AdminTool OPERATOR'S MANUAL |              |                |
| 19          | Apr. 27, 2021 | uwano         | Revised for MT-LINK <sup>i</sup> Version 3.11  |                          |                                                           |              |                |
| 18          | July 28, 2020 | uwano         | Revised for MT-LINK <sup>i</sup> Version 3.10  |                          |                                                           |              |                |
| 17          | Jan. 30, 2020 | uwano         | Revised for MT-LINK <sup>i</sup> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                    |              |                |
| 16          | July 22, 2019 | uwano         | Revised for MT-LINK <sup>i</sup> Version 3.8   |                          |                                                           |              |                |
| 15          | Mar. 29, 2019 | ogawa         | Revised for MT-LINK <sup>i</sup> Version 3.7.2 |                          |                                                           |              |                |
| 14          | Jan. 29, 2019 | ogawa         | Revised for MT-LINK <sup>i</sup> Version 3.7   | <b>FANUC CORPORATION</b> |                                                           |              | Sheet. 821/868 |
| <b>Ed.</b>  | <b>Date</b>   | <b>Design</b> | <b>Description</b>                             |                          |                                                           |              |                |
| <b>Date</b> | Nov.30,2015   | <b>Desig.</b> | <b>Ogawa</b>                                   | <b>Check</b>             | <b>Apprv.</b>                                             | <b>Inoue</b> |                |

(9) Update interval of signal information

The interval for updating the signal data display area of the [Monitoring - Signal Monitoring] screen can be customized.

If this item is unspecified, the default value will be set.

If a parameter out of range is specified for this item, the value will not be set.

|                      |                                 |
|----------------------|---------------------------------|
| <b>Name of item</b>  | SET_SIGNAL_UPDATE_INTERVAL_TIME |
| <b>Parameter</b>     | 500 to 60000 (msec)             |
| <b>Default value</b> | 500                             |

(10) Update interval of tool life information

The interval for updating the tool life information display area and the tool life information graph display area of the [Monitoring - Tool Life Monitoring] screen can be customized.

If this item is unspecified, or a parameter out of range is specified for this item, the default value will be set.

|                      |                                    |
|----------------------|------------------------------------|
| <b>Name of item</b>  | SET_TOOL_LIFE_UPDATE_INTERVAL_TIME |
| <b>Parameter</b>     | 60000 to 300000 (msec)             |
| <b>Default value</b> | 60000                              |

(11) Update interval of periodic maintenance information

The interval for updating the periodic maintenance information display area and periodic maintenance information graph display area of the [Monitoring - Periodic Maintenance Monitoring] screen can be customized.

If this item is unspecified, or a parameter out of range is specified for this item, the default value will be set.

|                      |                                               |
|----------------------|-----------------------------------------------|
| <b>Name of item</b>  | SET_PERIODIC_MAINTENANCE_UPDATE_INTERVAL_TIME |
| <b>Parameter</b>     | 60000 to 300000 (msec)                        |
| <b>Default value</b> | 60000                                         |

(12) The maximum number of rows of tool life information to display

Set this item to specify a maximum number of the rows and bar graphs that the [Monitoring - Tool Life Monitoring] screen can display in the tool life information display area and tool life information graph display area.

If this item is unspecified, or a parameter out of range is specified for this item, the default value will be set.

|                      |                          |
|----------------------|--------------------------|
| <b>Name of item</b>  | SET_TOOL_LIFE_DISP_COUNT |
| <b>Parameter</b>     | 20 to 5000               |
| <b>Default value</b> | 1000                     |

(13) The maximum number of rows of periodic maintenance information to display

Set this item to specify a maximum number of the rows and bar graphs that the [Monitoring - Periodic Maintenance] screen can display in the periodic maintenance information display area and periodic maintenance information graph display area.

If this item is unspecified, or a parameter out of range is specified for this item, the default value will be set.

|                      |                                     |
|----------------------|-------------------------------------|
| <b>Name of item</b>  | SET_PERIODIC_MAINTENANCE_DISP_COUNT |
| <b>Parameter</b>     | 20 to 5000                          |
| <b>Default value</b> | 1000                                |

|      |               |        |                                            |       |  |        |       |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        | Draw  | A-40829-00001EN                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                                                       |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       |                                                       |  |        |         |
| Date | Nov.30.2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue | FANUC CORPORATION                                     |  | Sheet. | 822/868 |

(14) Previous value retention setting for the [Monitoring - Equipment Monitoring] screen

Whether to retain the previous value or not can be customized for the following screens when moving to a screen shown below from the [Monitoring - Equipment Monitoring [equipment]] screen by selecting the corresponding screen selector. Users can move to the [Monitoring - Equipment Monitoring [equipment]] screen from the [Overlook] screen:

- Monitoring - Signal Monitoring
- Results - Operational Results
- Results - Production Results
- Diagnosis - Alarm History
- Diagnosis - Program History
- Diagnosis - Signal History
- Diagnosis - Macro Variable History
- Diagnosis - Report Output
- Utility - File Transfer
- Monitoring - Tool Life Monitoring
- Monitoring - Periodic Maintenance Monitoring
- Utility - Simultaneous File Transmission

If this item is unspecified, the default value will be set.

|                      |                                                                    |
|----------------------|--------------------------------------------------------------------|
| <b>Name of item</b>  | SET_EQUIPMENT_LOCALSTRAGE                                          |
| <b>Parameter</b>     | true: Retain previous value<br>false: Do not retain previous value |
| <b>Default value</b> | true                                                               |

(15) Show/hide [Home] screen

Whether to show/hide the [Home] screen when signing in to the Web screen can be customized.

If this item is unspecified, the default value will be set.

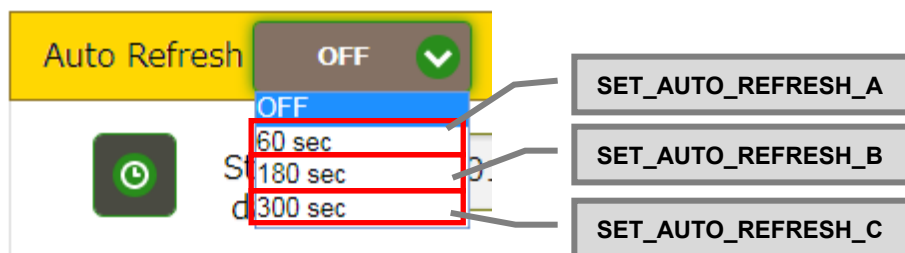
|                      |                           |
|----------------------|---------------------------|
| <b>Name of item</b>  | SET_IHMI_DISPLAY          |
| <b>Parameter</b>     | true: Show<br>false: Hide |
| <b>Default value</b> | true                      |

(16) Auto refresh interval on results screens

Auto refresh interval on the results screens ([Results - Group Results] screen, [Results - Operational Results] screen, and [Results - Production Results] screen) can be customized.

If this item is unspecified, or a parameter out of range is specified for this item, the default value will be set.

- Location for setting auto refresh interval and the name of attribute



|             |               |               |                                    |                   |                                               |        |         |
|-------------|---------------|---------------|------------------------------------|-------------------|-----------------------------------------------|--------|---------|
| 20          | Nov. 22, 2023 | uwano         | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |         |
| 19          | Apr. 27, 2021 | uwano         | Revised for MT-LINKi Version 3.11  |                   |                                               |        |         |
| 18          | July 28, 2020 | uwano         | Revised for MT-LINKi Version 3.10  |                   |                                               |        |         |
| 17          | Jan. 30, 2020 | uwano         | Revised for MT-LINKi Version 3.9   |                   |                                               |        |         |
| 16          | July 22, 2019 | uwano         | Revised for MT-LINKi Version 3.8   | Draw              | A-40829-00001EN                               |        |         |
| 15          | Mar. 29, 2019 | ogawa         | Revised for MT-LINKi Version 3.7.2 |                   |                                               |        |         |
| 14          | Jan. 29, 2019 | ogawa         | Revised for MT-LINKi Version 3.7   |                   |                                               |        |         |
| <b>Ed.</b>  | <b>Date</b>   | <b>Design</b> | <b>Description</b>                 | FANUC CORPORATION |                                               | Sheet. | 823/868 |
| <b>Date</b> | Nov.30,2015   | <b>Desig.</b> | Ogawa                              |                   |                                               |        |         |
|             |               | <b>Check</b>  |                                    | <b>Apprv.</b>     | Inoue                                         |        |         |

|               |                    |
|---------------|--------------------|
| Name of item  | SET_AUTO_REFRESH_A |
| Parameter     | 30 to 600 (sec)    |
| Default value | 60                 |

|               |                    |
|---------------|--------------------|
| Name of item  | SET_AUTO_REFRESH_B |
| Parameter     | 30 to 600 (sec)    |
| Default value | 180                |

|               |                    |
|---------------|--------------------|
| Name of item  | SET_AUTO_REFRESH_C |
| Parameter     | 30 to 600 (sec)    |
| Default value | 300                |

#### (17) Alarm history exclusion setting

This setting tells the [Diagnosis – Alarm History] screen to narrow down the search results by excluding from the search results the alarm history data which match the specified exclusion conditions in this setting. When this setting has one or more exclusion conditions, the [Diagnosis – Alarm History] screen displays the [Exclude] check box and [Exclusion Setting] button.

For the details on how to operate the [Diagnosis – Alarm History] screen, refer to the "DIAGNOSIS – ALARM HISTORY SCREEN" section of the "FANUC WebUI OPERATOR'S MANUAL."

The screenshot shows the 'Diagnosis - Alarm History' interface. At the top, there's a title bar with 'Menu', 'Filter', and navigation icons. Below it, a note states: '\* Calendar has the upper limit for time span (Upper limit [day]: 7).' The main area is divided into 'Search' and 'Summary' tabs. The 'Search' section includes a 'Start date' (2019/10/15), 'Start time' (00:00:00), and an 'Exclude' checkbox (highlighted with a red box and labeled 'Exclude CHECK BOX'). Below the search filters, there are 'Exclusion Setting' buttons (highlighted with a red box and labeled 'Exclusion Setting BUTTON') for various criteria: Period range (minute), Equipment name, Machine name, Alarm type, Alarm number, and Alarm message. Each criterion has a 'Min' and 'Max' input field. The 'Alarm list' section displays a table of alarm history data. At the bottom, there's a status bar showing the time (12:36:58), a 'FANUC MT-LINKi' logo, and a page indicator (11 / 16).

| Date and time of occurrence | Time span (minutes) | Equipment name | Machine name | Alarm type | Alarm number | Alarm message |
|-----------------------------|---------------------|----------------|--------------|------------|--------------|---------------|
| 2019/10/15 14:41:02         | 0.73                | Equipment0001  | Machine001   | EX         | 1001         | PRODUCT ALARM |
| 2019/10/15 14:41:02         | 0.73                | Equipment0006  | Machine001   | EX         | 1001         | PRODUCT ALARM |
| 2019/10/15 14:41:02         | 0.73                | Equipment0011  | Machine001   | EX         | 1001         | PRODUCT ALARM |
| 2019/10/15 14:41:02         | 0.73                | Equipment0014  | Machine001   | EX         | 1001         | PRODUCT ALARM |

|               |                                                                                         |
|---------------|-----------------------------------------------------------------------------------------|
| Name of item  | SET_EXCLUSION_ALARM                                                                     |
| Parameter     | Values for items (machine name, alarm type, and alarm number) in an exclusion condition |
| Default value | [] (no exclusion condition is defined)                                                  |

|      |               |        |                                    |                   |                                               |         |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw              | A-40829-00001EN                               |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                   |                                               |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | FANUC CORPORATION | Sheet.                                        | 824/868 |
| Ed.  | Date          | Design | Description                        |                   |                                               |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             | Apprv.                                        | Inoue   |

## ■ Defining exclusion conditions

An exclusion condition comprises three items in the following order:

- 1 Machine name
- 2 Alarm type
- 3 Alarm number

Example of an exclusion condition) Machine name: Machine001, Alarm type: PS, Alarm number: 3001

```
Var SET_EXCLUSION_ALARM = [
 ['Machine001','PS','3001']
];
```

### NOTE

For items to exclude, specify the exact combination of them in an exclusion condition because the [Diagnosis – Alarm History] screen performs an exact match search when searching for the alarm history data to exclude.

## ■ Specifying an item not to be excluded

To set an item not to be excluded, define it as a null ("). (\*enclosing it with single-quotes).

As for alarm types or alarm numbers, they can be defined not to be excluded by omitting them from an exclusion condition.

Example)

| Exclusion condition | Pattern                          | Alarm to be excluded                            |
|---------------------|----------------------------------|-------------------------------------------------|
| ","'25'             | Specify nulls for items          | Alarms with alarm number 25                     |
| 'Mac1','PS'         | Omit alarm number                | Alarms with machine name Mac1 and alarm type PS |
| Default value       | Omit alarm type and alarm number | Alarms with machine name Mac2                   |

## ■ Examples of exclusion conditions

- Specify a specific machine name  
['Machine001','"]
- Specify a specific alarm type  
["','PS','"]
- Specify a specific alarm number  
["','3001']
- Specify a specific alarm type and alarm number  
["','PS','3001']
- Specify all the items  
['Machine001','PS','3001']
- Define multiple exclusion conditions  
['Machine001','PS','3001'], [","MCR','3002']

|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | <b>A-40829-00001EN</b>                        |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 825/868 |

- Examples of exclusion conditions defined in the system setting file (Settin.js) and the [Exclusion Setting] window

Setting.js

```
var SET_EXCLUSION_ALARM = [
 ["PS"],
 ["EX","5001"],
 [Machine001,'MCR','3001'],
 [Machine001,'MCR','3002'],
 [Machine001,'MCR','3003']
];
```



| Exclusion Setting                   |            |              |
|-------------------------------------|------------|--------------|
| Machine name                        | Alarm type | Alarm number |
| <input type="checkbox"/>            | PS         |              |
| <input type="checkbox"/>            | EX         | 5001         |
| <input type="checkbox"/> Machine001 | MCR        | 3001         |
| <input type="checkbox"/> Machine001 | MCR        | 3002         |
| <input type="checkbox"/> Machine001 | MCR        | 3003         |

OK

## NOTE

- Maximum number of exclusion conditions  
Exclusion conditions can be defined up to 1000.
- Cases when an exclusion condition is handled as invalid  
An exclusion condition is handled as invalid in the following cases:
  - The machine name specified in the exclusion condition is not registered with [Machine Setting] window (in this case, this exclusion condition is not counted).
  - Items in the exclusion condition are not specified in a text string.
  - All the items in the exclusion condition are specified as a null ("") (in this case, this exclusion condition is not counted).
- Handling duplicated exclusion conditions  
Duplicated exclusion conditions are discarded.  
Example) When three identical exclusion conditions are defined, only one of them is counted as valid.
- If an exclusion condition has a forth or more items, the item(s) is discarded.
- Escaping  
When using the following special characters as normal characters, escape them with the escape character ("\") (placing a backslash in front of them).

| Special character | Value when using a special character as a normal character |
|-------------------|------------------------------------------------------------|
| Single-quote      | \'                                                         |
| Backslash         | \\                                                         |

- (18) Display format of real numbers for the [Diagnosis - Macro Variable History] screen  
Values (real numbers with fractional part) of macro variables in the [Diagnosis - Macro Variable History] screen will be rounded as specified with this item. This screen will refer to this item in the first session.

If this item is unspecified or a parameter out of range is specified for this item, the default value will be set.

| Name of item  | SET_MACROVALUE_DIGITS_AFTER_DECIMAL_POINT                                                                                                                                                                                                                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameter     | -1: to disable rounding (the value will be displayed as stored)<br>0: to round the value to an integer<br>1 to 14: to round the value to the <n>-th decimal place, where <n> is specified (i.e., <n> is the number of the digits the fractional part)<br>If the number of its fractional digits is less than <n>, trailing zeros will be concatenated to the fractional part. |
| Default value | -1                                                                                                                                                                                                                                                                                                                                                                            |

| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|----------------|
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |                   |                                               |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw              | A-40829-00001EN                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |                   |                                               |                |
| Ed.  | Date          | Design | Description                        | FANUC CORPORATION |                                               | Sheet. 826/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             | Apprv.                                        | Inoue          |

(19) Display color for [Results - Power Consumption Monitor] screen

The display color of table/graph for [Results - Power Consumption Monitor] Web screen.

#### NOTE

The display color of each equipment status (EMERGENCY, ALARM, OPERATE, MANUAL, WARMUP, SUSPEND, STOP, DISCONNECT, UNKNOWN) will be specified in item (3) "colorMap."

If this item is unspecified or a parameter out of range is specified for this item, the default value will be set.

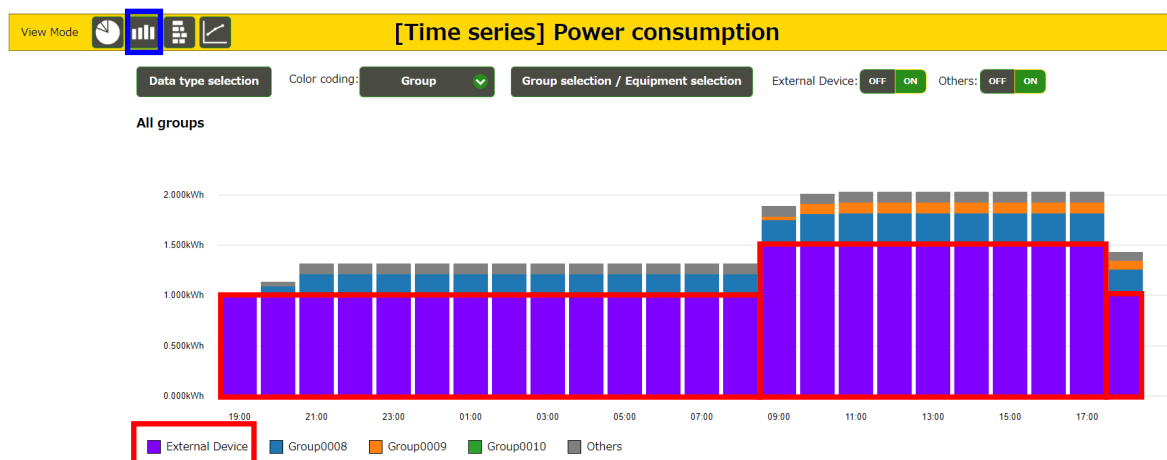
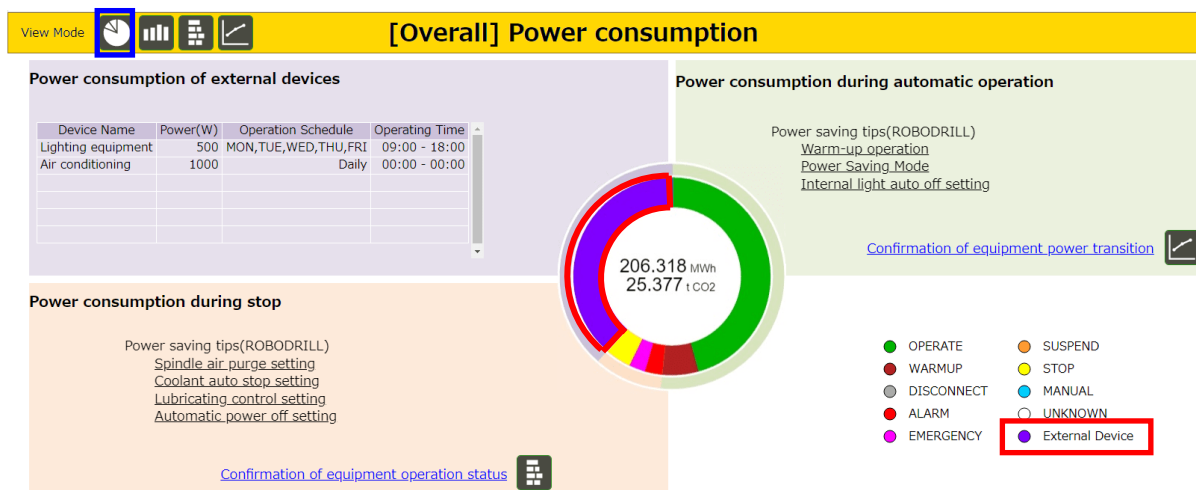
| Name of item  | powerConColorMap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------------------------------------------------------|-------------|------------------------------|-----------|-------------------------------------------------------------------|---------------|--------------------------------------------------------------------------|-------------|-----------------------------------------------------------|-----------|------------------------|----------|-----------------------|-------|------------------------|
|               | EXTERNAL: Graph and legend color for "External Device"<br>OTHERS: Graph and legend color for "Others"<br>E_SAVE_ON: Graph color for "Power Save Mode ON"<br>E_SAVE_OFF: Graph color for "Power Save Mode OFF"                                                                                                                                                                                                                                                                                                                                                                                                                          |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
|               | CATEGORY: <table> <tr> <td>OPERATE:</td><td>Overall power consumption pie chart (outside) color</td></tr> <tr> <td>STOP:</td><td>"During Automatic operation"</td></tr> <tr> <td>EXTERNAL:</td><td>"During Stop"</td></tr> <tr> <td></td><td>"External Device"</td></tr> </table>                                                                                                                                                                                                                                                                                                                                                      | OPERATE:      | Overall power consumption pie chart (outside) color           | STOP:       | "During Automatic operation" | EXTERNAL: | "During Stop"                                                     |               | "External Device"                                                        |             |                                                           |           |                        |          |                       |       |                        |
| OPERATE:      | Overall power consumption pie chart (outside) color                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
| STOP:         | "During Automatic operation"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
| EXTERNAL:     | "During Stop"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
|               | "External Device"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
|               | PIE_MODE: <table> <tr> <td>TABLE_HEADER:</td><td>Header color for "Power consumption of external device" table</td></tr> <tr> <td>BACKGROUND:</td><td></td></tr> <tr> <td>EXTERNAL:</td><td>Background color for "Power consumption of external device" table</td></tr> <tr> <td>OPERATE:</td><td>Background color for "Power consumption during automatic operation" area</td></tr> <tr> <td>STOP:</td><td>Background color for "Power Consumption during stop" area</td></tr> </table>                                                                                                                                               | TABLE_HEADER: | Header color for "Power consumption of external device" table | BACKGROUND: |                              | EXTERNAL: | Background color for "Power consumption of external device" table | OPERATE:      | Background color for "Power consumption during automatic operation" area | STOP:       | Background color for "Power Consumption during stop" area |           |                        |          |                       |       |                        |
| TABLE_HEADER: | Header color for "Power consumption of external device" table                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
| BACKGROUND:   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
| EXTERNAL:     | Background color for "Power consumption of external device" table                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
| OPERATE:      | Background color for "Power consumption during automatic operation" area                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
| STOP:         | Background color for "Power Consumption during stop" area                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
| Parameter     | "#000000" to "#FFFFFF" (write HTML color code in 6 digit hex character string)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
| Default value | EXTERNAL: #8000FF (purple)<br>OTHERS: #808080 (gray)<br>E_SAVE_ON: #4F81BD (blue)<br>E_SAVE_OFF: #808080 (gray)<br>CATEGORY: <table> <tr> <td>OPERATE:</td><td>#D7E4BD (light green)</td></tr> <tr> <td>STOP:</td><td>#FDDFC7 (orange)</td></tr> <tr> <td>EXTERNAL:</td><td>#CCC1DA (light purple)</td></tr> </table> PIE_MODE: <table> <tr> <td>TABLE_HEADER:</td><td>#8000FF (light purple)</td></tr> <tr> <td>BACKGROUND:</td><td></td></tr> <tr> <td>EXTERNAL:</td><td>#E6E0EC (light purple)</td></tr> <tr> <td>OPERATE:</td><td>#EBF1DE (light green)</td></tr> <tr> <td>STOP:</td><td>#FDEADA (light orange)</td></tr> </table> | OPERATE:      | #D7E4BD (light green)                                         | STOP:       | #FDDFC7 (orange)             | EXTERNAL: | #CCC1DA (light purple)                                            | TABLE_HEADER: | #8000FF (light purple)                                                   | BACKGROUND: |                                                           | EXTERNAL: | #E6E0EC (light purple) | OPERATE: | #EBF1DE (light green) | STOP: | #FDEADA (light orange) |
| OPERATE:      | #D7E4BD (light green)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
| STOP:         | #FDDFC7 (orange)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
| EXTERNAL:     | #CCC1DA (light purple)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
| TABLE_HEADER: | #8000FF (light purple)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
| BACKGROUND:   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
| EXTERNAL:     | #E6E0EC (light purple)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
| OPERATE:      | #EBF1DE (light green)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |
| STOP:         | #FDEADA (light orange)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |                                                               |             |                              |           |                                                                   |               |                                                                          |             |                                                           |           |                        |          |                       |       |                        |

#### NOTE

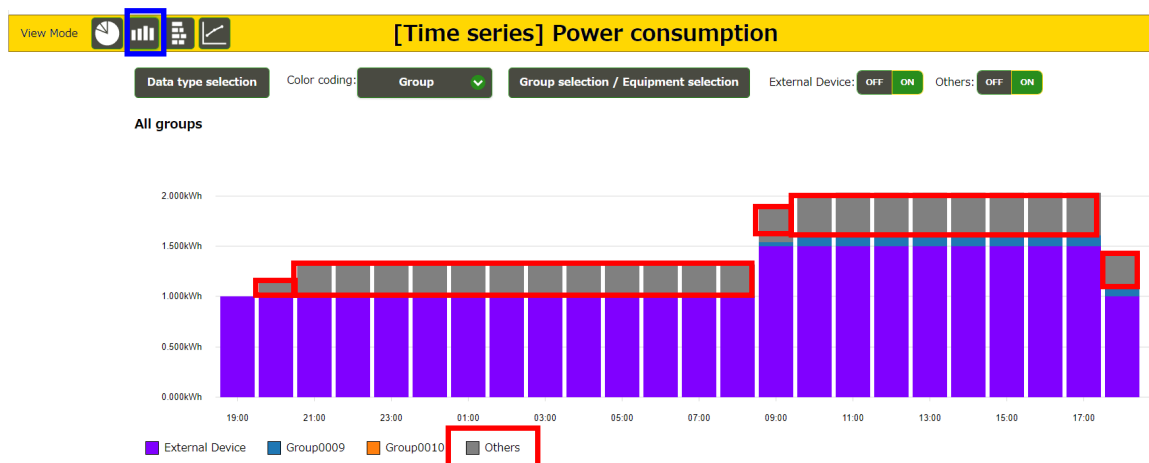
When setting the HTML color name, refer to the information on the Internet, etc.

|      |               |        |                                            |                   |                                                       |        |       |        |         |
|------|---------------|--------|--------------------------------------------|-------------------|-------------------------------------------------------|--------|-------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |       |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                   |                                                       |        |       |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                   |                                                       |        |       |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw              | A-40829-00001EN                                       |        |       |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                   |                                                       |        |       |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                   |                                                       |        |       |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | FANUC CORPORATION |                                                       |        |       | Sheet. | 827/868 |
| Ed.  | Date          | Design | Description                                |                   |                                                       |        |       |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check             |                                                       | Apprv. | Inoue |        |         |

■ Graph and legend color for "External Device" (View Mode:   )



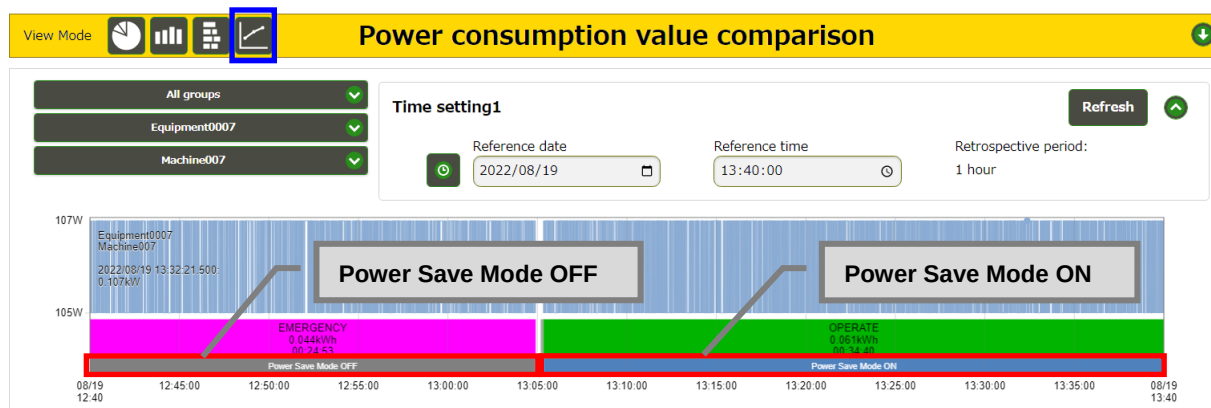
■ Graph and legend color for "Others" (View Mode:  )



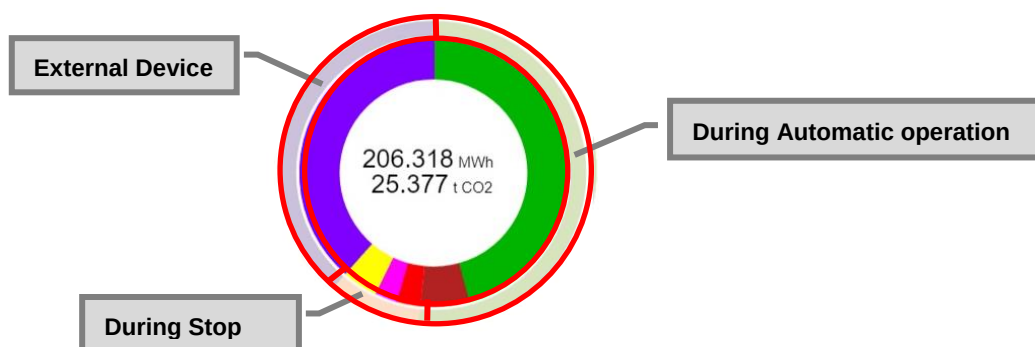
|      |               |        |                                    |       |                                               |       |                   |                |
|------|---------------|--------|------------------------------------|-------|-----------------------------------------------|-------|-------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |       |                   |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |       |                   |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |       |                   |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |       |                   |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | A-40829-00001EN                               |       |                   |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |       |                   |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |       |                   |                |
| Ed.  | Date          | Design | Description                        |       |                                               |       |                   |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue | FANUC CORPORATION | Sheet. 828/868 |



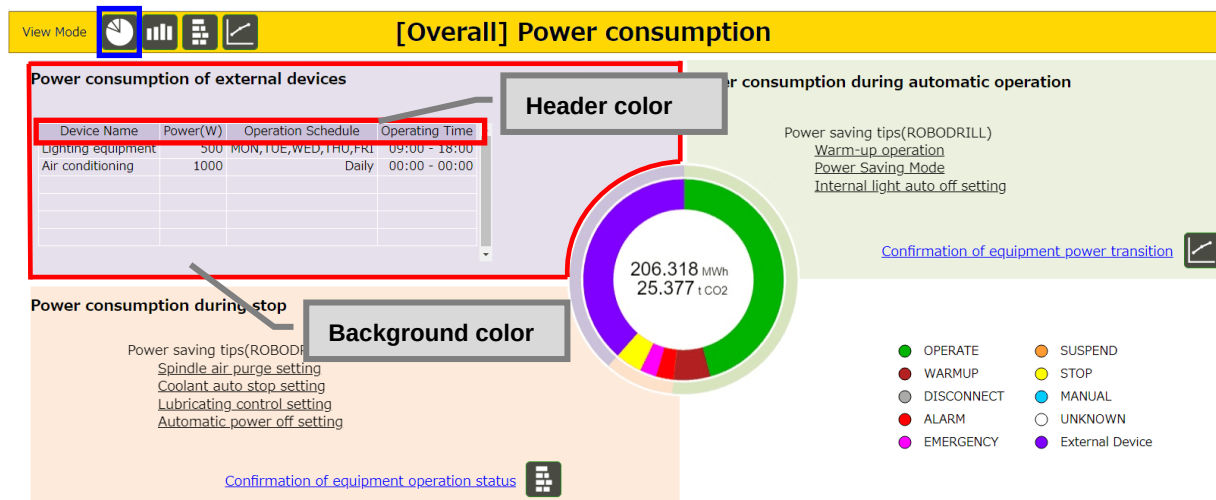
- Graph color for "Power Save Mode ON/OFF" (View Mode:  )



- Overall power consumption pie chart (outside) color (View Mode:  )



- Header and background color for "Power consumption of external device" table (View Mode:  )



|      |               |        |                                    |                   |                                               |       |                |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw              | A-40829-00001EN                               |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                   |                                               |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | FANUC CORPORATION |                                               |       | Sheet. 829/868 |
| Ed.  | Date          | Design | Description                        |                   |                                               |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             | Apprv.                                        | Inoue |                |

- 



- WARMUP
- DISCONNECT
- ALARM
- EMERGENCY
- STOP
- MANUAL
- UNKNOWN
- External Device

- 



- OPERATE
- WARMUP
- DISCONNECT
- ALARM
- SUSPEND
- STOP
- MANUAL
- UNKNOWN
- External Device

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 830/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         |

Following is a sample when the default values are defined for Setting.js.

```
var colorStatusMap = {
 EMERGENCY: '#FF00FF', //Pink Emergency stop
 ALARM: '#FF0000', //Red Alarm
 OPERATE: '#00FF00', //Green Operating
 MANUAL: '#000000', //Black Manual Operation
 WARMUP: '#000000', //Black Warmup
 SUSPEND: '#000000', //Black Suspended
 STOP: '#000000', //Black Stopped
 DISCONNECT: '#808080', //Grey Disconnected
 UNKNOWN: '#808080' //Grey Others
};

var colorMap = {
 EMERGENCY: '#FF00FF', //Pink Emergency stop
 ALARM: '#FF0000', //Red Alarm
 OPERATE: '#00B400', //Green Operating
 MANUAL: '#02C8FC', //Cyan Manual Operation
 WARMUP: '#B22222', //Brown Warmup
 SUSPEND: '#FF9B32', //Orange Suspended
 STOP: '#FFFF00', //Yellow Stopped
 DISCONNECT: '#A9A9A9', //Grey Disconnected
 UNKNOWN: '#FFFFFF', //White Others
 ALL: '#FFFFFF' //White All
};

var colorGrpMap = {
 axis: 'blue' // Scale
};

var UserSettingStr = {
 ja: { // Japanese User
 StatusName: {
 Emergency: '非常停止中',
 Alarm: 'アラーム中',
 Operate: '運転中',
 Manual: '手動運転中',
 Warmup: '暖機運転中',
 Suspend: '一時停止中',
 Stop: '停止中',
 Disconnect: '切断中'
 }
 },
 en: { // English User
 StatusName: {
 Emergency: 'EMERGENCY',
 Alarm: 'ALARM',
 Operate: 'OPERATE',
 Manual: 'MANUAL',
 Warmup: 'WARMUP',
 Suspend: 'SUSPEND',
 Stop: 'STOP',
 Disconnect: 'DISCONNECT'
 }
 },
};
```

|      |               |        |                                    |                          |                                               |        |        |         |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |        | Sheet. | 831/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    |                                               | Apprv. | Inoue  |         |

```

zh-Hans: { // Chinese User
 StatusName: {
 Emergency: '紧急停止中',
 Alarm: '发生中的警报',
 Operate: '运转中',
 Manual: '手动运转中',
 Warmup: '热机运转中',
 Suspend: '临时停止中',
 Stop: '停止中',
 Disconnect: '断开中'
 }
},
// local: { // Local User
// StatusName: {
// Emergency: 'EMERGENCY',
// Alarm: 'ALARM',
// Operate: 'OPERATE',
// Manual: 'MANUAL',
// Warmup: 'WARMUP',
// Suspend: 'SUSPEND',
// Stop: 'STOP',
// Disconnect: 'DISCONNECT'
// }
// },
};

var SET_STATUS_UPDATE_INTERVAL_TIME = 1000; // equipment status (ms)
var SET_SIGNAL_UPDATE_INTERVAL_TIME = 500; // signal info. (ms)
var SET_TOOL_LIFE_UPDATE_INTERVAL_TIME = 60000; // Tool life info. (ms)
var SET_PERIODIC_MAINTENANCE_UPDATE_INTERVAL_TIME = 60000; // Periodic Maintenance info. (ms)
var SET_TOOL_LIFE_DISP_COUNT = 1000;
var SET_PERIODIC_MAINTENANCE_DISP_COUNT = 1000;
var SET_EQUIPMENT_LOCALSTORAGE = true;

var SET_IHMI_DISPLAY = true;

var SET_AUTO_REFRESH_A = 60;
var SET_AUTO_REFRESH_B = 180;
var SET_AUTO_REFRESH_C = 300;

var SET_EXCLUSION_ALARM = [
 []
];

var SET_MACROVALUE_DIGITS_AFTER_DECIMAL_POINT = -1;

```

|      |               |        |                                    |                          |                                               |        |        |         |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |        | Sheet. | 832/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    |                                               | Apprv. | Inoue  |         |

```

var powerConColorMap = {
 EXTERNAL: '#8000FF', // Purple External device
 OTHERS: '#D3D3D3', // Grey Power Consumption of unselected groups
 E_SAVE_ON: '#4F81BD', // Blue eSave mode ON
 E_SAVE_OFF: '#808080', // Grey eSave mode OFF

 CATEGORY: {
 OPERATION: '#D7E4BD', // Light Green Power Consumption while machine is operating
 STOP: '#FDDFC7', // Orange Power Consumption while machine is stopped
 EXTERNAL: '#CCC1DA' // Light Purple Power Consumption of external device
 },
 PIE_MODE: {
 TABLE_HEADER: '#CCC1DA',
 BACKGROUND: {
 EXTERNAL: '#E6E0EC',
 OPERATE: '#EBF1DE',
 STOP: '#FDEADA'
 }
 }
};

```

|      |               |        |                                            |                          |                                                       |        |        |         |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |        | Sheet. | 833/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    |                                                       | Apprv. | Inoue  |         |

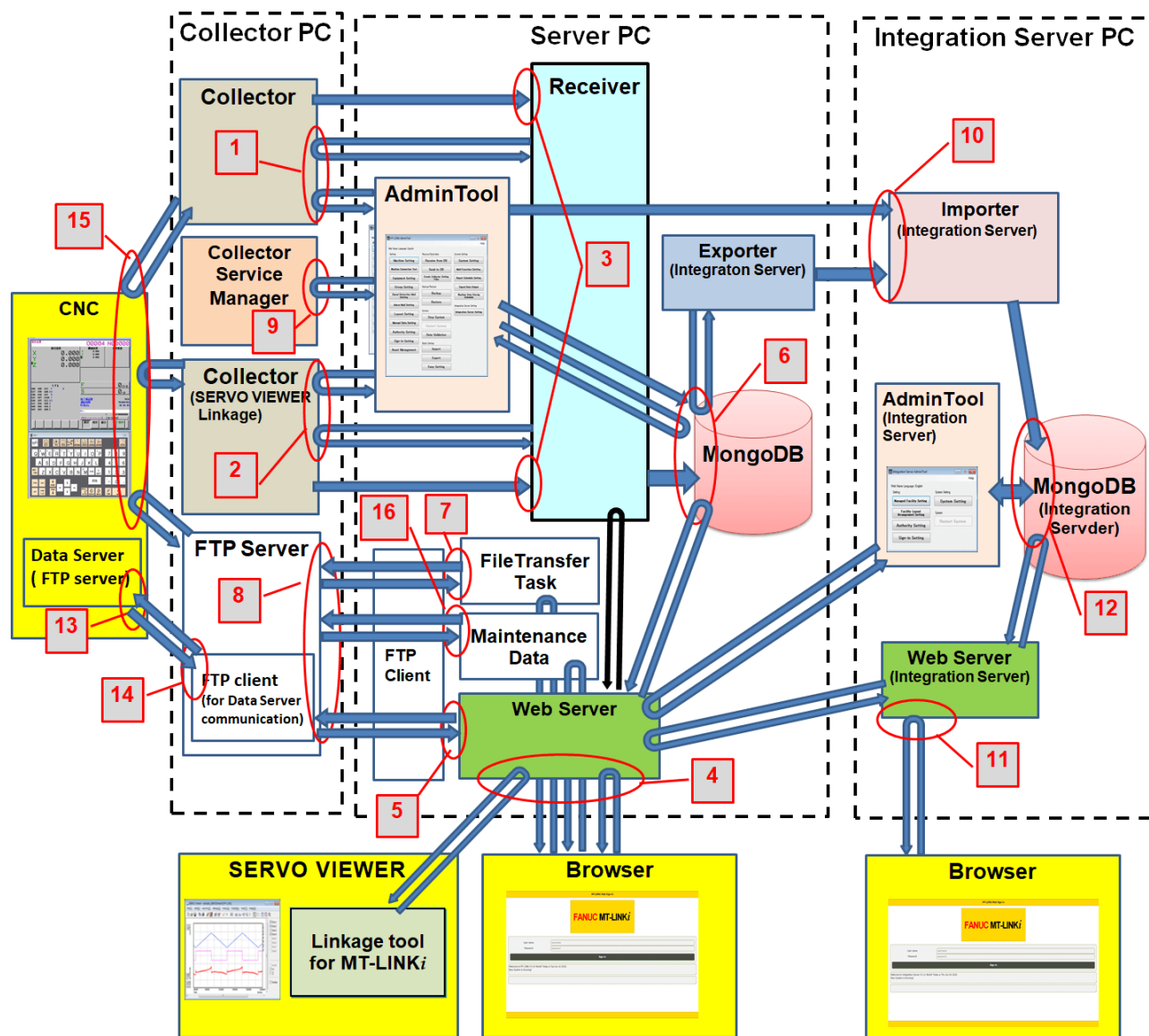
# 15.2 PORT NUMBERS USED IN MT-LINK*i*/INTEGRATION SERVER

This section explains the port numbers used in MT-LINK*i* and Integration Server.

## 15.2.1 System Configuration

The system configuration is shown in the following.

The numbers of the pop-ups correspond to the numbers (# column) in the next subsection "[List of Port Numbers in Use.](#)"



|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 834/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         |

### 15.2.2 List of Port Numbers in Use

## List of Port Numbers in Use (MT-LINK*i*/Integration Server)

| #  | Item                             | Port Number<br>(NOTE 1) | Editable<br>(NOTE 2) | Setting Window  | Description                                                                                                                                                                                                                                                                                                                                |
|----|----------------------------------|-------------------------|----------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Collector                        | 43321                   | YES                  | Machine setting | Port used by Collector<br><ul style="list-style-type: none"> <li>Communication with AdminTool (Check connection, Collector file transfer, etc.)</li> <li>Communication with Receiver (Diagnosis information collection, etc.)</li> </ul>                                                                                                   |
| 2  | Collector (SERVO VIEWER Linkage) | 43322                   | YES                  | Machine setting | Port used by SERVO VIEWER linkage on the Collector PC (NOTE 3)<br><ul style="list-style-type: none"> <li>Communication with AdminTool (Check connection, modification of Collector file, etc.)</li> <li>Communication with Receiver (Measurement data transfer, etc.)</li> </ul>                                                           |
| 3  | Receiver                         | 43320                   | YES                  | Machine setting | Port used by Receiver to receive data from Collector                                                                                                                                                                                                                                                                                       |
| 4  | Web Server (MT-LINK <i>i</i> )   | 3000                    | YES                  | System setting  | Port used by Web Server (MT-LINK <i>i</i> )<br><ul style="list-style-type: none"> <li>Communication with Web screen (browser)</li> <li>Communication with Linkage tool for MT-LINK<i>i</i> (SERVO VIEWER Linkage function)</li> <li>Communication with Web Server (Integration Server) (Integration Server Facility monitoring)</li> </ul> |
| 5  | Web Server (MT-LINK <i>i</i> )   | ephemeral port          | -                    | -               | Port used by Web Server (MT-LINK <i>i</i> ) for the following function<br><ul style="list-style-type: none"> <li>File Transport</li> </ul>                                                                                                                                                                                                 |
| 6  | Mongo DB (MT-LINK <i>i</i> )     | 27017                   | NO                   | -               | Port used by Mongo DB (MT-LINK <i>i</i> )                                                                                                                                                                                                                                                                                                  |
| 7  | File Transfer Task               | ephemeral port          | -                    | -               | Port used by File Transfer Task for the following function<br><ul style="list-style-type: none"> <li>Machine Data Storing Schedule</li> <li>Simultaneous File Transmission</li> </ul>                                                                                                                                                      |
| 8  | FTP Server                       | 21<br>(NOTE 4)          | NO                   | -               | Port used by the FTP Server for the following functions<br><ul style="list-style-type: none"> <li>File transfer</li> <li>Machine data storing schedule</li> <li>Simultaneous File Transmission</li> <li>Maintenance Data Aggregation</li> </ul>                                                                                            |
| 9  | Collector Service Manager        | 43323                   | YES                  | Machine Setting | Port used by Collector Service Manager (service which monitors/controls the services of the Collector PC) to communicate with AdminTool                                                                                                                                                                                                    |
| 10 | Importer (Integration Server)    | 50005                   | NO                   | -               | Port used by Importer (Integration Server)<br><ul style="list-style-type: none"> <li>Data receipt from MT-LINK<i>i</i></li> <li>Communication with MT-LINK<i>i</i> AdminTool (Integration Server Facility monitoring)</li> </ul>                                                                                                           |
| 11 | Web Server (Integration Server)  | 3001                    | NO                   | -               | Port used by Web Server (Integration Server)                                                                                                                                                                                                                                                                                               |

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 835/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. |                   |                                                       |  |        |         |

|    |                                            |                         |     |     |                                                                             |
|----|--------------------------------------------|-------------------------|-----|-----|-----------------------------------------------------------------------------|
| 12 | Mongo DB (Integration Server )             | 27020                   | NO  | -   | Port used by Mongo DB (integration Server)                                  |
| 13 | Data Server (FTP server)                   | 21                      | NO  | -   | Port used in communication with FTP clients (for Data Server communication) |
| 14 | FTP client (for Data Server communication) | ephemeral port (NOTE 4) | -   | -   | Port used in communication with Data Servers (FTP servers)                  |
| 15 | CNC (FOCAS communication)                  | 8193                    | YES | CNC | Port used in FOCAS communication                                            |
| 16 | Maintenance Data Aggregation               | ephemeral port          | -   | -   | Port used by the following functionality:<br>●Maintenance Data Aggregation  |

#### NOTE

- 1 In the case of editable port numbers, the default value set in installation is indicated.
- 2 Port with "YES" in the Editable column can be changed in the respective setting windows. The port with "NO" in the Editable column cannot be changed (port number is fixed).
- 3 Used when SERVO VIEWER linkage setting is enabled in the machine setting.
- 4 FTP Servers use port 20 to provide a data connection in FTP communication.

|      |               |        |                                    |                          |                                               |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               | Sheet. 836/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue          |



## 15.2.3 Windows Firewall Setting

If the MT-LINK*i* is not working properly, check the Firewall settings. If the ports used by MT-LINK*i* and the Integration Server is being blocked, configure the setting so that the connection is allowed (authorized).

When the Server PC and the Collector PC are not identical, they each should open the following ports:

| PC           | Service                                       | Protocol type<br>(NOTE 1) | Local Port<br>(NOTE 1) | This program path<br>(NOTE 1)                                                                                          |
|--------------|-----------------------------------------------|---------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------|
| Server PC    | Receiver                                      | TCP                       | 43320<br>(NOTE 2)      | <MT-LINK <i>i</i> installation folder><br>\\MT-LINK <i>i</i> Receiver\\MT-LINK <i>i</i> Receiver.exe                   |
|              | Web Server<br>(MT-LINK <i>i</i> )             | TCP                       | All Ports              | <Node.js installation folder>\\node.exe                                                                                |
|              | File Transfer Task                            | TCP                       | All Ports              | <MT-LINK <i>i</i> FTP Client installation folder><br>\\FileTransferTask.exe                                            |
|              | Maintenance Data<br>Aggregation               | TCP                       | All Ports              | <MT-LINK <i>i</i> FTP Client installation folder><br>\\MaintenanceDataAggregation.exe                                  |
| Collector PC | Collector                                     | TCP                       | 43321<br>(NOTE 2)      | <MT-LINK <i>i</i> Collector installation folder><br>\\MT-LINK <i>i</i> Collector.exe                                   |
|              | Collector<br>(SERVO VIEWER linkage)           | TCP                       | 43322<br>(NOTE 2)      | <MT-LINK <i>i</i> Collector installation folder><br>\\MT-LINK <i>i</i> CollectorSERVOVIEWERLinkage.exe                 |
|              | FTP Server                                    | TCP                       | 21                     | <MT-LINK <i>i</i> FTP Server installation folder><br>\\FTPServer.exe                                                   |
|              | FTP client (for Data Server<br>communication) | TCP                       | All Ports              | <MT-LINK <i>i</i> FTP Server installation folder><br>\\FTPServer.exe                                                   |
|              | Collector Service Manager                     | TCP                       | 43323<br>(NOTE 2)      | <Collector Service Manager installation folder><br>\\CollectorServiceMgr\\MT-<br>LINK <i>i</i> CollectorServiceMgr.exe |

When the Server PC and the Integration Server PC are not identical, they each should open the following ports:

| PC                          | Service                            | Protocol type<br>(NOTE 1) | Local Port<br>(NOTE 1) | This program path<br>(NOTE 1)           |
|-----------------------------|------------------------------------|---------------------------|------------------------|-----------------------------------------|
| Server PC                   | Web Server<br>(MT-LINK <i>i</i> )  | TCP                       | 3000<br>(NOTE 2)       | <Node.js installation folder>\\node.exe |
| Integration<br>Server<br>PC | Importer<br>(Integration Server)   | TCP                       | 50005                  | SYSTEM                                  |
|                             | Web Server<br>(Integration Server) | TCP                       | 3001<br>(NOTE 2)       | <Node.js installation folder>\\node.exe |

### NOTE

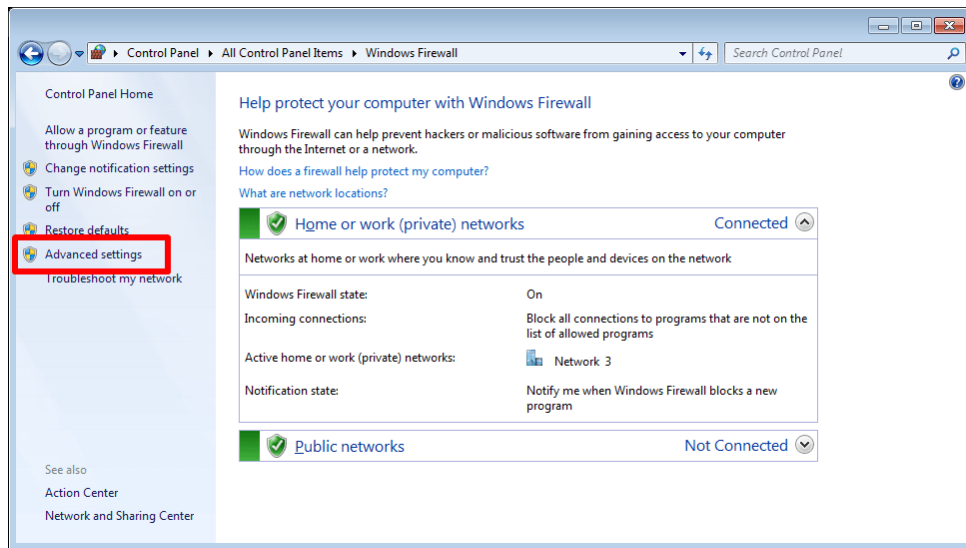
- 1 The column titles of the tables such as [Protocol type], [Local Port], and [This program path] show the values to set for the corresponding fields of the [New Inbound Rule Wizard] dialog box. For the details, refer to the setting steps (examples) written below in this subsection.
- 2 The [Local Port] columns of the tables show the default port number for the corresponding service that is specified at installation. If the port setting for the service is modified to another port number, set the newer port to open.

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 837/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |
| Date | Nov.30.2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue             |                                                       |  |        |         |

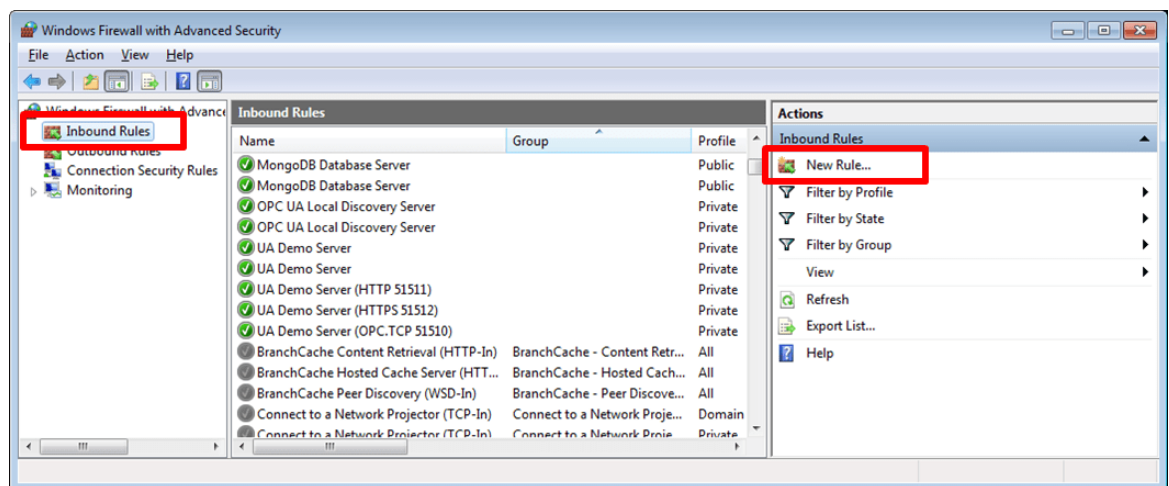
Following are the steps (examples) for adding a new rule to allow connection within the Windows Firewall.

■ In the case of the port used by the Collector

1. On the Collector PC, click [Control Panel] > [Windows Firewall], and click [Advanced settings].

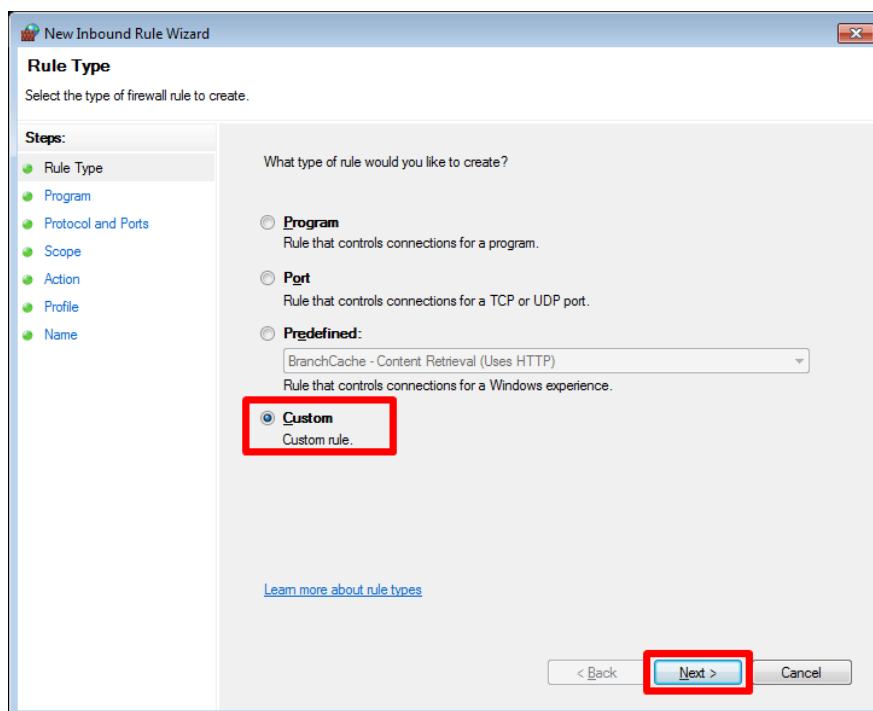


2. On the [Windows Firewall with Advance Security] window, click [Inbound Rules] > [New Rule...].

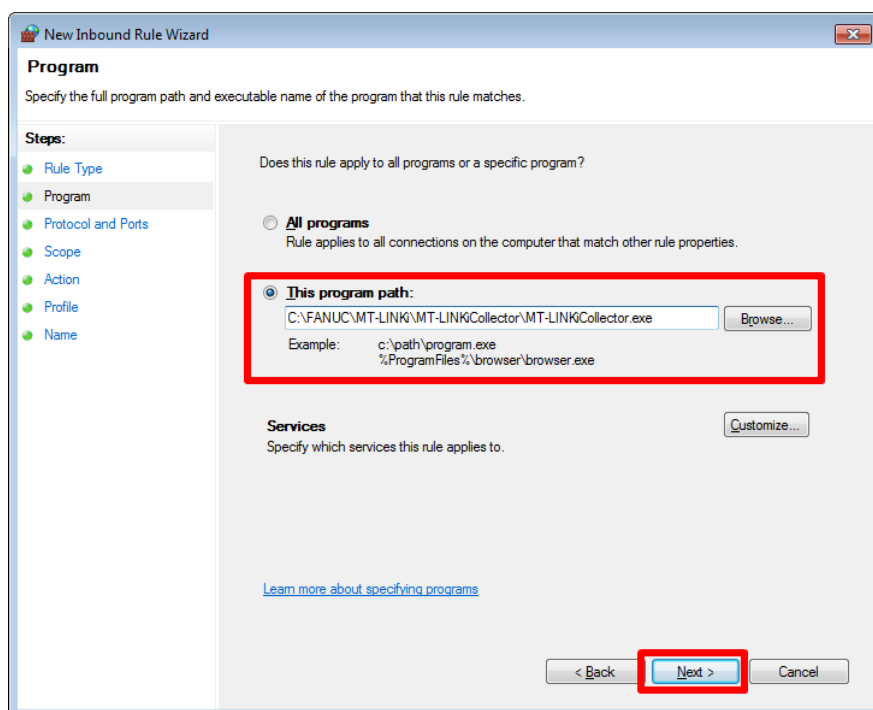


|      |               |        |                                            |                          |                                                       |       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |       | Sheet. 838/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue |                |

- On the [Rule Type] page of the [New Inbound Rule Wizard] dialog box, select [Custom], and click the [Next] button.



- On the [Program] page of the [New Inbound Rule Wizard] dialog box, select [This program path] and click the [Browse...] button. On the [Open] dialog box that appears, select [MT-LINKiCollector.exe] and click [Open]. After the [Open] dialog box closes, click the [Next] button.  
 <MT-LINKi Collector installation folder>\MT-LINKiCollector.exe



|      |               |        |                                    |                          |                                               |       |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |       | Sheet. 839/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue |                |

5. On the [Protocol and Ports] page of the [New Inbound Rule Wizard] dialog box, select [TCP] from the drop-down menu for the [Protocol type], select [Specific Ports] from the drop-down menu for the [Local Port], and in the entry column under [Specific Ports], enter and set the port number for the Collector (default port number is "43321"). After completing the settings, click the [Next] button.

**New Inbound Rule Wizard**

**Protocol and Ports**  
Specify the protocols and ports to which this rule applies.

**Steps:**

- Rule Type
- Program
- Protocol and Ports
- Scope
- Action
- Profile
- Name

To which ports and protocols does this rule apply?

Protocol type: TCP

Local port: Specific Ports  
43321  
Example: 80, 443, 5000-5010

Remote port: All Ports  
Example: 80, 443, 5000-5010

Internet Control Message Protocol (ICMP) settings: [Customize...](#)

[Learn more about protocol and ports](#)

< Back **Next >** Cancel

6. On the [Scope] page of the [New Inbound Rule Wizard] dialog box, click the [Next] button.

**New Inbound Rule Wizard**

**Scope**  
Specify the local and remote IP addresses to which this rule applies.

**Steps:**

- Rule Type
- Program
- Protocol and Ports
- Scope
- Action
- Profile
- Name

**Which local IP addresses does this rule apply to?**

☒ Any IP address

☐ These IP addresses:

Customize the interface types to which this rule applies: [Customize...](#)

**Which remote IP addresses does this rule apply to?**

☒ Any IP address

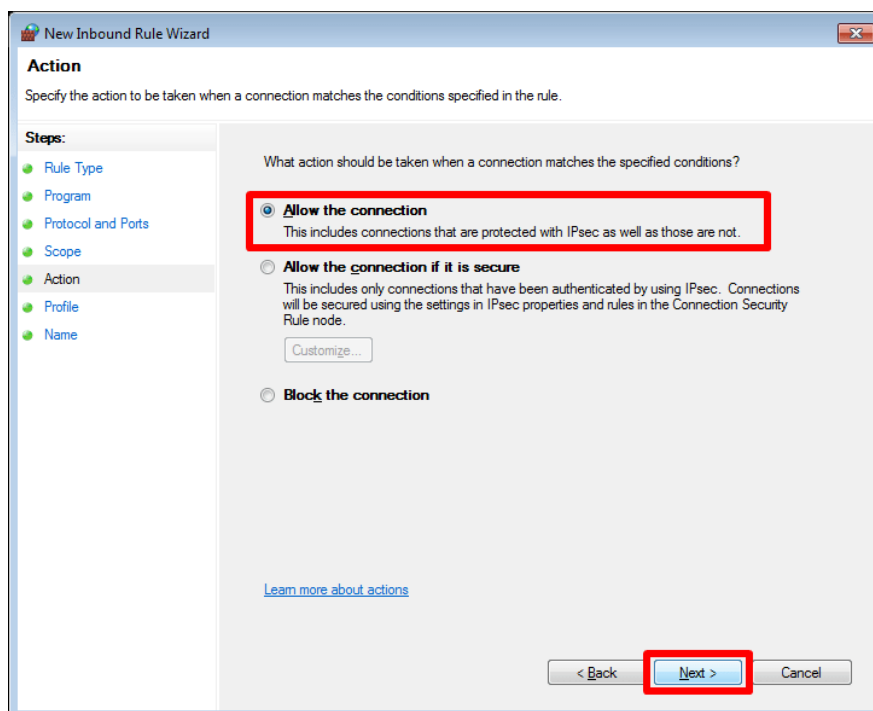
☐ These IP addresses:

[Learn more about specifying scope](#)

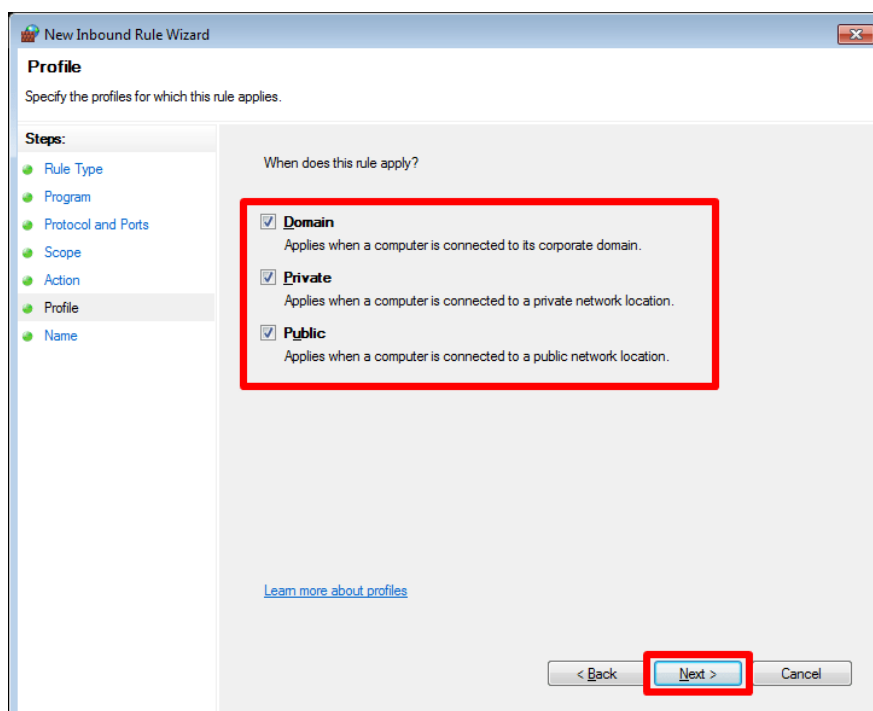
< Back **Next >** Cancel

|      |               |        |                                            |       |  |        |       |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        | Draw  | A-40829-00001EN                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                                                       |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue | FANUC CORPORATION                                     |  | Sheet. | 840/868 |

7. On the [Action] page of the [New Inbound Rule Wizard] dialog box, select [Allow the connection] and click the [Next] button.

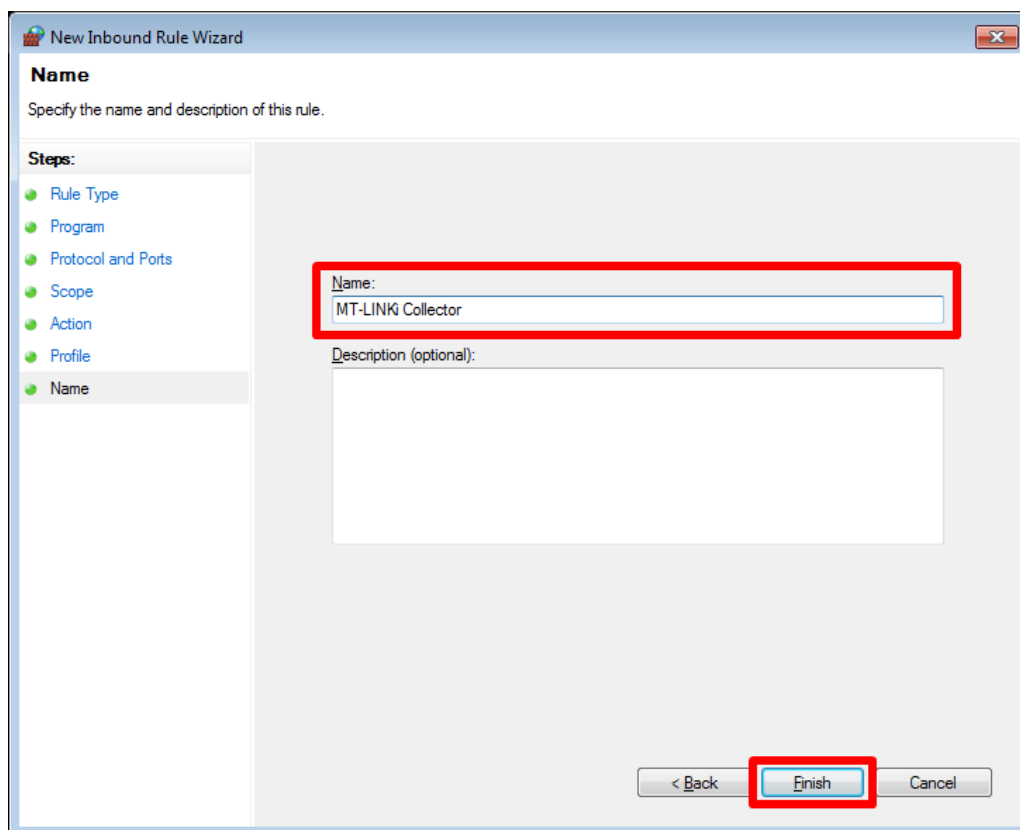


8. On the [Profile] page of the [New Inbound Rule Wizard] dialog box, confirm that [Domain], [Private], and [Public] are all selected. Click the [Next] button.

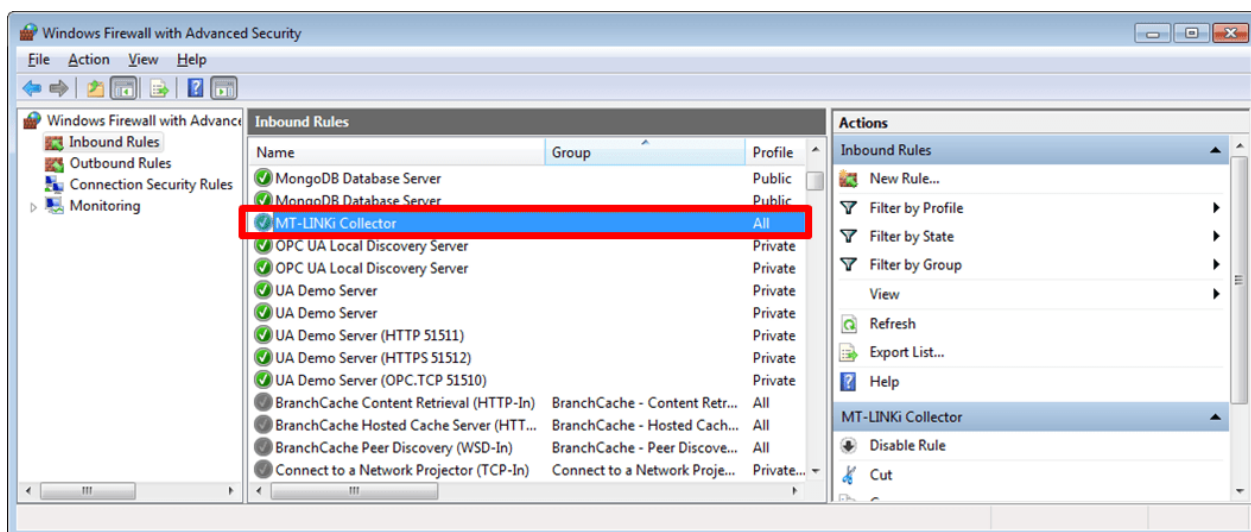


|      |               |        |                                    |                   |                                               |       |                |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw              | A-40829-00001EN                               |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                   |                                               |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | FANUC CORPORATION |                                               |       | Sheet. 841/868 |
| Ed.  | Date          | Design | Description                        |                   |                                               |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             | Apprv.                                        | Inoue |                |

9. On the [Name] page of the [New Inbound Rule Wizard] dialog box, enter [MT-LINKi Collector] in the column, if necessary enter comments in the [Description (optional)] column, and click the [Finish] button.



10. Confirm that [MT-LINKi Collector] is added in the list of the [Inbound Rules].



|      |               |        |                                    |                          |                                               |       |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |       | Sheet. 842/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue |                |

When adding rules for other ports, execute similar steps with the settings described as follows.

■ In the case of the port used by the Collector (SERVO VIEWER linkage)

Execute settings on the Collector PC.

- Step 4.

[This program path]: <MT-LINK*i* Collector installation folder>  
 \MT-LINKiCollectorSERVOVIEWERLinkage.exe

- Step 5.

[Protocol type]: Select [TCP]  
 [Local Port]: Select [Specific Ports]  
 Enter port number used by the Collector (default: 43322)

- Step 9.

[Name]: Enter "MT-LINKi CollectorSERVOVIEWERLinkage"

■ In the case of the port used by the Receiver

Execute settings on the Server PC.

- Step 4.

[This program path]: <MT-LINK*i* installation folder>  
 \MT-LINKiReceiver\MT-LINKiReceiver.exe

- Step 5.

[Protocol type]: Select [TCP]  
 [Local Port]: Select [Specific Ports]  
 Enter port number used by the Receiver (default: 43320)

- Step 9.

[Name]: Enter "MT-LINKi Receiver"

■ In the case of the port used by the Web Server (MT-LINK*i*)

Execute settings on the Server PC.

- Step 4.

[This program path]: <Node.js installation folder>\node.exe

- Step 5.

[Protocol type]: Select [TCP]  
 [Local Port]: Select [Specific Ports]  
 Enter port number used by the Web Server (MT-LINK*i*) (default: 3000)  
 \* Select [All Ports] when the Server PC and the Collector PC are not identical.

- Step 9.

[Name]: Enter "MT-LINKi WebServer"

■ In the case of the port used by Mongo DB (MT-LINK*i*) (\*NOTE)

Execute settings on the Server PC.

- Step 4.

[This program path]: <MT-LINK*i* installation folder>\MongoDB\bin\mongod.exe

- Step 5.

[Protocol type]: Select [TCP] and [UDP]  
 [Local Port]: Select [Specific Ports]  
 Enter port number used by Mongo DB (MT-LINK*i*) (default: 27017)

- Step 9.

[Name]: Enter "MT-LINKi MongoDB"

|      |               |        |                                            |                          |                                                       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       | Sheet. 843/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue          |

■ In the case of the port used by File Transfer Task

Execute settings on the Server PC.

- Step 4.

[This program path]: <MT-LINK*i* FTP Client installation folder>\FileTransferTask.exe

- Step 5.

[Protocol type]: Select [TCP]

[Local Port]: Select [All Ports]

- Step 9.

[Name]: Enter "MT-LINK*i* FileTransferTask"

■ In the case of the port used by Maintenance Data Aggregation

Execute settings on the Server PC.

- Step 4.

[This program path]: <MT-LINK*i* FTP Client installation folder>\MaintenanceDataAggregation.exe

- Step 5.

[Protocol type]: Select [TCP]

[Local Port]: Select [All Ports]

- Step 9.

[Name]: Enter "MT-LINK*i* Maintenance Data Aggregation"

■ In the case of the port used by the FTP Server

Execute settings on the Collector PC.

- Step 4.

[This program path]: <MT-LINK*i* FTP Server installation folder>\FTPServer.exe

- Step 5.

[Protocol type]: Select [TCP]

[Local Port]: Select [Specific Ports]

Enter port number used by the FTP Server (default: 21)

- Step 9.

[Name]: Enter "MT-LINK*i* FTP Server"

■ In the case of the port used by FTP client (for Data Server communication)

Execute settings on the Collector PC.

- Step 4.

[This program path]: <MT-LINK*i* FTP Server installation folder>\FTPServer.exe

- Step 5.

[Protocol type]: Select [TCP]

[Local Port]: Select [All Ports]

- Step 9.

[Name]: Enter "MT-LINK*i* FTPServer(Client)"

■ In the case of the port used by Collector Service Manager

Execute settings on the Collector PC.

- Step 4.

[This program path]: <Collector Service Manager installation folder >  
 \CollectorServiceMgr\MT-LINK*i*CollectorServiceMgr.exe

- Step 5.

[Protocol type]: Select [TCP]

[Local Port]: Select [Specific Ports]

Enter port number used by Collector Service Manager (default: 43323)

- Step 9.

[Name]: Enter "MT-LINK*i* CollectorServiceMgr"

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |  |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|--|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |  |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |  |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |  |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |  |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |  |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |  |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 844/868 |  |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |  |
| Date | Nov.30.2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue             |                                                       |  |        |         |  |



■ In the case of the port used by the Importer (Integration Server)

Execute settings on the Server PC.

- Step 4.

[This program path]: Enter "SYSTEM"

- Step 5.

[Protocol type]: Select [TCP]

[Local Port]: Select [Specific Ports]

Enter port number used by the Importer (Integration Server) (default: 50005)

- Step 9.

[Name]: Enter "MT-LINKi Integration Server Importer"

■ In the case of the port used by the Web Server (Integration Server)

Execute settings on the Server PC.

- Step 4.

[This program path]: <Node.js installation folder>\node.exe

- Step 5.

[Protocol type]: Select [TCP]

[Local Port]: Select [Specific Ports]

Enter port number used by the Web Server (Integration Server)  
(default: 3001)

- Step 9.

[Name]: Enter "MT-LINKi Integration Server WebServer"

■ In the case of the port used by Mongo DB (Integration Server) (\*NOTE)

Execute settings on the Server PC.

- Step 4.

[This program path]: <Integration Server installation folder>\MongoDB\bin\mongod.exe

- Step 5.

[Protocol type]: Select [TCP] and [UDP]

[Local Port]: Select [Specific Ports]

Enter port number used by Mongo DB (Integration Server)  
(default: 27020)

- Step 9.

[Name]: Enter "MT-LINKi Integration Server MongoDB"

**NOTE**

In the case of Mongo DB (MT-LINKi / Integration Server)

- 1 Add 2 rules by executing settings each for the selection of "TCP" and "UDP" on the [Protocol type].
- 2 Upon installation of MT-LINKi earlier than Version 3.7 / Integration Server earlier than Version 1.8, a Windows Security Alert saying "Windows Firewall has blocked some features of this program" appears. If "Allow access" has been selected, rule is set with the name of "MongoDB Database Server." In such a case, addition of rule for Mongo DB is not required.

|      |               |        |                                    |                          |                                               |        |       |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|--------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |        |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |        |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |        |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |        |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |        |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |        |       | Sheet. 845/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |        |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    |                                               | Apprv. | Inoue |                |

# 15.3 SIGNAL CHECK TOOL

This section explains the Signal Check Tool.

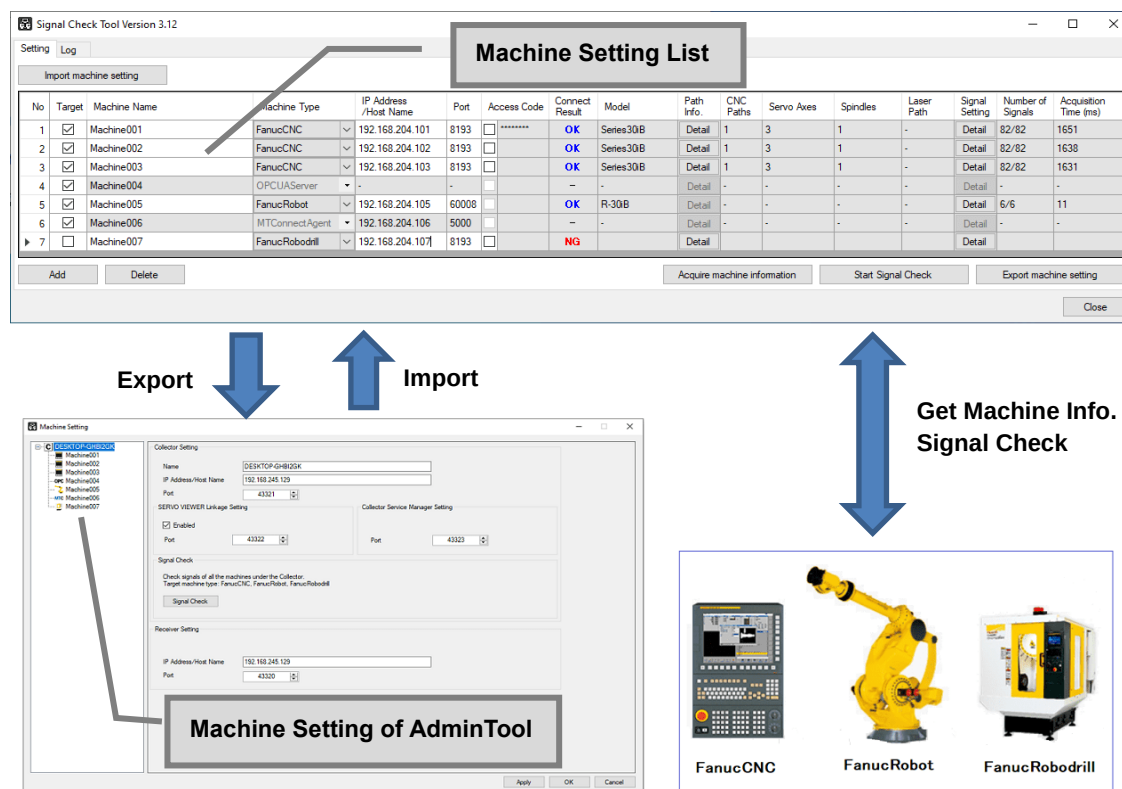
## 15.3.1 Overview of Signal Check Tool

Signal Check Tool is a tool that can check the connection with the machine or the signals independently, and is included in the installer of the MT-LINK*i* server components.

- Signal Check Tool outputs the error codes and processing time in addition to the features for checking the machine setting of AdminTool.
- Checks alarms and information acquired when connected to the machine, which are not checked by the features for checking the machine setting of AdminTool.
- Machine setting files can be imported from/exported to the machine setting window of AdminTool.
- Up to 100 machines can be registered on the Signal Check Tool.

Signal Check Tool can be used in the following cases.

- Checks the connection status and the signals of the machine you want to set in advance, to make use in the support for the machine setting.
- By importing and checking the existing machine setting, make use in analyzing the mistakes in the setting.



### NOTE

This tool is available for checking FanucCNC, FanucRobot, and FanucRobodrill type machines. However, FanucRobodrill type machines can be checked even if ROBODRILL Option is not installed.

|      |               |        |                                            |                   |                                                       |         |
|------|---------------|--------|--------------------------------------------|-------------------|-------------------------------------------------------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                   |                                                       |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                   |                                                       |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw              | <b>A-40829-00001EN</b>                                |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                   |                                                       |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                   |                                                       |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | FANUC CORPORATION | Sheet.                                                | 846/868 |
| Ed.  | Date          | Design | Description                                |                   |                                                       |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check             | Apprv.                                                | Inoue   |

## 15.3.2 Functionalities of Signal Check Tool

Shown below is a list of the functionalities of Signal Check Tool, and the targeted machine types.

| Functionality                        | Overview of Functionality                                                                                                                                                                                                                                                | Machine Type Supported (NOTE 1)          |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Acquire machine information (NOTE 1) | Connects to the machines based on the connection information input, and acquires information such as machine type / CNC path number / servo axis number / spindle axis number.                                                                                           | FanucCNC<br>FanucRobot<br>FanucRobodrill |
| Signal Check (NOTE 1)                | Connects to the machines based on the connection information input, and checks if the signals can be acquired or not.<br>- Signals enabled in the details of signal setting are targeted for checking.<br>- Check results are output to the log file for each execution. | FanucCNC<br>FanucRobot<br>FanucRobodrill |
| Import machine setting               | Displays in the machine setting list by selecting and importing the machine setting file.<br>- File with the same format as L0_Setting.json can be imported. (NOTE 2)<br>- Files of the older versions can be imported also.                                             | All machine types (NOTE 3)               |
| Export machine setting               | Exports the machine setting displayed on the machine setting list as the machine setting file.<br>- Exports in the same format as L0_Setting.json.                                                                                                                       | All machine types (NOTE 4)               |
| Details of Path Information          | Displays details of the path information of each machine.                                                                                                                                                                                                                | FanucCNC<br>FanucRobodrill               |
| Details of Signal Setting            | Displays details of signal setting of each machine. Can be changed.                                                                                                                                                                                                      | FanucCNC<br>FanucRobot<br>FanucRobodrill |

### NOTE

- 1 The machine targeted for machine information acquisition or Signal Check must have the power turned ON and must be in a connectable status.
- 2 The machine setting file (L0\_Setting.json) created in the [Create Collector Setting Files] dialog box is not supported.
- 3 When the machine setting file which has machine types OPCUAServer / MTConnectAgent registered is imported with the machine setting import process, these machines are also added to the machine setting list. However, machine setting acquisition, Signal Check, or signal setting cannot be executed for these machines.
- 4 In the machine setting export process, the machine settings of all machine types displayed on the machine setting list can be exported, including the imported machine settings of the machines with machine types OPCUAServer / MTConnectAgent.

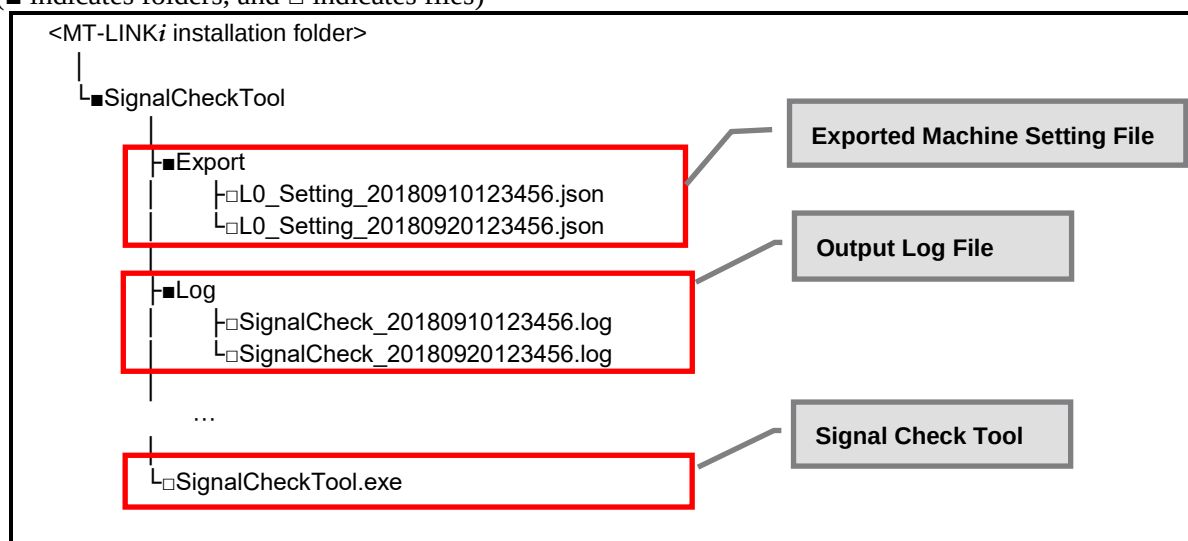
|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | <b>A-40829-00001EN</b>                        |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 847/868 |

### 15.3.3 Installation Folder of Signal Check Tool

Shown below is a structure of the installation folder and the files of Signal Check Tool.

Structure of folders and files for Signal Check Tool (example)

(■ indicates folders, and □ indicates files)



### 15.3.4 Steps for Operating Signal Check Tool

Following are the steps to operate the Signal Check Tool.

- (1) Start the Signal Check Tool (SignalCheckTool.exe), and display the [Signal Check Tool] window.  
For the details of the items on the [Signal Check Tool] window, refer to the "[Signal Check Tool Window](#)" subsection.
- (2) If there is a machine setting already registered by the machine setting of AdminTool, by clicking the [Import machine setting] button and reading the machine setting file, the existing machine setting can be added to the machine setting list. For the details, refer to the "[Machine Setting Import Process](#)" subsection.
- (3) Click the [Add] button to add a row in the machine setting list. Make entries in the [Machine Name], [Machine Type], [IP Address / Host Name], [Port], and [Access code] columns of the added row. When adding two or more machine settings, repeat these steps for each machine setting added.
- (4) Establish connection with the machine, and acquire machine information.  
Select the [Target] check box of the machine to acquire machine information, and click the [Acquire machine information] button. If there is a machine with "NG" displayed in the [Connect Result] column, check the detail of the error by displaying the [Log] tab of the [Signal Check Tool] window. For the details, refer to the "[Acquisition of Machine Information](#)" subsection.

|      |               |        |                                    |                   |                                               |        |         |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |                   |                                               |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw              | A-40829-00001EN                               |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |                   |                                               |        |         |
| Ed.  | Date          | Design | Description                        | FANUC CORPORATION |                                               | Sheet. | 848/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                              |                   |                                               |        |         |
|      |               | Check  |                                    | Apprv.            | Inoue                                         |        |         |

- (5) Checking the machine information acquired.  
 When checking the path information, click the [Detail] button of the [Path Info.] column to display the [Details of Path Information] window. For the details, refer to the "**Details of Path Information Window**" subsection.  
 When checking the signal setting, click the [Detail] button of the [Sign] column to display the [Details of Signal Setting] window. On the [Details of Signal Setting] window, the contents of the setting can be customized by the tab for each signal setting. For the details, refer to the "**Details of Signal Setting Window**" subsection.
- (6) Execute Signal Check  
 Select the [Target] check box of the machine to execute Signal Check, and click the [Start Signal Check] button. Confirm the result of the process on the [Log] tab which will be displayed.  
 For the details, refer to the "**Check Signal**" subsection.
- (7) Return to the [Setting] tab, and check the result of the Signal Check.  
 Click the [Detail] button of the [Signal Setting] column to display the [Details of Signal Setting] window.  
 Check if the signals are acquired or not in the [Check] column of each signal setting tab.  
 For the details, refer to the "**Details of Signal Setting Window**" subsection.
- (8) Export machine setting to the machine setting file.  
 Select the [Target] check box of the machine which will have the machine setting exported, and click the [Export machine setting] button to save the machine setting file. For the details, refer to the "**Machine Setting Export**" subsection.  
 The exported machine setting file can be imported on the [Machine Setting] window. For the details, refer to the "**Import Machine Setting**" subsection.
- (9) Click the [Close] button to close the [Signal Check Tool] window.

|      |               |        |                                    |                          |                                               |       |                |
|------|---------------|--------|------------------------------------|--------------------------|-----------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title                    | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                          |                                               |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                          |                                               |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                        |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                          |                                               |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                          |                                               |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | <b>FANUC CORPORATION</b> |                                               |       | Sheet. 849/868 |
| Ed.  | Date          | Design | Description                        |                          |                                               |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check                    | Apprv.                                        | Inoue |                |

## 15.3.5 Signal Check Tool Window

This subsection explains the items in the [Signal Check Tool] window.

### ■ [Signal Check Tool] window ([Setting] tab)

| No | Target                              | Machine Name | Machine Type   | IP Address / Host Name | Port  | Access Code | Connect Result | Model     | Path Info | CNC Paths | Servo Axes | Spindles | Laser Path | Signal Settings | Number of Signals | Acquisition Time (ms) |
|----|-------------------------------------|--------------|----------------|------------------------|-------|-------------|----------------|-----------|-----------|-----------|------------|----------|------------|-----------------|-------------------|-----------------------|
| 1  | <input checked="" type="checkbox"/> | Machine001   | FanucCNC       | 192.168.204.101        | 8193  | *****       | OK             | Series30B | Detail    | 1         | 3          | 1        | -          | Detail          | 82/82             | 1651                  |
| 2  | <input checked="" type="checkbox"/> | Machine002   | FanucCNC       | 192.168.204.102        | 8193  |             | OK             | Series30B | Detail    | 1         | 3          | 1        | -          | Detail          | 82/82             | 1638                  |
| 3  | <input checked="" type="checkbox"/> | Machine003   | FanucCNC       | 192.168.204.103        | 8193  |             | OK             | Series30B | Detail    | 1         | 3          | 1        | -          | Detail          | 82/82             | 1631                  |
| 4  | <input checked="" type="checkbox"/> | Machine004   | OPCUAServer    | -                      | -     |             | -              | -         | Detail    | -         | -          | -        | -          | Detail          | -                 | -                     |
| 5  | <input checked="" type="checkbox"/> | Machine005   | FanucRobot     | 192.168.204.105        | 60008 |             | OK             | R-30B     | Detail    | -         | -          | -        | -          | Detail          | 6/6               | 11                    |
| 6  | <input checked="" type="checkbox"/> | Machine006   | MTConnectAgent | 192.168.204.106        | 5000  |             | -              | -         | Detail    | -         | -          | -        | -          | Detail          | -                 | -                     |
| 7  | <input type="checkbox"/>            | Machine007   | FanucRobodrill | 192.168.204.107        | 8193  |             | NG             | -         | Detail    | -         | -          | -        | -          | Detail          | -                 | -                     |

### Machine setting list

On the machine setting list, the description of the settings on the imported machine setting, or the information about the added machine setting is displayed on a list.

- Target

A check box used for selecting the target machine for acquiring machine information, for checking signal, or for exporting the machine setting.

When the check box is selected, the machine will be the target machine to Acquire machine information / Start Signal Check / Export machine setting.

\* By selecting multiple rows, right-clicking, and selecting [ON] or [OFF] on the menu displayed, the [Target] check boxes on these rows can be selected or cleared at once.

- Machine Name (entry required)

Enter the machine name. A half-width alphabet can be used for the first character of the machine name. Half-width alphanumerics, hyphens ("-"), and underscores ("\_") can be used for the characters that follow. Existing machine name cannot be input. This field can be up to 32 bytes long.

In the case of machine types OPCUAServer / MTConnectAgent, the machine name already set will be displayed, and it cannot be changed.

- Machine Type (entry required)

Select machine type on the drop-down list. FanucCNC / FanucRobodrill can be selected.

In the case of machine types OPCUAServer / MTConnectAgent, the machine type already set will be displayed, and it cannot be changed.

- IP Address / Host Name (entry required)

Enter the IP address or the host name of the machine. In the case of a host name, half-width alphabet can be used for the first character. Half-width alphanumerics, hyphens ("-"), and dots (".") can be used for the characters that follow. This field can be up to 253 bytes long.

In the case of machine type OPCUAServer, "-" will be displayed, and it cannot be changed. In the case of machine type MTConnectAgent, the IP address or the host name already set will be displayed, and it cannot be changed.

|      |               |        |                                            |                   |                                                       |                |
|------|---------------|--------|--------------------------------------------|-------------------|-------------------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                   |                                                       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                   |                                                       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw              | A-40829-00001EN                                       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                   |                                                       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                   |                                                       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | FANUC CORPORATION |                                                       | Sheet. 850/868 |
| Ed.  | Date          | Design | Description                                |                   |                                                       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check             | Apprv.                                                | Inoue          |

- **Port (entry required)**  
Enter the port number used for the communication with the machine. A value between "0" and "65535" can be used for the port number.  
If the machine type is FanucCNC / FanucRobodrill, "8193" is set as the default value, and if it is FanucRobot, "60008" is set as the default value.  
If the machine type is OPCUAServer, "-" will be displayed, and it cannot be changed. In the case of machine type MTConnectAgent, the port already set will be displayed, and it cannot be changed.
- **Access Code**  
For FanucCNCs / FanucRobodrills on which an access code is set, select the check box and set an access code for this column. Enter less than nine numeric characters. The entered access code will be masked with asterisks ("\*").  
For machines of other machine types, this column populates the symbol "-" and does not allow to modify.

#### NOTE

To set an access code on a FanucCNC type machine, CNC system software must be installed that supports a security function to use an access code.

- **Connect Result (cannot be changed)**  
Result of connection with the machine when acquiring machine information, is displayed.  
"OK": State when the connection with the machine is confirmed  
"NG" (for "not good"): State when the connection with the machine cannot be established  
"-": When the machine is loaded by machine setting import
- **Model (cannot be changed)**  
If the machine type is FanucCNC / FanucRobodrill, the model will be displayed and if it is FanucRobot, its robot controller model will be displayed. For the other machine types, "-" will be displayed.  
\* For FanucCNC and FanucRobodrill type machines which are imported from the machine setting files, "-" may be displayed.
- **Path Info. (cannot be changed)**  
If the machine type is FanucCNC / FanucRobodrill, the [Detail] button will be displayed. By clicking the [Detail] button, the [Details of Path Information] window is displayed. For the other machine types, the column is disabled.  
For the details on the [Details of Path Information] window, refer to the "[Details of Path Information Window](#)" subsection.
- **CNC Paths (cannot be changed)**  
The number of CNC path for the machine type FanucCNC / FanucRobodrill is displayed. For the other machine types, "-" is displayed.
- **ServoAxes (cannot be changed)**  
The number of servo axis for the machine type FanucCNC / FanucRobodrill is displayed. Displayed in the order of the path, delimited by "-." (Example: 2-3-3)  
For the other machine types, "-" will be displayed.

|      |               |        |                                            |       |  |        |                   |                                                       |  |        |         |  |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------------------|-------------------------------------------------------|--|--------|---------|--|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |  |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |                   |                                                       |  |        |         |  |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |                   |                                                       |  |        |         |  |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        | Draw              | A-40829-00001EN                                       |  |        |         |  |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |                   |                                                       |  |        |         |  |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |                   |                                                       |  |        |         |  |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        | FANUC CORPORATION |                                                       |  | Sheet. | 851/868 |  |
| Ed.  | Date          | Design | Description                                |       |  |        |                   |                                                       |  |        |         |  |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue             |                                                       |  |        |         |  |

- Spindles (cannot be changed)  
The number of spindle axis for the machine type FanucCNC / FanucRobodrill is displayed. Displayed in the order of the path, delimited by "-." (Example: 1-1-2)  
For the other machine types, "-" will be displayed.
- Laser Path (cannot be changed)  
The paths of a FanucCNC type machine that has a laser oscillator connected are displayed. When multiple paths exist, they are displayed delimited by "/" (Example: 1/3)  
If the laser oscillator is not connected, or if the machine type is other than FanucCNC, "-" will be displayed.
- Signal Setting  
If the machine type is FanucCNC / FanucRobot / FanucRobodrill, the [Detail] button will be displayed. By clicking the [Detail] button, the [Details of Signal Setting] window is displayed. For the other machine types, the column is disabled.  
For the details on the [Details of Signal Setting] window, refer to the "[Details of Signal Setting Window](#)" subsection.
- Number of Signals (cannot be changed)  
If the machine type is FanucCNC / FanucRobot / FanucRobodrill, "<Number of acquirable signals> / <Number of signals set as enabled>" will be displayed.  
For the "<Number of acquirable signals>" before checking signals, "-" will be displayed. (Example: "-/100")  
For the other machine types, "-" will be displayed.
- Acquisition Time (cannot be changed)  
The total time (ms) required to acquire the signal during Signal Check, is displayed.  
Before checking signals, "-" will be displayed.  
For the other machine types, "-" will be displayed.

## Buttons

- [Import machine setting] button  
Adds the already registered (an existing) machine setting to the machine setting list, by loading the machine setting file.  
Following are the default input folder and the filename.  
Folder name : <MT-LINK*i* installation folder>\MT-LINKiAdminTool\SettingData\  
Filename: L0\_Setting.json
- [Add] button  
Adds a row in the machine setting list. Following items are set with default values when the row is added.  
[Target]: selected  
[Machine Name]: "NewMachine<Sequential number>"  
<Sequential number> starts from 001 (example: NewMachine001)  
[Machine Type]: "FanucCNC"  
[Port]: "8193"
- [Delete] button  
Deletes the row selected on the machine setting list. Multiple rows can be selected and deleted at once.  
When you click [OK] on the confirmation message, the selected row will be deleted.

|      |               |        |                                            |       |  |        |       |                                                       |  |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------------------------------------------|--|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        | Title | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                                                       |  |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                                                       |  |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       |                                                       |  |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        | Draw  | A-40829-00001EN                                       |  |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                                                       |  |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       |                                                       |  |  |        |         |
| Ed.  | Date          | Design | Description                                |       |  |        |       | FANUC CORPORATION                                     |  |  | Sheet. | 852/868 |
| Date | Nov.30.2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                                                       |  |  |        |         |



- [Acquire machine information] button  
Establishes connection for the machine with the [Target] check box selected based on the entered information (IP address / host name, port, access code), and displays "OK"/"NG (meaning 'Not Good')" in the [Connect Result] column.  
When "OK" is displayed in the [Connect Result] column, displays the acquired machine information in the machine setting list ([Machine Type] (see NOTE) / [Model] / [CNC Paths] / [Servo Axes] / [Spindles] / [Laser Path] / [Number of Signals]).
- [Start Signal Check] button  
Executes Signal Check for the machines with the [Target] check box selected.
- [Export machine setting] button  
Exports machine setting to the machine setting file for the machines with the [Target] check box selected.  
Following are the default export folder and filename.  
Folder name: <MT-LINK*i* installation folder>\SignalCheckTool\Export\  
Filename: L0\_Setting\_<yyyyMMddHHmmss>.json  
(<yyyyMMddHHmmss> is the date/time of export.)
- [Close] button  
Closes the [Signal Check Tool] window.  
All the entries on the machine setting list will be aborted.

#### ■ [Signal Check Tool] window ([Log] tab)



#### Log display area

When the [Acquire machine information] button in the [Setting] tab is clicked, the log of the result of machine information acquisition is displayed.

When the [Start Signal Check] button in the [Setting] tab is clicked, the log of the Signal Check process is displayed.

The log contents displayed in the Signal Check process is output in a text format file for each Signal Check process executed.

Following are the output folder and the filename.

Folder name: <MT-LINK*i* installation folder>\SignalCheckTool\Log\  
Filename: SignalCheck\_<yyyyMMddHHmmss>.log  
(<yyyyMMddHHmmss> is the date/time when the file was output.)

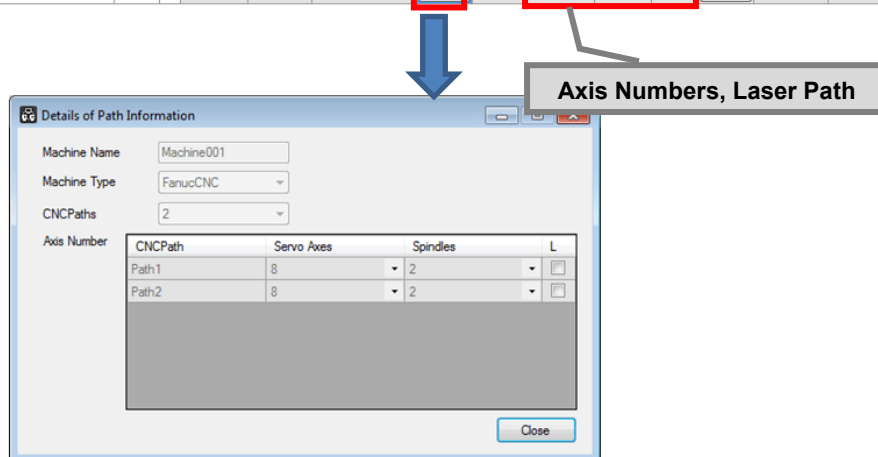
|      |               |        |                                            |                          |                                                       |       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |                          |                                                       |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   | Draw                     | <b>A-40829-00001EN</b>                                |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |                          |                                                       |       |                |
| Ed.  | Date          | Design | Description                                | <b>FANUC CORPORATION</b> |                                                       |       | Sheet. 853/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue |                |

### 15.3.5.1 Details of Path Information Window

For FanucCNC and FanucRobodrill type machines which have the machine information acquired, the number of axes and laser paths are displayed on the machine setting list. However, when the number of CNC paths increases, it becomes difficult to display all the information, and some of the information is omitted in the display. In such a case, by clicking on the [Detail] button in the [Path info.] column, the [Details of Path Information] window is displayed, enabling to confirm all of the information.

#### ■ [Details of Path Information] window

| No  | Target                              | Machine Name | Machine Type | IP Address / Host Name | Port | Access Code | Connect Result | Model      | Path Info. | CNC Paths | Servo Axes | Spindles | Laser Path | Signal Setting | Number of Signals | Acquisition Time (ms) |
|-----|-------------------------------------|--------------|--------------|------------------------|------|-------------|----------------|------------|------------|-----------|------------|----------|------------|----------------|-------------------|-----------------------|
| ▶ 1 | <input checked="" type="checkbox"/> | Machine001   | FanucCNC     | 192.168.200.1          | 8193 |             | OK             | Series30iB | Detail     | 2         | 8-8        | 2-2      | -          | Detail         | -/358             | -                     |



- Machine Name (cannot be changed)  
The machine name of the machine selected on the machine setting list is displayed.
- CNCPaths (cannot be changed)  
The number of CNC path is displayed.
- CNCPath / Servo Axes / Spindles / L (cannot be changed)  
Information about CNC path, numbers of servo axes and spindles, L for each CNC path are displayed. For L, check box will be selected in the case of the path with laser oscillator connected.
- [Close] button  
Closes the [Details of Path Information] window.

|      |               |        |                                            |                          |                                                       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |                          |                                                       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   | Draw                     | <b>A-40829-00001EN</b>                                |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |                          |                                                       |                |
| Ed.  | Date          | Design | Description                                | <b>FANUC CORPORATION</b> |                                                       | Sheet. 854/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue          |

## 15.3.5.2 Details of Signal Setting Window

In the [Details of Signal Setting] window, the signal setting can be customized by checking the signal setting created during machine information acquisition.

Also, whether the machine information is acquired or not can be confirmed for each signal after the Signal Check process is executed.

The [Details of Signal Setting] window is displayed by clicking on the [Detail] button in the [Signal Setting] column.

### ■ [Details of Signal Setting] window (In the case of machine types FanucCNC and FanucRobodrill)

\* The following figure shows a setting example of a FanucCNC type machine.

| No | Target                              | Machine Name | Machine Type | IP Address /Host Name | Port | Access Code | Connect Result | Model     | Path Info. | CNC Paths | Servo Axes | Spindles | Laser Path | Signal Setting | Number of Signals | Acquisition Time (ms) |
|----|-------------------------------------|--------------|--------------|-----------------------|------|-------------|----------------|-----------|------------|-----------|------------|----------|------------|----------------|-------------------|-----------------------|
| 1  | <input checked="" type="checkbox"/> | Machine001   | FanucCNC     | 192.168.200.1         | 8193 |             | OK             | Series30B | Detail     | 2         | 8-8        | 2-2      | -          | Detail         | /358              | -                     |

Details of Signal Setting

Name: Machine001 Type: FanucCNC CNCPaths: 1

Signal Setting

Narrow down

Description: Enable: Reading Cycle: Apply Discard

| No | Signal      | Description                         | Axis | CNCPath | Enable                              | Outlier Detection                   | Reading Cycle | Check |
|----|-------------|-------------------------------------|------|---------|-------------------------------------|-------------------------------------|---------------|-------|
| 1  | SigOP       | Automatic operation signal          | 1    |         | <input type="checkbox"/>            | <input type="checkbox"/>            | 500msec       | -     |
| 2  | SigSTL      | Cycle start lamp signal             | 1    |         | <input type="checkbox"/>            | <input type="checkbox"/>            | 500msec       | -     |
| 3  | SigSPL      | Feed hold lamp signal               | 1    |         | <input type="checkbox"/>            | <input type="checkbox"/>            | 500msec       | -     |
| 4  | SigAL       | Alarm signal                        | 1    |         | <input type="checkbox"/>            | <input type="checkbox"/>            | 500msec       | -     |
| 5  | EMG         | Emergency stop signal               | 1    |         | <input type="checkbox"/>            | <input type="checkbox"/>            | 500msec       | -     |
| 6  | Mode        | Mode                                | 1    |         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | 500msec       | OK    |
| 7  | MainProgram | Main program                        | 1    |         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | 500msec       | OK    |
| 8  | ActProgram  | Executing program                   | 1    |         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | 500msec       | OK    |
| 9  | Sequence    | Sequence number                     | 1    |         | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | 500msec       | -     |
| 10 | MainComment | Program comment (main program)      | 1    |         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | 500msec       | OK    |
| 11 | ActComment  | Program comment (executing program) | 1    |         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | 500msec       | OK    |
| 12 | ActS        | Speed S [actual]                    | 1    |         | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | 500msec       | -     |
| 13 | ActF        | Feed rate F [actual]                | 1    |         | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | 500msec       | -     |
| 14 | ActFdec     | Feed rate F [actual]                | 1    |         | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | 500msec       | -     |
| 15 | ModalS      | Speed S [command]                   | 1    |         | <input type="checkbox"/>            | <input type="checkbox"/>            | 500msec       | -     |
| 16 | ModalF      | Feed rate F [command]               | 1    |         | <input type="checkbox"/>            | <input type="checkbox"/>            | 500msec       | -     |
| 17 | ModalT      | T code                              | 1    |         | <input type="checkbox"/>            | <input type="checkbox"/>            | 500msec       | -     |
| 18 | ModalM      | 1st M code                          | 1    |         | <input type="checkbox"/>            | <input type="checkbox"/>            | 500msec       | -     |

☒ All enabled Default setting

OK Cancel

- Name (cannot be changed)  
The machine name of the machine selected on the machine setting list is displayed.
- Type (cannot be changed)  
The machine type of the machine selected on the machine setting list is displayed.
- CNCPaths (cannot be changed)  
CNC path number of the machine selected on the machine setting list is displayed.
- [Basic Signal] tab  
Signal setting of the basic signals for the machine selected on the machine setting list is displayed.  
For the description of display and setting method for each item, refer to the "[Set Signal of FanucCNC](#)" subsection.

|      |               |        |                                    |                   |                                               |                |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw              | A-40829-00001EN                               |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                   |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | FANUC CORPORATION |                                               | Sheet. 855/868 |
| Ed.  | Date          | Design | Description                        |                   |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             | Apprv.                                        | Inoue          |

- [Custom Signal] tab  
Signal setting of the custom signals for the machine selected on the machine setting list is displayed.  
For the description of display and setting method for each item, refer to the "[Set Custom Signal of FanucCNC](#)" subsection.
- [Tool Life Signal] tab (\* available only to FanucCNC type machines)  
Signal setting of the tool life signals for the machine selected on the machine setting list is displayed.  
For the description of display and setting method for each item, refer to the "[Set Tool Life Signal of FanucCNC](#)" subsection.
- [Laser control data signal setting] tab (\* available only to FanucCNC type machines)  
Signal setting of the laser data control signals for the machine selected on the machine setting list is displayed.  
[Laser control data signal setting] tab is displayed only when the laser oscillator is connected.  
For the description of display and setting method for each item, refer to the "[Set Laser Control Data Signal of FanucCNC](#)" subsection.
- [Robodrill Signal Setting] tab (\* available only to FanucRobodrill type machines)  
Signal setting of the ROBODRILL signals for the machine selected on the machine setting list is displayed.  
For the description of display and setting method for each item, refer to the "[Set FanucRobodrill Specific Signal](#)" subsection.
- [Macro Variable Getting Condition Definition] tab  
Macro variable getting condition definition for the machine selected on the machine setting list is displayed.  
For the description of display and setting method for each item, refer to the "[Register Macro Variable Getting Condition Definition of FanucCNC](#)" subsection.
- [Variable History Retrieval Condition Definition] tab  
Variable history retrieval condition definition for the machine selected on the machine setting list is displayed.  
For the description of display and setting method for each item, refer to the "[Setting Variable History Retrieval Condition Definition of FanucCNC](#)" subsection.
- All enabled  
A check box for switching all the signals (including mandatory signals) to enable/disable.  
When the [Enable] check box is selected, the [Enable] check boxes of all the signals displayed on the tab are selected. If the check box is cleared, the [Enable] check boxes of all the signals displayed on the tab are cleared.
- [Default setting] button  
A button that will set the default values.  
When the [Default setting] button is clicked, the default values will be set in the [Enable] / [Outlier detection] / [Reading cycle] check boxes on the tab displayed.
- [OK] button  
Confirms the contents of the settings, and closes the [Details of Signal Setting] window.  
If the check result of the signal is displayed in the [Check] column, it will all be "" (blank).  
When the settings are changed, the display in the [Number of Signals] and [Acquisition Time (ms)] of the machine setting list will be updated.
- [Cancel] button  
Aborts the contents of the setting, and closes the [Details of Signal Setting] window.

|      |               |        |                                    |       |  |        |                   |                                               |  |        |         |
|------|---------------|--------|------------------------------------|-------|--|--------|-------------------|-----------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |                   |                                               |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |                   |                                               |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        | Draw              | A-40829-00001EN                               |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        |                   |                                               |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |                   |                                               |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        | FANUC CORPORATION |                                               |  | Sheet. | 856/868 |
| Ed.  | Date          | Design | Description                        |       |  |        |                   |                                               |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check |  | Apprv. |                   |                                               |  |        |         |

## ■ [Details of Signal Setting] window (In the case of machine type FanucRobot)

| No | Target                              | Machine Name | Machine Type | IP Address /Host Name | Port  | Access Code | Connect Result | Model | Path Info. | CNC Paths | Servo Axes | Spindles | Laser Path | Signal Setting | Number of Signals | Acquisition Time (ms) |
|----|-------------------------------------|--------------|--------------|-----------------------|-------|-------------|----------------|-------|------------|-----------|------------|----------|------------|----------------|-------------------|-----------------------|
| 5  | <input checked="" type="checkbox"/> | Machine005   | FanucRobot   | 192.168.204.105       | 60008 |             | OK             | R-30B | Detail     | -         | -          | -        | -          | Detail         | 6                 | 11                    |



**Details of Signal Setting**

Name: Machine005    Type: FanucRobot    Robot Controller: R-30B

Basic Signal   Custom Signal   Variable History Retrieval Condition Definition   Position Register

Signal Setting

Narrow down

Description:     Enable:     Reading Cycle:        

| No | Signal        | Description                   | Enable                              | Reading Cycle | Check |
|----|---------------|-------------------------------|-------------------------------------|---------------|-------|
| 1  | REMOTE        | Remote signal                 | <input checked="" type="checkbox"/> | 500msec       | -     |
| 2  | BUSY          | Busy signal                   | <input type="checkbox"/>            | 500msec       | -     |
| 3  | HELD          | Hold signal                   | <input type="checkbox"/>            | 500msec       | -     |
| 4  | FAULT         | Alarm fault signal            | <input type="checkbox"/>            | 500msec       | -     |
| 5  | BATAL         | Abnormal battery alarm signal | <input checked="" type="checkbox"/> | 500msec       | OK    |
| 6  | TPENBL        | Teach pendant enable signal   | <input type="checkbox"/>            | 500msec       | -     |
| 7  | Mode          | Mode                          | <input checked="" type="checkbox"/> | 500msec       | OK    |
| 8  | RobotState    | Robot State                   | <input checked="" type="checkbox"/> | 500msec       | OK    |
| 9  | Alarm         | Alarm message                 | <input checked="" type="checkbox"/> | 500msec       | OK    |
| 10 | ProgramName   | Executing program             | <input checked="" type="checkbox"/> | 500msec       | OK    |
| 11 | ProgramLineNo | Sequence number               | <input type="checkbox"/>            | 500msec       | -     |

☒ All enabled   

- Name (cannot be changed)  
The machine name of the machine selected on the machine setting list is displayed.
- Type (cannot be changed)  
The machine type of the machine selected on the machine setting list is displayed.
- Robot Controller (cannot be changed)  
The robot controller of the machine selected on the machine setting list is displayed.
- [Basic Signal] tab  
Signal setting of the basic signals for the machine selected on the machine setting list is displayed.  
For the description of display and setting method for each item, refer to the "[Set Signal of FanucRobot](#)" subsection.
- [Custom Signal] tab  
Signal setting of the custom signals for the machine selected on the machine setting list is displayed.  
For the description of display and setting method for each item, refer to the "[Set Custom Signal of FanucRobot](#)" subsection.
- [Variable History Retrieval Condition Definition] tab  
Variable history retrieval condition definition for the machine selected on the machine setting list is displayed.  
For the description of display and setting method for each item, refer to the "[Set Variable History Retrieval Condition Definition of FanucRobot](#)" subsection.

|                   |               |        |                                            |       |                                                       |         |
|-------------------|---------------|--------|--------------------------------------------|-------|-------------------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |                                                       |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |                                                       |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |                                                       |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   | Draw  | <b>A-40829-00001EN</b>                                |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |                                                       |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |                                                       |         |
| Ed.               | Date          | Design | Description                                |       |                                                       |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                                      | Check | Apprv.                                                | Inoue   |
| FANUC CORPORATION |               |        |                                            |       | Sheet.                                                | 857/868 |

- [Position Register] tab  
The position register setting for the machine selected on the machine setting list is displayed.  
For the description of display and setting method for each item, refer to the "**Set Position Register of FanucRobot**" subsection.
- All enabled  
A check box for switching all the signals (including mandatory signals) to enable/disable.  
When the [Enable] check box is selected, the [Enable] check boxes of all the signals displayed on the tab are selected. If the check box is cleared, the [Enable] check boxes of all the signals displayed on the tab are cleared.
- [Default setting] button  
A button that will set the default values.  
When the [Default setting] button is clicked, the default values will be set in the [Enable] / [Outlier detection] / [Reading cycle] check boxes on the tab displayed.
- [OK] button  
Confirms the contents of the settings, and closes the [Details of Signal Setting] window.  
If the check result of the signal is displayed in the [Check] column, it will all be "" (blank).  
When the settings are changed, the display in the [Number of Signals] and [Acquisition Time (ms)] of the machine setting list will be updated.
- [Cancel] button  
Aborts the contents of the setting, and closes the [Details of Signal Setting] window.

|      |               |        |                                            |                          |                                                       |        |       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|--------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |        |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |        |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw                     | <b>A-40829-00001EN</b>                                |        |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                          |                                                       |        |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |        |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | <b>FANUC CORPORATION</b> |                                                       |        |       | Sheet. 858/868 |
| Ed.  | Date          | Design | Description                                |                          |                                                       |        |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    |                                                       | Apprv. | Inoue |                |

## 15.3.6 Machine Setting Import Process

This subsection explains the machine setting import process.

| No | Target                              | Machine Name | Machine Type   | IP Address / Host Name | Port  | Access Code | Connect Result | Model | Signal Setting | Number of Signals | Acquisition Time (ms) |   |
|----|-------------------------------------|--------------|----------------|------------------------|-------|-------------|----------------|-------|----------------|-------------------|-----------------------|---|
| 1  | <input checked="" type="checkbox"/> | Machine001   | FanucCNC       | 192.168.204.101        | 8193  | *****       | -              | -     | Detail         | 1                 | 0                     | 0 |
| 2  | <input checked="" type="checkbox"/> | Machine002   | FanucCNC       | 192.168.204.102        | 8193  |             | -              | -     | Detail         | 1                 | 0                     | 0 |
| 3  | <input checked="" type="checkbox"/> | Machine003   | FanucCNC       | 192.168.204.103        | 8193  |             | -              | -     | Detail         | 1                 | 0                     | 0 |
| 4  | <input checked="" type="checkbox"/> | Machine004   | OPCUAServer    | -                      | -     |             | -              | -     | Detail         | -                 | -                     | - |
| 5  | <input checked="" type="checkbox"/> | Machine005   | FanucRobot     | 192.168.204.105        | 60008 |             | -              | R-30B | Detail         | -                 | -                     | - |
| 6  | <input checked="" type="checkbox"/> | Machine006   | MTConnectAgent | 192.168.204.106        | 5000  |             | -              | -     | Detail         | -                 | -                     | - |
| 7  | <input checked="" type="checkbox"/> | Machine007   | FanucRobodril  | 192.168.204.107        | 8193  |             | -              | -     | Detail         | 1                 | 3                     | 1 |

- (1) By clicking the [Import machine setting] button, [Select machine setting file] dialog box is displayed. Select the machine setting file to import, and click the [Open] button.

\* Following are the selected default folder and the filename.

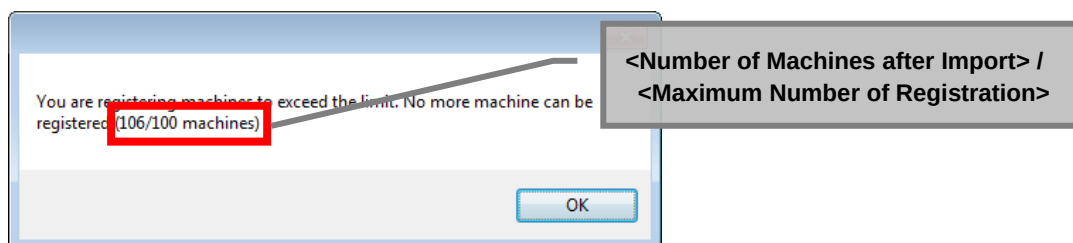
Folder name: <MT-LINK*i* installation folder>\MT-LINK*i*AdminTool\SettingData\

Filename: L0\_Setting.json

### NOTE

- 1 Files with the same format as L0\_Setting.json can be imported as the machine setting file.  
Files of the older versions can be imported also.
- 2 The machine setting file (L0\_Setting.json) created in the [Create Collector Setting Files] dialog box is not supported.

- (2) If the number of machines after the import process is executed exceeds the maximum number of machines that can be registered (100), following message is displayed, and the machine cannot be added.  
Click the [OK] button to return to the [Signal Check Tool] window.



- (3) When the import process is completed, the imported machine setting is added to the machine setting list.  
At this point, the [Target] check box of the imported machine will be selected.

### NOTE

When the machine setting of machine types OPCUAServer / MTConnectAgent is imported, it is added to the machine setting list. However, machine setting acquisition, Signal Check, or signal setting cannot be executed. Also, only the [Target] check box can be set.

|      |               |        |                                            |                   |                                                       |       |                |
|------|---------------|--------|--------------------------------------------|-------------------|-------------------------------------------------------|-------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |       |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                   |                                                       |       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                   |                                                       |       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw              | A-40829-00001EN                                       |       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                   |                                                       |       |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                   |                                                       |       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | FANUC CORPORATION |                                                       |       | Sheet. 859/868 |
| Ed.  | Date          | Design | Description                                |                   |                                                       |       |                |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check             | Apprv.                                                | Inoue |                |



## 15.3.7 Acquisition of Machine Information

This subsection explains the machine information acquisition process.

- (1) Select the [Target] check box of the machine targeted to acquire the machine information, and click the [Acquire machine information] button.

\* The default signal setting is created in the machine information acquisition process.

The detailed setting of the signal setting can be customized on the [Details of Signal Setting] window.

\* Machine types OPCUAServer / MTConnectAgent are not supported. Machine information will not be acquired even when the [Target] check box is selected.

- (2) If FanucRobot type machines are targeted, the following warning message will be appear. Read the content and follow the instruction if needed.

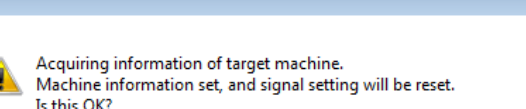
### NOTE

Before conducting Signal Check of the following FANUC Robot Series machines, stop applications that may communicate with those machines. If such an application is running, it may conflict with Signal Check function and experience a communication error:

- R-30iA / R-30iA Mate
- R-J3iB / R-J3iB Mate

|                   |               |        |                                    |       |                                               |         |
|-------------------|---------------|--------|------------------------------------|-------|-----------------------------------------------|---------|
| 20                | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |         |
| 19                | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |         |
| 18                | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |         |
| 17                | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |         |
| 16                | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | A-40829-00001EN                               |         |
| 15                | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |         |
| 14                | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |         |
| Ed.               | Date          | Design | Description                        |       |                                               |         |
| Date              | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue   |
| FANUC CORPORATION |               |        |                                    |       | Sheet.                                        | 860/868 |



- 
- Acquiring information of target machine.  
Machine information set, and signal setting will be reset.  
Is this OK?
- Yes: Reacquire information of all target machines.  
No: Information of machines already acquired will not be reacquired
- Yes No Cancel

By clicking the [No] button, among the machines with the [Target] check box selected, the machines which has already acquired the machine information are excluded from the target.

- Acquiring machine information
- 
- 2/3 units completed

- 
- Machine information cannot be acquired from a machine.  
Review connection or delete unnecessary row.
- Check error details of acquisition failure by referring to Log tab.
- OK

|      |               |        |                                    |       |  |        |                   |                                               |  |        |         |       |
|------|---------------|--------|------------------------------------|-------|--|--------|-------------------|-----------------------------------------------|--|--------|---------|-------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   |       |  |        | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |  |        |         |       |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |  |        |                   |                                               |  |        |         |       |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |  |        |                   |                                               |  |        |         |       |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |  |        | Draw              | A-40829-00001EN                               |  |        |         |       |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |       |  |        |                   |                                               |  |        |         |       |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |  |        |                   |                                               |  |        |         |       |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |  |        | FANUC CORPORATION |                                               |  | Sheet. | 861/868 |       |
| Ed.  | Date          | Design | Description                        |       |  |        |                   |                                               |  |        |         |       |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check |  | Apprv. |                   |                                               |  |        |         | Inoue |

- (6) If the [Connect Result] is "OK," the machine information is displayed in the machine setting list ([Model] / [CNC Paths] / [Servo Axes] / [Spindles] / [Laser Path] / [Number of Signals]).

## NOTE

When the selected machine type in the [Machine Type] column and the acquired machine type are different from each other, the column is set to as follows:

- If "FanucRobodrill" is selected and the acquired machine type is "FanucCNC," the column is set to "FanucCNC"
- If "FanucCNC" is selected and the acquired machine type is "FanucRobodrill," the column is left as "FanucCNC"

By clicking the [Detail] button in the [Path information] column, the [Details of Path Information] window is displayed, and the path information can be confirmed. For the details, refer to the "[Details of Path Information Window](#)" subsection.

By clicking the [Detail] button in the [Signal Setting] column, the [Details of Signal Setting] window is displayed, and the signal setting can be confirmed/changed. For the details, refer to the "[Details of Signal Setting Window](#)" subsection.

- \* When the [Detail] button is clicked for the machine whose machine information have not been acquired, a message saying "Path information / Signal information is not specified for this machine. Acquire machine information." is displayed, and the [Details of Path Information] / [Details of Signal Setting] window will not be displayed.

For the machines with "NG" displayed in the [Connect Result] column, select the [Log] tab to check the results of the acquisition process and the details of the error, and review or delete the settings.

| Setting Log            |                                     |              |                |                        |       |             |                |           |           |           |            |          |            |                |
|------------------------|-------------------------------------|--------------|----------------|------------------------|-------|-------------|----------------|-----------|-----------|-----------|------------|----------|------------|----------------|
| Import machine setting |                                     |              |                |                        |       |             |                |           |           |           |            |          |            |                |
| No                     | Target                              | Machine Name | Machine Type   | IP Address / Host Name | Port  | Access Code | Connect Result | Model     | Path Info | CNC Paths | Servo Axes | Spindles | Laser Path | Signal Setting |
| 1                      | <input checked="" type="checkbox"/> | Machine001   | FanucCNC       | 192.168.204.101        | 8193  | *****       | OK             | Series30B | Detail    | 1         | 3          | 1        | -          | Detail         |
| 2                      | <input checked="" type="checkbox"/> | Machine002   | FanucCNC       | 192.168.204.102        | 8193  |             | OK             | Series30B | Detail    | 1         | 3          | 1        | -          | Detail         |
| 3                      | <input type="checkbox"/>            | Machine003   | FanucCNC       | 192.168.204.103        | 8193  |             | -              | -         | Detail    | 1         | 0          | 0        | -          | Detail         |
| 4                      | <input type="checkbox"/>            | Machine004   | OPCUAServer    | -                      | -     |             | -              | -         | Detail    | -         | -          | -        | -          | Detail         |
| 5                      | <input checked="" type="checkbox"/> | Machine005   | FanucRobot     | 192.168.204.105        | 60008 |             | NG             | -         | Detail    | -         | -          | -        | -          | Detail         |
| 6                      | <input type="checkbox"/>            | Machine006   | MTConnectAgent | 192.168.204.106        | 5000  |             | -              | -         | Detail    | -         | -          | -        | -          | Detail         |
| 7                      | <input type="checkbox"/>            | Machine007   | FanucRobodrill | 192.168.204.107        | 8193  |             | -              | -         | Detail    | 1         | 3          | 1        | -          | Detail         |

Results of Machine Information Acquisition

|      |               |        |                                    |                   |                                               |                |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   | Draw              | A-40829-00001EN                               |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   |                   |                                               |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   | FANUC CORPORATION |                                               | Sheet. 862/868 |
| Ed.  | Date          | Design | Description                        |                   |                                               |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check             | Apprv.                                        | Inoue          |

Shown below are log examples of machine information acquisition.

Machine type FanucCNC: when [Connect Result] is "OK"

Check Process Start at 2018/08/08 8:08:08.

---

Check Machine : Machine001  
Machine Type : FanucCNC  
IP Address / Host Name : 192.168.0.101  
Port No : 8193  
Network type ... TCP/IP  
Ping executing [192.168.0.101] ... OK  
Connected to the CNC ... OK

Check Connection with Machine

< Machine Information >  
-- CNC Model = Series30iB  
-- Max Path = 1  
-- Laser Paths = 1/3  
-- O Number Digits = 4  
-- Registration Program = 74

< Path1 Information >  
-- CNC Type = M  
-- Axis = 3  
-- Spindle = 1

Acquisition of Machine Information

-----

Check Machine : Machine002  
...

-----

Check Process finished.

Machine type FanucCNC: when [Connect Result] is "NG"

Check Process Start at 2018/08/08 8:08:08.

---

Check Machine : Machine001  
Machine Type : Unknown  
IP Address / Host Name : 192.168.0.101  
Port No : 8193  
Ping executing [192.168.2.1] ... OK  
Connected to the CNC ... **NG**  
Investigate CNC power supply, Ethernet cable and I/F board.[cnc\_allclibhndl3(192.168.2.1, 8193, 30, ...)]  
(err=-16)(EW\_SOCKET)

Check Connection with Machine (ERROR)

-----

Check Process finished.

|      |               |        |                                            |                   |                                                       |                |
|------|---------------|--------|--------------------------------------------|-------------------|-------------------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                   |                                                       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                   |                                                       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |                   |                                                       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   | Draw              | <b>A-40829-00001EN</b>                                |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                   |                                                       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |                   |                                                       |                |
| Ed.  | Date          | Design | Description                                | FANUC CORPORATION |                                                       | Sheet. 863/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check             | Apprv.                                                | Inoue          |

Machine type FanucRobot: when [Connect Result] is "OK"

Check Process Start at 2021/10/05 9:09:09.

```
Check Machine : Machine002
Machine Type : FanucRobot
IP Address / Host Name : 192.168.0.102
Port No : 60008
Network type ... TCP/IP
Ping executing [192.168.0.102] ... OK
Connected to the Machine ... OK
```

### Check Connection with Machine

```
< Machine Information >
-- Connected Robot = R-30iB
-- Connected Robot Version = V9.10204
...
```

## Acquisition of Machine Information

Check Process finished.

|      |               |        |                                            |       |  |        |       |                   |                                                       |  |        |         |
|------|---------------|--------|--------------------------------------------|-------|--|--------|-------|-------------------|-------------------------------------------------------|--|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   |       |  |        |       | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |  |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |       |  |        |       |                   |                                                       |  |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |       |  |        |       |                   |                                                       |  |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |       |  |        |       | Draw              | A-40829-00001EN                                       |  |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |       |  |        |       |                   |                                                       |  |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |       |  |        |       |                   |                                                       |  |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |       |  |        |       | FANUC CORPORATION |                                                       |  | Sheet. | 864/868 |
| Ed.  | Date          | Design | Description                                |       |  |        |       |                   |                                                       |  |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check |  | Apprv. | Inoue |                   |                                                       |  |        |         |

## 15.3.8 Check Signal

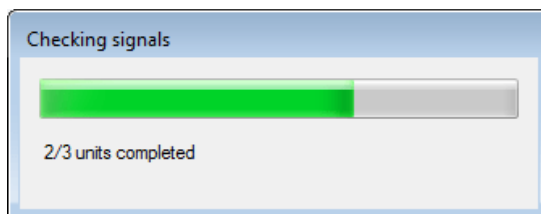
This subsection explains the Signal Check process.

For the machines which have acquired machine information, the Signal Check process enables to check whether the signals set as enabled in the signal setting can be acquired or not.

- (1) Make sure that the [Target] check box of the machine targeted for signal check is selected, and click the [Start Signal Check] button.

- \* If the machine which has not acquired the machine information is targeted, Signal Check cannot be executed.
- \* Machine types OPCUAServer / MTConnectAgent are not supported. Signal Check will not be executed even when the [Target] check box is selected.

- (2) During the Signal Check process, a progress bar indicating the status (<number of completion> / <number of targets>) is displayed.



- (3) When the Signal Check process is finished, the tab is switched to the [Log] tab on the [Signal Check Tool] window.

- (4) The log displayed for the Signal Check process is output in text format for each Signal Check process executed.

Following are the output folder and filename.

Folder name: <MT-LINK*i* installation folder>\SignalCheckTool\Log\  
 Filename: SignalCheck\_<yyyyMMddHHmmss>.log  
 (<yyyyMMddHHmmss> is the date/time of file output.)

- (5) Click the [Setting] tab to return to the [Setting] tab.

The results of signal acquisition are displayed in the [Number of Signals] / [Acquisition Time (ms)] column of the machine setting list.

- (6) The check results of the signal setting are displayed in the [Details of Signal Setting] window. Click the [Detail] button in the [Signal Setting] column of the machine which you want to check the results. On the [Details of Signal Setting] window displayed, check the display in the [Check] column on the tab of each signal setting.

For the details, refer to the "[Details of Signal Setting Window](#)" subsection.

|      |               |        |                                            |                          |                                                       |                |
|------|---------------|--------|--------------------------------------------|--------------------------|-------------------------------------------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title                    | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                          |                                                       |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                          |                                                       |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   |                          |                                                       |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   | Draw                     | <b>A-40829-00001EN</b>                                |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                          |                                                       |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   |                          |                                                       |                |
| Ed.  | Date          | Design | Description                                | <b>FANUC CORPORATION</b> |                                                       | Sheet. 865/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check                    | Apprv.                                                | Inoue          |

Shown below are log examples of Signal Check.

Log example: when machine type is FanucCNC ("NG" stands for "not good")

Check Process Start at 2020/09/10 11:12:13.

Check Machine : Machine001  
Machine Type : FanucCNC  
IP Address / Host Name : 192.168.0.101  
Port No : 8193  
Network type ... TCP/IP  
Ping executing [192.168.0.101] ... OK  
Connected to the CNC ... OK

Information of  
Connection  
with Machine

< Machine Information >  
-- CNC Model = Series30iB  
-- Max Path = 1  
-- Laser Paths = 1/3  
-- O Number Digits = 4  
-- Registration Program = 74

Machine Information

< Path1 Information >  
-- CNC Type = M  
-- Axis = 3  
-- Spindle = 1

Basic Signal,  
Custom Signal,  
Tool Life Signal,  
Laser Control Data Signal

< Signal Setting >  
[0001] -- SigOP\_path1 ... OK (Value:False, Elapse:0 ms)  
[0002] -- SigSTL\_path1 ... OK (Value:False, Elapse:4 ms)  
[0003] -- SigSPL\_path1 ... OK (Value:False, Elapse:0 ms)  
[0004] -- SigAL\_path1 ... OK (Value:False, Elapse:0 ms)  
[0005] -- EMG\_path1 ... OK (Value:False, Elapse:0 ms)  
[0006] -- Mode\_path1 ... OK (Value:MEMORY, Elapse:2 ms)  
...  
[0032] -- CncState\_path1 ... **NG** (Code:6, Message:No applicable CNC option.  
[cnc\_rddiaginfo(32772)](err=6,0,0))  
[0033] -- CncWarning\_path1 ... **NG** (Message:An unexpected error occurred.)  
...  
[0042] -- Alarm ... OK (Elapse:0 ms)  
[0043] -- OpMacroMsg ... OK (Elapse:0 ms)  
[0044] -- PMC\_A1\_path1 ... OK (Value:0, Elapse:10 ms)  
[0045] -- Diagnosis\_365\_path1\_spindle0 ... OK (Value:1, Elapse:10 ms)  
...  
[0100] -- ToolGroupNum\_path1 ... OK (Value:1, Elapse:7 ms)  
[0101] -- ToolLifeCount01\_path1 ... OK (Value:10, Elapse:10 ms)  
[0102] -- ToolLifeCount02\_path1 ... OK (Value:30, Elapse:7 ms)  
[0103] -- ToolLifeCount03\_path1 ... OK (Value:14, Elapse:8 ms)  
...  
[0060] -- LaserSequence\_path1 ... OK (Value:3000, Elapse:0 ms)  
[0061] -- SigBeamOut\_path1 ... OK (Value:False, Elapse:0 ms)  
[0062] -- SigLaserAlarm\_path1 ... OK (Value:False, Elapse:0 ms)  
...

< Macro Variable History Setting >  
[0145] -- Log ID : ID0001 (Elapse:0 ms)  
-- Trigger Signal : #500\_path1 ... OK (Value:0.0)  
-- Capture Signal(1) : #501\_path1 ... OK (Value:0.0)

Macro Variable Getting  
Condition Definition

|      |               |        |                                    |                   |                                               |        |         |
|------|---------------|--------|------------------------------------|-------------------|-----------------------------------------------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title             | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |                   |                                               |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |                   |                                               |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |                   |                                               |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw              | <b>A-40829-00001EN</b>                        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |                   |                                               |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |                   |                                               |        |         |
| Ed.  | Date          | Design | Description                        | FANUC CORPORATION |                                               | Sheet. | 866/868 |
| Date | Nov.30,2015   | Desig. | Ogawa                              |                   |                                               |        |         |
|      |               | Check  |                                    | Apprv.            | Inoue                                         |        |         |

< Variable History Setting >  
 [0146] -- ID : ID0001 (Elapse:0 ms)  
 -- Trigger : MacroVar\_501\_path1 ... OK (Value:0.0)  
 -- Variable(1) : MacroVar\_1\_path1 ... OK (Value:0.0)

Variable History  
Retrieval Condition  
Definition

< check result >  
 -- NG (144/146)

Check Result  
per Machine

< total time >  
 500 ms

Total Acquisition Time  
per Machine

-----  
 Check Process finished.

#### Log example: when machine type is FanucRobot

Check Process Start at 2021/10/05 10:10:10.

Check Machine : Machine002  
 Machine Type : FanucRobot  
 IP Address / Host Name : 192.168.0.102  
 Port No : 60008  
 Network type ... TCP/IP  
 Ping executing [192.168.0.102] ... OK  
 Connected to the Machine ... OK

Information of  
Connection  
with Machine

< Machine Information >  
 -- Connected Robot = R-30iB  
 -- Connected Robot Version = V9.10204

Machine Information

< Signal Setting >  
 [0001] -- BATAL ... OK (Value:False, Elapse:0 ms)  
 [0002] -- Mode ... OK (Value:Auto, Elapse:0 ms)  
 ...  
 [0006] -- EMG ... OK (Value:False, Elapse:0 ms)  
 [0007] -- IO\_DO100 ... OK (Value:False, Elapse:0 ms)  
 [0008] -- DATA\_NumReg5 ... OK (Value:0.0, Elapse:0 ms)  
 [0009] -- DATA\_G2\_PosReg1\_X ... OK (Value:0.0, Elapse:0 ms)  
 ...

Basic Signal,  
Custom Signal

< Variable History Setting >  
 [0030] -- ID : ID0001 (Elapse:0 ms)  
 -- Trigger : IO\_SO1 ... OK (Value:False)  
 -- Variable(1) : IO\_SO8 ... OK (Value:True)  
 -- Variable(2) : DATA\_NumReg1 ... OK (Value:0.0)  
 -- Variable(3) : DATA\_G1\_PosReg2\_X ... OK (Value:0.0)

Variable History Retrieval  
Condition Definition

< check result >  
 -- OK (30/30)

Check Result  
per Machine

< total time >  
 500 ms

Total Acquisition Time  
per Machine

-----  
 Check Process finished.

|      |               |        |                                    |       |                                               |       |                   |                |
|------|---------------|--------|------------------------------------|-------|-----------------------------------------------|-------|-------------------|----------------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINKi Version 4.0   | Title | FANUC MT-LINKi<br>AdminTool OPERATOR'S MANUAL |       |                   |                |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINKi Version 3.11  |       |                                               |       |                   |                |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINKi Version 3.10  |       |                                               |       |                   |                |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINKi Version 3.9   |       |                                               |       |                   |                |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINKi Version 3.8   | Draw  | A-40829-00001EN                               |       |                   |                |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7.2 |       |                                               |       |                   |                |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINKi Version 3.7   |       |                                               |       |                   |                |
| Ed.  | Date          | Design | Description                        |       |                                               |       |                   |                |
| Date | Nov.30,2015   | Desig. | Ogawa                              | Check | Apprv.                                        | Inoue | FANUC CORPORATION | Sheet. 867/868 |

## 15.3.9 Machine Setting Export

This subsection explains the machine setting export process.

| No | Target                              | Machine Name | Machine Type   | IP Address / Host Name | Port  | Access Code | Connect Result | Model     | Path Info. | CNC Paths | Servo Axes | Spindles | Laser Path | Signal Setting | Number of Signals | Acquisition Time (ms) |
|----|-------------------------------------|--------------|----------------|------------------------|-------|-------------|----------------|-----------|------------|-----------|------------|----------|------------|----------------|-------------------|-----------------------|
| 1  | <input checked="" type="checkbox"/> | Machine001   | FanucCNC       | 192.168.204.101        | 8193  | *****       | OK             | Series30B | Detail     | 1         | 3          | 1        | -          | Detail         | 82/82             | 1010                  |
| 2  | <input checked="" type="checkbox"/> | Machine002   | FanucCNC       | 192.168.204.102        | 8193  |             | OK             | Series30B | Detail     | 1         | 3          | 1        | -          | Detail         | 82/82             | 1016                  |
| 3  | <input checked="" type="checkbox"/> | Machine003   | FanucCNC       | 192.168.204.103        | 8193  |             | -              | -         | Detail     | 1         | 0          | 0        | -          | Detail         | -/19              | -                     |
| 4  | <input checked="" type="checkbox"/> | Machine004   | OPCUAServer    | -                      | -     |             | -              | -         | Detail     | -         | -          | -        | -          | Detail         | -                 | -                     |
| 5  | <input checked="" type="checkbox"/> | Machine005   | FanucRobot     | 192.168.204.105        | 60008 |             | OK             | R-30B     | Detail     | -         | -          | -        | -          | Detail         | 6/6               | 2                     |
| 6  | <input checked="" type="checkbox"/> | Machine006   | MTConnectAgent | 192.168.204.106        | 5000  |             | -              | -         | Detail     | -         | -          | -        | -          | Detail         | -                 | -                     |
| 7  | <input checked="" type="checkbox"/> | Machine007   | FanucRobodril  | 192.168.204.107        | 8193  |             | -              | -         | Detail     | 1         | 3          | 1        | -          | Detail         | -/150             | -                     |

Buttons: Add, Delete, Acquire machine information, Start Signal Check, Export machine setting

- (1) Select the [Target] check box of the machine setting to export, and click the [Export machine setting] button.

[Save As] dialog box appears. Specify the storage folder and the filename, and click the [Save] button.

\* Following are the default folder and filename of the export destination.

Folder name: <MT-LINK*i* installation folder>\SignalCheckTool\Export\

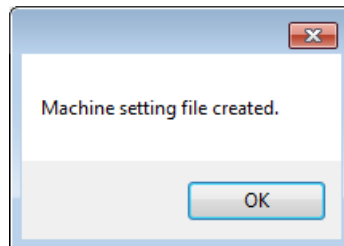
Filename: L0\_Setting\_<yyyyMMddHHmmss>.json

(<yyyyMMddHHmmss> is the date/time when the file is exported.)

### NOTE

- 1 Machine setting file is exported in the format of L0\_Setting.json.
- 2 Machine setting export process will not be executed if the machine selected as the target has not acquired machine information.

- (2) Following message is displayed when the machine setting export process is finished. Click the [OK] button to return to the [Signal Check Tool] window.



|      |               |        |                                            |                   |                                                       |        |        |         |
|------|---------------|--------|--------------------------------------------|-------------------|-------------------------------------------------------|--------|--------|---------|
| 20   | Nov. 22, 2023 | uwano  | Revised for MT-LINK <i>i</i> Version 4.0   | Title             | FANUC MT-LINK <i>i</i><br>AdminTool OPERATOR'S MANUAL |        |        |         |
| 19   | Apr. 27, 2021 | uwano  | Revised for MT-LINK <i>i</i> Version 3.11  |                   |                                                       |        |        |         |
| 18   | July 28, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.10  |                   |                                                       |        |        |         |
| 17   | Jan. 30, 2020 | uwano  | Revised for MT-LINK <i>i</i> Version 3.9   | Draw              | A-40829-00001EN                                       |        |        |         |
| 16   | July 22, 2019 | uwano  | Revised for MT-LINK <i>i</i> Version 3.8   |                   |                                                       |        |        |         |
| 15   | Mar. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7.2 |                   |                                                       |        |        |         |
| 14   | Jan. 29, 2019 | ogawa  | Revised for MT-LINK <i>i</i> Version 3.7   | FANUC CORPORATION |                                                       |        | Sheet. | 868/868 |
| Ed.  | Date          | Design | Description                                |                   |                                                       |        |        |         |
| Date | Nov.30,2015   | Desig. | Ogawa                                      | Check             |                                                       | Apprv. | Inoue  |         |